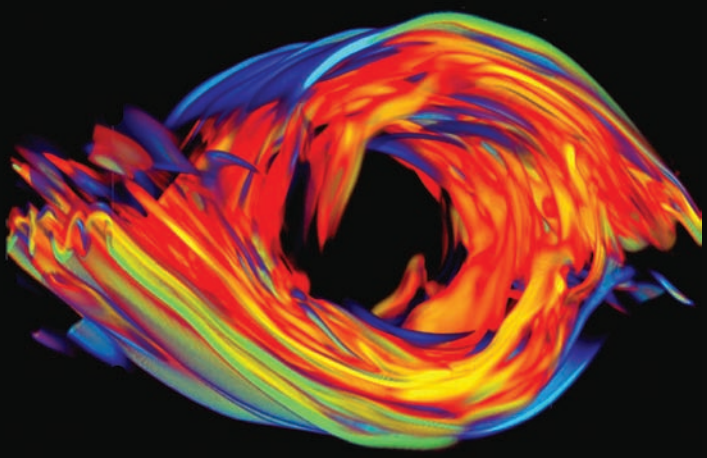
INTRODUCTION TO PLASMA DYNAMICS



A. I. Morozov

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Foreword

The enormous role played by the plasma in nature is now generally recognised, and along with it we have clearer understanding of its value now and in the future, especially for the daily practice of mankind. There is no doubt that the XXI century will be the century of not only computer science, biology and space exploration, but also the century of plasma technology. This will provide new sources of energy (thermonuclear fusion), methods of direct conversion (MHD generators and plasma thermocouples), effective means of ion-plasma processing of materials (in both microelectronics and heavy machinery), wave electromagnetic radiation sources (including 'eternal' household lamps with adjustable range and plasma TVs with the unlimited size of thin screens), space plasma thrusters, etc.

Parallel to the development of plasma technologies there have been intensive experimental and theoretical studies of the plasma state of matter and its hidden features. Such research not only accelerate technological development, but also can dramatically improve the understanding of natural phenomena, in particular the amazing processes in the magnetospheres of planets, stars and nebulae. The need for progress is evidenced by at least the fact that we still do not have a convincing model of the solar prominences and flares, which are so strongly influenced by the magnetosphere and many processes on the Earth.

Naturally, in view of this, it is necessary to write different books on plasma, both specialized and broadly covering this area of science. However, the familiarity with published books says that if narrowly themed books still appear on store shelves, the books on 'common plasma dynamics' are rare[1]. There is a demand for books written for graduate students, postgraduates and engineers. One of the first bricks to close this loophole should be, in my opinion, this 'Introduction to plasma dynamics'.

[1]Four introductory volumes of the Encyclopedia of Low-Temperature Plasma were published in Russia at the beginning of 2001. At the moment, this publication contains the largest amount of information on plasma but it is limited to the particle energy in the range 1–10 eV.

I would like to specify the features of this book.[2]

First of all, it is really an introduction. It should introduce the reader to the terms, the basic experimental facts and the main ‘basic’ models – the systems of equations that describe the broad areas of plasma dynamics processes. ‘Introduction ...’ uses specific examples to show how to use these models and, finally, to familiarize the reader with specific examples of modern plasma technology. Thus, we would like to, having mastered this "Introducing ...", to ensure that the reader has enough open-mindedness to start studying the literature.

What has been said is explained by the widespread lack of books on the subject of plasma and ignorance of the actual diversity of plasma technology and completely inadequate attention to two fundamental points: the creation of plasma and its loss on the walls of the working volume.

Indeed, in the laboratory conditions plasma is the ephemeral substance, limited in space and time. It is artificially created, passing the stage of a neutral substance (gas, liquid, solid) to a final state of the plasma with the required parameters. On the other hand, the plasma is surrounded by ‘plasma consuming’ walls. Naturally, the book on plasma dynamics should adequately reflect this specificity of plasma.

Although the above described general formulation by the author is unlikely to cause objections, however, there are clearly visible difficulties of the program in general. Here there are two main problems.

The first of them is the fact that the world of plasma processes has no natural physical boundaries. Once heating of the material and its ionisation start, it is hard to stop. It is natural to want to further increase the energy of the particles, increasing their density, lifetime, etc. As a result, from low-temperature plasma and cold walls we move to a system in which there is a plasma of medium to higher and higher energies, etc. to the black holes, which until recently (2000) were dealt with by astronomers only, and now their ‘germs’ are beginning to be established on the Earth in accelerators.

Thus, plasma physics is, in fact, unique. So, for example, solid state physics or electrodynamics can in many cases be studied without knowing anything about the plasma. But this does not work with the

[2]I believe the term ‘plasma dynamics’ is more suitable for the content of the book than the very general term ‘plasma physics’. This is explained by the fact that the book deals extensively with plasma flows. I also believe that the term ‘plasma electrodynamics’ is less suitable, although the book of H. Alfvén, the founder of modern plasma dynamics, had the title ‘Cosmic electrodynamics’. At the same time, H. Alfvén formally referred to the field (not corpuscular) plasma component and I believe this to be less accurate.

plasma did not work. In any experiment we have to deal with the interaction of the plasma with solid (liquid) bodies (electrodes, walls, probes) and the processes of radiation in a wide wavelength range, etc. Therefore, the general plasma dynamics should include more or less all sections of physics.

Thus, plasma is actually a link between different forms of matter from a dilute gas to extremely dense substances. It is not surprising that the analogues of plasma processes can be seen in other environments. Therefore, the terms like 'plasma-like environment' (semiconductors, metals, electrolytes) or 'quark–gluon plasma' have appeared, though in the latter case it is a qualitatively different substance of black holes.

Of course, some boundaries can be made between plasma processes and facilities, but they are conditional and temporary. This 'Introduction' will focus on the relatively rarefied, non-degenerate and non-relativistic plasma, with the emphasis in the applied sense placed on the 'laboratory' plasma dynamics, i.e. man-made plasma systems. These are discussed in detail in the final chapter of the book. As for 'cosmic" plasma dynamics', a few fragments associated with the Earth and the Sun, are discussed in Chapter 9.

As can be seen, the present book covers quite a wide range of issues on a fundamental level, but not all. To characterise the most pressing issues, it is necessary to prepare at least three more major reviews. Here is a brief description of them.

– Overview of plasma dynamics (PD) of essentially three-dimensional non-stationary processes. This must include questions about the stability of the equilibrium configurations and flows, turbulence, the dynamics of solar activity, etc.

– Overview of PD of dense ('quantum') plasma. It should include the dynamics of strongly coupled plasma, formation of the superdense states of laser targets, processes in white dwarfs, etc. Thus, this volume should be linked to plasma physics and the physics of solids.

– Review of relativistic PD. Relativism has long become a laboratory of PD. Of course, in 'classic' laboratories systems with relativistic electrons are studied. These are generators of high-current (mega-ampere) electron beams, and also a number of plasma systems (tokamaks, Z-pinches of different types) in which 'unwelcome' fluxes of relativistic electrons appear. However, the laboratories in which the processes with elementary particles with energies greater than 10^9 eV are studied, have been using for many years special storage devices for such particles (colliders) in which the particles collide with each other. Although the functions of the distribution of the relativistic ions

in the colliders are almost δ-shaped, nevertheless, here we are already concerned with ion relativistic plasma.

So the basis for the three volumes is already ready.

The second difficulty is the following. Confining ourselves to the ideal (in the kinetics sense) rarefied, non-relativistic plasma, we have not solved the problem of selecting the principles of systematization of the material. The usual courses of plasma physics as an analysis of the specific processes in arbitrarily chosen plasma configurations. But because the 'simple' plasma systems do not exist, then the attention of the authors is focused on a few systems, most often on homogeneous plasma (waves) or thermonuclear fusion systems (tokamaks, etc.). Clearly, for any general review of plasma phenomena (and technology) this way is unacceptable. The basis of the Introduction is the consideration of the principles of modelling or, in other words, the principles of the hierarchy of the 'basic' models and their relationships with one another, as well as a demonstration of their work.

So, we have a section dedicated to structures (morphology) of magnetic fields, one- and zero-dimensional (block) plasma models. Then we sequentially consider single-, two- and multi-component simulation models, and then the 'different kinetics' and ionisation processes, radiation transport, plasma interaction with the walls. Thus, we are able to show how to approach the analysis of rather complex plasma systems with a large number of degrees of freedom, and also take into account the great diversity of the environment in which it is located. Of course, each specific model, being a part of the hierarchy, has its own field of activity and its own characteristics. We illustrate these features, mainly on the example of the three types of plasma processes: static configurations, linear oscillations and stationary flows, such as shock waves, i.e. consider zero-dimensional, one-dimensional, two-dimensional ((x_1, x_2) or (x, t)) and three-dimensional (x_1, x_2, t) models. Where there are particularly interesting situations that are outside of this triad, we describe them as well.

At the end of the book, in chapter 8, we examine several general problems, associated with oscillations in PD systems. As already mentioned, in chapter 9 we discuss a number of cosmic PD systems, and chapter 10 is concerned with the examples of advanced plasma technologies.

The wide range of the subjects investigated in the book may result in a number of inaccuracies. Unfortunately, because of subjective and objective circumstances I could not have avoided them. Therefore, I would be grateful to the readers for their understanding.

In the book, I present the portraits of the scientists who contributed greatly to the discussed areas of science and technology. Of course, I could not have mentioned all of them. Therefore, the selection of the portraits is partly subjective. I hope that others will do this better.

Finally, the number of publications on plasma physics and related disciplines is very large. I have tried to mention easily available sources that are easy to read (for example encyclopedias) and early, fundamental publications.

In a sense, the prototype structure of the Introduction is my book, 'The physical basis of cosmic electroreactive thrusters', volume 1, 'Elements of flow dynamics in the ERT' (1978). Further improvement of this structure is based on lectures on plasma dynamics that I presented in a special course for about 20 years at the Physics Department of the Moscow State University. The then head of the Department of Mathematics, Prof. A.G. Sveshnikov, who invited me to read this special course, advised me to prepare and publish an expanded version of the lectures in the book form. I am very grateful to him for that. The appearance of this book owes much also to my old friend, from my school days, Nikolai Ivanovich Dolbin, one of the founders of a new section of mechanics – magnetoelasticity. Nikolai wrote and edited many of my lectures. His untimely death in 1995 deprived me not only of a friend but also of an assistant at the final stage of the Introduction. But the memory of his enthusiasm and his work were very important for me at the end of this book.

Finally, I am grateful to my colleagues from the Kurchatov Institute, M.V. Keldysh MIREA Research Centre, and also colleagues and friends from abroad who carried out many of the experiments described in this book.

Introduction

I.1. What is plasma?

Plasma is an ionised gas. However, not every cloud of ionised gas is plasma. The main special feature of the plasma state can be explained by the following hypothetical experiment (Fig. I.1.1).

It is assumed that a photon with the energy $\hbar\omega$ falls on the single stationary atom with ionisation energy I and ionises the atom (Fig. I.1a). As a result, an electron with the energy

$$\varepsilon_1^{(e)} = h\omega - I \tag{I.1.1}$$

separates from the resultant almost stationary ion. We now consider two identical atoms, situated at a certain distance a from each other and irradiated consecutively with a small delay one atom after the other (Fig. I.1b). From the first atom, the electron is emitted as previously with the energy $\varepsilon_1^{(e)}$ but the second atom leaves with a lower energy

$$\varepsilon_2^{(e)} = \varepsilon_1^{(e)} - (\delta\varepsilon)_1 \tag{I.1.2}$$

because this atom must overcome attraction to the first ion.

Evidently

$$(\delta\varepsilon)_1 \sim \frac{e^2}{a} \tag{I.1.3}$$

If we now consider three atoms at distances ~a, then in consecutive ionisation, the last (third) electron leaves with the energy

$$\varepsilon_3^{(e)} = \varepsilon_1^{(e)} - (\delta\varepsilon)_2$$

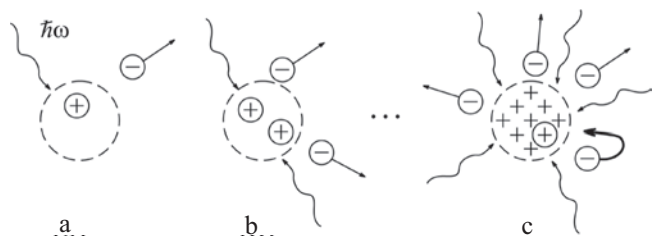


Fig. I.1.1. Transition from the set of ions and electrons to plasma.

and

$(\delta\varepsilon^{(e)})_2 \sim 2(\delta\varepsilon^{(e)})_1$ and so on

Evidently, at some number of atoms N_* (we assume that the atoms occupy some spherical volume with radius a), attraction to this volume of the ions will be such that the last electron can no longer leave to infinity (Fig. I.1.1c), since the kinetic energy of this electron at the moment of ionisation is not sufficient for overcoming attraction to the already formed ions, i.e.

$$\varepsilon_{N_*}^{(e)} < 0 \tag{I.1.4}$$

Increasing the number of atoms further and repeating the procedure of consecutive ionisation, it can be seen that at the given ionisation energy I and photon energy $\hbar\omega$ up to $N = N_*$, it is not possible to increase the charge of the cloud, and the increase of the total number of the charged ions N_i in the cloud will result in a decrease of the fraction of non-compensated charges

$$\delta \equiv \frac{N_i - N_e}{N_i} \sim \frac{N_*}{N_i} \to 0 \tag{I.1.5}$$

Thus, the cloud of charged particles becomes 'quasi-neutral', i.e. N_e becomes almost equal to N_i. This quasi-neutral gas is also referred to as plasma (Langmuir, Tonks, 1923).

The strength of bonding of the electrons and ions in plasma can be evaluated on the basis of the following example. Let the volume of air in 1 cm^3 in the form of a sphere under normal conditions be fully ionised. Consequently, the volume will contain $N \sim 5\cdot10^{19}$ ions and the same number of electrons. It is assumed that somebody (or something) is capable of taking all the ions in one 'hand' and all of the electrons in the other hand and we shall try to separate them from each other. Evidently, the maximum force of interaction of positive and negative 'spheres' is found at the distance between the spheres of $r \approx a$, where $a \sim 0.5$ cm is the radius of the spheres. This interaction force is determined by the Coulomb law:

$$F \approx \frac{N^2 e^2}{a^2} \tag{I.1.6}$$

Substituting into this equation $N = 5\cdot10^{19}$ cm^{-3}, $e = 4.8\cdot10^{-10}$ CGS (ESU unit), we obtain the gigantic force

$F \approx 2.5\times10^{12}$ tonnes

i.e. 2.5 trillion tonnes! However, the ionisation of this amount of gas required only ~100 J, and its mass is ~1 mg.

This unity of giant and tiny and does not contain any information on the huge possibilities offered by plasma.

Debye radius

The criterion of quasi-neutrality may be given a more constructive form. For this purpose, we estimate critical density n_*, assuming that N_* heavy particles are situated inside a sphere with radius a. In this case, the potential on the surface of the sphere will be:

$$\phi_* = \frac{N_* e}{a} \tag{I.1.7}$$

The last electron, which can travel to infinity, has the initial energy ε_1, and is determined by the condition

$$e\phi_* = \varepsilon_1 \tag{I.1.8a}$$

Or

$$\frac{N_* e^2}{a} = \varepsilon_1 \tag{I.1.8b}$$

Introducing the critical density of electrons n_*

$$N_* = \frac{4\pi}{3} a^3 n_*$$

equation (I.1.7) can be written in the form

$$\frac{\varepsilon_1}{4\pi e^2 n_*} = \theta a^2, \ \theta = \frac{1}{3} \tag{I.1.9}$$

The term on the left side of the equation has the dimension of the square of length. At any n and ε, this length is equal to

$$r_D \equiv \sqrt{\frac{\varepsilon}{4\pi e^2 n}} \tag{I.1.10}$$

and is referred to as the Debye radius.

Therefore, it may be assumed that the critical density at which the transition of the charged cloud to a quasi-neutral formation, i.e. to plasma, starts is the density at which the Debye radius becomes smaller than the radius of the cloud of the charged particles. Correspondingly, the cloud of the ionised gas becomes the ‘actual plasma’ is

$$r_D \equiv \sqrt{\frac{\varepsilon}{4\pi e^2 n}} \ll a \qquad \text{(I.1.11)}$$

where n and a are the characteristic values of the density and size of the cloud.

We estimate the value of n_* for a sphere with radius a = 1 cm at the electron energy ε = 1 eV (1 eV = $1.6 \cdot 10^{-12}$ erg). Taking into account equation (I.1.8) we obtain

$$n_* = \frac{3\varepsilon}{4\pi e^2 a^2} \approx 1.7 \times 10^6 \,\mathrm{cm}^{-3}$$

Two comments should be made here.

a. As indicated by the above considerations, the criteria of the plasma state (I.1.5) or (I.1.11) are also valid in the case in which we examine a cloud in which there are also neutral atoms and molecules, in addition to the charged particles. They simply do not have any role. However, the degree of ionisation is indicated by the terms: 'fully ionised plasma' and 'incompletely ionised plasma'.
b. The quasi-neutral medium is referred normally to as plasma, if it is gaseous. However, there are quasi-neutral media which are not gaseous. They include solid (metals, semiconductors) or liquid (melts, electrolytes) conductors with moving electrically charged particles. They are now referred to as plasma-like media.

Plasma is often referred to as the fourth state of matter. This is justified, but the equation (I.1.11) shows that the transition to the plasma state is determined not only by the properties of the substance (n, ε) but also by external factors – the size of the system a. By this plasma differs at first sight from the classic states of matter: solid, liquid, gaseous, and the transitions between the states do not appear to depend on the macrodimensions of the specimen. However, in the case of small specimens, the phase transition temperature is also not constant. For example, the melting point of tin is T_m = 505 K but if the size of a piece of tin is reduced 10 nm, T_m decreases to 480 K.

This can also be said of other phase transitions of the first kind. It is evident that the important role of the macrodimensions of the plasma is explained in the first instance by the long-range character of Coulomb interactions.

In some cases it is suggested that the plasma is not the fourth state of matter because of the large size of the temperature range within which the transition from the 'neutral' to 'completely ionised gas' takes place. However, this objection cannot be regarded as important

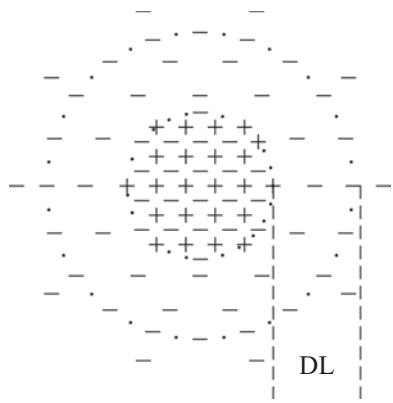


Fig. I.1.2. Distribution of charges in a plasma cloud, dispersed in vacuum, DL is the Debye layer.

because the criterion of the plasma state does not include the degree of ionisation.

Debye shells of 'free' plasma formations in the absence of the magnetic field[1] (**H** = 0)

Plasma formations – plasma blobs in vacuum, in a dense atmosphere, in vessels, usually have a 'shell' in which the quasi neutrality is violated. These layers will be referred as 'Debye layers'. They are often referred to as 'double' layers or 'Langmuir layers'.

To understand the nature of formation of these layers, it is assumed that a plasma cloud is located in vacuum. The cloud can be produced by, for example, evaporation of a piece of solid hydrogen with its subsequent ionisation as a result of radiation from all sides by laser radiation. What takes place subsequently? Assuming that the random energy of the ions is of the order or lower than the electron energy, it can be seen that the electrons, as the particles with higher mobility, start to leave the cloud and rapidly produce a deficit of electrons of the order N_* in the surface layer of the plasma volume. This deficit is determined by the equations (I.1.7) and (I.1.8) and further departure of the electrons is interrupted (Fig. I.1.2). Finally, the electrons will continue to be emitted but they will be pulled into the cloud by its positive charge, and only the faster electrons which form in a considerably smaller number of cases (as a result of collisions) will leave the volume.

Consequently, an almost vacuum field will form at a large distance from the cloud ($r \gg r_D$):

$$\phi \sim \frac{eN_*}{r}$$

[1]In the presence of magnetic fields, the situation is greatly complicated and this will be discussed in section 3.5.

In the area around the surface, within the limits of the layer with the thickness

$$r_D = \sqrt{\frac{\varepsilon}{4\pi e^2 n}}$$

there is a layer of non-compensated ions – this layer may also be referred to as the Debye layer, and in the depth of the cloud the plasma will be simply neutral.

Further evolution of the cloud, as shown in chapter 3, is identical with the expansion of a gas sphere with temperature $T_{ef} = T_i/\overline{z}_i + T_e$, where $\overline{z}_i$ is the mean ion charge.

The Debye layer at dielectric walls

It is well known that a discharge burns in the so-called daylight lamps and its main volume is occupied by the slightly ionised plasma formed in this case (for more details see section 10.2).

Its electrons have a temperature of ~*2* eV and rapidly move in all directions, including the direction to the wall. At the same time, because of collisions with neutral atoms, the ion temperature is close to room temperature and the velocity of the ions is more than four orders of magnitude lower than that of electrons. Consequently, the internal surface of the tube is rapidly negatively charged. At the same time, the velocity of travel of the electrons to the wall rapidly decreases down to the velocity of travel of the ions (Fig. I.1.3).

Thus, a regime is established in which:

$$j_n^{(i)} = j_n^{(e)}(1-\sigma) \qquad \text{(I.1.12)}$$

Here j_n is the density of the flux reaching the wall, and σ is the coefficient of secondary electronic emission of the wall, i.e. the number of electrons emitted by the wall under the effect of a single incident

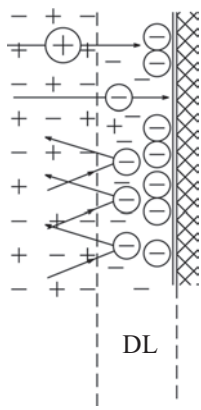


Fig.I.1.3. The Debye layer on a dielectric wall.

electron. The ions and electrons arriving at the wall recombine with each other. In (I.1.12) it is assumed that $\sigma < 1$. The case in which $\sigma > 1$ is investigated in chapter 7.

The field of the electrons 'sitting' on the wall penetrates into the plasma volume again to the thickness $\sim r_D$. This can be explained approximately as follows. The Maxwell equation for the electrical field around a flat wall has the form:

$$\frac{d^2\phi}{dx^2} = -4\pi e(n_i - n_e) \tag{I.1.13}$$

It is assumed that the electrons are described by the Boltzmann distribution

$$n_e = n_0 \exp\left\{\frac{e\phi}{kT_e}\right\}$$

and we restrict ourselves to calculations of only the long-range zone where it may be assumed that $\phi \ll kT_e$, and the ion density is constant. Consequently

$$n_i = n_0,\ n_e \approx n_0\left(1 + \frac{e\phi}{kT_e}\right)$$

and equation (I.1.13) is simplified to the maximum extent

$$\frac{d^2\phi}{dx^2} = \frac{4\pi e^2 n_0}{kT_e}\phi \tag{I.1.14a}$$

Consequently, the electrical field, generated by the wall, rapidly decreases in intensity with increase of the distance from the wall

$$\phi = \phi_0 \exp\left\{-\frac{x}{r_D}\right\},\quad r_D = \sqrt{\frac{kT_e}{4\pi e^2 n_0}} \tag{I.1.14b}$$

Therefore, in many cases, the Debye radius is regarded as the screening radius. In the above considerations, the daylight lamp acted as a suitable example. The resultant equation (I.1.14b) is valid always if $\sigma < 1$, and the wall is insulated.

Langmuir frequency

In addition to the Debye radius, another characteristic parameter of plasma is the so-called Langmuir or plasma frequency[2]. The simplest model which makes it possible to derive an equation for this frequency is constructed using the following procedure (Fig. I.1.4). We take a

[2]In fact, there may be large differences between these two terms, but they are often regarded as synonymous.

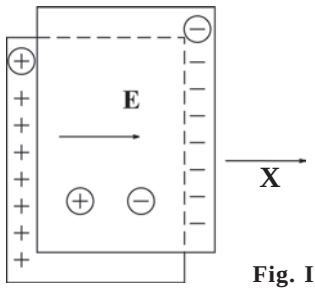


Fig. I.1.4. Block model of Langmuir oscillations.

plasma layer and treat it as a set of two non-deformed layers–blocks, electronic and ionic. If the electronic block is moved by x_e, and the ionic block by x_i, charges with the density $\pm q$ appear on the sides of the plasma sheet, where

$$q = en_0(x_i - x_e)$$

These charges generate an electrical field with the strength

$$E = -4\pi en_0(x_i - x_e) \qquad \text{(I.1.15a)}$$

in the volume.

It is assumed that the thicknesses of the projecting part of the electronic and ionic layers are negligibly small. The appearance of the field (I.1.15a) in the quasi-neutral volume of the layer results in the oscillation of the electronic and ionic blocks in relation to each other. Evidently, they are described by the equations (related to 1 cm^2 of the surface area of the layer)

$$(m_e n_0 l)\frac{d^2 x_e}{dt^2} = (en_0 l)4\pi en_0(x_i - x_e) \qquad \text{(I.1.15b)}$$

$$(Mn_0 l)\frac{d^2 x_i}{dt^2} = -(en_0 l)4\pi en_0(x_i - x_e). \qquad \text{(I.1.15c)}$$

Here l is the thickness of the layer, m is the electron mass, M is the ion mass. Shortening in these equations $n_0 l$ and deducting the second equation from the first one gives

$$\ddot{\xi} = -\omega_0^2 \xi,$$

where $\xi = x_i - x_e$, and

$$\omega_0^2 = 4\pi e^2 n_0\left(\frac{1}{m} + \frac{1}{M}\right). \qquad \text{(I.1.16)}$$

The frequencies ω_{0e} and ω_{0i} determined by the equations

$$\omega_{0e}^2 = \frac{4\pi e^2 n_0}{m},\ \omega_{0i}^2 = \frac{4\pi e^2 n_0}{M}, \tag{I.1.17}$$

are referred to as the electronic and ionic Langmuir frequencies, respectively, and the frequency

$$\omega_0 = \sqrt{\omega_{0e}^2 + \omega_{0i}^2} \tag{I.1.18}$$

is the plasma frequency. Evidently, with a high accuracy $\omega_0 \approx \omega_{0e}$.

We estimate the scale of ω_{0e} using the density $n = 10^{14}$ cm^{-3} characteristic of future thermonuclear reactors. Substituting into (I.1.17) we obtain

$$\omega_{0e} \approx 5 \cdot 10^4 \sqrt{n_0} \approx 5 \cdot 10^{11} \text{s}^{-1}.$$

Evidently, the electronic block oscillates around the almost stationary ionic block. The amplitudes of these oscillations relate to each other as M/m.

If the amplitude of the velocity of the electronic block is V_{em}, than because of the harmonic form of the oscillations its maximum displacement is

$$D \approx \frac{V_{em}}{\omega_{0e}} = \sqrt{\frac{mV_{em}^2}{4\pi e^2 n_0}}. \tag{I.1.19}$$

This value, with the accuracy to the multiplier ~1, coincides with the equation for the Debye radius (I.1.10).

I.2. Region of rarefied non-relativistic plasma in the coordinates *n*, *T*

We now examine how large the 'ocean' of plasmas in which we are interested is and the 'shores' which it sweeps. This area is defined by three conditions.

1. As already mentioned in the preface, in this book we examine only rarefied plasmas[3] in the sense that the kinetic energy of their particles $(3/2)kT$ is considerably higher than the potential energy of the pair of the particles at the mean distance $r_0 \sim n^{-1/3}$

$$\frac{e^2}{r_o} \ll \frac{3}{2} kT. \tag{I.2.1a}$$

[3]We shall use this term instead of the widely used term 'ideal' gas ('ideal plasma'), and the word 'ideal' will be used as the synonym of dissipation-free synonym.

If we introduce the Debye radius in the form (I.1.15b), the condition (I.2.1a) has the form

$$N_D \gg \frac{1}{20}, N_D = \frac{4\pi}{3} r_D^3 n_0; \; r_D = \sqrt{\frac{kT}{4\pi e^2 n_0}} \qquad \text{(I. 2.1b)}$$

where N_D is the number of particles in the Debye sphere. For example, at $n_0 = 10^{13}$ cm^{-3} and $kT = 10^4$ eV, we have $N_D \sim 3 \cdot 10^8$. It should be stressed that the number of particles in the Debye sphere N_D is proportional to $n_0^{-1/2}$, i.e. increases with the decrease of plasma density n_0.

2. Both the electronic and ionic components are assumed to be non-relativistic (c is the speed of light)

$$kT_e \ll mc^2, \; kT_i \ll M_i c^2 \qquad \text{(I.2.2)}$$

For the electrons this restriction means that the temperature in the case of, for example, Maxwell's distribution, in a tokamak should be considerably lower than 500 keV and, in particular, as shown by the calculations of radiation, should be lower than 50 keV.

3. Finally, it is assumed that the plasma is of the non-quantum nature, i.e. the mean distance between the particles is

$$r_0 \approx n_0^{-1/3} \gg \lambda_B \qquad \text{(I.2.3)}$$

where $\lambda_B = h/mv$ is the de Broglie wavelength.

Regardless of these restrictions, the parameters of the investigated plasma change in a very wide range. Information on this is provided by Fig. I.2.1, in which the graph in the coordinates T (characteristic substance temperature) and n (concentration of the substance particles) indicates the regions related to the rarefied classic plasma (it is crosshatched) and the adjacent media[4].

We characterise briefly the regions shown in Fig.I.2.1, moving from bottom to top. It can be seen immediately that the boundaries between the regions are conditional and depend on the specific properties of the matter and external conditions. The region I ($T \lesssim 0.2$ eV) is the region of neutral (non-ionised) gases and solids. For every specific substance the region is 'spotted', because the density of the saturated vapours usually differs by an order of magnitude from the density of the condensed liquid or solid. It should be mentioned that the entire set of the technologies of mankind from stone hammer to space technology originated from this region.

[4]After all, any plasma formation is characterised by a large number of parameters. However, these parameters cannot be listed in a convenient manner and, therefore, we must restrict ourselves to two coordinates.

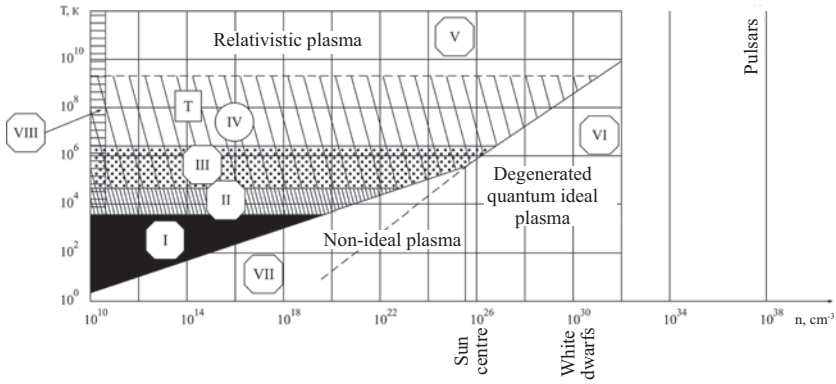


Fig. I.2.1. Diagram of the regions of rarefied classic plasma and of surrounding media in the temperature–density coordinates.

In this region, the energy of the particles is limited by the level of the energy of chemical bonds, and density by the density of the solid (n_{max} ~10^{23} cm^{-3}).

Above this region, there is region II – 'low-temperature plasma' ($T_i \sim T_e \lesssim 5$ eV) with the highest population in the temperature range ~0.3–0.6 eV and the degree of ionisation is not high (from fractions to units of percent). This region includes the plasma of the classic discharges (arc, glow, etc) and also of the surface layers of the Sun. When the density increases to 10^{18}–10^{19} cm^{-3}, the plasma becomes 'dense', i.e. the condition (I.2.1a) is violated and we transfer to the region VII.

This is followed by the region III ($5 \lesssim T \lesssim 1000$ eV). This is the region of completely ionised plasma with 'mean' energy. In specific cases, it is necessary to separate the two sub-regions. They will be referred to as III_z and III_0. Region III_0 is the region in which all the ions have lost their electronic shells so that they represent Coulomb centres. The simplest example are hydrogen ions. However, if the charge of the nucleus is $Z \gg 1$, inner shells are retained in most cases in the ions. They form the region III_z. Therefore, the Coulomb interaction between these ions takes place here only during their interactions at relatively large distances. At short distances, the nature of interaction becomes more complicated (chapter 6). The boundary between the regions III_z and III_0 depends on the type of substance. With the increase of density, especially in the case of relatively high values of z, we also arrive in the region of 'dense' plasma.

The next region is region IV ($10^2 \lesssim T \lesssim 5 \cdot 10^4$ eV). This is the region of 'hot' plasma. In the laboratory conditions, this plasma can be produced in the stationary form only on hydrogen and light elements

(He, Li) as a result of powerful bremsstrahlung. The plasma of the light elements in this energy range is of considerable interest for the problem of controlled thermonuclear fusion (see section 10.5). It should be mentioned that the temperature in the centre of the Sun is of the order of 1000 eV, but the density (mostly hydrogen) is of the order of 100 g/cm^3, i.e., $n_0 \sim 7 \cdot 10^{25}$ cm^{-3}.

Boundary regions will now be briefly discussed. Region V is the region of relativistic plasma. Initially, the electrons become relativistic here. This plasma in the presence of the magnetic field generates synchrotron radiation due to centripetal acceleration.

In the cosmic conditions, the synchrotron mechanism is responsible for radio emission of the stars, including pulsars. If the electron-relativistic plasma formation is large, and gamma radiation is suppressed to a large degree, then at $T_e \sim 1$ MeV we observe the birth of electron–positron pairs. Evidently, this atmosphere exists in the pulsars (neutron stars).

At the present time, the stationary formations, which can be regarded as ion-relativistic plasma, are produced in laboratories. In fact, as already mentioned in the preface, unique 'traps' are used in the technology of accelerators of charged particles, i.e. storage rings into which the particles, accelerated to high energies, are introduced. If particles and antiparticles are injected into such a ring with a transverse magnetic field, they will move against each other. Consequently, it is possible to examine the collisions of particles at ultra-relativistic energies. These storage rings are referred to as colliders. Evidently, a quasi-neutral ion–relativistic medium is produced in such a single-ring collider at equal concentrations of the particles and antiparticles. In particular, this assumption is valid if we use a collider consisting of two storage rings with ions of the same sign. Consequently, the density of the protons in the zone of intersection of the fluxes of the protons moving in the opposite directions (with the energy of up to $\sim 10^{12}$ eV) reaches 10^8 cm^{-3}.

This zone is a unique volume of the non-isothermal (i.e. $T_e \ll \varepsilon_i$) plasma with a highly anisotropic function of the distribution of the ions.

The region VI is the plasma with the quantum degeneration of the electrons. In other words, here in contrast to (I.2.3) the distance between the particles of the order of the de Broglie wavelength is:

$$r_0 \sim \frac{1}{n_0^{1/3}} \sim \lambda_B. \qquad \text{(I.2.4)}$$

Therefore, the electron energy is not characterised by temperature. but, because of the Pauli principle, it is characterised by the Fermi energy

$$\varepsilon_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}. \qquad \text{(I.2.5)}$$

In other words, if $kT < \varepsilon_F$, then in the rarefaction criterion in (I.1.18) we should use ε_F and not kT. Consequently, in the range of high n, the electronic component of the 'rarefied' plasma is ideal if:

$$\frac{e^2}{r_0} \ll \varepsilon_F. \qquad \text{(I.2.6)}$$

Region VII is the region of the classic dense plasma. It is characterised by high density and low temperature. In the laboratory conditions, in most cases the density $n_0 \gtrsim 10^{19}$ cm^{-3}, $T \lesssim 1$ eV. A suitable example of the systems with the non-ideal plasma are discharges at high pressures. As regards the properties (structure), the non-ideal plasma is closer to liquids than gases. Taking into account the non-ideal state criterion

$$e^2 \sqrt[3]{n} \sim \frac{3}{2} kT,$$

it may be seen that the boundary line between the classic rarefied and dense plasma in the graph I.2.1 is described by the straight line

$$\ln n = 3 \ln T + \text{const}.$$

The lower boundary of quantum plasma as regards density is determined by the condition ($r_0 = n^{-1/3}$):

$$\frac{e^2}{r} = e^2 \sqrt[3]{n} = \varepsilon_F$$

and depends only on density. It is the critical density $n^* \approx 5 \cdot 10^{25}$ cm^{-3} and corresponds to $\varepsilon_F \sim 30$ eV. It should be mentioned that the density $n^* \approx 5 \cdot 10^{25}$ cm^{-3} is at present produced in systems with laser compression of the targets. It may be seen that the transition to the quantum range takes place at a density considerably higher than the rarefied–dense plasma transition in the classic situation. Therefore, at $kT \lesssim 30$ eV, the boundary of rarefied plasma is determined by the classic criterion (I.2.3).

Finally, region VIII in this graph, close to the vertical axis, corresponds to the set of charged particles. In reality, there are no restrictions on energy (protons with the energy of $>10^{12}$ eV have already been produced in accelerators), but these sets are restricted in density because of the volume charge.

Outside the investigated regions there is the white dwarf plasma ($n_0 > 10^{30}$ cm^{-3}), the neutron matter of pulsars and quarks-gluon plasma of black holes. However, this is far away from the region of 'rarefied plasma' in which we are interested.

Figure I.2.1 shows the region T – thermonuclear plasma in stationary magnetic traps.

The figure shows how wide is the range of the parameters which relate to the classic rarefied plasmas. It is clear that the prospects for the practical application of plasma in this range are enormously encouraging.

I. 3. History of plasma investigations [50, 52]

I.3.1. Investigations up to the 30th of the 20th century

The origin of plasma investigations belongs in the middle of the 18th century. At that time, precursors of electrophoretic machines and Leyden flasks were produced and were used to entertain the public on squares or palaces[5] with different electrical phenomena and, in particular, spark discharges.

At the same time (1750–1752), B. Franklin 'tamed' the lightning in 1769 the Grand Duke of Tuscany ordered to put lightning rods at all about gunpowder stores of the Duchy. The attitude to electricity became serious.

The development of galvanic elements by Volta resulted in the discovery of the electrical arc at the very beginning of the 19th century (Petrov, Davy). In the second half of the 19th century, the electric arc was used in completely new practical applications: light source (Yablochkov, 1870), welding tool (Benardos, Slavyanov, 1880).

The development of technological applications of arc charges was accompanied, in the middle of the 19th century, by laying the foundations of plasma physics. This was associated with the examination of the electrical properties of materials, initially in the form of Faraday electrolysis laws (1830) and subsequently by influencing fluxes of particles (on the cathode and channel beams) of magnetic and electrical fields in electric discharge tubes with a low gas pressure. These are discharges were detected for the first time by Faraday and subsequently by Geisler in many variants (1850s), but detailed investigations of the detected 'cathode rays' started in the 1870s. In particular, W. Crookes proved that they represent particle fluxes. This

[5]In a series of portraits of students of the Smol'nyi Institute, produced by Levitskii in 1776, there is a portrait of E.I. Molchanova next to an electrical machine.

lecture, presented in 1879 at a meeting of the Royal Institute, had the name: 'On radiant matter or the fourth aggregate state'. Presenting convincing experimental data, indicating the corpuscular nature of cathode rays, Crookes concluded: 'In examination of this fourth state of matter it becomes apparent that we have finally at our disposal the 'final' particles which can be justifiably regarded as forming the base of the physics of the universe.... We have entered here the region in which the matter and energy appear to be merged into a single unit... I would like to assume that the main problems of the future will be to find 'solutions' in this region'.

Indeed, this was a truly brilliant foresight!

However, approximately 16 years were required for the final acceptance of the electrons and this took place in 1895 due to Thompson who determine the ratio e/m for cathode particles. In physics, the word 'electron' appeared in 1891, and after 1900 was given todays meaning. This resulted in the birth of electronics and the physics of plasma or, more accurately, gas discharges.

Subsequently, the events progressed very rapidly: the Planck equation was derived for radiation (1900); Rutherford experiments, describing the atomic nucleus[6], were formulated (1911); the Bohr atom model was constructed (1930), quantum mechanics by Heisenberg–De Broglie–Schrödinger–Dirac (the first half of the 1920s). Overall, the foundations of plasma physics were laid.

However, intensive work was also carried out on the development of the technology of charged particles: mass-spectroscopy (Thompson, 1912, Aston, 1920); the first oscilloscope ('Brown tube'), electronic and radio valves (De Forest), electrostatic accelerators of charged particles (van de Graaf, 1931), etc.

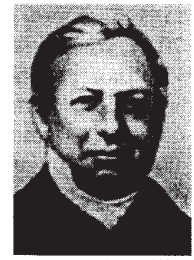
Work also started on the development of plasma technology. In 1908, 'neon tubes' – Geissler discharges, were used for advertising purposes. Work also started on plasma switches: mercury rectifiers, tyratrons, arc extinguishing chambers for breakers of powerful electrical circuits, development of lamps with high illumination capacity based on air discharge, etc.

All this development was important but, on the whole, occurred at the periphery of advances in physics. There is no adequate term for this new region, i.e. gas discharge physics: arc, glow, spark, etc. Up to the middle of the 1920s, only studies by Townsend (1910s) were regarded as fundamental and concerned with the mobility of charged particles in slightly ionised gases. Investigations carried out

[6]By the way, the spinthariscope with which Rutherford made his observations, was invented by the same Crookes.


D. Bernoulli


L. Euler


G. Liouville

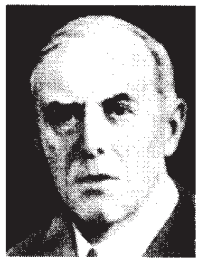

L. Boltzmann

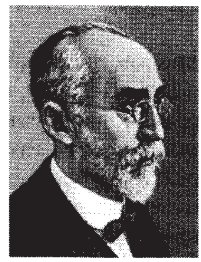

M, Faraday

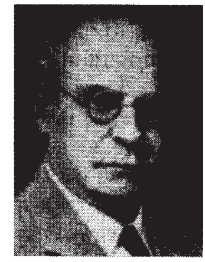

D. Maxwell


W. Crookes


G. Tomson

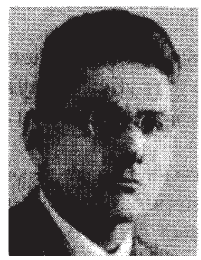

G. Townsend


H. Lorenz


E. Hall


I. Langmuir

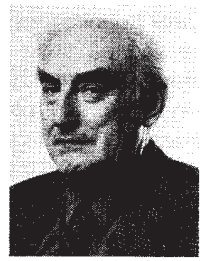

M. Saha

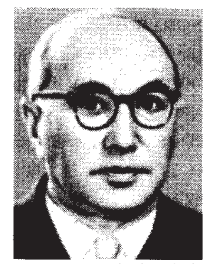

B.I. Davydov


A.A. Vlasov


H. Alfven


Ya.B. Fainberg


V.L. Ginzburg


Ya.B. Zel'dovich


A.L. Chizhevskii

Founders of hydro- plasma dynamics

by Townsend into the effect of the transverse magnetic field on the mobility of electrons in the electrical field proved to be very important for future investigations. Townsend showed, both theoretically and by experiments (in the case weak ionisation!) that the mobility of the electrons is inversely proportional to the square of the strength of the transverse magnetic field

$$\mu_{\perp}^{(e)} \sim \frac{1}{H_{\perp}^{2}} \tag{I.3.1}$$

This dependence is referred to as the 'classic' dependence. However, only in the 60s (see below) it was possible to apply this dependence in the experiments in the volume of the efficiently ionised and relatively dense plasma.

The separation of plasma physics from the physics of discharges is usually attributed to I. Langmuir. His fundamental studies in this area were carried out in the 1920s and were characterised by three fundamental achievements.

Firstly, he developed a completely new vacuum technology - diffusion pumps which made it possible to produce efficiently high and pure vacuum. Secondly, Langmuir perfected the theory and techniques of measuring the plasma parameters using electrostatic probes. These devices made it possible to determine, on a completely new level, the spatial distribution of electronic temperature (T_e), electrical potential (ϕ) and the density in plasma configurations. Therefore, it was not surprising that the electrostatic probes were referred to as 'Langmuir' probes, although the first probing experiments were carried out in 1887 by Lecher.

Thirdly, Langmuir and Tonks were the authors of a number of theoretical studies, in particular, the discovery of specific plasma oscillations with 'Langmuir' frequency

$$\omega_0 = \sqrt{\frac{4\pi e^2 n}{m}}$$

Finally, Langmuir introduced the concept of 'plasma' and this was used to define this state of matter which was defined previously.

1920s are also characterised by the development of the theory of thermodynamically equilibrium plasma based on Saha's equation (see section 6.5.7). Important achievements of this period were the discovery of the ionosphere of the Earth (Heaviside) and the initial attempts for simulation of the Northern polar lights (chapter 9).

I.3.2. Investigations and developments in 1930s and 1940s

The 30s of the previous century were regarded as the years of evolution development of experimental investigations and rapid development of theory. In astronomy, spectral methods were improved and the characteristics of cosmic objects were determined more accurately. Radio physicists described the structure of the ionosphere of the Earth and constructed models of propagation of radio waves in the ionosphere. In the laboratories, using mainly the Langmuir probe method (improved by D. Bohm and M.J. Druyvestein) and, to a lesser extent, spectroscopy, detailed investigations were carried out on different varieties of discharges. This was the basis for the development of a number of gas discharge devices. Of the completely new devices, developed in that period, it is necessary to mention the 'Penning' cell – one of the first gas discharge devices with the magnetic field which is still used at present, with the field magnetising the electrons (more details on this class of devices are presented in section 10.6). However, most significant advances took place in the theory. This included the construction of the effective models of fragments of classical discharges which are in good agreement with experiments (Schottky, Engel, Mecker, etc). Work also started on the general kinetics of plasma. In 1936, L.D. Landau, modifying the collisional term of the Boltzmann kinetic equation for Coulomb interactions, derived the collisional Landau term. These equations are discussed in chapter 5. At the same time, B.I. Davydov derived, on the basis of the kinetic Boltzmann equation, for slightly ionised plasma, the adequate Boltzmann–Davydov collisional term and derived the function of distribution of the electrons in the presence of a uniform electrical field. Finally, in 1938, A.A. Vlasov, ignoring the Liouville equation, formulated his system of equations with a self-consistent field which reflects most adequately the kinetics of completely ionised plasma. The Vlasov (or, more accurately, Vlasov-Maxwell) equations are the subject of chapter 4.

The comparatively smooth although quite rapid development of plasma physics in the 1930s was completely interrupted in the 1940s, mostly because of the Second world war. Its effect on the development of plasma physics and technology was huge. This was associated with the development of microwave technology for radio location and its basis – powerful generators of the waves in the centimetre and decimetre range, waveguides and super-sensitive (at that time) receivers. These achievements resulted in the development of radio-diagnostics of plasma, generation of microwave discharges and radio astronomy which provided invaluable information on the plasma processes taking place in the universe.

Of even greater importance in this area was the work on the development of the atomic bomb. It is important to mention three moments:

a. One of the stages of solution of this problem was the development of equipment for 'electromagnetic' separation of uranium isotopes U^{238} and U^{235}. For this purpose, it was necessary to develop powerful sources of ions and high-productivity separators capable of working with high ionic currents. These beams had to be quasi-neutral, i.e. represent plasma formations.

b. Investigations of the behaviour of plasma in the gas discharge source of ions (carried out using the Penning cell) and in the separator resulted in a conclusion that the Townsend law (I.2.6) does not work and, consequently, D. Bohm proposed – on the basis of processing experimental data – equations (the so-called scalings) for the transfer coefficient which differed from the classic equations by substituting the duration of the free path of the electron τ_e by the quantity

$$\tau_B = \left(\frac{2\pi}{\omega_H} \right) \tag{I.3.2}$$

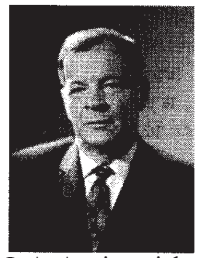
Thus, in particular, the diffusion coefficients – classic (Townsend) and Bohm's, have the form ($\omega_e \tau_e \gg 1$):

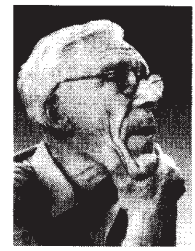
$$D^{(C)} = \frac{mc^2 k(T_i + T_e)}{e^2 H^2 \tau_e}, \quad D^{(B)} = \frac{ckT}{16eH} \tag{I.3.3}$$

It may be seen immediately that $D^{(B)}$, at relatively high T and H, is many orders of magnitude higher than $D^{(C)}$. If the 'Bohn diffusion' were fatal, it would not be possible to develop magnetic traps for controlled thermonuclear fusion (CTF) and the majority of plasma accelerators.

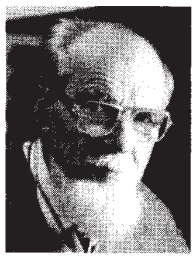
Fortunately, in the 60s, it was established that Bohm diffusion is determined by anomalous processes in the plasma, i.e. this type of diffusion is not universal, and the transfers can be reduced to the level determined by the classic equations, i.e. Townsend equations. However, ‫כ‬to determine and confirm this fact, more than 50 years of work of scientists was required because in the initial stage when the simplest plasma systems were examined, the Bohm equation reasonably corresponded to the experimental data obtained in greatly differing systems. In any case, this equation represented a unique 'reference point' and


O.A. Lavrent'ev

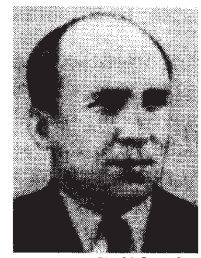

L.A. Artsimovich


M.A. Leontovich


L. Spitzer

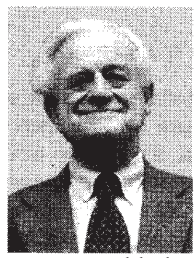

N.V. Fillippov


S.I. Braginskii


I.M. Gel'fand


A.B. Mikhailovskii

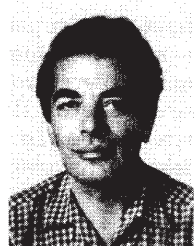

G.I. Budker


M. Rosenbluth

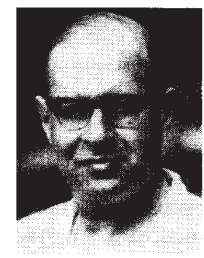

M.S. Ioffe

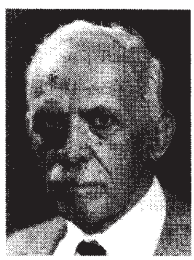

N.G. Basov


V.D. Shafranov


L.S. Solov'ev


R.Z. Sagdeev


M.S. Rabinovich


I.N. Golovin


N.A. Yavlinskii


M.I. Guseva


S.V. Mirnov

Pioneers of controlled thermonuclear synthesis

this obviously contributed to the systematisation of experimental facts.

c. Finally, the development and investigations of the electromagnetic method of separation of the isotopes proved to be an excellent school, and its principals in Russia later headed experimental investigations of controlled thermonuclear fusion (L.A. Artsimovich, etc).

Explosions of the atomic bombs resulted in the formation of a plasma with very high parameters. Outstanding scientists Ya.B. Zel'dovich, I.E. Tamm, A.D. Sakharov, and others) were invited to investigate the processes of explosion of the atomic bombs and the resultant shock waves and these investigations became a stimulus for the investigations of advanced low-temperature plasma systems with multi-electron atoms.

I.3.3. Investigations in the 1950s and 1960s. Problem of controlled thermonuclear fusion

The problem of the hydrogen bomb provided an impetus for the concept of controlled thermonuclear fusion[7] and, at the same time, the development of a large new area of plasma investigations – physics of 'hot plasma', i.e. plasma with a temperature of ~10 keV = 10^8 K, in 'peaceful' laboratory conditions.

These temperatures are essential to ensure that in a reactor two nuclei of light elements – in the simplest case the nuclei of the hydrogen isotopes deuterium (D ≡ ^{2}H) and tritium (T ≡ ^{3}H) can overcome the coulomb repulsion, coming together to distances of ~10^{-13} cm over which the effect of nuclear forces operates, and merge, forming a helium nucleus (α-particle) and a neutron:

$$D + T \rightarrow {}_2He^4 + n + 17\ MeV$$

The reaction is accompanied by the generation of a large amount of energy. It is very important to note that in any water from a puddle to an ocean every ~5000 hydrogen atoms are associated with 1 deuterium atom. In other words, in contrast to gas, oil, coal, the reserves of thermonuclear fuel are almost inexhaustible because synthesis can also take place by the D + D scheme (for more details see 10.5). At the same time (the end of the 1950s) a road to 'energy Eldorado' was also shown: it is necessary to avoid using energy-absorbing walls (dense gases, liquids, glass tubes, etc.) and transfer to sustaining the plasma by electromagnetic fields.

[7]Explosion – uncontrolled synthesis – takes place in the hydrogen bomb.

The problem of controlled thermonuclear fusion in Russia is described by Academician Yu.B. Khariton, the scientific supervisor of the Centre for the Development of Nuclear Weapons, who knew well events of that period: 'Igor' Evengen'evich (Tamm)...did not ignore a letter he received in the summer of 1950 ... from the unknown Oleg Lavrent'ev, who served as a sergeant in the Far East Sakhalin military area. The author, a self-taught scientist, proposed to use the system of electrostatic thermal isolation for producing high-temperature deuterium plasma. Tamm asked young Sakharov to examine Lavrent'ev's idea. Later, Sakharov wrote that this 'proactive and creative person raised the issue of colossal importance'[8]. Very soon, it became clear to Sakharov that there are real possibilities offered by the application of magnetic thermal isolation. He and Tamm started detailed calculations[9]. This shows clearly the role of O.A. Lavrent'ev in the births of the problem of controlled thermonuclear fusion [22].

In addition to the concept of controlled thermonuclear fusion, the end of the 1940s and the beginning of the 1950s were marked by another important event. This period was characterised by the publication by the Swedish astrophysicist H. Alfvén of 'Cosmic Electrodynamics', with the author completely unknown to a large number of plasma scientists [21]. In his book, the author investigated the self-consistent dynamics of ideally conducting plasma and magnetic field. This book showed the significant role of electromagnetic processes in the space, and the proposed model of plasma dynamics, referred to as magnetic hydrodynamics (MHD), was immediately regarded as highly attractive because of its novelty and 'beauty'. Consequently, the astrophysical processes on huge scales, taking place with the velocities measured in hundreds and thousands of km/s, including solar corona, protuberances, magnetosphere of the Earth, etc, became accessible for serious theoretical analysis. Therefore, it is not surprising that giants such as Fermi and Chandrasekhar also provided a contribution to the development of MHD models very soon after publication of Alfvén's book. MHD rapidly found its application also on the Earth because of investigations into controlled thermonuclear fusion and similar tasks. Magnetic hydrodynamics is the subject of the main volume of chapter 2 and part of chapter 9 of this book. In his book, Alfvén also proposed an extremely effective 'drift' approach (section 1.2) for describing the dynamics of single particles in slightly non-uniform fields.

[8]A. Sakharov, Memories. New York, Chekhov Publishing House, 1990, p. 186.

[9]Yu.B. Khariton, et al., I.E. Tamm through the eyes of physicists of Arzamas-16, Memories of I.E. Tamm, ed. E.L. Feinberg, Moscow, 1995.

Thus, at the very beginning of the 1950s, plasma physicists had at their disposal a basis for the kinetic and hydrodynamic description of classic discharges and plasma rapidly heated by 'nuclear' explosions, and also the ideology of the dynamics of strongly heated plasma of cosmic objects, with sustainment of this plasma in the laboratory conditions using electromagnetic fields. Consequently, the 1950s and 1960s became the years of inventions of the circuits of greatly differing devices and a large number of experiments for realisation of these phenomena were conducted.

This activity was especially strong as a result of the problem of controlled thermonuclear fusion, whose physical model of the end grandiose prospects[10] excited leading physicists and, most importantly, administrators on higher levels, dealing with financing. As a result, on May 5, 1951, I.V. Stalin signed the Decree of the Government on the Organization of a magnetic fusion reactor (MFR). This day is now regarded as the start of work on controlled thermonuclear fusion in the former USSR.

The leader in the area of investigations in the area of controlled thermonuclear fusion in the former Soviet Union was The Institute of Atomic Energy (IAE), directed by I.V. Kurchatov[11]. The direct supervisor of work on controlled thermonuclear fusion at the IAE at the beginning of the 1950s was L.A. Artsimovich. In his department, the theoretical sector, headed by M.A. Leontovich, was soon formed. The initial scheme of MFR – magnetic thermonuclear reactor (the name used by Sakharov) – was not capable of operation. Therefore, Artsimovich proposed to start investigations of high-current (J_p ~100 kA) electrode pulsed (τ_p ~ 1 μs) discharges in straight dielectric tubes, filled with deuterium at a pressure of ~1 torr. Later, these devices were referred to as *Z*-pinches because it was found that a rapid increase of current is accompanied by the formation of a skin layer with high conductivity and the magnetic fields surrounding it constrict this plasma 'tube' which collects the ionising gas (for more details see section 1.6).

As a result, when the pinching sheath reaches the axis, there is a short time (~ 0.1 μs) when a dense ($n \sim 10^{19}$ cm^{-3}) and hot ($T \sim 100$ eV) plasma forms. Moreover, in the summer of 1952 it was found in experiments in the system constructed by N.V. Filippov that neutrons form at a certain point in the pinch. It seemed that the problem of UTC was solved in principle. But it soon became clear that the generation

[10]Processing in a fusion reactor of the 0.02% deuterium contained in any natural water supplies generates 300 times more energy than an equal volume of gasoline.

[11]I.V. Kurchatov was the supervisor of the entire nuclear power program of the country since 1945. Under his leadership the former USSR was the first country in Europe to construct a nuclear reactor, develop the first atomic bomb, and launch the first nuclear power plant.

of neutrons is not due to randomly moving (heated) particles but to particles accelerated in electric fields arising in the process of discharge. And, more than that, all attempts to raise the density and temperature of the plasma at the time of maximum pinch, and thus increase the neutron yield, were unsuccessful[12]. Soon became clear reason for the failure. It turned out that in the compressed cord develops instability, limiting the compression ratio (see § 1.6).

For the first time there was a terrible scourge of all the schemes of magnetic confinement of the hot plasma – instability. Hopes to solve the problem by using *Z*-pinches of different variations were finally extinguished in 1957. At this time there was a literally whooping search for new schemes of plasma 'traps'. Most exotic schemes were proposed, but the clear favorite in the late 50's were the so-called 'mirror trap' ('mirror cells', 'open traps', 'mirror traps'), proposed in the USSR by G.I. Budker (1954) and in the U.S. by R. Post (1952–53). More details are described in section 1.7.

Although shortly theorists (Rosenbluth, Longmayer, Kadomtsev) predicted the inevitability of a strong convective instability in mirror traps, they also proposed methods to combat them (section 1.7), which was confirmed in the brilliant experiments by M.S. Joffe (IAE, 1961). But the still remaining 'weak' instability and 'openness' of these traps led to large losses. Expectations related to the mirror cells also started to melt.

After these failures the late 50's saw the beginning of serious work on toroidal discharges in a strong magnetic field. Thus, tokamaks[13] were born. Their differences from the early toroidal discharges in England, the USA and the USSR were associated with a strong longitudinal magnetic field (see section 10.5). The need for such a strong field to stabilise the helical instability was predicted by theorists (V.D. Shafranov, M. Kruskal).

Soon modelling experiments by N.A. Yavlinsky confirmed the Kruskal–Shafranov criterion and Yavlinsky also constructed the first tokamak together with L.A. Artsimovich and I.N. Golovin (1960). Great misfortune caused the death of N.A. Yavlinsky in 1962 in a plane crash, and then the direct control of studies of these systems was taken over by L.A. Artsimovich. After extremely hard and painstaking work, at the IAEA International Conference in Novosibirsk in 1968, he presented the plasma parameters achieved in the tokamak T-3: electron density $n_e = 5 \cdot 10^{13}$ cm^{-3}, the temperature of ions and electrons, respectively

[12]Later, in one of the versions of Z-pinch – the so-called 'plasma focus' - *n* and *T* were increased by an order of magnitude, but it was still not enough to ensure a positive energy output.
[13]Tokamak – Toroidal Camber with Magnetic Coils (first letters of the Russian name).

$T_i \approx 200$ eV, $T_i \approx 500$ eV and the energy confinement time of the plasma $\tau_E \approx 0.01$ s[14].

The impression of the report was stunning and this started the expansion of tokamaks – of course with both qualitative and quantitative changes – at virtually every laboratory in the world. This leading position of tokamaks still lasts.

Investigations of controlled thermonuclear fusion abroad

Until now, we have been discussing the main stages of the development of investigations of controlled thermonuclear fusion in the former USSR and then Russia. We shall now briefly mention the special features of the investigations carried out in the 50s and 60s abroad. At that time, the leading laboratories were those of Great Britain and the USA.

In Great Britain, investigations of discharge in toroidal chambers have been carried out since 1947–1948. Z-pinches were also studied but the results obtained did not match those at the Institute of Atomic Energy. Therefore, the document presented by I.V. Kurchatov in 1956 in the Harwell Atomic Centre describing the investigations of the Z-pinches in the Institute of Atomic Energy surprised English scientists, not only because of the physical achievements but also the fact that the results were presented which had been secret up to them.

In England, special attention was given to toroidal discharges with a weak longitudinal magnetic field. The largest equipment used for this purpose at the time was ZETA equipment, and the results of investigations obtained in this equipment were published in 1958. They were quite modest, nevertheless, this type of discharge has a number of interesting properties and under the name ‘the pinch with the inverted field‘ (RFP) is being studied even now.

In the U.S., research on CTF was inspired in the second half of the 40’s by Fermi, Teller, and other founders of nuclear technology. But the first systems appeared in 1950–51. Direct discharges were also studied, but most promising were stellarators (see section 10.5) – toroidal systems without current inside the plasma, in contrast to the tokamak.

Stellarators were proposed by the outstanding astrophysicist L. Spitzer Jr. He also became the head of experimental studies of these systems. The stellarators produced sufficiently high plasma parameters, but they conceded at the end of the 60s to the results obtained in tokamaks, so in America, researchers modified their stellarators to tokamaks.

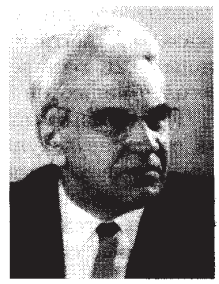
However, the revival of the stellarator program started soon in the Soviet Union where the theory of perturbations of toroidal magnetic

[14]This time was almost 30 times longer than that calculated by Bohm.


K.E. Tsiolkovskii

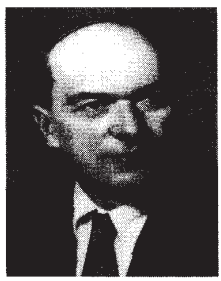

W. von Braun


S.P. Korolev

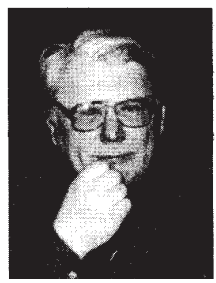

M.V. Keldysh

Founders of classical space exploration


H. Kaufman


A.M. Andrianov


G.Ya. Shchepkin


A.I. Morozov

Founders of space electroreactive thrusters

fields was developed (A.I. Morozov, L.S. Solov'ev, I.M. Gel'fand and their colleagues) and this stimulated the development of methods for the control of magnetic configurations directly in the systems[15]. Because of this, stellarators with the required structure of the magnetic field lines were built in the USSR. Here it is important to mention studieds at the FIAN (now the P.N. Lebedev Physical Institute, Russian Academy of Science) by M.S. Rabinovich, I.S. Shpigel', L.M. Kovriznykh, and others. This is currently the most advanced area of CTF research, after tokamak (see section 10.5).

We have described the formation and early evolution of the most popular magnetic plasma confinement systems, although there are many more. In addition to systems with magnetic confinement of a lot of effort has been invested in the idea of 'inertial fusion' enunciated in the late 60's by N.G. Basov and O.N. Krokhin. The gist of it is that in a very short time ($\sim 10^{-9}$) a small target (diameter ~1 mm) is hit with a powerful stream of laser or other light (particle fluxes can also be used); the target material is heated to thermonuclear temperatures and

[15]Until then the structure of the magnetic field in traps in US stellarators was not measured, and the magnetic field was evaluated only on the basis of calculations which naturally did not take into account the defects of manufacture of the magnetic circuit.

kept at these temperature for just the right time due to inertia (see Appendix B).

Ending this brief review of the early stages of CTF studies and taking into follow-up studies, we can sum up.

1. The foundations of the science of the properties of the hot plasma in a wide range of parameters and conditions have been laid. The progress made in the field of fusion would not be possible without the hard work of talented theorists. In the Soviet Union in the first place we must note Academician M.A. Leontovich, who created a powerful theoretical division at the IAE. A large role in the development of theoretical questions of hot plasma and also in original experimental studies was played by the team of Academician G.I. Budker (Novosibirsk Institute of Nuclear Physics), as well as scientists at the Khar'kov Physico-Technical Institute, The Physical Institute of the Russian Academy of Sciences, etc. A major contribution to the theory of fusion was made by mathematicians from the M.V. Keldysh Institute of Applied Mathematics, primarily the teams of Academicians I.M. Gel'fand, A.N. Tikhonov, and A.A. Samarskii. Fundamental contributions to the theory and numerical simulations have also been made by many foreign scientists (Rosenbluth, Kruskal, Bernstein, Furth).
2. An entirely new engineering of large magnetic systems, large vacuum chambers, and powerful electrical pulse has been developed. The methods for the production of hot plasma using pulsed high-current discharges, powerful ion sources and powerful generators of microwaves (gyrotrons) have been proposed.
3. Contactless methods of local diagnostics of the plasma density distribution functions of electrons and ions, magnetic and electric fields, and other variables have been developed.
4. Effective international cooperation has been organised (section 10.5) due to the efforts of I.V. Kurchatov, L.A. Artsimonovich, E.P. Velikhov, and B.B. Kadomtsev. In the framework of international collaboration, realisation of the ITER tokamak project was launched in 2007.

Now, despite all these impressive, but ultimately partial achievements, the CTF problem can not be considered as solved, although tokamaks JET (European Community) and JT-60 (Japan) produced over short periods of time the fusion power, compatible with the power going to heat the plasma (section 10.5).

It is important to emphasize that today this situation occurs in all areas of CTF research, related to both magnetic confinement and inertial confinement systems. This is despite the fifty years of focused and intense work, which cost more than 30 billion dollars and with the participation in these works in some periods to about 10 000 people. So now an intensive search for the 'alternative' schemes for plasma confinement has been launched.

Specifically, the approaches and CTF systems will be discussed below, but for now we emphasize that the research on fusion is not the only source of modern plasma dynamics.

I.3.4. Studies in the 50's and 60's. The problem of electroreactive thrusters

An important and, in many areas, controlling role in the development of plasma dynamics, especially middle energies (ε_i ~ 30–1000 eV), has been played by the problem of electric reactive (rocket) thrusters (ERT), which began to flourish in the years following 1958–59 when the Soviet Union launched into an orbit the Sputnik in 1957, although at a theoretical level, the idea of electric propulsion had been studied before this period.

The need to change to higher exhaust velocities is quite evident, since the thrust developed by the rocket engine is

$$\mathbf{F} = \dot{m}\mathbf{w},$$

where $\dot{m}$ is the mass flow rate, and **w** is the gas exhaust velocity. Therefore, for small **w** the supply of the working material for missiles becomes very large.

The exhaust velocity of the gases from the modern thermal thrusters is ~(3–4) km/s. At the same time, for most flights near the solar system (Mercury – asteroid belt) with the maneuvers we require exhaust velocities of ~20–40 km/s. However, this range of velocities with acceptable efficiency could not be produced with satisfactory efficiency by the available plasmatrons or sources of ion fluxes. It was natural to try to solve the problem using plasma accelerators, but for this it was necessary to learn to create epithermal electrostatic fields in the plasma volume. In the mid-60s, this problem was solved in principle at the IAE by A.I. Morozov and G.Ya. Shchepkin who developed in stationary plasma thrusters (SPT), in which the actual conductivity $\sigma_\perp$ was almost 1000 lower than Bohm' estimtate. Other types of electric reactive engines, ionic (Kaufman) and plasma (Andrianov, et al.) were also developed. However, the SPT are favourites even at the present time. This is discussed in section 10.4.

Thus, by joint efforts on the problem of controlled thermonuclear fusion, where the emphasis was on creating traps to confine the plasma, (i.e. 'magnetised' diffusion) and ERT where the focus was on generating flows (i.e. create epithermal electrostatic fields in plasma) made it possible to completely overcome Bohm's limit.

Of course, these fundamental achievements have opened a whole new world of plasma systems and a number of concrete are discussed below.

I.3.5. Other backgrounds of plasma dynamics

But not only the plasma systems of medium- and high-energy have been developed over the years. The physics and applications of low-temperature plasma were quite different and seemingly exhausted. The most important 'starter' achievement here was the research of powerful shock waves (SW) and flows resulting from the explosion of nuclear bombs. In these shock waves the temperature reaches many thousands and millions of degrees and there are intensive molecular reactions of excitation and ionisation of the air particles.

This cycle of research was summarized in a book by Ya.B. Zel'dovich and Yu.P. Raizer 'Physics of shock waves and high-temperature gas dynamic phenomena', two editions of which appeared in the first half of the 60s. The processes taking place at high temperatures are accompanied by powerful radiation, which largely determines the dynamics of the plasma. Specific features of the dynamics of dense plasma in the presence of intense radiation subsequently led to a scientific field known as 'radiation plasma dynamics' (N.P. Kozlov, Yu.S. Protasov, and others).

Besides this direction, an important role for the development of low-temperature plasma physics in the new phase was played by the invention of gas and plasma lasers (1960). Another great – but, unfortunately, not completed – series of papers in the late 50's – early 60's dealt with MHD generators, with which they tried to raise the efficiency of power plants. At that time (early 60s) the work on the creation of plasma chemical reactors sharply intensified. There is a new large area: plasma chemistry science (L.S. Polok, V.M. Smirnov, et al.).

It is also important to mention the development of plasma technology for surface treatment (V.M. Gusev and M.I. Guseva, V.G. Padalka and V.T. Tolok, and others).

These directions of the completely earth 'laboratory plasma dynamics' are continuing to develop rapidly, branch and find more and more applications. A number of studies to create instruments for plasma

technology will be discussed in later chapters and are summarized in chapter 10.

The achievements of laboratory plasma dynamics are finding more and more applications both in the practice of contemporary space exploration (including plasma thrusters) and the analysis of plasma dynamic processes in the magnetosphere of the Earth and planets, as well as the dynamics of stellar and galactic masses (chapter 9).

I.4. Features of plasma research

Let us look at the features of the theoretical analysis of plasma systems. Some features of plasma experimental studies have been described in detail in many books, for example, in [8], and we will not discuss this here. Only the following should be mentioned. Since the majority of the methods of plasma diagnostics are contactless, the plasma parameters can be determined also remotely, not in the immediate vicinity of experimental equipment. This can be carried out through the Internet using advanced network technologies[16]. See also the book by N.P. Norenkov and A.M. Zimin: Information technologies in education, N.E. Bauman Moscow State Technical University, Moscow, 2004.

It would seem that the behaviour of plasma dynamic systems should be close to that of conventional gas-dynamic systems. Indeed, in some cases it is so, but such cases are few, but even they have very non-trivial features. In general, however, there are fundamental differences between the two types of systems.

Somewhat arbitrary, these differences in relation to laboratory conditions can be combined into four groups.

I. First of all, in contrast to the classical three states of matter, plasma does not exist by itself on the Earth (we are now considering the magnetosphere). It should especially be created and is annihilated by contact with dense gas or walls. Therefore, analysis of processes in any real system must take into account its formation from a neutral gas (or evaporation products of condensed matter), and its destruction by contact with 'cold environment'. In other words, it is imperative to take into account the presence of the ionized and neutral components, at least in the initial stage of formation of the plasma configuration and around of the bounding walls, even if the bulk of the volume plasma is completely ionised. This explains the principal role of the 'low-temperature' plasma in any plasma system. It will be mainly discussed in chapters 6 and 7 of

[16]Such technologies have been applied in, for example, large tokamaks, and also at the N.E. Bauman Moscow State Technical University by A.M. Zimin in lectures on plasma spectroscopy (http://lud.bmstu.ru).

this book. Further, in the ordinary gas dynamics, the wall, for example, the aircraft wing, is a surface on which the boundary condition

$$\mathbf{v}\big|_{\Gamma} = \left(v_n, v_t\right)\big|_{\Gamma} = 0$$

Here v_n is the normal and v_t the tangential velocity component. This boundary condition is sufficient to calculate, using the Navier–Stokes equations, not only the flow, such as the flow around an airplane wing, but also the viscous boundary layer. If the speed of the wing is close to the speed of sound, it is still necessary to consider the heat exchange between the wing and the oncoming flow. Another situation is the one in which the plasma particles with energies measured in many units, tens, hundreds of electron volts interact with the wall. Here the recombination of ions and electrons takes place, as well as sputtering of the surface by ions, electron emission from the wall, charging of the wall, etc. These processes often play a large role and their consideration in the analysis of specific systems is essential. This will be discussed in Chapter 7.

II. The second difficulty of modeling is that the plasma, especially in the laboratory, is fundamentally multi-component. Even the simplest hydrogen plasma in fusion reactors, apart from a small number of impurities and neutral atoms, contains substantially different components: hydrogen ions, electrons, magnetic and electric fields, as well as radiation, which are all in the self-consistent dynamics. The presence of multi-charged ions drastically increases the number of possible situations.

The magnetic field is a powerful factor, which regulates the behavior of particles in the plasma volume. Therefore, it is the consideration of the features of its structure in chapter I where we start description of the foundations of plasma dynamics.

III. Conventional gas dynamics deals with neutral particles (atoms, molecules). These particles interact with each other only by direct contact. Figuratively speaking, they are ‘deaf and blind’. Quite a different picture arises in the plasma. The Coulomb fields of the ions and electrons relatively slowly ($1/r^2$) decrease with distance. Therefore, being even at a large distance, they disrupt each other’s path. Such a perturbation will be stronger, the smaller the particle velocity. The effective cross-section of paired collisions can be estimated by the formula[17]

$$\sigma^{(\mathrm{coul})} \sim \pi p^2, \qquad \text{(I.4.1)}$$

where p is the characteristic ‘impact parameter’ defined by the condition

[17]Coulomb collisions are examined in detail in chapter 5.

$$\frac{|e_1 e_2|}{p} = \varepsilon_{\text{rel}}. \qquad \text{(I.4.2)}$$

Here ε_{rel} is the relative kinetic energy in the reference system connected with the centre of masses

$$\varepsilon_{\text{rel}} = \frac{\mu u^2}{2}, \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2},$$

where μ is the reduced mass of the two particles, and $\mathbf{u} = \mathbf{v}_1 - \mathbf{v}_2$ is the relative velocity. In the case where the particle distribution is close to Maxwellian and $T_i \sim T_e = T$, we can write

$$\sigma^{(\text{coul})} \sim \frac{10^{-13}}{T^2}, \qquad \text{(I.4.3)}$$

where the cross section is measured in cm^2, and T is in eV.

This shows that at $T_e \sim 1$ eV,[18] and even more so when $T_e \sim 0.3$ eV the Coulomb cross section is very large compared to the gas-kinetic cross sections of neutral atoms or molecules ($\sim 10^{-16}$–10^{-15} cm^2). However, the Coulomb cross sections decrease rapidly with increasing temperature, and if we have a hydrogen plasma where the ions do not have electron shells, already at $T_e \sim 100$ eV the cross section is an order of magnitude smaller than the gas kinetic one, and at thermonuclear temperatures ($T_e \sim 10^4$ eV) the collision cross section decreases to about 10^{-21} cm^2 in a plasma with a density of air at atmospheric pressure ($n \sim 3 \cdot 10^{19}$ cm^{-3}) the mean free path λ of the Coulomb particles is ~30 cm. Even this simple example shows that by warming plasma we can 'almost imperceptibly' move from the environment well described by hydrodynamics (at $\lambda \ll L$) to the environment that require a kinetic description ($\lambda \gtrsim L$), where L is the scale of the system.

IV. Theoretical hydrodynamics is usually based on the macrostructures described quite efficiently by a system of hydrodynamics equations – laminar flows in pipes, the flow around thin profiles, in single vortices and shock waves, etc. In the real world, we have to deal with a spontaneously occurring mesostructure in the form of turbulence, which often rises and the macro level. In fact, only the emergence of powerful computers has made it possible to calculate the mesostructure of gas-dynamic flows. And we are talking about the media where the particles interact by the 'push method' with almost constant collision cross sections.

In plasma, the situation with the mesostructures is greatly complicated. Now the particles can interact over long distances, to

[18]Temperature 1 eV corresponds to ≈11 400 K.

create volumes with a predominance of particles of the same sign, break into groups with different velocities and, accordingly, different collision sections. We can say that plasma is charged particles immersed in an oscillating electric field. And not just oscillating but also leading to self-organisation of long-lived mesostructures. Today, even the powerful computers are not able to fully reproduce the real situation with the mesolevel taken into account.

We will draw conclusions from the above. As can be seen, there is no relatively simple and, at the same time, universal mathematical (theoretical) model. Only one thing remains: to consider the hierarchy of models consistently and more accurately describing the specific system under consideration. The most general models (single particle dynamics, the equations of hydrodynamic type, different kinetics), which will be called ‘base’ models, will be the focus of attention in chapters 1–6.

Mesostructures are dealt with in chapter 8. Cosmic systems are dsicussed in chapter 9. Chapter 10 (Plasma Technology) describes plasma devices used in practice in everyday life and technology. In addition to purely cognitive goals, the aim of chapter 10 is to give the reader to feel the pace and forn of the penetration of plasma technology in our lives and once again demonstrate how ‘slender’ plasma is connected with a rigid environment. Fragments of physics of the systems that are in chapter 10 are discussed in previous chapters, and the review is performed by step-by-step, namely in the transition to the next chapter the features of the model become more and more complicated. As a result, the reader, learning the basic material, becomes acquainted with the physics of many important plasma systems (traps, accelerators, classical discharges, etc.).

In view of this, it may be advisable after reading this introduction to skip to chapter 10, and only then begin the study of sequential chapters.

1

Fields, particles, blocks (point models)

In this chapter, we consider the basis of the plasma systems (**E**- and **H**-fields and the dynamics of single particles in these fields) which are described by the Maxwell and Newton equations or, more accurately, Newton–Lorenz equations.

The chapter starts with the Maxwell equations which should be known to the reader, and they are presented here mainly for demonstrating the selection of notations. Further, in contrast to conventional textbooks, special attention in this chapter is given to the morphology of the magnetic field, i.e. the structure of magnetic lines of force in a number of systems. This morphology plays a significant role in many advanced plasma dynamic systems (PDS). The PDS is a complex of plasma configuration, characterised by the required properties, a shell isolating to a certain degree the plasma from the destructive influence of the surrounding medium, and devices supplying power to the system (for the supply of working substances and electric power). In a general case, the PDS includes control and diagnostics systems.

The chapter is divided into two parts. The first part contains three paragraphs concerned with the general properties of electromagnetic fields, important for plasma dynamics (section 1.1), followed by the dynamics of single particles in electromagnetic fields of different structure (section 1.2) and finally, plasma systems in which the processes – in the first approximation – permit simulation by blocks and description using conventional (zero-dimensional) differential equations (section 1.3). The model of this type has already been described in the Introduction where the equation was derived for the frequency of Langmuir oscillations. The plasma volume was approximated by two

rigid blocks: ionic and electronic. The dynamics of these blocks was described by two differential equations for the centre of masses.

The second part of the chapter is concerned with the investigation of a number of models of plasma systems using methods described in the first part. Special attention is given to electromagnetic waves in cold plasma, systems of classic corpuscular optics (electromagnetic lenses and separators), plasma traps, used in investigations of controlled thermonuclear synthesis (CTS), etc.

1.1. Electromagnetic fields

1.1.1. Maxwell equations

In all PDS, electromagnetic fields are present and operate actively in the zones of ionisation, sustainment and acceleration of plasma. It is important to stress that in this case the electromagnetic fields are not simply some energy carriers (as is the case of, for example, heat), and they are a factor determining the entire plasma dynamics. A suitable example of these are ionic sources in which a system of electrons is used to produce the geometry of the equipotentials of the electrical field in the accelerating gap which ensures acceleration and focusing of the flux into a relatively narrow beam without affecting the electrodes (section 1.4).

Another example are magnetic traps for plasma sustainment (sections 1.6–1.7). Finally, the Debye fields have already been discussed.

It can be said without exaggeration that in these PDS the fields are 'constructional material'. It is therefore natural to start explanation of the fundamentals of the theory of processes in the PDS by mentioning the main properties of the electromagnetic field described by Maxwell equations [53].

In the Gaussian unit system, these equations have the form:

$$\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\mathbf{j} + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}; \tag{1.1.1a}$$

$$\operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \tag{1.1.1b}$$

$$\operatorname{div}\mathbf{H} = 0; \tag{1.1.1c}$$

$$\operatorname{div}\mathbf{E} = 4\pi q_e \tag{1.1.1d}$$

Here **E** and **H** are the strength of the electrical and magnetic fields in vacuum, respectively, **j** is the volume density of flows of all particles,

q_e is the volume density of all charges.

The system (1.1.1) can have different forms. We shall mention some of these forms which will be required in further considerations.

1. *Static fields*. In this case, the system (1.1.1) has the following form:

$$\operatorname{rot}\mathbf{E} = 0; \tag{1.1.2a}$$

$$\operatorname{div}\mathbf{E} = 4\pi q_e; \tag{1.1.2b}$$

$$\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \tag{1.1.3a}$$

$$\operatorname{div}\mathbf{H} = 0. \tag{1.1.3b}$$

This shows that in the statics the fields **E** and **H** are independent of each other, and the electrostatic field is potential

$$\mathbf{E} = -\nabla\phi, \tag{1.1.4a}$$

and potential ϕ satisfies at $q_e \neq 0$ the Poisson equation

$$\Delta\phi = -4\pi q_e. \tag{1.1.4b}$$

However, if $q_e = 0$, ϕ is governed by the Laplace equation

$$\Delta\phi = 0. \tag{1.1.4c}$$

The situation in the case of the magnetic field is slightly more complicated.

If we consider the field in the volume where $\mathbf{j} = 0$, then here, as in the case with the **E**-field, the magnetic field is potential (rot **H** = 0):

$$\mathbf{H} = \nabla\phi_m, \tag{1.1.5a}$$

and potential ϕ_m satisfies the Laplace equation

$$\Delta\phi_m = 0. \tag{1.1.5b}$$

However, if $\mathbf{j} \neq 0$, we should use the continuity equation (1.1.3b) which can also be satisfied by introducing the vector potential **A** and assuming that

$$\mathbf{H} = \text{rot}\ \mathbf{A}. \tag{1.1.5c}$$

The vector potential is ambiguous because without changing **H,** which also has a meaning, we can add the gradient of an arbitrary function f to the vector potential:

$$\mathbf{A} \sim \mathbf{A} + \nabla f. \tag{1.1.5d}$$

This circumstance often enables us to simplify the formal side in specific calculations.

Substituting (1.1.5c) into (1.1.3a) we obtain

$$\Delta\mathbf{A} = -\frac{4\pi}{c}\mathbf{j}. \tag{1.1.5e}$$

Here, we have taken into account the vector identity

$$\text{rot rot}\ \mathbf{a} = \nabla \text{div}\ \mathbf{a} - \Delta\mathbf{a},$$

and utilising the arbitratiness in the definition of **A**, we set div **A** = 0.

2. *Introduction of dielectric permittivity.* System (1.1.1) describes the field in vacuum generated by the currents and charges present there. In fact, this is a microscopic Lorenz model. **j** and q_e include both 'molecular' currents jmol and charges q_{mol} and also secondary 'macrocurrents' $\mathbf{j}_{sec}$ and 'macrocharges' $\mathbf{q}_{sec}$. In classic electrodynamics, the molecular current and charges are usually separated and their influence is described by means of the vectors of electrostatic induction **D** and magnetic induction **B**, retaining in the Maxwell equations only external (secondary) currents and charges in relation to the medium. There are a number of cases in which the plasma should be treated as a medium in which the displacement of the particles can be taken into account efficiently in the same manner. In most cases, it is sufficient to introduce single induction **D**. Consequently, the system of the Maxwell equations assumes the following form:

$$\text{rot}\,\mathbf{H} + \frac{4\pi}{c}\mathbf{j} = \frac{1}{c}\frac{\partial \mathbf{D}}{\partial t}; \tag{1.1.6a}$$

$$\text{div}\,\mathbf{H} = 0; \tag{1.1.6b}$$

$$\text{rot}\,\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t} \tag{1.1.6c}$$

$$\text{div}\ \mathbf{D} = 4\ \pi\ q_{sec} \tag{1.1.6d}$$

The vector of electrostatic induction **D** is connected with the strength of the field **E** by the relationship

$$\mathbf{D} = \mathbf{E} + 4\pi\mathbf{P}, \quad j_{mol} = \frac{\partial \mathbf{P}}{\partial t}. \tag{1.1.6e}$$

Here **P** is the polarisation vector of the medium. If polarisation is directly proportional to the strength of the electrical field

$$\mathbf{P} = \overleftrightarrow{\varkappa}\mathbf{E}, \tag{1.1.6f}$$

we can introduce dielectric permittivity of the medium $\overleftrightarrow{\varepsilon} = 1 + 4\pi\overleftrightarrow{\varkappa}$ and write

$$\mathbf{D} = \overleftrightarrow{\varepsilon}\mathbf{E}, \tag{1.1.6g}$$

The linear relationship between **P** and **E** is observed in particular in propagation of waves with a low amplitude in plasma. The letters with the two sided arrows above them indicate tensors. A specific example of calculation of $\overleftrightarrow{\varepsilon}$ and of the system (1.1.6) will be investigated in the section 1.5.

1.1.2. Conservation laws

Energy conservation law

In analysis of the operation of the PDS it is convenient to have general information on the dynamics of electromagnetic fields which can be extracted from the law of conservation of the energy of electromagnetic fields. If the equations (1.1.1a) and (1.1.1b) are multiplied in a scalar manner by **E** and **H**, respectively, and the second equation is deducted from the first one, we obtain

$$\mathbf{jE} + \frac{\partial}{\partial t}\left(\frac{E^2}{8\pi} + \frac{H^2}{8\pi}\right) = \frac{c}{4\pi}(\mathbf{E}\,\mathrm{rot}\,\mathbf{H} - \mathbf{H}\,\mathrm{rot}\,\mathbf{E}). \tag{1.1.7a}$$

Taking into account the identity **a** rot **b** – **b** rot **a** = div [**b**, **a**], equation (1.1.7a) can be presented in a new form

$$\mathbf{jE} + \frac{\partial}{\partial t}\left(w_E + w_H\right) = -\mathrm{div}\,\mathbf{\Pi}. \tag{1.1.7b}$$

Here w_E and w_H are the volume densities of the energy of the magnetic and electrical fields:

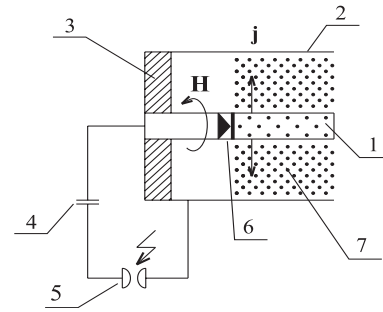


Fig. 1.1.1. Diagram of a pulsed plasma gun with an intrinsic magnetic field. 1) cathode, 2) anode, 3) insulator, 4) condenser batteries, 5) switch, 6) pulsed valve for letting in gas placed inside the cathode, 7) plasma bunch with the current flowing through it.

$$w_H = \frac{\mathbf{H}^2}{8\pi}; \quad w_E = \frac{\mathbf{E}^2}{8\pi} \tag{1.1.7c}$$

Π is the Umov–Pointing vector:

$$\Pi = \frac{c}{4\pi}[\mathbf{E},\mathbf{H}], \tag{1.1.7d}$$

which characterises the strength and the direction of the energy flow of the electromagnetic field.

It should be mentioned that in practice it is often convenient to use the integral form of the equation (1.1.7b). Denoting the volume element by *dV*, we obtain

$$\iiint \mathbf{j}\cdot\mathbf{E}dV + \frac{\partial}{\partial t}\iiint\left(\frac{\mathbf{H}^2}{8\pi}+\frac{\mathbf{E}^2}{8\pi}\right)dV = -\oiint \mathbf{\Pi}\, d\mathbf{S}. \tag{‘1.1.7e}$$

To illustrate the law (1.1.7e) we can use a coaxial high-current pulsed plasma accelerator – a ‘plasma gun’ – with an intrinsic magnetic fields (Fig. 1.1.1).

In this accelerator, there are azimuthal magnetic and radial electrical fields which ensure, in accordance with (1.1.7), the required supply of energy to the accelerated plasma.

Stress tensor

We now transfer to the law of conservation of the momentum for the electromagnetic field. We start with the ampere force, acting on the investigated volume:

$$\mathbf{F} = \frac{1}{c}\iiint[\mathbf{j},\mathbf{H}]dV\,. \tag{1.1.8}$$

Maxwell showed that integral (1.1.8) can be transformed in the general form to a surface integral. This means that the volume force (1.1.8) can be regarded as the result of the effect of some surface force, namely:

$$\mathbf{F} = -\oiint \vec{\mathbf{T}}^{(H)} d\mathbf{S}. \tag{1.1.9}$$

The minus sign here is associated with the selection of the vector of the surface element $d\mathbf{S}$ and the selection of the sign of $\vec{T}^{(H)}$ – the Maxwell strength tensor for the magnetic field.

In the Cartesian coordinates (see for example [53])

$$\vec{T}^{(H)} = \begin{vmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{vmatrix} = \begin{vmatrix} \frac{H^2}{8\pi} - \frac{H_x^2}{4\pi}; & -\frac{H_x H_y}{4\pi}; & -\frac{H_x H_z}{4\pi}; \\ -\frac{H_x H_y}{4\pi}; & \frac{H^2}{8\pi} - \frac{H_y^2}{4\pi}; & -\frac{H_y H_z}{4\pi}; \\ -\frac{H_x H_z}{4\pi}; & -\frac{H_y H_z}{4\pi}; & \frac{H^2}{8\pi} - \frac{H_z^2}{4\pi}; \end{vmatrix} \tag{1.1.10}$$

If the axis of the coordinates in the given point are selected in such a manner that the magnetic field has a single component of the field $H_Z = H$, then

$$\vec{T}^{(H)} = \begin{vmatrix} \frac{H^2}{8\pi} & 0 & 0 \\ 0 & \frac{H^2}{8\pi} & 0 \\ 0 & & -\frac{H^2}{8\pi} \end{vmatrix}. \tag{1.1.11}$$

This shows that if the normal to the surface element is normal to the z axis (Fig. 1.1.2a), the force is directed in the direction opposite to the normal and is equal to:

$$d\mathbf{F} = -\frac{H^2}{8\pi} d\mathbf{S} \equiv -p_M d\mathbf{S}, \tag{1.1.12}$$

i.e. the magnetic field, tangential to the boundary, exerts a pressure on this boundary. $p_H = H^2/8\pi$ is usually referred to as the magnetic

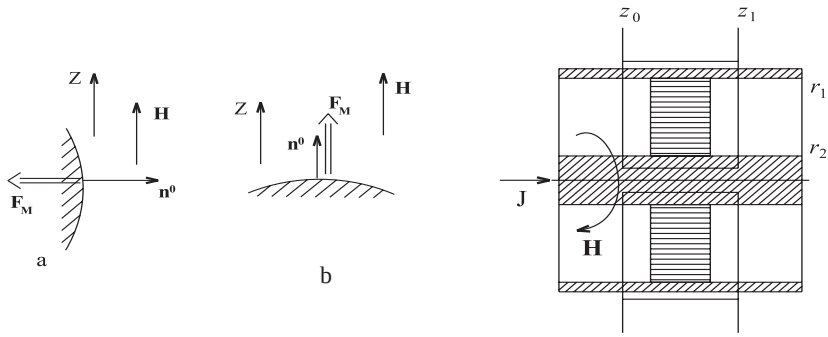


Fig. 1.1.2. (left) Effect of the magnetic field on the surface of the plasma volume; a) field parallel to the surface; b) field normal to the surface.
Fig. 1.1.3 (right). Scheme for calculating the force acting on the plasma bunch in a pulsed gun.

pressure[1]. However, if the normal to the surface element is directed along the *z* axis, then it is subjected to the effect of the tension force in the same direction (Fig. 1.1.2b), equal to:

$$d\,\mathbf{F} = \frac{H^2}{8\pi} d\,\mathbf{S}. \qquad (1.1.13)$$

Equations (1.1.12) and (1.1.13) can be 'felt' by holding two magnets.If the magnets make contact through the same poles they will be repulse each other and this corresponds to (1.1.2) if the surface is represented by the median plane *S*. On the other hand, if the magnets are brought together by the different poles, they will attract each other (1.1.13).

The effect of Maxwell tensions can be conventionally described if the magnetic lines of force are regarded as identical with the stretched and at the same time swollen rubber cords. These cords will, on the one hand, try to shorten (tension) and on the other hand pull away from each other (pressure).

If the boundary intersects the lines of force under an angle, then the force acting on the boundary will be determined by the composition of the ‘shortening’ and ‘repulsive’ capacities of the field.

As an example of application of the stress tensor, we consider acceleration of a plasma bunch in a coaxial pulsed gun (Fig. 1.1.3)

[1]It should be mentioned that the magnetic field H = 5000 Oe = 0.5 T generates a pressure of $p_M \approx 1$ kg/cm^2. However, the magnetic field of 100 kOe = 10 T – and such fields are required for a number of the PDS – generates a pressure of 400 kg/cm^2 acting on the coils of the PDS. The fields in the megaoersted range can be produced only in the pulsed regime.

which takes place as a result of the interaction of the azimuthal magnetic field H_θ with the radial component of the density of current j_r, flowing between the electrodes. Acceleration can be interpreted on the basis of (1.1.3a) also as the result of a decrease of magnetic pressure. The value of F_z will be calculated using a toroidal volume embracing the entire plasma bunch. In this case, only the azimuthal component of the fields differs from zero and, consequently

$$F_z = \iint n_z^0 \frac{H_\theta^2}{8\pi} dS = 2\pi \int_{r_i}^{r_2} \left(H_\theta^2 \Big|_{z_0} - H_\theta^2 \Big|_{z_1} \right) \frac{rdr}{8\pi}, \tag{1.1.14}$$

i.e. this includes the magnetic pressure $p_H = H^2/8\pi$ at the edges of the volume. If the entire current is closed through the plasma, then at $z = z_1$ the magnetic field converts to zero and therefore

$$F_z = \frac{1}{4} \int_{r_1}^{r_2} H_\theta^2 \Big|_{z_0} rdr. \tag{1.1.15}$$

Section z_0 was selected in front of the bunch, i.e. in the area where the current flows only through the electrodes. Consequently, in this section:

$$H_\theta = \frac{2J}{cr}, \tag{1.1.16}$$

where J is the total discharge current. Substituting (1.1.16) into (1.1.15) we obtain, irrespective of the detailed pattern of the flows in the plasma)

$$F_z = \frac{J^2}{c^2} \ln \frac{r_2}{r_1}. \tag{1.1.17}$$

These considerations also hold in the case of a stationary discharge in a coaxial system with cylindrical electrodes.

We now consider the effect of the electrical field on the volume charge

$$\mathbf{F} = \int q_e \mathbf{E} dV\,. \tag{1.1.18a}$$

The integral is taken over the entire space occupied by the charge. It appears that the Maxwell equations can also be used in this case to reduce the force **F** to surface force in accordance with (1.1.9)

$$\mathbf{F} = \iint \overleftrightarrow{\mathbf{T}}^{(E)} d\mathbf{S}. \tag{1.1.18b}$$

and the structure $\vec{\vec{T}}^{(E)}$ is completely identical with (1.1.10) if **H** is substituted by **E**.

The electromagnetic forces, applied to the volume, make the volume can rotate. The momentum of the electrondynamic force in relation to the *z* axis is:

$$M_z = \iiint f_\theta r dV = \frac{1}{c}\iiint r(j_z H_r - j_r H_z) dV . \qquad (1.1.19a)$$

Here f_0 is the azimuthal component of the ampere force.

If the field axially symmetric, then from the Maxwell equations (1.1.1a) in the stationary case we have:

$$j_z = \frac{c}{4\pi}\frac{1}{r}\frac{\partial r H_\theta}{\partial r}; \quad j_r = -\frac{c}{4\pi}\frac{\partial H_\theta}{\partial z}. \qquad (1.1.19b)$$

Using these relationships and the continuity equation div **H** = 0, equation (1.1.19a) can be transformed to the following form

$$M_z = \frac{1}{4\pi}\oiint (n_r^0 r H_r H_\theta + n_z^0 r H_z H_\theta) dS. \qquad (1.1.19c)$$

Here, as in (Int.1.9), we use the external normal to the external surface. Equation (Int.2.1b) shows that the momentum of the forces forms only when the azimuthal H_θ and poloidal (H_r, H_z) fields exist simultaneously.

1.1.3. Morphology of magnetic fields [54]

The processes in plasma systems depend very strongly on the morphology (geometry, structure) of the magnetic lines of force. From the physical viewpoint, this is associated with the high mobility of the particles, in particular, electrons, along the lines of force and the comparatively low mobility of these particles across this lines. Therefore, calculation of the morphology of the magnetic field is an essential stage of development of any plasma dynamic system with a magnetic field.

According to the definition, *magnetic lines of force* **r**(*s*) have at every point the tangent *d***r**, parallel to the strength of the field **H**

$$d\mathbf{r} \| \mathbf{H}. \qquad (1.1.20a)$$

In the Cartesian coordinate system this can be written in the following form

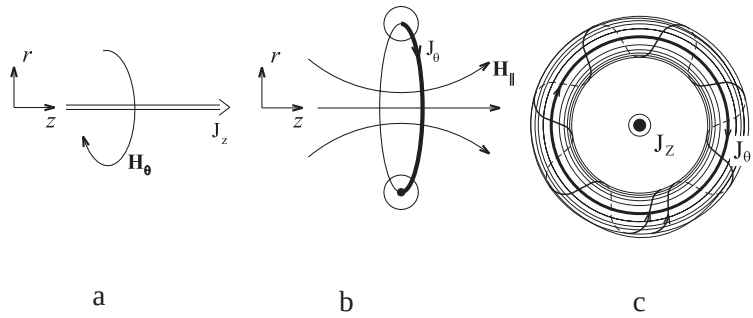


Fig. 1.1.4. The simplest axisymmetric field: a) azimuthal magnetic field of a straight filament with current; b) the poloidal magnetic field of a ring with current; c) the axisymmetric three-component magnetic field whose lines of force do not usually close with each other and do not extend to infinity.

$$\frac{dx}{H_x} = \frac{dy}{H_y} = \frac{dz}{H_z}. \qquad (1.1.20b)$$

Correspondingly, in the cylindrical coordinate system

$$\frac{dr}{H_r} = \frac{rd\theta}{H_\theta} = \frac{dz}{H_z}. \qquad (1.1.20c)$$

In physics courses, attention is usually given to two very partial cases of the magnetic field: the magnetic field of a straight filament with current (Fig. 1.1.4a) and the field of a ring with current (Fig. 1.1.4b). These fields are symmetric. The field with a single azimuthal component H_0 is referred to as the azimuthal field, and the field of the components H_r, H_z, at $H_\theta = 0$ is termed the poloidal field.

In a general case, the magnetic lines of force may be of different types. However, it should be mentioned here that the 'school' assumption according to which the magnetic force fields are either closed or extend to infinity is completely inaccurate. This holds only for the fields shown in Fig. 1.1.4a, b. This however does not apply to the symmetric field formed by combining the filament with current and a ring (Fig. 1.1.4c). Here, the main part of the lines of force does not close on itself, and only the line positioned along the axis of the system extends to infinity (I.E. Tamm, 1928, [53]).

Magnetic surfaces

In addition to the concept of the magnetic line of force, another important concept in morphology is the concept of the 'magnetic surface'. This is the surface formed by magnetic lines of force. Here

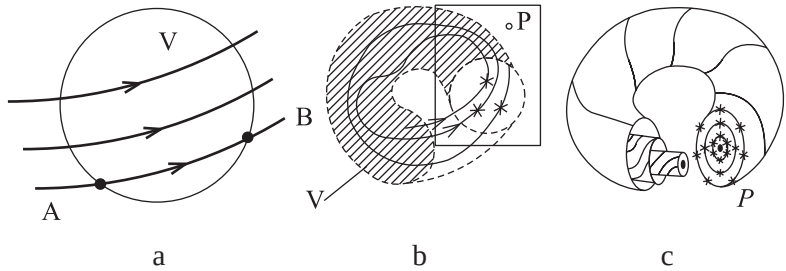


Fig. 1.1.5. The concept of the 'magnetic surface': a) working volume *V*, filled with 'segments' of the lines of force; b) the toroidal volume *V*, whose boundary is not intersected by the magnetic line; *P* is the 'imaging' plane; c) the toroidal magnetic field with magnetic surfaces.

we have two completely different situations, depending on whether the working volume V_0 contains only 'segments' of the lines of force (Fig. 1.1.5a) or whether the lines of force are completely situated in such a volume (Fig. 1.1.5b).

In the first case, we have no *apriori* criterion for construction (composition) of the magnetic surfaces from lines of force. They can be completely arbitrary. The situation in the second case is completely different. The structures of the magnetic field in the toroidal volume can differ greatly (see Appendix A at the end of this book). However, in most cases, fields consisting of well-defined magnetic surfaces are of special interest. This means (Fig. 1.1.5c) that on the 'imaging' plane *P*, intersecting the torus, there are (closed) lines μ which are characterised by the fact that the lines of force passing through them intersect the imaging plane *P* again at points of this line μ although, generally speaking, not at the points from which they originated.

Therefore, it appears that in many cases it is not possible to construct fields in which we could define the system of inserted well-defined (in the sense of the image on the plane *P*) surfaces on which we can place all the lines of force of the given toroidal volume (this problem is discussed in greater detail in Appendix A). As shown later, only the toroidal fields, consisting of such inserted surfaces[2], may act as efficient 'vessels' ('traps') for sustaining plasma (see section 1.5 of this chapter, and also section 2.4).

Two components of symmetric fields

An important group is formed by the axisymmetric fields whose components do not depend on the azimuth θ

[2]Or slightly 'eroded' surfaces, i.e. thin layers

$$\mathbf{H} = (H_r(r, z), H_\theta(r, z), H_z(r, z)). \tag{1.1.21a}$$

It may easily be seen that any such field should be treated as a superposition of two independent fields: poloidal $\mathbf{H}_{pol} = (H_r(r, z), 0, H_z(r, z))$ and the azimuthal field $\mathbf{H}_\theta(r, z)$, i.e. in a general case

$$\mathbf{H} = \mathbf{H}_{pol} + \mathbf{H}_\theta. \tag{1.1.21b}$$

In fact, every one of these fields satisfies separately the system of the Maxwell equations:

$$\operatorname{rot}\mathbf{H}_{pol} = \frac{4\pi}{c}\mathbf{j}_\theta; \quad \operatorname{div}\mathbf{H}_{pol} = 0 \tag{1.1.22a}$$

$$\operatorname{rot}\mathbf{H}_\theta = \frac{4\pi}{c}\mathbf{j}_{pol}; \quad \operatorname{div}\mathbf{H}_\theta = 0. \tag{1.1.22b}$$

Here $\mathbf{j}_{pol} = (j_r(r, z), 0, j_z(r, z))$. Attention should be given to the 'crossed' relationship $(\mathbf{H}_{pol}, \mathbf{H}_\theta) \leftrightarrow (\mathbf{j}_\theta, \mathbf{j}_{pol})$.

The 'function of the flux' can be introduced $\psi(r, z)$ taking into account the fact that the poloidal components of the field satisfy the equation $\operatorname{div}\mathbf{H}_{pol} = 0$. This function is connected with the components of the field by the relationship:

$$H_r = -\frac{1}{r}\frac{\partial\psi}{\partial z}; \; H_z = \frac{1}{r}\frac{\partial\psi}{\partial r}. \tag{1.1.23a}$$

Comparing (1.1.5c) and (1.1.23a), we obtain that at $A_r = A_z = 0$,

$$\psi = \mathbf{r}A_\theta. \tag{1.1.23b}$$

It may easily be seen that the magnetic flux Φ, passing through the ring restricted by the radii r_1, r_2 and located in the plane z = const, is connected with ψ by the relationship:

$$\Phi(r_1, r_2, z) = 2\pi\int_{r_1}^{r_2} H_z(r, z)\,dr = 2\pi\left(\psi(r_2, z) - \psi(r_1, z)\right). \tag{1.1.24}$$

It should be mentioned that from the equations (1.1.1) and (1.1.23a) we obtain the equation for the function of the magnetic flux (at $\partial\mathbf{E}/\partial t = 0$):

$$\Delta^*\psi = r\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial\psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2} = -\frac{4\pi}{c}rj_\theta. \qquad (1.1.25)$$

Here Δ^* is the modified Laplace operator[3].

At the same time, the azimuthal field, characterised by the component $H_\theta(r, z)$, as indicated by the Maxwell equation (1.1.1a), is reduced to the equation

$$\frac{\partial}{\partial r}rH_\theta(r,z) = \frac{4\pi}{c}rj_z(r,z). \qquad (1.1.26a)$$

Introducing the total current $J_z(r, t)$, passing inside a circle with radius r at the given value of z

$$J(r,z) = 2\pi\int_0^r j_z(r,z)rdr, \qquad (1.1.26b)$$

we can integrate the equation (1.1.26a) and obtain the equation, generalising the well-known equation for the field of the straight filament:

$$H_\theta = \frac{2}{cr}J(r,z). \qquad (1.1.26c)$$

It is easy to verify that the superposition of the two fields may be used to describe the general field also at other types of symmetry: flat, when the strength of the field does not depend on the Cartesian coordinate z, and the helical symmetry in the case in which the components of the strength of the field depend only on r and $\omega \equiv \theta - \alpha z$, where $\alpha = 2\pi/L$, L is the pitch of the helix (see below).

Lines of force of axisymmetric fields

The previously mentioned relationship (1.1.24) between ψ and $\mathbf{\Phi}$ shows that the equation ψ = const describes the magnetic surfaces or the lines of force of the poloidal field in the plane (r, z). This claim can easily be verified also formally.

For this purpose, we write the system of equations (1.1.20c), which determine the lines of force, in the form

[3]The conventional Laplace operator in the case of axial symmetry in the cylindrical coordinate has the form:

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}r\frac{\partial f}{\partial r} + \frac{\partial^2 f}{\partial z^2}.$$

$$H_z dr - H_r dz = 0; \qquad \frac{dz}{d\theta} = \frac{rH_z(r,z)}{H_\theta(r,z)}. \tag{1.1.27}$$

Substituting into the first equation the expression (1.1.23a) for H_r and H_z, we immediately obtain the first integral $\frac{\partial \psi}{\partial r} dr + \frac{\partial \psi}{\partial z} dz = 0,$ i.e.

$$\psi(r, z) = \text{const.} \tag{1.1.28}$$

This equation defines the toroidal surfaces on which the lines of force are situated. In order to determine completely the form of the force of lines, it is necessary to solve the second equation (1.1.27), substituting here the dependence $r = r\ (z, \psi)$, determined from the equations (1.1.28). The resultant equation can be solved easily by the numerical procedure, and for the analytical solution it is usually necessary to introduce the toroidal coordinates of some.

Previously, we have discussed and showed (Fig. 1.1.5c) that the toroidal magnetic field can often be regarded as a system of inserted surfaces, i.e. after passing around the torus, the line of force intersects the imaging plane P at the points situated on the same curve. In the axisymmetric case, the equations $\psi(r, z)$ describes strictly the magnetic surfaces which transfer into themselves after passing around the torus, if in addition to the poloidal field there is also azimuthal field H_θ. The azimuthal field does not disrupt the symmetry of these magnetic surfaces $\psi = \text{const.}$, but the lines of force now transform to helices wound onto the toroidal surface $\psi(r, z) = \text{const}$. Naturally, it is assumed that the line determined by the equation $\psi = \text{const}$ in the plane (r, z) is closed.

The helical lines of force, situated on the magnetic surfaces $\psi = \text{const.}$, are usually not closed and are continuously wound and cover densely 'their' surface. The density of the coating indicates that the line of force passes as close as possible to any surface point. However, this infinitely thin line of force does not cover the entire surface. This is due to the fact that the set of bypasses around the torus is countable, and to coat the entire surface we should have a continuum of the bypasses. In fact, this comment is of the formal nature because the 'real' thickness of the line of force is of the order of the electronic Larmor radius, i.e. it is finite. Such a 'thick line' completely covers the magnetic surface.

It should be stressed that in the investigated doubly connected axisymmetric case the system of the inserted magnetic surfaces exist *only in the presence of the azimuthal electrical current in the volume.* This situation is realised in tokamaks (section 1.7).

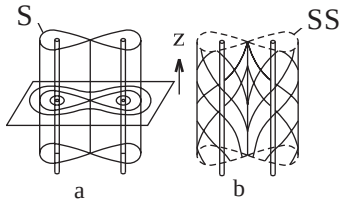


Fig. 1.1.6. Concept of the separatrix: a) the structure of the magnetic field of direct bars with the currents with the same direction, $H_z = 0$. *S* is the separatrix; b) the same for the superposition of the field H_z, *SS* is the separatrix surface.

Separatrix surfaces

Another fundamental concept of the morphology of the magnetic fields is the concept of the 'separatrix surface', i.e. the surface separating the surfaces (or the lines of force) which cannot be transferred into each other without disrupting the structure of the surfaces. This can be conveniently illustrated on the example of flat (i.e. independent of *z*) fields whose magnetic surfaces are formed by two straight parallel currents of the same direction (Fig. 1.1.6). The lines of force of this field are divided into three groups:

a) those which close around the left conductor,

b) those which close around the right conductor,

c) the lines covering both conductors.

The boundary between these groups is the self-intersecting line of force in the form of the number eight (lemniscate of Bernoulli). This line is referred to as the separatrix. The self-intersection of the separatrix is due to the fact that in the centre of the system both components of the field H_z, H_y convert to zero, and the equation of the line of force (1.1.20b) does not give an unambiguous solution, i.e. in this case

$$\left.\frac{dy}{dx}\right|_0 = \frac{H_x}{H_y} = \frac{0}{0}. \tag{1.1.29}$$

Since the magnetic field outside the conductor is vortex-free, the field satisfies the Laplace equation and it can easily be verified that, expanding the scalar potential ϕ_M in the vicinity of $x = 0$ with respect to the degrees *x* and *y*, the branches of the separatrix at zero are mutually perpendicular.

Now, a direct magnetic field $\mathbf{H}_z$ = const., oriented along the axis (this field is the analogue of the azimuthal field $\mathbf{H}_\theta$ in the axisymmetric system) is superposed on the field of two conductors. Consequently, we obtain smooth cylindrical magnetic surfaces everywhere, with the exception of the one which is formed by the separatrix. It is important to stress that the separatrix surfaces do not necessarily surround the regions containing conductors with current. A suitable example here are the helical fields, shown in Fig. 1.1.8 which will be discussed later.

The field of the system of parallel direct bars with current

The separatrices may have greatly differing forms, even in the symmetric field[4]. Here we note only the configuration of the magnetic lines of force of the field of the system of N straight bars with equal currents, distributed around a circle at equal distances from each other. This field can be described quite easily by a complex potential. We explain this procedure.

It was mentioned previously that in the region where $\mathbf{j} = 0$ the magnetic fiels can be described by both scalar potential ϕ_m and by vector potential $\mathbf{A}$. For a flat field, at $H_z = 0$ it is sufficient to assume that only component A_z, which corresponds to the direction of current which forms the field, differs from zero. The equalities $A_z(x, y) = \text{const}$. are the equations of the lines of force. From (1.1.5c) we obtain

$$H_x = \frac{\partial A_z}{\partial y}, \quad H_y = -\frac{\partial A_z}{\partial x}.$$

At the same time, $\mathbf{H} = \nabla\phi_m$. Therefore,

$$\frac{\partial \phi_m}{\partial x} = \frac{\partial A_z}{\partial y}; \quad \frac{\partial \phi_m}{\partial y} = -\frac{\partial A_z}{\partial x}. \tag{1.1.30a}$$

This is nothing else but the Cauchy–Rieman conditions for the complex function, known in the theory of the functions of the complex variable

$$w(z) = A_z + i\varphi_m. \tag{1.1.30b}$$

Finally, taking into account the equation for the field of the straight filament (1.1.26c)

$$H = \frac{2J}{c\,|\mathbf{r} - \mathbf{a}|},$$

it is quite easy to obtain the explicit expression for w for a single conductor

$$w = A_z + i\phi_m = \frac{2J}{c}\ln(z - a), \quad z = x + iy. \tag{1.1.30c}$$

Here a is the complex coordinate of the filament. The coordinates of the system of N bars symmetrically distributed around the circle have the form

[4]If there is no symmetry, the form of the separatrices is very complicated (see below).

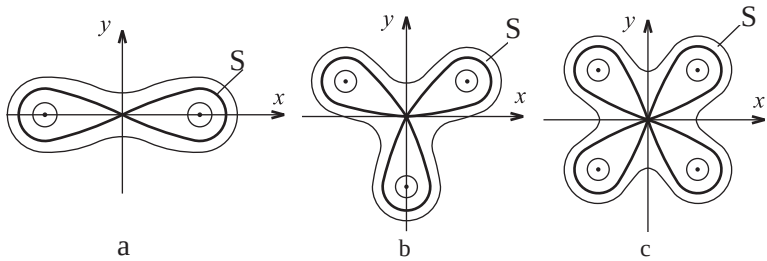


Fig. 1.1.7. Lines of force of the magnetic field of two parallel straight currents (a); lines of force of the field of three bars (b); the lines of force of four bars with a current flowing in the same direction (c), *S* are separatrices.

$$a_k = \mathrm{R}\ \exp\left\{\frac{2\pi}{N}k\right\}, \quad k = 0,\ldots,N-1. \tag{1.1.31a}$$

It may easily be seen that the total field is described by the equation:

$$w_N = A_{z,N} + i\phi_{m,N} = \frac{2J}{c}\ln(z^n - R^n). \tag{1.1.31b}$$

Consequently, transferring to the polar coordinates ($z = re^{i\theta}$),

$$A_{z,N} = \frac{J}{c}\ln\left(r^{2N} - 2r^N R^N \cos N\theta + R^{2N}\right), \tag{1.1.31c}$$

and, consequently, the equations of the lines of force of the field of *N* bars have the form

$$r^{2N} - 2r^N R^N \cos N\theta = \text{const}. \tag{1.1.31d}$$

The lines of force at $N = 2, 3, 4$ are shown in Fig. 1.1.7. The zero point at $N = 2$ is often referred to as the *x*-point.

Helical fields

Here we mention only one important group of fields having the characteristic feature that in this case the formation of the inserted magnetic surfaces does not require any current inside the volume of the field.

For this purpose, we examine symmetric 'straight' fields, formed by current-carrying helices wound on a straight circular cylinder (Fig. 1.1.8).

It is assumed that all the helices have the same pitch *L*. To simplify considerations, it is assumed that they are distributed at equal

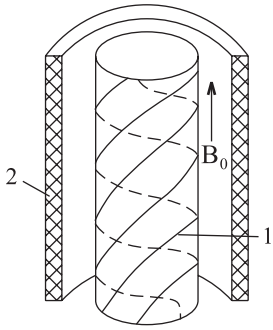


Fig. 1.1.8. A system of formation of the magnetic field of helical current-carrying conductors (1) and a homogeneous magnetic field (2).

distances around the azimuth. If the field is formed by N helices with the current with the same direction, these are 'N-pass' helical fields. This field is characterised by helical symmetry because it depends only on two coordinates (r, $\omega \equiv \theta - \alpha$, $\alpha = 2\pi/L$)

$$\mathbf{H} = (H_r(r, \theta - \alpha z), H_\theta(r, \theta - \alpha z), H_z(r, \theta - \alpha z)). \qquad (1.1.32a)$$

In a general case, the field can be described by two components of the vector-potential

$$\mathbf{A} = (0, A_\theta(r, \theta - \alpha z), A_z(r, \theta - \alpha z)). \qquad (1.1.32b)$$

The components of the magnetic field in this case are equal to:

$$\begin{aligned} H_r &= \frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z} = \frac{1}{r}\frac{\partial A_z}{\partial \omega} + \alpha\frac{\partial A_\theta}{\partial \omega} = \frac{1}{r}\frac{\partial}{\partial \omega}(A_z + \alpha r A_\theta) \\ H_\theta &= \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} = -\frac{\partial A_z}{\partial r} \\ H_z &= \frac{1}{r}\frac{\partial r A_\theta}{\partial r} - \frac{1}{r}\frac{\partial A_r}{\partial \theta} = \frac{1}{r}\frac{\partial r A_\theta}{\partial r}. \end{aligned} \qquad (1.1.33)$$

Substituting these expressions into (1.1.20c) we immediately determine the first integral

$$\Psi_{(r,\omega)} \equiv A_z + \alpha r A_\theta = \text{const}. \qquad (1.1.34)$$

This integral describes the magnetic surfaces of the helical field. The resultant morphology of the fields will be discussed for the case in which the field can be regarded as a sum of the homogeneous field H_0, oriented along the axis z, and the field of the n-th helical harmonics.

Such a vacuum field, satisfying the Laplace equation, is described by the scalar potential

$$\phi_m = H_0 z + \frac{h}{\alpha} I_n(n\rho)\sin n\omega, \quad \rho \equiv \alpha r, \tag{1.1.35a}$$

and the equation of the magnetic surfaces, corresponding to this field, has the form

$$\Psi = \frac{H_0}{2\alpha}\left(\rho^2 - \frac{2h}{H_0}\rho I_n'(n\rho)\cos n\omega\right) = \text{const.} \tag{1.1.35b}$$

Here I_n is the Bessell function of the imaginary argument.

Figure 1.1.9 shows the sections through the magnetic surfaces by the plane $z = 0$, i.e. the lines ψ = const for different n, obtained by the numerical solution of the equations (1.1.35b). Attention should be given to the fact that at all n there is a separatrix which separates the magnetic surfaces closed in the plane (r, ω), from the surfaces extending to infinity [54].

If these straight fields are now closed in a torus, we obtain magnetic configurations of stellarators proposed by L. Splitzer, as mentioned in the Introduction (section Int.3.3).

1.1.4. Metric characteristics of magnetic fields

Up to now, when discussing the morphology of the magnetic lines of force and surfaces, we have not mentioned the quantitative characteristics.

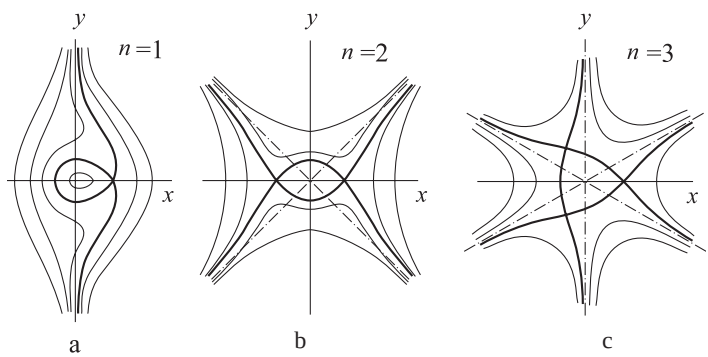


Fig. 1.1.9. Section of the magnetic helical surfaces at $n = 1$ (a); the same at $n = 2$ (b); the same at $n = 3$ (*c*). The ratio of the amplitude of the helical field to the homogeneous field is $h/H_0 < 1$.

1. *Angle of twist (rotational transform angle).* The angle of 'twist' or the 'rotational transform angle' is an important characteristic of toroidal magnetic surfaces. This angle is determined as the limit of the ratio of the sum of the angles of rotation $\Delta\theta_k$ of the imaging point in plane P (see Fig. 1.1.5) to N – the number of passes around the torus

$$\overline{\theta} = \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \Delta\theta_k . \tag{1.1.36}$$

The toroidal magnetic surface on which $\overline{\theta}/2\pi$ is a rational number, is referred to as the rational magnetic surface, and on this surface either all or some of the lines of force are closed. If not all lines are closed, and only some of them are, the latter are used as asymptotes for the non-closed lines. Correspondingly, the surface is referred to as irrational, if $\overline{\theta}/2\pi$ is irrational. If the angle $\overline{\theta}$ changes in transition from one magnetic surface to another, it is then concluded that the toroidal magnetic field has a 'shear'.

2. *The specific volume of the magnetic force tube*. It is important to mention another metric parameter which is often used in examining systems of either finite segments or closed magnetic lines of force. This is the so-called specific volume of the magnetic force tube (or, in other words, the 'line of force').

$$U = \lim_{S\to 0} \frac{V}{\Phi}. \tag{1.1.37a}$$

Here S is the cross-section of the magnetic tube, V is its volume, Φ is the magnetic flux inside the tube. Taking into account that the flux inside the tube is constant and equal to $\Phi = HS$, we can write

$$U = \lim_{S\to 0} \frac{1}{\Phi} \int_a^b S dl = \lim_{S\to 0} \int_a^b \frac{S dl}{\Phi} = \lim_{S\to 0} \int_a^b \frac{S dl}{SH} = \int_a^b \frac{dl}{H}, \tag{1.1.37b}$$

where a and b are the start and end of the tube. However, if the magnetic line of force is closed, then

$$U = \oint \frac{dl}{H}. \tag{1.1.37c}$$

Quantity U has a significant role in the criteria of stability of the static plasma configurations.

3. *The beta parameter*. In plasma systems with a magnetic field, the plasma generates its pressure p and the magnetic force its pressure

p_M. Consequently, we can introduce an important dimensionless 'beta parameter'

$$\beta = \frac{p}{p_M} = \frac{8\pi p}{H^2}. \tag{1.1.38}$$

1.1.5. Perturbation of the field morphology

We now examine the effect of perturbations on the morphology of the magnetic lines of force, in particular, the superposition of small random fields on the main field which is assumed to be symmetric[5].

Two completed different situations may form in this case.

(a) The working volume contains segments of the lines of force. In this case, if the strength of the perturbation field **h** is a small in comparison with the strength of the main field $\mathbf{H}_0$, the displacement of the line of force will be on the scale

$$\delta r \sim \int_A^B \frac{h}{H_0} ds \sim \frac{h}{H_0} L, \tag{1.1.39}$$

and at $h/H_0 \to 0$ $|\delta r| \to 0$. Here L is the length of the line of force.

(b) The situation greatly changes if we examine toroidal fields. This is due to the fact that in this case the lengths of the lines of force are usually infinite. Consequently, complete rearrangement of the morphology of the magnetic field can take place. If the amplitude of the pertubing field is small (or, more accurately, as small as possible), the rearrangement of the morphology is 'visually' regular, as shown clearly on the imaging plane. In particular, a peculiar resonance forms here between the harmonics of the perturbing field

$$\mathbf{A}_1 = \sum \mathbf{a}_n(r, z) \cos(n\theta + \alpha_n) \tag{1.1.40}$$

and the lines of force on the rational toroidal surfaces which are closed after n passages of the torus. As a result, the initially infinitely thin resonant surface is split and a chain of 'magnetic islands' forms on the imaged surface (Fig. 1.1.10) or a chain of magnetic fibres, if we are discussing the three-dimensional pattern. The width of the islands increases with the increase of the amplitude of perturbation

[5]This range of phenomena with special refernce fo the problems of controlled thermonuclear synthesis was studied initially by A.I. Morozov, L.S. Solov'ev and I.M. Gel'fand [54].

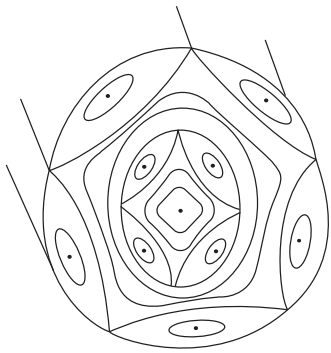


Fig. 1.1.10. Typical scheme of splitting under the effect of weak perturbations of magnetic surfaces, as seen on the representative surface.

and zones with stochastic deposition of the imaging points ('dynamic chaos') appear. Separatrix magnetic surfaces are especially sensitive to perturbations. The non-linear stage of the rearrangement can be plotted only by means of numerical calculations. An example of the evolution of the pattern on the imaging plane for a helical field, perturbed by a resonance 'corrugated' field

$$\phi_m = z + 3I_3(3r)\sin 3(\phi - z) + \varepsilon I_0(3r)\sin 3z \qquad (1.1.41)$$

at different perturbations ε is shown in Fig. 1.1.11.

It may be seen that a three-petal rosette initially appears in the centre and subsequently the morphology starts to be disrupted as a whole. As indicated by the images, the separatrix is characterised by the highest sensitivity to perturbations. It is shown that in the zone of disruption of the separatrix, the coordinates of the intersection of the lines of force with the imaging plane appear to be random. This phenomenon is referred to as 'dynamic chaos'. This will be discussed

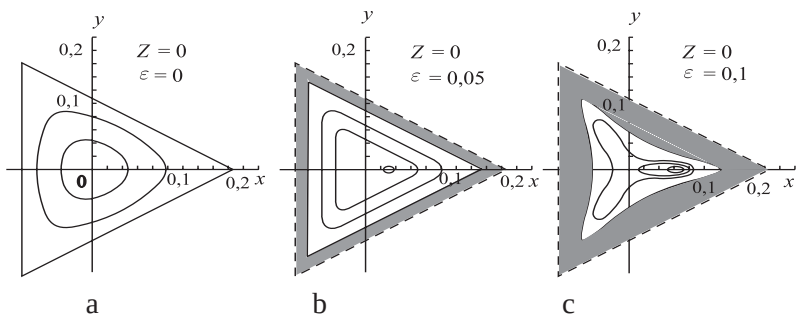
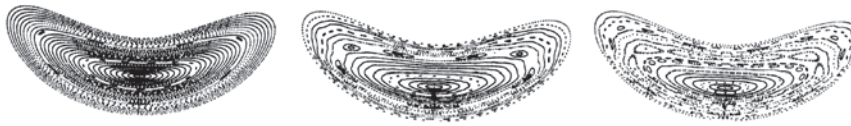


Fig. 1.1.11. Splitting of the surface of the field (1.1.41) at different perturbation amplitudes at ε = 0 (a), ε = 0.05 (b), ε = 0.1 (c); grey colour indicates the region of chaos.



a b c

Fig. 1.1.12. Splitting of the magnetic surfaces of the stellarator type field with the increase of plasma pressure: a) β = 0; b) β = 1.6%, c) β = 21%.

in greater detail in section 8.4. The appearance of perturbations of the symmetric fields may be associated with various circumstances, often unavoidable. A suitable example is the bending of a straight symmetric helical field into a torus or the perturbation of the magnetic field by the pressure of the plasma situated in it (Fig. 1.1.12). The splitting of the surfaces in a tokamak was detected for the first time in experiments by Mirnov [56].

1.2. Movement of particles in electromagnetic fields

Taking into account the fact that particles in many plasma dynamic systems are subjected to relatively rare collisions. The theory of movement of an individual particle in external electromagnetic fields is of fundamental importance for these systems. This movement is described by the Newton–Lorenz equation:

$$\frac{m d\mathbf{v}}{dt} = e\left(\mathbf{E} + \frac{1}{c}[\mathbf{v}, \mathbf{H}]\right); \quad \frac{d\mathbf{r}}{dt} = \mathbf{v}. \tag{1.2.1a}$$

The following Lagrange function corresponds to the above equation:

$$L = \frac{mv^2}{2} + \frac{e}{c}\mathbf{vA} - e\phi, \tag{1.2.1b}$$

where **A** is the vector potential of the magnetic field and ϕ is the scalar potential of the electrical field. The theory of the solutions of the equation (1.2.1a) has been described in many reviews (see, for example [55]). Here, we shall mention only several factors important for further considerations. The numerical solution of the equations (1.2.1a) for any **E**- and **H**-fields is in most cases quite easy but the analytical approach, although it is efficient only for special **E**(*x*, *t*), **H**(*x*, *t*), describes much better many features of the investigated movements.

1.2.1. Conservation laws

If the potentials of the fields (ϕ and **A**) not depend on one of the four coordinates *x*, *y*, *z*, *t*, then the conservation laws (Neter theorem) corresponds to this coordinate. In particular, if the fields are characterised by azimuthal symmetry, i.e. are independent of θ, then the generalised momentum of the amount of motion is conserved in movement of the particle:

$$mr^2\dot{\theta} + \frac{e}{c}\psi \equiv D \equiv \text{const.} \tag{1.2.2}$$

Here *r*, θ, and *z* are the polar coordinates of the particle; ψ is the function of the magnetic flux. It should be mentioned that the conservation of the momentum does not depend on the variability of the fields with time nor on the presence of the axisymmetric electrical field.

The application of the conservation law (1.2.2) reduces the problem of movement of the particle in the space of three measurements (*r*, θ, *z*) to the problem of movement of the particle in the two-dimensional space with the coordinates (*r*, *z*). These equations have the form

$$m\ddot{r} = -\frac{\partial U(r,z)}{\partial r}; \quad m\ddot{z} = -\frac{\partial U(r,z)}{\partial z}, \tag{1.2.3a}$$

where $U(r, z)$ is the so-called generalised potential:

$$U \equiv e\phi + \frac{e^2}{2mc^2r^2}\left(D - \frac{e}{c}\psi\right)^2. \tag{1.2.3b}$$

The law of conservation of energy is fulfilled if the fields are constant with time

$$\frac{mv^2}{2} + e\varphi = \text{const.} \tag{1.2.4}$$

It should be stressed that the magnetic field is not included in the expression for the law of energy conservation.

1.2.2. Movement of the particle in uniform constant electrical and magnetic fields

If **E** and **H** are constant with spect to magnitude and direction, equation (1.2.1a) is linear in relation to the sought function $\mathbf{v}(t)$. Its properties will be investigated.

Let **H** be parallel to the axis *z*, and **E** is situated, for example, in plane (*y*, *z*). It is natural to expand the movement of the particle into two movements – parallel and normal to the magnetic field

$$\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}.$$

Similarly, it is also convenient to expand **E** (Fig. 1.2.1):

$$\mathbf{E} = \mathbf{E}_{\parallel} + \mathbf{E}_{\perp}.$$

Only $\mathbf{E}_{\parallel}$ acts along the axis *z* on the particle, and uniform accelerated motion is observed along this coordinate:

$$z = z_0 + V_{\parallel 0} t + \left(\frac{\mathbf{e} E_{\parallel}}{m} \right) \frac{t^2}{2}. \tag{1.2.5}$$

The movement across the magnetic field at $\mathbf{E}_{\perp} = 0$ is reduced to the rotation around the circle with a frequency

$$\omega_H = \frac{eH}{mc} \tag{1.2.6}$$

and the radius

$$\rho_H = \frac{v}{\omega_H}. \tag{1.2.7}$$

This rotation is referred to as Larmor or cyclotron rotation.

Thus, at $\mathbf{E}_{\perp} = 0$ and $\mathbf{E}_{\parallel} \neq 0$ movement takes place along a helix with the pitch increasing along *z*.

If $\mathbf{E}_{\parallel} = 0$, and $\mathbf{E}_{\perp} \neq 0$, then in addition to Larmor rotation, there is also the 'electrical' drift of the particles along the *x* axis with the velocity

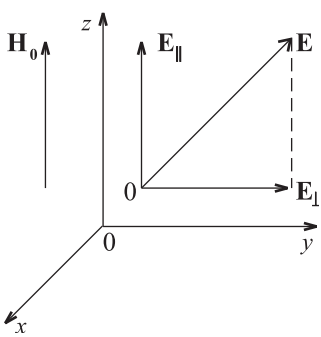


Fig. 1.2.1. Selection of the coordinate system for calculations in section 1.2.2.

$$\mathbf{u}_E = \frac{c[\mathbf{E},\mathbf{H}]}{H^2}. \qquad (1.2.8)$$

Consequently, as a result of the linearity of the equation (1.2.1) with the given assumptions, we can write $\mathbf{v}_\perp = \mathbf{u}_E + \mathbf{v}_1$, where $\mathbf{v}_1$ is the velocity of 'natural rotation' in the absence of the electrical field.

Attention should be given to the fact that the velocity of the drift does not depend on the particle charge nor on particle mass.

Equation (1.2.8) shows that the concept of the drift is rational (i.e. $\mathbf{u} < \mathbf{c}$) if $E_\perp < H$.[6] This always takes place if the magnetic field is strong. Otherwise, the presence of the magnetic field can be generally ignored. Evidently, if $H > E_\perp$, then only the rotation around the Larmor circle remains in the reference system moving with the drift velocity. The electrical field disappears in this reference system. This claim is general and directly follows from the Lorenz transform equations for the electromagnetic field. If $E > H$, we can transfer to a system in which the **H**-field disappears, and the particle moves only under the effect of the **E**-field.

Depending on the ratio of the drift velocity u_E and the velocity of movement around a circle with a radius $\rho_H = v_1/\omega_H$, the form of the trajectories of the particles differs (Fig. 1.2.2):

a) $u = 0$, $v_1 \neq 0$ – circle;
b) $u < v_1$ – trochoid;
c) $u = v_1$ – cycloid;
d) $u > v_1$ – hypocycloid, similar to a sinusoid.

However, if $\mathbf{v}_1 \to 0$, the trajectory degenerates into a straight line.

A particle which at some "initial moment" ceases to move (rest state) moves along the cycloid (Fig. 1.2.2c). The height of the cycloid is

$$h = \frac{2u}{\omega_H} = \frac{2c^2 Em}{eH^2}. \qquad (1.2.9)$$

In conclusion, it should be mentioned that if the electrical force is replaced by some other force, for example, the gravitational force, the movement of the charge particle will be identical to movement under the effect of the electrical field. In particular, the velocity of the drift is

[6] In order to transfer to the case in which $E_\perp > H$, it is necessary to consider the relativistic mechanics and, consequently, we obtain the general drift equation $\mathbf{v} = c\alpha\mathbf{K}$, where $\mathbf{K} = \frac{[\mathbf{E},\mathbf{H}]}{E^2 + H^2}$, $\alpha = \frac{2}{1+\sqrt{1-4K^2}}$.

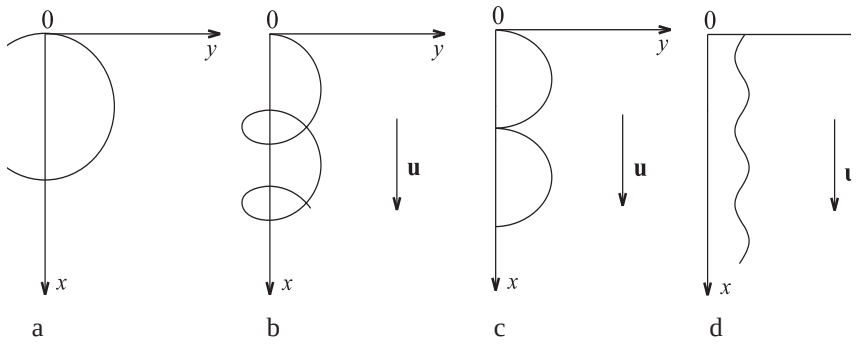


Fig. 1.2.2. Trajectory of the particles in relation to the ratio of the speed of Larmor rotation and the drift speed: a) circle, b) trochoid, c) cycloid; d) hypocycloid.

$$\mathbf{u}_F = \mathbf{c}\frac{[\mathbf{F},\mathbf{H}]}{\mathbf{e}H^2}. \qquad (1.2.10)$$

but now the direction of the drift depends on the sign of the charge.

1.2.3. Dynamics of particles in constant magnetic and alternating electrical field

A simple but important case is the movement of particles in uniform fields in which the magnetic field is constant and the electrical field depends on time *t* only

$$m\frac{d\mathbf{v}}{dt} - \frac{e}{c}[\mathbf{v},\mathbf{H}_0] = e\mathbf{E}(t). \qquad (1.2.11)$$

These equations are used for the analysis of linear waves in uniform cold plasma in a magnetic field (section 1.5). For this reason, we confine ourselves to the case in which the dependence of **E** on *t* is harmonic[7]

$$\mathbf{E}(t) = \mathbf{E}_1 e^{-i\omega t}, \qquad (1.2.12a)$$

and the process may be regarded as steady, i.e.

$$\mathbf{v} = \mathbf{v}_1 e^{-i\omega t}. \qquad (1.2.12b)$$

Here $\mathbf{E}_1$ and $\mathbf{v}_1$ are the amplitudes of oscillations.

[7]The real field $\mathbf{E}(t)_{real} = Re\mathbf{E}_1 e^{-i\omega t}$ is the real part of the field (1.2.12a).

Equation (1.2.11) with these assumptions can be solved efficiently in the Cartesian coordinates, using the direction of the magnetic field $\mathbf{H}_0$ as the *z* axis. The remaining axis are selected arbitrarily. Consequently

$$m\frac{dv_x}{dt}-\frac{e}{c}v_y H = eE_x;$$

$$m\frac{dv_y}{dt}-\frac{e}{c}v_x H = eE_y; \qquad (1.2.13)$$

$$m\frac{dv_z}{dt} = eE_z.$$

Substituting (1.2.12) into (1.2.13) we obtain a system of linear algebraic equations

$$-i\omega v_x - \omega_H v_y = \left(\frac{e}{m}\right)E_x,$$

$$-i\omega v_y + \omega_H v_x = \left(\frac{e}{m}\right)E_y, \qquad (1.2.14)$$

$$-i\omega v_z = \left(\frac{e}{m}\right)E_z,$$

From this:

$$v_x = \frac{e}{m}\frac{(-i\omega E_x + \omega_H E_y)}{\omega_H^2 - \omega^2};\ v_y = \frac{e}{m}\frac{(-\omega_H E_x - i\omega E_y)}{\omega_H^2 - \omega^2};\ v_z = -\frac{ie}{m\omega}E_z. \qquad (1.2.15)$$

or in the tensor form

$$\mathbf{v} = (v_x, v_y, v_z) = \frac{e}{m}\begin{pmatrix} -\frac{i\omega}{\Omega^2} & \frac{\omega_H}{\Omega^2} & 0 \\ -\frac{\omega_H}{\Omega^2} & -\frac{i\omega}{\Omega^2} & 0 \\ 0 & 0 & \frac{i}{\omega} \end{pmatrix}\begin{pmatrix} Ex \\ Ey \\ Ez \end{pmatrix}, \qquad (1.2.16a)$$

where $\Omega^2 \equiv \omega_H^2 - \omega^2$.

If we know the velocity, the displacement of the particle can be determined automatically from

$$\xi = \frac{i}{\omega}\mathbf{v} = \vec{\vec{S}}\mathbf{E}. \tag{1.116b}$$

Here $\vec{\vec{S}}$ is the displacement tensor:

$$\vec{\vec{S}} = \frac{ie}{\omega m}\begin{pmatrix} -\frac{i\omega}{\Omega^2} & \frac{\omega_H}{\Omega^2} & 0 \\ -\frac{\omega_H}{\Omega^2} & -\frac{i\omega}{\Omega^2} & 0 \\ 0 & 0 & \frac{i}{\omega} \end{pmatrix}. \tag{1.2.16c}$$

This equation will be used in the section 1.5.

1.2.4. Movement of a particle in a nonuniform high-frequency field

The plasma systems often contain two characteristic time or space scales. For example, in the Earth magnetosphere the electron rotates around a Larmor circle with a diameter of several centimetres and at the same time along a line of force, reflecting from some 'plugs' situated at a distance of ~10^4 km (see section 9.2). Evidently, in this case we are not interested in every Larmor circle; the displacement of the particles averaged out with respect to the rotation over a relatively long time t is more important.

The method of transformation of the exact equation of motion in the equation which contains only parameters of the averaged motion is referred to as the 'averaging method' [57].

In this section, we apply this method to the simplest example: to one-dimensional movement in an electrical field rapidly oscillating in time and non-uniform in space.

$$m\frac{d^2x}{dt^2} = eE_0(x)\sin\omega t. \tag{1.2.17}$$

This equation describes, for example, the movement of the electron in a flat high-frequency resonator (Fig. 1.2.3).[8] We are interested in the case in which the field carries out a large number of rotations during

[8]It should be mentioned that the modulation of density in the plasma volume can often play the role of a resonator Langmuir oscillations (chapter 8).

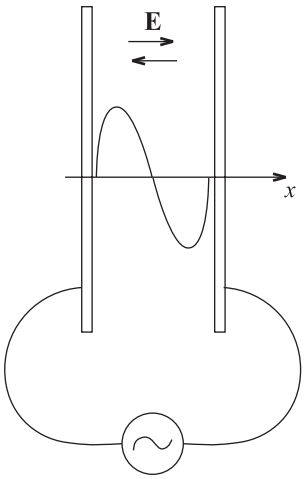


Fig. 1.2.3. Electron in a high-frequency resonator.

time τ, i.e. the time required for the electron to pass through the scale of the non-uniformity scale of the field:

$$\omega\tau \gg 2\pi. \tag{1.2.18}$$

To solve the equation (1.2.17) under the condition (1.2.18), the coordinate of the particle x is regarded as the sum of the coordinate of the average position $\bar{x}(t)$ and the high-frequency displacement $\xi(t)$:

$$x = \bar{x}(t) + \xi(t). \tag{1.2.19a}$$

ξ is selected accurately on the basis of the condition:

$$\int_0^T \xi(t)dt = 0, \quad T \equiv \frac{2\pi}{\omega}. \tag{1.2.19b}$$

Substituting (1.2.19a) into equation (1.2.17) and confining ourselves to 'linear' terms ~ξ:

$$\frac{d^2\bar{x}}{dt^2} + \frac{d^2\xi}{dt^2} = \frac{e}{m}\left[E(\bar{x}) + \frac{\partial E(\bar{x})\xi}{\partial(\bar{x})}\right]\sin\omega t. \tag{1.2.20a}$$

The formal integration of this equation with respect to t in the period T and subsequent division by T gives

$$\frac{d^2\bar{x}}{\partial t^2} = \frac{e}{m}\frac{\partial E_0}{\partial\bar{x}}\langle\xi\sin\omega t\rangle \tag{1.2.20b}$$

where

$$\left\langle \xi \sin \omega t \right\rangle = \frac{1}{T} \int_0^T \xi \sin \omega dt. \tag{1.2.20c}$$

The oscillating displacement ξ, included in this equation, if its smallness at $\omega \to \infty$ is taken into account, can be determined in the first approximation from the equation

$$\frac{d^2\xi}{dt^2} = \frac{e}{m} E_0(\bar{x}) \sin \omega t. \tag{1.2.21a}$$

assuming that $\bar{x}$ = const. Consequently, we obtain

$$\xi = -\frac{e}{m\omega^2} E_0(\bar{x}) \sin \omega t. \tag{1.2.21b}$$

This clearly shows that the value of ξ rapidly decreases with increasing ω.

Substituting this expression ξ into (1.2.20b) gives the required equation which determines the evolution of the averaged coordinate $\bar{x}$:

$$\frac{d^2\bar{x}}{dt^2} = \frac{e^2}{4m\omega^2} E_0 \frac{\partial E_0}{\partial x} = -\frac{\partial U_\omega}{\partial \bar{x}}, \tag{1.2.22a}$$

where the effective potential – referred to as the 'Miller potential' in this case, is equal to:

$$U_\omega = \frac{e^2}{4m\omega^2} E_0^2(\bar{x}). \tag{1.2.22b}$$

Thus, situated at an arbitrary point of the resonator, the electron oscillates and is displaced to the point with the minimum E_0^2.

The same method can be used to average out the trajectory of the particle in a strong magnetic field if the Larmor frequency $\omega \gg 1/\tau$, where τ is the flight time of the scale of the nonuniform magnetic field. Consequently, we obtain the highly effective 'drift approximation'.

1.2.4. Drift approximation

Let there be an electromagnetic field which slowly changes in space and with time. The slow rate of variation in space means that the values of the order of $(\rho_H/L)^2$ – the square of the ratio of the Larmor radius ρ_H

to the scale of non-uniformity of the magnetic field L, can be ignored. The slow rate of variation in time indicates that the terms of the order

$$\frac{1}{(\omega_H T)^2} \ll 1,$$

can be ignored; here T is the characteristic variation time of the field. In this case, in the first approximation, the movement of the particle is similar to the movement in uniform constant fields. Evidently, in the next approximation it appears that the centre of the Larmor circle has been displaced to some extent and the Larmor radius has changed. The equations describing the evolution of the parameters of the 'circle' are referred to as 'drift equations'. The strict derivation of these equations is carried by the method of averaging with respect to Larmor rotation. However, because of the three-dimensional form of the space, it is relatively cumbersome. Therefore, we shall not use it (see [55], [57]) and restrict ourselves to the analysis of the meaning of the terms of these equations. The drift equations are written for the components of velocity $v_{\|}$ and $v_{\perp}$, directed along and across the magnetic field respectively, and also for the radius vector **R** of some effective centre of the Larmor circle. Three equations for these quantities in the case of static **H** and **E**-fields – we need only this case for future considerations – have the form

$$\frac{v_{\perp}^2}{H} = \text{const}; \tag{1.2.23a}$$

$$\frac{m(v_{\|}^2 + v_{\perp}^2)}{2} + e\phi(\mathbf{R}) = \varepsilon = \text{const}; \tag{1.2.23b}$$

$$\frac{d\mathbf{R}}{dt} = v_{\|}\mathbf{h} + c\frac{[\mathbf{E},\mathbf{h}]}{\mathbf{H}} + \dot{\mathbf{R}}_{m,1} + \dot{\mathbf{R}}_{m,2}; \quad \mathbf{h} \equiv \frac{\mathbf{H}}{|\mathbf{H}|}. \tag{1.2.23c}$$

The first of these equations shows that the magnetic flux, passing through the Larmor circle, is preserved[9] regardless of the fact that the dimensions of the circle can greatly change during the drift process. It may easily be seen that this conservation law is nothing else but the conservation of the adiabatic invariant associated with the Larmor rotation

$$J_{\perp} = \oint p dq. \tag{1.2.24}$$

[9]In fact, $\Phi = \pi\rho^2 H = \pi\left(v_{\perp}Mc/eH\right)^2 H \sim v_{\perp}^2/H.$

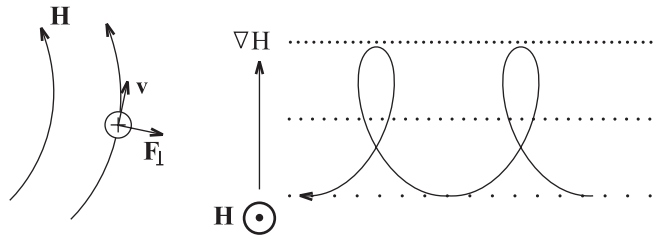


Fig. 1.2.4. Components of the magnetic drift: a) centrifugal drift; b) gradient drift.

Second equation (1.2.23b) is the law of energy conservation.

The first two terms in the third equation describe the movement along the magnetic field and the electrical drift which we both know.

The terms $\dot{\mathbf{R}}_{m,1}$ and $\dot{\mathbf{R}}_{m,2}$ are associated with the nonuniformities of the magnetic field and describe the 'magnetic drift'. For example, during movement of a particle along the distorted line of force (Fig. 1.2.4a), the particle is subjected to the effect of a centrifugal force causing 'centrifugal' drift, and the term $\dot{\mathbf{R}}_{m,1}$ describes this effect (see (1.2.10):

$$\mathbf{F}_{\text{centrifugal}} = \frac{mv_{\parallel}^2}{R_c}\mathbf{n}^0;$$

$$\dot{\mathbf{R}}_{m,1} = c\frac{\left[\mathbf{F}_{\text{centrifugal}}, \mathbf{h}\right]}{eH} = \frac{mc\mathbf{v}_{\parallel}^2}{eHR_c}\left[\mathbf{n}^0, \mathbf{h}\right]. \qquad (1.2.25)$$

Here R_c is the radius of curvature of the magnetic line of force; $\mathbf{n}^0$ is the vector of the normal to this line.

If the strength of the magnetic field varies in the direction normal to the lines of force, there is a gradient drift described by the term $\dot{\mathbf{R}}_{m,2}$ which is equal to

$$\dot{\mathbf{R}}_{m,2} = -\frac{\mathbf{v}_{\perp}^2}{2\omega_H^2}\left[\nabla\omega_H, \mathbf{h}\right]. \qquad (1.2.26)$$

The gradient drift is caused by the fact that the radius of curvature of the Larmor circle is smaller in the region of the stronger field and larger in the region of the weak field (see Fig. 1.2.4b).

If the magnetic fields is vortex-free (i.e. $\mathbf{j} = 0$), the radius of curvature ρ of the magnetic line of force is linked with the gradient $|H| = H$ by the relationship[10]

[10]This is evident for the magnetic field of direct current.

$$\nabla H = -\frac{H}{R_c}\mathbf{n}^0. \tag{1.2.27}$$

Substituting the equations (1.2.25), (1.2.26) into (1.2.23b) we finally obtain, taking (1.2.27) into account:

$$\frac{d\mathbf{R}}{dt} = v_{\parallel}\frac{\mathbf{H}}{H} + c\frac{[\mathbf{E},\mathbf{H}]}{H^2} + \frac{mc}{2eH^3}(2v_{\parallel}^2 + v_{\perp}^2)[\mathbf{H},\nabla H]. \tag{1.2.28}$$

The transition from the exact equation (1.2.1) to the system of the drift equations (1.2.23) denotes the transition from the three differential second order equations the three first order equations. This greatly simplifies the analysis of the particle dynamics.

In section 1.2.1 it was mentioned that in movement of particles each type of symmetry of the fields has its own strict law of conservation. The drift equations have the same properties. In the time-independent fields, as in the case of exact equations, the total energy of the particle is retained $\varepsilon = (m/2)(\mathbf{v}_{\perp}^2 + \mathbf{v}_{\parallel}^2) + e\phi$. If, in addition to this, the fields have axial symmetry, the analogue of the momentum of the amount of motion is preserved (Morozov, Solov'ev 1959 [55])

$$rA_{\theta}^2 = \psi(r,z) + \frac{mcv_{\parallel}(r,z)}{eH(r,z)}rH_{\theta}(r,z) = \text{const.} \tag{1.2.29}$$

Here $H \equiv |\mathbf{H}|$, $\mathbf{v}_{\parallel} = \sqrt{(2/m)(\varepsilon - e\phi) - \mathbf{v}_{\perp}^2}$, $\mathbf{v}_{\perp}^2 = \mathbf{v}_{\perp 0}^2(H/H_0)$. The zero denotes the values taken at some initial point. Equation (1.2.29) determines the trajectory of the movement of the particle in the plane *r*, *z*. Derivation of equation (1.2.29) and also examples of application of this equation are presented in [55]. This equation will be used in section 1.7.

1.2.6. Ion-optical approximation

It is assumed in the drift approximation that the Larmor radius is considerably smaller than the dimensions of the system. In most cases this is fulfilled for the electronic component of the plasma systems. However, the ionic component is often only slightly magnetised, i.e. the Larmor radius of the ions is greater or the same as the dimensions of the system. This case is referred to as the ion-optical case. In particular, if the Larmor radius ρ_H is large in comparison with the dimensions of the system (for example, engines with the azimuthal drift), then in the first approximation the effect of the magnetic field on the particles can be generally ignored and (1.2.1) can be replaced by

$$m\frac{d\mathbf{v}}{dt}=e\mathbf{E}. \tag{1.2.30}$$

This equation and also the overall equation (1.2.1) in a small section can be solved by the iteration method. The procedure may be described as follows. $\mathbf{r}_0$ and $\mathbf{v}_0$ denote the initial coordinates and velocity of the particle and the differential equation (1.2.30) has the form of the following integral equation:

$$\mathbf{r}(t)=\mathbf{r}_0+\mathbf{v}_0t+\frac{e}{m}\int_0^t dt'\int_0^{t'}\mathbf{E}(\mathbf{r}(t''))dt''. \tag{1.2.31}$$

In the theory of differential equations it is proved that the solution of the equation 1.2.31) is the limit of sequence of the functions $\{\mathbf{r}_n(t)\}$, where $\mathbf{r}_n$ is determined by the recurrent relationship:

$$\mathbf{r}_{n+1}(t)=\mathbf{r}_0+\mathbf{v}_0t+\frac{e}{m}\int_0^t dt'\int_0^{t'}\mathbf{E}(\mathbf{r}_n(t''))dt''. \tag{1.2.32}$$

If $\mathbf{E}(\mathbf{r})$ is the smooth function of the coordinates within the limits of the working volume, then it is sufficient to have a small number of iterations in order to obtain a solution with acceptable accuracy.

1.3. Block ('zero-dimensional') models of plasma systems

In many cases, the state of a plasma configuration can be described by one or several parameters which depend only on time $a(t)$. For example, we have already discussed the plasma pulse accelerators ('relsotrons'[11] and 'coaxial guns') in which a plasma bridge is accelerated. In this case, the main parameters which determine the process are the coordinates of the bridge $x(t)$ and the current in the electrical circuit $J(t)$ (Morozov, Artsimovich et al., etc.). Another similar example is the model of 'raking up' of a gas in *Z*-pinch proposed by Leontovich and Osovets. The model of this type will be referred to as the 'block' or 'zero-dimensional' model. They are examined in the sections 1.6 and 1.7.

In addition to the models in which the plasma acts as some conductor whose form does not change at al (relsotron bridges) or changes in the same fashion as in the *Z*-pinch, the zero-dimensional (block) models of plasma may consist of two or, in principle, a larger number of blocks. The situation was already discussed in section Intr.1 when deriving the

11 Relsotron is a rail-based gun (rail gun).

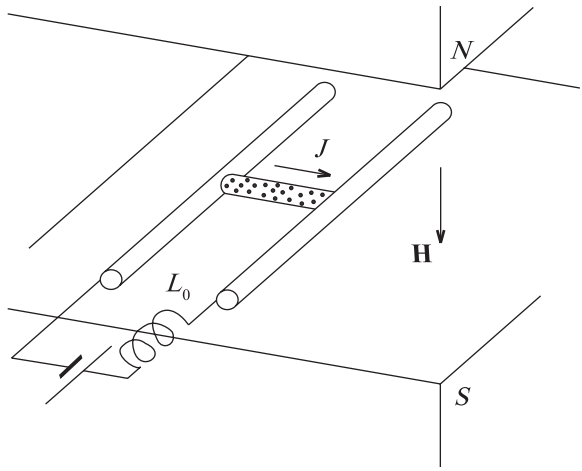


Fig. 1.3.1. Scheme of a magnetoelectric relsotron proposed by Morozov.

equation for Langmuir frequency. Two blocks – ionic and electronic – were investigated. Therefore, we shall discuss single-component and two-component block models.

The block models, in contrast to the one-particle models, make it possible in many cases not only (although only very approximately) to describe the plasma system as a whole but also determine clearly the principal value of quasi-neutrality.

In this section, attention will be given to two simple (methodical) examples of the zero-dimensional models which can be easily calculated to the end. One of the models is the acceleration of a single-component bridge with constant mass in a relsotron with an external magnetic field, the second model is an example of a two-component model, i.e. falling of a plasma bar in the gravity field in the presence of a magnetic field, parallel to the Earth surface. Section 1.6 and 1.7 discuss the relationship of these examples with more realistic models.

1.3.1. One-component model of a magnetoelectric relsotron [58]

Figure 1.3.1 shows the scheme of a relsotron with the external magnetic field and the electrical circuit, including the source of constant EMF ε and inductance L. It is assumed that the inductance of the external circuit is considerably higher than the inductance of the relsotron and, therefore, the latter can be ignored.

Denoting the mass and the length of the bridge by μ and ℓ, respectively we obtain:

$$\mu\frac{dv}{dt}=\frac{HJ\ell}{c},\quad \varepsilon=\frac{L}{c^2}\frac{dJ}{dt}+\frac{1}{c}Hv\ell. \tag{1.3.1a}$$

Excluding J, we obtain one equation

$$\frac{1}{\beta^2}\frac{dv^2}{dt^2}+v=\varepsilon_1, \tag{1.3.1b}$$

where

$$h=\frac{H\ell}{c};\quad \varepsilon_1=\frac{\varepsilon}{h};\ \beta^2=\frac{h^2c^2}{L\mu}=\frac{H^2\ell^2}{L\mu}.$$

Assuming that at the initial moment $x_0 = 0$; $V_0 = 0$, $J_0 = 0$, we obtain

$$J=c\sqrt{\frac{M}{L}}\sin\beta t;\ V=\frac{c\varepsilon}{H\ell}(1-\cos\beta t);\ x=\frac{c\varepsilon}{H\ell}\left(t-\frac{\sin\beta t}{\beta}\right). \tag{1.3.2}$$

This shows a strong analogy of between the movement of the particle in crossed fields (**E**, **H**) on the cycloid (Fig. 1.2.2c) and the movement of a bridge, if we compare $v_x \to v_x$, $J \to v_y$, $\beta \to \omega_H$. As indicated by equation (1.3.2), in this model which moves in a pulsed manner with the mean velocity:

$$u_E=\frac{c\varepsilon}{H\ell}.$$

However, the velocity is equal to the velocity of the electrical drift, since $u_E=cE/H$. The maximum velocity of the bridge is recorded at $\beta t_m = \pi$.

If a resistance is included in the investigated circuit, the oscillations attenuate and after some time the bridge moves with velocity u_E. The relsotrons will be also investigated in section 1.6.

1.3.2. Fall of a 'heavy plasma bar' in the magnetic field

The calculations the described below are interesting and at the same time they are superior from the procedure viewpoint in comparison with the number of calculations which will be carried out later.

A two-component model of the fall of a heavy plasma 'bar' in a transverse magnetic field is considered (Fig. 1.3.2).

We calculate the incidence process on the basis of the following scheme. At the initial moment, the blocks of electrons and ions are combined together and, for determinacy, they are stationary. Since

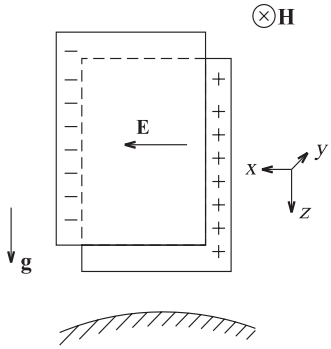


Fig. 1.3.2. Dynamics of a plasma bar in the magnetic field, $\mathbf{H} \perp \mathbf{g}$.

the gravitational force acts in the same manner on all the particles of the given type, and the magnetic field is assumed to be uniform and not 'sensing' plasma, the behaviour of all particles in a block can be described by a single coordinate of the centre of gravity. Of course, it is assumed that the plasma is cold. The initial plasma block is considered in the form of a flat layer whose thickness is considerably smaller than the transverse dimensions. The *z* axis is directed along the gravitational force, the *y* axis along the magnetic field, and the normal to the block is the *x* axis.

The movement of the particles will be investigated in the drift approximation, and its incompleteness is compensated later taking into account the dielectric permittivity of the plasma. When starting the calculations, it is initially assumed that the electrons and ions started to drift under the effect of the gravitational force in the presence of the magnetic field but because of the differences in the sign of the charge they drift in different directions (1.2.10)

$$u_{i,x}^{(g)} = \frac{c}{e}\left(\frac{[M\mathbf{g},\mathbf{H}]}{H^2}\right)_x = -\frac{c}{e}\frac{Mg}{H}, \quad u_{e,x}^{(g)} = \frac{c}{(-e)}\left(\frac{[m\mathbf{g},\mathbf{H}]}{H^2}\right)_x = \frac{c}{e}\frac{mg}{H}. \quad (1.3.3)$$

The displacement of the particles in the opposite directions results in the continuous increase of the charge density on the sides of the block

$$\dot{\sigma} = en_0(u_{i,x}^{(g)} - u_{e,x}^{(g)}) = -n_0 c\,\frac{g(M+m)}{H}. \quad (1.3.4a)$$

The density refers to the density on the side of the bar to which the drift of the ions is directed. Correspondingly, the rate of increase of the strength of the electrical field in the volume of the block is:

$$\dot{E}_x = \frac{4\pi\dot{\sigma}}{\eta} = \frac{4\pi n_0 c(M+m)g}{H_0\eta} = \frac{H_0 g}{c_A^2\eta}c. \quad (1.3.4b)$$

Here c_A is the Alfvén velocity, and η is the dielectric permittivity of the plasma in the direction normal to the magnetic field. These quantities are equal to

$$c_A = \frac{H}{\sqrt{4\pi n(M+m)}},\ \eta_0 = 1 + \frac{c^2}{c_A^2}.$$

More detailed discussion of c_A and η is described in the section 1.5.

Now, using the equation for the electrical drift (1.2.8), we determine the acceleration with which the bar falls across the magnetic field

$$\dot{u}_z = c\left(\frac{\left[\dot{\mathbf{E}},\mathbf{H}\right]}{H^2}\right)_z = c\frac{\dot{E}_x}{H} = g\frac{1}{\left(1+\frac{c_A^2}{c^2}\right)}. \qquad (1.3.5)$$

This shows that at $c_A^2 \gg c^2$, the acceleration tends to 0, and at $c_A^2 \ll c^2$, the acceleration approaches the normal freefall acceleration.

In the performed analysis, it is important to stress the 'two-stage' nature of the process of fall of the plasma, in particular, the particles initially moves under the effect of the gravitational force and the magnetic field and cause polarisation of the bar. Subsequently, after polarisation, the fall is characterised by equal acceleration. This is the result of quasi-neutrality.

1.4. Elements of classic corpuscular optics (CCO)

In the historical review, it was mentioned that plasma physics started with the examination of gas discharges. However, these are very complicated systems and the methods, described in the previous three paragraphs, are not sufficient for describing these systems. The physics of the charges will be investigated in the chapters 6 and 7.

As mentioned in Int.2, the significant advances in understanding the plasma processes were initially associated with studies of 'beam' structures – cathode and anode 'channel rays' – and their behaviour in the electrical and magnetic fields. However, these processes can already be described quite accurately by the previously mentioned methods. Therefore, in this section, we discuss these systems although they are formed in principle by particles of the same sign, i.e. they are not the plasma in the classic meaning of the word and, as is now often the case, form 'charged plasma'. However, the results obtained in section 1.4 will be used quite often later.

In the current stage, advances in science and technology in the 20th century were due to the achievements of corpuscular (electronic and ionic) optics. These are electronic radio valves – the basis of radio engineering and computing methods of the first half of the previous century, oscilloscopes and television tubes, electronic and ionic microscopes, accelerators of charged particles, ion cosmic engines, and many others. In the last couple of decades, it has become more evident that plasma offers completely new possibilities in controlling the particle fluxes and, most importantly, makes it possible to transfer (in most cases) from low-current CCO to high-current systems. Consequently, the ideology of 'plasma optics' has been developed and the efficiency of this work has been demonstrated on the example of stationary plasma engines for space systems and plasma lenses.

Therefore, acquaintance with the fundamentals of plasma corpuscular optics is essential for understanding advanced plasma dynamics.

In this section – as an introduction into this group of problems – we study the elements of classic (not plasma) CCO: namely, the production of ionic (electronic) fluxes and methods for focusing these fluxes by electrical and magnetic fields [59]. In classic – 'vacuum' CCO systems there are particles with the same sign. Therefore, the form of equations does not depend on whether we are discussing electronic or ionic fluxes. However, taking into account the fact that (chapter 5) we will discuss plasma optics, which studies the focusing of ionix fluxes, here we will discuss 'ionic' fluxes. This is also justified by the fact that classic ionic sources in particular are used widely in studies of controlled thermonuclear fuction and also as cosmic thrusters where they compete with plasma thrusters.

1.4.1. Ion sources

The simplest scheme of the ion source of positive ions is shown in Fig. 1.4.1. Here (1) is the ion generator. Usually, it is based on an electrical discharge of a specific type or can be represented by a heated sheet with the atoms of the working substance ionising after contact with the sheet (see section 7.1). The shaping electrode (1a) is placed in the immediate vicinity of the ion generator. The shaping electrode 'cuts out' narrow cylindrical or plate-shaped fluxes from a wide ion flux. The generator together with the shaping electrode is under a high positive potential U in relation to the external ('earthed') electrode and the cathode (3) – electron emitter (4), which have the potential $U_C \approx 0$. The accelerating electrode (2) has the potential $U_a < 0$. The reason for this arrangement is to prevent the electron flux travelling to

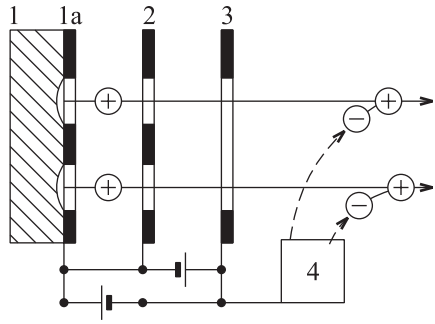


Fig. 1.4.1. Ionic source with a three-electrode ion–optical system: 1) ion emitter; 2) accelerating electrode; 3) body of the source; 4) neutraliser.

the accelerating and shaping electrodes from the side of the cathode. The selection of the shape of the electrodes and of orifices in the electrodes requires very detailed calculations. However, the relationship between the extracted ionic current J_i and the accelerating voltage can be deduced quite accurately with the accuracy to the multiplier ~1. We shall do this.

The Child–Langmuir equation

We calculate the simplest model of acceleration of the particles of the same sign in a gap between two flat electrodes where one of the electrodes (anode) is an ammeter of these particles (Fig. 1.4.2).

Of special interest for practice is the case in which the emissivity of the anode is not restricted and the current between the electrodes is restricted by the volume charge. The situation formed in the development of electronic valves with a heated cathode. At that time (1911) the volt–ampere characteristic of such a diode was calculated by Child and Langmuir. The dependence determined by them is often referred to as 'the 3/2 law'. Later (starting in the 30s), an identical

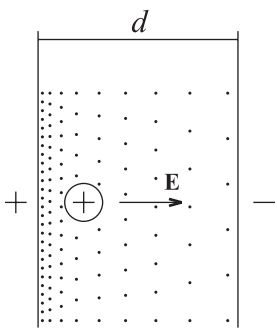


Fig. 1.4.2. Derivation of the Child–Langmuir equation.

scheme was introduced for producing ion beams in 'ion sources' ('ion injectors'). We shall discussed them later.

Langmuir considerations may be described as follows. In particular, since the particle flux is of low density, the collisions can be ignored and only the collective electrical field of the cloud of particles can be considered, i.e. the Maxwell equation is used in the form

$$\frac{d^2\phi}{dx^2} = -4\pi e n_i(x). \tag{1.4.1}$$

Assuming that the velocities of all the ions are the same, the density of the ions can be expressed through the density of the passing ionic current j = const and the velocity $v(x)$

$$\frac{d^2\phi}{dx^2} = -\frac{4\pi j}{v(x)}. \tag{1.4.2}$$

Ignoring the initial velocity of the ions on the anode, we obtain

$$v(x) = \sqrt{\frac{2e(U-\phi)}{M}}. \tag{1.4.3}$$

Here U is the potential of the anode in relation to the cathode.

The equation

$$\frac{d^2\phi}{dx^2} = -4\pi j\sqrt{\frac{M}{2e(U-\phi)}} \equiv -\frac{\partial W}{\partial\phi}. \tag{1.4.4}$$

is identical with the equation of the oscillations of a sphere in a potential well $W(\phi)$, where ϕ has the role of displacement, and x the role of time. This equation has the first integral (energy integral)

$$\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 - 2\cdot 4\pi j\sqrt{\frac{M}{2e}}(U-\phi)^{1/2} = \text{const}. \tag{1.4.5}$$

Assuming that the anode (because of the potential unboundedness of emission)

$$E = -\left.\frac{d\phi}{dx}\right|_{x=0} = 0, \quad \phi\big|_{x=0} = v, \tag{1.4.6a}$$

and on the cathode

$$\phi\big|_{x=d} = 0, \tag{1.4.6b}$$

after integration of the equation (1.4.5) and taking into account the boundary conditions (1.4.6) we obtain the '3/2 law'

$$j = \frac{1}{9\pi}\sqrt{\frac{2e}{M}}\frac{U^{3/2}}{d^2}. \qquad (1.4.7)$$

This conclusion and the resultant equation are interesting because of several reasons.

Firstly, here we introduce the 'self-consistent' field which is formed by the particles moving in this field. The concept of the 'self-consistent field' is the initial concept in the Vlasov kinetic model (chapter 4). Secondly, it is convenient to consider the relationship of the current density, defined by the ionic injector, with the density of the current defined by the plasma accelerator, where the current density is not restricted by the volume charge.

Thirdly, it may be seen that the flux of the pulse, carried by the ion flux, is determined by Maxwell tension

$$MnV^2 = \frac{E_C^2}{8\pi}, \qquad (1.4.8a)$$

where E_C is the strength of the electrical field on the cathode.

At the same time, the maximum thrust in an engine with a magnetic field, as shown in 1.1, is determined by the expression:

$$\left(MnV^2\right)_{\max} = \frac{H^2}{8\pi}. \qquad (1.4.8b)$$

Evidently, it is far easier to generate a magnetic field of $H = 10^4$ Oe than an electrical field with $E = 10^4$ absolute units = $3 \cdot 10^6$ V/cm. The ion and plasma accelerator is also discussed in section 10.4.

1.4.2. Examples of systems of vacuum corpuscular optics

Thus, we know the intrinsic volume charge of the particle beam, the fields are regarded as vacuum and the dynamics is investigated in the one-particle approximation.

Focusing of a particle flux by a uniform transverse magnetic field (Fig. 1.4.3)

Let *S* be a 'point ion source. The particles emitted from it have circular trajectories. If all particles are emitted in all directions, then there is no visible focusing, if we disregard the fact that the particles emitted at the same moment again return to the point *S* after time

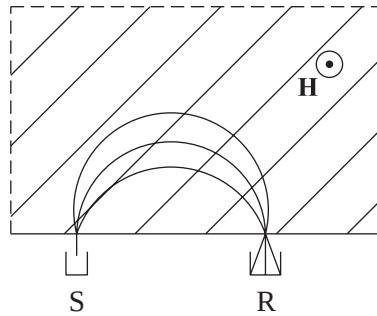


Fig. 1.4.3. '180°' focusing of particles in a uniform magnetic field; S) particle source; R) receiver.

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi Mc}{eH}. \tag{1.4.10}$$

However, there is no benefit from such focusing in stationary conditions. If a narrow beam with a small scatter with respect to the angles and velocity is emitted from *S*, then as indicated by Fig. 1.4.3, passing through half the Larmour circle, the beam is focused (compressed) and acquires a very small transverse size:

$$\delta \approx 2R\left(\frac{\delta v}{v} + \frac{\alpha^2}{2}\right), \tag{1.4.11}$$

where δv is the scatter of longitudinal velocity v, α is the half-angle of divergence of the flux emitted from point *A*.

Equation (1.4.11) shows that the blurring of the 'image' as a result of scatter of the velocity values is of completely different type, depending on whether the modulus of velocity or angular discontinuity is varied. In the former case, the relationship between δ and δv is linear, in the second case it is quadratic.

This shows that focusing in this case is carried out only with respect to the angles and not the modulus of the velocity. The modulus of the velocity and, more accurately, the amount of motion (mv), determines the position of the focus. This circumstance is clearly indicated by the equation of motion of the particles, if from dt we transfer to dl/v, where l is the length of the arc of the trajectory. Consequently, we obtain

$$v\frac{d}{dl}m\mathbf{v} = \frac{l}{c}\left[\mathbf{v},\mathbf{H}\right].$$

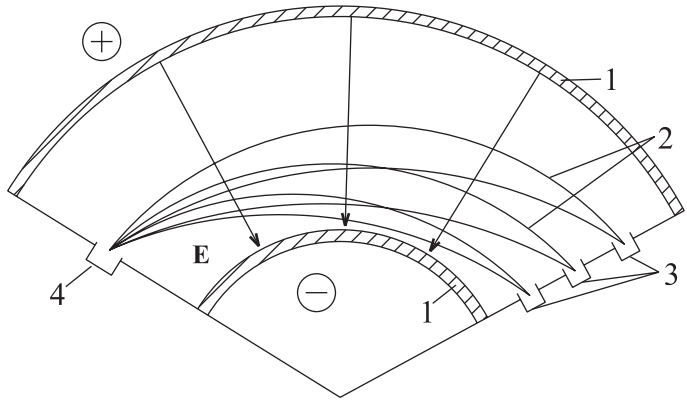


Fig. 1.4.4. Yuza–Rozhanskii energy analyser ('focusing' in the angle $\pi/\sqrt{2}$'): 1) electrodes; 2) particle beams with different energy; 3) beam receivers; 4) source

'Shortening' v in both parts of the equation we obtain

$$dl \sim \frac{c}{eH}\delta(Mv). \tag{1.4.12}$$

This is the common property of the systems with focusing by a single magnetic field; the field is focusing on the basis of pulses.

Focusing of a narrow beam by the electrical field

As an example of this method of focusing, we examine the so-called Yuza–Rozhansky energy analyser (Fig. 1.4.4).

The principle of this energy analyser is quite simple. The radial E-field is generated in the gap between two cylindrical electrodes. For some 'basic' energy of the particle $\varepsilon_0 = mv_0^2/2$ the strength E is selected such that these particles (ions are considered here) move along the circle with the radius R_0, passing between the electrodes. Evidently, in this case it should be that ($E_0 > 0$)

$$eE_0 = \frac{mv^2}{R_0} = \frac{2}{R_0}\varepsilon_0. \tag{1.4.13}$$

The behaviour of the ions whose energy differs from the basic energy and is equal to $\varepsilon = \varepsilon_0 + \delta\varepsilon$, where $\delta\varepsilon/\varepsilon_0 \ll 1$, will be studied. In the cylindrical coordinate system the equation of motion of the particles is determined by the Lagrange function (1.2.1)

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) - e\phi(r),$$

and taking into account that the **E**-field is independent of the azimuth, i.e. the momentum of the amount of motion remains unchanged, we can introduce the effective potential (1.2.3a)

$$U = e\phi + \frac{1}{2m}\frac{D^2}{r^2}, \quad D \equiv mr^2\dot{\theta}, \quad \phi = E_0 \ln\frac{r}{R_0} \tag{1.4.14a}$$

and write

$$m\ddot{r} = -\frac{\partial U}{\partial r} = -eE\,\frac{R_0}{r} + \frac{D}{m}\frac{1}{r^3}. \tag{1.4.14b}$$

The basic trajectory is determined from the condition $\partial U / \partial r = 0, r = R_0$,

$$eE = \frac{m^2 R_0^2 \dot{\theta}_0}{mR_0^3} = \frac{mv_0^2}{R_0},$$

which obviously coincides with (1.4.13). Searching for the trajectory in the vicinity of the base, assuming

$$r = R_0 + \xi. \tag{1.4.15a}$$

As a result, transferring from the independent variable t to θ

$$\frac{d^2\xi}{d\theta^2} + 2\xi = R_0\frac{\delta\varepsilon}{\varepsilon_0}. \tag{1.4.15b}$$

Here it is taken into account that $D_0^2 = 2mR_0^2\varepsilon_0$ and $\delta D^2 = 2mR_0^2\delta\varepsilon$. Assuming that at $\theta = 0$ $\xi = 0$ gives

$$\xi = \xi_1(1 - \cos(\sqrt{2}\theta)) + \xi_2 \sin(\sqrt{2}\theta), \tag{1.4.16a}$$

where

$$\xi_1 = \frac{R_0}{2}\frac{\delta\varepsilon}{\varepsilon}, \quad \varepsilon_2 = \frac{1}{\sqrt{2}}\left(\frac{d\xi}{d\theta}\right)_{\theta=0},$$

From (1.4.16a) it follows that the slightly diverging particle beam, leaving the source A ($\theta = 0$), if $\delta\varepsilon = 0$, is focused at $\theta = \theta_*$,where

$$\theta_* = \frac{\pi}{\sqrt{2}}. \qquad (1.4.16b)$$

However, if $\delta\varepsilon \neq 0$, the focus point is displaced along the radius by the value

$$\xi_* = R_0 \frac{\delta\varepsilon}{\varepsilon_0}. \qquad (1.4.16c)$$

Measuring ξ_* and knowing ε_0, we can determine $\delta\varepsilon$.

It may be seen that in the given case in focusing by **E** fields, the position of the focus is determined only by the particle energy[12].

Aberration-free separation with respect to mass

The different properties of focusing: by the magnetic field with respect to pulses

$$\mathbf{p} = m\mathbf{v} \qquad (1.4.17a)$$

and the electrical field with respect to the energy of the particles

$$\varepsilon = \frac{mv^2}{2} \qquad (1.4.17b)$$

offer a large number of possibilities for the construction of corpuscular-optical systems capable of focusing evidently with respect to any parameter

$$g \equiv m^{\alpha} v^{\beta}. \qquad (1.4.17c)$$

We describe an aberration-free system which regardless of the variation of the modulus and the direction of velocity $\mathbf{v}_0$ at exit from the point source A collects all the particles again into point focuses A_{m1}, A_{m2},..., whose position is determined only by the mass of the particles m_1, m_2,.... Of course, this aberration-free mass separator can be produced only by using the **E**- and **H**-fields.

The system (Fig. 1.4.5) contains two mutually perpendicular fields: electrical and magnetic. In paragraph 1.2.1 it was noted that movement

[12]This is natural. By analogy with (1.4.4) we can write that $m\mathbf{v} = \frac{d\mathbf{v}}{dl} = -e\nabla\phi$. Consequently $dl \sim \delta\frac{mv^2}{2}$.

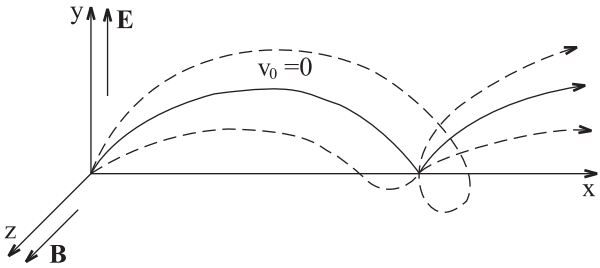


Fig. 1.4.5. Diagram of the aberration-free separator with respect to mass.

in these fields is the superposition of rotation around the Larmor circle with the angular frequency

$$\omega_H = \frac{eH}{mc} \tag{1.4.18a}$$

and the drift with the velocity

$$\mathbf{u}_E = \frac{c\left[\mathbf{E},\mathbf{H}\right]}{H^2}. \tag{1.4.18b}$$

This clearly indicates the principle of operation of this separator.

Since the period of Larmor motion and the velocity of the electrical drift are independent of the initial velocity of the particles, all the particles with the given mass, originating from point A at the moment t_0 has, in accordance with (1.4.18), circular trajectories during the time $\tau_H = 2\pi / \omega_H$ and arrive at the same point A_m whose distance from A is

$$\Lambda = u_E \tau_H = m\left(\frac{c^2}{eH^2}E\right). \tag{1.4.18c}$$

This scheme is not used widely in conventional vacuum optics, but the process described here is realised in movement of space systems with plasma or ion engines in the magnetosphere.

In this case, the role of velocity **u** is played by the velocity of the apparatus. In addition, 'focusing' of this type also takes place in the absence of an engine in the apparatus. By its movement, the apparatus disrupts the ionospheric plasma and, consequently, an odd tail forms behind the apparatus (Fig. 1.4.6). Of course, the described scheme does not focus of the component of the velocity of the particles, directed along the field **H**.

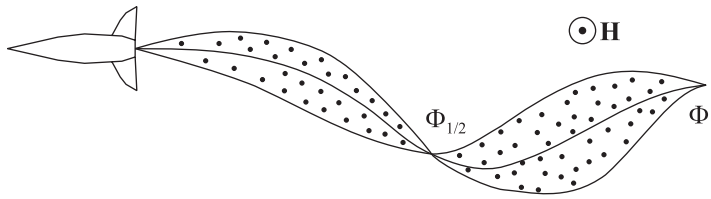


Fig. 1.4.6. Scheme of an ion beam, generated by an electroreactive engine in a space system, moving across the magnetic field of the magnetosphere.

Focusing properties of vacuum magnetic and electrical lenses

When examining plasma optics, it is useful to know the focusing properties of vacuum magnetic and electrical lenses.

We confine our considerations to examining thin lenses whose properties, as in 'school' optical lenses, for the near-axial (paraxial) beams, are determined only by the focusing distance. Therefore, these distances will be calculated.

The starting point is equation (1.2.3a), assuming that there is no azimuthal magnetic field H_θ, and as a result of paraxiality the momentum of the amount of motion is $D = 0$. Consequently, we can write

$$m\ddot{r} = -\frac{\partial U}{\partial r}, \quad U = e\phi + \frac{e^2\psi^2}{2mc^2r^2}, \quad m\ddot{z} = -\frac{\partial U}{\partial z}. \tag{1.4.19a}$$

Consequently, this yields the law of conservation of energy

$$\frac{1}{2}(\dot{r}^2 + \dot{z}^2) + \frac{U}{m} = \frac{v_0^2}{2} = \text{const}. \tag{1.4.19b}$$

Here $\mathbf{v}_0$ is the velocity of the incident particles away from the lens, with the velocity being parallel to the axis. Since we are interested only in the geometry of trajectories, the independent variable t should be replaced by coordinate z. Consequently, the equations of motion and the law of energy conservation have the form

$$\dot{z}\frac{d}{dz}\dot{z}\frac{dr}{dz} = -\frac{1}{m}\frac{\partial U}{\partial r}, \tag{1.4.20a}$$

$$\dot{z}^2(1 + r'^2) + \frac{2v}{m} = v_0^2, \quad r' = \frac{dr}{dz}. \tag{1.4.20b}$$

H- and **E**-fields are defined by expansion with respect to exponent r which can be easily calculated using (1.1.25) and (1.1.4c), knowing

the value of $\phi_0(z)$ in the field $H_0(z)$ on the axis. These expansions have the form

$$\psi = \frac{r^2}{2} H_0(z) - \frac{r^4}{96} H_0''(z) + \ldots \tag{1.4.21a}$$

$$\phi = \phi_0(z) - \frac{r^2}{4} \phi_0''(z) + \ldots \tag{1.4.21b}$$

To determine the focusing distance of the lens, it is sufficient to calculate in the first approximation the angle of deviation of the particle which initially moves in the direction parallel to the axis of the lens on the distance r_0. Consequently

$$\frac{1}{F} = \frac{-r'(z \to \infty)}{r_0}. \tag{1.4.22a}$$

The value $r'(+\infty)$ is determined by the method of ion–optical approximation (paragraph 1.2.6). After integrating the equation (1.4.19) with respect to z from $-\infty$ to $+\infty$:

$$v_0 r'(+\infty) = -\frac{1}{M} \int_{-\infty}^{+\infty} \frac{dz}{\dot{z}(r,z)} \left(\frac{\partial U(r,z)}{\partial r} \right). \tag{1.4.22b}$$

The value of F is calculated for the simplest thin lenses formed either by a ring with current (Fig. 1.4.7a) or by the same ring with a charge (Fig. 1.4.7b).

Magnetic lens

In this case, our considerations can be confined to the first term of the expansion (1.4.21a)

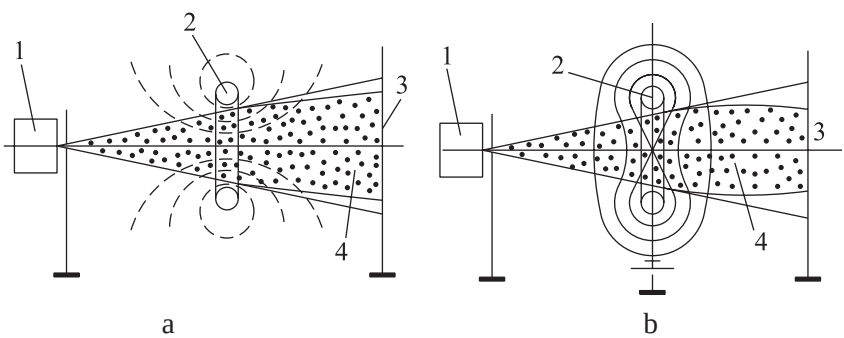


Fig. 1.4.7. Schemes of thin corpuscular lenses: a) magnetic lens and its lines of force; b) and electrostatic lens and its equipotentials. 1) ion source, 2) lens, 3) screen (target), 4) ion beam

$$\psi = \frac{r^2}{2} H_0(z), \quad U = \frac{e^2 r^2 H_0^2}{8c^2 m}, \quad \dot{z} = v_0, \quad r = r_0. \tag{1.4.23}$$

Consequently

$$r'(+\infty) = -\frac{r_0}{(Mv_0)^2} \cdot \frac{e^2}{4c^2} \int_{-\infty}^{+\infty} H_0^2 dz$$

and, therefore

$$\frac{1}{F} = \frac{e^2}{4(Mv_0)^2 c^2} \int_{-\infty}^{+\infty} H_0^2 dz. \tag{1.4.24}$$

On the axis of the ring with current J and radius R, the quantity

$$H_0(z) = \frac{2\pi J R^2}{(R^2 + z^2)^{3/2}}, \tag{1.4.25}$$

and the focusing distance is

$$F_H = R \frac{8}{3\pi^3} \left(\frac{c}{eJ} \right)^2 (Mv_0^2)^2. \tag{1.4.26}$$

In particular, in the case of focusing of Ar^+ ions with the energy ε = 10 keV at R = 5 cm, J = 7500 A, we have $F \approx 50$ m.

Electrostatic vacuum lens

In this case, if $r = r_0$, $\dot{z} = v_0$ is substituted into equation (1.4.22b) as in the previous case, the integral will be equal to 0. Therefore, it is important to take into account the terms of the second order of smallness

$$\phi = \phi_0(z) - \frac{r^2}{4} \phi_0''(z), \; \dot{z} = \sqrt{v_0^2 - \frac{2e\phi_0}{M}} \approx v_0 \left(1 - \frac{e\phi_0(z)}{M v_0^2} \right). \tag{1.4.27}$$

Consequently, we obtain the first approximation

$$r'_{(1)}(z) = \frac{r_0}{2Mv_0^2} \int_{-\infty}^{z} \varphi_0''(z) dz = \frac{r_0 e}{2Mv_0^2} \varphi_0'(z) \tag{1.4.28a}$$

$$r'_{(1)}(z) = r_0 \left(1 \frac{e\phi_0}{2M v_0^2} \right). \tag{1.4.28b}$$

Substituting these expressions into (1.4.22b), we obtain the final equation:

$$\frac{1}{F_E} = \frac{3}{16} \frac{e^2}{\left(\frac{Mv_0^2}{2} \right)^2} \int_{-\infty}^{+\infty} \left(\phi_0'(z)^2 \right) dz. \tag{1.4.29}$$

If the charge placed on the rings is Q, then

$$\phi_0(z) = \frac{Q}{\sqrt{R^2 + z^2}}.$$

and the focusing distance of such a lens is

$$F_E = R \frac{128}{3} \left(\frac{\varepsilon_0}{eU_F} \right)^2, \quad U_F = \frac{Q}{R}. \tag{1.4.30}$$

Assuming that the ion energy is ε_0 = 10 keV, U_F = 1 kV, we obtain $F_E \approx$ 200 m. Both F_H and F_E at the selected parameters J, U_F, ε_0 are very high. A completely different situation will be investigated in the section 5.7, concerned with plasma optics.

Attention should be given to the fact that the electrostatic and magnetic vacuum lenses are always collecting, irrespective of the sign of the focusing particles. However, this special feature is also independent of the details of design of the lenses.

Axisymmetric vacuum lenses have another considerable shortcoming. Spherical aberration is not removed in the lenses, i.e. the dependence of the focusing distance on the distance of the point on the object to the optical axis. Therefore, there are still no ‘natural’ schemes of the lenses which would make it possible to investigate the atom in details, although the deBroglie wavelength in electron microscopes is considerably smaller than the size of the atoms.

1.5. Dielectric permittivity and waves in homogeneous cold plasma [60, 61]

The most characteristic processes in any medium include its natural oscillations, i.e. oscillations which take place in the absence of external effects. There are many types of oscillations and are characterised by frequency, spatial structure and amplitude.

Waves with a small, formally infinitely small amplitude, are referred to as ‘linear waves’ because they are described by the equations

which include the terms which depend only in a linear manner on the amplitude. In turn, the theory of linear waves is especially simple in the case of a homogeneous medium. In this section, attention is given to the general theory of linear waves in a homogeneous cold plasma, situated in a homogeneous magnetic field, taking into account the solution of the one-particle problem (1.2.16). In chapter 3, section 3.5, it is shown that the model, constructed in this manner, relates to the cold (p_i, $p_e \to 0$) dissipation-free plasma.

It should be mentioned that the model of the homogeneous medium can in many cases (although not always) be also used for inhomogeneous media, if the wavelength is considerably smaller than the scale of inhomogeneity of the medium L:

$$\lambda \ll L.$$

1.5.1. Dielectric permittivity[13]

As indicated by the Maxwell equations (1.1.6), to describe the intrinsic waves, i.e. the waves at external **j** and q equal to 0, it is necessary to know the dielectric permittivity tensor

$$\mathbf{D} = \mathbf{E} + 4\pi\mathbf{P} \equiv \hat{\varepsilon}\mathbf{E}, \tag{1.5.1}$$

where **P** is the vector of polarisation of the medium which in the case of small perturbations can be written in the form

$$\mathbf{P} = \sum_{(k)} e_k n_{0k} \boldsymbol{\xi}_k = \sum_{(k)} e_k n_{0k} \boldsymbol{\xi}_k(E). \tag{1.5.2a}$$

Here $\xi_k(\mathbf{E})$ is the displacement of the particle of the k-th type under the effect of the electrical field **E**. These displacements were calculated in the one-particle approximation in section 1.2. These are equations (1.2.16). It is quite evident that they can be used for homogeneous cold rare plasma without collisions.

Substituting equations (1.2.16) into (1.5.2a), the following equation was obtained for the two-component (i, e) plasma

$$\hat{\epsilon}\,\mathbf{E} = \mathbf{E} + 4\pi e n_0 \left(\boldsymbol{\xi}_i - \boldsymbol{\xi}_e\right) = \mathbf{E} + 4\pi\left(\hat{S}_i - \hat{S}_e\right)\mathbf{E}. \tag{1.5.2b}$$

[13]Dielectric permittivity of plasma in hydrodynamic description was introduced by V. L. Ginzburg, and in kinetics by M. E. Gertsenshtein.

From this it follows that

$$\vec{\epsilon} = \begin{pmatrix} \eta & i\zeta & 0 \\ -i\zeta & \eta & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}, \tag{1.5.3a}$$

where

$$\eta = 1 - \sum_k \frac{\omega_{0k}^2}{\omega^2 - \omega_{Hk}^2}; \quad \varsigma = -\sum_k \frac{\omega_{Hk}\omega_{0k}^2}{\omega(\omega^2 - \omega_{Hk}^2)};$$
$$\varepsilon = 1 - \sum_k \frac{\omega_{0k}^2}{\omega^2}; \quad \omega_{Hk}^2 = \frac{e_k H}{m_k c}; \quad \omega_{0k}^2 = \frac{4\pi e_k^2 n_{0k}}{m_k}. \tag{1.5.3b}$$

The limiting cases of this tensor are:

(a) Let it be that $H \to 0$. Consequently, the tensor becomes a number, since

$$\eta = \varepsilon, \quad \zeta = 0, \quad \vec{\epsilon} = \varepsilon\vec{I} \tag{1.5.4}$$

where $\vec{I}$ is the unique tensor.

(b) If $\omega_H^2 \gg \omega^2$ and plasma is of the two-component type, then

$$\eta \approx 1 + \sum_{i,e} \frac{\omega_{0k}^2}{\omega_{Hk}^2} = 1 + \frac{c^2}{c_A^2}; \; \varepsilon = 1 - \sum_{i,e} \frac{\omega_{0k}^2}{\omega^2}; \; \zeta = +\sum_{i,e} \frac{\omega_{0k}^2}{\omega\omega_{Hk}^2}. \tag{1.5.5}$$

Here

$$c_A^2 = \frac{H_2}{4\pi n_0 (m + M)}, \; n_0 = n_0 i = n_{0e}$$

is the square of the Alfvén velocity. The quantity η and c_A have already been mentioned in section 1.3.

1.5.2 Equation for waves in homogeneous plasma

In analysis of the intrinsic waves, the system of the Maxwell equations can be conveniently reduced to a vector equation, containing only the electrical field **E**. Differentiating (1.1.6a) with respect to time and taking into account (1.1.6c) at $\mathbf{j}_{st} = 0$ we obtain:

$$-\mathrm{rot}(\mathrm{rot}\,\mathbf{E}) = \frac{1}{c^2}\frac{\partial^2 \vec{\epsilon}\,\mathbf{E}}{\partial t^2}. \tag{1.5.6a}$$

Using the vector identity

$$\mathrm{rot}(\mathrm{rot}\,\mathbf{a}) = \nabla\,\mathrm{div}\,\mathbf{a} \;\; -\Delta\mathbf{a}, \tag{1.5.6b}$$

gives

$$\Delta\mathbf{E} - \nabla\mathrm{div}\,\mathbf{E} = \frac{1}{c^2}\frac{\partial^2\vec{\epsilon}\,\mathbf{E}}{\partial t^2}. \tag{1.5.6c}$$

It should be mentioned that the Maxwell equation (1.5.6a) yields the equation:

$$\mathrm{div}\,\vec{\epsilon}\,\mathbf{E} = 0. \tag{1.5.7}$$

Since the tensor $\vec{\epsilon}$ is a quantity with constant components, the simple solution of (1.5.6c) can be obtained for harmonic fields

$$E \sim \exp\{-i\omega t + i\varkappa x\}. \tag{1.5.8}$$

Only harmonic fields will now be discussed.

In this case, the solution of the equation (1.5.6) and (1.5.7) has the form

$$\begin{aligned} &\left(\varkappa^2\delta_{ik} - \varkappa_i\varkappa_k\right)\mathbf{E}_k = \frac{\omega^2}{c^2}\epsilon_{ik}\mathbf{E}_k; \\ &\varkappa_i\epsilon_{ik}\varkappa_k = 0. \end{aligned} \tag{1.5.9}$$

To ensure that the system of linear equations (1.5.9) has a non-trivial solution, its determinant should be equal to 0

$$\mathbf{D}(\omega,\ \varkappa) \equiv \left| \varkappa^2\delta_{ik} - \varkappa_i\varkappa_k - \frac{\omega^2}{c^2}\epsilon_{ik} \right| = 0. \tag{1.5.10}$$

This equation, which links the frequency ω and the wave vector $\varkappa$, is the dispersion equation. In a homogeneous medium, vector $\varkappa$ is characterised by two parameters

$$|\varkappa| = \frac{2\pi}{\lambda} \text{ and } \cos\theta = \frac{\varkappa\mathbf{H}}{|\varkappa||\mathbf{H}|},$$

i.e. by the wavelength $\lambda = 2\pi/\varkappa$ and by the ‘pitch angle’, i.e. the angle between the wave vector and the strength of the magnetic field.

Knowing $\omega(\varkappa)$, we can determine two important quantities:

Phase velocity

$$\mathbf{V}_f = \frac{\omega}{|\varkappa|}\left(\frac{\varkappa}{|\varkappa|}\right), \tag{1.5.11a}$$

which shows the velocity and the direction of propagation of the fixed phase of the wave

$$\varkappa \boldsymbol{x} - \omega t = \text{const.} \tag{1.5.11b}$$

Group velocity

$$\mathbf{V}_{gr} = \left(\frac{\partial\omega}{\partial\varkappa_x}, \frac{\partial\omega}{\partial\varkappa_y}, \frac{\partial\omega}{\partial\varkappa_z}\right). \tag{1.5.11c}$$

This is the velocity of transfer of wave energy or the velocity of movement of the 'packet' of waves.

1.5.3. Waves in cold plasma without the magnetic field

The general dispersion equation (1.5.10) is investigated in sufficient detail in chapter 3. Here, this equation is used for the case in which there is no external magnetic field, i.e. when equation (1.5.4) holds.

Taking into account the isotropy of cold plasma in the absence of the magnetic field, the axis z is directed along the wave vector $\varkappa$, and the axis x in the plane ($\mathbf{E}$, $\varkappa$). Consequently, $\mathbf{E} = (E_x, 0, E_z)$, and the system (1.5.9) has the form

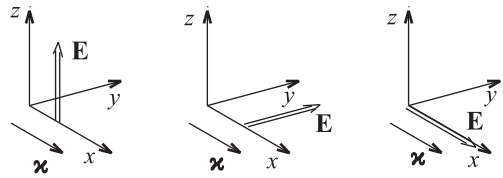
$$\varkappa^2 E_x - \frac{\omega^2}{c^2}\epsilon E_x = 0; \tag{1.5.12}$$

$$\frac{\omega^2}{c^2}\epsilon E_z = 0. \tag{1.5.13}$$

Its determinant is

$$\omega^2 \epsilon(\varkappa^2 c^2 - \omega^2 \epsilon) = 0. \tag{1.5.14}$$

Consequently, there are two types of linear processes in the absence of the magnetic field (Fig. 1.5.1)



a b c

Fig. 1.5.1. Three polarisations of waves at $H = 0$.

(a) Transverse waves (**E** $\perp$ $\varkappa$)

$$\omega^2\epsilon = \varkappa^2 c^2. \tag{1.5.15}$$

It should be mentioned that at the same values of ω and $\varkappa$ the transverse waves can, as is well known, have two polarisations. For the coordinate axes selected in our case, they correspond to the waves, for example, with **E**-fields along *x* and *y* (Fig. 1.5.1a, b).

(b) As shown by (1.5.14), the following equation correspondent to the longitudinal waves (**E** $||\varkappa$)

$$\epsilon = 0. \tag{1.5.16}$$

Further, substituting equation (1.5.4) for ϵ into (1.5.15) gives the relationship between ω and $\varkappa$ in transverse waves

$$\omega^2 = \varkappa^2 c^2 + \omega_0^2,\ \omega_0^2 = 4\pi e^2 n_0 \left(\frac{1}{M} + \frac{1}{m}\right). \tag{1.5.17a}$$

This shows that if the density of plasma is negligibly small, i.e. $\omega_0 \to 0$, we have the conventional electromagnetic waves (light, radiowaves), in which

$$\omega^2 = \varkappa^2 c^2, \tag{1.5.17b}$$

i.e. $\omega = \pm\varkappa c$. The ambiguity of the expression is associated with the fact that in reality (1.5.17a) describes two waves propagating in the directly opposite directions.

In the case of (1.5.17b), the phase and group velocities coincide and are independent of frequency. However, the presence of relatively dense plasma ($\omega^2 \sim \omega_0^2$) strongly influences the propagation of the field. For example, if $\omega < \omega_0$ the wave can propagate from the source (from the boundary into the depth of the plasma layer) only to the distance

$$\delta \sim \frac{1}{|\varkappa|} = \frac{c}{\omega_0}. \qquad (1.5.17c)$$

The wave is then reflected.

However, if $\omega > \omega_0$, then although the wave propagates in the plasma, the phase and group velocities now differ and depend on the wavelength

$$v_f = \frac{\omega}{\varkappa} = c\sqrt{1+\frac{\omega_0^2}{c^2\varkappa^2}}; \quad v_{gr} = \frac{c}{\sqrt{1+\frac{\omega_0^2}{c^2\varkappa^2}}}. \qquad (1.5.18)$$

The longitudinal waves (1.5.16) under the condition (1.5.4), i.e. in cold plasma at **H** = 0, lead to Langmuir oscillations

$$\omega^2 = \omega_0^2. \qquad (1.5.19)$$

In this case, the phase velocity can be of any magnitude

$$v_f = \frac{\omega_0}{\varkappa}. \qquad (1.5.20a)$$

and is determined by the wavelength of perturbation (Fig. 1.5.2). This is clearly indicated by the equation for the phase (1.5.11b). The group velocity of the Langmuir waves is equal to 0

$$v_{gr} = \frac{\partial\omega}{\partial\varkappa} = 0, \qquad (1.5.20b)$$

i.e. Langmuir waves 'stay in the same spot' and do not transfer energy. However, this takes place only at $T_i = T_e = 0$.

Two general comments should be made. Firstly, in further considerations the waves with $v_{gr} \neq 0$ will be referred to as 'signal'

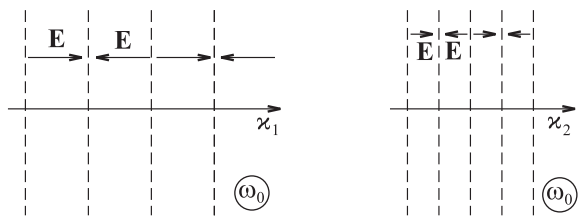


Fig. 1.5.2. Special features of Langmuir oscillations in cold plasma: the frequency of oscillations ω_0 is independent of wavelength $\varkappa$.

waves. Secondly, as already shown, in cold plasma at $H_0 = 0$ each frequency, with the exception of the single value $\omega = \omega_0$, is linked only with two signal waves – transverse waves in the present case, which differ by polarisation.

1.6. Block models of pulsed plasma systems (pulsed plasma guns and Z-pinches)

1.6.1. Two-component model of a magnetic-electrical railgun (A.I. Morozov) [58]

Analysis of the dynamics of a two-component block model of plasma (Fig. 1.6.1) in crossed fields **H** = (0, 0, H), $\mathbf{E}_0$ = (0, E_0, 0) is highly suitable for understanding the process of acceleration of the bridge investigated in section 1.3.1 and, therefore, also for explaining the difficulties in experiments arising in this case.

The equations describing the movement of the centres of gravity of the ion and electron blocks and also the x-component of the electrical field are determined by the shift of the blocks in relation to each other as a result of the difference of the masses and charges of the ions and electrons and have the form:

$$\begin{aligned} &M\frac{dV_{ix}}{dt} = eE_x + \frac{e}{c}H_0V_{iy};\ M\frac{dV_{iy}}{dt} = eE_0 - \frac{e}{c}H_0V_{ix};\\ &m\frac{dv_{ex}}{dt} = -eE_x - \frac{eH_0}{c}v_{ey};\ m\frac{dv_{ey}}{dt} = -eE_0 - \frac{eH_0}{c}v_{ey};\\ &E_x = 4\pi en_0\left(x_e - x_i\right), \end{aligned} \qquad (1.6.1)$$

Without carrying out relatively simple but quite time-consuming calculations of this system of linear equations, the result of the calculations is presented in the graphical form.

Figure 1.6.2a shows the trajectories of the centres of gravity of the blocks in the case of a very low density ($n_0 \to 0$), i.e. when quasi-neutrality is not observed, and the blocks move independently of

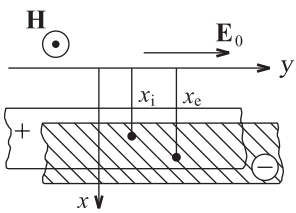


Fig. 1.6.1. The two-block model of the railgun: x_e, x_i are the coordinates of the centres of gravity of the blocks.

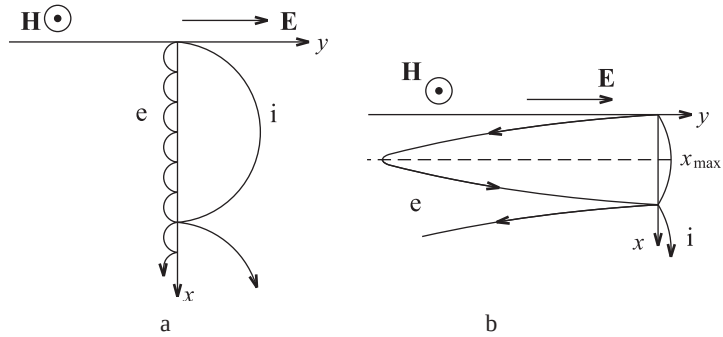


Fig. 1.6.2. Trajectories of ions and electrons in the two-block model of acceleration of the bridge of a railgun at $\omega_0 \to 0$ (a); $\omega_0 \to \infty$ (b); x_{max} is the coordinate at which the speed of the dense bridge is maximum.

each other. In this case, the periods with respect to x of the ions and electrons differ, and the displacement with respect to y in the case of the ions is large and in the case of electrons it is small. The trajectories of the particles at high n_0, where quasi-neutrality occurs, are completely different (Fig. 1.6.2b).

In this case, the electrons move faraway along y whereas the displacement of the ions along this axis is small. This means that the formation of current in the bridge is the role of electrons, and the ions are referred to as the 'purely inertial' components of the system. The ions are accelerated by the longitudinal (along x) electrical field E_x, formed in the volume of the bridge. At the start of the period, the velocity of the ions is low, the electrons move faster and pull the ions behind themselves. On the other hand, in the stage of deceleration of the bridge the electrons 'retreat' and pull the ions in the opposite direction.

However, it is evident that the mean velocities of drift u_E along x for the ions and electrons are approximately the same at any n_0.

Thus, the dynamics of ions in a dense (quasi-neutral) block is determined by the self-consistent longitudinal field. This field is referred to as the 'Hall' field. In real railguns, the bridges are in contact with highly conducting metallic electrodes. Naturally, in these conditions, the longitudinal field is disrupted in the vicinity of the electrodes and this effect interferes with obtaining higher velocities of plasma bunches.

1.6.2. Electrodynamic model of the railgun (L.A. Artsimovich) [62]

The model of acceleration of a plasma bridge in a railgun with an external magnetic field, discussed in paragraph 1.6.1, is formally

simple but quite difficult to realise. Pulsed electron accelerators with an intrinsic magnetic field, proposed by Artsimovich [62], proved to be simpler. Subsequently, in addition to the electrodes in the form of two rails (Fig. 1.6.3), experiments were carried out with a system of two coaxial electrodes, either cylindrical (Fig. 1.1.2) or conical. The coaxial pulsed accelerators were developed in approximately 1957 simultaneously in several countries. They have been used widely since.

In the zero-dimensional approximation with the plasma bunch regarded as a rigid bridge, the theory is the same for both the railgun and cylindrical coaxial electrodes. As in section 1.3.2, the process is described by two equations: for the coordinate z and the current J flowing through the contour:

$$\mu \frac{d^2 z}{dt^2} = F = \frac{J^2}{2c^2} \frac{\partial L}{\partial z}; \qquad (1.6.2)$$

$$\frac{1}{c^2}\frac{d^2}{dt^2} LJ + \frac{J}{C_0} = 0.$$

Here C_0 is the capacitance of the condenser, L is the total inductance, equal to

$$L = L_0 + lz, \qquad (1.6.3)$$

where L_0 is the initial inductance of the circuit, and the term lz takes into account its increase during movement of the current bridge. The

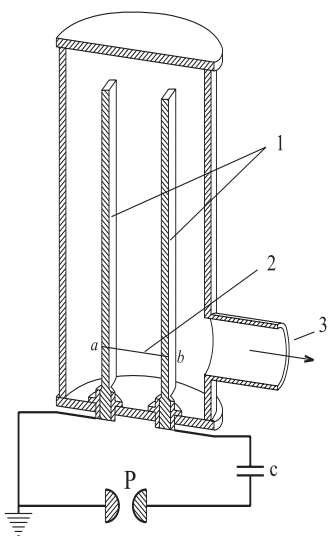


Fig. 1.6.3. Diagram of an electromagnetic railgun (L.A. Artsimovich); 1) rails, 2) bridge, 3) pumping.

system of equations is solved for the following initial conditions (t = 0):

$$U = U_0,\ z_0 = 0,\ \dot{z}_0 = 0. \tag{1.6.4}$$

Writing the equations (1.6.2) using the dimensionless variables

$$\bar{z} = \frac{b}{L_0} z,\quad \bar{U} = \frac{U}{U_0},\quad \tau = \omega_0 t,\quad \omega_0 = \frac{1}{\sqrt{L_0 C}},$$

and, introducing $J = -C_0 \frac{dU}{dt}$, the following system is obtained

$$\frac{d^2\bar{z}}{d\tau^2} = q\left(\frac{d\bar{U}}{d\tau}\right)^2;\ \bar{U} = -\frac{d}{d\tau}\left[\left(1+\bar{z}\right)\frac{d\bar{U}}{d\tau}\right] \tag{1.6.5}$$

with the boundary conditions

$$\bar{z}(0) = 0,\ \left.\frac{dz}{d\tau}\right|_0 = 0,\ \bar{U}(0) = 1,\ \left.\frac{dU}{d\tau}\right|_0 = 0. \tag{1.6.6}$$

The system (1.6.5) contains the single dimensionless parameter

$$q = \frac{\ell^2 C_0^2 U_0^2}{2\mu c^2 L_0}. \tag{1.6.7a}$$

The physical meaning of q is the square of the ratio of the scale of the oscillation period of the electrical circuit T_K to the duration of flight of the plasma bunch through the accelerator T_V

$$q \sim \frac{T_K^2}{T_V^2}. \tag{1.6.7b}$$

Figure 1.6.4a shows the time dependence of z for different values of q. Figure 1.6.4b compares the calculated and measured displacement of the bridge in the experiments carried out by L. A. Arsimovich. The difference between the calculated values and direct measurement of $z(t)$ is clearly visible, and this is natural for the displacement of the plasma bunch since it interacts ‘mechanically’ with the channel walls. However, the differences are not large for the oscillograms of current and voltage. Therefore, the electrodynamic model is used mainly for selecting the electrical circuit.

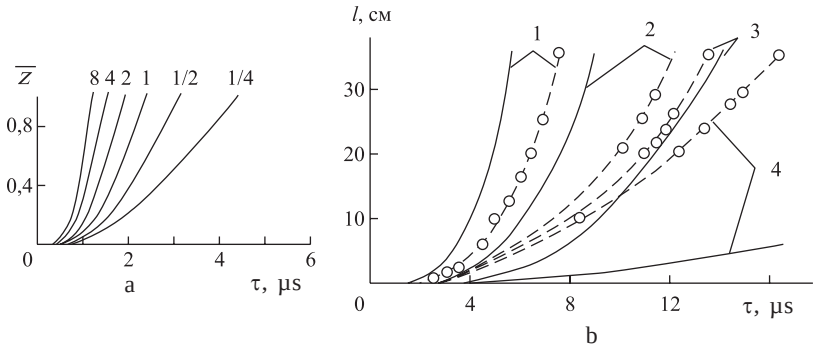


Fig. 1.6.4. Results of numerical solution of the equation of motion of the bridge for different values of parameter *q*: a) time dependence of the distance travelled by the bridge; b) theoretical and experimental (points) curves – time dependences of the distance travelled by the bridge and the plasma bunch. Burning-through of copper wires with a diameter of, mm: 1) 0.02; 2) 0.05; 3) 0.1; 4) 0.27.

1.6.3. Z-pinches

Z-pinches were already discussed in the historical review, i.e. direct electrode discharges, self-shrinking under the effect of the magnetic field, which were used for the first time to produce plasma with $T \sim$ 100 eV and also neutrons, although not of the thermonuclear origin. The main special features of this discharge, detected in experiments, were explained using the zero-dimensional model proposed by M.A. Leontovich and S. M. Osovets.

Classic Z-pinch

The diagram of the experimental equipment with *Z*-pinch, investigated at the beginning of the 50s of the previous century, is shown in Fig. 1.6.5a.

The largest amount of the experimental data is provided by the frames of superhigh-speed filming of the discharge in quartz thick wall pipes (Fig. 1.6.5b). Figure 1.6.5d shows the oscillograms of discharge current and voltage. The zero-dimensional model was constructed on the basis of the assumption which is in agreement with filming and local probe measurements.

In the initial moment when the discharge device closes the circuit and the electrodes receives a voltage, the effect of the electrical field stars to ionise the gas in the entire volume and, correspondingly, conductivity increases. At some moment, conductivity σ reaches the value at which the discharge current is concentrated in a thin (≤1 cm) skin layer. This skin layer with a high current density transforms into a non-transparent ‘piston’ which starts to ‘rake up’ (during its

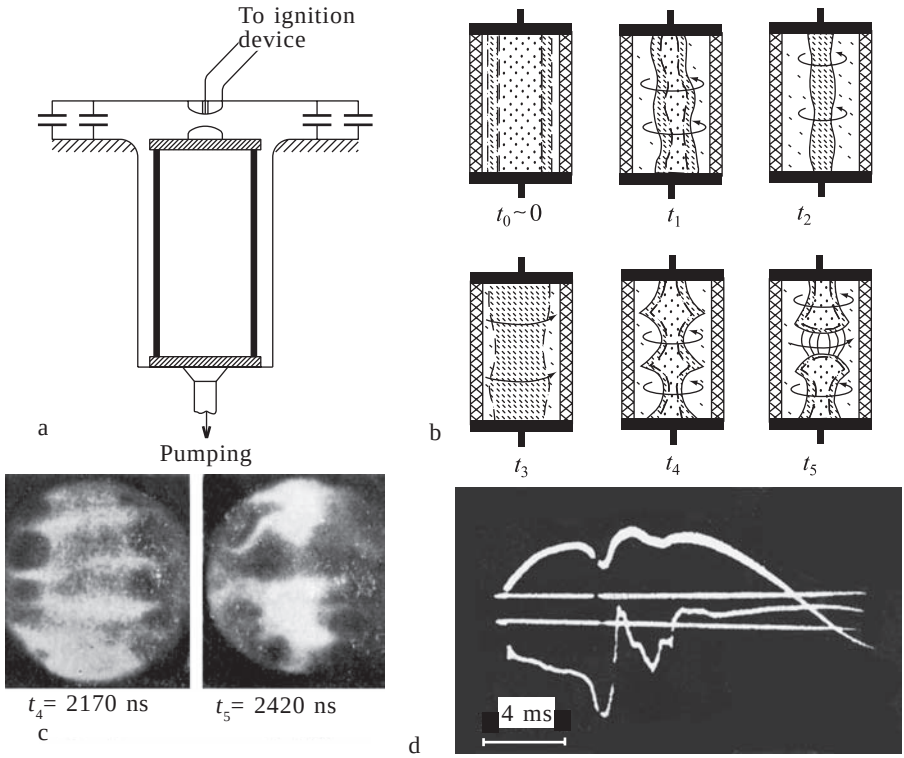


Fig. 1.6.5. *Z*-pinch equipment: a) principal diagram of equipment; b) kinematics of the frames of superhigh-speed filming of the process of constriction of the *Z*-pinch (the time in nanoseconds from the start of the discharge); c) selected frames of superhigh-speed films of *Z*-pinch (the moment of rupture of the waist t_5 and the preceding moment t_4); d) oscillograms of current and voltage, the special feature on the oscillograms corresponds to collapse of the pinch on the axis.

movement to the centre) the ionising gas. The constriction of the skin shell is explained by the fact that the pressure of the magnetic field is considerably higher than the plasma pressure in the skin layer and the pressure of the magnetic field considerably higher than the pressure of the slightly ionised gas inside the shell. The difference between the zero-dimensional model of the *Z*-pinch in comparison with the previously investigated models of the pulsed guns is that the variable mass moves in this instance, and the plasma volume is cylindrical.

Initial calculations were carried out in a simplified form, namely, only the equation of motion of the shell with the height h = 1 cm 'raking up' the gas with the density ρ:

$$\frac{d}{dt}\mu\dot{r} = F = \frac{H^2}{8\pi}2\pi r, \qquad (1.6.8a)$$

was used, where

$$\mu = \rho\pi(a^2 - r^2), \tag{1.6.8b}$$

and in accordance with the oscillograms of the current for the initial stages, the equation of the circuit was simply replaced by the equation

$$J = \alpha t,\ H = \frac{2J}{cr}. \tag{1.6.9}$$

If we introduce dimensionless quantities

$$\bar{r} = \frac{r}{a}, \quad \tau = \frac{t}{t_0},$$

the system (1.6.8) and (1.6.9) has the following form

$$\frac{d}{d\tau}\left[\left(1 - x^2\right)\frac{dr}{d\tau}\right] = \frac{\tau^2}{r}, \tag{1.6.10}$$

if t_0 is described by the equation

$$t_0 = \left(\frac{ac}{\alpha}\right)^{1/2} \mu_0^{1/4}. \tag{1.6.11}$$

This shows that in the variables $\bar{r}$; t equation (1.6.10) has the universal form, and at the initial conditions

$$\bar{r}(0) = 1; \quad \dot{\bar{r}}(0) = 0.$$

The results of the numerical solution of the equation (1.6.10) are shown in Fig. 1.6.6. It may be seen that at $\boldsymbol{\tau} \approx 1.5$, the calculation radius of the shell converts to 0. Of course, this is the result of neglecting the plasma pressure in the final stage of constriction. In fact, as shown by the experiments, the cylinder is constricted to $r_{\min}$ and subsequently expands and again starts to shrink.

But what is important is the equation for the moment of maximum compression which can be determined from

$$t_{\max} = 1,5 * t_0 = 1,5\left(\frac{ac}{\alpha}\right)^{1/2} \mu^{1/4}, \tag{1.6.12}$$

is confirmed quite satisfactorily by experiments.

The fundamental special feature of the constriction of the *Z*-pinches is the loss of stability of the cord $t \sim t_{\max}$. There are instabilities

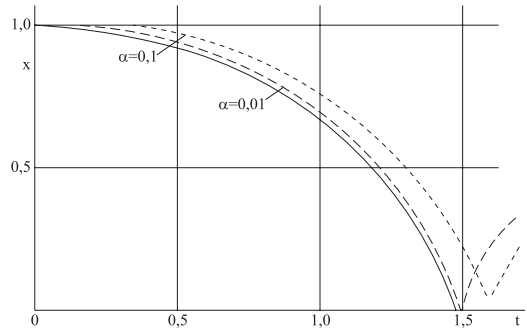


Fig. 1.6.6. The time dependence of the radius of *Z*-pinch calculated using the method proposed by Leontovich–Osovets with pressure not taken into account (solid line) and taken into account (broken line) [25].

of different types, but the axisymmetric 'waists' (Fig. 1.6.5b) and bending deformation with subsequent spreading of the cord are most distinctive. The appearance of deformation of the waist type may be easily understood if it is assumed that the region, occupied by current, is also 'corrugated'. In the area where the radius of the cord is smaller, the pressure of the magnetic field is higher ($P_M = H^2/8\pi$) and, consequently, the resultant 'waist' will progressively grow and in a number of cases the cord may be completely fracture[14]. At this moment, the flow of the particles in this gap is picked up by the bias current:

$$\mathbf{j}_{bias} = \frac{1}{4\pi}\frac{\partial \mathbf{E}}{\partial t}.$$

The very strong electrical fields, formed in this case, may be responsible for the acceleration of ions to the energy of the order of several hundreds of keV. This flux, formed at the fracture of the cord, bombards 'low-mobility' ions of the cord and causes nuclear reactions generating neutrons. However, it should be mentioned that the development of the 'waists' on the *Z*-pinch is also partially explained by the Rayleigh–Taylor instability[15]. The development of this instability is linked with the dynamic nature of the *Z*-pinch.

It may also be easily shown that random bending of the current-carrying cord will continuously become greater because of the resultant nonuniformity of the magnetic pressure on the cord surface.

[14]In Fig. 1.6.5c this is the moment at 2420 ns.

[15]Rayleigh-Taylor instability of the instability of the interface between two liquids in the field of gravity in the case in which a heavier liquid is on the top. In the *Z*-pinch, the role of the light liquid is played by the **H**-fields, and the role of acceleration of the gravitational force by the accelerated movement of the skin layer.

Types of Z-pinches

There are many different types of *Z*-pinch. Figure 1.6.7 shows non-classic pinches: (a) 'Filipov plasma focus' characterised by the force of a waist leading to the effective generation (depending on the conditions) of x-ray radiation of fast electron and also ion fluxes; (b) Triax – plasma shells of cylindrical geometry. The (quasi)-stationary Morozov compression flows will be discussed in sections 2.6.1 and 3.7.4. In addition to the pinches caused by the compressing effect of the azimuthal magnetic field, there are also plasma configurations

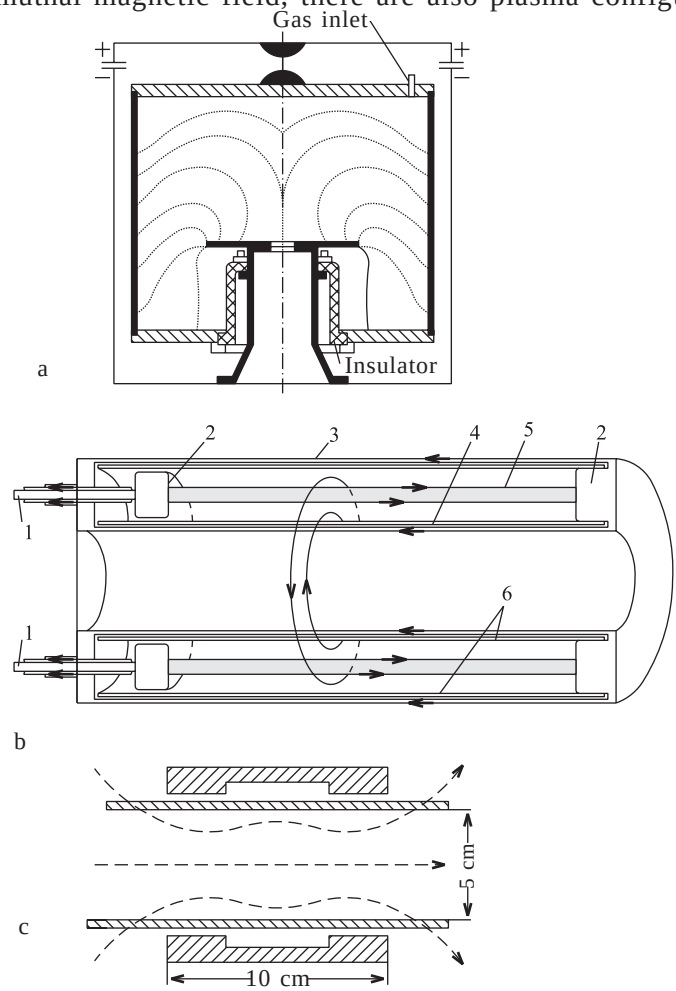


Fig. 1.6.7. Variety of pinches: a) 'plasma focus': the solid thin lines show that theconsecutive change of the front of current in the chamber; b) Triax equipment, 1) current supply, 2) electrodes, 3) external conductor, 4) internal conductor, 5) plasma, 6) insulators; c) θ-pinch (Scilla equipment), the dotted line indicates the form of lines of force of the magnetic field.

resulting from the effect of compression of plasma due to the effect of the longitudinal magnetic field (Fig. 1.6.7c). This type of configuration is referred to as the θ-pinch.

1.7. Simplest models of static magnetic traps

The problem of plasma confinement is multifaceted. The problem is also found in the formation of plasma and in transport of plasma along special plasma guides and, finally, as a result of the behaviour of plasma in 'traps' design for controlled thermonuclear synthesis where the temperature should be ~(100–500)·10^6 K. In this paragraph, we discuss only the traps.

Contact of plasma with the energy-absorbing 'walls' (solid, liquid, gas-like) can be avoided by either transferring to the very short term processes in which the plasma bunch forms in vacuum and does not manage to reach the walls (inertial confinement according to Basov and Krokhin) or in cases in which the plasma formation is stationary or quasi-stationary, the formation should be separated from the walls by electromagnetic fields (Lavrent'ev principle). At the moment we are interested only in (quasi) stationary magnetic confinement systems. The inertial systems will be discussed in section 10.5.

In this case, the procedure should be identical with that used in the formation of vessels (cylinders) for liquids or gases. In particular, to produce a magnetic or magneto-electrical comparatively thin wall 'cork' cylinder adequate to plasma, and fill it with the plasma (Fig. 1.7.1b).

However, because of historical reasons, which will be discussed in greater detail in section 10.5, the current approach to the magnetic plasma confinement is not based on the 'cork' schemes and it is based on 'seeding' of strong magnetic fields by the relatively undense plasma, i.e. on the system with (Fig. 1.7.1a)

$$\beta = \frac{8\pi p}{H^2} \ll 1.$$

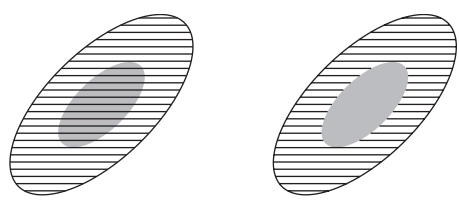


Fig. 1.7.1. Two schemes of plasma confinement by the magnetic field: a) magnetic 'sponge' (the field and plasma are mixed, $\beta \ll 1$); b) magnetic cylinder (the field and plasma are separated $\beta_0 \sim 1$); the region occupied by the magnetic field is cross-hatched, the plasma localisation region is grey.

At the same time, as in ‘cork’ traps

$$\beta_0 = \frac{8\pi p_0}{H_b^2} \sim 1.$$

Here p_0 is the pressure of plasma inside the balloon, and H_b is the characteristic strength of the barrier (‘cork’) field.

The existing and hypothetical stationary magnetic traps for plasma can be divided into three large groups on the basis of the topology connectedness of the plasma volume. They are:

– single-connected (open) traps;
– two-connected (toroidal closed) traps;
– multi-connected (‘cork’ and other galateas[16]).

Here, we examine the principles of the best-known traps, namely, from the group of open traps – the Budker–Post trap and some other modifications of this trap, and the two schemes of toroidal traps, i.e. tokamaks and two-pass stellarators [63]. As regards the galateyas, they will be described in section 10.5 where other representatives of the single-connected and two-connected traps will also be discussed.

1.7.1. The Budker–Post mirror trap

This trap can be schematically described as a system of two coils with currents of the same direction (Fig. 1.7.2). Here the magnetic field has minimum strength in the centre (H_0) and increases in the direction of ‘mirrors’, i.e. areas inside the coils (H_M). If a particle with velocity V_0, directed under the angle α_0 to the line of force, is considered in the medium plane, the particle will move in the direction to the mirror, rotating around the line of force. As a result of the retention of the transverse adiabatic invariant $J_\perp = \frac{V_\perp^2}{H} = \frac{V_0^2}{H_0}\sin^2\alpha_0$ (see (1.2.13), the transverse component of the velocity increases and the longitudinal component $V_{||}$ decreases as a result of energy conservation ($V_0^2 = V_\perp^2 + V_{||}^2 = \text{const}$):

$$V_{||} = \sqrt{V_0^2 - J_\perp H} = V_0\sqrt{1 - \frac{H}{H_0}\sin^2\alpha_0}. \qquad (1.7.1)$$

Here α is the angle between the vectors of the magnetic field **H** and velocity **v**.

This shows that if

16 The term ‘Galatea’ is explained in section 10.5.3.

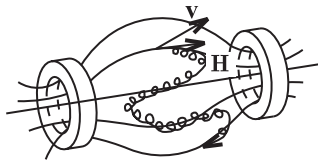


Fig. 1.7.2. Budker–Post mirror trap and trajectories of particles in the trap.

$$\sin\alpha \geqslant \sqrt{\frac{H_0}{H_M}}, \tag{1.7.2}$$

the particle does not leave the trap and, reflecting from the mirror, oscillates inside the trap.

In addition, as a result of the nonuniformity of the field, the particle also drifts along the azimuth of the trap. This indicated by the term proportional to $[\mathbf{H}, \nabla H]$ in equation (1.2.18). Figure 1.7.3 shows the space of velocities in the coordinates ($V_\perp$, $V_\parallel$) and in it the 'departure cone' into which the particle travels from the trap.

In reality, the dynamics of the particles in the mirror trap may be described as follows. Particles are introduced into the trap by the injector of neutral atoms[17] with the required energy and the fixed direction of velocity. Falling into the trap, fast neutral atoms are ionised. Colliding with the particles already situated in the trap, the 'new' particles gradually change the transverse adiabatic invariant and, travelling to the prohibited cone, they leave the trap. Calculations carried out initially by Budker and improved later by other authors show that the lifetime of the particles in the mirror trap is of the order of the time between the collisions, multiplied by the logarithm of the ratio H_M/H_0. For the ions this will be (see section 5.6.2)

$$\tau_{\text{life}} \approx \tau_{ii} \ln \frac{H_M}{H_0}. \tag{1.7.3}$$

Convective instability in the mirror trap with the sharp plasma boundary In the historical review, it was reported that soon after the formulation of the principle of the mirror trap, theoreticians Longmire and

[17] See section 10.1, where this injector is described.

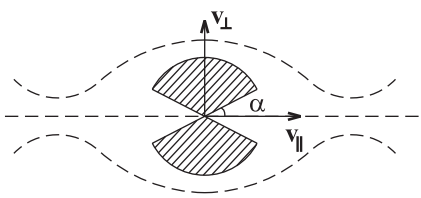


Fig. 1.7.3. Departure cones of the particles in the mirror trap at the minimum of the **H**-field.

Rosenbluth (USA) and slightly later Kadomtsev showed that the plasma in the mirror trap at relatively high pressure gradients (the criteria are discussed in section 8.1.2) should be unstable. This will be shown, in accordance with the first two authors, who used almost the same method as the one used in the present book for examining the incidence of an ion–electron bar in the gravitational field in the presence of the magnetic field.

The plasma boundary will be regarded as sharp: there is no plasma outside the boundary, and inside the boundary the plasma density is constant. It is assumed that because of random reasons, a magnetic pipe with the plasma has passed through the interface, and the magnetic pipe whose area is now occupied by the pipe with plasma descended inside the plasma volume (Fig. 1.7.4). To simplify the considerations, the cross-section of the investigated pipes is assumed to be rectangular. The coordinates (x, y, ℓ) are introduced in accordance with the figure. Here ℓ is the arc counted along the line of force in the centre of the cross-section of the projection. It is assumed that the ions and electrons are magnetised and their dynamics can be described by drift equations. Consequently, as a result of the nonuniformity of the magnetic field, the particles move along x with the velocity of the magnetic drift (1.2.18)

$$\dot{R}_{xe} = \frac{mc\left(2v_{\|e}^2 + v_{\perp e}^2\right)}{2(-e)H^3}(-H)\frac{\partial H}{\partial y} \equiv \frac{cQ_3}{eH^2}\frac{\partial H}{\partial y};$$
$$\dot{R}_{xi} = \frac{Mc\left(2v_{\|i}^2 + v_{\perp i}^2\right)}{2eH^3}(-H)\frac{\partial H}{\partial y} \equiv -\frac{cQ_i}{eH^2}\frac{\partial H}{\partial y}; \qquad (1.7.4)$$

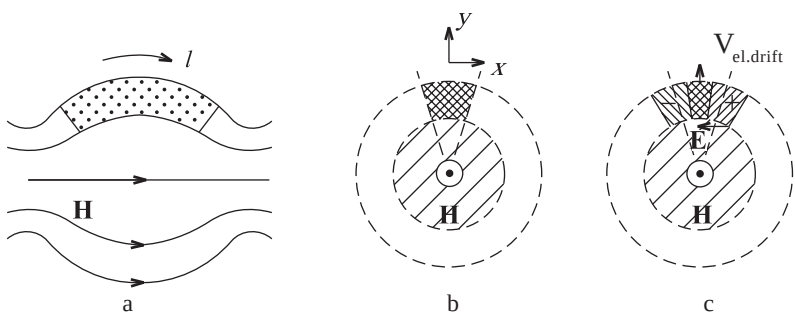


Fig. 1.7.4. Development of convective instability in a mirror trap with a sharp plasma boundary: a) projection on the surface of plasma in the coordinates (r, z); b) the same in the section z = const, at t = 0; c) the distribution of charges in the projection as a result of the gradient magnetic drift and the resultant electrical field leads to electrical drift.

The meaning of the introduced quantities Q_e and Q_i is evident – they are temperature analogues. In these equations, it is assumed that the direction H and the coordinate ℓ coincide.

The drift of the particles on the azimuth of the trap, i.e. along x, results in a flux of charges on the side surfaces of the projection and the depression. This flux is:

$$j_x = en_0(\dot{R}_{x,i} - \dot{R}_{x,e}) = cn_0(Q_i + Q_e)\frac{\partial}{\partial y}\frac{1}{H}. \qquad (1.7.5)$$

This shows that the direction of the flux of the drifting charges depends on the sign of $\frac{\partial}{\partial y}\frac{1}{H}$. If this derivative on the line of force is positive, i.e. magnetic field weakens with the increase of the distance from the plasma boundary, then $j_x > 0$ and the charge, accumulated on the side surfaces, generates the electrical field:

$$E_x = \frac{4\pi\bar{\sigma}}{\epsilon} < 0, \qquad (1.7.6a)$$

Here $\bar{\sigma}$ is the mean (along the line of force) value of the charge on the projection. The formation of the field with this orientation results in the electrical drift from the interface

$$u_{E,y} = \frac{e[\mathbf{E},\mathbf{H}]}{H^2} > 0, \qquad (1.7.6b)$$

i.e. the projection will grow and the depression will become deeper.

However, if the strength of the magnetic field increases with increase of the distance from the interface, E_x changes the sign and the projection is 'pushed into' the plasma volume. Consequently, in the first of the examined cases the interface is unstable, and in the second case it is stable.

In a simple mirror trap (Fig. 1.7.2) in the central part $\frac{\partial}{\partial y}\frac{1}{H} > 0$, and in the vicinity of mirrors $\frac{\partial}{\partial y}\frac{1}{H} < 0$. Therefore, the result depends on which of the sections provides a larger contribution to polarisation. The integral nature of the effect is explained by the fact that the charges easily flow along the lines of force.

If the plasma is concentrated in the central part of the trap, it is 'convectively' unstable. This was efficiently demonstrated by M.S. Ioffe (1960, The Institute of Atomic Energy). This instability can be easily suppressed by abandoning axial symmetry and taking into account the

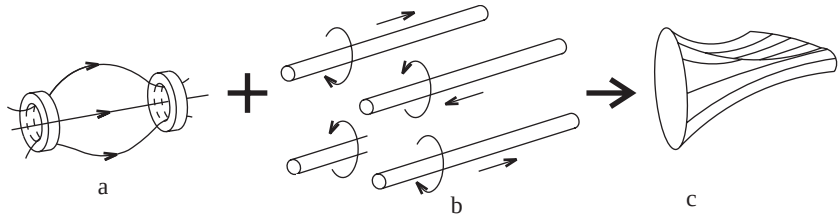


Fig. 1.7.5. Scheme of stabilisation of convective instability by the quadrupolar magnetic field (Ioffe 'sticks'): a) a system of two rings, forming the mirror trap field; b) Ioffe 'sticks', forming the quadrupolar field; c) the overall magnetic field.

fact that the stability criterion includes the integral along the section of the line of force occupied by the plasma.

The simplest stabiliser ('anchor') was proposed by Ioffe. This is simply a quadrupolar field, produced by four 'sticks' with alternating flux directions (Fig. 1.7.5). This field is almost normal to the field of the mirrop trap and therefore the square of the modulus of the resultant field is:

$$H^2 = H_q^2 + H_{\text{mirror}}\, a^2. \tag{1.7.7}$$

The quadrupolar field $\left|H_q\right| \sim r^2$, and it compensates the weakening of the strength of the mirror field with a radius.

The types of magnetic surfaces of the combined field are shown in Fig. 1.7.5c.

At present, there is a large number of 'anchors' capable of suppressing convection (Fig. 1.7.6). However, the problem of 'closing' of the ends remains unsolved. There has been a number of interesting proposals (see section 10.5) but no successes, sufficient for solving the problem of controlled thermonuclear synthesis have been reported.

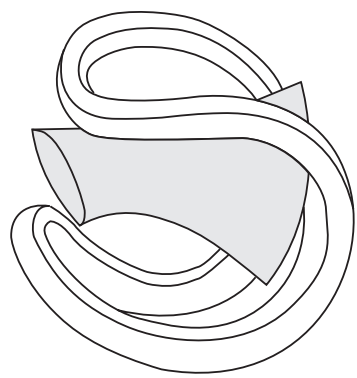


Fig. 1.7.6. 'Baseball' coil for stabilising plasma in a mirror trap.

1.7.2. Toroidal traps

Taking into account the fact that the particles in the magnetic field move freely along the magnetic lines of force, it is natural to use the magnetic fields whose lines of force remain within the limits of some two-connected toroidal volume in the simplest case.

From the group of the possible variants of the toroidal ('closed') traps we should mention tokamaks and stellarators (referred to in most cases as 'helicoidal' traps). These traps are characterised by a relatively strong azimuthal field. The fields of the traps have already been discussed in section 1.1.

Toroidal drift

The simplest magnetic configuration of this type is the field of a straight filament with current. It is natural to use it as a base and separate a torus in it, for example, with a right angled cross-section, fill it with sufficiently undense plasma in order to ignore collisions. Considering the direction of current as the z axis, the following equation can be written for the single azimuthal component of the field:

$$H_\theta = \frac{a}{r} H_a;\ \nabla|H| = -\frac{\mathbf{r}^0}{r^2} a H_a. \tag{1.7.8}$$

Here a is some 'reference' radius inside the ring at which $H_\theta = H_a$, $\mathbf{r}^0$ is the unit vector along the radius.

Since $\nabla H \neq 0$, a magnetic drift along the z axis forms (1.2.28):

$$(u_z)_e = -\frac{Mc\left(\left(v_{\perp e}^2 + 2v_{\|e}^2\right)\right)}{2eH_a a}. \tag{1.7.9a}$$

$$(u_z)_i = \frac{Mc\left(v_{\perp i}^2 + 2v_{\|i}^2\right)}{2eH_a^3} = \left[\mathbf{H}, \nabla H\right]_z = \frac{Mc\left(v_{\perp i}^2 + 2v_{\|i}^2\right)}{2eH_a a}; \tag{1.7.9b}$$

This shows that the ions and electrons drift in different directions, as in the previously examined cases of the fall of a plasma bar in the magnetic field, or in evolution of the projection of the plasma surface in the mirror trap. These drifts lead to the appearance of polarisation charges with opposite signs at the ends of the torus (Fig. 1.7.7)

The density of the polarisation charge increases in accordance with the law $(e_i = e,\ e_e = -e)$:

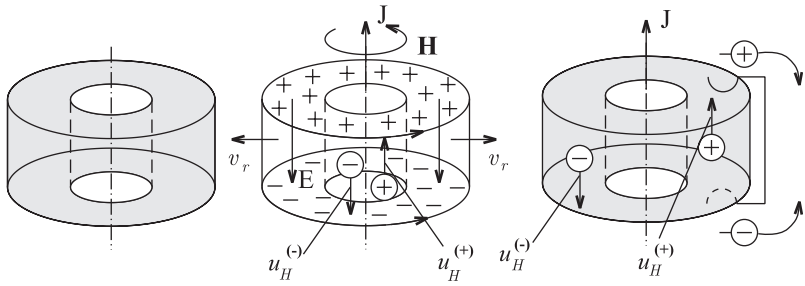


Fig. 1.7.7. Polarisation of the plasma torus as a result of toroidal drift: a) toroidal plasma volume, restricted by the magnetic fields, b) directions of toroidal drifts of the electrons and the ions, c) the principal diagram of removal of polarisation.

$$\dot{\sigma} = en(u_{H,i} - u_{He})$$

$$= \frac{cn}{2H_a a}\left[m(v_{\perp e}^2 + 2v_{\|e}^2) + M(v_{\perp i}^2 + 2v_{\|i}^2)\right]. \qquad (1.7.10)$$

The appearance of the electrical field along z results in the combined rest of the ions and electrons along the radius, i.e. results in the expansion of the ring. This expansion takes place with acceleration (see (1.3.4)):

$$\frac{dv_r}{dt} = -\frac{c\dot{E}_z}{H_\theta} = -c\frac{4\pi\dot{\sigma}}{H_\theta \eta}, \quad \eta = 1 + \frac{c^2}{c_A^2}. \qquad (1.7.11a)$$

This shows that at $c_A^2 \ll c^2$ and $r = a$, we have

$$\frac{dv_r}{dt} = \frac{4\pi n c_{A,a}^2}{H_a^2 a}\ \frac{3}{2}k\left(T_e^* + T_i^*\right) = \frac{3}{2}\ \frac{k\left(T_e^* + T_i^*\right)}{Ma}, \qquad (1.7.11b)$$

if it is assumed that

$$\left(\frac{m}{2}(v_\perp^2 + 2v_\|^2)\right)_\alpha = \frac{3}{2}kT_\alpha^*.$$

Attention should be given to the fact that the resultant equation for acceleration at $c_A^2 \gg c^2$ does not contain the magnetic field and individual characteristics of the particles. Equation (1.7.11b) includes only the effective speed of sound

$$c_T^2 = \frac{3}{2}\frac{k(T_e^* + T_i^*)}{M}.$$

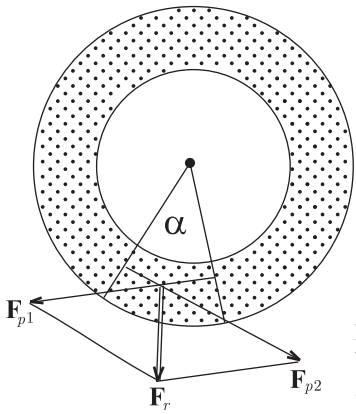


Fig. 1.7.8. Calculation of the expansion of a toroidal balloon with a gas: $\mathbf{F}p_1$, $\mathbf{F}p_2$ are the forces of pressure, acting on the cross-section of the balloon, $\mathbf{F}r$ is the resultant of these forces.

This indicates that the expansion of the ring of the plasma in the magnetic field is a conventional gas dynamic process. In fact, attention will be given to the expansion of a gas-filled ring with the cross-section S and radius a with absolute expansion in the direction of length (Fig. 1.7.8). A small sector with the angle α is defined. Its mass is

$$\mu = \alpha n M,$$

and the force of pressure acting on the sector in the radial direction from the side of the adjacent sections of the ring is:

$$F_r = Sp\alpha.$$

Consequently, the expansion of the ring takes place with acceleration

$$\dot{V}_r = \frac{F_z}{\mu} = \frac{p}{aM}. \qquad (1.7.12)$$

With the accuracy to the coefficient ~1, associated with the approximate nature of the calculations, the equations (1.7.11c) and (1.7.12) coincide.

Removal of polarisation in toroidal systems

Evidently, in order to form a toroidal configuration which will not be ejected to the walls of the chamber, it is necessary to prevent the formation of the electrical field in the plasma volume which would produce the 'ejecting' drift of the particles.

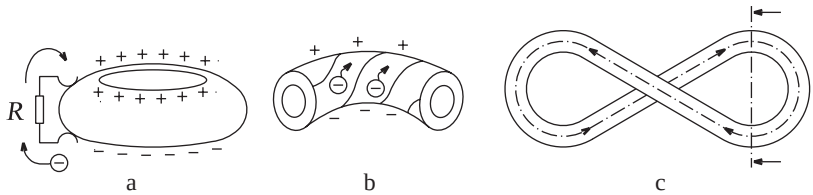


Fig. 1.7.9. Methods of removing polarisation: a) by shortening with a contact (for example, diaphragm), b) shortening with the lines of force of the 'top' and 'bottom' of the cord; c) by twisting of the torus (Spitzer's figure of eight, predecessor of a stellarator).

If the temperature of the plasma was low, polarisation could be removed by simply connecting the ends of the plasma torus in a short circuit using an efficient conductor (Fig. 1.7.9a). This stabilisation takes place in tokamaks in the stage of 'ignition' of the discharge when the poloidal field is small. Evidently, if the plasma volume, short-circuiting the conductor, is characterised by higher resistance, a low-strength electrical field is retained in the volume of the plasma cord and the latter will continuously travel in the direction of larger radia. This is the so-called Pfirsch–Schluter diffusion which takes place in the case in which the external resistance of the circuit is relatively low.

However, methods of removing polarisation are based on the application of magnetic fields whose lines of force form and wind around the magnetic surfaces (Fig. 1.7.9b). Consequently, the 'upper' and 'lower' sides of the surface are shortened.

The circuit of the magnetic reactor, proposed by Sakharov, used Tamm's magnetic configuration (Fig. 1.1.4), i.e. inside a toroidal chamber with the azimuthal magnetic field there is a hypothetical metallic ring rigidly connected with the power source, with the current flowing through the ring, i.e. a galatea was considered (Fig. 1.7.10). The poloidal field of this ring closed the 'top' with the 'bottom'. Later, the ring was not used and the generator of the poloidal field was the current flowing directly in the plasma. To ensure rough stability and produce a tokamak, it was necessary to select the correct ratio of the values of the azimuthal and poloidal fields. The Kruskal–Shafranov criterion was used in this case (section 10.5).

Evidently, the considerations relating to the removal of polarisation also hold for the stellarators because they are based on fields with inserted surfaces, formed by spiral lines of force. It should be mentioned that a clear understanding that the stellarators contain a system of inserted magnetic surfaces whose line of force pass around the magnetic axis, came slowly. The initial stellarators have the form of the figure 8 (Fig. 1.7.9) and were produced on the basis of the

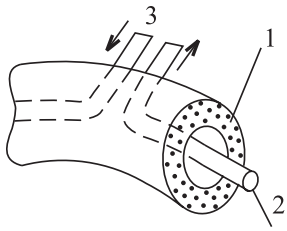


Fig. 1.7.10. Scheme of Sakharov's plasma configuration: 1) plasma, 2) central conductor (myxine) with current, 3) supply of current to the central conductor.

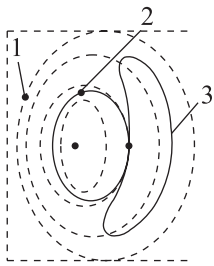


Fig. 1.7.11. Trajectories in a tokamak of passing (2) and trapped (3) particles (bannanas): 1) magnetic surfaces.

considerations that with this configuration, the toroidal drift in a single section this compensated by the toroidal drift in another section.

Passing and trapped particles in toroidal traps

The common feature of all toroidal traps are the complications associated with the fact that the strength of the magnetic field along the lines of force is not constant. We consider the simplest case of an axisymmetric field (Fig. 1.1.4). The strength of the field is greater in the sections of the lines of force which are closer to the centre of the system and smaller at the periphery in the case of large radii. It can be said that here we are concerned with the unique analogue of the mirror trap. This means that the particles, situated in such a trap, can be divided into two groups – those in which the longitudinal component is large and those in which it is small. More accurately, if (see (1.4.2))

$$\sin\alpha > \sqrt{\frac{H_{\min}}{H_{\max}}}, \qquad (1.7.13a)$$

where α is the angle between **V** and **H** on the external side of the magnetic surface, where the magnetic field is minimum ($H_{\min}$), the particle is reflected from the magnetic mirror in the region $H_{\max}$ and is trapped. However, if

$$\sin\alpha < \sqrt{\frac{H_{\min}}{H_{\max}}}, \qquad (1.7.13b)$$

the particle is of the passing-type, i.e. bypasses the entire magnetic surface. It appears that the trajectory of the passing and trapped particles greatly differs (Fig. 1.7.11). In particular, if the passing particles deviate as a result of drift from 'their' magnetic surface to the distance

$$\delta_{\text{trap}} \sim \frac{v_T}{\omega_{H_{\text{pld}}}}. \tag{1.7.14a}$$

where $\omega_{H_{az}}$ is the Larmor frequency, calculated from the azimuthal (entire) field, since the displacement of the trapped particles is considerably greater and as regards the order of magnitude is equal to

$$\delta_{\text{flight}} \sim \frac{v_T}{\omega_H}, \tag{1.7.14b}$$

Here $\omega_{H_{\text{pld}}}$ is the Larmor frequency calculated from the poloidal field, v_T is the thermal velocity of the particles.

The trajectories of the particles in the toroidal field follow directly from the integral of the drift equations (1.2.19) which is the equation of the drift trajectory. This difference of the displacements is of principal importance for the diffusion of plasma from the trap. This will be discussed in greater detail in 5.7.

In the scheme of the magnetic thermonuclear reactor, proposed by Sakharov, the poloidal field was assumed to be equal to only 200 Oe. It may easily be seen that with this field and at the particle velocity of $\sim 10^8$ cm/s, the displacement $\delta_{\text{trap}} \sim$ 1 m (!). Of course, this is not realistic.

2

One-fluid plasma models

Zero-dimensional models approximate plasma configurations by one or several blocks. Naturally, these models are characterised by greatly restricted possibilities. Incomparably greater possibilities are offered by the hydrodynamic models in which the medium is divided into small, formerly infinitely small, quasi-autonomous 'droplets'. The coordinates (**x**) of every droplet depend on time (t). A field of droplets is characterised by density $n(\mathbf{x}, t)$, velocity $\mathbf{v}(\mathbf{x}, t)$, temperature $T(\mathbf{x}, t)$, the strength of the fields $\mathbf{H}(\mathbf{x}, t)$, $\mathbf{E}(\mathbf{x}, t)$, etc. The equations for these quantities also form the 'basic' model of hydrodynamic approximation. In this chapter, attention will be initially given to the principal moments of classic gas-dynamics and this will be followed by examination of the one-fluid 'magnetic hydrodynamics'. The two-fluid plasma dynamics is discussed in chapter 3.

2.1. Special features of hydrodynamic models

2.1.1. Euler equations

The simplest model of the medium which can be considered in hydrodynamics[1] is a medium of fully autonomous, absolutely 'smooth' droplets which interact together only by a pressure on each other along the normal to the droplet surface (Fig. 2.1.1)

$$\mathbf{F}_p = -\int_{(S)} \mathbf{n}^0 p dS. \tag{2.1.1}$$

Here $\mathbf{n}^0$ is the external normal to the investigated 'droplet', dS is a scalar.

[1]Here and everywhere else the terms 'hydrodynamics' and 'gas dynamics' will be regarded as synonymous

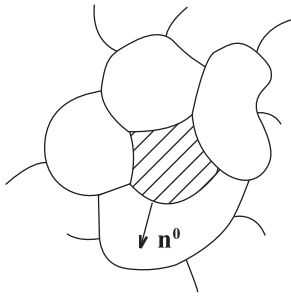


Fig. 2.1.1. Division of a continuous medium into 'droplets', $\mathbf{n}^0$ is the normal to the droplet surface.

This model, assuming isotropy of pressure, leads to Euler hydrodynamics. It is the prototype for the future plasma, hydrodynamic models and, consequently, we will be examined in greater detail.

The volume and intensity of the 'droplet " are denoted by letters V and ρ Consequently, the mass of the 'droplet' is $\mu = \rho V$ As a result of the assumptions made regarding the autonomous nature of the droplet, we can write the equation of conservation of the droplet mass, the Newton equation and the condition of conservation of entropy s

$$\frac{d\mu}{dt} = 0; \tag{2.1.2a}$$

$$\mu \frac{d\mathbf{v}}{dt} = \mathbf{F}_p + \mathbf{F}_{\text{bol}}; \tag{2.1.2b}$$

$$\frac{ds\mu}{dt} = 0. \tag{2.1.2c}$$

Here $\frac{d}{dt} = \frac{\partial}{\partial t} + (\mathbf{v}\nabla)$ is the substantional derivative with respect to time, taking into account displacement of the droplet in space; $\mathbf{F}_p$ is the pressure force (2.1.1), $\mathbf{F}_{\text{vol}}$ is the volume force, acting in the droplet volume (for example, gravitational force), s is the entropy of the unit mass.

The integral equations (2.1.2) are reduced gradually to the differential form, excluding the auxiliary value V – the droplet volume.

Continuity equation

The velocity of variation of the volume of the separated droplet in the flow is determined by the equation

$$\frac{dV}{dt} = \int \mathbf{v}\mathbf{n}^0 dS. \qquad (2.1.3a)$$

Using the Gauss–Ostrogradskii equation at a relatively small volume of the droplet, equation (2.1.3a) can be written in the following form

$$\frac{dV}{dt} = \int\limits_{(S)} \mathbf{v}\mathbf{n}^0 dS = \int \operatorname{div} \mathbf{v} dV = (\operatorname{div} \mathbf{v})V. \qquad (2.1.3b)$$

Consequently, taking into account the conservation of the droplet mass during movement of the droplet

$$\frac{d\mu}{dt} = \frac{d}{dt}\rho V = V\left(\frac{\partial \rho}{\partial t} + (\mathbf{v}\nabla)\rho + \rho \operatorname{div} \mathbf{v}\right) = 0,$$

we have

$$\frac{\partial \rho}{\partial t} + \operatorname{div} \rho \mathbf{v} = 0. \qquad (2.1.4)$$

This is also the standard form of expressing the law of mass conservation.

Dynamics equation

Using the Gauss–Ostrogradskii equation, we can determine the components of the force generated by pressure (2.1.1):

$$\mathbf{F}_{px} = -\int\limits_{(S)} p\mathbf{x}^0 d\mathbf{S} = -\int \frac{\partial p}{\partial x} dV = -\frac{\partial p}{\partial x} V \qquad (2.1.5a)$$

Here $\mathbf{x}^0$ is the projection of the normal $\mathbf{n}^0$ onto the axis *x*.

Consequently

$$\mathbf{F}_p = -V \nabla p.$$

Assuming that

$$\mathbf{F}_{\text{vol}} = V\mathbf{f},$$

and substituting into (2.1.2b) the expressions for μ, $\mathbf{F}_p$ and $\mathbf{F}_{\text{vol}}$ and shortening *V*, gives the Euler equation for the dynamics of the medium:

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t}+(\mathbf{v}\nabla)\right)\mathbf{v}=-\nabla p+\mathbf{f}. \qquad (2.1.5b)$$

The equation for the entropy has evidently the following form

$$\frac{\partial s}{\partial t}+(\mathbf{v}\nabla)s=0. \qquad (2.1.6a)$$

The system of the equations (2.1.4)–(2.1.6a) should be supplemented by the equation of the state of matter (gas)

$$p=p(\rho,s)=p_0\left(\frac{\rho}{\rho_0}\right)^{\gamma}\exp\left\{\frac{s}{c_v}\right\}. \qquad (2.1.6b)$$

Here $\gamma = c_p/c_V$ is the adiabatic exponent.

The equations (2.1.6a) and (2.1.6b) will be often replaced by the barotropic equation

$$p = p(\rho). \qquad (2.1.6c)$$

In particular, isothermal (T = const) and adiabatic $p=p_0\left(\frac{\rho}{\rho_0}\right)^{\gamma}$ flows are barotropic.

Using the equations (2.1.6) and (2.1.5) gives the energy equation [13]

$$\frac{\partial}{\partial t}\frac{\rho}{M}\left(\varepsilon+\frac{Mv^2}{2}\right)+\operatorname{div}\left(\frac{\rho}{M}\mathbf{v}\left(i+\frac{Mv^2}{2}\right)\right)=0. \qquad (2.1.7)$$

Here M is the particle mass, $i=\frac{5}{2}kT$ is enthalpy, ε is energy, $\varepsilon=\frac{3}{2}kT$.

If we take the heat conductivity into account, div $\varkappa_T\nabla T$, where $\varkappa_T$ is the heat conductivity coefficient, should be substituted into the right-hand part of the above equation.

2.1.2. Ensuring autonomous state of the droplet

The conditions of the 'autonomous state' of the droplets will now be discussed. Evidently, the autonomous state is associated in all likelihood

with the permissible upper size of the droplet when describing the process in the given system. As the variation of the parameters in the investigated volume of the medium becomes smoother and smoother, and as the rate of flow of the medium increases, the size of the droplets which can be used for describing the process also increases. This is especially evident when constructing the discrete schemes for numerical calculations. This is formally also confirmed by the example of classic hydrodynamics. As it is well known, the Navier–Stokes equation [13] is more general than the Euler equation

$$\frac{d\mathbf{v}}{dt} = -\nabla p + \Delta\mathbf{v} + \left(\zeta + \frac{\eta}{3}\right)_1 \operatorname{grad}\operatorname{div}\mathbf{v}. \tag{2.1.8}$$

Here η and ζ are the dynamic viscosity coefficients, and for conventional gases

$$\eta \sim \zeta \sim \nu\rho.$$

Here $\nu = \theta\lambda v_T$ is the kinematic coefficient of viscosity, λ is the free path, v_T is the thermal velocity of the particles forming the medium, $\theta \sim 1$, if the scales of the values are indicated by the braces, for example

$$\{x\} = L; \quad \{v\} = V$$

the ratio of the scales of the viscosity terms to the dynamic ones is

$$\frac{\left\{\eta\Delta\mathbf{v}\right\}}{\left\{\rho\dfrac{d\mathbf{v}}{dt}\right\}} = \frac{\nu}{LV} = \frac{1}{\mathrm{Re}}. \tag{2.1.9}$$

Here Re is the dimensionless Reynolds number. If in the entire investigated volume $\mathrm{Re} \gg 1$, then the viscosity can be ignored in the first approximation in the calculations. It should be mentioned that the appearance of viscosity terms in the Navier–Stokes equation is associated with taking into account the friction of the droplet on the droplet (coefficient η), and also friction in compression of the droplet (coefficient $\zeta + \eta/3$).

This can also be said of the equation for entropy (2.1.6a). Here, the sources of entropy are the heat generation in friction and heat transfer from droplet to droplet as a result of heat conductivity:

$$\rho T \frac{ds}{dt} = \mathrm{div}(\varkappa_T \nabla T) + \frac{\eta}{2}\left(\frac{\partial v_i}{\partial x_k} + \frac{\partial v_k}{\partial x_i} - \frac{2}{3}\delta_{ik}\frac{\partial v_i}{\partial v_i}\right)^2 + \zeta(\mathrm{div}\,\mathbf{v}^2). \qquad (2.1.10)$$

Here $\varkappa$ is the heat conductivity coefficient which is associated by the following relationship with the thermal diffusivity coefficient χ:

$$\varkappa_T = \chi\rho c_p.$$

It is also important to mention the dimensionless Prandtl's value with the order of one for gases

$$P = \frac{\nu}{\chi} \sim 1. \qquad (2.1.11)$$

However, we return to the general evaluation of the condition of the autonomous state of the droplet.

Equations (2.1.9) and (2.1.11) show that a sufficient condition of the smallness of the dissipative terms are: the large dimensions of the system, the relatively short time of stay of 'droplet' in the investigated volume, and the short free path, as efficiently reflected by the Reynolds number and its analog – the Prandtl number. However, the smallness of the free path is not a compulsory condition. In subsequent sections, attention will be given to the flow of plasma across the magnetic field. If the strength of the field is sufficiently high, the Larmor radii of the electrons and ions will be small and this will result in the quasi-autonomous nature of the 'droplets' which are in the given case elongated along the magnetic lines of force of the tubes with plasma. However, the free path in relation to collisions can be as long as necessary.

There is another case of hydrodynamic description (of course, with a very strong dissipation). Fine-scale oscillations can take place quite easily in plasma and they lead to the appearance of a strongly fluctuating electrical field on which the electrons and, in a number of cases, ions are scattered. Thus, this results in the formation of 'anomalous collisions' creating suitable conditions for hydrodynamic description.

All these factors confirm the importance of ideal hydrodynamics for a wide range of plasma flows. Evidently, this model does not describe the 'boundary' layer which forms in contact of a gas (liquid) with solid surfaces, and also does not describe turbulent processes accompanied by the formation of fine-scale vortices. These formations evidently depend on the values of the free path.

2.1.3. Two laws of conservation in the flow of an ideal gas

To derive the laws of conservation, the dynamics equation (2.1.5b) can be presented in the form:

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t}+\nabla\frac{v^2}{2}-\left[\mathbf{v},\mathrm{rot}\,\mathbf{v}\right]\right)=-\nabla p+\mathbf{f}_{\mathrm{vol}}. \tag{2.1.12}$$

Here, we used the vector identity

$$(\mathbf{v}\nabla)\mathbf{v}=\nabla\frac{\mathbf{v}^2}{2}-\left[\mathbf{v},\mathrm{rot}\,\mathbf{v}\right]. \tag{2.1.13}$$

The Bernoulli equation (law of energy conservation)
Let the flow be stationary and $\mathbf{f}_{\mathrm{vol}} = 0$, $p = p\,(\rho)$. Consequently, we can introduce generalised enthalpy

$$i\,(\rho)\equiv\int\frac{dp\,(\rho)}{\rho} \tag{2.1.14}$$

and write the equation (2.1.12) in the form

$$\nabla\left(\frac{v^2}{2}+i\,(\rho)\right)=\left[\mathbf{v},\mathrm{rot}\,\mathbf{v}\right]. \tag{2.1.15a}$$

Multiplying this equation in the scalar manner by V, we obtain[2]

$$(\mathbf{v}\nabla)\left(\frac{v^2}{2}+i\,(\rho)\right)=0. \tag{2.1.15b}$$

This equation indicates that the sum of the thermal and kinetic energies, normalised with respect to unit mass, is retained along the droplet trajectory (the trajectory is denoted by ψ):

$$\frac{v^2}{2}+i\,(\rho)=U\,(\psi). \tag{2.1.16}$$

If the entropy of the droplet is retained during the flow in accordance with (2.1.6a), then

$$s = s(\psi). \tag{2.1.17a}$$

[2]The same ratio follows directly from (2.1.7) if we assume that $\partial/\partial t = 0$ and the continuity equation div $\rho\mathbf{v} = 0$ is taken into account

Taking this fact into account and using the general equation of state $p = p(\rho, S)$, we can introduce the enthalpy for the given droplet

$$i(\rho, S(\psi)) = \int\limits_{\psi=\text{const}} \frac{d\,\rho(\rho, s(\psi))}{\rho} \tag{2.1.17b}$$

and we obtain an equation identical with (2.1.16)

$$\frac{v^2}{2} + i(\rho, s) = U(\psi). \tag{2.1.17c}$$

Finally, if the volume force, related to the mass unit, is potential (as, for example, the gravitational force), i.e.

$$\frac{\mathbf{f}_{vol}}{\rho} = -\nabla W, \tag{2.1.18}$$

then instead of (2.1.17c) we obtain a more general expression

$$\frac{v^2}{r} + i(\rho, s) + W = U(\psi). \tag{2.1.19}$$

Conservation of the circulation of the velocity on the 'liquid contour
In the 19th century, Kelvin and Helmholtz proved that the Euler equation for the incompressible liquid leads to the conservation of 'circulation'

$$C = \int\limits_{(\gamma)} \mathbf{v} d\,\mathbf{l} = \text{const}, \tag{2.1.20a}$$

where the integral is calculated along the 'liquid' contour γ, i.e., the contour formed by the labelled elements of the medium and deformed as a result of flow of the medium.

Actually, in accordance with the Stokes theorem, we can write

$$C = \int\limits_{(\gamma)} \mathbf{v} d\mathbf{l} = \iint\limits_{S_\gamma} (\text{rot}\,\mathbf{v}) d\mathbf{S}, \tag{2.1.20b}$$

Here S_γ is the area of the surface stretched on the contour γ.

Differentiating 2.1.20b with respect to t gives

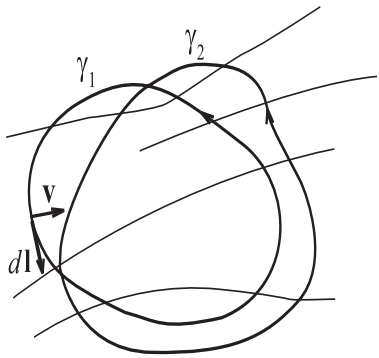


Fig. 2.1.2. Derivation of the Kelvin-Helmholtz theorem: γ_1 is the smooth contour which transforms into a contour γ_2 during movement of the medium, $d\mathbf{l}$ is an element of the contour, $\mathbf{v}$ is the velocity of the medium.

$$\begin{aligned}\frac{\partial C}{\partial t} &= \int\limits_{S_\gamma} \frac{\partial \operatorname{rot}\mathbf{v}}{\partial t} d\mathbf{S} + \int\limits_{(\gamma)} (\operatorname{rot}\mathbf{v})\left[\mathbf{v}, d\mathbf{l}\right] \\ &= +\iint\limits_{S_\gamma} \operatorname{rot}\left[\mathbf{v}, \operatorname{rot}\mathbf{v}\right] dS + \int\limits_{(\gamma)} \left[\operatorname{rot}\mathbf{v}, \mathbf{v}\right] d\mathbf{l} \\ &= \int\limits_{(\gamma)} \left(\left[\mathbf{v}, \operatorname{rot}\mathbf{v}\right] + \left[\operatorname{rot}\mathbf{v}, \mathbf{v}\right]\right) d\mathbf{l} = 0. \end{aligned} \tag{2.1.21}$$

The equation

$$\frac{\partial}{\partial t} \operatorname{rot}\mathbf{v} - \operatorname{rot}\left[\mathbf{v}, \operatorname{rot}\mathbf{v}\right] = 0, \tag{2.1.22}$$

was used in this case. This equation follows directly from (2.1.12) at $p = p(\rho)$ and $\mathbf{f}_{\text{vol}} = \nabla W$. In addition, it has been taken into account (see Fig. 2.1.2) that the differential of the variation of the area of the contour as a result of its deformation is equal to

$$\frac{d}{dt} d\mathbf{S} = \left[\mathbf{v}, d\mathbf{l}\right]$$

As a result, we use and generalise the Bernoulli equation and the Kelvin–Helmholtz theorem.

2.2. Examples of Euler hydrodynamics problems

Using the program described in the introduction, we examine here four classes of the Euler hydrodynamics problems: hydrostatics, linear waves

in a homogeneous gas, gas flow through thin tubes with a variable cross-section, and straight shock waves.

2.2.1. Hydro (gas)-statics

Static configurations are configurations in which $\partial/\partial t = 0$, $\mathbf{v} = 0$. In this case, the system of the Euler equations is reduced to a single equation

$$\nabla p = \mathbf{f}_{\text{vol}}. \tag{2.2.1}$$

a. If the volume forces (in particular, the gravitational force) can be ignored, the equation (2.2.1) has the following form

$$\nabla p = 0. \tag{2.2.2}$$

This is the Pascal law: in a stationary liquid or a gas in the absence of external forces the pressure in the entire volume is the same.

b. We now consider an incompressible liquid in the gravity field. In this case, $\mathbf{f}_{\text{vol}} = \mathbf{g}\rho$, and equation (2.2.1) is integrated in an elementary manner (the z axis is directed downwards)

$$p = p_0 + \rho g z .$$

This leads to the Archimedes law.

Barometric equation (Boltzmann equation)
We consider a gas column at a constant temperature. Consequently, the following equation can be written

$$p = n\kappa T; \quad \rho = nM.$$

Here n is gas density, k is the Boltzmann constant. Substituting these expressions into (2.2.1) and integrating gives the 'barometric formula'

$$n = n_0 e^{-\frac{Mgz}{T}}. \tag{2.2.3}$$

The problem of the stability of barometric and other distributions of the density in the direction of height is investigated in the section 8.1.

2.2.2. Linear waves in a homogeneous gas

These waves with a low amplitude in a homogeneous medium constant with time can easily be investigated using linearisation of the initial system of equations.

The system of the Euler equations:

$$\frac{\partial \rho}{\partial t} + \operatorname{div} \rho \mathbf{v} = 0\,; \quad \rho\left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v}\right) = -\nabla p; \quad p = p(\rho). \tag{2.2.4}$$

This is a very complicated non-linear system. However, at the same time, the system has a number of simple solutions. In particular, the uniform static distribution when

$\rho = \rho_0 = \text{const}, \quad p = p_0 = \text{const}, \quad \mathbf{v} = 0.$

If this equilibrium is now slightly disrupted, assuming that

$$\rho = \rho_0 + \rho_1, \quad p = p_0 + \frac{\partial p_0}{\partial \rho_0}\rho_1\,; \quad \mathbf{v} = \mathbf{v}_1, \tag{2.2.5a}$$

the system (2.2.4) can be 'linearised'.

The smallness of the perturbation indicates that $|\rho_1| \ll \rho_0$, $|v_1| \ll c_T$, where c_T is the velocity of sound. Later, it will be shown that

$$\frac{\partial p_0}{\partial \rho_0} = c_T^2, \quad c_T^2 = \frac{\gamma k T}{M}. \tag{2.2.5b}$$

Here γ is the adiabatic exponent.

Substituting (2.2.5a) into the Euler equations and taking into account the smallness of the perturbation, we can neglect the quadratic and cubic terms. Consequently, we obtain the following linear system of equations:

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \operatorname{div} \mathbf{v}_1 = 0\,; \tag{2.2.6a}$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial v} = -c_T^2 \nabla \rho_1. \tag{2.2.6b}$$

Sound waves

The resultant system, in particular, describes the sound waves. In fact, differentiating the first equation with respect to time and excluding (using the second equation) **v**, gives the classic wave equation

$$\frac{\partial^2 \rho}{\partial t^2} = c_T^2 \Delta \rho = c_T^2 \left(\frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 \rho}{\partial y^2} + \frac{\partial^2 \rho}{\partial z^2}\right). \tag{2.2.7}$$

In this equation, the required quantity is the perturbation of density ρ_1. As indicated by equation (2.2.6b), knowing ρ_1 we can determine the perturbation of pressure and the components of the velocity formed during propagation of the wave. It is also evident that the components of the resultant velocity also satisfy the wave equation.

The important properties of the sound waves in the homogeneous media can be explained on the basis of a simple example of a flat monochromatic wave. In calculations using the linear equations, it is convenient to use exponents and, if necessary, use the real part of the complex expression only in the final equations. In this case, the following equation can be used

$$\rho_1 = \mathrm{Re} A\exp\{-i\omega t + i\varkappa\mathbf{x}\} = A\cos(\omega t - \varkappa\mathbf{x}). \tag{2.2.8}$$

Here ω is the angular frequency, $\omega = 2\pi/T$, $\varkappa$ is the wave vector characterising the direction of propagation of the wave and the length of the wave ($|\varkappa| = 2\pi/\lambda$), T and λ is the period of the length of the wave. Substituting (2.2.8) into (2.2.7), we obtain the relationship of frequency with the wave vector

$$\omega^2 = c_T^2\varkappa^2. \tag{2.2.9a}$$

As already mentioned in section 1.5, the relationship between ω and $\varkappa$ is regarded as the dispersion ratio. In the case of a sound wave, the phase velocity for all wavelengths is the same and equal to c_T:

$$\frac{\omega}{\varkappa} \equiv v_\phi = c_T\,. \tag{2.2.9b}$$

It should be mentioned that the waves propagating in plasma usually do not have this property, i.e. phase velocity depends on $\varkappa$.

Sound waves are the longitudinal waves because the displacement of the particles of the medium **v** during their propagation is parallel to the wave vector (see (2.2.6b))

$$\mathbf{v} = \varkappa\frac{c_T\rho_1}{\omega\rho_0}. \tag{2.2.9c}$$

Transverse 'waves' in a gas

It may easily be seen that the transition from the complete linear system of the equations (2.2.6) to equation (2.2.7) for sound waves is accompanied by the loss of one type of structure. Actually, if (2.2.8) is substituted into (2.2.6), we obtain a system of uniform algebraic equations of the first order:

$$-\omega\rho_1 + \rho_0 \varkappa \mathbf{v}_1 = 0; \tag{2.2.10a}$$

$$\omega\rho_0 \mathbf{v}_1 - c_T^2 \varkappa \rho_1 = 0. \tag{2.2.10b}$$

The system has nontrivial solutions if its determinant is equal to 0

$$D(\omega, \varkappa) = \begin{vmatrix} -\omega & \rho_0 \varkappa_x & \rho_0 \varkappa_y & \rho_0 \varkappa_z \\ -c_T^2 \varkappa_x & \omega\rho_0 & 0 & 0 \\ -c_T^2 \varkappa_y & 0 & \omega\rho_0 & 0 \\ -c_T^2 \varkappa_z & 0 & 0 & \omega\rho_0 \end{vmatrix} = -\rho_0^3 \omega^2 \left(\omega^2 - c_T^2 \varkappa^2 \right) = 0. \tag{2.2.11}$$

Evidently, this determinant is also the general dispersion equation for the given case. It includes two greatly different co-multipliers

$$\omega^2 = 0, \quad \omega^2 - c_T^2 \varkappa^2 = 0. \tag{2.2.12a}$$

The second expression describes the previously investigated sound waves. The first one describes the stationary configuration. Substituting $\omega = 0$ into (2.2.10) gives the solution in the form

$$\varkappa \mathbf{v}_1 = 0, \quad \rho_1 = 0. \tag{2.2.12b}$$

This means that a standing ‘wave’ forms in this case. The fact that there are two roots ($\omega^2 = 0$) indicates that there can be two orientations of displacement of the medium in relation to the wave vector, i.e., two polarisations.

The nature of the ‘waves’ with $\omega = 0$ can be investigated more efficiently if we consider a planar case. Consequently, the single equation, describing these perturbations, has the form

$$\operatorname{div} \mathbf{v}_1 = \frac{\partial v_{1x}}{\partial x} + \frac{\partial v_{1y}}{\partial y} = 0. \tag{2.2.12c}$$

The procedure which will now be used is the same as that used previously when examining the symmetric magnetic fields. In particular, we introduce the function of the flow $\psi(x, y)$

$$v_{1x} = \frac{\partial \psi}{\partial y}, \quad v_{1y} = \frac{-\partial \psi}{\partial x}. \tag{2.2.12d}$$

This automatically satisfies the equation (2.2.12c) at any type of $\psi(x, y)$. Evidently, the trajectories of the ‘droplets’ of the liquid, like the lines of force of the magnetic field, are the lines

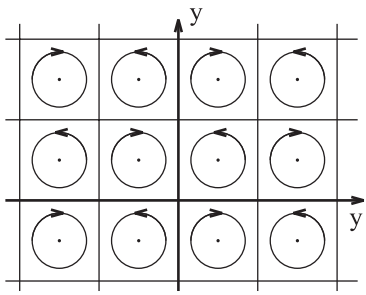


Fig. 2.2.1. 'Mosaic' of vortices.

$$\psi(x, y) = \text{const.} \tag{2.2.12e}$$

In particular, assuming that

$$\psi = A \sin\frac{\pi}{a}x \sin\frac{\pi}{b}y = \frac{A}{2}\left[\cos\pi\left(\frac{x}{a}-\frac{y}{b}\right)-\cos\pi\left(\frac{x}{a}+\frac{y}{b}\right)\right], \tag{2.2.13}$$

we obtain the mosaic of vortices (Fig. 2.2.1) which forms as the superposition of two 'oblique waves'.

The flow of a stationary uniform gas flow around thin profiles
Previously, attention was given to small perturbations of a uniform stationary gas. This model is automatically generalised for the case in which the non-perturbed state is represented by a uniform flow moving with a constant velocity

$$\rho = \rho_0 = \text{const}, \quad \mathbf{v} = \mathbf{v}_0 = \text{const}. \tag{2.2.14a}$$

In this case, the equations for the sound waves, as may easily be seen (either by direct linearisation of the system (2.2.4) on the condition (2.2.14a), using the Galileo transform for (2.2.7)), has the form

$$\left(\frac{\partial}{\partial t}+\mathbf{v}_0\nabla\right)^2 \rho_1 = c_T^2 \Delta \rho_1 \tag{2.2.14b}$$

This equation may be used in particular for analysing the flow of a uniform flow around 'thin profiles', for example, aircraft wings. If the process is investigated in a reference system connected with the wing (from the viewpoint of the passenger), the process is stationary and in this case $\partial/\partial t = 0$. Consequently, we obtain the equation

$$\left(\mathbf{v}_0\nabla\right)^2 \rho_1 = \Delta\rho_1. \tag{2.2.14c}$$

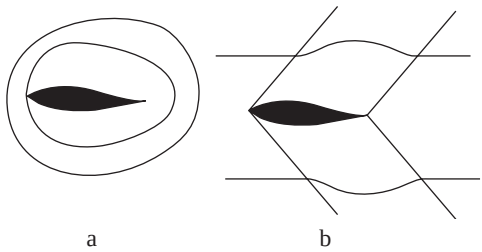


Fig. 2.2.2. A planar model of flow around an aircraft wing: a) subsonic flow; b) supersonic flow.

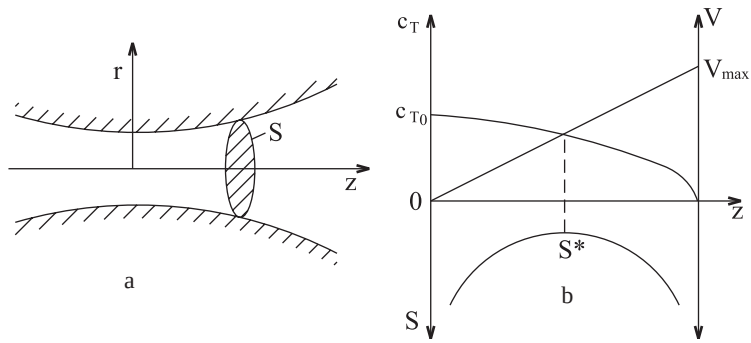


Fig. 2.2.3. Flow in a pipe with a variable section: a) scheme of the pipe, b) variation of the cross-section *S*, the velocity of flow *v* and the local velocity of sound c_T along the pipe.

In a planar case, directing the *x* axis along the velocity of the incident flow, the equation (2.2.14c) can be presented in the following form

$$\left(1-\frac{v_0^2}{c_T^2}\right)\frac{\partial^2 \rho_1}{\partial x^2}+\frac{\partial^2 p_2}{\partial y^2}=0. \qquad (2.2.15)$$

This shows that is the velocity of the incident flow is subsonic ($v_0 < C_T$), the resultant equation is of the elliptical type. This means that all the perturbations of the flow are concentrated around the aircraft. However, if $v_0 > C_T$, the equation becomes hyperbolic and the perturbations, described by this equation, are displaced to infinity (Fig. 2.2.2). These are the well-known shock waves generated by supersonic aircraft. The width of the perturbation zone is approximately equal to the width of the wing.

2.2.3. Flow of an ideal gas in a thin pipe with a variable section

Attention will now be given to the simplest model of a device transforming the thermal energy of a gas to its kinetic energy. It

is a 'thin' pipe with a variable cross-section (Fig. 2.2.3), and the input of the pipe receives the heated gas (temperature T_0) with a low velocity, and at the exit into the 'void' the gas has already cooled down ($T_{out} \to 0$) with the kinetic energy of the particles $\varepsilon_{out} \sim kT_0$. The 'thinness' of the pipe indicates that in the given approximation we can ignore the dependence of the longitudinal velocity v and density ρ on the distance r to the axis:

$$v_x(r, z) = v(z) + O(r^2);$$

$$\rho(r, z) = \rho(z) + O(r^2).$$

The process is regarded as stationary and adiabatic. This model was investigated in the 1860s by French scientist Gugonio[3].

With these assumptions, the dynamics of gas is described by a system of two algebraic equations: the law of conservation of matter and the Bernoulii equation

$$\rho v S = \dot{m} = \text{const}; \tag{2.2.16a}$$

$$\frac{v^2}{2} + i(p) = i_0 = \text{const}. \tag{2.2.16b}$$

Here S is the cross-section of the pipe, and in the second equation it is assumed that the kinetic energy of the gas at entry can be ignored in comparison with its thermal energy. In the case of the adiabatic process $p = p_0\left(\frac{\rho}{\rho_0}\right)^{\gamma}$, the enthalpy is:

$$i = \int \frac{dp(\rho)}{d\rho} = \frac{\gamma}{\gamma - 1}\left(\frac{p_0}{\rho_0}\right)\left(\frac{\rho}{\rho_0}\right)^{\gamma-1} = \frac{c_{T0}^2}{\gamma - 1}\left(\frac{\rho}{\rho_0}\right)^{\gamma-1}. \tag{2.2.17}$$

Equation (2.2.16b) shows that at exit, where $\rho \to 0$, and consequently also $i(\rho) \to 0$, the maximum velocity is:

$$v_{\max} = \sqrt{2i_0} = c_{T0}\sqrt{\frac{2}{\gamma - 1}}, \tag{2.2.18}$$

i.e. it is of the order of the velocity of sound at entry.

Taking into account that $p = nkT$, $\rho = Mn$:

[3]Profiled pipes, adequate to theory, were used in practice for the first time by Laval (~1860) in a steam turbine (Laval nozzle).

$$c_T = \sqrt{\gamma \frac{kT}{M}},$$

it may be seen that v_{max} increases with the increase of temperature at entry (in the combustion chamber) and the decrease of the mass of the resultant particles. Therefore, the highest discharge rates are produced by 'cryogenic' engines which use hydrogen and oxygen and water is produced as a result.

However, we return to the system of two equations (2.2.16) and explain the conditions at which there is a regular (without brakes) acceleration of gas as a result of thermal energy. This can be performed quite easily, writing the equations in the differential form

$$\frac{d\rho}{\rho} + \frac{dv}{v} + \frac{dS}{S} = 0; \quad vdv + c_T^2 \frac{d\rho}{\rho} = 0. \tag{2.2.19}$$

Excluding from the above equation $d\rho$, we obtain the relationship between dv and dS

$$\left(\frac{v^2}{c_T^2} - 1\right)\frac{dv}{v} = \frac{dS}{S}. \tag{2.2.20}$$

This shows that if the flow is subsonic ($c_T > v$), the velocity of the flow increases ($dv > 0$), if the cross-section of the pipe is compressed ($dS < 0$). The transition through the local velocity of sound ($v = c_T$), the cross-section should be minimum ($dS = 0$). Finally, the increase of the velocity of the supersonic flow requires expansion of the channel ($dS > 0$). Thus, the regular acceleration of the gas from $v \sim 0$ to $v \sim v_{max}$ may take place only in the droplet with a bundle ('critical cross-section') where the velocity of the flow transfers through the local velocity of sound (Fig. 2.2.3b).

However, if the cross-section of the pipe in the zone of the supersonic flow is reduced, a shock wave usually forms in this area.

Differences in the behaviour of the subsonic in supersonic flows in the pipe can be explained as follows. In the subsonic zone, the extent of expansion of the gas is small and the gas here behaves as an incompressible liquid. It is well known that the velocity of water increases when the cross-section of the flow decreases. On the other hand, in the supersonic zone, the interaction between the particles (droplets) weakens and this zone is characterised by the transition of the transverse energy of the particles into longitudinal energy as a result of collisions with walls. As a result of the conservation of the

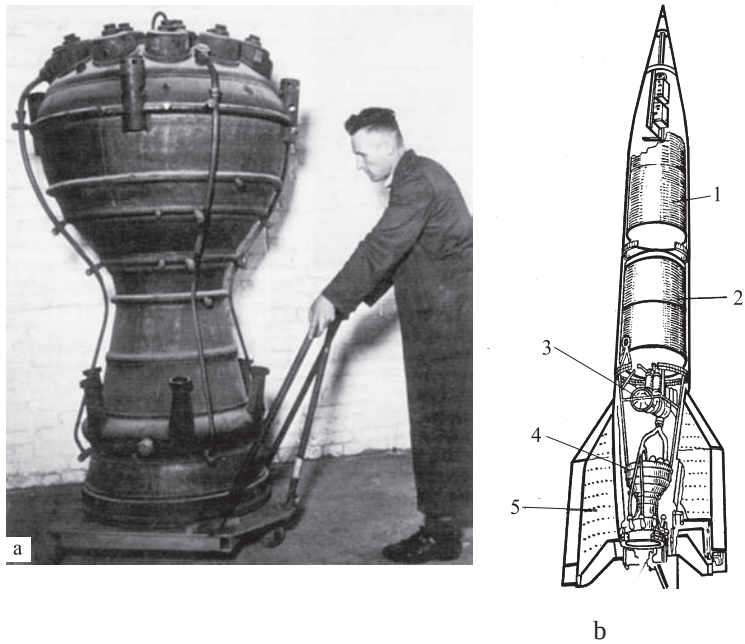


Fig. 2.2.4. Rocket A-4 (V-2): a) engines of the rocket, combustion chamber is on the top; b) the scheme of the rocket A-4; 1) a tank with alcohol, 2) the tank with liquid oxygen, 3) the turbine for supplying working substances, 4) combustion chamber, 5) stabiliser.

transverse adiabatic invariant ($J_\perp = v_\perp h$) the width of the channel h should increase because of this.

Comment. The second half of the 20th century was marked by technical achievements which rapidly penetrated into the entire area of human activity, including ideology.

This was the flight of humans to cosmos. This outstanding achievement was possible mainly due to the development of liquid-fuel rocket engines (LRE). The first powerful LRE, which produced a very high thrust (~ 20 t) for that period (the beginning of the 1940s) was the engine developed and applied in practice by the 30-year-old German engineer Werner von Braun who supervised work on the development of the V-2 rocket, with the working name A-4 (Agregat-4)[4]. The rocket weighed ~13 t, was ~14 m long, and its engine operated using a mixture of

[4]In her memories of her father, the daughter of S.P. Korolev who visisted with him the area where the A-4 was developed writes that to mark the first successful launch of a rocket: ''at the launch platform there was a huge rock with a bronze plaque saying: This stone fell from my heart October 3, 1942, Werner von Braun'' [65].

ethyl alcohol and liquid oxygen, ejecting approximately 100 kg of combustion products every second (Fig. 2.2.4) [64].

The engines, developed by von Braun were installed in the first ballistic rockets (P-I) in the former Soviet Union and in the USA (von Braun was sent to the USA after the war).

Later, the liquid-fuel rocket engines underwent extensive changes, with the main change being transition from low-pressure (~20 atm) to high-pressure combustion chambers ($\gtrsim$ 60 atm).

Working now in the USA, von Braun made another breakthrough marvel. Under his leadership, scientists constructed the engine Saturn-V with a thrust of ~1000 t (Fig. 10.4.1). These engines enabled the Americans to send three men to the Moon and return them to the Earth (July 1969).

The reader can naturally ask: how is all this connected with plasma dynamics? In fact, there is a direct connection.

Firstly, the liquid-fuel rocket engine is regarded as some reference device when we discuss (in section 10.4) the electric reactive engines (ERE).

Secondly, the temperature in the combustion chamber of the liquid-fuel rocket engine working with oxygen and hydrogen reaches ~4000 K and this is the temperature of solar spots. The concentration of electrons here is ~0.1% of the total number of the particles. Therefore, it is not surprising that the products of combustion of the rocket fuel are used in MHD generators as the conducting medium.

Thirdly, in return of space systems to the Earth, plasma clouds form around the systems as in the case of the meteorites. Therefore, the knowledge of rocket technology is essential for everybody who wants to work with plasma.

2.2.4. Shock waves in the ideal gas [66]

We examine another fundamental phenomenon of classic hydrodynamics – shock waves (SW).

These waves are also of considerable interest for plasma dynamics, because they are always accompanied by ionisation of the gas if the flows which form them move at high velocitys (entry of the space systems into the atmosphere, meteorites, atomic explosions, etc). Figure 2.2.5b shows high-velocity photographs of a propagating shock wave caused by the explosion of a block of an explosive substance in air ([67]).

One of the simplest schemes of generation of the shock waves may be described as follows. Let us imagine a pipe with a gas which is

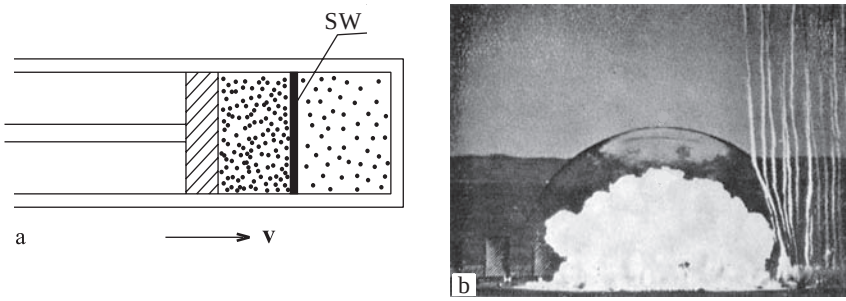


Fig. 2.2.5. Generation of a shock wave in a cylinder with a movable piston, **v** is the velocity of movement of the piston (a), the shock wave, generated by explosion of 20 t of trinitrotoluene (b).

closed hermetically on one side and on the other side it is closed with a movable piston (Fig. 2.2.5a). If the piston travels at a low velocity ($v_{pist} << c_T$), the pressure along the entire length of the pipe increases almost uniformly with the delay of $\tau \sim 2L/c_T$, where L is the length of the pipe.

With the increase of the velocity of movement of the piston, but at $v_{pist} < c_T$, the distribution of pressure in the closed part of the pipe becomes even less uniform. However, the image is greatly changed when the velocity of the piston v_{pist} exceeds the velocity of sound c_T in the gas. In this case, 'an infinitely fine' break forms, i.e. a jump in the parameters (ρ, T, v) – and this is the 'shock wave'. The gas in front of the shock wave remains stationary, and behind the shock wave it moves with a constant velocity and with a higher density and temperature.

If we transfer to the reference system, connected with the shock wave, it may easily be seen that the Euler equations can be used to determine the relationship of the parameters (ρ_0, T_0, v_0) in front of the wave and behind it (ρ_1, T_1, v_1).

Actually, since the flow is assumed to be stationary and uniform, the laws of conservation of mass, the amount of motion and energy are fulfilled:

$$\rho_1 v_1 = \rho_2 v_2; \tag{2.2.21a}$$

$$\rho_1 v_1^2 + p_1 = \rho_2 v_2^2 + p_2; \tag{2.2.21b}$$

$$\frac{v_1^2}{2} + i_1(\rho_1) = \frac{v_2^2}{2} + i_2(\rho_2). \tag{2.2.21c}$$

This shows that we have a system of three algebraic equations for three unknown quantities ρ_2, v_2, p_2. Of course, it is assumed that the parameters of the gas in front of the shock wave are known.

Analysis of the system (2.2.21) can be found in any textbook for gas dynamics (see, for example [66]). Therefore, referring the reader to these books, we present the final Renkin–Gugonio equations for the relationships of the parameters in front of and behind the shock wave for a polytropic gas with the polytrope exponent γ:

$$\begin{aligned}
&\frac{\rho_2}{\rho_1}=\frac{v_1}{v_2}=\frac{(\gamma+1)M_1^2}{(\gamma-1)M_1^2+2};\\
&\frac{p_2}{p_1}=\frac{2\gamma}{\gamma+1}M_1^2-\frac{\gamma-1}{\gamma+1};\\
&\frac{T_2}{T_1}=\frac{\left(2\gamma M_i^2-(\gamma-1)\right)\left((\gamma-1)M_1^2+2\right)}{(\gamma+1)^2M_1^2}.
\end{aligned} \tag{2.2.22}$$

Here $M_1 = v_1/c_{T1}$ is the Mach number of the incident flow.

In particular, this shows that at high Mach numbers ($M_1^2 \to \infty$) we have

$$\begin{aligned}
&\frac{\rho_2}{\rho}\to\frac{\gamma+1}{\gamma-1};\\
&\frac{T_2}{T}\to\frac{2\gamma(\gamma-1)}{(\gamma+1)^2}M_1^2.
\end{aligned} \tag{2.2.23}$$

In other words, at $M^2_1 \to \infty$ the gas density increases only four times in a monatomic gas ($\gamma = 5/3$) and up to six times in the diatomic gas ($\gamma = 7/5$)[5]. The relationships (2.2.22) leave two questions: what are the mechanisms ensuring the formation of the shock wave and what is their characteristic thickness. A partial answer to the first question is provided by the Gugonio adiabate which follows directly from the system (2.2.21)

$$i_2-i_1=\frac{1}{2}\left(\frac{1}{\rho_1}+\frac{1}{\rho_2}\right)(p_2-p_1). \tag{2.2.24}$$

From this it follows that the entropy in the shock wave increases. In particular, in the case of a small high-intensity shock wave,

[5]Expressing γ by the number of degrees of freedom κ we obtain a simple expression for the maximum compression in the shock wave $\rho_2/\rho_1 \to \kappa + 1$ (V.A. Belokon').

equation (2.2.24) may be used easily for determining the entropy jump [13]

$$s_2 - s_1 = \frac{1}{2T}\left(\frac{\partial^2(1/\rho)}{\partial p_1^2}\right)_s (p_2 - p_1)^3.$$

Consequently, dissipation is essential for the formation of a shock wave. Its specific nature determines the thickness and structure of the wave but in principle almost any dissipation (viscosity, heat conductivity, final conductivity, different types of turbulence, which is characteristic of plasma) may ensure the formation of a shock wave. In the case of a relatively strong shock wave in the 'conventional' gas, the thickness of the shock wave is of the order of the free path of the particles. However, this is observed in the case in which there is no excitation of the internal degrees of freedom of the particles.

2.3. One-fluid magnetic hydrodynamics (MHD) [12, 14]

Plasma is a conducting medium. Therefore, currents can be easily induced in the plasma. If a magnetic field appears in such a plasma formation, the plasma is affected by the ampere force with the density

$$\mathbf{f} = \frac{1}{c}[\mathbf{j} \times \mathbf{H}]. \qquad (2.3.1)$$

We have already discussed such an interaction, considering the pulsed accelerators and pinches. However, these systems were developed in the 50s of the previous century. Ten years prior to this stage, the astrophysicist H. Alfvén proposed the hydrodynamic theory of self-matched behaviour of highly conducting current-carrying plasma and the magnetic field [21].

Alfvén showed that in the first approximation dissipation is not significant for a large number of astrophysical phenomena. Therefore, these phenomena can be described by the combined system of equations of ideal hydrodynamics, the Maxwell equations and the Ohm's law at $\sigma \to \infty$. The processes can be regarded as non-relativistic and the bias currents can be ignored. Ignoring the dissipation greatly simplified the system and, consequently, it was possible to move forward in understanding of the magnetic hydrodynamic processes.

This simplified system of equations of 'magnetic hydrodynamics' (MHD) is very effective not only for modelling astrophysical processes

but also for describing a number of laboratory plasma systems[6]. Since MHD is relatively simple, it is used often, especially in the initial stage of modelling laboratory plasma-dynamic systems.

2.3.1. MHD equations

Formulation of a system of equations

In accordance with the above considerations, the starting point will be the following system of equations (σ conductivity, $\varkappa_T$ – heat conductivity of plasma):

$$\frac{\partial \rho}{dt} + \operatorname{div} \rho\mathbf{v}=0; \quad \rho\frac{d\mathbf{v}}{dt} = -\nabla p + \frac{1}{c}\left[\mathbf{j},\mathbf{H}\right]; \; \rho T\frac{ds}{dt} = \frac{\mathbf{j}^2}{\sigma} + \operatorname{div} \varkappa_T \nabla T, \tag{2.3.2a}$$

$$\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \quad \operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \quad \operatorname{div}\mathbf{H} = 0, \tag{2.3.2b}$$

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{c}\left[\mathbf{v},\mathbf{H}\right]. \tag{2.3.2c}$$

The first group of the equations (2.3.2a) is a system of Euler equations in which the volume force is represented by the ampere force and the entropy equation which takes into account Joule heat and heat conductivity.

As in section 2.1, the entropy equation is often replaced by the energy equation. In the MHD case, this equation is derived if the density of energy of the magnetic field $H^2/8\pi$ is added to the first term in (2.1.7), and the Umov–Pointing vector is added to the second term $\mathbf{\Pi}$:

$$\frac{\partial}{\partial t}\left(\frac{\rho}{M}\left(\varepsilon + \frac{Mv^2}{2}\right) + \frac{H^2}{8\pi}\right) + \operatorname{div}\left(\frac{\rho}{M}\mathbf{v}\left(\omega + \frac{Mv^2}{2}\right) + \mathbf{\Pi}\right) = \operatorname{div}\varkappa_T\nabla T; \tag{2.3.2d}$$

$$\mathbf{\Pi} = \frac{c}{4\pi}\left[\mathbf{E},\mathbf{H}\right].$$

[6]The flows of relatively low conductivity fluids (mercury, etc.) in the magnetic field was studied by G.Hartman ten years before H. Alfvén.

In the majority of cases, in this chapter we ignore detailed examination of the thermal processes and the equation for s can be simply replaced by the barotropic equation

$$p = p(\rho). \tag{2.3.2e}$$

The second group (2.3.2b) are the Maxwell equations without bias current[7]. These equations relate to the condition of closing (more accurately of the solenoidal form) of currents

$$\operatorname{div}\mathbf{j} = 0. \tag{2.3.3a}$$

Finally, the last equation (2.3.2c) is the conventional Ohm's law which takes into account the electromagnetic induction phenomenon. If we use rot from the Ohm's law and the equations (2.3.2b), **E** can be excluded. Consequently, at σ = const we obtain

$$\frac{\partial \mathbf{H}}{\partial t} = \operatorname{rot}\left[\mathbf{v},\mathbf{H}\right] + \nu_m \Delta \mathbf{H}. \tag{2.3.3b}$$

Here $\nu_m \equiv \dfrac{c^2}{4\pi\sigma}$ is 'magnetic viscosity'.

The right-hand part of this equation contains two terms: convective and dissipative. If we consider stationary conductors, the convective term is excluded. This leads to a conventional equation of the type of heat conductivity equation, and the equation describes the skin effect. The convective term can also be ignored if the 'magnetic Reynolds number' Re_m is small, i.e.

$$\mathrm{Re}_m = \frac{\left\{\operatorname{rot}\left[\mathbf{v},\mathbf{H}\right]\right\}}{\left\{\nu_m \Delta H\right\}} = \frac{L\upsilon}{\nu_m} \ll 1. \tag{2.3.4}$$

Here as previously the values in the braces indicate the scale of the quantity.

If, on the other hand, the magnetic Reynolds number is

$$\mathrm{Re}_m = \frac{\upsilon L}{\nu_m} \gg 1, \tag{2.3.5}$$

the ohmic term in (2.3.3b) can usually be ignored.

Below, we consider mostly dissipation-free processes. In this case, the system of MHD equations has the form:

---

[7]The fact that bias currents are ignored indicates that the value $d\mathbf{E}/dt$ is small in comparison with $4\pi\mathbf{j}$. In most cases, the condition $\{d\mathbf{E}/dt\} << 4\pi\{\mathbf{j}\}$ indicates the low frequency of the process.

$$\frac{\partial \rho}{\partial t} + \operatorname{div} \rho \mathbf{v} = 0; \tag{2.3.6a}$$

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\dot{\mathbf{v}}\right) = -\nabla p + \frac{1}{c}[\mathbf{j}, \mathbf{H}]; \tag{2.3.6b}$$

$$p = p(\rho); \tag{2.2.6c}$$

$$\operatorname{rot} \mathbf{H} = \frac{4\pi}{e}\mathbf{j}; \tag{2.3.6d}$$

$$\operatorname{div} \mathbf{H} = 0; \tag{2.3.6e}$$

$$\frac{\partial \mathbf{H}}{\partial t} = \operatorname{rot}[\mathbf{v}, \mathbf{H}]. \tag{2.3.6f}$$

This system of equations can be formally slightly simplified if **j** is excluded from equation (2.3.6b) by means of equation (2.3.6d).

Taking into account the vector identity

$$[\mathbf{a}, \operatorname{rot} \mathbf{a}] = \nabla \frac{\mathbf{a}^2}{2} - (\mathbf{a}\nabla)\mathbf{a},$$

gives:

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v}\right) = -\nabla\left(p + \frac{H^2}{8\pi}\right) + \frac{(\mathbf{H}\nabla)\mathbf{H}}{4\pi}. \tag{2.3.6g}$$

2.3.2. 'Freezing-in' of the magnetic field into plasma

Equation (2.3.6f), if it is taken into account that **H** = rot **A**, where **A** is the vector-potential, is formally completely equivalent to the equation (2.1.22) for **v**. However, the last equation gives the Kelvin–Helmholtz invariant – circulation of velocity. Evidently, the same invariant exists for **A**:

$$C = \oint_{(\gamma)} \mathbf{A} d\mathbf{l} = \iint_{S\gamma} \mathbf{H} d\mathbf{S} = \text{const.} \tag{2.3.7}$$

This means that the magnetic flux through an arbitrary liquid circuit γ is:

$$\Phi(\gamma) \equiv \iint_{S\gamma} \mathbf{H} d\mathbf{S} \tag{2.3.8}$$

is preserved in plasma with ideal conductivity (Fig. 2.3.1).

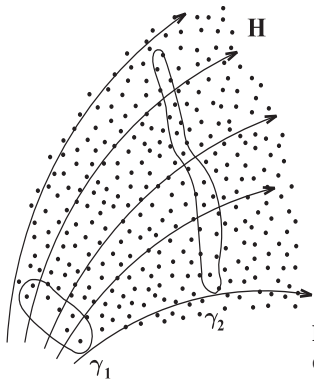


Fig. 2.3.1. 'Freezing-in' of the magnetic field in plasma: conservation of the magnetic flux, passing through the circuits γ_1, γ_2.

It is evident that in this case we are concerned with a slightly unique formulation of the well-known claim of classic electrodynamics on the conservation of the magnetic flux passing through an ideal conducting circuit at all its deformations without breaks and rejoining. This is shown by the law of electromagnetic induction proposed by Faraday and by the Ohm's law for the entire circuit. Actually, for a circuit having the resistance R we can write

$$-\frac{1}{c}\frac{d\Phi}{dt} = JR. \tag{2.3.9}$$

This shows that since the current cannot be infinitely high then at $R \to 0$

$$\frac{d\Phi}{dt}\Big|_{\gamma} \to 0,$$

i.e. Φ = const.

The integral equation (2.3.7) can be written in a form more suitable for application, if the system has axial symmetry and the field contains only the azimuthal component. For this purpose, we define an arbitrary magnetic toroidal surface with a small section S in the plasma flow (Fig. 2.3.2a). When this toroid moves, its mass $\mu = 2\pi rS\rho$ and the magnetic flux $\Phi = HS$ are preserved. Therefore, the 'freezing-in parameter' is also conserved

$$\varkappa \equiv 2\pi\frac{\Phi}{\mu} = \frac{H}{\rho r} = \text{const.} \tag{2.3.10a}$$

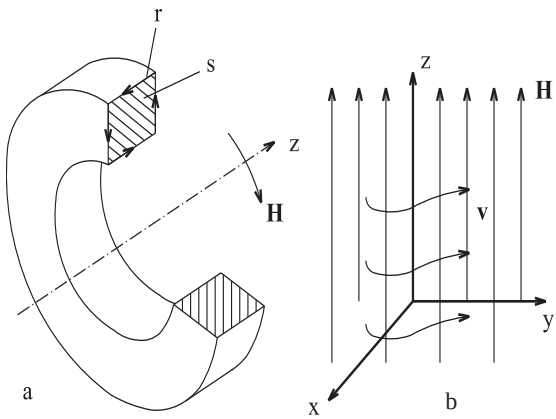


Fig. 2.3.2. Derivation of equation (2.3.10a) – ring-shaped plasma element in the azimuthal magnetic field (a). Flat flow of plasma in the transverse the magnetic field (b), **H** are the magnetic lines of force, **v** is the flow velocity.

The 'freezing-in parameter' parameter $\varkappa$ is preserved along the entire trajectory of the defined ring but its value can differ for different trajectories. If $\varkappa$ is constant in the entire flux, then this flux is referred to as 'isomagnetic'. If the flow is 'flat', i.e., depends only on (x, y), and the magnetic field has only the z-component (Fig. 2.3.2b), then this flow can be regarded as axisymmetric at $r \to \infty$. In this case, the 'freezing-in parameter' has a simpler form

$$\varkappa = \frac{H}{\rho}. \tag{2.3.10b}$$

These flat isomagnetic barotropic flows are formally completely identical with the Euler gas flows but with the modified barotropic relationship

$$p(\rho) \to P(\rho) \equiv p(\rho) + \frac{H^2}{8\pi} = p(\rho) + \frac{\varkappa^2}{8\pi}\rho^2. \tag{2.3.11a}$$

This claim follows directly from (2.3.6g) and (2.3.10b). In this case, the system of equations (2.3.6) has the explicitly 'Euler' form

$$\frac{\partial \rho}{\partial t} + \mathrm{div}\rho\mathbf{v} = 0; \quad \rho\frac{d\mathbf{v}}{dt} = -\nabla P(\rho). \tag{2.3.11b}$$

Isotropy of pressure in the MHD

The dynamics equation (2.3.2) includes the scalar quantity – pressure. This scalar form follows from the assumption on the isotropy of pressure.

In the presence of the magnetic field and in the absence of collisions the distribution is usually non-isotropic. For isotropisation to take place, the particles should be subjected to several collisions in the entire volume of the system. A suitable example of a plasma system with the anisotropic pressure is the mirror trap ('probkotron'), which also contains a loss cone in the space of velocitys. In addition, during movement of the particles towards the mirrors, the longitudinal energy in the particles transforms to transverse energy. Therefore, the mirror trap with the plasma, separated from the ends, cannot be described by the MHD models.

It should be mentioned that there have been attempts to construct gas-dynamics with anisotropic pressure (Chu, Goldberger, Low) but the results have not been used widely.

2.4. MHD statics

In accordance with the general plan of analysis of technical situations, for every basic plasma-dynamics model we start with the examination of the static ($\partial/\partial t = 0$, $\mathbf{v} = 0$) plasma configurations in MHD.

2.4.1. General properties of equilibrium MHD configurations

In this case, the system (2.3.6) is radically simplified and reduced to two vectors and a single scalar equation:

$$\nabla p = \frac{1}{c}[\mathbf{j}, \mathbf{H}]; \tag{2.4.1a}$$

$$\mathrm{rot}\,\mathbf{H} = \frac{4\pi}{e}\mathbf{j}; \tag{2.4.1b}$$

$$\mathrm{div}\mathbf{H} = 0. \tag{2.4.1c}$$

1. The first equation yields two conservation laws (by scalar multiplication by **H** and **j** in a scalar way)

$$(\mathbf{H}\nabla)p = 0; \tag{2.4.2a}$$

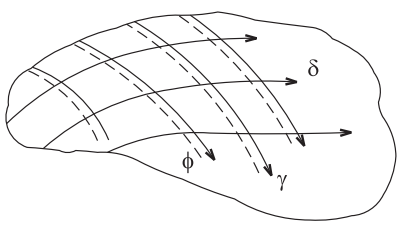


Fig. 2.4.1. Magnetic-current surfaces formed by magnetic lines of force (γ) and the lines of the electrical current (δ), ϕ-equipotentials at $T_e \to 0$.

$$(\mathbf{j}\nabla)p = 0. \qquad (2.4.2b)$$

This means that the pressure remains constant and on both the magnetic lines of force (γ) and on the lines of the electrical current (δ), i.e., on the 'magnetic current surfaces' (Fig. 2.4.1)

$$p = p(\gamma,\delta). \qquad (2.4.3)$$

The conclusion that the pressure in the MHD remains constant along the magnetic lines of force, i.e., there is no partial reflection from the regions with a high strength of the field, is the consequence of the assumption on the isotropic nature of the pressure determined, as mentioned in the previous section, by the frequent collisions of the particles. The role of rare collisions will be investigated in section 5.6.

The constancy of pressure on the magnetic-current surfaces $p(\gamma,\delta)$ leads immediately to two important conclusions.

a. For the magnetic field to confine the plasma in the presence of collisions of the particles (i.e. when p is isotropic), the lines of force of the field should remain completely within the limits of some finite volume.

b. If the plasma volume is surrounded by a cold wall, there should be a number of layers between the central zone with the highest temperature and the wall. Each of the layers is formed by the lines of force which do not extend outside the limits of 'their own' layer. Consequently, we obtain the pressure gradient in the system

$$\nabla p_{i,i+1} \sim \frac{p_{i+1} - \rho_i}{\delta}.$$

Here p_i is the pressure in the i-th case, δ is the thickness of the transition layer.

If the magnetic configuration has the structure in the form of the system of inserted toroidal surfaces[8] (Fig. 1.1.2), it is evident that the gradient may be without 'jumps'.

Evidently, because of (2.4.3), these magnetic surfaces are also current surfaces.

2. Prior to progressing further, attention will be given to the following analogy between the system (2.2.4) for the stationary flows of an incompressible ideal liquid and the system of equations (2.4.1) for equilibrium MHD configurations, noted by Shafranov [68]. The comparison is here

[8] i.e. the magnetic layers of extreme small thickness.

$$
\begin{array}{ll}
\text{(a)}\ \nabla p = \dfrac{1}{c}[\mathbf{j}x\mathbf{H}] & \text{(b)}\ \nabla P \equiv \nabla\left(\dfrac{p}{\rho_0}+\dfrac{v^2}{2}\right)=[\mathbf{q},\mathbf{v}] \\
\mathbf{j}=\dfrac{c}{4\pi}\operatorname{rot}\mathbf{H} & \mathbf{q}=\operatorname{rot}\mathbf{v} \\
\operatorname{div}\mathbf{H}=0 & \operatorname{div}\mathbf{v}=0.
\end{array}
\qquad (2.4.4)
$$

With the accuracy, limited to reservations associated with the sign of p and P, in both cases the systems (a) and (b) are formally identical.

3. We return to the magnetic-current surfaces and take into account the closed (toroidal) configurations and, correspondingly, toroidal magnetic-current surfaces. These surfaces are known to fill tightly the entire volume of the plasma configuration in the presence of axial symmetry, as in the vacuum case. However, there are also examples of non-symmetric toroidal MHD configurations filled with toroidal magnetic surfaces inserted in the ideal manner into each other[9].

2.4.2. One-dimensional equilibrium MHD configurations

The construction of the specific models of the plasma MHD configurations starts with the one-dimensional configurations using real prototypes represented by θ and *Z*-pinches. Only now they are static formations and not pulsed systems.

To exclude the current from the system of equations (2.4.1)

$$\nabla p + \frac{1}{4\pi}[\mathbf{H}, \operatorname{rot}\mathbf{H}] = 0,$$

we obtain (see (2.3.6g))

$$\nabla\left(p+\frac{H^2}{8\pi}\right)=\frac{(\mathbf{H}\nabla)j\mathbf{H}}{4\pi}. \qquad (2.4.5)$$

Planar case

Let all parameters depend only on the coordinate x (Fig. 2.4.2) Consequently, in a general case $\mathbf{H}$ = (0, 0, $H_z(x)$), and the right term in (2.4.5) is equal to 0. This means that the total pressure of the plasma and the field remains unchanged

[9]See appendix A

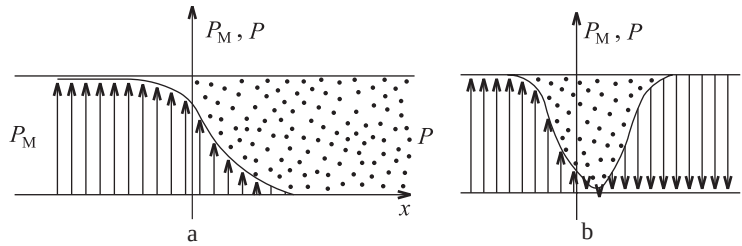


Fig. 2.4.2. Field and plasma in the one-dimensional equilibrium system: a) a single interface,b) a neutral current layer.

$$p + p_M \equiv p + \frac{H^2(x)}{8\pi} = P = \text{const.} \tag{2.4.6}$$

An important example of this type of flat configurations are the so-called neutral current layers in which the magnetic field from both sides of the plasma layer has the same strength but opposite direction (Fig. 2.4.2b). These layers play a significant role in cosmic objects. In particular, the 'tail' of the Earth magnetosphere (section 9.2) is such a neutral current layer[10].

Cylindrical configurations (pinches)

We now examine a *Z*-pinch with a single azimuthal magnetic field **H** = (0, H_θ, 0). In this case, the magnetic field not only presses on the plasma, as in the planar case, but also additionally compresses the plasma as a result of 'tensions " of the lines of force (section 1.1). Therefore, in this case the term

$$\frac{(\mathbf{H}\nabla)\mathbf{H}}{4\pi} = -\frac{H^2}{4\pi r}\mathbf{r}^0. \tag{2.4.7}$$

must be considered in (2.4.5). Here $\mathbf{r}^0$ is the unit vector along the radius. Consequently, equation (2.4.5) assumes the following form

$$r^2\frac{\partial p}{\partial r} + \frac{1}{8\pi}\frac{d}{dr}\mathrm{r}^2 H^2 = 0. \tag{2.4.8}$$

In contrast to the planar case, this equation in the general form is not integrated and, to solve this equation, it is necessary to define $p(r)$ or $H(r)$.

In particular, if the current density in plasma is constant j_0, then

$H = 2\pi j_0 r \equiv \alpha r$

[10] See section 9.2, and also Appendix B.

and, consequently,

$$\rho + \frac{H^2}{4\pi} = \text{const} = p_0. \tag{2.4.9}$$

Here p_0 is the pressure in the centre of the pitch, where $H = 0$. It may be seen that in this case the contribution of the magnetic field to the integral is greater than in the planar case.

Bennett equation

In a number of cases, it is convenient to have the integral equivalent of equation (2.4.8) at T = const. Consequently, as may easily be seen

$$NT = \frac{1}{4c^2} J_0^2. \tag{2.4.10}$$

Here $N = 2\pi r \int nr\, dr$ is the total number of particles per unit length of the pitch, J_0 is the current in the pinch.

Principal cylindricity of the pinch with a single azimuthal field

It will be shown that there are no static MHD configurations which depend on z in the presence of a single azimuthal field.

Actually, from (2.4.5) and (2.4.6), at $H = H_0(r, z)$, $p = p(r, z)$ we have

$$\frac{\partial}{\partial r}\left(p + \frac{H^2}{8\pi}\right) = -\frac{H^2}{4\pi r} \tag{2.4.11a}$$

$$\frac{\partial}{\partial z}\left(p + \frac{H^2}{8\pi}\right) = 0. \tag{2.4.11b}$$

The last equation shows that

$$p + \frac{H^2}{8\pi} = P(r).$$

Substituting this relationship into the first equation (2.4.11a) we obtain:

$$H^2 = -4\pi r \frac{dP(r)}{dr}.$$

i.e., H is only a function of r. However, from (2.4.11b) it now follows that also $p = p(r)$. This special feature of the Z-pinch in the MHD may

be associated with the toroidal drift of the particles in the azimuthal magnetic field, as discussed in 1.7.

2.4.3. Two-dimensional (symmetric) configurations. The Grad-Shafranov equation [69]

1. The initial system of the equations of MHD statics (2.4.1) contains seven scalar unknown quantities (j_x, j_y, j_z; H_x, H_y, H_z,; p). Excluding **j**, we obtain the system (2.4.5) containing four quantities. Using the integrals (2.4.11) in the general case, the last system can be reduced to two equations without making any special assumptions. However, this system has a very exotic form because of the possibility of splitting of the magnetic surfaces [70]. If it is assumed that there is a symmetry, then, as we know, magnetic surfaces can form and the system (2.4.1) can be reduced to a single equation. This will be shown for the symmetric case. This was carried out for the first time by the Fermi and Chandrasekhar and then independently by Grad and Shafranov (1957, [69]). This equation is usually referred to as the Grad–Shafranov equation.

Taking into account the axial symmetry, the equation div **H** = 0 can be identifically satisfied, as carried out in section 1.1, by introducing the function of the flux $\psi(r, z)$ for the poloidal field, assuming that

$$H_r = -\frac{1}{r}\frac{\partial \psi}{\partial z}; \quad H_z = -\frac{1}{r}\frac{\partial \psi}{\partial r}. \tag{2.4.12a}$$

Taking divergence from the first Maxwell equation gives

$$\text{div } \mathbf{j} = 0.$$

This equation makes it possible to introduce the function of the poloidal electrical current $I(r, z)$:

$$j_r = -\frac{1}{r}\frac{\partial I}{\partial z}, \quad j_z = -\frac{1}{r}\frac{\partial I}{\partial r}. \tag{2.4.12b}$$

Substituting (2.4.12) into (2.4.2) gives the equality to 0 of two Jacobians

$$\frac{D(p,\psi)}{D(r,z)} = 0 \quad \text{and} \quad \frac{D(p,I)}{D(r,z)} = 0. \tag{2.4.13a}$$

This means that pressure p can be regarded as a function of ψ and also of I, i.e.

$$p = \rho(\psi), \quad p = \rho(I). \tag{2.4.13b}$$

If the function of the magnetic flux $\psi(r, z)$ is used as the basis, then not only p but also I is the function of ψ:

$$I = I(\psi). \tag{2.4.13c}$$

This conclusion is in complete agreement with what was said in section 2.4.2a, namely that the magnetic surface ψ = const is also the current surface I = const and at the same time it is (ψ) the constant pressure surface.

We now consider two Maxwell equations containing the azimuthal component of the magnetic field:

$$-\frac{\partial H_0}{\partial z} = \frac{4\pi}{c} j_r = -\frac{4\pi}{c}\frac{1}{r}\frac{\partial I}{\partial z}; \quad \frac{1}{r}\frac{\partial r H_0}{\partial r} = \frac{4\pi}{c} j_z = \frac{4\pi}{c}\frac{1}{r}\frac{\partial I}{\partial r}.$$

Consequently,

$$rH_\theta = \frac{4\pi}{c} I + \text{const}. \tag{2.4.14}$$

Without loss of generality, it may be said that const = 0.

The third Maxwell equation, which contains the azimuthal component of current j_θ, as already mentioned previously (paragraph 1.1.3) has the following form

$$\Delta^* \psi = -\frac{4\pi}{c} r j_\theta, \tag{2.4.15}$$

where $\Delta^* \psi$ is the modified Laplace operator.

To complete derivation, it is necessary to find the expression for j_θ through ψ. The expression is found from any poloidal component of the equilibrium equation, for example, r-th:

$$\frac{\partial p}{\partial r} = \frac{1}{c}(j_\theta H_z - j_z H_\theta).$$

Using (2.4.12) and (2.4.13) we obtain:

$$p'(\psi)\frac{\partial \psi}{\partial r} = \frac{1}{c}\left(j_\theta \frac{1}{r}\frac{\partial \psi}{\partial r} - \frac{1}{r} I'(\psi)\frac{\partial \psi}{\partial r}\frac{4\pi}{c}\frac{I}{r}\right). \tag{2.4.16}$$

Here the prime is the derivative with respect to ψ.

Shortening $\partial\psi/\partial r$, we obtain the required expression:

$$\frac{j_\theta}{c} = rp'(\psi) + \left(\frac{4\pi}{c^2}\right)\frac{II'}{r}. \qquad (2.4.17)$$

Substituting this expression into (2.4.15), we obtain the Grad–Shafranov equation

$$\Delta^*\psi = -4\pi r^2 p'(\psi) - \left(\frac{4\pi}{c}\right)II'. \qquad (2.4.18)$$

The scalar equation resembles the Poisson equation. However, the right-hand part depends on ψ. Therefore, in the dependence on the type of $p(\psi)$ and $I(\psi)$ this equation can have one or several solutions for the given boundary conditions (K.V. Brushlinskii). The former case is encountered more frequently. We shall discuss in greater detail the properties of the resultant equation. In particular, it can be seen that the equation becomes determined if the distribution of pressure $p(\psi)$ and poloidal current $J_{pld} = 2\pi I(\psi)$ are given in advance. In addition, a corresponding boundary condition must be specified for the function of ψ. This may be described as follows. Let us imagine a set of thin toroidal absolutely stretched 'sacks' inserted into each other, with the poloidal magnetic fluxes $\delta\psi$ between them given. It is also assumed that a plasma with pressure $p(\psi)$ is produced in each such gap and poloidal electrical current $\delta J_{pld}(\psi)$ is generated. Now these 'sacks' should be placed in some toroidal box, with the value of $\delta J_{pld}(\psi)$ and pressure $\delta p(\psi)$ maintained constant during packing. The Grad–Shafranov equation can be used to determine the resultant geometry of the 'shells–sacks, assuming that they are absolutely deformable. It is necessary to assume that the model with the 'sacks' is not absolutely abstract. The current methods of injection of the neutral particles, and also UHF heating of the plasma, can be used to generate the pressure of the required magnitude in the vicinity of a magnetic surface and produce the required values of poloidal currents.

The Grad–Shafranov equation is linear if

$$p = p_0 + p_1\psi + p_2\psi^2;$$
$$I = I_0 + I\psi. \qquad (2.4.19)$$

This problem can be solved easily analytically in the case of the toroidal box with the right-angle section with the boundary condition ψ_B = const by the method of separation of variables. Computers were used for calculating the structure of MHD configurations for greatly differing dependences $p(\psi)$ and $I(\psi)$.

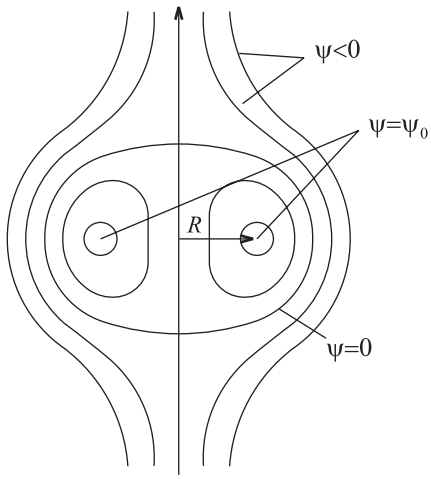


Fig. 2.4.3. 'Algebraic' sferomak. The figure shows magnetic-force lines and the appropriate values of the function of the magnetic flux.

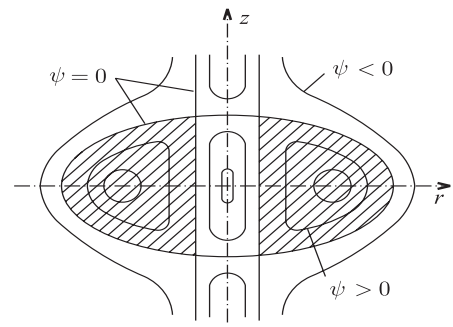


Fig. 2.4.4. Tokamak with the *D*-shaped section of the cord; sections of the magnetic surfaces $\Psi(r, z)$ = const. The internal region, restricted by the separatrix, is crosshatched.

For the special sets $p(\psi)$ and $I(\psi)$, the solution of the Grad–Shafranov equation is expressed by simple algebraic formulas. Two examples are presented here.

1. 'Sferomak' – a toroidal configuration without poloidal current (Fig. 2.4.3). In this case, assuming that $I = 0$, $p = a + b\psi$, we obtain (V.D. Shafranov)

$$\psi = \psi_0 \frac{r^2}{R^4}\left(2R^2 - r^2 - 4\alpha^2 z^2\right);$$

$$p = \frac{\psi_0}{2\pi^3 R^4}\left(1+\alpha^2\right)\psi. \qquad (2.4.20)$$

The meaning of ψ_0, R is indicated in Fig. 2.4.3.

The separatrix corresponds to $\psi = 0$. Parameter determines the stretching of the configuration along z. Outside the limits of the separatrix, the field is calculated using the equation $\Delta\psi^* = 0$.

2. *Tokamak with a D-shaped cross-section of the cord* (Fig. 2.4.4) Assuming that p' = const; $I^2(\psi) = \psi m + n$, m,n – const, we obtain (B.A. Trubnikov)

$$\psi = c_0\left(\frac{r^2}{r_{\min}^2} - 1\right)\left(1 - \frac{r^2}{r_{\max}^2} - \frac{z^2}{z_{\max}^2}\right). \qquad (2.4.21)$$

The meaning of r_{min}, r_{max}, and z_{max} is indicated by the graph. The azimuthal current in this model is:

$$j_\theta = \left(\frac{\alpha}{r} + \beta r \right) k; \quad \alpha, \beta, k - \text{const.} \tag{2.4.22}$$

Completing the examination of the axisymmetric configurations, it is important to mention two moments:

1. In many cases, the dependences $p(\psi)$ and $I(\psi)$ are not given directly and are determined from some additional conditions, for example, from stability considerations.

2. In the presence of translational (i.e., independent of z) or helical symmetry, we can derive equations identical to the Grad–Shafranov equation.

3. The derivation of the Grad–Shafranov equation assumes the presence of a poloidal field, i.e.,$\psi \neq 0$. There is no such field, the scheme of interaction of the equilibrium equation should be slightly different and leads to the natural equation

$$\frac{1}{I} \frac{dp(I)}{dI} = \frac{4\pi}{c^2 r^2}, \tag{2.4.23}$$

which can be obtained from (2.4.18), simply rejecting the left part. The resultant equation shows immediately that in this case the plasma configuration does not depend on z.

Parameter β in equilibrium configurations

When examining the plasma configurations, the dimensionless parameter is the ratio of the gas dynamic pressure p to magnetic pressure p_M which was discussed in section 1.1:

$$\beta = \frac{\rho}{p_M} = \frac{8\pi\rho}{H^2}.$$

This value is also regarded as both the local parameter β(x) and the integral parameter which is the ratio of the maximum plasma pressure in the system p_{max} to some characteristic strength of the field H_0: in this case, we can write β with the index '0':

$$\beta_0 = \frac{8\pi p_{max}}{H_0^2}. \tag{2.4.24}$$

Several numerical examples will be discussed (see also section 10.5).

The characteristic parameters of the plasma in the case of the (D, T)-reaction are $n_{DT} = n_e \approx 10^{14}$ cm^{-3}, $T_i \sim T_e \sim 10$ keV $\approx 10^8$ K. This corresponds to the pressure

$$p = (n_{DT} + n_e)kT \approx 2.7\text{atm}.$$

To sustain such plasma at $\beta_0 = 1$, the required external field is $H_0 \approx 8$ kOe. Similar parameters are characteristic of the θ-pinch or magnetic balloons–Galateas.

At the same time, in the currently available tokamaks in the centre of the cord, where n and T are maximum, $\beta_0 < 5\%$, which is determined by the boundary of stability of the plasma configurations. Therefore, assuming that $\beta = 4\%$, we obtain the field on the axis of the cord $H_{axis} \approx 40$ kOe and, correspondingly on the internal boundary of the cord (section 10.5), this field in the optimised formulations is three times stronger, i.e. ~100–120 kOe.

It should be mentioned that the field $H = 40$ kOe generates a pressure of ~ 60 atm, and the field $H \sim 100$ kOe already at a pressure of 400 atm. These estimates show that it is important to find schemes of the traps with $\beta \sim 1$. However, the situation is even further aggravated in transition from (DT) to other thermonuclear fuels (see section 10.5 and also Appendix E).

2.5. Linear MHD waves in homogeneous plasma

After investigating the static MHD configurations, we transfer to waves with a low amplitude (linear waves) in homogeneous ideal plasma with a homogeneous magnetic field. In contrast to conventional gas dynamics, in this case, we consider three pairs of signal waves and a single pair of entropic waves (see, for example [10], [14]).

2.5.1. Initial equations

We examine a homogeneous stationary ideal plasma (ρ_0, p_0, $\mathbf{v}_0 = 0$, $\sigma = \infty$) situated in a homogeneous magnetic field $\mathbf{H}_0$. In the presence of defects generated by the perturbations of the initial and boundary conditions[11], the homogeneity of the plasma is disrupted and its parameters appear to be 'perturbed', i.e. in this case

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{H}_1, \quad \rho = \rho_0 + \rho_1, \quad \mathbf{v} = \mathbf{v}_1, \quad p = p_0 + p_1. \tag{2.5.1}$$

[11]Here we do not consider the internal sources of perturbation – 'secondary forces' of different type because they result in the formation of 'right parts' in the linear MHD equations.

If the perturbations (H_1, ρ_1, $\mathbf{v}_1$, p_1) are sufficiently small, their dynamics can be described by the linear equations with constant coefficients which are obtained from the system (2.3.6) as a result of the linearisation procedure,

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \mathrm{div}\mathbf{v}_1 = 0;$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -c_{T_0}^2 \nabla \rho_1 - \frac{1}{4\pi}\left[\mathbf{H}_0, \mathrm{rot}\mathbf{H}_1\right];$$

$$\frac{\partial \mathbf{H}_1}{\partial t} = \mathrm{rot}\left[\mathbf{v}_1, \mathbf{H}_0\right]. \qquad (2.5.2)$$

This is a system of seven equations with seven unknown quantities and describes four different waves. It is quite simple to detect the special features of these waves, examining harmonic perturbations

$$(\rho_1, \mathbf{v}_1, \mathbf{H}_1) \sim \exp\{-i\omega t + i\varkappa\mathbf{x}\}. \qquad (2.5.3a)$$

Consequently, we obtain a homogeneous system of algebraic equations

$$-\omega\rho_1 + \rho_0\varkappa\mathbf{v}_1 = 0;$$

$$\rho_0\omega\mathbf{v}_1 - c_T^2\varkappa\rho_1 - \frac{1}{4\pi}\,[\mathbf{H}_0,[(\varkappa, \mathbf{H}_1)]] = 0; \qquad (2.5.3b)$$

$$\omega\mathbf{H}_1 + [\varkappa,[\mathbf{v}_1, \mathbf{H}_0]] = 0.$$

The condition of existence of a non-trivial solution, i.e., existence of 'self-sustained' waves, is the equality of the determinant of this system to 0

$$D(\omega, \varkappa, \rho_0, H_0, c_T) = 0.$$

This condition is the dispersion equation for the waves in the investigated medium. The simple but relatively time-consuming calculations show that

$$D(\omega, x) \sim \omega\left(\omega^2 - \omega_A^2\right)\left(\omega^2 - \omega_F^2\right)\left(\omega^2 - \omega_S^2\right). \qquad (2.5.4)$$

This shows that there can be four types ('modes', 'branches') of oscillations which could be termed as follows:

– entropy $\omega = 0$,
– Alfvén $\omega = \pm\,\omega_A$;
– fast magnetic sound $\omega = \pm\,\omega_F$;
– slow magnetic sound $\omega = \pm\,\omega_S$.

Each branch will now be investigated.

2.5.2. Entropy wave

($\omega = 0$)[12] This wave exists only if the wave vector $\varkappa$ is perpendicular to the external field $\mathbf{H}_0$. Actually, the system (2.5.3) at $\omega = 0$ has the following form ($\varkappa_3 \equiv \varkappa_{\|}$):

$$\varkappa \mathbf{v}_1 = 0; \tag{2.5.5a}$$

$$\varkappa_3 p_1 = 0; \quad \varkappa_\perp p_1 = -\frac{1}{4\pi} \varkappa_\perp (\mathbf{H}_0 \mathbf{H}_1); \tag{2.5.5b}$$

$$[\varkappa[\mathbf{v}_1, \mathbf{H}_0]] = \mathbf{v}_\perp (\varkappa \mathbf{H}_0) - \mathbf{H}_0 (\varkappa \mathbf{v}_1) = 0. \tag{2.5.5c}$$

Combining these equations MPs

$$\varkappa_\perp \mathbf{v}_1 = 0 \tag{2.5.6a}$$

$$\varkappa_3 = 0; \quad \left(p_1 + \frac{\mathbf{H}_0 \mathbf{H}_1}{4\pi} \right) = 0. \tag{2.5.6b}$$

Two structures correspond to the system of equations (2.5.6). Both are independent of *z*.

One of them (Fig. 2.5.1) corresponds to the plasma flows with the alternating velocity along the axis *z*, but without perturbation of density and field. This is the 'entropy' wave.

The second structure is a laminated equilibrium configuration, generally speaking, with the velocity $v \perp \varkappa$ different from zero but constant with time. The plasma density in the second structure changes along $\varkappa$ in the counter phase with the variation of the strength of the magnetic field. Taking into account the presence of 'freezing-in' of the magnetic fields into the plasma, in particular, in equation (2.3.10b), it may easily be seen that the second structure cannot form without dissipation in perturbation from the homogeneous state. Therefore, the first structure should be used as the mode with $\omega = 0$.

2.5.3. Alfvén waves

These are strictly transverse waves, propagating along the magnetic field without the perturbation of density (Fig. 2.5.2)

$$\frac{\partial \rho}{\partial t} = 0; \quad \mathrm{div}\,\mathbf{v} = 0 \tag{2.5.7a}$$

[12]Naturally, two entropy waves should be used. However, for this purpose, it is necessary to take into account the energy equation.

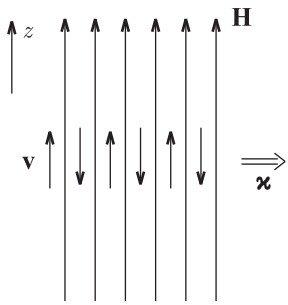


Fig. 2.5.1. Plasma flow in the entropy wave. The velocity **v** is directed against the strength of the magnetic field **H**, and the wave vector $\boldsymbol{\varkappa}$ is normal to **H**.

were described for the first time by H. Alfvén (1942). In this case, the system (2.5.3b) has the following form:

$$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\frac{1}{4\pi}\left[\mathbf{H}_0, \mathrm{rot}\,\mathbf{H}_1\right] = -\frac{1}{4\pi}\left(\nabla(\mathbf{H}_0\mathbf{H}_1) + (\mathbf{H}_0\nabla)\mathbf{H}_1\right); \qquad (2.5.7b)$$

$$\frac{\partial \mathbf{H}_1}{\partial t} = \mathrm{rot}\left[\mathbf{v}_1, \mathbf{H}_0\right] = \left(\mathbf{H}_0\nabla\right)\mathbf{v}_1 - \mathbf{H}_0 \mathrm{div}\,\mathbf{v}_1 = \left(\mathbf{H}_0\nabla\right)\mathbf{v}_1. \qquad (2.5.7c)$$

The z-components of these equations show that

$$v_{1z} = 0; \quad H_{1z} = 0. \qquad (2.5.7d)$$

Therefore, the remaining components **v** and $\mathbf{H}_1$ are described by two symmetric equations:

$$\rho_0 \frac{\partial \mathbf{v}_{\perp 1}}{\partial t} = \frac{\left(H_0\nabla\right)}{4\pi}\mathbf{H}_{\perp 1}; \quad \frac{\partial \mathbf{H}_{\perp 1}}{\partial t} = (\mathbf{H}_0\nabla)\mathbf{v}_{\perp 1}. \qquad (2.5.7e)$$

And, consequently,

$$\frac{\partial^2 \mathbf{H}_1}{\partial t^2} = (\mathbf{c}_A\nabla)^2\mathbf{H}_1; \quad \frac{\partial^2 \mathbf{v}_1}{\partial t^2} = (\mathbf{c}_A\nabla)^2\mathbf{v}_1. \qquad (2.5.8)$$

Here $\mathbf{c}_A$ is the Alfvén velocity

$$\mathbf{c}_A \equiv \frac{\mathbf{H}_0}{\sqrt{4\pi\rho_0}}.$$

Substituting into (2.5.8)

$$(\mathbf{H}_1, \mathbf{v}_1) \sim \exp\{-i\,\omega t + i\,\boldsymbol{\varkappa}\mathbf{x}\},$$

gives the dispersion equation

$$\omega^2 = (\mathbf{c}_A\boldsymbol{\varkappa})^2 = c_A^2\varkappa_3^2. \qquad (2.5.9)$$

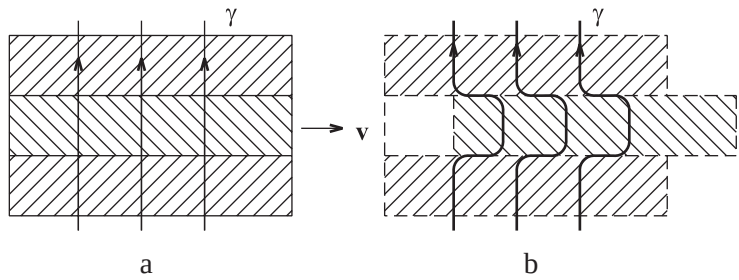


Fig. 2.5.2. Perturbation of the magnetic lines of force (γ) in the Alfvén wave: a) initial non-perturbed condition, b) deformation of the magnetic lines of force in displacement of plasma.

This shows that the phase velocity is equal to

$$\mathbf{v}_{ph} = \frac{\omega}{|\varkappa|}\left(\frac{\varkappa}{|\varkappa|}\right) = c_A \frac{\varkappa_3}{|\varkappa|}\left(\frac{\varkappa}{|\varkappa|}\right), \tag{2.5.10a}$$

and if $\varkappa = \varkappa_3$, the phase runs along z with the velocity c_A

The group velocity of the Alfvén waves is directed strictly on the magnetic field, irrespective of the orientation $\varkappa$:

$$\mathbf{v}_{\text{gr}} = \mathbf{c}_A\left(\frac{d\omega}{d\varkappa_x}, \frac{d\omega}{d\varkappa_y}, \frac{d\omega}{d\varkappa_z}\right) = \mathbf{c}_A(0,0,1). \tag{2.5.10b}$$

The similarity of the equations (2.5.8), which describes the behaviour of the perturbation of the magnetic field H1 and the velocity v1 enables us to regards the propagation of the Alfvén wave as transverse oscillations of the line of force, stretched with the force $T_0 = H^2_0$ /4π Actually, if we have a filament with density ρ_0 and tension T_0, the velocity of the wave running on the filament is equal to

$$c_\rho = \sqrt{\frac{T}{\rho_0}}. \tag{2.5.11}$$

The specifics of the magnetic fields are reflected here by the fact that the Maxwell tension $H^2/8\pi$ in equation (2.5.11) should be replaced by $H^2/4\pi$. This is caused by the fact that the magnetic lines of force are not only stretched but also exert lateral pressure on each other (section 1.1.2). The very fact of the sinphase condition $\mathbf{v}_1$ (t, $\mathbf{x}$) and $\mathbf{H}_1$(t, $\mathbf{x}$) is the evident demonstration of the 'freezing-in' of the

magnetic field in the plasma. The Alfvén waves play a significant role in the magnetospheres of the planets (including the Earth), and also in the upper layers of the Sun atmosphere (section 9.3). They are also evident in the tokamaks and can be reproduced quite easily in special experiments with liquid metals. For example, in a quartz pipe filled with mercury and positioned in a homogeneous magnetic field, the Alfvén waves are excited by placing a 'shovel' on the top which carries out oscillatory movement. Small coils, fixing the appearance of the transverse components of the magnetic field, are situated along the pipe. Experiments show that the periodic EMF is indeed induced in the coils, and the phase shift along the length corresponds to the Alfvén velocity.

In chapter 3, it will be shown that the Alfvén waves with the described properties exist at $\omega < \omega_{Hi}$, i.e. at the frequencies lower than the ionic Larmor frequency.

2.5.4. Magnetic sound waves (MSW)

We now consider MHD waves, accompanied by the perturbation of plasma density. For this purpose, the system (2.5.2) is reduced to a single equation for ρ, as was carried out previously in the investigation of sound waves in the Euler hydrodynamics.

Taking into account the equalities

$$\left[\mathbf{H}_0, \operatorname{rot}\mathbf{H}_1\right] = \nabla(\mathbf{H}_0 H_1) - (\mathbf{H}_0\nabla)H_1;$$
$$\operatorname{rot}\left[\mathbf{v}, H_0\right] = (\mathbf{H}_0\nabla)\mathbf{v} - \mathbf{H}_0 \operatorname{div}\mathbf{v},$$

and differentiating the first and third equation (2.5.2) with respect to *t* and substituting into these equations the second equation (2.5.2) we obtain:

$$\frac{\partial^2 \rho_1}{\partial t^2} = c_T^2 \Delta\rho_1 + \frac{H_0}{4\pi}\Delta H_{1z}; \qquad (2.5.12a)$$

$$\frac{\partial^2 H_{1z}}{\partial t^2} = -c_T^2 \frac{H_0}{\rho_0}\frac{\partial^2 \rho_1}{\partial z^2} + \frac{H_0}{\rho_0}\frac{\partial^2 \rho_1}{\partial t^2}. \qquad (2.5.12b)$$

Twice differentiating the first equation (2.5.12a) with respect to *t* and substituting (2.5.12b) into this equation gives the required equation

$$\frac{\partial^4 \rho_1}{\partial t^4} - \left(c_T^2 + c_A^2\right)\Delta\frac{\partial^2 \rho_1}{\partial t^2} + c_T^2 c_A^2 \Delta\frac{\partial^2 \rho_1}{\partial z^2} = 0. \qquad (2.5.13)$$

Knowing ρ_1, we can determine the perturbation of the velocity of plasma $\mathbf{v}_1$ and field $\mathbf{H}_1$. They are determined by the expressions:

$$\frac{\partial}{\partial t}\hat{L}_1\mathbf{v}_1 = -\frac{1}{\rho_0}\left\{\hat{L}\nabla\rho_1 - c_A\mathbf{c}_A\frac{\partial}{\partial z}\frac{\partial^2\rho_1}{\partial t^2}\right\};$$

$$\frac{\partial}{\partial t}\hat{L}_1 H_1 = -\sqrt{\frac{4\pi}{\rho_0}}\left\{c_A\frac{\partial}{\partial z}\hat{L}\nabla\rho_1 - \mathbf{c}_A\frac{\partial^4\rho_1}{\partial z^2}\right\}. \qquad (2.5.14)$$

Here $\hat{L}$ and $\hat{L}_1$ are operators

$$\hat{L} = \left(c_T^2 + c_A^2\right)\frac{\partial^2}{\partial t^2} - c_T^2 c_A^2\frac{\partial^2}{\partial z^2},$$

$$\hat{L}_1 = \frac{\partial^2}{\partial t^2} - c_A^2\frac{\partial^2}{\partial z^2}.$$

Substituting the exponent $\rho_1 \sim \exp\{-i\omega t + i\varkappa \cdot x\}$ into (2.5.13), we obtain the dispersion equation

$$\omega^4 - (c_T^2 + c_A^2)\omega^2\varkappa^2 + c_T^2 c_A^2\varkappa^2\varkappa_3^2 = 0. \qquad (2.5.15)$$

This biquadratic equation determines two types of waves with the frequency square

$$\omega_F^2 = \frac{\left(c_T^2 + c_A^2\right)\varkappa^2 + \sqrt{D}}{2} \qquad (2.5.16a)$$

$$\omega_S^2 = \frac{\left(c_T^2 + c_A^2\right)\varkappa^2 + \sqrt{D}}{2} = \frac{2c_T^2 c_A^2\varkappa^2\varkappa_3^2}{\left(c_T^2 + c_A^2\right)\varkappa^2 + \sqrt{D}}; \qquad (2.5.16b)$$

Here $D = (c_T^2 - c_A^2)^2\varkappa^4 - 4c_T^2 c_A^2\varkappa^2\varkappa_3^2$.

The waves with the frequency $\omega = \pm\omega_F$ are referred to as 'fast magnetic sound waves' and with frequency $\omega = \pm\omega_S$ as 'slow magnetic sound waves'. Their properties will be investigated.

1. In particular, it can be seen that the phase and group velocitys are independent of wavelength, i.e. independent of $|\varkappa| = \varkappa$, and depend only on the angle between the directions **H** and $\varkappa$.

2. If the waves propagate along the magnetic field, then

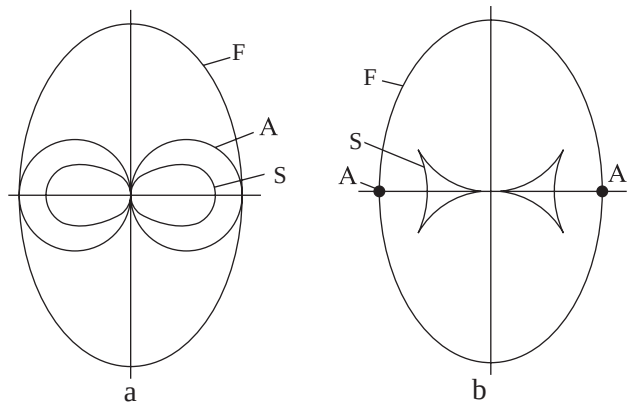


Fig. 2.5.3. Polar diagrams of the phase (a) and group (b) velocitys of the MHD waves. A) Alfvén wave, S – slow wave, F – fast magnetic sound wave.

$$\omega_F^2 = c_A^2 \varkappa^2,\ \omega_S^2 = c_T^2 \varkappa^2. \tag{2.5.17a}$$

3. If the waves propagate across the magnetic field, then

$$\omega_F^2 = \left(c_T^2 + c_A^2\right)\varkappa^2,\ \omega_S^2 = 0. \tag{2.5.17b}$$

4. If $c_A^2 \gg c_T^2$, then

$$\omega_F^2 \sim c_A^2 \varkappa^2,\ \omega_S^2 = c_T^2 \varkappa_3^2. \tag{2.5.17c}$$

In this case, the relationship between ω_S and $\varkappa$ as regards the structure is the same as in the case of Alfvén waves. At $T \to 0$, i.e., $c_T \to 0$, the slow wave disappears.

A large amount of information on the dependence of the phase velocity on the angle between **H** and $\varkappa$ is provided by the polar diagram (Fig. 2.5.3a). The diagram shows three lines, corresponding to the fast (F), Alfvén (A) and slow (S) MHD waves.

The difference in the behaviour of the fast and slow magnetic sound waves is explained by the fact that the elasticity of the medium, in the fast wave case, is the sum of the gas-kinetic and magnetic pressures (see (2.5.16)). At the same time, in the slow wave at $\varkappa \parallel \mathbf{H}_0$ the gas kinetic pressure is 'deducted' from magnetic pressure. This weakens the effective elasticity of the medium in the propagation of the slow magnetic sound wave. The expression for ω_F in the case $\varkappa \perp \mathbf{H}$ is completely natural because of the fact that we have discussed flat flows across the magnetic fields (2.3.11).

If the angle between $\varkappa$ and $\mathbf{H}_0$ is not equal to 0, the situation is relatively complicated, as indicated in particular by the polar diagram of the group velocitys plotted by Friedrichs (Fig. 2.5.3b).

In this diagram the Alfvén groups degenerate to two points, the slow magnetic sound waves correspond to curvilinear separate triangles which are compressed to points at $c^2_A \gg c^2_T$, and only fast magnetic sound waves are characterised by an oval similar to the oval of the phase velocitys. More details for this range of questions can be found in [10, 72].

The diagram of the group velocitys may be used for constructing the diagrams of perturbation of homogeneous plasma for various orientations and velocity $\mathbf{v}_0$ of the 'perturbing agent' in relation to the external magnetic field $\mathbf{H}_0$. This situation is simple even if a point object represents the 'perturbing agent'. Therefore, I refer the readers to the cited literature [10, 72].

2.6. Stationary plasma flows in the transverse magnetic field [73]

In section 2.2.3 we considered the flow of a gas in a Laval nozzle in the approximation of the 'narrow pipe'. This analysis showed three important moments.

– a continuous transition of thermal energy to kinetic energy takes place in the nozzle. The maximum velocity at the exit (v_m) is determined by the enthalpy (i_0) in the zone, where the heated gas is in the rest state

$$v_M = \sqrt{2i_0}.$$

– the characteristic parameter, which determines the evolution of the flow, is the velocity of sound

$$c_T = \sqrt{\frac{\partial p}{\partial \rho}}$$

– the cross-section of the nozzle, in the case of the flow without breaks, is not constant. The cross-section initially decreases and subsequently increases. In the minimum ('critical') cross-section, the velocity of the flow v_* is equal to the local velocity of sound $v_* = c_{T*}$.

As already mentioned, the velocity of discharge from thermochemical rocket engines is low for the optimum solution of the majority of space problems. New possibilities are offered by plasma engines.

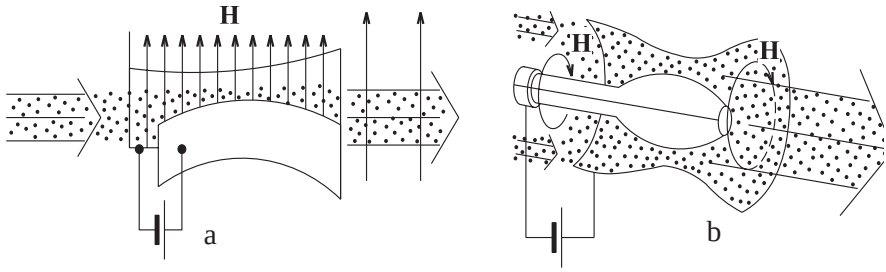


Fig. 2.6.1. Varieties of the magnetic-plasma nozzle: a) flat, b) axisymmetric nozzle.

2.6.1. Flows in narrow channels

We start with a flat plasma flow in a narrow profiled channel in a transverse magnetic field (Fig. 2.6.1a). In the planar case, all that was said of the Laval nozzle also remains qualitatively valid in this case since this flow is described by the system (2.3.11b) with only one difference. The role of the velocity of sound is now played by the fast magnetic sound

$$c_F = \sqrt{\frac{\partial P}{\partial \rho}} = \sqrt{c_T^2 + c_A^2}. \tag{2.6.1}$$

The maximum flow velocity at exit is correspondingly equal to

$$v_{\max} = \sqrt{2}\sqrt{\frac{c_{T0}^2}{\gamma - 1} + c_A^2}. \tag{2.6.2}$$

Thus, in this case, we can consider the 'magnetic plasma nozzle'. An important difference of this model in comparison with the gas dynamic nozzle is that in this case the exhaust velocity depends only slightly on the plasma temperature, because in the most interesting cases

$$c_A^2 >> c_T^2. \tag{2.6.3}$$

A numerical example will be discussed. We have a magnetic field H = 5 kOe and the pressure corresponding to this field (see section (1.1.2)) is $p_m \sim 1$ atm. Argon plasma with the density (at entry) of $n_0 \sim 10^{16}$ cm^{-3} is accelerated in the nozzle[13]. Therefore, taking into account that the mass of a single argon atom is $M \sim 6 \cdot 10^{-23}$ g, we obtain

$$V_M \approx 30 \text{ km/s}.$$

[13] In the normal conditions, this corresponds to the pressure of $p_0 \sim 0.3$ torr.

This velocity is considerably higher than the velocitys achievable in the gas rocket nozzles.

The flat model of the magnetic plasma nozzle which has already been discussed was important for methodical reasons because the exit of plasma from the magnetic system of such an accelerator is associated with difficulties. It is another matter if we discuss the axisymmetric model (2.6.1b). In this case, there are no problems with the exit of plasma from the magnetic fields because the plasma simply carries away the field 'frozen' in it, more accurately the magnetic flux, but at $H \to 0$.

As in the planar case, the magnetic field here is normal to the velocity of the flux because it is azimuthal and generated as a result of the discharge current between two profiled coaxial electrodes. A narrow ring-shaped tube is considered in calculations. The mean radius of the tube is regarded as variable $r(z)$, as is the gap between the 'electrodes', $f(z)$. In addition to these geometrical quantities which characterise the tube, the plasma parameters in the tube are density $\rho(z)$, longitudinal velocity $v(z)$ and the strength of the azimuthal magnetic field $H(z)$. Assuming there is no ohmic resistance, we can write the algebraic laws of conservation: magnetic flux and energy (Bernoulli equation):

$$\rho v\,(2\pi r f\,) = \dot{m} = \text{const}; \tag{2.6.4a}$$

$$\frac{H}{\rho r} \equiv \varkappa = \text{const}; \tag{2.6.4b}$$

$$\frac{v^2}{2} + i\,(\rho) + \frac{H^2}{4\pi\rho} \equiv U = \text{const}. \tag{2.6.4c}$$

In this case, the form of the Bernoulli equation is not immediately evident, in contrast to the planar case. However, this may easily be proven, assuming the isomagnetic nature of the flow (A.I. Morozov, 1959). In fact, the ampere force (after dividing by ρ), included in the right-hand part of the equation (2.4.2a), can be transformed by the following procedure (see identity (2.1.13))

$$\frac{1}{\rho c}\left[\mathbf{j},\mathbf{H}\right] = \frac{1}{\rho}\left(\nabla\frac{H^2}{8\pi} - \frac{(\mathbf{H}\nabla)\mathbf{H}}{4\pi}\right) = \frac{1}{\rho}\nabla\frac{H^2}{8\pi} + \frac{H^2\mathbf{r}^0}{4\pi\rho r} =$$

$$= \frac{\varkappa^2}{4\pi}\left[r\nabla(\rho r) + (\rho r)\nabla r\right] = \frac{\varkappa^2}{4\pi}\nabla(\rho r^2) = \nabla\frac{H^2}{4\pi\rho}. \tag{2.6.5}$$

This leads to (2.6.4c).

Acceleration and compression flow regimes [74]

The system (2.6.4) will now be investigated. In contrast to the gas dynamic and planar MHD cases where there were two equations and three parameters (p, v, f), in this case we have three equations for five parameters (ρ, v, H, r, f).

The resultant additional arbitrariness enables the formation of not one but two types of flows.

Actually, there are only two types of energy in the gas dynamics – kinetic and thermal, and here there is only one transition[14]

$$i_0 \to \frac{v_M^2}{2}. \tag{2.6.6}$$

In an MHD nozzle, the energy in a general case has three components: kinetic, thermal and magnetic. The case in which the contribution of the magnetic field is dominant is most interesting. Consequently, two extreme flow regimes can exist (see (2.6.4c))

$$U \equiv \frac{H_0^2}{4\pi\rho_0} \to \frac{v_M^2}{2}, \tag{2.6.7a}$$

$$U \equiv \frac{H_0^2}{4\pi\rho_v} \to i_{\max}. \tag{2.6.7b}$$

The flows of the first type are referred to as accelerating and those of the second type as compressive (A.I. Morozov [74]). Their properties will be investigated in greater detail.

For the accelerating regime to form, the magnetic fields and the density at the exit must decrease without limits

$$H \to 0,\ \rho \to 0.$$

Because of the 'freezing-in' condition, this takes place in the case in which the radius of the tube $r(z)$ remains more or less constant (Fig. 2.6.2a). If we consider $r = r_0 = \text{const}$, the system (2.6.4) becomes formally an equivalent gas dynamic system:

$$\rho v f = \text{const} = \frac{\dot{m}}{2\pi r_0},$$

$$H = \varkappa r_0 \rho,$$

[14] The reverse process $\frac{v_M^2}{2} \to i_0$ does not occur in the dissipationless regular flow. A shock wave forms in this case.

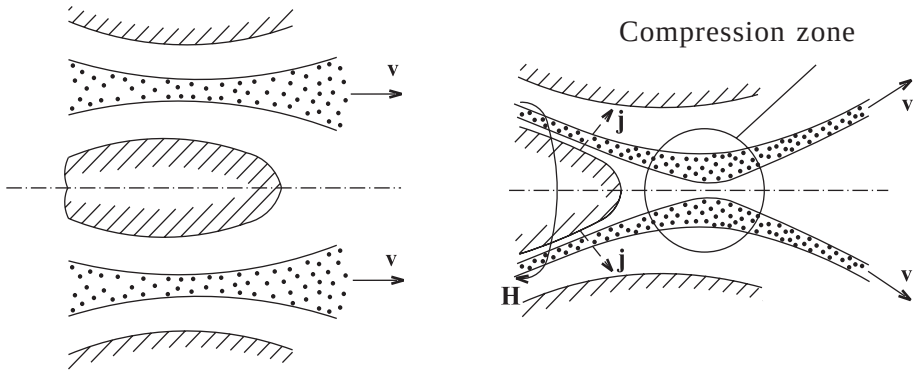


Fig. 2.6.2. Schemes of flows in a coaxial stationary system: a) accelerating flow, b) compression flow.

$$\frac{v^2}{2}+\left(i(\rho)+\frac{\varkappa^2 r_0^2 \rho}{4\pi}\right)=\frac{H_0^2}{4\pi\rho_0}=c_{A0}^2.$$

In this case (at $c_{A0}^2 \gg i_0$)

$$v_{\max}=\sqrt{2}c_{A0}. \tag{2.6.8}$$

For the compression regime to form, the following conditions (Fig. 2.6.2b) should be fulfilled at exit from the channel:

$$H \to 0, \quad v \to 0.$$

Consequently

$$i(\rho_{\max})=\frac{H_0^2}{4\pi\rho_0}. \tag{2.6.9}$$

The 'freezing-in' conditions show that these conditions can be satisfied if the radius of the tube of the flow decreases without limits $r \to 0$. It is evident that the flow of this type does not form in the planar case. Taking into account the condition (2.6.9) and assuming the polytropic relationship of pressure and the density

$$p=p_0\left(\frac{\rho}{\rho_0}\right)^{\nu},$$

we determine the maximum degree of compression

$$\frac{\rho_{\max}}{\rho_0} = \left((\nu - 1) \frac{c_{A0}^2}{c_{T0}^2} \right)^{1/(\nu-1)} . \qquad (2.6.10)$$

In this case, temperature increases to

$$kT_{\max} = (\nu - 1) c_{A0}^2 \times M . \qquad (2.6.11)$$

Thus, in the adiabatic compression of the ionised (monatomic) gas $\nu = \frac{5}{3} \equiv \gamma$ and if the velocity is $c_{A0} = 3 \cdot 10^7$ cm/s, the degree of compression will be

$$\frac{\rho_{\max}}{\rho} \approx 0.54 \cdot 10^3.$$

When ν approaches 1, the degree of compression rapidly increases. At high degrees of compression, the plasma is rapidly heated. This results in powerful radiation and efficiently reduces the polytrope parameter ν. This now causes further compression. This phenomenon is referred to as the 'radiation collapse'. Experimental investigations of accelerating and compression flows will be discussed in chapter 3.

2.7. Numerical modelling of MHD flows

The energy flows of non-viscous and heat non-conducting plasma in a channel was one of the first subjects of numerical modelling of plasma processes at the beginning of the 60s. They were stimulated by experimental investigations of the MHD flows started at that time at the Institute of Atomic Energy by A.I. Morozov and G.Ya. Shchepkin.

Numerical modelling of plasma flows in channels was carried out at the Institute of Applied Mathematics of the Academy of Sciences of the USSR by K.V. Brushlinskii, a pupil of I.M. Gelfandt, and his colleagues [75].

2.7.1. Calculations of the flow of ideal plasma in an axisymmetric channel

This is the simplest case which is described by equation (2.3.2). Calculation should be carried out in the dimensionless form, using the measurement units L – the length of the channels; ρ_0, T_0 – density and plasma temperature at the input; H_0 – the strength of the magnetic

field at input at some radius r_0^*. From these quantities it is possible to form the units of velocity $\bar{v}_0$, time $\bar{t}_0$, and pressure $\bar{p}_0$:

$$\bar{v}_0 = \frac{H_0}{\sqrt{4\pi\rho_0}}, \quad \bar{t}_0 = \frac{L}{v_0}, \quad \bar{p}_0 = \frac{H^2}{4\pi}.$$

In the dimensionless variable systems:

$$\frac{d\rho}{dt} = -\rho \operatorname{div} \mathbf{v}, \tag{2.7.1}$$

$$\frac{d\mathbf{v}}{dt} = -\frac{\nabla p}{\rho} + \frac{1}{\rho}[\operatorname{rot} \mathbf{H}, \mathbf{H}], \tag{2.7.2}$$

$$\frac{\partial \mathbf{H}}{dt} = \operatorname{rot} [\mathbf{v}, \mathbf{H}], \tag{2.7.3}$$

$$\frac{dT}{dt} = -(\gamma - 1)T \operatorname{div} \mathbf{v}. \tag{2.7.4}$$

Here $\beta = 8\pi\bar{p}_0 / H_0^2$. The last equation follows from the adiabate (2.1.6b) and the continuity equation.

The dimensionless boundary conditions at entry into the channel are described by the equation:

$$p = 1; \quad T = 1; \quad H = \frac{r_0}{r}.$$

The electrodes are assumed to be ideally conducting and non-permeable for plasma:

$$E_\tau = 0; \quad v_n = 0,$$

where $\mathbf{n}^0$ is the normal and τ is the tangent to the surface of the electrode. No boundary conditions are required at the exit, if the plasma is ideal and leaves the channel at a super signal velocity. The shape of the electrode differed (Fig. 2.7.1 and Fig. 2.7.2).

The results of these calculations of the main model may be summarised as follows.

1. The flow reaches in a stable manner the stationary or similar quasi-stationary regime during the flight time $t_0 \sim L/v_0$.

2. If the electrode profile changes relatively smoothly, the pattern of the flow is similar to that indicated by the approximation of the narrow channel (Fig. 2.7.1).

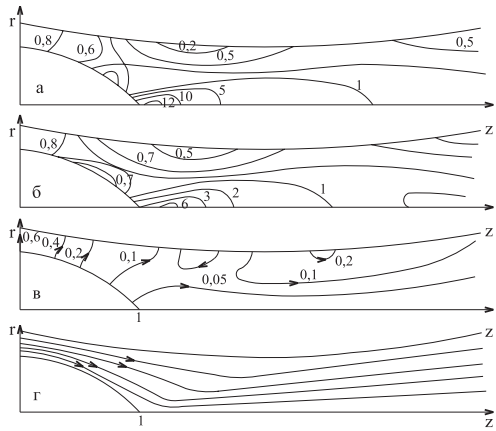


Fig. 2.7.1. Results of calculations of the compression flow of plasma in the channel: a) density, b) temperature, c) electrical current, d) the lines of flow (numbers of the curves are the dimensionless values of the appropriate parameters)

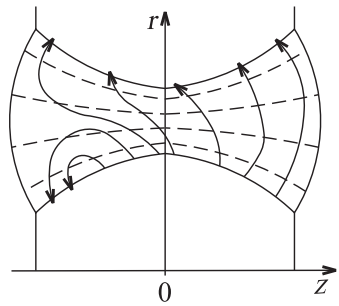


Fig. 2.7.2. Diagram of the plasma flow in the channel accompanied by the formation of an electric current vortex.

3. The dependence of the magnetic field on the radius causes that the velocity at exit from the channel depends on the radius.

If $\beta \equiv 8\pi\rho_0/H_0^2 \gg 1$, the flow is of the gas-dynamic type and similar to one-dimensional; in this case, the dependence $v(r)$ is very weak. The decrease of β increases the scatter of the velocitys at the exit. In the presence of relatively strong magnetic fields ($\beta < 0.3$), the qualitative pattern of the distribution of velocitys at the exit is almost completely independent of β. Finally, at all values of β, the dependence of the exit velocitys on r is weaker than r^{-1}.

4. In the channels with a relatively strong dependence of the cross-section on z and at high values of β electric current vortices are detected in the vicinity of the electrodes (Fig. 2.7.2). Theoretical analysis of this phenomenon showed that their formation is associated with the non-isomagnetic nature of the flow.

2.7.2. Shock waves in MHD [12, 76]

In the Euler gas dynamics, the gas flows are characterised by a single vector – velocity and by a single signal velocity – the velocity of sound c_T. If the flow meets a wall (it is assumed that the wall is flat), another vector appears, i.e., normal $\mathbf{n}^0$ to the plane. Depending on the mutual orientation of **v** and $\mathbf{n}^0$, shock waves of various configuration appear. In section 2.2.4 we investigated the case in which **v** and $\mathbf{n}^0$ were co-linear.

On the whole, the situation in MHD is far more complicated. Here in the volume of the flow there are two factors **V** and **H** in three types of signal velocity (A, FMS, SMS). Therefore, even in the case of the ideal MHD plasma, the systematic description of the shock waves requires a large space. If we are discussing the structure of MHD-SW, it is necessary to take into account not only viscosity and heat conductivity but also ohmic resistance. It should also be added that in addition to the volume nature of the vectors (**v**, **H**), it is also necessary to consider the interaction of flows with obstacles. A special situation forms here in many cases. In particular, an obstacle for the flow may be represented by the magnetic field which is deformed under the effect of the flow.

Taking the above considerations into account, we confine ourselves here to a single illustration of the numerical calculations (A.I. Morozov and V.V. Savel'ev). Namely, a supersonic plasma flow with finite conductivity around the magnetic field of a ring with current, as a dissipation mechanism [76]. To facilitate calculations, the ring was placed in a superconducting pipe ($H_n = 0$). Figure 2.7.3 shows the shock wave 'settled' on the ring, and a continuation of the wave encounters the pipe and generates the reflected wave. The lines in the figure are the lines of constant density.

In conclusion, it should be mentioned that the formation of the shock wave observed when the supersonic plasma flow meets the magnetic field can be seen most distinctively in the interaction of solar wind with the magnetic field of the Earth (section 9.2). It is also important to note the experiments which showed that the flow of a neutral gas, with the velocity $\gtrsim$ 10–15 km/s, meeting the magnetic field, is ionised and forms a glowing shock wave.

2.7.3. Role of ohmic resistance in the dynamics of highly conducting plasma

If the plasma has high conductivity, the magnetic viscosity is

$$\nu_m = \frac{c^2}{4\pi\sigma} \ll 1,$$

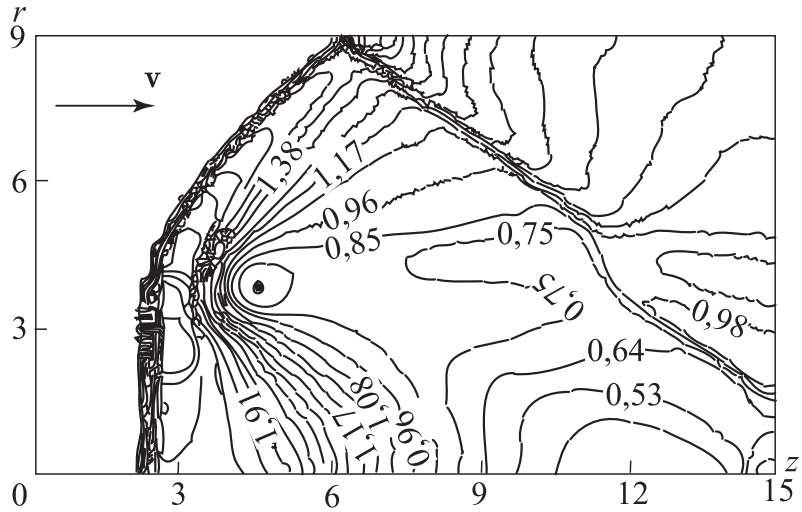


Fig. 2.7.3. The flow of a supersonic highly conducting plasma flow around the magnetic field of a ring with current. The lines on the figure are the lines of constant density at $\beta = 0.2$, $\nu = 0.1$, $M = 2.0$.

and in equation (2.3.3b) we reject the dissipative term $\nu_m \Delta \mathbf{H}$. However, the equation contains the second derivative with respect to the coordinates, and at shorter distances it may be very important. A suitable example of this is the previously examined example of the flow around the magnetic field accompanied by the formation of shock waves whose thickness is

$$\delta \sim \frac{\nu_m}{v}.$$

It is important to mention a number of important cases in which the dissipative layers form in a highly conducting plasma flow. This includes, in particular, the processes of reconnection of the magnetic lines of force in collision of highly conducting plasma volumes with the magnetic fields of different orientations (appendix B).

Dissipation must be taken into account when investigating the 'hydromagnetic dynamo' i.e. generation of the magnetic field in three-dimensional plasma flows (for example, on the Sun, see section 9.3.5), or in convective flows of the magma in the Earth interior. The patterns of these processes are relatively complicated and require further investigations. Some reports have been published in [77] to which the reader is referred.

Other examples include the tearing instability of current layers (section 8.2) and also entry of plasma into the magnetic field where it is most important to take into account dissipation, if it is required to obtain information on the structure of the layers formed here (see section 3.8).

3

Two-fluid hydrodynamic plasma models

The single-fluid MHD models represent a radical generalisation of the one-component block models. However, as shown in the introduction, even the fundamental characteristic of the plasma – Langmuir frequency – can be derived only from the two-component block model. A suitable demonstration of the special features of plasma processes is the analysis of the fall of a plasma bar in the magnetic field, carried out within the framework of the two-component block model in chapter 1 (section 1.3.2). It was shown that the quasi-autonomous nature of the dynamics of every component results in the formation of a strong electrical field in the plasma volume. This field is highly sensitive to the external conditions, and an example of this is the situation with the toroidal drift discussed in section 1.7. The examples show that it is often necessary to transfer to the two-fluid plasma model. This model can be regarded as a result of dividing each of the two blocks into $N \to \infty$ quasi-autonomous 'droplets'. The present chapter is concerned with the formulation of the two-fluid model on the basis of convincing 'physical' considerations and with also application of the model for calculating a number of plasma-dynamic structures. A strict scheme of the derivation of the equations for this model on the basis of the collisional kinetics will be described in chapter 5.

The two-fluid magnetic hydrodynamics is especially effective when examining the laboratory plasma systems with a low density and the relatively small size. This may easily be understood taking into account the simple equation for the difference of electronic and ionic velocities

$$\mathbf{v}_i - \mathbf{v}_e = \frac{j}{en}$$

In metallic conductors, for example, in mercury or liquid sodium, the density of conduction electrons is $n_e \sim 10^{22}$ cm^{-3}. Therefore, for the current density j = 1 A/mm^2

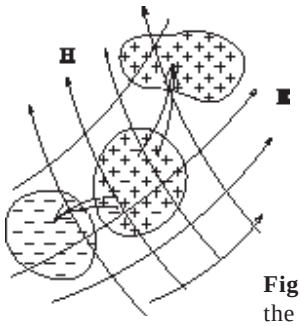


Fig. 3.1.1. Separation of ionic and electronic droplets under the effect of electromagnetic fields.

$$\left|\mathbf{v}_i - \mathbf{v}_e\right| \sim 0.06\,\text{cm/s}.$$

At the same time, at the same current density in the plasma, for example, a tokamak ($n \sim 10^{14}$ cm^{-3})

$$\left|\mathbf{v}_i - \mathbf{v}_e\right| \sim 6\cdot 10^{6}\,\text{cm/s}.$$

These estimates show clearly that the initial 'droplet' with the size of 1 cm contains 'marked' ions and electrons and is split into two non-intersecting droplets of ions and electrons already after $1.5\cdot 10^{-7}$ s (Fig. 3.11).

3.1. Equations of two-fluid hydrodynamics

3.1.1. Formulation of the equations

Two groups of 'droplets': ionic and electronic, are defined in the plasma volume.

The concentration of the particles, temperature, the velocity of the 'droplet' as a whole and its volume are denoted as n_i, T_i, $\mathbf{v}_i$, V_i in the case of ions, and n_e, T_e, $\mathbf{v}_e$, V_e for electrons. The ions are always regarded as singly charged ($+e$) with mass M. The electrons are characterised by the charge ($-e$) and mass m. As in the case of the single-fluid hydrodynamics, we shall not write here the equations for temperature, and the components will be regarded as isobaric

$$p_i = p_i(n_i),\ p_e = p_e(n_e), \tag{3.1.1}$$

in particular, isothermal (T_i = const, T_e = const) or adiabatic.

Assuming that there is no formation and annihilation of the particles and also assuming that the boundaries of the droplets of every component are impenetrable smooth surfaces for 'their own' particles and heat, the following equations can be written for droplets

$$\frac{dN_i}{dt}=0;\quad \frac{dN_e}{dt}=0;$$

$$MN_i\frac{d\mathbf{v}_i}{dt}=\mathbf{F}_{i,p}+\mathbf{f}_iV_i+\mathbf{F}_{i,e}; \tag{3.1.2}$$

$$mN_e\frac{d\mathbf{v}_e}{dt}=\mathbf{F}_{e,p}+\mathbf{f}_eV_e+\mathbf{F}_{e,i}.$$

Here $N = nV$, $F_p = -V\nabla_p$ is the pressure force on the given droplet from the side of other droplets of the same component[1], **f** is the density of the volume forces which will be regarded as caused only by the electrical and magnetic fields

$$\mathbf{f}_i=e\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]\right);\ \mathbf{f}_e=-e\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right]\right) \tag{3.1.3}$$

Finally, we take into account the friction force between the components in the form[2]

$$\mathbf{F}_{i,e}=-V_i\mu n_i n_e\left\langle\sigma v\right\rangle_\Sigma(\mathbf{v}_i-\mathbf{v}_e)\equiv-\alpha(\mathbf{v}_i-\mathbf{v}_e); \tag{3.1.4}$$

$$\mathbf{F}_{e,i}=-V_e\mu n_i n_e\left\langle\sigma v\right\rangle_\Sigma(\mathbf{v}_i-\mathbf{v}_e)\equiv\alpha(\mathbf{v}_i-\mathbf{v}_e).$$

Here σ_Σ is the cross-section of *e–i* collisions, μ is the reduced mass of the ion–electron pair.

Excluding auxiliary quantities – the volumes of the droplets V_i and V_e, as was the case in section 2.1, and introducing the Maxwell equations, we obtain a system of equations of two-fluid hydrodynamics

$$\frac{\partial n_i}{\partial t}+\mathrm{div}\,n_i\mathbf{v}_i=0; \tag{3.1.5a}$$

$$\frac{\partial n_e}{\partial t}+\mathrm{div}\,n_e\mathbf{v}_e=0; \tag{3.1.5b}$$

$$Mn_i\left(\frac{\partial\mathbf{v}_i}{\partial t}+(\mathbf{v}_i\nabla)\mathbf{v}_i\right)=-\nabla p_i+en_i\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]\right)-\alpha(\mathbf{v}_i-\mathbf{v}_e); \tag{3.1.5c}$$

$$p_i=p_i(n_i)$$

[1]See section 5.3.4.

[2]The need for taking into account the friction force between the components, with the viscous components ignored, is explained by the higher relative velocity of the components in comparison with the velocity gradient inside the components.

$$mn_e\left(\frac{\partial \mathbf{v}_e}{\partial t}+(\mathbf{v}_e\nabla)\mathbf{v}_e\right)=-\nabla p_e-en_e\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right]\right)+\alpha(\mathbf{v}_i-\mathbf{v}_e); \tag{3.1.5d}$$

$$p_e = p_e(n_e)$$

$$\mathrm{rot}\mathbf{H}=\frac{4\pi e}{c}(n_i\mathbf{v}_i-n_e\mathbf{v}_e)+\frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}; \tag{3.1.5e}$$

$$\mathrm{rot}\mathbf{E}=-\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \tag{3.1.5f}$$

$$\mathrm{div}\,\mathbf{H}=0; \tag{3.1.5g}$$

$$\mathrm{div}\,\mathbf{E}=4\pi e(n_i-n_e). \tag{3.1.5h}$$

In the Maxwell equations, it is taken into account that the density of current and charge has the following form:

$$\mathbf{j} = e(n_i\mathbf{v}_i - n_e\mathbf{v}_e),\ \rho_e = e(n_i - n_e). \tag{3.1.6}$$

In general, this is a cumbersome system of equations but, as shown later, in many cases analysis of the system is relatively simple.

3.1.2. Reduction of the system (3.1.5)

The system (3.1.5) can be reduced, under specific assumptions, to simpler systems.

Two-fluid equations for quasi-neutral plasma

If the characteristic dimensions of the systems L are sufficiently large ($L \gg r_D$), plasma can be regarded as quasi-neutral. In this case, assuming that

$$n_i = n_e = n, \tag{3.1.7a}$$

we have actually added another equation to the initial system and in order to avoid making the system overdeterminate it is necessary to reject equation (3.1.5h). However, taking into account that div from (3.1.5e) leads to the equation (3.1.5h), in the equation (3.1.5e) it is necessary to reject the bias currents. In this case, equation (3.1.5e) gives the natural equation

$$\mathrm{div}\ \mathbf{j} = 0. \tag{3.1.7b}$$

If it is assumed that $n_i = n_e = n$ in two continuity equations, then under the condition (3.1.7b) it is sufficient to leave one equation for the ions

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v}_i = 0$$

and the equation for the electrons can be rejected.

Thus, the form of the system (3.1.5) for the quasi-neutral plasma is:

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v}_i = 0; \tag{3.1.8a}$$

$$M\frac{d\,\mathbf{v}_i}{dt} = -\nabla p_i + en\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_i, \mathbf{H}\right]\right) - \alpha(\mathbf{v}_i - \mathbf{v}_e); \tag{3.1.8b}$$

$$m\frac{d\mathbf{v}_e}{dt} = -\nabla p_e - en\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right]\right) + \alpha(\mathbf{v}_i - \mathbf{v}_e); \tag{3.1.8c}$$

$$\operatorname{rot}\mathbf{H} = \frac{4\pi e}{c}\mathbf{j}; \tag{3.1.8d}$$

$$\operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \tag{3.1.8e}$$

$$\operatorname{div}\mathbf{H} = 0. \tag{3.1.8f}$$

The Ohm's law

If the characteristic scale L is considerably greater not only than the Debye radius r_D but also the electronic Larmor radius ρ_e, $L \gg \rho_e$, in (3.1.5d) we can ignore the inertia of the electrons and obtain the 'generalised Ohm's law'

$$\frac{\mathbf{j}}{\sigma} = \frac{\nabla p_e}{en} + \left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right]\right). \tag{3.1.9}$$

Here $\sigma = \dfrac{\alpha}{en}$ is the local conductivity of plasma. The generalised Ohm's law differs from the conventional Ohm's law (2.3.2c) in two respects. Here we have the term with ∇p_e which is responsible for thermal phenomena in classic physics, and the 'Hall effect' associated with the fact that (2.3.2c) does not include the velocity of the medium $\mathbf{v}$ (i.e. ions) and includes the velocity of the electrons $\mathbf{v_e}$. Here we carry out detailed analysis of the generalised Ohm's law and continue reduction.

Transition from the two-fluid model to the single-fluid model
Assuming that the plasma is quasi-neutral and $m \to 0$, the system (3.1.8) can be presented in the form

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v} = 0; \quad \mathbf{v} \equiv \mathbf{v}_i; \tag{3.1.10a}$$

$$Mn\left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v}\right) = -\nabla p_i + en\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right]\right) - \frac{\mathbf{j}}{\sigma} en; \tag{3.1.10b}$$

$$0 = -\nabla p_e - en\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right]\right) + \mathbf{j}\frac{en}{\sigma}; \tag{3.1.10c}$$

$$\operatorname{rot}\ \mathbf{H} = \frac{4\pi}{c} en(\mathbf{v} - \mathbf{v}_e); \tag{3.1.10d}$$

$$\operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \tag{3.1.10e}$$

$$\operatorname{div}\mathbf{H} = 0. \tag{3.1.10f}$$

Combining the equations (3.1.10b) and (3.1.10c) and introducing the mass density of the medium $\rho = Mn$ and the total pressure $p = p_i + p_e$, we obtain

$$\frac{\partial \rho}{\partial t} + \operatorname{div} p\mathbf{v} = 0; \tag{3.1.11a}$$

$$\rho\frac{d\mathbf{v}}{dt} = -\nabla p + \frac{1}{c}\left[\mathbf{j}, \mathbf{H}\right]; \tag{3.1.11b}$$

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right] + \frac{\nabla p_e}{en}; \tag{3.1.11c}$$

$$\operatorname{rot}\mathbf{H} = \frac{4\pi e}{c}\mathbf{j}; \tag{3.1.11d}$$

$$\operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \tag{3.1.11e}$$

$$\operatorname{div}\mathbf{H} = 0. \tag{3.1.11f}$$

Comparison of the system with the single-fluid MHD system shows that all the differences of these systems are found in the Ohm's law. If ∇p_e is ignored together with the difference between v_e and v_i, then both systems coincide. Evidently, it may be said that the single-fluid MHD describes the dynamics of 'bonded' ion–electron pairs.

Relationship of the two-fluid equations with the equations of dynamics of single particles

It would appear that this relationship is obvious. For this purpose, it is sufficient to reject terms with the gradient of pressures and the friction force. However, this is indeed the case if the trajectories of the particles, belonging to the same component, do not intersect. Otherwise, discontinuities should be observed in the hydrodynamics, i.e. the continuous medium (in hydrodynamics!) is unambiguous.

3.1.3. Relationship of the components of plasma with the magnetic field

In chapter 2, we discussed two mutually identical laws of conservation: conservation in the ideal gas of the circulation of velocity along the 'liquid' circuit, and the conservation of the magnetic flux through the liquid circuit

$$\oint_{\gamma} \mathbf{v} d\mathbf{l} = \text{const}; \tag{3.1.12a}$$

$$\iint \mathbf{H} d\mathbf{S} = \text{const}. \tag{3.1.12b}$$

The last equation can be transformed to the form identical with (3.1.12) if it is assumed that **H** = rot **A**. Consequently, we obtain

$$\iint_{\gamma} \mathbf{H}\, d\mathbf{S} = \int_{\gamma} \mathbf{A}\, d\mathbf{l} = \text{const}. \tag{3.1.13}$$

Both laws (3.1.12) follow from the proven claim that if some vector **a** satisfies the equation

$$\frac{\partial \mathbf{a}}{\partial t} + \left[\mathbf{v}, \text{rot}\, \mathbf{a}\right] = \nabla W, \tag{3.1.14a}$$

the circulation of **a** in the liquid circuit is conserved:

$$\oint_{\gamma} \mathbf{a} d\mathbf{l} = \text{const}. \tag{3.1.14b}$$

It may easily be seen that the dynamics equations (3.1.5c) and (3.1.15d) can be reduced to the form (3.1.14) in the absence of ohmic resistance. This will be shown on the example of the dynamics equation of the ions. Substituting the identities into this equation

$$(\mathbf{v}\nabla)\mathbf{v} = \nabla\frac{v^2}{2} - [\mathbf{v}, \operatorname{rot}\mathbf{v}]; \quad \mathbf{H} = \operatorname{rot}\mathbf{A}, \quad \mathbf{E} = \frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} - \nabla\phi,$$

we obtain

$$\frac{\partial}{\partial t}\left(M\mathbf{v} - \frac{e}{c}\mathbf{A}\right) - \left[\mathbf{v}, \operatorname{rot}\mathbf{v} - \frac{e}{c}\operatorname{rot}\mathbf{A}\right] = -\nabla\left(i + \frac{v^2}{2} + e\phi\right). \tag{3.1.15}$$

This shows that the following quantities are retained for the ions and electrons:

$$C_i \equiv \oint_{\gamma_i} dl\left(M\mathbf{v} - \frac{e}{c}\mathbf{A}\right) = \text{const}; \tag{3.1.16a}$$

$$C_e \equiv \oint_{\gamma_e} dl\left(m\mathbf{v} - \frac{e}{c}\mathbf{A}\right) = \text{const}. \tag{3.1.16b}$$

Equation (3.1.16) shows that if the electron inertia can be ignored, the magnetic flux is 'frozen' into the electron liquid circuit instead of the ion liquid circuit. The magnetic flux can be also 'frozen' into the ionic component only if the ion mass can also be ignored.

Evidently, in the presence of axial symmetry and a single azimuthal magnetic field, the equations (2.3.10) for 'freezing-in' remain unchanged but in this case they also hold for electronic trajectories, if $m \to 0$, and M is finite.

3.1.4. Law of conservation of energy in the stationary flows of two-component plasma

We introduce the concept of the trajectory of the ionic and electronic flows, and the tangents to these lines are parallel to the velocities of the appropriate components:

$$d\mathbf{r}_i \| \mathbf{v}_i; \quad d\mathbf{r}_e \| \mathbf{v}_e$$

These trajectories are denoted by ψ_i and ψ_e.

Assuming that in (3.1.5) $\frac{\partial}{\partial t} = 0$ and multiplying one equation by v_i and the identical equation for the electrons by v_e, we obtain the law

of conservation of energy

$$\frac{Mv^2}{2} + i_i + e\phi = \varepsilon_i(\psi_i); \quad \frac{mv^2}{2} + W_e - e\phi = \varepsilon_e(\psi_e). \tag{3.1.17}$$

3.2. Electron magnetic hydrodynamics. Generalised Ohm's law

The large difference in the masses of the ions and the electrons results in the relative independence of the processes in the electronic component on the processes in which the ions take part. If we are discussing oscillations, then the frequencies characteristic of the electrons are the electronic Langmuir and Larmor frequencies which differ by orders of magnitude from the appropriate ionic frequencies.

Thus, at the density of the electrons in the hydrogen plasma of a tokamak $n_e = 10^{14}$ cm^{-3} and the magnetic field $H = 50$ kOe, the electronic frequencies are equal to

$$\omega_0^e = 5{\cdot}10^{11}\,\mathrm{s}^{-1}, \quad \omega_{He} = 8{\cdot}10^{11}\,\mathrm{s}^{-1},$$

whereas for deuterium (the working substance of thermonuclear reactors using DT)

$$\omega_0^D \approx 8{\cdot}10^{9}\,\mathrm{s}^{-1}, \quad \omega_{HD} = 2.7{\cdot}10^{8}\,\mathrm{s}^{-1}.$$

Although the system selected for evaluation is in fact random, the values accurately reflect the tendencies in the scales of the characteristic times for the electronic and ionic processes. In chapter 5 it will be shown that the kinetic processes – formation of distribution functions – are also faster inside each of the components of the plasma than between the components. This shows that the electronic processes can often be investigated on a stationary ion background. Previously, such a model was referred to as 'electronic plasma' but now 'electronic MHD' is usually discussed.

If we are discussing the slow processes, then we can usually ignore the inertia of the electrons and the differential (in relation to **v**) equation (3.1.5d) becomes the algebraic 'generalised Ohm's law' (3.1.9) or the 'Ohm's law with the Hall effect taken into account'. In this section, we consider in detail this inertialess model of electrodynamics.

3.2.1. Dimensionless characteristics of the generalised Ohm's law

The generalised Ohm's law is presented in one of the two forms: either

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e \times \mathbf{H}\right]; \quad \mathbf{j} = en(\mathbf{v} - \mathbf{v}_e) \tag{3.2.1a}$$

or, excluding $\mathbf{v}_e$, in the form

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{c}\left[\mathbf{v} \times \mathbf{H}\right] - \frac{\left[\mathbf{j} \times \mathbf{H}\right]}{enc} \tag{3.2.1b}$$

The last equation shows clearly the 'Hall' term determined to different extent by $\mathbf{v}$ or $\mathbf{v}_e$. Equation (3.2.1a) contains four terms and the role of these terms greatly differs in different conditions. To evaluate this role, it is convenient to use dimensionless parameters. We consider the most interesting of these parameters.

The Ohm's number

The ratio of the scale of the Ohmic term $\{j/\sigma\}$ to the scale of the strength of the electrical field $\{E\}$ is referred to as the 'Ohm's number'

$$\Omega = \{\mathbf{j}/\sigma\} \Big/ \{\mathbf{E}\}. \tag{3.2.2}$$

Here the braces indicate the sign of the scale. In commercial electrical engineering in stationary conductors $\Omega = 1$.

At the same time, in plasma systems the quantity Ω can be very small. For example, in stationary plasma thrusters (SPT) which will be discussed in detail in section 10.4, the density of current in the working models is on the level $j \sim 0.2$ A/cm^2 = $6 \cdot 10^8$ absolute units. The electronic temperature $T_e \sim 13$ eV[3], the electrical field E ~0.3 kV/cm ~1 absolute unit and, consequently

$\Omega \sim 10^{-6}$.

The local exchange parameter is determined as the ratio of the scales of the relative velocity of the electrons and ions ('current' velocity $\mathbf{u} = \mathbf{v}_i - \mathbf{v}_e$) to the direction of the velocity of the ions in relation to the walls of the channel

$$\xi_{\text{loc}} = \frac{\{\mathbf{u}\}}{\{\mathbf{v}\}}. \tag{3.2.3}$$

In single-fluid magnetic hydrodynamics it is assumed that $\xi_{\text{loc}} \to 0$.

The Hall parameter $\omega_e \tau_e$

To introduce this parameter, we use the equation (3.2.1b) in which the

[3]This temperature corresponds to the classic (section 5.3.4) conductivity s ≈ 10^{13} $(T_e)^{3/2}$ ≈ $5 \cdot 10^{14}$ absolute units.

'Hall' term is explicitly separated

$$\mathbf{E}_{\text{hall}} = -\frac{[\mathbf{j} \times H]}{enc}. \tag{3.2.4a}$$

The parameter $\omega_e \tau_e$ is the ratio of the scales $\mathbf{E}_{\text{hall}}$ to the Ohmic term

$$\omega_e \tau_e = \frac{\left\{\dfrac{[\mathbf{j},\mathbf{H}]}{enc}\right\}}{\left\{\dfrac{j}{\sigma}\right\}}. \tag{3.2.4b}$$

Taking into account general equation for conductivity in plasma $\sigma = e^2 n_e \tau / m$, where τ is the free path time of the electron, gives

$$\left\{\frac{\tau H}{en_e c}\right\} = \left\{\frac{eH\tau}{mc}\right\} = \{\omega_e \tau_e\}. \cdot \tag{3.2.4c}$$

In the plasma dynamic systems the characteristic values $\omega_e \tau_e$ can be in any form. For example, in low-temperature MHD-generators ($T \sim 2000°C$) with a dense but slightly ionised plasma, $\omega_e \tau_e \lesssim 1$, whereas in tokamaks $\omega_e \tau_e \sim 3 \cdot 10^8$.

The thermalisation parameter is determined as the following ratio

$$\Pi_T = \frac{\left\{\dfrac{\nabla p_e}{en}\right\}}{\{E\}} \sim \frac{\{kT_e\}}{\{e\delta\phi\}}. \tag{3.2.5}$$

Here $\delta\phi$ is the period of the potential in the system. In most cases, $0 < \Pi_T \lesssim 1$.

In many cases, the temperatures of the components in the plasma systems greatly differ. These systems are referred to as non-isothermal. They may be of considerable interest, for example, as plasma accelerators with hot electrons ($T_e \gg T_i$) in which the **E**-field is generated almost exclusively as a result of ∇p_e (section 3.6).

The integral exchange parameter

There are a number of plasma systems with a flow of matter and with electrical current passing through the flow (Fig. 3.2.1)

These systems include quasi-stationary plasma accelerators and MHD generators. If the mass flow rate through this flow plasma dynamic system is equal to $\dot{m}$ and the matter is completely ionised once, this flow rate may be compared with the equivalent current

$$J_{\dot{m}} = \frac{e}{M}\dot{m}. \tag{3.2.6a}$$

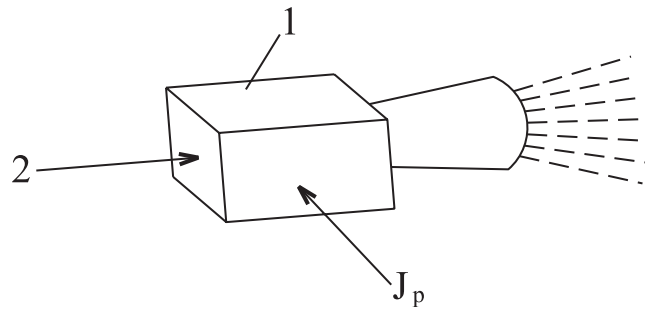


Fig. 3.2.1. Determination of the integral exchange parameter: 1) the flow plasma dynamic system; 2) the incoming flow of working matter; *Jp* – electrical current.

The ratio of J_p to discharge current J_m is referred to as the integral exchange parameter[4]

$$\xi = \frac{J_p}{J_{\dot{m}}}. \tag{3.2.6b}$$

Evidently, the integral exchange parameter is proportional to the local exchange parameter multiplied by the geometrical factor which depends on the design of the system.

3.2.2. Dissipationless electronic component: 'degenerate Ohm's law', 'Lorenz fields'

The previously described estimations show that the classic conductivity, i.e. the conductivity determined by paired collisions of the ions and electrons, is very low in the majority of plasma systems with medium and high energy. Therefore, in these cases, the Ohmic resistance can be ignored in the first approximation. Consequently, the Ohm's law (3.1.11c) degenerates and has the following form

$$0 = \mathbf{E} + \frac{1}{e}\left[\mathbf{v}_e, \mathbf{H}\right] + \frac{i\nabla p}{en}. \tag{3.2.7}$$

This equation is referred to as the 'degenerate Ohm's law'. If the thermal component can also be ignored, equation (3.2.7) becomes very simple:

$$\mathbf{E} + \frac{1}{e}\left[\mathbf{v}_e, \mathbf{H}\right] = 0. \tag{3.2.8a}$$

In this partial but very important case the electrostatic **E**-field is

[4]This parameter is also referred to as the Morozov parameter and denoted by Mo.

referred to as the 'Lorenz **E**-field'[5]. This shows that for a Lorenz electrical field to exist in the volume of plasma, the **H**-field and electron drift must be present.

Later, we examine the 'degenerate Ohm's' law only for the case of vortex-free fields, i.e. it will be assumed that

$$\mathbf{E} = -\phi\nabla. \tag{3.2.8b}$$

Equipotentialisation of magnetic lines of force in plasma. 'Thermalised potential'

Remaining within the limits of the hydrodynamics model, it may be assumed that the heat conductivity of the electronic component along the magnetic lines of force is high (section 5.3.4) and, consequently, the electronic temperature becomes the function of only of the number of the magnetic lines of force γ:

$$T = T(\gamma). \tag{3.2.9a}$$

Therefore,

$$p_e = nkT_e = p_e(n, \gamma). \tag{3.2.9b}$$

Taking into account the potentiality of the **E**-field (3.2.8b), equation (3.2.7) changes to:

$$\nabla\phi - \frac{1}{en}\nabla nkT(\gamma) = \frac{1}{c}\left[\mathbf{v}_e, \mathbf{u}\right]. \tag{3.2.10a}$$

Multiplying this equation by **H** gives

$$(\mathbf{H}\nabla)\left(\phi - \frac{kT_e(\gamma)}{e}\ln n\right) = 0. \tag{3.2.10b}$$

This means that the following quantity remains unchanged along the magnetic line of force γ:

$$\phi_T(\gamma) \equiv \phi(\mathbf{x}) - \frac{kT_e(\gamma)}{e}\ln n(\mathbf{x}). \tag{3.2.10c}$$

This quantity which differs from the conventional electrical potential ϕ by the 'thermal' additive is referred to as the 'thermalised potential' and its constancy along the magnetic line of force as the 'equipotentialisation of magnetic lines of force in plasma'. Evidently, the relationship (3.2.10c) denotes nothing else but the Boltzmann distribution of electrons along the line of force:

[5]However, in most cases this field is referred to as the 'Hall' field.

$$n = n_1 \exp\left(\frac{e\phi}{kT_e}\right); \quad n_1 \equiv n_0 \exp\left(-\frac{e\phi_T}{kT_e}\right) = \text{const.} \tag{3.2.10d}$$

The equipotentialisation of the magnetic lines of force (EPM) is a fundamental property of highly conducting plasma. Consequently, the **magnetic lines of force with electronic helices wound on them can be regarded as transparent magnet–electronic electrodes (MEE)**, (Fig. 3.2.2).

Several properties of the thermalised potential, following from equations (3.2.10), will be noted. In particular, it can be seen that in the case of cold plasma when

$$\left|kT_e \frac{\nabla n}{n}\right| << \left|\nabla\phi\right|,$$

the thermalised potential and the conventional electrostatic potential coincide at any rational distribution of density in the system so that (Fig. 3.2.2a)

$$\phi \approx \phi_T(\gamma). \tag{3.2.11}$$

However, if the electronic temperature is significant and the density gradients are high (such a situation is found in, for example, systems with ion beams where the transverse density gradients are large), the role of the thermal term in (3.2.10c) becomes considerable (Fig. 3.2.2b).

The difference between the thermalised potential and the electrical potential is associated with the fact that to ensure quasi-neutrality ($n = n_e \approx n_i$) the sections of the line of force in which the ion density is higher should also be characterised by a higher electrical potential in order to 'retain' the electrons. The existence of the higher potential in the range of high-intensity results in the additional expansion of the

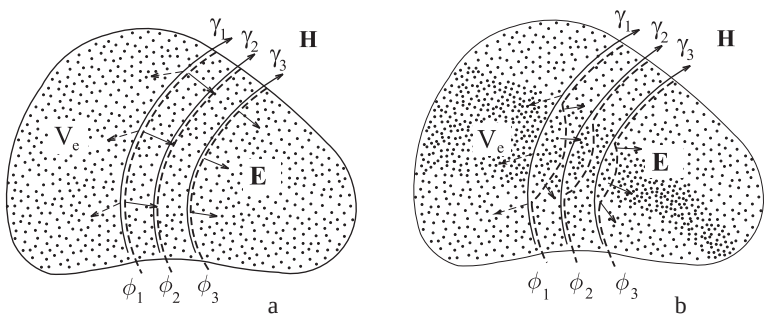


Fig. 3.2.2. Equipotentialisation of magnetic lines of force (γ) in non-uniform plasma: a) cold plasma, $T_e \to 0$; b) heated plasma, $T_e \neq 0$.

ion beam. This effect is nothing else but the expansion of the beam under the effect of electronic pressure. It is especially strong in the vicinity of the beam boundary layer where $|\nabla n|$ is particularly high (section 5.7.3).

The equations (3.2.10) for the thermalised potential are based on the Maxwell–Boltzmann distribution. However, in many plasma dynamic systems the actual distribution function of the electrons differs greatly from this distribution. Nevertheless, the thermalised potential can also be used in this case. In fact, it can be written in almost all cases that[6]

$$p(x) = p(n, \gamma) \tag{3.2.12a}$$

and the 'enthalpy of plasma on the line of force' can be introduced

$$i(\gamma, n) = \int \frac{dp(\gamma, n)}{n}\bigg|_{\gamma=\text{const}}. \tag{3.2.12b}$$

Consequently, the 'thermalised potential' which is in fact generalised enthalpy with the accuracy to the constant multiplier has the form

$$\phi_T(\gamma) = \phi - \frac{i(\gamma, n)}{e}. \tag{3.2.12c}$$

Dependence $\phi_T(\gamma)$

Equation (3.2.1c) describes the variation of the electrical potential only along the line of force. However, the equation does not determine how the potential varies from one line of force to another, i.e. the dependence of ϕ_T on γ remains unknown. In principle, we can examine two schemes of 'fixing the potential'. The first of them is associated with dissipative processes in the channel as a result of which the electronic flux penetrates the entire plasma volume in which we are interested and generates the distribution $\phi(\mathbf{r})$ in relation to the local resistance of the volume. A suitable example of this situation is the stationary plasma engine in which an electronic flux travels from the section to the anode as a result of the oscillations, Ohmic and near-wall conductivity[7]. Different sections of the channel offer different resistance to the flux and a specific distribution of the potential is therefore established in this channel.

The second scheme of potential fixation is based on the use of special, generally speaking, emitting electrodes (Fig. 3.2.3). This method is based on the fact that if the processes in the plasma volume

[6]If the pressure is not isotropic, the relationship of ϕ with p can be quite complicated. More accurately, this is the case if the strength of the magnetic field ($|\mathbf{H}|$) along the line of force changes and the adiabatic invariant remains unchanged.

[7]For more details see section 7.3.2.

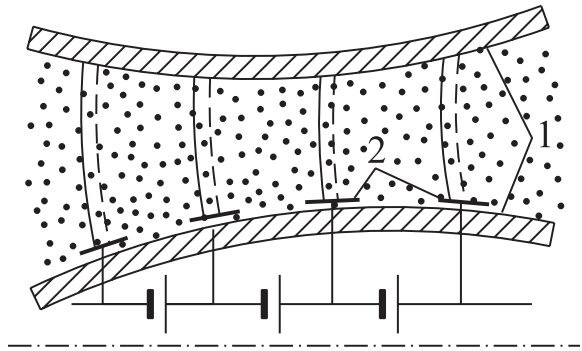


Fig. 3.2.3. Fixation of the potential of the magnetic lines of force by near-wall electrodes: 1) dielectric walls, 2) fixing electrodes.

do not lead to anomalously high ('turbulent') conductivity, the jump in the vicinity of the electrode is of the order of kT_e/e [8]. Therefore, in cases in which $\delta\phi$, i.e. the difference of the potentials in the plasma volume, is considerably greater than kT_e/e, i.e. $|\delta\phi| \gg kT_e/e$, the described fixation based on special electrodes appears to be realistic. At first sight, it may appear that any distribution of the potentials can be set in the same simple manner (or in the same complicated manner) using electrodes. However, this is not the case. This problem will be examined in detail later and now it is necessary to make another comment associated with the equipotential nature of the magnetic lines of force.

Magnetoelectrical systems

To make it more obvious, it is assumed that the thermal correction in (3.2.12) is small and, therefore, the magnetic lines of force are usually equipotential: $\phi = \phi(\gamma)$.

This shows that by selecting the magnetic fields of different configuration it is possible to generate superthermal potential reliefs of different form in the plasma and at the same time the 'magnetoelectrical' plasma devices of different types. Figure 3.2.4 shows schematically the main types of these devices with non-magnetised ions (A.I. Morozov [6]).

The accelerators of the type shown in Fig. 3.2.4a include engines with azimuthal drift, including stationary plasma thrusters (section 6.7).

However, if a high-energy quasi-neutral ion flux is directed against the electrical field, the flux slows down, transferring the energy to the source of the electrical field. These decelerating systems are also referred to as recuperators (see Fig. 3.2.4b).

[8]See section Int.1.

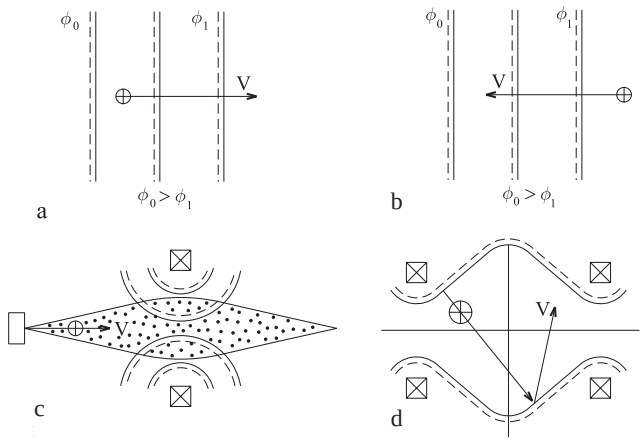


Fig. 3.2.4. 'Magnetoelectrical' plasma devices: a) accelerators of compensated ion fluxes (CIF); b) recuperators of the energy of CIF; c) the lenses for focusing the CIF; d) magnetoelectrical plasma traps; broken line – magnetic lines of force; solid line – equipotentials.

Figure 3.2.4c shows a 'plasma lens' (see section 5.7). Here the 'lens' refers to various different devices used for focusing and also for separating the quasi neutral high-current ion fluxes with respect to mass and energy. Such a 'lens' is a conventional image of future high-current corpuscular optics, i.e. 'plasma optics', which will efficiently supplement the current low-current 'vacuum' corpuscular optics. Finally, at present there are also plasma traps which relate to the type shown in Fig. 3.2.4d. They are referred to as 'magnetoelectrical'.[9]

Condition of autonomy of the ideal electronic component [6]
It was mentioned previously that if all the magnetic lines of force intersect the walls of the volume, then fixators can be used to produce almost any distribution $\phi_T(\gamma)$. However, it is necessary to solve the problem of the 'fee' required for producing a specific potential relief in the plasma system with the given $\mathbf{H}(r)$ and $n(r)$ and, in particular, the problem of the 'free of charge', in the given meaning of the word, distributions of the potentials. The 'free of charge' or 'natural' distributions of the potential refer to potential reliefs for which there is no (if dissipation is ignored) continuous exchange of the electrons between the fixators and the plasma. In these cases, the electronic component of the plasma volume may be regarded as 'autonomous', and the movement of the electrons is reduced to oscillations along the magnetic lines of force and drift transitions from one set of lines to

[9]They also include the first schemes of the traps, proposed by O.A. Lavrent'ev

others. Thus, the 'natural' distribution of the potential with the accuracy to low dissipative losses sustains itself. Here, we are interested in the equilibrium of the investigated systems and not in their stability.

The problem of the 'natural' and 'non-natural' stationary distributions of the potential will be investigated using the following system of equations for the electronic component

$$\operatorname{div} n\mathbf{u}_e = 0; \tag{3.2.13a}$$

$$\frac{\nabla p_e}{en} + \mathbf{E} + \frac{1}{c}\left[\mathbf{u}_e, \mathbf{H}\right] = 0; \quad \mathbf{E} = -\nabla\phi; \tag{3.2.13b}$$

$$c_v\left(n(\mathbf{u}_e\nabla)T_e - T_e(\gamma_a - 1)(\mathbf{u}_e\nabla)n\right) = \operatorname{div}\overleftrightarrow{\varkappa}\nabla T_e. \tag{3.2.13c}$$

Here $\overleftrightarrow{\varkappa}$ is the heat conductivity tensor, $c_v = (3/2)\,k$, γ_a is the exponent of the adiabate. In the case of ideal plasma it should be assumed that

$$\varkappa_\perp \to 0, \ \varkappa_\parallel \to \infty. \tag{3.2.14a}$$

In the system (3.2.13) under the condition (3.2.14a) n and $\mathbf{H}$ are assumed to be given, and ϕ and T_e should be determined. Without tending to generality, we confine ourselves to the case in which (3.2.13c) and (3.2.14a) hold, i.e.

$$T_e = T_e(\gamma), \quad \phi_T(\gamma) = \phi - \frac{kT_e}{e}\ln\frac{n}{n_0}. \tag{3.2.14b}$$

It is assumed that the magnetic field permits the introduction of an orthogonal system of coordinates in which one of the axis (x_3) coincides with the magnetic lines of force. Two other coordinates are denoted by x_1 and x_2.

In these coordinates we can write

$$T_e = T_e(\gamma) = T_e(x_1, x_2);$$

$$\phi^*(x_1, x_2) = \phi - \frac{kT(x_1, x_2)}{e}\ln\frac{n}{n_0}. \tag{3.2.14c}$$

The selection of the coordinates is shown in Fig. 3.2.5.

It is well known that the metric form, the gradient and divergence in an arbitrary orthogonal coordinate system have the form:

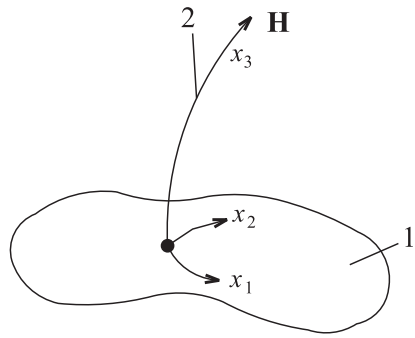


Fig. 3.2.5. Selection of the orthogonal coordinate system: 1) magnetic surface; 2) magnetic line of force **H** which coincides with the x_3 axis.

$$ds^2 = h_1^2 dx_1^2 + h_2^2 dx_2^2 + h_3^2 dx_3^2;$$

$$\nabla = \left(\frac{1}{h_1}\frac{\partial}{\partial x_1}, \frac{1}{h_2}\frac{\partial}{\partial x_2}, \frac{1}{h_3}\frac{\partial}{\partial x_3} \right);$$

$$\operatorname{div}\mathbf{a} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial x_1} h_2 h_3 a_1 + \frac{\partial}{\partial x_2} h_1 h_3 a_2 + \frac{\partial}{\partial x_3} h_2 h_1 a_3 \right].$$

Here h_1, h_2, h_3 are the Lamè parameters.

Using equation (3.2.13b) we obtain the following equations for the components of the electronic velocity across the magnetic field:

$$u_1 = -\frac{c}{H h_2}\left(\frac{\partial \phi^*}{\partial x_2} + \frac{k}{e}\left(\ln \frac{n}{n_0 e_0} \right) \frac{\partial T_e}{\partial x_2} \right); \tag{3.2.15}$$

$$u_2 = \frac{c}{H h_1}\left(\frac{\partial \phi^*}{\partial x_1} + \frac{k}{e}\left(\ln \frac{n}{n_0 e_0} \right) \frac{\partial T_e}{\partial x_1} \right).$$

Here e_0 is the base of the natural logarithms.

In order to determine the third component **u**, equation (3.2.15) is substituted into the continuity equation (3.2.13a). After transformations we obtain

$$u_3 = \frac{1}{h_1 h_2 h_3}\left[\frac{D(\omega, \phi^*)}{D(x_1, x_2)} + \frac{k}{e} \cdot \frac{D(\lambda, T_e)}{D(x_1, x_2)} \right], \tag{3.2.16a}$$

where

$$\omega(x_3) = \int_a^{x_3} dx_3 \frac{h_3 n}{H}; \quad \lambda(x_3) = \int_a^{x_3} \frac{n h_3 dx_3 L}{H}; \quad L = \ln \frac{n}{n_0 e_0}, \tag{3.2.16b}$$

and $a = x_3$ (0) is the coordinate of the 'beginning' of the line of force, where $u_3 = 0$.

Assuming that the velocity u_3 at the 'beginning' and at the 'end' of the magnetic line of force converts to 0, from (3.2.15) we obtain the equation linking ϕ^* and T_e i.e. the autonomy condition

$$\frac{D(\phi_T, W)}{D(x_1, x_2)} + \frac{k}{e}\frac{D(T_e, \Lambda)}{D(x_1, x_2)} = 0. \qquad (3.2.17a)$$

Here $\dfrac{D(\alpha,\beta)}{D(x_1,x_2)} \equiv -\begin{vmatrix} \dfrac{\partial\alpha}{\partial x_1} & \dfrac{\partial\alpha}{\partial x_2} \\ \dfrac{\partial\beta}{\partial x_1} & \dfrac{\partial\beta}{\partial x_2} \end{vmatrix}$ is the Jacobian.

$$W(\gamma) \equiv \int_a^b \frac{n dl}{H}; \quad \Lambda(\gamma) \equiv \int_a^b \frac{n}{H}\left(\ln\frac{n}{n_0 e_0}\right) dl, \qquad (3.2.17b)$$

where dl is the element of the length of the line of force, a, b is the beginning and end of the line of force.

If the plasma in the volume has a negligibly low temperature ($T_e \to 0$) or the system is isothermal (T_e = const), then the potential ϕ is constant on the surfaces with a constant value W = const, i.e.

$$\phi_T = \phi_T(W), \; W = W(\gamma). \qquad (3.2.18)$$

Quantity $W(\gamma)$ is referred to as the 'loading' of the magnetic line of force (Morozov [6]). An example of using the relationship (3.2.18) will be discussed later.

Magnetic-drift surfaces

All the considerations made in the previous sections were based on the equipotentialisation of the magnetic lines of force, i.e. on the integral (3.2.10) which was obtained after multiplying the total equation of the **E**-field (3.2.7) by **H**. The differential equations of the identical form can be derived by multiplying (3.2.7) in the scalar manner by $\mathbf{v}_e$. However, this equation can not be integrated at an arbitrary dependence $p = p\,(n, \gamma)$.

An exception from this is the case of the barotropic electronic component $p = p\,(n)$. Consequently, introducing the enthalpy with the same value for the entire plasma volume

$$i(n) = \int \frac{dp(n)}{n}$$

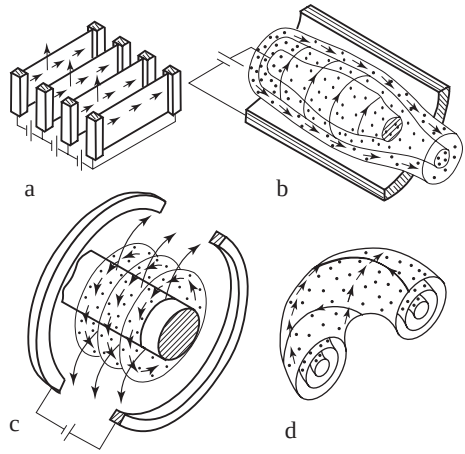


Fig. 3.2.6. Topology of magnetic-drift surfaces: a) in closing the drift through an external circuit; b) in the presence of only the azimuthal magnetic field; c) with azimuthal drift and the quasi-radial magnetic field; d) the set of toroidal magnetic drift surfaces, inserted into each other. The solid lines are the magnetic lines of force, broken lines are the lines of electron drift.

we obtain

$$(\mathbf{v}_e\nabla)\left(\phi-\frac{1}{e}i(n)\right)=0,$$

and, consequently, in this case the thermalised potential is maintained along the drift trajectories

$$\phi-\frac{1}{e}i(n)=\phi_T(\delta). \tag{3.2.19}$$

Of special interest for practice at the moment is a special case in which the system has axial symmetry. In this case, the concept of the 'magnetic-drift surface' is quite useful. These axisymmetric magnetic-drift surfaces are shown in Fig. 3.2.6.

Evidently, the magnetic-drift surfaces can also form in the case of cold electrons. In this case

$$\phi=\phi(\gamma,\delta). \tag{3.2.20}$$

3.2.3. Dynamics of the electronic component with $\sigma \neq \infty$, 'tensor conductivity'

The finite conductivity of the electronic component must be taken into account if the Hall parameter $\omega_e\tau_e$ is smaller than or is of the order of unity:

$$\omega_e \tau_e = \frac{\sigma H}{enc} \leqslant 1. \qquad (3.2.21)$$

In this case, we should start from the entire generalised Ohm's law:

$$\frac{j}{\sigma} = \frac{\nabla p_e}{en} + \mathbf{E} + \frac{1}{c}\left[\mathbf{v},\mathbf{H}\right] - \frac{1}{enc}\left[\mathbf{j},\mathbf{H}\right]. \qquad (3.2.22)$$

Taking into account finite conductivity greatly complicates the analysis of the behaviour of the electronic component: the concept of the thermalised potential loses its strictness, the equation of the equilibrium electronic configurations ceases to be valid, etc. Analysis should be carried out again.

Tensor conductivity

Equation (3.2.22) can be written in the form

$$\frac{j}{\sigma} = \mathbf{E}^* - \frac{1}{enc}\left[\mathbf{j},\mathbf{H}\right],\ \mathbf{E}^* \equiv \frac{\nabla p_e}{en} + \mathbf{E} + \frac{1}{c}\left[\mathbf{v},\mathbf{H}\right], \qquad (3.2.23)$$

where **E*** is the 'effective strength of the electrical field'.

Equation (3.2.23) contains current density **j** in the implicit form. In order to determine the explicit expression for **j**, it is rational to use the coordinate form of this equation. If the *z* axis is directed along the magnetic field, its components will have the form

$$\frac{j_x}{\sigma} + E_x^* - \frac{1}{enc} j_y H = 0;\quad \frac{j_y}{\sigma} + E_y^* + \frac{1}{enc} j_x H = 0;\quad \frac{j_z}{\sigma} = E_x^*. \qquad (3.2.24a)$$

Using the 'resistance tensor'

$$\overleftrightarrow{\Omega} = \frac{1}{\sigma}\begin{pmatrix} 1 & \omega_e\tau_e & 0 \\ -\omega_e\tau_e & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},\quad \omega_e\tau_e \equiv \frac{\sigma H}{enc}, \qquad (3.2.24b)$$

and the system of equations (3.2.24) can be written in the form

$$\mathbf{E}^* = \overleftrightarrow{\Omega}\mathbf{j}. \qquad (3.2.24c)$$

The inverse tensor to $\overleftrightarrow{\Omega}$ is the conductivity tensor $\overleftrightarrow{\Sigma}$. It is clear that

$$\overleftrightarrow{\Omega}^{-1} \equiv \overleftrightarrow{\Sigma} = \frac{\sigma}{1+(\omega_e\tau_e)^2}\begin{pmatrix} 1 & -\omega_e\tau_e & 0 \\ \omega_e\tau_e & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \qquad (3.2.25a)$$

Knowing the conductivity tensor, we can calculate the density of current through the strength of the effective electrical field

$$\mathbf{j} = \vec{\Sigma}\mathbf{E}^*. \tag{3.2.25b}$$

We consider two partial cases, assuming that the gradient of the electrical pressure (∇p_e) and the velocity of the plasma **v** can be ignored, so that **E***= **E**.

1. Let the electrical field **E** be directed along the magnetic field **H**. Consequently

$$\mathbf{E} = (0, 0, E_z);\ j_z = \sigma E_z .$$

This shows that the passage of current along the magnetic field takes place in the same manner as in the absence of the magnetic field.

2. Let the electrical field to be directed across the magnetic field, for example, along the *x* axis. Consequently

$$\mathbf{E} = (E_x, 0, 0);$$
$$i_x = \frac{\sigma E_x}{1+(\omega_e \tau_e)^2};\ i_y = -\frac{\sigma \omega_e \tau_e E_x}{1+(\omega_e \tau_e)^2}. \tag{3.2.26a}$$

Thus, the electrical field, directed along the *x* axis, generates the electrical current flowing under an angle in relation to the direction of the electrical field, and (Fig. 3.2.7)

$$\frac{j_y}{j_x} = \mathrm{tg}\,\theta = \omega_e \tau_e. \tag{3.2.26b}$$

If the time between the conditions of the electrons with heavy particles τ_e increases without limits (plasma transforms to ideal conductivity plasma) then, as indicated by (3.2.26b), the component of the electrical field j_x tends to 0. At the same time, since $\sigma = e^2 n \tau_e / m$, the component j_y tends to the limit

$$j_y \to -enc\frac{E}{H}, \tag{3.2.26c}$$

The physical interpretation of equation (3.2.26c) is very simple. In the absence of collisions, the electrons drift in the crossed fields along the *y* axis with the velocity $v_{\text{drift}} = c\frac{[\mathbf{E},\mathbf{H}]}{H^2}$ However, since it is assumed that the ions are in the rest state, this drift of the electrons also represents the investigated current.

In conclusion, it should be mentioned that in a general case, the

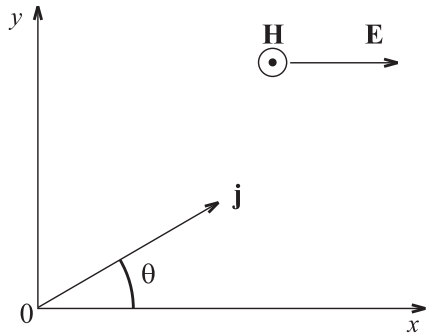


Fig. 3.2.7. Vectors **j**, **E** in the presence of the Hall effect; **H** = (0, 0, *H*).

equation (3.2.26a) can be presented in the form

$$\mathbf{j}_{\|E} = \theta_{\|}\sigma\mathbf{E}^{*};\ \mathbf{j}_{\perp E} = \theta_{\perp}\left(-enc\frac{[\mathbf{E},\mathbf{H}]}{H^2}\right), \tag{3.2.26d}$$

where

$$\theta_{\|} = \frac{1}{1+(\omega_e\tau_e)^2};\ \theta_{\perp} = (\omega_e\tau_e)^2\theta_{\|} = \frac{(\omega_e\tau_e)^2}{1+(\omega_e\tau_e)^2}. \tag{3.2.26e}$$

3.2.4. Electrodynamics of flat flows of incompressible plasma in a transverse magnetic field [79, 80]

One of the fundamental classes of the plasma dynamic processes is the flow of plasma in channels of different type in which all the walls or parts of them are electrodes. In a general case, the electrodes are walls through which the current flows. These systems are divided into two formally mutually reversible classes: 'accelerators', transforming the supplied electrical energy to the kinetic energy of the plasma flow, and 'recuperators', i.e. the generators of electrical energy as a result of the kinetic energy of the plasma flow. Their analogues in classic electrical engineering are electrical motors and electrical generators ('dynamo machines'). The recuperators in which we are interested are MHD-generators.

In this section, attention is given mainly to zero-dimensional models with a small number of figures explaining electrodynamics (i.e. the distribution of **E** and **j**) of the process. The density and velocity of the plasma are assumed to be constant. In reality, this approach is closer to subsonic MHD-generators, but we shall also mention special features of electrodynamics in the acceleration regime since these special features

are also found in the actual accelerators. The two-dimensional self-consistent dynamics of the electrons, ions and fields in the symmetric the case will be discussed in section 3.6.

In this section, we investigate electrical fields and currents, i.e. electrodynamics, on the basis of the Ohm's law, taking into account the Hall effect, in a flat flow moving in a transverse magnetic field with a constant velocity in the channel with a constant cross-section at $\sigma = \text{const} \neq 0$. Thus, the equation

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{e}\left[\mathbf{v}, \mathbf{H}\right] - \frac{\left[\mathbf{j}, \mathbf{H}\right]}{enc} \tag{3.2.27a}$$

will be solved for different boundary conditions.

Taking into account the Lorenz term enables us to examine two flow regimes: the regime of a plasma accelerator when

$$\frac{1}{e}\left[\mathbf{j}, \mathbf{H}\right] \mathbf{v} > 0 \tag{3.2.27b}$$

and the regime of the 'decelerator' – MHD generator where

$$\frac{1}{e}\left[\mathbf{j}, \mathbf{H}\right] \mathbf{v} < 0. \tag{3.2.27c}$$

The results of investigation of the first regime are useful in the analysis of plasma flows in the experiments with the accelerator which will be described later, and the properties of the second regime give information on the MHD generators. The preview consists of two stages. Initially, the processes are investigated under the condition that their characteristics are independent of the longitudinal coordinate *x* ('uniform' channel) and this is followed by discussing an example in which the dependence on *x* is strong ('non-uniform' flux).

Processes in uniform channels

The two previously mentioned regimes (acceleration and recuperation, i.e. generation of current) can be realised using different methods. Three methods will be considered here

$$(a)\ \mathbf{E} \perp \mathbf{v}; \quad (b)\ \mathbf{j} \perp \mathbf{v}; \quad (c)\ \mathbf{E} \parallel \mathbf{v}. \tag{3.2.28}$$

The realisations (*a*) and (*b*) are referred to as Faraday realisations, and (c) as the Hall realisation. Investigations will be carried out in the coordinate form. In this case, equation (3.2.27) is written in the form ($\chi \equiv \omega_e \tau_e$, $\mathbf{H} \parallel \mathbf{z}^0$, $\mathbf{v} \parallel \mathbf{x}^0$)

$$j_x = \sigma E_x - \chi j_y; \tag{3.2.29}$$

$$j_y = \sigma\left(E_y - \frac{v}{c}H\right) + \chi j_x$$

Each of the previously mentioned cases will now be studied.

a) Channel with $\mathbf{E} \perp \mathbf{v}$. Assuming that the parameters of the system are independent of x, we have $E_x = 0$, $E_y = E$ and, consequently, for the equipotential electrodes

$$j_x = -\chi\, j_y; \quad j_y = \frac{\sigma\left(E - \frac{v}{c}H\right)}{1+\chi^2}. \tag{3.2.30}$$

This shows that if $E > \frac{v}{c}H$, the channel operates as an accelerator, since $j_y > 0$ and

$$F_x = \frac{1}{c} j_y H > 0.$$

In this case, $j_x < 0$. This leads to the formation of force F_y, pressing the flow to the cathode (Fig. 3.2.8a). However, if $E < \frac{v}{c}H$, then $j_y < 0$ and the ampere force is

$$F_x = \frac{1}{c} j_y H < 0$$

and the channel operates as a 'decelerator', i.e. an MHD generator. In this case, the signs of j_x and force F_y change. The flow is now pressed to the anode.

b) The MHD channel with $\mathbf{j} \perp \mathbf{v}$. In this case, $j_x = 0$ and from (3.2.29)

$$E_x = \frac{\chi}{\sigma} j_y; \quad j_y = \sigma\left(E_y - \frac{v}{c}H\right), \tag{3.2.31a}$$

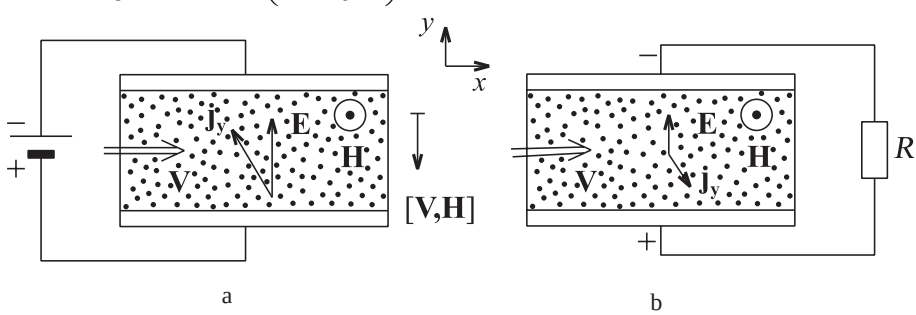


Fig. 3.2.8. MHD channel at $\mathbf{E} \perp \mathbf{v}$: a) acceleration regime, b) generation regime.

or

$$E_x = \chi\left(E - \frac{v}{c}H\right) \tag{3.2.31b}$$

In the previous case, the given scheme corresponds to the MHD generator if $E < \frac{v}{c}H$, and to the accelerator if $E > \frac{v}{c}H$. Here, the potential should change along the electrodes and the potential drop is directly proportional to the current flowing through the electrodes j_y. It is characteristic that the direction of the longitudinal (E_x) field changes its sign in transition from the accelerator to the generation regime. This means that in the first case the field E_x is directed along the velocity of the ions **v** and in the second case it is directed in the opposite direction.

In reality, the potential drop along the electrodes can be produced by one of the methods: either using electrodes having the appropriate resistance along which the current is passed, or by using sectional electrodes (Fig. 3.2.9a). In the former case, the process is not stable and is not used in practice. However, the second method has become the main method in the construction of MHD generators (Fig. 3.2.9 [79]). This method is not suitable for accelerators (see below).

c) the Hall channel with **E**||**v**. In this case, E_y = 0 and, consequently,

$$j_x = \sigma E_x - \chi\, j_y; \quad j_y = -\sigma\frac{v}{c}H + \chi\, j_x. \tag{3.2.32a}$$

Consequently

$$j_x = \frac{\sigma\left(E_x + \chi\frac{v}{c}H\right)}{1+\chi^2};$$

$$j_y = \frac{\sigma\left(\chi E_x - \frac{v}{c}H\right)}{1+\chi^2} \tag{3.2.32b}$$

Figure 3.2.10a shows the scheme with ‘contactors’ of the sectional electrodes, Fig. 3.2.10b shows a ring-shaped system with no contactors, and the role of the latter is played by the flux itself. Evidently, the coaxial scheme is preferred in this case. The scheme is used widely for accelerators but it has been studied insufficiently in application for MHD generators. However, in principle, this generator scheme is most efficient at $\chi \gg 1$.

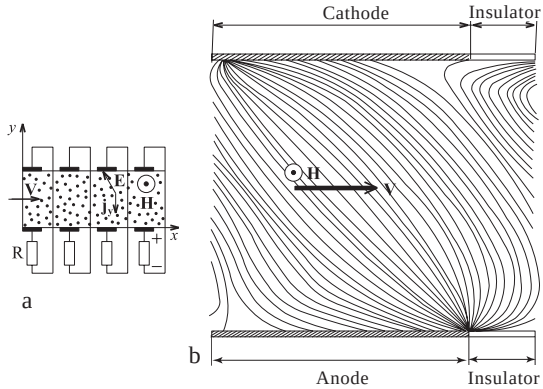


Fig. 3.2.9. MHD channel at **j**⊥**v** (non-equipotential electrodes): a) the scheme of the circuits and orientation of **E** and **j** in the energy generation regime; b) the scheme of the current lines within the limits of the same section of the channel with sectional electrodes ($\omega_e\tau_e$ = 3) [79].

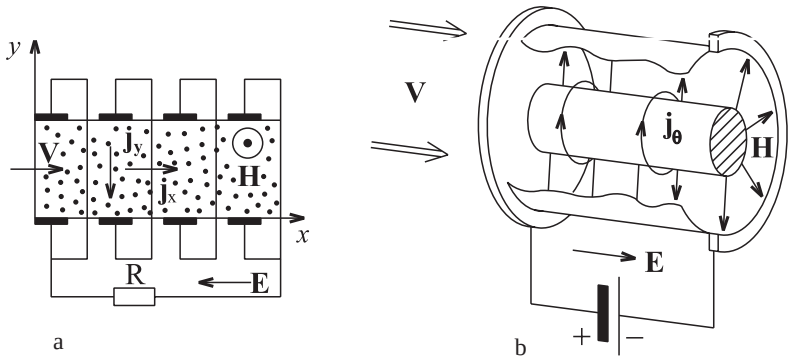


Fig. 3.2.10. Electrical circuits and the channels of Hall MHD systems (**E**||**v**): a) rectangular channel in the generation regime; b) a ring-shaped channel in the acceleration regime.

The special features of the Hall scheme are quite evident at $\chi\to\infty$. At this limit

$$j_x \to env, \quad j_y \to enc\frac{E}{H}. \tag{3.2.33}$$

Consequently, j_x is simply the current transferred by the ions, j_y is the drift electron current. The accelerators of this type are discussed in detail in sections 6.7 and 7.5.

The flow between equipotential electrodes of finite length

Previously, we examined the flow in an infinitely long channel. A more realistic model of the plasma flow between the equipotential electrodes (at least from the viewpoint of the distribution of current) is the model

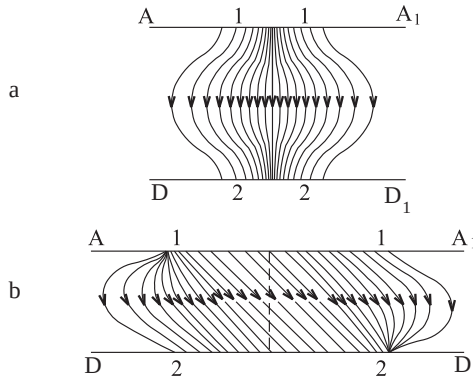


Fig. 3.2.11. The current lines in the MHD channel at a finite length of solid electrodes: a) in the absence of the Hall effect, $\omega_e\tau_e = 0$; b) in the presence of the Hall effect, $\omega_e\tau_e \neq 0$, acceleration regime. 1–2, 2–2) electrodes; AA_1, DD_1 are the walls of the dielectric channel. The magnetic field is perpendicular to the plane of the figure.

shown in Fig. 3.2.11.

Here, as previously, the length of the channel is infinite, the distance between the walls is constant, and the magnetic field is uniform along the entire length of the infinite channel. However, the channel walls are dielectric everywhere, with the exception of the comparatively short section which contains two electrodes with ideal conductivity, with a potential difference applied between the electrodes. If there is no Hall effect, then the distribution of current at a constant flow velocity is symmetric in relation to both electrodes (Fig. 3.2.11a). However, if the medium has a distinctive Hall effect, calculations lead to a non-symmetric flow pattern. In the accelerator regime (Fig. 3.2.11b), attention should be given to the current concentration at the inlet edge of the cathode and the outlet edge of the anode (near-anode current slipping). In the MHD generator regime, the situation is reversed.

Another model in which the plasma flow enters the zone with the magnetic field will be discussed in section 3.8.

3.3. Hall structures

Analogy of the degenerate Ohm's law and the MHD equilibrium equation

It was assumed that the magnetic field in the plasma volume is given. We now abandon considerations of the apriori given magnetic field and instead we use the Maxwell equations

$$\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \quad \operatorname{div}\mathbf{H} = 0.$$

To obtain specific results in this case, it is necessary to make certain assumptions regarding the ion dynamics.

The simplest assumption – the plasma is homogeneous, stationary and isothermal

$$n = \text{const},\ v_i = 0,\ T_e = \text{const}.$$

Regarding the electrical field as potential, the degenerate Ohm's law

$$\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right] = 0,$$

can be presented in the following form

$$\nabla P = \frac{1}{c}\left[\mathbf{j}, \mathbf{H}\right], \tag{3.3.1a}$$

where

$$P = -en\phi, \quad \mathbf{j} = -en\mathbf{v}_e. \tag{3.3.1b}$$

An interesting feature of equation (3.3.1a) is that it is completely identical with the MHD statics equation (2.4.1) [81]. In particular, in the presence of axial symmetry, the equation can be reduced to the Grad–Shafranov equation. The structures, described by (3.3.1a), can be referred to as Hall configurations. They can be realised not only in plasma but also in semiconductors. Another important feature of the Hall configurations is that they are not accompanied (in principle) by the perturbation of the density of the medium and are of the 'purely electrical' nature. Therefore, P can be both positive or negative, whereas in the MHD configurations $P > 0$.

The flow of current through a corrugated conductor at high $\omega_e\tau_e$ *[82]*
In the previous section we discussed the flow of current in plasma in a magnetic field with straight lines of force in the presence of a relatively strong Hall effect. It appears that the transition to the electrodynamics of axisymmetric conductors with the azimuthal magnetic field creates completely new situations. This will be discussed on the example of the flow of current through a stationary corrugated conductor (Fig. 3.3.1) with the distinctive Hall effect.

The density and conductivity of the conductor are assumed to be constant (n = const, σ = const). This problem helps in illustrating the previously mentioned analogy between the degenerate Ohm's law (**E**-field equation) and the MHD statics equation, and is reduced to the

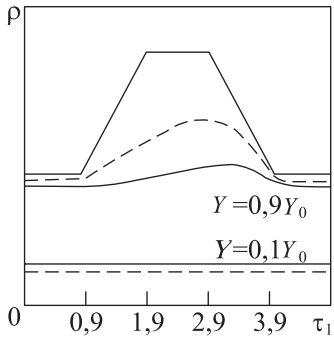


Fig. 3.3.1. Distribution of current in a corrugated conductor at different values of Yo. The broken lines correspond to Yo = 10, the solid lines to Yo = 50.

construction of the simplest non-trivial Hall structure.

It was mentioned in section 2.4.2 that in the presence of only a single azimuthal component there are only cylindrical equilibrium configurations independent of *z*. Consequently, because of this analogy, the region occupied by the current in the corrugated conductor should be cylindrical at $\omega_e\tau_e \to \infty$, i.e. the current should stop flowing into the corrugations. At the same time, in the absence of the Hall effect the distribution of current is governed by the Laplace equation and the current flows into the corrugations.

We derive the general (non-stationary) equation for the Ohm's law for the case of a single azimuthal field and show subsequently that at $\omega_e\tau_e \to \infty$ the current configuration tends to the cylindrical form. Under the given assumptions, the problem is reduced to solving the system of equations

$$\frac{j}{\sigma} = \mathbf{E} - \frac{1}{enc}[\mathbf{j},\mathbf{H}]; \quad \mathrm{rot}\,\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \quad \mathbf{j} = \frac{c}{4\pi}\mathrm{rot}\,\mathbf{H}. \tag{3.3.2a}$$

Using the axial symmetry, the third equation gives

$$j_r = -\frac{1}{r}\frac{\partial I}{\partial z}; \quad j_z = \frac{1}{r}\frac{\partial I}{\partial r}; \quad I = \frac{c}{4\pi}H_\theta r. \tag{3.3.2b}$$

the components in the first two equations (3.2.1) which differ from zero are:

$$\frac{j_r}{\sigma} = E_r + \frac{1}{enc}H_\theta j_r; \quad \frac{j_z}{\sigma} = E_z + \frac{1}{enc}H_\theta j_r; \quad \frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial z} = -\frac{1}{c}\frac{\partial H_\theta}{\partial t}. \tag{3.3.2c}$$

Excluding E_r and E_z and taking into account (3.2.1b) we obtain the main equation of electromagnetic processes at $H_r = H_z = 0$:

$$\frac{c^2}{4\pi\sigma}\left(\frac{\partial^2 I}{\partial z^2}+r\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial I}{\partial r}\right)=\frac{\partial I}{\partial t}-\frac{1}{enr^2}\frac{\partial I^2}{\partial z}. \tag{3.3.3}$$

If $\partial/\partial t = 0$ and $\sigma \to \infty$, the equation has the following form

$$\partial I^2 / \partial z = 0 \tag{3.3.4}$$

indicating the cylindrical form of the configuration. If the Hall effect is not distinctive (this is the case at $I/(en) \to 0$ or $r \to \infty$), i.e. the system becomes flat, the equation (3.3.3) is identical with the heat conductivity equation. If the process is stationary, then $\partial/\partial t = 0$ and the following equation is obtained:

$$\frac{c^2}{4\pi\sigma}\left(\frac{\partial^2 I}{\partial z^2}+r\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial I}{\partial r}\right)=-\frac{1}{enr^2}\frac{\partial I^2}{\partial z}. \tag{3.3.5}$$

It may easily be shown that the difference in comparison with the planar case is determined by the parameter $\omega_e\tau_e L/R$, where L is the perturbation scale, R is the curvature radius.

The most noticeable special feature of equation (3.3.5) in comparison with the identical equation for the planar case is the presence of the non-linear Hall member[10].

To investigate the evolution of the patterns with the increase of $\omega_e\tau_e$, the equation (3.3.3) was calculated numerically for different values of this parameter. After making the equations dimensionless, the required quantity was the dimensionless 'current' proportional to I:

$$Y(\rho,\zeta) = -\omega_e\tau_e\,\rho, \tag{3.3.6}$$

here ρ is the dimensionless radius and ζ is the dimensionless coordinate z. Equation (3.3.3) was solved numerically by the method of establishment in the region indicated in Fig. 3.3.1 with the following boundary conditions.

a. The 'total' current which increases from zero to Y_0 in accodance with the law

$$Y^*(\tau) = Y_0\left[1-\exp\left\{-\frac{\tau}{\theta}\right\}\right], \tag{3.3.7}$$

was defined on the surface of the conductor. Here τ is dimensionless time.

---

[10]In the planar case, when $\mathbf{H}$ = (0, 0, H_z (x,y)), the Hall term $\mathbf{E}_{\text{hall}} = -\frac{[\mathbf{j},\mathbf{H}]}{enc}$, at n = const is equal to $\mathbf{E}_{\text{hall}} = -\frac{1}{en}\nabla\frac{H^2}{8\pi}$. Consequently, the equation for H_z does not contain the non-linear term.

b. on the axis (ρ = 0) it was assumed that Y = 0.
c. The values at the ends were connected by the periodicity condition. The results of the calculations are presented in Fig. 3.3.1. Here the dotted line corresponds to Y_0 = 10 and the thin solid line to Y_0 = 50. The tendency for the region occupied by the current to become cylindrical is clearly evident.

3.4. Static configurations in the two-fluid hydrodynamics

After investigating a number of general relationships of two-fluid hydrodynamics, we can transfer to analysis of the hydrodynamics on the basis of typical plasma structures. We start with static configurations. The static MHD configurations, described by the equations

$$\nabla p = \frac{1}{c}\left[\mathbf{j},\mathbf{H}\right]; \quad \mathrm{rot}\,\mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \quad \mathrm{div}\,\mathbf{H} = 0. \tag{3.4.1a}$$

were investigated in paragraph 2.4.

Let the set of the functions

$$\mathrm{K} \equiv \left\{ p(\mathbf{x}), \mathbf{H}(x), \mathbf{j}(x) \right\} \tag{3.4.1b}$$

be some configuration – solution of the system (3.4.1a).

The two-fluid system of the equations which describes the equilibrium configurations in the absence of dissipation is obtained from (3.1.8) if we ignore the inertia of the ions and electrons. The system has the form

$$\mathrm{div}\, n v_i = 0, \tag{3.4.2a}$$

$$\nabla p_i = en\left(-\nabla\phi + \frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right] \right), \tag{3.4.2b}$$

$$\nabla p_e = -en\left(-\nabla\phi + \frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right] \right), \tag{3.4.2c}$$

$$\mathrm{rot}\,\mathbf{H} = \frac{4\pi}{c} en(\mathbf{v}_i - \mathbf{v}_e), \tag{3.4.2d}$$

$$\mathrm{div}\,\mathbf{H} = 0. \tag{3.4.2e}$$

Evidently, the configuration (3.4.1b) satisfies the system (3.4.2) because the sum of the equations (3.4.2b) and (3.4.2c) gives the first equation (3.4.1)

$$\nabla(p_i + p_e) = \frac{1}{c}[\mathbf{j}, \mathbf{H}]. \tag{3.4.3}$$

At complete analogy of (3.4.3) and (3.4.1) there is still some large difference between them. Here p is replaced by the sum $p_i + p_e$. This means that any suitable configuration differing in the ratio p_i/p_e can correspond to the same solution K (3.3.2b). This in turn results in the variation of ϕ and $\mathbf{v}_i$.

To ensure determinacy, we consider a toroidal configuration with inserted magnetic surfaces (ψ). Taking into account the fact that the total pressure p on the magnetic surfaces is constant ($p = p(\psi)$) and also assuming that the temperature of both ions and electrons on every surface is constant:

$$T_i = T_i(\psi), \quad T_e = T_e(\psi) \tag{3.4.4a}$$

we conclude that the difference of the pressures $p_i = n(\psi) kT_i(\psi)$ and $p_e = n(\psi) kT_e(\psi)$ is caused by the difference of the temperature of the components. In turn, the temperatures T_e and T_i are determined by the heating mechanisms of the components and by the mechanisms of losses of the energy by the component (see chapters 5 and 10). It should also be mentioned that in the static configuration under the condition (3.3.4a) every component is barotropic. Actually, since

$$n(\psi) = \frac{p_i(\psi)}{kT_i(\psi)} = \frac{p_e(\psi)}{kT_e(\psi)} \tag{3.4.4b}$$

then

$$p_i = p_i(n); \quad p_e = p_e(n). \tag{3.4.4c}$$

Taking into account the barotropic nature of the plasma in the configuration consisting of two components of the equations (3.4.2b, c) that are longitudinal with respect to **H**, we obtain two thermalised potentials

$$\phi_T^{(i)} = \phi - \frac{1}{e}\int \frac{dp_i}{n}, \quad \phi_T^{(e)} = \phi - \frac{1}{e}\int \frac{dp_e}{n}. \tag{3.4.4d}$$

which differ by the function which depends only on ψ. Consequently, the potentials ϕ on the magnetic surfaces are also constant.

Special features of the configurations at different p_e/p_i.
To progress further, the system (3.4.3) is transformed to the form

$$\operatorname{div} n\mathbf{v}_i = 0; \qquad \nabla p = \frac{1}{c}[\mathbf{j},\mathbf{H}];$$

$$\frac{1}{c}\frac{\nabla p_i}{n} = -\nabla\phi + \frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]; \qquad \operatorname{rot}\mathbf{H} = \frac{4\pi}{e}\mathbf{j};$$

$$\operatorname{div}\mathbf{H} = 0. \tag{3.4.5}$$

Here the first two equations reflect the specific features of the two-component nature of hydrodynamics and the three subsequent equations characterise the configuration as a whole. This shows that by defining the arbitrary dependence $p_i(n)$, i.e. the distribution of ionic temperature and also $\phi(\psi)$, we determine from the second equation (3.4.5) the component of ionic velocity transverse in relation to the field

$$\mathbf{v}_{i\perp} = -c\frac{\left[\nabla\phi + \frac{1}{e}\frac{\nabla p_i}{n},\mathbf{H}\right]}{H^2}, \tag{3.4.6}$$

and, substituting this expression into div $n\mathbf{v}_i = 0$, we determine $\mathbf{v}_{i||}$, i.e. the component of the velocity of the ionic component along H.

However, the problem of the general criteria of singularity of $\mathbf{v}_i$ on the magnetic surfaces of the toroidal systems, calculated in this manner, remains unsolved.

Two extreme cases are *apriori* possible: $p_e / p_i = 0$ and $p_i / p_e = 0$. We investigate them:

1. Let $p_e = 0$, $p = p_i$. In this case,

$$\nabla\phi = \frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right]; \quad \frac{\nabla p}{en} = -\nabla\phi + \frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]. \tag{3.4.7a}$$

These two equations can be satisfied in different fashion. Two extreme cases will again be considered.

a.The ions are magnetised, i.e. $\rho_{Hi} \ll L$. Consequently, they can be sustained by the same magnetic field and, consequently

$$\frac{\nabla p}{en} = \frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]; \quad \phi = 0, \quad v_e = 0. \tag{3.4.7b}$$

b. The ions are not magnetised. Consequently $\rho_{Hi} \gg L$ and

$$\nabla p = -en\nabla\phi; \quad \nabla\phi = -\frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right]. \tag{3.4.7c}$$

In this case, the ions are sustained by the electrical field generated by the electrons sustained by the magnetic field.

1. Let it be $p_i = 0$, $p = p_e$.

Consequently

$$-\frac{\nabla p}{e} = -\nabla\phi + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right]; \quad 0 = -\nabla\phi + \frac{1}{c}\left[\mathbf{v}_i, \mathbf{H}\right]. \tag{3.4.8a}$$

Two extreme cases again exist here.

a. $\phi = 0$. The electrons are sustained by the magnetic field and the ions are in the rest state

$$\frac{\nabla p}{en} = \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right], \quad v_i = 0. \tag{3.4.8b}$$

b. $\mathbf{v}_e = 0$. The electrons are sustained by the ions and the ions by the magnetic field

$$-\frac{\nabla p}{e} = -\nabla\phi; \quad 0 = -\nabla\phi + \frac{1}{c}\left[\mathbf{v}_i, \mathbf{H}\right]. \tag{3.4.8c}$$

Thus, the two-component plasma can form many different configurations which correspond to the same one-component configuration.

3.5. Linear waves in homogeneous plasma (two-fluid model) [83, 84]

In previous sections, we already investigated the linear waves in homogeneous plasma[11]. In section 1.5, the method of the one-particle approximation was used to calculate dielectric permittivity $\vec{\epsilon}$ for cold plasma at $\mathbf{H}_0 \neq 0$ (1.5.3a) and derive a general dispersion equation (1.5.10) which was analysed for a partial case with $\mathbf{H}_0 = 0$. It was shown that there is only one transverse wave for all frequencies ω, with the exception of one frequency ($\omega = \omega_0$), if polarisation is not taken into account. In contrast to gas dynamics where only one signal wave (sound) is detected, in the MHD model (2.5.4) of linear oscillations of homogeneous plasma there are three waves at $p \neq 0$ (fast and slow sound, Alfvén waves), and at $p = 0$ there are two types of signal wave: Alfvén and fast sound waves.

In two-fluid hydrodynamics which includes the model described in section 2.5, one can see the general pattern of the hydrodynamic waves in the ideal completely ionised plasma. Unfortunately, the overall pattern is very cumbersome. Therefore, we investigate in detail waves in plasma: (a) – without the magnetic field, but at $p_{i,e} \neq 0$, and (b) – waves in cold plasma at $\mathbf{H}_0 \neq 0$. As regards the general case, when $p_{i,e} \neq 0$ and $\mathbf{H}_0 \neq 0$, our considerations will be confined to several

[11]The waves in inhomogeneous plasma and also non-linear waves for a number of cases are discussed in chapter 8.

graphs of the dependence of the phase velocity on frequency ω.

3.5.1. Waves in the absence of the external magnetic field at p_i, $p_e \neq 0$.

As with cold plasma, we use here the equation (1.5.9) for the **E**-field of the wave. However, for this purpose it is necessary to calculate the dielectric permittivity of 'warm' plasma taking into account its definition

$$\overleftrightarrow{\epsilon}\mathbf{E} = \mathbf{E} + 4\pi n_0 e\left(\boldsymbol{\xi}_i - \boldsymbol{\xi}_e\right). \tag{3.5.1}$$

Here and later the unperturbed quantities are denoted by '0', and the perturbed ones are indicated without indexes.

Displacements $\boldsymbol{\xi}_{i,e}$ at propagation of the wave will be calculated for the planar harmonic case:

$$\left(\mathbf{E}, \mathbf{v}, n, \boldsymbol{\xi}\right) \sim \exp\left\{-i\omega t + i\varkappa \mathbf{x}\right\},$$

taking into account the linearised system of the equations (3.2.10d) at $\mathbf{H}_0 = 0$ and in the absence of collisions

$$\frac{\partial n}{\partial t} + n_0 \operatorname{div} \mathbf{v}_i = 0; \quad M\frac{\partial \mathbf{v}_i}{\partial t} = -\frac{Mc_{Ti}^2 \nabla n_i}{n_0} + e\mathbf{E}; \tag{3.5.2a}$$

$$\frac{\partial n_e}{\partial t} + n_0 \operatorname{div} \mathbf{v}_e = 0; \quad m\frac{\partial \mathbf{v}_e}{\partial t} = -\frac{mc_{Te}^2 \nabla n_e}{n_0} - e\mathbf{E}.$$

Consequently, for a harmonic wave

$$\begin{aligned} \omega \mathbf{v}_i &= -\frac{c_{Ti}^2 \varkappa\left(\varkappa \mathbf{v}_i\right)}{\omega} + i\frac{e}{M}\mathbf{E}; \\ \omega \mathbf{v}_e &= -\frac{c_{Te}^2 \varkappa\left(\varkappa \mathbf{v}_e\right)}{\omega} - i\frac{e}{m}\mathbf{E}. \end{aligned} \tag{3.5.2b}$$

Expanding the **E**-field and **v** into components $E_{\|}$ and $E_{\perp}$ and in the same manner **v** with respect to the wave vector $\varkappa$, taking into account the continuity equation we obtain

$$\omega \mathbf{v}_{i\perp} = i\frac{e}{M}\mathbf{E}_{\perp}; \quad \omega \mathbf{v}_{i\|} = -c_{Ti}^2 \varkappa^2 \mathbf{v}_{i\|} + i\frac{e}{M}E_{\|}.$$

Consequently,

$$\boldsymbol{\xi}_{i\perp} = \frac{i}{\omega}\mathbf{v}_{i\perp} = -\frac{e}{M\omega^2}\mathbf{E}_{\perp}; \quad \boldsymbol{\xi}_{i\|} = -\frac{e}{M}\frac{\mathbf{E}_{\|}}{\omega^2 - c_{Ti}^2 \varkappa^2}. \tag{3.5.3a}$$

The identical expressions for the displacement of the electrons have the form

$$\xi_{e\perp} = \frac{e}{m\omega^2} E_\perp; \quad \xi_{e\|} = \frac{e}{m} \frac{\mathbf{E}_\|}{\omega^2 - c_{Te}^2 \varkappa^2}. \tag{3.5.3b}$$

Substituting (3.4.3) into (3.4.1) we obtain two dielectric constants: for transverse waves

$$\epsilon_\perp = 1 - \frac{\omega_0^2}{\omega^2}, \quad \omega_0^2 = 4\pi e^2 n_0 \left(\frac{1}{m} + \frac{1}{M} \right) \tag{3.5.4a}$$

and for the longitudinal waves

$$\epsilon_\| = 1 - \frac{\omega_{0i}^2}{\omega^2 - \varkappa^2 c_{Ti}^2} - \frac{\omega_{0e}^2}{\omega^2 - \varkappa^2 c_{Te}^2}, \tag{3.5.4b}$$

Naturally, heating of plasma has no effect on the transverse waves, in contrast to the longitudinal waves.

Thus, the dielectric permittivity is a tensor even in isotropic plasma. If the *z* axis is directed along $\varkappa$, this tensor has the following form

$$\overleftrightarrow{\epsilon} = \begin{pmatrix} \epsilon_\perp & 0 & 0 \\ 0 & \epsilon_\perp & 0 \\ 0 & 0 & \epsilon_\| \end{pmatrix}. \tag{3.5.5}$$

The nature of the waves for which the dielectric permittivity $\overleftrightarrow{\epsilon}$ was calculated will now be studied.

Transverse waves

In this case, taking into account (1.5.9) and (3.5.4), we obtain a simple dispersion equation

$$\omega^2 = \omega_0^2 + \varkappa^2 c^2.$$

From this equation it follows, in particular, as already mentioned in section 1.4, that there is the boundary frequency $\omega^* = \omega_0$ below which the waves cannot propagate in plasma. This is clearly visible in irradiation of a plasma formation (for example, a layer) with a flux of microwaves with the frequency changing with time. If receiving antennae are placed behind and in front of the layer, then at some moment, with the frequency of the transmitter ω reaching $\omega^* \approx \omega_0$, the receiving antenna behind the layer ceases to receive the signal and the antenna in front of the layer receives the effectively reflected signal.

This situation is often used for determining the density of electrons n_e in plasma formations, on both laboratory and cosmic scales, for example, in probing of the ionosphere (for more details see section 9.2).

Longitudinal waves

In this case, the dispersion equation in accordance with (1.5.9) and (3.5.4b) has the form $\epsilon_{||} = 0$

$$1 = \frac{\omega_{0e}^2}{\omega^2 - c_{Te}^2 \varkappa^2} + \frac{\omega_{0i}^2}{\omega^2 - c_{Ti}^2 \varkappa^2} \equiv F(\omega). \tag{3.5.6a}$$

After removing the denominators, we obtain that ($a \equiv c_{Te}$, $b \equiv c_{Ti}$)

$$\omega^4 - \omega^2 \left(\varkappa^2 \left(a^2 + b^2 \right) - \omega_e^2 - \omega_i^2 \right) + a^2 b^2 \varkappa^4 + \varkappa^2 \left(\omega_e^2 b^2 + \omega_i^2 a^2 \right) = 0. \tag{3.5.6b}$$

This equation is biquadratic with respect to both ω and $\varkappa$. Being formally simple, this equation has roots that are very difficult to analyse. Therefore, the qualitative description of these roots can be obtained more efficiently by the graphical method. Figure 3.5.1 shows the graph of the function F (ω).

As indicated by the graph, the dispersion equation at all values of $\varkappa$ has four real solutions: two high-frequency solutions $|\omega| > |\varkappa| c_{Te}$, and two low-frequency solutions $|\varkappa| c_{Ti} < |\omega| < |\varkappa| c_{Te}$.

The first waves are Langmuir waves which have already been discussed, the second type of wave are ion-sound waves which will be described here.

Langmuir waves

In this case, assuming that $|\omega| \gg \omega_{0i}$ and $|\omega| \gg |\varkappa| c_{Ti}$, the equation (3.5.6a) can be presented in the form

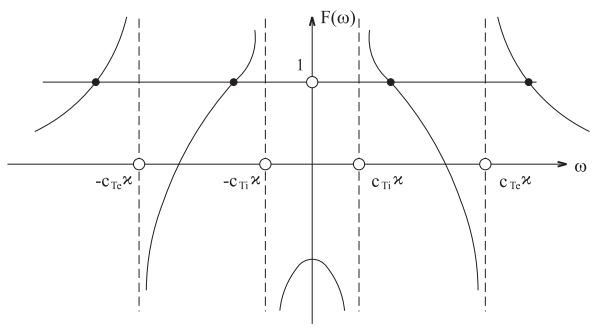


Fig. 3.5.1. The graph of the function $F(\omega)$.

$$1 = \frac{\omega_{0i}^2}{\omega_L^2} + \frac{\omega_{0e}^2}{\omega_L^2 - c_{Te}^2 \varkappa^2} \approx \frac{\omega_{0e}^2}{\omega_L^2 - c_{Te}^2 \varkappa^2}. \qquad (3.5.7a)$$

Consequently,

$$\omega_L^2 \approx \omega_{0e}^2 + c_{Te}^2 \varkappa^2. \qquad (3.5.7b)$$

Thus, if $\varkappa$ is real, then $\omega > \omega_{0e}$ and, consequently, the term rejected in (3.5.7) is indeed small:

$$\frac{\omega_{0i}^2}{\omega_L^2} < \frac{\omega_{0i}^2}{\omega_{0e}^2} = \frac{m}{M}.$$

Equation (3.5.7b) shows that if the electronic temperature is high and the wave number $\varkappa$ is so high that

$$\omega_{0e}^2 \ll \varkappa^2 c_{Te}^2. \qquad (3.5.8a)$$

then we have the conventional sound waves in the electronic component

$$\omega^2 \approx \varkappa^2 c_{Te}^2. \qquad (3.5.8b)$$

Evidently, the condition (3.5.8a) can be written in the form of comparison of wavelength λ with the Debye radius r_D

$$\varkappa^2 r_D^2 = \left(\frac{2\pi}{\lambda}\right)^2 r_D^2 \gg 1. \qquad (3.5.9a)$$

Here, the Debye radius is determined as the ratio

$$r_D \equiv \frac{c_T}{\omega_{0e}}. \qquad (3.5.9b)$$

Conversely, if $\lambda \gg r_D$ then $\omega \to \omega_0$ and we have Langmuir waves with a low group velocity

$$v_{gr} = \frac{c_{Te}^2}{\sqrt{\omega_0^2 + c_{Te}^2 \varkappa^2}} \approx \frac{c_{Te}^2}{\omega_0} \varkappa = c_{Te}\left(r_D \varkappa\right). \qquad (3.5.10)$$

The propagating (i.e. having the real values of ω and $\varkappa$) Langmuir waves exist only at $\omega > \omega_0$. However, the Langmuir branch of the oscillations can be also extended into the region $\omega < \omega_0$. This will be discussed later.

Ion-sound waves

Assuming that $\omega^2 \ll \varkappa^2 c_{Te}^2$, equation (3.5.6a) can be transformed to the form

$$1 = \frac{\omega_{0e}^2}{-\varkappa^2 c_{Te}^2} + \frac{\omega_{0i}^2}{\omega_s^2 - \varkappa^2 c_{Ti}^2}. \tag{3.5.11}$$

Consequently.

$$\omega^2 = \varkappa^2 c_{Ti}^2 + \varkappa^2 c_{Te}^2 \left(\frac{\omega_{0i}^2}{\omega_{0e}^2 + \varkappa^2 c_{Te}^2} \right). \tag{3.5.12a}$$

The Debye radius is described by equation (3.5.9b). Consequently, taking into account $\frac{\omega_{0i}^2}{\omega_{0e}^2} = \frac{m}{M}$, the dispersion equation (3.5.12a) can be presented in the following form:

$$\omega^2 = \varkappa^2 \left(c_{Ti}^2 + c_{Te}^2 \frac{m}{M} \frac{1}{1 + r_D^2 \varkappa^2} \right). \tag{3.5.12b}$$

It may be seen that the dependence $\omega(\varkappa)$ at relatively small $\varkappa \ll 1/r_D$ is linear

$$\omega = \varkappa \sqrt{\gamma \frac{k(T_i + T_e)}{M}}. \tag{3.5.13a}$$

This equation is completely identical with the equation for general sound (2.2.5b) but in this case the gas temperature is replaced by the sum of the temperatures of the ions and electrons[12]. In many plasma systems $T_e \gg T_i$ and, consequently, the phase velocity is

$$v_{ph} = \frac{\omega}{\varkappa} = \sqrt{\gamma \frac{kT_e}{M}}, \tag{3.5.13b}$$

i.e. the elasticity of the medium is determined by the electrons and the inertia by the ions. The group velocity coincides with the phase velocity.

Ion sound in the case of short waves when $\varkappa > 1/r_D$ is different. In this case, ignoring the unity in the denominator of (3.5.12), we obtain

$$\omega^2 \approx \varkappa^2 c_{Ti}^2 + \omega_{0i}^2, \tag{3.5.14a}$$

and for relatively cold ions ($|\varkappa| c_{Ti} \ll \omega_{0i}$)

$$\omega = \omega_{0i}. \tag{3.5.14b}$$

[12]Naturally, this follows from the dynamics equation (3.1.11).

In other words, the oscillations do not take place with the electronic Langmuir frequency, as would be expected, and they take place with the ionic Langmuir frequency. This is explained by the fact that in this case the perturbations are almost stationary ($v_{gr} = 0$), and the electrons which generate a periodic potential relief, with ions oscillating in the depressions of the relief, are also stationary.

Debye screening

The dependence of the frequency of oscillations of the medium at the given wavelength was investigated previously. However, in many cases, the frequency is specified and it is necessary to determine the spatial scale of the perturbations. A typical example of this situation is the perturbation of a plasma volume with a probe which receives and alternating potential. If the probe potential is constant, i.e. $\omega = 0$, the general dispersion equation has the form:

$$\varkappa^2 = -\frac{\omega_{0e}^2}{c_{Te}^2} - \frac{\omega_{0i}^2}{c_{Ti}^2} = -\frac{1}{r_{D0}^2}. \qquad (3.5.15)$$

Here r_{D0} is the Debye radius which takes into account the temperatures of both components. Consequently, in this case, quantity $\varkappa$ is imaginary and the perturbation of all the parameters is

$$(n, \phi) \sim e\left\{-\frac{x}{r_{D0}}\right\}, \qquad (3.5.16)$$

i.e. decreases with the increase of the distance from the probe or some object, including the walls restricting the volume. Therefore, in this case r_D is often referred to as the 'screening radius'. This fact has already been mentioned in Introduction, section 1. Evidently, if the frequency of modulation of the probe potential is increased, this results in particular in the excitation of the ion–sound oscillations at

$$\omega \sim c_S \frac{2\pi}{L},$$

where L is the length of the plasma column, and at $\omega \to \omega_{0e}$ the Langmuir oscillations are excited. All these conclusions have been satisfactorily confirmed by experiments.

Summarising this, it may be seen that for every value of ω in heated plasma, at $H_0 = 0$, we have three types of oscillations: transverse, Langmuir ($\varkappa$ is imaginary at $\omega < \omega_{0e}$), and ion–sound.

3.5.2. Linear waves in homogeneous plasma at $H_0 \neq 0$ (cold plasma)

In section 3.1.2 it was noted that the two-fluid equations of movement of plasma at $p_{i,e} = 0$ are identical with the equations of dynamics of single particles if there is no intersection of the trajectories of different particles. At a small amplitude of the oscillations of homogeneous plasma there are no such intersections and, therefore, the dielectric permittivity tensor $\overleftrightarrow{\epsilon}$, calculated in section 1.5 in the one-particle approximation, is also suitable for the two-fluid model at $p = 0$. For convenience, the equation is presented here

$$\overleftrightarrow{\epsilon} = \begin{pmatrix} 1-\sum\limits_{(\alpha)} \dfrac{\omega_{0\alpha}^2}{\omega^2-\omega_{\alpha H}^2} & -i\sum\limits_{(\alpha)} \dfrac{\omega_{\alpha H}^2\omega_{0\alpha}^2}{\omega(\omega^2-\omega_{\alpha H}^2)} & 0 \\ +i\sum\limits_{(\alpha)} \dfrac{\omega_{\alpha H}^2\omega_{0\alpha}^2}{\omega(\omega^2-\omega_{\alpha H}^2)} & 1-\sum\limits_{(\alpha)} \dfrac{\omega_{0\alpha}^2}{\omega^2-\omega_{\alpha H}^2} & 0 \\ 0 & 0 & 1-\sum\limits_{(\alpha)} \dfrac{\omega_{0\alpha}^2}{\omega^2-\omega_{\alpha H}^2} \end{pmatrix} \quad (3.5.17a)$$

It should be mentioned that this tensor has the form in the coordinate system in which the z axis is directed along the external magnetic field.

The general structure of the tensor is as follows

$$\overleftrightarrow{\epsilon} = \begin{pmatrix} \epsilon_\perp & ig & 0 \\ -ig & \epsilon_\perp & 0 \\ 0 & 0 & \epsilon_\parallel \end{pmatrix} \quad (3.5.17b)$$

As shown by conventional optics (and electrodynamics), the structure of $\overleftrightarrow{\epsilon}$ corresponds to the uniaxial gyrotropic crystal, i.e. the crystal with the detailed direction and (Iceland spar is a suitable example) rotating polarisation plane of the wave. The gyrotropicity is determined by the quantity g.

Tensor (3.5.17) is Hermitian, i.e.

$$\epsilon_{\alpha\beta} = \epsilon_{\beta\alpha}^*, \quad (3.5.18)$$

where the * is the sign of complex conjugation. The Hermitian form of the tensor $\overleftrightarrow{\epsilon}$ is the essential and sufficient condition of preserving the electromagnetic energy of the propagating wave [10].

To describe the propagation of the harmonic flat waves we use the equation (1.5.9) for the electrical field

$$\varkappa^2\mathbf{E} - \varkappa(\varkappa\mathbf{E}) = \frac{\omega^2}{c^2}\overleftrightarrow{\epsilon}\mathbf{E}. \quad (3.5.19a)$$

Equating to 0 the determinant of this linear system of equations for **E**, we obtain the dispersion equation

$$D(\omega, \varkappa) = 0. \tag{3.5.19b}$$

The dispersion equation (3.5.19b) becomes very cumbersome after substitution of the tensor (3.5.17). Therefore, this equation will not be considered in detail but we shall make several steps to the formulation of the equation which is suitable for calculations, either numerical or analytical.

First of all, we introduce the refractive index [83]

$$N = \frac{c}{\omega}\varkappa.$$

The general dispersion equation has the following form (θ is the angle between $\varkappa$ and **H**):

$$aN^4 + bN^2 + c = 0, \tag{3.5.20}$$

where

$$a = \epsilon \sin^2\theta + \eta\cos^2\theta; \tag{3.5.21}$$

$$b = g\sin^2\theta - \epsilon_1\eta - \epsilon_1(\epsilon_1\sin^2\theta + \eta\cos^2\theta);$$

$$c = \epsilon_1^2\eta - \eta g^2.$$

The roots of the equation (3.5.21) are equal to

$$N^2 = \frac{1}{2(\epsilon_1\sin^2\theta + \eta\cos^2\theta)}\Big\{(\epsilon^2 - g^2 - \epsilon\eta)\sin^2\theta + \\ +2\epsilon\eta \pm \sqrt{(\epsilon^2 - g^2 - \epsilon\eta)^2\sin^4\theta + 4\eta^2 g^2\cos^2\theta}\Big\}. \tag{3.5.22}$$

The components of the tensor (3.5.17) can be conveniently written in the form

$$\epsilon = 1 - A^2\frac{\mu(\alpha^2 - \mu)}{(\alpha^2 - 1)(\alpha^2 - \mu^2)};$$

$$g = A^2\frac{\mu(\alpha^2 - \mu)}{(\alpha^2 - 1)(\alpha^2 - \mu^2)}; \quad \eta = 1 - A^2\frac{\mu}{\alpha^2}. \tag{3.5.23a}$$

Here

$$A^2 = \frac{\omega_{0i}^2}{\omega_{Hi}^2} + \frac{\omega_{0e}^2}{\omega_{He}^2} = \frac{4\pi(M+m)nc^2}{H_0^2}, \quad \mu = \frac{M}{m}, \quad \alpha = \frac{\omega}{\omega_{Hi}} \tag{3.5.23b}$$

Substituting (3.5.23) into (3.5.22), we obtain the expression for $N(\omega,\theta)$:

$$N^2 = \frac{p_1 \pm \sqrt{p_2}}{Q} \equiv \frac{P}{Q}. \tag{3.5.23c}$$

Here

$$p_1 = \left\{\mu^3 A^4 - \mu A^2\left[(\mu^2 - \mu + 1)\alpha^2 - \mu^2\right]\right\}\sin^2\theta + 2(\alpha^2 - \mu A^2) * \left[(\alpha^2 - 1)(\alpha^2 - \mu^2) - \mu A^2(\alpha^2 - \mu)\right], \tag{3.5.23d}$$

$$p_2 = \left\{\mu^3 A^4 - \mu A^2\left[(\mu^2 - \mu + 1)\alpha^2 - \mu^2\right]\right\}\sin^4\theta + +4(\alpha^2 - \mu A^2)^2 A^4 \mu^2 (\mu - 1)^2 \alpha^2 \cos^2\theta, \tag{3.5.23e}$$

$$Q = 2\left[\alpha^2(\alpha^2 - 1)(\alpha^2 - \mu^2) - \mu A^2 \alpha^2(\alpha^2 - \mu)\sin^2\theta - -\mu A^2(\alpha^2 - 1)(\alpha^2 - \mu)\cos^2\theta\right]. \tag{3.5.23f}$$

The resultant equations are cumbersome. Therefore, prior to analysing the equations, it is convenient to present the graphs of the dependences $N^2(\omega, \theta)$ for different values of μ and A^2. These graphs, calculated numerically, are presented in Fig. 3.5.2. It is evident that for every frequency ω there are two values of N^2, i.e. two waves with different $\varkappa^2$.

In a general case at $p_{i,e} = 0$ there are five branches of oscillations restricted by three vertical asymptotes passing through $\omega_1^\infty \approx \omega_{Hi}$, $\omega_2^\infty \approx \omega_{He}$ and $\omega_3^\infty \approx \omega_0$. In the range $0 < \omega < \omega_1^\infty$ there are the Alfvén (A) and the fast magneto-sound waves (F). The absence of the slow magneto-sound wave is understandable because this wave disappears (as already reported in section 2.5.2) at $T \to 0$.

In the range $\omega_1^\infty < \omega < \omega_2^\infty$ the modified FMS wave is preserved, and it is referred to as the 'whistler' or 'whistling atmospherics'.

The second branch of the dispersion relationship cannot propagate in the plasma in the range $\omega_1^\infty < \omega < \omega_1^{(0)}$ and only at $\omega > \omega_1^{(0)}$ its phase velocity becomes real. The oscillations in the frequency range $0 \le \omega < \omega_2^{(\infty)}$ are referred to as low-frequency oscillations, if the Langmuir frequency is $\omega_0 \gg \omega_{eH}$ – the electronic Larmor frequencies. The oscillations with the frequencies $\omega > \omega_2^{(\infty)}$ are high-frequency

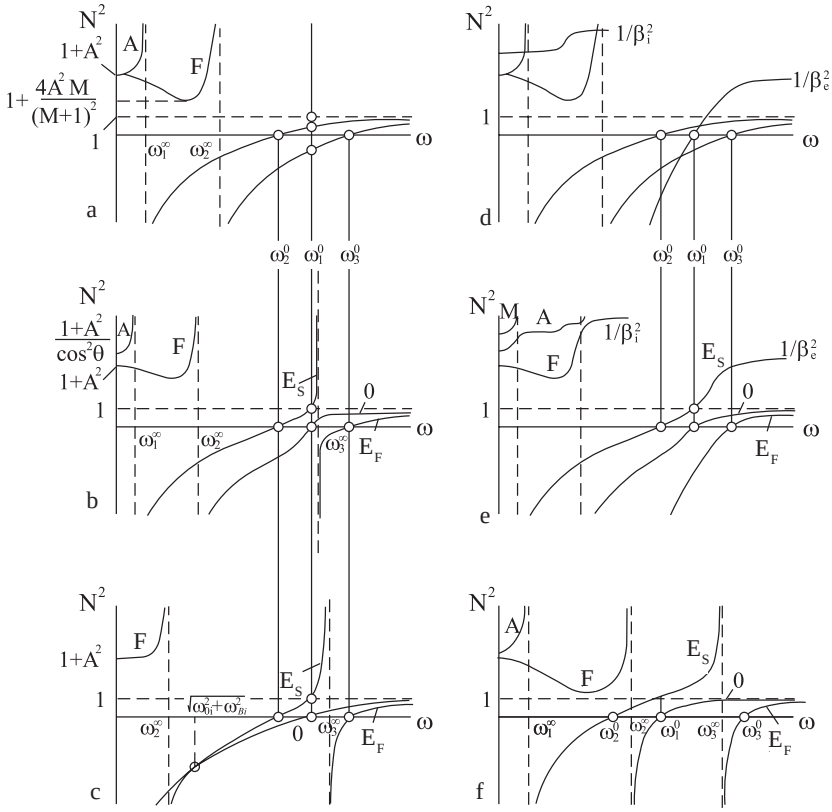


Fig. 3.5.2. Dependence of the square of the refractive index N^2 (ω, θ) on frequency: for cold plasma (a, b, c, f); for heated plasma in the two-fluid model (d, e), a) $\omega_0 > \omega_{He}$; $p = 0$; $\theta = 0$; b) $\omega_0 > \omega_{He}$; $p = 0$; $\theta \neq 0; \pi/2$; c) $\omega_0 > \omega_{He}$; $p = 0$; $\theta = \pi/2$; d) $\omega_0 > \omega_{He}$; $p \neq 0$; $\theta = 0$; e) $\omega_0 > \omega_{He}$; $p \neq 0$; $\theta \neq 0, \pi/2$; f) $\omega_0 < \omega_{He}$; $p = 0$; $\theta \neq 0, \pi/2$.

oscillations (or, in most cases, ultra high-frequency oscillations (UHF)).

Further, only one high-frequency wave exists in the range $\omega_2^{(0)} < \omega_2^{(\infty)}$.

Finally, in the frequency range $\omega > \omega_3^{\infty}$ there are Langmuir and conventional transverse waves.

The investigated special features of the branches of the oscillations show the importance of the critical frequencies $\omega_{1,2,3}^{(\infty)}$ and frequencies $\omega_{1,2,3}^{(0)}$. The former correspond to the waves at which the wavelength $\lambda = 2\pi|\varkappa|$ decreases without limits, whereas in the latter case the wavelength increases without limits.

Detailed analysis of the behaviour of the waves with the frequencies ω^{∞} shows that they are efficiently absorbed by plasma. Therefore, at present plasma is heated at the frequencies ω_1^{∞} and ω_2^{∞} using the electromagnetic waves. Heating at ω_2^{∞} is especially effective because

it can be carried out using high-efficiency gyrotrons and the short wavelength enables the entire equipment to be made compact.

Frequencies at which the refractive index converts to 0 and infinity will now be calculated.

The zero values of N correspond to $P = 0$. Regardless of the angle θ, N converts to 0, as indicated by (3.5.22) if one of the following conditions is satisfied:

$$\eta = 0;$$

$$\epsilon^2 - g^2 = 0. \tag{3.5.24a}$$

In the former case, as indicated by (3.5.22),

$$\omega_1^{(0)} = \sqrt{\omega_{0i}^2 + \omega_{0e}^2 = \omega_{0,}} \tag{3.5.25}$$

i.e. at the plasma frequency. In the second case

$$\omega_{2,3}^{(0)} = \sqrt{\omega_{0i}^2 + \omega_{0e}^2 + \frac{1}{4}(\omega_{He} + \omega_{Hi})^2} \pm \frac{\omega_{He} - \omega_{Hi}}{2} \tag{3.5.26a}$$

If $\omega_{0e} \gg \omega_{He}$ then, neglecting the ionic and Larmor frequencies, we obtain

$$\omega_{2,3}^{(0)} \approx \omega_0 \pm \frac{\omega_{He}}{2}. \tag{3.5.26b}$$

Infinite values of N correspond to $Q = 0$. Here there are also three roots. They are simple if the waves propagate along the magnetic fields ($\theta = 0$). Consequently

$$\omega_1^{(\infty)} = \omega_{Hi}; \quad \omega_2^{(\infty)} = \min(\omega_{Hi}, \omega_0); \quad \omega_3^{(\infty)} = \max(\omega_{He}, \omega_0). \tag{3.5.27}$$

For the wave vectors, almost completely perpendicular to the magnetic field

$$\left|\frac{\pi}{2} - \theta\right| < \frac{1}{\sqrt{\mu}}. \tag{3.5.28a}$$

with the accuracy to the values $\sim 1/\mu$.

$$\omega_1^{(\infty)} = \omega_{Hi};$$
$$\omega_{2,3}^{(\infty)} = \frac{1}{2}\left\{\sqrt{\omega_{0e}^2 + \omega_{He}^2 + 2\omega_{0e}\omega_{He}\cos\theta} \pm \sqrt{\omega_{0e}^2 + \omega_{He}^2 - 2\omega_{0e}\omega_{He}\cos\theta}\right\}. \tag{3.5.28b}$$

As regards the angles θ in the immediate vicinity of $\pi/2$, then

$$\omega_1^{(\infty)} \to 0; \quad \omega_3^{(\infty)} \to \sqrt{\omega_{oe}^2 + \omega_{He}^2};$$
$$\omega_2^{(\infty)} \Rightarrow \left(\omega_{Hi}\omega_{He} \frac{\omega_{Hi}^2 + \omega_{0i}^2}{\omega_{Hi}\omega_{He} + \omega_{0i}^2} \right)^{1/2}. \tag{3.5.28c}$$

In particular, if $\omega_{Hi}\omega_{He} \ll \omega_{0i}^2$, then

$$\omega_2^{(\infty)} \to \sqrt{\omega_{He}\omega_{Hi}}. \tag{3.5.29}$$

It should be mentioned that $\sqrt{\omega_{0e}^2 + \omega_{He}^2}$ is the upper hybrid frequency and $\sqrt{\omega_{He}\omega_{Hi}}$ is the lower hybrid frequency.

If $N^2 < 0$, the wave cannot penetrate to depths greater than

$$\delta \sim \frac{1}{\varkappa} = \frac{c}{\omega|N|}. \tag{3.5.30}$$

It may also be seen that the Alfvén wave (A) ceases to exist at $\omega = \omega_{Hi}$, since the fast magnetic sound disappears at $\omega_2^{(\infty)}$, i.e. at the lower hybrid frequency (3.5.29). Attention should also be given to the 'electronic' wave E_H which is 'arrested' at the upper hybrid frequency.

Plasma configurations always have nonuniform magnetic fields and a density distribution. Therefore, if a flux of electromagnetic waves with the frequency ω passes through the plasma volume and this frequency in some zone coincides with one of the frequencies where $N \to \infty$, the wave is 'arrested' and absorbed. These are the fundamentals of heating of plasma at the incidence of the waves in the direction normal to **H** using the ion–cyclotron resonance, lower hybrid and upper hybrid resonances. Heating based on electronic cyclotron resonance (ECR) using UHF waves, travelling along **H**, is also highly efficient.

3.5.3. Linear waves in homogeneously heated plasma at H≠0

If the final temperature of the plasma component is taken into account the situation is greatly complicated. We have already examined in detail the oscillations of heated plasma only at $H = 0$ and it was shown that only one wave exists at $T = 0$, i.e. Langmuir wave, whereas at $T \neq 0$ there are two waves and the ion sound is added. It has also been shown that a slow magneto-sound wave is added in the single-fluid hydrodynamics at $c_T \neq 0$. It appears that this also holds in a general case. The dispersion equation is now bicubic and, consequently, each value of ω has three corresponding values of N^2, i.e. there are six branches of oscillations. Information on them is shown in Fig. 3.5.2. The oscillations in the uniform medium at $\mathbf{H} \neq 0$ at T_i, $T_e \neq 0$ are investigated in detail in [85].

3.5.4. The simplest two-fluid model of beam instability

The specific pecularity of the plasma is that it often contains two or more components which move in a relatively free manner in relation to each other. It is natural to expect that when the relative velocity of these components approaches the characteristic velocity of the waves in the medium, the step-up of oscillations (waves) should start. The situation resembles the generation of sound waves at movement with the supersonic velocity of solids in air. However, in plasma the step-up is caused by the movement of the components of the plasma itself and not by a secondary object.

The increasing oscillations of this type are usually referred to as 'beam' instabilities because they are initiated in a simple manner at injection of beams – usually electrons, into plasma. The significant role of the instabilities, determined by the relative movement of the components, was stressed for the first time by the Ukrainian physicists Akhiezer and Fainberg (1949) who examined in detail the interaction of a low-density electron beam with the electronic component of plasma [86]. Almost simultaneously, in 1948–1949, a similar problem was investigated independently by Gaef, Bohm and Gross, although they used a large number of restrictions.

The above considerations will be illustrated by appropriate calculations assuming that the plasma is cold, the ions are infinitely heavy ($m/M \to 0$), and the beam is also cold but moves with the velocity v_0 along the axis x. Describing also the stationary electronic component and the beam by the hydrodynamics equations, we obtain a dispersion equation for longitudinal oscillations in the form

$$\epsilon = 1 - \frac{\omega_0^2}{\omega^2} - \frac{\omega_{0b}^2}{(\omega - \varkappa v_0)^2} = 0. \tag{3.5.31}$$

where ω_0 and ω_{0b} are the Langmuir frequencies of the electrons of the plasma and of the beam. Equation (3.5.31) can be derived directly by using the Fourier method for solving the linearised system of equations

$$\frac{\partial n}{\partial t} + n_0 \frac{\partial v}{\partial x} = 0; \quad \frac{\partial n_b}{\partial t} + \frac{\partial}{\partial x}(n_{0b} v_b + n_b v_0) = 0;$$

$$m \frac{\partial v}{\partial t} = -eE; \quad m\left(\frac{\partial v_b}{\partial t} + v_0 \frac{\partial v_b}{\partial x}\right) = -eE; \tag{3.5.32}$$

$$\frac{\partial E}{\partial x} = -4\pi e(n + n_b).$$

Here the quantities with no indexes relate to the perturbations of the plasma electrons and those with the index b to the beam electrons. The zero indicates the non-perturbed quantities ($v_0 \equiv v_{0b}$).

However, the dispersion equation (3.5.32) follows automatically from the equation (3.5.4) for ϵ, if the Doppler effect for the beam is taken into account

$$\omega \to \omega - \varkappa v_0 .$$

Equation (3.5.31) for the given the wave number $\varkappa$ is an algebraic equation of the fourth degree but its solution is rather cumbersome. However, it is quite easy to obtain information on these solutions, using the graphs of the function

$$F(\omega) \equiv \frac{\omega_0^2}{\omega^2} + \frac{\omega_{0b}^2}{(\omega - \varkappa v)^2}. \tag{3.5.33}$$

As indicated by Fig. 3.5.3, this equation has always two real roots, and the two other roots can be both real (Fig. 3.5.3a) and complex (Fig. 3.5.3b). The critical value $\varkappa \cdot \mathbf{v}_0$ at which the two 'second' roots coincide and, consequently, they are real, is:

$$(\varkappa v_0)^* = \omega_0 \left(1 + \left(\frac{\omega_b}{\omega_0} \right)^{2/3} \right)^{3/2} . \tag{3.5.34}$$

At $\varkappa v_0 < \varkappa v_0{}^*$, the roots of the equation (3.5.31) become complex

$$\omega_{3,4} = \Omega \pm i\gamma; \quad \gamma > 0 ,$$

and one of these roots ($\Omega + i\gamma$) results in an increase of the oscillation

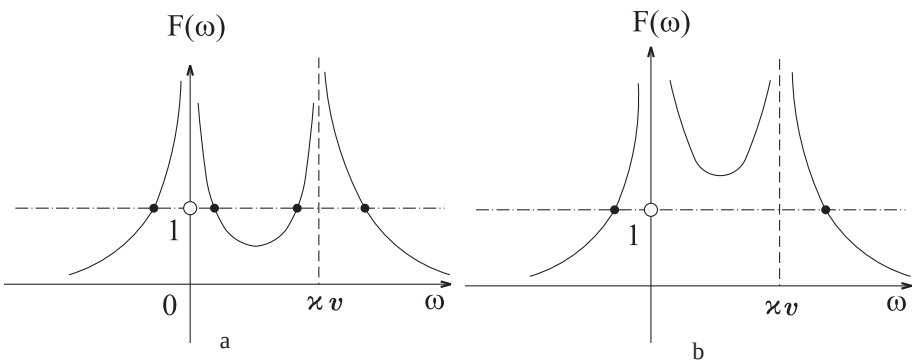


Fig. 3.5.3. Graphs of the function F (ω) for the two-beam system (for equation (3.5.33); a) stable regime, b) unstable regime.

amplitude in accordance with the law

$$A \sim e^{\gamma t}. \tag{3.5.35}$$

The instability criterion can be presented in the following form:

$$\frac{2\pi r_D}{\lambda} < \left(1+\left(\frac{\omega_b}{\omega_D}\right)^{2/3}\right)^{2/3} \approx 1. \tag{3.5.36}$$

This means that if the wavelength λ is considerably greater than the Debye radius r_D, the system (plasma electrons–electron beam) is unstable.

Solving the equation (3.5.31), we can determine the maximum increment of the increase of instability γ: it is equal to

$$\gamma = \omega_0 \sqrt{3}\frac{\alpha^{1/3}}{2^{4/3}}. \tag{3.5.37a}$$

at

$$\mathrm{Re}(\omega) = -\omega_0 \frac{\alpha^{1/3}}{2^{4/3}}. \tag{3.5.37b}$$

here $\alpha = n_b/n_0$, Re (ω) is the real part of the frequency.

It can be seen that $\gamma \sim \omega_0$ even at $\alpha = 1/10$, i.e. instability develops very rapidly, within several Langmuir oscillations.

It should be mentioned that the dispersion equation of the type (3.5.31) is also derived in the case in which the movement of electrons in relation to the ions is investigated.

It is necessary to replace ω_b^2 by

$$\omega_{0i}^2 = \omega_0^2 \frac{m}{M}.$$

To ensure that the resultant pattern is in agreement with the experiments, it is important to take into account T_e and T_i. The accurate analysis of this case will be described in section 4.4. A fundamental review of plasma instabilities has been published in a two-volume monograph by Mikhailovskii [87].

3.6. Dissipation-free axial-symmetric flows in the two-component hydrodynamics

The electrodynamics of plasma flows, investigated in section 3.2, provides an important but greatly restricted image of the processes in the plasma flow in a magnetic field. It is of course desirable to obtain more general and convincing information on the plasma dynamic

processes [88].

It was noted in chapter 1 that the simple structure of the magnetic fields exists only in the presence of a symmetry. Therefore, the first assumption which should be made is that we should confine ourselves to axisymmetric systems. The second assumption – plasma should be regarded as having ideal conductivity. Consequently, the energy, 'freezing-in', the momentum of the amount of motion, determined by axial symmetry, are preserved.

Thus, in this section we discuss hot plasma and high velocities at which the magnetic Reynolds number $\mathrm{Re}_m \gg 1$.

The main element of the method developed here is the application of the functions of the ion and electron fluxes resembling those used when deriving the Grad–Shafranov equations.

3.6.1. Derivation of the laws of conservation

Thus, it is assumed that the flow is stationary and can be described by the following system of equations:

$$\operatorname{div} n_e \mathbf{v}_e = 0; \quad \operatorname{div} n_i \mathbf{v}_i = 0; \tag{3.6.1}$$

$$M(\mathbf{v}_i \nabla)\mathbf{v}_i = -\frac{\nabla p_i}{n_i} + e\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_i, \mathbf{H}\right]\right); \tag{3.6.2a}$$

$$M(\mathbf{v}_e \nabla)\mathbf{v}_e = -\frac{\nabla p_e}{n_e} - e\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}_e, \mathbf{H}\right]\right); \tag{3.6.2b}$$

$$p_i = p_i(n_i, s_i); \quad (\mathbf{v}_i \nabla) s_i = 0; \tag{3.6.3}$$

$$p_e = p_e(n_e, s_e); \quad (\mathbf{v}_e \nabla) s_e = 0; \tag{3.6.4}$$

$$\operatorname{rot} \mathbf{H} = \frac{4\pi e}{c}\left(n_i \mathbf{v}_i - n_e \mathbf{v}_e\right); \quad \operatorname{div} \mathbf{H} = 0; \tag{3.6.5}$$

$$\mathbf{E} = -\nabla\phi; \quad \operatorname{div} \mathbf{E} = 4\pi e\left(n_i - n_e\right). \tag{3.6.6}$$

Assuming that the electromagnetic fields and the flow have axial symmetry, we can write the continuity equations (3.6.1) and the second of the equations (3.6.5) in the coordinate form

$$\frac{\partial}{\partial r} rnv_r + \frac{\partial}{\partial z} rnv_z = 0;$$
$$\frac{\partial}{\partial r} rHv_r + \frac{\partial}{\partial z} rHv_z = 0. \tag{3.6.7a}$$

These equations can be satisfied identically by introducing three flux functions ψ_i, ψ_e, ψ_H:

$$(rnv_r)_{i,e} = -\frac{\partial \psi_{i,e}}{\partial z}; \quad (rnv_z)_{i,e} = -\frac{\partial \psi_{i,e}}{\partial r};$$
$$rH_r = \frac{\partial \psi_H}{\partial z}; \quad rH_z = -\frac{\partial \psi_H}{\partial r}. \tag{3.6.7b}$$

It may easily be seen that the lines

$$\psi_i(r,z) = \text{const}; \ \psi_e(r,z) = \text{const}$$

depict correspondingly the projections of the trajectories of the ions and electrons on the plane *r, z*. The quantity $\psi(r, z)$ is associated with the number of particles *N* passing in unit time inside the axial-symmetric surface $\psi(r,z)$ = const, by the following equation:

$$\dot{N} = 2\pi\psi. \tag{3.6.7c}$$

As already mentioned in section 1.1, similar properties are also typical of the function of the magnetic flux ψ_H.

Taking into account the vector identity

$$(\mathbf{v}\nabla)\mathbf{v} = \frac{\nabla v^2}{2} - [\mathbf{v}\,\mathrm{rot}\,\mathbf{v}],$$

the equations of motion (3.6.2) can be presented in the form

$$\nabla U_t = \frac{e}{c}[\mathbf{v}_i, \mathbf{H}_i^*]; \quad \nabla U_e = -\frac{e}{c}[\mathbf{v}_e, \mathbf{H}_e^*]. \tag{3.6.8}$$

Here we introduce the term ‘effective magnetic fields’

$$\mathbf{H}_i^* = \mathbf{H} + \frac{Mc}{e}\mathrm{rot}\,\mathbf{v}_i; \quad \mathbf{H}_e^* = \mathbf{H} - \frac{mc}{e}\mathrm{rot}\,\mathbf{v}_e \tag{3.6.9}$$

the total particle energies

$$U_i = \frac{Mv_i^2}{2} + W_i + e\phi; \quad U_e = \frac{Mv_e^2}{2} + W_e - e\phi \tag{3.6.10}$$

and, taking into account (3.6.3) and (3.6.4) the functions proportional to the enthalpy:

$$W_{i,e} = \int \frac{dp_{i,e}}{n_{i,e}} \tag{3.6.11}$$

More accurately, introducing (3.6.11) it is assumed that the entropy of the unit mass of each component in the entire flux is the same (isoentropic flow).

Multiplying (3.6.8) in the scalar form by $\mathbf{v}_i$ and $\mathbf{v}_e$ taking into account (3.6.7) gives[13]

$$\frac{D(U_i, \psi_i)}{D(r,z)} = 0; \quad \frac{D(U_e, \psi_e)}{D(r,z)} = 0 \tag{3.6.12a}$$

Consequently

$$U_i = U_i(\psi_i); \quad U_e = U_e(\psi_e) \tag{3.6.12b}$$

This means that the energy of the electrons and ions is preserved along their trajectory of movement, as already mentioned in section 3.1.4. The direct verification confirms that

$$rH^*_{zie} = \frac{\partial a_{i,e}}{\partial r}; \quad rH^*_{rie} = -\frac{\partial a_{i,e}}{\partial z} \tag{3.6.13a}$$

where

$$a_i \equiv \psi_H + \frac{Mc}{e} r v_\theta^{(i)}; \quad a_e \equiv \psi_H - \frac{mc}{e} r v_\theta^{(e)} \tag{3.6.13b}$$

Using the expressions (3.6.7), (3.6.8), (3.6.12a) and (3.6.13a) gives the laws of conservation of moments

$$a_i = a_i(\psi_i); \quad a_e = a_e(\psi_e) \tag{3.6.14}$$

In turn, from the *r*-component (or *z*-component) of the equations (3.6.8) and the conservation laws (3.6.12a) we obtain the following equations for the functions ψ_i, ψ_e

$$\frac{c}{e} U_i' = \frac{v_{i\theta} a_i'}{r} - \frac{1}{r n_i} \left(H_\theta + \frac{Mc}{e} \mathrm{rot}_\theta \, \mathbf{v}_i \right); \tag{3.6.15a}$$

[13]It should be mentioned that the symbol $D(f, g)$ denotes the Jacobian: $\begin{vmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{vmatrix}$

The equality of the Jacobian to 0 indicates that the functions that *f* and *g* are linked by the relationship $f = f(g)$.

$$-\frac{c}{e}U'_e = \frac{v_{e\theta}a'_e}{r} - \frac{1}{rn_e}\left(H_\theta - \frac{Mc}{e}\mathrm{rot}_\theta \mathbf{v}_e\right); \tag{3.6.15b}$$

$$U' \equiv \frac{dU}{d\psi}; \quad a' \equiv \frac{da}{d\psi}. \tag{3.6.15c}$$

Finally, using (3.6.7) and the Maxwell equations (3.6.5) and (3.6.6) we obtain

$$I = 2\pi(\psi_i - \psi_e); \quad I = \frac{crH_\theta}{2};$$

$$\Delta^*\psi_H = \frac{4\pi e}{c} r(n_i v_{i\theta} - n_e v_{e\theta}); \tag{3.6.16}$$

$$\Delta\phi = -4\pi e(n_i - n_e); \quad \Delta^* \equiv r\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2}.$$

To obtain more information, all the mechanical equations are presented together excluding from the equations (3.6.5) the quantities $\mathrm{rot}_\theta \mathbf{v}_{i,e}$ using (3.6.7). Consequently, we obtain [88]

$$\frac{Mv_i^2}{2} + W_i + e\phi = U_i(\psi_i); \tag{3.6.17a}$$

$$\frac{mv_e^2}{2} + W_e - e\phi = U_e(\psi_e); \tag{3.6.17b}$$

$$\psi_H + \frac{Mc}{e} r v_{i\theta} = a_i(\psi_i); \tag{3.6.17c}$$

$$\psi_H - \frac{mc}{e} r v_{e\theta} = a_e(\psi_e); \tag{3.6.17d}$$

$$\frac{Mc}{e}\left(\frac{1}{rn_i}\frac{\partial}{\partial r}\frac{1}{rn_i}\frac{\partial\psi_i}{\partial r} + \frac{1}{rn_i}\frac{\partial}{\partial z}\frac{1}{rn_i}\frac{\partial\psi_i}{\partial z}\right) - \frac{H_\theta}{rn_i} + \frac{v_{i0}}{r}a'_i = \frac{c}{e}U'_i(\psi_i); \tag{3.6.17e}$$

$$\frac{mc}{e}\left(\frac{1}{rn_e}\frac{\partial}{\partial r} + \frac{1}{rn_e}\frac{\partial\psi_e}{\partial r} + \frac{1}{rn_e}\frac{\partial}{\partial z}\frac{1}{rn_e}\frac{\partial\psi_e}{\partial z}\right) + \frac{H_\theta}{rn_e} - \frac{v_{e0}}{r}a'_e = \frac{c}{e}U'_e(\psi_e); \tag{3.6.17f}$$

In these six equations for six quantities ψ_i, ψ_e, n_i, n_e, $v_{i\theta}$, $v_{e\theta}$, there are four functions: $U_i(\psi_i)$, $U_e(\psi_e)$, $a_i(\psi_i)$, $a_e(\psi_e)$, which in principle can be

defined arbitrarily.

Here, we can see the analogy with the arbitrary functions $P(\psi)$ and $I(\psi)$ which were discussed in section 2.4.3. The three-flux (ψ_e, ψ_i, ψ_H) system of equations (3.6.16) and (3.6.17) was derived by A.I. Morozov and L.S. Solov'ev.

Thermalised potential

The analogue of the thermalised potential (3.2.12c) follows from (3.6.17b), i.e. the law of conservation of energy, and from (3.6.17d), i.e. the law of conservation of the momentum of motion of the electronic component at $m \to 0$. In fact, at $m = 0$

$$W_e - e\phi = U_e(\psi_e); \quad \psi_H = a_e(\psi_e) \tag{3. 6.18a}$$

Excluding ψ_e, we obtain the law of conservation:

$$\phi_T(\psi_H) = \phi - \frac{W_e}{e} = -\frac{U_e}{e} \tag{3.6.18b}$$

The expression for ϕ_T is identical with the thermalised potential (3.2.12c).

Freezing-in

If $m \to 0$, equation (3.6.17f) also converts to the well-known law of conservation – the freezing-in of the magnetic field into the electronic component:

$$\frac{H_\theta}{rn_e} = -\frac{v_{e\theta}a'_e}{r} + \frac{c}{e}U'_e(\psi_e) \tag{3.6.19}$$

Equation (3.6.19) defines more accurately the well-known expression (2.3.10) and defines concretely the equation (3.1.16). Firstly, this equation takes into account the effect of the 'poloidal' field ψ_H and the presence of ordered azimuthal motion of the electron; secondly, the equation gives at $v_{e0} = 0$ the explicit expression for the constant $\frac{H_\theta}{\rho r}$, included in (2.3.10). This constant is equal to $\frac{c}{e}U'_e(\psi_e)$. If U'_e=const, these flows are isomagnetic.

3.6.2. Qualitative analysis of the system of equations (3.6.17)

The flow of cold plasma ($W_e = W_i = 0$ at $m = 0$). For a number of problems, including the problems of construction of plasma accelerators with medium and high energy, it is interesting to examine the electromagnetic acceleration of relatively cold plasma with a low degree of dissipation.

If it is assumed that $W_e = W_i = 0$, the system of equations (3.6.17) assumes the following form:

$$\frac{Mv_i^2}{2} + e\,\phi = U_i(\psi_i); \tag{3.6.20}$$

$$e\phi = -U_e(\psi_e); \tag{3.6.21}$$

$$\psi_H + \frac{Mc}{e} r v_{i\theta} = a_i(\psi_i); \tag{3.6.22}$$

$$\psi_H = a_e(\psi_e); \tag{3.6.23}$$

$$\frac{Mc}{enr^2}\left(r\frac{\partial}{\partial r}\frac{1}{nr}\frac{\partial \psi_i}{\partial r} + \frac{\partial}{\partial z}\frac{1}{n}\frac{\partial \psi_i}{\partial z}\right) = \frac{c}{e}U_i' + \frac{H_\theta}{rn}; \tag{3.6.24}$$

$$0 = -\frac{c}{e}U_e'(\psi_e) + \frac{H_\theta}{rn} - \frac{v_{e\theta}a_e'}{r}. \tag{3.6.25}$$

The equations (3.6.21) and (3.6.26) show clearly that the magnetic lines of force are equipotential and the electrons are 'tied up' with the magnetic lines of force.

It is necessary to distinguish between two classes of axial symmetry systems: the systems which contain only the azimuthal magnetic field H and the systems with all three components of the magnetic field. Each class will be investigated separately.

Flow in an intrinsic magnetic field ($\psi_H = 0$, $a_{i,e} = 0$). Thus, we consider a system shown in Fig. 3.6.1a. Equation (3.6.22) confirms that the electrons move on the equipotentials. This simple fact results in a very important conclusion according to which the system with solid conducting coaxial electrodes with no high-frequency oscillations or

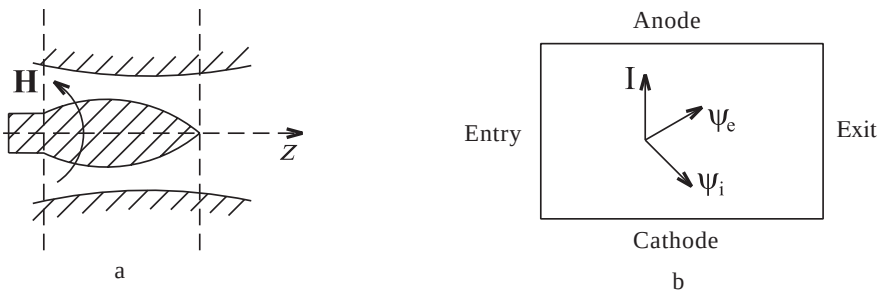


Fig. 3.6.1. A coaxial plasma accelerator with an intrinsic magnetic field (a) and its diagram (b).

dissipative processes cannot be characterised by the regular arrival of the electrons from the cathode into plasma nor from plasma into the anode. The most radical solution of this situation is to stop rgarding the electrons as carriers of current and apply this function to ions (‘ionic current transfer’). If we want to retain ‘electronic current transfer’, we should either accept the presence of dissipative processes in the system (for example, in the form of singularities – ‘attachments’ of the current on the electrodes of the type shown in Fig. 3.2.11) or transfer from the stationary regime to ‘machine-gun’ regimes or, finally, transfer to non-equipotential (for example, ‘sectioned’) electrodes.

The pattern of the processes in the gap between the electrodes of a coaxial accelerator becomes very convincing if we use the so-called ‘diagram of the accelerator’. For this purpose, the gap between the electrodes is imaged in the form of a rectangle (see Fig. 3.6.1b) whose sides correspond to entry, exit and two electrodes, and we draw the line I = const, ψ_i = const, ψ_e = const. It is assumed that the process is stationary and regular and there are no difficulties with ‘spreading’ of the current on the electrodes. As already mentioned, this can take place with non-equipotential electrodes in the electronic current transfer regime and when using the equipotential electrodes in the ionic current transfer regime. Both regimes will be investigated separately.

Electronic current between non-equipotential electrodes (Fig. 3.6.2)
In this case, the current lines I = const can be described by a system of equally spaced vertical lines. The lines ψ_i = const are imaged by the horizontal straight lines and the anode and the cathode correspond to the lines $\psi_i = 0$, $\psi_i = \psi_{i0}$. Of course, it is assumed that the ions do not settle on the electrodes. At the selected coordinates the equations (see (3.6.16))

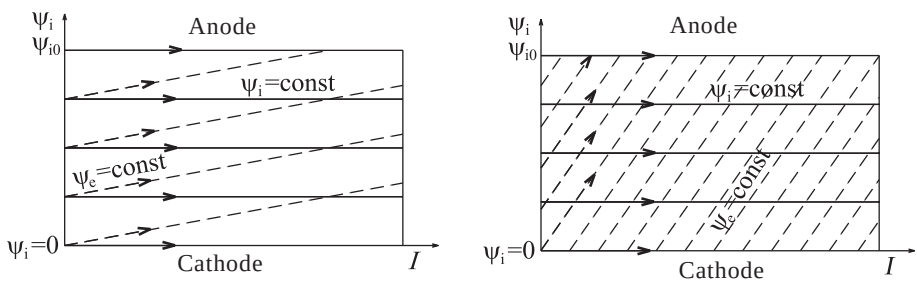


Fig. 3.6.2. Diagrams of the accelerator channel in electronic current transfer: exchange parameter $\xi \ll 1$, hydrodynamic regime (a); $\xi > 1$, Hall regime (b).

$$\psi_e = \psi_i - \frac{I}{2\pi e} = \text{const}$$

are imaged by inclined lines. It should be mentioned that the quantity in the lower left corner of the diagram $\psi_i = 0$, $I = I_0$

$$\psi_{e0} = -\frac{I_0}{2\pi e}$$

The fact that at this point of the diagram $\psi_e \neq 0$ is explained by the presence of the electronic flux in the cathode. In the lower right corner of the diagram we set $I_1 = 0$ and obtain $\psi_e = 0$ since here $\psi_i = 0$.

If the inertia of the electrons is ignored, the electrical potential

$$\phi = -\frac{1}{e} U_e(\psi_e)$$

and the freezing-in parameter

$$\frac{H_\theta}{rn} = -c\phi'(\psi_e)$$

remain unchanged along the lines ψ_e= const, which U_i remains unchanged along the lines ψ_i= const. Evidently, the ions are accelerated since they travel into the region with a lower potential

In principle, there are two types of diagram which can be called hydrodynamic and Hall (see Fig. 3.6.2). It is clear that in the first case the major portion of the electrons, which emerge from the accelerator, enter the latter together with the ions. In the second case the electrons, entering the accelerator, settle on the anode at the beginning of channel and everywhere else the compensation of the volume charge is accomplished by the electrons emitted from the cathode. Therefore, if in the first case the value of the freezing-in paremeter is almost constant along the length of the accelerator, in the second case it can easily be changed.

Transition from one regime to another is obviously, determined by the value $\frac{I_0 - I_1}{2\pi e \psi_{i0}}$, i.e. by the integral exchange parameter (see section 3.2.1)[14]:

[14]Here we assume that the current is not transferred from the accelerator and $I_1 = 0$.

$$\xi = \frac{I_0}{2\pi e \psi_{i0}} = \frac{J_p}{J_{\dot{m}}} \qquad (3.6.26)$$

Ionic current transfer between the equipotential electrodes
The diagram of the coaxial accelerator with the ionic current transfer is depicted in Fig. 3.6.3.

In this case I and ψ_e are ploted on the axes, whereas lines ψ_i = const are determined from the equation

$$\psi_i = \psi_e + \frac{I}{2\pi e}$$

Here at the left lower point of the diagram it is expedient to assign the following values to the quantities:

$$(\psi_i)_0 = 0; \quad (I)_0 = I_0; \quad (\psi_e)_0 = -\frac{I_0}{2\pi e}$$

This selection of the reference point is explained by the fact that the ions, after falling on cathode, perish (they are neutralized) and the current in the circuit is locked due to the electron flux in the cathode, with the electrons 'paired' with the incoming ions.

For the normal functioning of the mechanism of ionic current transfer the cathode should absorb the ions falling on it, and the anode should emit these ions. Actually, this means that ionised matter should be transferred through the anode. Such conditions are created in the QHPA described below.

Obviously, the regime of ionic current transfer can be of practical interest only with the sufficiently low exchange parameters ξ. Otherwise, too many ions will perish on the cathode.

Flow in the external magnetic field. Let all three components of

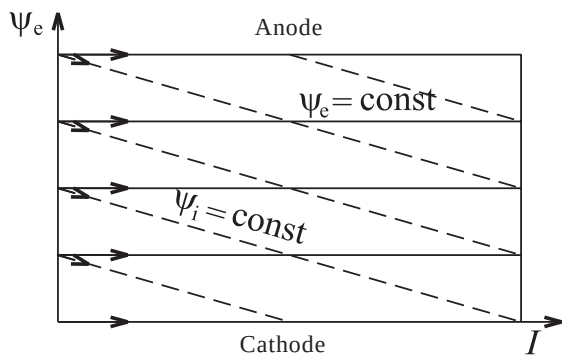


Fig. 3.6.3. Diagram of the accelerator channel with ionic current transfer.

the magnetic field be present in the system (external and intrinsic). Because of axial symmetry, the magnetic field can be considered as the system of magnetic surfaces ψ_H = const with 'attached' electrons ($\psi_H = a_e(\psi_e)$). In view of equations (3.6.21) and (3.6.23) these surfaces are equipotential. The potential distribution between them is determined either by dissipative processes in the volume or by some system of electrodes located on the channel walls. Since the electrons are attached to the equipotential surfaces, current transfer in the presence of external field can be achieved only by ions. Apparently, there are two versions. In the first version, magnetic surfaces are parallel to side walls of the accelerating channel; therefore, the walls can be made metallic, and the potential distribution between the magnetic surfaces is determined by its distribution on the 'rear' dielectric. This version of a non-dissipative high-current accelerator is practically very close to the previousy examined accelerator with the azimuthal magnetic field in the ionic current transfer regime. Here it is also necessary to supply part of the working substance through the anode, and carry it out through the cathode, and the electrons also move, without intersecting the electrode surface.

In the second version of accelerators with the external magnetic field all magnetic surfaces intersect the side walls of the channel. Therefore, they must not possess longitudinal conductivity. This accelerator is completely analogous to the Hall MHD systems, discussed in section 3.5, and also the stationary plasma engine which will be discussed in detail in section 6.7. Therefore, we will not discuss them here.

In the accelerators with the external magnetic field there is a problem of the exit of plasma from magnetic fields. In the first version of the accelerator in which the ions are magnetised, dissipation is essential for the exit from the magnetic field. In the second version the ions are not magnetised and it is possible to manage with the application of emitters of the electron–compensators 'which pour on' the outgoing ionic flux (see section 6.7).

Thermal acceleration in a 'magnetic nozzle' (Fig. 3.6.4)

Let us examine another limiting case when acceleration occurs only due to high temperature or, more precisely, due to the enthalpy of the plasma components. If we discuss obtaining comparatively low exhaust velocities to which the particle energy of the order of 1 eV corresponds (they are intended to be heavy particles), then it is not difficult to heat both components: both ions and electrons. But if the discussion deals with energies of tens of electron volts, or even hundreds and thousands of electron volts, then the so-called non-isothermal accelerators would

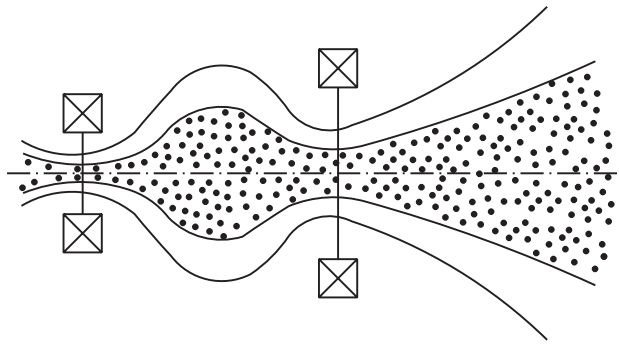


Fig. 3.6.4. 'Magnetic nozzle' for nonisothermal acceleration.

be of principal interest in which $T_e \gg T_i$. Heating electrons to high temperatures ($\geqslant$ 1 keV) does not present special difficulties at this time. This can be made both with the aid of the UHF field and with the aid of high-energy electron beams.

Therefore, as previously, the enthalpy of ions is assumed to be equal to zero and we will consider only electrons as hot. Let us also disregard the inertia of electrons. In this case we obtain a system of equations:

$$\frac{Mv_i^2}{2} + e\phi = U_i(\psi_i); \quad W_e - e\phi = U_e(\psi_e); \tag{3.6.27}$$

$$\psi_H + \frac{Mc}{e} r v_{i0} = a_i(\psi_i); \quad \psi_H = a_e(\psi_e); \tag{3.6.28}$$

$$\frac{Mc}{e}\left(\frac{1}{rn}\frac{\partial}{\partial r}\frac{1}{rn}\frac{\partial \psi_i}{\partial r} + \frac{\partial}{rn}\frac{\partial}{\partial z}\frac{1}{rn}\frac{\partial \psi_i}{\partial z}\right) = \frac{c}{e}U_i'(\psi_i) + \frac{H_\theta}{rn} + \frac{v_{i\theta}}{r}a_i'; \tag{3.6.29}$$

$$\frac{H_\theta}{rn} + \frac{v_{e\theta}}{r}a_e' = \frac{c}{e}U_e'(\psi_e) \tag{3.6.30}$$

This system of equations shows that in this case the electrons 'sit' on the magnetic surfaces ψ_H = const. However, now the magnetic surfaces are not equipotential. Making different assumptions about the entrance conditions into the system, i.e. about the functions $U_{i,e}$, $a_{i,e}$, we obtain different regimes.

One of the simplest regimes is the flow in which $\psi_i = \psi_e$, i.e. the trajectories of ions and electrons along the system coincide. In this case it is can be assumed that the azimuthal field is equal to zero ($H_\theta = 0$). If we now assume that the ions start with the zero azimuthal velocity, then it follows from the condition $\psi_i - \psi_e$ and equations

(3.6.28) that everywhere $v_{i\theta} = 0$. As a result, the system (3.6.28)–(3.6.30) can be written in the form:

$$\frac{Mv_i^2}{2} + W_e = U_i(\psi_i) + U_e(\psi_i);$$
$$\psi_H = a_i(\psi_i);$$
$$\frac{1}{rn}\frac{\partial}{\partial r}\frac{1}{rn}\frac{\partial \psi_i}{\partial r} + \frac{1}{rn}\frac{\partial}{\partial z}\frac{1}{rn}\frac{\partial \psi_i}{\partial z} = U_i'(\psi_i);$$
$$-\frac{v_{e\theta}}{r}a_e' = \frac{c}{e}U_e'(\psi_i) \qquad (3.6.31)$$

It is not difficult to ascertain that if we set $U_e(\psi_i)$ = const, then we obtain the model of purely thermal acceleration, described by the usual gas-dynamic equations:

$$\operatorname{div} n\mathbf{v}_i = 0; \quad M(\mathbf{v}_i\nabla)\mathbf{v}_i = -\frac{\nabla p_e}{n_i}$$

and existing in a 'magnetic nozzle'.

It should be noted that because of the high electronic thermal conductivity along the lines of force the hydrodynamic model of non-isothermic acceleration requires specific corrections. The problem of the exit of plasma from the magnetic field in non-isothermal accelerators is the same as in the cases examined above. Experiments with magnetic nozzles and non-isothermal plasma were described in, for example, [101].

3.6.3. Method of 'smooth' flows for system (3.6.17)

The system of equations (3.6.17) possesses that remarkable property that it does not contain the terms linear in relation to $\frac{\partial}{\partial z}$. The equations contain terms ~ $\frac{\partial^2}{\partial z^2}$, or ~ $\left(\frac{\partial}{\partial z}\right)^2$. Obviously, if the flow cross section changes sufficiently smoothly, then the role of these terms must be small. Therefore, the approximation of 'smooth' flows can be made after rejecting the terms which contain derivatives with respect to *z*. This yields a system of ordinary differential equations with only derivatives with respect to *r* (A.I. Morozov, L.S. Solov'ev [73]). However, it should not be assumed that the second measurement is unavoidably lost. The dependence on the coordinate *z* enters now integration constants.

In order to illustrate this method, let us examine the simplest case of flow of quasi-neutral plasma with $m = 0$ in its own azimuthal magnetic field ($H_r = H_z = 0$). In this case system (3.6.16) and (3.6.17) can be recorded in the form:

$$\frac{M}{2n^2r^2}\left(\frac{\partial\psi_i}{\partial r}\right)^2 + W_i(n) + e\phi = U_i(\psi_i); \tag{3.6.32a}$$

$$W_e(n) - e\phi = U_e(\psi_e); \tag{3.6.32b}$$

$$\frac{1}{rn}\frac{\partial}{\partial r}\frac{1}{nr}\frac{\partial\psi_i}{\partial r} = \frac{e}{Mc}\left(\frac{c}{e}U_i' + \frac{H}{rn}\right); \tag{3.6.32c}$$

$$0 = \frac{c}{e}U_e' - \frac{H}{rn}; \tag{3.6.32d}$$

$$rH = \frac{4\pi e}{c}(\psi_i - \psi_e) + Y_0. \tag{3.6.32e}$$

Obviously, the approximation of the slowly changing channel resembles the approximation of the boundary layer in the viscous flow theory[15].

In a number of cases system (3.6.32) has one integral and is reduced to quadratures. Let us for this purpose accumulate in pairs the first four equations (3.6.32):

$$\frac{M}{2n^2r^2}\left(\frac{\partial\psi_i}{\partial r}\right)^2 + W(n) = U_i(\psi_i) + U_e(\psi_e); \tag{3.6.33a}$$

$$\frac{M}{nr}\frac{\partial}{\partial r}\frac{1}{nr}\frac{\partial\psi_i}{\partial r} = U_i'(\psi_i) + U_e'(\psi_e); \tag{3.6.33b}$$

$$W = W_i(n) + W_e(n). \tag{3.6.33c}$$

Differentiating now the first equation with respect to r, and multiplying the second equation by $\frac{\partial\psi}{\partial r}$, let us deduct the obtained expressions from each other. As a result we find the connection:

[15]However, the equation of the viscous layer includes mainly the first derivative with respect to z. Therefore, rejecting terms $\sim\partial^2/\partial z^2$ does not lead in this case to conventional differental equations.

$$\frac{\partial W}{\partial r} = U_e'(\psi_e)\left(\frac{\partial \psi_e}{\partial r} - \frac{\partial \psi_i}{\partial r}\right) \tag{3.6.34}$$

This equation substitutes the second of equations (3.6.233), and the system of equations (3.6.33) and (3.6.34) is frequently more convenient, since it is the first order system of equations. However, the main advantage of equation (3.6.34) lies in the fact that it makes very simple physical sense and is integrated in a number of cases. Let us pause, first of all, to consider its physical sense. Taking into account (3.6.32d), (3.6.32e) and the determination of W:

$$W = \int \frac{dp}{n}$$

we obtain from (3.6.34) the equation of radial equilibrium in the slowly changing flow:

$$\frac{\partial}{\partial r}\left(p + \frac{H^2}{8\pi}\right) = -\frac{H^2}{4\pi r} \tag{3.6.35}$$

Equation (3.6.35) in the planar case is integrated in the general form:

$$p + \frac{H^2}{8\pi} = c(z) \tag{3.6.36}$$

In the axisymmetric case instead of (3.6.35) it is more convenient to write equation (3.6.34), which is easily integrated with one of two conditions: $W = 0$; or $U_e'(\psi_e) = \text{const}$. In these cases the task is completely reduced to the quadratures.

Let us illustrate the general diagram by a single calculation. Let the flow be isomagnetic:

$$U_e = k\psi_e \tag{3.6.37a}$$

and 'iso-Bernoullic'[16]

$$U_i = U_0 - k\psi_i \tag{3.6.37b}$$

In that case equations (3.6.34) and (3.6.33b), describing flows, take the form:

$$W(n) = k(\psi_e - \psi_i) + c_1(z); \tag{3.6.38a}$$

[16]In other words, it is assumed that the value of the Bernoulli integral is the same in the entire volume of the flow: $\frac{v^2}{2} + \frac{H^2}{4\pi\rho} + \int \frac{dp}{\rho} = \text{const}$.

$$\frac{M}{2n^2r^2}\left(\frac{\partial\psi_i}{\partial r}\right)=U_0-c_1(z)\equiv\frac{Mv_0^2(z)}{2} \tag{3.6.38b}$$

Hence it is apparent that the velocity of ions with this flow proves to be independent of r. To us it is more convenient to use $v_0(z)$ instead of $c_1(z)$. Assuming in (3.6.32e) that $Y_0 = 0, U_e' = k$, we obtain

$$n=\frac{4\pi e^2}{c^2k^2r^2}\left(U_0-\frac{Mv_0^2(z)}{2}-W(n)\right) \tag{3.6.39}$$

In a real case when $\gamma = 5/3$ solution of this equation relative to n is quite cumbersome. Therefore, let us take $\gamma = 2$. Then

$$W=\frac{2np_{00}}{n_{00}^2} \tag{3.6.40}$$

Here the p_{00} and n_{00} are some initial values of plasma pressure and density. With this assumption

$$n=\frac{n_0(z)}{1+\dfrac{r^2}{a^2}} \tag{3.6.41a}$$

where

$$n_0=\frac{1}{b}\left(U_0-\frac{Mv_0^2(z)}{2}\right);\quad a^2=\frac{4\pi e^2}{k^2c^2}b;\quad b=2\frac{p_{00}}{n_{00}^2} \tag{3.6.41b}$$

Substituting (3.6.41) in (3.6.38), we obtain an expression for the function of the ionic flow:

$$\psi_i=\frac{a^2n_0(z)v_0(z)}{2}\ln\frac{1+\dfrac{r^2}{a^2}}{1+\dfrac{r_0^2(z)}{a^2}} \tag{3.6.42a}$$

Here $r_0(z)$ is the integration constant. In order to solve this problem, we should assign conditions with the aid of which it will be possible to determine two functions $v_0(z)$ and $r_0(z)$. In the simplest case of electronic current transfer, when the ion flow on the walls can be disregarded, it is possible to assign geometry of both electrodes – cathode $r_k(z)$ and the anode $r_a(z)$ – with the aid of the conditions

$$\psi_i(r_k(z),z)=0;\quad \psi_i(r_a(z),z)=\psi_0=\text{const} \tag{3.6.42b}$$

Obviously, the first condition is satisfied automatically, if we assume that $r_k(z) = r_0(z)$. However, the substitution of the second condition

(3.6.42b) in (3.6.42a) leads to the solution of the cubic equation for v_0, since n_0 ~ const. The cubic equation always has one material root; therefore, the nonlinear boundary-value problem in question always has a solution.

Knowing ψ_i and n, (3.6.38a) can be used to find electron paths ψ_e.

The remarkable special feature of solution (3.6.42a) is the fact that it also describes the flow with the shortened central electrode. In this case with $z > 0$ it is necessary to assume that $r_0(z) = 0$. Obviously, we obtain here the compression region.

3.6.4. Analysis of system (3.6.17) by the narrow channel method

Above we calculated in the approximation of the 'smooth flow' the behavior of the isomagnetic iso-Bernoullian flow in the intrinsic azimuthal magnetic field. Unfortunately, to obtain the simple formula (3.6.42b) it was necessary to set $\gamma = 2$, and this somewhat masked the important special features of the obtained class of flows.

Let us examine this question in more detail, using the narrow flow approximation, analogous to the one which we used in the analysis of the flow in a Laval nozzle (see 2.2), and in magnetohydrodynamics (MHD), section 2.6. It should be noted that the narrow flow approximation and the smooth flow approximation are additional. Actually, in the first case we consider that the flow conditions slightly change in the transverse direction, whereas in the second case the 'weak' parameter is the longitudinal coordinate.

Considering the flow as before iso-Bernoullian and isomagnetic[17], we can write two laws of conservation:

$$\frac{v^2}{2} + i(\rho) + \frac{H^2}{4\pi\rho} = U_0 = const; \qquad (3.6.43a)$$

$$\frac{H}{\rho r} = k = const$$

Here the enthalpy

$$i(\rho) = \frac{W_i(n) + W_e(n)}{M} \qquad (3.6.43b)$$

These expressions with the assumptions made previously are not connected with the assumption of the narrowness of the flow. In order to develop the model of the narrow flow, it is necessary to reject the

[17]In the single-fluid model, the flow at the exchange parameter $\xi = 0$ in a narrow channel is automatically isomagnetic and isoBernoullian.

equation for the vortex (3.6.17) and to supplement system (3.6.43b) by the equation of conservation of the flow in a coaxial tube

$$\rho v r f = \delta\psi = \text{const}, \tag{3.6.43c}$$

where f is the width of the channel. Now all the terms in (3.6.43) should be considered as the functions of one longitudinal coordinate, for example z. The resultant system of equations is identical with the system (2.6.4) for the single-fluid flow. Therefore, the system (3.6.43) will not be discussed in detail. Let us note only that the Hugoniot relationships (see 2.2.20) can be written not only for gas dynamics but also for any narrow flows, in particular for equations (3.6.43). In this case for a velocity change we obtain

$$(v^2 - c_s^2)\frac{dv}{v} = c_T^2 \frac{d(rf)}{rf} + c_A^2 \frac{d(f/r)}{f/r} \tag{3.6.44}$$

Here c_s is the signal velocity (rapid sound):

$$c_s^2 = c_T^2 + c_A^2;\ c_A^2 \equiv \frac{H^2}{4\pi p}.$$

Hence it is apparent that thermal acceleration is connected with the change of the section of the channel fr, and electromagnetic – with a change in the relation f/r. It is not difficult to write an analogous equation for changes of other flow parameters.

3.7. Numerical and experimental studies of (quasi-) steady flows in coaxial systems with the intrinsic magnetic field

Theoretical analysis of plasma flows in transverse magnetic fields, described in sections 3.5 and 3.6, was stimulated by the experiments, carried at the beginning of the 60s by A.I. Morozov and his colleagues at the Institute of Atomic Energy, and then by the numerical calculations carried out later at the Institute of Applied Mathematics by K.V. Brushlinskii, et al [89]. In this section we examine only high-current accelerators and compressors with their own azimuthal magnetic field.

3.7.1. Numerical simulation of plasma flows in coaxials with continuous electrodes

Numerical calculations in comparison with the analytical simulation have a number of unquestionable merits. First of all, it is possible to describe the entire flow process from the gas supply into the channel to the output of the plasma flow into the vacuum chamber. Secondly,

numerical calculations make it possible to consider not only the Hall effect but also dissipative processes, especially Ohmic resistance with the different boundary conditions. Thirdly, and this is especially important, numerical simulation automatically yields information on flow resistance to axisymmetrical disturbances. This is connected with the fact that steady flows are usually calculated by the establishment method, i.e. taking into account the explicit time dependence of values.

Here we describe an axisymmetric flow of a completely ionised gas with the absence of ohmic resistance, calculated by the establishment method with the different values of the Hall parameter on the basis of the system of equations

$$\frac{\partial \rho}{\partial t} + \operatorname{div} \rho \mathbf{v} = 0;$$

$$\rho \frac{d\mathbf{v}}{\partial t} = -\nabla p + \frac{1}{c}[\mathbf{j}, \mathbf{H}]; \qquad (3.7.1)$$

$$\frac{\mathbf{j}}{\sigma} = \mathbf{E} + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right] - \frac{M}{e\rho c}\left[\mathbf{j}, \mathbf{H}\right];$$

$$\operatorname{rot} \mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \quad \operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial H}{\partial t}$$

The initial conditions were taken sufficiently arbitrary, the electrodes were assumed to be equipotential. Ionised plasma was considered at entry into the channel for the clearer manifestation of the Hall effect. The flow was assumed to be isothermal. Calculations were performed with the different local exchange parameters ξ_{loc}. It turned out that there are some critical values $\xi_{loc}^{(+)}$ and $\xi_{loc}^{(-)}$ for the external and internal anodes (polarities) respectively, which divide the established steady-

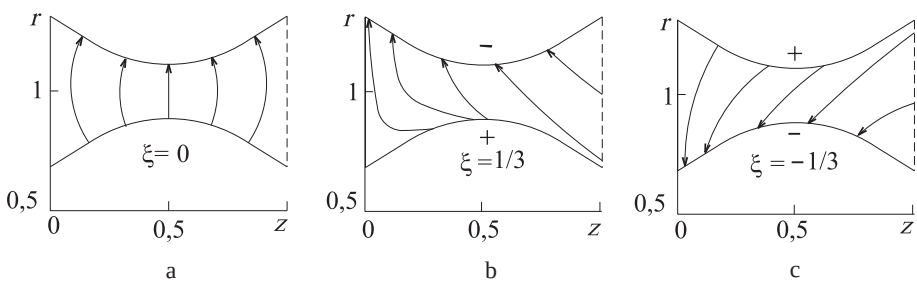


Fig. 3.7.1. Influence of the Hall effect on the current lines in an accelerator channel with continuous electrodes: a) $\xi = 0$; b) $\xi = 1/3$; $\xi = -1/3$.

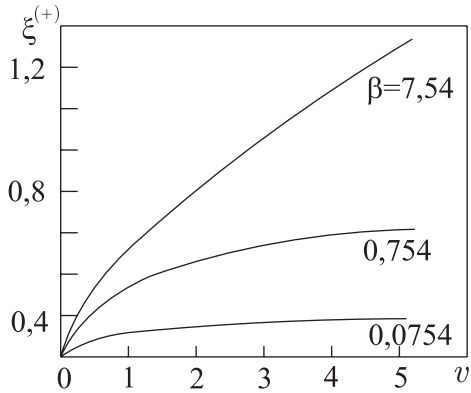


Fig. 3.7.2. Dependence of $\xi^{(+)}$ on ν for three different values of β.

state solutions and flows with 'the explosion' on the anode, with which the flow is completely destroyed.

Figure 3.7.1 shows current lines for different ξ_{loc} and polarities, while Fig. 3.7.2 is the dependence of critical values $\xi^{(+)}$ on the reverse magnetic Reynolds number – magnetic viscosity $\nu \equiv 1/Re_m$ with different β at entry. The graphs sufficiently clearly demonstrate transition from the radial currents when $\xi = 0$ to the slip of currents near the anode with $\xi \neq 0$, and then, at $\nu_m \to 0$, i.e. increase of conductivity, to flow breakdown. The last phenomenon is called usually 'the crisis of current'. The crisis of current will be discussed in greater detail below, while 'the explosions' – in chapter 8.

3.7.2. Experimental studies of accelerators with the continuous electrodes [90–92]

(Quasi-) stationary high-current (with the intrinsic magnetic field) coaxial plasma accelerators with the continuous electrodes will be referred to as simply 'coaxials', if the central electrode along the length is close to the external electrode and 'end high-current accelerators (EHA)', if the central electrode is substantially shorter. Systems generating compression flows are 'magnetoplasma compressors' (MPC) (Fig. 3.7.3).

The 'coaxial' KPU-1 is very similar to the previously discussed theoretical schemes. Moreover, it was the first subject studied in relatively detailed physical studies of plasma accelerators of this type at the beginning of the 60s at the Institute of Atomic Energy. Experiments were carried out in the installation shown in Fig. 3.7.4 which worked in the quasi-stationary regime. The duration of the operating pulse $\tau_{op} \sim 1$–10 ms, on one side, did not create difficulties with heating of

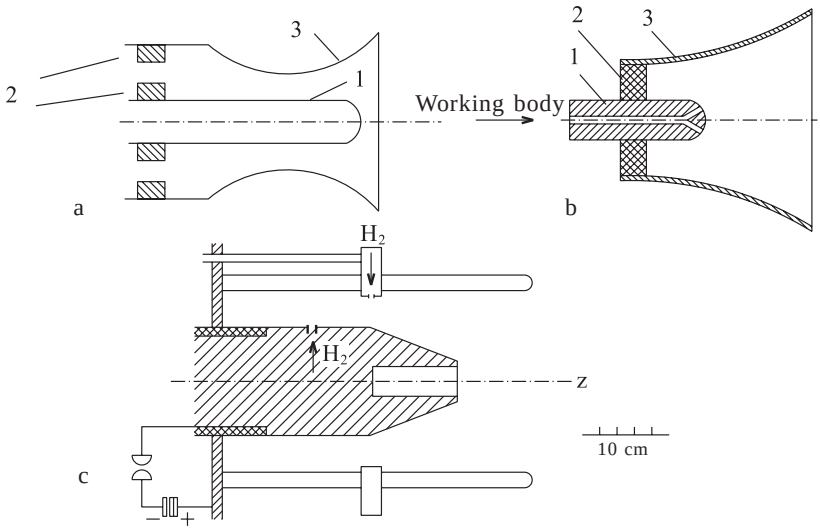


Fig. 3.7.3. Single-stage coaxial accelerators with an intrinsic magnetic field and with continuous electrodes: a) 'coaxial'; b) end high-current accelerator (EHA), 1) cathode, 2) dielectric, 3) anode, c) magnetoplasma compressor (MPC).

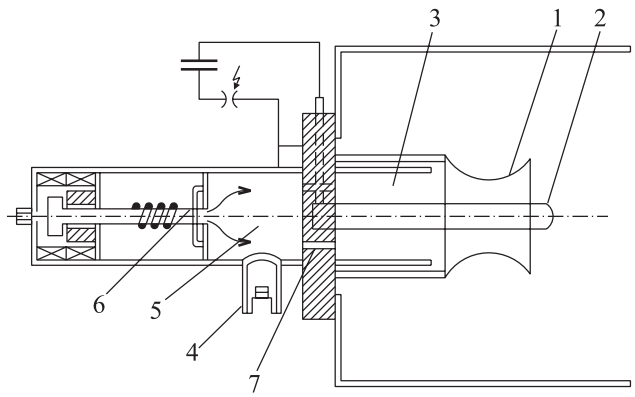


Fig. 3.7.4. KPU-1 installation (Institute of Atomic Energy): 1) outer electrode, 2) inner electrode, 3) buffer volume, 4) piezosensor for pressure measurement, 5) forward chamber, 6) valve, 7) hole in the diaphragm.

the coaxial and it did not require high energy (in this case, a capacitor bank) and, on the other hand, the selected time was almost two orders of magnitude longer than the flight time

$$\tau_0 \sim \frac{L}{v_M} \sim 3 \cdot 10^{-6} \text{s},$$

where L is the length of channel (~15 cm), and v_M is the expected discharge velocity (~$5 \cdot 10^6$ cm/s). With this selection of τ_{op}, the process

could be considered quasi-stationary.

The accelerator was placed in the vacuum chamber. The gas was supplied (N_2, Ar, H_2) from the buffer volume which was filled in pulses with the aid of a high-velocity valve. Discharge currents were ~20–100 kA.

However, what was shown in experiments?

Experimental data were quite surprising since in 1959 the special features of the manifestation of the Hall effect were not properly known and the concept of the single-fluid MHD model was in the initial stage.

It turned out that at low discharge currents ($H^2/8 \lesssim nk\,(T_i + T_e)$ the lines of electric current were close to radial, i.e. to those predicted by the single-fluid MHD.

This effect was especially evident when a long incandescent thermionic cathode was used in the coaxial. This regime was then studied in detail by many authors on 'high-current end accelerators' (Fig. 3.7.3b).

But if discharge current J increases at a constant flow rate of the substance $\dot{m}$, then at some 'critical' value of J_* the picture sharply changes. First of all this is evident on the voltage–ampere characteristic. Namely, if at $J < J_*$ voltage U of the discharge grows comparatively slowly with an increase in current

$$U \sim J^{\alpha},\ 2 < \alpha < 3, \tag{3.7.2}$$

Then at $J > J_*$ volatge starts to increase rapidly (Fig. 3.7.5).

In this case high-amplitude oscillations occur in the accelerator and a large part of the discharge current is transferred outside the outlet of the accelerator. The transferred current forms the region of compression, shown clearly in the photograph (Fig. 3.7.6).

The displacement of current from the channel becomes completely

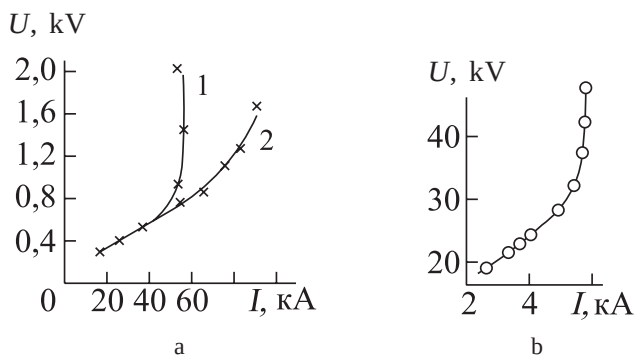


Fig. 3.7.5. Volt–ampere characteristics: a) discharge in KPU-1 (hydrogen: $\dot{m}$ = 1.5 g/s (1); $\dot{m}$ = 4.5 g/s) (2); b) end high-current accelerator (EHA) on lithium.

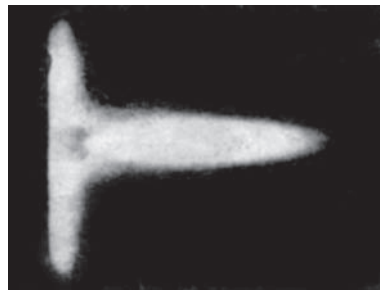


Fig. 3.7.6. External appearance of the discharged flow in the compression region in KPU-1 installation.

clear, if we recall the role of the Hall effect with the flow of plasma across the magnetic field in the channel with the equipotential electrodes, described in sections 3.2.4 and 3.7.1. It was shown there that the longitudinal 'Hall' component of current appears in this system together with the 'Ohmic' component of current which goes along the **E**-field. This appearance of the 'Hall's' component of current displaces plasma from the anode. A reduction in the concentration of the charged particles in this region increases $\omega_e\tau_e$ and even more greatly intensifies the current along the anode, etc. As a result, a further small change in the discharge current results in complete reconstruction of the discharge structure. Experiments showed that the critical discharge current J_{cr} is connected with the current equivalent of consumption

$$J_{\dot{m}} = \frac{em}{M}$$

where e is the ion charge, M its mass, by the relationship

$$\frac{J_{cr}^2}{J_{\dot{m}}} \approx K(\Gamma). \qquad (3.7.3)$$

Here K (Γ) is a practically universal value, which depends mainly on the geometry of system.

It follows from (3.7.3) that in the subcritical regime the maximum exhaust velocity of the working substance (see (1.1.1)) is

$$v_{max} \sim \frac{J_p^2}{\dot{m}} \sim \frac{J_p^2}{J_{\dot{m}} M / e} \sim \frac{K}{M} \qquad (3.7.4)$$

In other words, the velocity with this geometry of the accelerator increases with a decrease of the atomic mass of the working substance. Therefore, the maximum velocities in the coaxial (quasi-) stationary

plasma accelerators with the intrinsic magnetic field were achieved using hydrogen and lithium as the working substance.

The expression for the maximum discharge velocity can be written in another form, using Bennett's formula (2.4.10) for a cylindrical *Z*-pinch. This is especially reasonable in the case of end plasma accelerators. Then, assuming that the number of particles N per unit of the length of the pinch is equal ($\dot{m}$ is the mass flow rate)

$$N = \frac{\dot{m}}{v_{\max}} \tag{3.7.5}$$

we obtain, using (2.4.10) and (3.7.3),

$$v_{\max} = \theta \frac{J_{\text{cr}}^2}{c^2 \dot{m}} = \frac{2NkT}{\dot{m}} = \frac{2kT}{v_{\max} M} \tag{3.7.6}$$

i.e. as one would expect, the critical velocity is of the order of the thermal velocity of particles.

The physical sense of critical discharge current is simple. This is the current at which the Hall parameter $\omega_e \tau_e$ in a sufficiently large volume of the accelerator channel is comparable with unity as regards the order of magnitude

$$\omega_e \tau_e \big|_{J_{\text{cr}}} \sim 1. \tag{3.7.7}$$

Assuming that (see (5.3.16))

$$\sigma \sim T_e^{3/2}, \quad n \sim \frac{J_{\dot{m}}}{v_M}, \quad v_M \sim \frac{J_p^2}{J_{\dot{m}}}, \quad H \sim J_0 \tag{3.7.8}$$

we obtain

$$\omega_e \tau_e \frac{\sigma H}{enc} \sim T_e^{3/2} \frac{J_p^3}{J_{\dot{m}}^2} \tag{3.7.9}$$

Naturally, electronic temperature increases with an increase in the discharge current. Assuming

$$T_e \sim J_p^{\alpha} \tag{3.7.10}$$

we obtain the condition for the appearance of a 'crisis'

$$\omega_e \tau_e \sim \frac{J_p^{3+\frac{3}{2}\alpha}}{J_{\dot{m}}^2} \sim 1 \tag{3.7.11}$$

Hence it is apparent that after accepting $\alpha = 2/3$, which does not

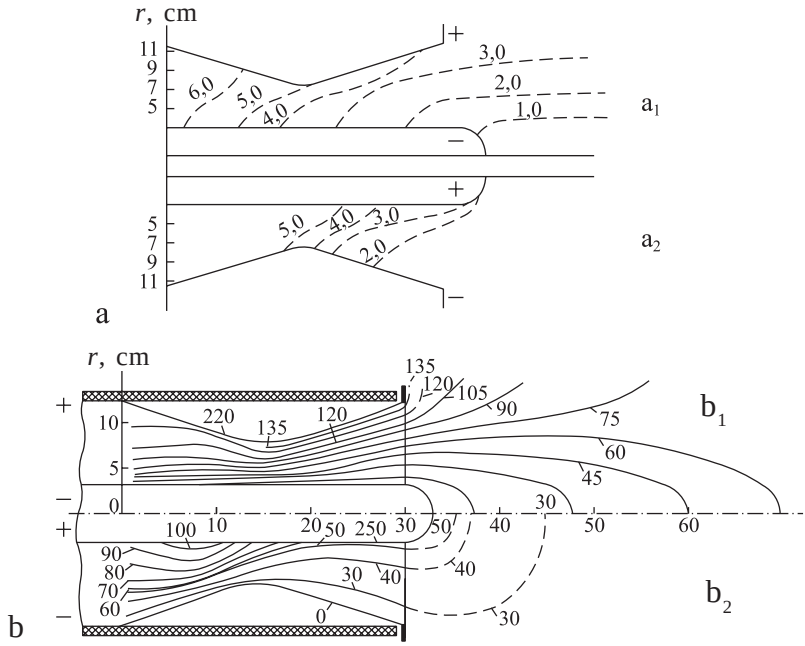


Fig. 3.7.7. Distribution of lines of current (a_1, a_2) and equipotentials (b_1, b_2) at different polarities in KPU-1 in supercritical regime. Working substance – nitrogen, $\dot{m} \approx 5$ g/s, $J_p = 35$ kA.

contradict clearly experience, we obtain criterion (3.7.3).

The charts of the lines of electric current and equipotentials at supercritical discharge currents were taken in the installation shown in Fig. 3.7.4. They are shown in Fig. 3.7.7 for two electrode polarities [90]. Here the slips of current along the anode and the formation of a potential jump near the anode are clearly visible. The effect of this pnenomenon on the surface condition of the anode will be discussed later (section 7.6.6) and here we only note a special feature in which the central electrode is an anode. In this case the slip of current is limited by the end of the anode, and a large amount of energy is generated at this end within a small spot with a diameter of ~1 cm. Consequently, this zone melts already after ~1 ms at a current of ~40 kA regardless of the anode material – copper or tantalum.

Concluding the brief survey of the work of the KPU-1 accelerator, let us note one additional fact. If we make a longitudinal slot in the external electrode and record glow with the aid of a high-velocity movie camera, then we can see the behaviour of the ionisation front. It fluctuates – oscilations in the logitudinal direction are well visible in hydrogen (Fig. 3.7.8a), and qualitatively the picture resembles that resulting from the calculations (see 6.9). However, another form of the

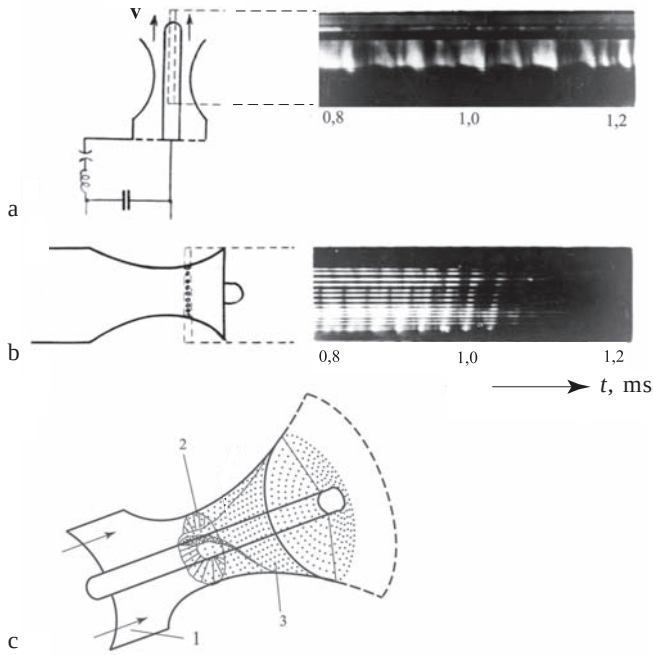


Fig. 3.7.8. Streak images of the oscillations of the ionisation front recorded through a transparent slit: a) longitudinal oscillationg at discharge in hydrogen; b) rotational instability of the ionisation front at discharge in nitrogen; c) flows in rotational instability; 1) neutral gas, 2) ionisation front, 3) region of a dense plasma flow.

instability of the ionisation front was found in nitrogen – 'rotational spin' instability (Fig. 3.7.8b) whose diagram is shownin Fig. 3.7.8c.

3.7.3. Coaxial quasi-steady high-current plasma accelerator with ionic current transfer (QHPA)

The schematic diagram of the QHPA is presented in [93–95]. It was shown in 3.2.4 that 'Hall' current transfer can be stopped using non-equipotential electrodes. This method is used in MHD generators with a dense weakly ionised gas and with the relatively low flow velocities ($\lesssim$1 km/s). The situation in the plasma accelerator is completely different.Density is low and velocities are high, measured in hundred and thousands of km/s. Here the splitting of electrodes is unacceptable since the electrodes must have magnetic shielding from the high energy flow in the channel and this is not possible with splitting. The analysis of the flows of ideal plasma in axisymmetrical channels, carried out in section 3.6, showed that it is possible to leave electrodes equipotential, but then it is necessary to ensure discharge current transfer by ions

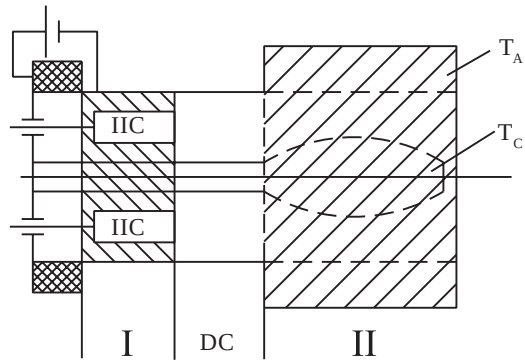


Fig. 3.7.9. Structural diagram of the QHPA: I – first stage, IIC – input ionisation chamber, DC – drift channel, II – second stage, T_A – anode transformer, T_C – cathode transformer.

(Fig. 3.6.3). To achieve this, the anode must be capable of emitting ions, and the cathode to accept these ions. Thus, the role of continuous metal electrodes should be played by fairly complicated plasma-dynamic systems which are capable of linking ionic current transfer in the channel of accelerator with the electronic current transfer in the external circuit. Such systems are referred to as 'transformers'.

There is another essential element of QHPA. Earlier, when describing the process in a simple coaxial quasi-stationary accelerator KPU-1 it was noted that the ionisation front was unstable. In order to avoid problems with this instability, a two-stage system was accepted in the construction of QHPA: the gas entering the accelerator first travels into the input ionisation chambers (IIC), where it is ionized and acquires the certain initial velocity. These IICs were also made from coaxial heavy current plasma accelerators which worked when $J_p < J_{cr}$.

4–6 IICs were used in the constructed QHPA. The plasma flows generated by them travelled into the 'drift' channel where they spread across the azimuth and were adjusted slightly with respect to the isomagnetic configuration. This was accompanied by the generation of current vortices. As a result, the overall diagram of QHPA takes the form shown in Fig. 3.7.9.

Transformers

The schematic diagrams of the functioning of transformers are depicted in Fig. 3.7.10. The anodic transformer (Fig. 3.7.10a) has four basic elements: 1 – gas-discharge devices anodic ionisation chambers (AIC), which generate plasma from the neutral gas entering them, 2 – the magnetic system which forms circuits for the motion of electrons and ions in the volume of the transformer, and also screening – where

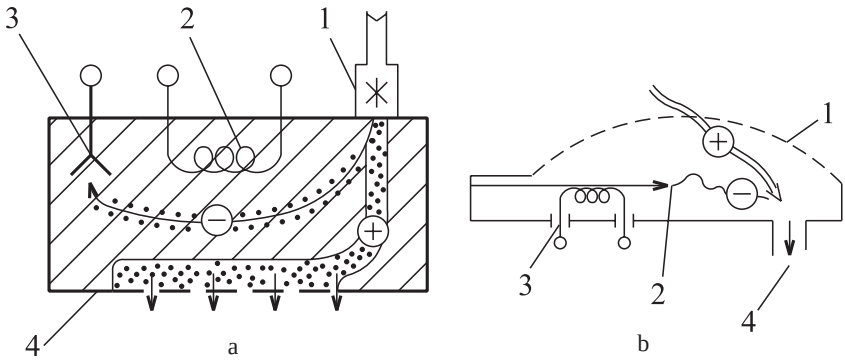


Fig. 3.7.10. Functional diagrams of transformers: a) anode transformer, b) cathode transformer.

necessary – the surface of the solid-state elements of the transformer, 3 – the receiver of the electrons, closing the volume of the transformer with the external electrical circuit, 4 – the magnetic surface, which separates the transfomer and the accelerating channel. Through this magnetic anodic surface (MAS) the current-carrying ions move from the transformer into the channel.

The functional diagram of a cathodic transformer is depicted in Fig. 3.7.10b. The transformer contains elements 'reciprocal' to the anodic transformer: 1 – the magnetic cathodic surface (MCS), accepting the ions falling from the accelerated flow, 2 – the emitter of electrons, connected with the external circuit and supplying electrons for compensation of the volume charges of the ions emitted from the accelerated flow, and thus for 'the interception' of the function of current transfer from the ions to the electrons. Finally, the composition of the cathodic transformer includes the magnetic system (3) and the

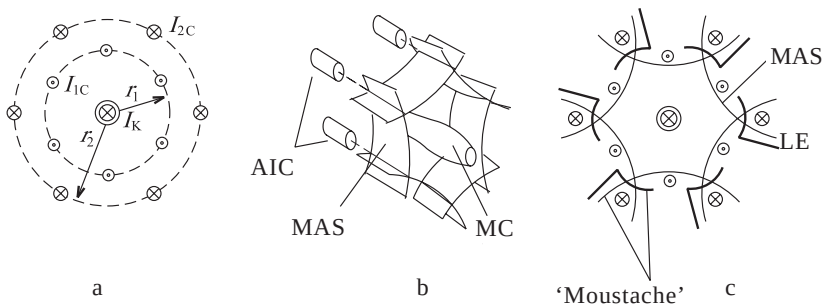


Fig. 3.7.11. Magnetic system of the anode *z*-transformer ($Z–T_A$); cross sections of the conductors, forming the magnetic configuration (a), the type of separatrix seurface (b), position of locking electrodes (LE) (c); AIC – anodic ionisation chamgers, MES – magnetic emitting surface, MC – cathodic ion-accepting surface.

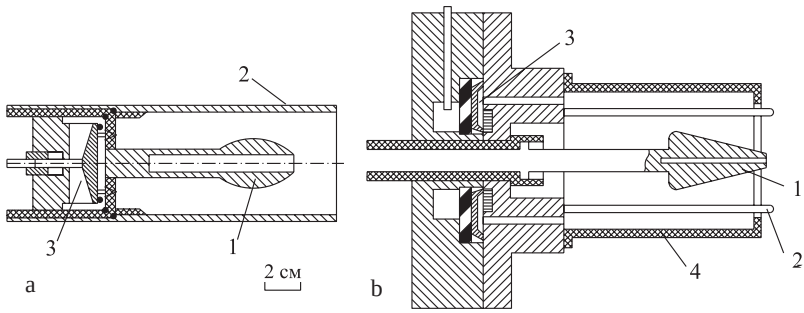


Fig. 3.7.12. Diagrams of the variants of anodic ionisation chambers (AIC) and input ionisation chambers (IIC): a) model of the Khar'kov Physico-Technical Institute; b) model IF, Minsk. The schemes show: 1) cathode, 2) anode, 3) valve, 4) jacket.

plasma discharge unit of the de-energized plasma.

The use of transformers, i.e. passage to ionic current transfer, even more accentuates the problem of the magnetic shielding of electrodes. This screening can be realised by the magnetic fields of greatly different configurations. Figure 3.7.11 shows the magnetic system of anodic *Z*-transformer (Z–T_A). This name is used because the magnetic configuration is formed here mainly by rods with current stretched along the axis. The figure shows magnetic surfaces: MAS – anodic, emitting ions, and MC – cathodic, accepting ions. The plasma, which spreads over the entire magnetic anodic surface, is injected in the anodic transformer along the zero lines of the magnetic field for special injectors – anodic ionisation chambers (AIC). The diagram of the AIC is shown in Fig. 3.7.12, and they have the form of small quasi-stationary plasma accelerators – QPA, working at subcritical discharge currents. The acceptors of the electrons have the form of bent sheets, intersecting the separatrix (Fig. 3.7.11c). The rods with current, forming the magnetic configuration of the anodic transformer, are efficiently screened by the magnetic field. In fact, the energy of the plasma in the volume and in the vicinity of the transformer is small. The schematic view of the QPA with the anodic and cathodic *Z*-transformers is shown in Fig. 3.7.13. The acceptor of the electrons in Z–T_A has the form of a multi-arched surface, situated under the earth potential. It is shown clearly in the photograph in Fig. 3.7.14a.

Attention will now be given to cathodic transformers. Like the anodic transformers, they can be produced by different methods. However, till now only the cathodic *Z*-transformer has proved to be capable of operation. Its external appearance is shown clearly in Fig. 3.7.14b. The *Z*-transformer is assembled from copper pipes bent as shown in the figures. Inside the pipes there are high-current cables

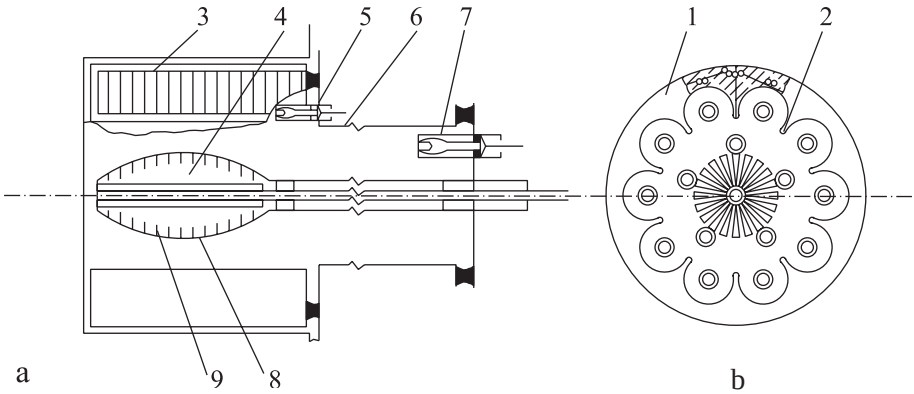


Fig. 3.7.13. General diagram of the QHPA: section by the plane (r, z) (a), section by the plane z = const (b): 1) anodic transformer, 2) anodic collector, 3) magnetic system of the anodic transformer, 4) cathodic transformer, 5) anodic ionisation chamber (AIC), 6) drift channel, 7) the input ionisation chamber (IIC), 8) copper pipes of the cathodic transformer, 9) needle cathodic emitters.

powered from an external source. The magnetic field of the cables facilitates the formation of a nozzle-like channel of the QPA and it also provides thermal protection of the pipes against the 'current-carrying' ions, falling out from the flow on the cathode. Needles (9) are secured on the internal (in relation to the cathode) side of the pipes. The needles are electron emitters because they ignite arc discharges. Thus, the ions and electrons, arriving into the transformers, form 'de-energised' plasma which travels through slits in the central tube into this tube and is discharged to the outside.

Construction of QHPA and experiments

The most advanced QHPA – its transformers are shown in the photograph in Fig. 3.7.14 – was constructed by the Institute of

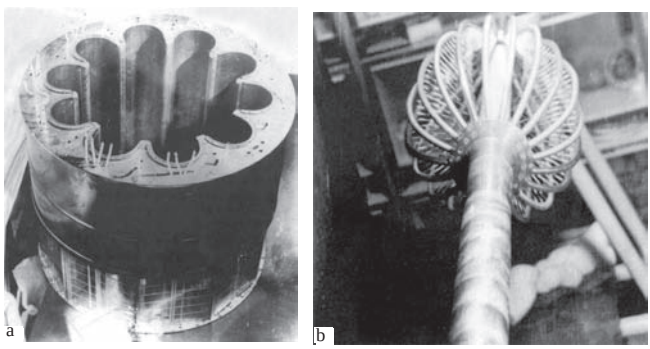


Fig. 3.7.14. QHPA Transformers K50: a) anodic, b, cathodic.

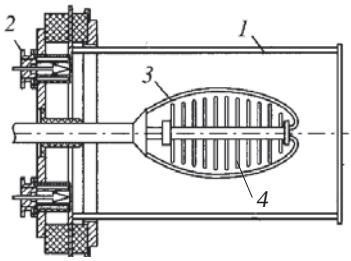


Fig. 3.7.15. Diagram of P-50 QHPA (Institute of Physics, National Academy of Sciences of Belorus, V.M. Astashinskii): 1) anode rods; 2) input ionisation chambers (IIC); 3) cathode transformer pipes, 4) needles.

Atomic Energy, the Scientific Research Institute of Electrophysical Apparatus (NIIEFA) and the Khar'kov Physico-Technical Institute (A.I. Morozov, V.I. Tereshin and colleagues) [95, 96]. The system parameters were as follows: total length ~1.5 m, the diameter of the MA approximately 0.5 m, the number of AICs 10, the number of IIC 5. At the discharge current of ~0.5 MA (in the regime of ionic current transfer) and the discharge voltage of ~10 kV over a period of ~0.2 ms (this time was determined by the energy reserve in the supplying condensers), the QHPA generated an ion flux with the equivalent current of $J_{\dot{m}} \sim 10$ MA and the hydrogen ion energy of ~1 keV. This energy was described by the equation (2.6.2)

$$v_{\max} = \sqrt{2} C_A$$

As indicated by theoretical considerations, the local velocity of the flows v in the critical cross-section is equal to the Alfvén velocity, since the thermal addition was small. The experimentally recorded distribution of current and potential in the QHPA channel, and also the integral characteristics are in good agreement with theory.

In conclusion, it should be mentioned that the efficiency of the simplified variant of QHPA (Fig. 3.7.15) was also sufficiently high over a short period of time (~0.1 ms). In this case, instead of the transformer, the anode is produced from rods, and the current-carrying ions are supplied into the channel from the side of the anode as a result of ionisation of the surrounding residual gas [97].

3.7.4. Quasi-stationary compression flows [98, 99]

In the previous chapter in section 2.6.1 it was mentioned that the plasma flows in the transverse magnetic fields are of two types: accelerating and compression. The investigations of accelerating flows have already been discussed. Attention will now be given briefly to the experiments with compression flows.

The principal special feature of these flows is the convergence of the flow to the axis and the formation of the compression region (CR) in the vicinity of the axis. The formation of this region takes place in almost all coaxial systems with an intrinsic magnetic field, although the plasma parameters in the compression region in the accelerators are usually far away from the limiting values, defined by the equations (2.6.10) and (2.6.11). The presence of these regions, as already mentioned, can be seen by the 'naked eye' on the basis of brighter glow (Fig. 3.7.6)

To ensure higher compression, the central electrode is produced with a smootly decreasing diameter (Fig. 3.7.3c). The degree of compression, as indicated by equation (2.6.10), increases with a decrease of the polytrope parameter or, in other words, with an increase of the intensity of release of energy as a result of radiation. Therefore, compression is especially high in operation with heavy gases. For example, at a relatively low discharge current (~50 kA), the degree of compression in nitrogen was ~500. In hydrogen it was correspondingly lower, by a factor of ~70 times. Unfortunately, regardless of the number of attempts, compression flows have not been sufficiently investigated, modelled or optimised. Information on the macrostability of these flows is provided by Fig. 3.7.16 which shows the frames recorded using a super high-

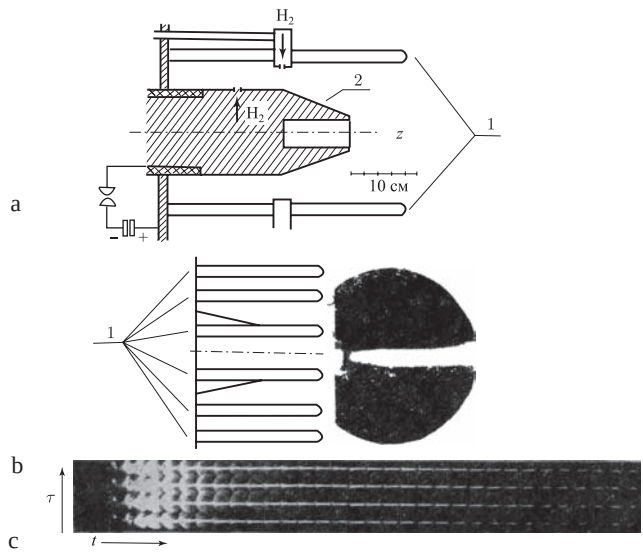


Fig. 3.7.16. Magnetic plasma compressor (MPC): a) schematic: 1) anode (outer electrode), consisting of a system of straight rods, 2) conical cathode with a central orifice, c) flow in the MPC, He, p_0 = 1 torr, U_0 = 3 kV, c) streak images, showing the evolution of the compression region with time. The very long and highly stable compression region can be seen.

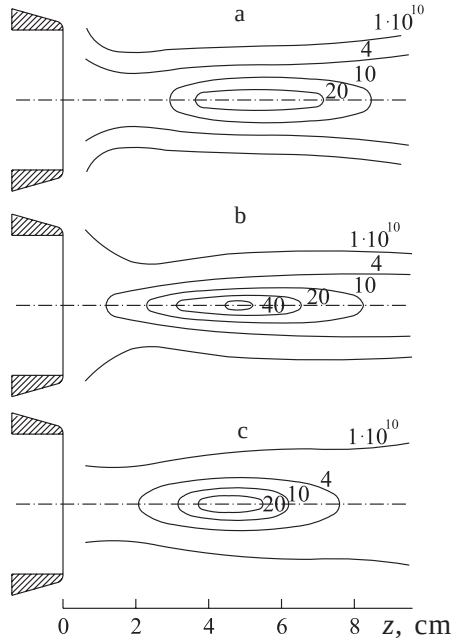


Fig. 3.7.17. Instantaneous distribution of dense plasma in the compression region of the flow, leaving the MPC (Institute of Atomic Energy), p_0 = 0.25 torr, a) t = 30 μs, discharge current J_p = 205 kA, b) t = 40 μs, J_p = 160 kA, c) t = 60 μs, J_p = 60 kA.

velocity camera, and in Fig. 3.7.17 which shows the distribution of the density of plasma recorded using a laser interferometer at different moments of time. It should be mentioned that the exit of neutrons from the compression region was detected at the previously mentioned discharge parameters using deuterium [100].

Figure 3.7.18 shows an example of the formation of the compression region (more accurately, the focusing region) of low-density current as a result of the radial component of the kinetic energy of the particles, leaving a pulsed gun (M.I. Pergament, [44]). Today, the erosion version of the MPC is usually used as a strong source of incoherent radiation.

3.8. Dynamics of plasma flows in magnetic fields

In previous sections we discussed plasma flows, 'pushed' into the gap between the walls of a channel. In practice, in addition to this phenomenon, it is often necessary to consider the 'free' (in a specific meaning of the word) plasma flows which move in the magnetic fields of different configuration. The following cases will be investigated in this section:

1. movement of a flow (bunch) separated from the walls in the

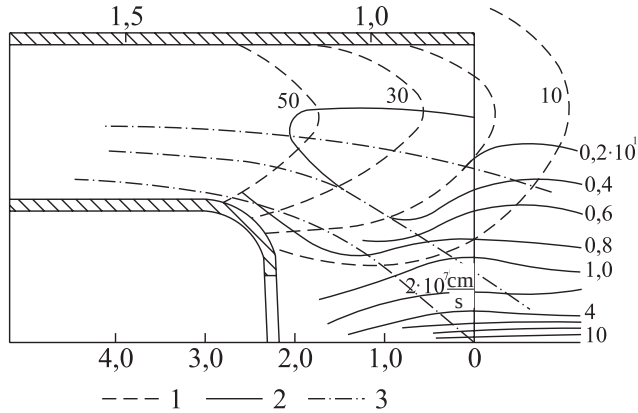


Fig. 3.7.18. Formation of the 'inertial' compression region: 1) current lines, 2) equal density lines, 3) the ion trajectory (TRINITI, M.I. Pergament).

transverse magnetic field and its auto-polarisation [102, 103];

2. one-dimensional diffusion of non-polarised plasma in a transverse nonuniform magnetic field;
3) special features of the entry of non-polarised plasma in the magnetic field [104, 105].

3.8.1. Movement of auto-polarised plasma flow (bunch) in a transverse magnetic field [102, 103]

Attention will be given to the movement of a plasma bunch in a transverse magnetic field or of a flow situated in such a field and restricted in the transverse direction. If the Larmor radii of the electrons are considerably smaller than the characteristic dimensions of the bunch b:

$$\rho_{He} \ll b, \tag{3.8.1}$$

then at first sight it can be assumed that this bunch cannot move in this magnetic field. However, this is not the case. It appears that if, for example, a gun placed in a magnetic field is used to generate a bunch, the latter will also move after leaving the gun. This is associated with the fact that 'auto-polarisation' of the bunch takes place (Fig. 3.8.1) and, consequently, a uniform transverse electrical field, referred to as the polarisation field, appears in the volume of the bunch. The auto-polarisation phenomena may be described as follows. Falling into the magnetic field, the ions and electrons are deflected to different sides under the effect of this magnetic field and, consequently, they cover the sides of the flow with positive and negative charges. Under the effect

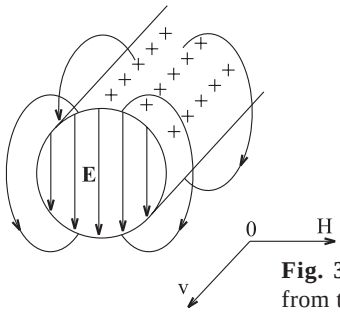


Fig. 3.8.1. Auto-polarisation of a plasma bunch, separated from the walls and moving in the direction across the uniform magnetic field.

of the magnetic field and the polarisation field the particles, situated in the plasma volume, drift with the velocity equal to the initial velocity of the bunch v_0:

$$v_0 = c\frac{[\mathbf{E},\mathbf{H}]}{H^2} \tag{3.8.2}$$

It can be seen that the equation (3.8.2) is equivalent to the condition (at $p_e \to 0$)

$$\mathbf{E} = -\frac{1}{c}[\mathbf{v}_0,\mathbf{H}] = 0 \tag{3.8.3}$$

If it is assumed that there is a vacuum outside the bunch, the electrical field satisfies the Laplace equation and has the form shown in Fig. 3.8.1. However, if low-density plasma is found outside the bunch, the pattern of the electrical field can be far more complicated.

This pattern of the movement of the plasma bunch leads to a number of very important consequences. In particular, it can be seen that if polarisation is removed by some method, the movement of the bunch is arrested. It may also be seen that the removal of polarisation indicates the passage of current through the bunch and some external circuit. The interaction of the current, flowing through the bunch, with the magnetic field generates an ampere force decelerating the movement of the bunch. It may be concluded that we have the model of an MHD generator in this case. Further, the transverse electrical field on the sides of the bunch where the polarisation charges are distributed is smaller than in the volume of the bunch. Therefore, the velocity of movement of the sides of the bunch is lower than the velocity of the centre of the bunch. Consequently, a plasma train is dragged behind the bunch. This is accompanied by the spreading of the plasma bunch, especially polarisation zones, along the magnetic field. The experiments confirm efficiently this considerations.

Several methods of removing polarisation will be investigated.

a. If the plasma bunch (or a flow restricted in the transverse direction) moves in a metallic tube, the plasma, spreading along the magnetic field, reaches the tube and the sides of the bunch with the opposite charges come into contact through the plasma. Consequently, depolarisation takes place.

b. Depolarisation can also take place during movement of the bunch in a dielectric tube with a considerably greater diameter than that of the bunch. Here, spreading again along the lines of force, the plasma reaches the pipe and if the **E**-field, produced by polarisation (3.8.1), is sufficiently strong then as a result of 'near-wall conductivity' (section 7.3.2) a current flows in the vicinity of the dielectric wall and this current carries out depolarisation. The value of voltage U between the side walls of the bunch is indicated by the following example. At $H = 3 \cdot 10^3$ Oe, $v = 10^7$ cm/s, $D = 5$ cm, we obtain

$$U_{AB} = D\frac{v}{c}H = 1500 \text{ V}$$

here D is the width of the bunch.

c. In section 1.7, attention was given to a toroidal drift, leading to polarisation, and to methods of preventing this phenomenon. The capacity of the toroidal configuration with the inserted magnetic surfaces to remove polarisation complicates the problem of injection of plasma into such systems. Therefore, high-energy plasma is usually generated in the toroidal systems by heating with Joule heat in the trap or by means of microwaves, or by injection of high-energy neutral atoms which penetrate the magnetic fields and are subsequently ionised in the plasma volume (for more details see section 10.1).

3.8.2. One-dimensional classic diffusion of plasma in a magnetic field

In the previous section we discussed the significant role played by the formation of the electrical field in plasma in the movement of the plasma in the **H**-field. However, if the **E**-field is not present, i.e. the conditions are such that all the polarisations are removed, there is comparatively slow diffusion of plasma across the magnetic field.

We examine the simplest case in which the strength of the magnetic field has only one component H_z which depends on the orthogonal coordinate x

$$\mathbf{H} = (0, 0, H(x, t)).$$

All other quantities will also be considered to be dependent on x and t.

Taking into account the fact diffusion is a slow process, in the equation for **v** we can ignore the inertial term and, because of the quasi-neutrality of the x-component, the velocity of the electrons and the ions should be equal:

$$v_{xi} = v_{xu} \tag{3.8.4}$$

Consequently, the continuity equation, the equilibrium equation and the generalised Ohm's law, taking into account the assumption that **E** = 0, can be presented in the following form:

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} n v_x = 0; \quad \frac{\partial p}{\partial x} = \frac{1}{c} j_y H; \quad \frac{j_y}{\sigma} = -\frac{1}{c} v_x H \tag{3.8.5}$$

Consequently

$$v_x = -\frac{cj}{\sigma H} = -\frac{c^2}{\sigma H^2}\frac{\partial p}{\partial x} \tag{3.8.6}$$

and therefore

$$\frac{\partial n}{\partial t} = \frac{\partial}{\partial x}\left(\frac{nc^2}{\sigma H^2}\frac{\partial p}{\partial x}\right) \tag{3.8.7}$$

If the plasma temperature is regarded as constant and we introduce the magnetic viscosity ν and the parameter β:

$$\nu = \frac{c^2}{4\pi t}, \quad \beta = \frac{8\pi p}{H^2}$$

equation (3.8.7) can be presented in the form of a conventional diffusion equation

$$\frac{\partial n}{\partial t} = \frac{\partial}{\partial x} D \frac{\partial n}{\partial x} \tag{3.8.8a}$$

where

$$D = \nu \frac{\beta}{2} \tag{3.8.8b}$$

The resultant equation for the diffusion coefficient of plasma in the magnetic field contains two characteristic features. Firstly, the equation includes the magnetic viscosity ν_m determined unambiguously by the plasma conductivity which, in turn, is determined by the frequency of ion–electron collisions and not by the frequency of ion–ion or electron–electron collisions. This is explained by the fact that at collisions of

identical particles in a homogeneous magnetic field the centre of the masses does not move.

The second special feature of equation (3.8.8) is that the diffusion coefficient of plasma differs from ν_m, the diffusion coefficient of the **H**-field in plasma, by the multiplier $\beta/2$, i.e. the rate of diffusion of the field into the plasma is higher than that of the plasma into the field.

Another comment should be made. The toroidal equilibrium configuration of the magnetic field, which represent a system of closed toroidal surfaces inserted into each other, have been discussed several times. In most cases, the strength of the field along the small azimuth ω is not constant (Fig. 3.8.2). Therefore, because of (3.8.7), the intensity of diffusion depends strongly on ω. Consequently, the process of plasma transfer between the surfaces consists of two stages. In particular, where $|\mathbf{H}|$ is lower, the density of the diffusion flow is higher. However, in addition to this slow process, there is also the equalisation of the density on the surface as a result of the rapid movement of the plasma along the lines of force. This two-stage nature of the process complicates specific calculations.

3.8.3. Entry of the plasma flow into the magnetic field

If the plasma flow without the magnetic field meets the magnetic field in its travel, there can be two variants of the process. Either the flow is inhibited by the field forming a layer of almost stationary plasma, or the flow suppresses the field and continues to move and gradually diffuses into the surrounding field. In the second variant, the situation depends strongly on the presence or absence of auto-polarisation.

In the first variant, the pressure of incident plasma is balanced by the magnetic pressure

$$nMv_{0i}^2 + p_e = \frac{H_0^2}{8\pi} \tag{3.8.9}$$

If the free path of the particle is sufficiently long then, according to the calculation, the plasma–field transition layer is of the order of the

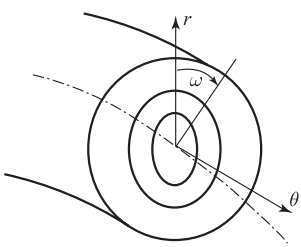


Fig. 3.8.2. A fragment of toroidal magnetic configuration in the toroidal coordinate system.

electronic Larmor radius calculated from the energy of the incident ions (usually $kT_e \ll Mv_{0i}^2/2$). An electrical field forms in the transition layer and the ions are decelerated within the limits of this layer. The theory and experiment (both in the laboratory and in space in investigating the boundaries of the magnetosphere) show that this transition layer is unstable and rapidly spreads to a thickness of the order of $\ell \sim \sqrt{\rho_i \rho_e}$, where is ρ the Larmor radius.

The presence of instability of the transition layer was most evident in the experiments carried out by Kartashev and Pistunovich in the form of microwave and x-ray radiation surges with the incidence of a plasma bunch on the magnetic field, and the energy of the x-ray quanta corresponds to the electron energy equal to the energy of the incident ions [104].

The experiments showed that, in many cases, the plasma flow penetrates into the plasma in the absence of auto-polarisation and at

$$\rho v_0^2 < \frac{H_0^2}{8\pi}.$$

This is associated with the fact that at contact of a quasi-flat plasma front with the field, the front is characterised by the formation of sharp projections which also penetrate into the field (V.F. Demichev, V.M. Strunnikov [105]). This greatly accelerates the penetration of the magnetic field into the plasma, Fig. 3.8.3.

The processes of entry of plasma into the magnetic field has been investigated in detail for dense collisional plasma at low magnetic Reynolds numbers, in connection with the problem of MHD generators and MHD flowrate meters [79,80]. The simplest case will be investigated. It is assumed that the magnetic field, oriented mainly along the *z* axis in a flat channel (*x*,*y*) with a constant cross-section increases from 0 to some constant value H_0 and subsequently decreases to 0, and a plasma flow enters this channel with constant velocity v_0.

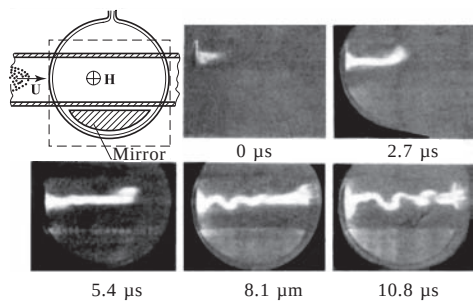


Fig. 3.8.3. Entry of the plasma jet into a transverse magnetic field.

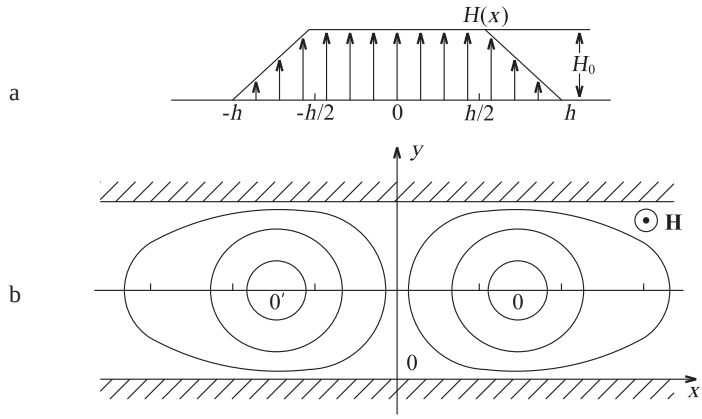


Fig. 3.8.4. Dependence of the strength of the magnetic field in the channels on the longitudinal coordinate x (a) and the current vortices in a dense mass flow at entry and exit from the magnetic barrier (b).

Consequently, as shown by theory and experiments, current vortices, closing outside the magnetic field (Fig. 3.8.4), formed at entry into and exit from the magnetic field. Evidently, as a result of the vortices the flow is decelerated both at entry and at exit. The current in the vortex and the dimensions of the vortices are determined by the conductivity of the medium and by the Hall parameter. If calculations are carried out taking into account the Hall effect, the electromagnetic problem can be solved even in the two-dimensional case (x, y) assuming that the magnetic field is of the 'stepped' nature ($H = 0$ at $x < 0$, $x > L$ and $H = H_0 = \text{const}$ at $L > x > 0$). Consequently, at $\sigma \to \infty$, the Joule losses, related to 1 cm of length along the z axis, are equal to a finite value which does not depend on plasma conductivity [80]:

$$Q_\infty \approx 0.4 h^2 v_0^2 en \frac{H_0}{c^2}. \qquad (3.8.10)$$

Attention should be given to the fact that $Q_\infty \sim h^2$ and not h.

The fact that the losses Q_∞ are not equal to 0 indicates that the dissipative processes at entry of plasma into the magnetic field are unavoidable and the proportionality of Q_∞ to the square of the width h^2, and not the first degree, confirms the previously mentioned fact that the one-dimensional model of entry of the plasma into the magnetic field can not be constructed.

4

Collisionless kinetic models of processes in plasma Vlasov-Maxwell equations

In the previously investigated block and hydrodynamic models of plasma, no account was made of the effect of peculiarities of velocity-dependent particle distribution functions on the plasma dynamic processes. Instead of this the dynamics equations containing macrocharacteristics: density, velocity, pressure; and the relationship of pressure with density was considered on the basis of formal simplicity considerations. This approach is justified in many cases and provides (not only qualitatively) the accurate scale of the process parameters.

However, in addition to the quasi-hydrodynamic systems, there are also plasma systems which can be described only by taking into account in the explicit form the special features of the distribution functions of either ions or electrons or of both particles at the same time. In the final analysis, this is associated with the longer free path and many plasma systems can operate only with long free paths.

Several characteristic examples will now be discussed.

1. A suitable example of the principal importance of the ion distribution function (IDF) may be the Budker–Post mirror trap discussed in section 1.7. It was mentioned that the trap can contain only particles with a relatively low longitudinal component of the velocity. However, as a result of the collisions, the particles move in the space of velocities and sooner or later they arrived at the ‘departure cone’. Evidently, the lifetime of the particle outside this cone is of great importance for practice.

2. The intensity of the processes of excitation, ionisation and radiation depends strongly on the electron distribution function (EDF). In turn, the distribution functions of the heavy particles with respect to

the energy levels ('population of the levels') determine many properties of the plasma, including the possibility of generation of coherent (laser) radiation (section 6.11.1)

3. It is quite easy to generate in the plasma volume the EDF consisting of several groups of the electrons moving in relation to each other[1]. For example: this can be carried out simply by injecting a beam of electrons into 'stationary' plasma. This takes place automatically in electrical discharges where the 'beams' of this type are injected by cathodes. Usually, the presence of a beam results in the acceleration of the Langmuir and other oscillations directly in the vicinity of the cathode thus resulting in fast 'Maxwellisation' of the flow leaving the cathode.

This list, demonstrating the role of the distribution function of the particles, can be extended as much as required.

However, analysis of the particle kinetics not only makes it possible to describe these processes but also determine more accurately the conditions of applicability of the hydrodynamic models and, most importantly, derive equations for the transfer coefficient, i.e. ohmic resistance of plasma, its heat conductivity and viscosity (section 5.3.4).

4.1. Initial concepts

4.1.1. Phase space and the distribution function (DF)

The advances in computing techniques may suggest that the plasma systems will be calculated in the future quite accurately, directly on the basis of the 'first principles', i.e. Newton–Lorenz equations for all N particles, and the Maxwell equations

$$
\begin{aligned}
&M_k \frac{d\mathbf{v}_k}{dt} = e_k\left(E + \frac{1}{c}\left[\mathbf{v}_k, \mathbf{H}\right]\right); \quad m_j \frac{d\mathbf{v}_j}{dt} = -e_j\left(E + \frac{1}{c}\left[\mathbf{v}_j, \mathbf{H}\right]\right);\\
&\text{rot}\,\mathbf{H} - \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c}\left(\sum_{k=1}^{Ni} e_k \mathbf{v}_k \delta(\mathbf{x}-\mathbf{x}_k) - e\sum_{j=1}^{Ne} \mathbf{v}_j\,\delta(\mathbf{x}-\mathbf{x}_j)\right);\\
&\text{rot}\,\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t};\\
&\text{div}\,\mathbf{E} = 4\pi\left(\sum_{k=1}^{Ni} e_k - e\sum_{j=1}^{Ne}\delta(\mathbf{x}-\mathbf{x}_j)\right); \quad \text{div}\,\mathbf{H} = 0.
\end{aligned}
\tag{4.1.1}
$$

and the system of this type, of course greatly simplified, is already solved by the method of 'large' particles. However, here we are interested in analytical approaches. For this purpose, it is necessary to tidy up the

[1]See section 3.5.3

'pack of midges', i.e. a set of randomly moving particles.

This is carried out efficiently by combining the particles into slightly spreading 'droplets'. Evidently, these droplets should consist of the particles distributed closely to each other with similar velocities **v**. At the same time, it is necessary to investigate the six-dimensional, so-called phase space (**x**, **v**) and determine in this space the distribution function f (t, **x**, **v**) as the density of particles in the unit phase volume:

$$f = \frac{\delta N}{\delta \Gamma} \tag{4.1.2}$$

Here $\delta\Gamma = \delta\mathbf{x}\delta\mathbf{v} = \delta x\ \delta y\ \delta z\ \delta v_x \delta v_y \delta v_z$.

Knowing the distribution function f, we can determine any macroscopic flow characteristic. In particular, the density of the particles at the given point of the coordinate space is determined by integration of f with respect to velocitys:

$$n(\mathbf{x}) = \int f d\mathbf{v} \tag{4.1.3a}$$

Similarly, the density of the particle flow is

$$n(\mathbf{x})\bar{\mathbf{v}}(\mathbf{x}) = \int \mathbf{v} f d\mathbf{v} \tag{4.1.3b}$$

Here $\bar{\mathbf{v}}$ is the mean ('hydrodynamic') flow velocity.

'Kinetic' temperature, which is the measure of random energy of the particles, can be determined from the equation

$$\frac{3}{2} nkT(\mathbf{x}) = \int \frac{M}{2} (\mathbf{v} - \bar{\mathbf{v}})^2 f d\mathbf{v} \tag{4.1.3c}$$

since the internal random energy of the unit volume in the thermodynamic equilibrium state is (3/2) nkT. It should be stressed that the kinetic temperature, in contrast to thermodynamic temperature, can be introduced for any system at any distance from equilibrium.

If the set of the particles is in thermodynamic equilibrium, the distribution function f of these particles is Maxwellian:

$$f = n\left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left\{-\frac{m(\mathbf{v} - \bar{\mathbf{v}})^2}{2kT}\right\} \tag{4.1.4}$$

Here T is the particle temperature. Evidently, this distribution is in agreement with the equations (4.1.3).

4.1.2. Liouville equation

In a general case, the particle distribution function f is determined as the solution of some kinetic equation. This equation should describe the spreading and deformation of 'droplets' in the phase space. The procedure used for deriving this equation is very simple. Assuming that the particles are not formed, do not disappear and do not 'jump' (as a result of collisions) from one point of the phase space to another, taking into account (4.1.2), we can write

$$\frac{d}{dt}\delta N = \frac{d}{dt} f \delta\Gamma = 0 \tag{4.1.5a}$$

If it is taken into account that in any space, including phase space, with the coordinates (ξ_k) the derivative of the volume $\delta\Gamma$ is

$$\frac{d}{dt}\delta\Gamma = (\delta\Gamma)\,\mathrm{div}(\dot{\xi}_k),\ \dot{\xi} = \frac{d\xi}{dt} \tag{4.1.5b}$$

in the present case we obtain when ξ = (**x**, **v**)

$$\frac{\partial f}{\partial t} + \mathbf{v}\frac{\partial f}{\partial \mathbf{x}} + a\frac{\partial f}{\partial \mathbf{v}} + f\,(div_x \mathbf{v} + div_v \mathbf{a}) = 0 \tag{4.1.6a}$$

Here div_x and div_v are the operators of the divergence in the conventional ('configuration') space and the space of velocities, respectively, and in addition to this, it is assumed that

$$\frac{d\xi_k}{dt} = \mathbf{v};\quad \frac{d\mathbf{v}}{dt} = \mathbf{a} \tag{4.1.6b}$$

As a result of elementary transformations, equation (4.1.6a) is converted to the form:

$$\frac{\partial f}{\partial t} + \frac{\partial f}{\partial \mathbf{x}}\mathbf{v}f + \frac{\partial f}{\partial \mathbf{v}}\frac{\mathbf{F}}{M} f = 0 \tag{4.1.7}$$

which takes into account the Newton equation

$$\mathbf{a} = \frac{\mathbf{F}}{M}$$

As indicated by these considerations, no assumptions regarding the nature of forces acting on the particles were made when deriving (4.1.7). The dynamics of the charged particles is governed by the Hamilton equations. However, the Liouville theorem regarding the conservation of the phase volume applies to the Hamiltonian equations. In particular, if we define

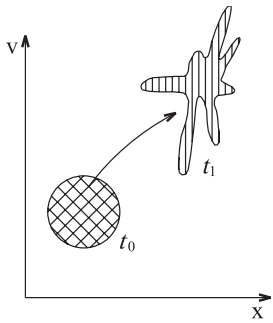


Fig. 4.1.1. Deformation of the droplet of the phase object with time.

an arbitrary region in the phase space (Fig. 4.1.1), surrounded by the particles, then, if the shape of this region does change with time, the volume of the region does not change.

$$\Gamma = \int d\mathbf{x}d\mathbf{v} = \text{const.} \tag{4.1.8}$$

Taking this fact into account, in equation (4.1.5) it should be set that $\delta\Gamma = 0$ and, consequently, we obtain the kinetic Liouville equation

$$\frac{Df}{Dt} \equiv \frac{\partial f}{\partial t} + \mathbf{v}\frac{\partial f}{\partial \mathbf{x}} + \frac{\mathbf{F}}{M}\frac{\partial f}{\partial \mathbf{v}} = 0. \tag{4.1.9}$$

This equation describe the dynamics of the cloud of independent particles moving in the potential and electromagnetic fields.

Example. A cosmonaut leaves a spaceship located far away from the Earth and tries to remove dust from a carpet. The dust particles move without collisions. The problem of the movement of the cloud of the dust particles is formulated as follows: the distribution function f_0 at the initial moment of time is given:

$$f_0 = f(0, \mathbf{x}_0, \mathbf{v}_0). \tag{4.1.10}$$

It is required to determine function f at any other moment of time.

If the spaceship travels far away from the Earth and gravity is negligible, the particles move by inertia, i.e.

$$\mathbf{x}_0 = \mathbf{x} - \mathbf{v}t; \quad \mathbf{v} = \mathbf{v}_0, \tag{4.1.11}$$

and the particle distribution function at an arbitrary moment of time t is:

$$f = f_0\,(\mathbf{x} - \mathbf{v}t, \mathrm{v}). \tag{4.1.12}$$

Evidently, this function satisfies the equation (4.1.9) at $\mathbf{a} = 0$:

$$\frac{\partial f}{\partial t} + \mathbf{v}\frac{\partial f}{\partial \mathbf{x}} = 0$$

General solution of equation (4.1.9)

We returned to the general case of quasi-linear differential equations:

$$\frac{\partial f}{\partial x_0} + a_1 \frac{\partial f}{\partial x_1} + ... + a_n \frac{\partial f}{\partial x_n} = g \qquad (4.1.13a)$$

where $a_k = a_k(x_0, x_1, ..., x_n, f)$. It is well-known that the general solution of equation (4.1.13a) can be reduced to solving a system of conventional differential equations

$$\frac{dx_0}{1} = \frac{dx_1}{a_1} = ... = \frac{dx_n}{a_n} = \frac{df}{g}. \qquad (4.1.13b)$$

In mathematics, these equations are referred to as the equations of the characteristics. In the case of the Liouville equation (4.1.9) they have the following form:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}; \quad \frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}}{M} \qquad (4.1.13c)$$

i.e. are the conventional Newton equations for the movement of the particles. These problems can be solved (see section 1.2).

In a general nonstationary case the system (4.1.13b) has six integrals with six constant ξ_{k0} which characterise the initial position and velocity of the particles:

$$\begin{aligned} &g_k(\xi_1,...,\xi_6,t) = \xi_{k0}; \quad k = 1,...,6; \\ &\xi_{1,2,3} = z, x, y; \quad \xi_{4,5,6} = v_z, v_x, v_y\,. \end{aligned} \qquad (4.1.14a)$$

If $f_0(\xi_{k0})$ is the distribution function f at $t = 0$, then at any other moment

$$f(\xi,t) = f_0(g(\xi,t)); \quad \xi = (\xi_1,...,\xi_6); \quad g = (g_1,...,g_6)\,. \qquad (4.1.14b)$$

In many cases, the stationary processes in which the plasma parameters are independent of t are of special interest. In this case, the independent variable can be conveniently represented by one of the coordinates, for example $\xi_1 = z$. Consequently, excluding t by means of the first equation (4.1.13b), we obtain five integrals

$$G_k(\xi_1,...,z) = \xi_{k0}; \quad k,i = 2,...,6 \qquad (4.1.15a)$$

and, consequently, if at $z = 0$ the distribution function is $f_0(\xi_{k0})$, then in any other cross-section

$$f(\xi,...,z) = f_0\left(G_0\left(\xi,z\right)\right). \qquad (4.1.15b)$$

4.1.3. Relationship between kinetic and hydrodynamic descriptions

Externally, the kinetic equation (4.1.9) greatly differs from the system of hydrodynamic equations. Therefore, it is important to find the relationship between them. We start with a simple example.

In a number of systems, for example, in ionic sources, the scatter of the particle velocities is small and, consequently, the distribution function f can be assumed to be proportional to the δ-function[2]

$$f(\mathbf{x},\mathbf{v}',t) = n(\mathbf{x},t)\delta\big(\mathbf{v}' - \mathbf{v}(\mathbf{x},t)\big). \tag{4.1.16}$$

In this case, the flow is characterised by four parameters: n, v_x, v_y, v_z which are functions of only four (and not seven!) independent variables t, x, y, z.

To derive equations for these functions, (4.1.9) is multiplied by 1, v'_x, v'_y, v'_0 and the resultant expressions are integrated with respect to the velocity, taking (4.1.16) into account.

Consequently, we obtain the hydrodynamic-type equations, assuming that $\mathbf{F} = e\left(\mathbf{E} + \frac{1}{c}\big[\mathbf{v},\mathbf{H}\big]\right)$

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v} = 0;$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v} = \frac{e}{M}\left(\mathbf{E} + \frac{1}{c}\big[\mathbf{v},\mathbf{H}\big]\right). \tag{4.1.17}$$

This shows that under the condition (4.1.16) the kinetic equation (4.1.9) is equivalent to the system (4.1.3).

We now investigate the arbitrary distribution function $f(t, \mathbf{x}, \mathbf{v}')$ and take into account the relationship of f with the hydrodynamic parameters (4.1.3)

$$\begin{aligned} n(t,\mathbf{x}) &= \int f(t,\mathbf{x},\mathbf{v}')d\mathbf{v}', \\ n\mathbf{v} &= \int \mathbf{v}' f(t,\mathbf{x},\mathbf{v}')d\mathbf{v}'. \end{aligned} \tag{4.1.18}$$

The kinetic equation is used without any relationship with the type of particle and without specifying the force acting on the particle, assuming that it is a function of t and $\mathbf{x}$[3]

[2]In contrast to (4.1.13), here the velocity, included in f, is indicated by the prime, and the hydrodynamic velocity is without the prime.

[3]Velocity is included in **F** as the Lorenz force, and if it is taken into account our derivations will not change significantly.

$$\frac{\partial f}{\partial t} + \mathbf{v}' \frac{\partial f}{\partial \mathbf{x}} + \frac{\mathbf{F}(t,\mathbf{x})}{M} \frac{\partial f}{\partial \mathbf{v}'} = 0 \tag{4.1.19}$$

If this equation is integrated with respect to $\mathbf{v}'$, we obtain

$$\frac{\partial}{\partial t}\int f d\mathbf{v}' + \frac{\partial}{\partial \mathbf{x}}\int \mathbf{v}' f d\mathbf{v}' + \frac{\mathbf{F}(t,\mathbf{x})}{M}\int \frac{\partial f}{\partial \mathbf{v}'} d\mathbf{v} = 0. \tag{4.1.20a}$$

Or, taking into account (4.1.18) and converting f to zero at $v \to \pm\infty$, we have

$$\frac{\partial n}{\partial t} + \frac{\partial f}{\partial x} v_x n + \frac{\partial}{\partial y} v_y n + \frac{\partial}{\partial z} v_z n = 0 \tag{4.1.20b}$$

i.e. we obtain the conventional hydrodynamic continuity equation

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v} = 0 \tag{4.1.20c}$$

The equation is now multiplied by v_x' and integrated with respect to $d\mathbf{v}$:

$$\frac{\partial}{\partial t}\int \mathbf{v}_x' f d\mathbf{v}' + \frac{\partial}{\partial x}\int \mathbf{v}_x' \mathbf{v}_x' f d\mathbf{v}' + \frac{\partial}{\partial y}\int \mathbf{v}_x' \mathbf{v}_y' f d\mathbf{v}' + \frac{\partial}{\partial z}\int \mathbf{v}_x' \mathbf{v}_z' f d\mathbf{v}' = \frac{nF_x}{M} \tag{4.1.21a}$$

Or

$$\frac{\partial}{\partial t} n\mathbf{v}_x + \frac{\partial}{\partial x_\beta} P_{x\beta} = \frac{nF_{x\alpha}}{M}, \quad (\alpha, \beta = x, y, z) \tag{4.1.21b}$$

The equations (4.1.20) and (4.1.21) are the hydrodynamic-type equations since the quantities n, $\mathbf{v}$, $P_{\alpha\beta}$ included in them are the functions of the spatial coordinates and time ($\mathbf{x}$,t). However, the equations have a single 'non-hydrodynamic' feature: they do not form a closed system. In fact, the change of n with time is determined by the flow of the particles $n\mathbf{v}$. In turn, the flow of the particles is determined by the first momentum of the kinetic equation, i.e. (4.1.21). However, the latter equation contains the tensor of the momentum flow $P_{\alpha\beta}$. To obtain the equations for determining $P_{\alpha\beta}$, it is necessary to calculate the second momentum of the kinetic equation multiplying this equation prior to integrating by $v_\alpha' v_\beta'$. But then we obtain the divergence of the tensor of the third order

$$q_{\alpha\beta\gamma} = \int v_\alpha' v_\beta' v_\gamma' f d\mathbf{v}'$$

and so on. This shows that the convective term ($\mathbf{v}\partial f/\partial \mathbf{x}$) in calculating n-th momentum of the equation generates the (n+1) momentum f, and

the chain of the momentum equations is infinite. Thus, in a general case, the single kinetic equation is equivalent to the infinite chain of the hydrodynamic-type equations.

However, at these or other assumptions the chain of the equations can be broken and, consequently, we obtain the 'conventional' or 'almost conventional' hydrodynamics. This will be discussed in detail in section 5.3.4.

We now return to equation (4.1.21b) and transform it to a more customary form. For this purpose, the total particle velocity **v** is regarded as the sum of the average ('hydrodynamic') velocity **v** and the random component **w**:

$$\mathbf{v}' = \mathbf{v} + \mathbf{w}, \quad \mathbf{v} = \mathbf{v}(\mathbf{x}, t)$$

The random component is determined by the condition

$$\int \mathbf{w} f d\mathbf{v}' = 0 \tag{4.1.22a}$$

Consequently

$$P_{\alpha\beta} = \int (v_\alpha + w_\alpha)(v_\beta + w_\beta) a(\mathbf{v} + \mathbf{w}) d\mathbf{w} = u_\alpha u_\beta n + p_{\alpha\beta} \tag{4.1.22b}$$

Substituting these expressions into (4.1.21b) we obtain

$$M\left(\frac{\partial}{\partial t} n v_a + \frac{\partial n u_\alpha u_\beta}{\partial x_\beta} \right) = -\frac{\partial p_{\alpha\beta}}{\partial x_\beta} + n\mathbf{F}_\alpha . \tag{4.1.23a}$$

If we take into account the continuity equations

$$\frac{\partial n}{\partial t} + \frac{\partial n u_\beta}{\partial x_\beta} = 0$$

then, differentiating the left-hand part, we obtain

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v} \right) = -\frac{\partial P_{\alpha\beta}}{\partial x_\beta} + \rho \frac{\mathbf{F}_\alpha}{M} . \tag{4.1.23b}$$

This shows that in the case in which the pressure tensor $\vec{\vec{P}}_{\alpha\beta}$ is diagonal and isotropic

$$\vec{\vec{P}}_{\alpha\beta} = p\delta_{\alpha\beta}, \tag{4.1.23c}$$

we obtain the standard Euler equation

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t}+(\mathbf{v}\nabla)\mathbf{v}\right)=-\nabla p+\mathbf{f}$$

$$\mathbf{f}\equiv\frac{\rho\mathbf{F}}{\mathrm{M}}$$

However, the entire infinite chain which we previously discussed has not disappeared, regardless of the assumption (4.1.23c). It is hidden in the quantity *p* which so far has not has as yet been determined.

4.2. Vlasov-Maxwell equations

4.2.1. Formulation of the system of equations

The main difference between the system of charged particles and the system of neutral particles is that the charged particles interact with each other and are situated at large distances from each other, i.e. there is no direct contact. It may be concluded that every particle interacts with the entire set of the charged particles as a whole, i.e. the collective (not paired) interaction is controlling in this case.

In many cases in undense plasma we can ignore in the first approximation the paired collision and retain only collective collisions in the form of the so-called self-consistent field. In fact, this approach was used in the section 1.4 when deriving the 3/2 law and in the equations of two-fluid hydrodynamics when the total particle flow $\mathbf{j}=e(n_i\mathbf{v}_i-n_e\mathbf{v}_e)$ and the total charge density $q=e(n_i-n_e)$ were substituted into the Maxwell equations.

The concept of the movement of the particle in the field generated by all other particles, used in deriving the two-fluid equations, can be easily generalised using the following procedure. Let there be a set of *N* types of particles characterised by the distribution functions $f_1,\ldots,f_{\mathrm{N}}$. In this case, the density of the electric charge, generated by all particles of all types, is the same (see (4.1.3))

$$q=\sum_{\alpha=1}e_\alpha\int f_\alpha d\mathbf{v} \tag{4.2.1a}$$

Similarly, the density of electrical current is described by the equation

$$\mathbf{j}=\sum_{\alpha=1}e_\alpha\int \mathbf{v}f_\alpha d\mathbf{v} \tag{4.2.1b}$$

Substituting these expressions into the Maxwell equation (1.1.1), we obtain the equations of the ‘self-matched’ field:

$$\left.\begin{aligned} &\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\sum_{\alpha}^{N} e_\alpha \int \mathbf{v} f_\alpha d\mathbf{v} + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}; \\ &\operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathrm{H}}{\partial t}; \\ &\operatorname{div}\mathbf{E} = 4\pi\sum_{\alpha}^{N} f_\alpha d\mathbf{v}; \quad \operatorname{rot}\mathbf{H} = 0. \end{aligned}\right\} \tag{4.2.2}$$

Since the collisions are ignored and the self-consistent field formally does not differ from the external field, the equation of motion of the particle is governed by the Newton–Lorenz equation:

$$m_\alpha \frac{d\,\mathbf{v}}{dt} = e_\alpha \left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right] \right)$$

Consequently, for every type of particle we can write the Liouville equation

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\frac{\partial f_\alpha}{\partial x} + \frac{e}{m_\alpha}\left(\mathbf{E}_\alpha + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right] \right)\frac{\partial f_\alpha}{\partial \mathbf{v}} = 0 \tag{4.2.3}$$

The set of the equations (4.2.2) and (4.2.3) was proposed by A.A. Vlasov [106].

4.2.2. Is the Vlasov equations system accurate?

Regardless of its simplicity and natural nature of its assumptions, the system of the Vlasov equations caused extensive discussions regarding its accuracy at the time. This discussion, having the form of specific investigations, is in fact still being continued. This is associated with the fact that with the system derived assuming that the particles are subjected only to the effect of the collective self-consistent field and the collisions of the particles are ignored, it appears unexpectedly that this equation may also describe absolutely accurately the dynamics of the particles. In fact, let the set of N identical point charged particles be given. We introduce the distribution function

$$f_N(\mathbf{x}, \mathbf{v}, t) = \sum_{(k)=1}^{N} \delta(\mathbf{x} - \mathbf{x}_k(t))\delta(\mathbf{v} - \mathbf{v}_k(t)) \tag{4.2.4}$$

Here $\mathbf{x}_k(t)$, $\mathbf{v}_k(t)$ is the coordinate and the velocity of the k-th particle. It may easily be seen that the function $f_N(\mathbf{x},\mathbf{v},t)$ satisfies the Vlasov equations. Since f_N is additive, this claim can be verified on the example of a single particle. We have

$$\frac{\partial f}{\partial t} = -\dot{\mathbf{x}}_k \delta'(\mathbf{x}-\mathbf{x}_k(t))\delta(\mathbf{v}-\mathbf{v}_k(t)) - \dot{\mathbf{v}}_k \delta(\mathbf{x}-\mathbf{x}_k(t))\delta'(\mathbf{v}-\mathbf{v}_k(t)) =$$

$$= -\mathbf{v}_k \delta'(\mathbf{x}-\mathbf{x}_k)\delta(\mathbf{v}-\mathbf{v}_k) - \frac{\mathbf{F}}{M}\delta(\mathbf{x}-\mathbf{x}_k)\delta'(\mathbf{v}-\mathbf{v}_k);$$

$$\mathbf{v}_k \frac{\partial f}{\partial \mathbf{x}} = \mathbf{v}_k \delta'(\mathbf{x}-\mathbf{x}_k)\delta(\mathbf{v}-\mathbf{v}_k);$$

$$\frac{\mathbf{F}}{M}\frac{\partial f}{\partial \mathbf{v}} = \frac{\mathbf{F}}{M}\delta(\mathbf{x}-\mathbf{x}_k)\delta'(\mathbf{v}-\mathbf{v}_k)$$

This shows the validity of the equation

$$\frac{\partial f_N}{\partial t} + \mathbf{v}\frac{\partial f_N}{\partial x} + \frac{\mathbf{F}}{M}\frac{\partial f_N}{\partial \mathbf{v}} = 0$$

Similarly, the validity of the Maxwell equations is verified

$$\text{rot}\,\mathbf{H} = \frac{4\pi}{c} e\int \mathbf{v} f_N d\mathbf{v} + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c}\sum \mathbf{v}_k \delta(\mathbf{x}-\mathbf{x}_k) + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} \text{ and so on}$$

Thus, it can be seen that the system of the Vlasov equations can describe accurately the dynamics of *N* particles. However, in this case, there are no averaging procedures and all is reduced to the exact solution of the Newton–Lorenz equations for *N* particles together with the Maxwell equations.

At the same time, the system describes collisionless plasma in which the particles appear to be spread over some plasma volume. We can estimate the spatial scale of the spreading for the case in which the plasma state is not too far way from thermodynamic equilibrium. The concept is used to estimate the distance to which the field of the given particle penetrates, i.e. to determine the screening length of the field of this particle.

In the introduction, it was shown that screening of the field of the particle takes place at the distance of the order of the Debye radius, and the number of particles in the Debye sphere N_D is:

$$N_D = \frac{4\pi}{3} r_D^3 n = \frac{4\pi}{3}\frac{(kT_{ef})^{3/2} n}{(4\pi e^2 n)^{3/2}} \sim \frac{(kT_{ef})^{3/2}}{n^{1/2}}. \qquad (4.2.5)$$

This shows that when the particle concentration decreases, the number of the particles with which the given particle interacts continuously increases. The Debye radius can be conveniently compared with the

mean distance between the particles:

$$a \approx \frac{1}{n^{1/3}} \tag{4.2.6a}$$

Consequently

$$\frac{r_D}{a} \approx \frac{\sqrt{kT_{ef}}}{n^{1/6}\sqrt{4\pi e^2}}. \tag{4.2.6b}$$

This ratio also increases with decreasing density. For example, for T_e = 100 eV we obtain

$$\frac{r_D}{a} \approx \frac{2*10^4}{n^{1/6}}$$

Assuming that $n = 10^{15}$ cm^{-3}, we get

$$\frac{r_D}{a} \sim 10^2$$

4.2.3. Hybrid approximation

In a general form, the system of the Vlasov-Maxwell equations is complicated. Therefore, as a rule, the system is solved with a large number of assumptions. Particular difficulties are encountered in calculating the dynamics of the electronic component in the plasma configurations with the dimensions considerably larger than the Larmor electron radius. However, there are a large number of situations in which the ions are 'weakly magnetised', i.e. their Larmor radius is

$$\frac{\rho_i}{L} \gtrsim 1,$$

i.e. greater or comparable with the dimensions of the system, whereas the Larmor electronic radius is:

$$\rho_e \ll L.$$

In this case, the ion dynamics can be described kinetically, and the dynamics of the electrons is described by the Ohm's law. Examples of these systems will be investigated later (sections 5.7 and 6.7).

The models where the ions are described kinetically and the electrons are described using hydrodynamics are referred to as hybrid.

4.2. 'Static' kinetic configurations

Assuming the 'static' nature of the configuration (or, more accurately,

stationarity, because in the kinetics everything moves), we can write, for example, the ionic kinetic equation (4.1.9) in the form

$$\mathbf{v}\frac{\partial f}{\partial \mathbf{x}} + \frac{e}{M}\left(-\nabla\phi + \frac{1}{c}[\mathbf{v},\mathbf{H}]\right)\frac{\partial f}{\partial \mathbf{v}} = 0 \tag{4.3.1}$$

Evidently, the system of the characteristic equations (4.1.30) at $\frac{\partial}{\partial t}=0$ always has the energy integral

$$\frac{Mv^2}{2} + e\phi = \varepsilon$$

Thus, in this case

$$f_i = f_i(\varepsilon,\ Y,\ Y_2,\ Y_3,\ Y_4). \tag{4.3.2}$$

Here Y_k are the residual integrals of the characteristic equations which can be determined by concretisation of the type $\phi(\mathbf{x})$ and $\mathbf{H}(\mathbf{x})$. In the case important for practice when the system is axially symmetric and the generalised moment of the amount of motion of the single particle (1.2.2) is also retained in the absence of collisions

$$mr^2\dot{\theta} + \frac{e}{c}\psi = D = \text{const.}$$

Consequently, the distribution function has the form

$$f = f(\varepsilon,\ \mathrm{D},\ Y_1,\ Y_2,\ Y_3) \tag{4.3.3}$$

The integrals Y_k are determined by the type of magnetic and electrical fields and correspond to the highly complex trajectories. However, if the particles are magnetised, i.e. $\rho \ll L$, then together with the accurate integrals ε and D, there are also adiabatic invariants which were discussed in section 1.2.5 and which remained unchanged for a very long period of time. Consequently, f cannot be usually averaged only on the basis of the 'intricacy' of the trajectory. However, in the presence of noise and collisions disrupting the adiabatic invariant, (4.3.3) retains only two 'local' integrals and the distribution function has the form

$$f = f(\varepsilon,\ \mathrm{D}). \tag{4.3.4}$$

This result will be used later.

4.3.1. One-dimensional static kinetic configurations [108, 109]

Theory is simplified to the maximum extent if we are concerned with flat one-dimensional configurations which depend on a single Cartesian coordinate x; in these configurations, the magnetic field has only

components (H_y, H_z) and does not have component H_x.

In this case, the system of the equations of the characteristics (4.1.13) has the following form

$$\frac{dx}{v_x}=\frac{dv_x}{\frac{e}{M}\left(-\frac{\partial\phi}{\partial x}+\frac{1}{c}(v_yH_z-v_zH_y)\right)}=\frac{dv_y}{\frac{e}{Mc}(-v_xH_z)}=\frac{dv_z}{\frac{e}{Mc}(v_xH_y)} \qquad (4.3.5a)$$

If we introduce the vector-potential with the components $A_y(x)$, $A_z(x)$, so that

$$H_y=-\frac{\partial A_z}{\partial x};\quad H_z=-\frac{\partial A_y}{\partial x}$$

the system (4.3.5a) is completely integrated

$$\varepsilon=\frac{Mv^2}{2}+e\phi;\quad Mv_y+\frac{e}{c}A_y=D_2;\quad Mv_z+\frac{e}{c}A_z=D_1. \qquad (4.3.5b)$$

As a result, the ion distribution function in this case has the following form

$$f_i=F_i(\varepsilon_i,D_{i1},D_{i2})=F_i\left(\frac{Mv^2}{2}+e\varphi,Mv_y+\frac{e}{c}A_y,Mv_z+\frac{e}{c}A_z\right) \qquad (4.3.6a)$$

and is determined with the accuracy to the type of the function $F(\varepsilon, P_1, P_2)$.

The identical result can also be obtained for the electrons:

$$f_e=F_e(\varepsilon_e,D_{e1},D_{e2})=F_e\left(\frac{mv^2}{2}-e\varphi,mv_y-\frac{e}{c}A_y,mv_z-\frac{e}{c}A_z\right). \qquad (4.3.6b)$$

The type of the functions $F_{i,e}$ (ε, P_1, P_2) can either be specified in advance or is determined from some additional considerations.

If F_i, F_e are known, then using the Maxwell equation we can determine the spatial distributions $\phi(x)$, $A_y(x)$, $A_z(x)$.

Substituting the functions (4.3.6a) into the Maxwell equations, we obtain

$$\frac{d^2A_y}{dx^2}=-4\pi e\int v_y\left[F_i\left(\frac{Mv^2}{2}+e\phi,Mv_y+\frac{e}{c}A_y,Mv_z+\frac{e}{c}A_z\right)-\right.$$
$$\left.-F_e\left(\frac{mv^2}{2}-e\phi,mv_y-\frac{e}{c}A_y,mv_z-\frac{e}{c}A_z\right)\right]d\mathbf{v};$$

$$\frac{d^2A_z}{dx^2} = -4\pi e\int v_z\left[F_i\left(\frac{Mv^2}{2}+e\phi, Mv_y+\frac{e}{c}A_y, Mv_z+\frac{e}{c}A_z\right)-\right.$$
$$\left.-F_e\left(\frac{mv^2}{2}-e\phi, mv_y-\frac{e}{c}A_y, mv_z-\frac{e}{c}A_z\right)\right]d\mathbf{v};$$
$$\frac{d^2\phi}{dx^2} = -4\pi e\int\left[F_i\left(\frac{Mv^2}{2}+e\phi, Mv_y+\frac{e}{c}A_y, Mv_z+\frac{e}{c}A_z\right)-\right. \quad (4.3.7)$$
$$\left.-F_e\left(\frac{mv^2}{2}-e\phi, mv_y-\frac{e}{c}A_y, mv_z-\frac{e}{c}A_z\right)\right]d\mathbf{v};$$

Thus, the problem for the known F_i, F_e is reduced to the system of three conventional differential equations: for the three unknown quantities A_y, A_z, ϕ:

$$\frac{d^2A_y}{dx^2} = I(A_y, A_z, \phi), \quad \frac{d^2A_z}{dx^2} = I_2(A_y, A_z, \phi), \quad (4.3.8a)$$

$$\frac{d^2\phi}{dx^2} = I_3(A_y, A_z, \phi)$$

Taking into account the law of conservation of the momentum, it may easily be seen that system (4.3.8) has the first integral:

$$\frac{H^2-E^2}{8\pi} = Mn_iv_i^2 + mn_ev_e^2. \quad (4.3.8b)$$

If the plasma in the investigated region is regarded as quasi-neutral, then assuming that $n_i = n_e$, we have the relationship of ϕ with A_y and A_z:

$$\int F_i(\varepsilon_i, P_{i1}, P_{i2})d\mathbf{v} = \int F_e(\varepsilon_e, P_{e1}, P_{e2})d\mathbf{v} \quad (4.3.9)$$

Finding the relationship

$$\phi = Q(A_y, A_z), \quad (4.3.10)$$

we reduce the system of three equations (4.3.8) to the system of two equations for A_y and A_z.

The resultant systems of equations (A.I. Morozov, L.S. Solov'ev [109]) can be used to calculate a large number of interesting one-dimensional structures. They include the plasma-field transition layers in traps, near-anode layers in high-current plasma accelerators with solid electrodes (the so-called magnetic insulation of high-voltage

vacuum current supplies), the Debye layer, and others. Some of these structures will be discussed later.

4.3.2. The inverse Bernstein–Green–Kruskal function [108]

In the absence of the magnetic field and the known values of F_i (ε_i) and $F_e(\varepsilon_e)$, equation (4.3.8)

$$\frac{d^2\phi}{dx^2} = -4\pi e\int (F_i(\varepsilon_i) + F_e(\varepsilon_e))dv \equiv 4\pi Y(\phi) \tag{4.3.11a}$$

can be used to calculate the distribution of the potential $\phi(x)$, i.e. find the *E*-structure. This is also supported by the presence of the first integral – the law of momentum conservation (4.3.8b)

$$\frac{1}{8\pi}\left(\frac{d\phi}{dx}\right)^2 = \int_{-\infty}^{+\infty} (Mv^2F_i(\varepsilon_i) + mv^2F_e(\varepsilon_e))dv = \int Y(\phi)d\phi + \text{const} \equiv Q(\phi). \tag{4.3.11b}$$

This leads to the quadrature

$$x + \text{const} = \frac{1}{\sqrt{8\pi}}\int \frac{d\phi}{\sqrt{Q(\phi)}}. \tag{4.3.11c}$$

Unfortunately, the type of F_i and F_e is usually very complicated and the solution of the system (4.3.11a) requires numerical calculations.

The determination of $\phi(x)$ at the known $F_i(\varepsilon_i)$ and $F_e(\varepsilon_e)$ is referred to as the 'direct' task of calculating *E*-structures[4]. However, equation (4.3.11a) can also be used to solve the 'inverse' problem, i.e. to determine the particle distribution function resulting in the equilibrium of the *E*-system for the given distribution of the potential $\phi(x)$. This was carried out for the first time by Bernstein, Green and Kruskal (1957). The results which they obtained can be described as follows.

Let it be that some one-dimensional volume contains several groups of electrons and ions with the known distribution functions F_{0i} and F_{0e}, at which the plasma in the volume is not in equilibrium. Consequently, as shown by the above authors, to this set of the particles we can add a group of electrons or ions trapped in the potential well of the given profile which ensures that the entire system is stationary. The distribution function (DF) of the 'stabilising' particles can be determined as the solution of the integral equation.

[4]The simplest task, solved using (4.3.11), is the derivation of the '3/2' Langmuir law (1.4.7).

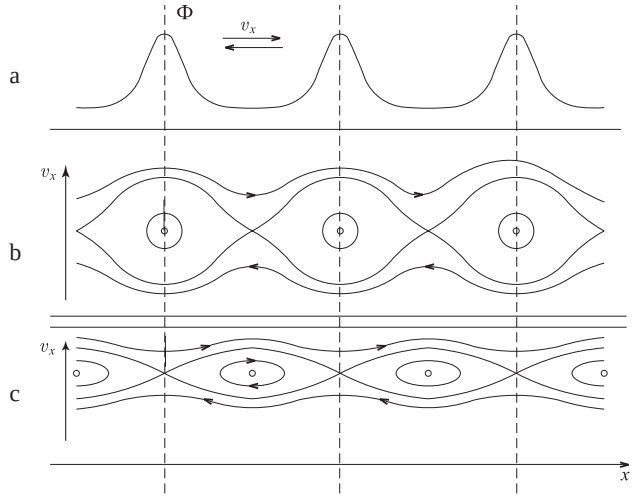


Fig. 4.3.1. Calculation of the direct Bernstein–Green–Kruskal problem: a) potential profile; b) the phase trajectories of the electrons; c) the phase trajectories of the ions.

We reproduce the scheme of solving the inverse problem. In order to avoid complicating analysis by formal moments, it is assumed that the dependence $\phi(x)$ is periodical and within the limits of the period there is a single maximum, i.e. the profile resembles a sinusoid (Fig. 4.3.1b,c). The trajectories of the electrons and the ions on the phase plane (x, v) greatly differ (Fig. 4.3.1b, c). Here, we can see the trajectories of the 'flying' particles which 'feel only slightly' the electrical field and the of 'trapped' particles with closed trajectories inside the appropriate potential wells.

Let us assume that we know, for example, distribution functions of all ions (F_i) and the distribution functions of only 'flying' electrons ($F_e^{(\mathrm{fl})}$) and also that the potential relief $\phi(x)$ is known. It is necessary to select the distribution function of the trapped electrons ($F_e^{(\mathrm{tr})}$ (at which they will be the stationary E-configuration, i.e. the equation (4.3.11a) will be valid).

Taking into account the fact that the configuration is one-dimensional, we can express the second derivative $d^2\phi(x)/dx^2$ by means of ϕ:

$$\frac{d^2\phi(x)}{dx^2} = A(\phi)$$

and since the density of all ions n_i and the density of the $n_e^{(\mathrm{fl})}$ electrons also depends only on ϕ, from (4.3.11a) we obtain the equations for the distribution functions of the trapped electrons:

$$n_e^{(\mathrm{tr})} = \int F_e^{(\mathrm{tr})}(v,\phi)dv = \frac{A(\phi)}{4\pi e} + n_i(\phi) - n_e^{(\mathrm{fl})}(\phi) \equiv 2S(\varphi). \qquad (4.3.12)$$

The resultant integral equations for $F_e^{(\mathrm{tr})}$ for the known function $S(\phi)$ is easily reduced to Abel-type equations. In fact, using the particle energy as the independent variable instead of the velocity v,

$$\varepsilon = \frac{mv^2}{2} - e\phi,$$

which changes in the range $0 \le \varepsilon < \varepsilon_{\max} = -e\phi_{\min}$ (here the zero ϕ is represented by the maximum potential), the equation (4.3.12) can be presented in the following form

$$\int_{-e\phi}^{e\phi_{\min}} \frac{F_{e0}^{(\mathrm{tr})}(\varepsilon)d\epsilon}{\sqrt{2m(\epsilon + e\phi)}} = S(\phi). \qquad (4.3.13)$$

Equation (4.3.13) takes into account that

$$dv = \frac{d\epsilon}{mv} = \frac{d\epsilon}{\sqrt{2m(\epsilon + e\phi)}},$$

and the coefficient 2 appears as a result of taking into account the particles with the velocity $v > 0$ and also $v < 0$. Finally, ϕ_m denotes the minimum value of the potential.

Equation (4.3.13) is the required integral equation.

The solution of this equation has the following form

$$F_{e0}(E) = \frac{\sqrt{2m}}{\pi} \int_{e\phi_m}^{-E} dU \frac{dS(U)}{dU} \frac{1}{\sqrt{-E-U}}, E + U < 0. \qquad (4.3.14)$$

The equation on the right-hand side contains an integrable singularity and this clearly indicates that $F_0(\varepsilon)$ does not depend in a linear manner on ϕ_m – the amplitude of the perturbing potential, if $|\phi_m| \to 0$. In fact, analysis shows that in the case in which the singularity in $F_0(\varepsilon)$ can be avoided by selecting $\phi(x)$ and $f_0(\varepsilon)$, the distribution function of the trapped particles at low values of ϕ_M is:

$$F_0(\varepsilon, \phi_M) = A_1(\varepsilon)(-e\phi_M - \varepsilon)^{1/2} + A_2(\varepsilon)(-e\phi_M - \varepsilon)^{3/2} + \ldots \qquad (4.3.15)$$

Thus, the expansion takes place by the exponents divided on 2. This leads to two important conclusions:

– there are no linear stationary waves in electron plasma, without trapped particles;

– the trapped particles make it possible to produce the equilibrium configuration from 'non-equilibrium' distribution functions[5].

These conclusions will now be confirmed by direct calculations of linear waves in the Vlasov model. The principal special features of the non-stationary nature of the linear waves were discovered by Vlasov, Landau and van Kampen.

The reason for the absence of the stationary linear *E*-waves is explained by Fig. 4.3.1.

If there is no *E*-field, the phase trajectories have the form of straight lines parallel to the *x* axis. Above this axis, the trajectories are directed to the right, below the axis to the left. The appearance of some small *E*-field radically changes the topology of the phase trajectories of the particles in the vicinity of the *x* axis: a chain of 'beads' forms. The separatrix, confining these 'beads' is determined by the equation

$$\frac{mv^2}{2} - e\phi = -e\phi_{\min}. \tag{4.3.16}$$

Thus, the square of perturbation of the velocity, and not simply the perturbation of the velocity, is proportional to perturbation ϕ.

4.3.3. Single-Larmor structures

The following problem (Fig. 4.3.2) will be discussed as an example of using equation (4.3.7) at $\mathbf{H} \neq 0$.

It is assumed that at $x \to -\infty$ the magnetic field $H \to H_0$, potential $\phi \to \phi_0$, whereas at $x \to \infty$ $H \to H_1$ and $\phi \to \phi_1$, respectively. The transition between these values of the parameters occurs within the limits of a narrow layer ($-d < x < 0$) with the thickness of the order of the electronic Larmor diameter. The electronic component, which is assumed to be cold and stationary at $x > 0$, $x < -d$, is penetrated by the ion flux, and in the region $x < -d$ the velocity of the ions is assumed to be very low so that the momentum can be ignored. On the other hand, in the range $x > 0$, the ions carry a large momentum, determined by the condition (4.3.8b):

$$\left(H_0^2 - H_1^2\right)/8\pi = Mn_1v_i^2 \tag{4.3.17}$$

Here, we ignore the term E^2 since the plasma is assumed to be quasi-neutral, and in this case $E^2/\delta H^2 \ll 1$. Without reducing generality, it may be assumed that

$$\Phi\ (0) = 0;\ A\ (0) = 0, \tag{4.3.18}$$

[5]Of course, the problem of the stability of the resultant configuration remains unsolved

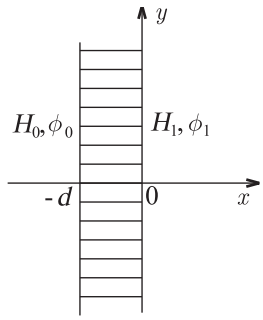


Fig. 4.3.2. The schematic of the accelerating layer.

and the effect of the magnetic field of the ions is ignored, i.e. instead of (4.3.6a) we use the equality v_y = const. Under these assumptions, the ion and electron distribution functions can be written in the following form:

$$f_i = N_i \delta\left(v^2 + 2\frac{e}{M}\left(\Phi - \Phi_0\right)\right)\delta(v_y)\delta(v_z) \tag{4.3.19a}$$

$$f_e = N_e \delta\left(v^2 - 2\frac{e}{m}\Phi\ \right)\delta\left(v_y - \frac{e}{mc}A\ \right)\delta(v_z) \tag{4.3.19b}$$

Here N_i and N_e are constants having the dimensions of the flux. They are determined from the initial or boundary conditions. The expression for f_i takes into account the start of the ions at the potential ϕ_0 and in the absence of the effect of the magnetic field on them. The selection of the distribution function for the electrons is determined by a special assumption according to which in the layer, i.e. in the region of the potential jump, all the electrons move along cycloids (Fig. 4.3.3a), and at x = 0, where = 0 and A = 0 the electrons are arrested. Of course, this is one of the possible variants of the movement of the electrons in the layer (see Fig. 4.3.3 b, c). Substituting the expression (4.3.19) into (4.3.7) the transfer to the following equations for the electrical and magnetic potentials:

$$\frac{d^2\phi}{dx^2} = -4\pi e\left(\frac{N_i}{2\sqrt{\dfrac{2e(\phi_0 - \phi)}{M}}} - \frac{N_e}{\sqrt{\dfrac{2e\phi}{m} - \left(\dfrac{eA}{mc}\right)^2}}\right);$$

$$\frac{d^2A}{dx^2} = \frac{4\pi e}{c}\frac{eA}{mc}\frac{N_e}{\sqrt{2e\Phi/m - (eA/(mc))^2}}.$$

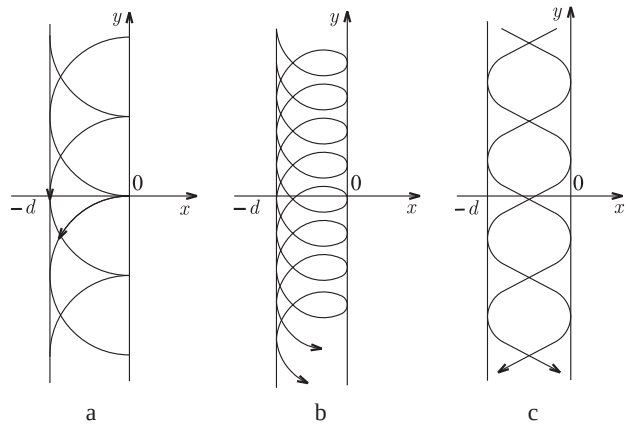


Fig. 4.3.3. Possible types of movement of the electrons in 'single-Larmor' layers: a) trachoid, b) cycloid, c) hypocycloid.

Introducing new variables:

$$2\frac{e}{m}\phi \equiv S;\ \frac{e}{mc}A \equiv \alpha;\ \frac{4\pi e^2}{mc^2}x^2 N_i \equiv \xi^2;\ \ \nu \equiv \frac{N_e}{N_i},$$

we obtain

$$\frac{d_1^2 S}{d\xi^2} = -c^2\left(\frac{1}{\sqrt{\frac{m(S_0 - S}{M}}} - \frac{2\nu}{\sqrt{S-\alpha^2}}\right); \tag{4.3.20a}$$

$$\frac{d^2\alpha}{d\xi^2} = \frac{\alpha\nu}{\sqrt{S-\alpha^2}}. \tag{4.3.20b}$$

If the plasma is non-the relativistic and sufficiently dense, it can be regarded as it is quasi-neutral, i.e. the right-hand part of equation (4.3.20) can be equated to 0. Consequently

$$\frac{m}{M}(S_0 - S) = \frac{1}{4\nu^2}(S - \alpha^2).$$

After obvious transformations we obtain:

$$S = \left(\frac{m}{M}S_0 + \frac{\alpha^2}{4\nu^2}\right)\left(\frac{m}{M} + \frac{1}{4\nu^2}\right)^{-1}. \tag{4.3.21}$$

Attention should be given to the fact that the equation (4.3.21) does not satisfy the condition (4.3.18). Equation (4.3.21) shows that at = 0, S ~ is not equal to 0, and

$$S_{\min} = \frac{m}{M} S_0 \left(\frac{m}{M} + \frac{1}{4\nu^2} \right)^{-1}.$$

Evidently, this discrepancy is explained by the fact that quasi-neutrality is disrupted in the immediate vicinity $x = 0$ and there is, all the way very small, but finite jump of the potential. At the same time, on the left boundary of the layer at $x = -d$, where

$$S = S_0;\ S = \alpha^2, \tag{4.3.22}$$

equation (4.3.21) after substitution of (4.3.22) gives the accurate value of the normalised potential $S = S_0$.

Substituting (4.3.21) into (4.3.20b) we obtain an equation which determines the structure of the layer

$$\frac{d^2\alpha}{d\xi^2} = \frac{\alpha}{2} \sqrt{\frac{(M/m + 4\nu^2)}{(S_0 - \alpha^2)}}.$$

Consequently, after integration

$$\left(\frac{d\alpha}{d\xi} \right)^2 = -\sqrt{(M/m + 4\nu^2)(S_0 - \alpha^2)} + const. \tag{4.3.23}$$

Taking into account (4.3.22) and assuming that the field H_0 at $x = -d$ is given, we determine the integration constant:

$$\text{const} = \left(d\alpha \big/ d\xi \right)^2 \Big|_{x=-d} = h_0^2;\ h_0 \equiv \frac{H_0}{\sqrt{4\pi m N_i}}.$$

Integral (4.3.23) is identical with (4.3.17). Writing this integral for the point $x = 0$, we obtain

$$\frac{H_0^2 - H_1^2}{2} = 2\pi N_i \sqrt{2e\phi_0 M \left(1 + \frac{4\nu^2 m}{M} \right)}. \tag{4.3.24}$$

Equation (4.3.19) shows that the ion flux is:

$$\int_0^\infty v_x f_i dv = \frac{1}{2} N_i. \tag{4.3.25}$$

Comparing (4.3.23) and (4.6.16) it can be concluded that the term

$$4v^2 \frac{m}{M} \ll 1 \tag{4.3.26}$$

and it should be rejected because the law of momentum conservation should be fulfilled:

$$\frac{H_0^2 - H_1^2}{8\pi} n_i M v_i^2 \Big|_{x>0} = \frac{MN_i}{2} \sqrt{\frac{2e\phi_0}{M}}.$$

In other words, the quantity $4v^2m/M$ is of the order of terms which are responsible for the disruption of quasi neutrality and which should be ignored.

Taking into account (4.3. 25), equation (4.3.23) can be presented in the new form:

$$\left(\frac{d\alpha}{d\xi}\right)^2 = h_0^2 - \sqrt{\frac{M\left(S_0 - \alpha^2\right)}{m}}. \tag{4.3.27}$$

In a general case, the solution of this equation is expressed by the elliptical integrals. However, in two limiting regimes, when

$$h_0^2 = \sqrt{\frac{MS_0}{m}} \quad \text{or} \quad h_0^2 \gg \sqrt{\frac{MS_0}{m}},$$

calculations are elementary. It can easily be seen that the first case corresponds to the layer which completely 'eats up' of the magnetic field, i.e. converts H_1 to 0, whereas the second term describes the regime in which, on the other hand, the magnetic field is only slightly disrupted by the layer. In the latter case, $\alpha \approx h_0\xi$, and, consequently, the thickness of the accelerating layer is

$$d \approx \frac{mc}{eH_0} \sqrt{2\frac{e}{m}\Phi_0}. \tag{4.3.28}$$

When deriving (4.3.28) the equation (4.3.22) was taken into account. As expected, the equation obtained for d is equal to the electronic Larmor radius calculated from the potential jump.

4.4. Kinetics of waves in plasma at $\mathbf{H}_0 = 0$ [106, 111]

We shall confine ourselves to investigating the kinetics of waves only in homogeneous plasma at $\mathbf{H}_0 = 0$. Naturally, this is associated with the fact that the theory at $\mathbf{H}_0 \neq 0$ is 'cumbersome'. More details can be found in [114].

4.4.1. Initial equations. Non-stationary Langmuir waves

The electronic longitudinal linear waves have already been investigated previously in the hydrodynamics model. Analysis of the stationary waves with a small amplitude, described in the previous section, shows that in kinetics of inmost deal with the problem of Particles. Indeed, this is the situation.

The starting point for describing the linear one-dimensional longitudinal waves in collisionless kinetics in the absence of the external magnetic field in homogeneous electronic plasma is the system of equations (electron charge e)

$$\begin{aligned} &\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} + \frac{e}{m}\frac{d\phi}{dx}\frac{\partial f}{\partial v} = 0; \\ &\frac{d^2\phi}{dx^2} = -4\pi e(n_0 - \int f dv). \end{aligned} \tag{4.4.1}$$

here n_0 is the concentration of ions which are assumed to be stationary. Assuming that in the non-perturbing condition

$$\int f_0 dv = n_0, \quad \phi_0 = 0$$

and in the presence of perturbations

$$f = f_0 + f_1, \quad \phi = \phi_1$$

Substituting these values into (4.4.1) and rejecting the values of the second order of smallness (1, f1 ~0), we obtain a linear system

$$\frac{\partial f_1}{\partial t} + v\frac{\partial f_1}{\partial x} + \frac{e}{m}\frac{\partial \phi_1}{\partial x}\frac{\partial f_0}{\partial v} = 0; \quad \frac{d^2\phi_1}{dx^2} = 4\pi e\int f_1 d\mathbf{v}. \tag{4.4.2}$$

The solution (4.4.2) is sought in the form of the exponent

$$f_1 = \overline{f}_1(v)\{-i\omega t + i\varkappa \cdot x\}; \ \phi_1 = \overline{\phi}_1 \exp\{-i\omega t + i\varkappa + i\varkappa \cdot x\}.$$

Consequently, we obtain

$$\left(\omega - \varkappa \cdot v\right) \bar{f}_1 = \varkappa \frac{e}{m} \bar{\phi}_1 \frac{\partial f_0}{\partial v}; \quad \varkappa^2 \bar{\phi}_1 = -4\pi e \int f_1 dv. \tag{4.4.3}$$

From this

$$\bar{f}_1 = \frac{e}{m} \frac{\partial f_0}{\partial v} \bar{\phi}_1 \frac{1}{\omega - \varkappa v}; \qquad \bar{\phi}_1 = \bar{\phi}_1 \frac{4\pi e^2}{m} \int \frac{\partial f_0}{\partial v} \frac{dv}{\omega - \varkappa v}. \tag{4.4.4}$$

Shortening ϕ_1 in the last equation, we obtain what appears to be a simple dispersion equation (Vlasov, 1938)

$$\epsilon(\omega, \varkappa) \equiv 1 - \frac{4\pi e^2}{m} \int \frac{\partial f_0}{\partial v} \frac{dv}{(\omega - \varkappa v)} = 0 \tag{4.4.5}$$

However, assuming that $\varkappa$ and ω are real, we have a peculiarity under the integral in 'kinematic resonance'

$$v_{ph} \equiv \frac{\omega}{\varkappa} = v \tag{4.4.6}$$

i.e. when the phase velocity of the wave is equal to the particle velocity[6].

A question arises: how to bypass this peculiarity? Physically, it is clear that under the condition (4.4.6) there is a strong interaction between the 'resonant' particles and the wave. However, not only strictly resonant particles interact with the wave. A strong interaction is also detected in the vicinity of the resonance condition. So that if

$$\left|e\phi_m\right| > \frac{m\left(v - \frac{\omega}{\varkappa}\right)^2}{2},$$

the particle may prove to be trapped in the potential well. If the velocity of the particle is $v < \omega/\varkappa$, the particle will be accelerated (Fig. 4.4.1a), and at $v > \omega/\varkappa$ it will decelerate (Fig. 4.4.1b). This means that in the first case the wave transfers its energy to the particle, and in the second case the particle to the wave.

Thus, the special feature of the processes taking place in the vicinity of kinematic resonance is quite clear. However, taking this into account, it is still not clear how to treat the subintegrand singularity. However,

[6]The condition (4.4.6) is evidently identical with the Mach condition (2.2.22). At present, there is no standard term for this type of resonance. It is often referred to as Cherenkov resonance, also as Landau resonance, etc. From the historical viewpoint, it would be more accurate to refer to it as 'Mach resonance' or, which describes the actual situation, 'kinematic resonance'. This term will also be used in this book.

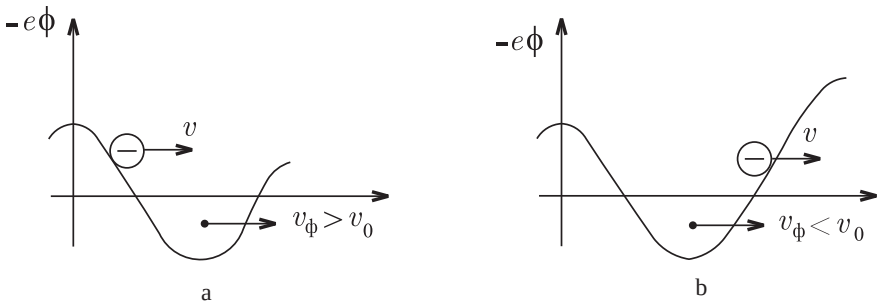


Fig. 4.4.1. Dynamics of the electron in the field of a running electrostatic wave: phase speed of the wave is greater than the initial speed of the particle (a); the phase speed of the wave is lower than the initial speed of the particle (b).

the situation becomes more understandable if it is taken into account that the system (4.4.3) has, in addition to (4.4.5), another solution:

$$f_2 = \overline{f}_2 \delta(\omega - \varkappa v),\ \phi_2 = 0. \tag{4.4.7}$$

The δ-shaped function f_2 describes evidently the strictly resonant particles, moving with the phase velocity of the wave. These are 'truly' trapped particles. These particles were described in the first time by the Dutch mathematician van Kampen [112] who proved that the waves f_1 and f_2 form a complete system of intrinsic waves in one-dimensional kinetics. The value of f_1 is calculated in the sense of the main value. Therefore, the 'waves' f_2, i.e. the fluxes of the particles with $v = \omega/\varkappa$, are often referred to as the van Kampen waves. They are also referred to as 'ballistic modes' or 'quasi-waves'.

4.4.2. Laplace transformation

Evidently, all the above considerations do not enable us to take into account singularity in (4.4.5). Of special interest in the group of the possible variants is the solution of the problem with the initial conditions. This problem can be solved using Fourier expansion for the dependence of ϕ and f on the spatial coordinate x:

$$\begin{aligned} \phi(x,t) &= \int d\varkappa\, \phi_\varkappa(t) \exp\{i\varkappa x\}; \\ f(x,v,t) &= \int d\varkappa\, f_\varkappa(v,t) \exp\{i\varkappa x\}; \end{aligned} \tag{4.4.8a}$$

and for $\phi_\varkappa(t)$ and $f_\varkappa(v, t)$ the Fourier transformation can be replaced by the Laplace transformation (Landau, 1946) [111]

$$f_p(v)=\int_0^\infty \exp\{-pt\}\ \ f_\varkappa(v,\ t)dt;\ \ \phi_p=\int_0^\infty \exp\{-pt\}\ \ \phi_\varkappa(t)dt. \tag{4.4.8b}$$

Here $p = \sigma + is$, and the real part Re $p = \sigma > 0$. The initial perturbation – the 'source' – is denoted by $g(v)\delta(t)$.

Multiplying the Fourier component (with respect to $\varkappa$) in equation (4.4.2) by exp $\{-pt\}$ and integrating with respect to t, we obtain

$$f_p=\frac{ig(v)}{ip-\varkappa v}+\frac{\varkappa}{ip-\varkappa v}\frac{e}{m}\frac{\partial f_0}{\partial v}\phi_p. \tag{4.4.9}$$

Here $g\ (v)$ is determined by the initial value $\partial f_0/\partial t$. It can be seen that this expression differs from (4.4.4) by the addition of the term with g and by replacement of ω by ip. Substitution of (4.4.9) into the second equation (4.4.4) and taking into account the expression (4.4.5) for ϵ makes it possible to determine ϕ:

$$\phi_p=-i\frac{4\pi e}{\varkappa^2\varepsilon(\varkappa,ip)}\int\frac{g(v)dv}{ip-\varkappa v},\quad \epsilon=1+\frac{4\pi e^2}{m\varkappa}\int\frac{\partial f_0}{\partial v}\frac{\partial v}{(ip-\varkappa v)}. \tag{4.4.10a}$$

From the known value of ϕ_p it is easy to determine $\phi_\varkappa(t)$:

$$\phi_\varkappa(t)=\frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty}\exp\{pt\}\phi_p dp\ . \tag{4.4.10b}$$

From p we transfer to the variable $\omega = ip$, and since the integration with respect to variable p is carried out in the right-hand half-plane, integration with respect to the complex variable ω should be carried out in the upper half plane. Therefore, when ϵ and other integrals with the kinematic resonance (4.4.6) are calculated, the quantity ω is regarded as having the positive imaginary part, and integration with respect to v is carried out over the real axis (Fig. 4.4.3a). Thus

$$\phi(t)=-i\frac{2e}{\varkappa^2}\int_{-\infty+i\sigma}^{+\infty+i\sigma}\frac{\exp\{-i\omega t\}}{\varepsilon(\varkappa,\omega)}\left[\int_{-\infty}^{+\infty}\frac{g(v)dv}{\omega-\varkappa v}\right]d\omega. \tag{4.4.11a}$$

The same procedure is used to find $f_\varkappa(v,t)$

$$f_\varkappa(v,t)=\frac{1}{2\pi i}\int_{-i\infty+\sigma}^{+i\infty+\sigma}\frac{\exp\{-i\omega t\}}{p+i\varkappa v}\left(g(v)+\frac{ie\varkappa}{m}\phi_p\frac{\partial f_0}{\partial v}\right)d\omega. \tag{4.4.11b}$$

The equations (4.4.11) solve fully the problem of the evolution of plasma oscillations caused by the initial perturbation $g(v)$. The equation shows that, generally speaking, there is no specific dependence of ω on $\varkappa$: at

the given $\varkappa$ integration in (4.4.11) is carried out over all ω. However, if $g(v)$ has no peculiarities, then in accordance with the theory of deduction of the function of the complex variables, the values of the integral (4.4.11) are determined by the zeros $\varepsilon(\varkappa, \omega)$:

$$\varepsilon(\varkappa, \omega_k) = 0. \tag{4.4.12a}$$

Thus, from the solution of (4.4.11) we separate the branches of plasma oscillations with natural frequencies ω_k and, consequently,

$$\phi(t) \sim \exp\{-i\omega_k t\}. \tag{4.4.12b}$$

Partial case: Maxwell distribution $f_0(v)$

If the electrons are characterised by the Maxwell distribution with temperature T, the dielectric permittivity, determined by the expression (4.4.5), can be calculated accurately and presented in the form [12]

$$\varepsilon = 1 + \frac{m\omega_{pe}^2}{k^2 T}\left\{1 + i\sqrt{\pi z}W(z)\right\} \tag{4.4.13}$$

where ω_{pe} is the Langmuir frequency, $z = \dfrac{\omega}{kv_{Te}}$, $v_{Te} = \sqrt{\dfrac{2T}{m_e}}$ is the mean thermal velocity, and the function

$$W(z) = \exp\left\{-z^2\right\} + \frac{2i}{\sqrt{\pi}}\int_0^z \exp\left\{-z^2 + \xi^2\right\} d\xi\,. \tag{4.4.14a}$$

At high and low z, the function $W(z)$ is approximated by simple expressions:

$$|z| >> 1$$
$$W(z) = \frac{i}{\sqrt{\pi}z}\left(1 + \frac{1}{2z^2} + \frac{3}{4z^4} + \ldots\right) + \exp\{-z^2\}; \tag{4.4.14b}$$

$$|z| \leqslant 1$$
$$W(z) = 1 + \frac{2iz}{\sqrt{\pi}}. \tag{4.4.14c}$$

Equating (4.4.13) to 0, we can determine the real and imaginary parts of the complex frequency. At low $k \ll 1/r_D$, where r_D is the Debye radius, $z \gg 1$, and after several transformations, we obtain that the expression for the real part of the frequency has the form

$$\omega^2 = \omega_{pe}^2 + \frac{3kT}{m}\varkappa^2, \qquad (4.4.15a)$$

and its imaginary parts (i.e. attenuation decrement γ) is:

$$\gamma = -\omega_{pe}\sqrt{\frac{\pi}{8}}\frac{1}{(kd)^3}\exp\left\{-\frac{1}{2k^2 r_D^2}\right\}. \qquad (4.4.15b)$$

It can be seen that at low $\varkappa r_D$ the attenuation decrement is exponentially small.

Formula (4.4.15a) differs from (3.5.7b) by coefficient 3 in front of the second term. This is associated with the fact that here we examine a one-dimensional problem, i.e. with a single degree of freedom, when the adiabate parameter $\gamma_a = 3$.

The general case f_0 at a low attenuation decrement $|\gamma/\omega| \ll 1$

We now study the case of the electron distribution function of an arbitrary type f_0, but at a low value of γ. Frequency ω is assumed to be located in the upper half-plane but close to the real axis and, consequently, the peculiarity can be bypassed as shown in Fig. 4.4.2b. This means that the integration path can be taken in the form of two sections of the axis v, connected by a half circle in the lower half plane with the radius $p \to 0$ and bypassing the singular point $v_* = \omega/\varkappa$. Integration along the sections determines the main value P of the appropriate integral[7]. This rule of bypassing the pole is referred to as the Landau rule. Taking this rule into account, the dielectric permittivity (4.4.5) is complex:

$$\varepsilon = 1 + \frac{4\pi e^2}{\varkappa k m_e} P\int \frac{dv}{\omega - \varkappa v}\frac{\partial f_0}{\partial v} - \frac{4\pi e^2}{\varkappa m_e}\frac{\pi_i}{|\varkappa|}\left.\frac{\partial f_0}{\partial v}\right|_{v=\omega/\varkappa}. \qquad (4.4.16)$$

Assuming $v_* \gg c_{Te}$, we can write

$$P\int_{-\infty}^{+\infty}\frac{\partial f_0}{\partial v}\frac{dv}{\omega - \varkappa v} = \frac{1}{\omega}\int_{-\infty}^{+\infty}\left(1 + \frac{\varkappa v}{\omega} + \ldots\right)\frac{\partial f_0}{dv} = -\frac{\varkappa}{\omega^2}n_0 + \ldots \qquad (4.4.17a)$$

Substituting 4.4.17a and $\omega = \omega_0 + i\gamma$ into 4.4.16, we obtain the explicit equation for γ:

[7]Langmuir oscillations when taking into account the main value of the integral were investigated by Vlasov [106]. They correspond to stationary waves with trapped particles.

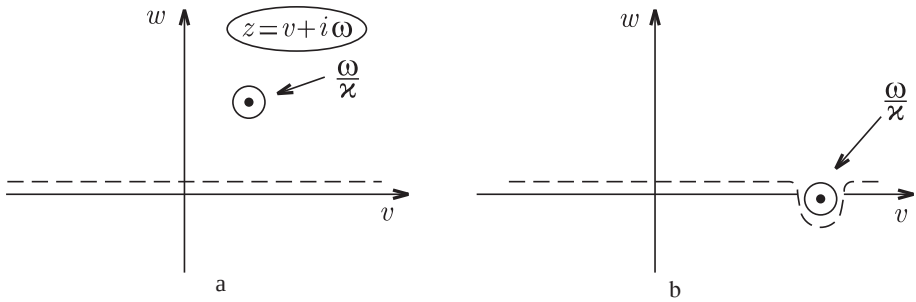


Fig. 4.4.2. Integration contours in calculation of integrals with a resonance denominator: a) general case, b) the case of slow attenuation (increase) of oscillations.

$$\gamma = \frac{2\pi^2 e^2}{\varkappa^2 m_0} \left. \frac{\partial f_0}{\partial v} \right|_{v_*} . \qquad (4.4.17b)$$

At very high t, a contribution to the integral (4.4.11b) is also provided by the pole $\omega - \varkappa v = 0$, corresponding to the free scattering of the particles from the initial perturbation, i.e. the van Kampen waves. This group of particles is described by the first term in the integral (4.4.11b). Therefore, the dependence of the potential ϕ on time can be complicated.

4.4.3. Attenuation and growth of Langmuir waves

The behaviour of the waves with time is determined by the sign of the imaginary additive γ to the frequency $\omega = \omega_0 + i\gamma$. As indicated by (4.4.17b), the sign of γ coincides with the sign of the arbitrary non-perturbed distribution function

$$\operatorname{sign}\gamma = \operatorname{sign} \left. \frac{\partial f_0}{\partial v} \right|_{v=\omega_0/x} . \qquad (4.4.18)$$

Therefore, if $f_0(v)$ is the Maxwell or any other monotonically decreasing function, then the Langmuir wave will attenuate in this electronic component. However, if the distribution function is non-monotonic (Fig. 4.4.3), i.e. there are velocity ranges in which $\partial f_0 / \partial v > 0$, in this case conversely $\gamma > 0$, i.e. the wave amplitude will increase. Previously, in chapter 3, when considering the two-fluid model, special attention was given to the beam instability. Evidently, the electronic component of the plasma plus the electron beam can be regarded as the single component with the non-monotonic distribution function. Thus, in this case, the kinetic and hydrodynamic models are in good agreement.

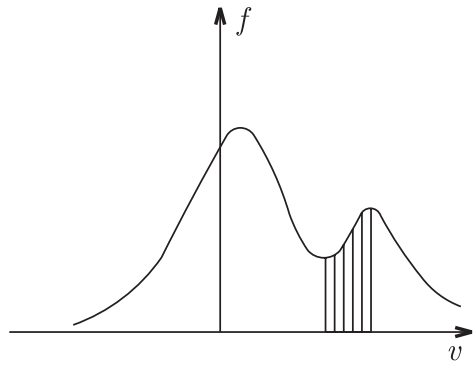


Fig. 4.4.3. Example of an unstable one-dimensional distribution function, the range of speeds, resulting in the growth of Langmuir oscillations, is crosshatched.

Equation (4.4.16) also shows that the wave does not attenuate and does not grow if $\partial f_0 / \partial v = 0$ at $v \sim \omega/\varkappa$. The mentioned relationship between γ and $\partial f_0 / \partial v$ can be efficiently interpreted by the considerations used for Figs. 4.3.1 and 4.4.1. Evidently, the running electrostatic wave interacts with the plasma electrons. The particle dynamics at a relatively low value of γ, i.e. in the quasi-stationary wave in the coordinate system connected with the wave, is efficiently described by the phase diagram shown in Fig. 4.4.1. If the velocity of the particle v is strictly equal to $\omega/\varkappa$, the particle remains stationary in the centre of the 'shells'. If the particle velocity is lower than the resonance velocity, the particle will be accelerated by the wave, and if it is greater, it will be decelerated. In turn, acceleration or deceleration should be accompanied by the removal of energy from the wave or, vice versa, by the supply of energy to the wave. In the former case, the wave will attenuate, in the latter case it will grow stronger.

In the plasma there are many particles which move in the absence of the wave, both slower than the phase velocity or faster than this velocity. Therefore, in the initial stage (i.e. during the time shorter than the time required by the imaging point ('particle') to bypass the phase trajectory), the wave will be intensified or attenuate depending on which type of particle prevails: $v > \omega/\varkappa$ or $v > \omega/\varkappa$. This difference in the number of the particles is also determined by the sign of $\left.\dfrac{\partial f_0}{\partial v}\right|_{v=v_\phi}$.

The final stage of evolution of the wave

What takes place further? Three situations can be realised here, taking into account the fact that the linear waves at the end should either attenuate or reach some stationary level with the phase diagram of the type shown in Fig. 4.3.1.

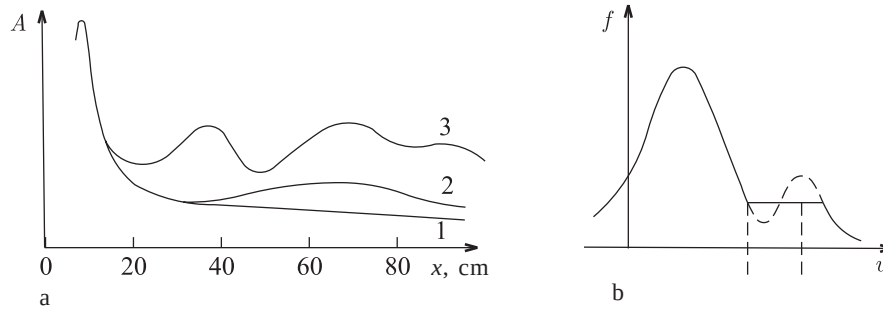


Fig. 4.4.4. Evolution of Langmuir oscillations in Maxwell in plasma: a) dependence of the oscillation amplitude on the distance between the probe-emitter and the probe-receiver with the variation of the perturbation amplitude, i.e. the probe potential V: V = 0.96 V (1); V = 2.58 V (2); V = 9 V (3); b) the final stage of variation of the distribution function – formation of a plateau.

a. $\gamma < 0$, $|\gamma| > |1/\tau_{1/2}|$, where $\tau_{1/2}$ is the half-cycle of bypass along the phase trajectory. In other words, the wave attenuates quite rapidly. In this case, perturbation remains linear, and the electrical field rapidly weakens (curve 1 in Fig. 4.4.4a)

b. $\gamma < 0$, $|\gamma| < |1/\tau_{1/2}|$. In this case, after time $t \sim \tau_{1/2}$, the particles on the phase trajectory complete half of the bypass and the interaction of the particles with the wave now changes its sign. The particles which receive energy from the wave now transfer their energy back to their wave, and vice versa. Attenuation of the wave, if $\partial f_0 / \partial v < 0$, is arrested and the amplitude starts to increase. The resultant situation is no longer described by the linear theory, and the wave reaches some stationary level. This confirms efficiently the results of experiments. The initial distribution function is characterised by the formation of a 'plateau' (curve 2 and 3 in Fig. 4.4.4a). The structures with the trapped particles discussed in the previous section form here.

c. Finally, $\partial f_0 / \partial v < 0$, the waves affect the entire velocity range $v_1 < \omega / \varkappa < v_2$, the amplitude increases and sooner or later reaches some stationary level.

Since the resonance attenuation of the waves, which is often referred to as 'Landau attenuation', is investigated, it is necessary to make another comment. This 'attenuation' is manifested in a clear form only at relatively high values of γ. However, only the electrical field attenuates rapidly, because the particles 'remember' about the wave during time τ_e for the wave, i.e. the free path time of the electron to a Coulomb collision. This is indicated not only by equation (4.4.11b) but also by direct experiments in which the trapped particles were detected. This is especially evident in the clarification of wave barriers, predicted

by V.N. Oraevskii and discussed below. Further confirmation of the presence of the memory is the 'plasma echo' effect [113].

4.4.4. Experimental investigations of resonance attenuation

Investigations of the attenuation of the electrical field [112]. In these experiments measurements were taken of the spatial attenuation of a longitudinal electron wave of the type $\exp\{-i\omega t + ik_r x - k_i x\}$, excited by means of high-frequency oscillations, supplied to a Langmuir probe. The plasma with a density of 10^8–10^9 cm^{-3} and a temperature from 5 to 20 eV was generated using a plasmotron. The oscillations were recorded with a second Langmuir probe situated at different distances from the first probe. It was shown that the attenuation does indeed take place, and the experimental points fit quite satisfactorily the theoretical curve, if the value of γ is high. However, if γ is low, the wave reaches a stationary level.

In order to verify whether attenuation is indeed associated with resonance electrons, the potential of the auxiliary electrode was changed in the experiments thus 'truncating the tail' of the Maxwell distribution, i.e. the fastest electrons were removed. As soon as the truncation boundary reached some specific value v_*, the attenuation mechanism greatly changed. The value v_* was very close to the phase velocity of the wave $v_{ph} = \omega/k$, and this confirms quite clearly the attenuation of the wave on the resonance electrons.

Detection of trapped particles

Previously, we discussed the role of trapped particles in the evolution of Langmuir oscillations, especially if these oscillations were generated without trapped particles. An efficient method of detection of these particles was proposed by V.S. Oraevskii [113]. The method may be described as follows. A plasma with a variable density is generated in a relatively long pipe (Fig. 4.4.5). Subsequently, Langmuir waves with a frequency $\omega > \omega_{0I}$ are generated in the zone I. The waves travel to the zone with a higher density, and $\omega_{0II} > \omega_{0I}$. The distance between the source of oscillations – the probe connected to a microwave generator and a barrier, and also the oscillation amplitude were selected to ensure the efficient formation of the structure of the set of the trapped particles. Consequently, in the zone III, where $\omega_{0II} = \omega_{0I}$, the probe starts to record oscillations with frequency ω. Evidently, in contrast to the harmonic wave which cannot overcome the barrier, the van Kampen waves overcome the barrier and generate Langmuir oscillations in the zone III.

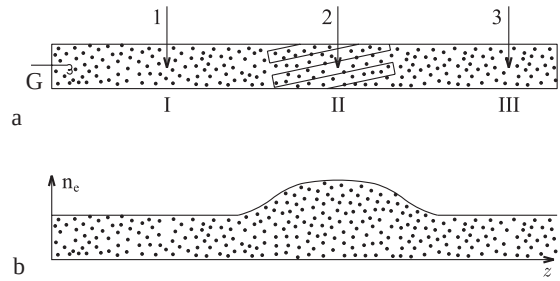


Fig. 4.4.5. Diagram of Oraevski's experiments with the investigation of the effect of clearing of a barrier for Langmuir waves: a) experimental setup, G – oscillation generator, 1, 2, 3 – Langmuir probes, I, II, III – plasma volumes; b) distribution of the concentration of the plasma electrons along the *z*-axis.

'Plasma echo'. Reversibility of wave attenuation

It may appear that the attenuation of waves is an irreversible process, but this is not so. The Vlasov equations are reversible in time. This is indicated by, in particular, the 'plasma echo' effect. In this effect, the effective attenuation of the **E**-field is due to the scatter of the particles close to resonance velocity and, therefore, the wave structure is 'eroded' as a result of this scatter with increase of the distance from the source. Nevertheless, the particles 'remember' their participation in the wave and this memory can be exposed by supplying a variable potential of the appropriate frequency to the probe immersed in the plasma in the area from which the **E**-field has already disappeared. Then, in the flow which appears 'to have forgotten' the wave, the **E**-field with Raman frequency appears again (Fig. 4.4.6).

A convincing model of the 'echo' was proposed by A.A. Vedenov and A.M. Dykhne [9]. Let there be an elongated light source. A grating with the wave number $k_1 = 2\pi/\Lambda_1$ is placed between the light source and a screen. Naturally, the screen situated at a relatively large distance is illuminated almost completely uniformly. Another grating is used - with a different wave number $k_2 = 2\pi/\Lambda_2$ – and is placed between the first grating and the screen. The screen is then moved without touching the grating. Consequently, a relatively sharp image of the grating with $k_3 = k_2 - k_1$ appears on the screen in some position.

Formally, this may be described as follows. At the distance ξ from each grating, the distribution of light in the angle contains a harmonics

$$\exp\{iky + ik\theta\xi\}\phi(\theta).$$

Here $k = k_1, k_2$, $\xi = x, x–d$ (Fig. 4.4.6), and $\Phi(\theta)$ is the distribution of light intensity in front of the grating. Evidently, the intensity of radiation

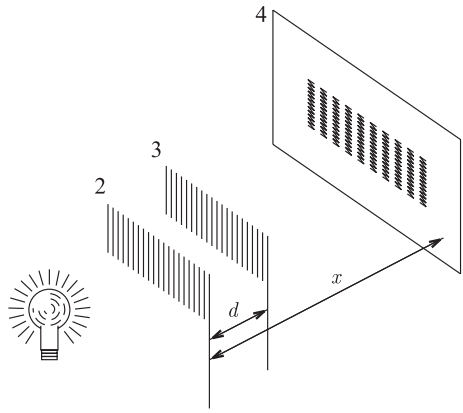


Fig. 4.4.6. Schematic of the 'echo' effect according to the Vedenov–Dykhne model: 1 – elongated light source; 2 – the grating modulating the light flux of the source; 3 – the grating 'showing' modulation; 4 – screen.

at the distance x from the first grating to the screen after passage through the second grating is proportional to

$$\int \exp\{ik_1 y + ik_1 x\theta \pm [ik_2 y + ik_2\theta(x-d)]\}\Phi(\theta)d\theta.$$

This shows that at the distance

$$x = k_2 d/k_2 - k_1$$

the exponent seizes to depend on θ and a 'moire' appears on the screen with $k_3 = k_2 - k_1$.

The transfer of the Vedenov–Dykhne model to plasma systems is quite evident. Therefore, no further attention will be given to this question, and the reader should refer to the monograph [112].

4.5. Oscillations of two-component plasma

4.5.1. Ionic sound

If the oscillations are longitudinal, then regardless of the model, the dispersion equation of the linear waves has the following form

$$\epsilon_0 = 0.$$

When we were discussing only the oscillations of the electronic component, then (4.4.5)

$$\epsilon_0 = 1 + \frac{4\pi e^2}{m\varkappa}\int_{\cup} \frac{\varkappa \dfrac{\partial f_0}{\partial v} d\mathbf{v}}{\omega - \varkappa \mathbf{v}}.$$

However, if we investigate the electron–ion plasma, then

$$\epsilon_0 = 1 + \frac{4\pi e^2}{m\varkappa}\int_{\cup}\frac{\varkappa\frac{\partial f_{e0}}{\partial v}dv}{\omega - \varkappa v} + \frac{4\pi e^2}{M\varkappa}\int_{\cup}\frac{\frac{\partial f_{i0}}{\partial v}dv}{\omega - \varkappa v}. \qquad (4.5.1)$$

It is shown here that the singularity is bypassed from the bottom.

On the example of the Langmuir waves in the electronic plasma it was shown that the dispersion equation in the kinetics, with the exception of the imaginary additive, coincides with the identical equation, derived in two-fluid hydrodynamics. Evidently, the same should apply to electron–ion plasma. However, in this case at $\varkappa r_D \ll 1$, the low-frequency ion–sound waves have the phase velocity

$$v_{ph} = \sqrt{k\left(\frac{T_i + T_e}{M}\right)}. \qquad (4.5.2)$$

Therefore, if $T_e \leqslant T_i$, then $v_{ph} \sim v_{Ti}$ and, consequently, ionic sound should rapidly attenuate as a result of the large number of the resonant particles. Therefore, in reality ionic sound only exists at $T_e \gg T_i$ (see [114, 115]).

4.5.2. Oscillations in current-carrying plasma (at $H_0 = 0$)

One of the simplest ion–electron systems with the relative movement of the particles is the current-carrying plasma in which the electrons move in relation to the ions. Naturally, suitable conditions are created in this system for growth of the oscillations. This situation will be studied in greater detail, assuming that the magnetic field can be ignored. As previously, the dispersion equation has the form $\epsilon = 0$ and in the case in which the non-perturbing distribution functions of the electrons and ions are Maxwellian, the equation is similar to the equation (4.5.1), but only with the shift of velocities

$$\mathbf{v}_e^1 = \mathbf{v}_i^1 + \mathbf{u}.$$

Thus

$$\epsilon = 1 + \sum_\alpha \frac{1}{\varkappa^2 r_D^2}\left(1 + i\sqrt{\pi}\frac{\omega - \varkappa\mathbf{u}^{(\alpha)}}{|\varkappa| v_T^{(\alpha)}} W\left(\frac{\omega - \varkappa\mathbf{u}^{(\alpha)}}{|\varkappa| v_T^{(\alpha)}}\right)\right).$$

Here $\mathbf{u}^{(\alpha)}$ is the current velocity of the α-component.

Analysis of the equation is quite cumbersome and, therefore, referring the reader to [31], it is convenient to present the diagram of the regions of excitation of different types of oscillations (Fig. 4.5.1).

The most interesting mechanism of the formation of oscillations is found in the case in which the current velocity is of the order of the

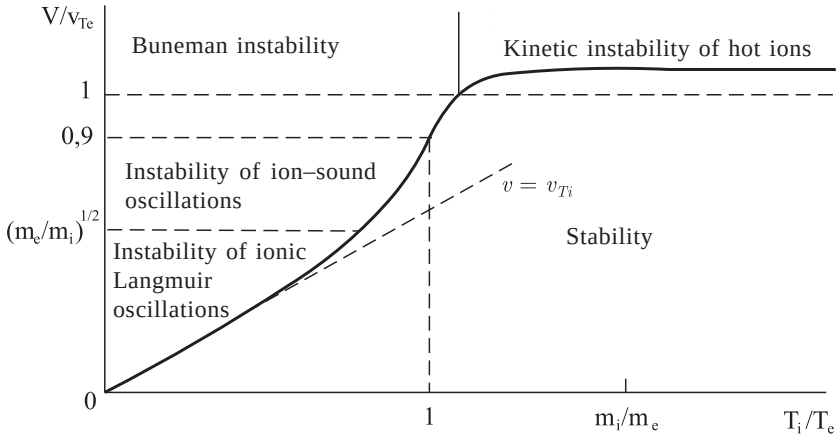


Fig. 4.5.1. Diagram of the regions of oscillations in current-carrying plasma.

thermal velocity of the electrons. This is the so-called Budker–Buneman instability. If it is attempted to generate a discharge in the *Z*-pinch configuration in a low density gas so that the condition

$$u \sim v_{Te},$$

is fulfilled, the developed small-scale oscillations cause rapid heating of the electrons. This process is referred to as 'turbulent heating'.

4.6. Quasi-linear approximation [7, 116]

4.6.1. Derivation of the main equation

In the case in which at the initial moment the distribution function $f(v, 0)$ contains the regions of instability ($\partial f/\partial v > 0$), and the amplitude of the resultant oscillations is relatively low, so that the resonant particles does not manage to travel around the 'eye' of the phase diagram, it is quite easy to derive the equation describing in the first approximation the reversed effect of the resultant waves on $f(v, t)$. This can be carried out by the averaging method using exactly the same procedure which was used in the section 1.2 in calculating the dynamics of particles in a heterogeneous high-frequency field.

For this purpose, assuming that the process is uniform and one-dimensional, $f(v, x, t)$, $E(x, t)$ is presented in the form

$$f = F(v,t) + \tilde{f}(v,x,t);\ E = E_0 + \tilde{E}(x,t). \tag{4.6.1}$$

Substituting (4.6.1) into the kinetic equation (4.4.1) and assuming that $E_0 = 0$, gives

$$\frac{\partial}{\partial t}\left(F+\tilde{f}\right)+v\frac{\partial}{\partial x}\left(F+\tilde{f}\right)=\frac{e}{m}\tilde{E}\frac{\partial}{\partial v}\left(F+\tilde{f}\right)=0. \tag{4.6.2}$$

In this equation, F is a slow function of time, and f oscillates with Langmuir frequency. Therefore, using averaging with respect to these oscillations, the equation (4.6.2) is divided into two equations:

$$\frac{\partial F}{\partial t}+v\frac{\partial F}{\partial x}=\frac{e}{m}\left\langle\tilde{E}\frac{\partial f}{\partial v}\right\rangle; \tag{4.6.3a}$$

$$\frac{\partial \tilde{f}}{\partial t}+v\frac{\partial \tilde{f}}{\partial x}=\frac{e}{m}\tilde{E}\frac{\partial F}{\partial v}. \tag{4.6.3b}$$

The solution procedure is now evident. Neglecting the dependence of F on t, we determine $\tilde{f}$ and $\tilde{E}$ of the system from the equation (4.6.3b) and the Poisson equation:

$$\frac{\partial}{\partial x}\tilde{E}=4\pi e\int \tilde{f}dv \tag{4.6.4}$$

in the form of the Fourier components with respect to space and time (~exp{$-i(\omega t-\varkappa x)$})

$$\tilde{f}_{\varkappa}=i\frac{e}{m}\frac{\tilde{E}}{\omega-\varkappa v}\frac{\partial F}{\partial v}. \tag{4.6.5}$$

Assuming that the intensity of oscillations is low, and the spectrum is wide[8] the electrical field can be written in the form of superposition of independent harmonics

$$\tilde{E}\left(x,t\right)=\int_{-\infty}^{+\infty}E_{\varkappa}\exp\left\{-i(\omega_{\varkappa}t-\varkappa\cdot x)\right\}d\varkappa. \tag{4.6.6}$$

Here $\omega_{\varkappa}=\omega_{\varkappa 0}+i\gamma_{\varkappa}$. Taking into account the reality $\tilde{E}$, we have $E_{(-\varkappa)}=E^{*}_{(\varkappa)}$, $\omega_{\varkappa 0}=-\omega_{(-\varkappa)0}$, $\gamma_{\varkappa}=\gamma_{(-\varkappa)}$.

Using the assumption on the low density of the beam, i.e. smallness of γ, we can apply the rule of passage around the contour of the Landau type and assume that

[8]In reality, this can be ensured by injecting into plasma a beam of electrons with a wide velocity distribution function.

$$\frac{1}{\omega_{\varkappa}-\varkappa v}=\frac{P}{\omega_0-\varkappa v}-i\pi v\delta\left(\varkappa v-\omega_{\varkappa 0}\right). \tag{4.6.7}$$

Consequently, we determine

$$\left\langle\tilde{E}\tilde{f}\right\rangle=\left(i\frac{e}{m}P\int_{-\infty}^{+\infty}\frac{E_{\varkappa}-E_{-\varkappa}}{\omega_{\varkappa}-\varkappa v}\exp\left\{-2\gamma_{\varkappa}t\right\}d\varkappa+\right.$$
$$\left.+\frac{e\pi}{m}\int E_{\varkappa}^2\exp\left\{-2\gamma_{\varkappa}t\right\}\delta\left(\omega_{\varkappa}-\varkappa v\right)d\varkappa\right)\frac{\partial F}{\partial v}. \tag{4.6.8}$$

Assuming that configuration is uniform with respect to x and taking into account only the resonance member in (4.6.8), we obtain the final system of equations of quasi-linear approximation (Romanov–Filippov, Vedenov–Velikhov–Sagdeev, Drummond–Pines)

$$\begin{aligned}
&\frac{\partial E_{\nu}^2}{\partial t}=2\gamma_{\nu}E_{\nu}^2; \qquad \gamma_{\nu}=\frac{\pi v^2\omega_{0e}}{2}\frac{\partial F}{\partial v};\\
&\frac{\partial F}{\partial t}=\frac{\partial}{\partial v}D_{\nu}\frac{\partial F}{\partial v}; \qquad D_{\nu}=\frac{\pi e^2 v}{m_e^2\omega_0}E_{\nu}^2;\\
&\frac{\partial E_{\varkappa}^2}{\partial t}=2\gamma_{\varkappa}E_{\varkappa}^2\Big|_{\varkappa=\frac{\omega}{v}} \qquad \nu\equiv\frac{\omega_0}{\varkappa}=v.
\end{aligned} \tag{4.6.9}$$

This shows that when the function $f\,(v,0)\equiv F(v,0)$ has an interval in which $(\partial f_0(v)/\partial v>0$, the evolution $F(v,\,t)$ leads to the formation of a plateau (Fig. 4.4.4b).

4.6.2. Several comments regarding the kinetics of Langmuir waves

1. It has been shown that the linear waves either attenuate relatively rapidly or reach a non-linear level. The interaction of the non-linear waves with each other generates a large variety of the processes. In particular, these interactions may result in the merger of the waves with the longitudinal type being retained, and may result in the transformation of longitudinal waves to transverse electromagnetic waves with subsequent radiation (section 8.1). In turn, the merger with the retention of the longitudinal waves leads, because of the Miller force, to the formation of the regions with the reduced ion concentration and to subsequent ‘collapse’ of these ‘Langmuir resonators’ or, in other words, ‘the collapse’ of Langmuir waves. This is the mechanism of collisionless transfer of the wave energy to the ‘thermal’ energy of the accelerating electrons.

2. The previously examined quasi-linear kinetic equation is suitable only for the case of weak ‘turbulence’ when the interaction of the field can be ignored, i.e. the perturbation of the medium can be regarded as sets waves with random phases and low amplitudes. Nevertheless, the interesting feature of this equation is that it is the simplest kinetic equation for particles with the right-hand part. However, in this case, in contrast to the Boltzmann and other similar equations, we take into account the interaction of particles with waves and not collisions of the particles with each other.

If the system is such that it is necessary to take into account the interaction of waves with each other, then we can do little here by analysis, with the exception of constructing semi-qualitative models with limited reliability. Here a significant role is now played by numerical calculations. Yu.S. Sigov (Institute of Applied Mathematics of the Russian Academy of Sciences) has contributed significantly to the procedure of these calculations [40].

5

Kinetics of two-component plasma in classic collisions

In chapter 4 we stressed the role of collective processes in plasma because in many cases, even in the case of relatively rare plasma, it is necessary to take into account paired collisions of the particles. This will also be discussed here.

However, prior to deriving the equations, we consider Fig. 5.0.1 which shows schematically the evolution of the region occupied by particles in the space of velocities at different types of collisions.

Figure 5.0.1a shows the initial region which is the same for all cases. Figure 5.0.1b shows its evolution at the moment $t_1 > 0$, in the absence of collisions. The shape of the region may become highly exotic, but the phase volume remains unchanged and the 'shell', containing these particles, is retained.

Figure 5.0.1c shows, for the same moment t_1, the region occupied by the particles in the presence of strong paired collisions ('Boltzmann case'). It can be seen that the particles may 'jump' outside the limits of

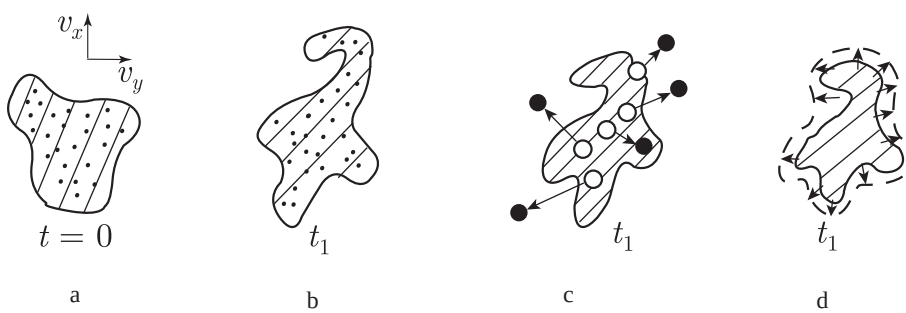


Fig. 5.0.1. Evolution of the cloud of particles in the space of velocities at different nature of collisions between them: a) initial form of the cloud; b, c, d) the form of the cloud at $t_1 > 0$, described by the Liouville equation (b); in the presence of strong, similar collisions (c); in the presence of weak (remote) collisions (d).

the shell, penetrate into the shell from the outside and move randomly inside the shell.

Finally, Fig. 5.0.1d shows schematically the evolution of the region occupied by the particles in the space of velocities in the case of weak (remote) but sufficiently frequent collisions. It can be seen that this region spreads and is governed by the diffusion law.

5.1. Introduction

The Boltzmann equation

The kinetic Liouville equation describes the regular dynamics on all levels of detailisation of the description of the plasma volume from the limiting smooth equilibrium configurations (f = f(ε)) to the system of individual particles. However, if the former – the models 'smoothed' to the maximum extent – are in many cases not sufficient for simulating actual processes, the latter are excessively detailised.

Therefore, it is often necessary to use intermediate models in which the fine-scale (with respect to space and time) processes would be taken into account at a sufficiently high 'smoothness' of the distribution function at some 'first approximation'. This is similar to the work carried out in the 19th century in hydrodynamics. i.e. generalisation of Euler equations by taking into account the viscosity, i.e.derivation of the Navier–Stokes equations (1821–1854) which take into account the effects $\sim \frac{\lambda}{L}$, where λ is the free path length of the particles of the medium.

This also happened in the case of the gas kinetics in the 19th century. In fact, in 1852, G. Liouville reported on the formulation of the kinetic equation[1]

$$\frac{df}{dt} = 0 \tag{5.1.1}$$

for the distribution function of non-interacting particles. In 1872, Boltzmann published his outstanding kinetic equation for a gas taking into account collisions [117]. In the simplest case, the particles of the same type

$$\frac{df}{dt}(t, \mathbf{x}, \mathbf{v}) = S \tag{5.1.2a}$$

where is the Boltzmann collisional term has the form ($d\sigma$ is the differential scattering section, see (5.2.1), $f_1 = f\,(t,\, \mathbf{x},\, \mathbf{v}_1)$

[1]The theorem regarding the retention of the phase volume was published by Liouville in 1838.

$$S = \iint (f f_1' - f f_1) |\mathbf{v} - \mathbf{v}_1| d\sigma d\mathbf{v}_1 = S_1 - S_2 \tag{5.1.2b}$$

Assuming that the reader is acquainted with the Boltzmann equation, we will discuss it only briefly. The second term on the right-hand side of (5.1.2b)

$$S_2 \equiv \iint f(t,\mathbf{x},\mathbf{v}) f(t,\mathbf{x},\mathbf{v}_1) |\mathbf{v} - \mathbf{v}_1| d\sigma d\mathbf{v}_1 = f(t,\mathbf{x},\mathbf{v}) n(t,x) \left\langle \sigma |\mathbf{v} - \mathbf{v}_1| \right\rangle \tag{5.1.3}$$

describes the 'knocking out' of the particle from its area in the space of velocities as a result of collision with any particle, moving with the velocity $\mathbf{v}_1$. The first term S_1 in the right-hand part (5.1.2b) characterises the 'penetration' of the particles into the given cell. A special feature of the functions included here f' and f_1' are only their arguments:

$$\begin{aligned} \mathbf{v}' &= \frac{\mathbf{p}_1 + \mathbf{p}}{2M} + \mathbf{e}_0 \left| \frac{\mathbf{p} - \mathbf{p}_1}{2M} \right|, \quad p_1 = M\mathbf{v}_1; \\ \mathbf{v}_1' &= \frac{\mathbf{p} + \mathbf{p}_1}{2M} - \mathbf{e}_0 \left| \frac{\mathbf{p} - \mathbf{p}_1}{2M} \right|, \quad p = M\mathbf{v} \end{aligned} \tag{5.1.4}$$

where $f' \equiv f(t,\mathbf{x},\mathbf{v}')$, $f_1' \equiv f(t,\mathbf{x},\mathbf{v}_1')$, $\mathbf{e}^0$is the unit vector of an arbitrary direction.

This selection of the arguments at collisions of the particles with the velocity $\mathbf{v}'$ and $\mathbf{v}_1'$ ensures the appearance of a single particle with the velocity $\mathbf{v}$. Of course, in this case we are concerned with elastic collisions at which the momentum and energy of the particles remain unchanged. It is assumed that the particle interaction forces are characterised by the small radius of action and the particles can be regarded as simple elastic spheres, situated at the distance $l \gg a$, where a is the particle radius.

It should be mentioned that the derivation of the equation (5.1.2a) by Boltzmann was purely 'physical'. The formal derivation of the equation was described in 1946 by N.N. Bogolyubov [14]. Bogolyubov's methods also improve the accuracy of this equation.

Special features of the kinetic equation for Coulomb particles

The kinetic equation for the particles with Coulomb interaction should take into account the long-range effect of the particle field. The smallness parameter which should be used for the expansion of the consecutive approximations of the collisional member is the quantity

$$\epsilon\frac{\epsilon^2 n^{1/3}}{kT}, \quad l \sim n^{1/3}.$$

Here, the numerator gives the potential energy of interaction of two particles situated at the mean distance $r_0 \sim n^{-1/3}$.

Formally, the equations (5.1.1) are easily transformed to the equation with the right-hand part – the collisional term, if f, $\mathbf{E}$, $\mathbf{H}$ are averaged out with respect to microfluctuations. For this purpose, following the classic Lorenz principle, it should be assumed that

$$f = f_0 + f_1, \quad \mathbf{E} = \mathbf{E}_0 + \mathbf{E}_1, \; \mathbf{H} = \mathbf{H}_0 + \mathbf{H}_1.$$

Without considering the details of the selection of the averaging procedure, it is assumed that

$$\langle f \rangle = f_0, \quad \langle f_1 \rangle = 0;$$
$$\langle \mathbf{E} \rangle = \mathbf{E}_0, \quad \langle \mathbf{E}_1 \rangle = 0.$$

and similarly for $\mathbf{H}$.

Consequently, averaging (5.1.1) we obtain

$$\frac{\partial f_0}{\partial t} + \mathbf{v}\frac{\partial f_0}{\partial \mathbf{x}} + \frac{e}{M}\left(\mathbf{E}_0 + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}_0\right]\right)\frac{\partial f_0}{dv} = -\left\langle \frac{e}{M}\left(\mathbf{E}_1 + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}_1\right]\right)\frac{\partial f_1}{\partial v}\right\rangle \equiv S \tag{5.1.5}$$

This shows that in order to determine S in the given scheme, f_0, $\mathbf{E}_0$, $\mathbf{H}_0$ should be used to express the correlation functions

$$R_1 = \langle \mathbf{E}_1 f_1 \rangle, \quad R_2 = \left\langle \left[\mathbf{v}, \mathbf{H}_1\right] \right\rangle \frac{\partial f}{\partial v}$$

However, these functions cannot be determined in the non-equilibrium case, although this is approximately possible in a number of cases.

The first term of expansion S was calculated by Landau [118], on the basis of the Boltzmann equation, ignoring short-range collisions and taking into account long-range ones. This imposes restrictions on the region of applicability of this kinetic expression. The scales of the calculated processes should satisfy the conditions

$$L \gg n^{-1/3} \equiv l_0; \; t \gg \frac{l_0}{v_{\text{rel}}}. \tag{5.1.6}$$

Here E and t are the spatial and time scales.

More accurate equations have also been proposed for the collisional term (Lenard, Balescu, Klimontovich, etc) but they are not sufficiently effective because they are very complicated.

Here, we derive the Landau collisional term using, as the basis, the specific calculation procedure proposed by V.A. Trubnikov. His approach is not only extremely simple but as a result of the derivation of the kinetic equation yields a number of highly suitable equations which are of separate interest [120].

Markovian processes. The Focker–Planck equation

The derivation of the Landau equation can be explained more efficiently if we take into account one of the aspects of the theory of random processes. In the group of the probability processes there is the class of 'maximally determined processes' – the so-called Markovian processes. One of their properties is that knowing the probability of distribution of some set of the parameters $\{a_1\}$ at t_1 we can determine unambiguously the probability distribution of the parameters $\{a_2\}$ at any other moment of time t_2. The formal determination of the continuous Markovian processes can be described as follows. We introduce the transmission function from the state 1 to state 2 during the time period (t_2, t_1):

$$W \equiv W(a_2;\ t_2|a_1;\ t_1), \tag{5.1.7a}$$

Evidently, function $W(1|2)$ can be regarded as the distribution function f $(t, \mathbf{x}, \mathbf{v})$ at moment t, normalised with respect to unity. However, in contrast to f, the distribution of the parameters f_0 $(t, \mathbf{x}, t_0)$ at the initial moment t_0 is clearly shown in W. The process is Markovian if the following integral equation is satisfied

$$W(a_2,t_2\,|\,a_1,t_1)=\int W(a_2,t_2\,|\,a_2,t_2)W(a,t\,|\,a_1,t_1)da\,. \tag{5.1.7b}$$

Naturally, all types of the kinetic equations are the equations for the transition functions of the appropriate Markovian process. For example, the Boltzmann equation corresponds to the processes at which the parameters of the system (particle velocity) change by 'large' jumps. The situation in the case of Coulomb particles is different. Here, if $\epsilon \ll 1$, the velocity 'jumps' are small and, therefore, the process is of the diffusion type and is described by the Focker–Planck equation resulting from (5.1.7a):

$$\frac{\partial W}{\partial t}+\sum_{j=1}^{n}\frac{\partial}{\partial a_j}\Big[A_j(t,a_1,a_2,...,a_n)W\Big]=\frac{1}{2}\sum_{j,l=1}^{n}\frac{\partial^2}{\partial a_j\partial a_l}\Big[B_{jl}(t,a_1,a_2,...,a_n)W\Big]=0, \tag{5.1.8a}$$

where coefficients A_j and B_j are determined by the equalities

$$\left.\begin{aligned} A_j(t,a_1,a_2,...,a_n) &= \frac{d}{dt}\left\langle \Delta a_j \right\rangle; \\ B_{jl}(t,a_1,a_2,...,a_n) &= \frac{d}{dt}\left\langle \Delta a_j, \Delta a_l \right\rangle. \end{aligned}\right\} \tag{5.1.8b}$$

Here $a_j(t)$ are random coordinates of the point moving in the phase space {a}, and a_j are the jumps; the angular brackets denote the averaging

$$\left\langle b \right\rangle = \int bW d\mathbf{a}. \tag{5.1.8c}$$

When deriving the Focker–Planck equation, it is assumed that Δa_i is such that the mean of the product of three or more jumps is negligibly small.

This shows that for the Coulomb systems it is necessary to calculate A_i, B_{jk} and prove that $C_{ijk} = 0$.

The phase space {**a**} at the relevant kinetics is (**x**, **v**), but we are interested in a more special case in which only the velocity ($\Delta\mathbf{v}$) changes in jumps, whereas the coordinates of the particles do not change. However, not changing in jumps, the particle coordinates can change as a result of the velocity of the particles, and the velocities can change, in addition to jumps, as a result of the external determined forces. Therefore, the following equation can be written

$$\Delta\mathbf{x} = \mathbf{v}\Delta t, \quad \Delta\mathbf{v} = \frac{\mathbf{F}_d}{m}\Delta t + \Delta\mathbf{v}_{st.} \equiv \Delta\mathbf{v}_d + \Delta\mathbf{v}_{st}. \tag{5.1.9}$$

Here $\mathbf{F}_d$ is the determined force, and $\Delta\mathbf{v}_{st}$ is the variation of velocity as a result of collisions. Substituting (5.1.9) into (5.1.8b), we determine the coefficients A_j and B_{jl}

$$\mathbf{A}_{(1-3)} = \left.\frac{d\Delta\mathbf{x}}{dt}\right|_{\Delta t\to 0} = \mathbf{v}; \quad \mathbf{A}_{(3-6)} = \left.\frac{d\Delta\mathbf{v}}{dt}\right|_{\Delta t\to 0} = \frac{\mathbf{F}_d}{m} + \frac{d}{dt}\Delta\mathbf{v}_{st}. \tag{5.1.10}$$

At the same time, the coefficient B_{jl} depends only on $(\Delta\mathbf{v})_{st}$, since the determined coefficients

$$\Delta v_{d,j}\Delta v_{d,l} \sim \Delta t^2$$

and

$$\frac{d}{dt}\Delta v_{d,j}\Delta v_{d,l}\Big| \sim \Delta t\Big|_{\Delta t\to 0} \to 0. \tag{5.1.11}$$

Thus, in the absence of collisions, we naturally obtain the Liouville equation but so far without taking into account the retention of the phase volume:

$$\frac{df}{dt}+\frac{\partial \mathbf{v}f}{\partial \mathbf{x}}+\frac{\partial}{\partial \mathbf{v}}\frac{\mathbf{F}_d}{m}f=0$$

By taking into account the Liouville theorem of 'soft' (long-range) collisions, the equation (5.1.8a) can be reduced, with certain concretisations, to the Focker–Planck equation

$$\frac{df}{dt}+\mathbf{v}\frac{df}{\partial \mathbf{x}}+\frac{\mathbf{F}_d}{m}\frac{\partial f}{\partial \mathbf{v}}+\frac{1}{m}\frac{\partial}{\partial \mathbf{v}}(\mathbf{F}_{st}f)=\frac{\partial^2}{\partial v_i \partial v_l}(D_{jl}f). \qquad (5.1.12)$$

Here D_{jl} is the diffusion tensor of the particles in the space of velocities. Summation is carried out with respect to the two repeating indexes. The derivation of equation (5.1.12) will also be our next task.

5.2. Kinetics of colliding charged particles

5.2.1. Main concepts

The paired collisions of the particles can be efficiently described using some effective particle diameter d and correspondingly the quantity – the collision cross-section

$$\sigma = \pi d^2. \qquad (5.2.1)$$

The cross-section has the dimension of the area. For a neutral relatively rarefied gas, when $d \ll l_0$ is the mean distance between the particles ($l_0 \sim 1/n^{1/3}$), the elastic collision cross-section is determined by the geometrical dimensions of the particles and depends only slightly on the gas temperature and density.

When the molecule travels the path L in the gas, the molecule should collide with all molecules whose centres are situated within the cylinder with the height L and the base $\sigma = \pi d^2$. The free path length λ is the section L in which the particle experiences a single collision. Simple evaluation of λ is based on the assumption that $\lambda\sigma n = 1$, from which[2]

$$\lambda = \frac{1}{n\sigma}. \qquad (5.2.2)$$

The time scale of the free path is

$$\tau = \frac{\lambda}{v} = \frac{1}{nv\sigma}. \qquad (5.2.3)$$

[2]If the velocity v_0 of the investigated particle is considerably lower than v_1 – the characteristic velocity of the particles within which this particle moves, the free path of this particle is expressed by the equation $\lambda \approx v_0/(n\sigma v_1)$. A suitable example of this is the movement of a neutral atom in the cloud of electrons ionising this atom.

Here v is the mean relative velocity of the investigated particles.

Knowing the velocity of the particles and the free path, we can determine the heat conductivity χ and viscosity η coefficients which determine the density of the momentum flow $\pi_{\alpha\beta}$ and the energy flux $\mathbf{q}$, respectively, determined by the collisions in the gas. In a conventional non-ionised gas

$$\mathbf{q} = -\chi \nabla(kT);$$
$$\pi_{\alpha\beta} = -\eta\left(\frac{\partial v_\alpha}{\partial x_\beta} + \frac{\partial v_\beta}{\partial x_\alpha} - \frac{2}{3}\delta_{\alpha\beta}\,\mathrm{div}\,\mathbf{v}\right). \qquad (5.2.4a)$$

Coefficients χ, η are referred to as the transfer coefficients. The kinetic theory of gases shows that these coefficients can be approximately determined using the equations

$$\eta = \frac{m v_T}{\sigma};\quad \chi = \frac{v_T}{\sigma}, \qquad (5.2.4b)$$

where $v_T = \sqrt{kT/m}$ is the thermal velocity of the molecules.

If the gas is ionised, another transfer coefficient – conductivity σ_E – appears

$$\mathbf{j} = \sigma_E \mathbf{E} \qquad (5.2.5a)$$

in the absence of the magnetic field

$$\sigma_E = \frac{e^2 n_e \tau_{e0}}{m_e}. \qquad (5.2.5b)$$

For detailed characterisation of individual processes it is necessary to introduce several sections and, correspondingly, several free path lengths and relaxation times.

Thus, for the elastic collisions of spherically symmetric particles, the following quantity is of special interest

$$\sigma^{(k)} = \int (1 - \cos^k\theta)\, d\sigma; \quad k = 1, 2, 3, \ldots \qquad (5.2.6)$$

Here (Fig. 5.2.1)

$$d\sigma = \rho(\theta) d\rho(\theta) d\varphi = \frac{\rho(\theta)}{\sin\theta}\frac{d\rho(\theta)}{d\theta} d\Omega; \quad d\Omega = \sin\theta\, d\theta\, d\varphi \qquad (5.2.7)$$

is the so-called differential scattering cross-section, ρ is the impact parameter as a function of the scattering angle θ in the system of the centre of inertia.

The cross-section

$$\sigma^{(1)} = \int (1 - \cos\theta) d\sigma \tag{5.2.8}$$

is referred to as the transport cross-section or diffusion cross-section. This is associated with the fact that the multiplier (1 – cos θ) determines the loss of the directional velocity of the particle in elastic scattering

$$\Delta v_z = -v_0 (1 - \cos\theta)$$

The quantity

$$\sigma^{(2)} = \int (1 - \cos^2\theta) d\sigma \tag{5.2.9}$$

should be referred to as the deviation cross-section because the multiplier $1 - \cos^2\theta = \sin^2\theta$ characterises the RMS of the transverse velocity of the particle at scattering of the flat flow on a stationary force centre. In conventional gases, the viscosity and heat conductivity of the gas are determined by the deviation cross-section $\sigma^{(2)}$, whereas diffusion is determined by the cross-section $\sigma^{(1)}$.

5.2.2. Coulomb collision: force acting on a test particle [120]

Collision cross-sections

The theory of paired elastic[3] collisions of charged particles is based on the Rutherford theory for scattering of a charged particle with the charge e_β and mass m on a stationary Coulomb centre with charge e_α (Fig. 5.2.1)

As a result of the central nature of the forces, the movement of a single particle can always be regarded as taking place on a plane. In this case, the particle moves along a hyperbola, and the scattering angle is associated with the impact parameter by the following relationship:

$$\operatorname{tg}\frac{\theta}{2} = \frac{\rho_0}{\rho}; \quad \rho_0 = \frac{e_\alpha e_\beta}{m v_0^2}. \tag{5.2.10}$$

Here ρ_0 is the value of the impact parameter at which the particle is deflected by the right angle ($\theta = \pi/2$). However, in plasma we are not interested in the deviation of the individual particle and would like to examine the scattering of the flow of identical particles 'falling' on the scattering centre. Let it be that the velocity of the particles in the initial

[3]Inelastic collisions, leading to excitation and ionisation, are investigated in chapter 6.

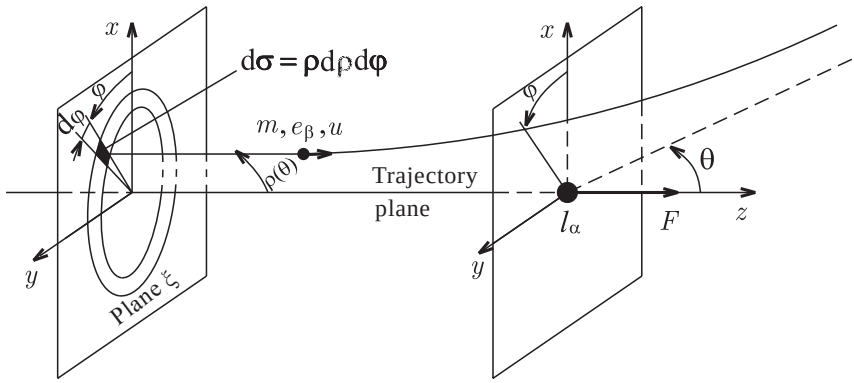


Fig. 5.2.1. Calculation of the mean force, acting from the side of the flow of charged particles on the moving centre.

stage is the same, v_0. Different particles in the flow have different impact distances and are correspondingly scattered under different angles θ. *dN* denotes the number of the particles scattered in unit time through the angles in the range between and θ and θ + *dθ,* and φ and φ + *d*φ. On its own, this number is not suitable for characterising the scattering process because it depends on the density of the incident beam (it is proportional to this density). Therefore, we introduce the differential cross-section (5.2.7)

$$d\sigma = \frac{dN}{\nu} = \rho d\rho d\varphi \tag{5.2.11}$$

where $\nu = n_0 v_0$ is the density of the particle flow.

This cross-section is determined by the type of scattering field. Using equation (5.2.10) and definition of the differential cross-section (5.2.1), we easily obtain that

$$d\sigma = \frac{1}{2}\left(\frac{e_\alpha e_\beta}{m v_0^2}\right)^2 \frac{\cos(\theta/2)}{\sin^3(\theta/2)} d\theta d\varphi = \frac{1}{2}\rho_0^2 \frac{\cos(\theta/2) d\theta d\varphi}{\sin^3(\theta/2)}. \tag{5.2.12}$$

Attention should be given to two moments: firstly, the scattering cross-section decreases in *inverse proportion* to the square of the energy of the scattered particle and, secondly, the total scattering cross-section at high target parameters (and correspondingly small θ) *diverges*:

$$\sigma(\rho \to \infty) = \int_{\theta \to 0}^{\pi} \int_{\varphi=0}^{2\pi} d\sigma \to \infty$$

This means that in calculating the total cross-section or the quantities associated with this cross-section, it is necessary to interrupt integration

at some 'maximum' distances ρ_{max}.

The force acting on a stationary centre

Knowing the expression for the differential scattering cross-section, it is quite easy to calculate the force with which the particle flow acts on a stationary centre. This force **F** is equal to (with the reversed sign) the variation in unit time of the total momentum of the particles of the flow. Thus

$$\mathbf{F} = n_\beta v_0 \int d\sigma(\mathbf{p}, \theta, \varphi) \Delta \mathbf{p}^\beta (\theta, \varphi). \tag{5.2.13}$$

Here β is the index of the incident particles, and $\mathbf{p}^\beta$ is the momentum of the β-th particle. Since the collision with a stationary scattering centre is elastic, the velocity of the particles after collision changes only in the direction but not in the absolute value. Therefore (5.2.14)

$$\Delta v_z^\beta = v_0(1-\cos\theta), \quad \Delta p_z^\beta = m\Delta v_z^\beta,$$

The number of particles β, equal to $n_\beta v d\sigma$, passes through the elementary area $d\sigma = \rho d\rho d\varphi$ of the plane ξ (Fig. 5.2.1) in the direction normal to the axis *z* in unit time. Multiplying this value by $\Delta p_z^\beta = m\Delta v_z^\beta$ and integrating over the plane ξ, from ρ = 0 to ρ = ρ_{max}, we determine the variation in unit time of the flow momentum and, consequently, force **F** acting on the stationary centre:

$$\mathbf{F} = \Lambda \frac{4\pi}{m} e_\alpha^2 e_\beta^2 n_\beta \frac{\mathbf{v}_0}{v_0^3} = \sigma^{(coul)} n_\beta v_0^\beta m \mathbf{v}_0^\beta. \tag{5.2.15a}$$

Here Λ is the so-called 'Coulomb logarithm', and $\sigma^{(coul)}$ is the effective cross-section of the coulomb collisions of the β-particles with the stationary centre

$$\Lambda = \frac{1}{2} \ln \left[\frac{\rho_{max}^2 + \rho_0^2}{\rho_0^2} \right] \approx \ln \frac{\rho_{max}}{\rho_0}, \quad 4\pi\rho_0^2 \Lambda = \sigma^{(coul)}. \tag{5.2.15b}$$

ρ_{max} is represented by the Debye radius r_D, calculated from the energy of the incident particles. This is explained by the fact that at the distance equal to the Debye radius, the fields of the individual particles, roughly speaking, merge into a common field[4]

[4]More detailed analysis shows that the charged centre, perturbing the movement of the surrounding particles, results in the change of the surrounding volume charge which screens the field of the centre at the distance of the order of the Debye radius. In fact, this indicates the correlation of the trajectories of the particles with each other.

Thus,

$$\Lambda \approx \left(\frac{r_D}{\rho_0} \right). \qquad (5.2.15c)$$

The characteristic values of the coulomb logarithm are usually distributed in the range $\Lambda = 10–20$. If the multiplier Λ is ignored in (5.2.15a), we obtain (as regards the order of magnitude) the force generated by the so-called short-range collisions at which the scattering angle is of the order of unity ($\theta \sim 1$).

The appearance of the coulomb logarithm is the manifestation of the long-range effect of the electrical field of the particles. It may be concluded that if the neutral particles react only to 'coarse' direct collisions, the charged particles are very 'delicate' and take leave with the counter particles already at large distances.

The force acting on the moving particle

Having the expression for the force (5.2.15a) acting on the stationary centre, it is quite easy to determine the mean force acting on the arbitrary charged particle α, moving with the velocity $\mathbf{v}$ in the medium consisting of the charged particles β. The velocity distribution of these particles is described by the function $f_\beta(\mathbf{v}')$.

In fact, if we have two particles, and the interaction potential between these particles depends only on the distance between them $|\mathbf{r}_1 - \mathbf{r}_2|$, the equations of motion of these particles have the form:

$$m_1 \ddot{\mathbf{r}}_1 = -\nabla_1 U\left(|r_1 - r_2|\right)$$

$$m_2 \mathbf{r}_2 = -\nabla_2 U\left(|r_1 - r_2|\right)$$

Here $\nabla_{1,2} = \left(\frac{\partial}{\partial x_{1,2}}, \frac{\partial}{\partial y_{1,2}}, \frac{\partial}{\partial z_{1,2}} \right)$.

Combining the resultant equations, we obtain the law of inertia – the movement of the centre of gravity

$$M\dot{\mathbf{R}} = m_1 \dot{\mathbf{r}}_1 + m_2 \dot{\mathbf{r}}_2 = \text{const}, \quad M = m_1 + m_2$$

with a constant velocity. At the same time, the difference of the coordinates

$$\boldsymbol{\rho} = \mathbf{r}_1 - \mathbf{r}_2$$

is governed by the equation of the type of motion of one particle

$$\mu_{1,2}\ddot{\boldsymbol{\rho}} = -\nabla U(\rho) \tag{5.2.16a}$$

Here $\mu_{1,2}$ is the reduced mass $\mu_{1,2} = \dfrac{m_1 m_2}{m_1 + m_2}$. Evidently

$$\mathbf{r}_1 = \mathbf{R} + \left(\frac{m_2}{m_1 + m_2}\right)\boldsymbol{\rho},\ \mathbf{r}_2 = \mathbf{R} - \frac{m_1}{m_1 + m_2}\boldsymbol{\rho}. \tag{5.12.16b}$$

This shows that the variation of the particle momentum as a result of collisions is equal to:

$$\Delta\mathbf{p}_1 = m_1\Delta\mathbf{v}_1 = \mu\Delta\mathbf{u};\quad \Delta\mathbf{p}_2 = m_1\Delta\mathbf{v}_2 = -\mu\Delta\mathbf{u} \tag{5.2.17a}$$

Here $\mathbf{u} \equiv \dot{\boldsymbol{\rho}}$. Thus, the transition from the equations for scattering on the stationary centre to the equations for scattering on the moving centre ('test particle') is reduced to substitution

$$m_1 \to \mu_{1,2} \tag{5.2.17b}$$

From the entire set of the particles β we separate an elementary particle flux moving with velocity **v**'. The density of the particles in this flux

$$dn_\beta(\mathbf{v}') = f_\beta(\mathbf{v}')d\mathbf{v}'. \tag{5.2.18a}$$

Consequently, the force acting on the particle α from the side of the elementary flux, taking equation (5.2.15a) and (5.2.6) into account, is described by the equation

$$d\mathbf{F}_\alpha = -\Lambda\frac{4\pi}{\mu_{\alpha\beta}}e_\alpha^2 e_\beta^2\frac{\mathbf{v}-\mathbf{v}'}{|\mathbf{v}-\mathbf{v}'|^3}f_\beta(\mathbf{v}')d\mathbf{v}'. \tag{5.2.18b}$$

Integrating (5.2.18b) with respect to $d\mathbf{v}'$, we determine the required mean force, acting on the particle α, moving through the medium from the particles β with the velocity **v**:

$$\mathbf{F}_\alpha(v) = \int d\mathbf{F}_\alpha = -\Lambda\frac{4\pi}{\mu_{\alpha\beta}}e_\alpha^2 e_\beta^2\int\frac{\mathbf{v}-\mathbf{v}'}{|\mathbf{v}-\mathbf{v}'|^3}f_\beta(\mathbf{v}')d\mathbf{v}'. \tag{5.2.19}$$

An important special feature of equation (5.2.19) is the presence of the reduced mass in the denominator of the right-hand part. If two particles with the same mass interact, then $m_{\alpha\alpha} = \dfrac{m_\alpha}{2}$ i.e. the reduced mass differs

by only a factor of two in relation to the mass of the single ion. However, if the ion and the electron interact, then $\mu_{ei} \approx m_e$. Correspondingly, at the same velocities force F_{ie} is considerably higher than force F_{ii}. This is associated with the fact that in collision of the ions with the electron the latter is deflected, at the given impact parameter p, to a considerably greater extent than the ion. However, if $T_e \sim T_i$, then $F_{i,e} \sim F_{i,i}$.

B.A. Trubnikov noted[5] that the integral with respect to the space of velocities in (5.2.19) has accurately the same form as the one which is used to express the electrostatic field of the system of charges continuously distributed in the conventional coordinate space with the density $q_e(\mathbf{r}) = f_\beta(\mathbf{r})$:

$$\mathbf{E}(\mathbf{r}) = \int \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3} q_e(\mathbf{r}')d\mathbf{r}' = -\nabla_r \Phi(\mathbf{r}) . \tag{5.2.20}$$

Here, the potential $\Phi(\mathbf{r})$ satisfies the Poisson equation $\Delta\Phi = -4\pi q$ and is equal to

$$\Phi(\mathbf{r}) = \int \frac{q(\mathbf{r}')d\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|} . \tag{5.2.21}$$

Taking into account this useful analogy, we introduce formally 'the potential' function $\Phi_\beta(\mathbf{v})$, which is such that

$$\Delta_v \Phi_\beta = f_\beta;$$
$$\Phi_\beta(\mathbf{v}) = -\frac{1}{4\pi}\int \frac{f_\beta(\mathbf{v}')d\mathbf{v}'}{|\mathbf{v}-\mathbf{v}'|}, \tag{5.2.22}$$

i.e.this function is identical with the electrostatic potential ϕ (multiplier, $1/4\pi$, as in the rationalised SI system, is selected because of the convenience considerations). Using the function $\phi_\beta(\mathbf{v})$, equation (5.2.19) for $\mathbf{F}_\alpha$ can be presented in the form:

$$\mathbf{F}_\alpha(\mathbf{v}) = -\Lambda \frac{m_\alpha + m_\beta}{m_\alpha m_\beta}(4\pi e_\alpha e_\beta)^2 \nabla_v \Phi_\beta(\mathbf{v}) = -\frac{q_{\alpha\beta}}{\mu_{\alpha\beta}} \nabla_v \Phi_\beta . \tag{5.2.23a}$$

$$q_{\alpha\beta} \equiv \Lambda(4\pi e_\alpha e_\beta)^2 . \tag{5.2.23b}$$

Function Φ_β for many specific types of $f_\beta(\mathbf{v})$ can be written directly by analogy with the electrostatic equations for the electric potential. Function Φ_β is referred to as the 'first Trubnikov function'.

[5]At approximately the same time a similar claim was made by M. Rosenblut, W. MacDonald and D. Judd [120].

5.2.3. Coulomb collisions: diffusion coefficients

To complete derivation of the kinetic equation, it is necessary to find the tensor

$$2D_{jk} = \frac{d}{dt}\left\langle \Delta v_j \Delta v_k \right\rangle. \tag{5.2.24}$$

These calculations will be carried out in two stages using the same procedure as in calculation of $\mathbf{F}_{st}$ in the previous section. In particular, the scattering on the stationary centre of the single velocity flow will be considered. Calculations are simplified if two coordinate systems are introduced: the Cartesian system for writing the components of the tensor, and the spherical coordinates for calculating these components. The axis z is selected along the flow. In contrast to the case of calculating the deceleration force when it was sufficient to calculate only the variation of the z-component of velocity, and therefore the azimuthal angle φ had no significant role, and here it is necessary to take into account all the components of velocity increment. Evidently, they are equal to

$$\begin{aligned} \Delta u_x &= u_0 \sin\theta\cos\varphi \\ \Delta u_y &= u_0 \sin\theta\sin\varphi \\ \Delta u_z &= u_0(1-\cos\theta). \end{aligned} \tag{5.2.25}$$

Now, by analogy with (5.2.13), we can write the following equation

$$g_{je} \equiv \frac{d}{dt}\left\langle \Delta u_j \Delta u_l \right\rangle = n_u^{(\beta)} u \int \Delta u_j \Delta u_l d\sigma \ . \tag{5.2.26}$$

Here $d\sigma = \rho d\rho d\varphi$, n_u is the density of the flow of the particles with the velocity u directed along the z axis. As previously, ρ and $d\sigma$ are determined by the equations (5.2.10), and $\Delta\mathbf{u}$ can be presented in the following form

$$\Delta u_x = 2u\frac{\rho\rho_0}{\rho^2+\rho_0^2}\cos\varphi; \quad \Delta u_y = 2u\frac{\rho\rho_0}{\rho^2+\rho_0^2}\sin\varphi; \quad \Delta u_z = -2u\frac{\rho_0^2}{\rho^2+\rho_0^2}. \tag{5.2.27}$$

Substituting these expressions into (5.2.26) and integrating with respect to φ, it can be seen that the tensor g_{jk} is diagonal

$$g_{jk} = \begin{pmatrix} B & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & B_1 \end{pmatrix}. \tag{5.2.28}$$

Here

$$B = g_{xx} = g_{yy} = n_\beta \int (\Delta u_x)^2 u d\sigma = \frac{n_\beta}{4\pi u}\left(\frac{4\pi e_\alpha e_\beta}{\mu_{\alpha\beta}}\right)^2 \int_0^{\rho_{\max}} \frac{\rho^3 d\rho}{(\rho^2+\rho_0^2)^2}. \tag{5.2.29}$$

This integral also diverges, and it should also be truncated at some ρ_{max} which can be represented naturally by the Debye radius. In this case:

$$\Lambda_{(2)} \equiv \int_0^{r_D} \frac{\rho^3 d\rho}{(\rho^2 + \rho_\perp^2)^2} = \Lambda - \frac{1}{2} \tag{5.2.30}$$

where Λ is determined by the equation (5.2.15c).

Taking into account that $\Lambda \gg 1$, it can be assumed that

$$\Lambda_{(2)} \approx \Lambda \tag{5.2.31}$$

Hence

$$B = \frac{1}{4\pi u}\left(\frac{4\pi e_\alpha e_\beta}{\mu_{\alpha\beta}}\right)^2 \Lambda \ . \tag{5.2.32}$$

It may easily be seen that the integral $B_1 = n\int (\Delta u_z)^2 u d\sigma$ does not diverge at $\rho \to \infty$ and therefore it can be ignored in comparison with B. Thus, in the final selected coordinate system and for the single velocity flow

$$g_{jl} = \begin{pmatrix} B & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & 0 \end{pmatrix} \ . \tag{5.2.33}$$

To transfer from the interaction of the incident flow with a stationary centre to the interaction with the moving particles, one should use the formula (5.2.17a)

$$\Delta \mathbf{v} = \frac{\mu_{\alpha\beta}}{m_\beta} \Delta \mathbf{u}. \tag{5.2.34}$$

In addition, it is necessary to transfer to an arbitrarily oriented system of Cartesian coordinates. Let the velocity $\mathbf{u}$ in the system have the component $\mathbf{u} = (u_x, u_y, u_z)$. Consequently, as can easily be verified, the diagonal tensor (5.2.33) with the zero zz-component has the form

$$g_{ik} = B\left(\delta_{ik} - \frac{u_i u_k}{u^2}\right) \ . \tag{5.2.35}$$

Taking into account (5.2.34) and (5.2.32) we obtain the final equation for D_{jk}

$$2D_{ik} = \int f_\beta(\mathbf{v}')\left(\frac{\mu_{\alpha\beta}}{m_\alpha}\right)^2 \int \Delta u_i \Delta u_k u d\sigma = \frac{q_{\alpha\beta}}{m_\alpha^2}\frac{1}{4\pi}\int f_\beta(\mathbf{v}')\left(\frac{\delta_{ii}}{u} - \frac{u_i u_k}{u^3}\right) d\mathbf{v}'. \tag{5.2.36}$$

However, this equation can be transformed to a more compact form if it is taken into account that $\mathbf{u} = \mathbf{v} - \mathbf{v}'$ and

$$\frac{\partial^2 |\mathbf{u}|}{\partial v_i \partial v_k} = \frac{\delta_{ik}}{u} - \frac{u_i u_k}{u^3} \quad . \tag{5.2.37}$$

Consequently, we obtain

$$D_{ik} = \frac{q_{\alpha\beta}}{m_\alpha^2} \frac{\partial^2}{\partial v_i \partial v_k} (-\psi(\mathbf{v})). \tag{5.2.38}$$

Here ψ is the 'second Trubnikov function':

$$\psi_{(\beta)}(\mathbf{v}) = -\frac{1}{8\pi} \int f(\mathbf{v}') |\mathbf{v} - \mathbf{v}'| d\mathbf{v}'. \tag{5.2.39}$$

Taking into account the first Trubnikov function (5.2.22)

$$\Phi_\beta = -\frac{1}{4\pi} \int \frac{f(\mathbf{v}')}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}'.$$

It can easily be seen that f, Φ and ψ and are connected by the relationships

$$\Delta\psi_\beta = \Phi_\beta, \quad \Delta\Delta\psi_\beta = \Delta\Phi_\beta = f. \tag{5.2.40}$$

It is important to mention two other equations, linking the variation of the momentum and energy of the test (α) particle, moving in the group of the field (β) particles.

Evidently (5.2.23a)

$$\frac{d\mathbf{p}_{st}}{dt} = -\frac{q_{\alpha\beta}}{\mu_{\alpha\beta}} \nabla_v \Phi_\beta. \tag{5.2.41}$$

The variation of the energy of the test particle is associated not only with the loss of the initial momentum but also with the deviation from the initial direction. Therefore

$$\frac{d\varepsilon_\alpha}{dt} = \frac{m_\alpha}{2} \frac{d}{dt} \overline{v_i v_i} = m_\alpha \left(v_i \frac{d}{dt} \langle v_i \rangle + \frac{1}{2} \frac{d}{dt} \langle \Delta v_i \Delta v_i \rangle \right). \tag{5.2.42a}$$

Using equation (5.2.39) we obtain

$$\frac{d\varepsilon_\alpha}{dt} = -q_{\alpha\beta} \left(\frac{\mathbf{v} \nabla_v \Phi}{\mu_{\alpha\beta}} + \frac{\Phi}{m_\alpha} \right) \cdot \tag{5.2.42b}$$

The equations (5.2.41) and (5.2.42a) will be used in section 5.4.

5.2.4. Landau collisional term

Taking into account that the variation of the phase coordinates of the cloud of the particles can take place not only under the effect of collisions but also under the effect of external regular forces $\mathbf{F}^{(t)}$, the equations (5.2.23a)–(5.2.38), substituted into the Fokker–Planck equation (5.1.8a) yield the kinetic equation for the given case:

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{\mathbf{F}^{(e)}}{m_\alpha}\frac{\partial f_\alpha}{\partial \mathbf{v}} = S^\alpha, \tag{5.2.43a}$$

where

$$S^\alpha = -div_v\left(\sum_\beta \mathbf{b}^{\alpha/\beta}\right);\ \mathbf{b}^{\alpha/\beta} = -\frac{q_{\alpha\beta}}{m_\alpha^2}\left[\frac{m_\alpha}{m_{\alpha\beta}} f_\alpha \nabla\phi_\beta - (\nabla f_\alpha \nabla)\nabla\psi_\beta\right]. \tag{5.2.43c}$$

Substituting (5.2.43b) and (5.2.43a), using the relationship (5.2.40) between f, ϕ and ψ, we obtain the kinetic collisional equation for the Coulomb plasma in the Trubnikov differential form:

$$\frac{D(\nabla\nabla\psi)}{Dt} = \frac{4\pi q^{\alpha/\beta}}{m_\alpha}\left((\nabla\nabla\psi)^2 - \frac{\partial^2\psi}{\partial v_i \partial v_k}\left(\frac{\partial^2 \nabla\nabla\psi}{\partial v_i \partial v_k}\right)\right). \tag{5.2.43c}$$

The expression (5.2.43b) for the flow can be reduced to a more convenient form, introducing

$$U_{ik} = \frac{\partial^2 |\mathbf{v}-\mathbf{v}'|}{\partial v_i \partial v_k} = \frac{\delta_{ik}}{u} - \frac{u_i u_k}{u^3},$$

Consequently, taking into account that $\Phi_\beta = \Delta\psi_\beta$, we obtain

$$\left.\begin{aligned} \frac{\partial}{\partial v_i}\Phi_\beta &= -\frac{1}{8\pi}\sum_{k=x,y,z}\int U_{ik}\frac{\partial f'_\beta}{\partial v_k}d\mathbf{v}';\\ \frac{\partial}{\partial v_i}\frac{\partial}{\partial v_k}\psi_\beta &= -\frac{1}{8\pi}\int U_{ik} f'_\beta d\mathbf{v}, \end{aligned}\right\} \tag{5.2.44}$$

where $f'_\beta = f_\beta(\mathbf{v}')$.

Substituting (5.2.44) into equation (5.2.43b), the expression for the flow $b_i^{\alpha/\beta}$ is presented in the symmetric form:

$$b_i^{\alpha/\beta} = 2\pi\Lambda\frac{e_\alpha^2 e_\beta^2}{m_\alpha}\int\sum_{k=x,y,z} U_{ik}\left(\frac{f_\alpha}{m_\beta}\frac{\partial f'_\beta}{\partial v'_k} - \frac{f'_\beta}{m_\alpha}\frac{\partial f_\alpha}{\partial v_k}\right)d\mathbf{v}'.$$

Consequently, we obtain the final expression for the Boltzmann–Landau kinetic equation:

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{\mathbf{F}^{(e)}}{m_\alpha}\frac{\partial f_\alpha}{\partial \mathbf{v}} = 2\pi\Lambda \sum_{(\beta)} \frac{e_\alpha^2 e_\beta^2}{m_\alpha} \cdot \frac{\partial}{\partial v_i} \int U_{ik} \left(\frac{f_\alpha}{m_\beta}\frac{\partial f'_\beta}{\partial v'_k} - \frac{f'_\beta}{m_\alpha}\frac{\partial f_\alpha}{\partial v_k} \right) d\mathbf{v} \ . \tag{5.2.45}$$

5.3. Transfer equations in two-fluid hydrodynamics

The Landau collisional term is complicated. Therefore, the equation (5.2.45) in its complete form was analytically solved only infrequently and either approximately or with a large number of reservations. Numerical calculations of the Landau equations were also carried out. Of considerable importance is the derivation of the equations of two-fluid hydrodynamics carried out by S.I. Braginskii, from the investigated equations at short free paths ($\lambda \ll L$). This yielded strict 'classic' formulae for the transfer coefficients in completely ionised plasma.

Taking into account the significant role of Braginski's equations in plasma dynamics, the derivation procedure and the equations will be investigated in the initial stage [121].

5.3.1. Scheme of solving the kinetic equation at frequent collisions

The relationship between collisionless kinetics and hydrodynamics was investigated in section 4.1.3. It was noted that in a general case the single kinetic equation is equivalent to an infinite chain of hydrodynamic equations. The situation is different in the case of collisional kinetics with short free paths of the particles. The single kinetics equation in this case is equivalent to three hydrodynamic equations: for density n, velocity $\mathbf{v}$ and temperature T.

If the free path of the particles is considerably smaller than the characteristic scales of heterogeneity of the flow L, the collisional term in the kinetic equation becomes controlling. Symbolically, this can be presented in the form:

$$\frac{Df}{Dt} = \frac{1}{\bar{\lambda}} S[f, f]. \tag{5.3.1}$$

The following notation has been introduced here

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + (\mathbf{v}'\nabla) f + \frac{\mathbf{F}}{M}\nabla_{\mathbf{v}'} f$$

and the small parameter $\bar{\lambda}$ is explicitly indicated and describes the free path related to the dimensions of the system.

If $\bar{\lambda}$ tends to 0, then in the absence of singularities, equation (5.3.1) has the following form:

$$0 = S\left[f^{(0)}, f^{(0)} \right] . \tag{5.3.2}$$

This equation has the unique non-singular[6] solution, corresponding to the state of the components situated in the thermodynamic equilibrium. The solution is the Maxwell function

$$f^{(0)}(\mathbf{v}') = \frac{n}{(2\pi T / m)^{3/2}} \exp\left\{ -\frac{m}{2kT}(\mathbf{v}' - \mathbf{v})^2 \right\} . \tag{5.3.3}$$

In a general case n, T and $\mathbf{v}$ are functions of the coordinates and time. To avoid confusion, in the following considerations the velocity, included as the argument in the distribution function f ($\mathbf{v}'$), will be indicated by the prime.

If we now take into account the finiteness of the free path, the solution of the equation (5.3.1) can be found in the form:

$$f = f^{(0)} + \bar{\lambda} f^{(1)} + \ldots \tag{5.3.4}$$

Substituting this expression into (5.3.1) and taking into account (5.3.2), we obtain a system of equations of successive approximations:

$$\left.\begin{aligned} & S\left[f^{(0)}, f^{(1)} \right] + S\left[f^{(1)}, f^{(0)} \right] = \frac{Df^{(0)}}{Dt}; \\ & S\left[f^{(2)}, f^{(0)} \right] + S\left[f^{(0)}, f^{(2)} \right] = \frac{Df^{(1)}}{Dt} - St\left[f^{(1)}, f^{(1)} \right] \\ & \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \end{aligned}\right\} \tag{5.3.5}$$

It can be seen that in each stage of calculations it is necessary to solve of the linear integral equation in relation to the unknown function. The procedure for solving these resultant equations for the case of the Boltzmann collisional term was developed by Enskog, Chapman and Cowling, and for the Landau collisional term by Braginskii, Spittser and others. It appears that the construction of these solutions leads to equation for the functions: n ($\mathbf{r}$, t); T ($\mathbf{r}$, t); v ($\mathbf{r}$, t), derived using the

[6]The singular solutions of equation (5.3.2) describes the configurations with point sources.

expressions (5.3.3). These are also the hydrodynamic approximation equations.

5.3.2. Transfer equations. General form

The kinetic equation of the general form is:

$$\frac{\partial f}{\partial t}+\frac{\partial}{\partial x_k}(\mathbf{v}'_k f)+\frac{\partial}{\partial v'_k}\left(\frac{F_k}{m} f\right)=S$$

This equation is multiplied successively by 1, $m\mathbf{v}'$, ($mv'^2/2$) and integrated with respect to velocity. Consequently, we obtain the 'transfer equation'

$$\frac{\partial n}{\partial t}+div(n\mathbf{v})=\int S d\mathbf{v}'$$

$$\frac{\partial}{\partial t}(mnv_j)+\frac{\partial}{\partial x_k}\left(mn\left\langle v_j v_k\right\rangle\right)-en\left(E_j+\frac{1}{c}\left[\mathbf{v},B\right]_j\right)=\int mv_j S d\mathbf{v}' \qquad (5.3.6)$$

$$\frac{\partial}{\partial t}\left(\frac{mn}{2}\left\langle v^2\right\rangle\right)+div\left(\frac{mn}{2}\left\langle v^2\mathbf{v}\right\rangle\right)-en\mathbf{E}v=\int\frac{mv^2}{2}S d\mathbf{v}'.$$

The following notations are used here:

$$\left.\begin{aligned}&\mathbf{v}(t,r)=\frac{1}{n}\int\mathbf{v}' f(t,\mathbf{r},\mathbf{v}')d\mathbf{v}'=\left\langle\mathbf{v}'\right\rangle;\\&\frac{1}{n}\int(v')^2 f(t,\mathbf{r},\mathbf{v}')d\mathbf{v}'=\left\langle\mathbf{v}^2\right\rangle\end{aligned}\right\} \qquad (5.3.7)$$

where j and k are the coordinate indexes, and if the single term contains two identical indexes, the summation with respect to this indexes is proposed.

If the transformation of the particles of the same type to particles of another type, i.e. ionisation, dissociation, etc., is not taken into account, then

$$\int S d\mathbf{v}'=0. \qquad (5.3.8)$$

Actually, the given integral describes the variation of the number of particles in the investigated volume as a result of collisions with other particles, but it is evident that at elastic collisions such a change does not take place.

Equations (5.3.6), by introducing new quantities and notations

$$\left.\begin{aligned}
&\frac{d}{dt}=\frac{\partial}{\partial t}+(\mathbf{v}\nabla);\quad \mathbf{w}=\mathbf{v}'-\mathbf{v};\quad \mathbf{R}=\int m\mathbf{u}Sd\mathbf{v}';\quad Q=\int\frac{m}{2}\mathbf{w}^2Sd\mathbf{v}';\\
&p=\frac{1}{3}nm\left\langle\mathbf{w}^2\right\rangle=nkT;\quad \pi_{jk}=nm\left\langle w_jw_k\right\rangle-\frac{1}{3}\delta_{jk}\left\langle w^2\right\rangle;\\
&q=\int\frac{m}{2}\mathbf{w}^2\mathbf{v}'f(t,r,\mathbf{v}')d\mathbf{v}'=nm\left\langle\frac{w^2}{2}\mathbf{v}\right\rangle
\end{aligned}\right\}\qquad (5.3.9)$$

are reduced to a more convenient form

$$\left.\begin{aligned}
&\frac{\partial n}{\partial t}+\operatorname{div}n\mathbf{v}=0;\\
&mn\frac{dv}{dt}=-\nabla p+\mathbf{f}_{\text{visc}}+en\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v},\mathbf{H}\right]\right)+\mathbf{R};\\
&\frac{3}{2}n\frac{dT}{dt}+p\operatorname{div}\mathbf{v}=-div\mathbf{q}+Q+Q_{\text{visc}}.
\end{aligned}\right\}\qquad (5.3.10)$$

When writing the equations (5.3.10), the following notations were used:

$$\left(f_{\text{visc}}\right)_j=\frac{\partial\pi_{jk}}{\partial x_k};\quad Q_{\text{visc}}=-\frac{\pi_{jk}\partial v_j}{\partial x_k}.\qquad (5.3.11)$$

The system of equations (5.3.10) was derived for one of the components, and $e > 0$ corresponds to the ions, and $e < 0$ to electrons. The law of momentum conservation shows that $\mathbf{R}_i = -\mathbf{R}_e$.

The equations (5.3.10) have the conventional hydrodynamic form and were derived without any restrictions on the distribution function. However, their universal nature is diminished as a result of the fact that the quantities π_{jk}, **R**, **q** and Q, included in this equations, are not expressed in a general case by the hydrodynamic parameters n, **v**, T. Therefore, the system of transfer equations is not closed in a general case.

5.3.3. Ideal plasma model

In the limit, when the free path length $\lambda \to 0$, the distribution function can be regarded as Maxwellian and in this case all the dissipative terms π_{jk}, **R**, **q** and Q convert to 0. Consequently, we obtain the equations of ideal hydrodynamics for each of the components:

$$\left.\begin{aligned}&\frac{\partial n_i}{\partial t}+divn_i\mathbf{v}_i=0;\\&\frac{\partial n_e}{\partial t}+divn_e\mathbf{v}_e=0;\end{aligned}\right\};\quad\left.\begin{aligned}&\frac{\partial s_i}{\partial t}+(\mathbf{v}_i\nabla)s_i=0;\\&\frac{\partial s_e}{\partial t}+(\mathbf{v}_e\nabla)s_i=0\end{aligned}\right\}$$

(5.3.12)

$$\left.\begin{aligned}&Mn_i\left(\frac{\partial\mathbf{v}_i}{\partial t}+(\mathbf{v}_i\nabla\mathbf{v})\right)=-\nabla p_i+en_i\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_i,\mathbf{H}\right]\right);\\&mn_e\left(\frac{\partial\mathbf{v}_e}{\partial t}+(\mathbf{v}_e\nabla\mathbf{v}_e)\right)=-\nabla p_e-en_i\left(\mathbf{E}+\frac{1}{c}\left[\mathbf{v}_e,\mathbf{H}\right]\right).\end{aligned}\right\}$$

5.3.4. Braginskii equation

We returned to the general case in which $\overline{\lambda}$ is small but finite. Using the methods described in section 5.3.1, we can determine the deviation of the distribution function from the Maxwellian function and the explicit equations for the dissipative terms in the equations (5.3.10). Unfortunately, these calculations are very complex (see [121]) and we confine ourselves only to the presentation of the final results.

Prior to presenting these results, we introduce the electronic and ionic time between the collisions:

$$\tau_e=\frac{3\sqrt{m}(kT_e)^{3/2}}{4\sqrt{2\pi}\Lambda e^4Z^2n_i}=\frac{3.5\cdot10^4}{\Lambda/10}\frac{T_e^{3/2}}{Z^2n_i};$$

$$\tau_i=\frac{3\sqrt{M}\,(kT_i)^{3/2}}{4\sqrt{2\pi}\Lambda e^4Z^4n_i}=\frac{3.0\cdot10^6}{\Lambda/10}\frac{T_i^{3/2}}{Z^4n_i}. \qquad (5.3.13)$$

Here M_p is the proton mass, Z is the ion charge ratio; Λ is the Coulomb logarithm, introduced previously. For calculating Λ at $T_e < 50$ eV, we can use the equation

$$\Lambda=23.4-1.15\lg n+3.45\log T_e. \qquad (5.3.14)$$

In 'practical' equations (5.3.13) and others, temperature is expressed in eV and all other quantities in the Gaussian unit system.

The values of the dissipative terms **R**, **q**, Q, π_{jk} depend strongly on the strength of the magnetic field, more accurately, on the Hall parameters $\omega_i\tau_i$ and $\omega_e\tau_e$, where $\omega_i = eH/(Mc)$, and $\omega_e = eH/(mc)$ are the Larmor frequencies.

Following Braginskii's study, we present here the values of the dissipative terms for the case in which the Hall parameters $\omega\tau \gg 1$, and the ion charge $Z = 1$.

We start with the friction force **R** which is included in (5.3.9).

The transfer of the momentum at collisions from the ions to the electrons $\mathbf{R} = \mathbf{R}_u + \mathbf{R}_T$ consists of two parts: friction force $\mathbf{R}_u$, determined by the presence of relative velocity $\mathbf{u} = \mathbf{v}_i - \mathbf{v}_e = \mathbf{j}/(en)$, and the thermal force $\mathbf{R}_T$ formed as a result of the gradient of the electronic temperature:

$$\left.\begin{aligned} R_u &= en\left(\frac{j_\parallel}{\sigma_\parallel} + \frac{j_\perp}{\sigma_\perp}\right); \\ R_T &= -0.7\ln\nabla_\parallel kT_e - \frac{3}{2}\frac{n}{\omega_e\tau_e}\left[h, \nabla(kT_e)\right]. \end{aligned}\right\} \tag{5.3.15}$$

Here the signs $\parallel$ and $\perp$ represent the components along and across the magnetic fields, and $\mathbf{h} \equiv \mathbf{H}/H$ is the unit vector oriented along **H**; the quantities $\sigma_\parallel$ and $\sigma_\perp$ are equal to respectively:

$$\left.\begin{aligned} &\sigma_\perp = \frac{e^2 n\tau_e}{m} = \sigma_l (kT_e)^{3/2}; \quad \sigma_\parallel = 1.96\sigma_\perp = 1.96\sigma_l (kT_e)^{3/2}; \\ &\sigma_l = \frac{0.9\cdot 10^{13}}{\Lambda/10}. \end{aligned}\right\} \tag{5.3.16}$$

The heat flow, transferred by the electrons, can also be represented in the form $\mathbf{q} = \mathbf{q}_u + \mathbf{q}_T$, where

$$\left.\begin{aligned} \mathbf{q}_u^e &= -0.7\ln kT_e \mathbf{u}_\parallel - \frac{3}{2}\frac{nkT_e}{\omega_e\tau_e}\left[\mathbf{h}, \mathbf{u}\right]; \\ \mathbf{q}_T^e &= -\varkappa_\parallel^e k\nabla_\parallel T_e - \varkappa_\perp^e k\nabla_\perp T_e - \frac{5}{2}\frac{cnk^2 T_e}{\omega_e\tau_e}\left[\mathbf{h}, \nabla T_e\right]. \end{aligned}\right\} \tag{5.3.17}$$

Here k is the Boltzmann constant, $n = n_e$:

$$\varkappa_\parallel^e = 3.16\frac{nkT_e\tau_e}{m}; \quad \varkappa_\perp^e = 4.66\frac{nkT_e\tau_e}{m(\omega_e\tau_e)^2}. \tag{5.3.18}$$

The heat flow, transferred by the ions at $\omega_i\tau_i \gg 1$, is:

$$\left.\begin{aligned} \mathbf{q}_i &= -\varkappa_{\parallel}^{i} k\nabla_{\parallel} T_i - \varkappa_{\perp}^{i} k\nabla_{\perp} T_i + \frac{5}{2}\frac{cnkT_i}{eH}[\mathbf{h}, \nabla T_i]; \\ \varkappa_{\parallel}^{e} &= 3.9\frac{nkT_i\tau_i}{M}; \quad \varkappa_{\parallel}^{e} = 2\frac{nkT_i\tau_i}{M(\omega_i\tau_i)^2}. \end{aligned}\right\} \tag{5.3.19}$$

The transfer coefficients at **H** = 0 followed directly from the presented equations because they coincide with the transfer coefficients along the magnetic field. The heat received by the ions at collisions with the electrons is:

$$Q_i \equiv Q_\Delta = \frac{3m}{M}\frac{n_e}{\tau_e}(T_e - T_i) \tag{5.3.20}$$

The heat, generated in the electronic component as a result of the collisions with the ions, is described by the equation

$$Q_e = -\mathbf{Ru} - Q_\Delta = \frac{j_{\parallel}^2}{\sigma_{\parallel}} + \frac{j_{\perp}^2}{\sigma_{\perp}} + \frac{1}{en_e} jR_T - \frac{3mn_e}{M\tau_e}k(T_e - T_i) \tag{5.3.21}$$

The tensor of viscous stresses in the absence of the magnetic field is determined by a single coefficient η_0:

$$\pi_{jk} = \eta_0 w_{jk} \tag{5.3.22}$$

where the shear tensor is

$$w_{jk} = \frac{\partial v_j}{\partial x_k} + \frac{\partial v_k}{\partial x_i} - \frac{2}{3}\delta_{jk}\,\mathrm{div}\,v \tag{5.3.23}$$

In a strong magnetic field ($\omega\tau \gg 1$), the component of the tensor π_{jk} has the following form in the coordinate system with the *z* axis parallel to the magnetic field:

$$\left.\begin{aligned} \pi_{xx} &= -\eta_0 w_{zz}; \\ \pi_{xx} &= -\eta_0\frac{1}{2}(w_{xx} + w_{yy}) - \eta_1\frac{1}{2}(w_{xx} - w_{yy}) - \eta_3 w_{xy}; \\ \pi_{yy} &= -\eta_0\frac{1}{2}(w_{xx} + w_{yy}) - \eta_1\frac{1}{2}(w_{yy} - w_{xx}) + \eta_3 w_{xy}; \\ \pi_{xy} &= \pi_{yx} = -\eta_1 w_{xy} + \eta_3\frac{1}{2}(w_{xx} - w_{yy}); \\ \pi_{xz} &= \pi_{zx} = -\eta_2 w_{xz} - \eta_4 w_{yz}; \\ \pi_{yz} &= \pi_{zy} = -\eta_2 w_{yz} + \eta_4 w_{xz}. \end{aligned}\right\} \tag{5.3.24}$$

The equations (5.3.24) are suitable for both the ions and electrons, but for every type of particles it is necessary after all to substitute its own tensor w_{jk} and own viscosity coefficient.

The viscosity coefficients for the ions and electrons are equal to respectively (Z = 1):

$$\left.\begin{aligned}
&\eta_0^i = 0.96 n_i k T_i \tau_i; \quad \eta_0^e = 0.37 n_e k T_e \tau_e;\\
&\eta_1^i = \frac{3}{10}\frac{n_i k T_i}{\omega_i^2 \tau_i}; \qquad \eta_1^e = 0.51 \frac{n_e k T_e}{\omega_e^2 \tau_e};\\
&\eta_2^i = 4\eta_1^i; \qquad\qquad \eta_2^e = 4\eta_1^e;\\
&\eta_3^i = \frac{1}{2}\frac{n_i k T_i}{\omega_i}; \qquad \eta_3^e = -\frac{1}{2}\frac{n_e k T_e}{\omega_e};\\
&\eta_4^i = 2\eta_3^i; \qquad\qquad \eta_4^e = 2\eta_3^e .
\end{aligned}\right\} \tag{5.3.25}$$

The energy equation (5.3.10) includes the term which takes into account the contribution of viscosity. Neglecting the quantities of the order $(\omega\tau)^{-2}$, we obtain

$$Q_{\text{visc}} - \pi_{jk}\frac{\partial v_j}{\partial x_k} = \frac{3}{4}\eta_0 w_{zz}^2 = \frac{1}{3}\eta_0\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} - 2\frac{\partial v_z}{\partial z}\right)^2 . \tag{5.3.26}$$

The equations presented for the dissipative terms should be substituted into the equations (5.3.10).

The review [121] presents equations for more general cases in which the Hall parameters $\omega\tau \leqslant 1$ and $Z \neq 1$. The same study also analyses the factors which determine the structure of the dissipative terms.

5.3.5. Comments on Braginski's equations

In chapter 3 equations were presented for two-fluid hydrodynamics using simple physical considerations. Now we have these equations in a sufficiently justified form. A number of consequences of these equations will be investigated later.

It can be seen that the equations are relatively cumbersome. If it is also taken into account that the assumption on the very short free path is usually not strict in many plasma systems, and the presence of microscale ($\sim r_D$) turbulent processes strongly influences the transfer coefficients, it becomes clear why the system of the Braginski' is equations is usually made greatly approximate in specific calculations (both analytical and numerical).

However, this does not indicate that the hydrodynamic equations are not interesting. In fact, since they are based on the laws of conservation

of mass, momentum and energy, they have a 'reserve of strength' and after accurate conversion they can provide the qualitatively accurate pattern even in cases in which the free path is greater than the dimensions of the system. At the same time, they are far simpler than the kinetic equations, and the calculations using these equations can in most cases be completed.

The system of Braginskii's equations should be supplemented with the boundary conditions, taking into account the equipotential nature of the electrodes, annihilation of ions at the boundary surfaces, etc. The problem of the boundary conditions will be investigated in greater detail in chapter 7. In conclusion, it should be mentioned that the hydrodynamic models of plasma dynamics are not exhausted by the Braginskii equations. In most cases, the literature mentions the Chu–Goldberger–Low equations for rare plasma in which the collisions are ignored and the pressure is a tensor [122].

5.4. Examples of collisional relaxation in Coulomb plasma

A number of examples of collisional relaxations in Coulomb plasma will be investigated.

5.4.1. The Belyaev–Budker paradox

This paradox may be described as follows. If the electrons and ions in the plasma are characterised by isotropic distribution functions, and the particles with low velocity do not exist, i.e. $v_i < v_i^*$ and $v < v^*$ and the functions f_i and f_e convert to 0, the flux of any charged particles, moving with the velocity

$$v_b < \min(v_i^*, v_e^*) \tag{5.4.1}$$

in the plasma with these distribution functions is not decelerated as a result of collisions.

This is immediately indicated by the Trubnikov electrostatic analogy (5.2.23a). In fact, there is no electrostatic field inside the charged spherical layer and, consequently, the flux is not decelerated, if the condition (5.4.1) is satisfied.

5.4.2. Relaxation of a rare flux of fast ions in isotropic plasma

At present, it is often necessary to deal with the fast ions (ε_i) decelerated in the plasma with the isotropic distribution functions of the electrons and ions, and the initial energy is considerably higher than the energy

of the plasma particles $\varepsilon_0 \gg (kT_i^{(pl)}, kT_e^{(pl)})$ and in this case $v_0 \gg v_e^{(pl)} \gg v_i^{(pl)}$. Specific examples in laboratory practice include the process of heating the plasma with 'neutral' (rapidly ionised) particle beams with the energy of hundred and thousands of keV, and also the deceleration of the charged products of thermonuclear reactions in nuclear synthesis equipment. For example, D + T $\rightarrow \alpha(_4He^2) + n$, and the energy of the α-particle is ~4 meV.

Here, we find a relatively unique pattern, determined by the strong dependence of the deceleration force on the reduced mass of colliding particles and special features of the velocity distribution functions of the ions and electrons (Belyaev–Budker defects).

In the initial stage, the deceleration of the fast particles is determined by the 'effective field' generated jointly by the electrons and the ions. However, since the reduced mass of the fast particle and of the electron $m_{I,e} \ll m_{I,i}$, in reality the deceleration is determined by collisions with the electrons. It is quite evident that the scattering of the fast particles is minimal and the particles, losing velocity, retain their initial direction of movement.

However, the velocity of the injected particles is now so low that the particles enter the level of electronic velocity. The Belyaev–Budker effect now starts to operate, and the electrons with the energy $\varepsilon_i^{(pl)} > \varepsilon_i(t)$ no longer have the decelerating effect. Therefore, the contribution of electrons to deceleration become smaller whereas the contribution of the ions to deceleration remains almost completely unchanged. However, the velocity of the injected ions continues to decrease and approaches the velocity of the plasma ions. Now, the collisions with the ions determine the deceleration process which is obviously accompanied by extensive scattering of the decelerated ions.

Discussing the qualitative aspects of the process of collisional deceleration, we carry out quantitative calculations assuming that the distribution of the electrons and ions in the plasma is Maxwellian.

Calculations start with the system of equations (5.2.41) and (5.2.42a). Index α behind the injected ions is retained:

$$\frac{d\mathbf{p}_\alpha}{dt} = -q_{\alpha i}\left(\frac{1}{\mu_{\alpha i}}\nabla_{\mathbf{v}}\phi_i(v_\alpha)\right) - q_{\alpha e}\left(\frac{1}{\mu_{\alpha e}}\nabla_{\mathbf{v}}\phi_e(\mathbf{v}_\alpha)\right); \qquad (5.4.2a)$$

$$\frac{d\varepsilon_\alpha}{dt} = -q_{\alpha i}\left(\frac{\mathbf{v}_\alpha}{\mu_{\alpha i}}\nabla_{\mathbf{v}}\phi_i + \frac{\phi_i(v_\alpha)}{m_\alpha}\right) - q_{\alpha e}\left(\frac{\mathbf{v}}{\mu_{\alpha e}}\nabla_{\mathbf{v}} + \frac{\phi_i(\mathbf{v}_\alpha)}{m_\alpha}\right). \qquad (5.4.2b)$$

The distribution functions of the plasma particles are used in the form:

$$f_{i,e} = n\left(\frac{m_{i,e}}{2\pi kT_{i,e}}\right)^{3/2} \exp\left\{\frac{-m_{i,e}v^2}{kT_{i,e}}\right\}. \tag{5.4.3}$$

The strength of the effective **E**-field, taking into account spherical symmetry in accordance with the Gauss theorem, is equal to:

$$\mathbf{E}_\beta^{\text{ef}} = \frac{n_\beta(\mathbf{v})}{v^2}\frac{\mathbf{v}}{v}, \quad n_\beta = \int_0^v f_\beta d\mathbf{v}. \tag{5.4.4a}$$

From this

$$E_\beta^{\text{ef}}(x) = nW(x_\beta), \tag{5.4.4b}$$

where

$$W(x_\beta) = \frac{2}{\sqrt{\pi}}\int_0^{x_\beta} e^{-t}\sqrt{t}dt; \quad x_\beta = m_\beta\frac{v^2}{2kT_\beta} \tag{5.4.4c}$$

is the 'Maxwell function'. Its values are:

$$W(x_\beta) \approx \begin{cases} \frac{2}{3}x^{3/2}\left(1-\frac{3}{5}x+...\right), & x<<1 \\ 1-\frac{\gamma}{\sqrt{\pi}}e^{-x}\sqrt{x}\left(1+\frac{1}{2x}+...\right), & x>>1. \end{cases} \tag{5.4.4d}$$

Knowing E_β^{ef} we determine ϕ_β:

$$\phi_\beta = \int_v^\infty E_{\text{ef}}^\beta dv = n\frac{W(x_\beta)+W'(x_\beta)}{v}; \; W'(x) = \frac{d\mu}{dx} \; . \tag{5.4.5}$$

Consequently, we obtain:

$$\frac{d\mathbf{p}_\alpha}{dt} = -nq\frac{\mathbf{v}}{v^3}\left(\frac{W(x_i)}{\mu_{\alpha i}}+\frac{W(x_e)}{\mu_{\alpha e}}\right) \; . \tag{5.4.6}$$

The variation of the deceleration factor

Equation (5.4.4d) shows that if $x_i \gg 1$, $x_e \gg 1$, then $W(x_i) \approx W(x_e) \approx 1$,

$$F = -nq\frac{\mathbf{v}}{v^3}\left(\frac{1}{\mu_{\alpha i}}+\frac{1}{\mu_{\alpha e}}\right) \approx -\frac{nq}{\mu_{\alpha e}}\frac{\mathbf{v}}{v^3} \; . \tag{5.4.7}$$

In this case, the fast injected ion is decelerated by the electrons. The decelerating effect of the electrons and ions are compared at v^* where

$$\frac{W(x_{i*})}{\mu_{\alpha i}} = \frac{W(x_{e*})}{\mu_{\alpha e}} \ . \tag{5.4.8}$$

Assuming that $x_{i*} \gg 1$, $x_{e*} \ll 1$, from equation (5.4 .8) taking (5.4 .4c) into account, we obtain

$$\frac{2}{3} x_{e*}^{3/2} = \frac{\mu_{\alpha e}}{\mu_{\alpha i}} \approx \frac{m_e}{\mu_{\alpha i}} \ . \tag{5.4.9}$$

In particular, in injection of the ions with the same mass as that of the ions in the plasma, $m_{\alpha i} = \frac{m_i}{2}$ and equation 5.4.9 can be presented in the new form

$$\left(\frac{\varepsilon_{\alpha *}}{kT_e} \right) = 3^{2/3} \left(\frac{M_i}{m_e} \right)^{2/3}. \tag{5.4.10}$$

Further deceleration at $\varepsilon_\alpha < \varepsilon_{\alpha^*}$ is mainly the result of the effect of the ions.

The variation of the kinetic energy of injected particles

Substituting into (5.4.2b) equation (5.4.4), (5.4.5) for ϕ and E_{ef}, in the case of the Maxwell distribution of the ions and the electrons in the 'decelerating' plasma we can write the equation for the variation of the energy of the injected particles in the form:

$$\frac{d\varepsilon_\alpha}{dt} = -\frac{q_1 n}{m_\alpha v_\alpha} \left\{ \left(\frac{m_\alpha}{m_i} W(x_i) - W'(x_i) \right) + \left(\frac{m_\alpha}{m_e} W(x_e) - W'(x_e) \right) \right\}, \quad q_{\alpha i} = q_{\alpha e} \equiv q,$$

$$x_i = \frac{m_i v^2}{2kT_i} = \frac{m_i}{m_\alpha} \frac{\varepsilon_\alpha}{kT_i}, \quad x_e = \frac{m_e v^2}{2kT_e} = \frac{m_e}{m_\alpha} \left(\frac{\varepsilon_\alpha}{kT_e} \right). \tag{5.4.11}$$

Here, energy is also transferred initially by mostly to the electrons and subsequently to the ions.

5.4.3. Runaway electrons

The strong dependence of the collisional cross-section of the charged particles on their relative energy leads to an important phenomenon referred to as the 'runaway electrons'. This effect is most evident in tokamaks in the form of a generation of x-ray radiation with the photon

energy of many hundreds of keV, whereas the voltage of the eddy electrical field at the pass is only of the order of hundreds of volts. Evidently, in order to acquire the energy of the order of the photon energy, the electron should perform thousands of passes along the magnetic axis of the plasma cord without losing any large amount of energy during this process.

The equation of the 'test' electron, moving amongst the ions in the presence of the electrical field follows directly from equation (5.4.7)

$$\frac{d\mathbf{p}}{dt} = -\frac{q_{ei}}{\mu_{ei}} n_i \frac{W}{v^2} - e\mathbf{E} \ . \tag{5.4.12}$$

Evidently, the collisions of the accelerated electron with the almost stationary ions can take the energy away from the electrons if

$$\frac{q}{\mu_{ei}} n_i \frac{\mu}{\nu^2} > eE, \quad \nu = v\sqrt{\frac{m_\beta}{2kT_\beta}} \ . \tag{5.4.13}$$

Otherwise, the electron will acquire energy continuously. The function

$$G_1 = \frac{W}{v^2} \ .$$

has a maximum at $v \approx 1$, and if at some moment of time the electron velocity exceeds this critical value, the acceleration can continue for an unlimited period of time, if the secondary factors do not interrupt this process.

In the actual conditions at the initial moment we have a specific velocity distribution of the electrons. The first candidates to 'runaway' are the fastest electrons. This takes place if the applied electrical field is stronger than the Dreicer field, determined by the equation

$$E_{\text{crit}} = 0.214 \frac{e\Lambda}{r_D^2} \sim \frac{e^2}{r_D^2}. \tag{5.4.14}$$

i.e. E_{crit} is of the order of the field of the elementary charge at the Debye distance. Thus, at T_e ~1 keV, n ~10^{15} cm^{-3}, Λ~15, the critical field E_{crit}~1 V/cm.

5.4.4. Relaxation time of distribution functions in the two-component plasma

Accurate calculations of the relaxation time for the arbitrary distribution function require application of the kinetic equations for the plasma components. However, in any case, it is sufficient to obtain a simple

estimate based on the cross sections of Coulomb collisions. Thus, it is assumed that we have a plasma which is not too faraway from equilibrium so that we can use the concept of the temperatures of the ions T_i and electrons T_e, representing the measures of the mean values of the energies of these particles.

In this case, the collision cross-section of all the charged particles is determined only by the energy of these particles. If $T_i \sim T_e$, we have

$$\sigma^{i,i} \approx \sigma^{i,e} \sim \sigma^{e,e} \sim \frac{1}{(kT)^2} \ . \tag{5.4.15a}$$

However, if T_i greatly differs from T_e, then

$$\sigma^{e,e} \sim \sigma^{e,i} \approx \frac{1}{(kT_e)^2}; \ \sigma^{i,i} \approx \frac{1}{(kT_i)^2}. \tag{5.4.15b}$$

The free path length behaves in a corresponding manner

$$\lambda \sim \frac{1}{n\sigma}.$$

However, the free path times greatly differ. We shall discuss this in greater detail. The free path time from collision to collision is almost the same for the collisions in which electrons take part:

$$\tau^{e,e} \sim \frac{1}{n\sigma^{e,e}v_e}, \quad \tau^{e,i} \sim \frac{1}{n\sigma^{e,i}v_e} \ . \tag{5.4.16}$$

The ion–ion free path time at $T_i \sim T_e$ is considerably longer

$$\tau^{i,i} \sim \frac{1}{n\sigma^{i,i}v_i} \sim \sqrt{\frac{M}{m}}\tau^{e,e}. \tag{5.4.17}$$

A single collision of electrons with each other results in the isotropisation of the distribution function of the electrons, and multiple collisions within the limits of the nonuniformity of the configuration result in Maxwellisation whose extent increases with increase of the number of collisions.

The same can be said of the collisions of ions with ions.

The collisions of the electrons with the ions have only a slight effect on the distribution function of the ions if the number of collisions is small. However, in the case of a large number of collisions as, for example, in plasma traps or low-temperature plasma, collisions may result in the equalisation of the temperatures of the electrons and the ions (5.3.20). The characteristic time of this process is

$$\tau_{e,i}^{(\varepsilon)} \sim \frac{M}{m}\tau_{e,e} \,. \tag{5.4.18}$$

Thus, we have three characteristic time scales

$$\tau_{e,e} : \tau_{i,i} : \tau_{e,i}^{(\varepsilon)} \sim 1 : \sqrt{\frac{M}{m}} : \frac{M}{m} \,. \tag{5.4.19}$$

This shows that the relaxation inside each component is faster than between the components.

5.5. Effect of the thermal force on equilibrium and heat transfer in plasma configuration

In section 2.4.3 we derive the Grad–Shafranov equation, describing the static axisymmetric MHD configurations. This equation includes to the derivative functions $p(\psi)$ and $I(\psi)$. In reality, these functions are determined by the initial conditions and/or dissipation processes. Therefore, in a general case, calculation of the configuration is complicated. Here we investigate the simplest model – the configuration of plasma in the magnetic shell of a myxine of a Galatea reactor (section 10.5) in which the distribution of density and temperature forms as a result of diffusion and heat conductivity. A special feature of the Galatea is the presence of levitating conductors inside the plasma volume[7] – usually rings with the current whose magnetic field screens the conductors (they are referred to as 'myxines') from contact with plasma.

This subject of the calculations described below is the determination of the plasma parameters in the magnetic shell of the myxine [123, 124].

5.5.1. Equilibrium configuration of the magnetic shell of the myxine

The magnetic shell of the myxine (MSM) is the region with the field and plasma which on one side is restricted by the solid surface of the myxine and on the other side by the separatrix of the magnetic field.

The equilibrium, stability and heat transfer in the MSM will be studied, assuming that the radius of the cross-section of the MSM $b_\mu < R_\mu$ is the large radius of the myxine.

The most important property of the plasma configuration of the MSM is the presence of a temperature dip between its outer boundary and the surface of the myxine. Consequently, as a result of thermal diffusion, the current leading to the decrease of pressure in the direction

[7] In reality, they are sustained by the magnetic field.

to the surface of the myxine is maintained in the MSM. Although this fact was already mentioned by A.D. Sakharov [23, vol. 1], its derivation will be discussed here because its generalised form will also be of interest to us.

The simplest situation is found in the presence of axial symmetry and a single poloidal magnetic field. Since this case is of special interest to us, we confine ourselves to this case only.

Assuming that plasma is quasi-static, we have two equations of motion (section 5.3.4)

$$\frac{\nabla P_i}{en} = \mathbf{E} + \frac{1}{c}\left[\mathbf{V}_i\mathbf{H}\right] + \mathbf{R}, \quad -\frac{\nabla P_e}{en} = \mathbf{E} + \frac{1}{c}\left[\mathbf{V}_e\mathbf{H}\right] - \mathbf{R}, \tag{5.5.1}$$

where

$$\mathbf{R} = \left(\frac{\mathbf{j}_\parallel}{\sigma_\parallel} + \frac{\mathbf{j}_\perp}{\sigma_\perp}\right) - 0.71n\nabla_\parallel kT_e - \frac{3}{2}\frac{n}{\omega_e\tau_e}\frac{\left[\mathbf{H},\nabla kT_e\right]}{H}\ . \tag{5.5.2}$$

Directing the axis x in the direction normal to the magnetic surfaces, the axis y along the direction of symmetry (azimuth), and the axis z along $\mathbf{H}$ (Fig. 5.5.1), we obtain the y-component of the second of the equations (5.5.1) in the form

$$\frac{j_y}{\sigma} = \frac{3}{2}\frac{n}{\omega_e\tau_e}\frac{\partial}{\partial x}kT_e\,. \tag{5.5.3}$$

Here, it is taken into account that $\partial P_e / \partial y = 0$, $E_y = 0$, $V_{ex} = 0$. Substituting (5.5.3) into the MHD equilibrium equation

$$\nabla P = \frac{1}{c}\left[\mathbf{j},\mathbf{H}\right]$$

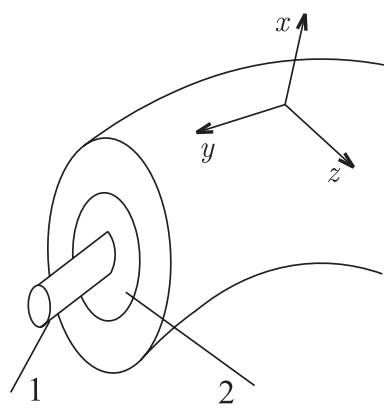


Fig. 5.5.1. Diagram of a ring-shaped myxine in the selection of the coordinate axes: 1) myxine; 2) MSM.

which follows from (5.5.1), we obtain

$$\frac{\partial}{\partial x} n\left(T_i + T_e\right) = \frac{3}{2} n \frac{\partial T_e}{\partial x} \quad . \tag{5.5.4}$$

If it is assumed that $T_i = T_e = T$, then

$$nT^{1/4} = \theta_m = \text{const.} \tag{5.5.5}$$

Accepting that on the external surface of the MSM $n_0 = 10^{14}$ cm^{-3}, $T_0 = 10$ keV, and that the temperature of the plasma in the vicinity of the surface of the myxine is $T = 1$ eV, the plasma density around the myxine is $n_\mu = 10^{15}$ cm^{-3}.

However, in a general case $T_i \neq T_e$, and if it is assumed that $T_e / T_i = \omega = \text{const}$, instead of (5.5.5) we obtain

$$nT^{(1-\omega/2)/(1+\omega)} = \text{const.} \tag{5.5.6}$$

Consequently in the case of cold electrons ($\omega = 0$)

$$P = \text{const.} \tag{5.5.7}$$

However, if the electronic temperature increases, the plasma density depends less and less on T and at $\omega = 2$ the plasma density becomes constant. At $\omega > 2$, the dependence of n on T_i changes its appearance qualitatively.

Two comments: the resultant relationship between n and T in the MSM conditions automatically ensures the stability of the magnetic–plasma configuration with respect to hydrodynamic convection. This is indicated by the Rosenbluth–Longmire–Kadomtsev stability criterion (section 8.1.2)

$$\frac{\partial}{\partial \psi} PU^{\gamma} \geq 0 \tag{5.5.8}$$

where $U = \int H^{-1} dl$ is the specific volume of the magnetic force tube, and ψ is counted from the surface of the myxine. Condition (5.5.8) is the condition of increase of the entropy of the unit mass of plasma with increase of the distance from the myxine.

Using the field of the direct conductor for estimation (J_μ is the current in the myxine), we obtain

$$U \propto \frac{c}{2J_\mu} \frac{r^2}{2} \tag{5.5.9}$$

and, combining (5.5.8), (5.5.9), for $\gamma = 5/3$ we obtain the stability condition

$$\frac{\partial}{\partial r} T^{3/4} r^{10/3} > 0 \ . \tag{5.5.10}$$

This condition is intentionally fulfilled with a large reserve because T increases with increase of the distance from the myxine.

5.5.2. Heat conductivity in the MSM

The classic heat conductivity in the MSM was investigated in [123]. The main results of this work will now be discussed.

The complete system of equations for the axisymmetric configuration of the MSM has the form

$$\Delta^* \Psi = -\frac{dP(\Psi)}{d\Psi}, \tag{5.5.11a}$$

$$T_i = T_i(\Psi), \quad T_e = T_e(\Psi), \tag{5.5.11b}$$

$$\text{div } \varkappa_{i\perp} \nabla T_i = Q_i, \tag{5.5.11c}$$

$$\text{div } \varkappa_{e\perp} \nabla T_e = Q_e - \frac{j_\theta^2}{\sigma}, \tag{5.5.11d}$$

$$\frac{j_\theta}{\sigma} = \frac{3}{2} \frac{1}{e\omega_e \tau_e} \left[\frac{\mathbf{H}}{H}, \nabla k T_e \right]_\theta \ . \tag{5.5.11e}$$

The first equation here is the Grad–Shafranov equation, ψ is the function of the magnetic flux. The equations (5.5.11b) denotes the constancy of the electronic and ionic temperatures of the magnetic surfaces. In addition, the heat conductivity coefficient (5.3.18) is also present here

$$\varkappa_{i\perp} = \frac{2nkT_i}{M\omega_i^2 \tau_i}, \ \varkappa_{e\perp} = 4.66 \frac{nkT_i}{m\omega_e^2 \tau_e}, \tag{5.5.12}$$

were

$$\tau_i = \frac{3}{4}\sqrt{\pi} \frac{\sqrt{M}}{\Lambda e^4} \frac{(kT_i)^{3/2}}{n}, \ \tau_e = \frac{3}{4}\sqrt{\pi} \frac{\sqrt{m}}{\Lambda e^4} \frac{(kT_e)^{3/2}}{n},$$

and the quantities Q_i, Q_e are equal to respectively (5.3.20)

$$Q_i = \frac{3m}{M} \frac{nk}{\tau_e} (T_e - T_i),$$

$$Q_e = -Q_i - \alpha n^2 \sqrt{kT_e}, \quad \alpha = \text{const.}$$

The last term in Q_e takes into account bremsstrahlung (see later in section 6.3.2)

The system (5.5.11) is very complicated. However, taking into account the evaluation nature of the calculated results, it can be assumed to the first approximation that the magnetic field is a vacuum field, ohmic heating and heat transfer by the electrons can be ignored, and the temperatures of the electrons and ions may be regarded as the same. Consequently, the problem is reduced to solving a single equation

$$\text{div}(\varkappa \nabla \text{T}) = \alpha n^2 \sqrt{T}\,. \tag{5.5.13}$$

Taking into account (5.5.5) the above equation can be written in the present form

$$\text{div}(\varkappa^{\perp \nabla T_i = Q_i} \text{T}) = \alpha \theta^2. \tag{5.5.14}$$

Flat model

Initially, we investigate a flat model in which n and T are the functions of the Cartesian coordinate x, and the magnetic field H = const. The characteristic dimensional parameter which determines the solution of the equation (5.5.14), is the quantity A_*:

$$A_*^2 = \frac{8\sqrt{\pi}\Lambda e^2 c^2}{3\alpha}\sqrt{M}\ . \tag{5.5.15}$$

Consequently

$$\varkappa_{i\perp} = \frac{A_*^2 \alpha \theta^2}{H^2 T}. \tag{5.5.16}$$

For deuterium plasma

$$A_* \approx 5 \cdot 10^5 \text{ Oe} \cdot \text{cm}. \tag{5.5.17}$$

Using this parameter, equation (5.5.14) in the one-dimensional case can be written in the form (5.5.18)

$$\frac{d^2}{dx^2}\ln T = \frac{H^2}{A_*^2} \equiv \frac{1}{L_*^2}\ \cdot \tag{5.5.18}$$

Consequently

$$\frac{T}{T_1} = \left(\frac{T_1}{T_0}\right)^{x/L} \exp\left\{\frac{x^2 - Lx}{2L_*^2}\right\}. \tag{5.5.19}$$

It is accepted here that the temperature at the boundary with the plasma volume is equal to T_0 and on the surface of the myxine ($x = L$) it is equal to T_1.

At arbitrary T_0 and T_1 the determined dependence $T(x)$ is not monotonic and has a minimum at:

$$x_{\min} = \frac{L}{2} - \frac{L_*^2}{L} \ln \frac{T_1}{T_0} \ .$$

Naturally, we are interested in the regime in which the heat flow is directed only to one side – to the myxine. Therefore, the inequality $x_{\min} \geq L$ should be fulfilled. The thickness of the MSM is assumed to be $L = x_{\min}$ and we obtain the relationship between L and T_1/T_0:

$$\ln \frac{T_0}{T_1} = \frac{L^2}{2L_*^2} \ .$$

Using this function, we determine L, assuming that $T_0 = 10^4$ eV, $T_1 = 10$ eV, $H = 10^4$ Oe and using the value (5.5.17) for A_*. Consequently

$$L = \frac{A_*}{H} \sqrt{2 \ln \frac{T_0}{T_1}} \approx 190 \text{ cm}. \qquad (5.5.20)$$

This is a relatively high value but it is obviously too high because here we consider the minimum value of the magnetic field for the reactor and do not take into account the quasi-cylindrical geometry of the field in the vicinity of the myxine. In addition to this, in the actual conditions in plasma there will always be some amount of impurities with $Z > 1$, for example, the same α-particles, and this greatly reduces A_* and, consequently, L.

It should be mentioned that the calculation of the axisymmetric model for the same conditions gives $L = 50$ cm.

5.6. Kinetics of departure of plasma particles from traps

It was mentioned in section 1.7 that in the case of a mirror trap there is a ‘departure cone’ in the space of velocity (moments) and if a particle falls into the cone in the absence of the electrical field, the particle is immediately removed from the trap. Thus, the flow of the particles from the injection zone to the ‘departure cone’ is established in the velocity space as a result of collisions. The problem of these flows was investigated for the first time by Budker in his study published in 1954 [125] when a probkotron was proposed. The special features of Budker's calculations will be described below.

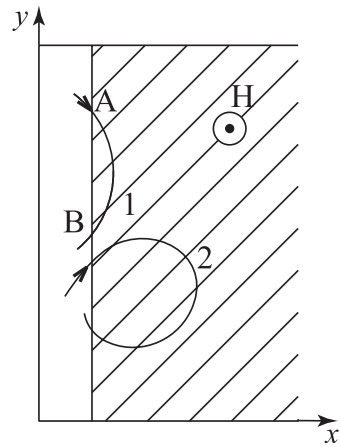


Fig. 5.6.1. Two types of colliding particles with a magnetic barrier: 1) co-collisions, 2) counter collisions.

However, the probkotron is not the only system with the zones of extensive departure of the particles in the velocity space. In the probkotron, the departure cone is in fact the 'black hole' of the system. However, in many systems there are no 'black holes' but in the velocity space there are 'grey' ('slightly holding') zones and the particles which fall into them leave the system considerably more rapidly. For example, in section 1.7.2 we discussed the drift trajectories in tokamaks and presence of passing and trapped particles in them. It was noted that the trapped particles described a trajectory in the form of wide 'sickles' (bannanas) and this greatly impairs the efficiency of retention of the particles. This will be discussed in greater detail in section 5.6.3.

In chapter 7 attention is given to near-wall conductivity and the 'trapped' and 'running away' particles will also be discussed. Therefore, the nonuniformity of the velocity space in relation to the retention of the particles is a very widely encountered property of the plasma systems and its role is always important.

5.6.1. The departure of particles from traps of the anti-mirror type

Very unique 'black holes' are found in the traps of the anti-mirror type [124]. Here, we are concerned with traps with $\beta_0 = 1$, i.e. anti-mirror traps and Galateas with counter fields (Galatea–A), shown in Fig. 5.6.2[8]. In the traps, the magnetic barrier – the plasma-field transition layer can be $\leq \rho_I$ – the Larmor ion radius. Therefore, in analysis of the departure

[8]These traps will be discussed in greater detail in section 10.5.

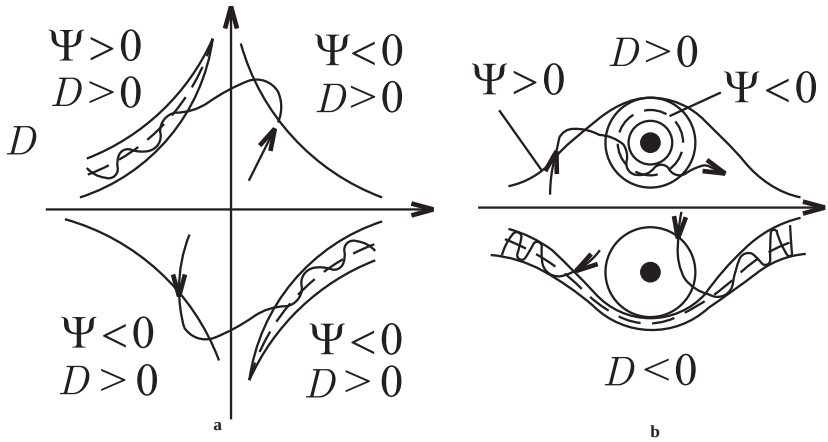


Fig. 5.6.2. Collisions of particles with a barrier: in the anti-probkotron (a), in Galatea -A (b).

of particles it is necessary to use the kinetics approach.

We consider a simple model of the magnetic barrier in the form of a step in which the field at $x > 0$ is directed along the z axis and is constant, and at $x < 0$ the field is equal to 0 (Fig. 5.6.1). It is assumed that a charged particle (for example, an ion) falls on this barrier at point A with the velocity $\mathbf{V}_0 = (V_{0x}, V_{0y})$. If $H_z > 0$, and $V_{0y} < 0$, the Lorenz force is directed into the region $x < 0$ and, describing an arc smaller than half the circle, the particle departs at point B from the region of the field with the velocity $\mathbf{V}_1 = (-V_{0x}, V_{0y})$.

If the same particle with the velocity $\mathbf{V}_{1*} = -\mathbf{V}_1$ is directed to the point B, the particle describes a completely different trajectory, passing an arc greater than the 'half-circle'. This difference of the trajectories is associated with the non-invariant nature of the Newton–Lawrence equation in the presence of the magnetic field in relation to the variation of the sign of time.

The collision of the first particle with the magnetic barrier is referred to as 'co-collision' and the particles undergoing this collision in this context are 'co-particles'. Collision of the second particle is referred to as 'counter collision', and such particles are 'counter particles'.

The presence of two types of collisions may have a strong effect on the retention of the particles in the traps. This is always clearly evident in open axisymmetric traps with a poloidal magnetic field: anti-probkotrons and Galatea-A. In these cases, the generalised moment of the amount of motion remains unchanged

$$MrV_\theta + \frac{e}{c}\Psi \equiv D = \text{const} \qquad (5.6.1)$$

and movement along r and z can be regarded as the movement in the field of forces with the effective potential (1.2.3a)

$$U = \frac{1}{2Mr^2}\left(\frac{e}{c}\Psi - D\right)^2 . \qquad (5.6.2)$$

It is assumed that the magnetic field in the central part is equal to 0 and here it can be assumed that $\Psi = 0$. Attention will now be given to some section of the magnetic barrier where for determinacy it is set that $\Psi > 0$. Consequently, the particle for which $D < 0$ cannot penetrate into the barrier because potential U again increases monotonically. However, if Ψ and D have the same signs, the potential shows a dip at $\Psi = D$ and particles may be trapped in this dip.

The anti-probkotron will be examined in greater detail (Fig. 5.6.2a). In this trap in the barrier surrounding the plasma the direction of the magnetic field changes and, the sign of Ψ also changes correspondingly. Therefore, regardless of the sign of the moment D the departure channel forms in the potential relief U (r, z, D) either on the left or right from the radial slit. To be captured in the channel the particle should have the longitudinal (along the boundary solid line) component of the velocity directed to the axial exit from the trap. With the decrease of r, as indicated by (5.2), the height of the channel walls increases and the particle appears to be trapped. Whether the particle leaves through the axial mirror or not, is now determined by the mirror ratio $\chi = H_{\max} / H_*$ ($H_{\max}$ is the magnetic field in the mirror, H_* is the field at the capture point.

Thus, there are two factors which lead to the departure of the particle from the anti-probkotron: the presence of the capture channel and of 'suitable' components of the velocity across and along the channel at the particle entering the channel. These factors cause that the areas of the effective axial orifices for the departure particles from this trap are unexpectedly large:

$$S^{an} \propto 2\pi R\rho_{\min} . \qquad (5.6.3)$$

Here R is the largest radius of the plasma volume, $\rho_{\min}$ is the ionic Larmor radius in the mirror.

The situation with Galatea-A is more complicated. Here the plasma boundary also consists of two elements – the surface of the magnetic shell of the myxine S_μ and the 'general' magnetic surface S_0 (Fig. 5.6.2b) on which the sign of Ψ differs. However, the surface S_μ does not extend outside the limits of the plasma volume. Therefore, for the particles for which the sign of D is the same (for example, positive),

as the sign of Ψ on the surface of the myxine, the presence here of the capture channel at the potential relief U does not lead to any loss of the particles. Consequently, the departure of co-particles takes place through the effective slit equal to

$$S^{co} \propto \pi\rho_{\min}^2. \tag{5.6.4}$$

A different situation arises for the particles with $D < 0$. These particles are co-particles for the magnetic shell of the myxine and counter particles for the general magnetic shell. Therefore

$$S^{cont} \propto 2\pi R\rho_i. \tag{5.6.5}$$

These considerations show that if the Galatea-A is filled with non-magnetised plasma then the particles with $D\Psi^0 > 0$ (Ψ^0 is the value of Ψ in the vicinity of the axis) should be the first to leave the trap. Subsequently, the plasma with the particles for which $D\Psi^0 < 0$ remains in the volume of the trap. This plasma starts to rotate and the rate of departure of the particles from the trap rapidly decreases. This is associated not only with the decrease of the effective cross-section of the orifice of departure by a factor of R/ρ_i, but also with the displacement of the particles from the axis as a result of rotation.

The collisions between the remaining particles generate the particles with $D\Psi^0 > 0$ and at the same time, the remaining plasma consisting of the particles with $D\Psi^0 < 0$ starts to rotate at a higher velocity if, after all, instabilities which could destroy all this do not appear. Unfortunately, these processes have not been studied sufficiently.

5.6.2. The departure of the particles from the mirror trap

The principal diagram of the mirror trap was shown in section 1.7 and it was noted that there is a 'departure cone' in the velocity space. Here we consider the stationary 'flow' of the particles only in the velocity space and ignore the time between the particle falling into the departure cone and its exit through the mirror. It is also assumed that there is no electrical field in the trap. All the information on the trap is in form of the aperture angle of the trapped cone θ_0 or the square of the sinus of this angle

$$\sin^2\theta_0 \equiv \alpha = \frac{H_M}{H_0}\ . \tag{5.6.6}$$

In addition, the 'start region' – the velocity of injected ions – is specified in the velocity space. Calculation of the flow from the start region to the cone is carried out using the Landau equation ignoring close collisions,

although this may be unjustified in the vicinity of the relatively thin cones. Under these assumptions, the initial equation is presented in the form (5.2.45)

$$\sum_{(i,e)} 2\pi e^2 e'^2 n^2 \frac{\partial}{\partial p_i} \int d\mathbf{p}' \left(\frac{\partial f'}{\partial p'_k} f - \frac{\partial f}{\partial p_k} f' \right) w_i = Q \,. \tag{5.6.7}$$

Here n is the density of the particles, and f is the distribution function (normalised with respect to unity) of the relevant ions in the velocity space (moments). This function $f = 0$ at $\theta = \theta_0$ and $\theta = \pi - \theta_0$, and $Q(\mathbf{v})$ is the injection intensity.

The analytical solution of equation (5.6.7) is very complicated and, consequently, we confine ourselves only to describing the procedure of the evaluation method used by Budker. Only the ion–ion collisions of the same particles are investigated [125].

Taking into account that the 'cross-hatched' particles are included only under the integral ($\int ...dp'$), without making a large error, in (5.6.7) it is set that

$$f'(p') = \frac{1}{(2mkT)^{3/2}} \exp\left\{ -\frac{(p')^2}{2mkT} \right\}. \tag{5.6.8}$$

Consequently, the equation for f is:

$$\frac{2\pi\Lambda e^4 n^2}{(2\pi mkT)^{1/2}} \frac{\partial}{\partial p_i} \int d\mathbf{p}' \left(\frac{p'_k}{mkT} f + \frac{\partial f}{\partial p_k} \right) \exp\left\{ -\frac{(p')^2}{2mkT} \right\} w_{ik} = Q. \tag{5.6.9}$$

Transferring to dimensionless variables

$$(\overline{p}_i, \overline{p}'_i) \to \frac{1}{\sqrt{2mkT}} (p_i, p'_i) \;\; \overline{u}_i \to \frac{u_i}{\sqrt{2kT/m}} \;\; . \tag{5.6.10}$$

instead of (5.6.9) we have

$$\frac{\partial}{\partial p_i} \left[\left(\frac{\partial f}{\partial p} + 2\overline{p}_k f \right) \overline{w}_{ik} \right] = -q \,. \tag{5.6.11}$$

Here

$$q = Q \frac{\sqrt{2\pi m}}{e^4 n^2 \Lambda} (kT)^{3/2} \tag{5.6.12}$$

$$\overline{w}_{ik} = \int \exp\left\{ -(\overline{p}')^2 \right\} \frac{\overline{u}^2 \delta_{ik} - \overline{u}_i \overline{u}_k}{\overline{u}^3} d\overline{\mathbf{p}'} = a(\overline{p}) \delta_{ik} - b(\overline{p}) \frac{\overline{p}_i \overline{p}_k}{\overline{p}^2} \cdot$$

The explicit form of $a(\bar{p})$ and $b(\bar{p})$ was published in [125] but this form is not required in the present case and, consequently, it will not be determined more accurately.

In the spherical coordinate system, the equation (5.6.11) is presented in the form

$$\frac{1}{p^2}\frac{\partial}{\partial \bar{p}}\left\{\bar{p}^2\left(a(\bar{p})-b(\bar{p})\right)\left(\frac{\partial f}{\partial p}+2\bar{p}f\right)\right\}+\frac{a(\bar{p})}{\bar{p}^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)=-q. \tag{5.6.13}$$

This equation can be easily solved if it is set that

$$q(\bar{p})=q_0\exp\left\{-\bar{p}^2\right\}\frac{a(\bar{p})}{p^2} \tag{5.6.14a}$$

Consequently, the solution is

$$f=\exp\left\{-\bar{p}^2\right\}\psi(\theta), \tag{5.6.14b}$$

and $\psi(\theta)$ satisfies the equation

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta)\frac{\partial\psi}{\partial\theta}=-q_0 \tag{5.6.15a}$$

Consequently, we obtain directly

$$\psi=q_0\ln\frac{\sin\theta}{\sin\theta_0} \tag{5.6.15b}$$

Finally, the approximate solution of the kinetic equation, satisfying the boundary condition $f = 0$ at $\theta = \theta_0$, has the form

$$f=q_0\exp\left\{-\bar{p}^2\right\}\ln\frac{\sin\theta}{\sin\theta_0}. \tag{5.6.16}$$

Taking into account that f should be normalised with respect to 1:

$$\int f d\,\bar{\mathbf{p}}=1,$$

we obtain

$$q_0=\frac{1}{\pi^{3/2}\left[-\ln\operatorname{tg}\frac{\theta_0}{2}-\cos\theta\right]} \tag{5.6.17a}$$

and at $\sin^2\theta_0=\alpha\ll 1$

$$q_0=\frac{2}{\pi^{3/2}(|\ln\alpha|+2\ln 2-2)}\approx\frac{2}{\pi^{3/2}(|\ln\alpha|-0{,}6)} \tag{5.6.17b}$$

The general flow of the particles to the forbidden cone is equal to the intensity of arrival of the particles in the trap

$$I = 4\pi\int_0^\infty Qp^2 dp = \frac{6.1}{|\ln\alpha| - 0.6}\frac{\Lambda e^4 n^2}{\sqrt{m}(kT)^{3/2}}. \qquad (5.6.18a)$$

If we introduce the effective departure cross-section σ_{ef} using the equation

$$I = n^2\sigma_{ef}v_{rel}, \qquad (5.6.18b)$$

then, assuming $v_{rel} = \sqrt{\frac{kT}{m}}$, we have

$$\sigma_{ef} = \frac{3}{|\ln\alpha|}\frac{\Lambda e^4}{(kT)^2}. \qquad (5.6.18c)$$

This shows that the effective departure cross-section is of the order of the duration of a single Coulomb collision and depends only very slightly on the mirror ratio H_{max}.

Budker proposed a number of models of particles falling into the departure cone – for different positions of the injection zone and taking into account the collisions of the ions with the electrons. The logarithmic dependence of σ_{ef} on $\alpha = \frac{H_{max}}{H_{min}}$ is retained everywhere.

This has also been confirmed by the numerical solution of equation (5.6.7)

The final comment. The initial experiments with the probkotrons shows that the plasma in a simple axisymmetric probkotron exist for a very short period of time (<1 μs) which is thousands of times shorter than that indicated by equation (5.6.18c). It has been shown, and this was already discussed in section 1.7.1, that poor retention is associated with the development of a convective instability. Therefore, the axisymmetric plasma configurations have been abandoned. Consequently, it was indeed possible to transfer to the classic 'retention' times determined by the equation (5.6.18c). However, the classic time at the temperatures required for the D–T reaction is an order of magnitude shorter than that required for a thermonuclear reaction with a positive yield. This difficulty was overcome, at least in principle, in the method proposed by Dymov (see section 10.5). The experiments have confirmed the accuracy of the main concept of this method but have shown new difficulties which have not as yet been studied.

5.6.3. Neoclassic diffusion in toroidal systems

Previously, the classic diffusion of plasma in a flat (one-directional) magnetic field was investigated in section 3.8.2. Classic diffusion of plasma in toroidal fields has a number of interesting points. Since the features in which we are interested are almost completely universal for the toroidal fields, they will be illustrated on the simplest symmetric configuration – tokamak.

These features are associated with two specific peculiarities of the toroidal configurations: the toroidal drift and the appearance of trapped particles which, as shown in section 1.7.2, described the sickle-shaped 'banana' trajectories in projection on the plane (*r, z*). The width of these 'bananas' is determined by the poloidal magnetic field:

$$\delta_{\text{ban}} \sim \frac{v_{\perp}}{eH_{\perp}}, \tag{5.6.19a}$$

whereas the passing particles have the Larmor width, determined by the strength of the total field (1.7.14)

$$\delta_{\text{pass}} \approx 2\frac{v_{\perp}}{eH}. \tag{5.6.19b}$$

In reality, the ratio

$$\frac{\delta_{\text{pass}}}{\delta_{\text{ban}}} \le 0.1 \div 0.3, \tag{5.6.19c}$$

and, therefore, the diffusion coefficient of the trapped particles is considerably higher than the diffusion coefficient of the drift of the particles

$$D_{\text{trap}} \sim \frac{\delta_{\text{ban}}^2}{\tau} \sim (10-25) D_{\text{pass}}.$$

These simple considerations indicate the presence of three diffusion regimes, depending on the value of the ratio of the time of travelling around the 'banana' (in the absence of collisions τ_{δ}) to the free path time τ_{fp}

$$\eta = \frac{\tau_{\delta}}{\tau_{\text{fp}}}.$$

If $\eta \gg 1$, the 'banana' trajectory does not form. If $\eta \sim 1$, a large piece of the 'banana' trajectory appears between the collisions and, finally, if $\eta \ll 1$, the particle manages to 'write' several banana trajectories.

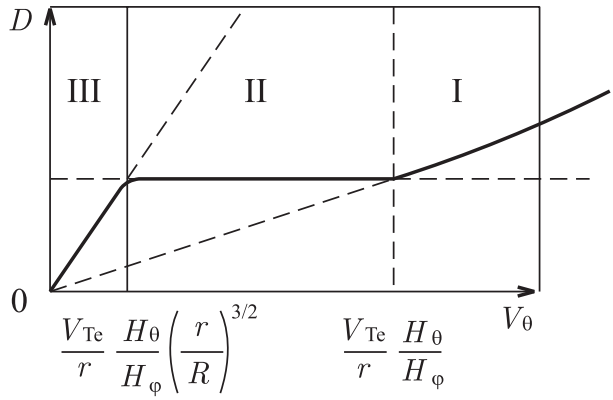


Fig. 5.6.3. Dependence of the coefficient of neoclassic diffusion on the frequency of collisions: I) quasi-MHD regime (Pfirsch–Schlütter); II) 'plateau'; III) 'banana' regime.

Consequently, $D_\perp$ depends on the frequency of collisions as shown in Fig. 5.6.3. Each regime will be investigated in greater detail [127].

Frequent collisions regime

It is assumed that the free path time $\tau_0 \ll \frac{2\pi}{\omega_H}$ of the Larmor rotation in the total field $H = |\mathbf{H}_0 + \mathbf{H}_\perp|$. Therefore, it would appear that the diffusion coefficient is simply equal to (section 3.8.2)

$$D^1 = \theta\frac{\beta}{2}\nu, \quad \theta \sim 1. \tag{5.6.20}$$

However, this is not so because of the toroidal drift. In fact, as already mentioned, the toroidal drift generates polarisation charges 'at the top' and 'at the bottom' of the rings. These charges flow along the lines of force and are neutralised. However, to ensure that this flow takes place at a finite resistance of the plasma between the 'top' and 'bottom' of the plasma ring, there should be a potential difference U_{tot}. In other words, the volume of the ring should contain the electrical field oriented mostly along the axis *z*. The presence of such **E**-field and of the azimuthal magnetic field results in the drift of the particles along the radius, and the drift is determined by finite conductivity, like the conventional drift. Calculations show (Pfirsch, Schlütter) that the radial diffusion coefficient in this case is equal to

$$D_\perp^{P-S} = D_{\perp 0}(1 + \theta_1 q^2), \tag{5.6.21}$$

where $D_{\perp 0}$ with the classic diffusion coefficient in the uniform field, θ~1,

and q is the 'stability reserve'

$$q = \frac{aH_{\|}}{RH_{\perp}}. \tag{5.6.22}$$

Here a and R are the small and large plasma pinch radii, $H_{\|}$ is the azimuthal and $H_{\perp}$ the poloidal field.

Since the characteristic values $q > 1$ are often ~3–5 [9], then taking into account the toroidal drift increases the classic diffusion coefficient, even at $\eta \gg 1$, by almost an order of magnitude.

In the range of the values $\eta \sim 1$ it is natural to expect that the diffusion coefficient depends only slightly on frequency ν_{ie} since it is not important where the collision took place at the top, in the centre of the external section or at the bottom of the banana. If it is also taken into account that the fraction of the trapped particles in relation to the total concentration $\sim \sqrt{a/R}$ [10], then the diffusion coefficient to the 'plateau' will be (Galeev, Sagdeev) [127]

$$D_{\perp} = 3.6\sqrt{\frac{a}{R}}\left(\frac{\rho_e^2}{\tau_{ie}}\right)\left(\frac{H_{\|}}{H_{\perp}}\right). \tag{5.6.23}$$

Finally, if $\eta \ll 1$, then

$$D_{\perp} \approx \frac{A^2}{\tau_{ie}}. \tag{5.6.24}$$

Here A is the Larmor radius calculated on the basis of the poloidal field. Diffusion at $\eta \sim 1$ and $\eta \ll 1$ is referred to as neoclassic diffusion [127]).

On the whole, the dependence of the coefficient of 'neoclassic' diffusion on the frequency of collisions has the form shown in Fig. 5.6.3. The three regions presented here are usually referred to as: MHD region, plateau, banana region.

The dependence of the heat conductivity coefficients for each component on ν_{ie} is the same as the dependence of the diffusion coefficient. But here we have the ionic and electronic conductivity. The neoclassic ionic heat conductivity in a specific range is in agreement with the experiment, and the electronic thermal conductivity differs by 1–2 orders.

[9] See appendix D and E.

[10] This follows directly from the relative value of the volume of the departure cone in the velocity space.

5.6.4. Confinement of energy and plasma in actual traps. Scaling

In the previous section we considered a number of classic, i.e. non-oscillation, departures of plasma particles from the traps. However, there is also a large number of factors resulting in the departure of energy and particles from the traps. They include the removal of energy by heat conductivity of the electronic and ionic components, and also a powerful factor, such as the radiation of impurities of highly ionised heavy ions which may also penetrate into the hydrogen plasma of the thermonuclear reactors. However, the shortcoming of these reactors is the 'instability' – spontaneous processes leading to destruction of the magnetic configuration.

In addition to large-scale oscillations, a significant role is usually played by microwave oscillations with plasma (ω_0) and electronic-Larmor frequencies caused by 'beam instabilities' of different types. The presence of these high-frequency, high-intensity **E**-fields ('strong turbulence') results in the situation in which the particles start to be scattered by the **E**-fields and this may greatly reduce τ_{ei} and, at the same time, greatly increase the diffusion coefficient.

Finally, it is also necessary to mention the 'purely technical' factors such as lack of alignment adjustment of the magnetic fields. In section 1.1.4 attention was given to the instability of the morphology of the magnetic fields, i.e. destruction of the magnetic surfaces, including the case of increasing β. This shows that analysis of plasma diffusion in plasma systems is determined by the properties of the entire system and not by some local parameter – 'diffusion coefficient' as, for example, in diffusion of a light gas in a heavy one. This already shows why now, almost 45 years later, there is no quantitative theory of the departure of energy and particles of plasma from tokamaks, and various 'scalings' are used instead.

Scalings

This term refers to semi-empirical equations obtained as a result of processing large sets of experimental data taking into account the laws of similarity. These scalings were adopted in plasma physics from hydrodynamics. A suitable example is the formula linking the characteristic parameter – the critical velocity of transition of the laminar Poiseuille flow into a turbulent flow, with the diameter of the pipe, viscosity and roughness of the wall.

The scalings are particularly natural when it comes to very complicated (at the present time) systems. With time, the actual situation will become clearer, and physically justified formulas, replacing the scalings, will be derived.

In the light of the above considerations regarding the complexity of the transfer processes in real traps, for example, tokamaks, it is understandable why the term scaling is used widely here. The currently used scalings for the tokamaks are presented in the appendix E and F.

To provide information on the nature of scalings, it is important to mention the so-called novoalcator scaling, based mainly on the experimental data obtained in a Alcator tokamak (USA). The scaling for the confinement time of energy during joule heating of the plasma has the form

$$\tau_E \sim \bar{n}_e R^2 a q. \tag{5.6.25}$$

Here $\bar{n}_e$ is the mean density of the electronic component in the cross-section, R is the large radius of the torus, a is the small radius, and q is the stability reserve (5.6.22).

The dependence (5.6.25) describes experiments quite efficiently. It is not clear what the physics of equation (5.6.25) is. It would appear that we should use the equation of the type

$$\tau_E \sim \theta \frac{a^2}{\chi_\perp} \tag{5.6.26}$$

Here χ is the coefficient of neoclassic thermal diffusivity.

However, the values calculated from equation (5.6.26) differ quite greatly from the experimental data, especially in the case of electrons. The reasons are the same as in the case of plasma diffusion - 'anomalous' (dynamic) processes.

5.7. Plasma optics (hybrid models)

It was noted previously that as a result of the large difference in the masses of the electrons and ions, the 'hybrid' models are often quite effective. In this case, the ions are described by the kinetic and electrons by hydrodynamic equations. In this section attention is given to this approach on the example of plasma optical systems.

Plasma is used in many applications. In particular, plasma is used as a medium forming unusual electromagnetic configurations. For example, Fainberg and Rukhadze have been developing for many decades and also using plasma waveguides, many investigators have studied the 'coherent' methods of accelerating heavy particles, including with the so-called ship's wake, etc.

In section 3.3.2 attention was given to the possibility of producing greatly differing reliefs of the electrostatic potential on the basis of equipotentialisation of the magnetic lines of force. The principal

schemes of various device classes which can be produced on this basis have already been mentioned. In this section, we discuss one of them – focusing plasma optical systems (POS). The section of the plasma dynamic, investigating the combinations of these systems, is referred to as plasma optics [129, 130][11].

Classic corpuscular optics will be referred to as 'vacuum optics'. With all the significant achievements, vacuum optics has two significant disadvantages. In particular, it uses low current, and the dimensions of the appropriate optical system, even the case of maximum current, are limited by the Debye radius, as indicated on the example of the diode. It is evident that the mastering of quasi-neutral fluxes by plasma optics, greatly changes the situation. The plasma optics systems, using high current, represent a powerful tool for industrial technologies.

A second disadvantage of the vacuum optical systems with the Laplace nature of the electrostatic field used in these systems, i.e. the vacuum electrostatic field is governed by the Laplace equation

$$\Delta\phi = 0. \tag{5.7.1}$$

This circumstance leads to a number of undesirable effects. For example, it is not possible to produce single axisymmetric lenses (both magnetic and electrical), scattering the parallel flux of the particles (these lenses are always collecting), and the spherical aberration in the lenses cannot be removed so that it is not possible to study atoms in electron microscopes, etc.

In plasma optics, instead of the Laplace equation, the potential is governed by the law of equipotentialisation of the magnetic lines of force

$$\phi_T = \phi_T(\gamma), \tag{5.7.2}$$

and the previously mentioned difficulties can be eliminated in this case.

In this section, we discuss the special features of plasma optical systems in a simplified hybrid approximation. In particular, we describe the dynamics of the ions in the one-particle approximation, the electrons are regarded as cold, and the thermalisation of the electrical potential is consequently ignored. It is quite easy to disregard these assumptions, and this will be carried out at the end of this section.

5.7.1. General principles of plasma optics

The previously mentioned low current nature of the vacuum systems can be illustrated by the following estimate. As a result of the intrinsic

[11]The principles of plasma optics were formulated by A.I. Morozov. A precursor of plasma optical systems were the 'lenses with the volume charge', proposed by D. Gabor.

electrical field, the ion beam with the particle energy ~1 keV and a current of 1 mA expands by a factor of 2 in the case of hydrogen on the length L ~5D, where D is the beam diameter, and in the same conditions, the argon ion beam expands by a factor of 2 at L ~12 D.

If only one magnetic field is used, such a system can focus the quasi neutral ion flux. A suitable example is the focusing in a homogeneous magnetic field, investigated in section 1.4.2. However, in this case we are interested in the systems with the volume electrostatic field.

It is desirable to ensure the autonomy of the electronic component in the sense of section 3.2.2. This is realised very simply in the axisymmetric systems using poloidal (**E**, **H**)-fields in which the drift is closed.

Two examples of plasma optical systems with quasi-neutral ion fluxes will be investigated.

'Electrostatic' plasma lens

In section 1.4, we discussed the focusing of the particles by the simplest ring-shaped lenses in vacuum.

Here, we discuss the focusing using a coil through which current flows and, in addition, this ring is charged. If the density of the ion beam n_b is low, so that the Debye radius, calculated from T_e and the beam density are considerably higher than the radius of the ring R

$$r_D \sim \sqrt{\frac{kTe}{4\pi e^2 n_b}} \gg R, \tag{5.7.3a}$$

then at $T_e \ll \varepsilon_i$, where ε_i is the ion energy, and at the previously mentioned parameters of the system, the focusing distance F_{EH}^{vac} is as previously very large, since according to the 'school' formula is

$$\frac{1}{F_{EH}^{\text{vac}}} = \frac{1}{F_E^{\text{vac}}} + \frac{1}{F_H^{\text{vac}}}. \tag{5.7.3b}$$

Now, we increase the beam density to ensure that the Debye radius is considerably smaller than the ring radius, and the magnetic fields is selected such that the electronic Larmor radius is also relatively small ($\rho_e \ll R$). The equipotentialisation of the magnetic lines of force takes place in these conditions and the completely rearrangement of the equipotentials occurs. Consequently, the sign-alternating radial component of the electrical field E_r, which existed in the vacuum case, transforms to the sign-constant quantity (Fig. 5.7.1). Without carrying out any calculations, it may be seen that the focusing distance which previously depended on the square of the

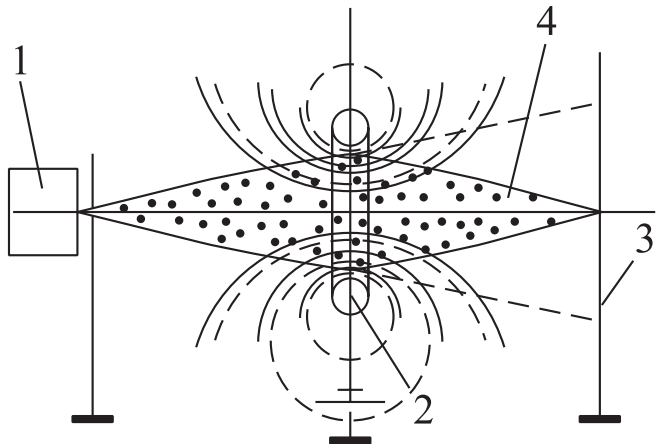


Fig. 5.7.1. Rearrangement of the equipotentials of the electrostatic lens at transition from the vacuum to quasi-neutral regime: 1) the source of the compensated ion beam, 2) the coil, generating the magnetic fields, combined with the electrode, 3) luminescence screen, 4) focused quasi-neutral beam.

strength of the **E**-field, now depends on the first degree of the strength of the field E_r. Consequently, the focusing length greatly decreases, and the lens can be both collecting and scattering.

The above considerations will be described by calculations. We consider the simplest case in which

$$\phi = k\psi. \tag{5.7.4}$$

The function ψ in the vicinity of the ring with current is equal to

$$\psi = \frac{r^2}{2}\frac{2\pi JR^2}{c(R^2+z^2)^{3/2}} + \ldots$$

If the relationship (5.7.2) is linear (phi ~ ψ), then

$$\phi = \frac{r^2}{2}a(z), \quad a(z) = \left(\frac{2U_a}{R}\right)\frac{R^2}{(R^2+z^2)^{3/2}} \tag{5.7.5}$$

Here U_a is the effective ring potential. Substituting (5.7.5) into (1.4.22b), we obtain

$$F_E^{\text{pl}} = R\frac{\varepsilon_{\text{kin}}}{2eU_a}. \tag{5.7.6}$$

Taking ε_{kin} = 10 keV, R = 5 cm, U_a = 1 keV, we obtain $F_E^{\text{pl}} \approx 25\,\text{cm}$ (!).

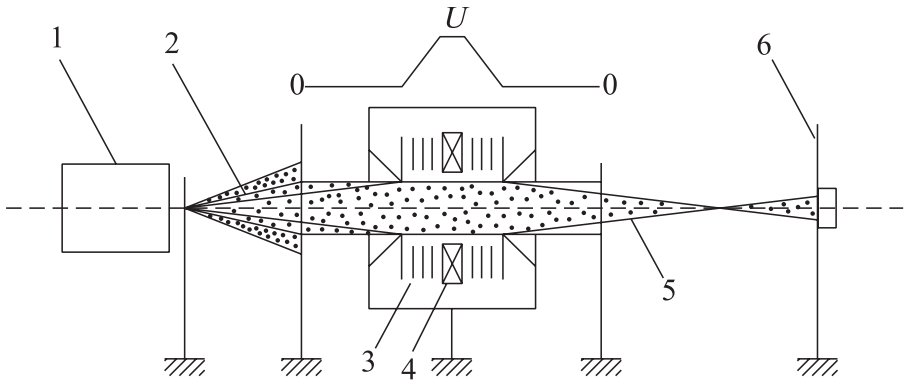


Fig. 5.7.2. Diagram of experimental equipment with the electrostatic plasma lens: 1) the ion beam source, 2) the compensated total beam, 3) the electrodes fixing the potentials of the lines of force in the lens, 5) the quasi-neutral ion beam, passing through the lens, 6) the screen.

The difference is huge in comparison with what was said in section 1.4.

The scheme of the plasma lens and of equipment as a whole, which was used for the first time for the rearrangement of focusing in 1967 by V.V. Zhukov, A.I. Morozov and G.Ya. Shchepkin at the Institute of Atomic Energy, is shown in Fig. 5.7.2 [131]. The investigations confirm the previously mentioned calculations.

However, it should be mentioned that the quality of focusing slightly decreases at high current (~1 A). There are no doubts that this shortcoming will be overcome.

The described lens is referred to as the electrostatic plasma lens, because the actual contribution of the intrinsic magnetic field remains as small as in the vacuum case and its role is actually reduced to ensuring the structure of the **E**-field which requires a relatively weak fields, provided the condition required for equipotentialisation

$$\rho_e << R$$

is fulfilled, together with the condition of partial deceleration of the flux

$$\frac{H^2}{8\pi} > Mnv_i^2 S \ .$$

Figure 5.7.3a shows the currently available plasma lens, developed and investigated by A.A. Goncharov et al [132]. This lens focused the ion fluxes from carbon to bismuth. The characteristic parameters of the ion fluxes: ε_i ~ 20–100 keV, I_i < 0.5 A, the beam diameter on the approach to the lens ~10 cm. The result of focusing is shown in Fig. 5.7.3b.

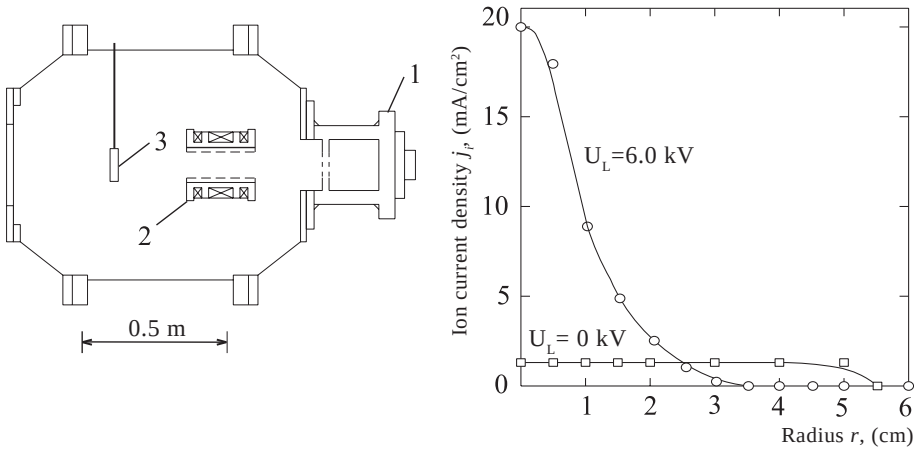


Fig. 5.7.3. Diagram of experimental equipment with the electrostatic plasma lens developed by A.A. Goncharov (a): 1) the ion beam source, 2) the plasma lens, 3) Faraday cylinder; radial distribution of the density of the compensated ion current at the distance of 0.5 cm from the lens, measured using the Faraday cylinder (b). The graph shows the voltage in the lens; the magnetic field in the lens 800 G, ion energy 22 keV.

Plasma optical mass separator [133]

Here, we describe the scheme of the simplest mass separator of quasi-neutral fluxes, containing ions with different masses but with the same charge. Work on the construction of such a separator has started. The concept of the separator differs to a certain extent from the principle of the energy analyser proposed by Yuz-Rozhanskii (Fig. 1.4.4). In particular, if the energy of the particles arriving in the cylindrical condenser depends only on the mass of the particles ($\varepsilon_{kin} = \varepsilon(m)$), then particles with different masses will be focused at different distances from the reference surface. Such a separator with the axisymmetric geometry and a homogeneous magnetic field, oriented along the axis of the system, and with the radial **E**-field is shown in Fig. 5.7.4. It is clear that the drift in this separator is closed. The figure shows that the ion flux travels into the separator from a ring-shaped relatively high-current source. It can be represented by an ion injector or a plasma accelerator with a closed drift. The requirement on the source is that the level of the azimuthally non-symmetric noise is low and the random azimuthal component of the velocity of the exit from the channel is also low

$$\frac{\delta v_\theta}{v_M} < \frac{\Delta M}{M} \qquad (5.7.7)$$

where ΔM is the separated difference of the ion masses. In addition to

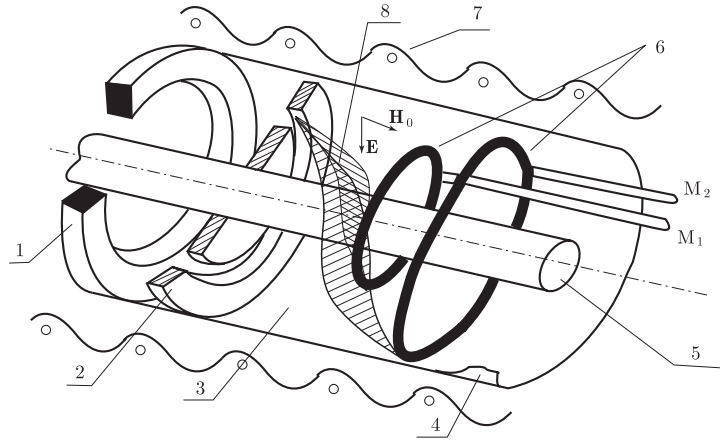


Fig. 5.7.4. Diagram of plasma-optical mass separator with *E*-focusing (POMS-E): 1) the ring-shaped plasma accelerator – ion source, 2) azimuthator, 3) separating volume, 4) the outer cylindrical positive electrode, 5) the inner negative electrode, 6) receivers of separated masses (M_1, M_2 – holders of the receivers), 7 – the coils of a weak magnetic field $\mathbf{H}_0$ in the separating volume, used for magnetisation of the electrons, 8) the focusing of ions, exiting from a single point of the ring-shaped slit.

this, it is assumed that the density of the particles in the flux varies only slightly.

The compensated ion ('plasma') flux travels to the input slit of the separator shut-off by the relatively strong transverse magnetic field. This block of the separator is referred to as the 'azimuthator'. The magnetic flux, intercepted by the particles, is equal to $2\pi\psi=2\pi RhH$, where h is the width of the 'azimuthator', R is the slit radius. Because of the law of conservation of the momentum of the amount of motion (at entry into the channel of the accelerator it is assumed that $\dot{\theta} = 0$, $\psi = 0$)

$$mr^2\dot{\theta}+\frac{e}{c}\psi = D \approx 0,$$

The particles, passing through the azimuthator, acquire the azimuthal velocity

$$v_\theta = \frac{e}{c}\frac{Hh}{m}$$

The velocity depends only on the particle mass and here we can use the equation (1.4.16c) for the displacement along the radius

$$\delta_r = R\frac{\delta_m}{m}$$

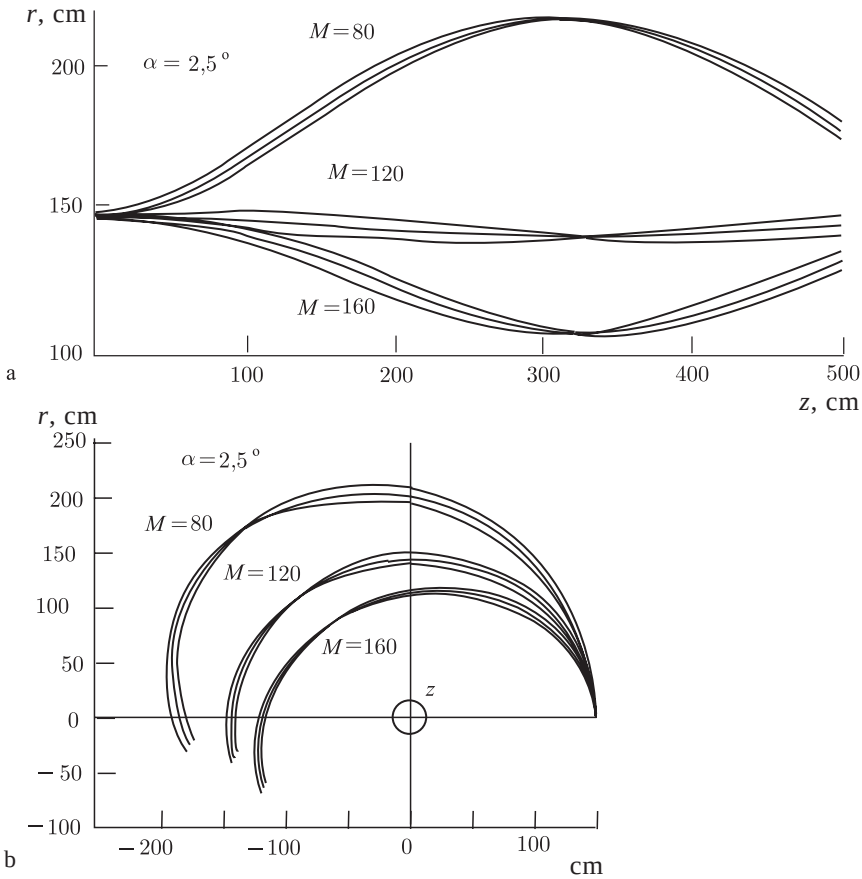


Fig. 5.7.5. Calculated trajectories of the ions in the POMS-E plasma-optical mass separator: projection on the plane (r, z) of the trajectories of ions of different masses, exiting from a single point of the source with the same energy, calculated in one-particle approximation (a); projections on the plane (r, θ) in the same conditions (b).

After passing through the azimuthator, the particle retains part of the longitudinal velocity

$$v_{\|,A} = \sqrt{v_0^2 - v_{\theta,A}^2}$$

Here the zero indicates the velocity of the particles in front of the azimuthator, and the letter A the components of velocity after passing through the azimuthator. An example of the calculated trajectories in such a separator is shown in Fig. 5.7.5.

Estimates show that a separator with the diameter of ~3 m can process ~20–30 t of material per annum with the mean atomic mass of ~50–100. Therefore, the plasma optical separators of this type are capable of providing a significant contribution to technology,

especially in atomic power engineering, which requires isotopically clean constructional materials, and it is also necessary to separate the waste from nuclear plants.

5.7.2. Some special features of plasma optical systems

Transitive and non-transitive systems

On the example of the electrostatic lenses it was shown that the transition from the vacuum to quasi-neutral regime (in the presence of an additional magnetic field) is accompanied by the radical rearrangement of the equipotentials. These systems are referred to as non-transitive.

On the other hand, the main volume of the previously described separator contains a uniform magnetic field and a radial (~1/*r*)electrical field. Here the condition $\phi = \phi(\psi)$ and the Laplace equation for ψ and ϕ are fulfilled at the same time. Therefore, such a separator with the rearrangement of the fields can operate both in the regime of single particles and in the regime of quasi-neutral fluxes. These systems will be referred to as transitive.

The 'trivially transitive' systems are regarded as optical systems containing only one magnetic field.

The conditions for equipotentialisation of the magnetic lines of force

In the case of the non-transitive systems, the equipotentialisation is associated with the appearance of electrical charges in the system. We determine the scale of these charges for the axisymmetric systems, taking into account the Poisson equation

$$\Delta\varphi(\psi) = -4\pi e\nu \tag{5.7.8}$$

Here $\nu \equiv n_i - n_e$ is the density of the non-compensated particles. Evidently, the critical density n^* at which the equipotentialisation has been almost completed should exceed ν by at least an order of magnitude. Therefore, we can write the following criterion of the plasma-optical regime

$$n > n^* \approx 10\nu \tag{5.7.9}$$

Using equations (5.6.10), (5.6.11) we obtain

$$\frac{n}{10} \geq \frac{1}{4\pi e}\left[\phi''\left[\left(\frac{\partial\psi}{\partial r}\right)^2 + \left(\frac{\partial\psi}{\partial z}\right)^2\right] + \phi''\frac{2}{r}\frac{\partial\psi}{\partial r}\right]. \tag{5.7.10}$$

Here it is taken into account that $\Delta^*\psi = r\dfrac{\partial}{\partial r}\dfrac{1}{r}\dfrac{\partial\psi}{\partial r} + \dfrac{\partial^2\psi}{\partial z^2} = 0$

When ϕ depends in a linear manner on ψ we obtain

$$\nu = -\frac{2}{r}\frac{\partial\phi}{\partial r}\frac{1}{4\pi e} = \frac{2E_r}{r}\frac{1}{4\pi e}. \tag{5.7.11}$$

As regards the order of magnitude, the above equation is equivalent to the estimate

$$\frac{\nu}{n} \approx \left(\frac{r_D}{L}\right)^2 \tag{5.7.12}$$

Here r_D is the Debye radius, calculated from the ion velocity, L is the characteristic scale of the system.

Arbitrariness of defining the **E***-field in plasma optical systems*

The most important special feature of the plasma-dynamic systems is the incomparably large arbitrariness in defining the electrical field in the system. Actually, in vacuum the distribution of the **E**-field in the volume is determined by the Laplace equation and for the given boundary condition

$$\phi = \phi(M)\big|_S$$

where M is the point on the boundary surface S, is expressed by the integral

$$\phi(\mathbf{x}) = \iint\limits_{(S)} dSG(\mathbf{x}, M)\phi(M)$$

Here G ($\mathbf{x}$,M) is the Green function for the given region. This shows that the value ϕ at the point $\mathbf{x}$ is determined by the entire distribution of the boundary of the region. The situation in the quasi-neutral case is different. Here the potential is brought along the line of force, connecting point $\mathbf{x}$ with point M on the surface.

The formally large arbitrariness in constructing $\phi(\mathbf{x})$ in plasma optics is also indicated by the fact that the equipotentialisation actually indicates the transition from the Laplace to Poisson equation.

A highly suitable specific example are the electrostatic fields in axisymmetric paraxial systems.

Taking into account that in this case $\psi\big|_{r\to 0} \approx \frac{r^2}{2}H_0(z) \to 0$, we can write

$$\begin{aligned}\phi &= \phi(\psi) = \phi'(0)\psi + \phi''(0)\frac{\psi^2}{2} + \ldots = \\ &= \frac{r^2}{2}\left(\phi'(0)H_0(z)\right) + \frac{r^4}{8}\left(\phi''(0)H_0^2(z) - \frac{\phi'(0)}{4}H_0''(z)\right).\end{aligned} \tag{5.7.13}$$

This shows that if the field in the vacuum case is determined by a single function $\phi_0(z)$, two functions $H_0(z)$ and $\phi(\psi)$ must be considered in the plasma case. This additional arbitrariness enables spherical aberration can be removed to some extent.

5.7.3. Expansion of the quasi neutral beam under the effect of electronic pressure

In the previous sections, the electrons were assumed to be cold and, correspondingly, the energy potential was constant on the lines of force (magnetic surfaces)

$$\phi = \phi(\psi)$$

It will be shown how to take into account the effect of the finite electronic temperature on the behaviour of the quasi-neutral beams, if this temperature is not high

$$\nabla(nkT_e) \ll \nabla\phi. \tag{5.7.14}$$

Assuming that the distribution of the electrons is Maxwellian, we take the thermalised potential in the form (3.2.10)

$$\phi = \phi_T(\psi) + \frac{kT_e}{e}\ln\frac{n}{n_0}. \tag{5.7.15a}$$

Correspondingly, the kinetic equation for the quasi-neutral ion beam has the form (at $\mathbf{H} = 0$)

$$\frac{\partial f_i}{\partial t} + \mathbf{v}\frac{\partial f_i}{\partial \mathbf{x}} - \frac{e}{m}\left(\nabla\phi_T(\psi) + \frac{kT_e}{e}\frac{1}{n}\nabla n\right)\frac{\partial f_i}{\nabla \mathbf{v}} = 0. \tag{5.7.15b}$$

Because of quasi-neutrality

$$n_e \approx n_i = \int f_i d\mathbf{v}. \tag{5.7.15c}$$

Taking into account the smallness of the thermal term, the system (5.7.15) can be solved by the method of successive approximations

$$\begin{aligned} f &= f_0 + f_1 + f_2 + \ldots\ldots, \\ n &= n_0 + n_1 + n_2 + \ldots\ldots \end{aligned} \tag{5.7.16a}$$

In reality, these expansions with respect to degrees

$$\epsilon = \frac{kT_e}{\varepsilon_i}. \tag{5.7.16b}$$

Here ε_i is the characteristic ion energy.

Introducing the notations

$$\frac{Df_0}{Dt} = \frac{\partial f}{\partial t} + \mathbf{v}\frac{\partial f}{\partial \mathbf{v}} + \frac{e}{M}(-\nabla\phi)\frac{\partial f}{\partial \mathbf{v}},$$

we can write the following chain of successive approximations:

$$\begin{aligned}
&\frac{Df_0}{Dt} = 0,\\
&\frac{Df_1}{Dt} = \frac{kT}{e}\frac{1}{n_0}\nabla n_0 \frac{\partial f_0}{\partial \mathbf{v}};\\
&\frac{Df_2}{Dt} = \frac{kT}{m}\frac{1}{n_1}\nabla n_1 \frac{\partial f_1}{\partial \mathbf{v}};....\\
&n_0 = \int f_0 d\mathbf{v}; \quad n_1 = \int f_1 d\mathbf{v}.
\end{aligned} \tag{5.7.17}$$

Thus, the problem is reduced to solving the linear equations in partial derivatives with the known right-hand part.

The latter method will be demonstrated using a simple example. Let the quasi-neutral flux travel into the free half space $z > 0$ and the ions of the flux at $z = 0$ have the distribution function

$$f_{(0)0}\Big|_z = n_{00}\exp\left\{-\frac{y^2}{b^2}\right\}\delta(v_z - v_0)\delta(v_y). \tag{5.7.18}$$

The temperature of the electrons T_e is assumed to be constant. In this case, the kinetic equation for the ions has the form

$$(\mathbf{v}\nabla) f - \frac{kT_e}{M}\frac{1}{n}\left(\nabla n \frac{\partial f}{\partial \mathbf{v}}\right) = 0; \quad n = \int f \, d\mathbf{v}.$$

The zero approximation of the distribution function satisfies the equation:

$$(\mathbf{v}\nabla) f_{(0)} = 0,$$

and the characteristics of this equation describes the uniform straight motion

$$v_{y0} = v_y; \quad v_{z0} = v_z; \quad y_0 = y - \frac{v_y z}{v_z}.$$

Consequently, at any x

$$f_0 = n_{00}\exp\left\{-\frac{y^2}{b^2}\right\}\delta(v_z - v_0)\delta(v_y). \tag{5.7.19}$$

We can now determine the distribution of the beam density in the zero approximation and the equation of the first approximation

$$n_0 = n_{00}\exp\left\{-\frac{y^2}{b^2}\right\};$$
$$v_z\frac{\partial f_{(1)}}{\partial z} + v_y\frac{\partial f_{(1)}}{\partial y} - \frac{kT_e}{M}\frac{2y}{b^2}\frac{\partial f_0}{\partial v_y} = 0. \qquad (5.7.20)$$

Its characteristic system

$$\frac{dz}{v_z} = \frac{dy}{v_y} = \frac{dv_z}{0} = \frac{dv_y}{\frac{kT_e}{M}\frac{2y}{b^2}} \qquad (5.7.21a)$$

has the solutions

$$v_{z0} = v_z; \quad v_{y0} = v_y\mathrm{ch}\left(\frac{qz}{v_z}\right) - qy\,\mathrm{sh}\left(\frac{qz}{v_z}\right);$$
$$y_0 = y\ \mathrm{ch}\left(\frac{qz}{v_z}\right) - \frac{v_y}{q}\mathrm{sh}\left(\frac{qz}{v_z}\right);\ q^2 = \frac{2kT_e}{Mb^2}. \qquad (5.7.21b)$$

Substituting (5.7.21b) into (5.7.19) we obtain

$$f_1 = n_{00}\exp\left\{-\frac{y^2}{b^2}\left(y\,\mathrm{ch}\left(\frac{qz}{v_z}\right) - \frac{v_y}{q}\mathrm{sh}\left(\frac{qz}{v_z}\right)\right)^2\right\}\times$$
$$\times\,\delta(v_{z0} - v_0)\,\delta\left(v_y\mathrm{ch}\left(\frac{qz}{v_z}\right) - qy\ \mathrm{sh}\left(\frac{qz}{v_z}\right)\right). \qquad (5.7.22)$$

Taking into account that

$$\delta(f(\alpha)) = \frac{\delta(\alpha - \alpha_0)}{|f'(\alpha_0)|}, \quad f(\alpha_0) = 0,$$

the equation is simplified

$$f_1 = n_{00}\frac{\exp\left\{-\frac{y^2}{b^2}\mathrm{ch}^2\frac{qz}{v_z}\right\}}{\mathrm{ch}\left(\frac{qz}{v_z}\right)}\delta(v_{z0} - v_0)\,\delta\left(v_y - qy\,\mathrm{th}\left(\frac{qz}{v_z}\right)\right). \qquad (5.7.23)$$

This shows that the quasi-neutral beam starts to expand under the effect of the electronic pressure forces. The beam density, corresponding to the first approximation (5.7.23), is

$$n_{(1)} = n_{00} \frac{\exp\left\{-\frac{y^2}{b^2}\mathrm{ch}^2\frac{qz}{v_z}\right\}}{\mathrm{ch}\left(\frac{qz}{v_z}\right)}. \qquad (5.7.24)$$

Knowing $n_{(1)}$, we can determine $f_{(2)}$, and so on. As indicated by the structure of the solution (5.7.23), the quantity

$$\frac{qz}{v_z} = \frac{c_T}{v_z};\ \ c_T = \sqrt{\frac{2kT}{M}}. \qquad (5.7.21)$$

is the dimensionless parameters of smallness.

5.8. Boltzmann–Davydov kinetic equation for electrons in weakly ionised plasma

In this section, attention is given to another modification of the collisional term of the Boltzmann kinetic equation, derived by B.I. Davydov in 1936[12]. The modification describes the electronic component in weakly ionised plasma (WIP) when the collisions of the electrons with each other and with the ions can be ignored. The WIP is the widely encountered form of plasma both in the laboratory (for example, glow discharge, various flames) and natural (ionosphere) conditions. In addition, the theory of WIP is relatively simple. In WIP, a highly nonequilibrium plasma forms in the presence of the electrical field (as in the glow discharge). Here, the electrons can acquire very high energies, regardless of partial collisions with the atoms, whereas the ions remain (in the presence of walls not too far away) almost completely cold, like neutral atoms. Evidently, this is explained by the low value of the mass ratio $m/M \leqslant 10^{-3}$–10^{-5}, i.e. by the fact that at $\varepsilon < e\phi_*$, where ϕ_* is the first potential of excitation of the heavy particles, and the collisions of the electrons with atoms are elastic and the electrons lose only a small amount of energy during these collisions, only $\delta\varepsilon_e \sim (m/M)\varepsilon_e$. At the same time, the ions colliding with the atoms, transfer the energy to the atoms on average during several collisions M_i/M_a, and the atoms transfer the energy to the walls.

Consequently, the atoms are regarded as stationary in the first approximation (at $m/M = 0$), and the entire kinetics is reduced only to the electron kinetics. The appropriate kinetic equation is derived from the Boltzmann kinetic equation for the mixture of three gases, where one of the gases is regarded as electrons:

[12]See also [134] and [135].

$$\frac{Df_e}{Dt} = S_1[f_e, f_a] + S_2[f_e, f_e] + S_3[f_e, f_i]. \tag{5.8.1}$$

Ignoring, because of the low concentration, S_1 and S_2 and taking into account that the ions are stationary, and also (at $m/M = 0$) the constant modulus of the velocity of the electrons $|\mathbf{v}|$ in collisions with the atoms, we obtain the intermediate kinetic equation

$$\frac{Df_a}{Dt} = n_a v \int d\sigma' [f_a(\mathbf{v}') - f_a(\mathbf{v})]. \tag{5.8.2}$$

Here $n_a = \int f_a^{(v)} dv'$. Since the modulus of the velocity of the electrons does not change during a collision, and only the direction of the velocity changes, it is convenient to introduce the unit vectors of the directions of the particle velocity

$$\mathbf{\Omega} = \frac{\mathbf{v}}{|\mathbf{v}|}; \mathbf{\Omega}' = \frac{\mathbf{v}'}{|\mathbf{v}'|}. \tag{5.8.3}$$

The value $d\sigma$ can also be calculated in the same manner as the procedure used for calculating the identical quantity in paragraph 5.2.2. It should only be taken into account that instead of the relationship (5.2.8), linking the impact parameter ρ with the angle of deviation θ, here, as indicated by Fig. 5.8.1

$$\cos\left(\frac{\theta}{2}\right) = \frac{\rho}{a}, \tag{5.8.4}$$

where a is the radius of the atom which is assumed to be spherical. Consequently, we obtain

$$d\sigma = rhod\rho d\phi = d\phi \cos\left(\frac{\theta}{2}\right) a^2 \cdot \frac{1}{2} \sin\left(\frac{\theta}{2}\right) d\theta = \frac{a^2}{4} \sin\theta \, d\theta \, d\phi = \frac{a^2}{4} d\mathbf{\Omega}. \tag{5.8.5}$$

From this, we determine the total cross-section

$$\sigma = \pi a^2. \tag{5.8.6}$$

The final form of the required collisional term is as follows:

$$S_{ea}^{(0)} = n_a v \sigma \frac{1}{4\pi} \int [f_e(\mathbf{\Omega}') - f_e(\mathbf{\Omega})] d\mathbf{\Omega}. \tag{5.8.7}$$

Taking into account the terms $\sim m/M$, i.e. elastic transfer of energy from the electrons to the ions, leads to the addition to the right-hand part (5.8.2) of the integral

$$S_{ea}^{(1)} = \frac{m}{M} \nu_{\text{col}} \frac{1}{v^3} \int \left[(v')^4 f_e(\mathbf{v}') - v^4 f_e(\mathbf{v}) \right] d\Omega' dv'. \tag{5.8.8}$$

Thus, the kinetic equation for the electrons in weakly ionised plasma has the following form:

$$\frac{\partial f_e}{\partial t} + \mathbf{v} \frac{\partial f_e}{\partial \mathbf{x}} - \frac{e}{m} \left(\mathbf{E} + \frac{1}{c} [\mathbf{v}, \mathbf{H}] \right) \frac{\partial f_e}{\partial \mathbf{v}} = S_{ea}^{(0)} + S_{ea}^{(1)} - Q(v). \tag{5.8.9}$$

Here $Q(v)$ is the term describing the inelastic processes (excitation and ionisation).

The distribution function of electrons in weakly ionised plasma at E = const

We present the results of calculations, using (5.8.9), of the electron distribution function in a uniform weakly ionised plasma in the presence of a constant electrical field.

If it is assumed that the free path length is constant, then

$$f_0 = C \exp\left\{ -\frac{3m}{M} \frac{\varepsilon^2}{\varepsilon_0^2} \right\}. \tag{5.8.10}$$

Here $\varepsilon_0 = cE\lambda$ is the energy collected by the electron in the free path. Constant C is determined by the processes of ionisation and annihilation of the particles.

We determine the scale ε_0 for a gas-discharge tube with Ar at p = 1 torr. In this case, $\lambda \sim 0.02$ cm and if it is taken into account that E = 1 V/cm, the characteristic electron energy is:

$$\varepsilon_* = \varepsilon_0 \sqrt{\frac{M}{3m}} \approx 3 \text{ eV}. \tag{5.8.11}$$

This value is close to $T_e \sim 2$ eV which is obtained in these devices.

6

Plasma processes with transformation of particles and radiation

6.1. Introduction

In previous chapters, plasma was regarded as a set of point electrons and ions, moving in the electromagnetic field. However, the actual situation is far away from such a simple scheme. This is explained by the presence in heavy particles, in addition to three[1] external (special) degrees of freedom, of an even larger number of internal degrees of freedom, associated with the electron shells (the only exception is the hydrogen ions) and nuclei. The nuclei will be discussed in section 10.5–6, and here we consider only the electronic shells and discuss indirectly the oscillations of the nuclei in the molecules. The 'untangling' of the internal degrees of freedom (transformation of particles[2]), i.e. as a result of the increase of the energy which the particles can have in the plasma. At the same time, in classic gas dynamics (kinetics) the particles remain unchanged because the energy of the particles is on the level $\lesssim$0.1 eV ~ 1000 K. The electron energy required for the single ionisation of the neutral atoms (ions) is ~5–15 eV. A slightly larger amount of energy is required for the formation of multi-charged ions (Fig. 10.6.1)

Evidently, a mixture of neutral particles, ions of different charge Z and electrons forms in the ionisation process. The identical situation is also recorded in the vicinity of energy-absorbing walls of high-temperature thermonuclear reactors, for example, tokamaks. This takes place taking into account that the energy in the centre of these systems reaches $\gtrsim$10 keV $\approx 10^8$K, resulting in the start of the reactions of

[1]If the particles had no spherical symmetry, the number of degrees of freedom can be 5 or 6

[2]The 'transformation' of the heavy particles refers to the processes in which their internal degrees of freedom take part.

merger of light nuclei (section 10.5 in this book). Examples of plasma formations with incompletely ionised plasma are the widely used glow and arc discharges which will be discussed in detail in section 6.10.

A suitable cosmic example will be considered. Our Sun, which can be regarded as a plasma formation, has the surface temperature of ~5800 K, whereas the corresponding degree of ionisation of hydrogen is only ≈0.1%. In the depth of the Sun, the matter is completely ionised.

Two parameters – Z_μ – the number of the heavy particle in the periodic table of elements, and Z – the degree of ionisation of the particle, do not exhaust the list of characteristics of the particle in the plasma. Actually, at $Z < Z_\mu$ the heavy particle retains part of the electrons and, consequently, there is a very large number of methods of distribution of these particles on the individual levels. These levels form another (and, possibly, several) degree of freedom which is described by the symbol ψ. This shows that in the ideal case when describing kinetically the dynamics of the actual plasma, each value of the complex $\alpha \equiv (Z_\mu, Z, \psi)$ should correspond to its own distribution function in the six-dimensional phase space

$$f_\alpha = f_\alpha(\mathbf{x}, \mathbf{v}, t),$$

or the generalised distribution function

$$F = F(\mathbf{x}, \mathbf{v}, \alpha, t)$$

in the generalised nine-dimensional phase space $\theta = (\boldsymbol{x}, \mathbf{v}_\alpha, \alpha)$, where α are discrete coordinates in a general case.

However, by introducing θ and F we have not yet completely formulated the problem of realistic description of plasma. Transitions between levels can take place as a result of different reasons: spontaneously, with the emission of a photon (if the initial energy of the level is $\varepsilon_m > \varepsilon_n$), as a result of collisions of particles with each other or absorption of photons. The transitions between the levels, accompanied by radiation, generate a very large number of types of photons with energy

$$\hbar\omega_{mn} = \varepsilon_m - \varepsilon_n$$

where $\hbar$ is the Planck constant, ω is angular frequency. The relative role of radiation in plasma dynamics can greatly differ. In many cases, for example, in powerful discharges, radiation removes from plasma more than 80% of supplied energy. Even in classic discharges the 'energy value of the ion'[3] ϵ_{ion} is usually on the level of three 'ionisation potentials'

$$\epsilon_{\text{ion}} \sim 3I.$$

[3]For more details see section 6.6

According to the traditional definition, the ionisation potential I is usually the ionisation energy expressed in eV. The difference between I and ϵ_{ion} is explained by the fact that collisions with the electron having the energy $\varepsilon > 1$ often lead to atom excitation instead of ionisation, and this atom then gets rid of of the 'excess energy' by radiation. This energy is lost in the system if the system is 'transparent' in the given frequency range. However, the photon can be absorbed in dense and relatively cold plasma, in particular, it can excite another atom, and so on, and the value of the ion ϵ_{ion} in these conditions can approach I.

The above considerations show that it is necessary to describe the 'transfer of radiation' of every type of photon taking into account the probability of emission (q_ω) and absorption ($\varkappa_\omega$) at the given (t,**x**).

It is evident that this requires the equation of the photon kinetics in the six-dimensional space (**x**,$\varkappa$), where $\varkappa$ is the wave vector. This equation has the following form:

$$\frac{dI}{dt} + c\,\mathrm{div}\,\mathbf{\Omega}\, I = q_\omega - \varkappa_\omega I, \tag{6.1.1}$$

where I (**x**,$\varkappa$, t) is the intensity of the light flux with frequency $\omega = c|\varkappa|$ in the direction $\mathbf{\Omega} = \varkappa/|\varkappa|$, c is the speed of light. The refractive index of plasma for the photons is usually equal to 1.

Considerable problems in calculating the dynamics of radiating plasma are associated not only with the large number of functions to be determined. The presence of a large number of discrete levels in the atoms, molecules and ions transform the functions q_ω, $\varkappa_\omega$ to 'ridges' of the very complex type (Fig. 6.11). However, this is not enough. In the actual conditions, the spectral lines 'widen' as a result of different mechanisms: the Doppler effect (resulting from the movement of the radiating particles), the Stark effect (as a result of the effect of electrical, in particular, fields between particles), etc. Thus, these functions q_ω, $\varkappa_\omega$, depend not only on the structure of the electronic shells of the particles in the ideal conditions but also on the random movement of the particles and the macroparameters of dynamics of the flow in which they take part, i.e. on the velocity of the flow, temperature and density, and also on the electromagnetic field.

The kinetic equations for particles in the presence of transformation are also found more complicated. They now have the form

$$\frac{Df_\alpha}{Dt} = S_\alpha + N_\alpha + R_\alpha \tag{6.1.2a}$$

here Df_α/Dt is the substantial derivative in the phase space, S_α describes the elastic interaction, N_α are the inelastic transformation collisions of

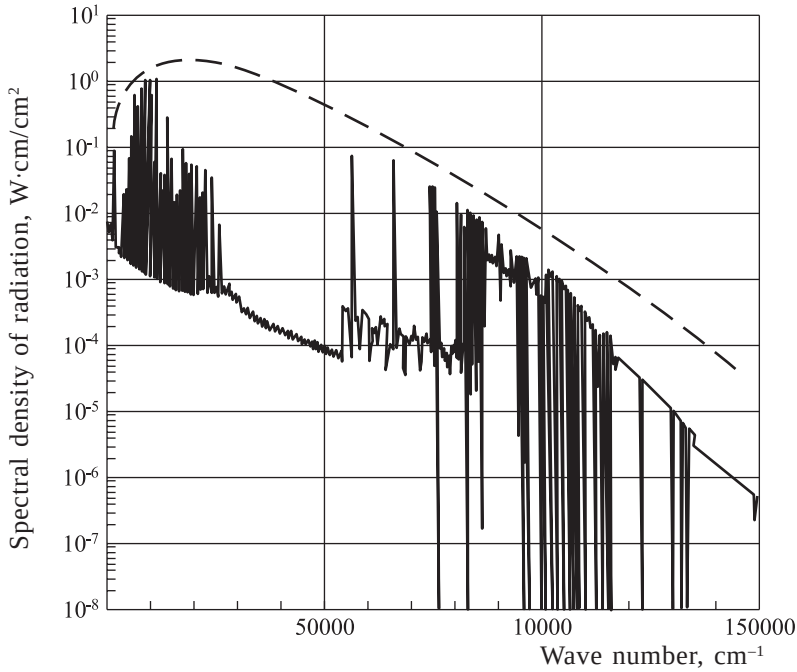


Fig. 6.1.1. Spectrum of the radiation flux to the surface of a space system at 500,000 points with respect to the wave number. The spectral intensity of the radiation of the absolutely blackbody for the same surface temperature is given for comparison.

the particles, R_α is the interaction of the particles with radiation.

In transition to the hydrodynamic description of the dynamics of the transforming plasma, the system of equations also becomes multicomponent with respect to α. In particular, the continuity equation has the following form

$$\frac{dn_\alpha}{dt} + \operatorname{div} n_\alpha \mathbf{v}_\alpha = \langle N_\alpha \rangle_f + \langle R_\alpha \rangle_f , \qquad (6.1.2b)$$

where $\langle ... \rangle_f$ is averaging with respect to f_α.

Thus, the transformation greatly complicates the theory and it is therefore not surprising that in this area there is a relatively small number of accurately solved problems taking into account many thousands of transitions, mostly in one-dimensional cases, and also the matter is restricted to relatively rough approximates. In the second half of this chapter attention will be given to the appropriate examples.

Naturally, the Maxwell equations should be added to (6.1.1) and (6.1.2a). The transition to investigating the dynamics of transforming plasma, as described previously, introducing a set of distribution functions $f_\alpha(\mathbf{x},\mathbf{v},t)$ instead of a single distribution function $f(\mathbf{x},\mathbf{v},t)$ – can

be used only for relatively low-density plasma when the mean spacing between the particles

$$r_0 \sim \frac{1}{\sqrt[3]{n}} \gg a,$$

is considerably greater than the characteristic dimensions of the atom a or residual electronic shells of the ions. In this case, we have two generally only slightly connected systems: natural plasma (classic plasma) and the microworlds of heavy incompletely ionised particles. In the final analysis, their relationship is characterised by some 'integral' characteristics, such as the excitation section, ionisation, the probability of spontaneous transition, etc.

Of course, when the distance between the particles is $d \sim$ a, the situation greatly changes. 'Mixing' of plasma and atomic electrons takes place in this case, leading in particular to the formation of clusters of different types. However, as stressed in the introduction, the physics of dense plasma is outside the framework of this book (see, for example, [18, 19].

The above considerations show that the world of the phenomena, associated with the transformation of electronic shells is virtually boundless, despite the fact that it is limited by the relatively low electronic temperature. Region II in Fig. I. 2.1. has been studied more successively up to now. This region includes all classic discharges, gas lasers, plasma chemistry and many other (chapter 10). One can hardly doubt that the region of 'low-temperature plasma' will bring many more surprises[4].

6.2. Velocity of transformation processes [137]

The dynamics of the internal degrees of freedom is described by quantum mechanics which will not be discussed here and we simply present the required data. Many of these data have been taken from the handbook [47] and are found very frequently in appendix E.

6.2.1. Energy levels

As indicated by quantum mechanics [137], the state of the electron in the atom (and, correspondingly, the levels of the atom energy) are determined by the set of four quantum numbers; n, l, m, s, where n is the main

[4]In most cases, it is necessary to investigate the processes of transformation in the systems with low-temperature plasma which is discussed in one of the volumes of the Encyclopedia of low-temperature plasma [40]. These volumes discuss in detail a wide range of such systems, including those described in the present chapter.

quantum number, l is the orbital quantum number, m is the magnetic quantum number, and s is the spin. The so-called degeneration, i.e. independence of the energy level on any quantum number, is encountered in a number of cases. The energy levels can be determined quite easily for the hydrogen atom or hydrogen-like systems, i.e. systems, formed by a nucleus with the charge Ze and a single electron. In this case, the energy of the level depends only on the main quantum number n (if relativistic effects are ignored) and is described by the formula

$$E_n = \frac{-13.6Z^2}{n^2}\ \text{eV}, \tag{6.2.1}$$

where Z is the charge of the nucleus, and the energy is measured from the ionisation boundary (i.e. negative energies correspond to bonded states, positive energies to the continuous spectrum).

There is no accurate analytical expression for the energy of the level for the atoms with two or more electrons. At present, the energy levels of many atoms (ions) have been determined by experiments and there is a large number of literature data concerned with the distribution of the levels [see, for example [47], pages 794–837]. Figure 6.2.1a shows the main levels and transitions for a lithium atom.

At relatively low temperatures (kT_e <1 eV) the controlling role in the systems in which we are interested can be played not only by the atoms and also molecules and molecular ions. The number of energy levels of the molecule is considerably larger than that of the atoms. This is associated with the fact that in addition to electron transitions, vibrational and rotational movement of the nuclei is also possible in the molecule (Fig. 6.2.1b). Taking into account the existence of the small parameter m/M (m is the electron mass, M is the nucleus mass) in the zero approximation, the energy of the diatomic molecule can be described by the sum of the electronic, rotational and vibrational energies:

$$E = U_e + \hbar\omega_e\left(n+\frac{1}{2}\right) + \frac{\hbar^2}{2I_M}\,j(j+1).$$

Here U_e is the electronic energy taking into account the energy of the Coulomb interaction of the nuclei; I_M is the inertia momentum of the molecule; ω_e is the frequency of oscillations; n = 0, 1, 2,... is the vibrational quantum number; j = 0, 1, 2,... is the rotational quantum number. Table 6.2.1 gives the parameter $\hbar\omega_e$ for a number of molecular ions.

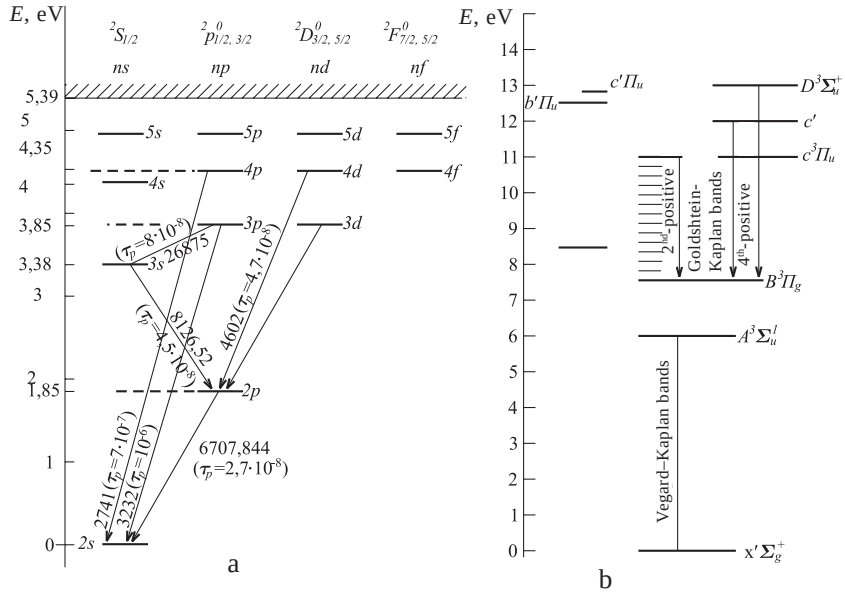


Fig. 6.2.1. The main levels and transitions for the lithium atom (wavelengths of the emitted waves (Å, Angstrom = 10^{-8} cm) and the lifetime (in seconds) of the excited states (a) and for the nitrogen molecule (b) are given.

Table 6.2.1.

Parameter	N_2	O_2	H_2	V_2	N_2^+	O_2^+
$\hbar\omega_e$, eV	0.29	0.19	0.54	0.09	0.27	0.23

6.2.2. Elastic scattering of electrons on atoms

The process of elastic scattering of the electrons on the atoms is described quite satisfactory by the existing theory. Figure 6.2.2 shows the typical dependences of the elastic scattering cross-section of the electrons on different atoms on energy. The so-caller Ramsauer is effect is detected in the scattering of the electrons with a low energy on some atoms (argon, krypton, xenon), i.e. the existence of a deep minimum in the scattering cross-section associated with a number of fine quantum effects. It should be mentioned that the minimum of the scattering cross-section of the atoms with the spin different from 0 is not so distinctive as in scattering of the electrons of the atoms with the zero spin (Ar, Xe, Kr).

The elastic scattering section in the region of high energies (ε>30 eV) of the electrons, when they penetrate into the electronic show, can be estimated using the Born approximation equation:

$$\sigma = 6\cdot10^{-15}\frac{Z^{4/3}}{\varepsilon}\text{cm}^2, \qquad (6.2.2)$$

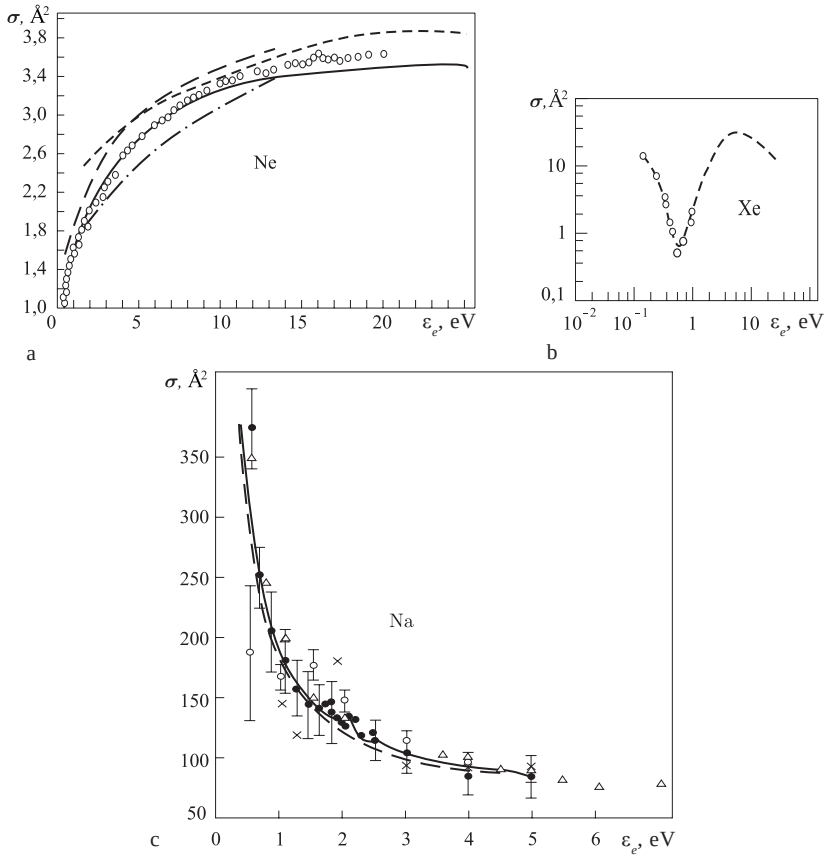


Fig. 6.2.2. Total scattering cross section of the electron on different atoms: Ne (a), Xe (b), Na (c).

where ε is the electron energy, eV; Z is the charge of the atom nucleus. The elastic scattering of low-energy electrons ($\varepsilon \leqslant 30$ eV) of the ions is efficiently described by the Rutherford equation, discussed in chapter 5.

The calculation of the elastic scattering of the ions of the neutral atoms and elastic collisions of the atoms gives the values of $\sim 10^{-15}$ cm^2 for the cross-sections (which are often referred to as the collision cross-sections). Thus, according to a number of studies, the collision cross-sections Ar–Ar$^+$ depend only slightly on velocity, and at $T = 10^3$ K, $\sigma \approx (0.6–0.9)\cdot 10^{-14}$ cm^2.

6.2.3. Excitation and ionisation of atoms by an electronic impact [138–140]

The processes of excitation and ionisation of the atoms during collisions of particles will now be investigated. Of greatest importance are the

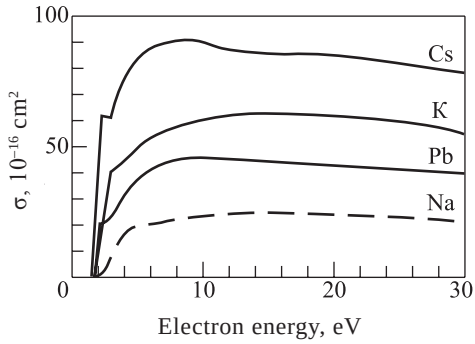


Fig. 6.2.3. Excitation cross sections of different atoms in relation to electron energy.

collisions with the electrons as the particles of plasma with the highest mobility. Evidently, the excitation cross sections of the atom in relation to the energy should have the form shown in Fig. 6.2.3.

Actually, the law of conservation of energy shows that the cross-section should convert to zero at $\varepsilon < \varepsilon_{threshold}$[5]. At high energies of the incident electrons the cross-section should also tend to 0 because the effective time of excitation decreases, thus the probability of excitation tends to 0. At high energies the asymptotics $\sigma \sim \ln \varepsilon/\varepsilon_*$, following from the Born approximation, should be taken into account. As regards the dependence of the cross-section on the energy in the vicinity of the threshold, it can be concluded with the accuracy sufficient for our purposes that in the vicinity of the threshold $\sigma \sim (\varepsilon - \varepsilon_{threshold})$. Figure 6.2.3 shows the excitation cross sections of the levels of a number of atoms by the electronic impact. No exact calculations of the cross sections of multi-electron atoms (ions) have been carried out. Therefore, all the equations presented here are semi–empirical. The excitation cross sections can be estimated using the Grizinskii equation

$$\sigma_{ik} = \frac{2.2 \cdot 10^{-14}}{(\Delta\varepsilon)^2}\left(\frac{2}{3}\frac{E_i}{\varepsilon} + \frac{\Delta E}{\varepsilon}\left(1 - \frac{\Delta E}{\varepsilon}\right) - \left(\frac{\Delta E}{\varepsilon}\right)^2\right), \quad \Delta\varepsilon = E_K - E_i \qquad (6.2.3)$$

Here the energy is measured in eV.

A partial case of excitation is ionisation in which the final state of the initially bonded electron is situated in the continuous spectrum. This situation in the calculations of the ionisation cross sections at the present time is the same as with the calculations of excitation cross sections. The general form of the dependence of the ionisation cross

[5]The electron energy in the hydrogen atom does not depend on the azimuthal number *l*. Therefore, this number can change without any energy loss, since in this case $\varepsilon_{threshold} = 0$.

section on energy is the same as in the case of the excitation cross section (Fig. 6.2.3). To obtain estimates, we present the Lotz equation which is often used because it is simple and relatively highly accurate. It has the form of a sum with $\sigma = \Sigma\sigma_k^{ion}$, where

$$\sigma_k^{ion} = 4.5 \cdot 10^{-14} a_k m_k \frac{\ln(\varepsilon / I_k)}{\varepsilon I_k}\left(1 - b_k \exp\left\{c_k + 1 - \frac{\varepsilon}{I_k}\right\}\right), \qquad (6.2.4a)$$

Here k is the number of the shell, m_k is the number of equivalent electrons in this shell, ε is the energy of the incident electron, I_k is the ionisation potential in eV, a_k, b_k, c_k are the constants (see [47] and Appendix G). Their values are in the range $0.6 < a_k < 1$, $0 < b_k < 1$, $0 < c_k < 0.5$. The values of the potentials of the first and second ionisation for a number of elements are presented in Table 6.2. Examples of the dissociation potentials I_D and the potentials of the first ionisation of the molecules are presented in Table 6.3.

Table 6.2

eV	Li	Na	Ar	Cs	Xe	Hg	Bi
I_1	5.39	5.14	15.70	3.89	12.13	10.44	7.29
I_2	75.6	47.3	27.6	23.4	21.2	18.73	16.6

Table 6.3

eV	H_2	N_2	O_2	H_2O	NH_3	S_2
I_D	4.47	7.3	5.08	5.2	4.8	4.35
I	15.42	15.6	12.2	12.6	10.3	8.3

Figure 6.2.4 shows examples of the experimental values of the ionisation cross sections of argon, xenon, lithium and sodium, and also the calculated data for hydrogen. For other ions and atoms see Appendix G.

Of some interest in a number of cases is the energy distribution of the electrons formed during ionisation. With the accuracy sufficient for our purposes, it can be assumed that the distribution function of a separated electron, normalised with respect to unity, is

$$f(x) \sim \frac{1}{(1+x)^2} \qquad (6.2.4b)$$

where $x = \varepsilon/I$, ε is the energy of the separated electron.

6.2.4. Ion recombination processes

Because of the reversibility of the equations of classic and quantum mechanics with respect to time, we must consider the 'principle of

detailed equilibrium', i.e. every 'direct' process has a reversed process. In particular, the ionisation of excitation of the atoms by the electronic impact corresponds to the recombination of electrons with the ions and the processes of de-excitation as a result of collisions with the electrons which are sometimes referred to as impacts of the second kind or superelastic collisions.

However, the special features of the recombination of the ions with electrons can be described more accurately using the following procedure. The recombination by the simple scheme

$$A^+ + e \rightarrow A \qquad (*)$$

cannot take place because of the laws of conservation of momentum and energy. This is indicated by conventional mechanics which describes the collisions of two 'adhering' spheres. In this inelastic collision, the kinetic energy of the system decreases by the value

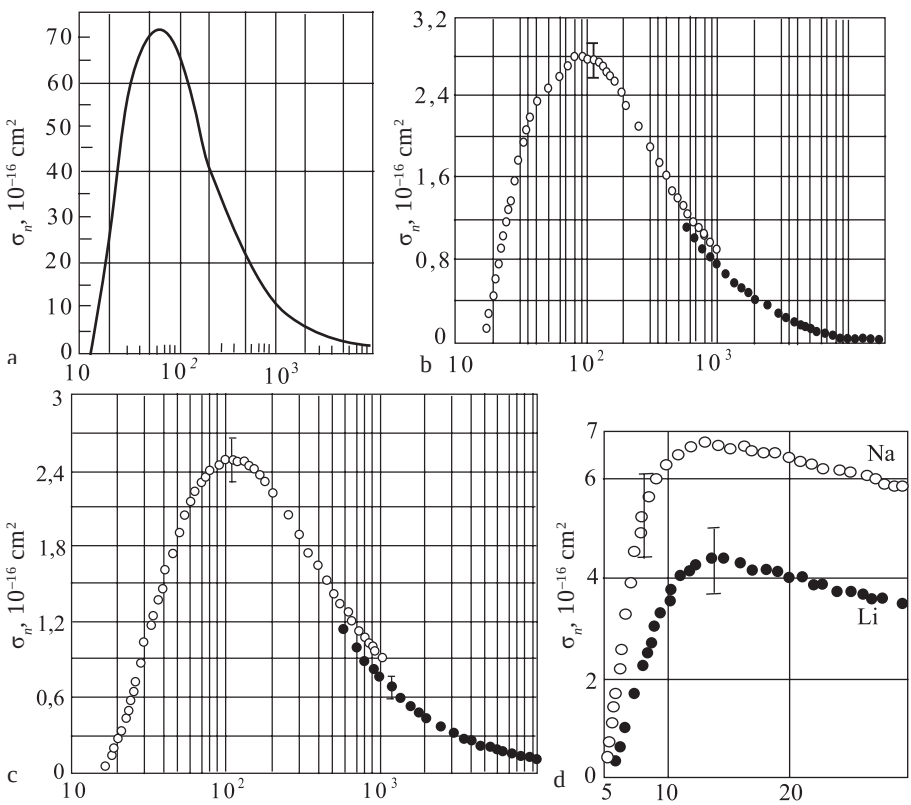


Fig. 6.2.4. Ionisation cross sections in relation to electron energy: hydrogen (a), argon (b), xenon (c), and lithium and sodium (d).

$$\Delta\varepsilon = \frac{M_1(\Delta \mathbf{V}_1)^2}{2} + \frac{M_1(\Delta \mathbf{V}_2)^2}{2},$$

which transforms to heat. Here $|\Delta V|$ is the difference of the velocity of the sphere prior to collision and the velocity of the centre of mass of the two spheres.

Similarly, in recombination the 'excess' energy can be released, i.e. the process should take place not according to the scheme (*) but according to the scheme

$$A^+ + e \rightarrow A + \mu(\Delta\varepsilon) \qquad (**)$$

Here $\mu(\Delta\varepsilon)$ is the energy carrier $\Delta\varepsilon$. If the role of μ is played by the photon flying away

$$A^+ + e \rightarrow A + (h\omega),$$

this process is referred to as 'photorecombination'. However, if the role of μ is played by the 'secondary' electron or another particle (ion, atom), this recombination is referred to as triple recombination or three-particle recombination. Usually, the secondary particle is represented by the electron. Therefore, the recombination scheme has the form

$$A^+ + 2e \rightarrow A + (e)_\mu.$$

However, the role of the μ-particle can also be played by 'internal' particles included in the composition of the recombined ion. Two processes of this type will be mentioned.

'*Dielectronic' recombination*' takes place in the case of many-electron ions. Here, under the effect of the electron falling on the ion, the energy excess is used for the excitation of the internal electrons of the ionic core

$$A^+ + e \rightarrow A_\mu^*$$

and this excitation is the removed either by the radiating photon or by the secondary electron colliding with A_μ^*.

Dissociative recombination

These processes take place in the molecular plasma in accordance with the scheme

$$(AB)^+ + e \rightarrow A + B.$$

Here, we confine ourselves to considering photorecombination and triple ($i + 2e$) recombination. The former dominates in the low-density plasma, the latter in dense plasma.

In order to describe more accurately the role of various transformation processes, the equation of the variation of the ion

concentration in the plasma taking into account the main process of ionisation by electronic impact and the previously mentioned types of recombination is written in the following form

$$\frac{dn_i}{dt} = n_a n_e \left\langle v\sigma_{\text{ion}} \right\rangle - n_i n_e \alpha_{\text{rad}} - n_i n_e^2 \alpha^{(ee)}. \tag{6.2.5}$$

Here n_i, n_e, n_a is the density of the ions, electrons and atoms, respectively; σ_{ion} is the ionisation cross section by electronic impact; $\alpha_{\text{rad}} \equiv \left\langle v\sigma_{\text{rad}} \right\rangle$ is the coefficient of paired (emissive) recombination; $\alpha^{(ee)}$ is the coefficient of triple recombination in the electron–ion–electron collisions. Here, we ignore the processes of photoionisation and also ionisation during mutual collision of the atoms.

The following equation can be presented for the coefficient of radiation recombination in rarefied plasma α_{rad}

$$\alpha_{\text{rad}} = 2{\cdot}10^{-13} \frac{Z^2 \phi(eI/kT)}{\sqrt{T_{\text{ev}}}}, \tag{6.2.6a}$$

where Z is the ion charge, $\phi(x)$ is the very slowly changing (in the range $x > 1$) function. At $x \approx 1$ $\phi \sim 1$ since for the estimates we can assume that $\phi = 1$. The characteristic time of variation of the electronic density as a result of radiation recombination is

$$\tau \equiv \frac{1}{\alpha_{\text{rad}} n_e} = \frac{5{\cdot}10^{12} \sqrt{T_{\text{ev}}}}{n_e Z^2}. \tag{6.2.6b}$$

In relatively dense and cold plasma with the recombination coefficient $\alpha^{(ee)}$ can be determined from the following equation

$$\alpha^{(ee)} = 8.75{\cdot}10^{-27} Z^3 n_e T_{\text{ev}}^{-9/2}. \tag{6.2.7}$$

Comparing (6.2.6) and (6.2.7) it can be seen that the increase of the plasma density increases the role of the triple recombination, since $\tau^{(ee)} \sim 1/n_e^2$.

The photorecombination time is equal to the three-particle recombination time when $\alpha_{\text{rad}} = n^*_e \alpha^{(ee)}$. Using the equations (6.2.6) and (6.2.7), we determine the characteristic density of the electrons

$$n_e^* = 2 \cdot 10^{13} \frac{T_{\text{ev}}^4}{Z}. \tag{6.2.8}$$

Correspondingly, at $n_e < n_e^*$ photorecombination is dominant, whereas three-particle recombination prevails at $n_e > n_e^*$ An important special feature of equation (6.2.8) is the weak dependence of n_e^* on the individual properties of the ions.

Figure 6.2.5 shows the regions of application of the equations (6.2.6), (6.2.7) and (6.2.8) for caesium and xenon plasma. Radiation recombination prevails in the plasma with the parameters situated below the appropriate curves, whereas three-particle recombination is dominant above the curve. In the cold plasma with a very low degree of ionisation recombination with participation of the ion, the neutral atoms and the electron can take place.

6.2.5. Negative ions

The negative ions have been attracting recently special attention of not only physicists but also applied scientists. The significant role of these ions in nature and technical devices has been unexpectedly explained.

The negative ions represent any atom or a molecule to which an electron 'sticks'

$$A + e \rightarrow A^{(-)}$$

Usually, the negative ions form atoms with unfilled electronic shells. Noble gases do not form negative ions whereas the binding energy $E^{M}(^{-})$ of the 'sticking' electron (amongst the atoms) is maximum in the halogens. In particular, for chlorine $E^{M}(^{-}) = 3.6$ eV. Table 6.4 shows the binding energies E^{M} $(^{-})$ for different elements.

The failure cross-section of the negative ions by the electronic impact is not large. For example, for the H^{-} ion, the cross-section

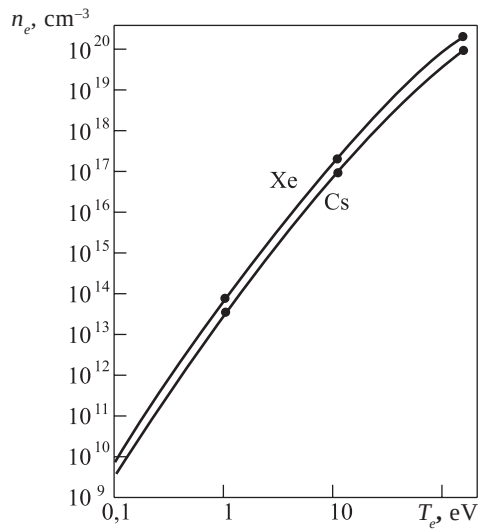


Fig. 6.2.5. Regions of applicability of the equations (6.2.6), (6.2.7) and (6.2.8) for caesium and xenon plasma.

Table 6.4.

Element	Li	Bi	Hg	S	Cs	O	C
Affinity energy, eV	0.6	0.7	1.54	2.1	0.23	1.46	1.25

at the maximum is $4.5 \cdot 10^{-15}$ cm^2 (Fig. 6.2.6). The negative ions form both as a result of volume and surface processes. The latter will be discussed in chapter 7, and it is only mentioned here that the formation of negative ions takes place by the same schemes as recombination, and their disappearance is similar to ionisation.

At present, the systems with the surface generation of negative ions are used most widely, whereas the volume processes operate most efficiently in the natural conditions.

Several examples will be mentioned. It has already been noted that the shining 'surface' of the Sun is a photosphere, with the temperature of ~5600 K and its radiation is close to Planck radiation with the same temperature. This is typical thermal radiation. The mechanism of generation of this radiation is associated with the emission of photons during photo-sticking of the electrons to the hydrogen atoms.

Another example. If the degree of ionisation of, for example, air, is not high and the air is relatively cold, then because of the relatively high binding energy $E^{M}(^{-})$ of the oxygen atoms ($E^{M}(^{-}) \approx 1.5$ eV) this plasma does not consist of ions or electrons and consists of positive and negative ions. Naturally, as a result of the relatively low mobility of the negative ions in comparison with the electrons, the conductivity of this plasma is low.

6.2.6. Excitation and ionisation of molecules by electronic impact

Excitation of the electronic levels of the molecule by the electronic impact is described by the same relationships as the excitation of atoms. The excitation cross section has a maximum in the vicinity of the threshold and reaches the value πa_0^2, where a_0 is the Bohr radius. Figure 6.2.7 shows the excitation cross sections of some electronic states in nitrogen. At relatively low temperatures ($T_e < 1$ eV) a significant role in molecular gases is also played by the excitation of vibrational and rotational levels. At low electron energies, the excitation cross section of the rotational levels of the molecule is described satisfactorily by the Stein equation

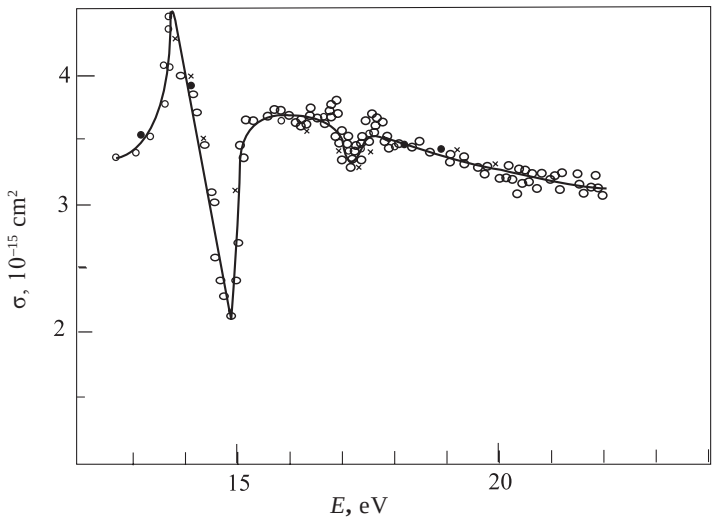


Fig. 6.2.6. Failure cross-section of the negative hydrogen ion in collision with the electron.

$$\sigma_{k,k+2} = \frac{8\pi a_0^2 Q^2}{15} \frac{(k+1)(k+2)}{(2k+1)(2k+3)} \left(1 - \frac{(2k+3)\hbar^2 / I_M}{\varepsilon}\right)^{1/2}, \qquad (6.2.9)$$

where ε is the energy of the incident electron; Q is the dimensionless quadrupolar moment of the molecule (for example for N_2 molecule Q = 1.10); I_M is the initial momentum of the molecule. At high electron velocities the results obtained using this equation are too low.

The excitation cross section by the electronic impact of the vibration levels of the molecules is usually considerably smaller than that of the rotational levels. However, in some cases larger cross sections can be obtained (~10^{-16} cm^2). This fact is associated with the formation of the unstable negative ion of the molecule and subsequent breakdown into the electron and the exciting molecule. Figure 6.2.8 shows the total excitation cross section of the vibration levels of the nitrogen molecule by the electronic impact. Figure 6.2.9 shows the ionisation cross sections of various molecules by the electronic impact.

For the molecules, in addition to the excitation and ionisation processes, the dissociation process is also very important. Dissociation may take place both under the effect of radiation and of electronic impact, as in collision with a heavy particle. Initially, we studied dissociation under the effect of electronic impact. This process can take place through several channels. Usually, the lowest energy threshold is typical of the area for the reaction which is referred to as dissociative capture: e + AB → A + B$^-$ at $E_{\text{threshold}}$ = 5–8 eV and σ = 10^{-18} cm^2. Impact

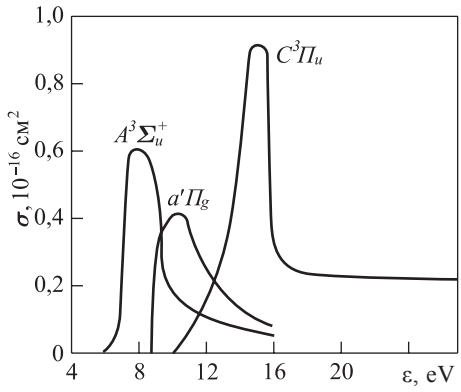


Fig. 6.2.7. Excitation cross sections of several electronic states of the nitrogen molecule.

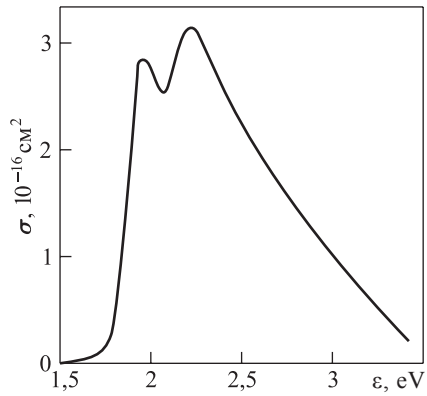


Fig. 6.2.8. Total excitation cross section of the vibration levels of the nitrogen molecules by the electronic impact.

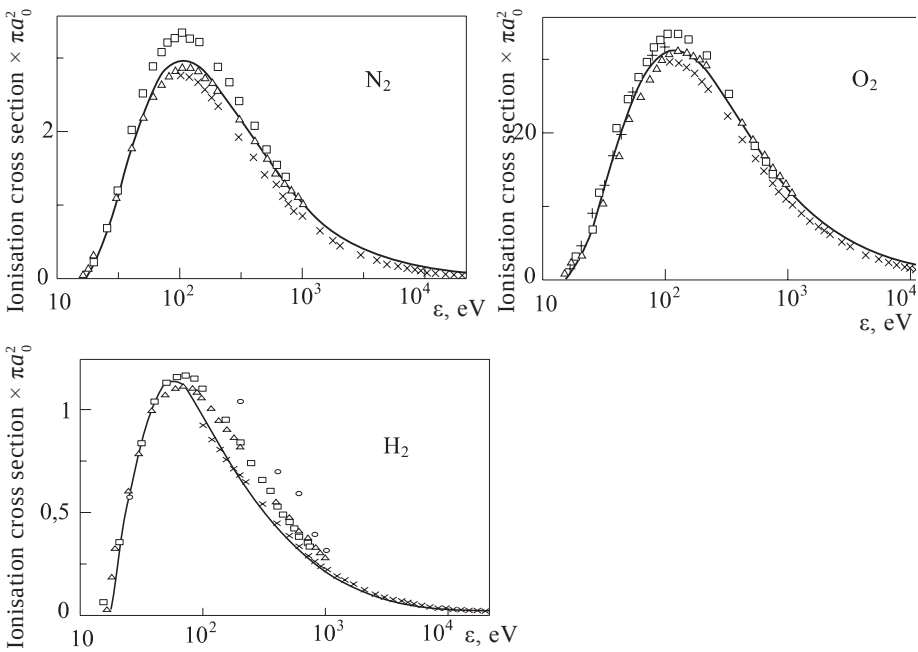


Fig. 6.2.9. Ionisation cross section of the nitrogen, oxygen and hydrogen molecules by electronic impact (solid lines – theory); a_0 is the Bohr radius.

dissociation takes place at a slightly higher energy: e + AB → A^- + B + e with $E_{\text{threshold}}$ = 5–10 eV and σ = 10^{-17} cm². When the energy is increased, the following reactions can take place: e + AB → A^+ + B^- + e with $E_{\text{threshold}}$ = 10^{-15} eV and σ = 10^{-18} cm² and e + AB → A^+ + B +2e with

$E_{\text{threshold}}$ = 20–25 eV and $\sigma = 10^{-16}$ cm^2. In principle, dissociation can take place at recombination of the molecular ion and the electron (this process is usually referred to as dissociative recombination) which take place by the scheme: $e + AB^+ \rightarrow A + B$. This process is important at low (room) temperatures.

6.2.7. Interaction of heavy particles

Elastic collisions

It is evident that the excitation and ionisation cross sections, generally speaking, should be smaller in this case in comparison with the collisions of the electrons with the atoms because the collisions of the heavy particles at $T < 1$ keV are adiabatic since the condition $a\Delta E/(\hbar v) \gg 1$ is fulfilled, where a is the characteristic particle size, ΔE is the characteristic transferred energy, v is the velocity of the heavy particles (the so-called Messi criterion). Actually, as can easily be seen, this condition is fulfilled even for hydrogen up to the energy of 1 keV. The experiments confirm this considerations.

Excitation and ionisation

To estimate the ionisation cross section in collision of the atom with the heavy particle, we can use the equation, proposed for the ionisation of a hydrogen atom situated on the level i:

$$\sigma_i^{(a)} = 4\pi a_0^2 \left(\frac{13.6}{E_i}\right)^2 \frac{m_a}{m_H} f_i \frac{2m_e}{m_e + m_a} \frac{w_i - 1}{\left(1 + \frac{2m_e}{m_a + m_e}(w_i - 1)\right)}. \qquad (6.2.10)$$

Here E_i is the energy of the level (measured from the ionisation boundary), eV; m_a is the mass of the incident atom, $w_i = \frac{(E_a - E_i)}{E_.}$; $E_a = \frac{m_a v_a^2}{2}$ is the energy of the incident atom in the system of the centre of inertia; m_H is the proton mass. Equation (6.2.10) contains a small multiplier m_e/m_a and, consequently, the value $\sigma_i^{(a)}$ is very small.

On the whole, it can be concluded that the role of the atomic processes may be comparable with the role of electronic processes only at very low degrees of ionisation (~0.1%) and for transitions with a low degree of energy transfer (~10^{-3}–10^{-2} eV). The appropriate cross sections are large (~10^{-15} cm^2) only in this case, for example, for the so-called Penning ionisation.

Recharging

The so-called recharging – electron exchange between the ion and the neutral is an important type of interaction of the heavy atomic particles.

Recharging can be of two types: resonant and non-resonant. Resonant recharging is the transition of the electron from the state of the given atom (ion) to the same state of the identical atom, for example $H(1S) + H^+ \rightarrow H^+ + H(1S)$. Non-resonant recharging is the process of transition of the electrons between different atomic particles, for example, the process $O^+(^4S) + H \rightarrow H^+ + O(^3P_2)$. It may be said that for resonant recharging the resonant defect (i.e. the difference of the energy terms of the system prior to and after collision) is equal to 0, whereas for non-resonant recharging it differs from zero. Evidently, the cross-section should be considerably larger in the case of resonant recharging. The characteristic values of resonant recharging in the region of the energies of the colliding particles (~1–10^3 eV) reach 10^{-14}–10^{-15} cm^2. For example, $\sigma(Li) = 2.7 \cdot 10^{-14}$ cm^2, $(Xe) = 7.2 \cdot 10^{-15}$ cm^2, $\sigma(Hg) = 1.0 \cdot 10^{-14}$ cm^2, $\sigma(Cs) = 3 \cdot 10^{-14}$ cm^2, and the dependence on velocity is relatively weak so that the value $\sigma_{res.rech} \approx 10^{-14}$ cm^2 can be used for approximate evaluation. The cross-section of non-resonant recharging at all energies of the colliding particles is considerably smaller.

Figure 6.2.10 shows the characteristic energy dependences of the cross-section of resonant and non-resonant recharging for a number of pairs.

6.3. Elementary radiation processes [141, 143, 144]

Radiation in plasma dynamic systems is caused mainly by electrons by the interaction with other atomic particles. As regards the radiation caused by the rotation of the electrons in the magnetic field (magnetically breaking' or 'synchrotron' radiation), these types of radiation in the energetics of the system are important only in the relativistic (for the electrons) region, in which $\varepsilon_e \sim mc^2$. It should also be mentioned that the collisions of the electrons with the electrons do not lead to dipole radiation. This is the common property of collisions of the particles with the same e/m.

If the initial and final states of the electron in the ion field are situated in the continuous spectrum, this transition is referred to as free–free (*f–f*-transitions). In principle, both the photon radiation (bremsstrahlung) and absorption (braking absorption or reversed bremsstrahlung) are possible. If the final state of the emitting electron is situated in the discrete spectrum, this transition is referred to as free-bonded (*f–b* transitions), and the process itself is referred to as

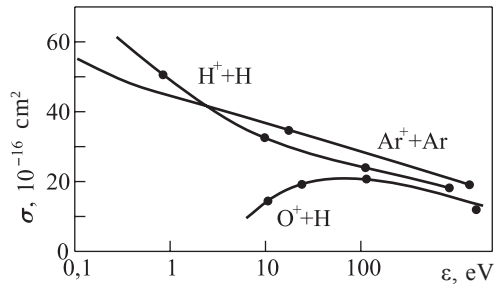


Fig. 6.2.10. Dependence of the recharging cross-section on particle energy.

photorecombination (or emissive recombination). In many cases, the transitions in the discrete spectrum (bonded–bonded or *b–b*-transitions) play a more significant role in many cases than the previously mentioned processes.

In principle, the radiation intensity can be calculated using quantum-mechanical theory, but in a number of cases this relates primarily to free–free transitions, i.e. bremsstrahlung, the calculations with acceptable accuracy can be carried out using classic schemes.

6.3.1. Linear radiation spectra

The radiation types include spontaneous radiation and forced radiation of the atoms (molecules), caused by the effect of fields already existing in the photon system. Therefore, there are three probabilities of transitions which are usually referred to as Einstein coefficients. They are linked by certain relationships. Let A_{ji} be the probability of transition of the atom from the upper *j*-level to the lower *i*-level as a result of spontaneous radiation; B_{ji} is the probability of transition for the forced radiation, and B_{ij} the probability of transition at photon absorption. Further, let the equilibrium gas be situated in a cavity penetrated by equilibrium radiation with density U_ω. Consequently, the number of absorption acts in 1 s in 1 cm^3 is determined by the equation

$$N_{ij} = n_i U_\omega B_{ij} \tag{6.3.1a}$$

where n_i is the density of the atoms on the lower-level, and the number of radiation acts

$$N_{ji} = n_j A_{ji} + n_j U_\omega B_{ji} \tag{6.3.1b}$$

In equilibrium, U_ω is expressed by the Planck equation:

$$U_{\omega}(T) = \frac{\hbar\omega^3}{\pi^2 c^3 \left(\exp\left(\frac{\hbar\omega}{kT}\right) - 1 \right)}. \qquad (6.3.2a)$$

In addition to this, in equilibrium with the density of the particles on the levels *i* and *j* are linked by the Boltzmann equation

$$n_j = n_i \frac{g_j}{g_i} \exp\left\{ -\frac{(E_j - E_i)}{kT} \right\} \qquad (6.3.2b)$$

and the intensities of the flows in the 'upward' and 'downward' directions are:

$$N_{ij} = N_{ji} \qquad (6.3.2c)$$

Substituting (6.3.2) into (6.3.1) we obtain

$$\frac{\hbar\omega^3}{\pi^2 c^3} \left(\exp\left\{\frac{\hbar\omega}{kT}\right\} B_{ij} g_i - B_{ji} g_j \right) = A_{ji} \left(\exp\left\{\frac{\hbar\,\omega}{kT}\right\} - 1 \right). \qquad (6.3.3)$$

For this relationship to be valid at any *j*, the Einstein equations should hold

$$A_{ji} = \left(\frac{\hbar\omega^3}{\pi^2 c^3}\right)\left(\frac{g_i}{g_j}\right) B_{ij} = \left(\frac{\hbar\omega^3}{\pi^2 c^3}\right) B_{ji}; \quad g_j B_{ji} = g_i B_{ij}. \qquad (6.3.4)$$

Here g_i is the statistical weight of the level *i*.

The transition probabilities are often replaced by a more formal quantity – the force of the oscillator at absorption *f*, which is linked with B_{ij} by the relationship (the oscillator force is convenient because it is a dimensionless quantity)

$$f_{ij} = \frac{m\hbar\,\omega B_{ij}}{\pi e^2} \qquad (6.3.5)$$

The oscillator force is linked with the square of the matrix element of the dipole moment of the atom taken between the final and initial states.

Since the exact analytical expression for the wave function is known only for the hydrogen atoms, the exact values of the oscillator forces (and, correspondingly, transition probabilities) are available only for the hydrogen atom.

Transition probabilities of other atoms can be calculated using approximate methods. Table 6.5 gives the oscillator forces for a number of transitions in Li (see Table 6.5) and Hg (Table 6.6), calculated by different methods.

Table 6.5.

Transition	$2^2P_{3/2}$ — $2^2S_{1/2}$	$3^2P_{3/2}$ — $2^2P_{1/2}$	$3^2S_{1/2}$ — $2^2P_{1/2}$	$3^2D_{5/2}$ — $2^2P_{3/2}$	$4^2D_{5/2}$ — $2^2P_{3/2}$	$4^2S_{1/2}$ — $2^2P_{1/2}$
λ, Å	6707	3232	8126	6103	4603	4971
f	0.498	0.003	0.11	0.582	0.109	0.0125

Table 6.6.

Transition	6^1S_0 — 6^1P_1	6^1S_0 — 6^3P_1	6^3P_2 — 7^3S_1	6^3P_0 — 7^3S_1	6^3P_1 — 6^3D_3	6^3P_1 — 6^3D_3	6^3P_0 — 6^3D_2
λ, Å	1849	2537	5461	4047	3650	3131	3126
f	1.2	0.03	0.16	0.21	0.3	0.2	0.1

6.3.2. Continuous radiation spectrum

In addition to linear radiation, continuous radiation is also often controlling. We discuss briefly the equations for the power of bremsstrahlung and recombination (Raman) radiation.

Bremsstrahlung

The spectral composition of the bremsstrahlung of the electron, flying around the ion, is described by very cumbersome equations. Therefore, we present only the equation for the radiation power of the unit volume of plasma with the Maxwell electron distribution:

$$Q^{\text{brem}} = 1.54 \cdot 10^{-23} n_e n_i Z^2 T_{\text{ev}}^{1/2} \quad \frac{\text{erg}}{\text{cm}^3\text{s}}. \tag{6.3.6}$$

Taking into account the relationship of the radiation and absorption processes, it may be shown that the absorption coefficient as a result of the reversed bremsstrahlung is

$$\varkappa_{\text{brem}} = 10^9 \frac{Z^2 n_e n_i}{T_e^{1/2} \omega^3} \text{cm}^{-1}. \tag{6.3.7}$$

Here n_e and n_i are presented in cm^{-3}, and T_e is in eV.

Raman radiation

This type of radiation is determined by the transition of the electron with velocity *v* from the continuous to discrete spectrum. The energy of the emitted quantum is evidently determined by the relationship $\hbar\omega = \frac{mv^2}{2} + \chi_n$, where χ_n is the binding energy of the electron on the level *n* on which recombination takes place. The problem of recombination and the reverse process – photoionisation, for the hydrogen atom can be solved quite accurately. However, as in bremsstrahlung, the resultant

equations are not very suitable for specific calculations because they are extremely long. A highly suitable approximation are quasi-classic equations, derived by Kramers using the correspondence principle:

$$\sigma_{\text{rec}}^{(n)} = 3.9\cdot10^{-20}\frac{Z^4}{n^3\varepsilon(\varepsilon+\chi_n)}, \text{cm}^2; \tag{6.3.8}$$

$$\sigma_{\text{photoion}}^{(n)} = 7.9\cdot10^{-18}\frac{n}{Z^2}\left(\frac{\omega_n}{\omega}\right), \text{ cm}^2. \tag{6.3.9}$$

Here $\sigma_{\text{rec}}^{(n)}$ is the recombination cross-section on the level n; $\sigma_{\text{photoion}}^{(n)}$ is the photoionisation cross-section from the level n; ε is the energy of the recombining electron, eV; $\hbar\omega_n = \chi_n$.

Equation (6.3.8) shows that the recombination cross-section rapidly decreases with increasing n since the main contribution to photorecombination is provided by the recombination to the ground level, and the characteristic values of the cross-section for the ion with $Z = 1$ are $\sim10^{-21}/\varepsilon$ cm^2. Knowing the equation for $\sigma_{\text{rec}}^{(n)}$ it is easy to find the total power of recombination radiation of the unit volume[6]. Actually, in the case of hydrogen-like plasma we have

$$Q_{\text{rec}} = n_e n_i \sum\left\langle v\sigma_{\text{rec}}^{(n)}\left(\varepsilon + \frac{13.6Z^2}{n^2}\right)\right\rangle. \tag{6.3.10}$$

The angle brackets indicate averaging with respect to the electron distribution function. For the Maxwell distribution, we obtain in the classic limit ($Ze^2/(\hbar v) \gg 1$)

$$Q_{\text{rec}} = 4.4\cdot10^{-24}\frac{Z^4 n_e n_i}{T_e^{1/2}}. \tag{6.3.11}$$

Here T_e is in eV, Q_{rec} in erg/(cm^3 s).

Knowing the expression for the photoionisation cross section, it is quite easy to calculate the coefficient of bonded-free absorption of the hydrogen-like gas. Taking into account the absorption of a quantum with frequency ω from the ground and excited levels, we evidently obtain

$$\varkappa_\omega = \sum_{n=n_0}\sigma_{\text{photoion}}^{(n)}(\omega)N_n, \tag{6.3.12a}$$

where N_n is the density of the atoms on the level n; n_0 is the smallest of the quantum numbers, satisfying the condition $\hbar\omega > \chi_n$. If the distribution of the atoms on the levels is described by the Boltzmann equation, then in the case of low energy ($\hbar\omega < \chi_n$) of the quanta

[6]It is assumed that plasma is optically thin, i.e. all the quanta leave the system.

$$\varkappa_\omega = 2.5\cdot10^{30}\frac{NT_e}{\omega^3}Z^2\exp\left\{-\frac{I-\hbar\omega}{kT_e}\right\}, \qquad (6.3.12b)$$

where N is the density of the atoms in the ground state, T_e is the electron temperature (eV), I is the ionisation potential (eV). For the coefficient of absorption of the photons with the energy higher than the ionisation potential

$$\varkappa_\omega = 5\cdot10^{13}\frac{NZ^2}{\omega^3}I. \qquad (6.3.12c)$$

Table 6.7 gives the recombination cross-sections for different levels of the caesium atom.

Table 6.7

Level	6S	6P	5D	7S	7P	8S
σ, 10^{22} cm^2	1.36	0.45	0.16	0.30	0.282	0.14

6.3.3. Mechanisms in broadening of spectral lines

The important characteristic of the spectrum line, in addition to the transition probability, is the profile of the line. Even in the ideal case of an isolated atom, the emitted line cannot be strictly monochromatic since the finiteness of the lifetime of the excited atom, according to the indeterminacy relationship, results in 'eblurring' of the level. The effect of thermal motion and electrical fields of other particles also causes line broadening. The main broadening mechanisms will now be investigated.

The natural width of the spectral line

Within the framework of classic electrondynamic, detailed examination of the problem of radiation attenuation of the oscillator leads to the following equation for the normalised $\left(\int_0^\infty P(\omega)d\omega = 1\right)$ Lorenz line profile:

$$P(\omega) = \frac{\gamma}{2\pi}\frac{1}{(\omega-\omega_0)^2+(\gamma/2)^2} \qquad (6.3.13)$$

where $\gamma = \frac{2e^2\omega_0^2}{3mc^3} = 1.7\cdot10^{-22}\omega_0^2$ is the natural width of the line with the central frequency ω_0; m is the electron mass. It can easily be seen that on the wavelength scale (used widely in practical spectroscopy), the natural width is independent of the wavelength and is equal to $\Delta\lambda_{nat}$ = $1.2\cdot10^{-4}$Å. Successive quantum mechanical investigations lead to the

same expression for $P(\omega)$ in which γ consists of the width of the initial and final stage, and γ should be replaced by Einstein coefficients for spontaneous transitions A_{ik}. Usually $A_{ik} \approx 10^8\ s^{-1}$. It should be mentioned here that the levels for which $A_{ik} \ll 10^8\ s^{-1}$ can exist since the lifetime is very long in comparison with the conventional duration of radiation transitions ($\sim 10^{-8}$ s). These levels are referred to as metastable and their natural width is very small. Usually, the metastable levels are the levels with a different spin (in contrast to spin of the ground state of the atom) since the most probable electrical dipole transitions are forbidden as a result of the sampling rule $\Delta S = 0$.

Doppler broadening of the spectral lines

This type of broadening is determined by the movement of the emitting particles. It is well known that the frequency of the oscillator whose velocity in the direction of the beam of vision is equal to v, is displaced by the value $\Delta\omega_D = \omega_0 v / c$ in accordance with the Doppler principle.

It can be seen that if f (v) is the velocity distribution function of the emitting particles, then

$$P(\omega) = \frac{\int f(\mathbf{v})\delta\left(\omega - \omega_0 - \frac{\omega_0}{c}\mathbf{\Omega}^0\mathbf{v}\right)d\mathbf{v}}{\int f(\mathbf{v})d\mathbf{v}}. \qquad (6.3.14)$$

here $\mathbf{\Omega}^0$ is the unit vector of direction of wave propagation. In particular, if the atoms are distribution according to Maxwell

$$f(v) = n\left(\frac{M}{2\pi kT}\right)^{3/2} \exp\left\{\frac{Mv^2}{2kT}\right\},$$

then

$$P(\omega) = \frac{1}{\omega_0}\sqrt{\frac{Mc^3}{2\pi kT}} \exp\left\{-\frac{Mc^2}{2kT}\frac{(\omega-\omega_0)^2}{\omega_0^2}\right\}. \qquad (6.3.15)$$

Equation (6.3.30) holds if the relationship $2\pi L \gg \lambda$ is fulfilled, where L is the free path of the emitting atom; λ is the wavelength. If the reverse conditions are fulfilled (this is possible in dense plasma or for very long wavelengths), we observe an interesting effect of the diffusion narrowing of the line. In this case, the line profile is of the dispersion type (i.e. it is defined by the equation of the type (6.3.13)) with the width considerably smaller than the width in the Doppler mechanism. For the optical wavelength range ($\sim 5\cdot 10^{-5}$ cm) the conditions $2\pi L \gg \lambda$ is equivalent to the condition $n < 10^{19}\ cm^{-3}$. The characteristic values of the Doppler width at $T \approx 1$ eV are as follows: for Li $\Delta\omega_D = 1.7\cdot 10^{-5}\ \omega_0$; for

Cs $\Delta\omega_D = 4 \cdot 10^{-6}\ \omega_0$. At $\omega_0 \approx 10^{15}\ s^{-1}$, $\Delta\omega_D$ (Li) $= 1.7 \cdot 10^{10}\ s^{-1}$, $\Delta\omega_D$(Cs)$= 4 \cdot 10^9\ s^{-1}$. On the scale of the wavelength $\Delta\lambda$(Li) = 0.34 Å, $\Delta\lambda$(Cs) = 0.08 Å. It can easily be seen that the Doppler width is considerably greater than the natural line width.

In the directional movement of plasma as an integral unit the the Doppler shift of the line takes place (consequently, the velocity of the atoms can be measured). For example, at an accelerating potential of ~100 V the Doppler shift of the line is 10 times greater than the previously mentioned values of the Doppler width, determined by random movement with T ~1 eV.

Line broadening in interaction with charged particles

Prior to discussing the important mechanism of line broadening in interaction of the emitting atom with the charged particles, it is natural to discuss the simplest case of broadening of the spectral line under the effect of external fields – broadening in a static electrical field. The exact quantum mechanical solution of the problem of the shift of the levels under the effect of static electrical field can be obtained by simple means only for the hydrogen atom (or a hydrogen-like ion). In this case, as a result of random Coulomb degeneration of the levels of with respect to l a linear Stark effect takes place, i.e. the shift of the level is proportional to the strength of the applied field, and the splitting of the lines (i.e. the distance on the frequency scale between the split components situated at the maximum distance) is defined by the expression

$$\Delta\omega = \left(\frac{3}{2}\right) n(n-1)\frac{\hbar E}{me}. \qquad (6.3.16a)$$

For all other atoms we obtain the quadratic dependence of the shift of the level on E, i.e. the so-called quadratic Stark effect.

The broadening of the atom (ion) levels in plasma is determined by the effect of many particles. At a high degree of ionisation it is interesting to investigate the interaction of the emitting particle with the electrons and ions. The electrical fields of the perturbing particles result in the displacement of the levels of the emitting particles and, consequently, in the displacement (and splitting) of the spectral lines. The theory of line broadening by the charged particles is being developed at the present time and here only the main assumptions and some results are discussed.

In particular, it should be mentioned that the strict and complete solution of the problem in the most general formulation is not yet available. Therefore, a number of assumptions must be made in

analysis. The most widely used assumptions are the so-called impact approximation and quasi-static approximation (only line broadening of the radiation of non-hydrogen like systems will be discussed). The boundaries of the region of applicability for each of these approximations can be estimated as follows using a simple procedure.

The presence of the perturbing particles situated at the distance R from the emitting atom (ion), leads to the following frequency shifts:

$$\Delta\omega(R) = \frac{C_m}{R^m} \tag{6.3.16b}$$

where m is some integer, C_m is a constant. For the quadratic Stark effect when the shift of the level is proportional to the square of the strength of the perturbing field, $m = 4$. Consequently, C_4 is referred to as the constant of the quadratic Stark effect. In the impact approximation, it is assumed that during the flight of the perturbing particle or some distance $\rho \leq \rho_0$ (ρ_0 is the Weisskopf radius) the emission phase failure takes place, i.e. the coherence of the oscillations is disrupted. The following expression can be obtained for 0:

$$\rho_0 = \left(\frac{\pi}{2}\frac{C_4}{v}\right)^{1/3} \tag{6.3.17}$$

where v is the velocity of the incident particles.

For the impact approximation to be used, the value ρ_0 should be smaller in comparison with the inter-particle distance which is the order of $n^{-1/3}$, i.e.

$$n\rho_0^3 << 1$$

where

$$n\left(\frac{\pi}{2}\frac{C_4}{v}\right) << 1 \tag{6.3.18}$$

Under this condition, a large part of the intensive lines concentrated in the region described by the impact theory. Since in most cases C_4 ~10^{-15}–10^{-12} cm^4/s, equation (6.3.18) shows that the impact theory can be used for many systems. With the framework of this model we obtain the Lorenz dispersion line profile:

$$P(\omega) = \frac{1}{\pi}\frac{1}{\tau_0}\frac{1}{(\omega-\omega_0)^2 + \frac{1}{\tau_0^2}} \tag{6.3.19}$$

where $1/\tau_0$ is the frequency of collisions with the perturbing particles. The width of the distribution (the distance between the symmetric points ω_1, ω_2 for which $P(\omega) = 1/2P_{max}$) is equal to $\gamma \equiv 2/\tau_0$. The concept of

the line width $\lambda \equiv c\tau_0$ is used in most cases. For γ the impact adiabatic theory (i.e. it is assumed that perturbation does not cause any transition between the different states of the atom) leads to the following equation

$$\gamma = 11.4nC_4^{2/3}v^{1/3}. \tag{6.3.20}$$

Here n, v relates to the perturbing particles.

The following estimate can be used for C_4

$$C_4 = 10^{-14}\frac{f}{(\Delta E)^2}\text{cm}^4/\text{s}. \tag{6.3.21}$$

Here ΔE is the distance between the levels, eV, f is the force of the oscillator of the given transition.

Broadening under the effect of neutral atoms

Finally, another type of broadening will be discussed – broadening of the lines as a result of interaction with neutral particles. Two mechanisms can operate here: resonant interaction of the emitting atom with the same atoms capable of absorbing the emitted quantum, and van der Waals interaction if the emitting and perturbing atoms differ. Both these mechanisms can provide a contribution comparable, for example, with broadening in the quadratic Stark effect if $n_e/n_0 \leq 10^{-3}$–10^{-2} (n_0 is the density of the neutral atoms).

Combined effect of different broadening mechanisms

Previously we investigated the broadening mechanisms of the spectral lines separately. However, in practice it is often necessary to consider the conditions in which two or more mechanisms operate at the same time.

If there are several broadening mechanisms of different physical nature and it can be assumed that the mechanisms act independently of each other, the resultant profile of the line $P_{1+2}(\omega-\omega_0)$ can be determined from the equation

$$P_{1+2}(\omega-\omega_0) = \int P_1(\omega-\omega')P_2(\omega'-\omega_0)d\omega' \tag{6.3.22}$$

6.3.4. Absorption coefficient in spectral lines

The processes of radiation and absorption of the photons are mutually reversible and, therefore, the factors responsible for the broadening of the spectral lines, have also a significant effect on the absorption coefficient of photons $\varkappa_\omega = 1/L_\omega$, where L_ω is the free path of the photon at the given frequency ω. Propagating in the medium, the photons can

transfer their energy to the electrons and, in particular, excite the atoms and ions. Consequently, the photons are absorbed 'in lines'. Using the relationship between the Einstein coefficients, we can easily obtain the expression for the absorption coefficient of radiation with frequency ω close to the frequency of the spectral line $\omega_0 = (E_k - E_i)/\hbar$:

$$\varkappa_\omega = n_i B_{ik} \frac{\hbar\omega_0}{c} P(\omega). \tag{6.3.23}$$

Here n_i is the concentration of the particles located on the lower level E_i; B_{ik} is the second Einstein coefficient. Using the concept of the oscillator force (6.2.4b), equation (6.3.23) can be presented in the following form

$$\varkappa_\omega = n_i f_{ik} \frac{1}{mc} \pi e^2 P(\omega). \tag{6.3.24}$$

Equation (6.3.24) shows that the maximum of $\varkappa_\omega$ is recorded at the frequency at which $P(\omega) = \max$. In most cases, the maximum P is found at the central frequency ω_0. If δω denotes the characteristic width of the spectral line then, taking into account the normalisation $\left(\int_0^\infty P(\omega) d\omega = 1\right)$, it can be assumed that $P(0) \sim 1/\delta\omega$. Consequently, the following equation can be used to estimate $\varkappa_\omega(\omega_0)$ – the absorption coefficient at the central frequency:

$$\varkappa_\omega(\omega_0) \approx n_i f_{ik} \frac{3\lambda_0^2}{4\pi^2} \equiv n_i i_{ik} \frac{\pi e^2}{mc} \frac{1}{\delta\omega}. \tag{6.3.25}$$

This shows that if $\delta\omega \to 0$, then $\varkappa_\omega \to \infty$. The smallest possible line width is determined by natural attenuation. Using (6.3.13), we obtain $P(0) = \dfrac{3mc^3}{\pi e^2 \omega_0^2}$ and, therefore:

$$\varkappa_\omega^{\text{ECT}}(\omega_0) = n_i f_{ik} \frac{3\lambda_0^2}{4\pi^2} \equiv n_i \sigma_{ik}^{\text{ECT}}. \tag{6.3.26}$$

Here $\lambda_0 = 2\pi c/\omega_0$ is the wavelength, and $\sigma_{ik}^{\text{ECT}} \equiv f_{ik} \dfrac{3\lambda_0^2}{(4\pi)^2}$ is the absorption cross-section of the photons with the central frequency ω_0. Substituting into (6.3.25) $f_{1k}=1$, $\lambda_0 = 5000$ Å, we obtain the estimate of the photon free path:

$$L_\omega = \frac{1}{\varkappa_\omega} \approx \frac{10^{10}}{2n_i}.$$

Thus, if the broadening is natural, it is sufficient to consider the atom concentration n of only 10^{10} cm^{-3} in order to have the photon path <1 cm. In Doppler broadening

$$\delta\omega = v_T \frac{\omega}{c} \tag{6.3.27}$$

where v_T is the mean value of the random velocity. Assuming that v_T = 10^5 cm/s, λ = 5000 Å, f_{ik} = 1, using (6.3.25) we obtain $\sim 2\cdot 10^{-12} n_i$, cm^{-1}.

These estimates show that the radiation, corresponding to the transitions to the ground level, is locked in the ionisation zones if the concentration of the neutral particles in these zones is $\geq 10^{30}$ in cm^{-3}.

Using the equations (6.3.15) and (6.3.25), it can be seen that the Stark broadening starts to approach Doppler broadening only at the densities of $\sim 10^{15}$–10^{16} cm^{-3} and then becomes dominant.

The systematic description of the behaviour of radiation in the conditions in which absorption plays a significant role can be carried out only by means of the kinetic equation for photons or its classic form – the radiation transition equation. This equation will now be discussed.

6.4. Radiation transition equation (photon kinetics) [140]

6.4.1. Formulation of the transfer equation

After discussing the main 'elementary' process of interaction of particles with each other and with the photons, we now transfer to describing the dynamics of a set of particles, and photons. We start with the description of the photon kinetics characterised by the simplest logic.

In section 3.5 we investigated the dispersion equation for the waves in two-component plasma and it was shown that at frequencies

$$\omega \gg \omega_0,$$

where ω_0 is plasma frequency, the waves cease to 'feel' the plasma and are governed by the equation

$$\omega^2 = c^2 \varkappa^2.$$

It is evident that this limiting transfer is general and does not depend on the specific plasma model. With increasing ω the photon energy increases and starting at some level which depends on the plasma composition, the photons start to play a significant role in the process of the particle transformation and also in energy transfer. To understand the scales of the parameters, it should be mentioned that the wavelength in the yellow doublet of sodium is

$$\lambda_{Na} \approx 600 \text{ nm}.$$

This corresponds to the frequency and energy

$$\omega_{Na} = 3\cdot 10^{15} s^{-1}, \quad \varepsilon_{Na} = \hbar \approx 2 \text{eV}$$

It should be mentioned that the Langmuir frequency ω_0 is equal to λ_{Na} at the density

$$n_{Na} \approx \frac{1}{3} 10^{22} \text{cm}^{-3}.$$

Thus, there is a wide range of values between the density range in which we are interested and the density of Na.

Since it is often necessary to take into account the electromagnetic radiation of the 'transformational' level of the particles, it is necessary to investigate the problem of the simplest description of these processes. Because of the relatively small wavelength of these types of radiation, the beam approximation, i.e. in fact, the photon kinetics equation, is used in most cases instead of the wave equation.

To describe the photon kinetics it is necessary, as in the case of the particles, to introduce six-dimensional phase space $\Gamma_\gamma = (\mathbf{x}, \mathbf{k})$, where $\mathbf{k}$ is the wave vector ($|\mathbf{k}| = \omega/c$) or, which is equivalent to this, the space $\Gamma_\gamma = (\mathbf{x}, \omega, \mathbf{\Omega})$, where $\mathbf{\Omega} = \dfrac{\mathbf{k}}{|\mathbf{k}|}$ is the unit vector along the direction of wave propagation.[7]

The speed of light (photons) is regarded here as constant because in the relevant density range the dielectric constant $\varepsilon_\perp \approx 1$, since the frequency of Langmuir oscillations $\omega_{0e} \ll \omega$ of the frequency of light. Similarly to (4.1.2), we can determine the distribution function of the photons in the phase space:

$$F(t, \mathbf{x}, \omega, \mathbf{\Omega}) = \frac{\delta N}{\delta \Gamma_\gamma}, \quad \delta \Gamma_\gamma = d\mathbf{x}\, d\omega\, d\mathbf{\Omega}. \tag{6.4.1}$$

However, in most cases the distribution function F is replaced by the concept of the intensity of the flux I_ω

$$I_\omega \equiv cF(t, \mathbf{x}, \omega, \mathbf{\Omega}) \tag{6.4.2}$$

This value can be determined without any quantum considerations as the amount of energy, penetrating the unit area in unit time inside the unit solid angle, perpendicular to the selected area (Fig. 6.4.1). The radiation intensity I_ω can be connected with the density of radiation energy of the given frequency U_ω which is identical with the density of the particles with the given energy. For U_ω we have

$$U_\omega = \frac{1}{c} \int I_\omega(\mathbf{\Omega})\, d\mathbf{\Omega} \tag{6.4.3}$$

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[7]Here the wave vector is denoted by $\mathbf{k}$ in order to avoid confusion with the absorption coefficient.

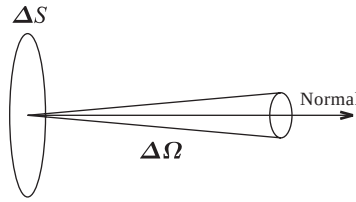


Fig. 6.4.1. Determination of radiation intensity.

where integration is carried out over the solid angle in which the radiation propagates. In addition to I_ω and U_ω another important characteristic of the radiation field is the radiant flux $\mathbf{S}_\omega$ determined from the following equation:

$$\mathbf{S}_\omega = \int\limits_{(4\pi)} I_\omega \boldsymbol{\Omega}\, d\Omega. \tag{6.4.4}$$

The optical characteristics of matter are suitable for further considerations. In fact, we have already discussed them when investigating the individual radiation mechanisms. Here, they can determine the most general form, without defining the radiation mechanism.

Let $q_\omega d\Omega d\omega$ be the amount of energy spontaneously irradiated by the medium in 1 s into 1 cm^3 in the solid angle $d\Omega$ and the frequency range $d\omega$. If the induced radiation is taken into account, the total energy, irradiated in 1 s into 1 cm^3 in the range $d\Omega d\omega$, is equal to (6.3.4)

$$q_\omega d\omega d\Omega\left(1+\frac{4\pi^3 c^2 I_\omega}{\hbar\omega^3}\right). \tag{6.4.5a}$$

When moving through substance radiation flux is attenuated. This attenuation is a result of either absorption or scattering. Taking into account the linear nature of electrodynamic equations (non-linear effects are not considered because they are not significant in the majority of cases) it can be assumed that the flux attenuation after passing through layer $dS = d\mathbf{x}\boldsymbol{\Omega}$ is proportional to the flux, i.e. we have

$$dI = -\varkappa_\omega I_\omega dS \cdot d\Omega d\omega \tag{6.4.5b}$$

The attenuation coefficient $\varkappa_\omega$ consists of the natural absorption coefficient $\varkappa_\omega^{(q)}$ and the scattering coefficient $\varkappa_\omega^{(s)}$. The attenuation coefficient $\varkappa_\omega$ is only the absorption coefficient, i.e. the scattering process is ignored because the scattering cross-section of the electrons (in fact, this is the main process in scattering) is extremely small ($\sigma \approx 6\cdot 10^{-25}$ cm^2).

We can now write to the so-called radiation transfer equation. This can be done conveniently regarding the radiation as a flux of ions and using the same considerations as those made in sections 4.1 and 5.1 when deriving the kinetic equations for particles. Taking (6.4.1) into account, in accordance with the above considerations we can write the law of conservation of the particles and the phase space (**x**, **k**)

$$\frac{\partial F_\omega}{\partial t} + \frac{\partial}{\partial \mathbf{x}}(\dot{\mathbf{x}} F_\omega) + \frac{\partial}{\partial \mathbf{k}}(\dot{\mathbf{k}} F_\omega) = \delta N_\omega \tag{6.4.6}$$

Here δN is the difference between the number of nucleation and the number of annihilation of the photons in unit time u in the unit volume of the phase space. Taking into account that $\dot{\mathbf{x}} = c\boldsymbol{\Omega}$, and the changes of the wavelength in the direction of propagation of the wave are ignored ($\dot{\mathbf{k}} = 0$), equation (6.3.6) is reduced to the form

$$\frac{\partial F_\omega}{\partial t} + c(\boldsymbol{\Omega}\nabla) F_\omega = \delta N_\omega \tag{6.4.7}$$

Returning to the flux intensity I_ω = cF_ω and taking into account the equations (6.4.5a) and (6.4.5b), we finally obtain

$$\frac{1}{c}\frac{\partial I_\omega}{\partial t} + \boldsymbol{\Omega}\nabla I_\omega = q_\omega\left(1 + \frac{4\pi^3 c^2}{\hbar\omega^3} I_\omega\right) - \varkappa_\omega I_\omega. \tag{6.4.8}$$

This equation is identical with the kinetic equation (5.1.1) in the absence of external forces.

Formally, the equation (6.4.8) at the known values of q_ω and $\varkappa_\omega$ is a simple linear equation in the partial derivatives of the first order. In most cases, even simpler stationary variants of the equation (6.4.8) are considered

$$(\boldsymbol{\Omega}\nabla) I_\omega = q_\omega\left(1 + \frac{4\pi c^2}{\hbar\omega^3} I_\omega\right) - \varkappa_\omega I_\omega \equiv \alpha + \beta I_\omega \tag{6.4.9}$$

This is associated with the high speed of light so that the time to establishment of the radiant flux, especially in the laboratory conditions, is very short. In other words, the dependence of I_ω on t is parametric and is determined by the time dependence of the characteristics of the plasma flow q_ω and $\varkappa_\omega$, i.e. $f_\alpha(\mathbf{x},t)$ or in the hydrodynamic approximation by the quantities:

$\eta_\alpha(\mathbf{x},t), T_\alpha(\mathbf{x},t), \mathbf{V}_\alpha(\mathbf{x},t)$

Assuming that $q_\omega(\mathbf{x})$, $\varkappa(\mathbf{x})$ are known functions of **x**, we obtain the explicit form of the solution of equation (6.4.9).

G denotes the surface restricting the radiating plasma volume (Fig. 6.4.2a) and we consider the light flux travelling in the direction $\mathbf{\Omega}$ through some point at the boundary of the volume $P_0(\mathbf{x}_0)$. This flux travels along a straight line intersecting surface G at the point P_1 $(\mathbf{x}_1)$. Evidently, if there is no radiation flux falling on G from the outside, the flux arriving at the point on the straight line $P(\mathbf{x})$ is generated (and in a general case partially absorbed) on the straight line $P_0(\mathbf{x}) - P(\mathbf{x})$. The points of the straight line have the coordinates

$$\boldsymbol{\xi} = \mathbf{x}_0 + \mathbf{\Omega} s,$$

where s is the distance counted from the point $P_0(\mathbf{x})$. This means that we assumed that at P_0 $s_0 = 0$, and at P_1 it is equal to s_1.

Therefore, the transfer equations (6.4.9) can be written in the form of the conventional differential equation [8]

$$\frac{d}{ds} I_\omega(\mathbf{x}_0 + \mathbf{\Omega} S) = q_\omega(\mathbf{x}_0 + \mathbf{\Omega} S)(1 + \Lambda I_\omega) - \varkappa_\omega(\mathbf{x}_0 + \mathbf{\Omega} S) I_\omega \equiv \alpha(\mathbf{x}_0 + \mathbf{\Omega} S) - \beta(\mathbf{x}_0 + \mathbf{\Omega} S) I_\omega(\mathbf{x}_0 + \mathbf{\Omega}^0 S).$$

It solution with the boundary condition $I_\omega(s_0) = 0$ has the form

$$I_\omega\left(\mathbf{x}_0 + \mathbf{\Omega} S\right) = \exp\left\{-\int_0^s \beta\, ds\right\}\left(\int_0^s \alpha \exp\left\{\int_0^{s'} \beta\, ds''\right\} ds'\right). \tag{6.4.10}$$

Example: let α and β be constant, then the flux leaving the plasma volume is

$$I_\omega = \exp\{\beta s_*\}\,\alpha \int_0^s \exp\{\beta s\}\, ds = \frac{\alpha}{\beta}\left(1 - \{\beta s\}\right). \tag{6.4.11a}$$

Assuming that $s = l$ is the length of the section P_1P_0, we obtain

$$I_\omega = \frac{\alpha}{\beta}\left(1 - \exp\{-\beta l\}\right). \tag{6.4.11b}$$

This shows that if the volume is transparent, i.e. $\beta l \ll 1$, we obtain $I_\omega = \alpha l$.

And, consequently, conventional volume clarification takes place.

However, if the plasma volume is large, so that $\beta l \gg 1$, and the system is close to the equilibrium, then (see below (6.4.13) and (6.4.14))

$$I_\omega = \frac{\alpha}{\beta} = I_\omega^{(pl)} \tag{6.4.11c}$$

and $I_\omega^{(pl)}$ is the Planck intensity. Consequently, in this direction the plasma radiates as an absolutely black body.

[8] $\Lambda = \dfrac{4\pi^3 c^2}{\hbar\,\omega^3}$.

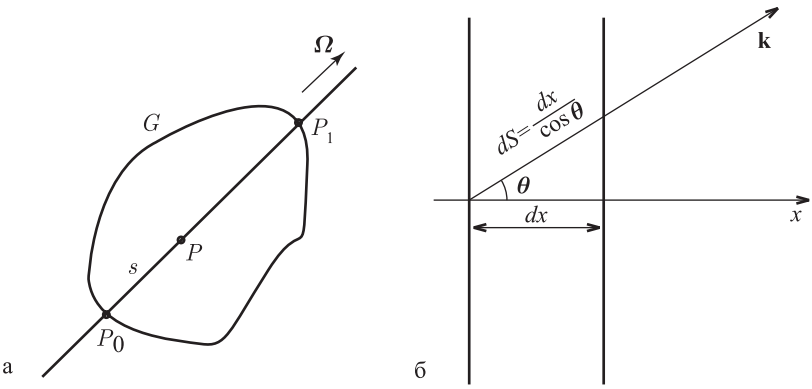


Fig. 6.4.2. Radiation transfer - in a general case (a), in a flat infinite layer (let about).

6.4.2. Radiation transfer in the conditions close to the equilibrium [140, 141]

If the plasma and radiation are in a state close to the thermodynamic equilibrium, the kinetic equation of radiation transfer (6.3.8)

$$\frac{1}{c}\frac{\partial I_\omega}{\partial t}+\mathbf{\Omega}\nabla I_\omega = q_\omega\left(1+\frac{4\pi^3 c_2}{\hbar\omega^3}I_\omega\right)-\varkappa_\omega I_\omega \qquad (6.4.12)$$

can be transformed to the hydrodynamic form. In other words, from function I_ω, determined in the six-dimensional space (ω,Ω,**x**), we can transfer to the functions U and S in the three-dimensional space (**x**) which were determined previously (see (6.4.3), (6.4.4)).

However, prior to this, assuming that the plasma is quasi-equilibrium, the equations (6.5.19) should be slightly transformed. Taking into account (6.3.2b) and (6.3.4) at $g_i = g_k$, we obtain

$$\frac{q_\omega}{\varkappa_\omega}=\frac{\hbar\omega^3}{4\pi^3c^2}\exp\left\{-\frac{\hbar\omega}{kT}\right\}. \qquad (6.4.13)$$

Equation (6.4.13) is valid for any radiation mechanism, regardless of the fact that the equation was derived by investigating a partial case of linear radiation since (6.4.13) is nothing else but the Kirchoff law. Substituting (6.4.13) into (6.4.12), we obtain the transfer equation in the form

$$\frac{1}{c}\frac{\partial I_\omega}{\partial t}+(\mathbf{\Omega}\nabla)I_\omega = q_\omega-\varkappa_\omega\left(1-\exp\left\{-\frac{\hbar\omega}{kT}\right\}\right)I_\omega. \qquad (6.4.14)$$

Finally, from this we derive the conventional form of the transfer equation for the quasiequilibrium conditions

$$\frac{1}{c}\frac{\partial I_\omega}{\partial t}+(\Omega\nabla)I_\omega = \varkappa'_\omega(I_\omega^{(p)}-I_\omega) \tag{6.4.15}$$

Here

$$\varkappa'_\omega = \varkappa_\omega\left(1-\exp\left\{-\frac{\pi\omega}{kT}\right\}\right), \tag{6.4.16a}$$

$$I_\omega^{(p)} = \frac{h\omega^3}{4\pi^3c^2}\frac{1}{\exp\left\{\frac{\pi\omega}{kT}\right\}-1} \tag{6.4.16b}$$

is the equilibrium (Planck) intensity of the radiant flux.

Now we can use the same concept of transition from the kinetics to hydrodynamics which was described for the first time in section 4.3.1. In particular, multiplying (6.4.15) by 1, Ω_α, $\Omega_\alpha\Omega_\beta$,... and integrating with respect to the solid angle Ω:

$$d\Omega = \sin\theta d\theta d\phi\,;\ \Omega_x = \sin\theta\cos\phi,\ \ \Omega_y = \sin\theta\,\sin\phi,\ \Omega_z = \cos\,\theta \tag{6.4.17a}$$

we obtain an infinite chain of the hydrodynamic type equations

$$\begin{aligned}
&\frac{1}{c}\frac{\partial}{\partial t}w_\omega+\frac{\partial}{\partial x_\alpha}w_{\omega\alpha} = \varkappa'_\omega(4\pi I_\omega^{(p)}-w)_\omega \\
&\frac{1}{c}\frac{\partial}{\partial t}w_{\omega\alpha}+\frac{\partial}{\partial x_{\alpha\beta}}w_{\omega\alpha\beta} = -\varkappa'_\omega w_{\omega\alpha} \\
&\frac{1}{c}\frac{\partial}{\partial t}w_{\omega\alpha\beta}+\frac{\partial}{\partial x_\gamma}w_{\omega\alpha\beta\gamma} = -\varkappa'_\omega w_{\omega\alpha\beta}+\varkappa'_\omega I_\omega^{(p)}\cdot\frac{4}{3}\delta_{\alpha\beta}
\end{aligned} \tag{6.4.17b}$$

Here $\delta_{\alpha\beta}$ is the Kronecker symbol,

$$w_\omega = \int I_\omega d\Omega,\quad w_{\omega\alpha} = \int I_\omega\Omega_\alpha d\Omega,...$$

The resultant chain can be interrupted by assuming that the intensity I_ω differs only slightly from equilibrium intensity $I_\omega^{(p)}$. However, this should be carried out using a slightly different procedure, using the equation (6.4.15) at

$$\frac{d}{\partial t}I_\omega = 0 \qquad (\boldsymbol{\Omega}\nabla)I_\omega = \varkappa'(I_{\omega n}^{(p)}-I_\omega) \tag{6.4.18a}$$

i.e.

$$(1+\hat{\chi})I_\omega = I_\omega^{(p)} \tag{6.4.18b}$$

Here $\hat{\chi}$ is the operator

$$\hat{\chi} \equiv \frac{1}{\varkappa'_\omega}(\Omega\nabla) \tag{6.4.18c}$$

The solution of the equation (6.4.18b) can be presented in the form

$$I_\omega = (1+\hat{\chi})^{-1} I_\omega^{(p)} = I_\omega^{(p)} - \hat{\chi} I_\omega^{(p)} + \hat{\chi}\hat{\chi} I_\omega^{(p)} - ... = \\ = I_\omega^{(p)} - \frac{(\Omega\nabla)}{\varkappa'} I_\omega^{(p)} + \frac{(\Omega\nabla)}{\varkappa'}\left[\frac{(\Omega\nabla)}{\varkappa'} I_\omega^{(p)}\right] - ..... \tag{6.4.19}$$

Equation (6.4.18) shows that expansion takes place with respect to the degrees

$$\theta = \frac{1}{\varkappa' L}. \tag{6.4.20}$$

where L is the scale of the nonuniformity of temperature and density in the plasma configuration. In other words, to ensure rapid convergence of the series, L should be considerably greater than the free path of the photon

$$\theta \ll 1.$$

Here knowing I_ω we can calculate all the characteristics of the flux. For example, the generation power in the unit volume is

$$q_S = \mathrm{div}\,\mathbf{S} = \mathrm{div}\int_0^\infty \mathbf{S}\omega d\omega = \mathrm{div}\int_0^\infty d\omega \mathbf{\Omega} = \int_0^\infty d\omega \int_{4\pi} (\mathbf{\Omega}\nabla) I_\omega d\mathbf{\Omega} = \tag{6.4.21}$$

$$= \int_0^\infty d\omega \;\; \varkappa'_\omega (I_\omega^{(p)} - I_\omega) d\Omega. .$$

This is a natural equation. It expresses the fact that the radiation yield from the element of the plasma volume is determined by radiation and absorption.

If expansion (6.4.19) is substituted into (6.4.21), then taking into account that

$$\int \mathbf{\Omega}\, d\mathbf{\Omega} = 0$$

the non-zero contribution is provided only by the second and subsequent even degrees of the expansion. Thus,

$$q_S = -\int_0^\infty d\nu - \frac{(\mathbf{\Omega}\nabla)}{\varkappa'}\left(\frac{(\mathbf{\Omega}\nabla)}{\varkappa'} I_\omega^{(p)}\right) d\Omega = -\int_0^\infty d\nu\, \omega_{ij} \frac{\partial}{\partial x_i}\frac{1}{\varkappa'_\nu}\frac{\partial I_\omega^{(p)}}{\partial x_j}. \tag{6.4.22}$$

Here

$$\omega_{ij} = \int_{4\pi} d\mathbf{\Omega}\,\Omega_i \Omega_j. \tag{6.4.23}$$

Using the expressions (6.4.17a), we obtain that

$$\omega_{xx} = \omega_{yy} = \omega_{zz} = \frac{4\pi}{3}, \quad \omega_{ij} = 0 \text{ at } i \neq j.$$

Taking into account that $\frac{1}{\chi_\omega} = \ell_\omega$ is the free path of the photon, we can write

$$q_S = \frac{4\pi}{3}\int_0^\infty d\omega \operatorname{div}\left(\frac{1}{\varkappa'_\omega}\right)\nabla I_{p\omega}^{(p)} = \frac{4\pi}{3}\operatorname{div}\int_0^\infty l_\omega d\omega \nabla I_\omega^{(p)}. \tag{6.4.24}$$

The last integral, assuming that $cU_\omega^{(p)} = 4\pi I_\omega^{(p)}$ can be presented in the form

$$\int_0^\infty d\omega l_\omega \nabla U_\omega^{(p)} = \ell_R \nabla U = 4\ell_R \sigma_{S-B} T^3 \nabla T, \tag{6.4.25a}$$

where $\sigma_{S\text{-}B}$ is the Stephan–Boltzmann constant. The Stephan–Boltzmann law links the equilibrium density of radiation energy U with temperature T:

$$U = \sigma_{S\text{-}B} T^4. \tag{6.4.25b}$$

(6.4.25a) includes some free path of the photons in the given plasma volume averaged out with respect to the Planck distribution

$$l_R = \frac{\int d\omega l_\omega \frac{d}{dT} U_\omega^{(P)}}{\frac{d}{dT} U^{(P)}} \tag{6.4.26a}$$

This value is referred to as the Rosseland path. It can be seen that we obtain the specific function

$$G(u) = \frac{\frac{dU_\omega^{(p)}}{dT}}{\frac{dU^{(p)}}{dT}}, \quad u = \frac{h\omega}{kT} \tag{6.4.27}$$

which uses the Stephan–Boltzmann Law.

Substituting the Planck equation into (6.4.27) we obtain the explicit expression for $G(u)$

$$G(u) = \frac{15}{4\pi^4}\frac{u^4 e^{-4}}{(1-e^{-4})^2} \tag{6.4.28}$$

Using the equations (6.4.25a) and (6.4.26) we can write

$$q_S = \mathrm{div}\left(\frac{4}{3}\ell_R \sigma_{S-B} T^3 \nabla T\right). \tag{6.4.29}$$

This equation has a more natural form if we introduce the effect of the light flux **S**. Consequently

$$\mathbf{S} = -\frac{4}{3}\ell_R \sigma_{S-B} T^3 \nabla T, \; q_S = -\mathrm{div}\,\mathbf{S}. \tag{6.4.30}$$

At first sight, it may appear that the Rosseland path is universal for all plasma volumes, optically thick and similar to the equilibrium state, and the Rosseland path is the same. However, this is not the case. The difference here is associated with different dependence of $\varkappa_\omega$ on frequency and temperature.

Approximation ℓ_R [158]. Previously, we stressed the complicated form of the dependence of the absorption coefficient $\varkappa_\omega$ on frequency and also on temperature and density. However, in a number of cases, we can use the approximating equations of acceptable accuracy. They relate to LTE (see section 6.5.7).

– for bremsstrahlung absorption at $\hbar\omega < kT$ (Kramers)

$$\varkappa_\omega^{(\mathrm{brem})} = 4.1\cdot 10^{-23}\frac{Z^3 n^2}{T^{7/2}}\left(\frac{kT}{\hbar\omega}\right)^3. \tag{6.4.31a}$$

– the absorption coefficient in plasma with the particles with the same ionisation level (lithium)

$$\varkappa_\omega^{(\mathrm{Li})} = 1.6\cdot 10^{-23}\left(\frac{kT}{\hbar\omega}\right)^3 \times \begin{cases} \exp\left\{\dfrac{\hbar\omega}{kT}\right\}, & \hbar\omega < I_Z \\ \dfrac{2I_Z}{kT}, & \hbar\omega > I_Z. \end{cases} \tag{6.4.31b}$$

– plasma with multi-charged ions (Kramers–Unzold)

$$\varkappa_\omega^{(\mathrm{Kr-Un})} = \sum_{(Z)} 0.96\cdot 10^{-7}\frac{n_Z (Z+1)^2}{T^2}\exp\left\{-\frac{I_{Z+1}}{kT}\right\}\times F\left(\frac{\hbar\omega}{kT}\right). \tag{6.4.31c}$$

Here

$$F_Z(x) = \begin{cases} \dfrac{e^x}{x^3} & \text{at } x < \dfrac{I_{Z+1}}{kT}, \; (\hbar\omega < I_{Z+1}); \\ \dfrac{2I_{Z+1}}{x^3 kT}\exp\left\{\dfrac{I_{Z+1}}{kT}\right\} & \text{at } x > \dfrac{I_{Z+1}}{kT}, \; (\hbar\omega > I_{Z+1}). \end{cases}$$

If we examine bremsstrahlung radiation at ions with the charge Z (section 6.3.2), though the Rosseland path

$$\ell_R \approx 4.8 \cdot 10^{24} \frac{(kT)^{7/2}}{Z^3 n_0^2}. \qquad (6.4.32)$$

Here n_0 is the ion concentration.

However, if the radiation is linear, then for the ions with $Z \gg 1$ the following equation is acceptable for evaluation by approximation

$$\ell_R = 4.4 \cdot 10^{22} \frac{(kT)^{7/2}}{Z(1+Z)^2 n_0^2}.$$

6.5. Schemes for describing the dynamics of the particles of transforming plasma

6.5.1. General characteristic of the models

In the previous section, we discussed the dynamics of radiation separately from the dynamics of the population of the corpuscular component of the plasma which is characterised only by *a priori* parameters q_ω and $\varkappa_\omega$.

In this section, we discuss mainly the transformation dynamics of the particles. Examples of self-consistent dynamic processes taking into account transformation will be presented later.

The main question which needs to be answered here is the possible methods of simplifying the description of the gigantic variety of the components in plasma in 'untying' of the internal degrees of freedom, as discussed in the introduction to this chapter.

The selection of the scheme of simplifying the actual set of the levels is determined by the properties of the calculated system, the aim of its functioning, and also the existing means for calculations. The most important parameters, which determine these schemes, are:

$$\alpha = \frac{T_e}{I}, \quad \gamma = \frac{\tau_{\text{rad}}}{\tau_{\text{coll}}}, \; \theta = \langle \sigma v \rangle n_e \tau_*, \; \xi = \frac{\tau_a}{\tau_0}.$$

Here α is the ratio of the electronic temperature to the characteristic difference of the energy of the levels prior to and after transition; γ is the ratio of the τ_{rad} time – radiation de-excitation (recombination) to τ_{coll} – the identical process, determined by collisions with the particles, θ is the ratio of the lifetime of the particle in the excited state τ_* to the excitation time (ionisation), ξ is the ratio of the duration of the effect (ionisation) τ_a to the lifetime of the particle in the system τ_0.

6.5.2. Two approaches to simplifying the actual situation

In most cases, the simplification procedure is based on two simple

concepts. The first of this concepts is based on the transition from the real particles with a large number of states (α-parameters) to some effective particles with the small number ψ and Z, i.e. with the small number of ionised (Z) and the small number of excited states. Regardless of the availability of computers, two- (Z = 0, 1, ψ = 0) and three- (Z = 0, 1, ψ = 0, 1) level models are now used for estimates.

In the first case, it was assumed that the 'effective particle' can be represented either by a neutral atom (Z = 0) in the ground state or by the ion (Z = 1) also in the ground state (ψ = 0). In the second case, the excited level (ψ = 1, Z = 0) is also taken into account.

Naturally, some averaged-out parameters were related to both the levels and characteristic transitions between them. A classic example is ionisation where we are not interested in the specific mechanism of the atom$\rightarrow$ion transition and are interested only in the energy loss in a single ionisation act, the so-called value of the ion.

The approximation of the small number of levels is used in the plasma almost in all cases, although this is not compulsory in extreme forms such as models with two and three levels. The models with a large number of the effective levels are used on an increasing scale with the development of numerical methods and the appearance of more and more powerful computers. At present, calculations are carried out of the models of the molecular plasma taking thousands of levels into account. This is important, for example, in calculating the glow of a meteorite or a space system entering the atmosphere of the Earth or Mars.

However, there is a number of plasma systems in which the accurate data can be obtained using models with a very small number of levels. These are mostly systems in which the main process is associated with one or several separated levels. These systems include lasers and a large number of plasmochemical reactors which will be discussed in section 6.11. In addition to the lasers, in plasma chemistry a small number of levels work in systems which can be regarded as short-term systems. These are systems in which the neutral particles are subjected to the effect of exciting and/or ionising factors for a short period of time and, therefore, only single levels can be excited ($\xi > 1$).

Here we are interested only in a partial case of short pulse systems, in particular, the ionisation fronts in plasma accelerators with relatively rarefied plasma. These fronts have the important property that excitation and ionisation can take place several times but recombination does not take place.

The structures will be referred to as passing structures. They will be discussed in greater detail in section 6.5.3 and 6.7.1.

The second method of simplifying the description of the state of the particles in the system is used when recombination must be taken into account and, in addition to this, the excitation of a large number of different levels may have to be considered. In these cases, the system can be regarded as quasi-equilibrium in the first approximation and we can determine the population of the levels using finite expressions derived from simple linear equations which depend on several parameters. These parameters can be determined from the conventional hydrodynamic equations.

Evidently, this is the same concept which was used previously in section 5.3 for transition from the kinetic to hydrodynamic equations. This was carried out using the Maxwell distribution with the locally variable parameters $n(\mathbf{x},t)$, $\mathbf{v}(\mathbf{x},t)$, $T(\mathbf{x},t)$. Now, in the same conditions ($\lambda/L \ll 1$) considerations will be based on the Maxwell–Boltzmann distribution. This is the approximation referred to as the 'approximation of the local thermodynamic process' (LTE) and will be investigated in section 6.5.5.

However, the LTE approximations are not the only quasi-equilibrium approximations. In principle, there should be many such approximations. Of these, quasi-equilibrium 'coronal equilibrium' approximation or the Elvert–Shklovsky equilibrium are encountered quite often.

The relationship between LTE and coronal equilibrium is indicated by the two-level hydrodynamic model of plasma (the atom and the ion in the ground state). The equation for the ion density can be presented in the form assuming for simplicity that $\mathbf{v} = 0$:

$$\frac{dn_i}{dt} = \beta n_a n_e - \alpha n_e^2 n_i - \gamma n_e n_i \tag{6.5.1}$$

here $\beta = \langle \sigma v \rangle_{\text{ion}}$, α is the triple recombination coefficient, $\gamma = 1/t_{\text{rad}}$, where t_{rad} is the lifetime of the excited state in relation to photo-recombination, more accurately photo- and dielectronic recombination. The equation of the type (6.5.1) was already discussed previously (6.2.5) and it is now assumed that the system is in equilibrium. Consequently

$$\beta n_a n_e - \alpha n_e^2 n_i - \gamma n_e n_i = 0 \tag{6.5.2a}$$

This shows that two extreme cases are possible

$$\beta n_a n_e - \alpha n_e^2 n_i \approx 0 \tag{6.5.2b}$$

$$\beta n_a n_e - \gamma n_e n_i = 0$$

In the first case, the process of ionisation by electronic impact is balanced by triple recombination. These are mutually reversible processes and the equality of these processes indicates thermodynamic equilibrium. In the

heterogeneous medium it is LTE.

In the second case, the ionisation by electronic impact is balanced by photo-and dielectronic recombination. If the plasma is sufficiently rarefied, ionisation takes place from the ground state, and photo-recombination either directly 'places' the electron on the ground level or does so as a result of a chain of transitions.

The population of the levels formed in the stationary conditions using this scheme is referred to as coronal equilibrium. It should be stressed that this equilibrium is characterised by two moments: ionisation by electronic impact from the ground state, and photo-recombination.

The following comments should be made regarding the quasi-equilibria:

Equation (6.5.1) is written for the case of ionisation of atoms (molecules, ions). However, similar equations can also be written for excitation. In the two-level model this is represented by the ground state ('0') and excited state ('1'), and the equilibrium equation has the following form:

$$\frac{dn_i}{dt} = \beta_1 n_0 n_e - \alpha_1 n_1 n_e - \gamma_1 n_1 = 0 \qquad (6.5.3)$$

This can also be said of LTE when

$$\beta_1 n_0 n_e - \alpha_1 n_1 n_e = 0 \qquad (6.5.4a)$$

and of coronal equilibrium, if

$$\beta_1 n_0 n_e - \gamma_1 n_1 = 0 \qquad (6.5.4b)$$

As indicated by the expressions (6.5.2) and (6.5.4), at the known values of n_e and n_0, the ion concentration n_i and the concentration of the excited atoms n_{i*} is determined by the algebraic linear equations. This fundamental fact holds not only for two-level models but also for any set of processes, if there are no collisions in these processes between the excited heavy particles. Therefore, in addition to the LTE and coronal models, other quasi-equilibrium models of population in the transforming plasma can also be constructed.

At present, the literature contains many articles dealing with the description of specific schemes of calculating the population of levels. The majority of them are combinations of the approaches mentioned here. However, the use of more and more powerful computers makes it possible to take into account explicitly many hundreds and thousands of α-components, and this will be discussed in detail in section 6.8.

The passing, coronal and LTE models will now be described in greater detail.

6.5.3. Passing systems

The characteristic special feature of the flight systems, as mentioned previously, it can be expressed by the inequality:

$$\tau_{ion} \le \tau_0 << \tau_{rec}. \qquad (6.5.5)$$

Here τ_0 is the duration of the effect of the transforming factors (energy electrons, photons, etc) on the neutral substance, and time τ_0 is longer or equal to time τ_{ion}, required for ionisation, but τ_0 is far shorter than the time required for recombination.

The condition (6.5.5) can be efficiently realised in, for example, the zone of ionisation of stationary plasma thrusters with relatively rare xenon plasma. This case will be discussed in greater detail. The thruster is described in section 6.7, and Fig. 6.7.1 shows the image of the thruster. The thickness of the ionisation zone is L ~1 cm, electronic temperature is T_e ~10 eV. The characteristic velocity of the xenon atoms prior to ionisation is ~10^4 cm/s, the ionisation coefficient is

$$\beta = \langle \sigma v_e \rangle_{ion} \sim 10^{-8} \text{cm}^3\text{s}^{-1}.$$

Using these parameters, we determine τ_0 and τ_{ion}, at $n_e \sim 10^{12}$ cm^{-3}

$$\tau_0 = \frac{L}{v_\alpha} \sim 10^{-4}\text{s}^{-1},$$
$$\tau_{ion} = \frac{1}{n_e \beta} = 10^{-4}\text{s}^{-1}. \qquad (6.5.6)$$

It is important to note that the quantities τ_0 and τ_{ion} are of the same scale. Since the electron density is $n_e < n_e^*$, and is determined by the equation (6.2.8), it is sufficient to confine considerations to the evaluation of the photo-recombination time (6.2.6b). Calculations for Z = 1 give

$$\tau_{rec} \approx 15\text{s}, \qquad (6.5.7)$$

i.e. the condition (6.5.5) is fulfilled with a large reserve in this case.

During passage through the investigated zone, the Xe atom can be excited several times prior to ionisation because the excitation and ionisation cross-sections are similar.

Evidently, the dynamics of every component in the passing regime is described by the equation (disregarding excitation)

$$\frac{dn_i}{dt} = \beta_{ion} n_0 n_e. \qquad (6.5.8)$$

It should be noted that secondary ionisation can also take place in the layer, but these processes do not occur often and will not be discussed here.

6.5.4. Coronal equilibrium [141]

The coronal systems are characterised by two moments: ionisation (excitation) of the atom (ion) by electron impact from the ground state, followed by return to the initial state as a result of radiation. It is assumed that the plasma volume is optically transparent. The first investigated object of this class is the solar corona.

Thus, the scheme of the coronal cycle with ionisation is

$$\begin{cases} A_0 + e \to A^+ + 2e \\ A_0^+ + e \to A + \gamma. \end{cases} \tag{6.5.9a}$$

Here the index 'zero' denotes the ground state.

The coronal processes also include the excitation of the particle from the ground state by electron impact followed by radiation

$$\begin{cases} A_0 + e \to A^* + 2e \\ A^* \to A + \gamma. \end{cases} \tag{6.5.9b}$$

The transfer of the particle to the initial state can take place directly from the ionised (excited) state into the initial (ground) state and can be of the stepped (cascade) nature. The transition mechanism is determined mainly by the specific nature of the levels: atom or ion.

Here, we investigate only 'single-transition' photo-recombinations, using the following notations: n_z is the concentration of the atoms ionised z times, $\beta_{z,z+1} \equiv \langle \sigma v \rangle_{z,z+1}$ is the velocity of transfer of the ions from z to the $(z+1)$-th degree of ionisation as a result of collision with the electron

$$A^{+z} + e \to A^{+(z+1)} + 2e \tag{6.5.10a}$$

$\gamma_{z,z-1}$ is the velocity of transfer $z \to z-1$ as a result of photo-recombination

$$A^{+z} + e \to A^{+(z-1)} + \gamma_{z,z-1} \tag{6.5.10b}$$

Consequently, the system of equations, describing the coronal equilibrium with the cascade-free transitions has the form

$$\frac{\partial n_0}{\partial t} = -\beta_{01} n_e n_0 + \gamma_{10} n_e n_0 = 0$$

$$\frac{\partial n_z}{\partial t} = -\beta_{z,z+1} n_z n_e - \gamma_{z,z-1} n_z n_e + \beta_{z-1,z} n_{z-1} n_e = 0$$

$$\frac{\partial n_{z^*}}{\partial t} = 0 - \gamma_{z^*,z^*-1} n_{z^*} n_e + \beta_{z^*-1,z^*} n_{z^*-1} n_e = 0$$

Here z^* is the maximum charge of the ion in the given conditions.

Simple calculations show that

$$n_z = n_{z-1}\frac{\beta_{z-1,z}}{\gamma_{z,z-1}} = n_{z-1}\left\langle \sigma v_e \right\rangle_{z-1,z}\cdot\tau_{z,z-1},\ \tau_{z,z-1} = \frac{1}{\gamma_{z,z-1}}, \tag{6.5.12}$$

i.e.

$$n_1 = n_0\left\langle \sigma v \right\rangle_{01}\tau_{10},\ n_2 = n_0\left\langle \sigma v \right\rangle_{01}\left\langle \sigma v \right\rangle_{12}\tau_{10}\tau_{21} \tag{6.5.11}$$

This shows the important special feature of the coronal distribution. It is clearly independent of n_e and is determined only by the particle concentration in the initial state and T_e.

Calculations using equation (6.5.11) are carried out using the quantities $\left\langle \sigma v \right\rangle$ and $\tau^{(\mathrm{rad})}$ of the type shown in the sections 6.1 and 6.2.

It is also important to mention another important though obvious property of the non-branching coronal equilibrium. Evidently, the radiation power of the unit volume is:

$$P_{z\to z-1} = n_e n_z h\omega_{(z_1\to z-1)}\left\langle \sigma v \right\rangle_{z-1\to z} \tag{6.5.13}$$

and, consequently, it does not depend on the probability of the emission transition.

The coronal model is limited at the bottom at low particle densities by the 'flyby parameter':

$$\theta_0 = \frac{\tau_0}{\tau_{\mathrm{rec}}^{(\mathrm{rad})}} \geqslant 1, \tag{6.5.14}$$

i.e. by the relatively high value of the lifetime (τ_0) of the particle in the system, and at the top by triple recombination

$$\theta_1 = \frac{\tau_0}{\tau_{\mathrm{rec}}^{(ee)}} \gg 1. \tag{6.5.15}$$

6.5.5. Dynamics of quasi-equilibrium transforming plasma

Attention will now be given to the behaviour of transforming plasma when, as previously, ionisation takes place as a result of electron impact, and recombination mainly as a result of triple collision. In this case, we have the local thermodynamic equilibrium (LTE).

Equilibrium population of energy levels of the atoms (ions, molecules)
In the presence of thermodynamic equilibrium any process, taking place in the system, is balanced by the strictly reversed process ('the principle of detailed equilibrium').

This shows that in thermodynamic equilibrium all the collisional terms in the kinetic equations for the particles convert to 0, and the distribution functions are the Maxwell–Boltzmann functions

$$f_\alpha^{(a)}(\mathbf{v},\alpha)=\frac{N_a g_\alpha}{(2\pi kT_a)^{3/2}Z_0}\exp\left\{-\frac{1}{2kT_a}\left[\frac{M_a(\mathbf{v}-\mathbf{v}_0)^2}{2}+E_\alpha\right]\right\} \qquad (6.5.16)$$

and radiation is governed by the Planck law (6.3.2a). Here $\alpha = (Z_\mu, Z, \psi)$ is the characteristic of the state of the particle[9], E_α is the energy of the internal degrees of freedom, N_a is the total concentration of the particles of the type a, Z_0 is their statistical sum with respect to discrete states

$$Z_0=\sum_\alpha g_\alpha^{(a)}\exp\left\{-\frac{E_\alpha^{(a)}}{kT_a}\right\}, \qquad (6.5.17)$$

$g_\alpha^{(a)}$ is the statistical weight of the level, $\mathbf{v}_0$ is the mean velocity of the particle.

If equation (6.5.16) is integrated with respect to velocity and the normalisation condition $\sum_k \int f_\alpha^{(a)}(\mathbf{v},\alpha)d\mathbf{v}=N_a$ is taken into account, we obtain the Boltzmann equation for the case of discrete levels ($\psi = k$):

$$n_k^{(a)}=\frac{N_a g_k}{Z_0}n_1\exp\left\{-\frac{E_k}{kT}\right\}, \qquad (6.5.18)$$

From this equation we obtain the conventional Boltzmann equation, linking the population of different levels (k and l):

$$n_k^{(a)}=\frac{g_k}{g_l}n_l\exp\left(-\frac{\Delta E_{kl}}{kT}\right);\quad \Delta E_{kl}=E_k-E_l \qquad (6.5.19)$$

In complete thermodynamic equilibrium, the radiation, strictly speaking, should also be in equilibrium with the particles. However, in dense plasma, even in non-transparent plasma, the radiation can be non-equilibrium since the role of collision of the particles in the formation of the population of the levels is dominant.

However, with the increase of temperature the photon density increases $\sim T^3$. This follows from the Stephan–Boltzmann law (6.4.25b) and the fact that the characteristic energy of the thermal radiation photons is:

$$\frac{h\omega}{2\pi}\sim kT.$$

Therefore, at the fixed density of the particles with appropriate optical

[9] Z_μ is the number of the element in the periodic table of elements, Z is the degree of ionisation.

thickness, with increasing temperature excitation and de-excitation take place mainly as a result of the photons. Thus, we transfer almost completely to the Einstein equilibrium model, investigated in section 6.3.1.

Using the equilibrium distribution functions, we easily obtain the relationship between the constants of the rate of direct and reversed processes. Actually, we investigate a simple two-level system in which transitions 1↔2 can take place under the effect of some factor γ (for example, electrons with concentration N_γ). Consequently, we obtain the balance equation in the stationary case

$$0 = n_1 N_\gamma \left\langle v\sigma_{12} \right\rangle - n_2 N_\gamma \left\langle v\sigma_{21} \right\rangle \tag{6.5.20}$$

where averaging is carried out with respect to the equilibrium function of distribution of the factor γ and N_γ is its concentration. Consequently

$$\frac{n_1}{n_2} = \frac{\left\langle v\sigma_{21} \right\rangle}{\left\langle v\sigma_{12} \right\rangle} \tag{6.5.21}$$

and, in accordance with (6.4.18) we obtain

$$\frac{\left\langle v\sigma_{21} \right\rangle}{\left\langle v\sigma_{12} \right\rangle} = \frac{g_1}{g_2} \exp\left(\frac{\Delta E_{12}}{kT} \right) \tag{6.5.22}$$

The resultant relationship is sometimes referred to as the Klein–Rosseland relation. Although this equation was derived when examining the equilibrium distribution function, the range of applicability of this equation is considerably wider. For the relation (6.5.22) to be valid, it is only necessary to ensure that the function of distribution of particles – factors, leading to the transitions between the levels 1 and 2, is equilibrium. In this case, this factor is represented by the electrons.

Evidently, this condition is essential but insufficient for the population of the levels to correspond to the Boltzmann equation (6.5.16)

it should be mentioned that the Boltzmann distribution with respect to the excitation levels can be established under the effect of different factors.

Saha equation [15]

Previously, we investigated the kinetic equations which take into account only paired interactions. However, three-particle processes, in particular, recombination with participation of two electrons, can play a significant role in the group of the transformation processes. This process is completely reversible in relation to ionisation by electronic impact. The thermodynamic equilibrium of the concentration of the atoms, ions and electrons, caused by the processes

$$A + e \rightleftarrows A^+ + e + e, \tag{6.5.23a}$$

is described by the Saha equation. Since we have not analysed the kinetic equations with triple interaction, we investigate the direct thermodynamic derivation of this equation.

Let there be a reversible reaction

$$\nu_k A_k = 0, \tag{6.5.23b}$$

where A_k is the symbol of the component, and ν_k are the stoichiometric coefficient (in this case $\nu = 1$, $\nu_e = 1$, $\nu_i = -1$; $\nu_{ee} = -2$). Consequently, the 'law of acting masses' follows from the thermodynamics

$$\sum \nu_i \mu_i = 0 \tag{6.5.23c}$$

where μ_i is the chemical potential of the i-th component of the reaction.

If every component of the plasma is an ideal gas, the following relationship can be written

$$\mu = T \ln n - T \ln Z_0 Z_1. \tag{6.5.24a}$$

Here Z_0 is the statistical sum with respect to the levels, and Z_1 is determined by the kinetic energy of the particles.

Substituting (6.5.24a) into (6.5.23c) we obtain

$$\frac{n_e n_i}{n_a} = \left(\frac{Z_0^i Z_0^e}{Z_0^a}\right)\left(\frac{Z_1^i Z_1^e}{Z_1^a}\right),\ Z_1^e = \left(\frac{mkT_e}{(2\pi\hbar)^2}\right)^{3/2}, \tag{6.5.24b}$$

Here $Z_{0,1}^{a,e,i}$ are the statistical sums of the appropriate components. The kinetic co-multipliers of the ions and atoms are equal to $z_1^a = z_1^i$ and $z_0^e = 2$, respectively to two spin orientations. The quantities z_0^i and z_0^a are taken from the two-level models. Therefore, it can be set – taking into account that (6.5.24b) includes their relationship – $Z_0^e = g^a, Z_0^i = g^i \exp\{-eI / kT\}$. Here I is the ionisation potential. Substituting these expressions into (6.5.24b) we obtain the Saha equation

$$\frac{n_e n_i}{n_a} = 2\frac{g_a}{g_i}\left(\frac{mkT}{(2\pi\hbar)^2}\right)^{3/2} \exp\left\{-\frac{eI}{kT}\right\}. \tag{6.5.25a}$$

To facilitate the calculations, the expression (6.5.25a) can be presented in the form (kT, eI – in eV, n in cm^{-3}):

$$\frac{n_e n_i}{n_a} = 2\frac{g_a}{g_i} 3{\cdot}10^{21} T_e^{3/2} \exp\left\{-\frac{eI}{kT}\right\}. \tag{6.5.25b}$$

Knowing the electronic temperature T_e and the total number of the particles of the given type N_0 in unit volume, we can use the recurrent Saha equation to determine the density of each component n_Z and the

total density of the electrons

$$N_0 = n_0 + n_1 + n_2 + ... = n_0 + \frac{n_0}{n_e} K_1 + \frac{n_0}{n_e^2} K_1 K_2 + ...$$

$$n_e = \frac{n_0}{n_e} K_1 + \frac{2n_0}{n_e} K_1 K_2 + ... \qquad (6.5.26)$$

Here K_Z (T) is the right-hand part of (6.5.25a), corresponding to the transition (Z–1)→Z.

Figure 6.5.1 shows the dependence of the concentration of ions of different degree for oxygen on T_e.

The effective charge of the ions and thermodynamic relationships in thermal plasma

If we examine the curves of the dependences of the ion concentration with given Z on temperature (see Fig. 6.5.1), attention should be given to the fact that ions of 2–3 types with similar values of Z usually prevail at every temperature. Consequently, it is rational to introduce the mean ion charge $\bar{Z}$. If the plasma contains ions with sufficiently large Z, it is possible to approximate the dependence of the ionisation potential I_{Z+1} of the ion with the charge Z – this results in the formation of an ion with the charge Z + 1 – some continuous function I (Z), determined by the interpolation of the values of the ionisation potential I_{Z+1} from integer argument Z + 1. Consequently, as shown in [145], the value $\bar{Z}$ can be determined as the solution of the transcendental equation:

$$I\left(Z + \frac{1}{2}\right) = kT \ln \frac{AT^{3/2}}{\bar{Z}n} \qquad (6.5.27)$$

Here on the left we have the half-sum of the ionisation potential of the ions with the charges Z –1 and Z. The quantity

$$A \equiv 2\left(\frac{2\pi m_e k}{h^2}\right) = 6{\cdot}10^{21}\,\mathrm{cm}^{-3}\mathrm{eV}^{-3/2}.$$

The expressions for the thermodynamic functions p (n, T) and the internal energy are quite evident (the plasma is assumed to be ideal)

$$p = n(1 + \bar{Z})kT \qquad (6.5.28)$$

To write the expressions for internal energy, it is taken into account that in the dynamics equation this energy is related to the unit mass and includes both the thermal energy and the losses through excitation and

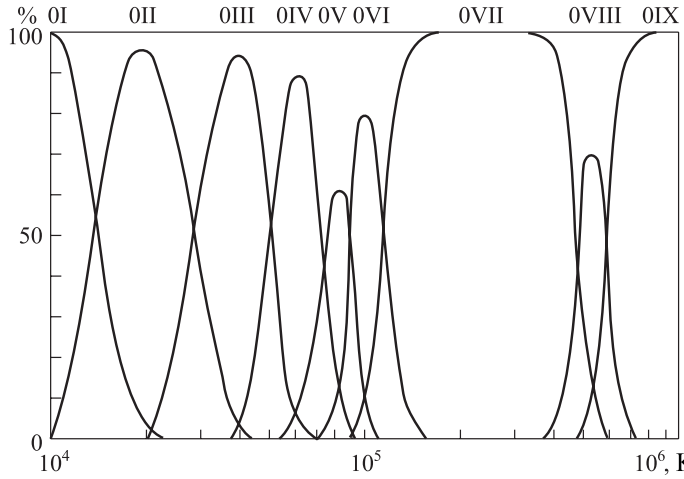


Fig. 6.5.1. Dependence of the concentration of the oxygen ions of different degree on electronic temperature at $n_e = 10^{17}$ cm^{-3}, calculated assuming that the Saha equilibrium is established.

ionisation. Therefore

$$\varepsilon = \frac{3}{2}\frac{(1+\bar{Z})}{M}kT + \frac{\bar{Q}(\bar{Z})}{M} + \bar{W} \tag{6.5.29}$$

Here $\bar{Q}(\bar{Z})$ is the sum of the losses through separation from the atom of $\bar{Z}$ electrons, and $\bar{W}$ is the excitation energy.

Electrical and heat conductivity of transforming plasma

In addition to the thermodynamic functions p (n, T) and $\varepsilon(T)$, we must know the dependence of electrical and heat conductivity of the plasma on density and temperature. The general equation for classic electrical conductivity has the form

$$\sigma = \frac{e^2 n_0 r_{ef}}{m}, \tag{6.5.30}$$

and

$$\frac{1}{r_{ef}} = \sum_{(\alpha)} \frac{1}{r_{e\alpha}}, \tag{6.5.31}$$

where summation is carried out with respect to α-types of particles.

At low temperatures, the conductivity id determined by the low concentration of the electrons and, in accordance with the Saha equation,

$$\sigma \sim \exp\left\{-\frac{I}{kT_e}\right\} \tag{6.5.32}$$

and this is confirmed quite efficiently by the experiments.

Starting at temperatures of T ~ 0.3–0.5 eV, Coulomb collisions of the electrons with the ions become an important factor. This transition period which extends to T~1.5 eV, and the conductivity then becomes 'Spitzerian'

$$\sigma \sim T^{3/2}. \tag{6.5.33}$$

In this case, $n_e\tau_{ei}$ is independent of n_e since $\tau_{ei} = \left(n_i \langle \sigma v \rangle_{\text{coul}}\right)^{-1}$, and also because of the quasi-neutrality $n_i \approx n_e$.

If hydrogen is not involved, ionisation starts subsequently. In this case, $n_e \sim Zn_i$, but the Coulomb cross-section $\sim Z^2/T^2$ and the dependence of σ on T become weaker

$$\sigma \sim T^{\ell}, \quad 0.5 < \ell < 1.5. \tag{6.5.34}$$

A suitable example of the dependence of σ on T_e, calculated using the Saha equation and formula (6.5.30), is shown in Fig. 6.5.2.

As regards heat conductivity, heat transfer in the absence of the magnetic field with efficient ionisation is carried out mostly by the electrons. The coefficient of this heat conductivity is

$$\chi_e = \frac{k^2 T}{e^2}\sigma$$

Here σ is Spitzer conductivity.

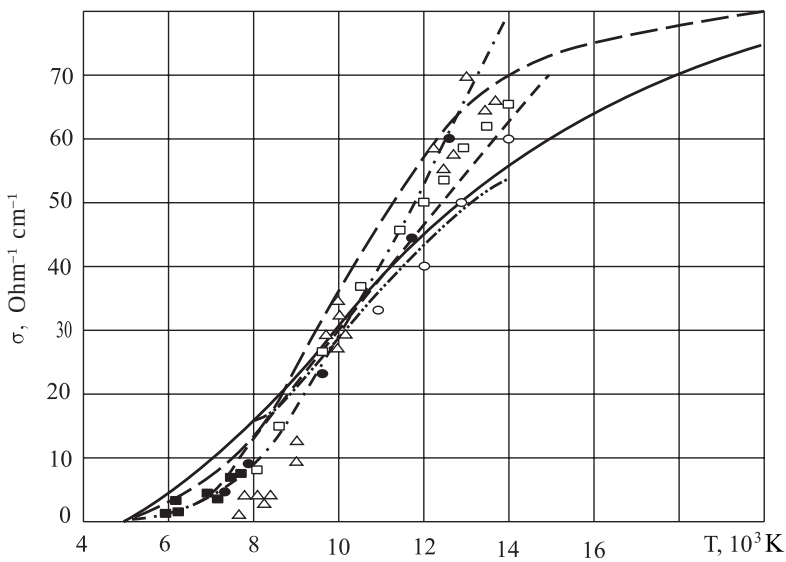


Fig. 6.5.2. Typical dependence of the coefficient of electrical conductivity of plasma on the electronic temperature: solid curve – calculated electrical conductivity of argon plasma; different symbols – experimental data published by different authors.

Application of the LTE model and Saha equation

Since this model relates to plasma with a relatively high electronic density, the criterion of applicability of LTE in a specific range of the atomic levels in the optically thin plasma is the condition for density

$$n_e \geqslant 10^{14} T^{1/2} (\Delta E)^3 = n_e^{\mathrm{cr}}, \tag{6.5.36}$$

where E is the maximum difference between the energy levels in the spectrum of the atom, eV; T is temperature, eV. Table 6.8 gives the values of n_e^{cr} for Li, Cs, Hg and Xe.

If the plasma is optically dense in the resonance lines, the criterion of applicability of local thermodynamic equilibrium has the following form:

$$n_e \geq \frac{n^{\mathrm{cr}}}{Q(\kappa_0 l)}, \tag{6.5.37}$$

here

$$Q(\kappa_0 l) = \begin{cases} \dfrac{\pi}{4} & \kappa_0 l \sqrt{\pi \ln(\kappa_0 l)} - \text{Doppler profile} \\ \dfrac{1}{3}\sqrt{\pi \kappa_0 l} - \text{dispersion profile} \end{cases}$$

At relatively high temperatures, the atoms and ions are excited in accordance with the Boltzmann equation. Here, it is necessary to answer the question of the number of the levels which should be taken into account because an increase of the main quantum number n results in an increase of the statistical weight $g \sim n^2$.

However, it can easily be seen that the highly excited levels do not usually have any significant role. The binding energy of the remote electron – in the same example with hydrogen – decreases by $\sim 1/n^2$.

Therefore, at the ionisation potential I ~15 eV at n ~10, the binding energy becomes equal to ϵ_{10} ~ 0.15 eV, which corresponds

Table 6.8

Element	T = 1 eV	T = 3 eV	T =5 eV	T = 10 eV	T = 30 eV
Li	$6.2 \cdot 10^{14}$	10^{15}	$1.4 \cdot 10^{15}$	$1.9 \cdot 10^{15}$	$3.4 \cdot 10^{15}$
Cs	$2.7 \cdot 10^{14}$	$4.7 \cdot 10^{14}$	$6 \cdot 10^{14}$	$8.5 \cdot 10^{14}$	$1.5 \cdot 10^{15}$
Xe	$5.6 \cdot 10^{16}$	9.710^{16}	$1.2 \cdot 10^{17}$	$1.8 \cdot 10^{17}$	$3.1 \cdot 10^{17}$
Hg	$3.5 \cdot 10^{16}$	$6 \cdot 10^{16}$	$8 \cdot 10^{16}$	$1.1 \cdot 10^{17}$	$1.4 \cdot 10^{17}$

to a temperature of only ~1700 K. However, it is often important to take into account the several first excitation levels. Therefore, the conversion using the Saha equation at very high densities should be carried out carefully.

The essential condition for the validity of the Saha equation is the inequality

$$n_e \alpha^{(el)} \gg \alpha_{\text{rad}}.$$

Thus, the equilibrium side equation is the zero term of expansion with respect to the parameter

$$\frac{\alpha_{\text{rad}}}{n_e \alpha^{(el)}} \ll 1.$$

This condition can be presented in the form

$$n_e^{\min} >\sim 10^{17}\,\text{cm}^{-3}. \tag{6.5.38}$$

If the reversed condition is satisfied (this is the case in the systems with low density), we also transfer to the stationary but nonequilibrium coronal distribution.

6.5.6. Diffusion approximation

We now transfer to the case of relatively dense plasma when the time between two collisions of a heavy particle with the electrons is considerably shorter than the radiation lifetime in the excited state. If it is also assumed that $kT \ll eI$, multi-stage processes will prevail in this plasma (Fig. 6.5.3). It is therefore natural to investigate in these conditions the processes of excitation, recombination (ionisation) as the process of diffusion of the electron in the energy space (Fig.

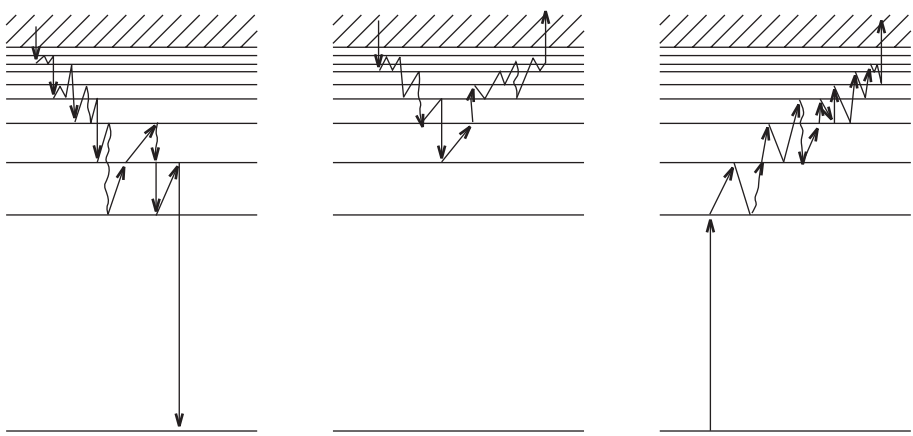


Fig. 6.5.3. Typical trajectories of wandering of the electrons on levels.

6.5.3). Evidently, this concept was accepted by Kramers and was later successfully applied by Budker and Belyaev in investigating the recombination process. The diffusibility of the process is caused by the fact that the probability of the collisional transition of the bonded electron with a large change of the quantum number is small, since $K_{nm} \sim (E_n - E_m)^{-4}$. Consequently, the electron energy changes only slightly on average during several collisions and, in these conditions, the kinetic equation can be reduced to the Fokker–Planck equation. Here, it is assumed that the spectrum of negative[10] energies is continuous. This approach can be used to calculate the recombination coefficient in a dense gas when the longest time, required for recombination, is spent by the electron on the highly excited levels. Since the structure of the levels of different atoms in this energy range does not differ greatly, the recombination coefficient is almost completely independent of the type of ion. For more accurate description, the authors of [142] propose the so-called modification diffusion approximation which describes the wandering of the electrons between the levels of the atom as the process of diffusion using the Fokker–Planck equation, but already in the discrete (not quasi-continuous) space.

6.5.7. Equations of dynamics of quasi-equilibrium plasma [27, 145]

If the LTE is established efficiently in the plasma volume and the relative velocities of the heavy components (in particular, with different Z) in relation to each other are negligible, the dynamics of such plasma can be described by the single- of two-fluid hydrodynamics, assuming that the temperatures of all the components of the same. Without taking into account the viscosity in the single-fluid approximation we have

$$\frac{\partial \rho}{\partial t} + \operatorname{div} \rho \mathbf{v} = 0;$$

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \frac{1}{c}[\mathbf{j}, \mathbf{H}];$$

$$\frac{\partial}{\partial t}\left(\varepsilon + \frac{\rho v^2}{2} + \frac{H^2}{8\pi} \right) + \operatorname{div}\left(\rho \mathbf{v}\left(\frac{v^2}{2} + w \right) + \mathbf{\Pi} + \mathbf{S} \right) = \operatorname{div} \varkappa \nabla T; \qquad (6.5.39)$$

$$\frac{\mathbf{j}}{\sigma} = \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right] + \mathbf{E} + \frac{\nabla p_e}{e n_e};$$

[10] i.e. corresponding to the bonded states.

$$\mathbf{\Pi} = \frac{c}{4\pi}\left[\mathbf{E},\mathbf{H}\right] = \frac{c}{4\pi}\left[\left(\frac{j}{\sigma} - \frac{1}{c}\left[\mathbf{v},\mathbf{H}\right] - \frac{\nabla p_e}{e n_e}\right),\mathbf{H}\right];$$

$$\operatorname{rot}\mathbf{H} = \frac{4\pi}{c}\mathbf{j}; \quad \operatorname{rot}\mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}; \quad \operatorname{div}\mathbf{H} = 0.$$

Here $\mathbf{S} = -\frac{16\sigma_{S-B}}{3}T^3\nabla T$ is the radiant flux, $\rho = \sum_{(\alpha)} M_\alpha n_\alpha$ is the total mass density, the mass of the electrons can be ignored; $\mathbf{\Pi}$ is the Umov-Pointing vector, p_e and n_e are the pressure and density of the electrons, $p = p_e + n_i kT$, $\varepsilon = \frac{3}{2}kT\left(n_i + n_e\right) + \sum_{(\alpha)} n_\alpha Q_\alpha$, Q_α is the total energy required for the transfer of the particle from the ground state to state α, $w = \varepsilon + (n_i + n_e)\, kT$.

The system of equations should be supplemented by the Saha and Boltzmann equations and also by the boundary conditions. These conditions for the particles will be discussed in chapter 7. As regards the radiant flux, it has its own special features associated in particular with the fact that there is no thermodynamic equilibrium in the vicinity of the boundary of the plasma volume (which is assumed to be optically thick). The problem is discussed in [27].

The difference between the system of equations (6.5.39) and the magnetic Alfvén hydrodynamics (section 2.3.1) is reduced mainly to the fact that here the equation of state and the equation for internal energy have the form of (6.5.28), (6.5.29), and the coefficients of electrical conductivity σ and heat conductivity are governed by the equations (6.5.32) and (6.5.31) taking the Saha equations into account. Finally, radiant energy transfer appears in the system (6.5.39). However, even for the LTE conditions, and examining the optically semi-transparent plasma, it is necessary to use the Rossland path, and this parameter should be continuously corrected because of changes in density and temperature at every point of the flux.

6.6. Radiation value of the ion in the coronal model

After investigating and describing the main special features of transformation processes, we can now transfer to calculating the specific characteristics of a number of plasma dynamics systems. Specific calculations with detailed consideration of the structure of the absorption coefficient are extremely cumbersome. Therefore, with the exception of one example (shock waves), we confine ourselves to investigating the simple models.

We start with the evaluation of the 'radiation value of the ion' ε_R – the minimum required at the given n, T_e and the degree of transparence of the volume of the energy losses for the formation of a single ion[11]. The value ε_R is determined by the work of ionisation I and unavoidable losses through radiation R

$$\varepsilon_R = I + R. \tag{6.6.1}$$

The resultant expressions for ε_R in a general case requires solution of the self-consistent problem of the dynamics of neutral atoms, ions, electrons and radiation transfer, as discussed in the previous sections. Here, we derive equations for R in the case of coronal equilibrium with respect to excitation.

Radiation is assumed to be not locked ($\varkappa L$ <1).

In other words, in this case the electrons collide with the non-excited atoms, and any inelastic collision leads either to ionisation of the atom or to excitation of the atom from the ground level to level k and subsequent emission of a photon with the energy $\hbar\omega_{0k} = E_R - E_0$

It can easily be seen that in these conditions

$$\varepsilon_R = I + \frac{1}{\langle \sigma_{\text{ion}} v_e \rangle} \sum_k^N \hbar\omega_{0k} \langle \sigma_{0k} v_e \rangle. \tag{6.6.2}$$

Here N is the number of 'excited' levels. This is often the first excited level.

The cross sections, included in (6.6.2) can be determined using, for example, equation (6.2.3).

Formula (6.6.2) follows from simple considerations. In fact, let all N be excited from the ground state and it is assumed that the transition coefficient $\langle \sigma v_e \rangle_{0k}$ is available for every level. Consequently, the probability of excitation of the k-th level is

$$w_k = \frac{\langle \sigma v \rangle_{0k}}{\sum_{(i)}^{N+1} \langle \sigma v \rangle_{0i}} \tag{6.6.3}$$

Similarly, the probability of ionisation is

$$w_{\text{ion}} = \frac{\langle \sigma v \rangle_{\text{ion}}}{\sum_{(i)}^{N+1} \langle \sigma v \rangle_{0i}}. \tag{6.6.4}$$

The fact that summation is carried out to N +1 and not to N is associated

[11]The actual value of the ion ε_0, is usually considerably greater than ε_R. This is explained by the losses of ions and electrons on the walls, the heat conductivity of the plasma components, the losses through heating of auxiliary elements, etc.

with taking ionisation into account. This results in the expression (6.5.8) for the radiation losses in a single ionisation act

$$R = \frac{\sum^{N} h\omega_{0k} w_k}{w_{\text{ion}}}.$$

Adding the ionisation potential I to R gived the equation (6.5.8).

Value ε_R depends strongly on the type of f_e via σ_{0i} and, primarily, on the mean electron energy. In the simplest case when f_e is Maxwellian, information on ε_R is provided by Table 6.9. The table shows that if T_e is sufficiently high ($\geq$10 eV), then ε_R ~20 eV. This value is a very rational estimate of the actual conditions in systems in which the density of the neutral atoms is low (~10^{13} cm^{-3}), and the electronic temperature is of the order of the temperature mentioned previously. However, for the optically transparent systems with low T_e, the value of the ion greatly exceeds the ionisation potential.

In systems with a dense gas it is important to take into account the locking of radiation and collisions of the electrons with the radiation-excited atoms. Consequently, the value ε_R at not too low T_e approaches I.

6.7. Volume processes in stationary plasma thrusters (SPT) and their similarity laws [146, 147]

Several specific plasma dynamic systems with the exact description of the transformation processes will now be discussed. The first suitable example is a stationary plasma thruster (SPT), Fig. 6.7.1, which was discussed in introduction (I. 3. 4). The SPT has a number of important special features and several of these features will be mentioned. In particular, it is a system with a simple circuit (but not simple physics!), Fig. 6.7.1, which, because of the low density of the atoms (n ~10^{13} cm^{-3}) and ions (n_i ~10^{11}–10^{12} cm^{-3}) is described by a hybrid model. Its working volume operates in the flyby regime in relation to ionisation (i.e. without recombination[12]), and excitation and radiation correspond to the coronal model. In particular, the previously mentioned estimates of the radiation value of the ion ε_R can be used in this case.

SPTs played a significant role in the development of general plasma dynamics. As already mentioned in the introduction, the SPTs were the first systems in which it was possible to generate in a stable manner a relatively large volume superthermal electrostatic field and,

[12]Not taking into account the particles which recombine, falling on the walls of the channel.

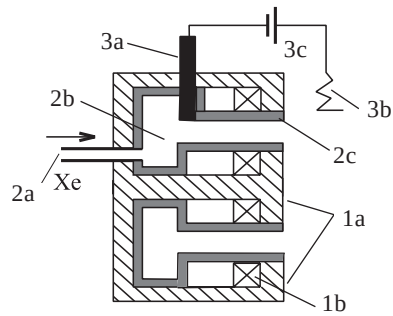


Fig. 6.7.1. Three blocks of the SPT.

consequently, realise almost completely Townsend classic conductivity and disprove the 'Bohm law' (section I. 3). This law was valid for about 20 years, or even longer, and it was believed by experts for more than 30 years (on the basis of primitive experiments) that volume superthermal **E**-fields cannot be produced in plasma.

Finally, it should be mentioned that the SPTs have been working for more than 30 years in systems in the USSR (Russia) and recently in France and the USA.

6.7.1. General characteristics of processes in the stationary plasma thruster

In section 3.2.2 it was noted that it is possible to produce plasma accelerators with poloidal (**E**, **H**) fields or, as is usually the case, 'accelerators with closed drift'. The design diagram of the first thruster of this type (SPT EOL) and the external appearance of a standard thruster M-100 can be found in section 10.4. Here, we discuss the main functional units of the SPT and describe the principal operating procedure. This will be followed by investigating the special features of the volume transformation processes in the SPT channel. The description of the physics of the processes in the SPT will be continued in chapter 7 in section 7.5.

Table 6.9

Element, I, eV	T = 1 eV	T = 3 eV	T = 5 eV	T = 10 eV	T = 15 eV	T = 30 eV
Li 5.4	1330	61	36	26	24	23
Cs 3.9	150	23	20	17	17,5	19
Hg 10.4	1167	100	39	25	21	18

The block scheme of the SPT

This scheme is shown in Fig. 6.7.1 and consists of three main blocks. The first block is the axisymmetric magnetic system, including the magnetic circuit (1a) with two poles: central and ring-shaped external, and also magnetisation coils (1b). The second block (gas-plasma) includes three elements: a gas supply pipe (2a), a gas distributor – buffer volume (2b) and the acceleration channel with dielectric walls (2c). The magnetic field, generated by the poles, intersects the accelerator channel, and the lines of force are almost radial.

The third block – the electrical circuit, contains three main elements: the cold anode (3a), thermal cathode (3b) with the flow of the particles of the working medium, and also the power source (3c) with the passive $Z = (L, C)$ elements, correcting oscillations in the discharge.

Functioning scheme

In the steady regime, the scheme may be described as follows (Fig. 6.7.2a). With the magnetic system and the electrical circuit activated, a quasi-radial magnetic field is generated in the channel and a longitudinal **E**-field forms between the anode and the cathode. In the crossed fields the electron drifts mainly in the azimuth, slowly travelling to the anode as a result of collisions with heavy particles and also with the walls of the channel, and also under the effect of random **E**-fields, formed as a result of some instabilities which are always present in plasma, especially during passage of a current.

The strength of the magnetic field increases from the anode to the outlet of the channel and at the maximum it is usually on the level of ~200–400. **The magnetic field with increasing strength is the fundamental property for operation of the SPT**. The field suppresses the strongest 'rotational' instability which leads to the formation of the azimuthal component of the **E**-field and, consequently, to the drift of the electrons from the cathode to the anode, i.e. to shorting. A rotational instability forms if the strength of the magnetic field decreases from the anode to the cathode, for more information see [44]

$$(\mathbf{v}\nabla)\frac{H}{n} > 0 \tag{6.7.1}$$

Neutral atoms (currently available thrusters use xenon) travel from the buffer into the acceleration channel where they are ionised in the cloud of the electrons drifting in the azimuth. The characteristic density of neutrals Xe at entry into the channel is $\sim 10^{13}$ cm^{-3} and of the ions in the channel $\sim 10^{12}$ cm^{-3}. At these values of the strength of the magnetic field and at the length of the channel $L \sim 1$–5 cm, strong inequalities usually

appear (ρ is the Larmor radius)

$$\rho_e \ll L \ll \rho_i, \tag{6.7.2}$$

i.e. the electrons are strongly magnetised and the ions almost do not feel the magnetic field. Therefore, the ion formed during ionisation, moves almost exclusively under the effect of the electrical field and its energy at exit from the channel is

$$\varepsilon = e\phi_*, \tag{6.7.3a}$$

where ϕ_* is the potential of the ion origination point.

In the single-stage SPTs which are discussed here, the mean ion energy at exit is

$$\overline{\varepsilon} = \theta e U_p \tag{6.7.3b}$$

where θ~2/3–3/4, and U_p is the voltage between the anode and the cathode.

The movement of the electrons, formed during ionisation – since $\rho_e \ll L$ – to the anode is a complicated multi-variant process. It will be discussed in chapter 7. The magnetic field whose strength increases in the direction to the outlet – since it is slightly distorted by currents in plasma and can be regarded as a vacuum field – has magnetic lines of force convex in the direction to the anode. Taking into account the equipotentialization of the magnetic lines of force (section 3.2.2) in plasma and the fact that the ions form with almost zero velocity, it can be seen that the ion flux, accelerated in the channel, is pushed away from the walls to the centre of the channel. The effect of 'thermalisation' of the electrical potential, described in section 3.2.2, inhibits the displacement of the ion flux from the walls but, nevertheless, the SPT can be optimised to ensure that the intensity of bombardment of the walls is minimum.

As regards the potential distribution along the channel, in the first approximation we can accept the 'isodrift' conditions, i.e. the constancy of u_E in the channel

$$u_E = c\frac{E}{H} = \text{const.} \tag{6.7.4}$$

This condition is efficiently confirmed by experiments outside the ionisation zone and indicates that the main part of the electronic component in the channel rotates as a solid unit. This fact is quite understandable since the strong dependence of u_E on z would support the development of the Rayleigh–Timofeev instability which leads to the 'isodrift' state.

To calculate the dynamics of the flux of the ionising heavy particles, it is also necessary to know the dependence $T_e(z)$. This value is required

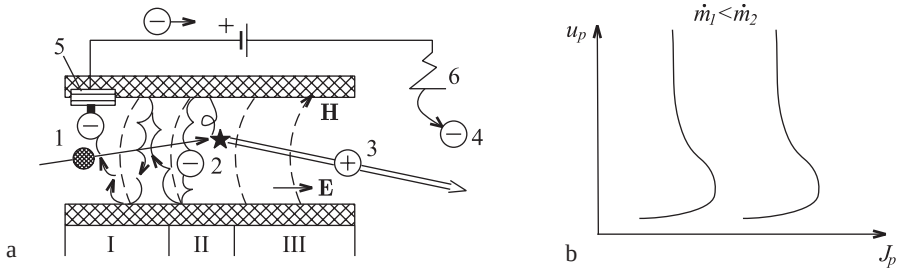


Fig. 6.7.2. Functioning scheme of the SPT (a): 1) neutral atom, 2) ionisation, 3) ion, 4) the electron, neutralising the charge of the departing ion, 5) the anode, I) the zone of weakly ionised plasma, II) the ionisation zone, III) the acceleration zone. The diagram of the volt–ampere characteristics of the SPT (b).

both for determining the ionisation and excitation coefficients and for taking into account the thermalisation effect of the potential. Unfortunately, it is not described by a simple finite equation and should be obtained from experiments. In chapter 7 we return to the experimental data and simulation of the dynamics of the electrons. It should only be mentioned here that in the currently available SPTs the characteristic value T_e is on average in the range $10 \le T_e < 20$ eV.

However, it should be remembered that the electron distribution function here is not Maxwellian (see paragraph 7.5.1). Knowing the scale of the concentration of the neutral atoms and the electrons, and also the electronic temperature, we can determine more accurately the special features of the transformation processes.

a. The free path of the Xe atom prior to ionisation is estimated from the equation

$$\lambda_{\text{ion}} = \frac{v_\alpha}{n_e \langle \sigma v_e \rangle_{\text{ion}}}.$$

Here $v_a \sim 10^4$ cm/s is the mean velocity of the xenon atoms along the channel at the temperature of the walls of the gas line $T \sim 300$°C; $n_e \sim 10^{12}$ cm^{-3}, and $\langle \sigma v_e \rangle_{\text{ion}} \sim 10^{-8}$ cm^3 s^{-1}, if $T_e \sim 15$ eV. Consequently, $\lambda_{\text{ion}} \approx 1$ cm.

Actually, in the experiments $\lambda_{\text{ion}} \sim 0.5$ cm. The difference between the estimated and experimental values of ion is explained by the arbitrariness in the numerical coefficients in estimating v_a, n_e, $\langle \tau v \rangle_{\text{ion}}$. The estimates presented in section 6.2.4 show that the recombination of the ions in the volume of the channel of the SPT is almost completely prevented.

b. Using these parameters, we can estimate the free path of the ion up to excitation during its movement in the acceleration channel. The estimates give $\lambda_{\text{exc}} \ge 10^2$ cm. The resultant values of λ_{ion} and λ_{exc}

unambiguously confirm the flyby regime of transformation of the particles in the SPT.

It is important to mention another relevant special feature of the passage of longitudinal current in the channel of the SPT. This is shown in Fig. 6.7.2a.

Three zones are indicated here: the vicinity of the anode (I), the ionisation zone (II) and the acceleration zone (III). The degree of ionisation in the vicinity of the anode is very small and the discharge current is transferred (between the ionisation zone and the anode) mainly by the electrons

$$J_p = J_I \approx J_e. \tag{6.7.5a}$$

Both the electrons and ions take part in the transfer of current in the ionisation zone

$$J_p = J_{II} = J_i + J_e. \tag{6.7.5b}$$

Finally, in the zone III, i.e. in the zone of main acceleration of the ions, the discharge current is transferred mainly by the ions

$$J_p = J_{III} \approx J_i. \tag{6.7.5c}$$

In an optimised thruster, the degree of ionisation of the working substance is close to 100%. This is efficiently confirmed by the experiments. Therefore, discharge current J_p is determined mainly by the flowrate and is equal to (Fig. 6.7.2b)

$$J_p = \frac{e\dot{m}}{M}(1+\alpha) + J_e^{(\mathrm{thr})}. \tag{6.7.6}$$

Here α is the fraction of the twice-ionised ions (α~10–20%), $\dot{m}$ is the mass of the working substance supplied into the channel in unit time.

The right-hand part of (6.7.6) also includes the 'through' electronic current which, passing continuously from the cathode to the anode, stabilises the discharge ($J_e^{(\mathrm{thr})} \sim 5\% J_p$).Evidently, for the normal operation of the SPT the path of the neutral atoms prior to ionisation should be relatively short. As this path decreases or, more accurately, as the dip of the potential along the free path become smaller, the monochromatic nature of the ions becomes more distinctive and the efficiency of the thruster increases.

The hybrid model of the SPT [146]

A suitable example of calculating the flyby regime with ionisation of the neutral particles is the hybrid one-dimensional model of the SPT with the feed electrical circuit for the latter, described below. In this model, the dynamics of the neutrals and the ions is described by the kinetic equations in which the right-hand part takes into account only

single ionisation

$$\frac{\partial f_0}{\partial t} + v\frac{\partial f_0}{\partial x} = -\beta(T_e) n_e f_0; \quad n_e = n_i = \int f_i dv;$$

$$\frac{\partial f_i}{\partial t} + v\frac{\partial f_i}{\partial x} + \frac{e}{M}E\frac{\partial f_i}{\partial v} = \beta(T_e) n_e f_0. \tag{6.7.7}$$

Taking into account the discussion of the functioning of the SPT, here we ignore the effect of the magnetic field on the ions. The dependence $\beta_e(T_e)$, selected with orientation to Xe, is simple

$$\beta(T) = \begin{cases} 0, & T < T_*; \\ \alpha(T - T_*), & T < T_*, \alpha = \text{const}. \end{cases} \tag{6.7.8}$$

The dynamics of the electrons is described in the ohmic approximation

$$\text{div } n\mathbf{v}_e = \beta n_0 n_e,$$
$$J - env_i = \sigma E. \tag{6.7.9}$$

The conductivity of the plasma was assumed to be classic with the accuracy to a single fitting coefficient (θ)

$$\sigma = \theta\frac{\sigma_0}{1 + \left(\frac{\sigma_0 H}{enc}\right)^2} \approx \frac{1}{H^2}. \tag{6.7.10}$$

This fitting coefficient θ is determined on the basis of the intensity of the electronic current between the ionisation zone and the outlet of the channel observed in the experiments.

Finally, the equations of dynamics of the heavy particles and the electrons are linked by the integral equation for the electrical circuit

$$\varepsilon = L\frac{dJ}{dt} + RJ + \int_0^L E\,dx. \tag{6.7.11}$$

Here ε is the EDF of the current source, L is the inductance of the circuit, R is the resistance of the circuit, and the integer describes the voltage applied to the thruster.

The system of the equations (6.7.7), (6.7.8), (6.7.9) was supplemented by the distribution of the magnetic field H (x) whose strength increases along the channel and is taken from the experiments, and also by the boundary conditions: the distribution functions of the neutrals and ions at entry into the channel:

$$f_0(v,0,t) = N\exp\left\{-\frac{v^2}{a^2}\right\}; \quad f_i(v,0,t) = 0. \tag{6.7.12}$$

The initial conditions were relatively arbitrary and after some time (time of pass scale) the process was no longer dependent on them. Calculations were carried out for different EDF (ε), flow rates ($\dot{m}$), conductivities of the channel (θ), and the strength of the magnetic field at the outlet of the channel H_0. The calculation results proved to be very interesting. The parameter having the strongest effect on the qualitative pattern of the process is the conductivity of the channel, i.e. the value θ.

If conductivity was high, the flow rapidly reached the stationary level, i.e. all the parameters ceased to depend on time. Figure 6.7.3a shows the dependence on x of the densities of the neutrals and ions, and also of the mean velocity of the ions.

With a decrease of conductivity (θ), the process in the channel (and, correspondingly, in the electrical circuit) changes from stationary to periodic, and the amplitude of the oscillations increases and the shape of the 'oscillograms' moves more and more away from sinusoidal (Fig. 6.7.4a). Finally, at even lower θ, the oscillations become random, although certain quasi-periodicity is retained. The form of the oscillograms becomes sensitive to the smallest variations of any parameter (Fig. 6.7.4b). The nature of evolution with time of the distribution of the plasma parameters in the channel for a single 'period' of the oscillations is shown in Fig. 6.7.5. The following should be added to this. With other parameters being constant, the variation of EMF results in a change of the discharge current which is similar to the experimentally detected change and, if the current oscillations are averaged out, the dependence $\overline{J}(\varepsilon)$ 'senses' the oscillations only slightly (within the limits of several percent). This can also be said of efficiency. The strong dependence of the nature of oscillations on the small variation of the parameters, detected in the calculations, is also efficiently confirmed by the experiments.

Similarity criteria of discharges in the SPT

The circuit of the SPT permits a large number of realisations. The dimensions of the thrusters can vary greatly, the thrusters can operate at different voltages and flowrate and also with different substances. It is therefore necessary to answer the question of the criterion (or criteria) which can be used in comparing the working processes in different accelerators (SPT).

The complete similarity of the processes in two accelerators can be naturally achieved only if the values of a relatively large number of dimensionless similarity criteria are identical. However, there is a single criterion which has a strong effect on the characteristics of the SPT. This is the Melikov–Morozov criterion which has been confirmed many

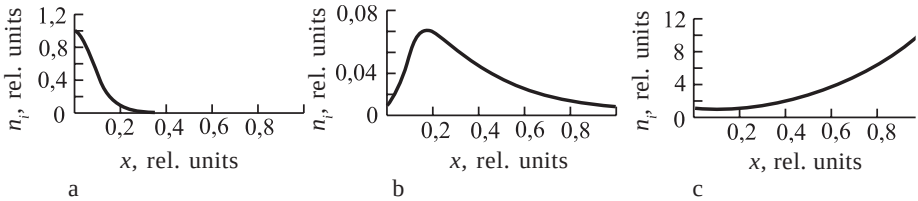


Fig. 6.7.3. Distribution of the parameters in the channel of the SPT in the steady regime.

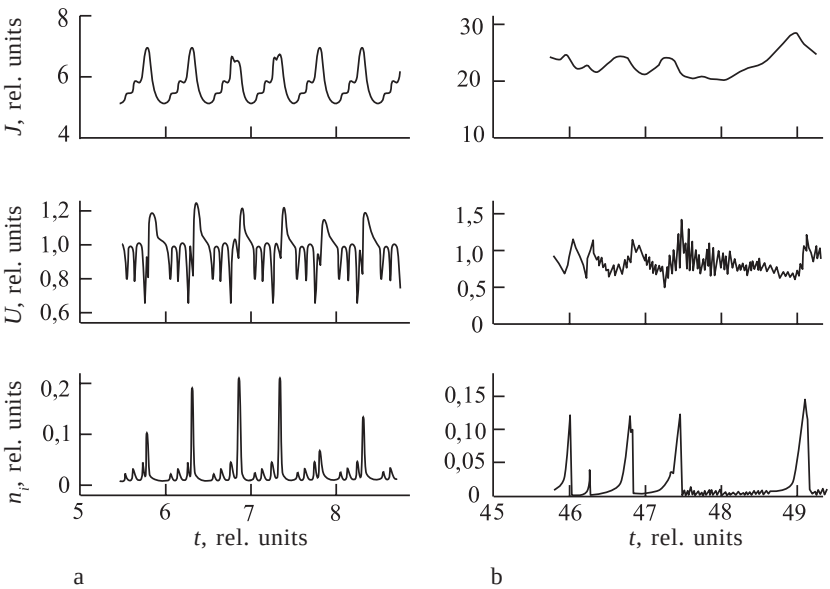


Fig. 6.7.4. Non-stationary flows in the SPT at low conductivity of the channel and different discharge currents (a) and (b).

times by experiments [6] and subsequently generalised by Bugrova and Maslennikov [146]. The criterion was derived from the analysis of kinetic equations describing the stationary dynamics of the neutrals and ions with ionisation taken into account. This system is identical with (6.7.7) and has the form (in the one-dimensional case)

$$v\frac{\partial f_i}{\partial x}+\frac{eE}{M}\frac{\partial f_i}{\partial v}=\beta n_e f_0$$

$$v\frac{\partial f_0}{\partial x}=-\beta n_e f_0 \qquad (6.7.13)$$

Here $\beta \equiv \langle \sigma v \rangle_{\text{ion}}$ is the ionisation coefficient.

The similarity criteria can be obtained if we make system 6.7.13 dimensionless. The length of the channel L, the density of incoming gas n_{a0}, its velocity v_{a0}, quantity $E_0=U_p/L$ and characteristic value β_0

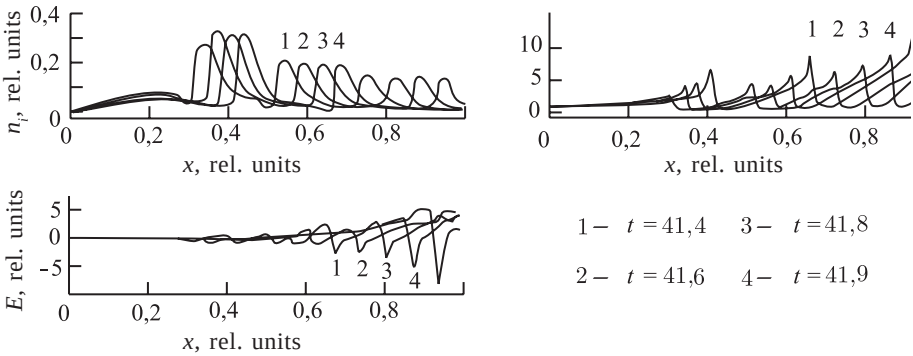


Fig. 6.7.5. Evolution of the plasma parameters along the length of the channel during a single oscillation 'period'.

are taken as the scale of length, density, electric field and ionisation correspondingly, then system 6.7.13 can be written as follows:

$$v\frac{\partial f_i}{\partial x}+\frac{1}{\xi}E\frac{\partial f_i}{\partial v}=\frac{1}{\varsigma}\beta n_e f_0;$$
$$v\frac{\partial f_0}{\partial x}=\frac{1}{\varsigma}\beta n_e f_0. \qquad (6.7.14)$$

Here, the dimensionless quantities are denoted by the symbols used previously. The system contains two dimensionless similarity parameters: the analogue of the Froode number and the analogue of the Knudsen number ζ:

$$\xi=\frac{Mv_{a0}^2}{eU_p},\quad \varsigma=\frac{v_{a0}}{\beta_0 n_0 L} \qquad (6.7.15)$$

Analysis of the experiment shows that the combination ($q_0 \equiv n_0 v_{0a}$)

$$\mu\equiv\frac{\varsigma^2}{\xi}=\frac{eU_p v_{a0}^2}{M\beta_0^2 q_0^2 L^2} \qquad (6.7.16)$$

has the strongest the effect on the efficiency of the thruster. This parameter is in fact the square of the ratio of the scale of the free path prior to ionisation λ_{ion} to the length of the working zone of the channel L

$$\mu=\left\{\frac{\{\lambda_{ion}\}}{L}\right\}^2. \qquad (6.7.17)$$

Here, the braces indicate the scale of the appropriate quantity.

In fact, the value λ_{ion} is determined by the characteristic values of the velocity of the neutrals v_a, the electron density n_e and the ionisation coefficient β:

$$\lambda_{ion} = \frac{v_a}{\beta n_e}. \tag{6.7.18}$$

Taking into account that $eU_p \sim Mv_i^2$, and $q = n_0 v_{0a} \sim n_i v_i$, $n_i = n_e$ and we obtain the estimate (6.7.18).

Evidently, the following conditions can be accepted as the criterion of normal burning of the discharge:

$$\mu \lesssim \mu_{max} = 1/10, \tag{6.7.19}$$

which shows that the path length prior to ionisation is the value λ_{ion} which is at least three times smaller than the length of the region of the strong **E**-field. This also shows that the transfer to the normal regime ($\mu < 1/10$) at constant discharge voltage U_p is determined in the first approximation by the mass flow rate $\dot{m}$ irrespective of the type of working substance because at a relatively high temperature (energy) of the electrons, the values of β for different gases are similar. In fact, the ratio (6.7.16) can be presented in the following form

$$\mu \sim \frac{eU_p kT_{0a}}{\left(\frac{\dot{m}}{2\pi R}\right)^2 \left(\frac{L}{b}\right)^2 \beta^2}. \tag{6.7.20}$$

Here b is the width of the channel, and usually $b/L \sim 1$, R is the mean radius of the channel.

Thus, for estimates it can be accepted that the minimum flowrate at which the SPT reaches the normal regime is given by the expression

$$\dot{m}_{min} = 2\pi R \sqrt{\frac{eU_p kT_{0a}}{\beta^2}}. \tag{6.7.21}$$

This estimate, with the accuracy $\leqslant 2$, is in agreement with the experiments carried out with different working substances and models.

6.8. Shock waves with radiation

6.8.1. Special features of shock waves with radiation

In chapter 2 we investigated a shock wave in conventional hydrodynamics using the conservation laws.

It was also noted that that the analysis of the structure of the shock wave using the Navier–Stokes equations, i.e. hydrodynamics with viscosity taken into account, and also using the Boltzmann equation shows that the width of the shock wave is of the order of several free path lengths. However, with increase of the velocity of the flow coming towards the obstacles, the situation with the structure of the shock wave starts to change greatly. The increase of temperature in the shock wave leads initially to excitation and dissociation and subsequently to the ionisation of the gas and the region of the shock wave starts to glow (at $T > 3000$ K). The fact that this glow forms already indicates that the region of the shock wave should expand and rearrange itself.

Up to the middle of the 1940s, strong shock waves, formed at high velocities of the flows hitting the obstacles, were only the subject of scientific interest as a result of the arrival of meteorites but the situation then changed. Atomic explosions, the movement of space systems in the atmospheres of the Earth, Venus and Mars and a number of other situations required detailed data on the properties of the strong shock waves.

The fundamentals of the theory of emitting shock waves in Russia were developed by Zel'dovich and Raizer [27] who developed an analytical model of such shock waves. In particular, they derived two dimensionless parameters, characterising the 'real' and 'visible' strength of the shock wave and, more accurately, the role of radiation. The first of them has the form

$$\chi = \frac{\sigma_{S-B} T_S^4}{q_h}. \qquad (6.8.1)$$

Here σ_{S-B} is the Stephan–Boltzmann constant, T_s is the temperature behind the front on the Gugonio adiabate, $q_h = \frac{1}{2}\rho_0 D U_s^2$ is the density of the hydrodynamic flow through the front of the shock wave, D is the velocity of the shock wave, U_s is the mass flow velocity of the gas behind the front, ρ_0 is the density of the gas in the rest state in front of them. If $\chi \ll 1$, the shock waves are referred to as subcritical, and if $\chi \leqslant 1$ they are critical, and at $\chi \gg 1$ they are supercritical.

However, the spectral composition of radiation inside and outside the shock wave and, even more so, at the distance of the order of one or several metres, greatly differs (see below). Therefore, another dimensionless parameter has been introduced

$$\chi_0 = q_0/q_h. \qquad (6.8.2)$$

Here q_0 is the density of the radiation flux, travelling to 'infinity', and q_h is the density of the radiation flux at the front of the shock wave.

The parameter

$$\chi = \chi_h - \chi_0$$

characterises the fraction of radiation absorbed ahead of the front and forming the so-called heated layer.

At present, the radiation shock waves are calculated in most cases by numerical methods [145].

Here, we describe the first numerical models of supercritical shock waves at high velocities of the incident flow, measured in tens of km/s (I.V. Nemchinov, et al, 1981 [149]). The interesting feature of these models is that they are relatively simple and were constructed 'almost completely accurately', i.e. taking into account ~4000 transitions between the levels and with sufficiently accurate calculations of radiation transfer. Of course, the calculations were performed for Xe because in this manner it was possible to avoid the large number of chemical reactions and the associated large increase of the number of transitions which would have to be taken into account. Although the shock wave in xenon is a relatively exotic object, nevertheless, in supercritical regimes the processes in Xe and in air are qualitatively similar. This is also supported by the fact that the structure of the wave on xenon, calculated in the 80s, is qualitatively close to the structure of the shock wave in air, approximately calculated (with the facilities available at that time) by Zel'dovich, et al in the 60s. The sufficiently universal structure of the shock wave in the atomic and simple molecular gases at high flow velocities is also indicated by experimental and numerical investigations carried out at a later date.

6.8.2. Results of calculations [149]

The calculations of the radiation-gas dynamics were carried out at piston velocities of $3 < U < 40$ km/s and at the density of Xe equal to 1/30 of the density in the normal conditions.

At the piston velocity of 3–5 km/s, the parameter $\chi = 0.3–0.45$, at the velocity $U = 10$ km/s $\chi \approx 2$. Thus, both the subcritical and supercritical shock waves were analysed in the calculations.

Calculations were carried out using the detailed tables of the spectral coefficients of absorption[13] and the thermodynamic properties of Xe.

One of the first results of the calculations was more accurate termination of the absorption of ultraviolet radiation in the actual conditions in front of the shock wave. Previously, it was assumed that the front of the shock wave emits 'forward' photons with the energy $\varepsilon < \varepsilon_* = I_1$ as the absolutely blackbody. Here I_1 is the first ionisation

[13]Taking into account the dominant Stark mechanism of broadening of spectral lines.

potential of a cold noble gas. At the same time, the photons with the energy $\varepsilon > \varepsilon_*$ are rapidly absorbed and are not visible on the side of the cold gas already at small distances from the shock wave.

However, these calculations show that not only the photons with $\varepsilon > \varepsilon_*$ but also the 'cut out' part of radiation $\varepsilon < \varepsilon''_*$ are not found in the spectrum of the radiation leaving the shock wave. This is associated with the broadening of the absorption profile of the line in the region of the gas heated in front of the shock wave. But the calculation showed that the effective transmission boundary ε_* for helium is 21 eV, for neon it is 20 eV, whereas $I_1 \approx 24.6$ eV and 21.6 eV, respectively.

In order not to review of the results obtained by them, we shall simply cite the text from [149], omitting individual fragments and transfer to our numeration of the figures.

Figure 6.8.1 shows the distribution of temperature T and the density of radiation flux q at distance x, counted from the piston at the piston velocity $U_1 = 10$ km/s for different moments of time t. In this case, the parameter $\chi = 2.2$.

The hydrodynamic discontinuity propagates through the gas already heated by radiation. Therefore, a temperature peak forms directly behind the front.

In deeper layers, the gas rapidly cools down and the temperature reaches some constant value T_1 (in this case, $T_1 = 2.5$ eV). The maximum temperature at the peak T_+ (in the present case 5.1 eV) is considerably higher than the temperature on the shock adiabate T_S (3.9 eV). The temperature before the front T_- (3.6 eV) already reached almost T_S. At the investigated moment of time (20–30 μs), the values T_+, T_-, T_1 are almost constant. They greatly change only in the first 2–3 seconds after the start of movement when the shock-compressed gas was transparent. Here, the plasma behind the front is optically thick.

For the 'critical' amplitude wave ($U_1 = 5$ km/s) calculations give $T_- = 0.70$ eV, i.e. considerably smaller than $T_S = 1.8$ eV.

Table 6.10 gives the main parameters of the shock waves in Xe at $\rho_0 = 1/30\ \rho_L$ for different velocities U_1. Here t is time up to which calculations were carried out, ρ_L is the density of Xe in the normal conditions.

As shown by Table 6.10, the values of T_1 are approximately half the values of T_S at both $U_1 = 20$–30 km/s where the energy losses at infinity do not exceed 3–10% and also at $U_1 = 40$ km/s where they are generally negligible. The radiation flux q_0 at velocities of 20–40 km/s is approximately constant and equals 9–10 MW/cm^2, which corresponds to the temperature of an equivalent black emitter of 1.7–1.8 eV, whereas

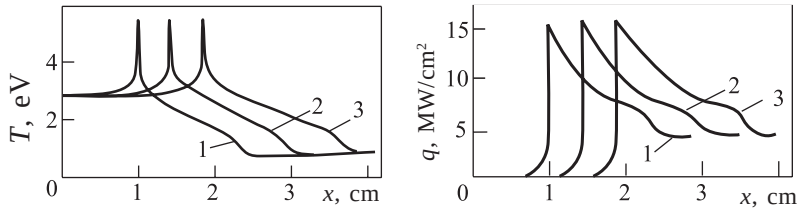


Fig. 6.8.1. Distribution of temperature and density of radiation flux in a supercritical shock wave: 1) t = 10 μs, 2) t = 23 μs, 3) t = 27 μs. The velocity of the piston 10 km/s.

Table 6.10. Values of parameters behind strong shock waves

U_1, km/s	T_S, eV	T_1, eV	T_-, eV	T_+, eV	g_h, MW/cm²	g_0, MW/cm²	χ_1	t, μs
3	1.15	1.0	0.10	1.15	0.28	0.13	0.63	200
5	1.8	1.4	0.70	1.8	1.5	0.45	0.8	70
10	3.9	2.5	3.6	5.1	11	3.6	2.2	35
15	6.2	4.3	5.5	8.4	40	5.8	4.5	20
20	10.8	5.3	6.7	11.1	100	9.0	7.4	10
30	15	7.5	11.7	19.7	270	9.0	21	6
40	21	9.0	12.3	27	600	10	34	2

the true temperature even for the piston is 5–9 eV. Therefore, a large part of the radiation emitted through 'the peak' belongs to quanta with the energies higher than the ionisation potential I_1 = 12.1 eV.

This is caused by the fact that although the parameters T_+, T_-, T_1 are almost constant with time, the thickness of the heated leayer rapidly increases – the problem is on the whole non-stationary. The radiation energy of the shock-compressed layer is used for the movement of the ionisation wave and for the expansion of the heated layer. This is shown in Fig. 6.8.2 which presents the dependence on time t of the thickness x_T of the heated layer at different piston velocities U_1. The boundary of the heated layer is represented by a point with the temperature of 1 eV. Thus, there is a relatively long stage in which the 'cork' of the shock-compressed layer is optically thick, the plasma parameters at the piston and in the vicinity of the shock wave front are almost constant, but the dimensions of the heated layer increase almost linearly with time.

In accordance with this description of the distribution of radiation temperature, the shock wave contains both low-energy photons with energy T and high-energy photons with energy T_1. The integral part of the flux of low-energy photons is close to the total power of the 'hard' photons.

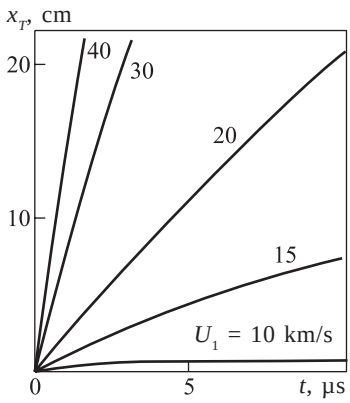


Fig. 6.8.2. Thickness of the heated layer as a function of time for different piston velocities.

However, the calculation show that approximately 30% of the emitted quanta belongs to the range $\varepsilon > 8$ eV, i.e. originated at the peak. The radiation spectrum, emitted by the temperature peak, is determined not only by the peak temperature but also by the behaviour of the spectral absorption coefficients – lines play a significant role.

The calculations determined the spectrum of exiting radiation for different velocities U_1. Figure 6.8.3 shows the dependence of the spectral brightness temperatures T_ε on the quantum energy ε, corresponding to the radiation emitted at a distance of 100 cm from the front for U_1 = 3; 10; 40 km/s at times of 200; 35; 2 μs, respectively, when the 'cork' of the shock-compressed gas is large.

It may be seen that at low velocities U_1 = 3–10 km/s and when T_ε is on average constant, 'valleys' form as a result of the absorption lines in the heated layer. At U_1 = 10 km/s 'avalanches' are already evident in infrared and ultraviolet ranges already at $\varepsilon < I_1$. At U_1 = 40 km/s, the screening of the radiation, emitted by the front, by the heated layer is efficient in very large areas of the spectrum. On the other hand, the radiation travelling large distances, is determined to a large extent by the heated layer.

The results of numerical investigations of the problem of the propagation of the emitting shock waves in air have confirmed the validity of the main assumptions presented previously and, at the same time, have made it possible to propose new qualitative special features of the shock waves which are not found during propagation of waves in Xe (and other inert gases). Thus, a two-region structure of the heated layer was found for the air which is a molecular gas. Analysis of the radiation spectrum and of the nature of variation of the group and integral (with respect to spectrum) one-sided radiation fluxes shows that the formation of the two-region structure is caused by the

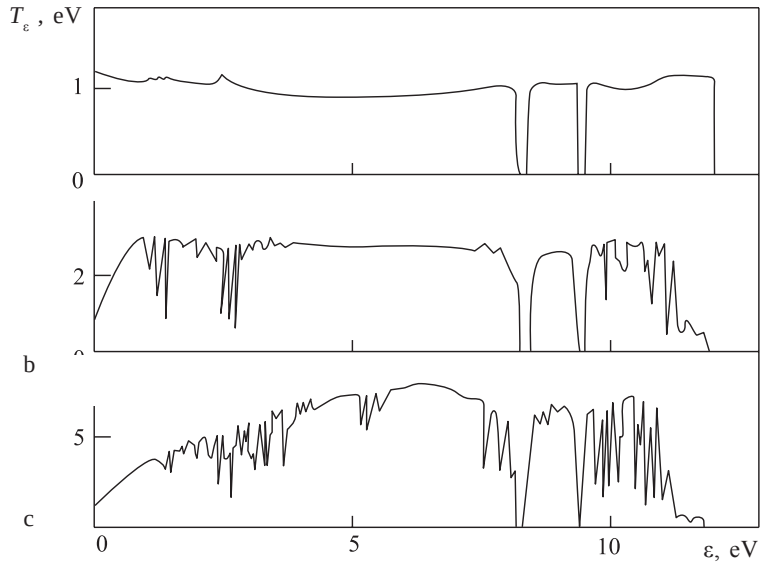


Fig. 6.8.3. Spectral brightness temperature at a distance of 100 cm from the shock front. The piston velocities: 3 (a), 10 (b), 40 km/s (c).

differences in the behaviour of the absorption coefficient in different regions of the spectrum with changes of temperature. The decrease of the absorption coefficient for the photons with the energy of 6.5–11 eV at temperatures of 0.7–0.9 eV, associated with the dissociation of the air molecules, leads to the formation of heating and clarification waves. The radiation with high quantum energies which is capable of photoionisation forms the hotter region adjacent to the front of the shock wave. A long, comparatively cold zone forms between the ionisation and dissociation waves.

This ends the investigation of the calculations of the shock waves, based on taking into account the large number of transitions and propagation of radiation associated with these transitions. In conclusion, it should be mentioned that we have calculated a large number of plasma systems with such definition of the process – this includes both the flight of meteorites, landing of space systems on the Earth and planets, laser CTS and atomic explosions.

6.8.3. Methods of calculating shock waves with the real spectrum [145]

There is a number of methods of calculating the dynamics of the emitting gas accompanied by the formation of the shock wave. However, all

these methods are based on the following 'reciprocating' scheme. Assuming LTE, the equations of state and the energy equation are initially calculated and, using quantum mechanics, the coefficient of absorption of radiation in the relevant range of density, temperature and frequency is also calculated

$$\rho(T,n),\ \in(T,\ n),\ \chi(\omega,\ Tn).$$

The results of calculations are transferred into a database.

Further, more or less rational distribution $n(\mathbf{x},\ 0)$, $T\ (\mathbf{x},\ 0)$ is defined at the initial moment and the equation of the radiant transfer and the appropriate boundary conditions are used to calculate the radiation field $I(\mathbf{x},\ 0;\ \omega,\ \Omega)$.

These calculations are carried out using a group method, i.e. the frequency range is divided – taking into account the specific features of $\varkappa\ (\omega)$ to the same or different but sufficiently large number N_ω of groups[14]. The same procedure is used for the angular distribution. Here, we introduce N_k groups. After determining distribution $I\ (\mathbf{x},\ 0;\ \omega_k,\ \mathbf{\Omega})$, we calculate that the contribution of radiation to plasma energetics (div $\mathbf{S} = q$). This is followed by the selection of the appropriate steps with respect to time and space and by calculations of the dynamics of the medium n, T, $\mathbf{v}$, taking into account absorption of radiation. After extensive rearrangement of the flow, we again solve the equation of transfer of radiation and determine

$$I_1 = I\ (\mathbf{x},\ t,\ \boldsymbol{\omega}_k,\ \mathbf{\Omega}_\kappa),$$

and this is again followed by the calculation of the hydrodynamics of the flow, etc.

6.9. Flows of ionising plasma in the coaxial [150]

The 'discontinuous' flows, caused by transformation, are encountered quite extensively in plasma dynamics systems. In the previous section we investigated one of such flow – strong shock waves. Another example are the ionisation fronts in dense plasma. Flows with such a front will now be discussed.

We also indicate the possibilities of considerable simplification of the model.

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[14]In the previously examined examples of the shock waves Xe $N_\omega \sim 600$. At present, calculations are carried out with $N \sim 10^4$–10^5 groups.

In chapter 2 we investigated the quasi-one-dimensional scheme of a stationary coaxial plasma accelerator. It was assumed that we are concerned with the acceleration of completely ionised ideal plasma. However, experiments are usually carried out using single-step accelerators of this type (Fig. 3.7.4) and, therefore, the neutral gas is supplied to the input into the channel. This gas can be slightly ionised by discharge radiation. Initial experiments in this equipment already show (Fig. 3.7.4) that a distinctive ionisation front forms in the flow and up to this time there was no glow from the flow and after this time the glow of the flow was quite strong, and the glow spectrum corresponds mostly to the singly-ionised nitrogen ions (if nitrogen was used as the working substance). In addition, high-velocity photography shows that the ionisation front is unstable. These instabilities are highly regular and can be of two types. The first type is associated with longitudinal oscillations of the front during discharge in nitrogen, and is shown in Fig. 3.7.8a. The second type (hydrogen) is the rotational ('spin') instability (Fig. 3.7.8b). The modelling of spin instability is the solution of a relatively complicated three-dimensional plasma dynamics problem. At the same time, the longitudinal instability can be modelled in quasi-one-dimensional approximation.

Formulation of the task

Here, we use the simplest method of describing the ionisation process – simply the rapid increase of conductivity with temperature. The flow of the gas and subsequently of plasma is investigated in the approximation of the narrow channel, as already mentioned in section 2.6.1. The channel is assumed to be flat and the magnetic field normal to the flow plane. Consequently, the system of equations for dimensionless quantities has the form:

$$\frac{\partial}{\partial t}(\rho f)+\frac{\partial}{\partial x}(\rho f v)=0;\quad \rho\frac{\partial}{\partial t}=-\frac{\partial}{\partial x}\left(p+\frac{H^2}{8\pi}\right);\quad p=\beta\rho T/2; \tag{6.9.1}$$

$$\frac{\beta\rho f}{2(\gamma-1)}\frac{dT}{dt}+p\frac{\partial}{\partial x}(vf)=f\nu\left(\frac{\partial H}{\partial x}\right)^2+q;\quad \frac{\partial}{\partial t}(Hf)+\frac{\partial}{\partial x}(Hvf)=\frac{\partial}{\partial x}\left(\nu f\frac{\partial H}{\partial x}\right).$$

Here, we have neglected viscosity and heat conductivity and also the Hall effect. As previously, f is the width of the investigated channel, ν is magnetic viscosity ($\nu \sim 1/\sigma$).

The ionisation process in this model is described by the following discontinuous dependence of conductivity on temperature

$$\sigma = \begin{cases} \sigma_0 & \text{at } T < T^*; \\ \sigma_1 + \sigma_2 T^{3/2} & \text{at } T > T^*, \end{cases} \tag{6.9.2}$$

The form of (6.9.2) shows that up to $T = T^*$ the gas remains slightly ionised and then it is rapidly ionised. Heat is used up during ionisation and, in order to reflect this in calculations, the term

$$q \equiv -I_0 \rho \exp\left\{-\alpha (T - T^*)^2\right\} \tag{6.9.3}$$

is added to the right-hand part of the energy equation. The exponent in (6.9.3) with relatively high α influences the process only in the vicinity of the value $T = T^*$, corresponding to ionisation. The equations (6.9.2) and (6.9.3) also restrict consideration of ionisation in the given model.

The calculations of the flow were carried out for a flat channel whose square section was given by the equations:

$$f(x) = \begin{cases} 0.3 - 0.8x(1-x) & \text{at } 0 \le x \le 1; \\ 1.5x - 1.2 & \text{at } 1 \le x \le 5. \end{cases} \tag{6.9.4}$$

Calculation results

In the cases in which $\sigma_0 \gg 1$, and the conductivity of the ionised gas is not high, i.e. the coefficients σ_1 and σ_2 in the equation (6.9.2) are of the order of unity, the flow is stabilised during the time of the order of flyby time. The stationary regime is characterised by relatively smooth functions (instead of discontinuous functions): T and v monotonically increase, and ρ and H monotonically decrease along the channel. Temperature continuously passes through the critical value T^* in the vicinity of the minimum channel cross-section.

The magnetic field H shows the strongest response to the ionisation process. The strength of the field rapidly decreases immediately after temperature T passing through T^*, which corresponds to the high density of electrical current in this area. Figure 6.9.1 shows the dependence of ρ, T, v and H on the distance x along the channel in the steady flow at the values of the parameters $\beta = 0.2$; $\sigma_0 = 0.05$; $\sigma_1 = \sigma_2 = 1$; $\varkappa = 0$; $T^* = 1.5$; $I = 0.3$.

With a decrease of σ_0 (in the calculations from 0.05 to 0.01) the flow starts to lose its stability and at $\sigma_0 = 0.01$ there are strong oscillations of temperature in the outlet part of the channel, i.e. on the falling section of the graph T (x). However, if the conductivity of the ionised gas is sufficiently high, the flow is generally not established and is of the quasi-periodic pulsing nature. At the same time, there are large

temperature variations. The position of the ionisation front varies with time in the vicinity of the minimum cross-section of the nozzle ($x \approx 0.5$).

Figure 6.9.2 shows the behaviour of temperature T at point $x = 0.7$ and the position of the ionisation front x_{fr} with time. Both figures (upper and lower) correspond to the shape of the channel (6.9.4) and the same parameters as in Fig. 6.9.1 with the exception of conductivity behind the ionisation front; here $\sigma_1 = \sigma_2 = 100$. The graph shows clearly the pulsating nature of the flow with a period $\Delta t \sim 0.6$. The amplitude of temperature oscillations increases with increase of σ_1 and σ_2 and also with a decrease of σ_0 at entry. Evidently, the formation of oscillations of the ionisation front is associated with the large difference between the velocities of the gas moving to the front and the plasma leaving the front. The state ahead of the front is close to stationary. Behind the front there is a narrow heated region with the oscillating temperature maximum. The heated conducting layers of plasma are periodically separated from the front and move to the right, cool down and spread in the direction of width. The strength of the magnetic field decreases in the direction from left to right to almost zero, and almost the entire drop of H is observed in the pulsating heating region behind the ionisation front. This corresponds to the situation in which the electrical current 'pierces' the channel in the heated area, i.e. the area with the highest conductivity.

This problem was also calculated by A.N. Kozlov [151] without introducing a conductivity discontinuity using simply the steep dependence of σ on T. Oscillations, identical with those shown in Fig. 6.9.2, also form here at the appropriate parameters.

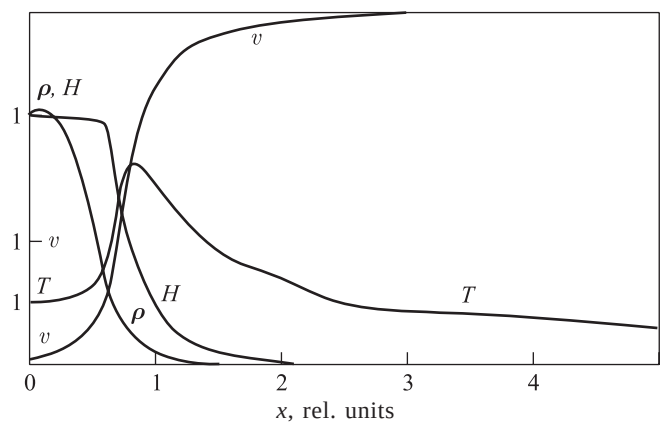


Fig. 6.9.1. Dependence of the flow parameters on the coordinate along the channel in the steady regime.

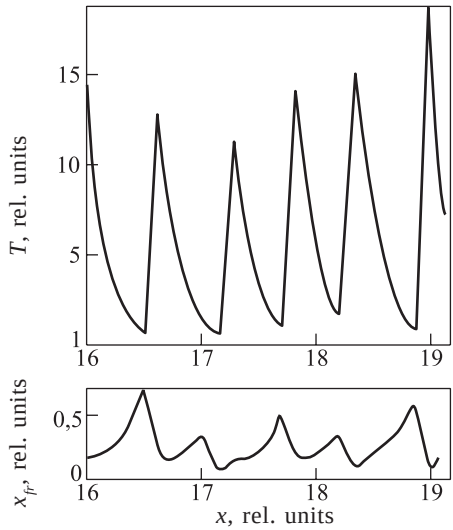


Fig. 6.9.2. Variations of temperature and the position of the ionisation front in the channel with respect to time in the non-stationary regime.

Effect of impurities on the stability of compression flows

Calculating two-dimensional compression flows, A.N. Kozlov showed[15] that the oscillations in the coaxial can be caused not only by the ionisation zone but also by strong volume radiation. For example, this takes place if heavy impurities (N_2, O_2, Cu, etc.), whose radiation power is $\sim Z^2$, where Z is the ion charge, are added to the slightly radiating hydrogen plasma. Consequently, the compression zone starts to cool down but this stimulates further compression of the flow and consequently increases the intensity of radiation. All this generates an instability - a discontinuity of the compression zone for a short period of time, with the zone later restored, etc.

6.10. Glow and arc discharges

The investigations of the plasma dynamics of space objects, the development of thermonuclear reactors and plasma accelerators, and also the construction of other completely new plasma systems may indicate that the classic discharges such as glow, arc, spark, etc, have been studied sufficiently and their possibilities have been exhausted.

[15]A.N. Kozlov, Calculation of two-dimensional flows of the ionised gas and plasma in a coaxial accelerator taking into account heat conductivity, radiation processes in the electrical circuit, Preprint No. 32, M.V. Keldysh IPM, Academy of the Sciences of the USSR, 1986.

However, this view is completely wrong. It can be suggested that any type of plasma system is inexhaustible both with respect to its physical organisation and applications. A suitable confirmation of the importance of the 'good old discharges' is the frequent publication of books dealing with the subject (see, for example, [5], [152–158]), without mentioning a large number of articles in journals.

Therefore, it is justified to pay special attention to the fundamentals of the physics of the most widely encountered types of discharge. Here, we consider the glow (GD) and arc (AD) discharges which together with the spark represent the basis for the development of the physics of gas discharge and, subsequently, until the beginning of the 1940s, plasma physics.

Although the arc discharge was discovered in the first decade of the 19th century and the glow discharge in the 30s of the same century, a large number of their most important features was explained only in the second half of the 20th century. This relates primarily to near-electrode regions and the stability of their plasma configurations. These advances were possible as a result of the increase of the general level of 'plasma thinking', the development of powerful computing procedures, which enabled solution of the systems of complicated integro-differential equation describing the dynamics and transformation of the particles and radiation transfer.

The second factor because of which there has been special interest in the old types of discharge is the development of new areas of applications, such as lasers, plasma chemistry, thermoemission converters, plasma technologies of surfaces, etc. These directions have imposed new requirements on the working parameters of the gas-discharge systems, such as the scale, power and stability of the plasma. The construction of these devices requires a large number of investigations.

In the following section we discuss the general properties of the glow and arc discharges and especially the positive columns of these discharges. Near-electrode processes in these discharges will be discussed in section 7.6 of chapter 7.

6.10.1. General characteristics of glow and arc discharges

Both these discharges are based on the electrodes, with the voltage between the electrodes being constant, and this will also be assumed in our considerations.

1. The glow discharge (GD) is a discharge formed in most cases at pressures of ~0.1–10 torr, cold electrodes[16], and the voltage between the electrodes of the order of several hundreds of volts. Characteristic current density on the cathode of the 'normal' GD is ~10 mA/cm^2. In classic studies, the glow discharge is usually investigated in cylindrical glass tubes with a diameter of ~2 cm which have greatly differing shapes if used for advertising, and now glow discharge is used as the active medium of lasers, using vessels in the form of parallelepipeds. Here we consider the discharge in straight tubes.

In most cases, the glow discharge shows three distinctive zones (Fig. 6.10.1a): the near-cathode zone, the bright 'positive' column (PC), and the near-anode layer. The key physical factor which determines the specific features of the glow discharge is the mechanism of passage of current through the near-cathode layer. This problem is discussed in greater detail in chapter 7, and here we only mention that the current is transferred mainly by the ions which acquire energy in the near-cathode potential gradient and bombard the cathode, causing electron emission.

A special feature of the glow discharge is the unique 'autonomy' of the near-electrode layers. In particular, if the distance between the electrodes is reduced, this results only in a change of the length of the PC, whereas the near-electrode layers remain unchanged. The discharge parameters also remain unchanged. It should be mentioned that the positive column of the glow discharge in the case in which noble gases represent the working substance is often laminated (Fig. 6.10.1b), and these layers are referred to as stratas. The theory of the stratas is relatively delicate. After many unsuccessful attempts in a large number of investigators, its fundamentals were defined by A.V. Nedospasov in [154].

In the following section we examine in greater detail the properties of the stationary PC of glow discharge.

2. The arc discharge (AD) – the electrode discharge at relatively high pressures ($p \geqslant 10$ torr) up to tens or more atmospheres, and its external features are also the presence of three zones: near-cathode, positive column and near-anode layer. However, the organisation of the processes in these zones in the arc discharge is completely different in comparison with the glow discharge and the plasma parameters and VAC also differ correspondingly. Actually, in contrast to the glow discharge, the arc discharge is a low-voltage discharge. The characteristic voltage is 15–30 V. The arc discharge is a high

[16]In luminescent lamps ('daylight lamps) the 'hybrid' local discharge is utilised: the cathode is heated but the discharge current is low and the positive column is the same as in glow discharge.

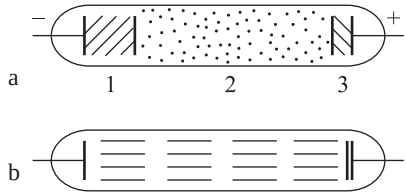


Fig. 6.10.1. Glow discharge zones (a), 1 – near-cathode zone, 2 – positive column, 3 – near-anode zone; the scheme of glow discharge with a stratified positive column (b).

current discharge, its discharge current is $J_p \geqslant 1$A and may reach many thousands of amperes.

The main special feature of the arc discharge is the electronic emission from the cathode. This is realised in two forms. Either the cathode is heated in some manner (i.e. as a result of external factors or by the discharge itself) to high temperatures at which the cathode becomes a thermal emitter, or highly constricted plasma appears on the cathode and this takes place on cold cathodes so that auto-emission starts to operate, or 'explosion emission' occurs in most cases (section 7.2.4).

The positive columns also differ greatly. In glow discharge, the PC is thermodynamically non-equilibrium, and in AD it is in thermodynamic equilibrium with a high degree of accuracy.

Information on the ranges of the parameters (p, U_p) in which glow discharge forms, will be presented in chapter 7 when discussing near-electrode regions of the glow discharge.

The positive column of the glow discharge
Main experimental data

The properties of the PC of the GD are almost completely independent of the length of the PC (we consider a fragment of the PC – a cylinder with the radius equal to the internal radius of a glass tube of more or less arbitrary length; the length of he cylinder is many times greater than the free path);

- electronic temperature in the PC is $T_e \sim 2$ eV, whereas the ionic temperature t_i is of the order of room temperature ($T_i \sim 0.03$ eV);
- the density of the electrons $n_e \lesssim 10^{10}$ cm^{-3} and, correspondingly, the degree of ionisation $\alpha = n_e/n_0 \lesssim 10^{-6}$ at the density of the atoms of $n_0 \gtrsim 10^{16}$ cm^{-3}.

Thus, it can be seen that the plasma in the TC is greatly non-equilibrium because there is a large discontinuity in the values of the

electronic and ionic temperatures and, in addition to this, the degree of ionisation does not correspond, at the given value of T_e, to the value obtained using the Saha equation, with all the reservations which can be made here. This is determined by three main processes:

– heating of the electrons in the electrical field takes place with frequent scattering on the neutral atoms (described by the Boltzmann–Davydov kinetic equation);
– ambipolar diffusion of the electrons and ions from the volume to the tube walls where they also recombine;
– effective (as a result of the fact that the masses are similar) transfer of the energy of the heavy particles of the ions and atoms to each other and transfer of energy to the wall.

The generally accepted term 'ambiolar diffusion' used here refers to the diffusion in which the ions and electrons are transferred at the same rate, i.e. it is 'currentless' diffusion[17]

$$\mathbf{j} = \mathbf{j}_i - \mathbf{j}_e = 0$$

The formal aspects of this matter will be now clarified.

A special feature of the dynamics of the electrons and ions in the case of low ionisation and comparatively high concentration of the neutrals is the short free path of these particles and the possibility of hydrodynamic description of their dynamics. 'Strong' friction of these components on the neutrals enables us to ignore the inertial members and the Euler equations have the form ($n_i \approx n_e = n$)

$$\begin{aligned} 0 &= -\nabla p_i + en\mathbf{E} - \nu_{ia}\mathbf{v}_i Mn \\ 0 &= -\nabla p_e - en\mathbf{E} - \nu_{ea}\mathbf{v}_e mn. \end{aligned} \qquad (6.10.1)$$

Here $\mathbf{v}_i$ and $\mathbf{v}_e$ are counted from the velocity of the neutrals, ν_{ia} and ν_{ea} are the frequencies of collisions of the charged particles with the atoms.

This shows that to fulfil the condition $\mathbf{v}_{ir} = \mathbf{v}_{er} = \mathbf{v}_r$ it is necessary to generates a radial 'ambipolar' electrical field determined from the equation

$$-\frac{\nabla_r p_i}{Mn\nu_i} + \frac{en}{\nu_i}\frac{\mathbf{E}_{am,r}}{Mn} = -\frac{\nabla_r p_e}{mn\nu_i} - \frac{en}{\nu_e}\frac{\mathbf{E}_{am,r}}{mn},$$

and equal to

[17]Any flux of quasi-neutral plasma can be referred to as ambipolar or almost ambipolar. A suitable example is the expansion of a plasma cloud in vacuum which was discussed in the Introduction.

$$\mathbf{E}_{am,r} = -\frac{\left(\frac{\nabla_r p_e}{m\nu_e} - \frac{\nabla_r p_i}{\nu_i M}\right)}{en\left(\frac{1}{\nu_i M} + \frac{1}{\nu_e m}\right)} \tag{6.10.2}$$

where

$\nu_e \equiv \nu_{ea}$, $\nu_i \equiv \nu_{ia}$.

In the literature on gas discharge, the system (6.10.1) is written in different notations. This form is presented here ($n_i = n_e = n$)

$$\begin{aligned} n\mathbf{v}_i &\equiv \Gamma_i = -D_i\nabla n + \mu_i n\mathbf{E}; \\ n\mathbf{v}_e &\equiv \Gamma_e = -D_e\nabla n - \mu_e n\mathbf{E}. \end{aligned} \tag{6.10.3}$$

Evidently, assuming that the temperature gradients are not steep, we can write

$$\nabla p_i = kT_i\nabla n; \quad \nabla p_e = kT_e\nabla n$$

and, correspondingly, the diffusion coefficient will be equal to

$$D_i = \frac{kT_i}{M\nu_i}, \quad D_e = \frac{kT_e}{m\nu_e}, \tag{6.10.4a}$$

and ‘mobility “

$$\mu_i = \frac{e}{M\nu_i}, \quad \mu_e = \frac{e}{m\nu_e} \tag{6.10.4b}$$

Comparison of (6.10.4a) and (6.10.4b) yields the Einstein equations

$$\frac{D_i}{\mu_i} = \frac{kT_i}{e}; \quad \frac{D_e}{\mu_e} = \frac{kT_e}{e} \tag{6.10.5}$$

Using these notations, we can write

$$\mathbf{E}_{am,r} = -\frac{(D_i - D_e)}{n(\mu_i + \mu_e)}\nabla_r n \tag{6.10.6}$$

and, consequently

$$n\mathbf{v}_{i,r} = n\mathbf{v}_{e,r} \equiv \Gamma_{i,r} = \Gamma_{e,r} = -\frac{(D_e\mu_i + D_i\mu_e)}{\mu_i + \mu_e}\nabla_r n$$

The quantity

$$D_{am} \equiv \frac{D_e \mu_i + D_i \mu_e}{\mu_i + \mu_e} \tag{6.10.7}$$

is the ambipolar diffusion coefficient. This coefficient depends, through D_e, D_i, on the characteristic energy of the particles which can be referred to as the temperature with usual postulations. Therefore

$$n\mathbf{v}_{i,r} = n\mathbf{v}_{e,r} \equiv \Gamma = -D_{am}(T_i, T_e)\nabla_r n \tag{6.10.8}$$

It should be mentioned that here $n = n_e \approx n_i$.

Having the expression for the particle flux and using the continuity equations for the stationary regime, we obtain an equation for $n(r)$:

$$\mathrm{div}_r\, D_{am} \nabla_r n + \beta n_a n = 0. \tag{6.10.9}$$

Here $\beta \equiv \langle \sigma v_e \rangle_{\mathrm{ion}}$ is the ionisation coefficient which depends on T_e, and recombination the volume of the tube is ignored. Equation (6.10.9) must be solved with the appropriate boundary conditions.

The PC GD model

Engel and Steenbeck derived and studied for the first time the equation (6.10.9) with special reference to glow discharge (1930). They solved this equation for cylindrical geometry with the boundary conditions

$$\left.\frac{\partial n}{\partial r}\right|_{r=0} = 0; \quad n\big|_{r=R} = 0. \tag{6.10.10}$$

In this case, the solution of equation (6.10.9) is the Bessel function

$$n = n_0 J_0(\xi) \tag{6.10.11}$$

where $\xi = r\sqrt{\frac{\beta n_a}{D_{am}}}$, and n_0 is the concentration of the electrons on the tube axis. At $r = R$, i.e. on the walls, $n = 0$ and, consequently

$$\xi_R = R\sqrt{\frac{\beta_0 n_a}{D_{am}}} = \alpha_1. \tag{6.10.12}$$

Here $\alpha_1 = 2, 4$ is the first root of the Bessel function $J_0(\xi)$. Thus, the plasma parameters are linked with a radius R of the discharge tube by the relationship

$$\frac{D_{am}}{\beta_0} = n_a \cdot (2,4)^2\, R^2, \tag{6.10.13}$$

where n_a is the density of the neutral gas. The quantity in the left-hand

part of equation (6.10.13) depends on T_i and T_e. Since T_i is determined by 'room' temperature, the relationship (6.10.13) can be used to find T_e. It is assumed that the electron distribution is Maxwellian[18], and Engel and Steenbeck determined the relationship of T_e/I with Rp. Raizer modified this relationship to the universal form shown in Fig. 6.10.2, introducing, for every type of particle, the coefficient (c), equal to $4 \cdot 10^{-3}$ for He, $4 \cdot 10^{-3}$ for Ar, and $4 \cdot 10^{-2}$ for N_2 [153]. Here I is the ionisation potential. We can now also find the relationship of the strength of the longitudinal **E**-field in the tube with the electronic temperature T_e. This relationship is determined from the energy balance condition.

In fact, since every fragment of the PC can be regarded as autonomous, the energy generated in this fragment as a result of the **E**-field should be dissipated in the same fragment. In the simplest case, this dissipation takes place on the tube walls.

Therefore, for a 1 cm PC we can write the energy balance equation

$$EJ_p = 2\pi R (n_e v_r)\big|_{r=R} \cdot \varepsilon_R \qquad (6.10.14)$$

Here ε_R is the energy of the electrons hitting the walls, and J_p is the discharge current

$$J_p = e\int_0^R n\, v_z 2\pi r dr \sim E \qquad (6.10.15)$$

since $\frac{\partial n}{\partial z} = 0$, E = const, then because of (6.10.3) we have $nv_z = \mu_e nE$

Since the movement of the electronic component along the radius is diffusion movement, the quantity

$$(n\, v_r)\big|_{r=R} = -D \left.\frac{\partial n}{\partial r}\right|_{r=R}.$$

From (6.10.14) and (6.10.15) at the known values of T_e, T and $n(z)$ we determine the E values of the fields and the strength of the positive column of glow discharge.

The resultant equation for E is strange because it does not depend on the passing current and depends only on the pressure of the neutral gas p_a or, more accurately, on density n_a. This conclusion is satisfactorily confirmed by experiments (Fig. 6.10.3).

The behaviour of glow discharge with the change of the voltage between the electrodes in a wide range is non-trivial, as discussed in chapter 7.

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[18]It is more adequate to use the distribution function of the electrons determined by Davydov's kinetic equations but this has no qualitative effect on the conclusions

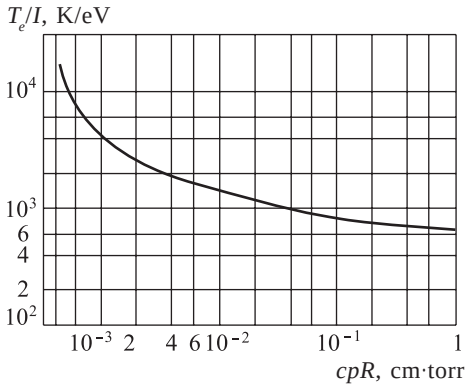


Fig. 6.10.2. Engel–Steenbeck–Raizer curve for the dependence of T_e/I on cpR.

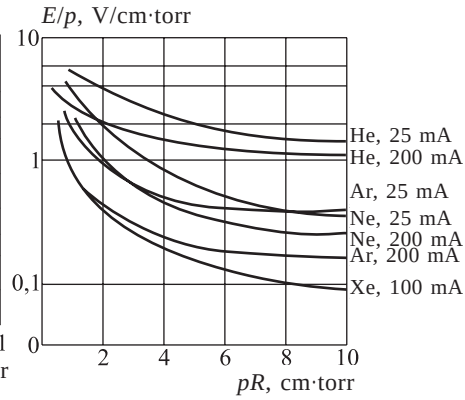


Fig. 6.10.3. Experimental dependence of E/p on pR in the positive column of glow discharge at different discharge current for He, Ne, Ar, Xe.

Radiation of PC GD

The structure of the radiation of PC GD is complicated. This is explained by the fact that the temperature of the neutral atoms is almost equal to room temperature and their fraction is considerably greater than that of the ions and, consequently, they are excited from the ground state and de-excited by radiation. The width of the appropriate spectral lines is close to the natural width and they are almost completely locked at $p \geqslant 0.1$ torr which usually occurs in the glow discharge. Therefore, to ensure the effective radiation of the glow discharge to the main gas, for example argon, mercury vapours are added. This was carried out, for example, in daylight lamps (section 10.2).

6.1.2. Positive column of arc stationary discharge

The increase of the gas density in the discharge gap or increase of voltage results in transition from normal glow discharge to anomalous glow discharge in subsequent arc discharge (AD). There are many types of arc discharge. They are associated with the difference in the cathodic processes, pressure, the form and dimensions of the vessels surrounding the arc, etc. However, the principle of PC AD can be expressed by a single phrase: ‘positive column of the arc discharge is the volume of thermally equilibrium plasma’.

Therefore, if the PC GD is described by a system of two equations: equation of the collection of energy by the electron from the external fields and the diffusion equation of the electrons and ions to the wall –

the recombination area, the theory of arc discharge with a low-intensity plasma flux is based on the energy (temperature) equation

$$c_v \rho \frac{\partial T}{\partial t} = \operatorname{div} \varkappa \nabla T + \frac{j^2}{\sigma} - I_R. \qquad (6.10.16)$$

Here I_R is the radiation energy yield. This equation, in its stationary variant, was used by Elenbass and Geller as a basis for analysis of the processes in arc discharge. Therefore, this discharge is often named after them. Everything else is determined by the laws of thermodynamic equilibrium (6.5.7) which have been verified by experiments for arc discharge. Therefore, the title of a classic book dealing with the arcs, written by Finkel'burg and Mekker, 'The electric arc and thermal plasma' [156] is not surprising.

Initially, we will not describe in detail the positive column of arc discharge because, on the one hand, the main aspects have already been mentioned and, on the other hand, the arcs are described in a large number of relatively available literature. It is necessary to mention several universal factors relating to the arcs burning in an unlimited atmosphere.

- The VAC (volt–ampere characteristic) of these arcs is drooping. Therefore, a ballast resistance is connected to the circuit to stabilise the discharge;
- at relatively high discharge current the intrinsic magnetic field of the arc starts to constrict the arc (pinch effect);
- in good agreement with the experiments, the calculations show that: the channel radius increases with increasing discharge current J_p, and current density increases only in the logarithmic manner; the plasma temperature also depends slightly (increases) on J_p, and the strength of the **E**-field decreases.
- the temperature in the arc is usually 5000–10 000 K and changes especially rapidly in the vicinity of the periphery of the plasma volume (Fig. 6.10.4)

Arc discharges are used in a large number of applications and will be described briefly in chapter 10.

6.10.3. Quasi-stationary strongly emitting Z-pinches [158]

In many applications it is important to develop powerful radiation sources with a wide range of parameters (wavelength, power, the duration of the light pulse, dimensions of the source). In the case of radiation sources with the effective temperature of $\leqslant 1$ eV it is often sufficient to use arc discharge open in the atmosphere. However, to generate a vacuum ultraviolet and soft x-ray radiation is necessary to increase greatly the

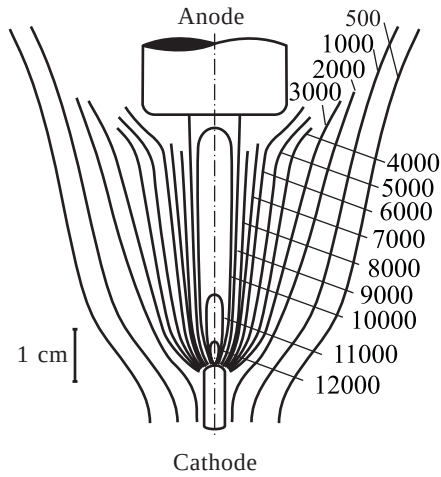


Fig. 6.10.4. Typical temperature distribution (K) in the electric arc, burning in the atmosphere (J_p = 200 A, cathode diameter 0.4 cm).

specific discharge power related to the unit volume of the plasma.

The radiating discharges of this type can be produced by many methods, but the most suitable methods are the *Z*-pinches and compression flows in the vacuum conditions[19] since glass, quartz and other materials effectively absorb radiation with a short wavelength. Thus, the temperature can be increased by several tens of eV or even higher. Here, we shall not describe their models, based on the system of the equations (6.5.39), taking into account the fact that this should be carried out using computers as in the case discussed for both the optically transparent plasma column and the non-transparent column. These problems have been studied in detail for discharges with the duration of ~1–10 μs (Aleksandrov and Rukhadze [158]). It should be mentioned that nanosecond superpowerful energetics procedures are used to investigate analogues of *Z*-pinches generating powerful x-ray radiation fluxes (see appendix B).

6.11. Systems using separated excitation levels of particles

6.11.1. Gas discharge and plasma lasers [162]

In the previous section we discussed the discharges in which the plasma is in the local thermodynamic equilibrium. In these media the particle concentration monotonically decreases with increase of the energy level of the media

[19]N.P.Kozlov and Yu.G. Protasov [153]

$$n_0 \sim \exp\left\{-\frac{\varepsilon_\alpha}{kT}\right\}.$$

These media are stable. This is not the case if we produce an inverse population in some energy range at which the concentration of particles on one or several excited levels is higher than on the underlying levels. In section 4.4.3 it was shown that if the section with $\partial f/\partial v_x > 0$ appears in the electron distribution function $f(v_x)$, oscillations start to take place in plasma. This is also observed in the case of disruption of the equilibrium distribution with respect to the energies of the exciting particles. Impressive possibilities existing here were demonstrated by the lasers – coherent radiation generators constructed originally by Basov and Prokhorov. The absolute majority of the lasers can be schematically described by Fig. 6.11.1. Here 1 is the active transparent medium which is also a generator of coherent induced radiation, 2, 3 is the optical resonator, usually in the form of two parallel mirrors one of which is semi-transparent for emitting generated radiation (3) and the other one (2) is 'dead end', and finally 4 is the system for 'pumping' the energy into the active medium. This may be simply heating of the gaseous active medium as in gas-dynamic lasers, electrical discharge, electron flux, powerful 'secondary' non-coherent radiation, etc.

The most important property of the active medium is the presence of two levels (Fig. 6.11.2) of which the upper level (ε_1) has, in the absence of induced radiation, a comparatively low probability of transition to some lower-level (ε_2); this applies even more so to other lower levels. At the same time, the lower level is characterised by higher probability for 'cleaning' and it is therefore always empty.

The lasers can operate both in the pulsed and stationary regimes. Here is the operating procedure in the pulsed laser.

'Pumping' is activated. Filling of the upper level ε_1 starts. Single transitions from the upper to lower level do not generate sufficiently intensive radiation in the active medium in order to stimulate coherent radiation. Therefore, laser radiation in this stage is spontaneous and stochastic. However, further pumping of the upper level at some moment results in spontaneous radiation which is so strong that the harmonics, generated by the resonator, starts to induce the radiation of excited particles (atoms, molecules) and the process becomes avalanche-like. As a result, the upper level is cleared (5) and the radiation stored in the resonator gradually leaves it because of the semi-transparency of the mirror. Of course, the ejection of the electrons from the lowest level is the process with the highest velocity in the scheme. To make this obvious, taking into account the pulsed regime, we can write the

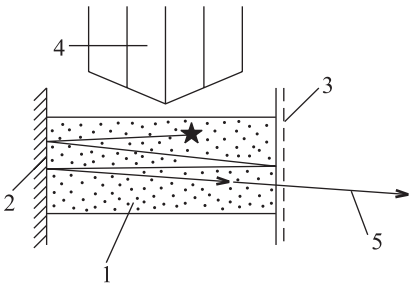


Fig. 6.11.1. Principal diagram of laser: 1 – active medium, 2 – non-transparent mirror, 3 – semi-transparent mirror, 4 – pumping, 5 – laser beam.

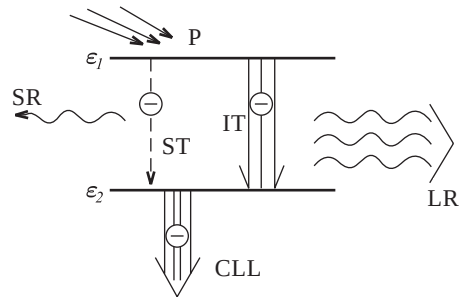


Fig. 6.11.2. Functional diagram of the lasers: (P) – pumping of the active medium, (ε_1) – upper energy level, (ε_2) – lower-level, (ST) – spontaneous transition of the electron from the upper to lower level, (SR) – spontaneous radiation, (IT) – induced electron transfer, (LR) – laser coherent radiation, (CLL) – cleaning of the lower level.

following maximally simplified equation for the particle concentration in the upper level, after pumping of this level:

$$\frac{\partial n}{\partial t} = -n\alpha(1+\gamma I_\omega), \quad \alpha \equiv \frac{1}{\tau^{(\text{spon})}}. \tag{6.11.1a}$$

The radiation transfer equation – taking into account the fact that as a result of the presence of the resonator the intensity of radiation increases with time and not in space – should be written in the form

$$\frac{\partial I_\omega}{\partial t} = \beta n_1(1+\gamma I_0) - \frac{I_\omega}{\tau_\omega} \tag{6.11.1b}$$

The last term on the right describes the departure of radiation from the resonator. We introduce the subpopulation of times

$$\tau_0 \ll \tau_H \ll \tau_\Lambda \ll \tau_\omega.$$

Here τ_0 is the time of depletion of the lower level, τ_H is pumping time, τ_Λ is the excitation time of the active medium, τ_ω is the time at which the radiation leaves the resonator.

We consider the processes of depletion of the upper level after completing pumping. Since the value of I at the initial moment is low and non-coherent, the system (6.11.1) has the following form (n_0 is the concentration of the excited particles after pumping)

$$\frac{\partial n}{\partial t} \approx -n_0\alpha; \quad \frac{\partial I_\omega}{\partial t} \approx \beta n_0 \tag{6.11.2}$$

Correspondingly

$$n_1 \approx n_0(1-\alpha t), \quad I_\omega \approx \beta n_0 t \tag{6.11.3}$$

When $I_\omega\gamma$ becomes greater than unity, induced radiation starts to play a more and more important role and the system (6.11.1) can now be written in the following form

$$\frac{\partial n_1}{\partial t} = -n_1 a I_\omega; \quad \frac{\partial I_\omega}{\partial t} = b n_1 I_\omega \tag{6.11.4}$$

Here $a \equiv \alpha\gamma$; $b = \beta\gamma$. The system (6.11.4) has the obvious integral

$$n_1 + hI_\omega = N_1; \quad h \equiv \frac{a}{b} \tag{6.1.5}$$

where N_1 is the initial concentration of the excited atoms (molecules).

Substituting (6.11.5) into (6.11.4) we obtain

$$\frac{\partial n_1}{\partial t} = -an_1\left(\frac{N_1 - n_1}{h}\right) = -bn_1(N_1 - n_1)$$

From this

$$\frac{N_1 - n_1}{h} = e^{-Nbt} \tag{6.11.6a}$$

i.e.

$$n = N\frac{1}{1+e^{Nbt}}; \quad I_\omega = \frac{N}{h}\frac{e^{Nbt}}{1+e^{Nbt}} \tag{6.11.6b}$$

Thus, the depletion of the upper level takes place along the exponent and its rate increases with the increase of the initial population.

This is followed by the final stage – departure of radiation from the resonator:

$$\frac{\partial I_\omega}{\partial t} \approx -\frac{I_\omega}{\tau}, \tag{6.11.7}$$

i.e. $I_\omega \sim \exp\{-t/\tau_{res}\}$. To construct a model of a continuously working laser, it is sufficient to add the constant term Y = const to the right hand part of equation (6.11.1a) to. This term describes the continuous pumping of the upper level.

In conclusion, we should mention the principal diagrams and active media of three widely used laser types.

A. Helium–neon lasers

These well-known gas-discharge lasers use the two-stage method of pumping of the active medium – neon (Fig. 6.11.3). Initially, pumping (population) takes place on the metastable level of helium in which the

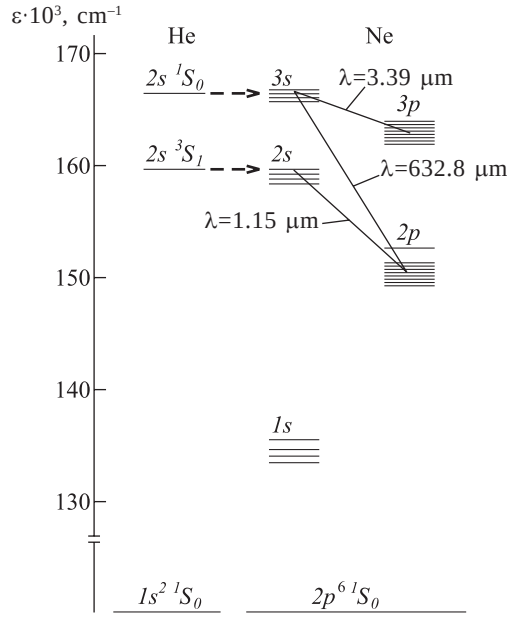


Fig. 6.11.3. Diagram of helium and neon levels. Three types of induced transitions in the laser are shown.

electrons with the energy close to the energy of the level 3*s* of neon, suitable as the upper level of lasing. The transition from this level of He to the level 3*s* of neon is quite fast. At the same time, the lower level 2*p* of neon on which the electrons fall, is easily 'cleaned' and this also confirms the continuous coherent generation of radiation.

The helium–neon lasers are low power systems. The typical power in radiation does not exceed several hundreds of milliwatt.

B. CO_2 lasers

These lasers use excitation of vibrational levels of the molecules. Gas-dynamic CO_2 lasers are of special interest in the group of these lasers.

As regards the circuit, these lasers are gas-dynamic nozzles, with the optical resonator placed in the supersonic section.

The main active component of the gas dynamic lasers is carbon dioxide which is heated in the preliminary chamber. In the simplest case this may take simply as a result of the combustion of carbon-containing fuel in air. The vibration levels are excited as a result of high temperatures (up to ~4000 K). With further flow, the gas rapidly cools down. An inverse population forms and, passing between the resonator mirrors, the gas generates infrared coherent radiation. The power of the gas-dynamic CO_2 lasers may reach tens of kilowatts at the efficiency, in relation to fuel of up to ~2%. At present, CO_2 lasers

are produced with different pumping systems, including with a gas-discharge system. These are stationary lasers with the beam power of tens of kilowatts, the wavelength of ~10 μm and the efficiency of 10% .

C. Excimer lasers

The lasers, using excimers[20] as the active medium, have high characteristics.Xe*Cl and Kr*F excimers (the asterisks indicate the excited atoms) are used most widely. This laser is a gas-discharge device in which the excimers are continuously synthesised (see the following section). Since the excimers are the excited molecules the active medium is also excited. When excitation is removed, the molecules break up and this results automatically in effective depletion of the lower level.

These lasers operate with short pulses (~30 and 50 ns) but their mean power in the periodic regime can reach kilowatts and the efficiency seveal percent.

6.11.2. Special features of plasma chemistry [159, 60]

Up to now we have been discussing the interaction of heavy particles with free electrons and radiation. Now, we investigate the interaction of heavy particles with each other with participation of electrons, paying special attention to chemical reactions at temperatures of the order of several thousand degrees. This area of science is usually referred to as plasma chemistry. The number of plasma chemical processes is extremely large. Here, we consider only three examples and more examples are discussed in section 10.3. Plasma chemical reactions can take place in highly non-equilibrium conditions and also in quasi-equilibrium conditions. Oxidation of nitrogen is a suitable example of the latter.

One example of non-equilibrium plasma dynamic reactions which we discuss here is the synthesis in discharges of ‘excimers’ – metastable compounds with participation of noble gases (Ar_2^*, Kr*Cl, etc). Another example is the synthesis of fullerenes[21] discovered in 1985 having the form of remarkable multiatomic carbon structures, Fig. 6.11.4.

A specific feature of plasma chemical processes is the presence of the electrons and/or of heavy particles with a high-energy (superroom) ensuring activation of the particles.

The reactions between two heavy particles may lead in the final analysis either to the generation of energy (exothermic reactions) or

[20]Excimers are molecules which exist only in the excited state

[21]These molecules carry these names in honour of the American architect Fuller who developed the structure of buildings are based on the spatial lattice structure

to absorption of energy (endothermic reactions). A suitable examplesof exothermic reactions is the process of burning of carbon in oxygen

$$C_n + nO_2 \rightarrow nCO_2 + \varepsilon.$$

Dissociation of an oxygen molecule is an example of endothermic reactions

$$O_2 \rightarrow 2O.$$

It is evident that the endothermic reactions can occur only in the presence of energetic particles with the energy level $>\varepsilon_{end}$. However, in the case of the exothermic reactions activation is usually required. This is associated primarily with the fact that the valence bonds in the particles, entering the reaction, are usually close. Consequently, in the normal conditions in the free state we have O_2 molecules and not oxygen atoms O, etc. Therefore, for the particles to enter the reaction, it is necessary to use the so-called activation energy ε_{act} (Fig. 6.11.5). In simple cases this is dissociation energy. However, subsequently as a result of the reaction – in the case of the exothermic reaction, the energy $\varepsilon_p = \varepsilon_{act} + Q$ is released. It is also evident that is a reversed reaction can take place, such a reaction is endothermic.

Naturally, all reactions (or almost all reactions) take place at a higher rate at elevated temperatures, and if the products of these reactions are important (for example, formation of nitrogen oxides), and the reaction itself is reversible, it is necessary to 'freeze' the composition formed at high temperature. This can be done by rapid cooling ('quenching') of the medium which contains the reaction products. If these products are unstable at room temperature, it is necessary to bond them in stable complexes.

Oxidation of nitrogen

A highly suitable example of the specific features of the quasi-equilibrium plasma chemical reactions is oxidation of nitrogen with the glocal equation of this process having the form:

$$\frac{1}{2}O_2 + \frac{1}{2}N_2 + 0.93\text{eV} \rightarrow NO.$$

This process, which takes place in the arc discharge at atmospheric pressure, has been used since the end of the 19th century in Norway which has large hydroenergy resources, for the production of 'Norwegian saltpetre' $Ca(NO_3)_2$. In particular, the airflow, passing through the arc and enriched with NO is directed into a tank with a solution of slaked lime in which the reaction, generating the saltpetre, takes place.

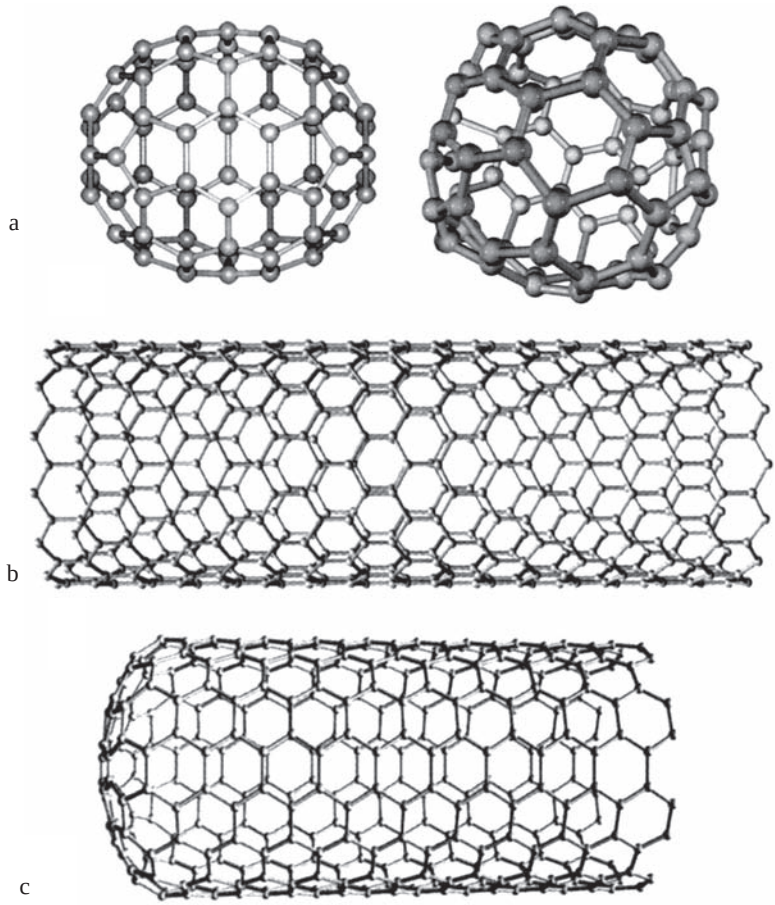


Fig. 6.11.4. The fullerenes C_{70}, C_{72} (a), opened nanotube (b), closed nanotube (bc).

If we consider the experimental data on the probability of direct transformation of two molecules O_2 and N_2 into two NO molecules, it appears that the equilibrium concentration of NO at T = 2000 K is 5–6 orders of magnitude lower than that detected in the experiments. This indicates that in reality the process consists of several stages. The maximally simplified scheme of the reaction which, however, provides quantitative data and is in satisfactory agreement with experiments has the following form:

$$O + N_2 \rightleftarrows NO + N - 3.3\,eV,$$
$$N + O_2 \rightleftarrows NO + O + 1.4\,eV.$$

These schemes indicate that the oxygen atom plays the role of a catalyst, enters intermediate reactions and again appears in the final reaction.

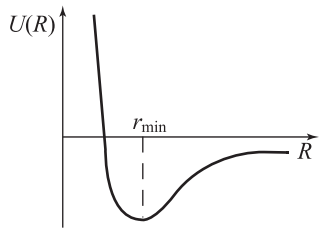


Fig. 6.11.5. Typical interaction potential of two atomic particles in relation to the distance between them.

The first reaction is slow because the reaction requires a large amount of energy (in comparison with the typical temperature of 3000–5000 K). However, the actual situation is even more complicated. The first reaction is of the 'multipass' nature. In reality, this reaction includes the stage of dissociation of N_2 which requires 9.8 eV and then only part of the energy, 6.5 eV, is returned in the formation of NO.

The stage of dissociation of N_2 takes place through a number of successive excitations of the oscillatory levels.

Excimers

It took 60 years to produce only several compounds, including noble gases, by classic chemical methods. Canadian chemist Neil Bartlett was the first scientist to the compound $XePtF_6$ by this procedure in 1962.

From approximately 1970, a large number of reports were published on the formation of compounds of noble gases with each other (Ar_2, Kr_2, Xe), with halogens (ArF, KrCl, KrF, XeBr, XeCl, XeF) and with oxygen (ArO, KrO, XeO). However, the majority of these components are short-lived and exist only in the excited state (excimers).

The halogen-containing excimers form in tubes with a discharge at a pressure of the order of 10^{-2}–1 atm. At a pressure of $\geqslant$10 atm, the effect of high-energy electron beams results in the formation of dimers. The monohalogens form by the following mechanism. The atom of the noble gas R*, excited by a discharge, interacts with the halogen molecule (X_2)

$$R^* + X_2 \rightarrow RX^* + X$$

Or

$$R_2^+ + Xe^- \rightarrow RX^* + R,$$

The dimer usually forms by the following reaction

$$R^+ + 2R \rightarrow R_2^+ + R,$$

and, subsequently

$$R_2^+ + e \rightarrow R + R^*$$

and

$$R^* + 2R \rightarrow R_2^+ + R.$$

The energetics of the formation of the excimers is explained quite convincingly by Fig. 6.11.6 which shows the binding energy ε in relation to the distance between the components of the excimer. It can be seen that ε(*r*) for the non-excited particles has almost no wells, whereas a well does exist for the excited polymer.

The excimers were used for the construction of lasers described previously.

The existence of the excited excimers can be explained quite simply. Excitation of, for example, the Xe atom transfers the electrons with the filled shell to the adjacent unfilled level. Consequently, Xe* transforms to the analogue of the alkali atom, but with 'hole' inside. This quasi-alkali atom reacts with the halogen without paying any attention to the hole. However, at some moment, the hole pulls in the electron which travelled upwards and the excimer disperses.

Fullerenes and nanotubes

Until recently, carbon was known in two basic allotropic modifications – in the form of graphite with a laminated structure and in the form of diamond. The binding energy of graphite is relatively low and, consequently, the arc discharge with graphite electrodes is already characterised by almost complete dissociation of the carbon (graphite) clusters arriving into the discharge gap into the individual atoms with free valences and, leaving the hot zone, they start to actively bond with each other, forming not only flat but also three-dimensional configurations, the so-called special analogue of the benzol ring.

In 1985, the condensate of carbon vapours of the arc discharge (Fig 6.11.7) were found to contain, in addition to soot, also spherical molecules C_{60} consisting of 60 carbon atoms (Fig. 6.11.4a). They were separated because the fullerenes dissolve in organic fluids and soot does not.

The formation of the fullerenes is a stochastic process. Therefore, the molecules with different numbers of the carbon atoms are produced. Later, qualitatively different structures were produced, in particular, in the form of tubes. These structures are referred to as nanotubes (Fig. 6.11.4b, c).

At present, the details of the kinetics of formation of the fullerenes are not completely clear. Therefore, we shall not pay special attention to this

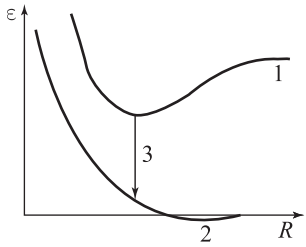


Fig. 6.11.6. Potential energy of the interaction of two atomic particles in relation to the distance between them in the formation of the excimer (1); potential energy of the non-interacting atoms (2); 3) emission transitions between (1) and (2).

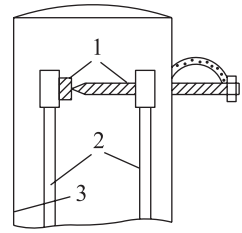


Fig. 6.11.7. Diagram of one of the first systems for synthesis of fullerenes: 1) graphite electrode; 2) watercooled current supply; 3) cooled surface of the hood on which the soot with fullerenes settles, 4) spring.

subject. It should only be mentioned that the technology of introduction of atoms of different type into the molecule has been mastered. In particular, when adding alkali metals, superconductors with the critical temperature T_{cr} up to 33 K have been produced. The fullerenes are characterised by a large number of other important properties and their formation is an outstanding illustration of the fantastic possibilities of plasma chemistry.

7

Interaction of plasma with the surface of solids

7.1. Introduction

In previous chapters, we discussed the 'basic' systems of equations which describe with some degree of idealisation the processes in plasma volumes. However, no attention was given to the problem of boundary conditions which usually exist on the surface of a solid body (SB). These boundary conditions should formally express the nature of interaction of plasma with the surface of the solid body. A number of these processes are investigated in this chapter.

The role of the boundary conditions ('walls') in the formation of the macrostructure of flows of gases and liquids is quite evident. In section 2.1 these conditions were investigated within the framework of Euler equations: flow around fine profiles and flows in nozzles. It was shown how the walls form the general structure of the flow. However, the effect of the walls is not restricted to the formation of the macrostructure. They organise in their vicinity their own dynamic structure – the boundary layer – within which the complex of the parameters, characterising the wall (its velocity, temperature, roughness, etc) is matched with the appropriate parameters of the main flow or, as often is the case, the 'core' of the flow.

In most cases, the properties of this layer greatly differ from those of the main flow. For example, in conventional hydrodynamics the flow in the core can be almost dissipation-free, whereas dissipation in the form of viscosity is of primary importance in the boundary layer. These viscosity layers, being often very thin, in comparison with the main flow – such a situation is found in, for example, the flow around an aircraft wing, can nevertheless strongly affect the flow as a whole. For example, in the example with the wing, the dissipative boundary layer, disrupting the Kelvin–Helmholtz law of the conservation of the vortex, results in circulation around the wing being different from zero thus ensuring the lifting force of the wing.

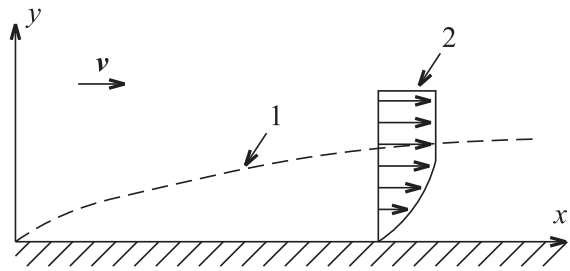


Fig. 7.1.1. Flow of a viscous liquid around a half-plane (Prandtl–Blasius model): 1) the conventional boundary of the boundary layer; 2) the dependence of velocity on distance to the half-plane.

The 'fineness' of the boundary layers in comparison with the dimensions of the flow core is explained usually by the fact that they are formally defined by a small parameter with an 'older' derivative in the equations. The boundary layers are very similar to other fine-scale perturbations of the investigated continuous medium. The laminar boundary layers and their relationship with the waves are especially evident in cases in which they form in homogeneous flows during flow-around or in contact with relatively smooth profiles. In these cases, the behaviour of the boundary layer at a large distance from the solid body producing this layer can be regarded as a small perturbation, i.e. by means of the linearised equations of the dynamics of the given medium[1].

7.1.1. Prandtl–Blasius boundary layer (Navier–Stokes hydrodynamics)

The above considerations will be explained on an example of the first investigated boundary layer, namely the viscous boundary layer in an incompressible liquid described by the Navier–Stokes equations.

The simplest example of this layer is a layer formed in the flow of a uniform (in the core) flow around a semi-infinite plane (Fig. 7.1.1). The layer was calculated quite accurately in 1908 by Blasius on the basis of general equations for viscosity boundary layers produced previously by Prandtl using simplified Navier–Stokes equations. This solution has the following form (see, for example [13,163]

$$v_x = v_0 f'(\xi);\ v_y = \frac{1}{2}\sqrt{\frac{w_0}{x}}(\xi f' - f);\quad \xi = y\sqrt{\frac{v_0}{\nu x}}; \tag{7.1.1}$$

[1]A boundary layer always forms in plasma at the boundary with the solid surface because of the mobility of the charged components and stationary SB.

where v_0 is the speed of the incident flow, ν is kinetic viscosity, $y = 0$ corresponds to the half-plane, and the coordinate x is counted from the boundary of the half-plane. Function $f(\xi)$ is governed by a very exotic equation

$$ff'' + 2f''' = 0 \tag{7.1.2a}$$

and by the boundary conditions

$$f(0);\ f'(0) = 0;\ f'(\infty) = 1 \tag{7.1.2b}$$

The dependence of the derivative $f'(\xi)$ on ξ is shown in Fig. 7.1.2. In the vicinity of $\xi = 0$, function $f(\xi)$ is approximated by the equation

$$f(\xi)\ \big|_{\xi\to 0} = \frac{\alpha}{2}\xi^2 + O\left(\xi^5\right). \tag{7.1.3a}$$

Here $\alpha = 0.332$. This shows that two boundary conditions are fulfilled at $y = 0$

$$v_x = 0 \text{ and } v_y = 0. \tag{7.1.3b}$$

Thus, the investigated solution corresponds to the boundary condition of the 'impermeability' of the plane, as in the case of the Euler equation,

$$v_x\big|_{y=0} = 0.$$

but it also corresponds to the condition of 'sticking' of the flow to the half-plane:

Evidently, this is the viscosity effect.

The following approximation holds at $\xi\to\infty$:

$$f(\xi)\big|_{\xi\to\infty} = \xi - \beta - \int_0^{\xi}(1-\Phi(\xi))d\xi = -\beta + \int_0^{\xi}\Phi(\xi)d\xi \tag{7.1.4a}$$

Constant $\beta = 1.72$, $\Phi(s)$ is the error integral

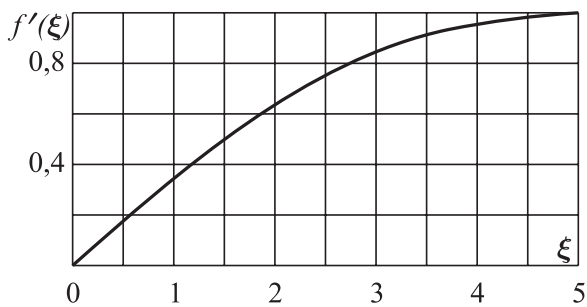


Fig. 7.1.2. The graph of function $f'(\xi)$.

$$\Phi(s)=\frac{2}{\sqrt{\pi}}\int_0^s \exp(-s^2)ds,\ \ \Phi(s)\big|_{s\to\infty}\to 1. \tag{7.1.4b}$$

From (7.1.1) and (7.1.4a) it follows that at $\xi \gg 1$

$$v_x \to v_0\Phi(e)\to v_0; v_y \to 0.$$ [2]

Further, accepting some value ξ_* as the boundary of the boundary layer, it can be seen that the width y_* of the boundary layer increases with increase of x, like $\sqrt{x}$, because of (7.1.1)

$$\xi_* \approx \frac{y_*}{\sqrt{x}}\sqrt{\frac{v_0}{\nu}}=\text{const.} \tag{7.1.5}$$

This behaviour of the thickness of the boundary layer is associated with the dissipative nature of viscosity responsible for this boundary layer.

Finally, should be mentioned that at $\xi \gg 1$ the first equation (7.1.4b) can be written in a more convenient form taking into account that

$$v_x - v_0\left(1-\frac{1}{\sqrt{\pi}}\frac{\exp\left\{-\xi^2\right\}}{\xi}\right)$$

The Prandtl–Blasius theory in linear approximation

In linear approximation, the situation becomes easier to understand if the flow of a compressible gas is investigated.

In this case, the system of the Navier–Stokes equations has the following form (if the 'second viscosity' is ignored)

$$\frac{\partial\rho}{\partial t}+\operatorname{div}\rho\mathbf{v}=0,\ \rho\left(\frac{\partial\mathbf{v}}{\partial t}+(\mathbf{v}\nabla)\mathbf{v}\right)=-\nabla p+\nu\rho\Delta\mathbf{v},\ p=p(\rho). \tag{7.1.6a}$$

The linearisation of the system results in the following equations

$$\left(\frac{\partial}{\partial t}+v_0\frac{\partial}{\partial x}\right)\rho_1+\rho_0\operatorname{div}\mathbf{v}_1=0,\ \rho_0\left(\frac{\partial}{\partial t}+v_0\frac{\partial}{\partial x}\right)\mathbf{v}_1=-c_T^2\nabla\rho_1+\nu\rho_0\Delta\mathbf{v}_1 \tag{7.1.6b}$$

It can easily be seen that the system splits into two systems ($\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$):

$$\rho_1\neq 0;\quad \left(\frac{\partial}{\partial t}+v_0\frac{\partial}{\partial x}\right)^2\boldsymbol{\rho}_1=c_T^2\Delta\rho_1+0(\nu); \tag{7.1.7a}$$

$$\rho_0\left(\frac{\partial}{\partial t}+v_0\frac{\partial}{\partial x}\right)\mathbf{v}_1=c_T^2\nabla\rho_1+0(\nu)$$

[2] $\int_0^\infty (1-\Phi(\xi))d\xi=\beta$

$$\rho_1 = 0; \quad \operatorname{div} \mathbf{w}_1 = 0; \left(\frac{\partial}{\partial t} + v_0 \frac{\partial}{\partial x} \right) \mathbf{w}_1 = \nu \Delta \mathbf{w}_1. \tag{7.1.7b}$$

The first of these systems (hyperbolic) describes the flow $\mathbf{v}_1$ in the centre of the flow which was discussed in section 2.2. The second system (parabolic) describes the flow $\mathbf{w}_1$ in the boundary layer. The general solution in the investigated region has the form

$$\mathbf{v} = v_0 x^0 + \mathbf{v}_1 + \alpha \mathbf{w}_1. \tag{7.1.8}$$

The coefficient α should be determined from the boundary conditions. However, taking into account the fact that the boundary layer is actually described by the non-linear equation ($0 < \mathbf{v} < \mathbf{v}_{max}$), the determination of α is relatively conditional.

As regards the stationary flow around the infinite half-plane, it may be assumed that $\mathbf{v}_1 = 0$, and the linear equation (7.1.7) can be written in the following form

$$v_0 \frac{\partial \mathbf{w}_{1x}}{\partial x} = \nu \frac{\partial^2 \mathbf{w}_{1x}}{\partial y^2}. \tag{7.1.9}$$

Here, we ignore the term $\partial^2 \mathbf{w}_{1x}/\partial x^2$, because it is small (at low ν) in comparison with $\partial^2 \mathbf{w}_{1x}/\partial y^2$.

If it is assumed that on the plane $v_x = 0$, which is 'rough'for the linear approximation but rational for asymptotics, then the following equation should be used as the boundary condition for w_{1x}

$$w_{1x}\Big|_{y=0} = -v_0. \tag{7.1.10}$$

The equation (7.1.8) with the boundary condition (7.1.8) gives for $v_x = v_0 + w_{1x}$ the expression which is completely identical with the classic solution of the problem of the cooling of a semi-bounded bar, uniformly heated to temperature T_0 as a result of activating at $t = 0$ a cooling system maintaining constant temperature $T = 0$ at the end of the bar. In this case

$$T = T_0 \Phi \left(\frac{x}{2\sqrt{\kappa t}} \right). \tag{7.1.11}$$

Here t is time, x is the coordinate of the cross-section of the bar, and κ is the thermal diffusivity coefficient[3]. Replacing T by v_x, T_0 by (v_0), x by y, κ by ν/v_0, and t by x, we obtain the required solution of (7.1.4).

Three consequences follow from all these considerations.

1. The linear approximation describes quite accurately the

[3]The appropriate heat conductivity equation is $\partial T/\partial t = \kappa \partial^2 T/\partial x^2$.

asymptotics of the perturbation of the flow, caused by the solid.

2. The number of types of linear waves, formed in the model, is equal to the number of the boundary conditions which must be formulated in this case on the surface of the solid.

3. The width of the boundary layer is determined by the particle kinetics, because of (7.1.5)

$$\xi_* \sim \sqrt{\lambda x}, \tag{7.1.12}$$

where λ is the free path of the particle, since (section 5.2.1)

$$\nu \sim \nu_T \lambda.$$

Thus, the boundary layer is connected by classic viscosity with the internal scale of the medium. Evidently, these special features of the boundary layer are general and they can be applied, with appropriate stipulations, to the plasma systems.

However, we now return to the hydrodynamic boundary layer. Evidently, in a general case, the heat conductivity equation should be added to the Navier–Stokes equations (7.1.6) and this would enable us to formulate another boundary condition on the surface of the solid – for example, its temperature T_0

$$T\big|_\Gamma = T_0 .$$

However, outside the limits of the hydrodynamic description of the boundary layer, there is another very important moment. It is associated with the fact that the Navier–Stokes equation does not describe what takes place in the immediate vicinity of the solid, within the limits of the so-called Knudsen sublayer whose width δ_K is of the order of the free path of the flow particles

$$\delta_K \sim \lambda.$$

Two important processes occur here. Firstly, as a result of collision of the flow particles with the surface, the distribution function of these particles $f^{(-)}(\mathbf{v}, \mathbf{x})$ transforms to another distribution function of the reflected particles $f^{(+)}(\mathbf{v}, \mathbf{x})$

$$f^{(-)}(\mathbf{v},\mathbf{x}) \to f^{(+)}(\mathbf{v},\mathbf{x})$$

whose properties are determined by the microstructure of the surface of the solid. In particular, the conversion to zero of the macrovelocity of the flow on the surface of the solid may be the consequence of the isotropic diffusion scattering of the incident particles. However, reflection is often

more complicated. The initial collisions of the particles with each other, taking place in the Knudsen sublayer, prepare Maxwellisation of the particles in the Navier–Stokes region. This all can be taken into account only by transferring to the kinetic Boltzmann equation and the boundary conditions appropriate for this equation (see below).

Briefly summarising the regions of interaction of the flow with the solid in gas dynamics, the latter is represented graphically in the form of five layers (Fig. 7.1.3). Here I is the thickness of the solid which can fail at a high particle energy and flow density, II is the 'atomic layer' of the solid subjected directly to the collisions with the flow particles, III is the Knudsen sublayer, IV is the viscosity boundary layer, V is the core of the flow.

We have examined the simplest case of a laminar boundary layer. In the actual conditions, the situation is often non-stationary but the main qualitative features, described here, remain usually unchanged.

7.1.2. General structure of the plasma–solid transition layer

The general pattern of interaction of the plasma with the solid is far more complicated than in gas dynamics. This is associated both with the large number of the plasma components (electrical and magnetic fields, electrons, ions, atoms, etc) and with the extremely large variety of the forms of interaction of the component with each other and with the solid, and also with the properties of the solid which do not differ in conventional gas dynamics (dielectrics, metals, electrons, etc) but they lead to different effects in interaction with plasma. Therefore, it will not be attempted here to present a relatively general classification of the transition structures in the region adjacent to the solid and we confine ourselves to the more general characteristics of such structures and to a number of specific examples.

In particular, we present the scheme of the type Fig. 7.1.3 for the 'plasma–solid' system. In the system we now define seven sub-regions,

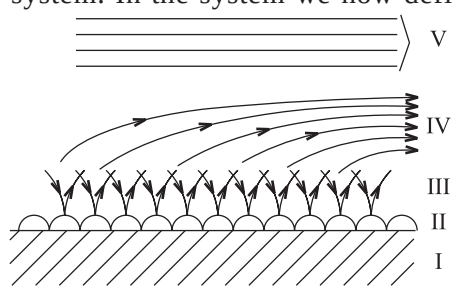


Fig. 7.1.3. Characteristic layers of the hydrodynamic region in the vicinity of the wall: I) the solid, II) the outer 'atomic' layer of the solid; III) the Knudsen sublayer, IV) the viscosity boundary layer, V) the core of the flow.

but the most important factor is not the number but the content (Fig. 7.1.4) of the sub-regions III, IV, which do not differ by the description models (they can be any (kinematic and hydrodynamic)) and it is the controlling role of the individual components in the formation of the structures in the given region. In particular, region III relates to the electronic structures with the scales – Debye and electronic–Larmor radii; the region IV – the structures of electronic–hydrodynamic scales (of the skin layer type) to the regions $III_{i,n}$ and V_{in} – the structures of the ion–atomic scales. Although the regions I and II retain on the whole the same meaning as in gas dynamics, their functions are far more comprehensive in this case.

It should be stressed that the scheme in Fig. 7.1.4 is highly ideal, and in many cases these subdivisions can be efficiently mixed.

In addition to the previously described phenomena, the near-wall region may be characterised by the formation of dynamic structures of various types with high energy concentration and relative autonomy. Some of them are produced, for example, by a sliding high-current discharge along the dielectric, as in pulsed erosion plasma thrusters, others are found in constriction of the discharge on the electrodes. Typical examples of the structures of the second type are cathode spots of the arcs burning between the cold electrodes (section 7.6).

In the first part of this chapter we describe the main processes of interaction of plasma particles with the surface of the solids, i.e. in subregion II. The second part is concerned with the boundary layers. Because of the large number of types of these layers, special attention will be given to the electronic layers in relatively rarefied plasma (r_D, $\rho_e \ll \lambda$ is the collisional free path of the particles). The third part

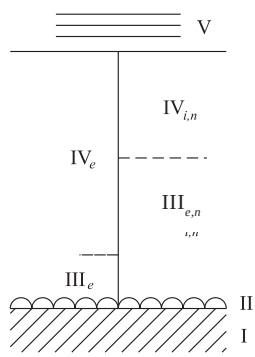


Fig. 7.1.4. Characteristic layers in the near-wall region of plasma: I) the solid, III) the outer layer of the solid, III_e) the region of the electronic structures of the Debye and electronic Larmor scales; $III_{e,n}$) the region of the structures of the electronic–atomic scales; IV_e) the structures of the electronic–hydrodynamic scales; $IV_{i,n}$) the region of the structures of the ion–atomic scales; V) the core of the flow.

describes a number of plasma systems in which the near-wall processes are non-trivial and play a significant role in the general pattern of the discharge.

Previously, we mentioned the effect of the walls on processes in plasma. However, solids themselves can be extensively modified and this is utilised in increasing numbers of applications, as discussed in section 10.3. Here, in section 7.4.3, we investigate the simple model of the ionic ('cathodic') sputtering of the surface by fast ions and atoms. The final part of the chapter discusses briefly the so-called 'dusty' plasma. The name itself describes the characteristics of these system.

Transferring to specific problems associated with this range of phenomena, we start by improving the terminology.

In particular, the entire set of the processes, formed in interaction of the flow with a wall and situated between the regions I and VI in Fig. 7.1.4, will be referred to as the near-wall phenomena processes. Later, we shall discuss only 'intrinsic' near-wall processes, referring to the processes taking place at the distance $d \ll D$, where D is the 'diameter' of the flow. In cases in which the word 'intrinsic' cannot cause confusion, it will be omitted.

In the group of the intrinsic near-wall processes, special attention is given to the surface phenomena taking place in region II, i.e. the phenomena which take place either on the surface of the wall or at a distance of several atomic sizes from the wall. These phenomena differ greatly: sputtering of the surfaces with fast particles, photo- and thermoemission, adsorption, surface ionisation, evaporation, etc.

Between the surface and the core of the flow there is a large number of different boundary layers, relating to the regions III, IV and and V. As already mentioned, they are caused by the characteristic scales of different plasma components.

7.1.3. Emission functions

All the elementary emission processes, taking place on the wall, can be divided into three groups.

The first group includes spontaneous emission processes: thermoemission, surface ionisation and auto-emission of the electrons, and also evaporation of matter under the effect of high temperature. These processes can be characterised by the functions $g_\alpha(\mathbf{v}, \Gamma)$. Index α indicates the type of emitted particles (electrons, neutral atoms, neutral molecules, ions). Symbol Γ relates to the set of the parameters, such as the wall temperature, the presence or absence of films on the wall, the strength of the electrical field around the wall, etc. Function

$g_\alpha(\mathbf{v},\Gamma) \equiv v_n G_\alpha(\mathbf{v}, \Gamma)$ characterises the density of the particle flow, and G_α is the density of the particles emitted 'spontaneously' by 1 cm^2 with the velocity in the range $\mathbf{v}$, $\mathbf{v} + d\mathbf{v}$. Therefore, the number of particles emitted by the area ds in time dt is dN_α:

$$dN_\alpha = g_\alpha(\mathbf{v},\Gamma) dS\, d\mathbf{v}\, dt. \tag{7.1.13a}$$

If we are not interested in the angular and energy distribution of the emitted particles, our considerations can be confined to the density of the emission flow $\nu_\alpha(\Gamma)$ equal to

$$\nu_\alpha(\Gamma) = \int g_\alpha(\mathbf{v},\Gamma)\, d\mathbf{v}. \tag{7.1.13b}$$

If we are concerned with the spontaneous emission of charged particles, then in addition to ν_α it is often necessary to use the emission current density $j_\alpha = e_\alpha \nu_\alpha$. A suitable example of the dependence j_α (T) is the Richardson–Dashman equation.

The second group of the elementary processes includes the processes of particle emission under the effect of radiation, i.e. photoemission, described by the function

$$P_\alpha(v, \Gamma, \gamma),$$

Here γ are the characteristics of the light flux incident on the surface element (frequency, incidence angle, polarisation). If I_ω denotes the intensity of the light flux, incident on the walls, the density of the particle flow is

$$\frac{dN_\alpha}{dt} = P_\alpha(v, \Gamma, \gamma) I_\omega d\gamma\, dS\, \mathrm{d}\mathbf{v}. \tag{7.1.14a}$$

Here $d\gamma = d\omega\, d\Omega$, where ω is frequency, and $d\Omega$ is the solid angle within which radiation propagates. If intensity I_ω, expressed in the energy units, is replaced by the quantum flux $I_\omega^{(q)}$, instead of (7.1.10) we obtain

$$\frac{dN_\alpha}{dt} = Y_\alpha(v, \Gamma, \gamma) I_\omega^{(q)}\, d\gamma \mathrm{d}S\mathrm{d}\mathbf{v}. \tag{7.1.14b}$$

Quantity Y_α is referred to as the quantum yield.

Finally, the third group of the elementary processes on the surface includes the processes taking place under the effect of incident particles. Let the flow of the particles of type β to fall on the surface. The velocity distribution of these particles can greatly differ and, consequently, it is natural to characterise them either by the kinetic distribution function f_β ($\mathbf{x}'$, $\mathbf{v}'$, t') or the kinetic function of the flow:

$$g_\beta(\mathbf{x}',\mathbf{v}',t') = v_n' f_\beta(\mathbf{x}',\mathbf{v}',t'), \tag{7.1.15a}$$

where v_n' is the component of the velocity of the particles normal to the given surface element. The incident particles generate flows of greatly differing particles α. These flows can also be characterised by the kinetic functions of the flow:

$$g_\alpha = g_\alpha(\mathbf{x},\ \mathbf{v},\ t)\ . \tag{7.1.15b}$$

We have marked the arguments of the functions g_β with the prime in order to stress the fact that the velocity of the incident particles differs from that of the knocked-out particles, so that they leave the surface with a delay in relation to the moment of incidence of the particles β and, finally, they can leave the surface in the area different from that of arrival of the primary particles. The largest difference between $\mathbf{x}$ at $\mathbf{x}'$, t and t' is found in cases in which the incident particle is adsorbed and remains on the surface for a relatively long time. During this time, as a result of thermal motion, the particle manages to travel a large distance (surface diffusion of the particles). Processes of this type are very important, for example, in the surface ionisation of caesium on tungsten.

From the formal viewpoint, the surface of the solid can be characterised by some operator $\hat{S}$ which transforms function g_β into g_α:

$$g_\alpha = \hat{S}_{\alpha/\beta}\left[g_\beta, \Gamma\right]. \tag{7.1.16}$$

The macroscopic parameters Γ, characterising the surface, are also determined to a large extent by the incident flows and usually change with time. Therefore, for the detailed description of the processes taking place on the surface, in addition to the availability of the operators $\hat{S}_{\alpha/\beta}\left[g_\beta, \Gamma\right]$it is also necessary to have equations for the parameters $\varkappa$, Γ:

$$\frac{\partial \Gamma(\mathbf{x},t)}{\partial t} = \hat{K}\left[g_\beta(\mathbf{x}',t'), \Gamma(\mathbf{x}',t');\ \mathbf{x},t\right]. \tag{7.1.17}$$

The system (7.1.16), (7.1.17) is non-stationary and non-linear in a general case. As an example, we can mention the extensive variation of the properties $\hat{S}$ during covering the surface with adsorbed atoms. For example, surface ionisation of caesium on tungsten is almost completely interrupted if the density of the adsorbed caesium layer is ~0.1 of that of monomolecular caesium. This circumstance restricts the permissible flow density $\int g_\beta d\mathbf{v}$ of the neutral layer on the surface of the emitter.

Operator $\hat{S}$ can be written in the convenient form

$$S_{\alpha/\beta} = \int d\xi' S_{\alpha/\beta}(\xi \mid \xi'; \Gamma) \tag{7.1.18a}$$

Here $\xi \equiv (\mathbf{x},\mathbf{v},t)$, and $S_{\alpha/\beta}(\xi|\xi;\Gamma)$ is the function which will be referred to as the function of forced emission or scattering function[4]. Taking (7.1.18a) into account, we can write the equation (7.1.16) in the form

$$g_\alpha(\mathbf{x},\mathbf{v},t) = \int_{-\infty}^{t} dt \int d\mathbf{x}' \int d\mathbf{v}' S_{\alpha/\beta}(\mathbf{x},\mathbf{v},t \mid \mathbf{x}',\mathbf{v}',t;\Gamma) g_\beta(\mathbf{x}',\mathbf{v}',t) \tag{7.1.18b}$$

If the characteristic lifetime of the particle in the bonded state is considerably shorter than the duration of variation of the macroscopic surface parameters (Γ), and the characteristic migration distance in this period is considerably shorter than the macroscopic heterogeneity, the following equation can be considered

$$S_{\alpha/\beta}(\xi \mid \xi') = S_{\alpha/\beta}(\mathbf{v} \mid \mathbf{v}'; \mathbf{x} - \mathbf{x}'; t - t') \tag{7.1.19}$$

In practice, it is interesting to take into account differences in t and t', x and x' in cases in which these differences are comparable with the time and spatial scales in the investigated processes. If the time delay and diffusion can be generally ignored, then

$$S_{\alpha/\beta} = S^{(0)}_{\alpha/\beta}(\mathbf{v} \mid \mathbf{v}'; \mathbf{x}, t)\delta(\mathbf{x} - \mathbf{x}')\delta(t - t') \tag{7.1.20}$$

and

$$g_\alpha(\mathbf{v}'; \mathbf{x}, t) = \int d\mathbf{v}' S_{\alpha/\beta}(\mathbf{v} \mid \mathbf{v}'; \mathbf{x}, t) g_\beta(\mathbf{v}') \tag{7.1.21}$$

Here and later when writing $S^{(0)}_{\alpha/\beta}$, the index (0) is ignored.

It may easily be seen that the function $S_{\alpha/\beta}(\mathbf{v} \mid \mathbf{v}'; \mathbf{x}, t)$ shows how many particles of the type α are emitted with the velocity in the range $\mathbf{v}$, $\mathbf{v} + d\mathbf{v}$ by 1 cm^2 of the surface area in incidence on the surface of a single particle β with the velocity in the range $\mathbf{v}'$, $\mathbf{v}' + d\mathbf{v}'$:

$$dN_\alpha = S_{\alpha/\beta}(\mathbf{v} \mid \mathbf{v}'; \mathbf{x}, t) d\mathbf{v}\, d\mathbf{v}' \tag{7.1.22}$$

It is often not necessary to know in in detail the distribution of the emitted ('secondary') particles in respect of all velocities. In many cases, the considerations are confined to various types of 'convolutions' of the function $S_{\alpha/\beta}(\mathbf{v}/\mathbf{v}')$. In particular, if we are interested in the total number of the particles of type β, we use the secondary emission coefficient

[4]In gas dynamics, the function $S(\mathbf{v}/\mathbf{v}')$ is sometimes referred to as the scattering core or interaction function.

$$\sigma_{\alpha/\beta}(\mathbf{v}') = \int S_{\alpha/\beta}(\mathbf{v}/\mathbf{v}')d\mathbf{v} \qquad (7.1.23)$$

In theoretical calculations, the functions *S* and are often simulated by simple equations, and this will also be done in a number of cases in the following sections.

7.2. Processes on the surface of the solid [167]

7.2.1. Adsorption of 'warm' particles [164, 165]

Adatoms

The surface of any solid or liquid is characterised by adsorption activity. This is explained by the fact that the surface layer of the atoms (molecules) contains unsaturated bonds in comparison with the atoms in the volume of the body. Therefore, falling with a low energy on the surface, the atom is captured and transforms to the so-called adatom. There are usually two extreme cases of interaction.

Physical adsorption

In this case, the atom is maintained as a result of the polarisation of the adatom by imaging forces or van der Waals forces (Fig. 7.2.1a). The interaction of this type is relatively weak because it is not accompanied by any significant rearrangement of the shell of the adatom. A suitable example of physical adsorption is the adsorption of noble gases by metals, and also adsorption of gases (O_2, N_2, etc), activated by coal, silica gel, and other sorbents.

The measure of bonding of the adatom with the surface is the so-called heat of adsorption *l*. The characteristic values of the heat of physical adsorption are presented in Table 7.1.

Chemical adsorption

The so-called chemical adsorption (chemisorption) is encountered more extensively than physical adsorption. In chemisorption, the electron shell of the adsorbed atom or molecule is subjected to significant rearrangement of the bonds and some of the bonds used for bonding with the surface, are released.

The chemisorption processes can be conventionally divided into two types depending on the nature of the controlling atom–solid bond.

Homopolar bond

In this case, despite the extensive exchange of the electrons between the adatoms and the solid, the adatom does not usually 'give away' its electron and does not accept a 'strange' electron and, consequently,

Table 7.1.

Absorbing substance	Adsorbent	Adsorption heat l, eV	Surface migration energy u_w, eV
Ar	Tungsten	0.08	0.02–0.05
Xe	"	0.35–0.4	~0.1
Kr	"	0.2	>0.05
Xe	Molybdenum	0.35	
Xe	Tantalum	0.2	

remains neutral. In this case, the binding energy may be comparable with the energy in physical adsorption. A suitable example is the adsorption of hydrogen on metals (for example nickel) that are relatively neutral in relation to hydrogen.

Heteropolar bond (strong chemisorption)

Adsorption in the presence of a heteropolar bond is accompanied by a change of the charge state of the particle. There are two types of heteropolar bond. The first type is the so-called strong acceptor bond in which the electrons of the solid are localised in the vicinity of the adatom[5], forming a negative complex (Fig. 7.2.1b). A suitable example is the adsorption of chemically active gases O_2, N_2 and halogens on the surfaces of metal that react with them. The second possibility is a strong donor bond in which the electron of the sort particle 'dissolves' in the solid, leaving a positively charged core (Fig. 7.2.1c). This is detected in the most convincing form in the adsorption of metal vapours whose ionisation potential I is considerably lower than the work function of the metallic surface. A suitable example of this situation is the interaction of the vapours of alkali metals K, Rb, Cs with the surface of refractory metals such as Pt, W, Mo.

In the case of formation of a layer of negative adatoms (see Fig. 7.2.1b), the work function increases. For example, if the work function of pure W is $\varphi = 4.54$ eV, then in the case of tungsten oxide WO_3 $\varphi = 6.15$ eV.

However, if $I < \varphi$ and a layer of positively charged adatoms forms on the surface, the work function decreases because the positively charged ions partially compensate the volume charge of the electron cloud surrounding the metal. Thus, if for tungsten $\varphi = 4.54$ eV, the work function of tungsten coated with caesium approaches that of caesium ($\varphi = 1.81$ eV).

[5]More accurately, the electron from the conduction band or the valence zone are transferred to the local level of the chemisorbed particle.

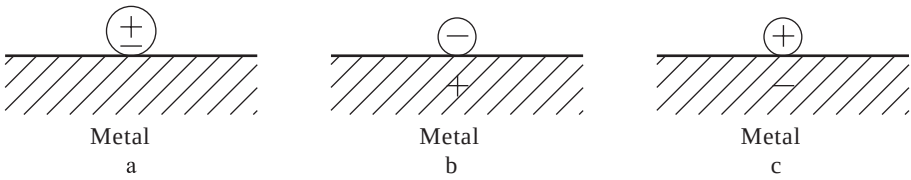


Fig. 7.2.1. Bonding of the adatom with the solid: a) polarisation bond, b) acceptor bond, c) donor bond.

Migration of adatoms

The adatom moves continuously at least owing to the fact that the surface on which it is situated continuously oscillates as a result of thermal oscillations of the particles forming the surface, and the potential energy of the adatom U in addition to the constant component l contains the oscillating function of the coordinates $\tilde{U}$:

$$U = -(l + \tilde{U})$$

Here l is the heat of adsorption, and the quantity $\tilde{U} > 0$ is linked with the local potential wells. If the depth of the local potential well ($\max \tilde{U}$), which is denoted by U_w, is considerably greater than $kT/2$, then the adatoms spends a long period of time in the vicinity of some 'host' and only rarely 'jumps' from one to the other. The characteristic lifetime of the particles in the bonded state is determined by the Frenkel function:

$$\tau' = \tau_0 \exp\left\{-\frac{U_w}{kT}\right\}, \qquad (7.2.1)$$

here $\tau \sim h/kT$; h is the Planck constant. If the amplitude $\tilde{U} \sim kT$, the adatom slides at the thermal velocity along the surface, scatters on its heterogeneities and collides with other atoms. Quantity U_w is often referred to as the heat of migration. Its value is usually ~1 eV.

The propagation of a set of adatoms on the surface is a diffusion process which can be characterised by the diffusion coefficient D_S. If the degree of covering is $\theta \ll 1$, the set of the atoms can be regarded, taking their properties into account, as a two-dimensional gas close to the ideal gas. However, if the value θ is comparable with unity, the gas starts to 'condensate' with the formation of dense clusters – spots.

Desorption of adatoms

Experience shows that in addition to the neutral atoms, the positively and negatively charged atoms can also evaporate. The physical reason for desorption is simple and is associated with the fact that, being on

the surface, the adatom is subjected to random pushes not only along the surface (as discussed previously) but also along the normal to the surface. At some moment, the force of normal effects may reach the level at which the adatom receives the energy sufficient for overcoming the binding energy *l*. It may be shown that the lifetime of the adatom should also be governed by the Frenkel equation:

$$\tau = \tau_0 \exp\left[-\frac{l}{kT}\right]$$

Here quantity τ_0 is the same as in (7.2.1). The scale of the quantities is indicated by the following estimate of the lifetime τ, calculated for room temperature (Table 7.2).

Thus, it may be seen that at the heat of adsorption of $\geqslant 1$ eV, the lifetime of the adatom becomes very long.

Surface ionisation

It is important to mention another specific form of the interaction of atoms (and not only of atoms) with the surface of the solid body. It is the so-called surface ionisation in which the adatom falling on the surface of the solid body is reflected in the form of an ion.

This phenomenon has various applications. Ion engines have been developed in which the working body is ionised at contact with a heated sheet. It is usually caesium which, falling on the surface of tungsten, is desorbed preferentially from the surface in the form of ions. This is explained by the fact that the work function of tungsten ($\varphi \approx 4.5$ eV) is higher than the ionisation potential of the caesium atoms ($I = 3.89$ eV). Therefore, the emission of positive ions Cs^+ is more advantageous from the energy viewpoint. At surface ionisation, the energy required for separating the electron from the atom is taken from the heated sheet. Surface ionisation or, more accurately, positive surface ionisation takes place in principle at any ratio of I and φ, but when $I \geqslant \varphi$ its efficiency is low.

In addition to the positive surface ionisation there is also negative surface ionisation which may be considerable if the affinity of the atom for the electron S is comparable with the work function.

The efficiency of positive surface ionisation may be estimated by the data shown in Table 7.3 which gives the work function (in eV) and ionisation potentials I.

The degree of ionisation α of the flux travelling from the surface of the solid body (emitter) is determined by the Saha–Langmuir equation

Table 7.2

Adsorption heat l, eV	0.004	0.065	0.174	0.65	0.87	1.3
τ, s	$1\cdot10^{-13}$	$1\cdot10^{-12}$	$1\cdot10^{-10}$	$2\cdot10^{-2}$	$1\cdot10^{2}$	$4\cdot10^{9}$

$$\alpha = \frac{n_i}{n_a} = \frac{1}{2}\exp\left\{\frac{e(\varphi - I)}{kT}\right\}.$$

Here T is the emitter temperature. This shows that to increase α it is necessary to reduce the emitter temperature. However, this will reduce the rate of desorption of, for example, Cs atoms (ions) situated on W.

The presence of Cs results in a decrease of α and, consequently, ionisation may be interrupted. The optimum value of T for the W–Cs pair is ~1500–1700 K.

7.2.2. Interaction of particles with superthermal energy with surfaces

The properties of the functions $S_{\alpha/\beta}(\mathbf{v}, \mathbf{x}, t|\mathbf{v}', \mathbf{x}', t)$ depend greatly on the energy of the incident particles. Therefore, the range of the possible values of energy can be naturally divided into a number of ranges. Three ranges will be discussed here: low energy ($\varepsilon_i \leqslant 0.1$ eV), medium (1 eV $\leqslant \varepsilon_i \leqslant$10 eV), and high energy ($\varepsilon_i >$ 50–100 eV).

The low energy particles are not likely to cause irreversible damage in the lattice and, consequently, the functions $S_{\beta/\beta}$ are of special interest here. In the given range these functions depend only slightly on particle energy, although they depend strongly on surface temperature. In most cases, the incident slow particle interacts immediately with a large population of the particles of the solid. In other words, the process in this energy range is of mostly collective nature.

Very interesting phenomena are detected here. For example, regardless of the apparently unavoidable roughness of the surfaces of single crystals on the atomic level, it has been possible in some cases to realise almost mirror reflections of the incident particles. These

Table 7.3.

Substance	Nb	Mo	Ta	W	Re	Pt	$MoSi_2$	H_2O
φ, eV	3.99	4.2	4.13	4.54	4.98	5.4	5.0-6.0	6.1

Substance	Li	Na	K	Rb	Cs	La	Hg	Ba	Al
I, eV	5.39	5.14	4.34	4.18	3.89	5.61	5.5	5.81	5.98

reflections are detected, for example, in interaction of the helium flows with single crystal surfaces of different metals (Fig. 7.2.2a). The appearance of the surface of adsorbed particles and of defects of different types and also the increase of surface temperature 'flattens' the reflected beam. In addition to helium, beams of hydrogen molecules can also be reflected in the almost mirror fashion. However, the atoms and molecules are reflected in a more diffuse nature (Fig. 7.2.2b).

To characterise the interaction of particles with the surface measurements are often taken of the so-called coefficient of energy accommodation α_E which can be determined by simple experiments. This slightly 'archaic' at the present time parameter was introduced by Knudsen using the relationship:

$$\alpha_E = \lim_{\varepsilon_i \to \varepsilon_r} \frac{\varepsilon_i - \varepsilon_r}{\varepsilon_i - \varepsilon_{wall}}. \qquad (7.2.2)$$

Here ε_i is the mean energy of the incident particles; ε_r is the mean energy of the reflected particles; ε_{wall} is the mean energy of the same particles, corresponding to the wall temperature. Table 7.4 gives the values of α_E for different gases on tungsten.

The table shows that the value α_E can be very small and this is in agreement with the low probability of sticking of He in the given temperature range. On the other hand, the gases (Kr and Xe) whose probability of 'sticking' in the given temperature range is close to unity, are also characterised by the accommodation coefficient close to unity.

Adsorption and reflection of fast ions and atoms ($\varepsilon_i \gg E_0$)

In the medium energy range (1 eV $\leqslant \varepsilon_i \leqslant$10 eV) where the kinetic energy ε_i of the incident particle is comparable with the binding energy of the atom (ion) in the lattice E_0, it is necessary to take into account a huge number of factors. Therefore, we shall not discuss these factors and transfer to high energies at which $\varepsilon_i \gg E_0$[6], i.e. ε_i > 100 eV. The general features of the processes taking place here are explained using 'the dynamic' method. In this method, the surface is irradiated with short pulses of ions and this is accompanied by analysis of the particles travelling away from the surface. Investigation of the delay of the arrival of the particle shows that there are three groups of particles:

1. Naturally reflected particles, which arrive with a minimum delay and a relatively high energy comparable with ε_i;
2. Evaporated particles, representing the results of desorption of

[6]It should be mentioned that the terms 'low', 'medium' and 'high' energies do not coincide with those used in investigating the interaction of ion fluxes with the surface. Therefore, we must use different terms in this and subsequent sections.

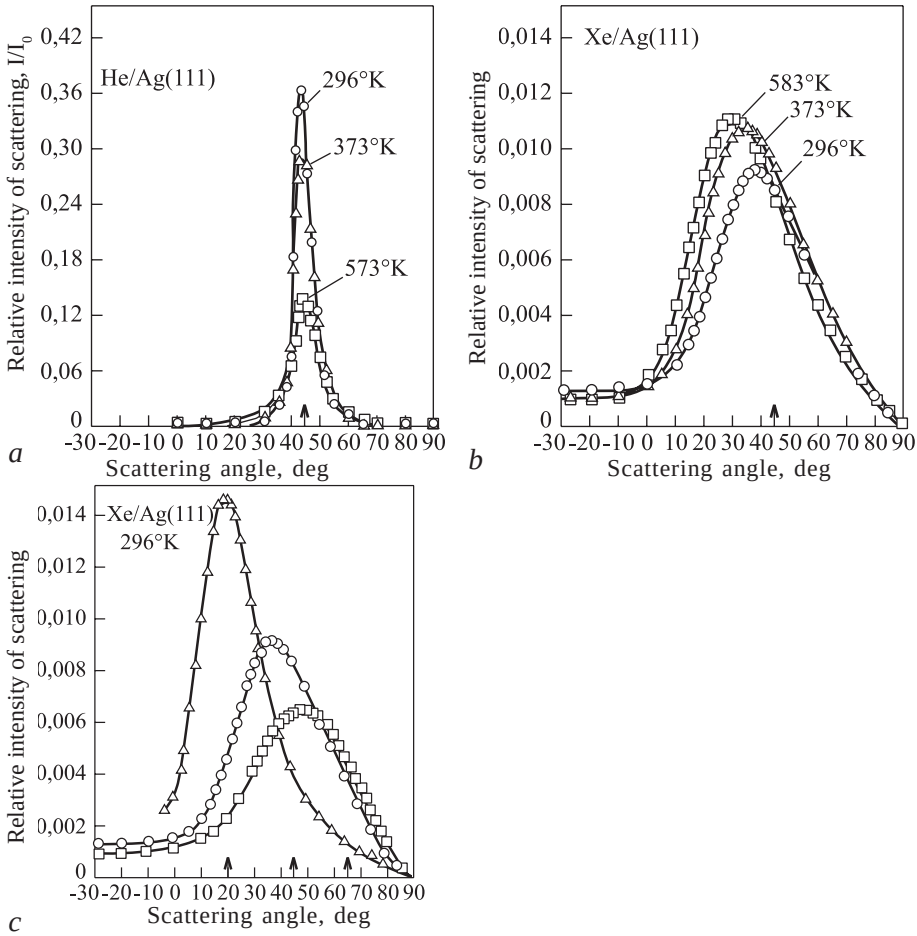


Fig. 7.2.2. The angular distribution of the atoms scattered by crystal surfaces: a) the scattering of helium atoms at different surface temperatures (face 111) of silver single crystals; c) the scattering of the xenon atoms at different incidence angles on the surface of single crystal silver (the arrows indicate the mirror reflection angles).

Table 7.4

T_{wall}, K	He	Ne	Ar	Kr	Xe
113	0.0137	0.049	0.47	0.74	0.93
193	0.0148	0.042	0.34	0.57	0.85
303	0.0166	0.045	0.27	0.46	0.77

the bombarding particles ‘sticking’ to the surface;

3. Diffused particles which formed as a result of the penetration of the primary particles into the depth of the subsurface layers and

subsequently travelled to the surface as a result of diffusion and were desorbed.

The results of experimental and theoretical analyses show that at $\varepsilon_i \geqslant 200$ eV the process of reflection of light ions from the surface can be investigated in a first approximation as the paired collision of these ions with the atoms (ions) of the surface. The word 'light ions' indicates that their mass $M_i < M_0$ are the masses of the particles of the solid. If the incident ion is 'light', then, colliding with the 'heavy' surface particle, it 'jumps away' from it. However, if on the other hand the mass $M_i > M_0$, the incident ion drags behind itself a surface particle and, in this case, the interaction process becomes very rapidly collective because many particles of the solid body start to take part in it. Therefore, in particular, the reflection coefficient of the light ions is considerably higher than that of the heavy ions.

In collision of the particle with mass M_i and energy ε_i, falling on the surface along the normal, with the stationary particle M_0, the maximum energy of the reflected particles $\varepsilon'_{\max}$ is

$$\varepsilon'_{\max} = \frac{M_0 - M_i}{M_0 + M_i}\varepsilon_i = \mu\varepsilon_i. \tag{7.2.3}$$

This case is realised at the impact parameter at which the angle between the initial and final velocities of the incident particles is equal to $\pi/2$.

A suitable example of the experimental data obtained for different ratios M_i/M_0 are the energy spectra of the Rb and Cs ions, reflected from the surface of Mo[7] (Fig. 7.2.3a). It can be seen that the spectrum of the Rb^+ ions extends to the energy range greater than 200 eV. This is in good agreement with (7.2.3) if it is taken into account that for the Mo + Rb^+ pair $\mu = 0.12$, and E_i in this case is equal to 2 keV. At the same time, the energy spectrum of Cs^+ is compressed completely in the region of low energies $\lesssim$50 eV because the incident ion interacts with the entire group of the atoms of the solid. If it is assumed that the incident particles are reflected only from the first layer of the surface atoms, then it may be shown that the distribution function of the reflected particles has the form:

$$g(\varepsilon') = \begin{cases} \text{const} & \varepsilon'_{\min} < \varepsilon' < \varepsilon'_{\max}; \\ 0 & \varepsilon' < \varepsilon'_{\min},\ \varepsilon' > \varepsilon'_{\max}. \end{cases} \tag{7.2.4}$$

Thus, the distribution function should have the form of a 'table' in

[7]It should be mentioned that the relative atomic masses of Re, C and Mo are equal to 85.5, 132.9 and 95.5, respectively.

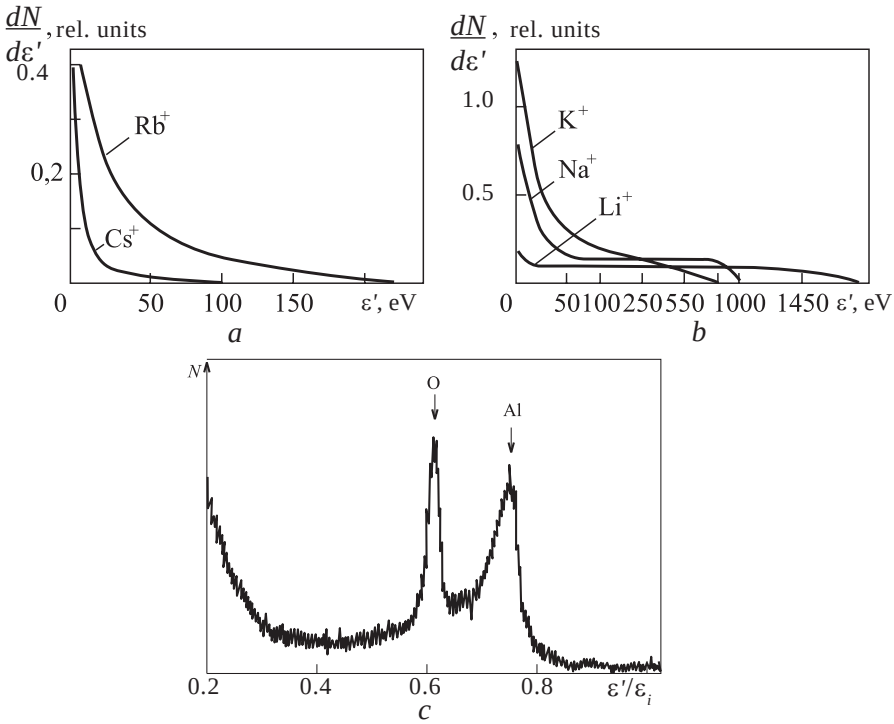


Fig. 7.2.3. Energy distribution of the reflected ions: a, b) reflection from the molybdenum surface of primary ions with the energy of 2 keV, incident along the normal; c) reflection of He^+ ions with the energy of 1.5 keV from the surface of aluminium oxide.

the model of the paired collisions, and the tendency for this is clearly indicated by the experiments carried out at relatively high values of μ (Fig. 7.2.3b). However, the presence of a large number of low-energy particles in these spectra indicates unambiguously the role of the collective processes even at $M_i < M_0$.

At present, the investigations of the energy spectra of scattered primary ions are regarded as a powerful tool for investigating the chemical composition of surfaces (scattered ion spectroscopy, SIS).

Computer modelling and direct experiments show that, depending on the type of ions and the irradiated solid, reflection takes place from different numbers of the crystal layers. Table 7.5 shows the characteristic data for bombardment with ions with the energy of 2.2 keV of the (001) plane of copper.

Attention should be given to the penetration of He^+ ions to a large depth. As a result of this, He can accumulate inside the irradiated specimens in microbubbles (blistering), with separation of the flakes

Table 7.5

Number of layer n	Incident ion			
	Cu^+	Ar^+	Ne^+	He^+
1	1	0.23	0.27	0.1
2	0	0.39	0.27	0.1
3	0	0.24	0.21	0.04
$\sum(n>3)$	0	0.13	0.25	0.76

of the bombarding material (flaking)[8].

Up to now, when discussing the reflected heavy particles, we have not mentioned their charge state. As shown by experiments and qualitative considerations, the fraction of the ions in the reflected flux of the heavy particles depends strongly on the ratio of the ionisation potential of the reflected atom I and the work function of the target φ. If $I > \varphi$, which is observed, for example, in bombardment of metals with ions of noble gases, the fraction of the charged ions is very small ($\leqslant$1%). On the other hand, if $I < \varphi$, the reflected flux consists almost completely of the positively charged ions. A suitable demonstration of these are the experiments in which the ions of alkali metals (for example, K or Cs) bombarded the surface of heated tungsten. In this case, the total flux of the reflected charged particles was almost equal to the incident ion flux.

7.2.3. Sputtering of surfaces [166]

If the energy of the incident particles is higher than the binding energy E_0 of the solid body particles, then in collision with the surface they start to 'sputter' the surface, i.e. knockout atoms, ions or small complexes. Only the sputtering of pure surfaces by ions is initially considered. This type of sputtering was detected for the first time in the cathode of a gas discharge tube (Grove, 1852) and is it is therefore often referred to as cathodic sputtering. The sputtering products are in most cases the neutral atoms whereas the fraction of the ions usually does not exceed 1%. The sputtering of the surfaces with the ions is of considerable importance both for technology and for plasma dynamic systems because in many cases this process controls the service life of these systems.

The main characteristic of sputtering by ions is the sputtering

[8]The characteristic temperature range from blistering is $T \leqslant 0.1\ T_{sur}$, and for flaking it is 0.1–0.4 T_{sur}.

coefficient

$$S = \frac{N_n}{N_p}, \tag{7.2.5}$$

which characterises the number of the emitted sputtered atoms[9] per one incident ion. The typical form of the dependence of S on energy ε_i of the incident ions is shown schematically in Fig. 7.2.4. The graph shows five regions. There is no sputtering in region I. Region II is the vicinity of the sputtering threshold, $\varepsilon^* \sim 30$ eV. In region III coefficient S increases linearly with a high degree of accuracy, and the characteristic values are $\varepsilon_1 \sim$ (30–50) eV, $\varepsilon_2 \sim 0.5$ keV. The rate of increase then drops, passes through the maximum in region IV and in region V ($v_i \sim 10^8$ cm/s) the value of S usually starts to decrease.

Prior to transferring to specific characteristics of sputtering in each energy range, it is necessary to make one general comment. It may appear at first sight that the physical pattern of the processes responsible for sputtering is simple and can easily be analysed theoretically. However, this is not so, and the sputtering mechanism is very complicated. Sputtering itself is to some extent a secondary and very weak effect in the general pattern of interaction of the energy particles with the surface, which is indicated by the fact that, firstly, the energy carried away by the knockout atom is usually less than 1% of the energy of the incident particle. However, the main process in interaction of the fast ions in the surface of the solid is the destruction of the lattice in the subsurface layer.

General information on the phenomena taking place can be obtained by examining Fig. 7.2.5 which shows the model pattern of movement of the iridium ion with $\varepsilon_i = 50$ keV in the two-dimensional crystal lattice calculated in a computer. The graph indicates that sputtering is actually a secondary process. Attention should also be given to the presence of long straight sections of movement of knocked-out particles. These sections form as a result of the movement of the particles between the atomic layers (channelling effect). The irradiated surface layer is a highly nonequilibrium system with a high mobility of the particles resulting in the formation in long-term (dose $> 10^{23}$ helium ions) of especially exotic structures of the type shown in Fig. 7.2.6[10]. Attempt to develop a method for calculating the sputtering coefficient S have

[9]This definition is understandable in the sputtering of simple substances. When sputtering complicated substances (insulators), it is more convenient to use the mass characteristics of the type $S = \Delta m/(j_i)$, where j_i is the ion flux density, Δm is the mass, sputtered in unit time.

[10]However, in addition to this, sputtering in the appropriate conditions (in particular, at high incidence angles) can be applied to polish the surface of the solid body (see later).

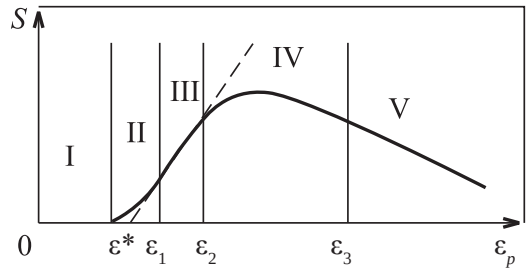


Fig. 7.2.4. Dependence of the sputtering coefficient on the energy of bombarding ions. I) no sputtering, II) transition region, III) linear region, IV) the region of the maximum sputtering coefficient, V) the region of decrease of the sputtering coefficient with increasing particle energy.

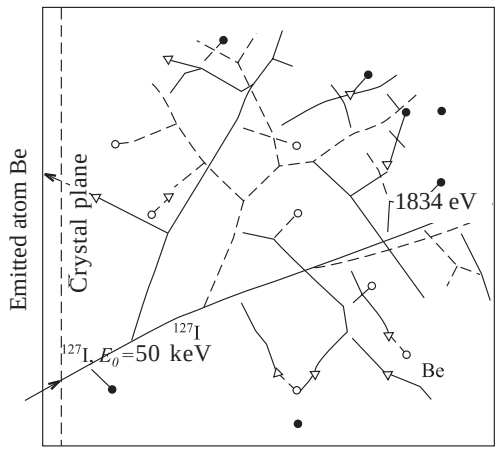


Fig. 7.2.5. Calculated trajectories of movement of an iridium ion with the initial energy of 50 keV and cascades of displaced atoms, formed in a beryllium oxide crystal: ○ – the area of arrest of the oxygen atom; ● – the area of arrest of the beryllium atom, △ - substituting collision.

been made many times but up to now their accuracy is limited because they do not take into account the entire complexity of the processes leading to sputtering. The most efficient results in this area are provided by systematisation of the experimental data and numerical calculations in computers. An example of the calculation of the formation of the surface in sputtering with medium energy particles is shown in section 7.4.3.

Sputtering in the vicinity of the threshold

Such a threshold exists because the sputtering coefficient for every ion–surface pair rapidly decreases on approach to some energy ε_0. For a number of pairs it can be assumed that in the vicinity of the sputtering

Table 7.6

Ion / Target	Ar^+	Xe^+	Hg^+	Cs^+
Be	15	15	–	–
Al	13	–	18	–
Ti	20	–	25	–
Cu	17	15	20	16
Mo	24	–	–	–
W	33	30	30	16
Re	35	30	35	–

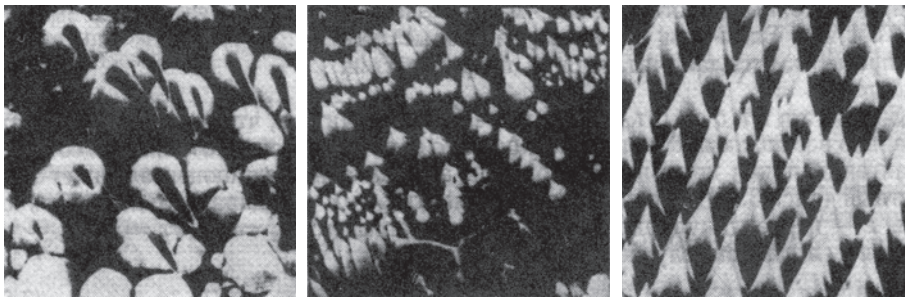


Fig. 7.2.6. Photographs of the surface of a copper single crystal bombarded with Ar ions with an energy of 40kleV.

threshold

$$S \sim (\varepsilon_i - \varepsilon_0)^n, \qquad (7.2.6)$$

where $n \sim 2$–3. In most cases, for the same incident ions, ε_0 increases with increasing sublimation heat of the target. The experimental estimates of ε_0 for a number of pairs are presented in Table 7.6.

The sputtering thresholds of the dielectrics are considerably higher than those of metals.

The medium energy range ($\varepsilon_i \sim (0, 1$–$5)$ keV). Figure 7.2.7 shows the characteristic dependence of the sputtering rate for different ion–target pairs. Attention should be given to the following moments:

– the dependence $S(\varepsilon_i)$ is similar to the linear dependence in almost all cases, if 50 eV $< E_i <$ 500 eV;

– in the region $E_i >$ 800 eV function $S(\varepsilon_i)$ is approximated quite accurately by the dependence $S \sim \sqrt{\varepsilon_i}$;

– the sputtering coefficient of the dielectrics is usually lower than that of the metals.

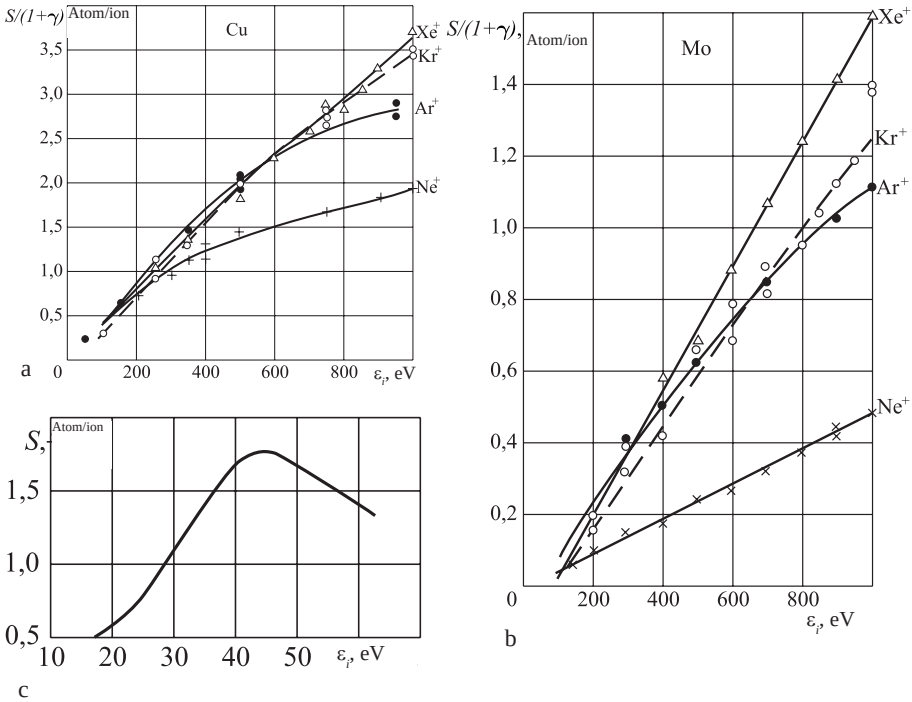


Fig. 7.2.7. Sputtering coefficient of copper (a) and molybdenum (b) by the ions of noble gases. The sputtering coefficient of molten quartz by xenon ions (c).

As already mentioned, the processes leading to sputtering are very complicated. This can be seen clearly in particular in the dependence of S on the atomic numbers of the incident ions Z_i and the target Z_0 in Fig. 7.2.8. It can be seen that $S(\varepsilon_i, Z_i, Z_0)$ is a quasi-periodic function of Z_i and Z_0. The periodicity of S in relation to Z_0 is determined in particular by the periodicity of the binding energy of the surface atoms.

The sputtering coefficient also depends greatly on a number of other parameters. The characteristic dependence of the sputtering coefficient on the incidence angle of the ions is shown in Fig. 7.2.9. It may be seen that the sputtering coefficient initially rapidly increases with increasing incidence angle:

$$S \sim (\sec\varphi)^n; \quad 1 < n < 2,$$

and subsequently starts to decrease, approaching zero. The angle at which S starts to decrease increases with increasing ion energy. This region of sliding incidence of the beam is also characterised by efficient polishing.

In sputtering of single crystals the yield of sputtered atoms depends on the direction of the crystallographic axes. The sputtering coefficient also depends on the crystallographic plane on which the ions fall.

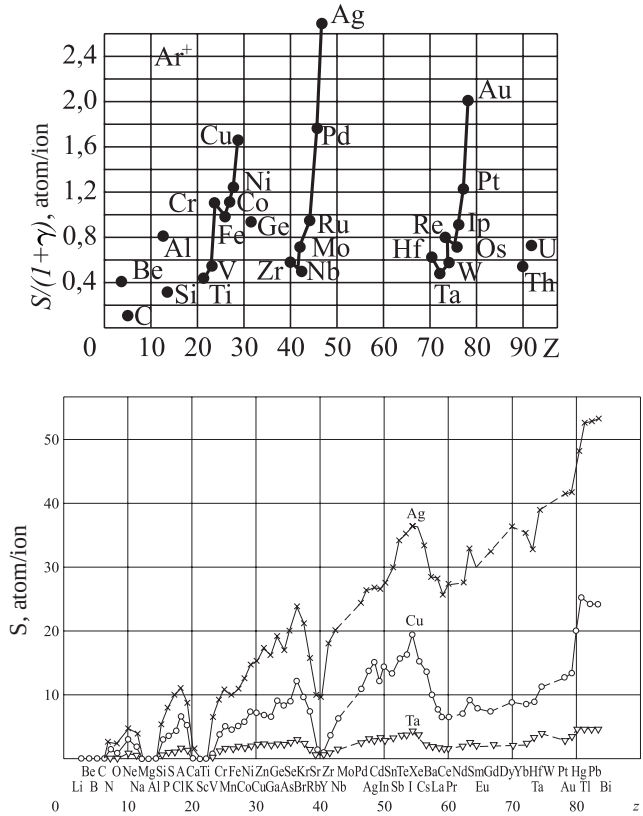


Fig. 7.2.8. Dependence of the sputtering coefficient of the target on the atomic number of the sputtered element in irradiation with argon ions with the energy of 45 keV (a). The same, in irradiation of targets made of Ag, Cu and Ta with the ions of 68 elements (b).

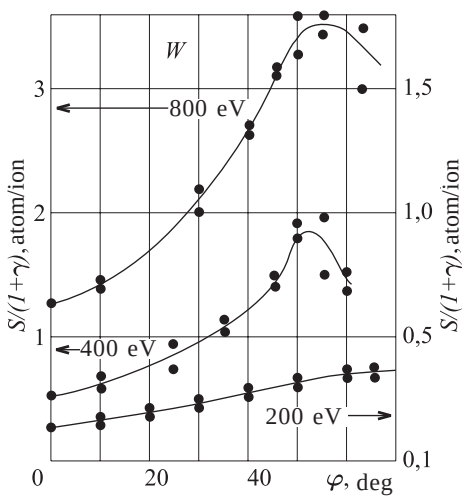


Fig. 7.2.9. Typical dependence of the sputtering coefficient on the incidence angle (sputtering of tungsten by mercury ions).

Figure 7.2.10 shows the energy spectra of the sputtered atoms knocked out from a copper single crystal by krypton ions with the energy of 80–1200 eV. The energy distribution function of the atoms has the form

$$\frac{dN}{d\varepsilon'} \sim B\sqrt{\varepsilon'}\exp\left\{-\frac{\varepsilon'}{W}\right\} \qquad (7.2.7)$$

where B and W are constant coefficients. The angular distribution of the sputtered particles at relatively low energies ($\varepsilon_i \leqslant 1$ keV) is 'flattened' and greatly differs from the cosine law (Fig. 7.2.11). The flatness of the polar diagram decreases with increasing particle energy.

7.2.4. Emission of electrons from surfaces [168, 169]

The emission of the electrons can be spontaneous (thermoemission) and can be caused by greatly differing factors: electrical fields (auto- or cold emission), electromagnetic radiation (photoemission), bombardment of the surface with the electrons (secondary electron emission) and ions (ion–electron emission), and passing current (explosion emission). In the actual conditions, the efficiency of emission depends greatly on the surface condition which in turn is controlled by the surrounding medium. Consequently, the emission of the electrons in the investigated system may greatly differ from the tabulated values.

Thermoemission. Effective cathodes

It is well known that thermoemission is explained by the fact that the

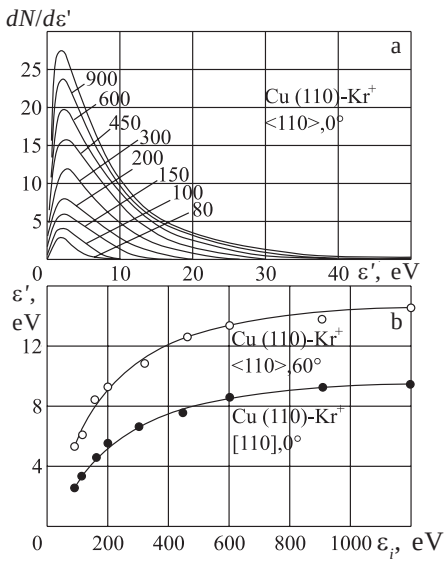


Fig. 7.2.10. Energy spectra (a) and mean energy (b) of the atoms knocked out from a copper single crystal by krypton ions.

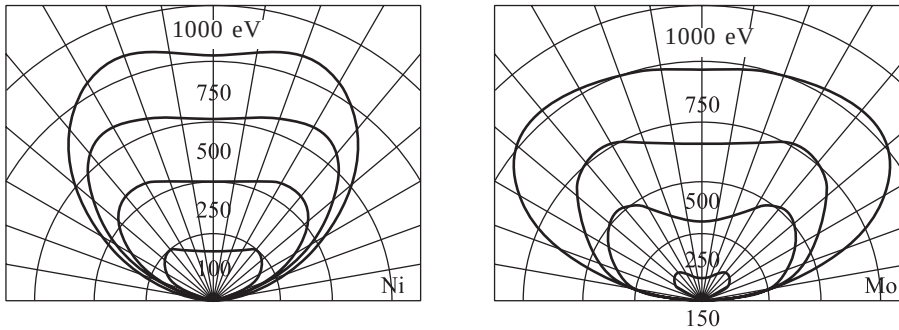


Fig. 7.2.11. Angular distribution of the atoms knocked out by Hg ions with different energy from Mo and Ni.

electrons which are situated in the volume of the metal in the vicinity of the surface and which travel at relatively high velocities in the direction normal to the surface as a result of collisions, are capable of overcoming the attraction forces to the metal and can leave the metal.

Thermoemission of a metal is described by the Richardson-Dashman law

$$j = AT^2 \exp\left\{-\frac{e\varphi}{kT}\right\}. \qquad (7.2.8)$$

Here A is the Richardson constant, φ is the work function. This equation can be derived on the basis of thermodynamic considerations investigating the equilibrium between the degenerated electron gas inside the metal and the classic electron cloud in the vicinity of the metal.

In the case of barium, φ = 2.5 eV, A = 60 A·cm^2/deg^2, for molybdenum φ = 4.3 keV, A = 60 A·cm^2/deg^2, for tungsten φ = 4.6 eV, A ~ 100 A·cm^2/deg^2.

The attempts to produce efficient emitters with a low work function have resulted in the construction of four types of cathodes.

Cathodes with a complicated composition and with super low work function

They include the so-called oxide–barium cathodes which are complicated systems of the oxides of Ba, Sr, Ca. The characteristic work function of these cathodes is 1.0–1.2 eV. At the same time, owing to the fact that Ba and other metals are bonded by oxygen, the evaporation rate of these cathodes is not high. At a working temperature of ~900°C they can operate for thousands of hours, emitting currents with the density of up to 1 A/cm^2. Unfortunately, they are easily ‘poisoned’, especially in air. The physical processes in these cathodes are complicated and we will not discuss them. The lowest value of the work function is recorded

for the cathodes of photoelements produced on the basis of Ag–O–Cs. The work function of these cathodes is 0.6 eV. However, these are very 'gentle' surfaces.

Resistant cathodes with complicated composition and low work function

A suitable example of emitter of this type is the well-known lanthanum hexaboride LaL_6 with high mechanical characteristics. The work function of LaB_6 in the working temperature range (up to 1500°C) is 2.6 eV, the Richardson constant ~50 A/cm^2. The cathode made of lanthanum hexaboride can operate in air and retain its properties in storage in corrosive media. The characteristic values of current density j are equal to respectively: 0.3 A/cm^2 at T = 1300°C; 1.0 A/cm^2 at T = 1400°C, and 10 A/cm^2 at T = 1600°C. The service life at 1600°C is not shorter than 5000 h.

Cathodes with the 'reduced' surface (film cathodes)

The principle of film cathodes has been utilised most efficiently in the so-called metallo-capillary cathodes (L-cathode). During desorption, the active material (for example, Ba) diffuses from a special box through a porous diaphragm onto the emitting surface. This group also includes the cathodes produced from thoriated or lanthanised tungsten. However, in this case the film is maintained as a result of diffusion of the active addition (La, Th) from the cathode volume.

Hollow cathodes

This type of cathode is already regarded as some gas-discharge device. Its distinguishing feature is the cathodic block, having the form of a cavity with a small orifice (Fig. 7.2.12). If the internal part of the cavity is an active emitter, then in operation with a very low evaporation rate it is possible to obtain high currents as a result of the electrical field 'falling' into the cavity resulting in the formation of plasma in the cavity, collecting the electron flux from the entire surface of the cavity.

Autoelectronic and explosion emission

If a strong electrical field, pulling out the electrons, appear in the vicinity of the investigated surface, the field influences the emission properties of the surface. The mean scale fields (~10^4–10^5 V/cm) resulted in quantitative changes of emission current (Schottky correction), and strong fields ($\geqslant 10^6$ V/cm) generate a qualitatively new phenomenon of cold or autoelectronic emission. Reasons for these defects can be easily

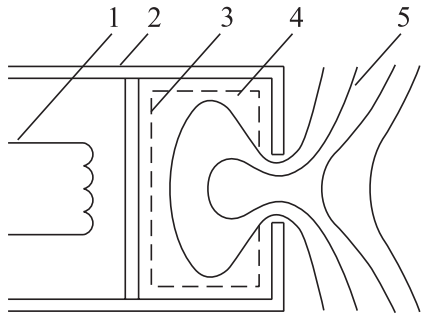


Fig. 7.2.12. The diagram of a hollow cathode: 1) heater, 2) the body, 3) the active emitting layer, 4) the cavity, 5) equipotential **E**-fields.

understood if we consider the distribution of the potential energy of the electron in the system (Fig. 7.2.13).

It can be seen that the appearance of the external electrical field is capable of exerting a qualitative effect on the exit of the electrons from the metal. Actually, if at $E = 0$ the thickness of the potential barrier surrounding the 'well with the electrons', is unlimited, then when the field appears ($E < 0$) the barrier thickness decreases with increasing E. Consequently, this may cause the 'tunnelling soaking-through' of the electrons directly from the entire 'well' (and not only from the vicinity of the Fermi level) into the external part of the space. This 'soaking-through' can evidently take place at $T = 0$ and, therefore, this type of emission is often referred to as cold emission, although the term 'autoelectronic emission' is used most extensively in this case. The equation for the density of the current of autoelectronic emission was derived by Fowler and Nordheim. The equation has the form

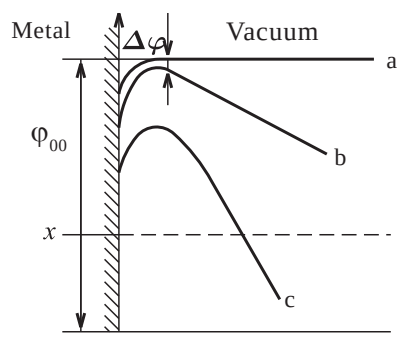


Fig. 7.2.13. Potential barrier at the metal–vacuum interface in the absence of the **E**-field (a) and in the presence of a weak (b) or a strong field (c).

$$f_F = a\frac{E^2}{\varphi}\exp\left\{-\frac{b\varphi^{3/2}}{E}\theta(\alpha)\right\}. \qquad (7.2.9)$$

Here φ is the work function; $\alpha = \frac{\sqrt{eE}}{\varphi}$; $a = \frac{e^2}{8\pi m}$; $b = \frac{8\pi}{3n}\sqrt{2m}$. The function $\theta(\alpha)$ is the Nordheim function which can be approximated by the equation (I.I. Beilis)

$$\theta(\alpha) = 1 - \alpha^2\left\{1 - 0.85\sin\left(\frac{1-\alpha}{2}\right)\right\}.$$

The orders of magnitude of the values defined by equation (7.2.9) can be estimated on the basis of the following values: at φ = 4.5 eV and $E = 3\cdot10^7$ V/cm, the current density is j = 0.03 A/cm^2).

Experimental verification of the Fowler–Nordheim equation shows that in experiments extensive emission from the cold electrodes is detected in the fields ~10^5–10^6 V/cm, whereas the theory requires the fields 10–100 times stronger for the formation of the same current density. This difference is associated to a large degree with the presence of microscopic sharp edges, explosion of these edges under the effect of high current and with the effect of the resultant plasma on both the increase of the intensity of the field around the emitter and on the emission of electrons by plasma. In this regime, the emission is referred to as 'explosion emission' (G.A. Mesyats). Micrographs also show a unique phenomenon of the growth of thin 'whiskers' on the surface of the cold cathodes. This growth was caused by the 'fluidity' of the surface of metal under the effect of the strong electrical field. By taking into account the sharp edges on the electrodes and the formation of plasma it is possible to match the Fowler–Nordheim theory and experiments.

If an electrical field is generated at the heated thermoemitter, the effect of the field on emission current can also be detected at relatively low values of E. This phenomenon is referred to as the Schottky effect, and is taken into account by means of the so-called Schottky correction in the Richardson–Dashman equation:

$$j = AT^2\exp\left\{-\frac{e(\varphi - \sqrt{eE})}{kT}\right\}. \qquad (7.2.10)$$

Thus, this correction is reduced to the effective decrease of the work function by the value $\left(-\sqrt{eE}\right)$.

Photoemission

Until now, we have been discussing the effect of electrostatic fields on electron emission. Attention will now be given to another extreme case – electron emission under the effect of visible and shorter wave radiation.

The photoemission of electrons (external photoeffect) still plays only a small role in plasma dynamics system. However, it is evident that it will become more important with time, both in the laboratories because of the increase of the power of light sources and in the cosmic space conditions where photoemission under the effect of solar laser radiation may become the source of electrons for solving greatly differing problems. The main laws of photoemission are well known and, consequently, we confine ourselves to presenting the experimental values of the quantum yield Y for a number of materials.

It should be mentioned that $h\nu = 1$ eV at $\lambda = 12\,400$ Å. The quantum yield of pure metals in the visible range is relatively small ($\sim 10^{-3}$–10^{-4}). At the same time, using complex cathodes (Cs + O + Ag; Sb + Cs + O, etc) it is possible to increase greatly the quantum yield. For example, for an oxygen–caesium cathode $Y \approx 2$–3%, and for an antimony–caesium cathode it is up to 20–30%.

Secondary electron emission

Emission under the effect of incident electrons: secondary electron-electron emission (SEE)

We shall not discuss the problems of the theory of these interactions and shall restrict our considerations to the main experimental results. The largest number of experimental data relate to σ – the so-called total coefficient of secondary electron emission measured at normal incidence of the beam of primary electrons on the surface. The flux of secondary electrons (i.e. the electrons emitted by the surface under the effect of the incident electron beam) is often divided into three groups: elastically reflected, inelastically reflected, and truly secondary electrons [170].

If σ is the total number of the electrons emitted by the surface under the effect of a single incident electron, the appropriate amounts for the elastically reflected, inelastically reflected and truly secondary electrons are denoted by r, η, δ, respectively. Evidently

$$\sigma = r + \eta + \delta \ . \tag{7.2.11}$$

The coefficient of inelastic reflection η is often determined as the ratio of the number of secondary electrons with the energy higher than 50 eV to the number of the incident primary electrons. Thus, at $\varepsilon_i < 50$ eV it is assumed that the inelastically reflected electrons do not exist.

The typical dependences of σ on the energy of the primary electrons ε_i for metals of insulators are shown in Fig. 7.2.14a. It may be seen that the value of σ initially increases with increasing energy and then starts to fall. The absolute values of σ for the dielectrics are considerably higher than for the metals (Fig. 7.2.14b). In the low energy range, the curves $\sigma(\varepsilon_i)$, $r(\varepsilon_i)$ and $\delta(\varepsilon_i)$ for different metals and dielectrics are quite similar (Fig. 7.2.15).

For the majority of the plasma systems associated with the interaction of the electrons with the surfaces, the following condition is of considerable importance: $\sigma < 1$ or $\sigma > 1$, i.e. whether the surface absorbs electrons or emits them. Therefore, every surface can be characterised by two energies ε_{i1} and ε_{i2} at which $\sigma = 1$. These values are referred to as the first and second threshold of multiplication, respectively. Table 7.7 shows the values of ε_{i1} and ε_{i2} for a number of substances, and also the maximum values σ_m and corresponding energies.

Attention should be given to the fact that in the dielectrics, the first multiplication threshold $\varepsilon_{i1} \sim 20$ eV, which is considerably lower than the first multiplication threshold of the metals. This is also the reason why the value of σ_m for the dielectrics is considerably higher than for the metals.

The dependence of σ and r on the energy of the incident electrons for a number of surfaces is shown in Fig. 7.2.16. The first maximum on the curve corresponds to the truly reflected particles.

It should be added here that the investigation of the interaction of the electron beams with the surfaces has yielded a large number of advanced methods of controlling the structure and chemical composition of the surfaces. Special attention is given to the so-called Auger electron spectroscopy, based on the examination of the spectrum of secondary electrons so that it is possible to determine the atomic composition of the surface of the solid body, Fig. 7.2.17.

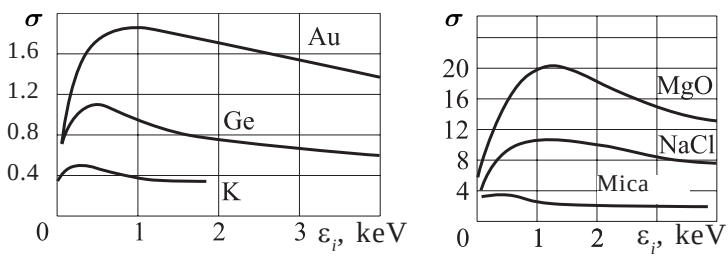


Fig. 7.2.14. The total coefficient of secondary electron emission at medium energies.

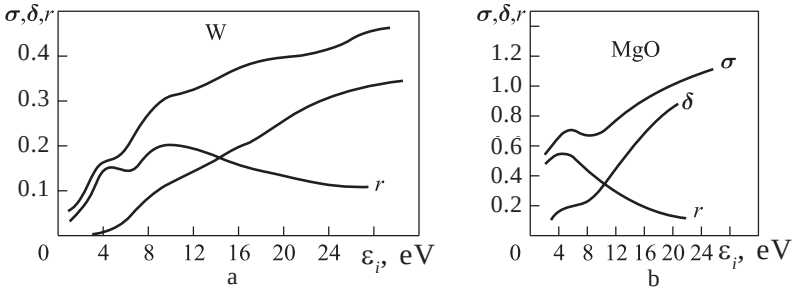


Fig. 7.2.15. Dependence of the components of secondary electron emission on the energy of the incident electrons E_i in the low energy range.

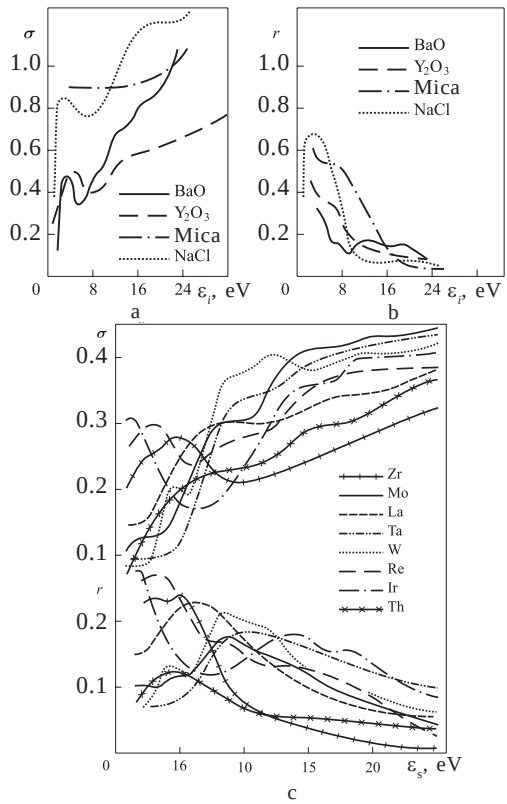


Fig. 7.2.16. Coefficient of SEE for different dielectrics (a, b) and metals (c) at low energies.

Emission of electrons under the effect of ions

If the ion flux falls on the surface this usually results in the emission of electrons referred to as secondary electron–ion emission. It has been shown that there are two mechanisms of separation of the electron from the surface of matter: potential and kinetic.

Table 7.7

Substance	σ_m, rel. units	ε_{pm}, eV	ε_{p1}, eV	ε_{p2}, eV
Be	3.4	2000	16	4000
MgO	20.0	1500	20	4000
Al_2O_3	1.54.8	350–1300	30–50	4000
quartz	2,1–4.9	400–440	20	2300
NaCl (20°C)	17.0	1000	–	3200
Li	0.5	75	–	–
C (graphite)	1.0	300	200	–
Cu	1.4	700	200	3000
Mo	1.2	600		3000
La	1.0	500	200	
Ta	1.3	700	200	2000
W	1.4	700	200	2000
Hg	1.6	700		2400

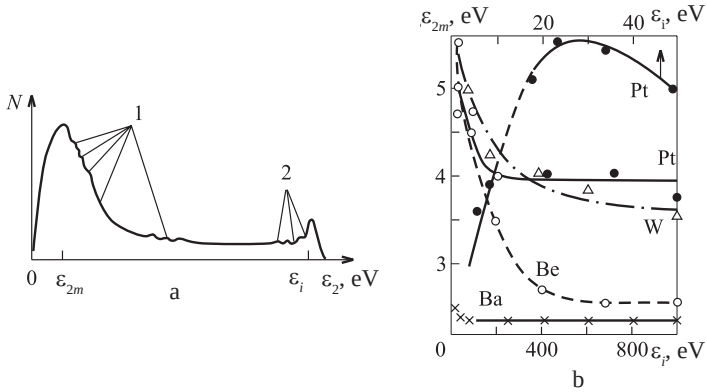


Fig. 7.2.17. Typical form of the spectra of secondary electrons: a) schematic of the curve of energy distribution of secondary electrons (1 – Auger electrons, 2 – the peaks of characteristic losses); spell b) dependence of ε_{2m} on ε_i.

In potential emission, the energy required for separation of the electrons is generated during the recombination of the ion on the surface. If I is the ionisation potential and φ is the work function, the essential condition for potential emission is the condition:

$$I > 2\ \varphi.$$

Actually, to ensure extensive emission, the ion travelling to the surface should pullout not one but two electrons. One of these electrons is used to neutralise the ion (resulting in the generation of energy eI), and the second electron remains free. Therefore, potential emission can be

detected only if I is relatively high or φ is very small. Figure 7.2.18 shows the experimentally recorded dependences of the potential emission coefficient $\gamma(\varepsilon_i)$. Attention should be given to the fact that the electron yield is almost completely independent of the energy of the incident ions ε_i.

The energy of the majority of secondary electrons is lower than $I–\varphi$.

Potential emission from the surface can be caused not only by ions but also metastable atoms. If excitation energy is $E \sim I$, the coefficient of electron emission may be close to the coefficient of electron–ion emission.

Kinetic emission is determined by the kinetic energy of the particles. Therefore, this type of emission can be detected only if $I < 2\varphi$. Kinetic emission has a distinctive threshold. Figure 7.2.18 shows the experimental curves of the secondary emission coefficient in bombardment of a clean molybdenum surface with the Ar^+ ions and Ar atoms. It can be seen that the curves differ by a constant component, equal to the potential emission coefficient. The energy spectrum of the secondary electrons is situated in the range $\varepsilon < 10$ eV.

7.3. Electron boundary layers

Up to now, we have been dealing with the processes taking place directly on the surface of a dielectric or a conductor which is in contact with a

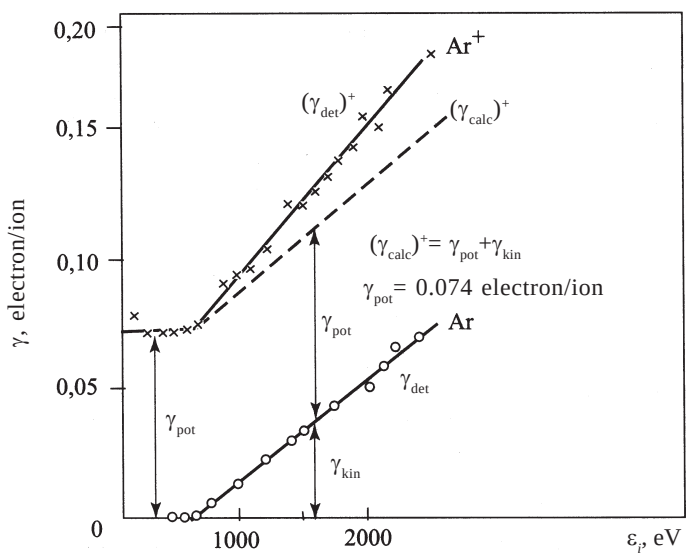


Fig. 7.2.18. Energy dependences of the determined values of the total coefficient of secondary electron emission γ_{det}, and also coefficients γ_{pot} and γ_{kin} for irradiation of the clean polycrystalline surface of molybdenum with Ar^+ ions and Ar atoms.

plasma or ion flux. Attention will now be given to examples of a number of specific structures of the boundary layers matching the parameters on the wall with the parameters in the flux.

In this section, we investigate the boundary layers of the electron scale and take into account the dynamics of the ions only to the degree which is important for understanding the electron layers. The electron component has tree internal scales: Debye radius r_D and electron Larmor radius ρ_e and also the free path λ_e[11]

$$r_D = \frac{v_{Te}}{\omega_0}; \quad \rho_e = \frac{v_{Te}}{\omega_{He}}. \tag{7.3.1}$$

In addition to the scales, there is also another scale – the scale of the system or, more accurately, the scale of the nonuniformity of the flux L. In order to be able to discuss more generally the electron boundary layers as quasi-autonomous objects, it is necessary to fulfil at least one of the criterial inequalities

$$\frac{r_D}{L} \ll 1; \quad \frac{\rho_e}{L} \ll 1; \quad \frac{\lambda_e}{L} \ll 1. \tag{7.3.2}$$

In analysis of specific situations it is useful to know the ratio of the Debye radius to the Larmor radius. Taking into account (7.3.1) we have

$$\frac{r_D}{\rho_e} = \frac{\omega_{He}}{\omega_0} = \frac{1}{c}\frac{H}{\sqrt{4\pi mn}}.$$

The quantity on the right-hand side is the ratio of the electronic analogue of the Alfvén velocity to the velocity of light. In most cases, this ratio is small. For example, even at $n = 10^{11}$ cm^{-3} $r_D < \rho_e$, if $H < 10^3$ Oe. It is assumed that

$$r_D < \rho_e,$$

In addition to the three internal scale parameters there is another parameter which is important in many respects, is a unique analogue of the Mach number and can be referred to as the ‘external’ parameter of the layer. This parameter can be determined as the ratio of the scale of the current velocity $u_\perp = \frac{j_\perp}{en}$ across the layer to the thermal velocity of the electrons $v_{Te} \sim c_{Te}$. In honour of Buneman and Budker (see section 4.5.2), it will be referred to as the *Bu*-parameter

[11]In many cases it is necessary to differentiate the free path in relation to collisions of different types, but here we shall not clarify the meaning of λ_e.

$$Bu = \frac{\{u_\perp\}}{c_{Te}}.$$

Thus, it can be seen that there are at least four important dimensionless parameters and, consequently, the number of types of boundary electron layers is very large. Therefore, the previously examined specific layers form a small part of the existing variants of the layers and the selection of these layers for these sections of the book is highly subjective.

In this section, we examine only 'currentless' layers ($Bu \to 0$). Examples of the current layers (near-electrode) will be presented when discussing specific systems – arc and glow discharges, and also discharges in coaxial accelerators with solid electrodes in section 7.6.2 and 7.6.6.

In addition, in this section we ignore the volume collisions. Consequently, the electron layers which will be studied are characterised by the Debye radius r_D and by the electron Larmor radius.

Regardless of the relatively strict restrictions which we have imposed, there are three different types of layers. They differ in the orientation of **E**- and **H**-fields in relation to each other and in relation to the 'wall'.

1. If the **H**-field does not exist (or, more accurately, if it is very weak so that $\rho_e \gg r_D$) or **H**||**E**, then the Debye layer (DL) mentioned previously many times forms in the vicinity of the wall. This layer can form in the vicinity of both the dielectric or conducting wall (Fig. 7.3.1).
2. If **H** is normal to the wall and **E** is parallel to the wall: a uniform layer can form only around the dielectric wall. This layer will be referred to as diffusion or 'the layer of near-wall conductivity', because it generates a new form of neoclassic conductivity, determined by scattering of the electrons on the wall (Fig. 7.3.5).
3. If **H** is parallel to the wall and **E** normal to the wall, then a 'drift layer' forms. This layer is often referred to as the layer of 'magnetic insulation'. Models of layers of this type were discussed in section 4.3.3. Usually, this layer forms around conductors through which the current generating a magnetic field tangential to the surface (section 7.3.3) flows.

7.3.1. Debye layers on dielectric walls [171]

The Debye layers (DL), which appear in almost all cases around the dielectric elements which are in contact with plasma, are such 'reliable'

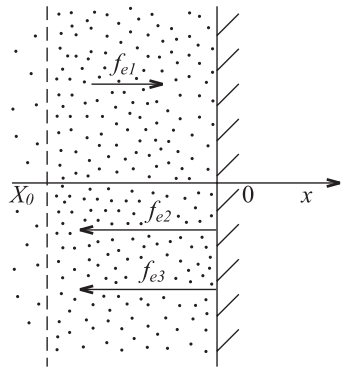


Fig. 7.3.1. Fluxes in the vicinity of the Debye layer: x_0 is the initial position of the electrons, f_{e1} is the distribution function of the incident electrons, f_{e2} is the distribution function of the elastically reflected electrons, f_{e3} is the distribution function of the secondary emission electrons.

structures that in many cases they are disregarded by investigators. As a result of this we know far less about the behaviour of the Debye layers on the dielectric surfaces in the actual conditions than about the behaviour of many highly exotic formations.

It would appear that the problem of the Debye layer for the dielectric surfaces is solved by Fig. I.1.3 and estimates obtained using equation (I.1.1)

$$j_{in} = j_{en}(1-\sigma)e^{-\frac{eU_D}{kT_e}} \tag{7.3.3}$$

which follows from the condition of equality of the ionic and electronic fluxes 'sticking' to the wall. In (7.3.3) j_{in} and j_{en} are the densities of the normal fluxes of the ions and electrons on the surface, U_D is the modulus of the near-wall ('Debye') jump of the potential; σ is the integral coefficient of secondary electronic emission.

However, the situation in many plasma dynamic systems (for example, in stationary plasma thrusters) is far more complicated. Firstly, the distribution function of the electrons, falling on the surface, is far from Maxwellian. Secondly, the surfaces of the insulators are usually rough. This is associated with both the large grain size of the materials of the insulators and with erosion of the materials under the effect of high-energy particles leading to the formation on the surface of the macrostructures which are an order of magnitude thicker than the thickness of the DL estimated on the basis of v_{Te} and ω_0. To this it should be added that all the previously mentioned irregularities usually contain sharp edges and projections. Finally, thirdly, the electronic component may contain a large number of particles for which

$\sigma(\varepsilon) > 1$ in equation (7.3.3) simply loses its meaning.

Here, we confine ourselves to examining one-dimensional Debye layers on flat dielectric surfaces. It is assumed that they form as a result of collisionless ionic and electronic fluxes falling on the surface. These fluxes originate from the quasi-neutral (plasma) volume with the known distribution functions f_{i0} (**v**) and f_{e0} (**v**). In addition to this, we also take into account the electronic flux travelling from the wall caused by secondary electron emission (Fig. 7.3.1).

Evidently, the theory of the Debye layer should be developed on the basis of the system of Vlasov equations if, of course, the collisions with the limits of the Debye layer can be ignored:

$$\frac{\partial f_e}{\partial t} + \frac{\partial f_e}{\partial \mathbf{r}} + \frac{e}{m}\left(\nabla\phi - \frac{1}{c}[\mathbf{v},\mathbf{H}]\right)\frac{\partial f_e}{\partial \mathbf{v}} = 0, \quad e > 0; \tag{7.3.4}$$

$$\Delta\phi = -4\pi e(n_i - \int f_e d\mathbf{v})$$

and the boundary conditions of the type (7.1.21).

Stationary Debye layers around dielectric surfaces

The Debye layers will be investigated starting with the stationary Debye layers around the dielectric surfaces. More accurately, we consider the Debye layers which are not penetrated by electric current because in the stationary conditions the sum of the normal components of the ionic and electronic fluxes to the surface of the solid body is equal to 0

$$j_{in} + j_{en} = 0 \tag{7.3.5}$$

The formation of the Debye layers on the dielectrics is interesting because of a number of reasons. In particular, in the dielectrics, the coefficient of secondary electron emission (SEE) is, as already mentioned in section 7.2, considerably higher than in the metals and, consequently, the critical value $\sigma = 1$ is reached here quite easily. As regards the stationarity, this is the most widely encountered case.

In the following sections, we consider the Debye layers on the flat smooth surfaces, i.e. it is assumed that the thickness of the layer h_D satisfies the conditions:

$\delta \ll h_D \ll R$.

Here δ is the scale of surface roughness, and R is the radius of curvature of the surface.

Figure 7.3.1 shows the functional diagram of the investigated Debye layers. The coordinate of the left boundary of the Debye layer is denoted by x_0, the coordinate of the surface of the solid body by x =

0. Three electronic fluxes form within the limits of the Debye layer: 1) the flux, originating from x_0, and its potential $\phi_0 = 0$, which generates the flux of secondary electrons 3; flux 2 is the flux of low-energy electrons which do not reach the walls if

$$\frac{mv^2}{2} < eU_{\min}.$$

Here $U_{\min}$ is the negative potential within the limits of the Debye layer (if such layer exists).

The phase diagrams (x,v), shown in Fig. 7.3.2 are more adequate than Fig. 7.3.1. Figure 7.3.2a shows the trajectories for $\sigma < 1$. In this case, the charge of the surface of the solid body is negative and some of the electrons (II) do not reach this surface. The dashed line separates the trajectories of the particles reaching the surface from those that do not reach the wall. Figure 7.3.2b shows the phase diagram for $\sigma > 1$. In this case, the charge of the wall is positive and the situation is greatly complicated. We consider the simplest case in which the ion energy is relatively high and all the ions reach the surface of the solid body and recombined there. In addition, the potential in the volume of the layer changes monotonically from 0 to $\phi_\Gamma = U_{AD} > 0$. The layers with these properties will be referred to as 'anti-Debye' layers (ADL). It may easily be shown why the three electron fluxes also form at $\sigma > 1$ (in the ADL regime with the monotonic course of the potential). In addition to the two fluxes I and III, which were detected at $\sigma < 1$, there is also a flux of slow secondary electrons, returning to the positively charged surface of the solid.

Calculations of both the DL and the ADL in the stationary regime can be carried out using the scheme described in section 4.3. In particular, the distribution function of the ions and the electrons, originating from the quasi-neutral plasma (Fig. 7.3.2), is known. For the case with $\sigma < 1$ it is assumed that three functions

$$f_i(v, x_0),\ f_{e1}(v, x_0),\ f_{e2}(v, x_0), \tag{7.3.6a}$$

are known, and the equations (4.3.6) can be used immediately to determine the dependence of these functions on v and ϕ in the entire volume of the layer:

$$f_i(v, \phi),\ f_{e1}(e, \phi),\ f_{e2}(v, \phi), \tag{7.3.6b}$$

Knowing the dependence of the first flux on the wall on v

$$f_{e1}(v, u_D)$$

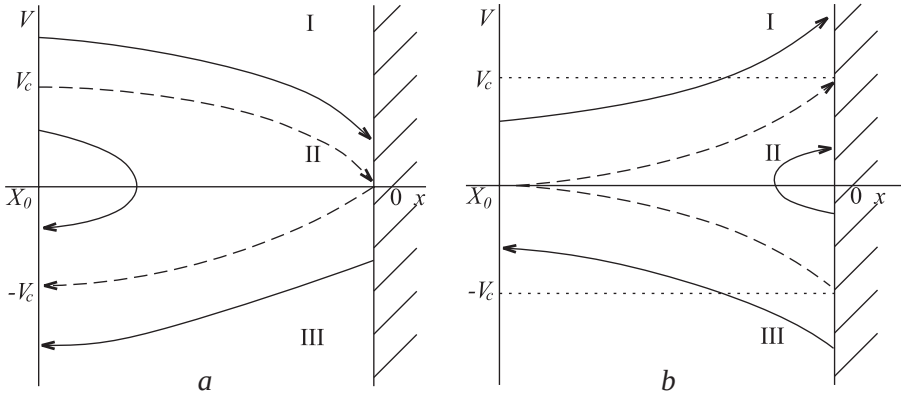


Fig. 7.3.2. Phase diagram of particle dynamics at σ <1 (a) and σ > 1 (b).

and the scattering operator $\hat{S}\left(v|v1\right)$, we determine the distribution function of the electrons (group 3) travelling from the wall

$$f_{e3} = \hat{S} f_{e1}\left(v, u_D\right),$$

and, at the same time, the dependence f_{e3} (**v**, ϕ). This will be when σ < 1. However, if σ > 1, the flux, travelling from the wall, splits into two fluxes f_{e3} and f_{ey}, and the first of them leaves the layer at $x = x_0$, and the second returns to the wall. Evidently, these fluxes are separated at

$$\frac{mv^2}{2} \lessgtr eU_D.$$

Determining all the functions f_e, including f_i (v, ϕ), we can calculate the density of the charged particle and write the Poisson equation

$$\frac{d^2\phi}{dx^2} = -4\pi e(n_i - \sum n_e) = q(\phi), \tag{7.3.7}$$

and taking into account what was said in section 4.3, determine the dependence of ϕ on x and three constants C_1, C_2 and the near-wall jump U_D

$$\phi = \phi(x, C_1, C_2, U_D) \tag{7.3.8}$$

These constants are determined by the boundary conditions

$$\phi(x_0) = 0;\ \phi'(x_0) = 0;\ j_{in} + \sum j_{en}\big|_{x=0} = 0. \tag{7.3.9}$$

The calculations of ϕ(x) and determination of U_D will not be

described here. These are usually extremely cumbersome equations and require a computer.

Examples of the results of calculations of $n(x)$ and $\phi(x)$ at low and high T_e are shown in Fig. 7.3.3 and Fig. 7.3.4.

It should be mentioned that if f_{e1} (v, x_0) and f_{e3} (v, 0) are the Maxwellian functions with the same temperature, and σ = const <1, we obtain the balance equation in the traditional form (7.3.3). However, if at least one of these conditions is not fulfilled, the balance equation is not expressed directly through the boundary conditions, i.e. the values j_{in} and j_{en} and the particle energy at the layer boundary ($x = x_0$). In this case, it is necessary to solve completely the previously formulated task.

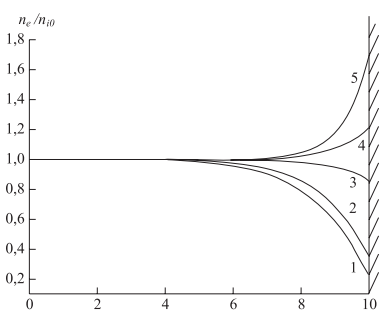
Conditions of disappearance of the DL in contact of Maxwellian plasma with a dielectric wall

In the previous subsection, the SEE coefficient σ was assumed to be constant. However, as indicated in Fig. 7.2.1, the value of σ for dielectrics depends quite strongly on ε_i, i.e. the energy of the incident electrons. This dependence in the range $0 < \varepsilon_i \lesssim \varepsilon_1^*$, where ε_1^* is the first multiplication threshold ($\sigma(\varepsilon_1^*) = 1$) can be approximated by a linear function

$$\sigma(\varepsilon) = \sigma_0 + (1-\sigma_0)\frac{\varepsilon}{\varepsilon_1^*} \tag{7.3.10}$$

Taking into account (7.3.6), the electron distribution function f_Γ of the incident electrons immediately in front of the wall has the form

$$f_\Gamma(v) = F_{0e}\left(\sqrt{v^2 + \frac{2eU_D}{m}}\right) \tag{7.3.11}$$



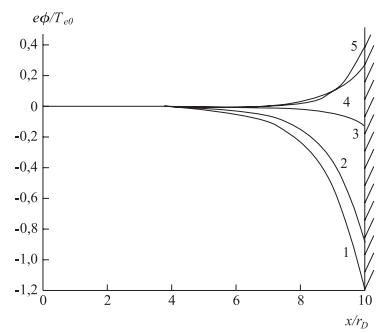


Fig. 7.3.3. Distribution of electron density in the near-wall layer at σ<1 (curves 1, 2, 3) and at σ >1 (curves 4, 5).

Fig. 7.3.4 (right). Distribution of the plasma potential in the near-wall layer: at σ < 1 (curves 1, 2, 3) and at σ >1 (curves 4, 5).

Consequently, the flux of the electrons, settling on the wall is

$$j_{e\Gamma} = -e\int (1-\sigma(\varepsilon))v f_{\Gamma} dv \tag{7.3.12}$$

or

$$= -e(1-\sigma_0)\int_{v>0}^{\infty} v\left(1-\frac{\varepsilon}{\varepsilon_1^*}\right)F_{0e}\left(\sqrt{v^2+\frac{2e}{m}U_D}\right)dv \tag{7.3.13}$$

It should be mentioned that for the classic DL to exist, this value should be equal with respect to the modulus to the ion current j_{in} arriving at the wall

Assuming that F_0 is the one-dimensional Maxwell distribution

$$F(v) = n_0\left(\frac{m}{2\pi kT_e}\right)^{1/2}\exp\left\{-\frac{(\varepsilon-e\phi)}{kT_e}\right\},\ \ \varepsilon=\frac{mv^2}{2}, \tag{7.3.14}$$

we obtain

$$j_{e\Gamma} = n_0\left(\frac{m}{2\pi kT_e}\right)kT_e\left(1-\frac{kT_e}{\varepsilon_*}\right)\exp\left\{-\frac{eU_D}{kT_e}\right\}. \tag{7.3.15}$$

Thus, the classic DL in the one-dimensional case can exist only at

$$kT_e < \varepsilon *. \tag{7.3.16}$$

In the three-dimensional case the condition is more ‘severe’

$$kT_e < \frac{1}{2}\varepsilon *. \tag{7.3.17}$$

The energy, transferred by the electrons to the wall in the stationary regime

The electron, falling on the wall, brings its kinetic and potential (work function $e\phi$) energies. In turn, the knocked-out particles carry away the work function and some kinetic energy.

Simple calculations show that for the Maxwellian distributions of the electrons and the ions, the energy flux into the wall is

$$q = \nu_e\left\{\left[(2kT_e+e\varphi_e)+(2kT_i+e\varphi_i)\right]-\sigma\left[(\overline{\varepsilon}_e+e\varphi_e)\right]+(1-\sigma)I\right\}. \tag{7.3.18}$$

Here

$$\nu_e = \frac{1}{4} n\bar{v} \exp\left\{-\frac{eU_D}{kT_e}\right\}, \qquad (7.3.19)$$

is the flux of the primary electrons, reaching the walls, $e\varphi_e$ and $e\varphi_i$ are the work functions of the appropriate particles, $\bar{\varepsilon}_e$ and $\bar{\varepsilon}_i$ are the kinetic energies with which the electrons and ions leave the Debye layer, I is the energy of recombination of the electron and the ion on the surface of the solid.

Numerical simulation of the DL

Numerical calculations were carried out of the non-stationary system of the equations (7.3.4) with the appropriate boundary conditions at both $\bar{\sigma}(T_e) < 1$ and $\bar{\sigma}(T_e) > 1$[12]. The ions falling on the wall were regarded as having high energy so that their velocity does not change direction. The calculations confirmed that at both $\bar{\sigma} < 1$ and $\bar{\sigma} > 1$ stationary layers form around the walls. The potential and density distributions at the Maxwellian distribution of the electrons falling on the wall at $\bar{\sigma}(T_e) < 1$ and $\bar{\sigma}(T_e) > 1$ are shown in Fig. 7.3.3 and 7.3.4. It may be seen that the dependences $n_e(x)$ and $\phi(x)$ in these regimes are quite different. If at $\bar{\sigma} < 1$ the wall is charged negatively and the electron density around the wall decreases, then at $\bar{\sigma} < 1$ the wall is charged positively and the electron density around the wall increases. Therefore, if in the first case we have the classic Debye layer (DL), in the second case the resultant structure may be referred to as the 'anti-Debye layer' (ADL).

The investigation of the stability of the DL and ADL was yielded curious results. These investigations were also carried out numerically by superposition of the periodic perturbations of different amplitude and frequency on the flux coming from 'infinity'. It was shown that at $\bar{\sigma} < 1$, the amplitude of perturbations decreases on the approach to the wall irrespective of frequency and amplitude. However, if $\bar{\sigma} > 1$, the amplitude of oscillations increases on approaching the wall and also with increasing $\bar{\sigma}$. Nevertheless, the oscillations take place around the stationary state.

7.3.2. Diffusion boundary layer. Near-wall conductivity [172]

General characteristics

The diffusion boundary layer forms as a result of the scattering of the particles on the wall which restricts the plasma volume, on the condition

[12]A.I. Morozov, V.V. Solov'ev [49, section 9], [146].

that there is a magnetic field intersecting the wall and an electrical field parallel to the wall (Fig. 7.3.5). If the investigated particles are magnetised, i.e. the movement of the particles in the plasma volume has the form of a drift in the direction perpendicular to **E** and **H**, the collapse of the drift for some reason enables the particles to move across the magnetic field and along the direction of the **E**-field.

This collapse of the drift can be – at a low noise level – be caused either by collisions of the electrons with the heavy particles (classic conductivity) or by collisions with the walls. As already mentioned, the second type of transfer is referred to as near-wall conductivity (NWC)[13]. Near-wall conductivity is in fact a classic type effect. It is accompanied by collisions with the 'superheavy particle' – the surface of the wall. There is another variety of neoclassic conductivity. In principle, near-wall conductivity can be caused by both ions and electrons. In this subsection the ions will be regarded as a homogeneous background.

In the absence of the magnetic field, the volume conductivity, equivalent to near-wall conductivity, is equal to

$$\sigma_{ef,0} = \frac{e^2 n \tau_{ef}}{m}, \tag{7.3.20}$$

where $\tau_{ef} = b/v_T$ is the characteristic time between two electron collisions with the walls, b is the transverse dimension of the plasma volume, v_T is the thermal velocity of the particle. This shows that in the presence of the magnetic field near-wall conductivity leads to the classic expression for the conductivity across the magnetic field:

$$\sigma_{\perp} = \frac{\sigma_{ef,0}}{1+\left(\frac{\sigma_{ef,0} H}{enc}\right)^2} \propto \frac{1}{H^2}. \tag{7.3.21}$$

The expressions (7.3.20) and (7.3.21) are identical to the classic equations (sections 3.2.3). However, while sharing some properties with classical transfer, NWC has some unique features.

a. Firstly, the distinctive spatial localisation of the areas of collisions of the electrons with the 'superheavy' wall. If in the classic case the areas of disruption of the drift are distributed randomly throughout the entire volume, then in the case of near-wall conductivity the drift is disrupted on the surface. Space synchronisation of disruptions emerges, leading to the formation of near-wall current layers with the thickness in most cases of the order of the electronic Larmor radius

[13]Near-wall conductivity in plasma was described for the first time by A.I. Morozov [172].

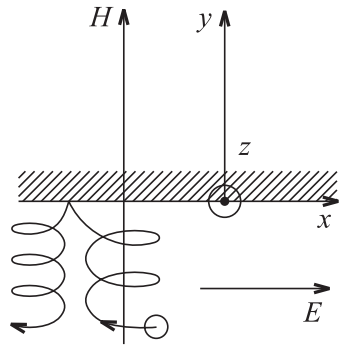


Fig. 7.3.5. Orientation of **E**- and **H**-fields and the trajectory of the electrons prior to and after collision with the wall in the diffusion boundary layer.

$$\delta \propto \rho_e \propto \frac{V_{Te}}{\omega_H}. \tag{7.3.22}$$

This is explained by the fact that the restoration of the electrical drift takes place on the length of the order of this scale.

b. Secondly, the density of the current, determined by near-wall conductivity, usually changes non-monotonically with increase of the distance from the scattering surface. The special feature of near-wall conductivity, which is its diagnostic feature, can be illustrated as follows. Let us imagine a channel formed by two parallel scattering planes, coated with infinitely thin Debye layers. Let the secondary electronic emission coefficient be $\sigma = 1$ and the accommodation coefficient $k = 1$. The magnetic field is assumed to be uniform and directed along the x axis, the electric field – also uniform and directed along the z axis. Consequently, the electrons, falling on the upper plane, overcome the Debye layer, losing almost the entire energy and, reflecting from the surface, they again penetrate through the Debye layer and subsequently travel with the same energy (with velocity $v_{\|} = \sqrt{2eU_D / m}$) to the lower plane, writing a cycloid during this movement. Figure 7.3.6a shows the projection of the trajectory on the plane (x, z). It can be seen that the gap between the planes is divided into layers with the alternating directions of electronic current $j_{ez} = - env_{ez}$. The thicknesses of the layers with the current with the same direction are:

$$\Lambda = v_{\|} \frac{\pi}{\omega_e H}. \tag{7.3.23}$$

Depending on the distance L between the planes, and also depending on $L/v_{\|}$, the number of current layers fitted between the planes can differ.

If the number of layers with the current with the opposite directions is the same, the total current between the layer will be equal to 0 (Fig. 7.3.6b). However, in a general case, the total current may have any sign, but as regards its absolute value it does not exceed the current of the single layer. Of course, this pattern is ideal but clearly indicates the possibility of formation of the distribution of the electronic current oscillating in space in the vicinity of the walls. However, in most cases a non-monotonic structure $j_{ez}(r)$ forms, and the structure can be interpreted as the superposition of distribution of the type shown in Fig. 7.3.6a but with different $v_{||}$.

Near-wall conductivity in semi-infinite plasma in the vicinity of a microrough plane

The simplest model is the near-wall conductivity in the vicinity of a flat microrough surface (Fig. 7.3.5), restricting the semi-infinite plasma volume in the presence of a uniform magnetic field (axis x), normal to the plane, and a uniform electrical field parallel to this plane (axis z). It is also assumed that the Debye layer is considerably smaller than the thickness of the layer determined by near-wall conductivity, and in calculation the thickness of the Debye layer can be ignored. In this case, both the movement of the incident particles and the movement of the reflected particles are described by the equations (section 1.2)

$$\begin{aligned} &x = V_x t\,, \quad V_x = V_{x0} = \text{const}\,, \quad u = \frac{cE}{H}\,; \\ &y = b + ut + \frac{A}{\omega_e}\sin(\omega_e t + \alpha)\,, \quad V_y = u + A\cos(\omega_e t + \alpha)\,; \\ &z = a + \frac{A}{\omega_e}\cos(\omega_e t + \alpha)\,, \quad V_z = A\sin(\omega_e t + \alpha)\,. \end{aligned} \tag{7.3.24}$$

When investigating the static problems in (7.3.24), the time should be excluded using the equation $x = V_x t$. If the values of the distribution functions of the incident particles f^+ (**v**', 0, y, z) and the reflected particles f^- (**v**', 0, y, z) on the surface of the wall are available, then

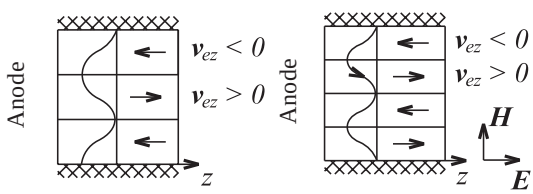


Fig. 7.3.6. Current layers between two walls determined by NWC: a) current along **E** is maximum; b) current along **E** is equal to zero.

using (7.3.25) it is quite easy to determine the values of these functions at any x, replacing **v**', y, z by

$V_x' \to V_x$;

$$\begin{aligned}
V_y' &\to u + (V_y - u)\cos\left(\omega_e \frac{x}{V_x}\right) + V_z \sin\left(\omega_e \frac{x}{V_x}\right);\\
V_z' &\to V_z \cos\left(\omega_e \frac{x}{V_x}\right) - (V_y - u)\sin\left(\omega_e \frac{x}{V_x}\right);\\
y' &\to y - u\frac{x}{V_x} + \frac{V_z}{\omega_e} - \frac{1}{\omega_e}\left[V_z \cos\left(\omega_e \frac{x}{V_x}\right) - (V_y - u)\sin\left(\omega_e \frac{x}{V_x}\right)\right];\\
z' &\to z + \frac{V_y - u}{\omega_e} - \frac{1}{\omega_e}\left[V_z \sin\left(\omega_e \frac{x}{V_x}\right) + (V_y - u)\cos\left(\omega_e \frac{x}{V_x}\right)\right]
\end{aligned} \tag{7.3.25}$$

The distribution function of the incident particles f^+ is regarded as Maxwellian with the shift in the space of velocities, equal to the drift velocity u:

$$f^+ = \frac{1}{N^+}\exp\left\{-\frac{1}{c_{T0}^2}\left[v_x^2 + (v_y - u)^2 + v_z^2\right]\right\} \tag{7.3.26}$$

The distribution function of the reflected particles (at x = 0) is used in the form of the 'stationary' Maxwellian function

$$f^- = \frac{1}{N^-}\exp\left\{-\frac{1}{c_{T1}^2}\left[V_x^2 + V_y^2 + V_z^2\right]\right\}, \quad c_T^2 = \frac{2kT_e}{m} \tag{7.3.27}$$

The normalisation coefficients are equal to

$$N^+ = \frac{\pi^{3/2} c_0^3}{2n^+}, \quad N^- = \frac{\pi^{3/2} c_0^3}{2n^-} \tag{7.3.28}$$

where n^+ and n^- are the intensities of the incident particles and of the particles reflected from the wall. If σ_e is the coefficient of secondary electronic emission, then

$$n^+ c_{T0} \sigma_e = n^- c_{T1}. \tag{7.3.29}$$

Replacing (7.3.25) in (7.3.26), (7.3.27) it can be seen that the function (7.3.27) does not change, whereas (7.3.26) has the new form

$$f^- = \frac{1}{N^-}\exp\left\{-\frac{1}{c_{T1}^2}\left[V_x^2 + V_z^2 + (V_y - u)^2 + u^2 + 2u(V_y - u)\cos\left(\omega_e \frac{x}{V_x}\right) + 2uV_z \sin\left(\omega_e \frac{x}{V_x}\right)\right]\right\}$$

It is interesting to note that $n^- \int f^- d\mathbf{v}$ does not depend on x.

In the present case it is interesting to consider the z-component of current and, therefore, we calculate only this component. Because of the $f^+(v_z)$ is even, its contribution to j_z is equal to 0 and, consequently, we have (the minus sign in front of the integral is ignored):

$$j_z = e\int_{V_x=-\infty}^{0} \int V_z f^- d\mathbf{v}$$

Substituting (7.1.29) into the above expression we obtain

(7.3.30)

$$j_z = e\frac{2n^-}{\sqrt{\pi}} c \frac{E}{H} \int_{-\infty}^{0} \sin\left(\frac{x\omega_e}{c_{T1}\alpha}\right)\exp\left\{-\alpha^2\right\} d\alpha. \qquad (7.3.31)$$

This shows that the distribution of current density in the vicinity of the wall is determined by the function

$$Y(k) = \int_0^{\infty} \sin\left(\frac{k}{a}\right)\exp\left\{-\alpha^2\right\} d\alpha,\ k \equiv \frac{x\omega_e}{cT_1} \equiv \frac{x}{\rho_e},\ \alpha = \frac{V_x}{c_{T1}}. \qquad (7.3.32)$$

The current density converts to 0 at $x = 0$ and increases at a slightly higher rate than the rate given by the linear law in movement away from the wall ($Y \sim k \ln |k|$).

At $k \to \infty$, the function $Y(k)$ can be approximated by the function

$$Y(k) \approx k\exp\left\{-\frac{3}{2^{5/3}}|k|^{2/3}\right\}\cos\left(\frac{3\sqrt{3}}{2^{5/3}}|k|^{2/3} + \text{const}\right). \qquad (7.3.33)$$

The general form of the dependence $j_z(x)$ is shown in Fig. 7.3.7.

As expected, the current is concentrated mostly in the layer $\sim\rho_e$. The total current, passing in the layer with a thickness 1 cm along the axis z, is

$$i = \int_{-\infty}^{0} j_z dx$$

Substituting the equation (7.3.31) into the above equation and using the expressions for the density of the electronic current on the wall $j_{e0} = en^+ c_{T0}/\sqrt{\pi}$ and $u = cE/H$, we obtain (σ_e the coefficient of SEE)

$$i = \sigma_e j_{e0} \frac{u}{\omega_e} = \sigma_e j_{e0} \frac{h}{2} \tag{7.3.34}$$

Here $h = 2u/\omega_e$ is the height of the cycloid. The physical meaning of the resultant equation is obvious.

Equality (7.3.34) (for the electron distribution close to Maxwellian) can be presented in the form of the Ohm's law [172]

$$i = \sigma_e \frac{n^+ c_{T0}}{\sqrt{\pi}} \frac{mc^2}{H^2} E \tag{7.3.35}$$

We now estimate of the scale i, using the parameters similar to parameters of the flow travelling to the wall (their accurate values are not known at the present time): $\sigma = 1$, $n^+ = 0.5 \cdot 10^{11}$ cm^{-3}, $c_{T0} = 2 \cdot 10^8$ cm/s, which corresponds to $T_+ = 10$ eV, $E = 100$ V/cm, $H = 150$ Oe. Substituting these values into (7.3.35) gives $I = 0.14$ A/cm which corresponds quite closely to the experimental data for stationary plasma thrusters (see Ref. 173).

The near-wall conductivity of electronic plasma for a flat layer

A natural generalisation of the investigated task is the problem of the near-wall conductivity of the plasma in the form of a flat layer. Assuming that the thickness of the layer is $2b$, the distribution function of the electrons – Maxwellian, with the parameters n^+, T^+, n^-, T^-, reflected from the wall, we obtain the following expression for the longitudinal current

$$j_z(x) = \frac{eu}{\sqrt{\pi}} \left[n^+ Y\left(\frac{b}{\rho_L^+} \left(1 + \frac{x}{b} \right) \right) + n^- Y\left(\frac{b}{\rho_L^-} \left(1 - \frac{x}{b} \right) \right) \right]. \tag{7.3.36}$$

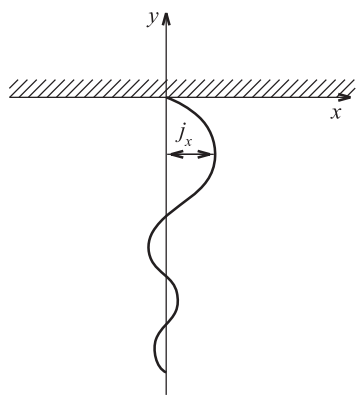


Fig. 7.3.7. Distribution of current density j_z(y) in the vicinity of the wall.

Here $\rho_L^{\pm} = \left(2T^{\pm}/m\right)^{1/2}/\omega_e$ are the Larmor radii, calculated from the temperatures of the upper and lower walls.

We have investigated examples with the simplest reflection model in which the secondary electrons are governed by the Maxwell law. Evidently, identical calculations can be carried out for any scattering function S (**v**, **v**').

Near-wall conductivity determined by surface macroheterogeneities
It was stressed in the previous considerations that near-wall conductivity does not appear on the 'mirror' planes, normal to **H** and parallel to **E**. However, if the surface is macro-heterogeneous and disrupts the drift velocity, then near-wall conductivity appears in this case, even if the surface is mirror-like, i.e. coated with a reflecting Debye layer.

7.3.3. Drift electronic boundary layers

'Magnetic isolation'
At the end of 50s and beginning of 60s special attention was given to the electronic layers formed in the vicinity of smooth metallic conductors in the presence of a magnetic field tangential to the layer and an orthogonal electrical field (Fig. 7.3.8).

Initially, these layers were detected at the boundary of a plasma cylinder in a longitudinal magnetic field in the so-called ion magnetron by M.S. Ioffe and E.E. Yushmanov (Fig. 7.3.9).

Later, they were observed by a team headed by A.I. Morozov in coaxial accelerators with solid anodes described in section 3.7.2 (Fig. 7.11.6), and American scientists detected them in the construction of equipment for the generation of relativistic electron beams (REB). These layers appeared on coaxial and 'strip' lines, connecting an energy storage device with a diode – a generator of such beams. Since the currents in these lines reach hundreds of thousands and millions of amperes strong magnetic fields form there. As a result of this, regardless of the megavolts applied to the strips, the electrons do not 'jump over' from one strip to another during the working pulse. This phenomenon is referred to as 'magnetic isolation'. The electronic structures form here are in fact variants of the 'single-Larmor structures', discussed in section 4.3.3.

The high energy, stored in these layers at high voltages of the transferred energy, makes them highly unstable. Sooner or later they 'stumble' on some surface and cause a local explosion. Therefore, in the actual conditions, magnetic ionisation in the megavolt range is sustained for ~1 μs. Recently, magnetic isolation in the appropriate devices has

been used for switching currents of extremely high powers [41].

Ion magnetrons

Previously, we mentioned the Ioffe–Ushmanov magnetrons [23, T.IV]. These are coaxial systems with a longitudinal magnetic field. In the centre of the system there may be a plasma cylinder (Fig. 7.3.9a) under the positive potential in relation to the periphery. The central 'rod' in equipment is generated by the plasma exiting from the chamber with a burning arc. In subsequent experiments carried out by Zharinov in which the ion magnetrons were studied in detail, this could have been a simple solid-state cylinder–anode (Fig. 7.3.9b). In this case, a low gas pressure of ~10^{-3} torr was maintained in the system. A discharge formed as a result of the potential difference applied between the anode and the external ring-shaped electrode. The thickness of the ionisation and acceleration zones in the systems is several electronic Larmor radii. Therefore, only a relatively small number of collisions is required for the electron, formed in the layer as a result of ionisation, to be transferred to the anode. At the same time, the resultant ions leave the layer without collisions and are accelerated by the electrical field. The dynamics of the neutral particles and ions in the ion magnetrons is described completely by a simple kinetic model, whereas the following hydrodynamic system of equations can be written for the electrons:

$$\frac{\partial}{\partial x}(n_e u_x) = \beta n_e n_0; \quad \frac{\mathbf{j}}{\sigma} = \frac{\nabla p_e}{en} + \frac{1}{c}\left[\mathbf{u}, \mathbf{H}\right] + \mathbf{E}; \tag{7.3.37}$$

A number of important conclusions on the system can be made as a result of simple estimates. It is taken into account that the applied potential difference is relatively large and equals many hundreds of volts. As indicated by (7.3.37), the electronic temperature T_e in the layer is comparable with the applied potential difference, i.e. at least T_e ~100 eV. At these values of T_e, the ionisation coefficient $\beta = \langle \sigma v \rangle$ becomes almost completely independent of electron energy. Averaging the first of the equations (7.3.37) with respect to the thickness of the layer δ gives the number of the electrons formed in the layer and reaching

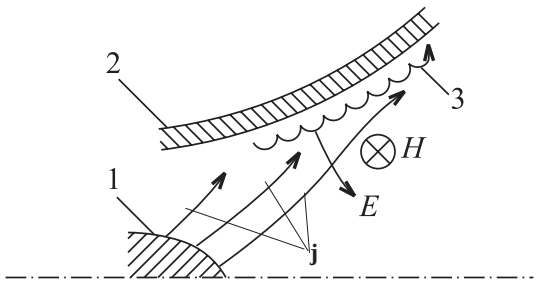


Fig. 7.3.8. Orientation of the fields in the drift electron boundary layer: 1) the cathode, 2) the anode, 3) the drift electron boundary layer.

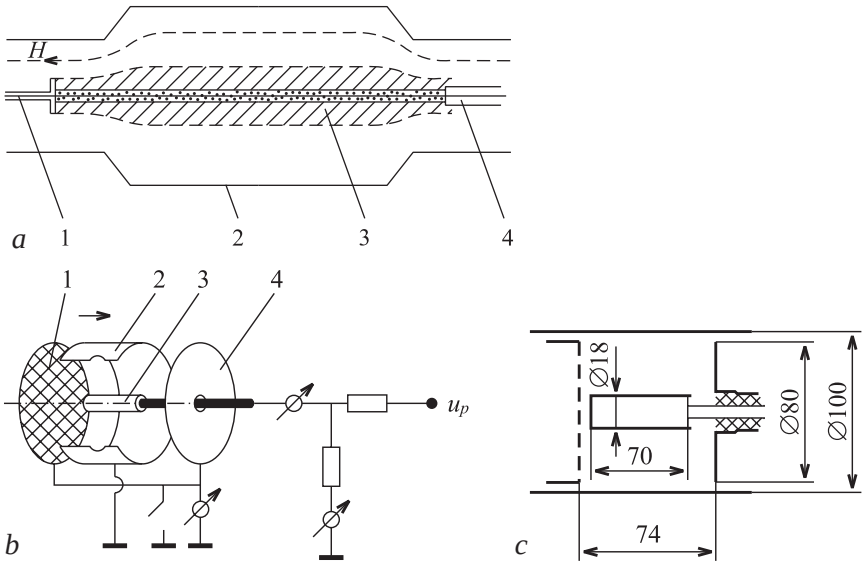


Fig. 7.3.9. Ion magnetrons: a) Ioffe–Yushmanov magnetron: 1) plasma source; 2) chamber walls; 3) plasma beam; 4) the reflector (U_p = 5 kV, H = 2 kOe, $p = 5 \cdot 10^{-5}$ torr); b) Zharinov magnetron: 1) the end mesh cathode; 2) the anode; 3) the cylindrical cathode; 4) the end solid cathode (U_p = (1–10) kV, $H \leqslant 1.3$ kOe, $p = (1 \cdot 10^{-5}–1 \cdot 10^{-3})$ torr.

finally the anode

$$(n_e u_x)_{\text{anode}} = \langle \sigma v_e \rangle_{\text{ion}} \, \overline{n}_e \overline{n}_a \delta, \tag{7.3.38}$$

where n_a is the atom concentration. At the same time, the x-component of the second equation (7.3.37), if ∇p_e and the unity in the Ohm law are ignored, has the form

$$j_{ex} \approx \frac{\sigma E_x}{(\omega_e \tau_e)^2} \tag{7.3.39}$$

Equating (7.3.38) and (7.3.39) and taking into account that

$$E = \frac{U}{\delta}; \quad \sigma = \frac{e^2 n_e \tau_e}{M}; \quad \nu_{\text{ion}} \equiv n_a \langle \sigma v \rangle_{\text{ion}}; \quad \nu_0 = \frac{1}{\tau_e},$$

where ν_{ion} is the frequency of ionisation collisions; ν_0 is the frequency of scattering of the electrons, it gives the expression for the thickness of the near-anode layer (A.V. Zharinov [44])

$$\delta = \rho_e^{(U)} \sqrt{\frac{\nu_0}{\nu_{\text{ion}}}}. \tag{7.3.40}$$

Here $\rho_e^U = \sqrt{\frac{2eU}{m}}\frac{mc}{eH}$ is the Larmor radius, calculated from the total applied voltage. Equation (7.3.40) shows that at classic conductivity the thickness of the near-anode layer is of the order of the electron Larmor radius. More detailed investigations of the properties of the layer were carried out on devices of the type shown in Fig. 7.3.9b. They show the agreement between the calculated value of the thickness of the layer and the experimental data.

The ion magnetrons were used as prototypes of plasma 'accelerators with the anode layer' (AAL) or, as is usually the case, the cosmic 'engines with the anode layer' (EAL). For more details see appendix D.

7.4. Examples of boundary processes with heavy particles taking part

Near-wall plasma processes with heavy particles taking part are very frequently encountered and are usually complicated. Several simple cases characteristic of plasma systems will be mentioned here. It would also be useful to describe the emitting viscous boundary layers which form around meteorites and launched space systems and also in treatment of various components with high-energy plasma fluxes. However, this is very complicated, as indicated by the theory of one-dimensional emitting shock waves, described in chapter 6.

7.4.1. Recycling

In the majority of plasma systems 'recycling' takes place around the walls. The principle of this phenomenon is often very simple and leads to the following scheme (Fig. 7.4.1a).

The ion flux falls from the plasma volume onto the wall, it is greatly neutralised there, and in the form of neutrals returns to the plasma volume. Here, it is again ionised, mostly as a result of recharging, then under the effect of the positive potential of the plasma volume (in relation to the wall) again starts to travel to the wall, and so on.

Recycling is a frequently encountered process and its appearance is almost unavoidable in the presence of a plasma volume and a wall. This phenomenon is also found in the near-electrode zones of conventional gas discharges and in the vicinity of 'passive' walls, such as, for example, the positive column of glow discharge and in thermonuclear traps.

The thickness of the recycling zone is of the order of the free path of the neutral to recharging. In some cases, this length can be of the order of the dimensions of the vessel.

Without discussing the details of calculations of the recycling processes despite the fact that there is a very large number of these processes, it is interesting to discuss the figure from the first article by A.D. Sakharov, concerned with the problem of controlled thermonuclear synthesis (Fig. 7.4.1b). The figure shows the distribution of the plasma parameters in the vicinity of the wall of a magneto-thermonuclear reactor (MTR). The recycling calculations on which the figure is based were carried out jointly by N.E. Tamm and A.D. Sakharov [23]. However, as shown by later experiments, the recycling processes in tokamaks are far more complicated. They develop not only in space but also with time. This is how the situation is described by S.V. Mirnov in his fundamental review of the physics of tokamaks [175]: ‘‘The main function of the discharge chamber of a tokamak is the formation of a toroidal vacuum volume, essential for the organisation of the plasma filament. The preliminary stage of this organisation is the procedure of vacuum cleaning of the chamber walls and pumping of the wall to the residual pressure of (10^{-7}–10^{-9}) mm Hg. The next operation is filling the chamber to the pressure of (10^{-4}–10^{-3}) mm Hg using spectropure hydrogen.

This is followed by the activation of the toroidal magnetic field B_θ and then by the induction electrical and controlling transverse field B_y. The activation of the electrical field E_θ (3–30 V/m) causes a discharge in hydrogen. After completing ionisation only 20–40% of ions from the initial number of the atoms[14], situated in the chamber, remain in the plasma volume, i.e. the supplied hydrogen is bonded mostly on the walls of the chamber. This relationship is not convincing. As shown by the measurements of the intensity of the hydrogen spectral lines, a strong hydrogen flow slowly returns from the wall to the plasma filament throughout the entire discharge...

After completing the discharge, the hydrogen ions, situated on the wall, recombine and most of them return to the chamber volume. If the wall is made of stainless steel, return takes place in two stages. The first, smaller portion is returned quite rapidly (~0.1 s), and a large one slowly, during several hours’’.

We have mentioned this in order to stress the non-trivial nature of the interaction of plasma with the wall. However, this is not the end of the matter. If the energy in the trap is not high ($\gtrsim$100 eV), then, falling on the walls, the hydrogen ions can knock-out heavy particles from the wall. After multiple ionisation, these particles greatly increase the intensity of bremsstrahlung (~z^2) and support rapid cooling of the plasma.

[14]Relatively small laboratory tokamak models are referred to here.

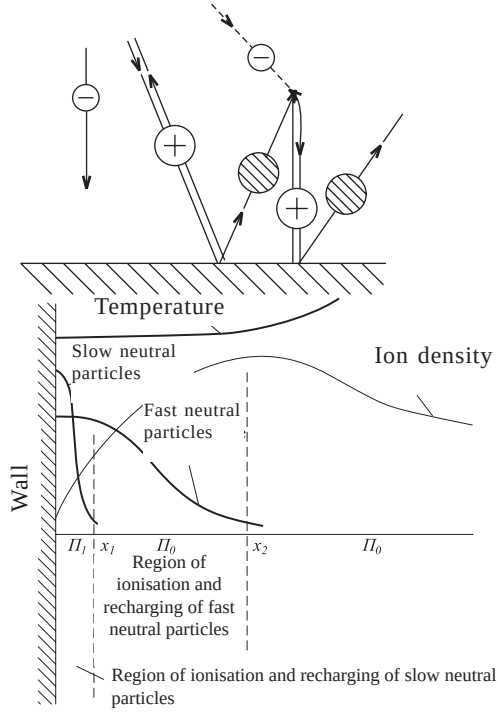


Fig. 7.4.1. Recycling: the principal diagram of recycling (the neutral atom is cross-hatched) (a); the model of recycling in a magnetic thermonuclear reactor according to A.D. Sakharov (b).

However, even this does not exhaust the list of the processes on the tokamak wall. Blistering (section 7.2.3) and ignition of 'unipolar' arcs also take place.

7.4.2. Discharges, sliding on the dielectric surface

There is a number of plasma dynamic systems based on the discharge sliding on the dielectric surface. The discharge causes erosion of the dielectric (more accurately ablation) and, consequently, generates plasma which may cause discharge.

There are two families of such plasma dynamic system. The first family includes the pulsed erosion plasma accelerators and magnetic plasma compressors in which the Ampere force accelerates the plasma. Three variants of these systems are shown in Fig. 7.4.2. The first of them, the 'buttonhole-like' force (Fig. 7.4.2a) was constructed by the American physicist W. Bostick (1952). It has the form of a small cylinder made from organic glass with a diameter of ~1.5 cm containing two inserted hydrogen-saturated titanium electrodes–bars connected to tan external circuit, containing a capacitance and a discharger (switch).

The ‘button’ is placed in the vacuum chamber. Closing the circuit results in the development of a plasma discharge with the duration of ~0.5 μs which caused desorption of hydrogen from the bars and evaporation of organic glass. Under the effect of the Ampere force generated by the intrinsic magnetic field of the discharge, the plasma formation at the initial moment resembles Z-pinch and is then extended and at the end of the discharge separates in the form of a toroid. Bostick referred to these toroids, flying at speeds measured in many tens of kilometres/second, as ‘plasmoids’ [179]. Figure 7.4.2b shows the diagram of a plasma erosion engine for light space systems (section 10.4), used for correcting of the trajectory of the systems. It has the form of a ‘rail gun’, with plasma formed as a result of evaporation of an insulator. Under the effect of the heat flow from the discharge the resultant plasma is accelerated by gas-kinetic pressure and ampere force. The exhaust velocity, depending on the conditions, is in a wide range from ~10 km/s to ~100 km/s. The energy input in a single pulse for the engines is ~1–100 J. Working substances for these engines often include Teflon. Regardless of the fact that the circuit of the erosion engine is quite simple, there is still no accurate quantitative theory of these systems. This is explained not only by the complicated chemism of the transition from a solid dielectric to plasma consisting of relatively simple particles, but also by the significant role played by the heterogeneities of the Teflon, formed on the surface, since the discharge has no symmetry and has the form of streamers. Therefore, only the experimentally recorded characteristics of the engines are reliable. The LFS-6 engine maintained a communication satellite at a specific point of the geostationary orbit for many years (see also [179]).

Finally, Fig. 7.4.2c shows an erosion magneto-plasma compressor (MPC). In contrast to the engine with a low energy input which we have just discussed, the MPC is a quasi-stationary high-current device in which the discharge current reaches hundreds of kA at a voltage of U_p ~(1–10) kV. The characteristic duration of the pulsed here is ~1 ms, the channel diameter ~10 cm.

At present, the erosion MPCs are produced mostly as powerful sources of non-coherent radiation. The main emitter in this case is the compression zone which emits at these discharge parameters almost as a blackbody with the temperature of T ~(3-5) eV. The MPC is a highly efficient converter of electrical energy to light energy (more accurately, into ultraviolet radiation). The surface radiation power is $\sim 3 \cdot 10^7$ W/cm^2. From the energy supplied to the MPC, up to 70% of radiation is emitted in this direction. Erosion of the insulator takes place under the effect of discharge radiation.

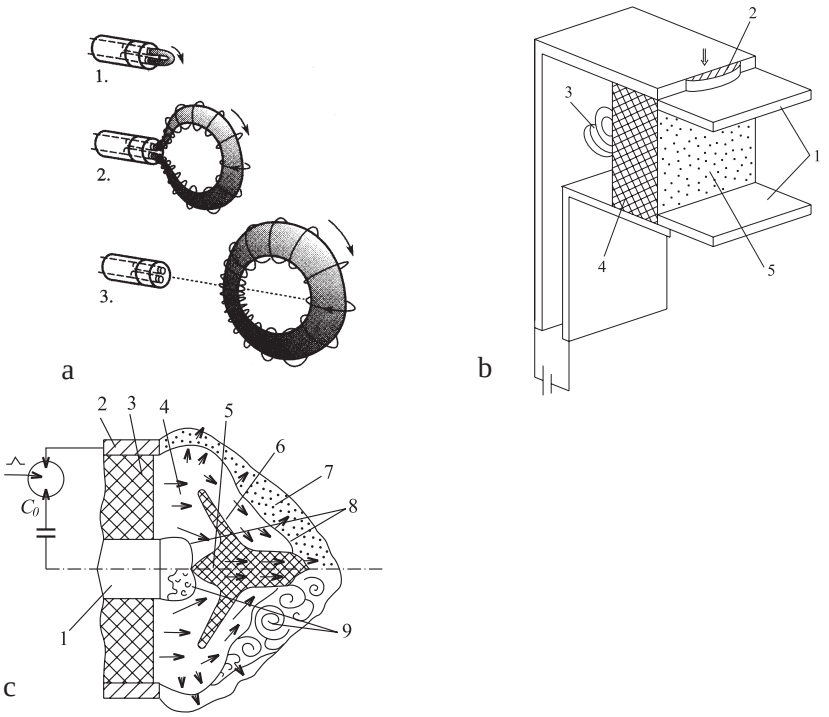


Fig. 7.4.2. Erosion accelerators and compressors (MPC): a) mechanism of the formation of the plasma blob with toroidal configuration in the Bostick gun; b) pulsed erosion plasma thruster for space systems LFS-6: 1) electrodes; 2) discharge ignition; 3) the spring moving the eroded insulator (4) during erosion; 5) discharge; c) calculated structure of the erosion MPC in gas; 1, 2) electrodes, 3) plasma-forming dielectric (fluoroplastic); 4) plasma flow; 5) compression zone; 6) the conical shock wave; 7) the zone of impact-compressed gas; 8) contact boundary; 9) the zone of turbulent mixing. Discharge in helium, $\rho = 2 \cdot 10^{-2}$ kg/m^3; $J_{max} = 0.24$ MA, $T/2 = 28$ μs, time $t = T/4$. Emissivity ~10^{10} W/m^3 ([49], volume 3, section).

In the previously described systems, the Ampere force accelerated the plasma formed as a result of erosion. However, there is a number of plasma-dynamic systems in which the Ampere force presses the discharge to the dielectric (Fig. 7.4.3). These ‘pressed’ discharges, pulsed and high current, are also of interest as the sources of high power radiation. The main advantage of these discharges is the high stability of the plasma configuration.

Ending this brief review of the specific boundary plasma layers of the dielectrics, it should be stressed that here, as also in the case of recycling, there is a flux of particles of the dielectric from the main plasma volume (this is especially evident in low-power discharges) but, in addition to this, the role of the radiation causing the evaporation of the wall in powerful discharges may also be significant.

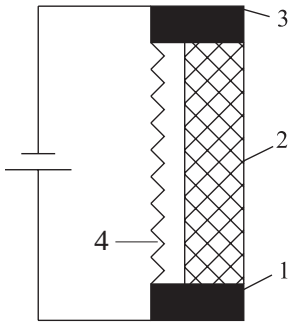


Fig. 7.4.3. Diagram of 'compressed' discharge: 1, 3) electrodes, 2) dielectric, 4) discharge.

7.4.3. Kinematics of dispersion of the surface and by a monospeed ion flux

Previously, it was mentioned that the type of change of the subsurface layers of the solid under the effect of plasma (and, in particular, as a result of the effect of its ionic component) may greatly differ (Fig. 7.2.6). One of the simplest and, at the same time, most widely encountered structures formed on the solid surface, irradiated with an inclined ionic flux ($0 < \theta < \theta_{max}$, where θ_{max} is the angle of maximum sputtering), is the configuration of the 'pike's tongue' (Fig. 7.4.6). This configuration can be calculated quite easily analytically in the approximation of a rare ('quasi-one-particle) flux [180].

Equation of kinematics of iron sputtering

The initial equation is the equation for δ – speed of displacement of the flat surface along the normal as a result of sputtering

$$\dot{\delta} = N_n S_0(\varepsilon)\Phi(\alpha,\varepsilon) \tag{7.4.1}$$

Here N_0 is the normal density of the particle flux, S_0 is the sputtering coefficient at normal incidence, $\Phi(\alpha, \varepsilon)$ is the angular factor ($\Phi(0, \varepsilon) = 1$), α is the incidence angle, ε is the energy of the incident particles.

It is also assumed that the profile depends only on one coordinate x. The profile is denoted by

$$z = z(x, t). \tag{7.4.2a}$$

If ω is the angle between the direction of incidence of the particles and the plane z = const, then the angle of incidence of the particles on the surface element (Fig. 7.4.4) is

$$\alpha = \omega - \beta, \; tg\beta = \frac{\partial z}{\partial x} = z' \tag{7.4.2b}$$

Taking into account that

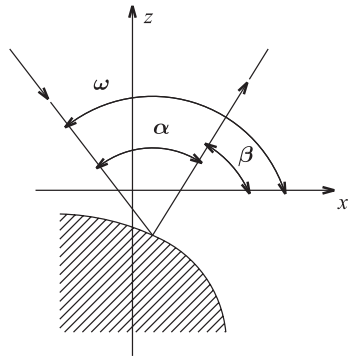


Fig. 7.4.4. Mutual positioned of the angles α, β and ω at incidence of the particles on a non-flat surface (explanation of the equation (7.4.2b)).

$$\dot{\delta} = \frac{\partial z}{\partial t}\cos\beta \tag{7.4.2c}$$

equation (7.4.1) can be written in the following form:

$$\dot{z} \equiv \frac{\partial z}{\partial t} = -N_0 S_0 \frac{\cos(\omega-\beta)}{\cos\beta}\Phi(\omega-\beta) \tag{7.4.3}$$

If N_0, S_0, Φ, ω does not depend on x, the equation (7.4.3) transforms to

$$\dot{z} = F(z') \tag{7.4.4a}$$

Otherwise

$$\dot{z} = F(x, z') \tag{7.4.4b}$$

If the equation (7.4.3) is differentiated with respect to x, we obtain the equation started by many mathematicians

$$\frac{\partial u}{\partial t} + \frac{\partial \varphi(u)}{\partial x} = 0 \tag{7.4.5}$$

Here $u \equiv \frac{\partial z}{\partial x}$. The simplest variant of this equation is the equation of motion of the gas with zero temperature

$$\frac{\partial V}{\partial t} + \frac{\partial}{\partial x}\frac{V^2}{2} = 0$$

It is well known (see, for example, [9]) that the solutions of this equation even for the smooth initial conditions can have singulatiries

(discontinuities). Suitable conditions are created for this in the case of the nonmonotonic dependence $\Phi(u)$.

The general solution of the equation (7.4.4a) is found by the method of characteristics determined by the equations

$$\frac{dx}{dt}=\frac{dF(q)}{dq},\quad \frac{dz}{dt}=-F+q\frac{\partial F}{\partial q},\quad \frac{dq}{dt}=0, \tag{7.4.6}$$

$$F(q)=F(z')\big|_{z'}$$

Integrating the system (7.4.6), we obtain

$$x=\frac{\partial F}{\partial q}t+x_0,\quad z=-\left(F-q\frac{\partial F}{\partial q}\right)t+z_0(x_0),\quad q=q_0=z'(x_0) \tag{7.4.7a}$$

Here x_0 and z_0 are interpreted as the coordinates of the surface at t = 0. This leads to the parametric representation of the surface to be determined

$$z=-\left(F(q)-q\frac{\partial F(q)}{\partial q}\right)t+z_0\left(x-\frac{\partial F(q)}{\partial q}t\right)\ q=z_0'\left(x-\frac{\partial F(q)}{\partial q}t\right) \tag{7.4.7b}$$

Excluding q from the system, we obtain the equation of the surface

$$z=z(x,t)$$

Equation (7.4.7a) shows that here we can consider the displacement during erosion of the points of the profile with the velocities

$$W_x=\frac{\partial F}{\partial q},\quad W_Z=-\left(F-q\frac{\partial F}{\partial q}\right) \tag{7.4.8}$$

Taking into account the complicated nature of the general solution, we confine our considerations here to a brief examination of only stationary processes whose partial case is the formation of the ‘pike's tongue’

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Stationary structures

This term will be defined more accurately. Stationary structures are the modification of the surface which do not change their form with time and only move as an integral unit.

Equation (7.4.7) shows that different stationary changes of the surface are possible.

a. The surface may uniformly ‘fall down’. This case is described by

$$q = 0\,, \quad z_0 = 0 \;\; z = F(0)t \tag{7.4.9}$$

A more general solution of this type is the erosion of a homogeneous inclined area

$$z_0 = qx\,, \quad z_0' = q = \text{const.} \tag{7.4.10a}$$

In this case

$$z = -\left(F - q\frac{\partial F}{\partial q} \right) t + q\left(x + \frac{\partial F}{\partial q} t \right) = qx - Ft \tag{7.4.10b}$$

b. As indicated by the general solution of (7.4.8), for the formation of a non-trivial self-similar erosion surface there should be several values of q_k satisfying the condition

$$\left(\frac{\partial F}{\partial q} \right)_1 = \left(\frac{\partial F}{\partial q} \right)_2 = \ldots \tag{7.4.11a}$$

$$\left(F - q\frac{\partial F}{\partial q} \right)_1 = \left(F - q\frac{\partial F}{\partial q} \right)_2 = \ldots \tag{7.4.11b}$$

These conditions can be transformed to a very convincing form as the ‘rule of the tangent’. In particular, let the characteristic velocities at the points q_k be identical. Consequently, the condition of equality of the *x*-components of this velocity (7.4.11a) is the condition of equality of the angle of inclination of the tangent to the curve $F(q)$ (Fig. 7.4.5). The condition (7.4.11b) shows that all values of q_k to which the same characteristic velocity corresponds, are the coordinates of the points F_k on the curve $F(q)$ which are distributed on the same straight line tangent at all points to $F(q)$. This geometrical criterion, equivalent to the conditions (7.4.11), is referred to as the ‘rule of the tangent’.

Formulation of the configuration of the ‘pike's tongue’

Using this tool, it can be easily shown that in the case of the two-hump angular factor $\Phi(\alpha)$ (Fig. 7.2.9), the only stationary and periodic profile is the ‘pike's tongue’ (Fig. 7.4.6 and 7.2.6c). In fact, equation (7.4.3) has been derived for an arbitrarily oriented incident beam. However, if the *z* axis is represented by the direction of the beam, equation (7.4.3) has the following form

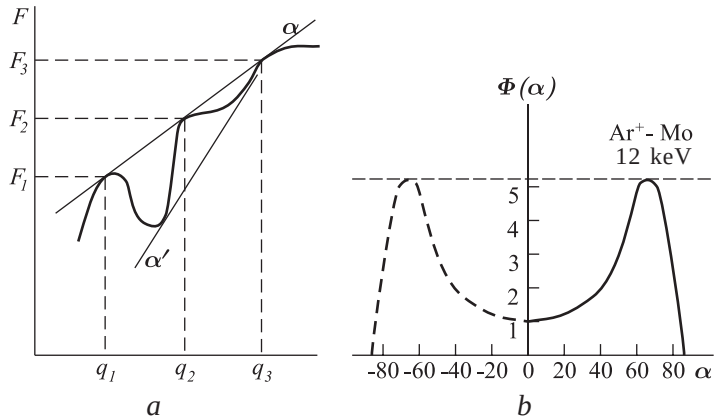


Fig. 7.4.5. Formulation of the 'rule of the tangent': a) in the case of a multi-hump function $F(q)$; b) the same, in the case of a two-hump function.

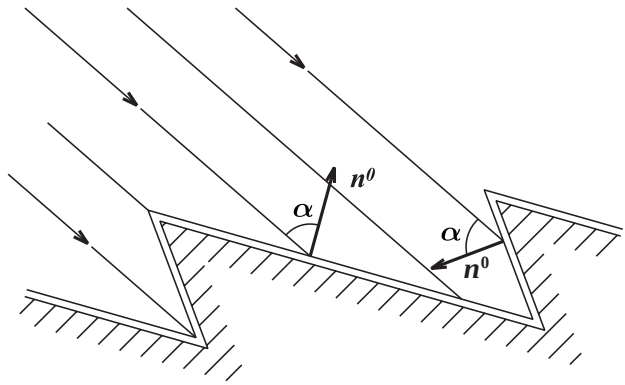


Fig. 7.4.6. Diagram of the 'pike's tongue' at inclined incidence of the flux.

$$\frac{\partial z}{\partial t} = -S_0 \Phi(\beta) \qquad (7.4.12)$$

In this case, the function $F(q)$ coincides with the function of the angular dependence of the sputtering factor $\Phi(\beta)$ with the accuracy to the constant multiplier and selection of the argument. It can be seen (Fig. 7.4.5b) that in the two-hump case there are only two incidence angles (they correspond to the horizontal tangent, parallel to the axis α)

$$\beta_{1,2} = \pm\beta^*$$

which determine the angles of inclination of the straight sections of the surface having the same characteristic velocity. The function $\Phi(\beta)$ reaches the maximum at these points. Knowing that the angle between the planes in the stationary configuration is equal to $2\beta^*$, and the

direction of the beam coincides with the direction of the bisectrix, it is quite easy to construct the 'pike's tongue' also in the case in which the flux hits the surface at an angle. If the angle of incidence of the beam ω and the 'step' of the structure L are given, then the sides of the projections will be (Fig. 7.4.6)

$$a = L\frac{\sin\beta_2}{\cos 2\beta^*}, \quad b = L\frac{\sin\beta_1}{\cos 2\beta^*},$$

$$\beta_1 = \omega - \beta^*, \quad \beta_2 = \pi - \omega - \beta^*$$

This completes the analysis of the analytical model of the 'pike's tongue'. It should be mentioned that the numerical calculations of the non-stationary equation (7.4.3) have confirmed the stability of the investigated configuration of the sputtered surface.

7.5. Surface-determined discharges (using the stationary plasma thruster as the example [181]

In the large majority of plasma dynamic systems, especially with dense plasma ($\lambda/L \ll 1$), the formation of the electron distribution function (EDF) and the nature of transfer are determined mainly by volume processes. In these conditions, the EDF 'gravitates' to the Maxwellian function, and the transfer coefficients become similar to those discussed in Chapter 5.

However, in relatively rare plasma (at $\lambda/L \gtrsim 1$), the electron kinetics can be determined by the collisions of the particles with the walls, restricting the plasma volume.

The situations of this type form in the near-electrode layers (using the examples of glow and arc discharges, these layers will be discussed in section 7.6), and also in the vicinity of the walls, as discussed previously, describing the Debye layers and the near wall conductivity. However, the plasma dynamic systems, in which the EDF is determined by the walls in the entire volume, together with the transfer processes, are of considerable interest. A suitable example of this system are the stationary plasma thrusters which were already discussed in Chapter 6

Examining the Debye layer and near-wall conductivity on the dielectric walls, we obtain date for the analysis of the processes in the stationary plasma thruster. This explains the 'jump' from Chapter 6 to this chapter. Here we investigate in a relatively considerable detail only the electron distribution function in the acceleration channel of the stationary plasma thruster, and other special features of the electron dynamics will be mentioned only briefly.

7.5.1. Electron distribution function in the channel of the stationary plasma thrusters (SPT) [182]

In the stationary plasma thruster, the volume processes are controlling from the viewpoint of the formation of the electron distribution function only in the ionisation zone and in the vicinity of the channel outlet. In both cases, a significant role is played by collective microvave oscillations, which are discussed in detail in section 7.5.4. At the same time, the role of the oscillations in the acceleration zone and in the near-anode zone is relatively small and the main factor affecting the electron distribution function are the classic collisions with the walls. Consequently, simple analytical models of the electron distribution function can be constructed for these zones.

Qualitative characteristics of the electron distribution function
General considerations show that in the sections of the channel of the stationary plasma thruster in which a significant role is played by the collisions with the walls, there should be three groups of electrons which will be referred to 'locked', 'escaping' and 'intermediate' (Fig. 7.5.1). The appearance of the three groups is explained by the fact that the

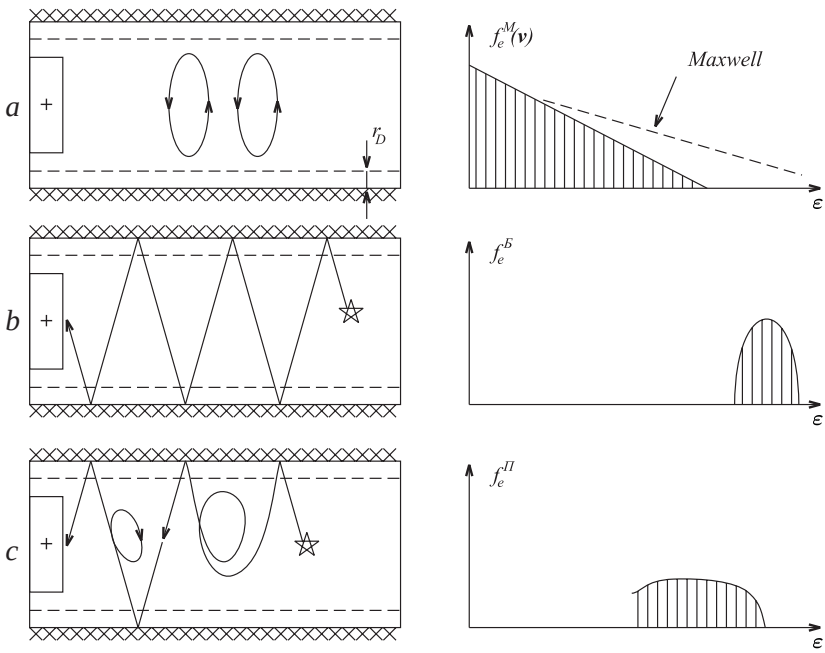


Fig. 7.5.1. Trajectories and the distribution of the three groups of the electrons, formed by collisions with the wall: a) the trajectory of the 'locked' electrons and their distribution function (right); b) the same, for the escaping electrons; c) the same, for the electrons in the intermediate group.

central part ('core') of the flow has a higher potential than the surface of the walls. This is also supported by the Debye layers around the walls.

The group of the 'locked' electrons is formed by the electrons prevented by the radial electrical field from leaving for the wall. Of course, this prevention is not absolute. The collisions of the electrons with each other, with the heavy particles, and also the oscillations of different frequencies ensure constant renewal of this electron group. The particles which are systematically subjected to elastic collisions with the walls enter the group of the escaping electrons. Evidently, the energy of these particles should increase with approach to the anode in accordance with the law $\varepsilon_p \approx (\phi - \phi_k)$, where ϕ_k is the potential at which 'birth' takes place. However, on the approach to the anode, the amplitude of the distribution function of these particles naturally decreases as a result of the losses on the walls due to the 'impurity' of inelastic collisions.

Finally, the intermediate group is formed by the electrons whose energy is sufficiently high for the electrons to reach the walls, but the collisions with the walls are inelastic. Correspondingly, the energy of the intermediate particles lies between the maximum energy of the particles $\varepsilon_{\max}^{\mathrm{Loc}}$ and the minimum energy of the escaping electrons $\varepsilon_{\max}^{\mathrm{Esc}}$

$$\varepsilon_{\max}^{Loc} < \varepsilon^{Int} < \varepsilon_{\min}^{Esc} \tag{7.5.1}$$

The maximum of the distribution function of the intermediate electrons is situated in the vicinity of $\varepsilon_{\max}^{\mathrm{Loc}}$.

Experimental data on the electron distribution function in the channel of the stationary plasma thruster

A.I. Bugrova *et al* carried out detailed investigations of the electron distribution function in a number of models of stationary plasma thrusters [182]. The experiments were conducted using directional Langmuir probes with reception surfaces of ~0.2 × 0.2 mm^2 secured to alundum tubes with the diameter of ~1 mm. Measurements were taken at different points of the channel (a total number of points was 20) in different conditions and at six orientations of the reception surfaces. The signals were recorded with static devices and subsequently averaged out with respect to time. Using the probe characteristic (*P* is the point at which the measurement was taken)

$$J_{x,y,z}(P,\phi) = \int V_{x,y,z} \boldsymbol{f}_e(P,\mathbf{V},\phi) dV S_{PR} \tag{7.5.2}$$

the Druwestein method was used to restore the form of the electron

distribution function $f_e = f_e$ (**v**, P). Consequently, the following results were obtained:

1. In the zone of generation of 'channel' electrons (the ionisation zone and the zone of exit of the electrons from the cathode-compensator to the channel) there is a 'single-hump' electron distribution function. This fact correlates with the high level of microwave noise in these zones.

2. With movement of the electrons in the direction to the anode from generation zones, the single electron distribution function starts to split into three components corresponding to the locked, escaping and intermediate particles (Fig. 7.5.2).

3. The distribution function of the locked particles is efficiently approximated by the linear spline[15]

$$f_e^{Loc}(\mathbf{v}) = \frac{1}{N}\begin{cases}\mu - \varepsilon_*, & \varepsilon_* < \mu \\ 0, & \varepsilon_* > 0.\end{cases} \tag{7.5.3}$$

Here

$$\varepsilon_* = \frac{m}{2}(v_r^2 + (v_0 - u)^2 + v_z^2), \quad u = \frac{cE}{H} = \Omega r, \ \Omega = \text{const.} \tag{7.5.4}$$

This shows that the locked electrons rotate as a solid ('isodrift of the electronic component').

4) The distribution function of escaping particles is characterized by the following features:

a) at the experiment accuracy f_e^{Esc} depends only on the total energy

$$f_e^{Esc}(\mathbf{v}, \mathbf{r}) = F(\varepsilon) \tag{7.5.5a}$$

b) the maximum of this function accounts for

$$\varepsilon = e(\phi - \phi_*) \tag{7.5.5b}$$

where ϕ_* is some mean potential zone corresponding to the zone of 'birth' of the electrons;

c) the amplitude f_e^{Esc} decreases almost exponentially on approach to the anode;

d) the energy width of the distribution function Δ is almost completely independent of r.

Thus, on the basis of these experimental results we have

[15]A similar electron distribution function forms in particular in induction plasmatrons without the magnetic field.

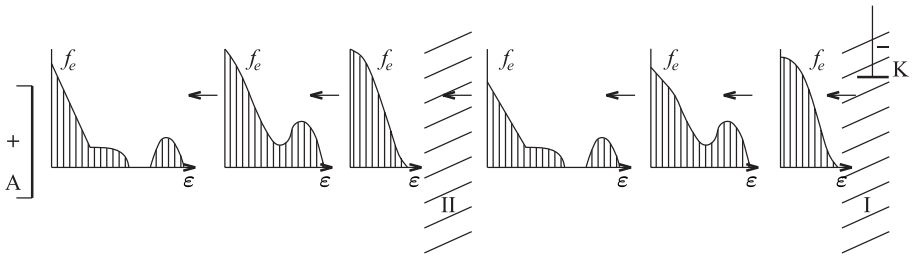


Fig. 7.5.2. Scheme of evolution of the electron distribution function in the channel of the stationary plasma thruster: I) the region of exit of the electrons, emitted by the cathode, into the magnetic field of the channel; II) the region of main ionisation of the working substance. Between I and II and also between II and the anode the distribution function gradually transform from the single-hump to a two-hump and three-hump function.

$$f_e^{Esc}(v,r)=\frac{\exp\left\{-\frac{x}{\Lambda}\right\}}{N}\begin{cases}1-\left(\dfrac{\varepsilon+e(\phi^*-\phi}{\Delta}\right)^2, & \Delta>\left|\varepsilon-e(\phi-\phi^*)\right| \\ 0, & \Delta<\left|\varepsilon-e(\phi-\phi^*)\right|\end{cases} \tag{7.5.6}$$

Here Λ and Δ are constants. The approximation (7.5.6) is in full compliance with the qualitative characteristic which was given above (Fig.7.5.2).

5) Intermediate electrons, which mainly provide near-wall conductivity, are most difficult to image by probe diagnostics. This is explained by their relative scarcity, and by the fact that their zone of localization is closely pressed against the outer insulator, and the energy distribution function is pressed to the high-energy part of the locked particles.

7.5.2. Analytical models of fragments of the EDF

Kinetics of escaping electrons

In this and the following points we discuss the semi-phenomenological conclusions for distribution functions of escaping and locked components of the electrons.

Let us start with the escaping particles. Key elements of the formula (7.6.7) can be obtained from the stationary one-dimensional kinetic equation

$$V_z\frac{\partial f^F}{\partial z}-\frac{e}{m}\frac{\partial\phi}{\partial z}\frac{\partial f^F}{\partial V_z}=-\frac{f^F}{\tau_c}. \tag{7.5.7}$$

Here τ_c is the effective time between the collisions at which the amplitude decreases *e* times. Quantity τ_c is naturally expressed by:

$$\tau_c = \frac{b}{V_\perp} \sim \frac{b}{\alpha V_z} \tag{7.5.8a}$$

Here b is the distance between the walls, $V_\perp$ is the 'radial' component of the velocity of the electron which, because of the isotropic nature of scattering on the walls, can be regarded as proportional to V_z:

$$V_\perp \approx \alpha V_z, \alpha = \text{const.} \tag{7.5.8b}$$

In this case, the general solution of (7.5.7) has the form

$$f_e^F = A(\varepsilon - e\phi)e^{-\frac{z}{b}} \tag{7.5.9}$$

The function $A(\varepsilon)$ is determined by the conditions in the zone of generation of the fast electrons. The experiments show that $A(\varepsilon)$ is efficiently approximated by a quadratic spline.

The distribution function of locked electrons f_e^{Loc}

The locked component forms from the intermediate and escaping components as a result of rare collisions and not very intensive noise. Therefore, it is difficult to calculate. However, there is one important circumstance which enables us, with the accuracy to the type of the function of a single argument, to take this function from the experiment, determining f_e^{Loc}. This key circumstance the already mentioned 'isodrift' nature of the electronic cloud (i.e. the rotation of the cloud in the channel of the stationary plasma thruster as a solid body)[16]

$$u_\theta = \frac{c[E_T, H]}{H^2} = \Omega r, \quad \Omega = \text{const.} \tag{7.5.10a}$$

Here

$$E_T = -\nabla\phi_T = E + \frac{\nabla P_e}{en} \tag{7.5.10b}$$

From (7.5.10a) we obtain automatically the relationship of the thermalised potential ϕ_T with the function of the magnetic flux Ψ

$$\phi_T = \Omega(\Psi + \text{const}). \tag{7.5.11}$$

The experiments show clearly that the isodrift is the fundamental property of the locked component of the electrons. The reasons for isodrift are the viscosity and buildup of various types of oscillations

[16]Evidently, the stretched systems with the poloidal field are isodrift in a wide range of the parameters. The isodrift nature of the plasma lenses has been confirmed by experiments

of the diocotronic type.

We investigate how to utilise the fact of the isodrift nature for determining f_e^{Loc}. If collisions are rare, f_e^{Loc} then satisfies the Liouville kinetic equations to the first approximation[17]

$$v\frac{\partial f}{\partial \mathbf{r}} - \frac{e}{m}\left(\mathbf{E} + \frac{1}{c}\left[\mathbf{v}, \mathbf{H}\right]\right)\frac{\partial f_e}{\partial \mathbf{v}} = 0 \tag{7.5.12}$$

In the axisymmetric case, this equation has five integrals

$$\varepsilon = \varepsilon_k - e\phi = \text{const}, \quad P = mrV_\theta - \frac{e}{c}\Psi = \text{const}, \tag{7.5.13a}$$

$$Y_\alpha(r, z, \mathbf{v}) = \text{const}, \quad \alpha = 1, 2, 3$$

since the general solution (7.5.12) it can be presented in the form

$$f_e = F(\varepsilon, P, Y_1, Y_2, Y_3) \tag{7.5.13b}$$

Of course, the functions $\phi(\mathbf{r})$ and $\Psi(\mathbf{r})$ are known. The integrals $\varepsilon(\mathbf{r}, \mathbf{v})$ and $P(\mathbf{r}, \mathbf{v})$ are local in the sense that they depend only on the type of the phase space ($\mathbf{r}$, $\mathbf{v}$). At the same time, the integrals Y_α are determined by the entire type of the function $\phi(\mathbf{r})$ and $\Psi(\mathbf{r})$, i.e. by the trajectory of the particles in this space. This shows that the distribution function of escaping and intermediate electrons, which start to move along a new trajectory after every collision with the wall, should be described by the function F which depends on all five integrals (ε, P, Y_α). In particular, this is the reason why the equation (7.5.9) for f^{Esc} contains the explicit dependence on z which effectively shows that the integrals Y_α must be taken into account.

The situation in the case of the locked electrons is different. They exist for a long time in the same space, subjected to the effect of the relatively weak but mixing influences which, after all, should result in 'forgetting' of the initial parameters of the trajectories[18]. This means that after some time after the formation of the given group of the locked particles, the distribution function of these particles will depend only on the two local integrals

$$f_e = F(\varepsilon, P) \tag{7.5.14}$$

If the isodrift nature of (7.5.11) is taken into account, we obtain a

[17]Index Loc is omitted

[18]Evidently, the largest timescale of 'forgetting' is the time of 'forgetting' of the transverse adiabatic invariant.

single-argument function

$$f_e = F(\chi), \quad \chi = \varepsilon - \Omega P + A \tag{7.5.15}$$

and A is a constant, because the functions $\phi(\mathbf{r})$ and $\Psi(\mathbf{r})$ are determined with the accuracy to the constant.

Argument χ can be written in a different form

$$\chi = \vartheta - \overline{\varepsilon} \tag{7.5.16a}$$

where

$$\overline{\varepsilon} \equiv \frac{m}{2}\left[V_r^2 + V_z^2 + (V_\theta - u)^2\right], \quad \vartheta = e\left(\phi - \frac{1}{c}\Psi\right) + \frac{m}{2}\Omega^2 r^2 + A, \quad u = \Omega r \tag{7.5.16b}$$

The interpretation of the quantities ε, u is quite evident.

It is now necessary to link the general analysis results with the specific conditions in the stationary plasma thruster. As already mentioned, here we are concerned with the isodrift regime outside the ionisation zone and relatively thin near-wall layers. To determine the type of function $F(\chi)$, it is necessary either to carry out experiments or detailed analysis of the given situation in the channel. Thus, it is much more convenient to use the previously mentioned experimental data which show that the function $F(\chi)$ can be approximated by the linear spline (7.5.2)

$$f = \frac{1}{N}\begin{cases} \mu - \overline{\varepsilon} & \overline{\varepsilon} < \mu \\ 0 & \overline{\varepsilon} > 0. \end{cases} \tag{7.5.17a}$$

Quantity μ on the basis of (7.5.16) is equal to

$$\mu = \mu_0 + e\left[\left(\phi - \frac{1}{c}\Psi\right) - \left(\phi_0 - \frac{1}{c}\Psi_0\right)\right] + \frac{m}{2}\Omega^2(r^2 - r_0^2) \tag{7.5.17b}$$

has the meaning of the maximum electron energy at the point of the channel with the given ϕ, Ψ, r.

7.5.3. Experimental investigations of near-wall conductivity in the stationary plasma thruster [173]

The first arguments in favour of existence of near-wall conductivity in the stationary plasma thruster were of the integral nature. Some of them are here:

– if a ceramic bar (for example, a probe holder) is moved along the channel on the side of the outlet, discharge current rapidly increases;

– the discharge current, thrust and the efficiency of the engine are highly sensitive to the material of the insulator and contamination of the surface;

– if longitudinal scratches are made on the surface of the insulator, discharge current J_p and electronic temperature T do not change greatly, if the depth of the scratch does not become greater than some value of the order of the Debye radius, calculated from $\langle \varepsilon_e \rangle$ and n_e.

Systematic experimental investigations of the near-wall conductivity were carried out by A.I. Burgova *et al* [173] using oriented probes for recording the distribution of the density of electronic current $j_z(r)$ along the radius of the *z*-component. The importance of these characteristics for identifying near-wall conductivity was also mentioned in section 7.3.2, because in contrast to volume conductivity, the dependence of j_z on the radius is non-monotonic ('oscillating'), Fig. 7.5.3.

The distributions $j_z(r)$ where recorded at different discharge voltages in the magnetic fields, and also at different decelerating potentials on the probe. The results show the spatial oscillations of $j_z(r)$ and the expected effects, for example, the decrease of the distance between the oscillations with increase of the intensity of the magnetic field, and also unexpected effects. The latter include, in particular, the non-symmetry of the distribution $j_z(r)$ with respect to the external and internal insulators and also the presence of a sharp peak determined by 'excessively' low-energy electrons, with the peak situated very close to the external insulator. The fact that the peak is caused by these electrons is indicated in particular by its disappearance when a decelerating potential of 1–2 V is added to the probe, and also by the following estimate. Evidently, the maximum density of the current, generated by the group of the electrons with similar values v_{II} – velocity along **H**, is found at the distance from the wall of

$$\delta r_{\max} = v_{II} \frac{\pi}{2\omega_{He}}$$

Knowing $\delta r_{\max} \approx 2$ mm, $H \approx 50$ Oe, we determine $v_{II} \approx 7 \cdot 10^7$ cm/s. This speed corresponds to $\varepsilon_{II}^{peak} \sim 1$ eV which is lower than the characteristic energy of the same locked electrons ($\varepsilon_{II}^{(Loc)} \sim 15$ eV). In particular, this difference between these two energies led to the hypothesis on the 'collapse of the Debye layer'[19] at relatively high T. This was also discussed in section 7.3.1. Later, experimental indications

[19] i.e. appearance of the 'anti-Debye' layer.

(in the investigation of microwave noise) of the collapse of the Debye layer also appeared.

As regards the non-symmetry of the distribution $j_z(r)$ in relation to the surface of the external and internal insulators, this is explained by the calculations of the trajectories of the electrons which indicate the important role of radial heterogeneity of the magnetic field and the reflection of a large number of electrons travelling to the internal insulator as a result of the 'mirror' effect, i.e. retention of the transverse adiabatic invariant.

7.5.4. Microwave oscillations in the channel of the stationary plasma thruster

Investigations of the behaviour of the electron distribution function in the channel of the stationary plasma thruster using conventional Langmuir probes show that, as discussed previously, the ionisation zone is characterised by the 'mixing' of all three components of the electron distribution function, including those travelling from the outlet. This can be explained only by accepting the development of strong high-frequency ($f \geqslant 10^6$ Hz) oscillations. The determination of the areas of generation of microwave oscillations and of their frequency and amplitude is an efficient indicator of the processes taking place in the channel of the engine.

Here, we shall mention the investigations carried out by K.P. Kirdyashev [184] into the longitudinal microwave oscillations from the viewpoint of diagnostics of electron processes in the channel. These investigations were carried out using a 'two-wire' probe, inductively connected with the recording device. The distance between the conductors was $\Delta l \sim 0.8$ mm. Naturally, the selection of this distance influences the strength of the recorded signal in the vicinity $\alpha \lesssim 1$, where $\alpha = k\Delta l$. Here k is the wave number. The values were corrected with respect to α but the correction was relatively conditional. Therefore, we shall pay attention in particular to the qualitative results which cannot be doubted.

Characteristic frequencies of microwave oscillations

The measurements were carried out in the following frequency range

$$0.5 \text{ GHz} \leqslant f \leqslant 30 \text{ GHz}. \tag{7.5.18}$$

Electronic Langmuir frequencies for magnetic fields typical of the channels of the stationary plasma thrusters in the range 100 erg < H < 300 erg are in the range:

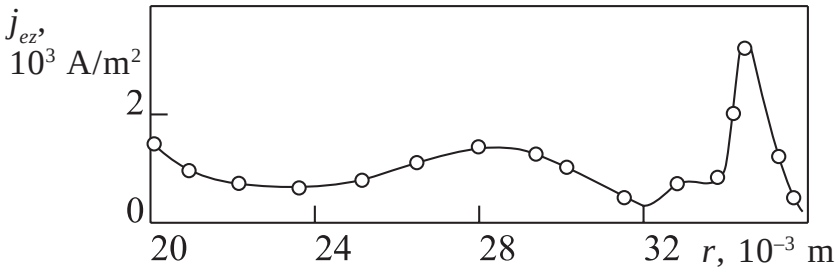


Fig. 7.5.3. Experimental measured distribution of the longitudinal electronic current in the channel of the stationary plasma thruster.

$$280 \text{ MHz} \leqslant f \leqslant 840 \text{ MHz}. \tag{7.5.19}$$

It can be seen that the ranges (7.5.18) and (7.5.19) do not fully overlap and, therefore, no final conclusion can be made regarding the intensity of noise at the Langmuir frequencies. However, in the overlapping region, the probe does not sense the oscillations of the electronic Langmuir frequencies. At the same time, the Larmor frequencies, corresponding to the characteristic range of the electron concentration 10^{11} cm$^{-3} \leqslant n_e \leqslant 10^{12}$ cm^{-3} are found in the range:

$$2.5 \text{ GHz} \leqslant f \leqslant 8 \text{ GHz} \tag{7.5.20}$$

and can be clearly seen on the spectrograms.

Localisation of noise sources

Figure 7.5.4b shows the typical distributions of the noise level along the channel at two frequencies f_1 = 1.9 GHz and f_1 = 4.0 GHz. It can be seen that the noise amplitude rapidly increases in the ionisation zone and in the vicinity of the channel outlet.

Figure 7.5.4c show identical distributions along the radius of the noise amplitude for the frequencies of f_1 = 1.7 GHz and f_1 = 4.5 GHz. The noise level in the vicinity of the outer insulator is considerably higher than that of the inner insulator. The details of distribution of the noise amplitude and the maximum values of these amplitudes for different models and operating conditions of the stationary plasma thruster vary but the qualitative form does not change. Figure 7.5.4a shows the crosshatched zones of noise generation. It may be seen that the zones are associated with the areas of formation of groups of electrons with different velocities. Actually, the newly formed electrons interact in the ionisation zone with the electrons which have

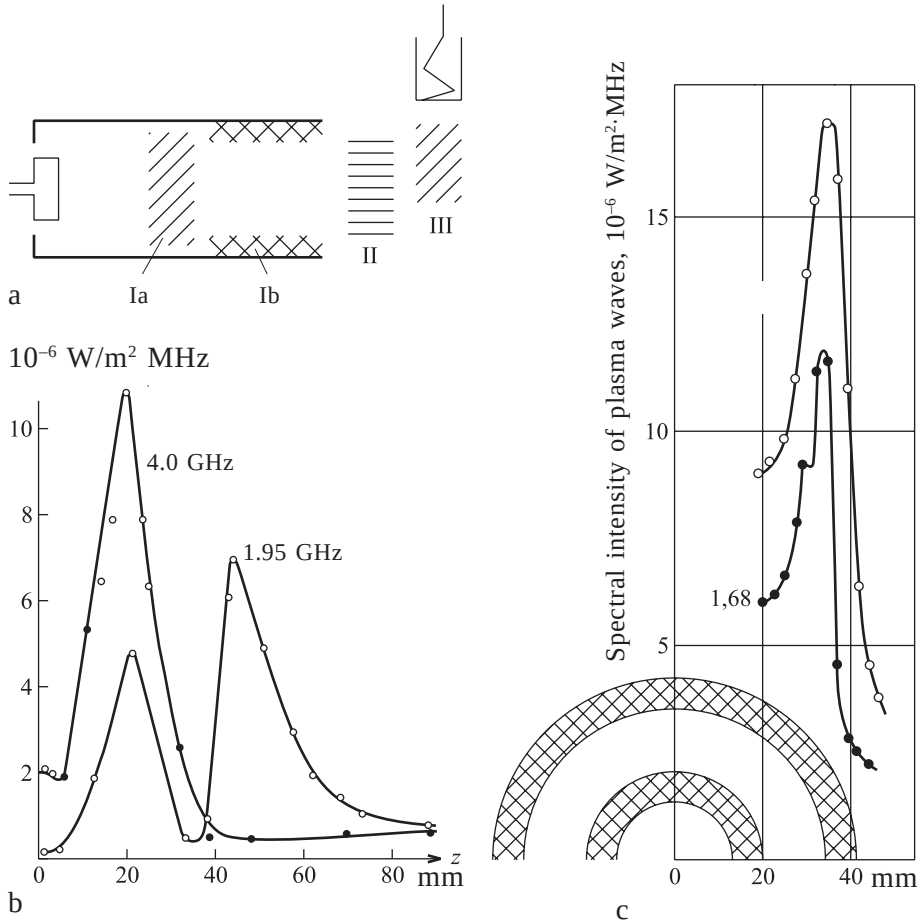


Fig. 7.5.4. The level of microwave noise in the channel of a stationary plasma thruster: a) the zone with the *apriori* high noise level, I – the cathode, Ia – the ionisation zone, Ib – near-wall zone, II – the zone of entry of the electrons from the cathode into the magnetic field of the channel, III – the zone of capture of the electrons by the outgoing ion flux; b) the distribution of the level of noise of different frequency along the length of the channel: first maximum corresponds to the ionisation zone, the second maximum to the the vicinity of the cathode; c) the distribution of noise intensity along the radius.

been drifting in the azimuth for a long period of time. In the vicinity of the insulators, the electrons, drifting on the azimuth, 'collide' with the decelerating flux arriving from the wall. In addition, as shown previously (section 7.3.1), buildup of the oscillations at the Langmuir frequencies can also form when the temperature T_e of the electrons, falling on the insulator approaches the first multiplication threshold ε*.

The characteristic feature of the radial distribution of noise intensity (Fig. 7.5.4) is the distinctive asymmetry between the vicinities of the internal and external insulators. However, this asymmetry is in excellent

agreement with both the special features of the electronic trajectories and the special features of near-wall conductivity.

Noises are also generated in the zone of exit from the channel. The interpretation of the processes taking place in this zone, leading to the generation of light at the exit from the channel, is more complicated. In principle, there are two completely different processes. The first process is the 'quasi-ohmic' collisionless capture in the magnetic field of the electrons, travelling from the cathode into the channel. The second process is the capture of the electron from the same source into the outgoing plasma jet. The first of these processes is closely linked with the electron–ion and interaction and, consequently it should have a relatively low frequency. In fact, noises with the frequencies of ~0.5–0.8 GHz prevail in this case. The second process can be of the electro-electron type, i.e. takes place at the Langmuir frequency.

Indication of the possibility of collapse of the Debye layer

Collapse of the Debye layer or the absence of the collapse should have a strong effect on the nature of microwave oscillations in the channel of the stationary plasma thruster. It has been shown that the envelope of the microwave radiation is characterised by transients randomly distributed in time with the duration of up to ~100 μs and spaced at ~100–500 μs. The interpretation of the transients of the envelope as the result of collapse of the Debye layer appears to be completely justified.

7.5.5. Erosion of insulators in the stationary plasma thruster

The focusing of the ion flux in the stationary engine is far from ideal, and part of the accelerated ions falls on the walls leading to sputtering of the latter. In addition to this, in the vicinity of the walls the crossed fields result in the flow of the electron near-wall current and this results in new effects (B.A. Arkhipov, R.Yu. Gnizdor, N.A. Maslennikov and A.I. Morozov).

The models of the stationary plasma thruster of the first generation in which the divergence of the ion flux is large, erosion in the first ~500 hours is reflected only in the form of a 'pike's tongue' having the form of sharp projections oriented along the azimuth and placed at an angle in relation to the incident flux, with a distance between them (along z) being of the order of 0.1–0.2 mm. The appearance of the structure – the structure will be referred to as the classic structure – visually produces a moire image. ~2.5–3 mm of the insulator is worn down in this period, and the profile of the insulator (in the (r, z)-plane) is 'adjusted' with respect to the profile of the ion flux. Consequently, the intensity of ion sputtering rapidly decreases and 'anomalous' erosion

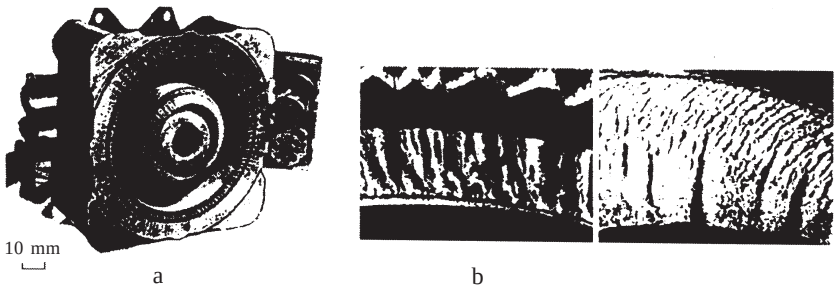


Fig. 7.5.5. Nature of anomalous erosion of the insulators of the stationary plasma thruster: a) the general view of M-100 stationary plasma thruster after operation for 5000 h; b) erosion of the output edge of the external insulators, c) erosion of the outlet edge of the internal insulator.

starts to take place. Its initial features are the appearance of deeper and deeper 'scratches', oriented mostly along the channel.

In the SPR M-10 these 'scratches' become ~5 mm long after ~500 hours, ~1 mm wide and ~1 mm deep. The side surfaces of the 'scratches' are 'ragged', and are spaced at more or less constant distances of ~1–2 mm. This is the distance of the scale of the electronic Larmor radius. With time, this pattern evolves and after ~1500 hours assumes its 'final' form (Fig. 7.5.5). The erosion rate now decreases and is stabilised at a level of ~2–3 μm/h. The resultant macrostructure has a number of special features. In particular, the asymmetry of erosion of the external and internal insulators is quite evident. Erosion is very prominent on the surface of the external insulator, whereas the internal insulator is characterised by less extensive erosion formations, and the contours of these formations are relatively smooth as if blurred. Another important special feature is the presence of traces of erosion in the sections of the insulator 'screened' from the direct effect of the ions. The periodicity of the structure of anomalous erosion remains the same – equal to the scale of the electronic Larmor radius ρ_e. It is surprising that the integral characteristics (discharge current, thrust, efficiency) at constant consumption m and discharge voltage U_p remain almost constant for >7000 h of operation, regardless of the fact that the output part of the insulator is completely 'worn down' and wear of the magnetic poles also starts to take place. At the same time, the behaviour of microwave noise is completely different. Initially, the microwave noise in the new model is on the quasi-equilibrium level, determined by T_e, and subsequently rapidly increases, exceeding the thermal level ~10^2–10^3 times and reaching the maximum value in the stage of development of the 'scratch' of anomalous erosion and subsequently decreases by approximately a factor of 10 in transition

to the steady profile of erosion.

Although the details of this phenomenon are not yet completely clear, anomalous erosion is obviously associated with the electronic component. This is indicated by the following facts:

a) stronger erosion of the outer insulator;
b) the electronic scale of the period of the structure;
c) the nature of 'gullies', oriented along the flux;
d) the behaviour of microwave oscillations, associated with the rearrangement of the surface structure, etc.

A suitable analogue is the erosion of insulators in erosion cosmic engines (section 7.4.2).

It can be assumed that it is difficult to find a better argument in favour of the important role played by direct collisions of the electrons with the walls than the phenomenon of anomalous erosion.

7.6. Examples of near electrode processes

In the previous three sections we discussed the processes taking place around the dielectric (more accurately, currentless) walls. Attention will now be given to several examples of near-electrode processes, i.e. the processes taking place during the passage of current through the plasma-solid interface.

We start with a simple model which may be used as some reference point in the case of thermal cathodes.

7.6.1. The McCowan equation

The system of the Vlasov equations can be used to describe not only the Debye layers of the insulated (and, in particular, dielectric) surfaces but also near-electrode, 'current-conducting' layers. The simplest case of this layer was investigated by McCowan. This is a layer between the cathode with unlimited emission and the plasma column. To simplify considerations, it is assumed that we have a flat cathode (plane $x = 0$) which emits into plasma a flux of cold electrons with current density j_e.

The flux of cold ions with the density j_i travels from the plasma to the cathode. The current density in the investigated interval is $j = j_i + j_e$. The initial velocities of both the ions and electrons are equal to 0 (Fig. 7.6.1).

It is necessary to calculate the distribution of the electrical field in the near-cathode layer on the condition that the strength of the electrical field in the depth of the plasma (formally at $x \to -\infty$) can be assumed to be equal to 0 and the potential is given, i.e.

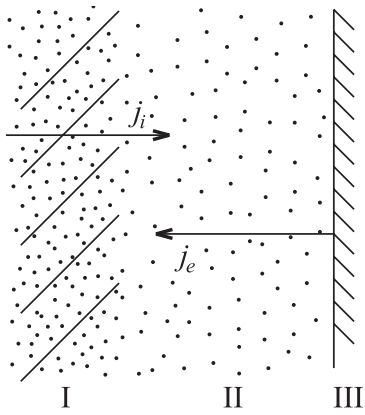


Fig. 7.6.1. Derivation of the McCowan equation: I) quasi-neutral plasma, II) the region of the volume charge ($n_e \neq n_i$), III) emitting thermal cathode.

$$E\big|_{x\to-\infty} \to 0; \quad \phi\big|_{x\to-\infty} \to U \tag{7.6.1}$$

The cathode potential is assumed to be equal to ($\phi(0)=0$). To simplify calculations, they are carried out in the hydrodynamic approximation.

Because of the continuity equations and the assumptions made in this case, we can write the following equations for the density of the charged particles

$$n_i = \frac{j_i}{e\sqrt{2e(U-\phi)/M}}; \quad n_e = \frac{j_e}{e\sqrt{2e\phi/m}}.$$

Substituting these values into the Poisson equation gives

$$\frac{d^2\phi}{dx^2} = -4\pi e\left(\frac{j_i}{e\sqrt{2e(U-\phi)/M}} - \frac{j_e}{e\sqrt{2e\phi/m}}\right). \tag{7.6.2}$$

Consequently, we obtain

$$\frac{1}{8\pi}\left(\frac{d\phi}{dx}\right)^2 = \sqrt{\frac{2}{e}}\left\{j_i\sqrt{M(U-\phi)} + j_e\sqrt{m\phi}\right\} + C. \tag{7.6.3}$$

The integration constant C is determined from the condition (7.6.1) and is equal to

$$C = -j_e\sqrt{2mU/e} \tag{7.6.4}$$

Using (7.6.3), we determine the strength of the field around the cathode

$$E_{\text{cat}}^2 = 8\pi\sqrt{2/e}\,(j_i\sqrt{MU} - j_e\sqrt{mU}). \tag{7.6.5}$$

This expression is also referred to as the McCowan equation.

In a partial case when the electronic current from the cathode is less intensive than the Richardson–Dashman saturation current, it can be assumed that on the cathode $E_{\text{cat}} = 0$. Consequently

$$\frac{j_e}{j_i} = \sqrt{\frac{M}{m}} \tag{7.6.6}$$

This ratio between the densities of the current in the actual conditions is fulfilled with acceptable approximation only in a small number of cases. However, in most cases, the fraction of the ionic current is incomparably greater than that indicated by (7.6.6). There is a number of reasons responsible for this phenomenon. In particular, this is explained by the need to have a sufficiently strong ion flux for compensating the energy losses by the cathode as a result of its heat conductivity, thermal radiation, and also the losses, caused by the work function of the electrons. Specific reasons, causing a strong ion flux to the cathode, include the fact that E_{cat} differs from zero, surface roughness of the cathode, instabilities, etc.

7.6.2. Near-electrode layers in glow and arc discharges

In section 6.6, attention was given to two types of classic discharge: glow discharge (GD) and arc discharge (AD). Only the positive columns were discussed in some detail: non-equilibrium in glow discharge and equilibrium in arc discharge. In this section, attention is given to the near-electrode regions of these discharges. Our considerations will be restricted to discharges at relatively high pressures ($p > 10^{-2}$ torr) when the free path of the particles is relatively short in comparison with the dimensions of the gas discharge volumes. Consequently, we can use the hydrodynamic models.

Near-cathode region of glow discharge [184]

The cathode in glow discharge is usually cold and a specific plasma structure forms around it. The distribution of the parameters in the near cathode zone is shown schematically in Fig. 7.6.2. The fact that 5 (!) layers were found between the cathode and the positive column was unexpected. These layers are:

– Aston dark space (ADS);
– cathode glow (CG);
– Crookes dark space (CDS);
– negative (or glow) glow (NG)
– Faraday dark space (FDS).

These layers are then followed by the column discussed in section 6.6. Externally, the structure depends only slightly on pressure.

The transfer of current through the 'plasma volume'–solid-state cold electrode' transition takes place by the flux of ions travelling from the plasma and by the 'secondary' electrons, knocked out by the ions from the electrode.

As indicated by the potential distribution, and even more accurately by the strength of the electrical field, there are two distinctive regions in the near-cathode region: the region of a sharp potential drop, including ADS, CG, CDS, and the region with a low strength of the **E**-field, including the section in which the **E**-field can change its sign. This includes mainly NG and part of the FDS.

The first of the mentioned regions, which is often referred to as simply the cathode layer, is the main block of the glow discharge and disruption of this layer, for example, during the movement of the electrodes with each other, results in the extinction of the discharge. This region is characterised by the main drop of the voltage, applied to the glow discharge electrodes. Therefore, it is useful to investigate its volt–ampere characteristic

The distributions of the densities of the electrons and ions in the cathode layer, shown in Fig. 7.6.2, indicate that the quasi-neutrality is disrupted here and the ions prevail. Therefore, for the one-dimensional case, the Maxwell equations yield the relationship between E and n_i

$$\frac{dE}{dx} = 4\pi e n_i. \tag{7.6.7}$$

The experiments show that the decrease E (x) in the cathode layer takes place in accordance with the law close to linear

$$E_x = -E_0\left(1-\frac{x}{d}\right) \tag{7.6.8}$$

Here d is the thickness of the layer, E_0 is the strength on the surface of the cathode. Substituting (7.113) into (7.112), we obtain the relationship between E_0, d and n_i:

$$E_0 = 4\pi e n_i d. \tag{7.6.9}$$

For the next step, it is taken into account that to maintain the stationary discharge, each electron, knocked out from the cathode, should move in the plasma volume and carry out the number of ionisation acts equal to the number of ions required for the exit of a single electron. If γ is the number of secondary electrons knocked out by a single ion, actually $\gamma \ll 1$, every electron should carry out $1/\gamma$ ionisation acts.

For this purpose, the process should have the cascade form. In particular, the electron, flying from the cathode, acquires the energy in the **E**-field and, colliding with the neutral, causes the 'first' ionisation. This results in the formation of two electrons which again acquire energy sufficient for ionisation and generate further two electrons, and so on. The dimensions of the cathode layer and the strength of the field in the cathode are automatically adjusted in order to ensure the required number of the ionisation acts. An avalanche develops in accordance with the equation

$$\frac{dN}{dx} = \alpha N \quad \text{or} \quad N = \exp\left\{\int_0^x \alpha\, dx\right\}.$$

Here N is the number of the electrons, generated by a single ('cathodic') electron at distance x from the cathode. α is referred to as the 'first ionisation Townsend coefficient' and its value depends on the pressure and strength of the electrical field. Evidently, $\alpha \sim n_0 \sim p$ is the concentration of the neutrals and, in addition to this, α should depend on the energy required by the electron in the free path

$$\varepsilon = E\lambda \sim \frac{E}{n} \sim \frac{E}{p}$$

Thus, the dependence of α on p and E in a first approximation should be expected in the form

$$\alpha = pf\left(\frac{E}{p}\right)$$

In fact, both successive theoretical analysis and experiments confirm efficiently this similarity law.

Taking into account some guiding consideration of experimental data, Townsend proposed the following equation[20]

[20]The strange at first sight argument of the exponent becomes obvious if it is taken into account that $T_e \sim E\lambda$.

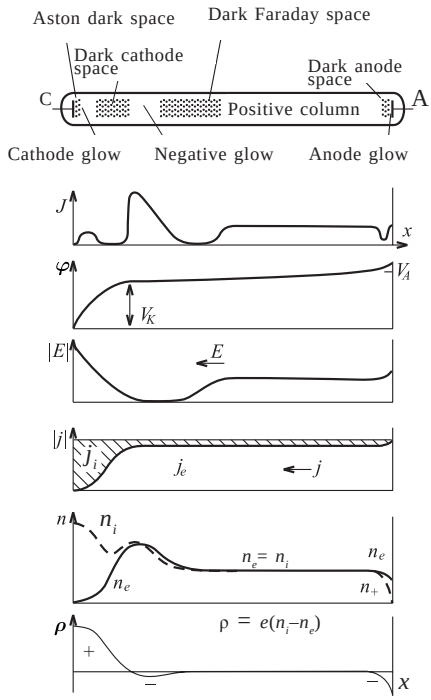


Fig. 7.6.2. Glow discharge in a tube together with near-electrode regions: distribution of glow intensity J, potential φ, longitudinal field E, the density of electronic and ionic current j_e and j_i and charges n_e and n_i, volume charge $\rho = e\,(n_i - n_e)$.

$$\alpha = pA\exp\left(-\frac{B}{E/p}\right)$$

The constants A and B depend on the nature of gas filling the pipe. Several characteristic examples are given below in the table.

	A, (cm · torr)$^{-1}$	B, V/cm · torr	E/p, V/cm · torr
1. Argon	12	180	100–600
2. Nitrogen	8.8	275	27–200
3. Hydrogen	5	139	22–1000
4. Xenon	26	350	200–800
5. Mercury	20	320	200–600

In accordance with the above considerations, the following equation for the discharge equilibrium is obtained for the cathode layer

$$\frac{1}{\gamma} = \exp\left\{\int_0^d \alpha\, dx\right\},\quad \alpha = Ap\exp\left\{-\frac{\beta}{E/p}\right\}. \qquad (7.6.10a)$$

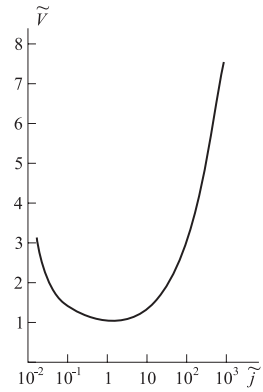


Fig. 7.6.3. 'Theoretical' VAC of glow discharge.

These equations for the known γ, A, B, p yield another relationship between E_0 and d.

In order to close the system of equations, it is necessary to determine n_i. However, if we accept the relationship of n_i with the discharge current j_p which is equal to[21]

$$j_p \approx j_i = \theta e n_i \sqrt{\frac{2E_0 d}{M}}, \quad \theta \sim 1 \tag{7.6.10b}$$

we obtain the volt-ampere characteristic of the cathode layer

$$j = F(U_k), \quad U_k = E_0 d$$

Unfortunately, the relationship between E_0 and d, following from the equalities (7.6.10), is difficult to express in the analytical form but it is easy to calculate numerically. The volt–ampere characteristic (VAC), shown in Fig. 7.6.3, is obtained. The special feature of these curves is the presence of a minimum.

Here, the physicists face a paradoxical situation. It appears that only the regimes, corresponding to the minimum VAC, are stable. More accurately, if it is attempted, by changing the voltage, to increase or reduce the current density in the conditions in which the cathode spot occupies only a part of the cathode, the current density on the cathode does not change and simple corresponding splitting of the area of the cathode spot takes place. The voltage at the discharge remains almost unchanged. The decrease of discharge current is accompanied by a decrease of the size of the area of the cathode spot. The regime of this type is referred to as the normal glow discharge. However, if the cathode spot occupies the entire surface as a result of the increase of discharge current, a further increase of current results in a

[21]The current in the cathode layer is transferred mostly by ions (Fig. 7.6.2).

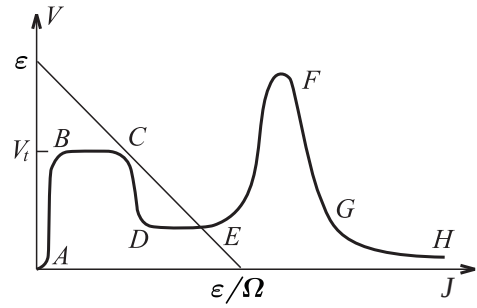


Fig. 7.6.4. The VAC of the discharge between the electrodes in a wide range of current and the loading straight line: A) the region of a non-independent discharge, BC) the dark Townsend discharge, DE) the normal glow discharge, EF) the anomalous glow discharge, FG) transition to arc discharge, GH) arc discharge.

rapid increase of discharge voltage. This regime is referred to as the 'anomalous glow discharge' and at some voltage, which depends on the special design features of the system, the discharge changes from glow to arc (Fig. 7.6.4). The mentioned special feature of the cathode spots of normal glow discharge was difficult to understand for a long period of time and was explained only at the beginning of the 80s of the previous century after carrying out numerical two-dimensional calculations of the cathode regions. It was shown that this phenomenon is explained by the structure of the equipotentials in the vicinity of the cathode which makes other distributions of current on the cathode in the stationary regime unstable [184].

The second quasi-neutral part of the cathode region will now be discussed briefly. In this sub-region, more accurately the negative glow (NG) region, high-intensity ionisation of the gas by the electrons, accelerated and heated in the cathode layer, takes place. This region is also characterised by the formation of the ions which bombard the cathode, and of the electrons which later transfer the current at the start of the positive column. In the NG region the plasma density is higher than in other regions and the spreading of the plasma, like the current transfer, takes place mostly as a result of Δp_e. Therefore, the electrical field in the main part of the NG region is weak and its direction may even be negative. Consequently, the NG layer becomes a trap for the ions. The transfer of current from the NG layer to the positive column as a result of Δp_e causes that the value of T_e in this region greatly decreases and, consequently, the intensity of glow rapidly drops. This leads to the formation of the Faraday dark space. This is followed by the appearance of the **E**-field and the positive column also appears. The accurate description of the near-cathode region on the whole requires

a large system of two-dimensional equations taking into account the dynamics of electron energy and transformation of heavy particles. Therefore, it cannot be said even now that the investigations in this area have been completed.

7.6.3. Vicinity of the arc thermal cathode

The arc discharge cathodes can be divided slightly conventionally into two large classes: thermal ('hot') cathodes and 'cold' cathodes. The processes taking place around the cold cathodes will be discussed in the next section, and here we discuss briefly the special features of processes taking place around the thermal cathode.

At first sight, it could appear that the processes in the cathode regions of the arc discharge with the thermal cathodes are extremely simple and described by the McCowan equation (7.6.5). However, this is not so. There are several reasons for this. One of them is the small thickness of the near-cathode layers in the arcs at normal pressure and, even more so, at high pressures. For example, at the normal pressure of the surrounding air and complete ionisation of the air $n \sim 10^{18}$ cm^{-3} and, consequently,

$$\omega_{0e} \approx 5 \cdot 10^{13} \text{ s}^{-1,}$$

At $T_e \sim 5$ eV, the characteristic velocity of the electrons is $v_e \sim 2 \cdot 10^8$ cm/s and, consequently, the Debye radius is

$$r_D \sim \frac{2 \cdot 10^8}{5 \cdot 10^{13}} = 4 \cdot 10^{-6} \text{ cm.}$$

This is the value of the order of 0.1 of the wavelength of the yellow line.

Therefore, the structure of the layer should be sensitive to the surface roughness of the cathode and its 'blotchiness', i.e. the presence of different films on the cathode surface.

The second reason which complicates the analysis of the region in which we are interested, is the relatively high level of energy generation leading to melting and evaporation of the cathode and also to quite intensive thermal-electro-chemical reactions, including 'poisoning' of the cathode. However, the reverse situation may also occur. For example, hafnium (Hf), used as an arc discharge cathode in air, is activated during discharge as a result of the formation of hafnium nitride (HfN) on the surface of the cathode and has the form of an attractive gold colour compound with high emissivity.

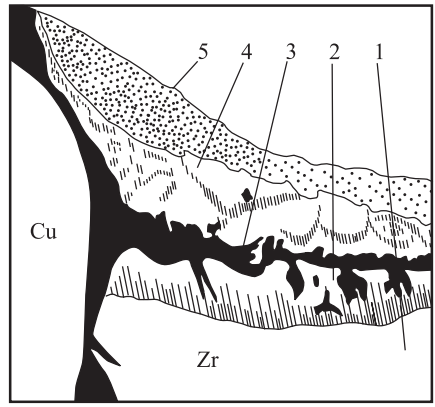


Fig. 7.6.5. The multilayer structure in the volume of the thermochemical cathode made of zirconium (Zr) in a copper sheath (Cu) after operation in air for ~1 h, J = 100 A. 1) undamaged Zr, 2) the melt zone, 3) the zone of solid phase reactions, 4) the zone of liquid phase reactions, 5) the active surface of the cathode.

In a general case, as a result of unavoidable impurities in the electrode materials used in practice and impurities in the surrounding gases (air, argon), multilayer structures usually form at the end of the electrode (Fig. 7.6.5). In addition to this, in this region the single crystals, oriented along the current direction, may also grow in the cathode. This list of complicated circumstances can also be extended. For this reason, we will not describe relatively individual models of the thermal cathodes of arc discharge and will answer two important questions.

The thermal cathodes usually operate – depending on the discharge current – in two regimes: 'with the cathode spot' and in the 'diffusion' regime'. In the former case, the specific erosion of the cathode is greater than in the latter case.

As a result of the above considerations, the theoretical models of the near-cathode layer of the arc discharge are qualitative. The two-layer model of the near-cathode zone is discussed in most cases (Fig. 7.6.6).

The figure shows the distribution of the plasma parameters in the near-cathode region, corresponding to the two-layer model. The first layer is the volume charge layer which is smaller than the free path of the ions and electrons[22]. As a result of the relatively large fraction of ionic current, this layer is characterised by the formation of a surplus positive spatial charge, resulting in the formation of a potential jump at the cathode surface. The fraction of ionic current in the first layer is constant. The second layer separates the first layer and the arc column is referred to as the ionisation column; in this layer, the quasi-neutrality

[22]In the ideal situation, it is the McCowan layer.

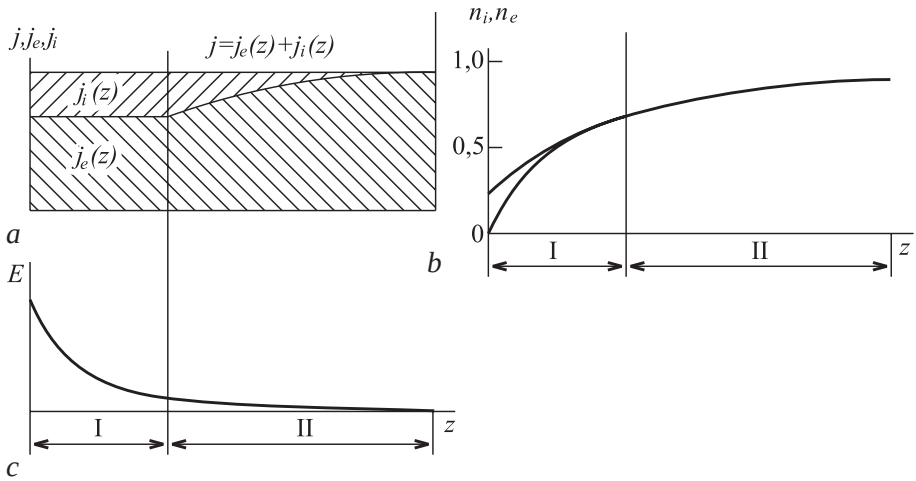


Fig. 7.6.6. The two-layer model of the near-cathode zone in the arc discharge with the thermal cathode. The distribution of current j (a); the concentration of the electrons and ions (b) and of the electrical field (c) in the near-cathode region.

condition is fulfilled and charged particles are produced at a high rate as a result of the energy acquired by the electrons in the first layer, as in the negative glow discharge.

The fraction of the electronic current S (z) increases from $S \approx 0.7$–0.9 at the cathode surface to $S = b_e/(b_e + b_i)$ in the discharge column (b_e, b_i is the mobility of the electrons and ions). The high value of the ionic current, sustained by the processes taking place in the second layer, results in the formation of a surplus of the positive charge and in a potential jump in the first layer. Thus, the processes in both layers are closely linked with each other.

The energy, brought to the surface of the cathode by the ions, is used for heating to the temperature required for emission.

These considerations show clearly that the processes in the near-cathode region of the arc should depend strongly on the properties of the surface and cathode material, its geometry and cooling conditions. For example, if the cathode material is characterised by a high work function, then the required electron emission can be ensured only by increasing the surface temperature. However, for this purpose it is necessary to increase the heat flow on the surface from the near-cathode region. This can be achieved by the rearrangement of the processes taking place in this region and leading to a larger cathode potential drop.

The minimum value of the near-cathode potential jump – without taking into account the heat conductivity and the radiation of the cathode – can be estimated from the equation

$$\min eU_k = \frac{\varphi_e}{1-S}.$$

Here φ_e is the work function of the electron which is ~3–5 eV and, consequently, at S ~0.8 V we have $\min U_k$ ~ (15–25) V.

In the studies carried out at the N.E. Bauman Moscow State Technical University (A.M. Zimin, N.P. Kozlov, V.I. Khvesyuk) the results were used for the first time to formulate the general problem of investigating the cathode processes of arc discharges in the conditions with thermal emission. A model based on the closed system of equations describing self-consistent processes taking place in three spatial regions: near-cathode region, on the surface of the cathode and inside the cathode, was constructed.

The equations include the description of the main processes: formation of ionic current on the cathode, formation of a spatial charge in the vicinity of the cathode surface, the kinetics of ionisation-recombination in the conditions of large deviations from the equilibrium and recycling of heavy particles, relaxation of the emission electron beam (in the near-cathode region), energy exchange, neutralisation of the ions, electron emission, removal of the atoms of the electrode material (on the cathode surface), current transfer, Joule heat generation, heat dissipation, diffusion of the activating addition, decreasing the work function of the electron from the cathode (in the solid).

The resultant numerical models were used for investigating the relationships governing the cathode processes in plasma generators of different types (plasmatrons, high-intensity light sources, plasma furnaces), working in inert gases in a wide current range from 100 to 10 000 A[23].

7.6.4. Spots on the cold cathode

The arcs with the cold cathodes are usually ignited by short-term contact of a moving anode (or a secondary electrode–igniter) with a cold cathode. A suitable example is arc ignition in electric arc welding. The contact area – with sufficiently high currents – is characterised by rapid heating of the electrodes, start of evaporation of the contacting regions. The resultant vapours are subsequently ionised and an arc is ignited.

If the discharge is long-lasting, the evolution of the cathode spot

[23]Detailed mathematical description of the processes taking place on the thermal cathodes, the results and comparison of the results with the experiments are presented in the monographs: I.G. Panevin, et al., *Theory and calculation of near-electrode processes*, Nauka, Novosibirsk, 1992; A.M. Zimin, et al., *Mathematical modelling of cathode processes*, Nauka, Novosibirsk, 1993.

depends strongly on whether the cathode can be heated to temperatures at which thermal emission ensures generates discharge current, or the boiling temperature of the cathode is too low and the required intensity of thermal emission is not reached. In the former case, a stationary more or less homogeneous emitting zone with a low erosion rate forms after some time, and in the latter case small spots, existing for the entire period of time, form on the cathode. When discussing long-term discharges in the following sections, we refer to the cathodes with a low boiling point.

These spots are detected in a very wide range of discharge current – from fractions of ampere to thousands of amperes or more, both in the classic devices (for example, in mercury rectifiers for breaking hig-current electric contacts) and in plasma accelerators with cold cathodes (see below). The spots cause extensive erosion of the cathode material and the erosion products are ejected in the form of ionised vapour and droplets with the characteristic parameter of the order of several microns[24]. Generation of plasma by the cathode spot causes that the characteristics of the spot depend only slightly on the pressure of the surrounding medium, at least in the range from 0 to 1 atm. As a result of the extensive use of mercury rectifiers, the processes taking place on the cold cathodes were investigated in most devices using a mercury cathode.

Depending on the discharge current and the speed of movement of the cathode zone of the electrode (or the time passed from the start of the discharge), it has been suggested that there are three types of spots which can be detected on the cathode [185].

The ‘fast’ spots (‘cells’ according to I.G Kesaev) – these are structures with the characteristic size of ~1–10 μm and the current in every spot of ~0.1–1A. They move rapidly on the cathode surface with the speed of several tens of metres per second, leaving specific shallow (~0.1–0.3 μm) damage in the form of ‘herringbones’ or ‘twigs’ (Fig. 7.6.7b, c).

The ‘slow’ spots appear at discharge currents higher than 100–300 A. Theis size is ~0.1 mm, and they move at a speed of several metres per second. Figure 7.6.7a shows the traces of slow spots. At discharge currents of ~1000 A, the characteristic values of current in a single spot are ~200–500 A (copper).

The ‘stationary’ spots appear in the conditions in which the arc at a current of ~100 A and higher burns in a stationary manner between the two closely spaced electrodes. The existence of the stationary arc is accompanied by the formation of a molten spot. These spot types

[24]Here we refer to explosive emission

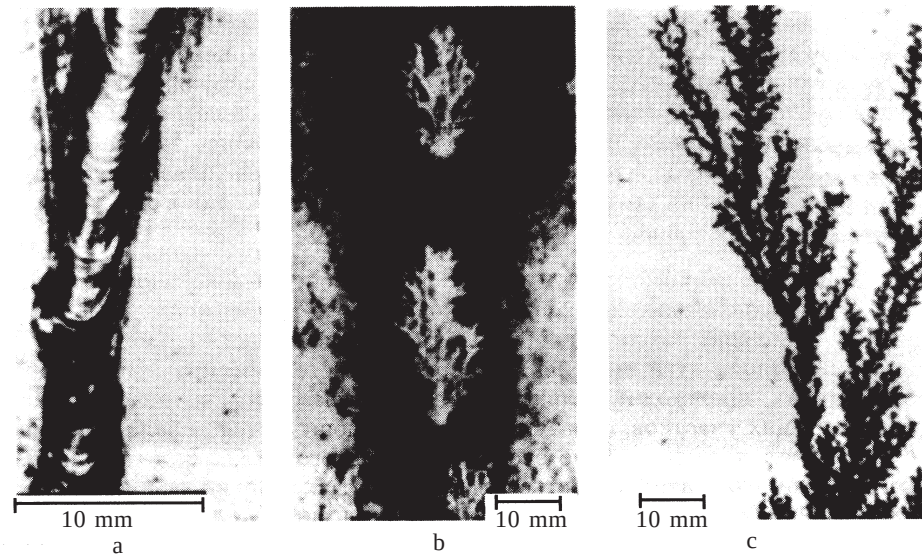


Fig. 7.6.7. Different types of traces of erosion of a cold cathode of the arc discharge at different speeds of movement of the cathode spot: a) v = 0.2 m/s, J_p = 400 A; b) v = 20.2 m/s, J_p = 40 A; c) v = 60 m/s, J_p = 100 A.

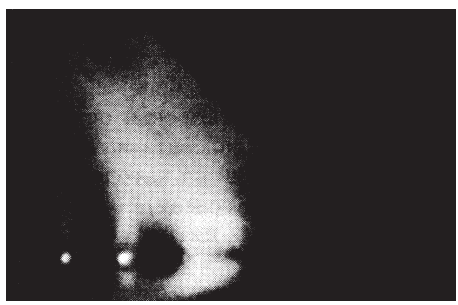


Fig. 7.6.8. Photograph of slow cathode spots on a zinc electrode at a current of $J_p \approx 10^3$ A. Interelectrode gap 2 mm (A.I. Bushin, et al).

are shown clearly in Fig. 7.6.8 [186].

As regards the properties, the fast spots are most interesting. Some of their special features are mentioned below:

1. The size of the spots according to different estimates is in the range 1–10 μm.

2. The movement of the spots is chaotic and their displacement is governed by the stochastic law $\overline{x}^2 \sim t$.

3. 'Quantisation' of the current. In a wide range of discharge current the current in one cell changes in a relatively narrow range from some minimum value J_0 to the maximum value $\leqslant 3\ J_0$. If the discharge current becomes lower than J_0, the discharge is extinguished, and if it is greater

than $3J_0$, the spot splits into two spots. The value of J_0 depends on the boiling point and heat conductivity of the cathode material. The characteristic values of J_0 are as: Hg – 0.07 A, Al – 1 A, Fe – 1.5 A, Cu – 1.5 A.

4. The estimates show that the characteristic current densities in the cells are ~10^5–10^6 A/cm^2. These high current densities indicate the explosion nature of the emission process.

5. The lifetime of cells (the holding time in the same area), ~10^{-4}–10^{-6}.

6. The cathode voltage drop is usually in the range ~10–15 V.

7. It is paradoxical but it is a fact: if a magnetic field is applied in the direction parallel to the cathode surface the cathode spots move in the opposite direction, not in the direction of the ampere force.

7.6. 5. Near-anode layers in glow and arc discharges

These layers form in contact with the positive column of the arc and glow discharges with the current-receiving metallic anode. The characteristic manifestation of these layers is the presence of the anode potential drop (APD), ΔV_A. This drop can be both positive and negative.

The appearance of ΔV_A of the same sign is determined by a large number of different factors. The first criterion of the sign of ΔV_A was formulated by Langmuir and Mott-Smith. They connected the sign of ΔV_A with the value Λ of the ratio of the density of the random electrical current $j_e^{(x)}$ to the density of discharge current j_p, i.e.

$$\Lambda = \frac{j_e^{(x)}}{j_p}.$$

If $\Lambda > 1$, then according to Langmuir and Mott–Smith $\Delta V_A < 0$, and if $\Lambda < 1$, then $\Delta V_A > 0$. In many cases, this criterion can be used to predict the sign of ΔV_A.

However, this criterion has a single uncertainty: where to obtain the values of $j_e^{(x)}$ because the presence of the anode results in the rearrangement of the positive column at large (of the order of the column diameter) distances. Therefore, the sign of ΔV_A, especially at $\Lambda \sim 1$, calculated from the parameters of the positive column, is influenced by the electrode material, electrode geometry, pressure, etc.

The absolute values of $\Delta V_A < 0$ are of the order of keT_e/e, and $\Delta V_A > 0$ is of the order of ionisation potential. Theoretical analysis became possible only after the development of sufficiently powerful computers. However, at the moment only the simplest models of axisymmetric geometry can be calculated.

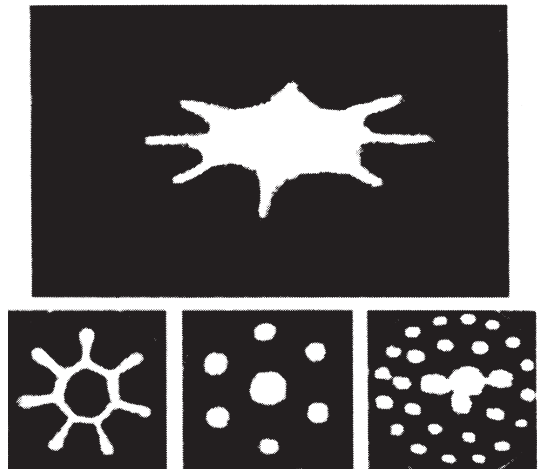


Fig. 7.6.9. Examples of figure anode spots.

The fact that these models are urgently required is indicated by the interesting photographs (Fig. 7.6.9) of the anode spot in a glow discharge. The experimental data relating to the anode layers are also very scarce. The complicated nature of the processes in the near-anode zone is due to a large extent to the fact that the electrode here does not emit particles, in contrast to the situation on the cathode.

Nevertheless, there are a number of relatively general features of the near-anode processes. In particular, it is evident that irrespective of the sign of ΔV_A a non-quasi-neutral layer forms around the anode, where $n_i \neq n_e$. If $\Delta V_A > 0$, the layer forms due to the fact that the ions cannot travel to the positively charged anode. On the other hand, if $\Delta V_A < 0$, the electrons can not travel to the anode.

The case with $\Delta V_A > 0$ has been studied more extensively. This regime (it is detected in most cases in low current discharges) has the following characteristic features. In particular, the positive column transfers to the dark anode space and a bright film forms on the anode and covers its. With the increase of pressure the film becomes non-uniform with convex formations, and the film breaks up into the individual spots – 'pearls'. Usually, the latter form regular figures. In some cases, these spots move quite rapidly forming shiny rings.

At negative ΔV_A or low $\Delta V_A > 0$, the bright film does not form on the anode.

The formation of the bright film at $\Delta V_A > 0$ is easy to understand. This is the result of the fact that the electrons, travelling to the anode, acquire the energy sufficient for ionisation of the neutrals. The formation of the spots is the result of the loss of stability by the film.

The resultant spontaneous thickening increases the plasma potential in the given area and this results in the additional attraction of the electrons travelling to the anode and in the increase of the size of the initially small projection.

7.6.6. Erosion of electrodes and the processes in coaxial (quasi) stationary accelerators

We now examine erosion of electrodes in 'supercritical' conditions in high-current coaxial plasma accelerators (CPA and others) which were discussed in section 3.7.2. The data presented in 3.7.2 show that the processes on the anode and the cathode greatly differ. In particular, in the supercritical regimes the anode is covered with a thin layer not 'drawn' by the probes, with the sharp potential drop while the drop around the cathode is small ($\leqslant$10 V). Since the structure of these layers could not as yet be drawn, we restrict our considerations to examining only the traces of electrode erosion.

In contrast to the conventional arc, the rate of erosion of the anode is considerably higher than that of the cathode. Figure 7.6.10 shows the traces of erosion of the central electrode in relation to polarity. It can be seen that at negative polarity the electrode shows traces of erosion in the form of 'rime' (Fig. 7.6.10c), typical of the arc on a cold cathode. A completely different situation is observed in the case of positive polarity. There are three types of failure of the electrode. In particular, the surface of the electrode evaporates and, consequently, figures and intercrystalline gaps are clearly evident under a microscope. Further, in most cases, craters with a diameter of ~0.1–1 mm form in these gaps and some of these craters develop into wide molten grooves (Fig. 7.6.10d). Finally, a circular molten spot with a diameter of the order of 1 cm (Fig. 7.6.10a) forms at the tip of the anode at every discharge. Etching and the presence melted grooves and fine spots indicate the existence of a thick not very stable layer in the vicinity of the anode. The presence of the spot at the tip of the anode indicates the constriction of the discharge in this area.

If the outer electrode (nozzle) is the anode, the output edge of the electrode in operation in hydrogen shows burning and partial failure. It can hardly be doubted that the discovered traces of anode erosion are caused by the layer of magnetic insulation whose thickness is $\sim\rho_e$.

7.7. Dusty plasma

'Dusty plasma' is a plasma which contains, in addition to conventional microparticles (atoms, molecules, ions and electrons), also macroparticles

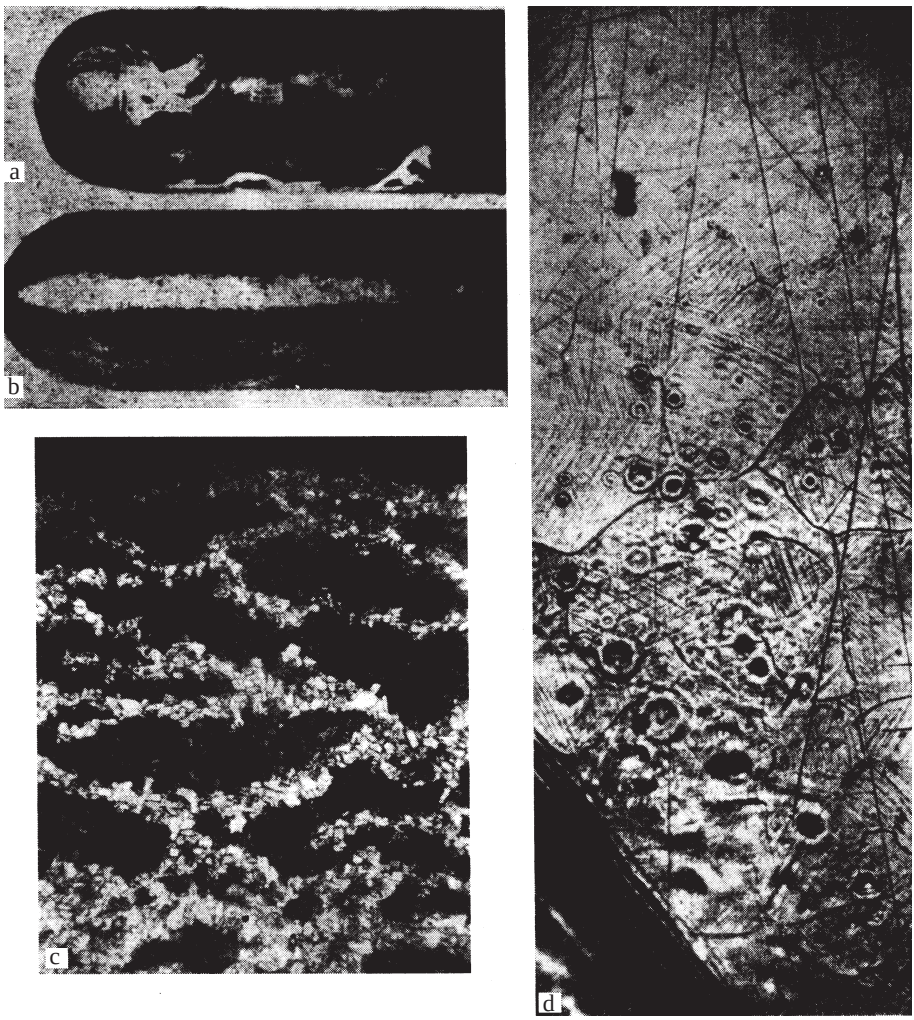


Fig. 7.6.10. Photographs of erosion of the central electrode CPA-1 in relation to its polarity: a) anode, b) cathode, c) cathode grooves on the etched surface of the electrode (magnification ×60), d) erosion structure of the anode in the region of the molten groove (magnification ×130).

– specks of dust, droplets, etc. At the moment, special attention is given to 'specks' of dust with the size of ~(1–10) μm. The 'range of main interest' at present is:

$$b \ll r_d \ll l_d, \quad l_d \sim \frac{1}{\sqrt[3]{n_d}}. \tag{7.7.1}$$

Here n_d is the concentration of the macroparticles, b is the radius of these macroparticles, r_D is the Debye radius of the plasma.

In particular, theory assumes existence – experiments produce – spherical macroparticles. The gaseous component of the plasma is regarded as 'background' plasma. This plasma is generated by conventional means: direct discharges (arc, glow), high-frequency discharges of specific types, ultraviolet radiation, etc. If the specks of dust do not emit electrons, and the electron temperature is low

$$kT_e \ll \varepsilon_*, \tag{7.7.2}$$

i.e. smaller than the first multiplication threshold in secondary electron emission, then the speckle dust has a negative charge and in the typical conditions its charge is

$$q = Z_d e \sim (10^3 \div 10^4)e. \tag{7.7.3}$$

Here e is the elementary charge. Evidently, the appearance of these 'giants' in conventional (background) plasma has a strong effect on the plasma properties.

Of course, dusty plasma can be regarded as a unique variety of the transforming plasma which was discussed in Chapter 6.

Evidently, the dusty plasma was discussed for the first time in 1920 by Langmuir. However, serious interest in dusty plasma appeared in the 60s as a result of the development of MHD generators working with the products of combustion of solid fuel. Slightly later, the problem of dust in plasma (in the purely utilitarian aspect) was studied at the end of 60s and the beginning of 70s as a result of the development of technology of plasma spraying. However, the dusty plasma became the object of extensive scientific investigations only in the second half of the 80s. In Russia, the leading role in the investigation of dusty plasma has been played by the team headed by V.E. Fortov. A significant contribution to the development of the theory of these systems has been provided byV.N. Tsytovich.

The specks of dust discussed here can be brought in from the outside but they can also form spontaneously in the volume as a result of the condensation of background plasma particles. The dust is widely encountered in cosmos. The solar system has also probably formed from a dust cloud, and this is also typical of the evolution of star systems. As a result of the investigation of the central cluster, the gas was ionised to a considerable extent. Dust also appears around space systems and is also ionised under the effect of the sun, solar radiation and solar wind.

Undoubtedly, the physics of dusty plasma is in its initial stages at the moment; extensive work in this area in near future is unavoidable.

7.7.1. Charging of macroparticles and their interaction with each other

Charging of macroparticles

Previously, we mentioned only one variant of charging of the macroparticle: the macroparticle does not emit, and the electronic temperature of background plasma is low. In these conditions, the charge of the macroparticle is negative. However, if the particle starts to emit electrons under the effect of some factor (thermal emission, photoemission, secondary electron emission at $kTe > \varepsilon_*$), the charge of the macroparticle will be positive. It can be seen that the difference in the electrical state of dusty plasma is large, especially if it is taken into account that the particles with different emission properties can be combined.

In particular, the condition of quasi-neutrality of the dusty plasma has evidently the form

$$(\pm)Z_d en_d + en_i - en_e = 0.$$

This shows that if the charge of the dust is positive, the dust can bond the main part of the electrons and, consequently, greatly reduce conductivity, and vice versa.

If $q < 0$, the value of q can be estimated by simple considerations: the 'typical electron' (the last one!) should have enough kinetic (thermal) in energy in order to catch up with the already charged macroparticle. In other words, since $kT_e \sim qe/b$, then

$$Z_d \sim \frac{kT_e b}{e^2} \tag{7.7.4}$$

This estimate also yields the previously mentioned values of Z_d.

The field of every particle is surrounded by a screening layer which at the distance $r \gtrsim r_D$ is described by the conventional 'Yukawa potential'

$$\phi = -\frac{q}{r} e\left\{-\frac{r}{r_D}\right\}; \quad r_D = \sqrt{\frac{kT_e}{4\pi e^2 n_e}}. \tag{7.7.5a}$$

However, at large distances, because of the nonuniform distribution of the ions, the rate of decrease of the potential can be smaller, for example

$$\phi \sim \frac{-q}{r^2}. \qquad (7.7.5b)$$

Interaction of macroparticles

The situation with the interaction of the macroparticles is relatively complicated. The point is that, because of large dimensions, the macroparticles screen quite efficiently the fluxes of ions, travelling to the adjacent particles and, consequently, attraction forces form between the distant macroparticles. Therefore, the potential energy of two particles forms a well. Because of this special feature, we can consider here different phase states of the set of the macroparticles:

– 'gaseous', when the position of the particles is not correlated with each other;

– 'liquid' – when there is strong correlation only at short distances;

– 'solid' ('crystalline') – when a correlation is found at large distances.

The possibility of existence of different phase states of the cloud of the macroparticles in plasma can be linked with the high values of the non-ideality factor which has the following form in the dusty plasma

$$\Gamma \equiv \frac{q^2 / l_d}{kT_d}, \quad K = \left(N_d^{1/3}\lambda_D\right)^{-1}. \qquad (7.7.6)$$

Numerical modelling shows that 'crystallisation' starts at $K \cdot \lg \Gamma \geqslant 4$.

In experiments, the behaviour of the macroparticles can be easily studied in a capacitance high-frequency discharge (Fig. 7.7.1). It has been shown that crystallisation is especially reflecyed in the distribution of the particles in the horizontal plane. As regards the distribution along the vertical (gravitational force is very important here), the particles are aligned strictly one after the other. With increasing plasma temperature the 'crystal' changes to the 'liquid' state and, subsequently, to the 'gaseous' state.

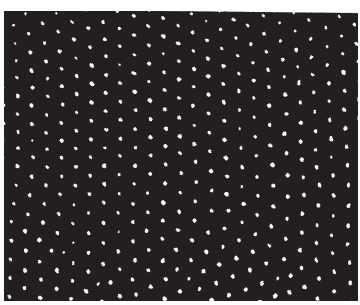


Fig. 7.7.1. The 'crystal' formed from dust particles: typical video image of the horizontal section of the ordered structure of the particles produced in the near-electrode region of the high-frequency discharge. The area 6.1 × 4.2 mm^2 containing 392 particles with a diameter of 6.9 μm is shown. The structure has a hexagonal lattice.

7.7.2. Linear oscillations in uniform dusty plasma at H = 0

The appearance of charged dust in background plasma naturally changes the spectrum of its oscillations. At present, only numerical analysis can be carried in the entire range of the non-ideality parameter Γ_d. Therefore, we confine ourselves to cases in which $\Gamma_d \ll 1$, i.e. the dusty component can also be regarded as ideal.

Therefore, the hydrodynamic system of equations has the form (for the isothermal case and $q < 0$)

$$\begin{cases} \dfrac{\partial n_d}{\partial t} + \operatorname{div} n_d \mathbf{v}_d = 0 \\ M_d \left(\dfrac{\partial \mathbf{v}_d}{\partial t} + (\mathbf{v}_d \nabla)\mathbf{v}_d \right) = -eZ_d \nabla\phi = \dfrac{1}{n_d} \nabla p_d \\ p_d = k n_d T_d \end{cases} \tag{7.7.7}$$

$$\begin{cases} \dfrac{\partial n_\alpha}{\partial t} + \operatorname{div} n_\alpha \mathbf{v}_\alpha = 0 \\ m_\alpha \left(\dfrac{\partial \mathbf{v}_\alpha}{\partial t} + (\mathbf{v}_\alpha \nabla)\mathbf{v}_\alpha \right) = -e_\alpha \nabla\phi - \dfrac{\nabla p_\alpha}{n_\alpha}; \quad \alpha = i, e \\ p_\alpha = k n_\alpha T_\alpha \end{cases}$$

$$\Delta\phi = -4\pi e(n_i - n_e - z_d n_d).$$

Here, it is assumed that the effects of sticking (emission) to the macroparticles are not important.

Linearising the system in a conventional fashion and assuming that the non-perturbed plasma is stationary, and the waves are flat $\sim\exp\{-i\omega t + i\varkappa x\}$, the following dispersion equation is obtained for longitudinal waves

$$\epsilon = 1 - \frac{\omega_{0e}^2}{\omega^2 - c_{Te}^2 \varkappa^2} - \frac{\omega_{0i}^2}{\omega^2 - c_{Ti}^2 \varkappa^2} - \frac{\omega_{0d}^2}{\omega^2 - c_{Td}^2 \varkappa^2} = 0. \tag{7.7.8}$$

The roots of this equation in the limiting cases have the form

$$\omega_1^2 \approx \omega_{0e}^2 + c_{Te}^2 \varkappa^2, \tag{7.7.9}$$

$$\omega_2^2 \approx \varkappa \sqrt{\frac{kT_e}{m_i}} \left(\frac{n_{i0}}{n_{e0}} \right)^{1/2},$$

$$\omega_3^2 \approx \omega_{0d}^2 \varkappa^2 \frac{r_D^2}{1+\varkappa^2 r_D^2}, \quad \frac{1}{r_D^2} = \frac{1}{r_{De}^2} + \frac{1}{r_{Di}^2}.$$

The first group corresponds to the Langmuir waves, the second to the ion sound at $\varkappa r_{De} \ll 1$, the third to the dust sound. Evidently the difference between n_{i0} and n_{e0} is associated with the presence of negatively charged particles and not with the disruption of quasi-neutrality.

8

Instabilities and self-organisation of plasma dynamic systems

This chapter is concerned with the non-stationary processes in plasma systems. These processes were already investigated in some instances in the previous chapters. This included the waves in homogeneous media (Chapters 2–4), oscillations, associated with the ionisation process (Chapter 6), and also with insufficient conductivity in the channel of the stationary plasma thruster (Chapter 6).

In this chapter, the non-stationary phenomena will be investigated from a more general viewpoint. However, it should be stressed immediately that is the number of non-stationary processes in plasma systems is very large, not only owing to the fact that the number of these systems is large. Even in individual systems the variety of the 'oscillations' is extremely large and in most cases have not been studied. A suitable example in laboratory practice are tokamaks in which many dynamic phenomena have not been explained despite gigantic expenditure (>US$30 billion) and more than 50 years of investigations.

In this chapter, we do not consider the linear perturbations in homogeneous media (with the exception of section 8.1.1) and we investigate linear perturbations in heterogeneous plasma which will be the subject of the first two sections[1]. The third section is concerned with the quasi-autonomous non-linear structures using modelling equations. We shall also discuss dynamic chaos and note several important special features of turbulent flows, both hydrodynamic and plasma. The chapter will be ended by a brief description of the stabilisation of oscillations using feedbacks.

Special features of plasma instabilities

One of the fundamental properties of any system – natural or artificial

[1]Reviews of a large number of linear instabilities are described in [191] and [192].

– is their stability of instability, i.e. the capacity to retain or not retain its functioning (state) throughout a long period of time in the presence of unavoidable internal (for example, thermal oscillations) or external weak effects. If the system is unstable, then after some time the system either decays or transfers to some stable state and in this case we are concerned with self-organisation[2].

Plasma has a very large number of degrees of freedom and, consequently, shows almost always a tendency for self-organisation. This is associated to a large extent with the long-range effect of electromagnetic interaction of the plasma particles with each other.

In the final analysis, the entire dramatic (lasting more than 50 years) history of investigations of controlled thermonuclear synthesis is in fact the battle with the self-organisation of plasma. However, even if the produced plasma dynamic systems more or less corresponds to the calculation scheme, it is always 'noisy'. This occurs not only in laboratory but also in the natural plasma systems. A suitable example of this is the Sun. It would appear that the Sun is just a sphere and nothing else. However, the gigantic corona, large spot, protuberances – all this is the result of self-organisation of plasma dynamic systems which nobody 'touches' from the outside.

Thus, the plasma dynamic systems are almost always full of self-organised movements – 'noise' in the entire range of the scale from the dimension of the system (L) to the Debye radius (r_D)

$$r_D \leqslant \lambda_{\text{ef}} \leqslant L.$$

Here λ_{ef} is the spatial scale of the 'oscillations'. In this case, the structures of the oscillations are usually very fanciful, non-linear and sensitive to the variation of the external conditions. The concept of 'self-organisation' describes highly adequately the most important property of the plasma systems. Therefore, it is always difficult 'adjust' for the plasma 'temper' and develop a device carrying out the required operations.

The term 'oscillations', which is usually employed despite the fact that it is inadequate in many cases, will also be used in the present book. A large amount of work has been carried out from the initial stage of theoretical investigations of plasma systems in which only the linear perturbations of the systems were investigated. Special attention was given to finding perturbations increasing with time, i.e. instability of plasma configurations.

[2]In the last couple of decades that have been attempts to develop a general science of self-organisation, referred to as synergetics.

The range of the investigated perturbations has been constantly increasing. Special attention has been paid to non-linear perturbations and this was accompanied by the appearance of the term 'collective processes'. This term can also be accepted only with difficulties because in any solid medium a large part of the processes is of the collective nature and is not associated with specific features of plasma and its perturbations. However, we shall leave this terminological discussion and transfer to the crux of the matter.

The oscillations (including here also the self-organisation processes) can be divided slightly conventionally into three groups.

The first group includes the oscillations with the characteristic scale

$$\Lambda_1 \sim (10^{-2} \div 1)L ,$$

where L is the controlling dimension of the plasma formation as a whole. These oscillations will be referred to as large-scale oscillations or, in the appropriate cases, low-frequency oscillations.

The third group includes oscillations with scales

$$\Lambda_3 \sim (1 \div 10)\max(r_D, \rho_e) .$$

Here ρ_e is the characteristic Larmor radius of the electrons, and r_D is the Debye radius. The oscillations of the group will be referred to as fine-scale or high-frequency oscillations. The frequencies of these oscillations are usually in the range $\geq$ 0.1–1000 GHz. They are generated as a result of different 'beam instabilities' or transformation of waves (see below).

Finally, the second group of the oscillations whose scales are in the range

$$\Lambda_3 < \Lambda_2 < \Lambda_1 ,$$

will be referred to as the oscillations of medium scales. These oscillations are usually not very important but may represent a linking member between the large-scale and fine-scale oscillations.

Only the large-scale oscillations will be now be discussed. This is explained by the fact that the oscillations of the second group are usually highly individual, and examples of oscillations of the third group were discussed in Chapter 3 and 6.

In most cases, the large-scale oscillations should be calculated together with the 'main' configurations because they change rapidly to the non-linear level, transforming into self-organised dynamic structures which are sensitive to the boundary conditions and the nature of the system as a whole. A suitable example are the results of numerical calculations of a hybrid one-dimensional model of a stationary plasma thruster discussed in Chapter 6.7. However, regardless of the high degree of individuality of many large-scale instabilities, they include

a number of sufficiently universal instabilities and some of them will be mentioned here.

These 'universal' instabilities can be divided into two groups: 'non-specific' and 'specific'. The first group includes instabilities associated slightly with the specific features of plasma and manifested in other media. One of these instabilities is, for example, thermal (Joule) instability which forms in any medium – solid, liquid, gaseous, if its conductivity increases with temperature.

The instabilities of this type will be the first instabilities to be investigated. This will be followed by examination of instabilities specific to plasma.

8.1. Examples of identical hydrodynamic and plasma instabilities

8.1.1. Superheated ('Joule') instabilities

The simplest systems in which these instabilities can be found have the form of some conducting column with the electrodes placed at the opposite ends of the column (Fig. 8.1.1). Suitable examples include the glow and arc discharges, semiconductor bars, etc. For the instability in which we are interested to develop, the conductivity of the medium should depend quite strongly on temperature.

The equation for the temperature is used in the simplest form

$$\rho c_V \frac{\partial T}{\partial t} = \sigma(T)E^2 + \mathrm{div}\left(\varkappa \nabla T\right). \qquad (8.1.1)$$

For the arc discharge it is the Elenbass–Heller non-stationary equation, which was already discussed in section 6.10.2.

It will be assumed that in the stationary state, the Joule energy generation ($\sigma_0 E^2$) is balanced by the removal of heat ($\mathrm{div}(\varkappa \nabla T_0)$)

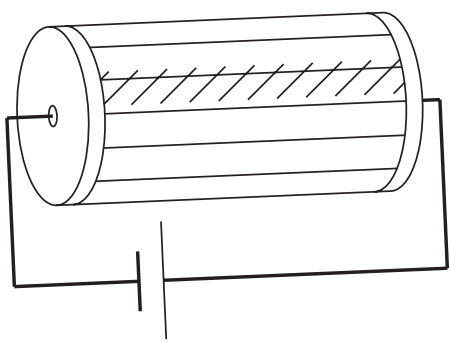


Fig. 8.1.1. Example of a superheated instability (pinch instability of a powerful glow discharge.

$$\sigma(T_0)E^2 + \operatorname{div}(\varkappa_0 \nabla T_0) = 0. \qquad (8.1.2)$$

However, let us now assume that the stationary state is disrupted for some reason. We shall investigate what takes place in the simplest case when the strength of the electrical field E does not change and heat conductivity $\varkappa$ is constant throughout the entire volume. For this purpose, the equation (8.1.1) is made linear in the vicinity of equilibrium temperature T_0:

$$T = T_0 + T_1, \quad |T_1| << T_0$$

Consequently, taking into account (8.1.2) at ρ = const, we obtain

$$\rho c_V \frac{\partial T_1}{\partial t} = \sigma_1 T_1 E^2 + \varkappa \Delta T_1, \; \sigma(t) = \sigma_0 + \sigma_1 T_1, \qquad (8.1.3)$$

Again, to simplify considerations, it is assumed that the boundary conditions can be used to find the solution in the form of flat waves

$$T_1 = A \exp\{qt + i\mathbf{kx}\}. \qquad (8.1.4)$$

Consequently, we obtain

$$q = \frac{1}{\rho c_V}\left(\sigma_1 E^2 - \varkappa k^2\right). \qquad (8.1.5)$$

This shows that there are two conditions.

Stable temperature distribution. In this case, the temperature modulation formed for some reason attenuates with a decrement:

$$\gamma \equiv -q = \frac{1}{\rho c_V}\left(\varkappa k^2 - \sigma_1 E^2\right) > 0, \qquad (8.1.6a)$$

$$\text{i.e. } T_1 = A \exp\{-\gamma t + i\mathbf{kx}\} \qquad (8.1.6b)$$

This regime is realised if the heat conductivity has a significant role and the conductivity depends only slightly on the temperature or the strength of the electrical field is low.

The unstable temperature distribution ($p > 0$) is observed in the conditions opposite to those in the previous case.

Of course, the exponential increase of the perturbation of temperature takes place only in the initial stage. Subsequently, the rate of this increase decreases.

A suitable example of the manifestation of this instability is the 'lacing' of the distributed glow discharge or the very fact of burning of the arc in the form of a relatively thin plasma column between the extended electrodes. It is well known that the dependence of conductivity on temperature can be determined not only by the increase of electron mobility but also by the increase of the degree of ionisation.

8.1.2. Convective instabilities

One of the universal forms of movement of the solid media is the convection determined by the presence of the temperature gradient. Convection has a significant role in the atmosphere, in water volumes and in plasma dynamic systems in which the temperature gradients are usually especially steep. At this point, we examine initially the stability criterion of the layer of the non-uniformly heated conventional gas in the field of gravity and subsequently, the conditions of plasma stability with $\beta \ll 1$ in magnetic fields of open traps and traps with closed magnetic lines of force.

Gas convection conditions (Fig. 8.1.2)
We define a small volume V_1 at a height z_1 and denote the pressure and density in the volume by p_1 and ρ_1 and it is assumed that the volume slowly rises upwards to the distance δz and acquires the volume V_2. The pressure and density of the surrounding gas at this height are denoted by p_2 and ρ_2. The gas density in the investigated volume is now determined by the adiabate equation

$$\rho_{11}^{\gamma} = \frac{p_2}{p_1}\rho_1^{\gamma} \tag{8.1.7}$$

Here it is taken into account that the pressure in the volume V_2 is equal

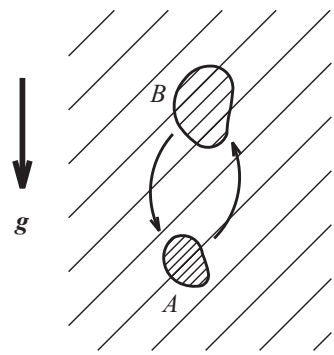


Fig. 8.1.2. Derivation of the stability condition in relation to the convection of a non-uniformly heated gas.

to the external pressure. If

$$\rho_{11} > \rho_2$$

then the gas is stable since the specific weight of the volume V_2 is greater than that of the surrounding gas. However, if

$$\rho_{11} < \rho_2$$

V_2 is lighter and 'rising' will continue.

Taking into account that

$$\frac{p}{\rho^{\gamma}} \sim e^{S} \tag{8.1.8}$$

It is taken into account that there is no convection in the medium if the entropy increases with increasing height [13].

Convection of rare plasma in the MHD model with closed magnetic lines of force

This convection was already discussed in section 1.7 where it was investigated with the help of drift equations [195, 196]. Here, we describe convection by the hydrodynamic procedure ([197), B.B. Kadomtsev).

Figure 8.1.3 shows schematically the configuration with closed lines of force – for determinacy of the field of a ring with current. An arbitrary magnetic pipe with the rare plasma is defined ($\beta \ll 1$). Assuming that the specific volume of the magnetic pipes is known (1.1.37)

$$U = \int \frac{dl}{\boldsymbol{H}} \tag{8.1.9}$$

the pressure of plasma along the lines of force is constant because

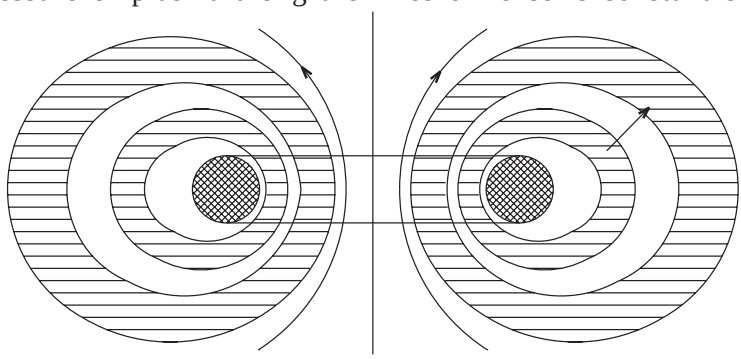


Fig. 8.1.3. Derivation of the stability condition of a plasma configuration with closed lines of force.

(section 2.4) the pressure in the entire plasma volume can be represented as the function of two arguments

$$p = p(U, \alpha) \tag{8.1.10}$$

Here α is some parameter, for example, the azimuth which fixes, together with U, the magnetic line of force. If the system is axisymmetric, then a single parameter U is sufficient for the coordination of the magnetic line of force. We consider this case only. The generalisation to the three-dimensional case is evident. It is assumed that U increases in the direction to the periphery of the trap. If it is now assumed that there has been rearrangement between the positions A and B of two magnetic pipes with the same magnetic fluxes but different U, then the mass in each pipe does not change because of freezing-in, and the pressure changes in accordance with the adiabatic equation

$$p_1 U_1^{\gamma} = p_{11} U_{11}^{\gamma} = \text{const}$$

However, if the changed pressure $p_1 \to p_{11}$ is lower than the pressure in the equilibrium configuration (P_{2B}) in the area of the new position of the pipe, the pipe will no longer expand and the configuration is stable.

This stability criterion is usually presented in the form (B.B. Kadomtsev [195])

$$\frac{dp(U)}{dU} > -\frac{\gamma p}{U} \tag{8.1.11}$$

This shows that to ensure stability the entropy of the configuration should increase with increasing U.

Example 1. The Z-pinch

As already mentioned, the resultant criterion (8.1.11) holds only for rare plasma. Therefore, this criterion is suitable only for the periphery of the stationary diffusion Z-pinch. In this region

$$U = \int \frac{dl}{H} = \frac{2\pi r}{2J/cr} = \left(\frac{\pi c}{J}\right) r^2 \equiv \alpha r^2 \tag{8.1.12}$$

Substituting this expression into (8.1.11) we obtain

$$p_0\big|_{r\to\infty} > A r^{-2\gamma}. \tag{8.1.13}$$

In particular, for the monatomic gas $\gamma = 5/3$ and the law of decrease at the stability boundary

$$p_0 \sim r^{-\frac{10}{3}} \tag{8.1.14}$$

Thus, even in the case of a relatively rapid decrease of pressure, the Z-pinch can be stable in relation to convection. However, in the case of a 'sharper' plasma–field boundary the Z-pinch is not stable.

If, using the energy method (paragraph 8.2.1) we solve the problem of the stability of the Z-lunch without restriction with respect to the plasma density, we obtain the stability criterion in the form (B.B. Kadomtsev)

$$\frac{d\ln p}{d\ln r} > -\frac{4\gamma}{2+\gamma\beta}. \tag{8.1.15}$$

If $\beta = 8\pi p/H^2 \to 0$, this criterion coincides with (8.1.14). If we now take into account the equilibrium equation of the Z-pinch (section 2.4.2), we obtain the following parametric equations for p and r at which the pinch is still stable with respect to the constriction [197]

$$p = p_0\left(\frac{\beta}{0.8+\beta}\right)^{5/2},\ r = a\frac{0.8+\beta}{\beta^{5/4}}. \tag{8.1.16}$$

Here $\beta \equiv 8\pi p/H^2$, p_0 is the plasma pressure in the centre of the pinch, α is some characteristic radius of the plasma pinch. It can easily be seen that at $\beta \to 0$ the radius r increases without bounds, and the pressure in the presence of equilibrium is

$$p \sim \frac{1}{r^2} \tag{8.1.17}$$

2. The dipole

The magnetic field of the Earth at short distances from the surface of the Earth is similar to the dipole field. The zone is characterised by the presence of radiation belts (these belts will be discussed in greater detail in Chapter 9). These plasma formations, although they are subjected to periodic perturbation under the effect of solar wind or as a result of overfilling with the trapped particles, can nevertheless be regarded as stable.

The stability of these belts was demonstrated efficiently in the 80s when a small comet crashed into Jupiter. Jupiter also has a magnetic field of the dipole type. Observations from the Earth showed that the plasma configurations of the belts and of the magnetosphere were not affected on the whole. The stability of the belts made the Japanese astrophysicist Hasegawa to propose a plasma trap in the form of a levitating superconducting ring. This trap, under the name 'Dipole'

was constructed in the USA (Fig. 10.5.9)

8.1.3. Rayleigh–Timofeev hydrodynamic resonance

Investigating the Langmuir waves in the Vlasov model, we encountered the kinematic resonance in the space of velocities:

$$\omega - \varkappa \mathbf{v} = 0. \tag{8.1.18}$$

As a result of this resonance, depending on the sign $f_0'(v_\phi)$, where $v_\phi = \omega/\varkappa$, the waves either attenuate or buildup. In this case, since the electronic component is assumed to be stationary on the whole, and v_ϕ is usually considerably greater than v_{Te}, the condition (8.1.20) is fulfilled for a small part of the distribution function $f_0(\mathbf{v})$ in the velocity space. A completely different situation may exist in the case in which the movement of plasma is non-uniform. In this case, the condition (8.1.20) can be satisfied by large spatial areas of the plasma with satisfactory accuracy.

The theory of this phenomenon for the flat flow of an incompressible liquid was constructed for the first time by Rayleigh in 1880 and in the last decades of the 20th A.V. Timofeev applied this theory to the non-uniform plasma flows, referring to this phenomenon as 'hydrodynamic resonance' [198].

Prior to reproducing formally the Rayleigh model, we describe initially its principle qualitatively. For this purpose, we consider two parallel sheets spaced at h (Fig. 8.1.4), with an ideal incompressible fluid flowing between them. If the velocity of the fluid in the volume is constant, there are no propagating waves in this case. However, if the non-perturbing flow velocity depends on the transverse coordinates

$$v_0 = v_0(y), \tag{8.1.19}$$

then volume waves can propagate in the fluid. If a small perturbation (for example, $\sim e^{-i\omega x}$) is introduced into this flow, there could be two scenarios in principle: either the perturbation attenuates or its intensity increases and reaches some non-linear level. In the second case, as shown by analysis, the perturbation concentrates in the vicinity of the ordinate y_* where the maximum $|\partial v_0/\partial y|$ is found, i.e. $\partial^2 v_0/\partial y^2 = 0$

In the reference system, where $v_0(y_*) = 0$, the frequency of the wave with increasing intensity is equal to

$$\omega \sim \nabla v_0(y_*) \tag{8.1.20}$$

and the phase velocity will be like always $\varkappa v_\phi(y_*) = \omega/\varkappa$. Therefore, if

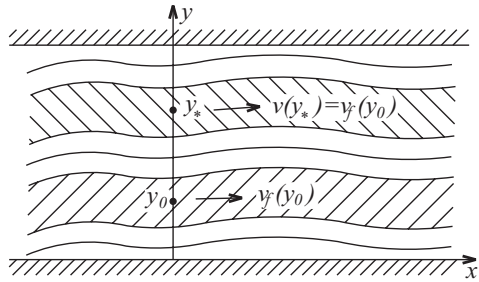


Fig. 8.1.4. Concept of hydrodynamic resonance.

the flow contains the point y_s, where the velocity $v_0(y_s) = v_\phi(y_*)$, then the wave at y_* will either oscillate or be extinguished. This is also the hydrodynamic resonance. Thus, two conditions are essential for the oscillations of the perturbations:

– the existence of non-attenuating waves,

– the presence of the resonance zone of the flow.

However, it should be mentioned that the problem of the resonance extinction of the wave has been studied less extensively.

The Rayleigh model will now be examined briefly.

The model is based on the flat flow of an incompressible ideal fluid between two parallel walls, and in the non-perturbed state the lines of flow are straight. However, the velocity depends on the transverse coordinate y, i.e.

$$v_{0x} = v_0(y) \tag{8.1.21}$$

It is necessary to determine the dependences $v_0(y)$ at which the flow is stable.

The initial equations are represented by two Euler equations

$$\operatorname{div}\mathbf{v} = 0, \quad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v} = -\frac{\nabla p}{\rho}$$

The perturbations are assumed to be small

$$\mathbf{v} = v_0(y) + v_1(t,x,y) + ..., \quad p = p_0 + p_1(t,x,v) \tag{8.1.22a}$$

Consequently

$$\operatorname{div}\mathbf{v}_1 = 0, \quad \frac{\partial \mathbf{v}_1}{\partial t} + (\mathbf{v}_0\nabla)\mathbf{v}_1 + (\mathbf{v}_1\nabla)\mathbf{v}_0 = -\frac{\nabla p_1}{\rho} \tag{8.1.22b}$$

The first equation is identicall satisfied by introducing the function of the flow ψ:

$$v_{1x} = \frac{\partial \psi}{\partial y}; \quad v_{1y} = -\frac{\partial \psi}{\partial x} \tag{8.1.23}$$

Substituting (8.1.23) into (8.1.22b) we obtain

$$\frac{\partial}{\partial t}\frac{\partial \psi}{\partial y} + v_0 \frac{\partial}{\partial x}\frac{\partial \psi}{\partial y} - v_0' \frac{\partial \psi}{\partial x} = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} \tag{8.1.24}$$

$$-\frac{\partial}{\partial t}\frac{\partial \psi}{\partial x} - v_0 \frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial x}\right) = -\frac{1}{\rho_0}\frac{\partial p_1}{\partial y}$$

Differentiating the first of these two equations with respect to y, and the second equation with respect to x and deducting one equation for the other, we obtain the required equation for ψ – the Rayleigh equation:

$$\left(\frac{\partial}{\partial t} + v_0 \frac{\partial}{\partial x}\right)\Delta\psi - v_0'' \frac{\partial \psi}{\partial x} = 0 \tag{8.1.25a}$$

The solution of this equation has the form

$$\psi = f(y)\exp\{-i\omega t + i\varkappa x\}, \tag{8.1.25b}$$

where $\varkappa$ is an arbitrary constant, ω is the quantity determined from the regularity conditions and boundary conditions on the solid walls ($y = 0$, $y = h$):

$$f(0) = f(h) = 0 \tag{8.1.26}$$

Substituting (8.1.25b) into (8.1.25a) gives the conventional equation of the second order

$$(\omega - v_0\varkappa)(f'' - \varkappa^2 f) + v_0''\varkappa f = 0. \tag{8.1.27}$$

It should be mentioned that the requirement for the regularity of the solution is important because the equation (8.1.27) contains a singularity at the coordinate $y = y_*$, determined by the kinematic resonance condition

$$v_0(y_*) = \frac{\omega}{\varkappa}. \tag{8.1.28}$$

To process the singularity associated with the resonance, it is necessary to introduce into (8.1.22a) low viscosity as in the analysis of the Langmuir waves it is convenient to introduce rare collisions.

The equations of the type

$$(1 - \alpha(y))Z''(y) + (a + \beta(x))Z(y) = 0 \tag{8.1.29}$$

is solved analytically for many $\alpha(x)$, $\beta(x)$. However, we are interested in the conditions on $v_0(y)$ of the general type at which the flow is stable, i.e. ω is real.

The following artificial measure will be used for this purpose.

Equation (8.1.27) is multiplied by ψ^* and from the resultant complex equation we subtract its completely conjugate expression. Consequently,

$$\frac{1}{2}\frac{d}{dy}\left(f\frac{df^*}{dy} - f^*\frac{df}{dy}\right) = \frac{i\gamma |f|^2 v_0''}{(\omega_1 - \varkappa v_0)^2 + \gamma^2}; \quad \omega = \omega_1 + i\gamma. \tag{8.1.30}$$

Integrating (8.1.30) from the wall ($y = 0$) to the wall ($y = h$), and taking into account (8.1.26), we obtain

$$0 = \gamma \int_0^h \frac{v_0''(y)|f|^2 dy}{(\omega_1 - \varkappa v_0(y))^2 + \gamma^2}. \tag{8.1.31}$$

This shows that if there is no inflection point, i.e. $v_0''(y)$ retains its sign everywhere, then

$$\gamma = (\mathrm{Im}\omega) = 0, \tag{8.1.32a}$$

and the flow is stable. This is also the Rayleigh theorem. However, if there is an inflection point, then $v_0''(y)$ is a quantity whose sign changes, the integral can convert to 0 and when

$$\gamma = (\mathrm{Im}\omega) \neq 0, \tag{8.1.32b}$$

i.e. the flow can be unstable. Thus, the essential although insufficient condition of instability of the flow with $v_{0x}(y)$ is the presence of the inflection point

$$v_0''(y_*) = 0 \tag{8.1.32c}$$

The importance of the hydrodynamic resonance will be stressed.

These resonances have a strong effect on the stability of the non-uniformly moving flow, leading in particular to the generation of vortices. A suitable example of such a vortex, caused by the inflection of the dependence $v_0(y)$, is the gigantic large red spot of the Jupiter (see section 9.1). In section 8.3 it will be shown that the drift flows in plasma are the formal analogue of the investigated hydrodynamic flows. Taking into account the analogy between the kinetic theory of the Langmuir waves (section 4.4) and the Rayleigh equation, it may be asserted that the hydrodynamic resonances can oscillate, or attenuate the oscillations, depending on the specific conditions and, evidently, the

initial stabilising effect of the non-uniform flow on the oscillations of plasma was already detected in the 60s in the experiments carried out in open traps OGRA-1 and Alice where the plasma was artificially brought into non-uniform rotation. Similarly, this mechanism of suppression of anomalous transfers was also detected in tokamaks.

The concept of the hydrodynamic resonance makes it possible to determine the most general relationships determining the oscillatory properties and the stability of the non-uniform flow of arbitrary 'solid' media.

8.1.4. Transformation of waves

Previously, carrying out linearisation, we rejected non-linear terms, assuming them to be small and of a higher order. However, this can not be done all the time. The non-linear terms, although they are small, may cause oscillations to high amplitudes of both the wave of the same type (the simple Riemann waves often evolve into shock waves as a result of this mechanism) and waves of other types. Later, we examine the 'merger' of two longitudinal linear Langmuir waves into a transverse electromagnetic wave.

The capacity of the small non-linear terms to have a strong effect on the wave processes is based on the resonant phenomenon. We shall describe this phenomenon.

In the simplest case of a point oscillator, the behaviour of the latter under the effect of the harmonic force is described by the equation

$$\ddot{x} + \omega_0^2 x = A\sin\omega t \tag{8.1.44a}$$

If $\omega \neq \omega_0$, some time after the start of the effect the oscillations reach the stationary level

$$x = \frac{A\sin\omega t}{D(\omega,\omega_0)};\ D \equiv \omega_0^2 - \omega^2 \tag{8.1.33b}$$

This shows that the amplitude of steady oscillations continuously increases when ω approaches ω_0. However, if $\omega = \omega_0$, the steady oscillations no longer exist because the amplitude now continuously increases:

$$x = -\frac{At}{2\omega_0}\cos\omega_0 t \tag{8.1.33c}$$

The same situation is also observed in the case in which the solid medium is subjected to the effect of some 'running perturbation', for

example, the air is perturbed by a flying aircraft.

For formal simplicity, the perturbation will be assumed to be harmonic

$$q = A\sin(\omega t - \varkappa \mathbf{x}) = A_1 \exp\{-i(\omega t - \varkappa \mathbf{x})\} + \text{c.t.} \tag{8.1.34}$$

Here $A_1 = A/2i$, and c.t. is the term complexly conjugate with the previous term. It will be shown how the situation with the resonances looks in multidimensional systems using the example of the waves in the plasma in which we are interested, with the equation (1.5.6c) used as the starting point

$$\Delta\mathbf{E} - \nabla div\mathbf{E} = \frac{4\pi}{c}\mathbf{j} + \frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} \tag{8.1.35a}$$

Here **j** is the extraneous flow-disturber. Using **E** in the form (8.1.34)

$$\mathbf{E} = \mathbf{E}_0 \exp\{-i\omega t + i\varkappa\,\mathbf{x}\},$$

we obtain the system of algebraic equations generalising the system (1.5.9)

$$\varkappa^2 E_k - \varkappa_k(\varkappa_\ell E_\ell) - \frac{\omega^2}{c^2}\epsilon_{k\ell}E_\ell = \frac{4\pi}{c}j_k. \tag{8.1.35b}$$

The solution of this system has the form

$$E_k = \frac{M_k(\mathbf{j}, \varkappa, \omega)}{D(\varkappa, \omega)}. \tag{8.1.36}$$

Here M_k is the 'Kramer' determinant, and $D(\varkappa, \omega)$ is the determinant of the left-hand part of the system (8.1.35)

$$D(\varkappa,\omega) = \left|\varkappa^2\delta_{k\ell} - \varkappa_k\varkappa_\ell - \frac{\omega^2}{c^2}\epsilon_{k\ell}\right| = a\left(\frac{\omega^2}{\varkappa^2} - c_1^2\right)\ldots\left(\frac{\omega^2}{\varkappa^2} - c_N^2\right), \tag{8.1.37}$$

where $c_1,\ldots,c_N$ is the velocity of intrinsic waves in the medium – the roots of the equation $D(\omega, \varkappa) = 0$.

Having the equations (8.1.36) and (8.1.37), we can now almost completely repeat all the considerations made in relation to the equations (8.1.33). In particular, if the phase velocity of the perturbation $v_\phi = \omega/\varkappa$ approaches the velocity of the intrinsic wave of the medium, the amplitude increases, and at $v_\phi = c_\kappa$ there is no stationary regime in the linear approximation with respect to **j**.

Here, we are interested in the simplest case in which the resonance factor in equation (8.1.35) is the quadratic (in relation to the linear term) term:

$$\mathbf{j}_{(2)} = e(n_{1\alpha}\mathbf{v}_{1\beta} + n_{1\beta}\mathbf{v}_{1\alpha}) \tag{8.1.38}$$

Here α corresponds to (ω_α, $\varkappa_\alpha$), and β corresponds to respectively (ω_β, $\varkappa_\beta$), and the index '1' indicates that these are the values of the linear approximation, i.e.

$$\begin{aligned}&\left(n_{1\alpha}, \mathbf{v}_{1\alpha}\right) \sim \exp\left\{-i\omega_\alpha t + i\varkappa_\alpha \mathbf{x}\right\} + \text{c.t.}\\ &\left(n_{1\beta}, \mathbf{v}_{1\beta}\right) \sim \exp\left\{-i\omega_\beta t + i\varkappa_\beta \mathbf{x}\right\} + \text{c.t.}\end{aligned} \tag{8.1.39a}$$

In this case

$$\mathbf{j}_{(2)} \sim \exp\left\{-i\left((\omega_\alpha + \omega_\beta)t - (\varkappa_\alpha + \varkappa_\beta)\mathbf{x}\right)\right\} + \text{analogues} = \exp\left\{-i(\omega_\gamma t + i\varkappa_\gamma \mathbf{x})\right\} + \text{analogues.}$$

The analogues referred to

$$\begin{aligned}&\omega_\gamma = \omega_\alpha + \omega_\beta;\ -\omega_\alpha + \omega_\beta;\ \omega_\alpha - \omega_\beta;\ -\omega_\alpha - \omega_\beta,\\ &\varkappa_\gamma = \varkappa_\alpha + \varkappa_\beta;\ -\varkappa_\alpha + \varkappa_\beta;\ \varkappa_\alpha - \varkappa_\beta;\ -\varkappa_\alpha - \varkappa_\beta.\end{aligned} \tag{8.1.39b}$$

Evidently, the resonance and, consequently, pumping of the energy from the linear waves α and β to the wave γ will take place in the case in which the determinant at ω_γ, $\varkappa_\gamma$ is

$$D(\omega_\gamma,\ \varkappa_\gamma) = 0 \tag{8.1.39a}$$

In addition to the processes of merger of two waves into a single wave, the inverse process of 'decay' of a single wave into two waves ($\gamma \to \alpha + \beta$) can also take place. This is shown schematically in Fig. 8.1.5. Evidently, the processes with participation of more than three waves can take place [199, 200].

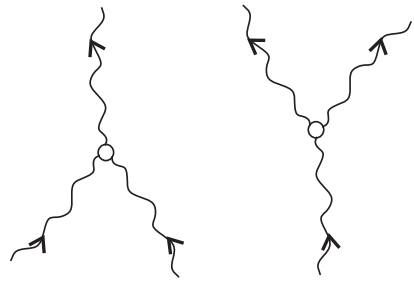
We investigate a specific example of the merger of two longitudinal (Langmuir) waves into a transverse electromagnetic wave. This is a highly universal process in different types of plasma. In particular, this process is of considerable interest for the diagnostics of Langmuir oscillations by the contactless method using external antennae. This is explained by the fact that there are no longitudinal waves emitted in this case. However, the merger of two longitudinal waves results in the formation of a transverse electromagnetic wave which leaves the plasma volume and carries information on the Langmuir oscillations.

In the hydrodynamic approximation, the wave frequencies are:

$$\omega^2 = \omega_0^2 + c_T^2 \varkappa^2.$$

Therefore, the resonance conditions have the form

$$\varkappa_\alpha + \varkappa_3 = \varkappa_\gamma; \tag{8.1.40a}$$



a b

Fig. 8.1.5. Merger of two waves into a single one (a) and the reversed process – decay of the wave (b).

$$\sqrt{\omega_0^2 + c_T^2 \varkappa_\alpha^2} + \sqrt{\omega_0^2 + c_T^2 \varkappa_\beta^2} = \omega_\gamma. \tag{8.1.40b}$$

In equation (8.1.40b) the value ω_γ can be excluded, using the relationship $\omega = c|\varkappa_\alpha + \varkappa_\beta|$ for the electromagnetic waves:

$$\sqrt{\omega_0^2 + c_T^2 \varkappa_\alpha^2} + \sqrt{\omega_0^2 + c_T^2 \varkappa_\beta^2} = c\left|\varkappa_1 + \varkappa_2\right|. \tag{8.1.40c}$$

Thus, we have obtained one equation for six unknown ($\varkappa_\alpha$, $\varkappa_\beta$), assuming that ω_0 and c_T are given. It should be added here that in most cases the thermal addition in (8.1.40) is small in comparison with the Langmuir frequency and, therefore, equation (8.1.40c) can be presented in the following form

$$\omega_\gamma \approx 2\omega_0; \left|\varkappa_\gamma\right| = \frac{2\omega_0}{c}. \tag{8.1.41}$$

However, this equation does not yet determine the transverse wave for the general case. For this purpose, the following transversality condition must be satisfied

$$(\mathbf{E}_\alpha + \mathbf{E}_\beta)(\varkappa_\alpha + \varkappa_\beta) = 0. \tag{8.1.42}$$

Taking into account the longitudinal form of the linear waves, we can write

$$\mathbf{E}_\alpha = \beta_1 \varkappa_\alpha,\ \mathbf{E}_\beta = \beta 2 \varkappa_\beta, \tag{8.1.43}$$

Here β_1 and β_2 are the scalars proportional to the amplitude of the appropriate waves. They should be determined from the condition (8.1.42). Substituting (8.2.1) into (8.1.42), we obtain equations for β_1 and β_2:

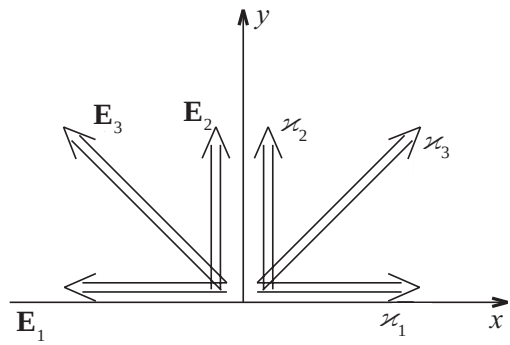


Fig. 8.1.6. Merger of Langmuir waves and the formation of a transverse wave.

$$\beta_1(\varkappa_\alpha)^2 + \beta_2 (\varkappa_\beta)^2 + \beta_1\varkappa_\alpha\varkappa_\beta + \beta_2\varkappa_\alpha\varkappa_\beta = 0 \qquad (8.1.44)$$

In the simplest case when $\varkappa_\alpha$ and $\varkappa_\beta$ are perpendicular and equal in the modulus, equation (8.1.44) is reduced to a simple relationship: $\beta_1 = -\beta_2$.

This is indicated by the scheme of the transverse wave, shown in Fig. 8.1.6.

The merger conditions (8.1.40) acquire a specially transparent meaning if they are multiplied by the Planck constant $\hbar$. It can be seen that they are equivalent to the laws of conservation

$$\varepsilon_1 + \varepsilon_2 = \varepsilon_3, \ \mathbf{p}_1 + \mathbf{p} = \mathbf{p}_3. \qquad (8.1.45)$$

Here $\varepsilon = \omega\hbar$, $\mathbf{p} = \varkappa\hbar$, are respectively the energy and the moment of the quasi-particle – plasmon.

Quantum analogy is used widely in the theory of plasma waves and plasma turbulence.

It should be mentioned that the conditions (8.1.45), which are often referred to as the condition of merger or decay of waves, are not fulfilled for many types of plasma waves.

8.2. Examples of specific MHD perturbations of plasma systems

Previously, it was mentioned several times that there is a huge number of plasma instabilities, especially specific instabilities. Here, we investigate a number of specific perturbations of plasma systems.

They are:

– the criterion of stability as a whole of static plasma MHD configurations, based on the ‘energy method’;

– tearing instabilities, accompanied by the rearrangement (reconnection) of magnetic lines of force of MHD configurations;

– the criterion of non-evolution of symmetric configurations with

the 'orthogonal' magnetic field, manifested in the ideal two-fluid hydrodynamics in the form of 'explosions' of the entire flow.

All these processes are investigated in the linear approximation.

8.2.1. Analysis of the stability of MHD configurations by the energy method

The principle of this method is very simple and identical with the analysis of the stability of equilibrium states of the material point in the potential field (Fig. 8.2.1). In particular, if the potential increases in possible directions, originating from the rest point, the position of the particle is stable. Conversely, if there are directions where the potential decreases, then the rest state of the particle is unstable. This method of analysis of stability does not require the solution of the dynamics equations and this often greatly simplifies the problem. To make sure that further considerations are easy to understand, we shall mention the simple fact from the linear algebra in the *n*-dimensional space.

If *n* linear forms are given

$$\lambda_i \equiv \sum_{(k)} a_{ik} x_k, \quad 1 \leqslant i, k \leqslant n, \tag{8.2.1}$$

then for the 'potential' $U(x_1 \ldots x_u)$ which generates these forms to exist

$$\lambda_i \equiv \sum_{(k)} a_{ik} x_k = \frac{\partial U}{\partial x_i}, \quad U = \sum_{(i,k)} \frac{1}{2}(x_i a_{ik} x_k) \tag{8.2.2a}$$

it is necessary and sufficient, as can easily be seen, that the matrix $||a_{ik}||$ is symmetric

$$a_{ik} = a_{ki}. \tag{8.2.2b}$$

This equality leads to the relationship of the self-conjugation' of the matrix $\hat{A}$

$$y_i(\alpha_{ik} x_k) = x_i(\alpha_{ik} y_{ik}) \tag{8.2.3a}$$

Or in the vector form

$$\mathbf{y}(\hat{A}\mathbf{x}) = (\hat{A}\mathbf{y})\mathbf{x}. \tag{8.2.3b}$$

the property of self-conjugation of the operator $\hat{A}$ is automatically transferred to the space of functions in the form

$$\int \phi(\hat{A}\psi)\, dx = \int (\hat{A}\phi)\psi\, dx. \tag{8.2.4}$$

If now we have the system of linear differential equations

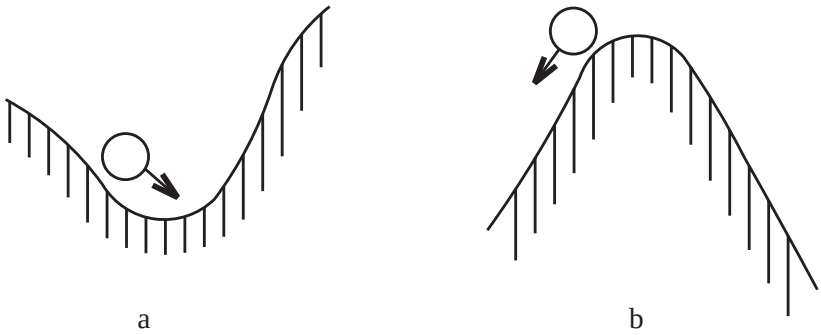


Fig. 8.2.1. Stable (a) and unstable (b) equilibrium positions of a sphere in a potential field.

$$\rho_0 \frac{\partial^2 \overline{\psi}}{\partial t^2} = \hat{L}\overline{\psi}, \quad \overline{\psi} = (\psi_i, ...\psi_2) \tag{8.2.5a}$$

and the operator $\hat{L}$ is self-adjoint, the system can be obtained from the Lagrangian

$$\mathcal{L} = \int d\mathbf{x} \frac{\rho_0}{2} \left(\frac{\partial \psi}{\partial t} \right)^2 - w[\psi], \tag{8.2.5b}$$

varying the action

$$S = \int \mathcal{L}\, dt.$$

Here $w[\psi]$ is the functional playing the role of the potential energy of the system, equal to

$$w[\psi] = \frac{1}{2} \int \psi \hat{L} \psi d\mathbf{x}. \tag{8.2.6}$$

Knowing the form of the functional $w[\psi]$ and confirming its positive determinacy for any $\psi(\mathbf{x})$, we obtain the sufficient condition of stability of the static configuration.

In principle, the conditions of positive determinacy w for all possible ψ are not required because in reality it is sufficient to restrict our considerations to only those ψ which satisfy the equation (8.2.5a). However, this requires solving this equation and in this case the energy approach loses its meaning.

Because of the problem of controlled thermonuclear synthesis, investigations were carried out into the stability of the systems consisting of plasma and a magnetic field, Fig. 8.2.2. Here V_i is the volume occupied by the plasma and the field, S is the boundary between the plasma and the field, V_e is the volume occupied only by the magnetic field, Γ is the

external boundary surface. The following conditions are usually specified for this boundary

$$v_n = 0; \quad \boldsymbol{E}_t = 0 \tag{8.2.7}$$

and at the boundary between the plasma and the field

$$H_n = 0. \tag{8.2.8}$$

Linearising the conventional ideal system of the MHD equations (2.3.6), we obtain

$$\begin{aligned}
&\frac{\partial \rho}{\partial t} + div(\rho_0 \mathbf{v}) = 0; \\
&\rho_0 \frac{\partial \mathbf{v}}{\partial t} + \nabla p = \frac{1}{4\pi}\left[\operatorname{rot}\mathbf{H}_0, \mathbf{H}\right] + \frac{1}{4\pi}\left[\operatorname{rot}\mathbf{H}, \mathbf{H}_0\right]; \\
&\frac{\partial p}{\partial t} + \mathbf{v}\nabla p_0 + \gamma p_0 \operatorname{div}\mathbf{v} = 0; \\
&\frac{\partial \mathbf{H}}{\partial t} = \operatorname{rot}\left[\mathbf{v}, \mathbf{H}_0\right]
\end{aligned} \tag{8.2.9}$$

Here, the perturbations of the parameters are presented without the indexes.

Without discussing the relatively cumbersome calculations, we present the equation for the displacement $\xi = \int \mathbf{v}\, dt$ following from (8.2.9)

$$\begin{aligned}
\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} &= \nabla\left\{\boldsymbol{\xi}\nabla p_0 + \gamma p_0 \operatorname{div}\boldsymbol{\xi}\right\} + \frac{1}{4\pi}\left[\operatorname{rot}\mathbf{H}_0, \operatorname{rot}\left[\boldsymbol{\xi}, \mathbf{H}_0\right]\right] + \\
&+ \frac{1}{4\pi}\left[\operatorname{rot}\operatorname{rot}\left[\boldsymbol{\xi}, \mathbf{H}_0\right], \mathbf{H}_0\right] \equiv \hat{L}\{\boldsymbol{\xi}\}.
\end{aligned} \tag{8.2.10}$$

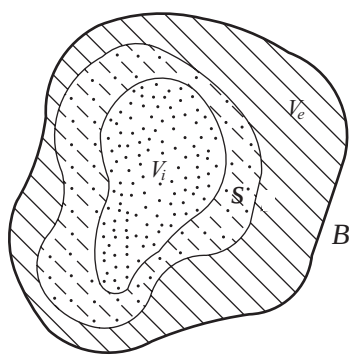


Fig. 8.2.2. General scheme of a magnetic trap for plasma: V_i – the volume occupied by the plasma; V_e – the volume occupied only by the magnetic field; S – the boundary between the plasma and the field; B – the external boundary surface.

Finally, the equations for the potential energy of the perturbati on [201]

$$w=\frac{1}{2}\int_{V_i}\left\{\gamma\, p_0\left(\operatorname{div}\boldsymbol{\xi}\right)^2+\frac{1}{4\pi}\left(\operatorname{rot}\left[\boldsymbol{\xi},\mathbf{H}\right]\right)^2+\left(\boldsymbol{\xi}\nabla\rho_0\right)\operatorname{div}\boldsymbol{\xi}-\frac{1}{4\pi}\left[\boldsymbol{\xi},\operatorname{rot}\mathbf{H}_0\right]\operatorname{rot}\left[\boldsymbol{\xi},\mathbf{H}_0\right]\right\}dx+$$

$$+\frac{1}{8\pi}\int_{V_e}\left(\operatorname{rot}\mathbf{A}\right)^2\,d\mathbf{x}-\frac{1}{2}\oint_{S_a}\left(\frac{\partial p_0}{\partial n}+\frac{1}{8\pi}\frac{\partial H_{0i}^2}{\partial n}\frac{\partial H_{0e}^2}{\partial n}\right)\xi_n^2\,ds. \qquad (8.2.10)$$

Here $\boldsymbol{\xi}$ is the displacement and **A** is the perturbation of the vector-potential.

To determine whether the given equilibrium configuration (p_0, ρ_0, H_0) is stable or unstable, it is necessary to verify the sign of w for possible $\boldsymbol{\xi}()$. Of course, this is not simple, and this type of analysis can be carried out only in a relatively small number of cases and usually for a specific class of $\boldsymbol{\xi}$ regarded as 'most dangerous' for some reasons. Nevertheless, this method was used to obtain a number of very important stability criteria. Here, we describe the method of deriving one such criterion.

Stability of a sharp (thin) plasma–field boundary

The plasma systems in which the plasma and the field are separated in space are encountered in a relatively large number of cases. These cases include anti-mirror traps and galateas, Z- and θ-pinches, and also dynamic systems in which the plasma flux without the magnetic field flows around the magnetic configuration. A suitable example of the latter is the magnetosphere of the Earth with the solar wind flowing around it (section 9.2).

In this case, the equation for the potential energy (8.2.11) is greatly simplified and has the following form:

$$w=\frac{1}{2}\int_{V_i}\gamma(\operatorname{div}\boldsymbol{\xi})^2\,p_0 d\mathbf{x}+\frac{1}{8\pi}\int_{V_e}(\operatorname{rot}\mathbf{A})^2 d\mathbf{x}+\frac{1}{16}\oint_{S_0}\frac{\partial H_0^2}{\partial n}\xi_n^2\,ds. \qquad (8.2.12)$$

This equation shows that if the strength of the magnetic field increases everywhere with increase of the distance from the interface, i.e.

$$\frac{\partial H_0^2}{\partial n}>0, \qquad (8.2.13)$$

then $w > 0$ and the configuration is stable at any displacement $\boldsymbol{\xi}$.

However, if there are areas of the surface S_0 where $\partial H_0^2 / \partial n < 0$, these areas will be characterised by plasma discharges, regardless of the fact that the first two terms in (8.2.12) are always positive. Actually, if we consider the perturbation not accompanied by plasma constriction,

and the selection is not contradicting at all, we have div $\xi = 0$ and the first term in (8.2.12) disappears.

It is not possible to remove completely the second term, but if we consider the perturbation which is extended along the line of force and and is relatively narrow, i.e. in the form of a 'tongue', passing between the lines of force (Fig. 8.2.3), then the second term can be made smaller than the negative surface integral. Consequently, the configuration with $\frac{\partial}{\partial n} H_0^2\Big|_{S_0} < 0$ are unstable if the plasma–field interface is relatively thin. However, if the plasma and the field are mixed, then as indicated in the previous section the configuration can be stable.

Since the magnetic field outside the plasma is vortex-free, then equation (8.2.13) shows

$$\left(\mathbf{n}^0 \nabla\right) H = \frac{H}{R}, \tag{8.2.14}$$

where R is the radius of curvature of the line of force. In this case, $R > 0$, if the centre of curvature is situated along the extension of the normal and $R < 0$ in the opposite case (Fig. 8.2.4).

Naturally, we would like to produce an ideal trap in which the plasma and the field are separated. However, this is not so simple.

Particular, if we want to ensure that the field increases everywhere when moving away from the interface, because of the stability criterion (8.2.13) we obtain an acute-angled boundary surface and galateas. This will be discussed in detail in section 10.5.

In conclusion, it is important to make one comment. Criterion (Fig. 8.2.13) was derived for the MHD statics configuration. This means, in particular, that the distribution function of the particles, falling on the interface, must be 'semi-isotropic'. Otherwise, if we have a plasma (ion) flux with the ion Mach number $M_i > 1$, the equilibrium interface is determined by the condition

$$2\rho_0 v_n^2 + p_0 = \frac{H^2}{8\pi} \tag{8.2.15}$$

However, this interface may be unstable even if condition (8.2.13) holds.

This is explained by the fact that the incident plasma flux may show the 'collapse' of the particle trajectories, forming 'wedges' which will generate dynamic pressure

$$p_{\text{dyn}} = \rho_1 v_n^2 + p_1 \gg \rho_0 v_n^2 + p_0.$$

Here ρ_0 and p_0 are the parameters in the flight to the boundary, and ρ_1 and p_1 – in the wedges (see section 3.8.3).

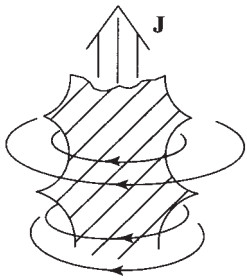


Fig. 8.2.3. Plasma 'tongue' penetrates into the magnetic field of the Z-pinch.

8.2.2. Reconnection of magnetic lines of force in plasma. Tearing instability

The rearrangement of the topology of the magnetic field in the plasma can take place only if the freezing-in is disrupted. However, this requires participation of ohmic resistance. This means that if powerful vibrational processes, leading to high 'anomalous' resistances do not operate (see section 9.3.4), and the dissipation is restricted only by the classic resistance, the size of the reconnection region will be of the order of the conventional skin layer.

In section 1.1, we investigated the magnetic fields with different morphology. For example, depending on the number of the direct conductors with current which took part in the formation of the field, different separatices are obtained. Applying different perturbations to the toroidal fields produced different split magnetic surfaces. The transition from one structure to another at the same effect is accompanied, as is now accepted, by reconnection (rejoining) of the magnetic lines of force [43]. One of the simplest and most suitable examples of reconnection of this type is the formation of a separatrix in the form of figure 8 during the approach of two direct currents with the same direction. This does not take place if the field in vacuum are

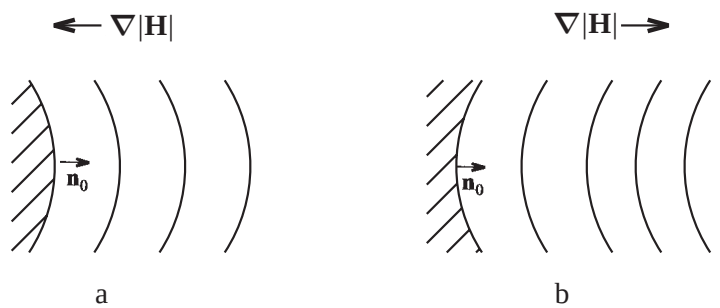


Fig. 8.2.4. Relationship of the direction of the gradient of the strength of the vacuum magnetic field with the curvature of lines of force.

considered. The process of reconnection of the magnetic lines of force is completely different in the case in which the fields are located in a highly conducting plasma.

The reconnection theory is cumbersome even in the linear MHD approximation because this process takes place in systems with steep gradients and a 'small coefficient at the old derivative', proportional to the low resistance of the plasma. Therefore, calculation of the reconnection in the actual systems are carried out for specific cases and numerically in most instances.

The simplest example of the system in which reconnection can take place is a neutral current layer compressed from two sides by a homogeneous magnetic field with the opposite direction (see section 2.4.2 and Fig. 8.2.5a).

As mentioned in section 2.4.2, this equilibrium configuration is described by the equation

$$p + \frac{H^2}{8\pi} = P = \text{const.}$$

A special feature of the current layer is a small thickness and high current density in the vicinity of the plane $x = 0$, where $H = 0$. In the MHD approximation, the ideal conductivity plasma is stable. However, if we take into account the finite value of conductivity, and in most cases conductivity is always finite, then in certain conditions the layer is divided into a chain of magnetic islands – current-carrying pinches (Fig. 8.2.5b). This is also the example of a specific instability of the 'tearing mode'.

The linear hydrodynamic theory of the tearing mode was proposed in 1963 by M. Rosenbluth et al. Subsequently, this study was continued by a number of researchers, both in hydrodynamics and in the kinetic approximation.

The characteristic structure of the formula for the incremental development of instability has the form

$$\gamma = \theta \tau_R^{-\alpha} \tau_A^{-(1-\alpha)}, \theta \sim 1; \tag{8.2.16a}$$

$$\tau_R \sim \frac{(\Delta L)^2}{\nu_m} \; \tau_A \sim \frac{1}{\varkappa c_A}; \; \nu_m \sim \frac{c^2}{4\pi\sigma}. \tag{8.2.16b}$$

Here $\varkappa$ is the wave number along the current layer (axis y), σ is conductivity, $(\Delta L)^{-1} = \frac{1}{H_x}\frac{\partial H_x}{\partial z}$ is the characteristic thickness of the current layer. The times τ_R and τ_A are referred to as the dissipative and Alfvén

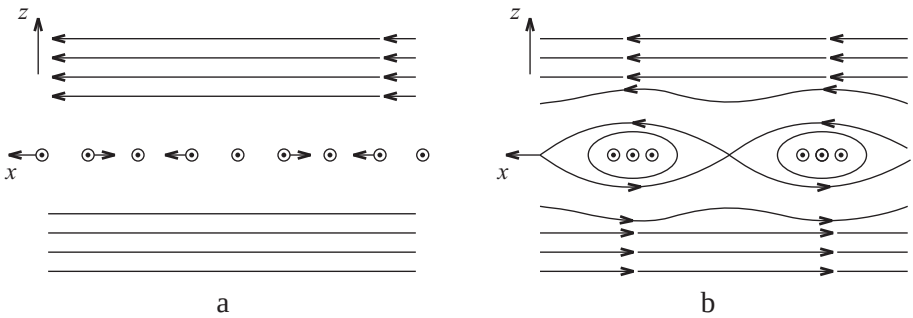


Fig. 8.2.5. Diagram of tearing instability: a) initial configuration; b) splitting of the neutral current layer; ⊙ currents **J**.

times. The value of α in different conditions changes in the range 1/2-3/5.

The resultant system of the current pinches – fibres usually continuous its evolution – the process of merger of thin fibres into large ones starts, but the situation in further stages depends on the specific features of the system.

Initial investigations into the magnetic reconnection in plasma were carried out by astrophysicists (Parker 1957, 1963; Danjee 1958, 1961; Sweet 1958; Petchek 1964). In Russia, these investigations were continued by Syrovatskii [203]. In the group headed by A.G. Frank at the Institute of Physics of the Academy of Sciences [204] experimental investigations of magnetic reconnection with special reference to astrophysics problems started at the end of the 60s. Later, the problem of reconnection, mostly in the form of splitting of the magnetic surfaces, was studied on tokamaks by Mirnov [175] on the basis of the theoretical studies by Morozov, Solov'ev and Gel'fand of reconnection in vacuum (section 1.1).

At the beginning of 2005, the monograph 'Magnetic reconnection. Magnetohydrodynamic Theory and Application' was published in the Russian language [43]. This book considered in detail the problem of magnetic reconnection. The experimental investigations of this problem, carried out at the Institute of Physics of the Academy of Sciences (IOFAN) are described in appendix C.

8.2.3. Hall non-evolution of flat flows of ideal plasma

The extensive erosion of the anode in coaxial high-current accelerators was discussed in section 7.6.5. These experiments stimulated numerical modelling of flows in such systems and the investigations revealed 'near-anode explosions', i.e. the parameters started to increase in the presence of a relatively strong Hall effect at some moment in a small area, and

the calculations were interrupted. This effect, which was correlated with the couplings on the anode, stimulated the analysis carried out by Brushlinskii and Morozov and described below. We shall reproduce this analysis.

Flat two-dimensional flows will be investigated in the Cartesian coordinates (x, y), assuming that the magnetic field is directed along the z axis. Thus, we have $v_z = 0$; $H_x = H_y = 0$; $\partial/\partial z = 0$. Also, it is taken into account that at $\sigma\to\infty$ the Ohm law with the Hall effect taken into account, can be written in the form

$$0 = \mathbf{E} + \frac{1}{c}[\mathbf{v},\mathbf{H}] - \frac{1}{enc}[\mathbf{j},\mathbf{H}]. \tag{8.2.17a}$$

Taking rot from this equation leads to:

$$\frac{\partial H}{\partial t} + \frac{\partial v_x H}{\partial x} + \frac{\partial v_y H}{\partial y} = \xi\left[\nabla\frac{1}{\rho}, \frac{\nabla H^2}{8\pi}\right]_z,$$

Here, it is taken into account that for the given assumptions

$$\frac{1}{c}[\mathbf{j},\mathbf{H}] = -\frac{\nabla H^2}{8\pi};\ \mathrm{rot}\left(\frac{1}{n}[\mathbf{j},\mathbf{H}]\right) = -\left[\nabla\frac{1}{n}, \nabla\frac{H^2}{8\pi}\right]. \tag{8.2.17b}$$

In this case $\xi = M/4\pi e$. This notation is introduced as the indicator of the Hall term. As a result, the required system of equations has the form:

$$\frac{\partial\rho}{\partial t} + \frac{\partial\rho u}{\partial x} + \frac{\partial\rho v}{\partial y} = 0;$$

$$\rho\frac{du}{dt} = -\frac{\partial}{\partial x}\left(p + \frac{H^2}{8\pi}\right); \tag{8.2.18a}$$

$$\rho\frac{dv}{dt} = -\frac{\partial}{\partial y}\left(p + \frac{H^2}{8\pi}\right); \tag{8.2.18b}$$

$$\frac{\partial H}{\partial t} + \frac{\partial Hu}{\partial x} + \frac{\partial Hv}{\partial y} + \xi\frac{H}{\rho^2}\left(\frac{\partial\rho}{\partial x}\frac{\partial H}{\partial y} - \frac{\partial\rho}{\partial y}\frac{\partial H}{\partial x}\right) = 0, \tag{8.2.18c}$$

where

$$u = v_x;\ v = v_y;\ H = H_z;\ \frac{d}{dt} = \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} \tag{8.2.18d}$$

A special feature of the two-dimensional flow across the magnetic field is the degenerated Hall effect (8.2.18d): in the above equations, the

terms determined by the Hall effect, contain only the first derivatives with respect to the coordinates, whereas in the general three-dimensional case they contain the second derivatives. It should also be mentioned that the whole terms are always non-linear with respect to the first derivatives in contrast to all other terms of the equations (8.2.18). Both these circumstances have a significant role in further considerations: degeneration is the reason for the simple procedure of obtaining the main result, and non-linearity is the reason for the non-evolution nature of the equations.

These equations are linearised to investigate the stability of the flow. The coefficients of the resultant system of linear equations are assumed to be constant because in this case we are concerned with small-scale perturbations which can be investigated locally, in the small vicinity of any point, 'freezing' at this point the values of the coefficient. In linearisation of the non-linear Hall terms in (8.2.18d), the derivatives of the non-perturbing solution play the role of coefficients, and they also should be regarded as constant.

If the small perturbations are denoted by the index '1' (ρ_1, u_1, v_1, H_1), then after linearisation of the equations (8.2.18), we obtain

$$\frac{d\rho_1}{dt}+\rho\left(\frac{\partial u_1}{\partial x}+\frac{\partial v_1}{\partial y}\right)=0;\quad \frac{du_1}{dt}+\frac{c^2}{\rho}\frac{\partial \rho_1}{\partial x}+\frac{H}{4\pi\rho}\frac{\partial H_1}{\partial x}=0;$$

$$\frac{dv_1}{dt}+\frac{c_T^2}{\rho}\frac{\partial \rho_1}{\partial y}+\frac{H}{4\pi\rho}\frac{\partial H_1}{\partial y}=0;\quad c_T^2=\gamma\rho/\rho; \qquad (8.2.19a)$$

$$\frac{dH_1}{dt}+H\left(\frac{\partial u_1}{\partial x}+\frac{\partial v_1}{\partial y}\right)+\xi\frac{H}{\rho^2}\left(\frac{\partial \rho}{\partial x}\frac{\partial H_1}{\partial y}-\frac{\partial \rho}{\partial y}\frac{\partial H_1}{\partial x}+\frac{\partial H}{\partial y}\frac{\partial \rho_1}{\partial x}-\frac{\partial H}{\partial x}\frac{\partial \rho_1}{\partial y}\right)=0$$

The solution of the system of equations is obtained in the form of flat waves $\exp\{i\omega t + i\varkappa_1 x + i\varkappa_2 y\}$. A special feature of the system (8.2.19) is the presence in its terms of only derivatives of the first order with respect to the parameters of the perturbations (ρ, v_1, u_1, H_1). Consequently, the frequency ω depends in a linear manner on the modulus of the wave number

$$\omega = W\varkappa. \qquad (8.2.19b)$$

The direction of the flat wave is determined by the unit vector $\boldsymbol{\varkappa}^0 = \boldsymbol{\varkappa}/|\boldsymbol{\varkappa}|$ and, consequently, we can write

$$v_x=\left[\boldsymbol{\varkappa}^0,\mathbf{v}\right],\left[\boldsymbol{\varkappa},\nabla\right]_z=\frac{\partial}{\partial\tau},$$

where τ is the direction along the front wave, orthogonal to $\varkappa^0$. Finally, if it is assumed that $W + v_{\varkappa} \equiv z$, the dispersion equation, linking ω, $\varkappa_1$, $\varkappa_2$ has the following form:

$$z\left(z^3 - \frac{\xi H}{\rho^2}\frac{\partial \rho}{\partial \tau}z^2 - c_m^2 z + \frac{\xi H}{\rho^2}\frac{\partial \mathrm{P}}{\partial \tau}\right) = 0, \tag{8.2.20}$$

where $c_m^2 = c_T^2 + \dfrac{H^2}{4\pi e}$; $\quad \mathrm{P} = p + \dfrac{H^2}{8\pi}$.

The condition of correctness of the Cauchy problem for the systems (8.2.18) and (8.2.19) is evidently the inequality

$$\operatorname{Im}\omega = k \operatorname{Im}\lambda = k \operatorname{Im} z \geq \text{const} \tag{8.2.21}$$

for any real $\varkappa_1^0$, $\varkappa_2^0$ and $\varkappa > 0$. At finite values of $\varkappa$, the solutions Z are always finite and, therefore, in practice it is necessary to fulfil the condition (8.2.21) at $\varkappa \to \infty$. The coefficients of equation (8.2.20) are real and do not depend on $\varkappa$. Therefore, if the equation has two complex conjugate roots, then for one of them ImZ < 0 and Imω$\to -\infty$ at $\varkappa \to \infty$, i.e. the condition (8.2.21) is not fulfilled. This shows that the correctness of the Cauchy problem is equivalent to the reality of all three roots of the equation (8.2.20)

It is well known that all the roots of $z^3 + pz^2 + qz + r = 0$ with the real coefficients are real only and only when the discriminant

$$D \equiv p^2q^2 + 18pqr - (4p^3r + 4q^3 + 27r^2) \geq 0 \tag{8.2.22a}$$

When applied to the equation this means that

$$D = 4gY^2 + (c_m^4 + 18c_m^2 g - 27g^2)Y + c_m^6 \tag{8.2.22b}$$

where

$$Y = \left[\left(\frac{\xi H}{\rho^2}\right)\left(\frac{\partial \rho}{\partial \tau}\right)\right]^2; \quad g = \left(\frac{\partial \mathrm{P}}{\partial \tau}\right) / \left(\frac{\partial \rho}{\partial \tau}\right). \tag{8.2.23a}$$

The elementary investigation of the quadratic trinomial (8.2.22) shows that the essential condition of stability, i.e. $D \geqslant 0$ at $\partial\rho/\partial\tau \to 0$ is the condition

$$-27g^2Y^2 + 4c_m^6 \geqslant 0. \tag{8.2.23b}$$

Actually, at $\partial\rho/\partial\tau \to 0$ the quantities

$$Y \sim \left(\frac{\partial\rho}{\partial\tau}\right)^2 \to 0;\ g \sim \left(\frac{\partial\rho}{\partial\tau}\right)^{-1} \to \infty$$

and, consequently, all the terms in the right-hand part (8.2.22b), with the exception of (8.2.23b), disappear. Substituting (8.2.23a) into (8.2.23b), we obtain the necessary stability condition

$$27\left(\frac{\partial P}{\partial\tau}\right)^2\left(\frac{\xi H}{\rho^2}\right)^2 < 4c_m^6. \qquad (8.2.24)$$

This condition is always fulfilled in the absence of the Hall effect and if $\nabla P \parallel \nabla\rho$. The last case is observed for example in equilibrium configurations.

However, if this condition is not fulfilled, dissipation-free configuration is of the non-evolution type, i.e. it explodes immediately after it appears [205]. These conclusions were initially encountered in the numerical calculations of two-dimensional flows taking into account the Hall effect and sufficiently high conductivity in the vicinity of the anode where the angle between ∇P and $\nabla\rho$ is especially large as a result of near-anode slip (see Chapter 7). Later, it was shown that these explosions can be produced in any area of the channel in which the non-parallelity of ∇P and $\nabla\rho$ is more evident.

The non-evolution character disappears if we investigate three-dimensional configurations or if $\sigma \neq \infty$. In these conditions we have 'normal' instabilities. They are reflected not only in the coaxials with solid electrodes but also in the magnetosphere of the Earth (section 9.2).

8.2.4. Drift flows across the magnetic field

Special attention was already given in Chapter 1 to the importance of the systems in which the movement of the particles is of the drift type, i.e. their Larmor radii are

$$\rho_e,\ \rho_I << L,$$

where L is the scale of the inhomogeneity of plasma configurations. In the hydrodynamic models, the drift flows are described by two-fluid equations in which the inertia terms are ignored not only in the equation of the dynamics of the electrons but also the equations of the dynamics of the ions. Consequently, assuming that the flow is barotropic, one obtains ($\mathbf{v} \equiv \mathbf{v}_i$, $\mathbf{u} \equiv \mathbf{v}_e$)

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{v} = 0; \tag{8.2.25a}$$

$$0 = -\frac{\nabla p_i}{en} - \nabla\phi + \frac{1}{c}[\mathbf{v}, \mathbf{H}]; \tag{8.2.25b}$$

$$\frac{\partial n}{\partial t} + \operatorname{div} n\mathbf{u} = 0; \tag{8.2.25c}$$

$$0 = -\frac{\nabla p_e}{en} + \nabla\phi - \frac{1}{c}[\mathbf{u}, \mathbf{H}]. \tag{8.2.25d}$$

With the accuracy to the derivatives $\partial n/\partial t$ in the continuity equations, the system is identical with the equilibrium equations of two-component plasma (section 3.4).

Introducing the thermalised potential

$$\phi_T = \phi - \frac{1}{e}\int \frac{dp_e}{n},$$

the dynamics equation (8.2.25b, d) can be presented in the form

$$\frac{\nabla(p_i + p_e)}{en} + \nabla\phi_T = \frac{1}{c}[\mathbf{v}, \mathbf{H}],\ \nabla\phi_T = \frac{1}{c}[\mathbf{u}, \mathbf{H}]. \tag{8.2.26}$$

The system (8.2.25) and also the system (8.2.26) together with the continuity equations are always characterised by a rich content and, therefore, we restrict our considerations to the case in which $V_z = u_z = 0$, and the magnetic field has only one z-component. Consequently, for the electronic component we obtain

$$u_x = -\frac{c}{H}\frac{\partial\phi_T}{\partial y},\ u_y = -\frac{c}{H}\frac{\partial\phi_T}{\partial x} \tag{8.2.27}$$

substituting these equations in to (8.2.25c) we obtain after simple transformations

$$\frac{\partial n}{\partial t} + c\frac{D\left(\phi_T, \dfrac{n}{H}\right)}{D(x, y)} = 0 \tag{8.2.28}$$

Similarly, the dynamics equation of the ionic component (8.2.26) can be transformed to the form

$$v_x = -\frac{c}{H}\left(\frac{\partial \phi_T}{\partial y} + \frac{1}{en}\frac{\partial}{\partial y}(p_i + p_e)\right);$$

$$v_y = \frac{c}{H}\left(\frac{\partial \phi_T}{\partial x} + \frac{1}{en}\frac{\partial}{\partial x}(p_i + p_e)\right). \qquad (8.2.29)$$

Substituting these equations into the continuity equation (8.2.25a) gives

$$\frac{\partial n}{\partial t} + c\frac{D\left(\phi_T, \frac{n}{H}\right)}{D(x,y)} + \frac{c}{e}\frac{D\left(p_i + p_e, \frac{1}{H}\right)}{D(x,y)} = 0. \qquad (8.2.30)$$

This shows that if the magnetic field and the pressure of the plasma $p_i + p_e$ are linked by the equilibrium condition (2.4.6)

$$p_i + p_e + \frac{H^2}{8\pi} = \text{const},$$

then

$$\frac{D\left(p_i + p_e, \frac{1}{H}\right)}{D(x,y)} = 0 \qquad (8.2.31)$$

In this case, the equations (8.2.28) and (8.2.29) coincide.

Assuming that the condition (8.2.31) is fulfilled), the equation of the drift flows (8.2.28) for the layer

$$\frac{\partial n}{\partial t} + c\frac{D\left(\phi_T, \frac{n}{H}\right)}{D(x,y)} = 0$$

will be examined in greater detail. If it is assumed that the fixators of the thermalised potential cause that the quantity $\phi_T(x)$ is independent of time, whereas n depends on t, x, y, then at $H = H(x)$ we can write

$$c\frac{D\left(\phi_T, \frac{n}{H}\right)}{D(x,y)} = \frac{c}{H}\frac{\partial \phi_T}{\partial x}\left(H\frac{\partial}{\partial y}\frac{n}{H}\right) = u_y\left(\frac{\partial n}{\partial y}\right) \qquad (8.2.32)$$

Substituting (8.2.32) into (8.2.28), we obtain the equation for the drift waves in the time-independent magnetic field

$$\frac{\partial n}{\partial t}+u_y\frac{\partial n}{\partial y}=0\,,\,u_y=u_y(x) \tag{8.2.33}$$

If the initial static configuration has no electrical field T_e = const

$$\phi_T=-\frac{kT_e}{e}\ln\frac{n}{n_0}$$

and the drift velocity is

$$u_y=-\frac{c}{H}\frac{kT_e}{e}\frac{1}{n_0}\frac{\partial n_0}{\partial x}. \tag{8.2.34}$$

Quantity u_y does not contain the imaginary addition and, consequently, does not change with time. To the investigations of the drift waves taking into account the inertia of the ions in the second approximation with respect to the perturbation amplitude we shall return in the sections 8.3 and 9.1.

8.3. Modelling equations of 'autonomous' plasma structures ('auto-structures')

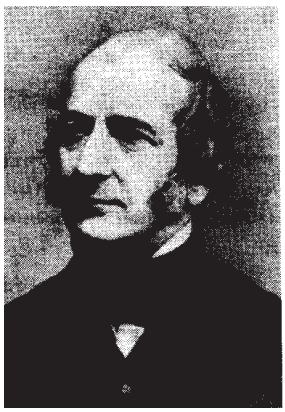
Previously, we discussed the linear waves. However, non-linear dynamic structures have often a significant and even controlling role. The dimensions of these waves can be similar or considerably smaller than the characteristic scales of the system *L*.

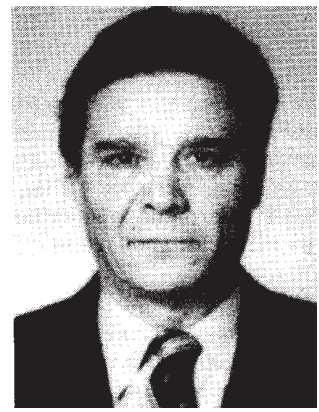
Suitable examples of the relevant structure here is the network of localised perturbations, formed on a thin layer of water flowing on asphalt, the development of a waist in a *Z*-pitch, cyclones and anticyclones in the atmosphere, including the large red spot on Jupiter which will be discussed in detail in Chapter 9.

The non-linear formations of this type, observed in the hydrodynamics, gas-dynamics and plasma[3] are adequately described by modelling equations which are not only far simpler than the total initial equations but also as simple analytical solutions in many cases.

Attention will be given to the modelling equations: Korteweg–de Vries (KdV), Charney–Obukhov (CO) (and its Hasegawa–Mima equivalent (H–M)) and a modelling system of two equations – Chaplygin–Trubnikov (C–T).

[3]This also includes less evident objects – for example, nerve pulses, propagating through neurons, etc.





G.S. Russell A.M. Obukhov B.A. Trubnikov

8.3.1. Korteweg–de Vries (KdV) equations. Solitons [206-208]

Examples of the structures described by the KdV equation, include the previously mentioned network of 'waves' on the surface of a thin water layer.

Making a number of assumptions, the KdV equation can be derived formally from the Euler equations, but it can also be derived on the basis of simple considerations. The second method will be used here. In particular, we restrict our considerations to the one-dimensional case, i.e. it is assumed that the required function u depends on two variables (t, x) and its velocity whose meaning will be described more accurately later. The equation for $u(t, x)$ should describe the lonely non-spreading formations, i.e. the solution of the following type

$$u = u(x - at). \tag{8.3.1}$$

should also be available amongst others. The solutions of this type are automatically realised in media without dispersion, i.e. when

$$\omega^2 = a^2 \varkappa^2, \tag{8.3.2a}$$

and all the harmonics have the same phase velocity. In this case, the equation for u has the following form

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0 \tag{8.3.2b}$$

However, if there is a slight dispersion but no dissipation, the dispersion equation for u should be presented in the form

$$\omega^2 = a^2 \varkappa^2 + \alpha \varkappa^4 + \ldots, \tag{8.3.3a}$$

i.e. the square of the frequency should be expanded with respect to for even powers of the wave number. Physically, this means that the behaviour of the perturbation should not depend on the direction of propagation of these perturbations.

Equation (8.3.3a) shows that

$$\omega = a\varkappa + b\varkappa^3 + \ldots, \; b \equiv \frac{\alpha}{2a}. \tag{8.3.3b}$$

This dispersion equation is described by the differential equation

$$\frac{\partial u}{\partial t} - a\frac{\partial u}{\partial x} + b\frac{\partial^3 u}{\partial x^3} = 0 \tag{8.3.4}$$

If the harmonics of these equations are used to produce a packet, the latter will spread. To prevent this process on taking place, we need some 'shoveling' factor. This factor is well known in hydrodynamics. It is associated with the fact that the left-hand part of the Euler equation includes the non-linear term $v\partial v/\partial x$[4]. Therefore, replacing the constant value a by u in (8.3.4) we obtain, as shown later, the equation with the required property

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \beta\frac{\partial^3 u}{\partial x^3} = 0 \tag{8.3.5}$$

This equation was derived in 1895 by Korteweg and de Vries from the Euler equations, with special reference to perturbations of the thin water layer, already mentioned previously. The Korteweg-de Vries (KDV) equations been derived for many cases, in particular, ion-sound, Langmuir and Afvén perturbations in the case of plasma and similarly for other media. It should be mentioned that if $\beta = -\overline{\beta}$, where $\overline{\beta} > 0$, then after substitution $u \to -u$, $x \to -x$, we obtain the equation (8.3,5) but with β substituted by $\overline{\beta}$. This substitution corresponds to the transition from the wave, propagating in the right-hand direction, are to the wave propagating to the left.

Stationary solutions of the KdV equation

The solution of (8.3.5) will be found in the form $u = u\,(x - at)$.

Taking into account that in this case

$$\frac{\partial u}{\partial x} = -\frac{1}{a}\frac{\partial u}{\partial t}$$

[4]Because of this term in the Euler equation, the harmonic sound wave with a sufficienly high amplitude transforms after a certain period of time to a chain of shock waves [13].

we can write the equation (8.3.5) in the form of the conventional differential equation

$$(u-a)\frac{\partial u}{\partial x}+\beta\frac{\partial^3 u}{\partial x^3}=0 \tag{8.3.6a}$$

After another integration, we obtain

$$\beta\frac{\partial^2 u}{\partial x^2}+\frac{u^2}{2}-au=C_1 \tag{8.3.6b}$$

Without restricting generality, simply adding a constant term to u, we can assume that $C_1 = 0$, and the equation (8.3.6b) can be presented in the following form

$$\beta\frac{\partial^2 u}{\partial x^2}=-\frac{\partial w}{\partial u} \tag{8.3.7a}$$

where

$$w=-\frac{au^2}{2}+\frac{u^3}{6} \tag{8.3.7b}$$

As regards its form, equation (8.3.7) is completely identical with the equation of movement of the material point with the mass β in a potential well w (u). The form of this 'well' is shown in Fig. 8.3.1. Evidently, the oscillations here are of the harmonic nature, if u is close to $2a$. However, with increase of the amplitude, the shape of the oscillations will differ more and more from sinusoidal. Finally, when the amplitude becomes so high that the point $u = 0$ is reached, the graph u (x) degenerates into a single hump – 'soliton' (from the word 'solitary wave'), Fig. 8.3.2. This is understandable because, starting moving in the direction $u > 0$ from the point close to $u = 0$, the particle will gather velocity for a long period of time and, subsequently, rolling down, will be reflected from the opposite side of the well and again travel infinitely to $x = 0$. The equation of the profile of the soliton has the form

$$u=\frac{u_0}{ch^2\left[\dfrac{x-at}{b}\right]} \tag{8.3.8}$$

Here b is the characteristic width of the soliton. The amplitude of the soliton u_0 is connected with its velocity and width by the relationships:

$$u_0 = 3_a,\ b^2u_0 = 12\beta. \tag{8.3.9}$$

Properties of the solitons

At first sight, it may appear that the solitons are exotic and not interesting objects. However, this is not the case. In particular, attention should be given to the autonomy of the solitons. They exist in principle for an infinitely long period of time on the background of a homogeneous medium. In this sense, they are identical with the vortices in the ideal conditions. This is clearly indicated by the stable network of perturbations in shallow water. In advanced optical communication fibres, a light soliton can travel the distance of the order of 5000 km without any serious loss of its characteristics.

The results of collisions of the solitons, detected in the process of numerical calculations, proved to be unexpected. If the velocities of the solitons greatly differ, one soliton passes through the other one. More accurately, this moment is characterised by the formation of a 'two-soliton' state which slightly later decays into two solitons completely identical with the initial ones. Therefore, externally, the situation appears as if the solitons have passed through each other. At a low relative velocity when the faster soliton (with the higher amplitude (8.3.9)) catches up with the slower soliton, the faster soliton 'pours over' part of the perturbation to the slower soliton at a small distance through the bridge connecting them. As a result, the slow soliton is accelerated and the fast soliton slows down (Fig. 8.3.3).

The decay of the initial arbitrary perturbation in the KdV model is also quite interesting. The perturbation transforms to a chain of solitons with decreasing amplitude and a small quasi-harmonic 'tail' (Fig. 8.3.4).

The solitons have been studied in a very large number of investigations. A very large amount of information on the solitons can be found in the books [206–208].

History of discovery and examination of solitons

This history is very unusual and is discussed in all books on solitons. One of the examples will now be mentioned.

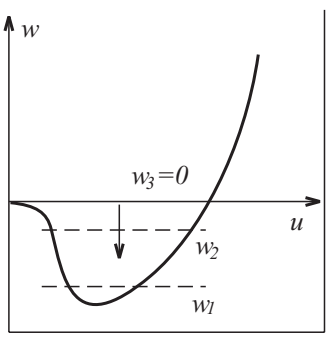


Fig. 8.3.1. Graph of the effective potential $w(u)$.

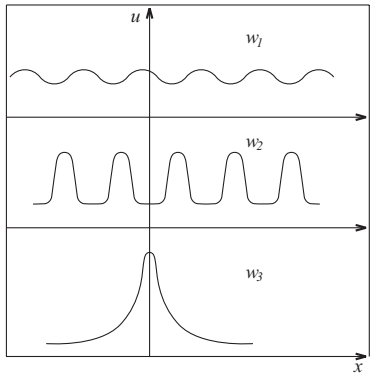


Fig. 8.3.2. Graph of the oscillations $u(x)$ at different effective energies.

This history started in August 1834 when John Scott Russell, 26, a graduate of the University of Glasgow (1808–1882) paid attention to this phenomenon. He was an outstanding person, as indicated by two factors. During his studies at Glasgow, he also attended lectures at St Andrews and Edinburgh universities. Subsequently, in the 1860s he supervised the construction of the largest ship of the 19th-century Great Eastern which was used for laying the first cable between Europe and America.

Let us return to the 1830s. After graduating from the University, J.S. Russell was given a task to investigate the transmission capacity of the Union channel included in the system of channels connecting the Eastern and Western shores of Scotland.

This is how the soliton on ‘shallow water’ was discovered.

‘‘I studied the movement of a barge which was rapidly pulled through a narrow channel by a pair of horses, when the barrage unexpectedly stopped but the mass of water brought into motion by the barge did not stop; instead, the water gathered around the nose of the vessel in the state of ‘mad’ movement, then unexpectedly left the barge behind, and rolled forward with a very high vecloty and acquiring the shape of a large single elevation, i.e. a circular smooth and distinctive water mound which continued its travel along the channel with no changes in the shape and no reduction in velocity. I followed the mound on the shore and when I caught up with it, it rolled forward with the velocity of approximately 8 or 9 miles/hour, retaining its initial elevation profile with the length of approximately 30 foot and height of 1-1.5 feet. Its height gradually decreased and after one or two miles of chasing it I lost it in the bends of the channel’’. Further: ‘‘this is the most beautiful and unusual phenomenon; the day when I observed it was the best day of my life’’. So this is it.

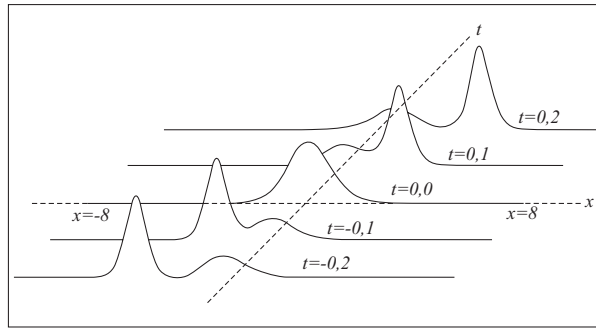


Fig. 8.3.3. Two-soliton interaction.

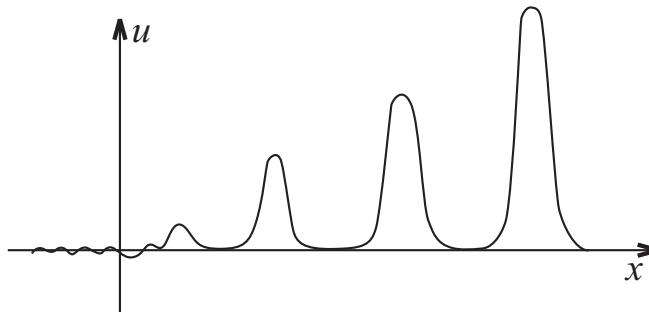


Fig. 8.3.4. Decay of an arbitrary initial perturbation of a relative high amplitude.

Russell believed deeply in the future of solitons and discussed it many times. However, the article by Korteweg–deVries appeared only in 1895. In this article the Korteweg–deVries equation was derived from the general hydrodynamics equations.

This equation was not regarded as interesting and remained known to only a small number of experts. This 'almost complete indifference' to the solitons and the KdV equation is continued up to the 60s of the 20th century. The first impetus in this case were the investigations by the numerical method of the relaxation of oscillations in anharmonic chains, carried out by Fermi, Pasta and Ulam, who detected an unusual - for that time – effects of 'non-relaxing' oscillations.

This paradox stimulated investigations of the properties of the KdV equation, and a study by Gardner, Green, Kruskal and Miura appeared in 1967 in which it was shown that there is an analytical solution of the Cauchy problem for this equation based on the close relationship between the investigated equation and quantum mechanics, and that the task of solving the equation can be reduced to solving the Gel'fand–Levitan–Marchenko linear integral equation for the inverse problem of quantum theory of scattering. The inverse problem here is

the restoration of the form of the potential in the Schrödinger equation on the basis of the scattering data at this potential by the 'probing particles'. This discovery was used not only to solve completely the KdV equations but also a number of other non-linear equations playing a significant role in the greatly differing areas of physics.

Subsequently, disregarding a study by Kruskal et al, it was possible to develop an efficient and highly general science of integrable non-linear equations in partial derivatives. Soviet physicists and mathematicians V.E. Zakharov, S.P. Novikov, L.D. Faddeev and A.B. Shabat played a significant role in this area.

As regards the problems of a plasma physics, it is very important to mention the development of the theory of strong Langmuir turbulence on the basis of the concept of Langmuir solitons (A.S. Kingsep), and also the discovery of the role of solitons in many other plasma processes.

8.3.2. Taking into account attenuation in the KdV equation

The propagation of solitons is naturally accompanied by their attenuation which was already noted by Russell. In different conditions, the attenuation mechanism can greatly differ: viscosity, classic resistance in dense plasma, 'anomalous' resistance in rare plasma[5], etc.

The natural and simplest explanation of dissipation is reduced to the addition to the classic KdV equation of the dissipative member of the viscosity type

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \beta\frac{\partial^3 u}{\partial x^2} = \nu\frac{\partial^2 u}{\partial x^2}. \tag{8.3.10}$$

Previously, we investigated only stationary solutions of the type:

$$u = f(x - at)$$

and obtained either only periodic solutions or a single soliton. If a stationary solution is found for equation (8.3.10), after single integration we obtain

$$\beta\frac{d^2u}{dx^2} - \nu\frac{du}{dx} = -\frac{\partial w}{\partial u} = au - \frac{u^2}{2}. \tag{8.3.11}$$

If we compare $x \to \tau$, $u \to \xi$, then this equation can be regarded as the equation of oscillations of the material point with the mass $c_0/2k_0^2$ in the potential well $w(\xi)$ in the presence of friction. Starting movement at $\tau \to -\infty$ at the point $\xi = 0$, the particle rolls into the well, gradually

[5] 'Anomalous' resistance is described in 8.4.4.

losing energy, and is arrested at $\tau \to +\infty$ at:

$$\xi_* \equiv u_* = 2a. \tag{8.3.12}$$

Figure 8.3.5a shows the movement of the effective particle in the well, and Figs. 8.3.5b and 8.3.5c show the changes of u in relation to x. It can be seen that we obviously obtain the structure of the shock wave. Attention should be given to the oscillatory nature of the transition from one state to another. In this case, depending on the sign of the dispersion correction in (8.3.8), the oscillations can precede the main reduction of u and can outstrip this decrease. Figure 8.3.6 shows the experimentally determined variation of density in the 'collisionless' ion–sound shock wave. The correspondence between Fig. 8.3.5 and Fig. 8.3.6 is clearly evident.

8.3.3. Chaplygin–Trubnikov instabilities [209]

Previously, we derived a non-linear modelling equation for the

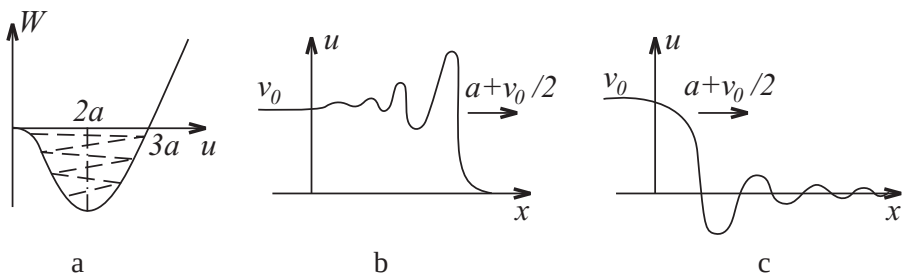


Fig. 8.3.5. Attenuation in a potential well (a) and shock waves in media with negative (b) and positive (c) dispersions.

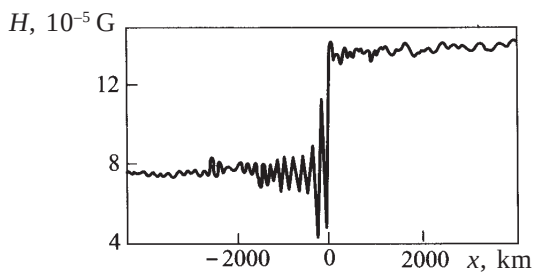


Fig. 8.3.6. Profile of the magnetic field at the front of an oblique ($\theta \approx 60°$) interplanetary shock wave with the Mach number $M = 2.5$ according to the measurements taken on the board of the ISP satellite, October 26, 1977. The ratio of the gas-kinetic pressure and the magnetic pressure $\beta \approx 3$. The thickness of the shock front $\Delta \sim 90$ km, which equals approximately $2c/\omega_{pi}$ [Russell C.T., Greenstadt E.W., Report of Inst. of Geophys. and Planet Phys., 1978, No. 1847).

perturbations in a slightly dispersing medium, based on the dispersion equation for the linear perturbations of the hyperbolic type

$$\omega^2 = c_0^2 \varkappa^2 + o(\varkappa^4).$$

Trubnikov and Zhdanov constructed and investigated a modelling nonlinear system of equations on the basis of the second dispersion equation for the linear perturbations of the elliptical type

$$\omega^2 + c_0^2 \varkappa^2 = o(\varkappa^4). \tag{8.3.13}$$

If the KdV equation, corresponding to (8.3.3), describes not only the dynamic but also stationary configurations, the equations, corresponding to (8.3.13) describe only the dynamic configurations. The model used by these authors was a system of two[6] hydrodynamic equations with the negative pressure

$$\frac{d\rho_*}{dt} = -\rho_* \operatorname{div} \mathbf{v}\,; \frac{d\mathbf{v}}{dt} = c_0^2 q \nabla \rho_*^{\frac{1}{q}}. \tag{8.3.14}$$

Here * at ρ is used to stress that in the specific investigated instabilities, the role of density can be played by greatly differing parameters.

The system of equations of the type (8.3.14) was derived for the first time by Chaplygin in 1896 in investigations of the dynamics of the ideal gas with the anomalous dependence of pressure on density

$$p = p_0 \left(\frac{\rho_0}{\rho} \right) \sim \frac{1}{\rho}. \tag{8.3.15}$$

Evidently, this system corresponds to the system (8.3.14) at $q = -1/2$, $c_0^2 = p_0/\rho_0$.

Trubnikov and Zhdanov showed in their studies that a large number of instabilities are reduced to the system (8.3.14) under specific assumptions. Some of them are presented in Table 8.1.

An important special feature of the system of Chaplygin–Trubnikov equations is that in the two-ordinate cases (t, x) or (x, y) the system can be reduced by means of the Legendre transformations, i.e. transition from the required functions $\rho_*(t, x)$, $v(t, x)$ to the required functions $t = t(\rho_*, v)$. $x = x(\rho_*, v)$ to a single linear second order differential equation.

This equation for some function

$$\psi(\rho, v, \phi) \sim \rho^m t(\rho, v, \phi), \tag{8.3.16}$$

which satisfies the Laplace equation

[6]The possibility of describing hyperbolic perturbations by a single KdV equation is explained by the expansion of $\omega^2 - c_0^2 \varkappa^2$ to real multipliers, corresponding to the waves, travelling in the opposite directions.

Table 8.1.

m	Quasi-Chaplygin medium
–2	Liquid cylinder with surface tension
–1	Constriction in a compressed scanned pinch
–1/2	One-dimensional non-stationary Chaplygin gas
–1/2	Two-dimensional stationary flow of Chaplygin gas
–1/2	Buneman plasma instability
–1/2	Tearing instability of the plasma layer with current
–1/2	Parametric instability of plasma in an alternating field
–1/2	Perturbations of sinus-Gordon solitons
1/2	Perturbations of Kadomtsev-Petviashvili solitons
1/2	Perturbations of solitons of the nonlinear Schrödinger equation
1/2	Kelvin-Helmholtz instability
1/2	Heating-emissive instability of slightly ionised plasma
1	Toppled shallow water (droplets on the ceiling)
1	Rayleigh-Taylor longwave instability
1	Self-focusing of light in the cubic medium
1	Modulation instability of Langmuir waves in plasma
1	Self-focusing of transverse waves in plasma
1	Collapse of Langmuir waves in plasma
1	Self-focusing the wave packets according to Lighthill
1	In particular, waves in deep water (the theory of the 9th wave)
1	Instability of the gravitational gas layer
1	Instability of the tangential discontinuity of velocity
1	'Immersed' cylindrical stream in a rigid pipe
1	Stratified 'fine' atmosphere
1	Bunching of the electron beam in plasma
1	Separation of the beam into layers and filaments
–1/2	Bunching of ions, accelerated on beats of the waves in plasma
1	Perturbation of Benjamin–Ono solitons
1	Filamentation of the laser beam in the cubic medium
3/2	Perturbation of Korteveg-de Vries solitons
3/2	Soliton of the Schrödinger nonlinear equation with defocused non-linearity
3/2	Wells of the density of slightly non-ideal Bose gas

$$\Delta\psi = 0. \tag{8.3.17}$$

The parameter ϕ is purely auxiliary and it is assumed that

$$\psi = \psi_1(\rho, v)\cos m\phi. \tag{8.3.18}$$

Solving equation (8.3.17) for the appropriate boundary and the initial

conditions, we can, moving away from ψ, determined using the algebraic procedures

$$\rho = \rho(t,x),\ v = v(t,x).$$

The possibility of this type of linearisation is well known in gas dynamics (see, for example, [13]) and was described for the first time by Chaplygin.

Specific examples of the Trubnikov–Zhdanov calculations will not be examined in detail, mainly owing to the fact that it is necessary to overcome a comparatively cumbersome formal barrier in the initial stage and refer the reader to original publications. In particular, one of the recommended books is [34] which is interesting because of its novelty and generality. It is therefore important to mention another general assumption, formulated by the authors of [34]. Usually, the perturbations are investigated using harmonic representations. However, studies of the KdV equation and solitons indicate that in the case of non-linear situations it is more adequate to consider localised structures and specifically the solitons in the previously examined case. It was found, and this was confirmed by Zhdanov and Trubnikov for the system (8.3.14), that the localised structures are also more adequate in this case.

A suitable example here is the instability of a single liquid film on a bath ceiling, Fig. 8.3.7. At some thickness, the film loses its stability and starts to sag. The liquid from the surrounding areas of the film flows into the sagging section at and, consequently, a droplet is separated. This process, can ignore in the final stage, in which it is necessary to take into account surface tension, is described by the system (8.3.14) with $m = 1$. This can also be said of the formation of constrictions in the *Z*-pinch, which may form alone, but even if several of them appear, they are not directly linked with each other.

It is also important to mention that the plasma contains only four universal types of localised perturbations, corresponding to $m = -1, -1/2, 1/2, 1$.

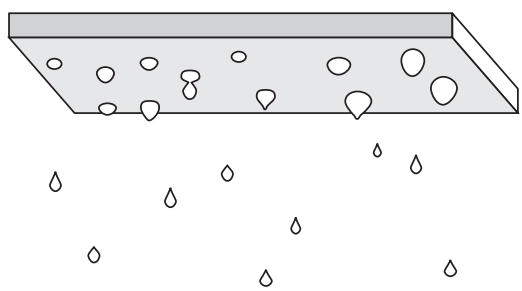


Fig. 8.3.7. Instability of a thin liquid film on a smooth bathroom ceiling.

8.3.4. The Charney–Obukhov equation [32, 210, 211]

The KdV equation and the Chaplygin–Trubnikov equations do not describe various well-known autonomous structures such as vortices. However, the equation describing many types of vortices is available and we shall now consider it. This equation relates to the class of drift flows which were already discussed in section 8.2.4. However, there we discussed only inertialess perturbations. Here we also investigate drift perturbations in plasma taking into account the inertia of the ions, and also the global dynamics of the planet atmospheres. We shall discuss the equation which was derived for the first time by geophysicists Charney (1948) and Obukhov (1949). Later, in 1978, in the development of the theory of non-linear drift processes in plasma, the identical equation was published by Hasegawa and Mim.

Here, we derive the Charney–Obukhov equation in the simplest form. Subsequently, without derivation, we present the generalised Charney-Obukhov and Hasegawa–Mim equation.

Later, in section 9.1, attention will be given to the analogy between the two-dimensional plasma dynamics in the transverse magnetic field and the dynamics of 'shallow water' with the free boundary in a rotating vessel, i.e. in the presence of a Coriolis force. In the latter case, the system of equations has the form (9.1.5)

$$\frac{\partial h}{\partial t} + \operatorname{div}_{\perp} h\mathbf{v} = 0, \tag{8.3.19a}$$

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \left(\mathbf{v}\nabla_{\perp}\right)\mathbf{v}\right) = -\rho g \nabla_{\perp} h + 2\rho\left[\mathbf{v}, \boldsymbol{\Omega}\right]. \tag{8.3.19b}$$

Here $h = h(\mathbf{x}, t)$ is the thickness of the liquid layer, ρ is the density of the incompressible liquid, $\boldsymbol{\Omega}$ is the angular velocity of rotation of the vessel, g is the gravitational acceleration, index $\perp$ denotes the two-dimensional form of the operators.

It may be seen that the equations (8.3.19) are identical with the equations of the dynamics of the ions in the two-fluid model of plasma in the presence of a magnetic field.

Here, we are interested in the case in which the Coriolis force is considerably higher than the inertia force, i.e. the term in the left-hand part of the equation (8.3.19b), and is balanced mostly by the hydrodynamic pressure force. This means that the parameter

$$R_0 = \frac{1}{\tau\Omega} \ll 1, \tag{8.3.20}$$

where τ is the characteristic time scale of the process, for example, the duration of rotation of the vortex. This geophysical parameter is often referred to as the Rossby–Kibel parameter. Evidently, it indicates that the period of circulation in the vortex should be considerably longer than the duration of revolution of the vessel or a planet around the axis. Taking into account this analogy between the dynamics of the 'shallow' water and plasma in the magnetic field into account, it may be concluded that the Rossby–Kibel criterion for it corresponds to the criterion of magnetisation of the particles, i.e. the applicability of the drift approximation (paragraph 1.2.5).

In order to avoid making the equations more complicated, we consider the case in which the non-perturbing medium is homogeneous and stationary. Consequently, restricting our considerations to the velocity perturbations quadratic with respect to the amplitude, it can be assumed that

$$h = h_0 + h_1 + h_2,\ h_0 = \text{const} \qquad (8.2.21)$$
$$v_x = v_1 + v_2;\ v_y = u_1 + u_2 .$$

Because of the assumption (8.3.20), the quantities ~ $\frac{\partial v_1}{\partial t}$ can be regarded as values of the second order of smallness. Substituting the expansion (8.3.21) into (8.3.19), and grouping the terms of the same order, we obtain

$$\frac{\partial h_1}{\partial t} + \frac{\partial}{\partial x} h_0 v_1 + \frac{\partial}{\partial y} h_0 u_1 = 0; \qquad (8.3.22)$$

$$0 = -g\frac{\partial h_1}{\partial x} + 2\Omega u_1$$
$$0 = -g\frac{\partial h_1}{\partial y} - 2\Omega v_1 \qquad (8.3.23)$$

$$\frac{\partial h_2}{\partial t} + \frac{\partial}{\partial x}(h_0 v_2 + h_1 v_1) + \frac{\partial}{\partial y}(h_0 u_2 + h_1 u_1) = 0$$

$$\frac{\partial v_1}{\partial t} + v_1\frac{\partial v_1}{\partial x} + u_1\frac{\partial v_1}{\partial y} = --g\frac{\partial h_2}{\partial x} - 2\Omega u_2 \qquad (8.3.24)$$

$$\frac{\partial u_1}{\partial t} + v_1\frac{\partial u_1}{\partial x} + u_1\frac{\partial u_1}{\partial y} = -g\frac{\partial h_2}{\partial y} - 2\Omega v_2$$

Substituting (8.3.23) into (8.3.22) gives

$$\frac{\partial h_1}{\partial t} = 0 \tag{8.3.25}$$

i.e. in the first approximation, the density of the medium in the presence of the drift perturbations remains constant. In the first approximation, the velocity of drift motions is

$$\mathbf{v}_1 = -\chi\left[\nabla h_1, \mathbf{\Omega}^0\right] \tag{8.3.26}$$

Here $\chi = \frac{g}{2}\Omega$, and $\mathbf{\Omega}^0$ is the unit vector along the axis of rotation of the vessel.

Taking into account the expression (8.3.23) for $\mathbf{v}_1 = (v_1, u_1)$, it may easily be shown that

$$v_1\frac{\partial v_1}{\partial x} + u_1\frac{\partial v_1}{\partial y} = \chi^2\frac{D(h_{1y}, h_1)}{D(x,y)}; v_1\frac{\partial u_1}{\partial x} + u_1\frac{\partial u_1}{\partial y} = -\chi^2\frac{D(h_{1x}, h_1)}{D(x,y)} \tag{8.3.27}$$

Substituting (8.3.27) and (8.3.26) into (8.3.24), we determine the expressions for v_2 and u_2:

$$\begin{aligned} v_2 &= -\chi h_{2y} - \frac{\chi}{2\Omega}h_{1tx} + \frac{\chi^2}{2\Omega}\cdot\frac{D(h_{1x}, h_1)}{D(x,y)} \\ u_2 &= \chi h_{2x} - \frac{\chi}{2\Omega}h_{1ty} + \frac{\chi^2}{2\Omega}\cdot\frac{D(h_{1y}, h_1)}{D(x,y)}. \end{aligned} \tag{8.3.28}$$

Here the indexes x, y, t denote the appropriate partial derivatives.

Substituting (8.3.28) into (8.3.24) and taking into account (8.3.26) gives

$$\frac{\partial}{\partial t}(2\Omega h - \chi\Delta h) + \chi^2 h_0\frac{D(h, \Delta h)}{D(x,y)} = 0. \tag{8.3.29}$$

This is the simplified case of the Charney–Obukhov equation. In some cases, it is referred to as the Larichev–Reznik equation, in accordance with the view of the scientists who investigated the properties of this equation in detail.

When deriving these equations, the non-perturbing configuration was stationary. However, the propagation of waves (vortices, solitons) of the Rossby type (and also of drift waves) usually takes place on the moving background. Therefore, in (8.3.21), when deriving the Charney-Obukhov equation, it can be set that

$\mathbf{v} = \mathbf{v}_0 + \mathbf{v}_1, \quad \mathbf{v}_0 = (v_0, 0).$

This yields the equation which has the following form in the stationary dimensionless form

$$(\Delta h - h)_t + v_0 h_x + v_0 h h_x + \frac{D(h, \Delta h)}{D(x, y)} = 0 \tag{8.3.30}$$

Taking into account the geophysical application of this equation, the quantity v_0 can be regarded as the function of the transverse (width) coordinate

$$v_0 = v_0(y) \approx v_{00}\left(1 - \frac{2y}{R}\right) \tag{8.3.31}$$

where R is the radius of the meridial curvature.

The Hasegawa–Mim equation, corresponding to equation (8.3.30), derived for the perturbation of the potential in the plasma, has the following form

$$(\Delta\phi - \phi)_t + v_0(\phi)\phi_x + v_1\phi\phi_x + \frac{D(\phi, \Delta\phi)}{D(x, y)} = 0. \tag{8.3.32}$$

Here v_0 is the drift velocity.

The main special features of equation (8.3.30) will be mentioned:

a. it can be seen that if we ignore the non-linear terms, we obtain the linear perturbations with the dispersion equation

$$\omega = \varkappa_x (1 + \varkappa^2)^{-1}. \tag{8.3.33}$$

This equation coincides with the equation (8.2.33) if we ignore here $\varkappa^2$ in the denominator which is determined by taking into account the inertia of the media in the terms of the second order of smallness.

b. In the equation (8.3.30) there are two non-linear terms: 'scalar' $\sim h h_x$, and 'vector' $\sim \frac{D(h, \Delta h)}{D(x, y)}$. If the vector term is small, we obtain, in the region of the small wave numbers, the KdV equation with its waves and solitons. Actually, in the absence of vector nonlinearity in the one-dimensional case we have

$$\frac{\partial^3 h}{\partial x^2 \partial t} - \frac{\partial h}{\partial t} + v_0 h + v_0 h \frac{\partial h}{\partial x} = 0$$

Taking into account that in the linear approximation (at $\varkappa \to 0$)

$$\frac{\partial h}{\partial t} - v_0 h = 0$$

we can write

$$\frac{\partial^3 h}{\partial x^2 \partial t} \approx v_0 h \frac{\partial^3 h}{\partial x^3}$$

However, if the vector nonlinearity prevails, then in addition to the linear waves with the previously mentioned dispersion law we also have here the vortices and vortex-like solitons. The assumption that (8.3.30) contains vortices can be confirmed most efficiently by the example of the stationary equation (8.3.29)

$$\frac{D(h, \Delta h)}{D(x, y)} = 0 \qquad (8.3.34a)$$

From this equation we obtain the following solution

$\Delta h = -q^2 n$,

where

$h = A \sin \varkappa x \cdot \sin \varkappa_2 y; \quad \varkappa_1^2 + \varkappa_2^2 = q^2.$ (8.3.34b)

The vortex structure of this equation was already previously shown in Fig. 2.2.1. More details of the vortex structures will be published in section 9.1.

8.4. Stochasticity of the processes in plasma [212]

8.4.1. Stochasticity and turbulence

The high mobility of the particles in the plasma and the existence of a very large number of different types of wear processes may cause the characteristics of plasma systems to become stochastic quantities. In particular, this relates to the local parameters of the plasma volume. Usually, these instabilities are referred to as 'noise' if the process with respect to the macroparameters is relatively regular. However, if the macroparameters also show a non-regular behaviour, we are usually concerned with the 'turbulence' of the process.

Up to the beginning of the 60s of the previous century it was assumed that only the systems with a large number of degrees of freedom can be stochastic. For example, the equal probability of a coin landing on both sides is a result of a complex systems of vortices, formed during flow around the falling coin. However, in the 60s there was a considerable change in general views.[7] Although there

[7]Individual assumptions on the possible stochastic nature of the system with a small number of degrees of freedom were also expressed previously but they did not 'resonate' with the general considerations.

were several events leading to this, the study by E. Lorentz (1963), concerned with the modelling description of thermal convection in a ring-shaped gap between two pipes in the presence of a temperature difference between the lower and upper end covers, proved to be most interesting. Using the method proposed by Galerkin, Lorentz reduced the problem to a system of three conventional non-linear differential equations

$$\dot{x} = \alpha(y - x); \quad \dot{y} = -y + \beta x - xz; \quad \dot{z} = xy - \gamma z \tag{8.4.1}$$

Here $x(t)$ is the amplitude of the speed of movement of the liquid, y (t) is the symmetric component of temperature, z (t) is the non-symmetric component of temperature, taking into account the difference in the temperatures of the covers, α is the Prandtl number ($\beta = R_a/R_{ac}$ is the reduced Rayleigh number[8], γ is the parameter which determines the wave number of the perturbation. The system (8.4.1) is solved numerically.

In the calculations, it is usually assumed that $\alpha = 10$, $\gamma = 8/3$, and the parameter β is variable. Depending on β, the nature of the processes taking place changes as follows.

At $\beta < 1$ the liquid is stationary. At $1 < \beta < \beta_2 = 13.92$, the liquid circulates at a constant speed whose direction is determined by the initial conditions. At $\beta > \beta_2$, the flow becomes sensitive to small changes of the initial conditions, the flow speed becomes irregular, once directed to one side and then to the other. This is the transition from the stable rest state to 'dynamic' chaos which was previously referred to as 'turbulence' of the process.

Previously, similar changes of the dynamics of the liquid, poured in the form of a thin layer on a pan were detected by Benar ('Benar cell', Fig. 8.4.1).

Without discussing detailed reasons for this behaviour of the Lorentz and other models, it should only be mentioned that in the system with two parameters (and not more), the chaotisation of the solution is not possible[9].

The second cycle of investigations, playing a significant role in the investigation of the fundamental role of dynamic chaos, i.e. chaos in the systems with the small number of degrees of freedom

[8]The Prandtl number: $\mathrm{Pr} = c_p\mu/\kappa$, where c_p is the heat capacity, μ is dynamic viscosity, κ is heat conductivity. This number characterises the ratio of the heat, generated as a result of viscosity, to the heat taken away by heat conductivity. For conventional air Pr = 0.7. The Rayleigh number: $\mathrm{Ra} = gL^3\beta\Delta T/\nu a$, here β is the coefficient of volume expansion of the medium, ν is kinematic viscosity, a is the thermal diffusivity coefficient, L is the scale of heterogeneities (wavelength), ΔT is the temperature gradient.

[9]This is associated with the formation in the three-dimensional space of the parameters (x,y,z) of the specific structure – the 'strange attractor' (see [212]).

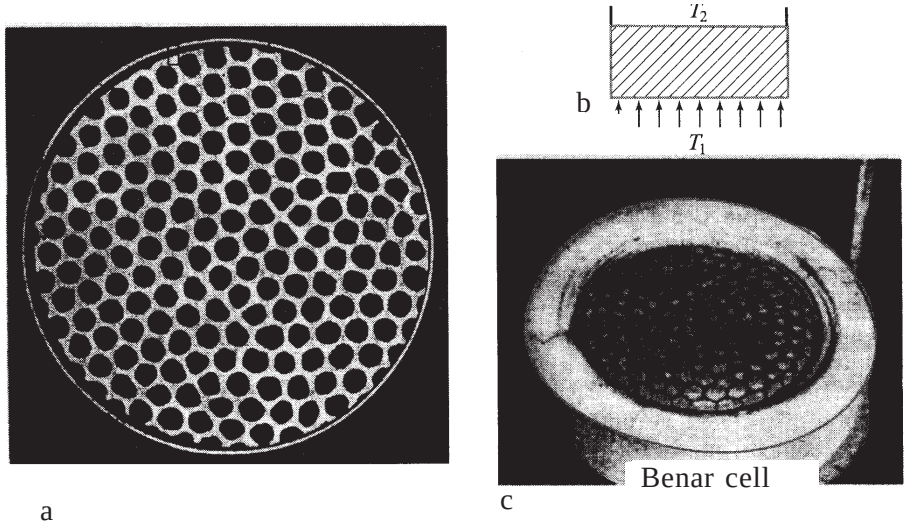


Fig 8.4.1. Ordered structure of Benar convective cells formed in the liquid layer in heating from the bottom (a); the experimental setup (b), experimental equipment (c).

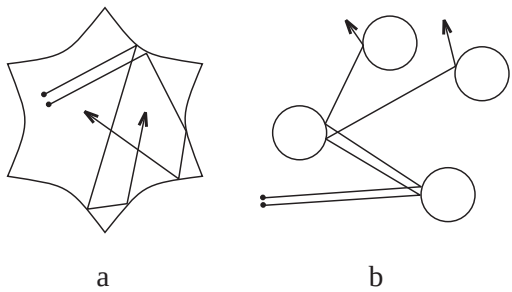


Fig. 8.4.2. Movement of a ball in billiards with curved walls (a) and between balls (b).

($N \geqslant 3$) was the 'guiding of science' to the movement of the particle 'in billiards' with curved walls on the condition that the reflection from the sides takes place in mirror fashion (Fig. 8.4.2). The studies of the Soviet mathematician Ya.G. Sinay were of considerable importance in this area. Evidently, from the historical viewpoint, the initial studies were those by N.S. Krylov (1930) who investigated the irreversibility of kinetics and stressed the stochastic nature of dynamics of balls. Their predecessor was obviously A. Poincaré who investigated the topology of curves, determined by the conventional differential equations.

Finally, the third cycle of the investigations was associated with the investigations by the author of the present book and his colleagues of the magnetic lines of force of vacuum fields at the beginning of the 60s. The structures formed in this case were already discussed in section 1.1. It was also reported there that a disruption of symmetry

the system of the magnetic surfaces starts to be disrupted itself, and the highest sensitivity to the effect is exerted by the vicinity of the separatrix and, as already mentioned, a chaos forms here.

An interesting fact should be mentioned here. If prior to perturbations some volume has the form of a system of magnetic pipes separated with separatrices, then in the presence of a perturbation the resultant layers in the vicinity of the separatrix with chaos represent a united system and the line of force from the vicinity of any area of the 'chaotic web' sooner or later (in the sense of the travelled path) falls into the vicinity of any point of the web. This phenomenon is referred to as 'Arnold diffusion' in accordance with the name of the Russian mathematician who established this fact.

Thus, it can be seen that only the simplest differential equations create chaos. It is clear that the transition to more complicated models is accompanied by formation in most cases of even more refined structures with stochastic regions. In section 6.7 attention was given to the one-dimensional model of a stationary plasma thruster showing the transition from the regular to chaotic flow with the variation of only one parameter – the effective resistance of the channel.

On the basis of these considerations we can transfer to the problem of turbulence.

8.4.2. Turbulence

The term turbulence is used usually quite extensively almost as the synonym of any stochastic process. Here we do not try to define exactly the turbulence, and it is only mentioned that the characteristic feature of turbulence is the development in the medium of a relatively large number of instabilities, leading to the formation of a hierarchy of non-linear dynamic structures and dissipation of the energy of macro-motion.

The first stages of investigating turbulence

Investigations into the turbulent processes started in the second half of the 19th century by Reynolds (1883). He investigated the transition of laminar flows in cylindrical pipes to turbulent flows (chaotic flows) and established that this transition is determined by the quantity

$$\mathrm{Re} = \frac{vl}{\nu}, \tag{8.4.2}$$

which was later referred to as the Reynolds number. In (8.4.2) v, l are the characteristic values of the flow velocity and the scale, and ν is the

kinematic viscosity. The critical value Re_{cr} at which this transition takes place in a circular smooth pipe is equal to Re ≈ 1800, if $l = D/2$, where D is the inner diameter of the pipe.

Subsequently, over many years, the most popular model was the probability model of stationary turbulence in a uniform hydro (gas) dynamic medium. This model was developed on the basis of similarity considerations by Kolmogorov and Obukhov (1941). It was based on the assumption on the existence of a cascade of vortices of different scale through which the energy is pumped from large scale vortices ('starting' the movement of the medium) to small scale vortices. It is assumed that the Reynolds number Re_0, starting the cascade of the vortices is

$\mathrm{Re}_0 = \frac{v\lambda_0}{\nu} \gg 1$, and small scale $\mathrm{Re}_{\min} = \frac{v\lambda_1}{\nu} \sim 1$.

With these assumptions, the above authors derived an universal dependence of the characteristic pulsations rate on the wave number k in the range $\mathrm{Re}_{\min} < \mathrm{Re} < \mathrm{Re}_0$ [13]:

$$\overline{v} \sim k^{-\frac{5}{3}}. \tag{8.4.3}$$

This dependence is efficiently confirmed by experiments in the range of variation of k by three orders of magnitude. Nevertheless, numerical modelling shows that the local dependence $\overline{v}(k, \mathbf{x})$ is of the 'fractal' type and at the individual points of the flow the difference between v and $\overline{v}$ may reach high values.

We do not discuss here the specific features of turbulence in gas-hydrodynamics and mention only the most general properties of turbulent processes.

Approaches to the investigation of turbulence

Usually, there is weak and strong turbulence.

Weak turbulence is the turbulence of wave fields, slightly interacting with each other (in the form of decay, mergers, scattering), in which the hypothesis of random phases of the waves can be applied (section 4.6).

Previously, we described the dynamic chaos in a stationary plasma thruster in one of the elementary processes forming the basis of the turbulence of the fields: merger of the waves.

An important special feature of weak turbulence is the relatively low size N of its phase space ($N \leqslant 10$). In the same Lorentz model $N = 3$.

Strong turbulence can form for different reasons. They include:

- hydrodynamic turbulence at $\mathrm{Re}_0 \gg 1$. In particular, the Kolmogorov–Obukhov turbulence;
- the dynamics of the systems with strongly non-linear waves (for

example, shock waves), which at the same time strongly interact with each other;

- turbulence in the presence of many solitons, vortices and other quasi-autonomous structures, with the dimension of the phase space or the number of independent excited modes $\geqslant 100$.

With the variation of the 'controlling parameter' the transition from the regular regime – reproduced under the same initial and boundary conditions – to turbulence which cannot be produced under the same conditions at the start and at the boundaries, can take place both smoothly (as indicated by Lorentz and the model of the stationary plasma thruster) and in the form of 'jumps'.

The high complexity and existence of a large number of types of turbulent processes were the reasons for the development of a number of approaches to their theoretical simulation. There are three main approaches: statistical, structural and dynamic.

The statistical approach was developed by O. Reynolds. It is based on the Navier–Stokes equations which in the case of an incompressible liquid have the form (at a constant viscosity)

$$\operatorname{div}\mathbf{v} = 0;$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v} = -\frac{1}{\rho}\nabla p + \nu\Delta\mathbf{v}.$$

The velocity and pressure are presented in the form of regular quantities **U** and P, and in addition also as chaotic additions **u** and q

$$\mathbf{v} = \mathbf{U} + \mathbf{u}; \quad p = P + q$$

We can also write

$$\operatorname{div}(\mathbf{U} + \mathbf{u}) = 0;$$

$$\frac{\partial}{\partial t}(\mathbf{U} + \mathbf{u}) + \big((\mathbf{U} + \mathbf{u}), \nabla\big)(\mathbf{U} + \mathbf{u}) = -\nabla(P + q) + \nu\Delta(\mathbf{U} + \mathbf{u}) \qquad (8.4.4)$$

Assuming that there is a distribution function[10] and, averaging with respect to this distribution, we obtain for the condition $\langle u \rangle = 0$, $\langle q \rangle = 0$

$$\operatorname{div}\mathbf{U} = 0;$$

$$\frac{\partial}{\partial t}\mathbf{U} + (\mathbf{U}\nabla)U = -\nabla P + \nu\Delta U - \big\langle(\mathbf{u}\nabla)\mathbf{u}\big\rangle. \qquad (8.4.5)$$

[10]In fact, this should be the functional $\Phi[\eta, \mathbf{x}, t]$, where $\eta = (\mathbf{v}, P)$ are the solutions of the Navier–Stokes equations. Subsequently, the method for describing turbulence by means of the probability functional was developed by Hopf.

To determine $\langle(\mathbf{u}\nabla)\mathbf{u}\rangle$, it is necessary to multiply (8.4.5) by $\mathbf{u}$ and again carry out averaging, etc.

This procedure is identical with the transition from the kinetic equation to the system of hydrodynamic equations. Here, as in the previous case, we obtain an infinite system of equations for specific moments. In reality, this system should be interrupted in some manner but there are no *apriori* rules for this. Therefore, regardless of the fact that this approach in combination with either theoretical or experimental data is useful, it is becoming less and less interesting, especially as a result of the development of powerful computers.

The structural approach to describing turbulence is based on the separation (mostly on the basis of experimental investigations) of some structures whose dynamics is described by relatively simple equations. This is why the approach was used by Lorentz.

As regards hydrodynamic turbulence, this concerns the system of vortices with different scales. This approach was developed by Richardson in 1922 who proposed a specific model of transfer of energy from large vortices to smaller ones. In his model, the initial flow with $\mathrm{Re}_0 = vl_0/\nu$ is unstable and generates a vortex with the scale $l_1 < l_0$, and so on, to $\mathrm{Re}_{\min}$, at which the vortices are stable and are disrupted only by viscosity.

Evidently, the previously described Kolmogorov–Obukhov model of hydrodynamic turbulence is based on the two approaches: statistical and structural.

It is important to mention another of the facts, determined on the basis of experiments. It appears that if a small-scale turbulence is anisotropic or a medium is compressible, the developed turbulence may prove to be unstable and this is characterised by the inversion of the energy flow: the energy travels from small to large scales. In particular, this process responsible for the formation of planetary vortices which will be discussed in section 9.1.

The dynamic approach is based on the direct solution of the Navier-Stokes equations, although other approximate methods can also be used for this purpose.

The dynamic approach is now dominant owing to, in particular, the development of powerful computers and efficient computing programs.

8.4.3. Some special features of plasma turbulence [43, 44]

This type of turbulence naturally differs from the turbulence of a viscous liquid or gas by the presence of an incomparably large number of types of waves and vortices determined by the presence of electromagnetic

fields in the plasma. Consequently, there are different types of turbulence: MHD, ion-sound, Langmuir and drift.

They will now be briefly described.

MHD turbulence

At high conductivity, the displacement of the plasma takes place together with the magnetic field as a result of 'freezing-in'. However, the random nature of motion generates a large number of x-points where magnetic lines of force are re-connected. This type of turbulence is especially important in cosmic conditions, in particular on the Sun. Here, as in other objects, this type of turbulence causes the high-intensity transfer of mass, energy and also generation and transfer of the magnetic field. The small parameter forms of MHD turbulence are evident in large traps - tokamaks, stellarators.

Ion-sound turbulence

This type of turbulence at low frequency resembles gas-dynamic turbulence, and at high frequencies it is characterised by the strong interaction of ion-sound waves, leading to the anomalous resistance of plasma.

Langmuir turbulence

The predecessor of this type of turbulence in the linear stage are Langmuir oscillations generated by, for example, an electron beam. These oscillations rapidly reached the non-linear level generating Langmuir solitons[11] which in the three-dimensional case are unstable in relation to collapse accompanied by the transitions of the energy of electrical oscillations to the kinetic energy of accelerated electrons. This phenomenon is referred to as the 'collapse of Langmuir waves'. Evidently, this collapse plays the main role in the 'cleaning' of the lower level in a cascade of turbulent transitions (V.E. Zakharov).

Drift turbulence

Previously, we derived an equation for non-linear drift waves (8.3.32) which describes all perturbation types, if the condition of low inertia is satisfied. Therefore, this equation also describes the drift turbulence. Here, we also have a transition from the laminar flows, and solitons and vortices to chaos. This problem will be discussed again in section 9.1.

Self-organisation of processes in plasma

[11]A specific role in the formation of Langmuir solitons is played by the displacement of ions and the effect of the Miller force (section 1.2.4), caused by Langmuir oscillations.

The existence of the large number of types of interaction in plasma not only complicates the general pattern of the possible turbulent movements but also in many cases simplifies this pattern as a result of the self organisation of highly stable structures. In particular, the transition from the initial perturbation of the final one – the static state, does not usually take place by monotonic degradation and passes through the stages of formation of large-scale long-life structures, such as the ones indicated on the example of the Pasta–Fermi–Ulam numerical model, which passes through the stage of formation of the solitons, as shown in section 9.1 on the example of planetary vortices.

We have restricted our considerations to describing briefly the types of turbulent processes in plasma. Of course, the current state of this area of plasma physics contains many more features. However, in this area there are many more uncertainties than in the case of hydrodynamics where in many cases it is only necessary to improve the accuracy and define more accurately the already indicated general schemes. In the case of plasma we are far away from this because at the moment we are only capable of describing the attempts to construct the theory of plasma turbulence but not the theory itself.

8.4.4. Anomalous resistance of plasma [7]

Previously, we discussed the three types of transfer processes: classic, neoclassic ('banana-shaped', near wall) and anomalous, determined by small-scale oscillations.

In this section, attention is given to a specific case of plasma turbulence, leading to anomalous plasma resistance.

In the group of individual mechanisms of creating anomalous condutivity, a significant role is played by the mechanisms determined by the formation in plasma of 'chaotic' small-scale electrostatic oscillations with relatively high amplitudes. The appearance of these oscillations shortens the effective time of free path τ_{ef} and, consequently, the conductivity coefficient

$$\sigma = \frac{e^2 n \tau_{ef}}{m}$$

The value of τ_{ef} can be calculated from the equation

$$\tau_{ef} \sim \nu^2 \left(\frac{m^2}{e^2 \left\langle \tilde{E}^2 \right\rangle} \right) \omega \sim \frac{kT_e}{\left\langle \tilde{E}^2 \right\rangle} \left(\frac{m\omega}{e^2} \right). \qquad (8.4.6)$$

Here ω is the characteristic frequency of pulsations, $\tilde{E}$ is the strength of the chaotic electrical field. This equation follows from the considerations identical with those leading to the Einstein equation for Brownian motion

$$\tau_{ef} = \frac{\langle x^2 \rangle}{D}, \quad D = \frac{\lambda^2}{\tau}$$

and λ and τ is the free path and time.

In the present case, the role of x is played by v - the thermal velocity of the particle, $\tau \sim \omega^{-1}$ is the value reciprocal to frequency, and the role of λ is played by the velocity required from the chaotic field during its 'oscillation' period ($\lambda = \Delta v \sim eE/m\omega$).

As indicated by these considerations, the problem of τ_{ef} is reduced to the determination of the characteristic frequency of pulsations ω and the mean value of the square of the pulsating electrical field $\tilde{E}$. To determine these values, it is necessary to carry out relatively complicated theoretical constructions. Therefore, the readers interested in the theory of small-scale turbulence, leading to anomalous resistance, should refer to the books [7,10]. Here we describe only briefly the simplest estimates of the detected dependence of current density j on the strength of the electrical field $j(E)$ and describe experiments with turbulent heating.

The Ohm law at high strength of the electrical field ($H_0 = 0$ *or* $\mathbf{H}||\mathbf{E}$)
If the strength of the electrical field is low and the electron which acquires the energy from the field between collisions is capable of transferring this energy to heavy particles, then the density of the current is directly proportional to the strength **E**:

$$j = \sigma E\big|_{E-\text{small}}.$$

However, with an increase of E the rate of collection of the energy and the flow velocity increase and when the velocity u_0 approaches the speed of ionic sound, the dependence of j on E weakens and its value in the first approximation becomes equal to $\sim\sqrt{E}$. At $u_0 \sim v_{Te}$, the current density changes only slightly with a further increase of E.

On the whole, the situation is made more complicated, the plasma column becomes non-uniform (in particular, the so-called double layers with a very high anomalous-type resistance appear) and the fraction of running away electrons increases.

Both the experiments and theoretical analysis show that the general form of the dependence $j(E)$ is similar to that shown in Fig. 8.4.3. The main type of oscillations in the presence of strong electrical fields are

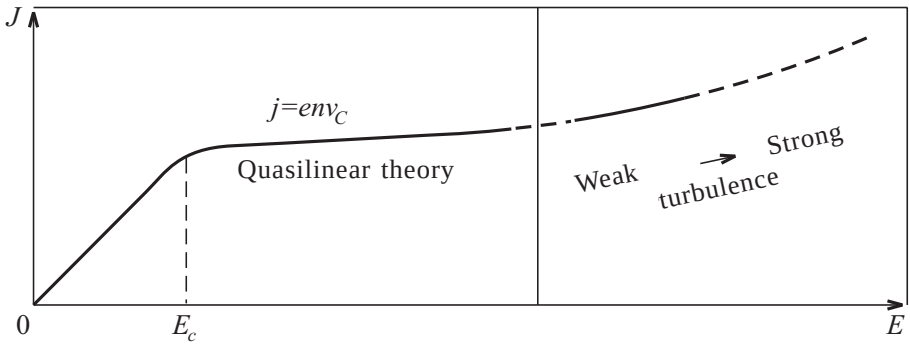


Fig. 8.4.3. General view of the Ohm law in the case of ionic-sound turbulence [7].

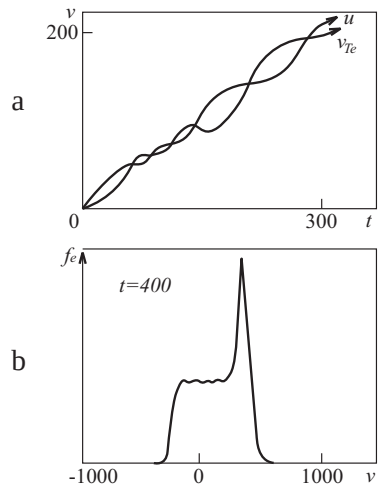


Fig. 8.4.4. Dynamics of electrons in a discharge with increasing E: a) the dependence of current velocity u and thermal velocity v_{Te} on t; b) the typical electron distribution function in the turbulence stage [7].

short-wave ionic-sound oscillations with the wavelength of the order of the Debye radius. The characteristic form of the electron distribution function, determined by calculations, is shown in Fig. 8.4.4. It can be seen that the distribution is tabular, determined by the high level of the oscillations $\varkappa D \sim 1$ and the 'regrowth' of the running away electrons [40].

Experiments with turbulent heating (E.K. Zavoiskii, L.I. Rudakov)
Small-scale turbulence is easily realised in greatly differing systems. In particular, this can be achieved in a direct electrical discharge of the *Z*-pinch type with the applied longitudinal magnetic field. In hydrogen, the working pressure range is 10^{-4}–10^{-2} torr, the duration of the discharge

pulse ~0.2–0.6 μs, current density ~10^2 A/cm^2. During the discharge, the electronic temperature increases to several keV, and ionic temperature to 100–300 eV. It can be seen that this is in good agreement with the value of the threshold of the increment of development of the Buneman–Budker instability.

8.5. Active methods of stabilising plasma instabilities [216]

Previously, we discussed the behaviour of plasma configurations, restricted by some walls and left to one's devices. However, in the surrounding world we always have to face special systems controlled by some processes. This control may be carried out by the man, animals (for example, the bees maintain a specific temperature in a beehive), automatic systems (an autopilot fulfils a specific flight plan of the aircraft), etc.

The general tendency in the development of technology is that larger and larger parts of the blocks of a complex system are controlled by special diagnostics and control organs. It is clearly evident that the plasma systems can become more and more saturated with the automatic information-control systems, and the classic plasma 'ameboid' volumes transform to 'cyber plasma' structures.

The simplest system for the stabilisation of the plasma object appeared as early as in the 19th century and ensured stable burning of an electric arc. The need for this stabilisation is explained by the drooping volt–ampere characteristic (VAC) of the arc which makes the discharge unstable. Therefore, a special resistance is placed close to the arc which also ensures the stability of the discharge as a result of the rising VAC. However, this is 'passive' stabilisation which in contrast to 'active' stabilisation does not contain any special sensors of the state of the system.

One of the first systems for the stabilisation of the plasma configuration was a system for the stabilisation of the position of the plasma pinch in a tokamak. This plasma ring tries to expand as a result of the current flowing through it. This can be counteracted if the plasma is placed in a superconducting box[12]: consequently, the magnetic field between the highly conducting pinch and the wall impermeable to the field is compressed during expansion of the ring (intensifies), and this arrests the process. However, the superconducting jackets are not convenient and the copper ones are not suitable because of the increase in the thickness of the skin layer at long holding times.

Therefore, the active system for stabilising the position of the ring

[12]The role of the superconductor in first tokamaks with a relatively short lifetime was played by a thick copper 'jacket' with a relatively thin skin layer.

with a feedback (Fig. 8.5.1) is used widely at the present time. The system includes magnetic coils – the sensors of the position of the ring and a system for processing information in controlling the current in the power coils, generating a magnetic field with the preferential orientation along the axis of the system. Changing the strength of the magnetic field, it is possible to change the intensity of the ampere force $\sim J_0 H_\perp$ and, consequently, control the dimensions of the ring. It should be mentioned that the advantages of the plasma pinches with the extended sections have been demonstrated, and they are unstable without feedbacks.

Some integral parameter was stabilised in these examples. Evidently, in principle it is possible to control only a fragment of the system (instead of the entire system), having the appropriate sensors and means of exerting the effect – the working organs of the stabiliser.

The theoretical model of stabilisation of the distributed instabilities of the plasma pinch was investigated for the first time by A.I. Morozov and L.S. Solov'ev in 1964 [216], using a system with feedbacks. The principal diagram of the system is shown in Fig. 8.5.2.

The actuating organs of the system included a mosaic of the coils changing locally the strength of the magnetic field around the pinch in accordance with the signal from a local sensor.

In the above study it was also mentioned that, using the beams of neutral atoms and directed microwave fluxes, it is also possible to influence the inner regions of the pinch. The publication of this article led to a large number of independent theoretical investigations and also the experimental realisation of the concept.

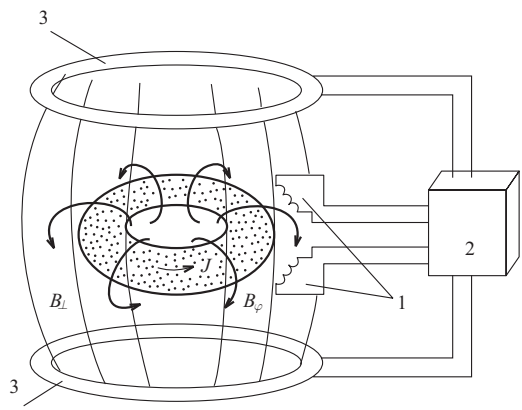


Fig. 8.5.1. Diagram of active stabilisation of the diameter of the plasma pinch in a tokamak: 1) the sensors of the size of the pinch; 2) the regulator of current in controlling coils; 3) controlling coils.

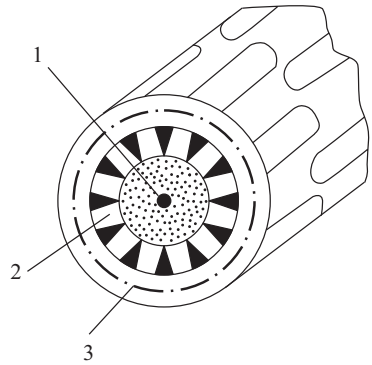


Fig. 8.5.2. Diagram of active stabilisation of local perturbations of the plasma configuration (1) using a layer of sensors (2), and a layer of locally excited currents of the contours (3).

Suppression of convective instability in axisymmetric mirror traps

Attention will now be given to the experimental investigations of the stabilisation of the oscillations by feedbacks in a mirror trap with rarefied plasma (V.V. Arsenin, V.A. Chuynov, et al., 1967–1976 [217]). The experiments were carried out in an Ogra-3 trap, in which the length of the plasma volume was ~8 cm, and the diameter ~14 cm. The magnetic field in the centre of the trap was ~20 kOe, the density of plasma without stabilisation was ~10^8 cm^{-3}.

The stabilisation system was developed for suppressing fulte-shaped instabilities, formed at a relatively sharp boundary of the plasma. According to the stabilisation concept, the external electrode-panels are used to produce the **E**-field in the volume of the plasma projection extended along **H**, with the **E**-field being opposite to the polarisation field, determined by the magnetic drift. This concept was based on the low plasma density in the trap or, more accurately, a low value of the dielectric constant $\epsilon_\perp \sim 1$.

Since the development of instability is a random process, and the instabilities can have different forms, the stabiliser circuit was oriented to the perturbations, which can be described as a sum of 6 harmonics

$$\Phi = \sum_{k=1}^{b} \Phi_k \sin k(\theta - t_k).$$

Consequently, a system of six pairs of probes-sensors, placed in the peripheral zone of the plasma, and six pairs of the panels, was used. The signal from the sensors travelled to the unit for spatial Fourier analysis and subsequently to the units, generating the control signals for each mode and for subsequent supply to the panels. The simplified circuit of

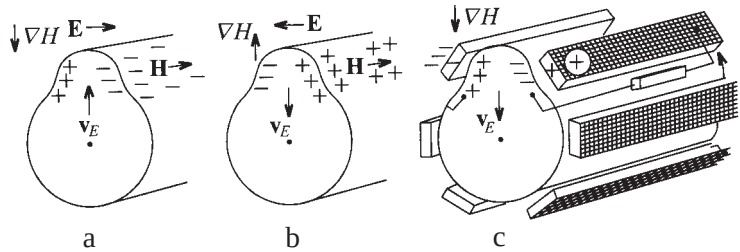


Fig. 8.5.3. Stabilisation of a convective instability in a mirror trap using external electrodes.

the stabiliser (where 6 panels, not 12 panels, for better understanding) is shown in Fig. 8.5.3[13].

It should be mentioned that the same Ogra-3 trap was used for the experiments with the suppression of cyclotron instability.

The suppression of oscillations using the feedbacks was investigated in a stationary plasma thruster by Nevrovskii et al [218]. A number of authors successfully stabilised the drift instabilities of low-temperature plasma in the magnetic field [219, 220].

The suppression of instabilities in tokamaks using feedbacks is described in appendix *E*.

[13]Application of this stabilisation system increased the plasma density in the Ogra-3 by almost 50 times. The initial density was ~10^8 cm^{-3}.

9

Processes in cosmos and plasma dynamics

Some typical plasma and not only plasma space formations will be briefly described in this chapter. Special attention will be paid, on one side, to the specific character of a number of space phenomena and, on the other side, to the presence of the non-trivial analogies between the space processes and 'classical' plasma dynamics under laboratory conditions.

The clear example of such similarities is the straight (formal) analogy of the non-linear drift vortices in inhomogeneous plasma with cyclones and anticyclones in the atmosphere of the Earth and planets (in particular with the large red spot on Jupiter) and further with the dynamics of a number of the spiral nebulas (section 9.1).

This and other numerous facts confirm the fruitfulness 'of the overflow' of the achievements of space plasma dynamics into laboratory plasma dynamics (a suitable example is the influence of Alfvén's book Cosmical Electrodynamics) and vice versa. An example of the latter is the rapid interpretation of Earth radiation belts based on the viewpoint of the physics of mirror traps or simulation of the magnetospheres of planets in the laboratory (I.M. Podgorny [221]).

But an example of a more distant analogy can also be mentioned. So, taking into account that Newtonian gravitational interaction is similar to Coulomb's – but without 'the charges' of two signs – much of plasma dynamics can be transferred to describing the accumulations of the gravitating masses, such as nebula or rings of planets. This approach led in particular A.M. Fridman to the prediction of existence of Uranus satellites which were soon discovered by the American interplanetary probe Voyager in 1986 [222].

Another example. Earlier in Chapter 1 discussion centered on the generation of fast particles in the *Z*-pinch. Recently, B.A. Trubnikov developed the relativistic model of this process and it turned out that the energy spectrum of fast particles takes the form

$$\nu \sim \left(1-\beta^2\right)^{\alpha/2}, \quad \beta = \frac{v}{c},$$

where $\alpha = 1+\sqrt{3}$ [223]. However, this exponent is in excellent agreement with the observed value $\alpha = 2.7$ for the galactic cosmic rays. Additional studies are of course necessary in order to estimate this agreement, but Trubnikov's claim is impressive.

The travel of humanity to cosmos makes the solar system an object of experiments on a completely new level. This includes not only the examination of its components 'near' the spacecraft but also direct visits to planets and satellites, active experimental studies of their magnetospheres, or solar wind. Thus, the framework of terrestrial laboratories has expanded into cosmos.

But the matter is not limited to purely scientific interests in cosmos. The work of man on the space stations requires the prognostication of 'space weather', in particular, the appearance of powerful ejections by the Sun of plasma and radiation. Such ejections can lead to catastrophic consequences for the cosmonauts and also to failure and malfunction of communication and power systems not only in space but also on the Earth (see, for example [224]).

Even modest solar ejections noticeably agitate the magnetosphere ('magnetic storms') and this affects the health of people. Moreover, they are reflected in social processes in the human society. This connection was revealed and described by the outstanding Russian scientist A.L. Chizhevskii in his book 'Terrestrial echo of solar storms' [238].

Finally, plasma engines with the high exhaust velocities have been developed for spacecraft. They are incomparably more adequate for long-distance flights (section 10.4). It is obvious that the mastery, at least, of the circumsolar space will be in many respects due to the availability of these engines.

In this chapter attention is given to planetary vortices and their connection with the drift waves in plasma. The nearest space MHD formation – the Earth magnetosphere – is then discussed. At the end of the chapter, MHD-configurations and MHD-hurricanes on our priceless Sun are discussed.

9.1. Planetary vortices. Spiral nebulas

9.1.1. Cyclones and anticyclones. Zone flows

Cyclones and anticyclones are relatively frequent phenomena in our Earth's atmosphere. They are large-scale ($\geqslant$1000 km) vortices in the

atmosphere. In this case the rotation of air in a cyclone coincides with the direction of rotation of the Earth whereas in an anticyclone the air rotates in the opposite direction. Usually in the Earth's atmosphere there are simultaneously several cyclones and anticyclones (Fig. 9.1.1a). It was established long ago that the properties of cyclones and anticyclones are very different.

So, the pressure in the cyclone centre is lower than in the environment, and in the anticyclone centre it is higher. The lifetime of cyclones is usually short, whereas anticyclones live, as a rule, considerably longer (they are steadier). Other special features will be discussed later.

We have become accustomed to cyclones and anticyclones in the Earth's atmosphere. However, similar swirl flows are also observed in the oceans (Fig. 9.1.1b) where they are, however, smaller.

Especially impressive are vortices on the giant planets, in particular on Jupiter (Fig. 9.1.2). The photograph shows the 'Large Red Spot' (LRS) which was already observed in 1660 by R. Hooke and J. Cassini. We now know that the LRS is a gigantic anticyclone in Jupiter's atmosphere.

Together with the vortices, the photograph shows the so-called zone flows – the atmospheric flows which flow along geographic parallels. The direction and, in particular, the velocity of these flows change quite chaotically with the latitude and time. The appearance of these latitudinal flows is determined by a complex chain of processes which

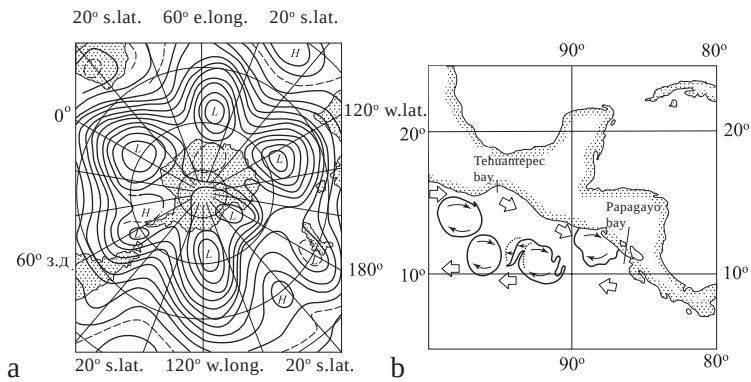


Fig. 9.1.1. Planetary vortices on the Earth: the system of cyclones (L) and anticyclones (H) in the Earth atmosphere above the Antarctic (the figure shows the equidistant levels of the isobaric surface in the southern hemisphere) (a); the distribution of anticyclones and jets of large-scale flows in the part of the Pacific Ocean close to the central America at the end of February 1976. (The contours of the anticyclones are drawn along the boundary separating the warm surface water on the outside of the vortices from the cold water inside the vortices. For one of the vortices the figure shows its drift to the west) (b) [225].

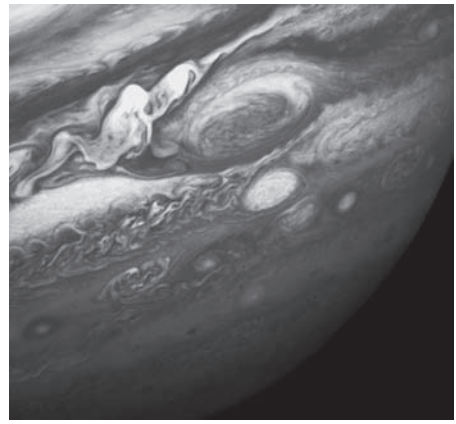


Fig. 9.1.2. The large red spot of Jupiter: the large dark oval – the large red spot of the Jupiter, south and southwest – the vortices White Ovals; the zonal structure of the rings blowing along parallel directions is shown.

include turbulent processes and confluence of small-scale disturbances into larger ones.

The noted irregular mechanism of their generation is responsible for the time-dependent 'fanciful' dependence of azimuthal velocity on latitude shown in Fig. 9.1.3. The presence of the latitude-dependent variation of the velocity of zone flows leads to the appearance of vortices as a result of the Rayleigh mechanism (section 8.1).

The observations of large-scale vortices showed:

1. The cyclone–anticyclone asymmetry mentioned above.
2. Hierarchy of scales and times of the life of planetary vortices. Vortices have the dimensions of the order of several thousand kilometers and even more, and they are observed in the more or less stationary form for a long time that reaches tens or hundreds of years. The larger the vortex the greater its lifetime. The large red spot of Jupiter has been visible for more than 300 years. Somewhat south of it there are Large White Ovals which formed in 1938 as a result of a strong disturbance of Jupiter's atmosphere.
3. All large vortices drift around the axis of the planet. The direction of drift can vary.
4. The spin time τ_{vort} in all large vortices is considerably longer than the period of revolution of the planet T_{pl}. In Jupiter $T_{pl} \sim 10$ h, $\tau_{LRS} \sim 7$ days.
5. Linear velocities of particles in the vortices ~10 m/s and exceed their drift velocity by approximately an order of magnitude.

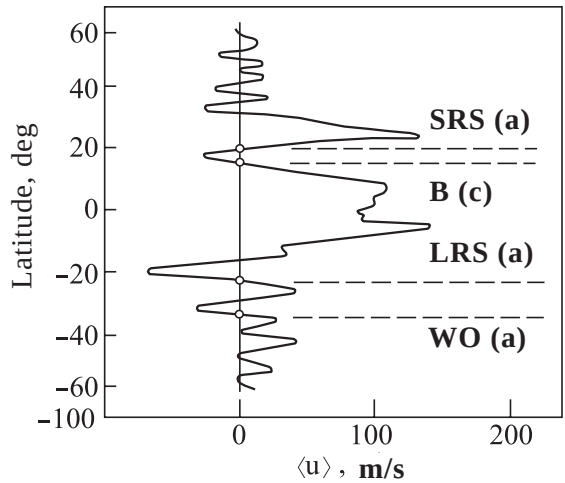


Fig. 9.1.3. Zonal flows in the upper atmosphere of Jupiter: average velocity <*u*> of the wind, blowing along the parallel (the positive direction of the wind is to the east), in relation to planetary latitude. The coordinates of the main large long life vortices are shown (LRS – the large red spot, SRS – the small red spot, WO – white ovals, B – 'barges') and their signs (a-anticyclone, c-cyclone).

6. All anticyclones are observed in those regions of the zone flows where the vorticity of flows is anticyclonic.
7. All large vortices are localized in the region of parallels where the meridian gradient of the velocity of zone flows passes through a local maximum and the Rayleigh type criterion of instability is satisfied.

9.1.2. Larmor's analogy

In his time, Larmor focused attention on the fact that the motion of a charged particle in a magnetic field greatly resembles particle motion in a revolving coordinate system, moreover the Coriolis force is the analogue of the Lorentz force.

Actually, the equation of motion of a particle without the magnetic field in a stationary fixed coordinate system takes the form

$$M\frac{d\mathbf{v}}{dt}=\mathbf{F} \tag{9.1.1a}$$

Passing to the revolving coordinate system, we assume

$$\mathbf{v}=\mathbf{w}+\left[\boldsymbol{\Omega},\mathbf{r}\right]=\frac{d\mathbf{r}}{dt} \tag{9.1.1b}$$

Here $\boldsymbol{\Omega}$ is the angular velocity vector, $\mathbf{r}$ is the radius-vector of particle

and we obtain

$$M\frac{d\mathbf{w}}{dt} = M\Omega^2\mathbf{r}_\perp + 2M\left[\mathbf{w},\mathbf{\Omega}\right] + \mathbf{F}. \tag{9.1.2a}$$

Hence, it is apparent that if $\mathbf{F} = -\nabla U_1(\mathbf{r}_\perp)$, then

$$M\frac{d\mathbf{w}}{dt} = \left[\mathbf{w}, 2M\mathbf{\Omega}\right] - \nabla U(\mathbf{r}) \tag{9.1.2b}$$

where

$$U(\mathbf{r}) = U_1(\mathbf{r}) - M\frac{\Omega^2 r_\perp^2}{2}$$

Equation (9.1.2b) is analogous to the equation

$$M\frac{d\mathbf{v}}{dt} = \left[\mathbf{v}, \frac{e\mathbf{H}}{c}\right] - e\nabla\phi$$

Thus, we have the correspondence[1]

$$\mathbf{\Omega} = \frac{1}{2}\frac{e\mathbf{H}}{Mc} \tag{9.1.3}$$

The adequacy of two initially developed independent scientific directions was established on this basis. These directions are associated with the drift waves – the Hasegawa–Mim equation and with the flow in the 'thin atmosphere', described by the Charney–Obukhov equation, was discussed in Chapter 8.

Let us examine the flow in the thin layers in more detail.

9.1.3. Two-dimensional hydrodynamics of thin layers

Let us begin with the well known (in hydrodynamics) analogy between the planar (two-dimensional) isothermal flows of gas and the dynamics of incompressible 'shallow water', poured as a thin layer on a flat surface. In this case, the role of internal pressure is played by hydrostatic pressure (Fig. 9.1.4)

$$p = \rho g h \tag{9.1.4}$$

where h is the variable height of the liquid layer. Then, obviously, the system of equations describing the dynamics of the layer will be

[1]In fact, this frequency which is half the cyclotron frequency, was referred to as 'Larmor' frequency.

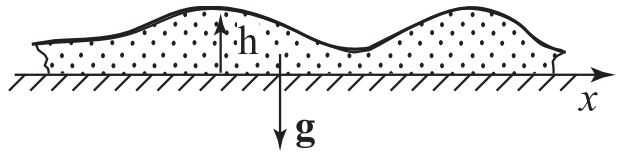


Fig. 9.1.4. Hydrostatic pressure as an analog of gas-kinetic pressure.

$$\frac{\partial h}{\partial t} + \operatorname{div} h\mathbf{v} = 0; \qquad (9.1.5)$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\nabla)\mathbf{v} = -g\nabla h$$

The analogy of this system with the gas dynamics equations is obvious. It is also clear that this system can be used for describing long-wave disturbances in the atmosphere if the length of disturbance Λ is much greater than the effective thickness of the air layer h:

$$\Lambda \gg h.$$

However, in order to describe the process from the viewpoint of the terrestrial observer, system (9.1.5) must be written in the revolving coordinate system. This revolving coordinate system is determined by the relations (9.1.1b) between the velocities **v** and **w**:

$$\frac{\partial h}{\partial t} + \operatorname{div} h\mathbf{w} = 0; \qquad (9.1.6a)$$

$$\rho h\left(\frac{\partial \mathbf{w}}{\partial t} + \left((\mathbf{w}\nabla)\mathbf{w}\right)\right) = -\rho g\nabla_{\perp} h + 2h\rho[\mathbf{w},\boldsymbol{\Omega}]. \qquad (9.1.6b)$$

Here g is gravitational acceleration, ρ is the density of the liquid, and when we discuss the atmosphere then it is some average density, $\nabla_{\perp}$ is the two-dimensional gradient. When writing equation (9.1.6b), the term in the right-hand side

$$\rho h\Omega^2\mathbf{r}^0$$

describing centrifugal action was rejected. This was connected with the fact that under the terrestrial conditions (and not only under these conditions) this term is much smaller than the term $\rho g\nabla_{\perp} h$.

The comparison of equation (3.1.10b) for the ionic component of the two-fluid model with equation (9.1.6) shows that they are identical.

9.1.4. Rossby parameter [225]

Among the plasma configurations there are many that are described in the first approximation by the drift system of equations, i.e. the scale of these configurations is greater than the Larmor ionic radius.

It was indicated in section 8.3 that the large-scale vortex flows are the analogue of this type of configurations in the atmosphere of planets or in large reservoirs (seas, oceans) (zone flows, cyclones and anticyclones). Such vortices were thoroughly investigated for the first time in connection with the Earth's atmosphere by Swedish geophysicist Carl Gustav Rossby in the 40s of the 20th century. He showed their large role in the general atmospheric circulation. Their specific role in the oceanic currents (Gulf Stream, Kuroshio Current, and others) was established somewhat later. Assuming for simplicity and with respect to the essence that the flow is stationary, equations (9.1.6b) can be written in the form

$$\text{div } h\mathbf{w} = 0$$

$$h(\mathbf{w}\nabla)\mathbf{w} = -g\nabla h - 2h\left[\Omega, \mathbf{w}\right]. \tag{9.1.7}$$

As already mentioned, the relative role of the flow inertia is determined by the Rossby–Kibel parameter (8.3.20)

$$R_0 = \frac{\left\{(\mathbf{w}\nabla)\mathbf{w}\right\}}{\left\{2\left[\mathbf{\Omega}, \mathbf{w}\right]\right\}} = \frac{\left\{w\right\}}{2\Omega\left\{L\right\}}. \tag{9.1.8}$$

Here { } indicates the scale of the quantity, and $\{L\}$ is the scale of the heterogeneity of the flow. The parameter R_0 is especially low for the atmosphere of giant planets (Jupiter, Saturn) or for the large water basins on the Earth. Here are the characteristic numbers for terrestrial reservoirs

$$\Omega = \frac{2\pi}{T} \approx 7.3 \cdot 10^{-5}\,\text{s}^{-1}, \quad v \sim 10\ \text{m/s}, \quad L = 150\ \text{km}.$$

we have $R_0 = 1$. The characteristic velocity for the air flow is ~30 m/s and then $R_0 = 1$ with $L \sim 500$ km. It can be seen that the considered values of v are quite high. However, one must take into account that here we consider the difference in the velocities over a large distance L.

We estimated the flow parameters for $R_0 = 1$. Actually, to be able to consider R_0 as small, its value must be ~0.1–0.2. At the same values of v the scale L must be increased 5–10 times. If $R_0 \ll 1$, then, as we saw in 8.3, system (9.1.7) can be reduced to a single Charney–Obukhov scalar equation (8.3.4).

Let us note that the most essential types of Rossby's structures have been analytically constructed and very thoroughly investigated as regards stability. It was shown, in particular, that the cyclones in contrast to the anticyclones are unstable and criteria were also obtained for the appearance of vortex structures under the action of a Rayleigh-type instability.

9.1.5. Nezlin's analogue experiments [225]

It is possible to create an installation in which processes will be described by the system of equations (9.1.7). The processes taking place in the installations of this type were studied most extensively by M.V. Nezlin with colleagues (Kurchatov Institute of Atomic Energy). Such analogue simulation was a highly useful addition to the observations in the natural conditions, analytical analysis and numerical calculations.

The basis of the described installation is a rotating parabolic[2] 'saucepan', in which a thin (~ several mm) layer of water of a constant height is produced. In order to cause zone flows or vortices in this layer, the part of the walls of this 'saucepan' can move (Fig. 9.1.5a). The detailed analysis of the work of installations was described in the previously mentioned review of the authors. Let us mention here only the following results of observations.

1. If we create a single vortex with the aid of a small revolving disk placed on the lateral side of the 'saucepan', then during a cyclone-like rotation the vortex rapidly decomposes during the time corresponding to the dispersion of linear waves. On the contrary, if the vortex resembles an anticyclone, then it exists for a long period of time without spreading until it is 'extinguished' by viscosity. In other words, we have here a soliton whose prolonged existence – in the language of the Charney-Obukov equation – is explained by the compensation of dispersion spreading by a non-linear term not found in the cyclonic regime.

2. Experiments with 'markers' (coloured specks) on the water surface showed that Rossby's soliton seizes the particles of the medium and makes them rotate in contrast to the KdF solitons. Another large difference between Rossby's and KdFF solitons lies in the fact that the collision of two solitons of the first type leads to their confluence, whereas in the second case they do not merge.

3. If two rings, placed at the wall and creating zone flows, are placed at a small distance fron each other (~1 mm), i.e. they model tangential rupture, then vortices of both the cyclonic and anticyclonic type are generated. However, if the distance between the generating

[2]More accurately, almost parabolic, in order to suppress a number of secondary effects, in particular the role of viscosity.

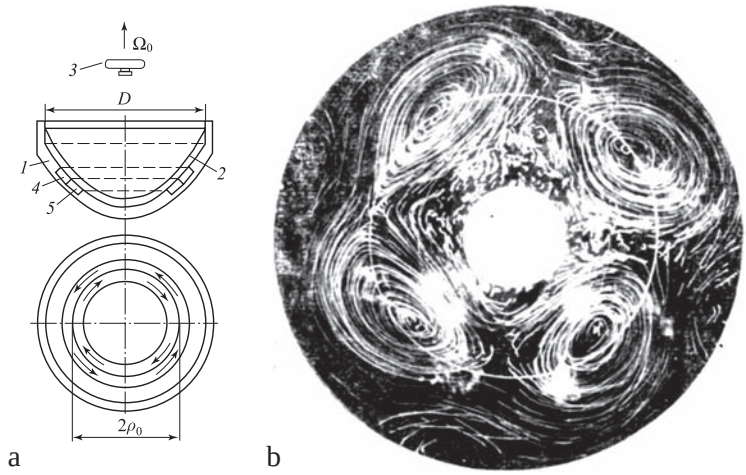


Fig. 9.1.5. Experimental equipment for investigating the instability of the tangential break of velocity in shallow water with a free surface: a) diagram of equipment, 1) a vessel with a parabolic bottom, 2) the surface of the liquid which spreads during the rotation of the vessel on the bottom of the vessel in the form of a flat layer, 3) photographic camera, 4, 5) ring-shaped sections of the bottom generating counter zonal flows, Ω_0 is the angular velocity of rotation of the vessel and the camera, ρ_0 is the radius of the tangential break in the velocity, ρ_0 = 10 cm, D = 28 cm. In the top view, the arrows indicate the direction of flows in relation to the vessel at a cyclonic jumping velocity; b) the pattern of flows of the shallow water layer with the free surface; the white circle, passing through the centres of the cyclones is the line of the tangential break of velocity [225].

rings is large, then only the generation of anticyclonic configurations is steady (Fig. 9.1.5b).

4. The number of vortices at the perimeter of the chain is determined by a drop in the velocity in the system of counterflows. Up to 10 anticyclones are observed at a relatively small drop. Their number decreases with an increase in the velocity drop.

Moreover, only one vortex is generated at a sufficiently high velocity – Rossby's anticyclone. The vortex is the result of self-organizing of the sequestered structure in the system of counterflows. The profile of flows radically changes with the appearance of the vortex and is adjusted slightly under that which is self-consistent with the vortex. In this case, the profile of the flow changes not only in the place where the vortex forms but also on the opposite side of the vessel.

One additional important property of the described sequestered vortex is that its maximum vorticity (velocity rotor) is several times greater than that of the circumfluent flow – it is analogous to what occurs in large vortices in the atmosphere of large planets.

9.1.6. Spiral structures in the galaxies

The majority of galaxies (including our Galaxy) have a distinctive spiral structure (Fig. 9.1.6). There are three main components in the structure of such galaxies: disk, almost spherical halo and corona (Fig. 9.1.7a). We are interested in the disk component – flat (in its major portion), round formation whose thickness is two orders of magnitude smaller than its radius. The contribution of the disk to the overall mass is small. The small part of its mass (for the Galaxy it is about 10%) is related to the gas – this is practically the entire gas existing in the Galaxy.

Specifically, the spiral arms are arranged in the disk which occupy the substantial part of its area. Arms are visible, first of all, as the regions of increased luminosity. They shine vividly in comparison with the remaining part of the Galaxy because the stars exist in them, and in the rest parts of the Galaxy contain no stars: the difference in the general concentration of stars in the arms and between them is comparatively small – only of the order of 10%. The contrast of the glow of arms is determined as by the formation of young massive stars in them which are considerably brighter than old ones. The brightness of the arms is determined by the glow of excited hydrogen, and also

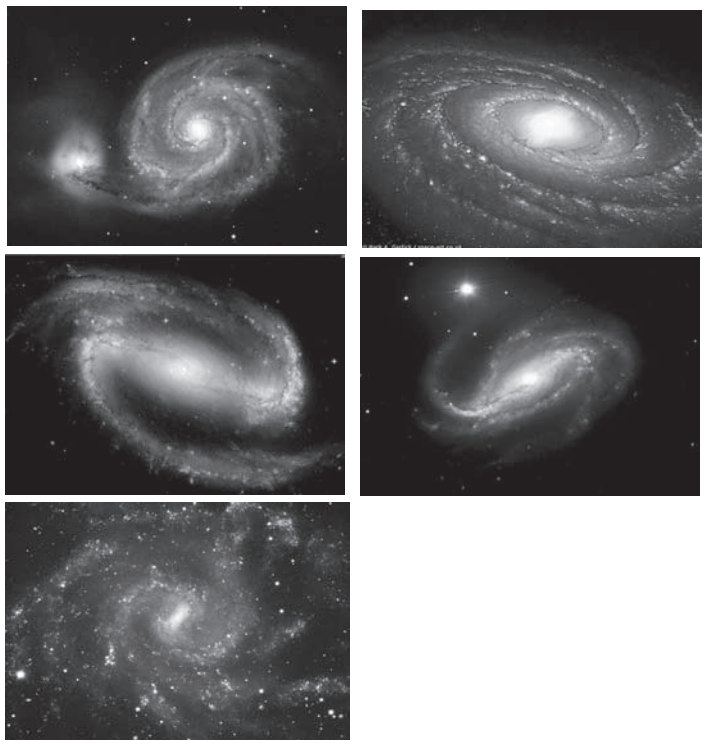


Fig. 9.1.6. Examples of typical spiral galaxies.

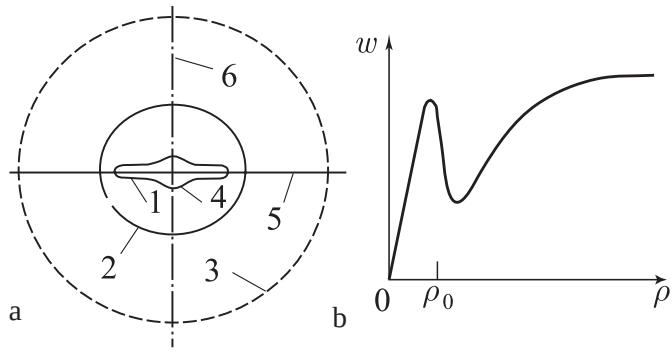


Fig. 9.1.7. Characteristics of a spiral galaxy: a) the axial section of the helices of the galaxy (kinematic); 1) the disk, 2) the halo, 3) the corona, 4) the bulge, 5) the central plane, 6) the axis of rotation, b) dependence of the velocity of rotation of the spiral galaxy w on the radius ρ ($\rho = \rho_0$ is the radius of the 'jump' in velocity).

by the glow of 'warm' molecular clouds of H_2. Observations show that almost entire gas is concentrated in the arms of the Galaxy – increased density of gas and ensures constant star formation in them.

Galactic disks are in the state of rotation, this rotation showing the clearly expressed differential nature: the angular rate of rotation of the matter of the disk very strongly depends on the distance to the centre of the Galaxy. The typical example of the radial profile of rotation, characteristic for the spiral galaxies, is shown in Fig. 9.1.7b. The profile of rotation of the galaxy is determined by the distribution of the gravitating masses. This distribution is such that the centre section of the disk is comparatively compact with dense thickening – the so-called bulge.

The gravitational potential, created by this region, increases from the centre to its edge beyond which it rapidly diminishes on the background a comparatively slow growth of the gravitational potential caused by the action of the remaining disk, halo and corona. As a result, the total gravitational potential is such that the graph of the dependence of the equilibrium linear velocity on the radius has the form of a N-shaped curve: the practically solid-state revolving centre section adjoins the relatively narrow section with the sharp decrease of velocity[3]. The rotation of the disk of the spiral galaxy in the first approximation can be considered stationary: the centrifugal force of rotation is balanced by the radial component of gravitational force. In other words, gravity is the factor of the equilibrium of the galaxy. The observations based on the Doppler effect show that the arms of a typical spiral galaxy are

[3]Typical values of ρ_0 are expressed in units of kiloparsec, w (at the maximum in the centre and on the plateau away from it) to hundreds of kilometres per second.

bent out back – down the side, reverse to the direction of motion of the matter of the disk. Such spiral arms are called trailing spirals. The opposite situation – leading of spirals – are not observed in the isolated galaxies. They are encountered extremely rarely: only in the systems of the interacting galaxies and even then only under specific conditions.

At first glance, the spiral form of galaxies corresponds to our intuition: any radial displacement of matter must twist into a spiral in accordance with the law of conservation of momentum. But if spiral arms were created as a result of the radial displacements of particles, then because of the strong differentiality of the rotation of matter they would disappear already during several revolutions of the galaxy. In fact, any such arm must be washed away in the time indicated since its part, closer down to the centre of the galaxy, revolves at a considerably higher velocity than its periphery. Astronomical observations show that the time of one revolution of a spiral galaxy is of the order of hundreds of millions of years, whereas the spiral pattern exists two orders longer. These data lead inevitably the conclusion that the spiral structures of galaxies revolve as solid-state formations. The difficulty of the interpretation of the solid-state rotation of the spiral structure against the background of the differential rotation of the matter of the galaxy led Jeans to the assumption that the phenomenon of the spiral structure of galaxies is determined by new, yet unknown physical laws. This resulted in the problem of the spiral structure solved by Heisenberg and Weizsacker, Chandrasekhar, Fermi. The fundamental step to the solution of this problem was made by Lindblaad in 1941. He advanced the concept, according to which the spiral arms are density waves, and the stationary (with time) form of the arms is explained by the fact that the density wave propagates around the axis of the galaxy with a constant angular velocity.

9.1.7. Installations for the simulation of the spiral structures of galaxies and experimental results

In principle, these installations were similar to those (described above), which were adapted for the simulation of the generation of atmospheric and plasma eddies by counterflows.

The basis of the experimental installations (Fig. 9.1.8a) is a round vessel, whose central section of the bottom (‘core’, angular rate of rotation Ω_1) revolves faster than the external (‘the periphery’, angular velocity Ω_2). The diameter of the core of the vessel was $2\rho_0 = 8$ cm, the diameter of the periphery $D = 30$ cm. The exterior of the vessel was either generally fixed or it revolved with a relatively low velocity – such that the Rossby–Obukhov radius r_R was only four times less than

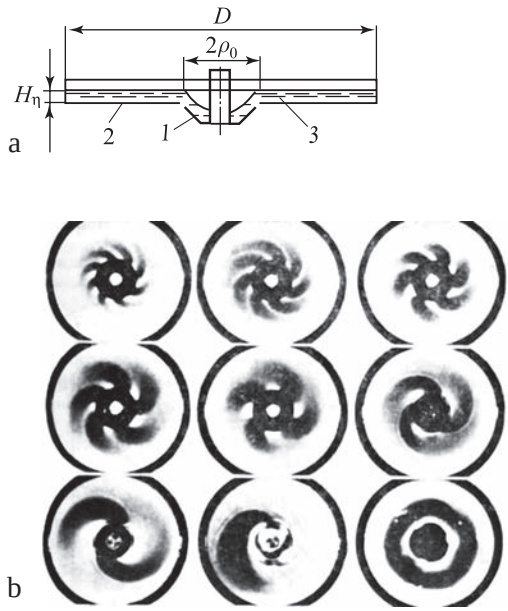


Fig. 9.1.8. Simulation of a spiral galaxy: a) diagram of experimental equipment for modelling the mechanisms of generation of spiral structures in gas disks of galaxies: 1) the core, 2) the periphery, 3) the layer of shallow water, b) spiral waves, excited in the differentially rotating water; modes m = 8–0 at $\Omega_0 = 0$; the core and the spiral pattern rotate in the clockwise direction.

the radius of the vessel. With this low rotational velocity the parabolic free surface of the liquid slightly differed from flat; therefore, the periphery of the vessel of this modification was made in the form of a hollow cone whose surface was inclined down to the horizontal at angle near 2°. The internal part of the vessel (‘core’) was also conical with the inclination of the generatrix to the horizon of 45°.

The following results were obtained in the experiments.

1. Differential rotation of shallow water with the velocity discontinuity between the rapidly revolving core and the comparatively slowly revolving periphery is unstable as with the low values of the number $M = (\Omega_1 - \Omega_2)\ \rho_0(gH_\eta)^{-1/2} < 1$, so with large $1 \leqslant M \leqslant 12$ (here Ω_1 and Ω_2 are the angular rates of rotation of the core and the periphery).
2. Development of instability leads to the appearance of spiral waves of surface density intersecting the break (region of the generator) and extending from it along the periphery and the core. In the steady-state regime, the waves form the symmetrical pattern of arms – elevations on the surface of liquid (mode $m \neq 0$). The pattern stationarily revolves in the same direction as the core.

These results obtained by M.V. Nezlin et al are in agreement with Lindblad's assumptions.

9.2. Magnetosphere of the Earth

Studies of the magnetosphere are of interest because of two reasons. On one side, this closest to us 'magnetoplasma sea' strongly influences our terrestrial matters while on the other side it is an impressive example of the complexity of the real plasma system which is formed, it would seem, by the simplest components: by the dipole field of the Earth, by solar wind and by the Earth's atmosphere under the magnetosphere. Of course, it is also necessary to consider the luminous flux (especially its ultraviolet component) emitted by the Sun.

From the atmosphere to the magnetosphere

The atmosphere – we will now speak only about the Earth – is a very complex laminar system (Fig. 9.2.1). In the previous paragraph we had in mind its lowest part – the troposphere. It would seem that above we should have expected a monotonic change of all its parameters and, first of all, density and temperature. Density actually monotonically falls, but temperature behaves non-monotonically. These are the results of absorption at specific altitudes by other air molecules of 'resonance' bands of the solar spectrum. But, nevertheless, to altitudes of ~100 km the elementary air composition remains almost the same as near the Earth's surface. In the range of 50–120 km, due to the absorption of ultraviolet radiation and partially due to the corpuscular flux from the Sun, two zones with the noticeable (~ 1%) ionization appear. These are the so-called *D*- and the *E*-layers of the ionosphere. New ionized layers appear at the altitude of 150–300 km appear: F_1 (medium altitude ~of 170

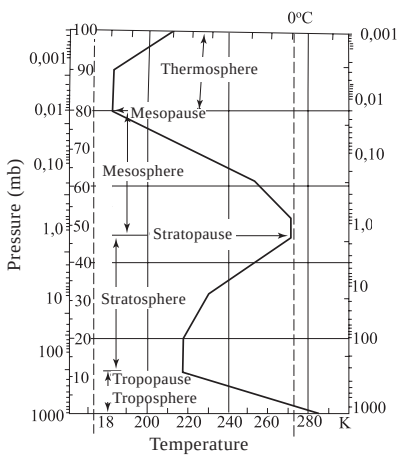


Fig. 9.2.1. Structure of the atmosphere and the vertical distribution of temperature in the atmosphere.

km) and F_2 (medium altitude of ~250 km). Electron density for the quiet Sun in the D-layer is $n_e^{(D)}$~300 cm^{-3}, in the E-layer $n_e^{(E)} \sim 1.3 \cdot 10^5$ cm^{-3}, in the F_1-layer $n_e^{(F1)} \sim 2.4 \cdot 10^5$ cm^{-3}, and in the F_2-layer $n_e^{(F2)} \sim 6 \cdot 10^5$ cm^{-3}.

Layers E and F_1 reflect radio waves in accordance with $f \leqslant 4$ MHz and $f < 10$ MHz.

At altitudes of more than 100 km the gas composition of the atmosphere begins to change: the fraction of atomic oxygen begins to grow rapidly. Above 110 km the fraction of atomic oxygen is greater than that of molecular O_2, and at the altitude of ~300 km ionized hydrogen becomes dominant (first on the night side). At altitudes of the order of thousands of kilometres the ionosphere changes to the magnetosphere – a large region in which the fraction of the ionized component rapidly grows, approaching 100% of the total number of heavy particles.

9.2.1. Concept of the 'magnetosphere'

The high degree of ionization in the magnetosphere is due to UV radiation of the Sun and also to the plasma of the solar wind which penetrates partially into the magnetosphere, in spite of the magnetic 'shell' of the Earth. It is natural that the practically completely formed ionized plasma begins 'to obey well' the magnetic field, and the currents formed in the plasma start to influence the configuration of this field. Thus, the special self-consistent magnetoplasma configuration forms. This is the magnetosphere of the Earth, streamlined by the solar wind[4].

Prior to describing in detail the physics of the magnetosphere, let us note the role of magnetospheric (and the closely related ionospheric) phenomena in the life of earthlings.

– One of the most colorful and impressive celestial spectacles are aurorae polares[5].
– 'Magnetic storms' (they develop in the magnetosphere and the ionosphere) strongly influence the organisms.

[4]Magnetospheres are also found on planets with no intrinsic magnetic field. They form due to the intrinsic magnetic field of the solar wind.

[5]The sky glowed. An endless transparent veil covered the entire sky. Some invisible force shook it. All of it was burning soft purple light. Here and there appeared bright flashes and then turned pale, as if for a moment clouds, woven from a single light, were born and then scattered. Through the veil the stars shone brightly. Suddenly the fog disappeared. In some places, the purple clouds again broke out. A split second later it seemed that the glow was extinguished. But then long rays gathered in the bright beams, fluttering pale green. Here they were getting away on all sides, quick as lightning, rushed to the zenith, for a moment stood still in the sky, formed a huge solid crown, flared and dimmed. (G.A. Ushakov) [228].

– Presence of the ionosphere makes possible radio communications almost on the entire Earth. The magnetic storms, agitating the ionosphere, can not only interrupt, but can also cause in exceptional cases the failure of electric power lines.
– The magnetosphere contains the so-called radiation belts from the energetic protons (ε_p to 800 MeV) and electrons (ε_e to 500 keV). This excludes prolonged stay in them of automatic spacecraft (with semiconductor electronics) and of course of spacecraft with the people in these belts. The influence of the energetic electrons, which generate γ-rays in collisions with equipment is especially strong.
– At present, the magnetosphere is the test range for active plasma experiments (see below).
– The magnetosphere is very sensitive – as this is not paradoxical – to the deep processes in the Earth. A procedure of the prediction of earthquakes on the basis of the fixation of specific disturbances in the magnetosphere is being developed.

And this list can be continued for a long time.

The magnetosphere is thus far the only space scale object where the developed MHD processes can be studied. This is very important for the progress of plasma dynamics and astrophysics.

The basic stages of studies of the ionosphere and magnetosphere

The Earth was referred to as a large magnet, as is known, by the court doctor of Queen Elizabeth I W. Gilbert in the book 'About the magnet, magnetic bodies and the large magnet – the Earth' (1600). In this case, he considered the Earth as a magnetized iron sphere. In the 19th century it was suggested that the Earth is a 'turn with current', but models of the generation of this current by some hydromagnetic dynamo were built only in the middles of the 20th century. However, today (2005) there is still no conventional model of terrestrial hydrodynamo.

The presence in upper layers of the atmosphere of the ionized component was proven at the beginning of the 20th century as a result of the development of radio communications and, first of all, in connection with the epochal achievements of G. Marconi who established in 1901 radio communication between Europe and America. The main aspect in this achievement was the proof of the fact that comparatively short (~1 km) waves could go around almost a quarter of the globe. The scientists of that time, when Marconi was preparing for his communications proved that diffraction in this case – and this was the only known factor to them which worked in favour of Marconi, cannot

ensure success. However, the enthusiast Marconi did not listen to them and completed this miracle at that time.

This phenomenon unexpected from the viewpoint of optics was explained by A. Kennely and O. Heaviside (1902) who assumed the existence above the Earth of a conducting layer – screen, which reflects radio waves. This semi-hypothetical layer was called the Heaviside layer for a long time. By the way, the scientific society did not accept Heaviside's idea almost to the middle of the 1920s, until direct experiments showed that short waves do not travel along the Earth's surface but they come from the top (!).

Heaviside's proposal stimulated the study of the properties of the ionosphere and with it of the magnetosphere with the aid of sounding them by radio waves with different frequencies. It is possible to say that towards the end of 1940s the science of radiowave propagation in these spheres was developed to a high level of perfection, and these studies were in many respects the base of the subsequent general wave theory in plasma[6].

In parallel with this, the properties of the magnetosphere were naturally clarified. By the way, the term 'magnetosphere' was introduced in 1959 by T. Gold.

Together with radiophysics, another stimulus for studying the magnetosphere was the aurorae polares. Already at the beginning of the 1910s Swedish scientists attempted to understand in nature, in particular, by directing an electron beam to the magnetic dipole. Somewhat later two theorists, Chapman and Ferraro constructed a simple model of reflection by the magnetic field of the plasma flow dipole since the existence of the solar wind, at least as a factor swinging Earth's magnetic field in the periods of the growth of Solar activity and causing intensive aurorae polares was acknowledged. However, up to until 1950 the question remained open: does the solar wind blow continuously or only during the flashes. In 1950 L. Birman, analyzing the dynamics of the tails of comets, gave convincing arguments in favour of continuous existence of the solar wind. In 1957 E. Parker theoretically proved the fundamental dynamicity of the solar corona and thus supported the Birman's conclusions.

The new stage of studies of the magnetosphere of the Earth began in the 60's, first of all, in connection with the launch of apparatuses into cosmos. Such apparatuses, equipped with different diagnostic means (electrical and magnetic probes, antennas, spectroscopic instrumentations and the like) made it possible to draw in detail the structure of the

[6]It is important to mention here studies by V.M. Ginzburg and his monograph 'Propagation of electromagnetic waves in plasma' [28].

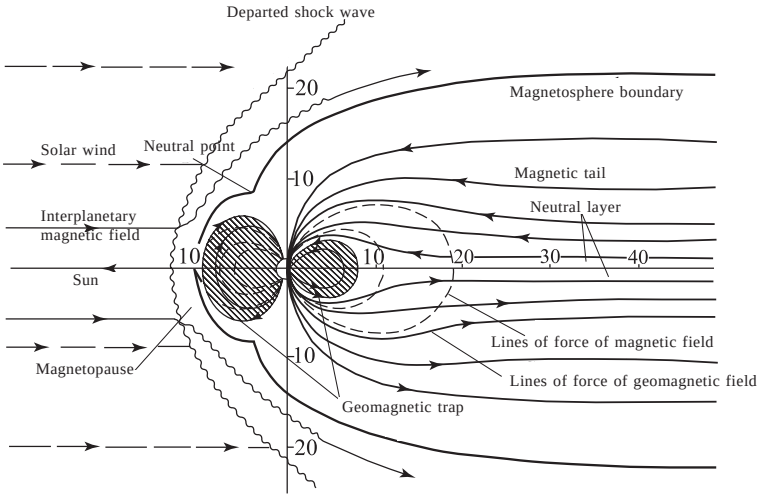


Fig. 9.2.2. General scheme of the magnetosphere.

magnetosphere and its connection with the Earth's atmosphere and the solar wind.

Together with the appearance of space technology, the rapid progress in studies of the magnetosphere was connected with two more circumstances. First of all, it was necessary to note the beginning of rapid development of plasma physics, connected, as it was mentioned earlier, with Alfvén's book 'Space electrodynamics', and also studies of the problem of controlled thermonuclear fusion. The second circumstance is the practical requirement for reliable knowledge of the 'new living environment' of apparatuses and man – the magnetosphere.

And the last observation. The basic content of this paragraph is devoted to the magnetosphere of the Earth.

9.2.2. Characteristic properties of the magnetosphere [227, 229]

The magnetosphere of the Earth is a cavity in the solar wind determined by the Earth's magnetic field. The exterior view of this cavity is schematically depicted in Fig. 9.2.2. On the side facing the Sun the distance of the magnetosphery boundary from the Earth is ~60 000 km, the side of the cavity is found at a distance to ~500 000 km, and the tail stretches to many millions of kilometers.

In the first approximation, the boundary of the magnetosphere (it is call 'magnetopause') is determined from the condition of the equality of the gas kinetic pressure of the solar wind[7]

[7]Here we do not take into account the force effect of the magnetic field frozen in the solar wind since it is relatively small.

$$p^{(SW)} = (\rho v_n^2 f + p)_{SW} \qquad (9.2.1a)$$

and the pressure of the magnetic field in the magnetosphere

$$p^{(M-S)} = \left.\frac{H^2}{8\pi}\right|_{M-S}. \qquad (9.2.1[b])$$

In (9.2.1a) ~v_n is the component of the velocity of solar wind normal to the magnetopause, f = 1–2 is the elastic ratio of reflection, p_{SW} is the pressure is the solar wind. In the part facing the Sun the strength of the field is ~10^{-3} Oe, which is noticeably more than the magnetic field of the terrestrial dipole at the given distance. It is obvious this is the result of the currents which flow in the magnetopause. The parameters of solar wind in the Earth orbit in the case of the quiet Sun are given in Tables 9.1 and 9.2.

The structure of the magnetosphere is complex, and it always changes both due to the rotation of the Earth around the axis and the orbital movement.

Therefore, let us examine the simplest case, when the magnetic axis of the Earth and the Sun–Earth radius-vector are perpendicular to each other. In this case the picture of magnetic lines of force in the plane passing through the magnetic axis and the Sun has the form shown in Fig. 9.2.2, and the volumetric structure of the field is shown in Fig. 9.2.3. As is evident, 4 dividing surfaces are separated on the day side. They are 1 – a shock wave decelerating the solar wind (SW), then 2 – 'magnetopause' where the Earth's magnetic field, compressed by SW,

Table 9.1.

Velocity	400 km/s
Proton concentration	~6 cm^{-3}
Proton temperature	$5 \cdot 10^4$ K
Electron temperature	$1.5 \cdot 10^5$ K
Magnetic field strength	$5 \cdot 10^{-5}$ Oe
Proton flux density	$2.4 \cdot 10^5$ cm^{-2} s^{-1}
Kinetic energy flux density	0.3 erg cm^{-2} s^{-1}

Table 9.2.

Components of the solar wind	%
$^1H_1^+$	96
$^4He^+$	4
$^3He^+$	$1.7 \cdot 10^{-3}$
O^+	$5 \cdot 10^{-2}$

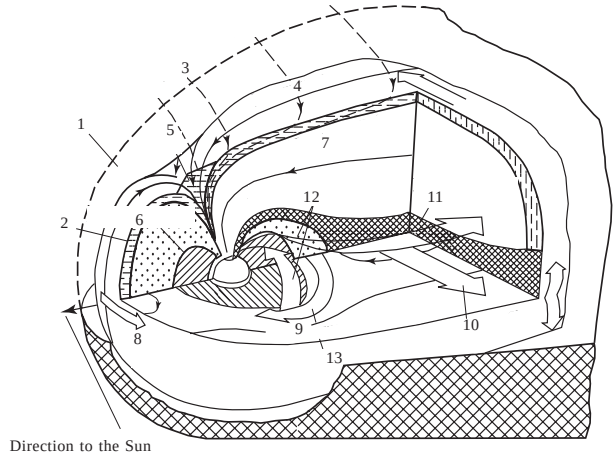


Fig. 9.2.3. Volume scheme of the magnetic field and fluxes in the magnetosphere. Light wide arrows indicate the fluxes in the magnetopause and in the plasma layer, ring-shaped and longitudinal fluxes. 1) magnetic transition layer; 2) entry layer; 3) interplanetary magnetic field; 4) magnetopause; 5) cusp; 6) plasmosphere; 7) plasma mantle; 8) flux to magnetopause; 9) plasma convection; 10) flux across the tail; 11) plasma layer; 12) longitudinal flux; 13) ring-shaped flux.

ends. Boundary 2 adjoins the input layer, which penetrates the plasma directly from SW. The following interface is the 'plasma pause' (3). This is the boundary between the 'almost empty' zone of the magnetic field and the plasmasphere, which is supplemented by particles mainly from the ionosphere, and finally surface 4 – the boundary between the plasmasphere and the ionosphere. The transition from the weakly ionized plasma of the ionosphere to the highly ionized plasma of the plasmosphere takes place at this boundary.

The region between the shock wave (1) and the magnetopause (2) is called ' the magnetic transition layer', in which the magnetic field of SW changes to the field of the magnetosphere.

Its tail is the characteristic property of the magnetosphere. It is formed by two magnetic fluxes with opposite directions, divided by a neutral layer, in which the current flows as shown in Fig. 9.2.3.

The magnetic lines of force on both the day and evening side enter the Earth's atmosphere predominantly in the region of magnetic poles. Here are expanding funnels (cusps), formed by the lines of force, which divide them into those which pass across the day side and those which depart down the night side. Thus, this is the separatrix zone.

The figures show that the separatrix zone on the day side extends to the magnetopause and solar wind. In turn, near the Earth this separatrix at the quiet Sun is situated in the vicinity of ~70° of the geomagnetic

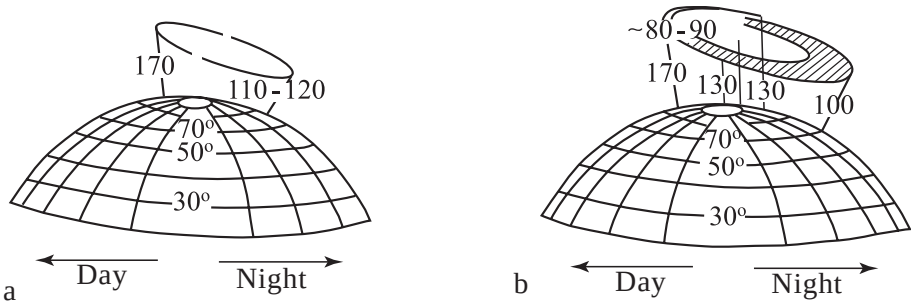


Fig. 9.2.4. The Northern auroral oval of the Earth: a) in the form of a narrow ring in the magnetically quiet periods and b) in the form of the crosshatched region in the magnetically disrupted periods. The numbers indicate the altitude of the oval above the surface of the Earth.

latitude (i.e. near the magnetic pole). First, this was discovered via the analysis of observations from different points of the Earth, and then it was directly photographed from cosmos, showing that aurorae polares were concentrated in the zone of the separatrix at a distance of ~100–150 km above the Earth's surface (Fig. 9.2.4). This beautiful in every respect fact is understandable. Particles from the solar wind penetrate ino the magnetosphere through the cusps and due to the drift and convection in the non-uniform field they spread, although quite chaotically (see Ushakov's story, page 479) across the entire azimuth. Zones of aurorae polares were called 'auroral ovals'.

The sides of ovals can be comparatively narrow – of the order of a kilometer – during the calm magnetic situation, and they can be enlarged to thousand of kilometers during strong magnetic storms and descent down to the south, seizing Scandinavia and more southern regions. The altitude at which the oval is situated is connected with the atmospheric density with the atmosphere which glows.

On the dynamics of plasma around and in the magnetosphere[8]

The plasma flows in the vicinity and inside the magnetosphere are varied. Even if we limit ourselves to the periods of the quiet Sun, even then the picture is very complex. Therefore, without attempting to draw any complete picture, let us note a number of the specific special features of the dynamics of the plasma flows, generated directly by solar wind.

(a) External part of the magnetosphere

The plasma flow, which goes from the Sun, is of the super-signal type, i.e., supersonic and super-Alfvén (Mach number ~7); therefore, braking

[8]It should be mentioned that section 2.7.2 contains the results of numerical calculations of the two-dimensional pattern of flow around of the magnetic filed by a supersonic flow.

of the flow in the magnetic field of the Earth is connected with the formation of a shock wave. The thickness of the front of this wave in the region facing the Sun is 100–200 km. This wave is collisionless, since the mean free path of the particles before and after the passage of front is ~100 million km, which is incommensurable with the observed thickness of the front. The distance between the shock wave front and the magnetopause – in the region facing the Sun – is ~20000 km and approximately in the layer of this scale the braked plasma of solar wind flows around the magnetosphere. In the region facing the Sun, the distance of the magnetopause from the Earth is ~70000 km.

After passing through the shock wave, the plasma is heated to temperatures T of ~100–300 eV. Together with the inflow of plasma into the cusps from the transitional magnetolayer there is also 'direct' penetration of plasma into the magnetic field of the magnetosphere, forming the so-called input layer. As a whole, this is an irregular process significantly determined by interaction of the magnetic field, brought in by solar wind, with the field of the magnetosphere. The mechanism of the reconnection of magnetic lines of force plays an essential role here.

Let us discuss in greater detail the role of the process of reconnecting the magnetic lines of force at the seizure of plasma by the magnetosphere. This mechanism, equivalent to the tearing mode, was proposed in 1961 by astrophysicist Dungey. Its idea in essence is simple (Fig. 9.2.5) and is reduced to the following. Solar wind carries the magnetic field. Therefore, after the passage of the shock wave the magnetic field of

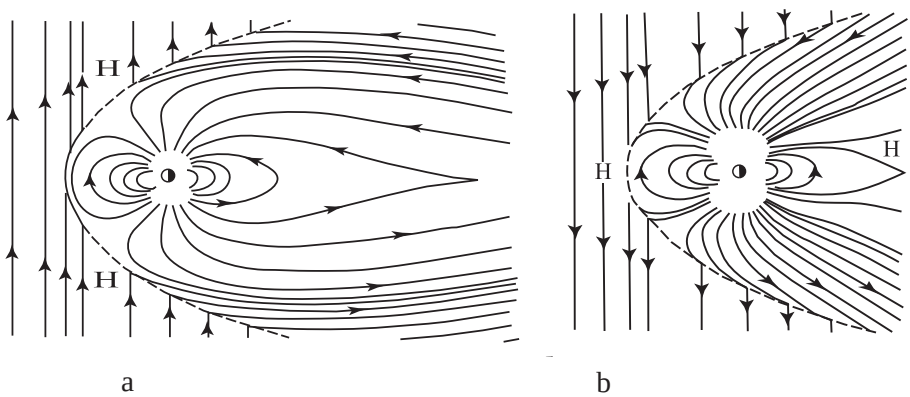


Fig. 9.2.5. The Dungey model of the interaction of the magnetic field of solar wind with the magnetic field of the Earth: a) the fields have the same direction; b) the directions of the fields differ. H denotes the regions in the vicinity of the central lines of the magnetic field.

solar wind enters the zone of the magnetic field of the magnetosphere and the two fields interact. For clarity, we will consider that the field lines of force in the incident flow lie in the same plane as in Fig. 9.2.3 and are directed strictly either from top to bottom (Fig. 9.2.5b), or vice versa (Fig. 9.2.5a).

As is evident in Fig. 9.2.5b, an x-point appears in this case, and the solar wind zone is connected with the inner zone of the magnetosphere. In this case, the depth of penetration of the external field (i.e. the position of the x-point) in the magnetosphere increases with increase of the size of the wind field. However, if the magnetic field of the wind is directed from bottom to top (Fig. 9.2.5b), then there is no rearrangement of the morphology of the lines of force in the zone of interaction of different magnetic fields. This confirms well observations made in space (motion of magnetic field in the solar wind) and simultaneously on the Earth (disturbance of the magnetic field of the magnetosphere, aurorae polares).

(b) Flow of plasma in the volume of the magnetosphere and in the tail

Entering the field of the magnetosphere – outside the cusps, on the day side – plasma moves down the night side in the near-surface layer under the action of a number of factors and including viscous forces on the side of the circumfluent solar wind. It goes without saying that the entrance of plasma into the magnetosphere and its further motion are influenced by the electrostatic fields which appear because of the finite dimensions of the magnetosphere in the direction perpendicular to both $\mathbf{H}$ and $\mathbf{v}_{SW}$. This means that the situation, described in section 3.8.3, can form here

But, one way or another, the plasma falling on the night side of the magnetosphere, as shown by the observations, only partially departs into the tail, and its part together with the frozen-in magnetic field drifts to the day side. It would seem natural to expect the tranquil flow of plasma, at least in the tail. However, it is justified to assume that this bipolar magnetic flux is unstable with respect to the tearing mode, and that reconnection, as shown in Fig. 9.2.6, also takes place in this flux. As a result, due to the tension of magnetic lines of force part of the plasma from the tail is ejected towards the side of the Earth, and part into more distant zones of the tail, contributing to its lengthening. This model was also proposed by Dungey.

However, in addition to this diagram another mechanism of the overflow of plasma from the night to the day side was also discussed – simply due to the pressure in the transferred plasma and the frozen-in field connected with it. This is the model proposed by Hines. The

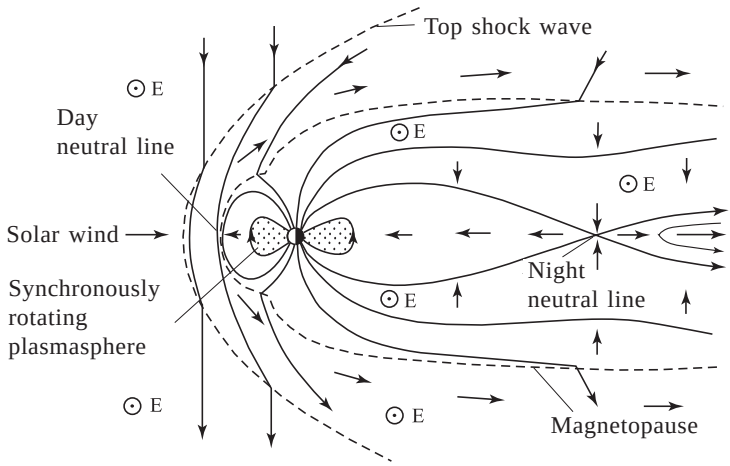


Fig. 9.2.6. Reconnection of the magnetic lines of force in the tail of the magnetosphere.

assumption about work of both mechanisms is very probable. The already given quick overview of several fragments of the dynamics of plasma and the field in the magnetosphere shows that things are not simple there [230].

9.2.3. Radiation belts

There is another additional special feature which proved to be completely unexpected and was discovered when the satellites were fitted with the devices for recording fast particles. This special feature are radiation belts which were discovered in the plasmasphere in 1985 by the Soviet researcher of cosmic rays Vernov and by American Van Allen independently but practically at the same time.

These observations, augmented by the subsequent thorough studies, showed that the magnetosphere is a very effective particle accelerator and also a retaining mirror trap for the particles (Fig. 9.2.7a). In this case, if earlier we discussed the plasma which fills the magnetosphere and considered particle energies of $\leqslant$100–1000 eV and densities of $\leqslant$10–12 cm^{-3}, then in the radiation belts, although the density of energetic particles was small ($n^{(p)}_{\max} \sim 10^{-3}$ cm^{-3} for the protons and $n^{(e)}_{\max} \sim 10^{-4}$cm^{-3} for the electrons, with the quiet Sun), the energy of protons reaches many hundreds of MeV, and that of the electrons many hundreds of keV. In addition to the protons and electrons, heavy ions (He^+ and other) were also detected.

The given values of the densities of energetic particles in the belts can seem to be very small in comparison with the particle density in the solar wind or the magnetosphere itself. But this impression is deceptive

since these densities must be compared with the density of cosmic rays, which is ~10^{-10}–10^{-11} cm^{-3}. A more adequate characteristic is the flux N of particles in 1 cm^2 in 1 s. So, if for the cosmic rays $N \sim 1$ cm^{-2} s^{-1}, then in the radiation belts this value for the megavolt equation is 4–6 orders of magnitude higher. The zones of the relatively high concentration of high-energy particles actually form belts. The internal belt, predominantly proton, contains particles with the energy ~30–800 MeV. It is restricted by the magnetic surface whose lines of force intersect the Earth's surface at ±40° of the geomagnetic latitude. The basic volume of this belt is located at a distance 600–1500 km from the Earth's surface. Figure 9.2.7b show the lines with one and the same flux (in 4π) of particles with energy $\varepsilon_p > 30$ MeV. The numbers near these equilines indicate the fluxes [N] particles/cm^2·s in the calm conditions. Analysis showed that the more energetic component is located nearer to the Earth. This is natural since the greater the energy the greater the Larmor radius and the greater the diffusion coefficient. This means that the energetic particles are retained much less efficiently than the less energetic particles in the regions remote from the Earth, where the strength of the magnetic field is lower. The external radiation belt is sometimes called 'electronic'. This belt is found between the magnetic surfaces formed by lines intersecting

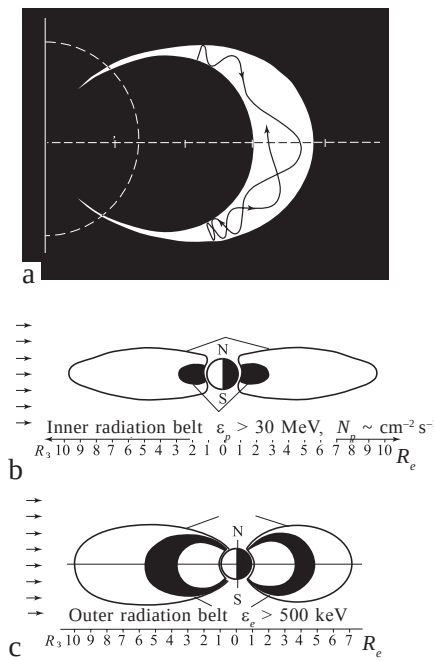


Fig. 9.2.7. Radiation belts of the Earth: a) the trajectory of the particles in a radiation belt, b) the inner (proton) belt, c) the outer (electron) belt. The distance is shown in the radii of the Earth R_e.

the Earth's surface on the geomagnetic latitudes ± 50–70° (Fig. 9.2.7c). The intensity maxima of the proton fluxes with the energy $\varepsilon_p > 1$ MeV and $\varepsilon_p > 30$ MeV were located at the distances of respectively of $2 \cdot 10^4$ and 10^4 km (Fig. 9.2.7b)

The retention of particles in these belts is typically of the 'mirror' nature and is explained by the increase of the strength of the magnetic field with the approximation along the line of force to the Earth. The calculation of the dependence of the plasma pressure in the magnetosphere on 'the number' of the magnetic surface shows that criterion (8.1.10) of convective stability is fulfilled here.

The question about the mechanisms of collection of energy by the particles forming the belt is very complicated. Schematically, the present-day picture of acceleration can be described as follows.

The particles (first of all, protons) as a whole are found in circulation which begins in the upper layers of the atmosphere. Here, acquiring energy of the order of several eV, the particles move through the polar regions into the plasma tail of the magnetosphere and because of the continuous pulsations in this zone increase their energy to several keV. From the tail they then return to the outer boundary of radiation belts where, because of the large nonstationary *E*-fields, their energy still increases by 1–2 orders of magnitude, and because of the scattering on the particles and the fluctuations, protons enter the region of the strong Earth's magnetic field and here an increase in the energy continues due to the induction electric field. There the proton energy reaches many hundreds of MeV. However, the particles gradually fall into a 'withdrawal cone' and fall onto the Earth. The rate of withdrawal of particles sharply grows during magnetic storms. Taking into account that the belts are restored within several days, it is possible to assert that this is also the scale of lifetime of the particles in the belts. It is clear that they travel enormous distances during this time. However, it should be emphasized that the described multistage scheme of filling of the radiation belts is hypothetical in many respects.

Moreover, the recently described model cannot be considered as the only one. Long ago it had been indicated that the fastest protons form by the spontaneous decay of the neutrons

$$n \rightarrow p + e + \nu,$$

in the collisions of cosmic particles with the particles of the magnetosphere. The collisional capture of cosmic particles and so forth can also take place.

9.2.4. Active experiments in the magnetosphere

Active experiments are usually the experiments carried out by introduction into the investigated system of specific disturbances and observations of the reactions of the system to these disturbances.

To study the magnetosphere, the latter was subjected to the action of radiowaves, electronic and plasma beams, plasma ('baric') clouds, moving solid objects: meteorites and space systems. To this it is necessary to add observations of the reaction of the magnetosphere under the influence of variable space factors: variation in the solar wind, fall of the meteorites and the like.

The explosions of A-bombs were the most immense active experiments in the upper layers of the atmosphere. The first explosion was produced by Americans in August 1958 above the Johnson coral atoll in the Pacific Ocean. One of the unexpected results of this and similar explosions was the overcrowding of radiation belts which lasted for more than a year. However, 'calm' experiments provide a large amount of information for science. We will now examine them briefly.

Radiosounding. It has already been mentioned that the special features of radiowave propagation in particular led to the discovery of the ionosphere ('Heaviside layer') and yielded the primary observation data about the inhomogeneity and variations in the magnetosphere. The same data stimulated theoretical works on the propagation of waves in plasma.

Electron beams in the magnetosphere. Ideas about particle dynamics in the magnetosphere were refined in a number of experiments with the injection of electron beams into the magnetosphere. Let us describe briefly some of them – the Franco–Soviet experiment by the name 'Arax' [239] (on the Soviet side, the head of the experiments was R.Z. Sagdeev, and on the French side F. Kambu). Its diagram is shown in Fig. 9.2.8. The line of force of Earth's magnetic field, which fixes two magnetically coupled points on the Earth's surface, is shown here: one – on the French island Kerguelen in the Indian Ocean, and the second – in the Arkhangel'sk province near the Sogra settlement. A rocket was launched from the Kerguelen island which carried an electron beam gun to an altitude of ~100 km. The gun began to send electron pulses with the energy of ~12 keV and the total current ~1 A. After being 'held' at the point of the line of force, these electrons rose to an altitude of ~36 000 km and, after describing an arc with the overall length of ~100 000 km, 'they were poured out' in the upper layers of the atmosphere, after producing a shining strip flickering with the frequency of the gun and radio noises. True, radiance was weak, but reliably fixed

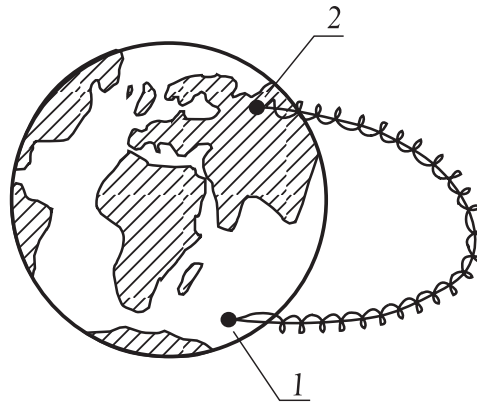


Fig. 9.2.8. Diagram of Arax experiment: 1, 2) the magnetocoupled points; 1) French island Kerguelen, 2) Arkhangel'sk region.

with the aid of a light amplifier. This result speaks about the relatively weak 'failure' or 'fall' of electrons from the calculated trajectory under the action of collisions and different fluctuations. The Arax experiment was carried out in 1975.

'*Baric clouds'*. Very valuable information was obtained in the experiments with the so-called 'baric clouds', proposed by I.S. Shklovskii. These clouds are created via explosions at the relevant altitude of the containers with metallic barium transported into the magnetosphere by geophysical rockets. During the explosion, a cloud of barium vapours forms and is rapidly ionized by solar radiation and converted to plasma clouds. The resultant Ba^+ ions are again excited by Sunlight and they begin to emit the characteristic green line which is clearly visible from the Earth. Experiments with such clouds make it possible to observe the orientation of the magnetic lines of force (luminous plasma is pulled apart along the lines of force of the magnetic field), the special features of the diffusion of plasma across the magnetic lines of force (i.e. the explanation of the role of classical and different kinds of anomalous processes), the presence of a transverse electric field (on basis of the drift velocity of the cloud across the magnetic field) and so forth.

Artificial plasma flows. A large number of processes are initiated in operation of electrically powered propulsion systems of different types in the magnetosphere. Such engines emit a compensated (quasi-neutral) ion flux with the energy of hundreds and thousands of eV.

It was noted in section 3.8 that the quasi-neutral plasma flow can move across the magnetic field because of the phenomenon of polarization, i.e. appearance on 'lateral' sides of the flux of the positive and negative charges which create an electric field in the volume of the

flux whose value is self-consistently established on this level so that the velocity of the electrical drift is equal to the 'initial' velocity of the flux

$$c\frac{E}{H} = v_{\text{flux}}.$$

The same situation appears when the quasi-neutral (plasma) flow travels from the EPPS into the magnetosphere. However, it is easy to see that in the magnetosphere the field of polarization charges either attracts (to the positively charged side) electrons from those sections of the magnetosphere which lie near the magnetic surface which passes near the positively charged side of the flow, or drops them, if the negatively charged side is considered. The result of these interactions is weakening – with movement away from the source of polarization of beam, i.e. retardation of the flow as a whole and its transformation to the beam of 'independent' ions, the space charge, which is now already compensated by the electrons of the magnetosphere. Here, this capture of magnetospheric electrons and disintegration of the beam is accompanied by the buildup of intensive fluctuations whose spectrum makes it possible to determine the local characteristics of the magnetosphere.

Another matter is if the plasma flow is injected along the magnetic field. In this case, the degree of polarization of the flow is considerably lower and fluctuations swing more slowly.

The aforesaid has been confirmed quite well by numerous observations of fluctuations in the magnetosphere with the aid of antennas, installed both on the automatic spacecraft with EPPS and on the Earth. A special study program was also prepared for the dynamics of the flow and fluctuations, caused by EPPS in international experiments Porcupine[9]. The processes in the jet and its environment were observed by several space apparatuses with a large number of probes.

However, the matter did not end by 'the scattering' of the beam and a burst of HF emission. The formed flow of free ions moved under the action of one magnetic fields and, as is easy to see, the beautiful phenomenon of the focusing of ionic fluxes, which was already discussed in section 1.4 (Fig. 1.4.6), was observed here.

Planned active experiment [232]

One active experiment, which was prepared by a number of Russian institutes at the beginning of 1990 will be described here. However, in 1992 studies they were ended because of financial difficulties. It was

[9]The supervisor of the experiments on the Soviet side was R.Z. Sagdeev, on the German side H. Haerendel [240]. A large contribution to data processing was made by V.N. Oraevskii and his colleagues [241].

planned using a powerful source of ultraviolet radiation to illuminate a large volume of the ionosphere and greatly increase the number of excited atoms and molecules.

Taking into account the interesting scheme of the planned experiment, let us describe it in more detail. It was proposed to lift a radiating complex by a standard meteorological rocket MR-12 to an altitude of 100–140 km. The source of UV emission was a short-pulse erosional MPC (magnetoplasma compressor) with a discharge current of $\sim 2 \cdot 10^6$ A and a pulse time of ~10 μs. The special features of the rocket imposed limitations on the mass of the radiating complex (it had to be $\leqslant$125 kg) and its overall dimensions (diameter $\leqslant$150 mm, length $L \sim 700$ mm). Naturally, neither capacitive nor inductive accumulators were suitable because of their mass, since the general reserve of energy must be of the order of several hundreds kJ.

Here the sole energy source can be only the explosive magnetic generator (EMG)[10]. Let us describe its principle and the circuit which was assumed to be the basis of the constructed system.

Explosive magnetic generator (EMG)

The idea of the EMG is simple (Fig. 9.2.9). Let us visualize two cylinders with sufficiently high conductivity and assume that the internal cylinder is filled with an explosive substance (ES), and the external one is securely fastened. Then, if a magnetic field H_0 is generated in the gap, then after explosion the gap will decrease and to maintain the flow of field $H(t) = H_0 \dfrac{R^2 - a^2}{R^2 - r^2}$ will increase. Here R is the radius of the external cylinder, a is the initial radius of the internal cylinder, $r(t)$ is the current radius of the expanding internal cylinder.

If the external cylinder is sectioned along the helixes, we obtain a powerful current generator.

Let us note that record magnetic field strengths $H \sim 20 \cdot 10^6$ Oe were obtained using the EMG.

Radiating complex as a whole

The scheme of the complex as a whole is depicted in Fig. 9.2.11. The EMG block is on the left, and on the right is the erosive MPC. Each of these blocks was thoroughly developed both separately and in the complex. The fundamental characteristics of the radiating complex are

- mass of the explosive 1.5 kg
- maximum current 1.8×10^6 A
- electrical power scattered in plasma 15 GW
- discharge emission 100 kJ

[10]Generators of this type were proposed by A.D. Sakharov.

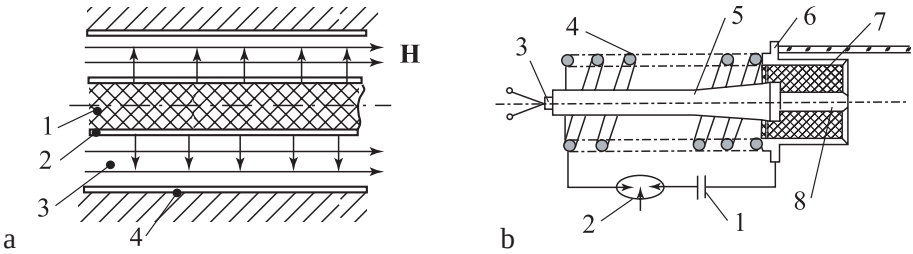


Fig. 9.2.9. Principal diagram of an explosion magnetic generator (EMG) (a): 1) the explosive; 2) conducting shell of the explosive, 3) the gap with the magnetic field, 4) the outer conducting cylinder. The diagram of the energy emitting complex EMG–MPC (b): 1) the condenser for powering EMG, 2) discharger, 3) electric detonator, 4) the spiral of the EMG, 5) the cassette with the explosive, 6) the MPC anode, 7) the plasma-forming dielectric (fluoroplastic), 8) MPC cathode.

- energy input in plasma 200 kJ
- the velocity of the plasma flow at the maximum 60–65 km/s.

The energy-radiating complexes with such parameters have a diameter of 120 mm and a length of 600 mm.

9.2.5. Simulation of the magnetosphere

A large number of studies of the simulation of the processes in the magnetosphere by experimental, numerical and analytical methods have been carried out. Both the magnetosphere as a whole and its separate fragments were simulated.

Here we will note only simulation as a whole.

As mentioned earlier, the first attempts to experimentally study interaction of plasma with the dipole type magnetic field were undertaken as early as in the 30s of the 20th century. However, the setup of the first experiments was not adequate to the full-scale situation.

Realistic experiments were carried out in the 60's of the 20th century byI.M. Podgornyi et al [224]. The source of the moving plasma – analogue of solar wind – here was a quasi-stationary coaxial gun. The parameters of the plasma flow and magnetic field strength were selected in such a way that the majority of the dimensionless parameters, which characterize interaction of the flow and the field in the experiment and on the full scale, would be close to each other. The thoroughly prepared work studies gave their results. The adequate configuration of the magnetic field with a long tail and the leading shock wave were obtained, and the seizure in the magnetic field of particles from the flux, etc, was described.

The advantages of these experiments were: low cost, the possibility of rapid analysis of the magnetospheres of other planets and, it goes without saying, accessibility for measuring any parameters at any points of system.

Numerical simulation required a number of important assumptions for overcoming the computational difficulties. Therefore, in the initial stage the first models were hydrodynamic and two-dimensional, of the type described in section 2.7.1. Gradually, models became more adequate and three-dimensional models are now available, although with the strong hydrodynamic 'component'.

9.3. The Sun

This paragraph is dedicated to the star nearest to us, the Sun. We will see how everything is complicated there. This is not simply 'a thermonuclear reactor overhead', as is frequently assumed, but it is the most complex MHD generator of currents and magnetic fields. If the atmospheric vortices examined in 9.1 are described analytically and they are simulated in a 'the saucepan' with water, if the appearance of the magnetosphere is simulated in computers and in laboratory experiments and, furthermore, the detailed analysis of the properties of the magnetosphere can be explained at any point of it with the aid of automatic spacecraft and the active experiments, then the situation with the Sun is completely different.

But, nevertheless, as mentioned above, the knowledge of the mechanisms of the functioning of the Sun has great practical value, in particular for predicting space weather and adoption of measures for protecting cosmonauts and automatic spacecraft, protection of the ground-based systems of communications and power circuits, and also warning about the harmful effects of the magnetic storms on the organisms of people. At the same time, the study of processes in the Sun is the high-power engine of the scientific progress of many fields of knowledge, up to the common world view.

In the middle of the 20th century it was believed that in essence the physics of the Sun was clear: the concept of thermonuclear nature of energy release inside the Sun ($4H \rightarrow {}^{2}_{4}He$) was created, and the 'space electrodynamics' made it possible to explain everything in a simple manner.

However, approximately 50 years have passed and it is evident that, although the base ideas about thermonuclear power engineering and the role of the magnetic field did not change, that there is a whole series

of principally important events which have not been unexplained. The most important of them are:

- the 11-year period of the activity of the Sun, including the period of the change of the polarity of its overall magnetic field;
- the differential rotation of the Sun because of which the time of revolution at different depths and latitudes is different;
- the mechanisms of generation and the structure of the magnetic field in the depth of Sun and special feature of its exit to the Sun surface in the form of spots, faculae and other formations;
- the mechanisms of the ejection of plasma in the form of protuberances;
- the fibrous 'structure' of all observed formations in the Sun.

And this list can be continued and continued.

The reasons for the poor knowledge of physics of the Sun are well known. They are, first of all:

- The complexity of the processes taking place on the Sun and the impossibility of constructing comparatively simple adequate self-consistent models of the Sun. This prevent three-dimensional turbulent MHD processes, enormous range of densities and temperatures, and also nuclear energy release;
- The distance of the Sun not only from the Earth but also from special automatic spacecraft. Because of this it is not possible to examine the objects of less than ~300 km at the same time observation show the universality of the fibrous structure of all units of those seen before the Sun, but the insufficient permission of image does not make it possible to determine the minimal size of fibers;
- The absence of the methods to glance beside the depths of the Sun, although the 11-year cycle of the magnets of field and much other are determined as far as the precisely inner zone of the Sun.

But, fortunately, situation now begins to sharply change. To begin based on the fact that the automatic spacecraft increasingly nearer occur to the Sun and make it possible to continuously observe the Sun, removing films in many instances from the beginning of process to its end, and in this case without the distortions caused by the Earth's atmosphere and by its rotation. The absence of the atmosphere between the automatic spacecraft and the Sun made it possible for the observations to use the entire spectrum of electromagnetic waves, and this gave much new material.

The second fundamental achievement of recent decades is, it would seem completely improbable, the creation of the methods of diagnostics of the interior of the Sun. There are two such methods.

The first of them is reduced down to the fixation of the solar neutrinos, generated as a result of thermonuclear fusion

$$4\,{}_1^1\mathrm{H} \rightarrow {}_2^4\mathrm{He} + 2e + 2\nu.$$

It is known that the neutrino possesses a very small interaction cross section with other particles and, therefore, it is almost free to leave the depths of the Sun. But, naturally, it is very difficult to trap the neutrinos arriving at the Earth. Nevertheless, such detectors were created first in Japan and Canada and then in other countries. Information about the scale of these installations is given by the parameters of the Japanese detector Superkamiokande. Its basis is a tank with very pure water with a diameter of ~35 m and height 45 m. On the walls of this tank there are $13 \cdot 10^3$ sensors for recording of emission formed during neutrino capture.

These experiments led down the fundamental discovery ‘of the oscillations of the neutrino’, but it is important for us that with the aid of these detectors it was possibel to confirmed the calculated parameters of the solar thermonuclear reactor.

Another trend in development of diagnostics of the interior of the Sun is the so-called helioseismology, i.e. the study of the fluctuations of the solar sphere, similar to usual seismology, which made it possible ‘to X-ray’ the depths of the Earth. Let us note that helioseismology originated in 1960, when R. Leyton, analyzing local Doppler shifts on the images of the Sun, showed that the surface of the Sun rhythmically pulsates with the period of ~5 min. However, a comprehensive study of this type of the fluctuations over a wide range of frequencies began in 1990. These studies, in particular, confirmed the prediction of the presence of a nucleus in the Sun, where the energy is transferred by radiation, and of an outer shell, where the energy transfer occurs because of convection (Fig. 9.3.1).

But there is one additional circumstance which radically contributes to the progress before the study of the Sun. This is the appearance of powerful computers. Obviously, the enormous information flow caused by space helioseismology and neutrino observations would be significantly depreciated if an increase in the power of computers and refinement of the simulation of solar processes in parallel did not occur. Astrophysicist R. Stein concludes: ‘In the study of the Sun computer models played not smaller role than satellites and telescopes’ [237].

As a result, according to astrophysicist, ‘the Golden Age of the study of the Sun has began’. One additional observation. It is believed that the time when astrophysicists understand the main fundamentals of the solar processes will be the time when on physicists and engineers create industrial thermonuclear reactors on the Earth.

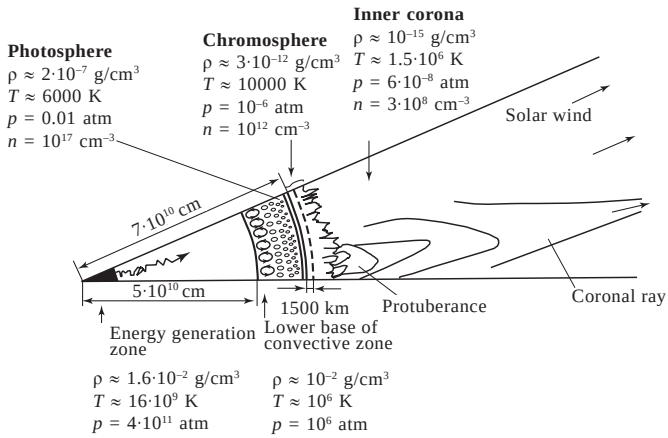


Fig. 9.3.1. Distribution of the plasma parameters along the radius from the centre of the Sun to the corona according to the standard model of the Sun (section 9.3.5). Here ρ is density, T temperature, n is the number of particles in 1 cm^3, p is pressure. The dimensions of the photosphere and the chromosphere on the figure are magnified [235].

But let us return to our book. Processes on the Sun are exceptionally varied. Therefore, in this paragraph we briefly describe only the surface (visible) of the Sun in the 'calm state' (granulation, Sun spots and quiescent protuberances), and then well visible stormy dynamic processes (eruptive protuberances, flashes, ejections of coronal mass).

At the end paragraph we describe the 'standard' model of the Sun as a whole and the place of the Sun among other stars.

9.3.1. Integral characteristics of the Sun [234,235,237]

For the best orientation in the following description of processes on the Sun, let us give its integral characteristics.

To this, it is necessary to add that the rotation of the surface of the Sun, according to provisional observations, is differential. The highest velocity of rotation is detected in the equatorial zone (14.5° in 24 hours). In the vicinity of the poles, the extent of rotation in 24 hours is 10°. The data obtained in solar seismology show that different internal layers also rotate with different velocities. For example, the core of the rotates at the velocity almost twice the velocity of the Sun surface.

Figure 9.3.1 shows schematically the structure of the Sun. More details of the calculated distributions of the parameters of plasma in the depth of the Sun will be discussed in 9.3.5. Let us note here only that the emission in the centre of the Sun lies in essence in the X-ray

range, and the photons born there, being repeatedly re-emitted, reach the surface approximately after 1 million years and pass in this case into the visible region.

1. Average distance based on the Earth	~ $1.5 \cdot 10^{13}$ cm
2. Radius	$6.96 \cdot 10^{10}$ cm
3. Mass	$1.99 \cdot 10^{33}$ g
4. Mas density	1.4 g/cm^3
5. Gravitational acceleration on the surface	$2.74 \cdot 10^4$ cm/s^2
6. Orbital velocity	$4.4 \cdot 10^7$ cm/s
7. Escape velocity	$5.7 \cdot 10^7$ cm/s
8. Energy of proton at escape velocity	1600 eV
9. Complete emission	$3.86 \cdot 10^{33}$ erg/s
10. Emission from 1 cm^2 of surface	$6.35 \cdot 10^{10}$ erg/s
11. Effective surface temperature	5830 K
12. Characteristic temperature in the centre the large spots (diameter ≥ 10 000 km)	4100–4500 K
13. Estimation of the temperature in the centre of the Sun	$1.6 \cdot 10^7$ K
14. Estimation of density in the centre of the Sun	~160 g/cm^3
15. Composition of the Sun as a whole (estimation)	
hydrogen	68%
helium	30%
other elements	2%
16. Solar constant	136.9 MW/cm^2

9.3.2. Structure of the visible range of the Sun

Three regions are visible: photosphere, chromosphere and corona.

Photosphere – basic radiation source of the Sun. This is a relatively dense gaseous envelope with a low degree of ionization. It emits (with exception of X-ray and radio ranges) as a blackbody with the temperature of ~5800 K. The thickness of the photosphere is ~350 km.

Table 9.3, where h is the distance from the surface of the Sun, gives information about the parameters of the photosphere.

Table 9.3 shows that the degree of ionization of plasma with $h = 0$ is of the order 10^{-4}, i.e. is very small. Hence, it is apparent that conditions in the upper layers of photosphere are close to the conditions of low-pressure arc discharge.

Conductivity of this plasma is $\sigma \sim 3 \cdot 10^{12}$ abs. units and, therefore, magnetic viscosity is $\nu_m = c^2/4\pi\sigma \sim 2 \cdot 10^7$ cm^2 s^{-1}.

Due to these conditions, the classical skin-layer in 1 hour ‘grows’ to a depth of ~ 3 km.

Table 9.3.

h, km	T, K	P, dyne/cm^2	P_e, dyne/cm^2
+50	4626	8000	0.9
200	5090	40 000	6.8
–335	6460	125 000	69
–450	8260	250000	2000

Chromosphere is a layer with a thickness of ~1500 km, which most accurately fits the term 'atmosphere of the Sun'. It is located between the dense and comparatively cold photosphere and the corona – discharged and hot. The chromosphere in essence is transparent for the continuous spectrum of the photosphere, but it absorbs individual sections, forming dark Fraunhofer lines. The chromosphere is a very dynamic system because of its intermediate position.

Corona and the solar wind. As evident in Fig. 9.3.1, at the chromosphere – corona transition there is a sharp jump of density and temperature. In this case, the thickness of the transition layer is only 10–100 km. The corona actually does not have any outer boundary. It sufficiently continuously passes into the solar wind, which is observed not only near the Earth but also reaches the boundary of the solar system, where its pressure becomes similar to the pressure of interstellar gas. This size of the corona or, more precisely 'solar magnetosphere', is explained by the fact that it is principally a dynamic formation. By the way, the glow of the corona with an accuracy to several percentages, is a result of the Thomson scattering of the emission of the photosphere on the electrons of comparatively dense parts of the corona.

Its high temperature is the characteristic property of corona. This temperature is created due to the energy of magnetic fields both in the form the waves, which travel from the photosphere, and due to the dissipation (classical and anomalous) of quasi-static fields in the structures of the type of current layers (section 8.2) with the reconnection of magnetic lines of force. However, as far as heating of the corona by waves is concerned, a special role is placed here by transverse Alfvén waves. They are absorbed by the mechanism well known from the general wave theory. Namely, with the propagation of a wave in an inhomogeneous medium with increasing group velocity, the particles of the medium begin to move increasingly more rapidly, and their velocity can become supersignal. But this leads to rapid dissipation of wave energy.

Heating plasma to temperatures of the millionth (in Kelvins) scale in turn leads to the fact that its many particles acquire the escape velocity

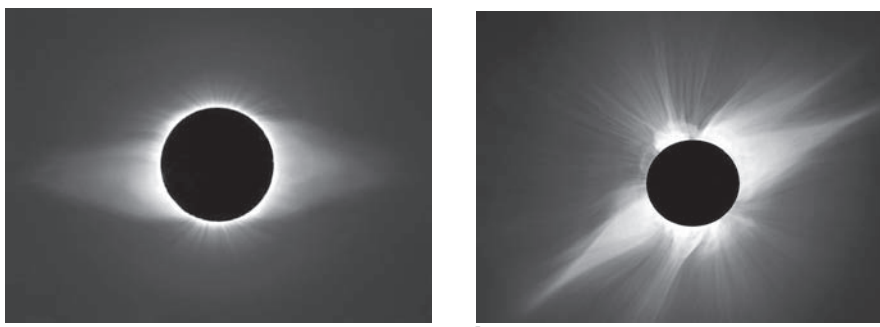
and they leave the environment of the Sun. The low-energy component of solar wind forms by this mechanism. The magnetic field existing in the remote corona cannot hold the flux both because of its size and configuration and, moreover, is itself dragged by the solar wind.

Together with the energetic particles, which exit from the Sun in the form of wind, there is a flux of sufficiently energetic particles, which go from the corona to the chromosphere. After passing in the chromosphere the distance of several mean free paths, the fast particle from the corona is impeded, transmitting its energy to the particles of the chromospheres, which can then pass into the corona. Knowing particle density in the chromosphere ($n \sim 10^{12}$ cm^{-3}) and the energy of ions from the corona (~1000 eV), we find the mean free path of $\lambda \sim 1$ km, which to a considerable degree also explains the small thickness of the chromosphere – corona transition layer.

As show by eclispe observations (Fig. 9.3.2), the form of the corona is not constant during an eclipse. This is understandable, since the form is determined, first of all, by the places of the entering of plasma from the chromosphere, i.e. by the places of its heating, and they are not constant.

'Vents' through which the main mass of the corpuscles is extracted in the interplanetary space were recently discovered. They are the so-called coronal holes. They easily are noted on X-ray photographs of the Sun showin the coronal structure. These are the dark regions, which sharply contrast with extensive bright areas. The matter density in them is three times less than in a normal calm corona. The temperature is also lower – does not exceed $1.0 \cdot 10^6$ K. The coronal holes appear as the regions of a unipolar magnetic field, i.e. (in the photosphere and the chromosphere) the field opened in the interplanetary space.

An intensive corpuscular flux appears as a result along the open magnetic field and this is also confirmed by observations of the solar



a b

Fig. 9.3.2. Solar corona: during the minimum solar activity (a) and in the period of maximum solar activity (b).

wind. The coronal holes around the poles of rotation of the Sun usually exist for a period of many months, but their corpuscular fluxes do not reach the Earth.

9.3.3. Quasi-equilibrium structures associated with the photosphere

If we describe by several words the contemporary state of our knowledge about the local structures of the Sun, then, with small exceptions, it consists of the enormous mass of the photographs, which, however, are not 'compressed' into clear diagrams, described by mathematical models. Therefore, here we will confine ourselves to the description of the three types of the structures connected with the photosphere, which, with the certain reservations, can be named thermal, magnetic and current. These are granules, sunspots and 'quiescent protuberances', and their accompanying 'substructures'. Here the discussion will deal with the quasi-stationary states of the indicated structures, and in the following section – with the explosive processes associated with the sunspots and the quiescent protuberances.

Granules

Figure 9.3.3 shows the typical section of the photosphere with small sunspots with good resolution. It is evident that the surface is not uniform and has a cellular structure. These convection cells are called usually granules. They are similar to Benard convection cells. The characteristic granule size is 500–1000 km. The granules do not live long – usually ~5–7 min.

The careful examination of the granulation of the photosphere shows that these granules merge into some complexes – supergranules, which most clearly are manifested in the chromosphere located above the photosphere.

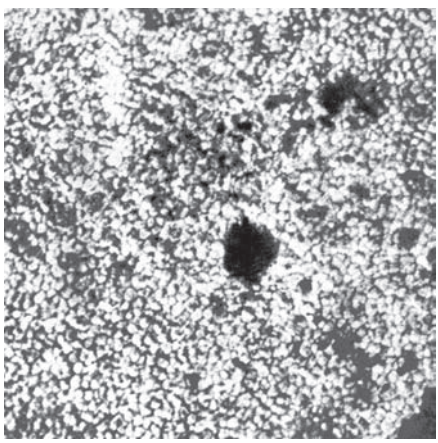


Fig. 9.3.3. Solar granulation and small spots, July 5, 1885, scale 1 mm = 1".

Magnetic fields in the region of granules – if sunspots are distant – amount to ≤ 10 Oe.

On the spectroheliograms the chromospheres are also visible under the favorable conditions for the survey of the granules in the photosphere. However, the 'grid' from the merger of the granules ('supergranules'), which cover entire solar disk, is more frequently visible. Supergranules are the exterior of the of larger (~30000 km) and deeper convective cells located in the thickness of the Sun. Thus, we encounter here the hierarchy of the convective structures: from the large in the depth to relatively small in the photosphere of the Sun.

Studies of Doppler shift in the different sections of supergranules show that in the center the flux of solar substance is directed from the solar depths down to the surface, and on the periphery vice versa – into the depth of the Sun.

The supergranules sufficiently tightly cover the surface of the Sun and, therefore, their boundaries take the form of polygons. So the chromosphere surface of the Sun as a whole resembles a pineapple.

The supergranules live comparatively for long, up to ~10 hours. The flow of plasma from the centre of the supergranules to their periphery is accompanied by the transfer of the magnetic field. As a result, a magnetic field 'builds up at the boundary between them and the size of this field reaches hundreds of oersteds, and in the corners ~ $(1–2)\cdot 10^3$ Oe. The magnetic field, which exists between the supergranules, penetrates the upper layers of the chromosphere and the corona. Short-term plasma flows pass along the lines of force of these layers – 'spicules' (Fig. 9.3.4). They form in many respects due to the fibrous structure of the chromosphere. The spicules are well visible in the form of frequent comparatively thin and not long ejections near the edge of solar disk.



Fig. 9.3.4. The chromosphere with spicules.

Sunspots

Figure 9.3.5a, b shows photographs of a typical single sunspot and the typical group of spots. The spots are distinguished the dark unstructured spot – 'shadow' and 'the penumbra', formed by the large number of elongated 'fibers'. The sizes of spots are very different. A spot with the diameter ~ 180 000 km was once observed, however, the average sizes of the spots are ~ 40 000 km. In comparison with the granules, the spots are long-life objects. Although the lifetime scatter is very great, the mean time for the small spots is of the order of week, and for the large ones several months. With the appearance of a spot, when its penumbra is still small, teh spot shows the same photosphere granules which are then replaced by fibers with the long lifetime. As far as the visible depth of spots is concerned, they are frequently depressions (in the form of small plate) with the conically compressive walls, which exit into the depth of the Sun.

In the zone of spot there is a fairly complicated picture of flows with significant velocities (~3 km/s). In this case, the dominant process is the discharge of substance from the spot in the region of its walls and the inflow into the central section of the spot. The available data clearly indicate that the temperature in the spot is located ~4500 K, whereas in the photosphere it is ~5800 K. Thus, the sunspots are comparatively cold formations. Therefore, the degree of ionization of hydrogen and, especially, helium is here small.

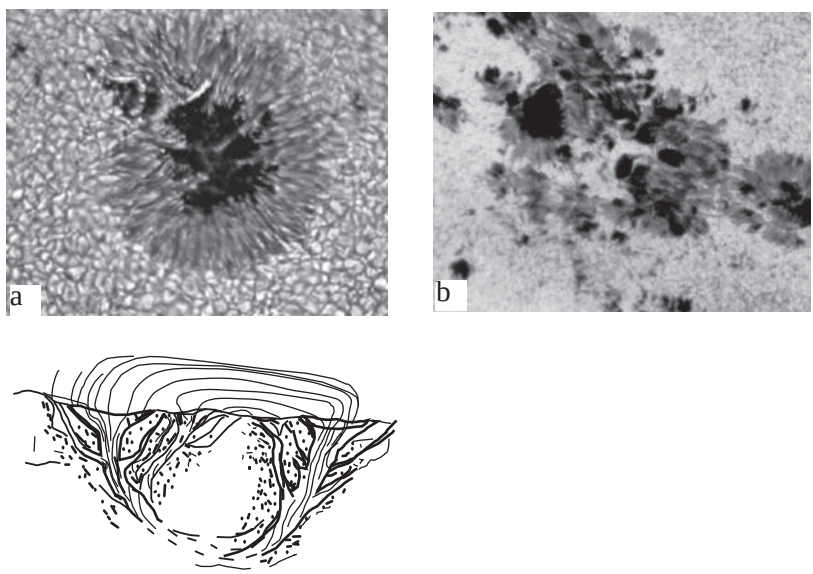


Fig. 9.3.5. Solar spots: a) a single spot, clearly visible granulation of the Sun photosphere; b) a group of spots; c) a possible structure of the subsurface region of a group of spots.

It is characteristic that in many instances we observe comparatively closely spaced spots of the opposite magnetic polarity forming a bipolar group together. This gives grounds to assume that the observed spots are sections by the surface of the Sun of the magnetic tubes partially lowered into the Sun. The dark central section of the spot corresponds to that deep part of the tube where the gas kinetic pressure of the surrounding plasma $P_{\text{s.p.}}$ becomes equal to magnetic pressure, i.e.

$$P_{s.p.} \approx \frac{H^2}{8\pi} \equiv P_H.$$

The magnetic field in the spot is ~4.5 kOe and its pressure (~1 atm) is higher than the pressure of the photosphere (~0.01 atm) surrounding plasma, and that of the more adjacent chromosphere (~10^{-5} atm). Therefore, the magnetic flux expands here, forming 'the penumbra' of the spot. In this region the photosphere takes the form of the well formed fibers.

It is evident from the photographs that near the spots there are (especially near the edge of the disk) bright fibrous formations of irregular shape. They are called 'torches'. In most cases these torches cover considerable areas, forming the so-called torch fields. The torches and their fields penetrate, spreading sufficiently deeply in the chromosphere and during photographing in the narrow regions of the spectrum in the chromosphere they can screen large areas. The characteristic size of the single torches is of the order of the granule size, the time of their existence is expressed in months, their temperature ~6500 K. The torches can be interpreted as the channels of more intensive discharge of energy from the internal volumes of the magnetic tube compensataing for its weak discharge from the sunspots.

The structure in the form of a group of tightly located spots mentioned above can be interpreted as the result of splitting due to the instability of the united initial magnetic flux.

In accordance with what has been said, the structure of typical spots similar to Fig. 9.3.5c seems probable.

Quiescent protuberances [224,233]

In contrast to the granules and the sunspots, the quiescent protuberances are more complex and even mysterious formations. In the photographs in Fig. 9.3.6a they are depicted at a low spatial resolution against the background of the solar disk as a long thin and dark formation, whose length is commensurate with the radius of the Sun. In the second photograph (Fig. 9.3.6b) the quiescent protuberance is located at the

edge of the disk and is visible on the side, very resembling an arched bridge. The characteristic dimensions of such protuberances are immense: their length reaches ~200 000 km, height above the chromosphere to 50 000 km, and the thickness of the horizontal part ~10 000 km. The horizontal part rests on the vertical posts – 'the bases' which depart into the photosphere. The typical density of the protuberances is $\sim 10^{10}$–10^{11} cm^{-3}, whereas the density of the corona into which they are submerged is two orders less ($\sim 10^{8}$–10^{9} cm^{-3}). The temperature of the plasma, which forms a protuberance is ~10 000 K, whereas the temperature of the adjacent corona is two orders higher (~ 1–$2 \cdot 10^{6}$ K). Thus, pressures in the protuberance and the corona are close.

Let us note several other special features of the structures in question. Their lifetime changes over wide limits: from several days for the small and to several months for the large protuberances. They are arranged in essence in the polar zones of the Sun and are elongated predominantly in the latitudinal direction.

The photographs (Fig. 9.3.6b and 9.3.6c) with the higher spatial resolution show that these protuberances are characterised by the intermittent-fibrous structure. The width of fragments is ≤ 350 km, although it can seem that they, in turn, consist of smaller subfibers.

In spite of visible 'rough' appearance of protuberance, the most probable factor preventing its heavy horizontal part from the drop beyond the photosphere is the presence of electric current, which flows along this configuration. The magnetic field necessary for creating the

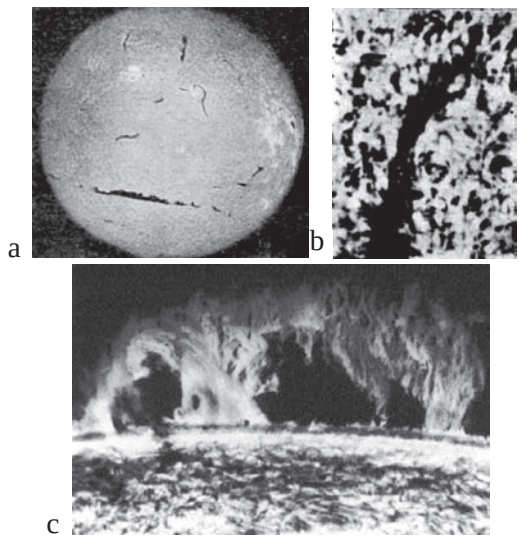


Fig. 9.3.6. Quiescent protuberances: a protuberance on the background of the disk (a), the structure of a protuberance, taken at a high resolution, on the background of the disk (b) and the same at the edge of the disk (c).

supporting ampere force must, one way or another, be generated in the photosphere and the chromosphere.

Numerous observations show that the quiescent protuberances are arranged predominantly near the boundaries between the supergranules. In the zone of the horizontal sections of such protuberances the magnetic field is parallel to the solar surface and has a significant component along its axis. Thus, the lines of force wind around protuberances like spirals. Finally, it is established that the protuberance is arranged, as a rule, above the zero line in the photosphere. In the section perpendicular to the protuberance the lines of force possibly appear in the manner shown in Fig. 9.3.7, i.e. plasma lies on the lines of force, which form a 'cradle'.

In the spaces between the bases the plasma can enter a protuberance both from the chromosphere and from the corona ('condensation' of coronal plasma) and flow into the base with a velocity of ~5 km/s. This velocity is lower than the velocity of sound and Alfén velocity.

9.3.4. Catastrophic processes seen on the Sun (flashes, ejections of coronal mass)

The aforesaid by no means covers all which occurs in the environments of the photosphere. For example, 'the active protuberances' appear quite frequently and can rise above the surface of the Sun by many tens and even hundreds of thousands of kilometers.

They are very diverse (Fig. 9.3.8). They are characterised by rapid development and brief existence, sometimes about one hour. The active protuberances can appear both due to the ejection of substance from the chromosphere (photosphere) and by the condensation of the coronal plasma, cooled by emission and thermal conductivity.

The motion of matter in the active prominences occurs at high velocities, which reach ~700 km/s. With the upward lift spasmodic

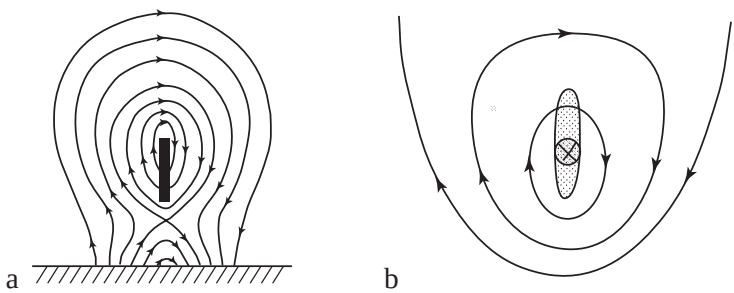


Figure 9.3.7. Probable magnetic configurations of a quiescent protuberance in the plane normal to the protuberance: a) the configuration with the line of the zero magnetic field, b) the configuration, formed in the presence of image flows in the photosphere.

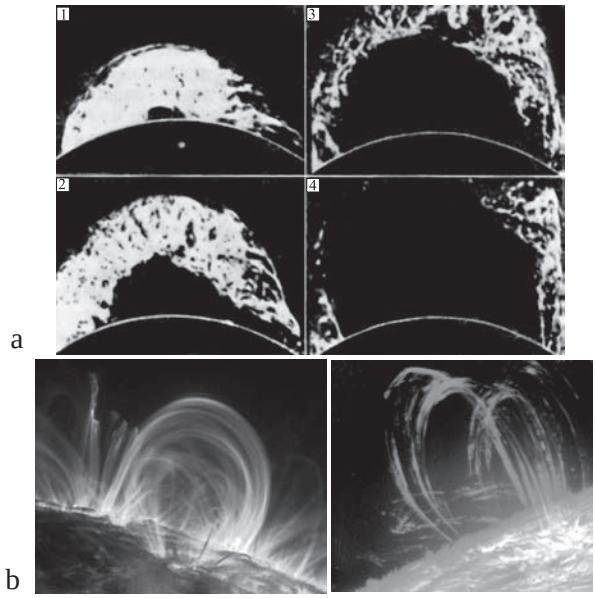


Fig. 9.3.8. Protuberances: a) the evolution of a short-lived protuberance (1, 2, 3, 4 – the time sequence of the frames. Frame 1 in the form of a bright spot shows for comparison the approximate size of the Earth; b) examples of active (eruptive or hinge) protuberances.

accelerations of flows due to obscure factors are sometimes observed. The plasma flows usually take the form which resembles the lines of force of magnetic fields. Frequently they are pulled into the photosphere and disappear.

All such protuberances were previously called ‘eruptive’ (‘explosive’).

However, this is a standard manifestation of the activity of the Sun. But, together with such processes, immense catasrhophes occur one or several times in the period of the active Sun (see next section). They have two substantially distinguished forms and they are respectively called ‘flashes‘ and ‘ejections’ of coronal mass (ECM) [203,204].

For a long time them they could not be separated, especially as sometimes they are occur together. However, only two parameters re common here: enormous energy release (to 10^{32} ergs) and the characteristic time of ejection (~ 10^3 s).

Flashes are connected with the sunspots and are accompanied predominantly (~ 50% of energy) by X-ray (ε ~100 eV) and gamma-radiation. The remaining energy is taken away energetic protons (ε_p ~ 1 MeV).

A large flash resembles outwardly a very bright torch field. The radiation spectrum of the powerful flashes is often continuous, which indicates the significant optical thicknesses of the volume of the flash. The basic/major energy of flash is taken away into the corona and then into the interplanetary space. Usually less powerful flashes are observed as a sharp increase in the brightness of the chromosphere in strong lines, covering the area up to 10^{-3} of the visible hemispheres of the Sun.

A number of the models of the flash mechanisms has been proposed.

Thus, it is believed that these considerations, according to the authors of this book and a number of other researchers, make very plausible the assumption that the flash is some 'super-torch' or 'super-spicule', based on the breakthrough of emission and plasma from the sufficiently deep layers of the Sun along certain 'magnetic plasma-light guides', i.e. in this model, flashes are processes resembling the eruptions of terrestrial volcanoes. High-temperature solar depths here come out as the energy source. As a whole the process of a flash appears as follows. Because of some random reasons – and the interior of the Sun is in a turbulent state – individual magnetic fluxes form a chain penetrating the convective zone of the Sun and forming a magnetic lightguide. The most powerful 'shaft' of the plasma, 'saturated' by radiation and sweeping way all obstacles appears along the lightguide from the depth to the surface. The scale of its intensity there is

$$I = \sigma_{S-B} T_*^4,$$

where σ_{S-B} is the Stefan–Boltzmann constant, and T_* is the temperature of a layer on the Sun, from which the radiant flux starts to 'break out' to the surface. Assuming $T_* = 10^6$ K, we obtain the estimate

$$I \sim 5 \cdot 10^{12} \text{ W/cm}^2 = 5 \cdot 10^{19} \text{ erg/cm}^2 \text{ s}.$$

Assuming the emission time is $\tau = 10^3$ s, we obtain the ejection of energy of $Q = 5 \cdot 10^{22}$ erg. Consequently, 1 km^2 is sufficient for the general energy release to be $\geq 10^{32}$ erg. Thus, the essence of hypothesis is maximally simple: the flash, it is an eruption along a randomly formed magnetic plasma lightguide.

But now preference is given to the models in which the flashes are considered as a chromospheno–coronal process based on gradual energy accumulation in the magneto-plasma structures which then become unstable. The reconnection of lines of force occurs and the field releases its energy in the form of a flash. In this case the basic discharge of energy occurs in the corona. This explanation is preferred

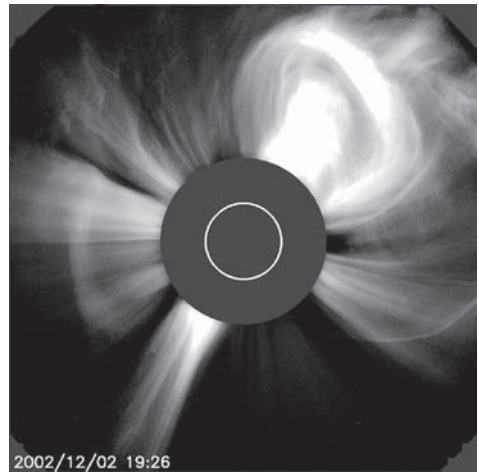


Fig. 9.3.9. Photograph of the ejection of coronal mass taken by the Soho satellite. European Space Agency and NASA.

due to the fact that in many instances a flash develops near the zero lines [203].

Now briefly about the coronal ejections of mass (CEM). Today their mechanism seems more transparent. Namely, filming shows that the development of CEM begins with the loss of stability of the horizontal part of a quiescent protuberance. This part rapidly is moved away from the surface of the Sun, seizing larger and larger part of the corona. In this case, the base of the protuberance remain fixed. With the removal of plasma mass to the distance of the order of the radius of the Sun or more, magnetic lines of force break up and enormous 'lump' of plasma with the seized magnetic field, having the mass of the order of trillions of tons, leaves the Sun with a velocity of ~50 km/s. Specifically, this ejection put out of action an enormous power system in Canada on March 13, 1989.

But the ejections of small power can also occur in the periods of the quiet Sun.

9.3.5. Cyclic recurrence of solar activity [234]

Dynamics of sunspots

It was noted above that the lifetime of sunspots on the surface of the Sun is limited. Therefore, the number of the sunspots always changes. In 1845, Schwabe, analyzing the data about the number of spots in different years, established that their number substantially changes with the period of ~11 years. Figure 9.3.10 shows the dependence of the Wolf number on time. The Wolf number W is determined by the formula

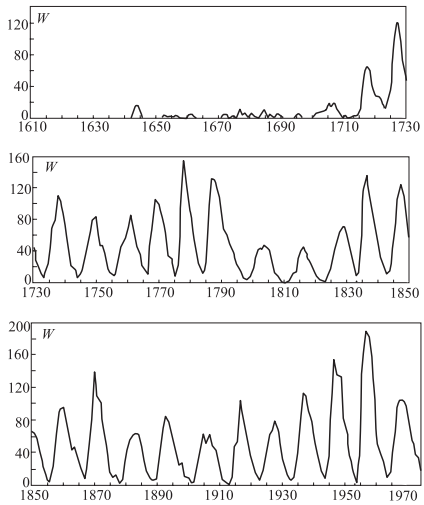


Fig. 9.3.10. Variation of the Wolf number in individual years.

$$W = 10g + f. \tag{9.3.1}$$

Here g is the number of groups of spots, and f the number of single spots, including those which enter into groups.

Thus, for instance, if there is only one spot on the Sun, then $W = 10 + 1 = 11$. The Wolf number characterises with good accuracy the fraction F of the solar surface covered by the spots

$$F = 16.7W. \tag{9.3.2}$$

Here F is expressed in the millionth portions of the solar surface.

Figure 9.3.10 clearly indicates the periodicity of the appearance of the minimums on the curve $W(t)$ with the interval of ~11 years, moreover either they disappear entirely during the minima of the solar activity of the spot or a single spot appears. At the same time, the maxima W in different periods are different. In this connection it is possible to note a curious fact. During the reign of Louis XIV in France from 1643 to 1715, who was called as the 'Sun King', there were almost no sunspots, and a group of spots appeared only once (Maunder minimum).

Let us note several important special features of the behavior of spots.

a. Spots most frequently appear in pairs with the opposite magnetic polarities.

b. When a spot begins to diminish, the strength of its field does not change and only its area decreases. Observations show that the magnetic field does not disappear after the disappearance of the

spot, but it continues to exist and provokes the appearance of new spots in the same region.

c. Sunspots usually drift to the West. If they appeared in pairs, then one of them (the one which is located to the west) is called leading. Observations showed that the polarity of the leading spot and of the following one changes through the period. Thus, the true period is equal to ~22 years.
d. In the mean, the region with spots out is limited to 5° and 35° of both the northern and southern heliographic latitude. From the beginning of the period they are appear near 30° latitude, and in the course of time their places of generation approach the equator. It is significant that the polarities of the leading spots in the northern and southern hemispheres are opposite. The noted periodicity of spots is the manifestation of the general periodicity of solar activity, including the intensity of solar wind, and, as a result, the disturbances of the terrestrial magnetosphere.
e. The general magnetic field of the Sun is small (~1 Oe) and is sufficiently clearly observed during the minimum of solar activity and at the latitudes > ±40°, where the spots are absent. The general field is also subjected to a 11-year cycle and, in particular, its polarity changes in this period. The last reversal of the polarity of the general field was in 2000.
f. In addition to the indicated period of magnetic activity there are also indications – in particular according to the data of the analysis of the yearly rings of trees or the content of isotopes in the layers of polar ices, and also in the historical materials, that there are the fluctuations of solar activity with the periods of ~ 80200 years.

From the aforesaid one can see the global role of the magnetic fields on the Sun. At the same time, magnetic structures are divided sufficiently clearly into two classes: with the weak fields ($\leqslant$10 Oe), connected with the granules and a number of other formations, with strong fields ($\geqslant$1 kOe) observed in the sunspots.

At present, there are no convincing data about the behavior of strong fields in the depth of the Sun. It is usually assumed that the generation of these fields is due mainly to the differential rotation of the Sun, during which in the velocity of rotation in the equatorial plane is higher than near the poles. As a result, the lines of force (frozen in plasma) are drawn out in the manner shown in Fig. 9.3.11 leading to energy storage in the magnetic field. This is explained by the fact that plasma conductivity in the depth of the Sun is high[11] and, therefore, the magnetic flux in the

[11]It is usually assumed that the generation of a field takes place at the boundary of

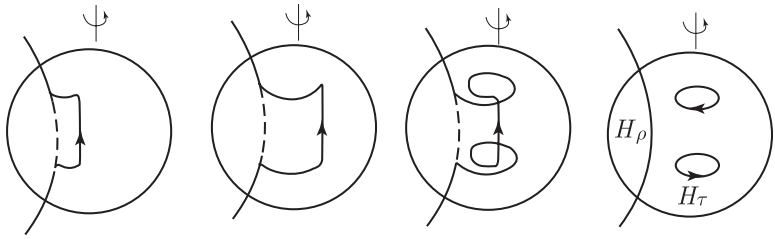


Fig. 9.3.11. Generation of a toroidal field during differential rotation of a conducting medium with a poloidal magnetic field frozen in it.

tube does not change, but the length of the tube increases. 'Sticking' of the turns of the flow can take place in this case.

However, this diagram does not explain the 11-year cycle and many other aspects.

The construction of a convincing model of a solar hydromagnetic dynamo requires taking into account a large number of factors, such as the differential rotation of the Sun, Coriolis force, generation and transport of energy, convection, turbulent pulsations.

Evidently, the construction of the real MHD-dynamics of the Sun is one of the impressive problems of plasma physics of the 21st century.

9.3.6. Standard model of the macrostructure of the Sun

All these consideration actually concern the observed outer shells of the Sun. The state and dynamics of internal – invisible directly – subphotospheric zone can be judged only on the basis of the observations of surfaces in models of the internal structure of the Sun, which are in agreement with the data of Table 9.1. These models are based on schemes of thermonuclear reactions which ensure the necessary energy release. The greatest acknowledgment for the Sun is at present the 'thermonuclear' reaction in the process of which hydrogen is converted to helium

$$2({}_1^1\mathrm{H} + {}_1^1\mathrm{H}) \rightarrow {}_2^4\mathrm{H} + 2\beta^+ + 2\nu + \gamma. \qquad (9.3.3)$$

Here β^+ is the positron, ν the neutrino, γ the photon. The real chain of reactions leading to the transformation of four ${}_1^1\mathrm{H}$ to ${}_2^1\mathrm{H}$ appears as follows:

$$\begin{aligned} &{}_1^1\mathrm{H} + {}_1^1\mathrm{H} \rightarrow {}_1^2\mathrm{D} + \beta^+ + \nu \ (+1.44\ \mathrm{MeV});\ \tau = 8\cdot 10^9\ \text{years},\\ &{}_1^1\mathrm{H} + {}_1^2\mathrm{D} \rightarrow {}_2^3\mathrm{He} + \nu \ (+5.49\ \mathrm{MeV});\ \tau = 4\ \mathrm{s},\\ &{}_2^3\mathrm{H} + {}_2^3\mathrm{He} \rightarrow {}_2^4\mathrm{He} + 2{}_1^1\mathrm{H} \ (+12.85\ \mathrm{MeV});\ \tau = 5\cdot 10^4\ \text{years}. \end{aligned} \qquad (9.3.4)$$

the zones of radiation and convective energy transfer.

The above-indicated times are calculated for $T = 1.5 \cdot 10^7$ K, $\rho_n = 10^2$ g/cm^3, which is characteristic for the central regions of stars of the type of the Sun.

Among the different models of the internal internal structure of the Sun most frequently is used the simple so-called 'standard' model of the Sun. In this model, the Sun is considered a spherically symmetric static configuration[12], described by a system of two fundamental equations.

The hydrostatic equilibrium equation

$$\frac{\partial \rho}{\partial M_r} = -G \frac{M_r}{4\pi r^4} \tag{9.3.5a}$$

and the heat balance equation

$$\frac{\partial L}{\partial M_r} = \epsilon - T \frac{dS}{\partial t}. \tag{9.3.5b}$$

Here the independent space coordinate is not radius r but the mass of the Sun inside a sphere with radius r:

$$M_r = 4\pi \int_0^r \rho r^2 dr$$

Heat flux L is the sum of the radiant and convection currents

$$L = 4\pi r^2 \left(-4\pi r^2 \rho K \frac{\partial T}{\partial M_r} - Nu \cdot K \frac{\Delta T}{l} \right). \tag{9.3.5c}$$

Here radiative transfer is described in the diffusion approximation and $K \equiv 16\sigma T^3/3\varkappa\rho$, and $\varkappa$ is the effective absorption coefficient. The convective transfer term is proportional to the Nusselt number Nu[13], ΔT is the temperature difference in convective cells, l - mixing length. The remaining values p, S, ϵ make usual sense: pressure, entropy, power manufactured by the unit of mass. To equation (9.3.5) are added the equations of state $\rho = \rho\ (p, T, x_i)$, $s = s\ (p, T, x_i)$, the expression for the coefficient of absorption $\varkappa = \varkappa\ (\rho, T, x_i)$ and the velocity of the generation of energy $\epsilon = \epsilon\ (\rho, T, x_i)$. Here x_i is the relative content of the mass of elements with the atomic number i. Of course when the evolution of the Sun is studied, the equations for changes of x_i are used.

[12]As mentioned previously, convection is taken into account here although it is introduced simply as some additional heat conductivity

[13]$Nu = \alpha l/\kappa$, $\alpha = q/(T_1 - T_2)$, q is the density of the heat flow, κ is the heat conductivity coefficient.

Table 9.4. Parameters of the Sun according to the standard model (Bachall et al, 1982)

Luminosity (L)	$3.86\cdot10^{33}$ erg/s
Mass ($M_\odot$)	$1.99\cdot10^{33}$ g
Radius ($R_\odot$)	$6.96\cdot10^{10}$ cm
Age ($t_\odot$)	$4.7\cdot10^{9}$ years
Density in centre (ρ_c)	156 g/cm^3
Temperature in centre (T_c)	$15.5\cdot10^{6}$ K
Hydrogen content by mass on surface (X_1)	0.732
Hydrogen content by mass in centre ($X_{1,c}$)	0.355
Effective surface temperature (T_e)	$5.78\cdot10^{3}$K
Initial helium content by mass (X_4)	0.25 ± 0.01
Initial content of heavy elements by mass (Z)	0.018
Depth of convective zone	$0.27R_\odot(M_r-0.22M)$
Fraction of energy of hydrogen cycle	0.985
Fraction of energy of the carbon-nitrogen cycle	0.015
Neutrino flux from pp-reaction	$6.1\cdot10^{10}$ cm^{-2}s^{-1}
Neutrino flux from nuclear decay reaction ^{3}B	$5.6\cdot10^{6}$ cm^{-2}s^{-1}

The system of equations is supplemented by four boundary conditions: given pressure and radiant flux on the surface of the Sun and two obvious conditions in the centre of the Sun: $M_r\big|_{r=0}=0,\ L\big|_{r=0}=0.$ The parameters of the Sun, calculated by the standard model, are represented in Table 9.4

The calculations of the standard model, and also the systematization of the existing facts lead to the diagram of the structure the internal – invisible – part of the Sun, which is depicted in Fig. 9.3.1. The nucleus with the radius $r_n \approx 0.25\,R_\odot$ is given here in particular, where $R_\odot$ - radius of the Sun, in which $\rho \sim 160$ cm^{-3}, $T \sim 1.6\cdot10^{7}$ K and where basic energy release takes place due to the reaction $4\,{}^1_1\mathrm{H}\rightarrow{}^4_2\mathrm{He}$. It is followed by the layer of radiation energy transfer from the nucleus to the surface. It lies in the interval

$$0.25\mathrm{R}_\odot < r < 0.7\mathrm{R}_\odot.$$

This is then followed by the convective layer with a hierarchy of convective cells in the form of supergranules and granules. The layer

is within the limits of $0.7R_{\odot} < r \leqslant 1R_{\odot}$ – in reasonable agreement with helioseismology data.

9.4. On the evolution of the stars of the main sequence [234]

In conclusion, briefly let us pause on the evolution of the stars close to the Sun as regards their parameters and the origin.

Such stars, as considered on the basis of observational data and their systematics, appear from the cold gas-dust accumulations, in which contraction of matter starts under the effect of gravitation for various reasons. This 'contraction' leads to warming-up of matter. A 'protostar' appears as a result; its dimensions continuously decrease, and the temperature and pressure in its center grow. At some moment conditions are created in its centre in which the thermonuclear reactions first begin on Li, Be, B, which are present in small quantities, and then in the form of the hydrogen cycle. Now this is already a 'present' star. The evolution of the luminosity of the star, including of the Sun, occurs slowly after transition to combustion of hydrogen.

One additional general comment. It is not necessary to think that the birth of stars occurred only in very ancient times. It is shown that protostars and new stars also form nowadays. The characteristic mapping of the typical evolution of stars is the so-called Hertzsprung-Russell diagram. In this diagram the axes give the luminosity of the star and its 'spectral class'[14] (Fig. 9.4.1). Typical are the stars which lie on 'the main sequence'.

On the whole there are many versions of 'posthydrogen' evolution of stars. This depends on the mass of star, its chemical composition and the like.

Let us simply note that the cases when evolution concludes by a flash with the formation of a 'white dwarf', neutron star (pulsar) or, finally, black hole, are most spectacular.

White dwarfs

White dwarfs are formed after the burning out of hydrogen in stars with a mass of $M \leq M_{\odot}$ – the mass of the Sun. This is confirmed by both the changes in the mass of stars by the gravitational shift to the red side and by theoretical calculations. It was also established that the radius of white dwarfs is not less than R_{min}~1000 km, and density does not exceed $\rho_{max} \sim 2 \cdot 10^{10}$ g/cm^3.

[14]Spectral class – complex of characteristics of the spectrum of a star closely linked with its effective temperature.

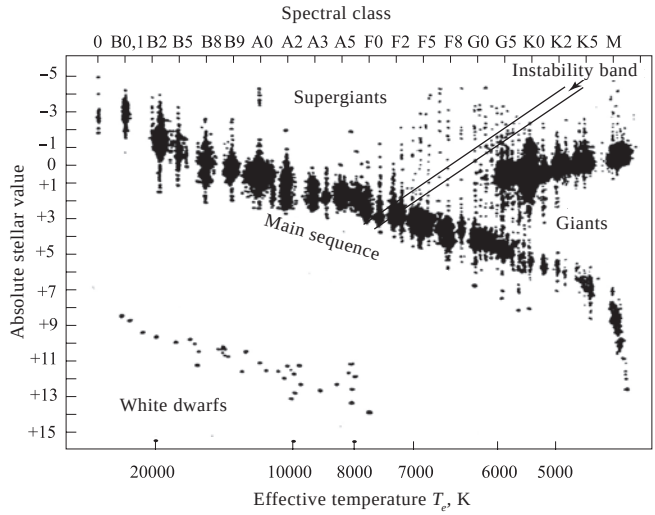


Fig. 9.4.1. The Hertzsprung–Russell diagram of the position of stationary stars situated at the known distances.

The white dwarfs are characterised by the degeneracy of the electronic component, in spite of the high temperature $T \sim 10^7$ K. However, this temperature is insufficient for any intensive nuclear reactions, and they shine due to the previously accumulated thermal energy.

The white dwarfs possess a magnetic field with the strength which reaches 10^8 Oe. This field forms during the compression of the star, since the initial magnetic flux remains.

Pulsars

If the mass of the collapsing star lies within the limits

$$1.5M_\odot < M < (2 \div 3)M_\odot ,$$

the compression ges further than the compression of white dwarfs and it leads to the formation of neutron stars. In this case the electrons in the suppressing mass connect with the protons, forming neutrons – this reduces the pressure of the medium and leads to further compression – and makes neutrons steady with respect to the decay $n \rightarrow {}^1_1\mathrm{H} + e$. The neutrons inside the star are degenerated and it is assumed that they form a superfluid liquid, and the outer layer is thin and solid. In this case the radius of such stars is ~10 km, the density of neutron matter reaches $\rho \sim 5 \cdot 10^{14}$ g/cm^3, and temperature $\geqslant 10^{10}$ K. The neutron star was discovered for the first time by J. Bell, a collaborator of group led by A. Hewish (Cambridge) completely unexpectedly because of the sequence

of radiobursts which possess the improbably constant period (with an accuracy to 12 signs). The sources of this form of radio-frequency pulses were called 'pulsars'. The periods of radiobursts in the currently known pulsars vary within the limits ~0.0016–4.3s.

Soon after the discovery of radiobursts the stars responsible for them were identified with the previously forecast neutron stars. The radio emission of pulsars is synchrotron electron emission in a strong magnetic field of the pulsar. In this case, the periodicity of splashes is explained by the fact that there is an angle between the rotational axis of the pulsar and its magnetic axis, and the flux of radio emission propagates in a narrow cone around the magnetic axis and once per revolution 'strikes a match' for the observers on the Earth. The high rotational velocity is explained by the conservation of the angular momentum with the compression of the 'normal' star into a pulsar. The 'permissible' scales of the masses of pulsars were noted above. However, they can be formed also from stars that are substantially more massive. This occurs when an exploding star ejects sufficiently thick outer layers. The flash of a 'supernova' is detected in this case. The flash observed in 1054 in the Taurus constellation, which was visible even in the daytime, became chrestomathic. The result of this explosion was the expanding (velocity to ~1000 km/s) Crab Nebula, and a pulsar was discovered in its centre. At present, the generation of radio emission pulsars is in the form of a following chain of the processes, generated in the final analysis by the magnetic field which reaches $\sim 10^{12}$ Oe[15]. The rotation of a pulsar leads to the appearance in its environment of a strong electric field; in this field, the particle is accelerated to the ultrarelativistic energies; moving along the bent magnetic lines of force such particles generate γ radiation; in turn, these γ-quanta give birth to electron–positron pairs. The development of instabilities in the electron–positron plasma leads in particular to the generation of the epithermal radio emission of pulsars.

This is the overall scheme of energy transfer of the rotation of pulsars to radio emission. It is natural that many components of this bond of processes are obscure. But the fact that the primary radiation source is rotation of the pulsar is confirmed by the slow increase of the period between the radiobursts of reversed retarded rotation of the pulsar.

These estimations should be added: the concentration of electron-positron plasma near the surface of the pulsar $\sim 10^{13}$–10^{19} cm^{-3}. The energy of the plasma particles $\sim(10–10^4)\ m_e c^2$. This plasma is pierced by the beams of electrons and/or positrons with the energy

[15]Einstein equation ($E = mc^2$) shows that the mass of 1 cm^3 of such a field is ≈40 g.

~$(10^6–10^7)$ $m_e c^2$, but with the density ~$10^3–10^4$ times lower than the density of plasma near the surface of the pulsar.

The black holes

If hydrogen burns in a star with the mass much larger than the mass of the Sun $M > 3M_\odot$ then it either explodes and converts to a ragged expanding shell or this is accompanied by the formation of a super-massive fundamentally new formation – a black hole, as named by J.A. Wheeler.

The possibility of existence of the objects which possess this strong gravitational field capable of trapping light was for the first time indicated at the end the 18th century (Laplace), on the basis of the Newton ideas the light as a corpuscular flux. Then – in connection with the considerable successes of the wave theory of light – the idea of the black holes disappeared from the field of interest of scientists. This was revived after the discovery of the quanta of light and universal connection of energy and mass. Using the classical Newton's law of gravitation, one can write

$$G\left(\frac{\pi\omega}{c^2}\right)\frac{M}{r_*} = h\omega.$$

This leads to the 'gravitational' radius of the black hole with mass M

$$G\frac{M}{c^2} = r_* \tag{9.4.1}$$

Here G is the gravitational constant.

The accurate analysis of the special features of spacetime of the static gravitational field with point mass was carried out on the basis of general theory of relativity by Schwarzschild. He showed that critical value is not r_*, determined by formula (9.3.7), but the value two times larger

$$r_* = 2\frac{GM}{c^2}.$$

The black holes appear in two ways. First, it is all the mass, which influences the motion of bodies in close vicinity. Second, it is the retraction of the matter of the adjacent objects that are accompanied by powerful radiation, including X-ray radiation, on approach to critical surface r_*. This takes place because the trajectory of bodies in the Schwarzschild field is spiral and not conic sections as in Newton's field.

The black holes have been studied insufficiently. However, it seems that they are a quark–gluon medium (QGM), which is an analog of the

plasma in which the role of particles is played by the quarks, and the role of photons is played by gluons.

But all this already far exceeds the range of our book. Let us simply note that QGM was produced in CERN on the microlevel, after colliding two multiply-charged ions accelerated to high energies (see 10.6).

And the last observation. The recently mentioned QGM was apparently generated in large quantities during the Big Bang and was the material used for building our visible world.

10

Examples of modern plasma technologies

Any discovery eventually finds practical application and creates a new product. However, the transition from discovery to application is usually not simple. A good example is the transition from the magnetic needle deflection discovered by Oersted when a current flows in a conductor situated next to electric motors of different scales, including motors for rolling mills, electric trains and electric razors.

In previous chapters we have talked about the basic physical laws that determine the dynamics of the plasma, and examined a number of plasma systems which are of little interest because of their applied value. Here, on the contrary, we will focus on the application side. In the first four sections it would be a system in the 'commodity' form, including the plasma in the home and medicine, in technology, including synthesis of diamond jewelry, and electric reactive space engines. In the fifth section, we outline the basic modern solutions for fusion systems, and in section 10.6 shall dwell on the possibilities offered by the charged ions in fundamental physics.

Speaking of 'commodity' plasma systems, we will not touch those which came into our lives and have become accustomed. These include gas discharge light sources – we will focus only on simple tubular fluorescent lamps – and will not talk about arc welding and spark plugs in the car engines. We will not touch a wide variety of electrical plasma devices such as arcing camera switches, power switches, mercury rectifiers, gas-discharge rectifiers, thyratrons etc.

This enumeration, as well as what is said in the following paragraphs, suggest that in the XXI century the plasma will become a fundamental component of the 'technology sum' of humanity. All these developments represent only the first steps in plasma Eldorado.

10.1. Plasma generators [242]

The plasma technologies are of course bases on plasma generators, i.e.

systems in which the ionization of the working medium and the transfer of energy to the produced plasma take place.

Plasma confinement

As repeatedly noted above, the ground conditions of plasma generation are closely related to its confinement and the fundamental question is what is the lifetime of the plasma and its energy in the system. In those cases where plasma forms in an environment of dense gas or inside a container, this time is the order of the flight time of ions to the environment.

If it is necessary to hold the plasma for much longer times, electromagnetic fields are used. This problem is especially acute while keeping the high-energy plasma in connection with the tasks of controlled thermonuclear synthesis (CTS). This will be discussed in section 10.5. However, the isolation of plasma from the energy-absorbing annihilation 'walls' is needed not only here. So, in section 3.7.3 we described QHPA with magnetic shielding of electrodes from the damaging effects of powerful plasma streams.

The problem of isolation of the plasma from the walls also arises in the systems of more modest proportions. This applies, in particular, to gas-discharge chambers (GDC) of ion sources for various purposes, including the ion engines. It uses different ways, but particularly progressive proved to be magnetic mosaic 'walls' first proposed by Moore in 1969 for the CDC of ion engines (Fig. 10.1.1). These 'walls' are assembled from the permanent magnets situated outside the body of the GDC that can form different configurations [243]. Under the

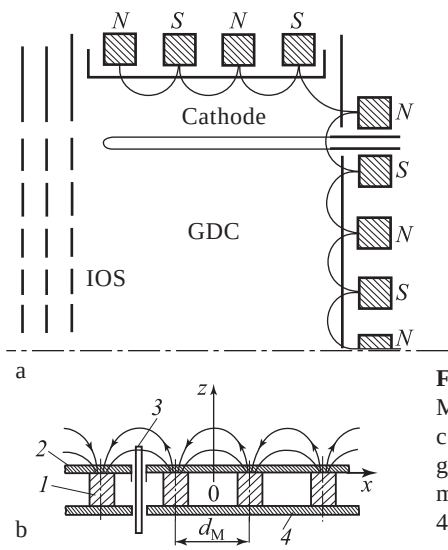


Fig. 10.1.1. Diagram of the ion source of the Moore type (a): GDC – the gas discharge chamber, IOS – the ion-optical system; the geometry of the magnetic system (b): 1) the magnet; 2) the wall of the GDC; 3) the probe; 4) magnetic circuit.

arches of the lines of force the chamber body has a positive potential in relation to plasma. As a result, the ions are reflected by the electric field from the casing, and the electrons can not directly enter the casing due to the magnetic field.

Thus, the lifetime of the particles in the chamber is increased many times over. An important advantage of the peripheral magnetic wall is the ability to create large volumes of homogeneous plasma without a magnetic field.

10.1.1. Types of plasma generators

Plasma can be produced under a variety of energy: electromagnetic fields, streams of electrons, ions and neutral gas fluxes, if they have the necessary parameters. Below, in accordance with the general content of the book, we will speak mainly about the stationary and quasi-stationary generators, and pulsed plasma sources and generators powered by HF will be discussed only briefly.

Plasma generators on the basis of their design features and character of the processes can be somewhat arbitrarily divided into 5 types, each of which is in turn divided into substantially different subtypes.

These are the types.

A. *Plasma heaters in confined spaces*. It is natural to distinguish two subtypes: low-temperature plasma heaters and heaters of high-temperature plasma. The first subtype includes classic discharges in confined spaces (e.g. different discharge lamps based on a glow and arc discharges, which were discussed in Chapters 6 and 7).

The second subtype – ohmic regimes in tokamaks and stellarators, as well as regimes in which plasma is heated by microwave radiation and the fluxes of fast neutral atoms. The second subtype will be discussed below. Specific generators of multicharged ions will be described in Section 10.6.1.

B. *Generators of plasma flows*. It is also natural to distinguish two subtypes:

− plasma torches (plasmatrons);

− plasma accelerators.

Plasma torches are generators of low-energy fluxes, typically operating at the atmospheric pressure. If the fluxes at the output are subsonic, the plasma torches can be regarded as generators-heaters, but according to the operation of the scheme they are ‘streaming’ systems.

The plasma torches are now very diverse. They work in a constant flow mode, under the influence of HF and microwaves, and also laser radiation. The plasma torches are one of the most common plasma dynamic systems. They exist in various versions and have a capacity

of hundreds of watts to megawatts. In Section 10.2.3. we describe quite unexpected application in medicine. Their main applications in plasma chemistry, engineering and scientific research.

Plasma accelerators (PA) – generators of plasma blobs (pulsed guns) or streams (stationary or quasi-stationary) with enough high-energy particles from a dozen of eV to hundreds of keV and higher.

Among the pulsed PAs in the microsecond range the coaxial electrode guns, which are discussed in Section 1.1.2, are used most extensively. They vary in size and energy input in the discharge: from hundreds of joules to hundreds of kilojoules. Here, the most complicated section is a valve which must form a gas 'puck' during ~0.03–0.1 ms between the electrodes; current then flows through the 'puck', ionizing it and accelerating under the action of ampere force. Currently, to solve a number of problems, high-current nanosecond accelerators have also been developed [244]. But they are outside of our subjects.

Stationary plasma accelerators and systems almost identical to them – the quasi-stationary accelerators – differ, first of all, in the acceleration mechanism. In these accelerators, the plasma is quasi-neutral, and in the hydrodynamic language the accelerating force acting in them is

$$\mathbf{f} = -\nabla p + \frac{1}{c}(\mathbf{j} \times \mathbf{H}).$$

Here the pressure p is equal to the sum of pressures of ionic p_i and electronic p_e components

$$p = p_i + p_e.$$

This shows that one can distinguish the following subtypes of PA:

− isothermal ($p_i \sim p_e$, and $\mathbf{j} \times \mathbf{H}$ – is small). These include the usual gas-dynamic nozzles and many plasma torches;

− Ampere (p is low). This includes most of the PAs;

− non-isothermal (($p_i \ll p_e$, and $\mathbf{j} \times \mathbf{H}$ is small);

− 'thermo-magnetic" $\left(|\nabla p| \sim \frac{1}{c}|\mathbf{j} \times \mathbf{H}|\right)$.

Ampere accelerators are naturally divided into three classes:

− High current PAs with their own magnetic field. These include QHPA (coaxial high-current plasma accelerator) described in Section 3.7. They are similar to magnetoplasma compressor (MPC), which were discussed in Sections 2.6 and 3.7. A special feature of the latter is the formation at the output under the effect of the magnetic field and the

incident flux, of a zone with higher density and temperature. This zone is called the compression zone or 'stationary flowing *Z*-pinch'. The compression zone is a powerful source of radiation when a working medium with high *Z* is used. When working with deuterium the compression zone can be a source of neutrons.

– Low current ($J_p \leqslant 100$ A) with an external magnetic field. These are mainly 'Hall' PAs. One of them – a stationary plasma thruster – will be discussed in Section 10.4, but the principle and elements of the physics of these systems were discussed in Sections 6.7 and 7.5.

– PAs with a combined magnetic field in which the magnetic field in the working area is a superposition of external and intrinsic fields.

C. *Ion sources as plasma generators*. The scheme of one of the options of a powerful ion source with a gas discharge chamber (GDC) is shown in Fig. 10.1.2. This ion source was developed in the 70's for injectors of neutral hydrogen atoms. The parameters of this source are as follows. The GDC has a rectangular shape measuring 20 cm × 12 cm. Along the long side it has two cathode blocks (1) with LaB_6 plates secured in tantalum holders and heated by radiation of tungsten spirals (2). Discharge current in the GDC is ~700 A. The ion optical system (IOS) source (3,4,5) has three electrodes (emitting, accelerating, and earthed electrodes). In the IOS there are three slots. The power of the ion beam at the output of the source varies up to 3 MW, ion energy ~40 keV, pulse duration ~1.5 sec.

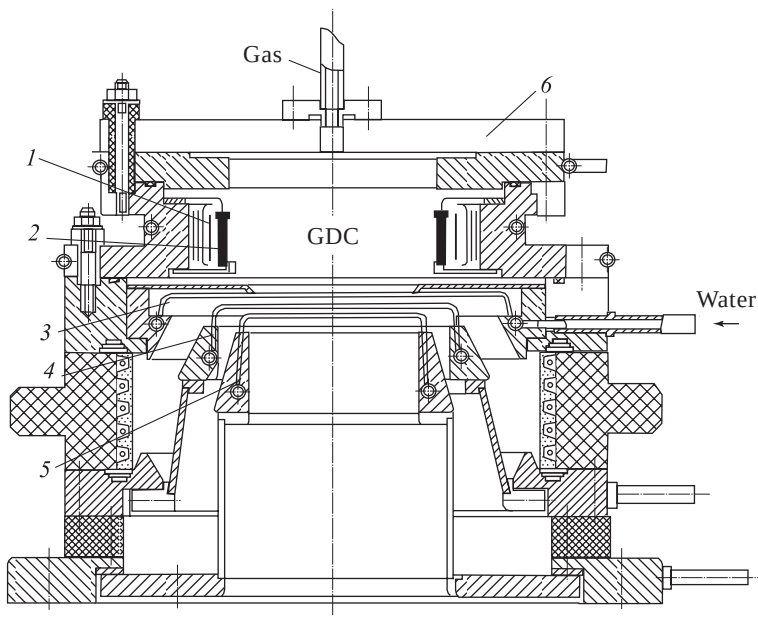


Fig. 10.1.2. The ion source without the magnetic field designed by N.N. Semashko.

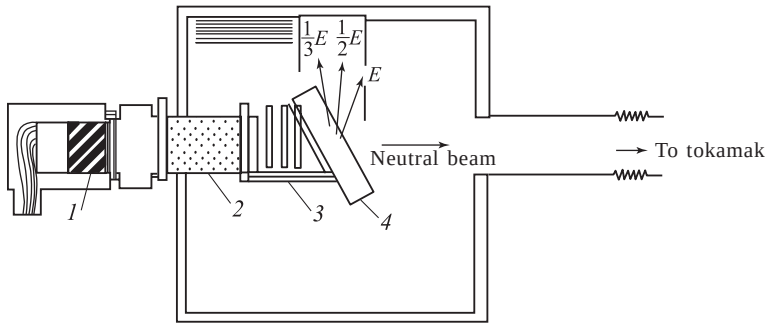


Fig. 10.1.3. The injector of high-energy neutral atoms: 1) the ion source, 2) the gas target for recharging ($A^+_* + A \rightarrow A_* + A^+$,* is the sign of high energy), 3) magnetic screen; 4) the magnet deflecting non-recharged ions.

In this source, the GDC gas no magnetic field, which allows virtually noiseless ion beams. This is very important to maintain the quasi-neutrality of the beam after its release from the source and the subsequent passage of magnetic systems (see Fig. 10.1.3).

The ion flux leaving the source vigorously 'looks for' an easily accessible source of electrons to neutralize its space charge and become quasi-neutral, i.e. 'plasma'. If it is a cosmic engine, then a special source of electrons – the neutralizer (the 'cathode-compensator') is attached to it.

The ion sources are especially useful when it comes to high-energy low-mass ion fluxes (see the Langmuir equation). A specific form of application of high-energy ion beams is to convert them, by recharging, to a flux of neutral atoms with subsequent entry into thermonuclear traps with a magnetic field, where they are ionized (Fig. 10.1.3). Note that the main contribution to the development of powerful ion sources for CTS in Russia was made by a team headed by N.N. Semashko [245].

E. *Erosion plasma sources*. In the preceding paragraphs it was assumed that in the initial state the working substance is in the form of gas and then it becomes ionized. However, in many cases the source is the solid state. Under the influence of the discharge the substance evaporates and the pairs are ionized. Specific areas of erosion may be different. In arc discharges they are usually cathodes (welding, mercury rectifiers, cathode spots on the cold cathode). In recent years increasing attention has been paid to discharges in which the dielectric separating the anode and the cathode is eroded. They have already been mentioned in Section 7.4 and will also be mentioned in Section 10.4.

D. *Hypersonic flows that are decelerated.* The most prominent representatives of this type of plasma generators are meteors, spaceships returning to the Earth, as well as nuclear explosions. We shall not discuss them and only note that the shock waves with high amplitudes were considered in Section 6.8.

10.2. Plasma in the home

Plasma in the home today occupies a prominent place. This started with gas-discharge tubes used for advertising that appeared in 1906, and sparkplugs in cars[1].

Here we briefly describe the four systems: fluorescent lights, plasma screens, plasma scalpels and Chizhevskii chandeliers (lamps).

10.2.1. Daylight lamps (fluorescent)

In Russia, these lamps were developed by S.I. Vavilov. They are usually straight glass tubes (length ~50 cm and diameter 25 mm) at the ends of which there are periodically heated electrodes. Inside the tube is argon at a pressure $p \sim 1$ Torr and ~ 0.1 g of mercury, which evaporates in a few tens of seconds after the start of the discharge. In the steady state, the usual glow discharge burns in the tube which, due to the presence of mercury, mainly emits in the UV region, and light efficiency is five times higher than that of incandescent lamps. To convert this radiation to 'daylight', the inner walls of the gas discharge wall are covered with special luminophor that converts the radiation of the discharge into visible light (Fig. 10.2.1a).

These lamps are powered directly or, more accurately, through the inductance – from the AC mains voltage. Therefore, both electrodes in the lamp are the same (Fig. 10.2.1b).

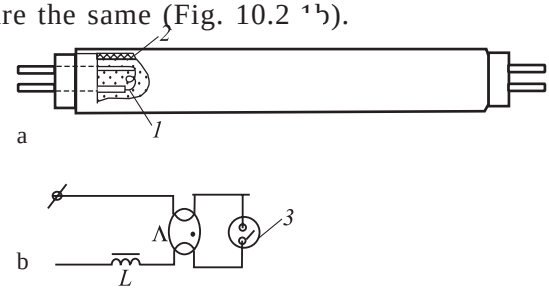


Fig. 10.2.1. The diagram of a luminescent lamp (2) and the lamp power supply (b): 1) one of the electrodes; 2) the luminophor layer on the internal side of the pipe; 3) a neon lamp with a bimetallic sheet.

[1]Arc discharges – 'Yablochkov candles' – were used even earlier for lighting

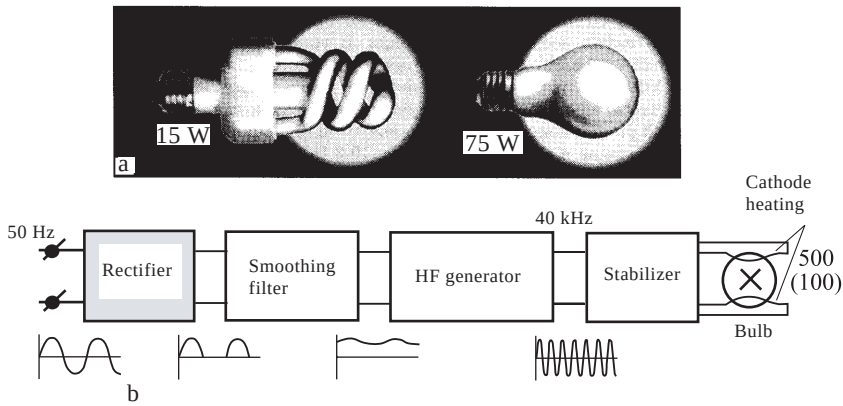


Fig. 10.2.2. Aladin luminescent lamp: a) lamps: luminescent and incandescent with identical light fluxes; b) the flow chart of the electronic start regulator.

A very original electrical system is made of these tubes with alternating current, which ensures their ignition. This happens as follows. After closing the contact voltage is applied to the lamp and a neon bulb (3) that lights up. Current flow in the auxiliary circuit heats the spiral electrode lamps which start to emit electrons. A bimetal plate in the neon tube is heating in parallel with the coils and this opens the contact in the neon tube circuit and thereby extinguishes its discharge. Current interruption due to the presence of inductance in the circuit produces a voltage pulse between the lamp coils and thus ignites a glow discharge in the lamp. Alternating voltage, applied to the system, leads to a change of electrode polarity of the lamp. The lamp makes noise and the light flickers with a frequency of 100 Hz, although the latter is usually little noticeable.

The initial models of the recently developed fluorescent lamps were referred to as the Aladin lamp and can be screwed into the traditional cartridges (Fig. 10.2.2a). They are fitted with a voltage frequency converter (Fig. 10.2.2b), thus eliminates the disadvantages of conventional fluorescent lamps (noise and flicker). These lamps are referred to as ‘energy-saving’ and they are now used on a very large scale.

10.2.2. Plasma TV panels

The sale of TVs and computer monitors with ‘plasma’ screens (panels) started at around 2000. These panels have a number of advantages. Here are the characteristics of one of them (ROS-42 IV): size 961 × 512 mm^2, the number of pixels 853 × 480 pixels, 300:1 contrast ratio and

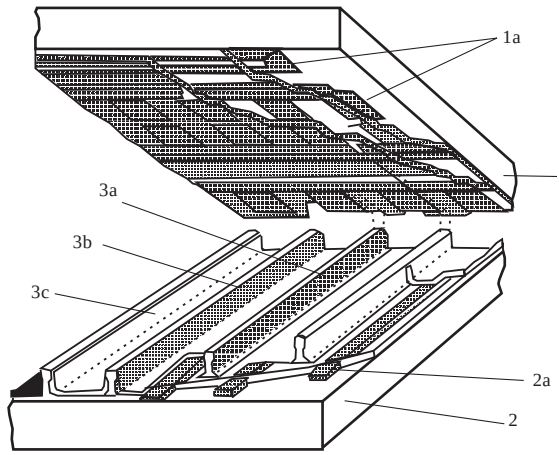


Fig. 10.2.3. The cell of the plasma display: 1) the external glass sheet with a semi-transparent electrode (1a); 2) the internal sheet with three 'boats' for each cell; 2a) the electrodes for each boat; 3) the basic lumihophors; (3a) – blue, (3b) – green, (3c) – red.

25 gradations of brightness, angle of view of 160º, panel thickness 99 mm. Progress has been rapid and in a short time the characteristics of the panels have increased significantly

The panel consists of ~3 × 05·10^6 luminescent light bulbs of the three basic colors: red, green and blue. The scheme of a three-color cell is shown in Fig. 10.2.3. These cells are formed by two glass plates on which transparent electrodes are deposited. On the outer plate of each cell there is a rectangular electrode and on the inside – three rectangles. The inner surface of the plate is curly. Within one cell, it contains three 'boats' the side walls of which are opaque. The space between the glass plates is usually filled with xenon, and the entire volume is sealed. When a voltage is supplied each boat shows a glow discharge which is the source of UV. Feeding voltage pulses of different duration to the electrodes of different 'boats', one can control the brightness of the glow of their luminophors and thus get the desired colour of the cell as a whole. The relatively complicated switching system manages the potentials on the electrodes and provides the desired image on the screen.

10.2.3. The plasma scalpel [247]

Plasma as a surgical instrument at first sight seems unnatural. However, researchers at the N.E. Bauman Moscow State Technical University, led by N.P. Kozlov, together with a number of medical institutions developed a miniature plasma torch with a ~1 mm hole in the anode, operating in

air, which gives a narrow (diameter ~2 mm) and relatively long (about 3 cm) plasma jet, which can be used as a surgical scalpel.

Figure 10.2.4a shows the serial 'Plazmon' installation and Fig. 10.2.4b a section of the scalpel. As can be seen, the Plazmon device is small enough. Its size is 19 × 23 × 33 cm and the weight with accessories around 8.5 kg. At work it consumes ~500 watts. Discharge current and voltage of the plasma torch are respectively 15 A and 30 V. The energy efficiency of the plasmatron is ~40%, the temperature at the jet axis near the exit of the plasma torch ~3000 K, and the degree of ionization ~1%.

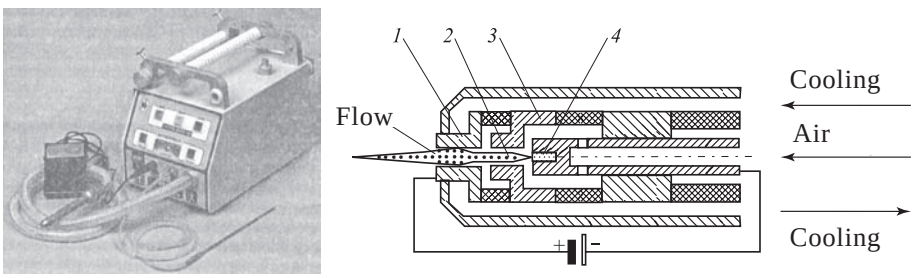
Compared with traditional metal scalpel, the Plazmon scalpel has a number of fundamental advantages.

First of all, the burnt plasma cut surface shows virtually no bleeding, because the blood coagulates and closes the blood vessels. This is particularly important in the surgical treatment of large wounds in combat, which was confirmed in mobile military hospitals.

But there are other advantage of Plazmon.

In 2000, the molecule of the year – this is a competition of the world community of chemists – was announced to everyone's amazement as the molecule of nitrogen monoxide NO. In the same year, the Nobel Prize was awarded for discoveries of remarkable physiological properties of this molecule. It turns out that it is a potent stimulator of many important organs, including the heart. The laureates, in particular, found that the long-known beneficial effects of nitroglycerin on the heart is based on the generation of NO in its use. Now the interest in NO in the world is big. Special international conferences, etc, are organized.

So the plasma flow causes the formation of NO in the surrounding air and thereby stimulates the reparative processes in the treatment of wounds, trophic ulcers, bedsores, and others by blowing their gas flow



a b

Fig. 10.2.4. Plazmon equipment, a) the external view of equipment, b) the diagram of the generator part of the manipulator: 1) the anode; 2) the electrical arc; 3) the inter-electrode insert; 4) the cathode.

with a temperature of 40ºC, containing nitrogen monoxide. Therefore, the full name of the device is the ‘scalpel–coagulator–air-plasma stimulator’.

10.2.4. Chizhevsky chandelier [248]

Everybody knows that in different places there are different ways of breathing. One case in a closed room and another on the street after the rain, in the woods or in the mountains. Studies have shown that to a significant degree this difference is due to the presence of negative ions in the air. The atmosphere of fresh air was artificially created by the remarkable scholar A.L. Chizhevsky. The principle of his ‘chandelier’ is simple. It is a low-current high-voltage discharge between the cathode tip and a smooth anode (Fig. 10.2.5). When applying a voltage of ~300 V to this electrode system, the main voltage drop occurs near the tip where a dark discharge burns under a high atmospheric pressure. The negative ions formed during the discharge are carried away from the cathode due to electrostatic repulsion. Outside the sharp tip zone the conductivity is low, so the ions move in a wide flux to the anode. At the same time positive ions are positioned on the developed smooth part of the cathode near the tip.

The therapeutic effect of the ‘chandelier’ has been studied many times and is still studied. The positive effect of chandeliers has been repeatedly confirmed and the device is used not only in Russia but also in many countries: Germany, Japan, France, etc. Here are some

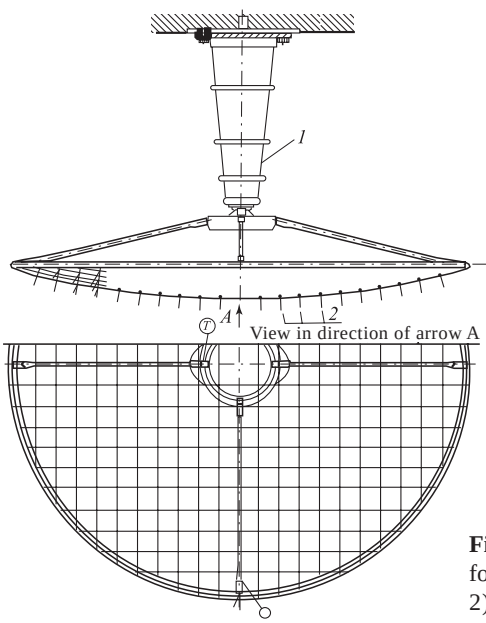


Fig. 10.2.5. Chizhevsky ‘chandelier’, original for industrial areas: 1) high-voltage insulator, 2) needles–cathodes.

Table 10.1

Disease name	Full healing	Considerable improvement	Unknown results	Deteriotation
Angina	35	59	6	0
Hypertensive heart disease	83	15	2	0
Asthma	69	24	7	0
Initial stage of TB	94	6	0	0
Insomnia	66	20	14	0
Influenza	65	25	10	0

data obtained by Chizhevsky together with doctors in the Karaganda Hospital in 1950–57.

Similar results were obtained by other researchers.

But the Chizhevsky chandelier not only generates the desired ions, it also plays the role of an electric filter. Namely, dust and germs in the air acquire a charge and are attracted to electrodes and are deposited on them. Thus, an additional effect is cleaning of the air to remove small particles. In recent years the manufacture of conditioners with a built-in air ionizer started in many countries..

10.3. Formation of structures on solids by plasma technology

10.3.1. General characteristics of the plasma processing of materials [249,250]

The extent of the use of plasma, especially ion–plasma systems, in engineering, microelectronics, coating films of various nature objects, tools, etc. is increasing continuously. The transition to plasma technology allows to replace labor-intensive or polluting practices by easier and cleaner procedures.

In many cases, plasma methods open up entirely new possibilities unattainable using conventional technologies. Obviously, these features are associated with relatively high energy of the particles in comparison with conventional (non-plasma) thermal energies which do not exceed $(3\div4)\cdot10^3$ K, i.e. 0.3÷0.4 eV. At the same time, the energy of plasma particles is unlimited. Of particular interest for treatment of surfaces is the ion energy of ~1÷1000 eV. The lower limit is the heat level and

the upper limit in the vicinity of the maximum of the sputtering factor of the surface.

Processes occurring in the surface layers during the interaction with the plasma are very diverse. They can be divided into six groups.

The first of these is the thermal effect, which leads to melting of the surface. This includes welding and electric arc melting and metal cutting with torches.

The feature of the second group, called implantation, is the saturation of the surface layers with target ions of different nature. The typical thicknesses of saturated layers are measured typically of the order of several microns.

Implantation in the case of semiconductors is used to produced *p–n* junctions in thin surface layers but can be also used to harden the surface layers of metals. For implantation of ions the energy of ~10 ÷ 100 keV is required. The fundamentals of this technology were laid by V.M. Gusev and M.I. Gusev in 1960 [251] (Fig. 10.3.1).

The figure shows the ion-beam unit (ILU-4) designed for implantation of ions in any semiconductor materials, metals and insulators. Controlled implantation in solids can modify their electrical, structural, chemical, optical and other properties. ILU equipment been applied in electronics, metallurgy, chemical and other industries. Specifications of ILU equipment is as follows:

1) ion current to 30 mA;
2) ion energy:
   − single-charge to 100 keV
   − thriple-charge to 300 keV;
3) the area processed by a scanning ion beam at the receiver (20–50 cm)^2.

Under the supervision of M.I. Guseva, the Kurchatov Institute of Atomic of Energy developed an original implantation-plasma accelerator

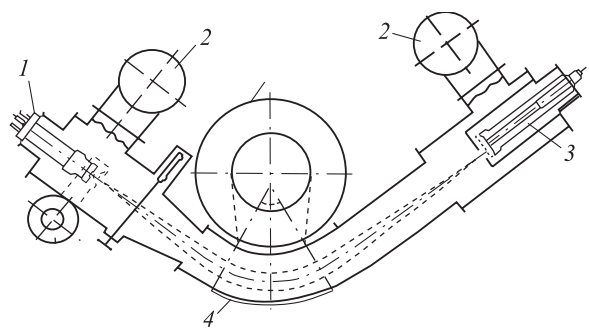


Fig. 10.3.1. Ion-beam equipment (ILU-4) designed by V.M. and M.I. Gusev: 1) the ion source; 3) diffusion pumps; 3) the ion acceptor; 4) the electromagnetic analyser; 5) the coil of the electromagnet of the analyser [251].

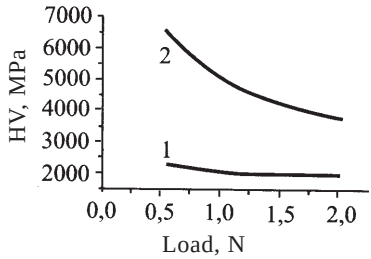


Fig. 10.3.2. Dependence of Vickers microhardness of a ferritic–martensitic steel on load: curve 1 – after implantation of N^+ ions with the energy of 30 keV, dose 10^{21} m^{-2} at a temperature of the specimen of T = 300°C. Curve 2 – successive irradiation with ions N^+ (E = 30 keV, D = 10^{21} m^{-2}, T = 300°C) and the nitrogen plasma beam (E = 300 eV, D = $2 \cdot 10^{23}$ m^{-2}, T = 300°C).

VITA, which includes, in addition to the source of high-energy ions (E = 50 keV), a plasma accelerator of the SPT type. The high density of the ion flux, generated by SPT, results in the formation in the target of a unique 'turbulence' of the dislocations resulting in the penetration of alloying ions into a large depth (up to 1 mm).

The implantation-plasma irradiation of the surface of metals hardens the surface. This is shown in Fig. 10.3.2 which shows the curves of the dependence of Vickers microhardness of a ferritic-martensitic steel on load after implantation of a beam of high-energy N^+ ions (curve 1) and, as a result of subsequent bombardment with the same beam of N^+ and, in addition to this, a plasma flux of nitrogen (curve 2).

The ion implantation method has been introduced to the manufacture of blades of compressors and turbines of aviation engines at the Ufa Motor Engineering Production Association where there is a section consisting of six VITA accelerators for the final processing of the blades in the technological line. The service life of the aviation engine has increased 2.5–3 times.

The third group includes various cases of the deposition layers on the kind of ‘substrate’ without changing the chemical composition of the sprayed substance. The simplest example is the coating of window glass with a thin metal foil, allowing the glass to become a ‘mirror’ when observed through the glass a room darker than the one where the observer is situated.

The fourth group consists of the processes in which reactions occur on the surface of the substrate by which a film is formed which has a different chemical composition than the plasma component incident on the surface. Such coatings include the well-known ‘gold’ coatings of titanium nitride (TiN) or molybdenum nitride (MoN). Another example is different polymer coatings on the metal surfaces. In particular, the Teflon coating of the cookware.

The fifth group consists of ‘compaction’ of the surface, i.e. radical change in the structure of the surface layers. For metals, this can be performed either by rapid (~1 μm) heating to the melting point and

the equally rapid cooling. An amorphous layer of metal with a very high hardness, low erosion and high corrosion resistance often forms in this case.

Similar results can be obtained in relatively low-power stationary plasma accelerators. For example, irradiating a metal plate with the flow of Ar ions with an energy of particles about 200 ÷ 300 eV and current density of about ~100 mA/cm^2 for a few hours can dramatically reduce the size of the crystals forming the surface layers and bring their properties to the properties of amorphous metal.

The sixth group is formed by special methods of sputtering surfaces. Here are typical examples. It is known that the maximum accuracy achieved by the mechanical grinding of sufficiently large optical surfaces is around 10 nm. However, using plasma accelerators such as SPT we can easily reduce surface errors by almost 10 times (i.e. up to the limit permitted by the control system) and the time spent is reduced to tens of minutes.Another example – forming by sputtering of large scale integrated circuits – will be considered in section 10.3.3. There are many such examples.

Below, we consider examples of technologies that apply to groups 2–5 according to our classification.

10.3.2. Examples of coating technology

Deposition of metal films. Deposition of metal films on the surface of various products has been known for many thousands of years. At the present time many new methods have been added to the old ones, but we will describe only one here: formation of films by sputtered atoms. The schematic diagram of the method is as follows: a generator of sufficiently fast ions, which bombard the target, is used. The resulting 'atomized' particles, as a rule neutral, fly to the processed plate and settle on it. This method in comparison to the classic one not require 'wet' processes nor rather sophisticated methods of 'firing' of powders or thin foils to the surface. And yet, it provides very good adhesion of the coating to the substrate. This is due to the fact that the energy of sputtered particles is about 3÷10 eV, which is incomparably higher than the conventional thermal energy and energetic particles are strongly 'bonded' to the surface.

Today, these deposition processes widely use the so-called 'magnetron sputterer (MS)[2], shown in Fig. 10.3.3, which is as one can see, a specialized version of the plasma accelerators with closed drift.

[2]It is also often referred to as 'magnetron', but it is far away from the electronic device – magnetron.

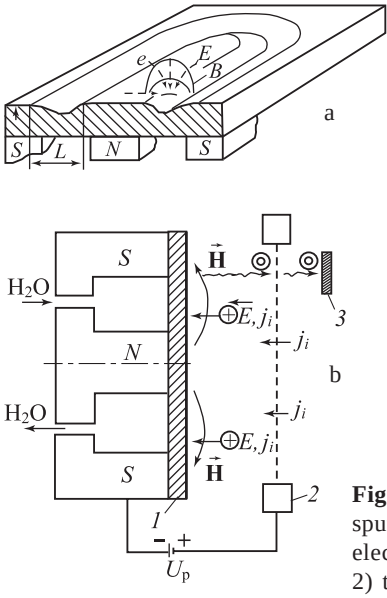


Fig. 10.3.3. Magnetron sputterer: a) the diagram of the sputtering device (half is shown); b) the functional and electrical circuits: 1) the sputtered material (cathodes); 2) the anode; 3) the component–substrate.

We can say that MS is an SPT, in which the outlet of the channel is closed and the entrance is open.

This magnetron also uses the magnetic field, growing in the direction of motion of the accelerated ions or as they say in relation to this accelerator – 'arch' magnetic field that is necessary to ensure the stability of the discharge with respect to asymmetric disturbances, such as the azimuthally asymmetric perturbations in the SPT.

Ion sputtering was used much earlier and for practical problems, however, conventional ion sources used for this purpose were low-power and energy-consuming as the maximum output of the sputtered particles is represented by the energy inaccessible for ion sources ε_i ~300 ÷ 500 eV[3]. At the same time, this range of ion energies is optimal for the SPT, the EAL with the anode cavity and their derivatives – MS.

In this energy range, MS can provide ythe ion current density up to $(1\div2)\cdot10^3$ A/m^2. The upper value of the ion flux is usually limited by permissible heating of the workpiece. In the MS the ion-bombarded target must also be cooled, because heating power reaches a value of ~ $(3\div10)\cdot10^5$ W/m^2. Ionization and acceleration of the ions occur in a layer of thickness ~ 0.5÷1 cm, which accounts for ~ 0.9 U_p, where U_p is the voltage between the target – the cathode – and the body of the installation – the anode.

[3]Usually, the energy of the ions accelerated in the ion source is $\varepsilon_i > 1$ keV.

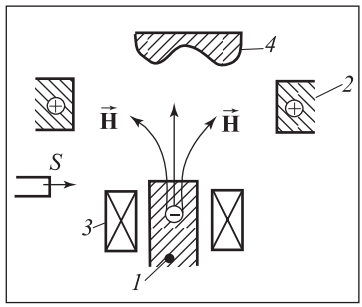


Fig. 10.3.4. Bulat equipment for the deposition of TiN coatings based on the Hall end accelerator: 1) Ti cathode; 2) the anode of the accelerator; 3) the coils of the magnetic field; 4) the processed component; 5) the nitrogen supply pipe.

The gas pressure in the chamber (e.g. Ar, N_2), at which effective sputtering starts is 0.13–0.4 Pa. The magnetic field at the peak is ~0.1 T. The performance of MS can be judged as follows: during the deposition of the sun- and heat-shielding coatings on glass windows one sputterer treats several thousands of square metres per anuum.

The deposition of titanium nitride[4]. Titanium nitride (TiN) – a dielectric – has many advantages. Its colour resembles the colour of gold. It has high strength, hardness and low abrasion. Therefore, TiN coatings have found a variety of applications as both decorative (dishes, the Dome of Christ the Savior, etc.) and as reinforcement materials, such as crowns for teeth, bits for lathe work, milling cutters, drills.

Figure 10.3.4 shows the standard setup of Bulat equipment for spraying TiN. The coating forms as follows. In the end Hall accelerator, titanium is the cathode, and the vacuum chamber Is filled with nitrogen. Ti atoms, knocked out from the cathode, are partially ionized in its neighborhood, and together with the nitrogen atoms form a ‘golden’ finish on the workpiece.

Diamond and diamond-like films. The properties of diamond are unique: the beauty of transparent faceted (‘jewelry’) diamond, hardness (150 times harder than corundum), low abrasion and incredible thermal conductivity (4 times greater than that of copper). Until the mid XX century, the only source of diamonds were rare deposits (South Africa, Yakutia − basic). The overwhelming mass of mined diamonds (small and with defects) are used in the industry for the manufacture of various cutting and grinding tools. Jewelry diamonds are rare, and each of the large diamond creates a chain of historical events.

When at the beginning of the XIX century it was shown that at ~700°C diamond transforms into graphite, this started attempts to transform graphite into diamond. However, this proved very difficult.

[4]First industrial systems (Bulat) for these studies were constructed at the Khar'kov Physico-Technical Institute under the supervision of V.G. Padalka and V.T Tolok [242].

Why is this so is indicated by the phase diagram shown in Figure 10.3.5a. It is seen that at low pressures and low temperatures diamond is metastable and can in principle transform to stable graphite. But this transition is incredibly slow and does not represent practical danger. Similarly, in static conditions at low p and T, as seen in the phase diagram (region 1) the graphite → diamond transition, though possible, is also very difficult. This transition is effective only at $p \sim 2 \cdot 10^{10}$ Pa and $T \sim 2000$°C. Under these conditions, the transition takes ~1 hour.

For the first time reproducible synthetic diamond production technology (small and dark) started only in the middle of the XX century (Sweden, USA, USSR).This synthesis occurred in the region 2 of the phase diagram.

In the 70s of the last century a significant breakthrough took place. It was discovered possible to create diamond films on metals and insulators irradiated with a carbon plasma with ion energies of ~1÷

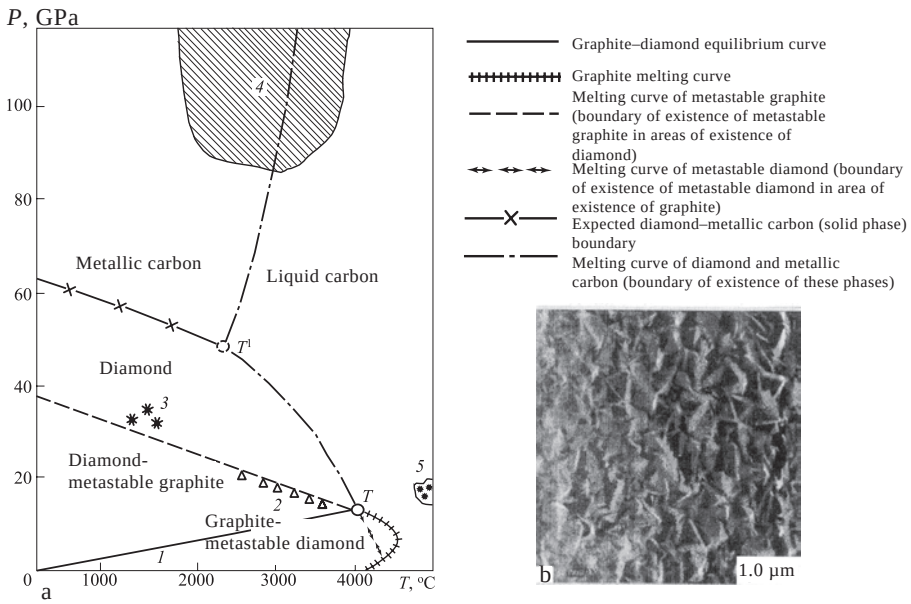


Fig. 10.3.5. A) the phase p–T diagram of carbon: 1) the region of synthesis of diamond using metals-solvents-catalysts; 2) the region of experimental studies to investigate the transformation of graphite into diamond by the static method in direct transition; 3, 4) the region of experimental studies to investigate the transformation of graphite to diamond by the dynamic method; 5) the regional experimental studies of crystallisation of diamond from molten carbon; T – the triple point graphite–diamond–liquid carbon; T^1 – the expected triple point liquid carbon–diamond–metallic carbon; the points on the equilibrium diagram correspond to the temperatures and pressures at which the temperature starts to decrease (quenching of the specimens) to conserve the resultant phase; b) the image of the surface of the polycrystalline iron diamond film produced in a scanning electron microscope.

10 eV. The essence of the process taking place lies in the fact that the impact of ions with the corresponding energy onto a suitable target at the atomic scale creates the conditions close to those listed above.

Thus, it is possible to obtain diamond-like films whose macrocharacteristics are close to those of diamond crystals. This is enough for many practical purposes, although the structure of the films is polycrystalline (Fig. 10.3.5b). Therefore, a number of ion-plasma and gas discharge synthesis methods of films to improve stability of surfaces of parts to wear and corrosion has been itroduced into the industry. Unfortunately, at that time the thickness of the films was on the micron scale. Thicker films fractured.

Later, an intensive search for methods of obtaining cheap large ‘jewelry‘ (i.e. single-crystal) diamonds continued. By 2003, this work culminated in success, and almost immediately on two fronts [255].

One of them was linked to the development of relatively simple chambers, which created a pressure of > 50000 atm and temperatures of ~1200 K, at which graphite is no longer thermodynamically stable and diamond is stable instead. Diamond growth begins with a small ‘seed’ crystal, and after three days a ‘jewelry’ stone with the mass of ~3 carats forms.

Another method is based on plasma. This is an extremely optimized previously mentioned method of diamond film production. Growth also takes place on a seed diamond surface.

The diamonds produced by the plasma method are especially clean and correct. The method does not require high pressures and high temperatures and can, in principle, create large diamond plates. This is very important for microelectronics. The point is that a further increase in computer processing speed is associated with an increase of the density of mounting elements on silicon wafers. This leads to large thermal loads. At a temperature of ~100°C silicon wafers loose the desired properties. At the same time the temperature ceiling of performance of the diamonds is many times higher. This is due to the fact that diamond has a uniquely high thermal conductivity.

Of course, the large size and low cost of diamond crystals have resulted in a technological revolution also in other areas of technology. So what the future that awaits us is in many ways ‘brilliant’.

Deposition of polymer coatings. This includes a large number of different processes. We are interested in devices with the plasma component in which the process of polymerization of simple organic molecules take place on the surface. The formation of solid deposits on the walls of electric devices in the presence of hydrocarbons was described as early as 1874 (A. Tenard, P. de Wilde). But they are

mostly perceived simply as dirt, although they have been known long been for their toughness, poor solubility in many cases, as well as good adhesion to surfaces. In the 1960's in connection with the development of microelectronics the situation began to change. These plasma polymerisation films became the subject of serious research. The following scheme of polymerization on the surface can be described. It begins with the appearance here of the active centres, which may be free radicals or ionized molecules. The emergence of these centers is due to discharges – usually glow. The active centre captures a monomer molecule, and the latter the next molecule, etc. This is the process of chain polymerisation. Here is an example of a chain that begins with ionization

$$CH_2 = \underset{|}{\underset{R}{CH}} \rightarrow CH_2 = \underset{|}{\underset{R}{CH^+}} + e;$$

$$CH_2 = \underset{|}{\underset{R}{CH^+}} + CH_2 = \underset{|}{\underset{R}{CH}} \rightarrow CH_2 - \underset{|}{\underset{R}{CH}} - \underset{|}{\underset{R}{CH}} - \underset{|}{\underset{R}{CH_2^+}} \quad \text{etc.}$$

Further, the chain molecules is constructed from both a positive radical, and an electron.

The growth of the chains may be interrupted for various reasons, in particular, at a meeting of two chains of different signs

$M^+ + M^- \rightarrow$ breakage.

The typical pattern for a reactor for producing these coatings is as follows. The plasma source is a glow discharge. A polymerisable substance and a buffer gas, such as Ar, are introduced into the reactor. Free atoms and radicals, formed in the discharge, diffuse to the surface and cause a chain polymerisation.

10.3.3. The formation of microelectronic circuits

Advances in modern electronics are to a large extent associated with the miniaturization of electronic components, the transition from large-scale integrated circuits (LSIC) to ultra-large integrated circuits (ULIC). The so-called Moore's Law has been valid for nearly 20 years, which states that the number of transistors on a chip doubles every 18 months. If in

the late 80's the minimum size of the transistors was ~2 ÷ 3 μm, in the late 90's it was ~0.2 μm, and by 2007 ~0.07 μm. This is on the area of a 1400 mm^2 chip. Naturally, such fantastic settings can be achieved by using highly sophisticated technology.

Recall that the integrated circuits are now produced on silicon crystal wafers, which are gradually coated with SiO_2 layers or metal, and in which 'patterns' are then cut to ensure the formation of the desired circuit element. In principle, the implementation of the above is as follows. Take for simplicity the case when it is required to cut the pattern directly on silicon. To do this, the silicon surface is covered with a thin sensitized film – 'photoresist'. A 'mask' with the necessary 'pattern' is then projected on the photoresist. Exposure to light – preferably short-wave – changes the properties of the photoresist. Now – as with 'wet' technology – the irradiated plate with the film is treated with a suitable solution. As a result, the photoresist remains only in those areas which should remain unchanged. After that, the plate is immersed in another solution, which dissolves silicon to the desired level. Finally, the plate is placed in a third solution, which washes away the protective photoresist, and the etched pattern remains on the quartz plate. Then, to get the following elements of the integrated circuit, the process is repeated, but with a different program.

As can be seen from the above, the 'wet' technology is associated with the large amounts of reagents. This not only creates environmental problems with their disposal, but also leads to a large percentage of rejects. However, wet processing has a fundamental flaw. It is the 'isotropy of of the effect of reagents' explained in Fig. 10.3.6. In other words, the profile of the depression is relatively difficult to control.

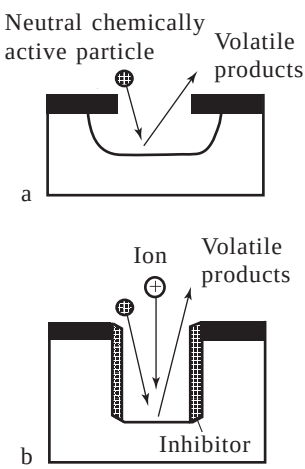


Fig. 10.3.6. The form of 'trenches' (grooves) formed as a result of etching: a) chemical isotropic etching; b) ion-stimulated etching with the inhibitor of particles (atoms, molecules, etc), deposited on the surface from plasma and protecting the walls against the effect of isotropic chemically active particles.

To circumvent this difficulty, it is necessary to use factors with the anisotropic (directed) action. Such possibilities are offered by beam and plasma systems: electron guns, ion sources, plasma accelerators of different types. In addition to the directional flow from the source, a number of other qualities is required. A high etching rate, a strong dependence of the etching rate on the chemical composition of the irradiated surface, as well as minimum radiation damage to the surface layers should be ensured.

10.3.4. Modification of metal surfaces under the influence of energetic plasma blobs

Surface treatment without making any 'foreign' substances, i.e. modification of the structure of the surface layers to increase hardness, corrosion and abrasion resistance, etc, becomes increasingly attractive. Lasers, electron and ion sources, plasma accelerators, low-temperature discharges and the combination of the above are used for this purpose.

Because of its relative simplicity and effectiveness, much attention is paid to surface modification by energetic plasma blobs generated by pulsed guns, short-pulse MPC and a quasi-stationary accelerator. The following gases are used as working substances: H_2, He, Ar, N_2, O_2. For example, consider the impact on the Fe–Ni alloys. The nature of surface modification depends primarily on the specific energy. The 'hard' mode, which is accompanied by melting the surface with exposure time of ~5 μs is realized when the flow is

$$q \geq (3 \div 5)\cdot 10^6 \text{ W/cm}^2.$$

A universal structure of the subsurface layer, shown in Fig. 10.3.7, forms during this process. There are clearly visible three modified layers, not counting the unchaged layer (4).

The thickness of the first layer – solidified melt – is ~3÷5 μm. It has a significant number of discontinuities (cavities, pores, etc.).

The second layer is very homogeneous and has a uniform thickness of up to 15 μm, and usually contains regular structures, formed mainly by thermal stresses associated with both rapid heating and rapid cooling. These structures are similar either to cylinders or hexagonal

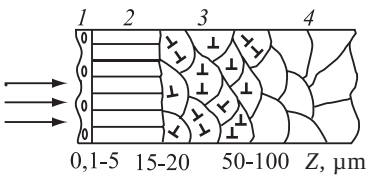


Fig. 10.3.7. Modification of the structure of the subsurface layer of the metal under the effect of a high-energy plasma blob: the diagram of the layers with different rearrangement of the structure.



Fig. 10.3.8. Structure of a silicon wafer after the effect of a plasma blob originated from MPC ('Astashinsky logs').

columns. These differences are determined by the preferred orientation of the crystal structure.

In the third layer, which sharply differs from the second layer, the structure of the starting material is undamaged. However, in each grain, which lies at a depth of up to 100÷150 μm from the outer surface, there is a large number of slip bands. This indicates that due to thermal stresses, deformation exceeds the elasticity limit of the material.

Next is the unmodified material. As can be seen from the above, the restructuring of the surface layers during a brief vigorous action is rather non-trivial.

An additional confirmation of the possibility of a non-trivial adjustment of the surface during the plasma 'blow' is the micrograph of the surface of a silicon wafer (Fig. 10.3.8), exposed to the plasma of the MPC. The wafer was placed in the compression zone. The nitrogen plasma flow parameters were: speed ~$5 \cdot 10^6$ cm/s, the ion density in the compression zone ~$7 \cdot 10^{17}$ cm^{-3}, electronic temperature ~2 eV, the lifetime of the flow ~80 μs. Examination of the wafer gives the impression that 'micro-logs' were piled up on this surface (V.M. Astashinsky, et al [256]).

10.4. Ion and plasma space propulsion engines

10.4.1. Fundamental shortcoming of thermochemical engines

The reactive principle of motion, referred to in section 2.2.3, still remains a major principle in astronautics. And so, this is what it leads to today. Figure 10.4.1 shows the U.S. space complex Apollo-Saturn (the engines

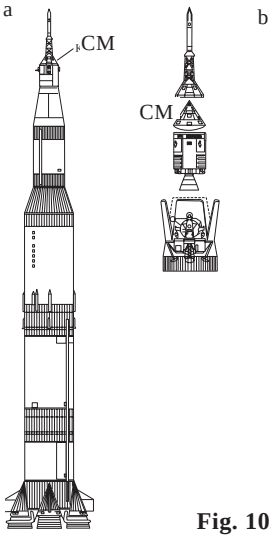


Fig. 10.4.1. The Apollo-Saturn complex; CM is the command module.

for this complex were designed by a team headed by W. von Braun), which first landed men on the Moon and then returned them to the Earth [257]. The launch weight of the system was 2 900 tons, and the mass of the 'command module' (CM) which returned to the Earth was only about 5 tons. This difference is explained by the fact that the engines of this complex were of the jet type. They create thrust through mass ejection, located in the rocket itself, and this thrust is

$$F = \dot{m}w, \tag{10.4.1}$$

here $\dot{m}$ is the mass flow rate, w is the velocity of flight of this mass in relation to the apparatus. From (10.4.1) it immediately implies that, while maintaining thrust, one can reduce the consumption of mass weight only by increasing the exhaust velocity.

A more obvious connection of the starting mass M_0 of the spacecraft with its finite mass M and the exhaust velocity appears in the Tsiolkovsky's formula

$$v = w \ln \frac{M_0}{M}. \tag{10.4.2}$$

Here v is the so-called characteristic velocity of the trip. This is the velocity which a spacecraft moving in the absence of external forces in a straight line would acquire provided that the engines operate in the same mode as in a real flight.

These are the characteristic velocities of a number of optimum flights from the reference orbit around the Earth (h = 140 km).

- reference orbit – the geostationary orbit: $v \sim 5$ km/s.
- reference orbit – an orbit around the Moon: $v \sim 7$ km/s.
- reference orbit – an orbit around the Moon – landing on the Moon – return to Earth: $v \sim 14$ km/s.

From the values of characteristic velocities we can see that to build the spacecraft for which in flights within the limits of the Mars orbit

$$\frac{M_0}{M} \leq 10$$

the required discharge velocity is

$$10 \text{ km / s} \leqslant w \leqslant 50 \text{ km/s}. \tag{10.4.3}$$

At the same time, the discharge velocity from the very advanced modern engines using kerosene and liquid oxygen (RD-107, F = 102 t)

$$w = 3010 \text{ m/s}, \tag{10.4.4a}$$

and using oxygen and hydrogen

$$w \sim 4500 \text{ m/s}. \tag{10.4.4b}$$

Hence it is clear that in any complex flight the giant initial mass is inevitable, even if multi-stage systems are used[5] with hydrogen and oxygen, the given discharge velocity of the water vapours is in general close to the limit for the products of chemical reactions. Therefore, the velocity of the required range (10.4.3) can be obtained only by supplying energy to the accelerated flow from an external source. This can be done differently, but most flexible are electrical propulsion engines (EPE), powered by the onboard generator, the primary source for which is the Sun, a nuclear reactor or a power generator with a radioactive isotope.

It should be noted that in every real flight of the spacecraft there is some optimal value of discharge velocity w. This is because the higher the velocity, the higher the energy cost of propulsion

$$\zeta \equiv \frac{N}{F} = \frac{v}{2\eta}. \tag{10.4.5}$$

[5]The first stage of the Apollo-Saturn engine worked with kerosene and liquid oxygen, and two others with hydrogen and oxygen.

Here, N is the power supplied to the engine, F is its thrust, η is 'traction' efficiency. Increasing v, we reduce the consumption of the mass but increase the weight of the power plant. This also determines the optimum exhaust velocity.

10.4.2. On the varieties of electric propulsion

The basis of the specific EPE may be almost any system that transforms electrical energy into the kinetic energy of the flux. For almost every mission of a spacecraft we can pick up a scheme of the accelerator with the most appropriate characteristics. We emphasize that we are talking about the concept, because for an engine to be constructed, a lot of work is required to optimize the design. It is therefore not surprising that the front of the search for EPE development, especially at the initial stage (50–60-ies of XX century) was very wide [258]. About 30 different versions of accelerators were investigated. But gradually emerged patterns of having a set of important advantages which would ensure in future rather diverse applications. It is natural that these accelerators were paid special attention [259].

The EPEs can be divided into three types:

- preheating (with a solid or arc heat source and the acceleration of the working substance due to ∇p);
- ion, including colloidal (in the accelerating gap there are only positively charged particles accelerated by an electrostatic field);
- plasma (with a quasineutral well ionized flow and preferential acceleration under the Amperé force).

Each of these types is splits into several subtypes. Obviously, the simplest EPEs are preheating ones with a solid state heater. They are now working on more than 30 communication satellites connecting geostationary systems. They support the location of the satellites, compensating for the effect of solar wind, the non-spherical Earth's gravitational field, etc. The main advantage of these engines is due to the fact that at very small sizes, they provide fluxes of light neutral gases (e.g. H_2 or decomposition products of ammonia, etc.) with velocities of $\geqslant$4 km/s and efficiency $\geqslant$0.5.

Several dozen (around 30), geostationary satellites are equipped with electric arc engines. These engines have a relatively low efficiency ($\leqslant$40%), but using hydrogen as a working body, can provide a discharge velocity of w ~12 km/s.

We now consider the ion and plasma thrusters, which have already worked in the cosmos. There are four EPE types. The first of them – ionic and pulsed EPE – to date (end of 2003) worked with single

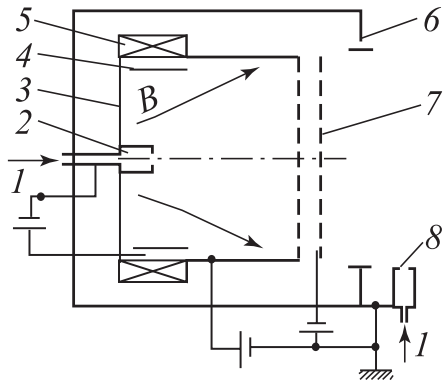


Fig. 10.4.2. Multi-beam ion thruster designed by H Kaufmann: 1) supply of working substance to the gas discharge chamber and the neutraliser; 2) the hollow cathode; 3) the body of the gas discharge chamber; 4) the anode; 5) the coil of the magnetic field; 5) the jacket of the engine; 7) the accelerating intermediate electrode; 8) the neutraliser; B – the lines of force.

samples or small batches for a total of about 20 times each. The engines of the third type – stationary plasma thrusters, commonly known by their abbreviation (SPD) – have become full-time systems and have worked on more than 40 satellites. To them we dedicate a special section 10.4.3. Finally, the engine of the fourth type, called the engine with anode layer (EAL) – has so far worked on only one spacecraft. This thruster is similar to the SPT and, therefore, it will be discussed in Appendix D.

Ion engines. For the discharge velocities $\geqslant$40 km/s and not very large powers ($\leqslant$(10–30) kW), a number of advantages is offered by multi-beam (multi-aperture) ion thrusters with a single gas discharge chamber (GDC) first proposed by G. Kaufmann (USA). Figure 10.4.2 is a diagram of his first multi-aperture ion thruster. The average density of thrust, developed by ion thrusters, is limited by the space charge in the acceleration gap (‘3/2’ law) and equals

$$\overline{f} = \frac{1}{8\pi}\left\langle E^2 \right\rangle.$$

Thus, at $E \sim 1$ kV/mm the value

$$\overline{f} \sim 4\cdot 10^{-2}\,\text{g/cm}^2.$$

Therefore, such thrusters are of the high voltage type, and to reduce the exhaust velocity to the optimum value, the ion thrusters use a substance of a high atomic weight.

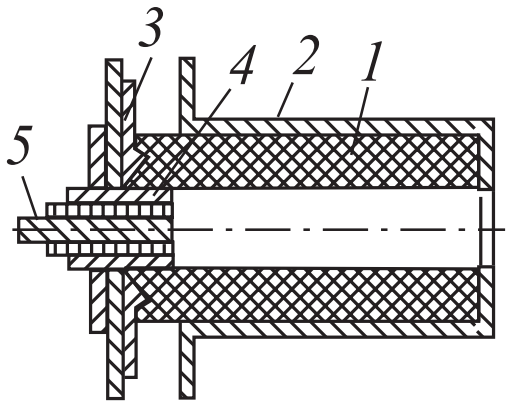


Fig. 10.4.3. Ablation pulsed plasma engine of the thermal type working in space (Zond-2, 1964, USSR): 1) Teflon; 2, 3) the electrodes, 4, 5) the discharge initiator (ignition).

The first ion thrusters successfully operating in the U.S. satellites, were withdrawn in 1970, and on ballistic missiles in 1964. They were designed by G. Kaufman. By the year 2005 there were about 20 flights of ion thrusters on the satellites.

Pulsed ablative engines. These erosional (ablative) pulsed plasma accelerators and, accordingly, thrusters (APPT) are of two types: electrothermal (Fig. 10.4.3) and electrodynamic (Fig. 7.4.2b). In both cases, the working medium is a plasma formed from the insulator under the action of the sliding discharge. The characteristic mass delivered per pulse from an insulator to the discharge is usually in the engines of the order of several tens of milligrams. In electrothermal APPT the plasma heated by Joule heat and is accelerated by the usual gas kinetic pressure. On the contrary, in the electrodynamic APPT erosion plasma is accelerated mainly by the Amperé force, using usually a railgun circuit, i.e. a system of two planar electrodes.

APPT is a low-thrust engine. They were used either for altitude control systems or to maintain the position of small spacecraft in orbit. The first APA (thermal type), shown in Figure 10.4.3, was launched into space on a spacecraft Zond-2 in 1964 in the USSR. It was created under the leadership of A.M. Andrianov (IAE). The first APPT of the electrodynamic type was launched in the U.S., where it worked for about two years, maintaining the position of the spacecraft LES-6 on the geostationary orbit.

10.4.3. Stationary plasma thrusters (SPT)

Today, SPTs are attracting particular attention of experts.

The principles of the SPT have already been described in 6.7 and 7.5. Let us briefly recall that according to their functional scheme SPTs resemble ion engines, but the role of solid-state electrodes here is played by what might be called 'electron–magnetic electrodes' (EME). This is because the plasma magnetic lines of force are equipotential up to relatively weak thermal perturbations. Therefore, the lines of force with electronic trajectories wound on them can be regarded as clear filamentous electrodes. The schematic diagram and physical features of these engines were discussed above, but here we describe three models: the first propulsion system (PS) with SPT was launched into space (SPT EOL-1), a modern engine SPT M-100 and a promising model SPT-Aton.

The first propulsion system with SPT EOL-1 [260]. By mid-1968 The Institute of Atomic Energy in the laboratory of G.Ya. Schepkin on the basis of the proposal and guidance of A.I. Morozov produced a working model of SPT model of radiative cooling and efficiency $\geqslant$35%. Created in the same place, a year later the designed propulsion system was transferred to the bureau 'Fakel' (Kaliningrad), where it was brought to the required technical level under the leadership of R.K. Snarsky and delivered in late 1971 to the All-Union Scientific Research Institute of Electromechanics to be installed on the satellite 'Meteor' (Chief Designer A.G. Iosif'yan).

The basic setup options include: power of PS 400 W, thrust ~0.02 N; weight: ~18 kg.Structural scheme of an SPT-EOL and his appearance are depicted in Figure 10.4.4.December 31, 1971 satellite "Meteor" was launched into space. At the beginning of 1972, tests of the EOL propulsion system were successful. The main results were:

a) demonstration of the reliability of the PS with the SPT and its compatibility with the spacecraft;

b) no significant effect of the plasma jet on connection with the Earth:

c) a good correspondence of the traction characteristics defined on the stand and on the basis of changes in the trajectory of the satellite. Rather than the claimed 100 hours of operation, the EOL systems worked about 150 hours, used the entire stock of xenon and has provided a useful orbit correction of the system, putting it on the quasi-synchronous orbit, lifting the altitude of the orbit by 15 km.

The SPT EOL efficiency is ~30%. Shortly after starting EOL-1, the researchers at he IAE showed that the efficiency can be significantly (up to ~50%) raised by transfer to the 'thin' poles, i.e. increase the magnetic field gradient along the length of the chamber. This was implemented in the engine EOL-2. As a result, the Fakel Design Bureau

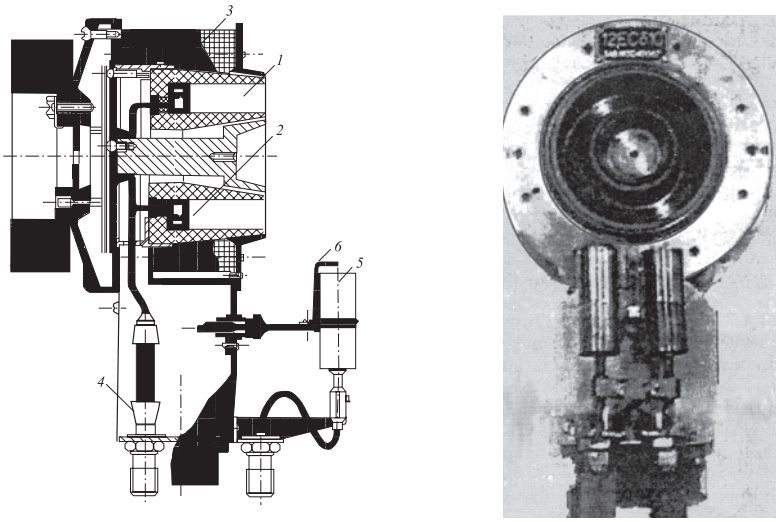


Fig. 10.4.4. Stationary plasma thruster EOL-1: a) the diagram of the structure; 1) the discharge chamber; 2) the anode-gas distributor; 3) the magnetic circuits; 4) high-voltage inlet; 5) the hollow cathode; 6) the ignition electrode; 6) external view.

has been designing for more than 30 years propulsion systems with the SPT.

SPT M-70 and SPT-100 [262]. In late 1970s, the Design Bureau Fakel designed a standard module SPT M-70[6] on the basis of the EOL SPT. It was designed for a power of ~1 kW, traction ~4 g and an efficiency of ~45–50%. The SPT M-70 was installed on 15 satellites.

In the 1990's a new engine SPT M-100 started opertion; the thruster was also created in the Fakel Bureau and was designed for a power ~1.5–2 kW. It was similar in design to the M-70, but in the M-100 the thickness of the insulator near the cutoff was increased. Its appearance is shown in Figure 10.4.5. The operating voltage of the M-100 was ~250–400 V, 6–10 g thrust, efficiency ~50%, and the divergence of the flow $\alpha_{eff} \approx \pm 45°$.

The most remarkable feature of the M-100 is its life. Despite the fact that the ion flux 'ate' in 8000 hours not only an insulator, but also part of the poles (see Fig. 7.5.5), the traction performance of the thruster remained unchanged. Now the M-100 is working on many satellites.

The success of the SPT has stimulated the development of a number of firms in Russia, as well as in the U.S., France and other countries. In December 2005 two French spacecraft and one U.S. spacecraft were

[6]The number indicates the internal diameter of the external insulator in mm

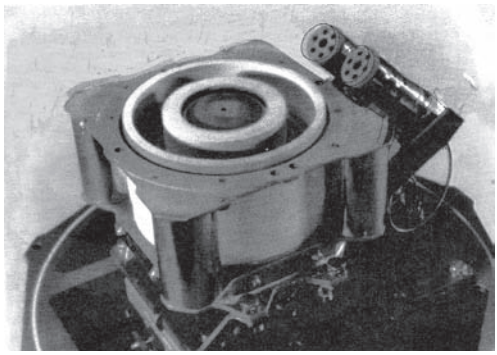


Fig. 10.4.5. External view of the M-100 stationary plasma thruster (Fakel Company, Kaliningrad).

equipped with a satellite SPT and work in space. In total, the M-100 has been installed on more than 40 spacecraft.

Note that the SPT and similar systems are attracting increasing attention, as well as new tools for beam technology for surface treatment.

10.4.4. Promising schemes of EPE

In the future, electric propulsion engines will cover the power range from a few watts to many megawatts. They will be of the multi-mode type, have thrust vector control, high efficiency up to $\gtrsim$80%. The future tasks will be discussed briefly.

1. The above described SPT had a relatively low efficiency and greater beam divergence. It was clear that the divergence of the flux can be radially reduced only by placing the ionization zone not near the channel outlet, but in the depth of the channel – an area where the curvature of the lines of foce is greatest, using magnetic fields with the zero point [263].

The laboratory model of this thruster was called SPT ATON. The size of the device was approximately the same as that of the M-70. After optimization, this model showed unique characteristics. Its efficiency (taking into account the flow of xenon in the cathode-compensator) approached 65%, the effective angle of divergence was $<\sim 15°$. Direct observations showed that under the optimal conditions the flux inside the channel was practically separated from the walls, while the output nozzle has a cylindrical geometry ([49], section 9, page 466).

This type of SPT was created at MIREA (the laboratory headed by A.I. Bugrova with the idea of A.I. Morozov (IAE) and financial support of the French company SEP).

Modern SPT schemes allow significant expansion of the range of the operating parameters. For example, the IAE laboratory were successfully tested the model with an external insulator diameter of

300 mm. The thruster power reached 30 kW, and the thrust, depending on the discharge voltage, was in the range of 100–150 g (1–1.5 N). In this case, efficiency was> 60% (working substance – xenon). But this is not the limit.

Finally, it should be noted that the tests of the SPTs were conducted with different working substances, ranging from hydrogen to xenon. The results obtained corresponded well to the previously established criteria for such discharges (section 6.7).

2. A number of laboratories (USSR – Russian Federation, Germany, USA, Japan) quite intensively studied and optimized prototypes of possible future 'end' engines: high-current (EHTC) and Hall (EHT), referred to in section 3.7.2. The first is based on the intrinsic magnetic field of the discharge, and the second uses as an external magnetic field. In the English literature EHT is usually called 'magnetoplasmadynamic' thruster (MPDT). A substantial contribution to the development and optimization of these engines in the formed USSR was provided by A.A. Porotnikov, V.B. Tikhonov, I.N. Ostretsov [259].

EHT and, in particlar, EHCT have suitable characteristics (efficiency $\gtrsim$ 50%) at a sufficiently high power ($\gtrsim$100 kW) even when working with light substances, especially lithium. These thruster also showedthe current crisis, (section 7.6.6) and for this reason they should be called 'thermo-magnetic'.

The just discussed end thrusters are of the two-electrode type. They contain a cathode and a solid anode. However, a three-electrode TSD reminiscent QHPA was also constructed. Its remarkable feature is the absence of the current crisis, at least for those parameters (w, J_p), where the crisis was observed for the two-electrode models [259].

3. At the beginning of section 10.4.2 it was stated that many different schemes of possible prototypes of electric propulsion engines were tested at that time. Without describing them in any detail, we only recall the most attractive schemes:

− thruster with an anode layer, single-stage;

− thruster with an anode layer, two-stage;

− 'colloidal engine' thruster (similar to the ion thruster), but with the acceleration of charged particulates, possibly nanoparticles;

− laser thruster (similar to the arc thruster, but with an optical discharge).

10.5. The problem of controlled thermonuclear fusion (CTF)

One of the main factors determining the situation in the world today is the production of energy. Some of these energy sources are renewable

(solar energy and its derivatives: wood, wind, ocean currents, etc., and, with reservations, geothermal), and the others non-renewable (coal, oil, gas, uranium, thorium, etc.). Today, the main ones are the non-renewable energy sources, but their sources are unevenly distributed, and this gives rise to all known conflicts. But there is an energy carrier that is everywhere, and its reserves are almost inexhaustible. It is water, plain water. More precisely, 0.02% of the total amount of hydrogen contained in water. Such a role in natural water is played by the heavy isotope of hydrogen – deuterium $D \equiv {}^{2}_{1}H$.

The energy here is released at the confluence of two deuterium nuclei. As a result, as we shall see below, 1 liter of tap water, due to the presence of deuterium, is equivalent to 300 liters of gasoline! The problem of controlled thermonuclear fusion is the problem of extracting this energy in 'quiet' conditions, in contrast to the hydrogen bomb, which also is the fusion reaction of light nuclei [264].

10.5.1. Underlying principles [265], [266]

'Thermonuclear' reactions of nuclear fusion

The mass of the nuclei m_{nuc}, as is well known, is less than the sum of the masses of the constituent protons m_p and neutrons m_n

$$m_{nuc} - (Zm_p + Nm_n) = \Delta m > 0.$$

The value of Δm is called the mass defect, and it gives a measure of the energy $\varepsilon = \Delta mc^2$, which is released during the formation of the given nucleus from protons and neutrons. Calculated per nucleon, the maximum $(\Delta m)_1$ falls in the middle of the periodic table. Therefore, light elements emit energy in nuclear synthesis, and heavy one – in decay. The nuclear fusion reaction of the lightest elements is called 'thermonuclear'.

Below are the equations of these reactions with the energy released in these reactions, and also the plots of the dependence of the synthesis sections, depending on the relative energy of the particles.

The reactions are ranked according to decreasing cross sections:

1. Deuterium–tritium (DT)

$$D + T \rightarrow {}^{4}_{2}He + n + 17.6\ \text{MeV}.$$

2. Deuterium – an isotope of helium ${}^{3}_{2}He - (D^{3}He)$

$$D + {}^{3}_{2}He \rightarrow {}^{4}_{2}He + p + 18.4\ \text{MeV}.$$

3. Deuterium – deuterium (DD)

This reaction can go through two channels

$$\mathrm{D+D}\rightarrow\begin{cases}\mathrm{T}(\equiv {}^{3}_{1}\mathrm{H})+p+4.0\ \mathrm{MeV}\\ {}^{3}\mathrm{He}+n+3.3\ \mathrm{MeV}.\end{cases}$$

The tritium produced in this reaction rapidly reacts with deuterium. The efficiency of the reaction of the nucleus ${}^{3}_{2}\mathrm{He}$ with the deuteron is slightly lower.

It is easy to verify, using the specified energy release, that a liter of ordinary water, due to deuterium, has the energy content equivalent to the specified amount of gasoline.

4. Neutron–lithium (nLi)

$$ {}^{1}_{0}n+\ {}^{6}_{3}\mathrm{Li}\rightarrow {}^{4}_{2}\mathrm{He}+\mathrm{T}+4.8\ \mathrm{MeV}.$$

This reaction is intended to be used to produce tritium at work on DT.

5. Deuterium – lithium (D^6Li)

Here the synthesis takes place in two channels

$$\mathrm{D}+\ {}^{6}_{3}\mathrm{Li}\rightarrow\begin{array}{l}\rightarrow 2\,{}^{4}_{2}\mathrm{He}+22.4\ \mathrm{MeV}\\ \rightarrow {}^{7}_{3}\mathrm{Li}+{}^{1}_{1}\mathrm{H}+5.0\ \mathrm{MeV},\ \text{etc.}\end{array}$$

As seen in Fig. 10.5.1, the largest reaction cross-section has a pair of DT. The cross-section is large even for ε ~ 10 keV. Therefore, DT

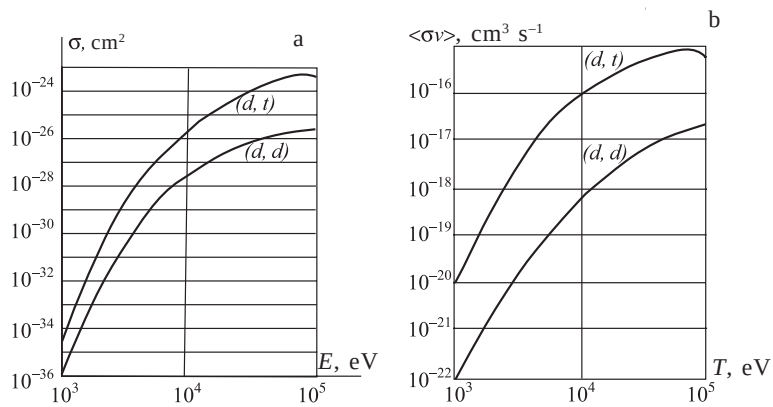


Fig. 10.5.1. Dependence of the cross-section σ on energy (a) and the rate of thermonuclear reactions ⟨σv⟩ on temperature (b) for different mixtures.

is now seen as the main working substance in the initial stage of the thermonuclear reactor.

Because in the reactor there will be particles with different velocities, then the basic role in the calculations is played by the reaction rate $\langle \sigma v \rangle$, and averaging is usually performed by a Maxwellian distribution of reacting particles (Fig. 10.5.1).

10.5.2. Lawson curves

The knowledge of the reaction cross-section at various energies of the particles and the magnitude of the energy output is still not enough to answer the question as to what properties should have the reactor itself and the system of conversion of the kinetic energy of the particles formed during the reaction into electrical energy – if heating is carried out outside the reactor by precisely the electrical energy so that system as a whole had a positive yield. Better yet, if the operation of the reactor would be self-sustaining.

In principle, the kinetic energy of the α-particles and protons can be converted directly into electricity through a variety of systems of 'direct conversion' – recuperators. At the same time, the kinetic energy of the neutrons can at present be converted into electrical energy only through the thermal cycle. Note that currently the neutrons are regarded as an undesirable product of the reaction, not only because of the need for the heat cycle, but also because of the large radiation-activating ability of the neutrons.

The reaction cross-section, as mentioned above and depicted in Fig. 10.5.1, is very small. Hence, it is impossible to achieve a positive energy output, if the fast particles irradiate any target. Indeed, since the cross-section of elastic collisions of the fast particle with target nuclei is much greater than the cross-section with a nuclear reaction, the fast particle loses energy prior to the occurrence of the reaction. It is therefore necessary to create some 'traps', where the particles will live long enough, many times colliding with each other, until the desired reaction occurs. This assertion can be called the basic principle of thermonuclear fusion or the 'Lavrent'ev principle'.

So, we should have a trap in which the released energy is greater than the energy used for preheating the particles to the desired 'temperature', on the unavoidable losses due to radiation, thermal conductivity and simply 'slipping' of particles from the trap.

In other words, the following condition should be satisfied

$$\eta \langle \sigma v \rangle_{nuc} \epsilon_r \frac{n^2}{4} > \gamma n + \frac{n\bar{\varepsilon}}{\tau_*} \equiv \frac{\alpha n}{\tau_E}. \qquad (10.5.1)$$

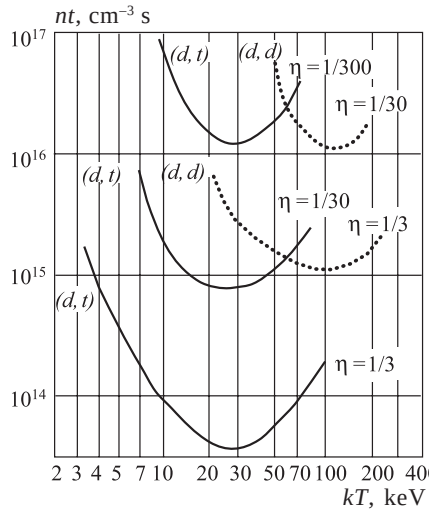


Fig. 10.5.2. Dependence of the Lawson constant (parameter) nTE on temperature for different mixtures.

Here η is the coefficient of conversion of the released nuclear energy into electrical energy (usually taken as η~1/3)[7], ϵ_r is the energy released in a single reaction, $\langle\sigma v\rangle_{\text{nuc}}$ is the reaction rate averaged over the distribution functions of heavy particles, coefficient 1/4 allows for participation of two particles in the reaction of two particles, γn is the intensity of bremsstrahlung, $n\varepsilon/\tau_*$ is the energy loss due to 'slipping' of the particles from the trap and thermal conductivity, τ_E is the 'lifetime of energy in the trap'.

Equation (10.5.1) can be written as

$$n\tau_E > A. \tag{10.5.2}$$

Here $A \equiv \dfrac{4\alpha}{\eta\langle\sigma v\rangle_{nuc}\,\epsilon_r}$. The condition (10.5.2) is referred to as (not very fairly) the Lawson criterion, although in some form it was formulated earlier by other researchers.

Figure 10.5.2. shows the dependence of the Lawson constant A on the plasma temperature for different mixtures. It is seen that the lowest value $(n\tau)_{\text{min}} = A \approx 10^{14}$ cm^{-3} s is achieved for a DT mixture at a temperature $T \sim 10$ keV $\approx 10^8$ K. For the pure deuterium and mixtures D^3–He the values $(n\tau)_{\text{min}}$ are an order of magnitude larger, and the optimal temperature lies in the vicinity of $T_{\text{opt}} \sim 50 \div 60$ keV. For other 'thermonuclear' mixtures the values $(n\tau)_{\text{min}}$ and T_{opt} are even higher.

[7]In 2005, electric power stations started to used extensively gas turbines whose efficiency is twice that of steam turbines. Therefore,b the value $\eta = 2/3$ can now be used.

The obtained values ($n\tau$) comply with the regime, when the 'fuel' with the required operating temperature is placed in the reactor. For the reactor to go to the 'ignition', i.e. when it is enough to use cold fuel, it is necessary that:

$$(n\tau)_{\min}^{(\text{ign})} \geq 2A.$$

Here we recall that in the case of the DT reaction 4/5 of the released energy is carried away by neutrons, and the cold working substance is heted solely by α-particles.

We estimate the plasma parameters for the simplest – and now the basic case – when the reaction is a mixture of DT $(n\tau)_{\min} = A$, and the energy trap time $\tau_E = 1$ s. Under these conditions, the charged particle density $n_i = n_e = n = 10^{14}$ cm^{-3}. From the equilibrium conditions and definition of the parameter β

$$2nkT = \frac{H^2}{8\pi}\beta, \tag{10.5.3a}$$

we find

$$H = \frac{9\cdot 10^3}{\sqrt{\beta}}\text{Oe}.$$

If $\beta = 5\cdot 10^{-2}$, which corresponds to the maximum value of β in tokamaks, we have $H = 4\cdot 10^4$ Oe. The energy of the electrons and ions, contained in 1 m^3 is equal to

$$W = 2n\frac{3}{2}kTV = 0.5 \text{ MJ}. \tag{10.5.3b}$$

And, consequently, if $\tau_E = 1$ s, then the fusion power generated by 1 m^3 of such a plasma, is equal to 0.5 MW.

10.5.3. Trap circuits

As mentioned in the introduction (I.3), experiments with CTF started in 1950 and are still being continued and what has been obtained – a few seconds of work with DT $(n\tau_E) = 0.4$ cm^{-3} s, $T \sim 10$ keV. This was done at the end of the 90's in a tokamak JET and a Japanese tokamak JT-60 (see Appendix E).

The central problem of CTF is to create a fusion of traps for plasma satisfying the Lawson criterion, i.e. retention of plasma energy for a long enough time τ_E at the appropriate temperature.

A variety of ways have been tried for the implementation of CTF:

− Fixed traps using magnetic fields;

− Fixed traps using combined magnetic and electric fields;

− Pulsed magnetic systems (*z*- and θ-pinches of varying configurations);

− Heavy-duty, short-pulse systems with inertial confinement (heating and compression of small $\leqslant$1 mm targets with the help of coherent and incoherent radiation, electron beams, fluxes of heavy ions, etc.);

These systems are very different from each other. It suffices to say that the density of hydrogen plasma (DT) in the proposed and implemented systems ranges from $5 \cdot 10^{-10}$ g/cm^3 to ~100 g/cm^3, and the duration τ_E from ~1 ns to seconds or longer (the latter – in perspective)[8].

Below, the focus will be paid to stationary traps with magnetic confinement. However, for the full picture, we first briefly discuss the systems with inertial confinement, the study of which has been the subject of muct attention around the world with a number of outstanding achievements.

Parameters of the system with inertial confinement [265]

The main elements of such systems are:

− spherical 'target' with a diameter of 1 mm, containing D + T;

− system of powerful generators of energy flows, focused on the target. In principle, energy carriers can be particle fluxes, but now the most convenient are the fluxes of coherent radiation of short-wave lasers on neodymium glass and non-coherent X-ray radiation of the 'two-stage' *Z*-pinch (Appendix B).

Both the target and generators of the radiation fluxes are very complicated systems and their features will not be discussed here.

At thermonuclear temperatures, the expansion velocity of the heated target is ~10^8 cm/s. Therefore, the working pulse at the specified target size should be ~10^{-9} s = 1 ns. The energy deposition in the target to obtain a positive energy balance, as the calculations and experimental data show, should be the order of several megajoules. Accordingly, the laser pulse power should be ~10^{15} W.

[8]There are also non-plasma approaches to the problem of CTF. The besk know is the 'mu-catalysis' which uses μ-mesons, but these are short-lived and energy losses in producing them are large.

If we consider that the efficiency laser systems on Nd-glass $\lesssim$0.3%, the Lawson constant will be $A \gtrsim 10^{16}$ cm^{-3} s. Therefore, the particle density of hydrogen isotopes must be

$$n \sim \frac{A}{\tau_E} \sim 10^{25} \text{ cm}^{-3}.$$

At the same time, the density of solid hydrogen under normal conditions $n_T \sim 5 \cdot 10^{22}$ cm^{-3}. This shows that the target must be compressed almost 1000 times. More careful calculations increase this estimate 3÷5 times, and then we arrive at the mass densities of ~200÷300 g/cm^3.

Compression of this magnitude can be obtained (and now compression only one order of magnitude smaller have been obtained) due to the jet pressure from the mass evaporating and departing by the radiation from the target surface.

The most impressive results were obtained in 1990 by the American installation Nova (Fig. 10.5.3). There the laser pulse with a duration of ~1 ns had an energy of 100 kJ. In this case, the following values were obtained $n\tau_E \approx (2–5) \cdot 10^{14}$ cm^{-3} s, $T \approx$ (1.5–2) keV, $\rho \sim$ (20–30) g/cm^3. These are very high settings, and a new project of a laser system with inertial confinement (project NIF), capable of a positive energy yield, was subsequently developed in the USA.

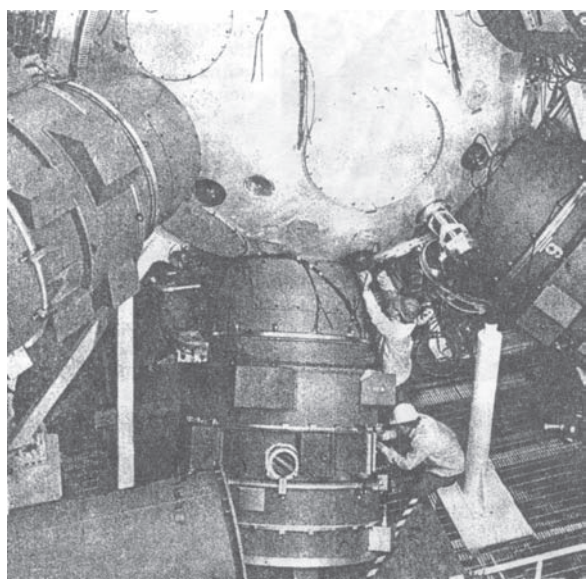


Fig. 10.5.3. Chamber of Nova equipment in which the target is irradiated.

However, the physics of inertial confinement refers mainly to the physics of dense non-ideal plasma. Therefore, we will not dwell on it.

Let's move on to the stationary magnetic traps which look better for industrial purposes.

Three types of magnetic confinement systems

On the basis of the topology of the plasma volume, the stationary magnetic systems, as mentioned in section 1.7, can be divided into three types:

− simply connected;
− doubly connected;
− multiply connected.

The first type includes θ- and *Z*-pinches and 'open' traps with a longitudinal magnetic field whose edges are covered with a variety of 'stoppers' (Figure 10.5.4). This is primarily a solenoid without stoppers – 'solenoidal trap' (proletotron in Russian) (Fig. 10.5.4a). Today, it is not used, primarily because of the high energy consumption. The concept of the solenoidal trap was proposed by A.I. Morozov in 1968 to create a kind of benchmark for assessing the scale of reactors [275]. The length of the solenoidal trap on DT was of the order of several kilometers.

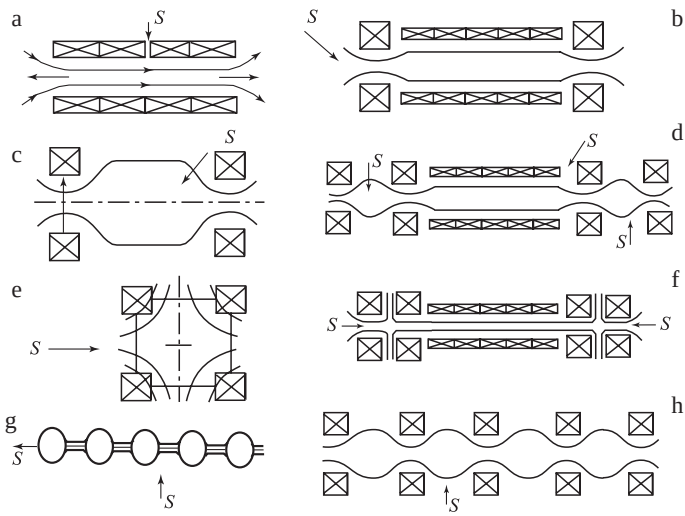


Fig. 10.5.4. Diagram of the open traps: a) solenoidal trap; b) the gas-dynamic trap; c) the simple Budker–Post mirror trap; d) the three-section Dimov trap with electrostatic closure of movement of the particles through the lens; e) the simple anti-mirror trap; f) the long anti-mirror trap DAL; g) Morozov diffusion (ESPL); h) Budker corrugated trap (GOL); S) the injection of the working substance.

But the role of the concept of the solenoidal trap was more significant. Starting from this idea, G.I. Budker [275] proposed a corrugated trap with dense plasma (see below), and D.D. Ryutov − a hydrodynamic trap (Fig. 10.5.4b). The latter scheme is based on the fact that the length of the solenoidal trap satisfying the Lawson criterion, in the case of DT should be an order of magnitude greater than the mean free path relative to the ion–ion collisions. Therefore, as a first approximation, the solenoidal trap (proletotron) can be regarded as gas-dynamic system and the losses can be reduced by increasing the field at the ends. Then the loss of particles decreases H_{max}/H_0 times. True, because of the higher value H_0 the gain is really in the length $\lesssim 2$.

Further, in Fig. 10.5.4c there are 'mirror' or 'adiabatic' traps. These traps were referred to in sections 1.7.1 and 5.6.2. Retention in them of the particles is due to conservation by the particles of the transverse adiabatic invariant ($J_\perp = v_\perp^2/H$). The retention time of these traps is of the order of the ion–ion collision time. To increase the retention time of particles in a mirror trap, G.I. Dimov proposed three-section trap, shown in Figure 10.5.4g. Dimov's idea was as follows. In the end traps (sections) hot-ion plasma with $T_i \sim 1$ MeV is created. At this temperature, the time between collisions increases and becomes longer than τ_E in the middle section, where most of the nuclear reactions take place. Plasma density n_{max} in the end sections is higher than n_0 in the central section, and so the potential there in accordance with section 3.2.2 is higher than at the edges:

$$\delta\phi \sim \frac{kT_e}{e}\frac{n_{\max}}{n_0}.$$

These potential 'humps' also keep the ions in the main section. Selecting n_{max}, n_0, $T_{i,\,max}$, $T_{e,\,max}$ $T_{e,0}$ and the volumes of the sections, one could count on a positive energy output of the reactor. But ... the emergence of new factors − large electric fields and high T − has led to new difficulties which have not yet been overcome.

The drawings in 10.5.4e and 10.5.4f show the circuits of 'anti-mirror' traps. These traps provide the hydrodynamic stability of the plasma, situated in the centre of the trap, since the magnetic field grows in all directions with increase of the distance from the centre, but large gaps do not allow us to hope to achieve a positive energy output.

The principle of diffusion traps is simple (Fig. 10.5.4 g,h). They are chains of some elementary 'stochasticizing' cells. Such cells have two openings through which particles can enter and exit. At the same time, entering the cell, the particle "forgets" where it entered and is equally

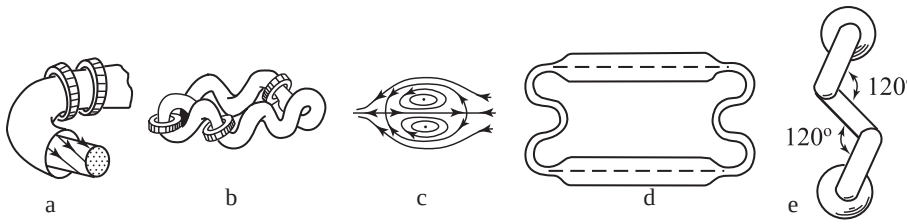


Fig. 10.5.5. Diagrams of toroidal traps: a) tokamaks; b) stellarators; c) compact tori; d) Drakon trap (two projections).

likely to go through any of the two. As a result, the walk starts along the chain, and the lifetime of the particle in the trap of N cells will be

$$\tau_N \sim N^2. \tag{10.5.4}$$

The concept of these traps was proposed by A.I. Morozov at about 1961, and later by G.I. Budker (1969). Such traps have been implemented in multimirror (corrugated) fields. To ensure stochasticization, cells are represented either by short mirror traps with length $\sim\rho_i$ – the ion Larmor radius (ESPL trap, A.I. Morozov)[9], or cells with very dense plasma, where the mean free path $\lambda \sim L$ – the length of the cell (Gol trap, Budker).

Traps with a doubly-connected plasma volume. These are toroids of different configurations, the magnetic fields of which are composed of nested magnetic surfaces (Fig. 10.5.5.). They were proposed by L.A. Artsimovich, I.N. Golovin, N.A. Javlinsky, V.D. Shafranov, L. Spitzer, etc.

With axial symmetry, as discussed in detail in section 1.1, such magnetic configuration in the chamber can only be constructed through the coils producing an azimuthal magnetic field from azimuthal electric current. If the current flows through the plasma in the chamber, the system is referred to as a tokamak and to date remains subject to the most intensive investments and the most detailed design-engineering studies, although we cannot claim that they will be included in future fusion power engineering systems. Nevertheless, we will devote the next section 10.5.4 to them.

The occurrence of strong currents in the toroidal plasma pinch creates a number of difficulties with the stability of plasma configurations and the need to maintain for years (in the case of industrial plants) the current in the plasma torus. Therefore, more interesting are traps in

[9]See [267].

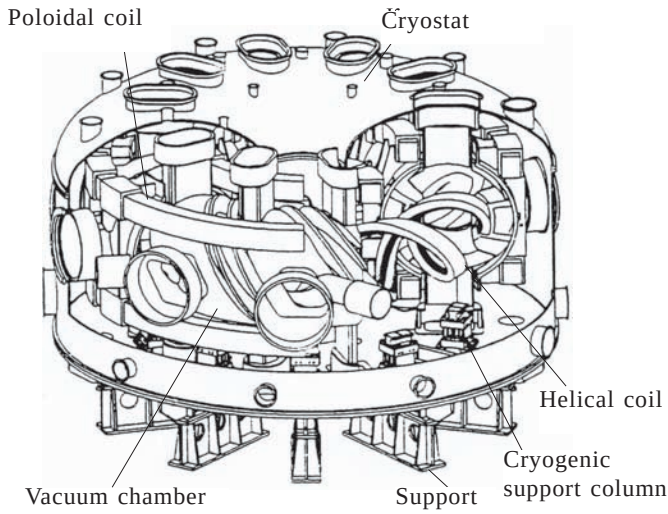


Fig. 10.5.6. Diagram of the LHD stellarator.

which the configuration of nested magnetic surfaces is created external to the plasma 'screw' conductors. However, these systems lack axial symmetry and magnetic 'surfaces' are slightly blurred. These systems have a collective name 'stellarators' or 'helicoidal traps'. Some aspects of their theory were discussed in section 1.7.2. The largest installation of this type is the Japanese LHD (Large Helicoidal Device) with a superconducting magnet system. The scheme of the working area of this trap is shown in Fig. 10.5.6.

Here are some parameters of this system. Diameter of the circular magnetic axis D = 7.3 m, diameter of the cross section of the plasma pinch d = 1.2 m, the magnetic field on the circular axis of $H_0 \leqslant 3$ T. This system produced in different experiments plasma with $n_e = 5 \cdot 10^{13}$ cm^{-3}, $T_e > 10$ keV, $\tau_E < 0.4$ s, $T_i \leqslant 5$ keV, $\beta = 8\pi p/H^2 \sim (2–3)\%$.

Drakon[10] trap. Figure 10.5.5d shows a doubly connected 10.5.5g trap with a long equilibrium configuration representing a hybrid mirror trap and a stellarator. It was proposed and developed by B.B. Kadomtsev, B.A. Trubnikov and V.M. Glagolev. Curved elements connecting the mirror traps, selected with such a geometry that 'gradient currents' ($\mathbf{j} \sim \nabla p$) do not pass into the mirror traps [9].

Multiply connected traps (Galatea) [267]

A Galatea trap is characterized by the fact that parts of the conductor generating a magnetic field are immersed in the plasma (Fig. 10.5.7).

[10]Drakon is an abbreviation of the Russian name Dlinnaya RAvnovesnaya KONfiguratsiya (Long Equilibrium Configuration).

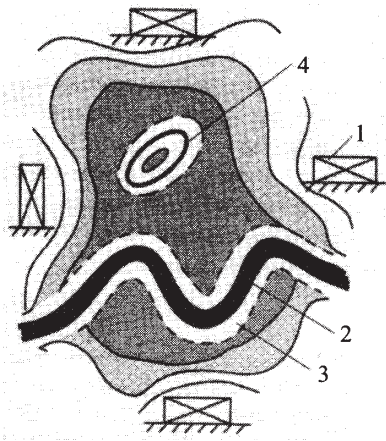


Fig. 10.5.7. General diagram of the galatea: 1) simply supported coils, 2) myxines.

These conductors are called submerged myxines[11]. Using myxines one can create gapless 'crust traps' or 'magnetic cylinders'. The principal difference between the magnetic cylinders from the classical simply- connected and doubly-connected traps is that in the latter the magnetic field and plasma are mixed, while in the cylinder the field is concentrated at the periphery (in the form of a 'crust') and the plasma is in the centre of the trap where the magnetic field is not found (Fig. 1.7.1). This distinction is fundamental and leads to a number of significant advantages of the galatea traps. Here are the some of them:

− plasma is diamagnetic. Therefore, placing the plasma in the volume of the magnetic field is not desirable from the point of view of physics, since diamagnetics are pushed out by the magnetic field. Of course, ways are now available of overcoming the 'rough' pushing, but more subtle forms of manifestation – in the form of various 'weak' instabilities − still lead to a reduction in the lifetime of the plasma in the trap. In Galateas, diamagnetism, conversely promotes retention as well as prevents the penetration of the plasma into the 'crust'.

− Modern orientation to use as a working substance the D–T mixture is clearly of a temporary nature. It is most natural to use the D–D reaction, i.e., rely only on the isotope in the water. But then, as seen from the graph 10.5.2 the required operating temperature is $T \sim 50$ keV and $n\tau \sim 10^{15}$ cm^{-3} s. At such high temperatures the loss due to synchrotron radiation rapidly increases as a result of centripetal acceleration of electrons in a magnetic field. To radically overcome this difficulty, it is necessary to move to the crust traps – magnetic cylinders. Synchrotron radiation is unlikely to work effectively in

[11]It should be mentioned that the trap proposed by A.D. Sakharov was a galatea (1.7.10).

tokamaks and stellarators with DD, as maximum β in the first case is 5%, and in the second ~3%. If a lot of rings-myxines are used, the area occupied by the magnetic field will be small compared with the area occupied by the plasma, i.e. we really have the cylinders. This configuration of the magnetic system is obviously the most economical.

However, to ensure stability of the sharp boundary between the plasma and the field in the cortical configuration, the boundary should be 'pushed' into the plasma volume (see section 8.2.1). This produces 'acute-angled configuration'. To avoid formation of slits, a projection ('cusp') should be shorted to the other. As a result, current-carrying conductors are surrounded on all sides by the plasma. A thin plasma layer that connects the cusp is called the 'mantle'. Thus, the conductors that create a magnetic field should be levitating, i.e. suspended in a magnetic field (Fig. 10.5.8).

At first glance it might seem that such a suspension creates insurmountable technical difficulties. Such statements can be found in the literature. However, simple estimates show that it is not so [267]. Consider in detail the three main objections to the magnetic suspension [267–269].

The first of these is the very possibility of suspension. It is obvious that to ensure that the current-carrying myxine 'does not fall", it is necessary to create an external magnetic field H_r that is radial in relation to the myxine. If it is a trap with no nuclear reactions, the value of H_r can be estimated only by taking into account the weight of the superconductor. In this case we have

$$\rho gls = 0.1\, jlsH_r$$

Here ρ is the density of the superconductor (g/cm^3), s is its cross section (cm^2), l is the length of the myxine (cm), j is current density

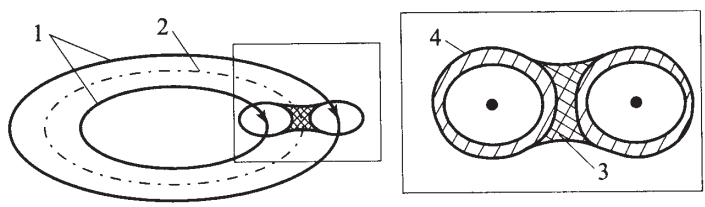


Fig. 10.5.8. General view (a) and cross section (b) of 'Dublon' trap: 1) levitating conductors with current; 2) line of the zero magnetic field; 3) acute-angled 'main plasma volume'; 4) 'mantle'.

(A/cm^2). Taking typical values for superconductors (e.g. NbSn) $\rho \sim 10$ g cm^{-3}, $j \sim 10^4$ A/cm^2, we find

$$H_r = 10\frac{\rho g}{j} \sim 10 \text{ Oe. (!)}$$

This field is very small compared to typical fields for CTF traps $H \sim 10^4$ Oe. Once again, to ensure there are no problems with the proper magnetic suspension, we note that in Japan, Germany, China high-speed passenger trains ($\gtrsim$500 km/h) on magnetic suspension are already in service [268].

No fundamental technical problems arise in the case of Galatea reactors with CTF, in which the myxines are exposed to neutrons and other energy-carrying fluxes with a density of ~(1–2) mW/m^2. In this case a three-layer protection is placed on the superconducting 'chord'. These layers are sometimes called: external − red, next − gray, and the third (cryozone) − blue. They must be separated by insulating spacers. The red layer is made of high-temperature materials (graphite, tungsten, etc.). Under the action of the incoming energy the layer is heated up to temperatures of ~ 2000–2300 K and releases the above-mentioned power in the form of heat radiation. The gray layer mainly partitions off the penetrating flux of neutrons and γ-quanta on the way to cryozone. In the gray layer and the cryozone there should be coolers, maintaining the desired temperature. A system reminiscent of household refrigerators is formed here. The energy for them can be supplied from the outside by, for example, microwaves. The emergence of high-temperature superconductors will obviously radically simplify the problem of maintaining the chord in the superconducting state.

Examples of experimental units − galateas are depicted in Figure 10.5.9, and Fig. 10.5.9 shows the scheme of the fields of these traps. Figure 10.5.9a shows the galatea of S. Yoshikawa with a three-magnetic field, which has one superconducting myxine [269]. It is similar to the toroidal trap by A.D. Sakharov and is not, obviously, a magnetic cylinder. Figure 10.5.9b shows the octupolar galatea–cylinder of the University of Wisconsin with four warm myxines. The plant operates in pulsed fashion. This ensures the levitation and stability of the myxines by the skin effect in the surrounding casing [267]. Figure 10.5.9c provides insight into the Tornado trap (A.I. Ioffe LPTI) proposed by G.V. Skornyakov and realised by B.P. Peregud [267]. Finally, Fig. 10.5.9d shows the galatea Dipole, proposed by Hasegawa. This trap is in essence a model of the radiation belts of planets.

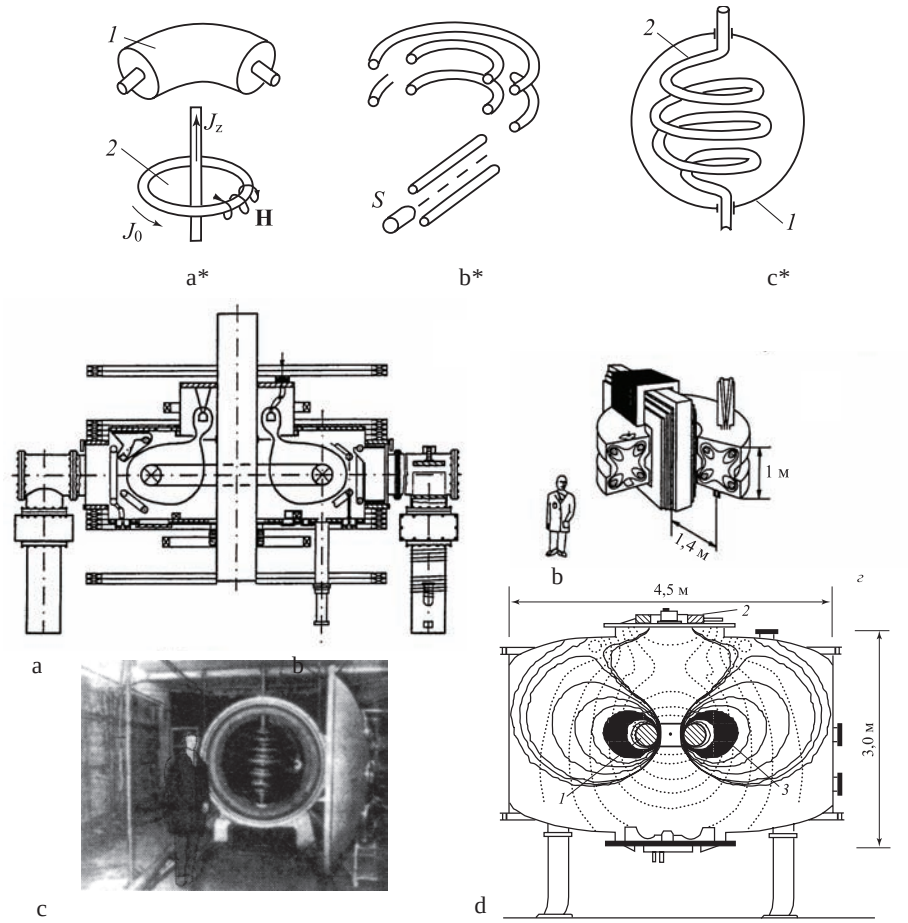


Fig. 10.5.9. a) Yoshikawa galatea; b) Wisconsin University galatea; c) Tornado galatea; d) the laboratory model of the Dipol trap-galatea: 1) levitating ring with the current (1.24 MA); 2) 'sustaining' turn (0.3 MA), 3) hot electron plasma; a*, b*, c*) schematic images of the appropriate systems.

Without giving the detailed data obtained at different galateas, we emphasize the main result of this research. Here modes were achieved **with classical transfers!**

Begun in the 1960's, intensive research into galateas was, unfortunately, discontinued in the early 70's because of the belief in an unconditional promise and simplicity of the tokamaks. But the difficulties in the further development of tokamaks stimulated in the 1990's renewed interest in galateas.

10.5.4. Tokamaks (see, for example, [270,271])

The tokamaks have been through many stages of development. This changed both the design of installations and their operation. Figure 10.5.10a shows the first toroidal chamber constructed at the Institute of Atomic Energy (IAE), Fig. 10.5.10b shows the scheme of the first tokamaks, and Fig. 10.5.10c the internal view of the JET tokamak. Figure 10.5.10d shows the international tokamak ITER constructed at Cadarache (France) (IAE). This tokamak is described in details in Appendix E and F. Typical parameters of these and other tokamaks are listed in Table 10.5.1.

Common to all these system are: 1 – a toroidal vacuum chamber, which is filled with working gas (usually hydrogen or deuterium) at a pressure of ~10^{-3} torr); 2 – coils, creating in the chamber a longitudinal (azimuthal) magnetic field of ~(20–50) kOe; 3 – a reversing a iron

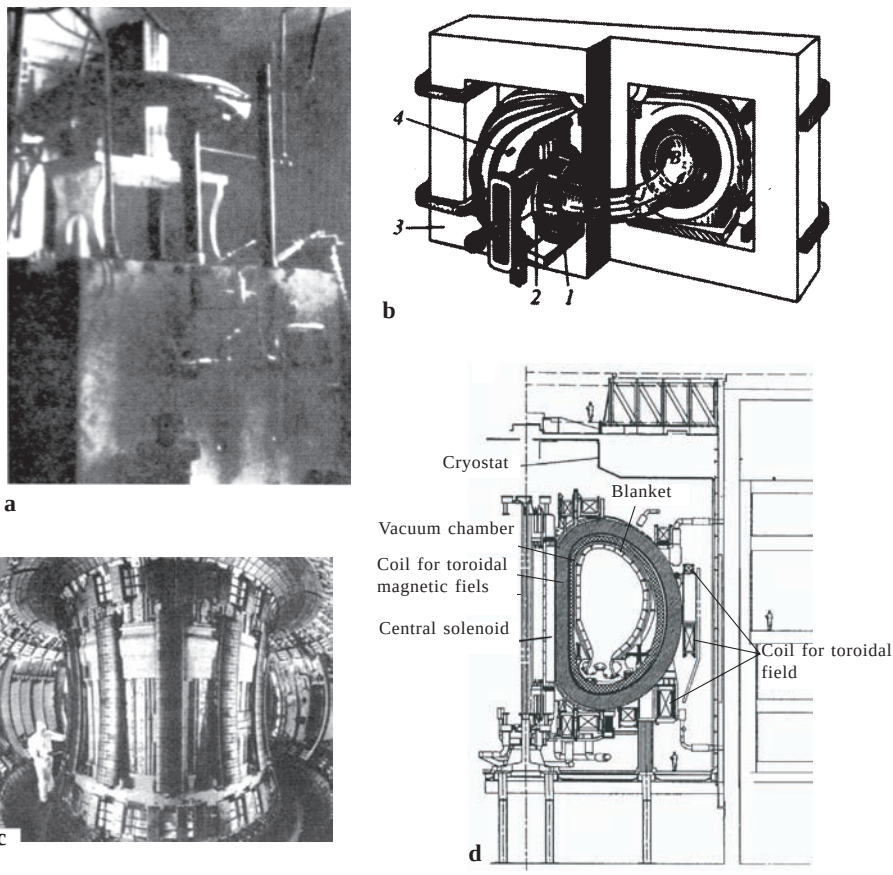


Fig. 10.5.10. Diagrams and photographs of tokamaks: a) the first toroidal system, constructed by the Institute of Atomic Energy; b) T-3 tokamak; c) JET tokamak; d) the design of the ITER tokamak.

Table 10.2

	Name of the installation,country	Year built	R_m, m	a/b, cm	H_0, T	Plasma heating
1	Tokamak T-3 (USSR)	~1965	1	25/25	....	ohmic
2	Tokamak T-10 (USSR)	1975	1.5	78/78	5	neutral injection
3	JET (European Community)	1983	3	125/210	3.5	– // –
4	JT-60 (Japan)	1987	3	190/190	4.5	– // –
5	Project ITER (International)	–	8.1	240/450	5.7	self-sustaining

R is the radius of the magnetic axis of the system; *a* and *b* are the diameters of the cross-section of the magnetic pinch with respect to *r* and *z*; H_0 is the maximym strength of the azimuthal field at the magnetic axis of the chamber.

core that creates a vortex electric field in the chamber that provides a breakdown in the cold gas in the chamber and formation of a current, which performs two important functions. Firstly, thanks to the current gas ionization continues together with heating of the resultant plasma to $T_i \sim T_e$ of the order of several hundred eV. Secondly the magnetic field of this current has poloidal (H_r, H_z) components, allowing closed – ideally – magnetic lines of force of the azimuthal field to 'break', then transformed into spirals and form nested toroidal magnetic surfaces. Thus, a magnetic configuration capable of removing 'toroidal polarisation' and sustaining the plasma is formed.

The possibility of a simple method of implement both heating and confinement of the plasma was demonstrated in 1969 by L.A. Artsimovich at the IAEA International Conference in Novosibirsk and resulted in extensive 'tokamisation' of nuclear fusion research worldwide.

However, subsequent studies, now worldwide, showed that all is not so simple. All this was reflected in designs and circuits of functioning tokamak in subsequent years.

The first thing that attracts attention is the significant increase in the size of plants. The JET, the largest of existing tokamaks, built by the European Community in Britain, is an experimental installation which did not reach the stationary thermonuclear parameters, even when working on DT. Even bigger is the international project ITER with the permanently[12] continuing thermonuclear reaction on DT, where

[12]Or, more accurately, short working stages with a duration of ~1000 s (for mode detail see Appendix E)

no ignition is planned, i.e. heating of cold fuel supplied to the reactor by α-particles formed during the reaction.

The second fundamental difference is the way of obtaining the required fusion temperature. In early tokamaks it was hoped to achieve heating with Joule heating with current flowing in the plasma. However, this method becomes ineffective at a sufficiently large thickness of the plasma volume and high plasma conductivity which increases $\sim T_e^{3/2}$ and already at $T_e \approx 1$ keV is equal to the conductivity of copper. Evaluation and experiments show that the limit of Joule heating is just $T_e \sim 1$ keV, and for the DT reaction we must have $T_i \sim T_e \geqslant 10$ keV.

Clearly, non-Joule methods of plasma heating are required. As noted in section 10.1, two methods are used at present: electron cyclotron resonance (ECR) and the injection of fast neutrals.

The next very important distinction of early and advanced tokamaks is the time of maintaining the azimuthal current in them. Of course, without the appropriate EMF in the plasma toroid, the current will attenuate within a relatively short skin time ($\lesssim$1 hour for industrial tokamaks). But this cannot be allowed in an industrial reactor, since in this case the 'first' wall of the tokamak will operate under variable thermal loads that will dramatically reduce the life of a tokamak reactor, not to mention other negative consequences of such intermittent operation. Therefore, great effort has been made to find the most simple and effective methods of maintaining the current. These methods were the same as those needed for plasma heating. This is primarily tangential injection of neutrals which, ionizing, provide ionic current, because the electrons are 'entangled' in a magnetic field and generally stop. Another method of electron cyclotron resonance (ECR) is based on the use, as a means of formation, of an electron flux. Both these methods have been tested and proven effective.

It should be noted that there is another method of maintaining the current in the filaments, although this is achieved to the desired extent, only in the conditions of classical transfers. The point is that, thanks to diffusion, the plasma from the pinch axis moves to the periphery across the magnetic field and thus creates the EMF, which maintains the current. Thus, it is sufficient to inject the working substance on the axis of the pinch and the substance goes to the periphery and maintains current. In experiments on tokamaks it was thus possible (this effect is called bootstrap) to increase the retention time of the current in the pinch by ~30%.

The principal possibility of self-sustainment of current in the plasma pinch due to diffusion of plasma in the intrinsic magnetic field of the pinch can be explained by a simple one-dimensional model. Let a layer

of thickness L at $x = 0$ be continuously fed with the plasma with the flux density

$$\dot{\nu} = n\nu = \text{const.} \tag{10.5.5a}$$

Because of the stationarity and the continuity equation, the flux $\dot{\nu}$ remains in the entire volume of the layer. Three other equations (Euler, Ohm's law and Maxwell's equation) at $T_e = 0$ and $T_i = \text{const}$ have the form:

$$kT_i n + \frac{H^2}{8\pi} = \frac{H_0^2}{8\pi};\ \frac{j_y}{\sigma} = -\frac{\nu H}{c};\ j_y = -\frac{c}{4\pi}\frac{\partial H}{\partial x}. \tag{10.5.5b}$$

Here it is taken into account that there is no electric field. Expressing n, v, i through H, we obtain a simple equation

$$\nu_m \frac{\partial H}{\partial x} = \frac{\dot{\nu} H 8\pi k T_i}{H_0^2 - H^2} \equiv \nu_m \theta \frac{H}{H_0^2 - H^2};\ \theta \equiv \frac{\dot{\nu} 8\pi k T}{\nu_m}. \tag{10.5.6a}$$

Hence, we find

$$H_0^2 \ln\frac{H}{H_1} + \frac{H_1^2 - H^2}{2} = \theta x. \tag{10.5.6b}$$

The existence of the stationary solution in this case means the existence of a bootstrap.

Another important difference between the early and advanced tokamaks id related to the problem of contamination of the working (hot) zones of the plasma volume by heavy particles coming from the walls due to exposure to the plasma and radiation. In the early models this interaction was weakened due to installation in the chamber of diaphragms which limited the transverse size of the pinch, and except for a narrow zone of contact of the pinch with the thin diaphragm, separated from from the walls.

In modern tokamaks, contact of the working area with the wall is prevented by the so-called divertator proposed in the 50's of last century by L. Spitzer (USA). Its essence boils down to using a conductor, which is located outside the chamber, with the current in the conductor coinciding with the direction of current in the plasma pinch, and the magnetic lines of force, adjacent to the first wall, leave the chamber, where they cross the 'divertor' plate (Fig. 10.5.10d). Impurities coming from the wall, as well as plasma, diffusing from

the working area outside, entering the magnetic flux of the divertor travelling out, quickly flow onto the divertor plate, located outside the chamber, where they become somehow linked. In particular, the gaseous products are pumped away with pumps. The divertor is clearly visible in the diagram of the ITER in the lower part of the chamber.

The important distinction between methods of stabilizing the plasma pinch in the chamber of the old and new tokamaks is described in section 8.5, and here we do not dwell on it.

Another significant difference between the old and new tokamaks is associated with the choice of material for the 'first' wall of the chamber directly in contact with plasma. Originally, it was stainless steel, but it had a significant disadvantage. Its atoms of iron and nickel, getting into the working volume, were ionized in hot plasma to $Z \sim 20$ and thus dramatically increased the losses due to bremsstrahlung because they increased $\sim Z^2$. So, the level of impurities in the working area should be $\leqslant 0.1\%$ not to exceed the bremsstrahlung of hydrogen. Therefore, the surface of the first wall should be made of Li or Be. In particular, the first wall of the JET is covered with beryllium.

All the above gives some idea of the variety of problems that must be addressed. To this we must add the development of adequate systems of plasma heating and introduction of the working substance, diagnostic systems, capable of working in a high-power neutron flux and γ-rays. It is also necessary to create a set of robots that can repair such a reactor, and a system of generation of tritium, which is not present in nature in appreciable quantities, etc.

The tritium is expected to be generated using a 'blanket' which surrounds on all sides the chamber with plasma. The thickness of the blanket is ~1 m, and inside it there is lithium (Li) in some form, which captures neutrons produced at the confluence of D and T. As a result, the reaction of the formation of tritium takes place in the blanket

$$^{6}_{3}\mathrm{Li} + {}^{1}_{0}n = {}^{3}_{1}\mathrm{T} + {}^{4}_{2}\mathrm{He}.$$

Tritium is pumped out and travels to the reaction zone.

We have listed the principal structural designs in recent years. Of course, an enormous number of experimental, theoretical and numerical studies has been carried out. However, at the moment it is impossible to say that everything is clear. This is indicated by the fact that the design of large systems is based on scalings, which were discussed in Chapter V, and not on indisputable theoretical models, Appendix E and F).

Ending this brief overview of the major achievements over the past nearly 50 years of studying and improving the tokamaks, we note what these works have yielded in terms of the ultimate goal – a stationary industrial synthesis reactor on the DT-mixture.

There are currently no problems with the heating of electrons up to several tens of keV. Injection of neutrals can produce fluxes of particles with energies up to (1–2) MeV.

However, at the desired densities ($n \sim 10^{14}$ cm^{-3}) and temperature $T_i \sim T_e \sim 10$ keV it is not yet possible to obtain the desired retention times τ_E.

Nevertheless, the possibility of obtaining a sufficiently powerful nuclear energy in the tokamaks has already been demonstrated. The best that has been done was achieved in the JET, where a reaction power of ~17 MW was obtained for several seconds.

What next?

We assume that further study of tokamaks, including ITER, will not bring fundamental difficulties, and therefore the time spent on the development of the tokamak power engineering will be minimal. According to preliminary estimates, together with the final selection of sites for the construction of ITER the actual construction would take about 10 years.About 5 years will be needed to master this installation and commissioning it. It is assumed that full-scale tests using tritium will take about 10 years. However, these tests will be conducted in relatively short pulses of ~ 1000 sec, i.e. ITER is not the task of continuous long-term work. Shortly after finding out the main features of the ITER, it is necessary to design and create the DEMO plant whose parameters are close to ITER, but it should be run in a continuous burning deuterium-tritium mixture, heated by α-particles formed by the following reaction. This reactor will operate in a cyclic mode, with continuous operation time of ~1–10 days. The total operating time of DEMO is assumed to be 8 years without changing the elements. Note that the DEMO fusion power is assumed to be 2.44 GW. Taking 10 years to construct and study DEMO, we see that it would take 35-40 years to begin to start production of industrial plants based on the tokamak. This still requires a minimum of 10 years. So take about 50 years, and it is natural that a number of countries that did not belong at that time in the ITER program, will seek their own ways of solving the problem of fusion, but there are also great opportunities. First of all, it is the continuously growing power of computers, thereby significantly reducing the experiment time. Next: tokamaks have − especially in the perspective – two significant drawbacks. Firstly, tokamaks, with a small β, about 5%, are now focused on working with the D–T mixture, which corresponds, as we have said,

to a temperature of ~10 keV and $n\tau_e \sim 10^{14}$ cm^{-3}s. At the same time, the CTF program at the time was developed under the banner of the D–D reaction, based on deuterium, which is in ordinary water. But the transition to the D–D reaction requires a temperature of ~50 keV and $n\tau_e \sim 10^{15}$ cm^{-3}s. If we consider that the desired settings have not as yet been, even for the D–T reaction, it is clear that the probability of a tokamak using the D–D reaction is negligible. But this is not enough. In tokamaks and stellarators, the plasma is immersed in a strong magnetic field. At an electron temperature of ~ 50÷60 keV and the presence of a strong magnetic field the intensity of synchrotron radiation dramatically increases and this is likely to be an insurmountable obstacle to the traps with $\beta \ll 1$. Therefore, the D–D fusion reactor must have $\beta \sim 1$, i.e. in the bulk of the plasma volume the magnetic field should be practically absent. In other words, if we are talking about the magnetic confinement traps, it must be magnetic cylinders, rather than a trap with a small β.

The second reason, enabling the search for new circuits of fusion reactors, is a clear discrepancy between the parameters of ITER and principally possible parameters of traps with $\beta = 1$, and classical transitions for the same T and $n\tau_e$. It is known that sooner or later the evolution of technical devices will reach the physical limit. It is natural to expect that the same transition to the limit ($\beta \approx 1$, the classical transitions) will happen in the case of CTF systems before the moment of transformation of the DEMO to industrial systems.

10.6. From generators of multiply-charged ions to the island of stability and black holes in the experiment

In this last section of the book, we consider methods for generating heavy ions with $Z \gg 1$ up to hydrogen-like and generally devoid of electrons, and describe them with the help of two major steps in the knowledge of our world, which will eventually lead to literally fantastic technologies. It would be a synthesis of far transuranium elements and the creation of tiny black holes. Both these achievements are associated with acceleration to high velocities of multiply-charged ions in special particle accelerators. Transuranic elements (with $Z > 92$) were created mainly in Berkeley (California, USA) and Dubna (USSR–Russia)[13] using very sophisticated technologies for production and identification of single atoms of new elements. Synthesis of ‘microholes’ was carried out at the beginning at CERN (Switzerland), and then in Brookhaven (USA).

The accelerators, used for these purposes, are not included in the range of plasma systems considered here, but their ‘hearts’,

[13]Scientists at Hessen (Germany) also contributed to these achievements

as mentioned by Academician Georgy N. Flerov – the creator of transuranium physics in Russia – is a source of multiply-charged ions.

10.6.1. Sources of multiply-charged ($Z \gg 1$) ions

Above, as a rule, we considered singly-ionized ions, which could be 'accidentally' mixed with the doubly-ionized particles, such as in the stationary plasma thrusters. In the previous section, devoted to the tokamaks, we mentioned ions of heavy elements with $Z \gg 1$ (up to a complete loss of electrons). There they appeared as 'dirt', as an unwanted impurity.

However, there are cases where the ions with a high degree of ionization ($Z \gg 1$) are essentially needed. Here, we briefly discuss the scheme for such ions. These ions acquire, in the respective accelerators, an energy proportional to Z, which allows for fundamentally new results. This will be discussed in the following paragraphs of this section. In the meantime, let us say briefly how to obtain these ions.

The existing methods for producing multiply-charged (up to complete stripping of electrons) ions are naturally divided into two groups.

In the first group stripping takes place during the passing of a sufficiently fast particle flying through a foil of suitable thickness. This process begins in the plasma volume, where the neutral atoms acquire a small charge Z. They are then accelerated to the MeV scale, and usually pass through the foil. Some of them are lost, and the rest get a charge $Z_2 > Z_1$. 'Surviving' ions are accelerated again and again pass through the foil. Ions with $Z_3 > Z_2$ appear, etc. This method is used to generate droplets of quark-gluon plasma (QGP), as discussed in section 10.6.3.

The other group consists of more traditional methods that use relatively rare plasma. These methods are more versatile, and we pay them more attention. In particular, they apply to systems of synthesis of transuranium elements.

Features of multiple ionization in relatively rare plasma

The first thing is to face the high ionization potentials $I_{z \to z+1} \sim Z^2$. Figure 10.6.1 shows the values of ionization potentials for all degrees of ionization for all elements. Here, each broken line corresponds to a certain degree of ionization, indicated at the left end of the line.

As can be seen, the potential of a fixed degree of ionization I_z decreases with increasing Z_μ – the number of the element in the periodic table. However, the value of $I_{z \to z+1}$ increases rapidly with increasing Z for Z_μ = const. Therefore, the energy required to uranium U^{+92} is $\varepsilon_i \sim I_{91 \to 92} \sim 200$ keV.

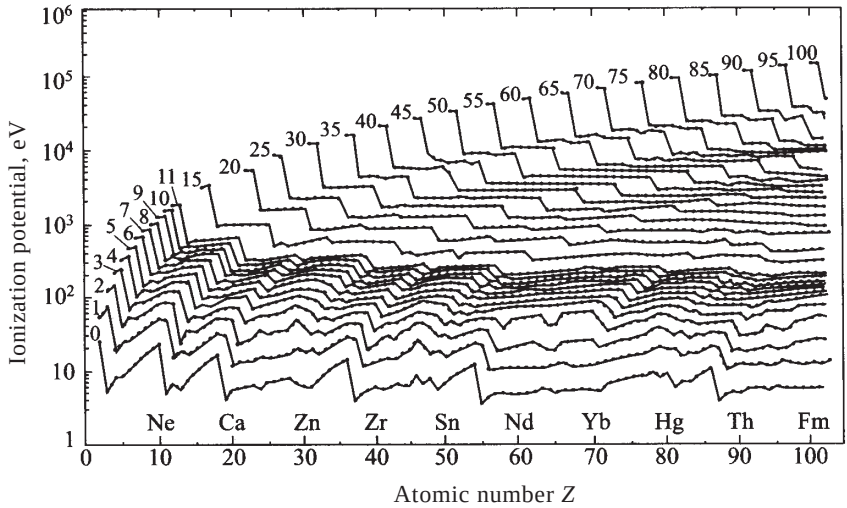


Fig. 10.6.1. Ionization potentials of multiply-charged ions of all elements [46].

But these difficulties do not stop there. The ionization cross-section decreases with increasing Z

$$\sigma_i \sim Z^{-4}$$

and, consequently, the ionization rate

$$\beta \sim Z^{-3}$$

since $v_e \sim \varepsilon_e^{1/2} \sim I_{z\to z+1}^{1/2}$. Further, the generation of ions with $Z > 1$ basically goes through successive ionization. Therefore, the time required to achieve the desired degree of Z is determined by an obvious expression

$$n_e \tau_z = \sum_{k=0}^{z} \frac{1}{\langle \sigma v_e \rangle_{Z_k \to Z_{k+1}}} \equiv A(Z_\mu, Z, \varepsilon_e). \tag{10.6.1}$$

Equation (10.6.1) is similar to the Lawson criterion. The value of A is the minimum for the given Z_μ, Z, ε_e. The real requirement on the source should be written as

$$n_e \tau_z > A.$$

Given that $\sigma \sim Z^{-4}$, it is easy to see that A is a large quantity, if $Z \gg 1$. This is indicated by the curves (Z, ε) for uranium (Fig. 10.6.2).

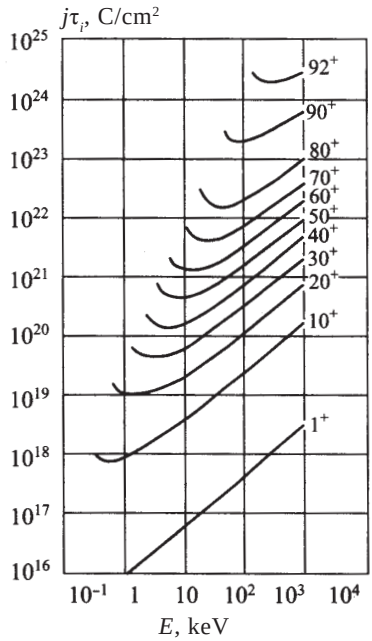


Fig. 10.6.2. Dependence of $j\tau_i$ on electron energy at different degrees of ionization [46].

True, they are not defined for $n\tau_E$, and for $A_1 = j_e\tau_Z$, i.e. $A_1 \equiv ev_e A$. The expediency of the transition on the density of the electron flux is associated with the specificity of the ion source ISEB, which will be discussed below. As seen in Fig. 10.6.2, curves $A_1(Z, \varepsilon_e)$ as well as Lawson's curves also have a minimum of $A_1^*(Z, \varepsilon_e)_{\min}$ for each Z. It is evident that to achieve the desired charge multiplicity Z, ions must stay sufficiently long at the source. This is also indicated in Fig.10.6.3. Thus, the sources of multiply-charged ions should be relatively good traps. Figure 10.6.4 shows in the coordinates $(n_e\tau_i - \varepsilon_e)$ the characteristics of different types of sources attained so far..

Below we describe the detailed diagrams of three types. One of them – the scheme source with oscillating electrons – has as its prototype the Penning 'cell', developed in 1930 for ionization gauges. Later, having undergone a number of significant changes, these cells have become very popular ion sources for various purposes. Sources of this type have a collective name the PIG-source (PIG – Penning Ionization Gauge). This scheme proved to be convenient also for ions with $Z \leqslant 10$–15.

The second interesting type of sources of multiply-charged ions, the so-called ISEB – the ion source with an electron beam, was created by E.D. Donets in 1967.

Finally, the third type of sources of multiply charged ions was developed under the influence of studies of magnetic traps for CTF, in which well 'stripped' ions of heavy impurities were produced 'by

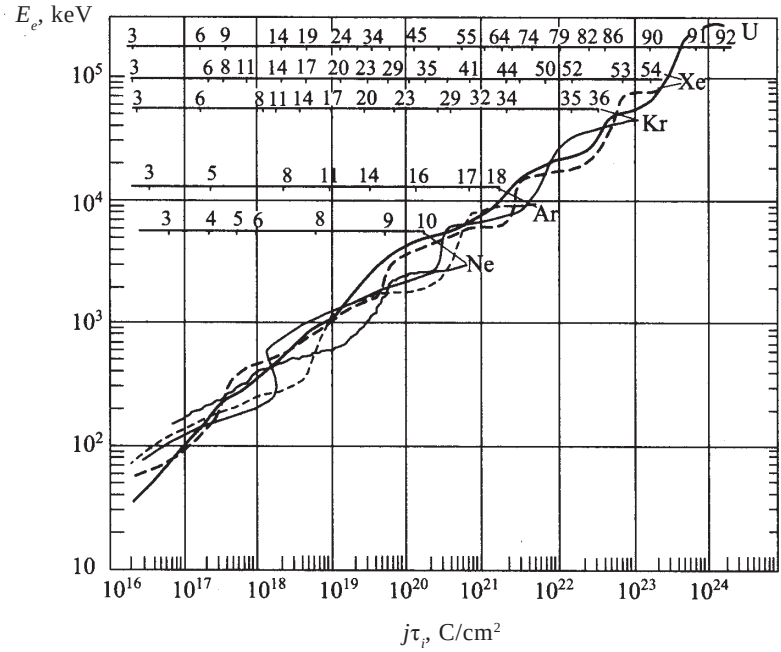


Fig. 10.6.3. Dependence of $j\tau_i$ on electron energy for five elements: Na, Ar, Kr, Xe, U [46].

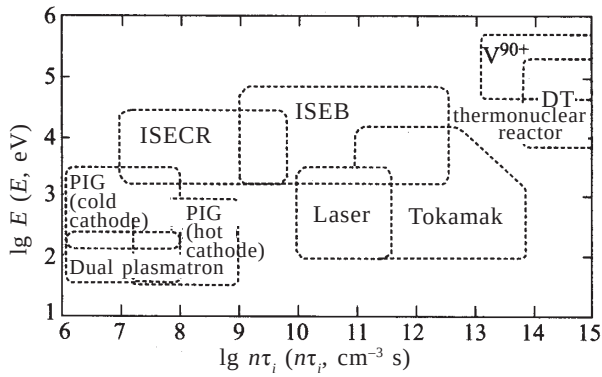


Fig. 10.6.4. Possibilities of different ion sources in the coordinate ($n\tau_i$ – ε_e). ISECR is the ion source at electron cyclotron resonance [46].

themselves'. This is natural, since such traps achieved high T_e and large retention times.

These sources were developed in two versions. The first one was designed to produce ions with Z = 8 out of gaseous $^{20}_{10}$Ne, the second – ions with Z = 13 from solid calcium $^{20}_{10}$Ca.

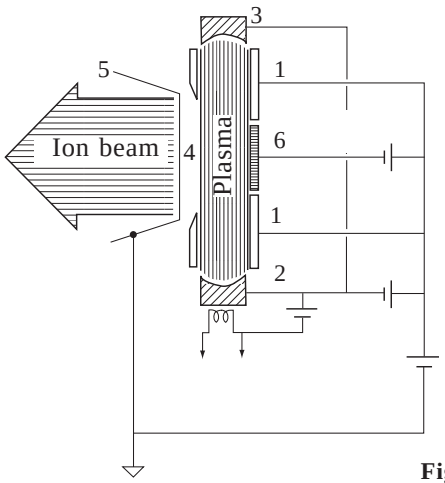


Fig. 10.6.5. The diagram of PIG ion source.

PIG-sources [272]

Neon ions were used for producing transuranic elements with the number Z_{μ} to 113, and calcium ions – for the synthesis of transuranium elements up to Z_{μ} = 118.

The typical PIG circuit is shown in Fig. 10.6.5. Here 1 – cylindrical anode, 2 – heated cathode, 3 – 'reflector', 4 – the gap in the anode, through which positive ions are drawn by the ion–optical system 5, 6 – sputtering insert. This source is placed in a sufficiently strong longitudinal magnetic field. The peculiarity of this source is the presence of a reflective electrode 3, which is under the cathode potential. Therefore, the electrons emitted by the cathode and accelerated in the cathode jump in passing along the magnetic field are reflected by the reflector and oscillate in the discharge for a long period of time, slowly diffusing through the magnetic field to the cylindrical anode. With respect to the anode the central part of the plasma volume has a negative potential, and therefore it keeps the ions during the time of the order of the free path time of the ions with respect to collisions with each other.

Figure 10.6.6 shows the source of multiply-charged ions designed by B.N. Makov. The design feature of this source is a cathode, heated by bombardment of electrons emitted by the auxiliary incandescent filament 1. The use of such heating scheme provides longer service life without service. This source can operate on both gaseous working fluids and in the solid state. To this end, a protruding piece of metal is inserted into the anode. The plasma column 'licks' this piece and provokes 'sputtering' of the metal. The resulting atoms enter the plasma where they are ionized.

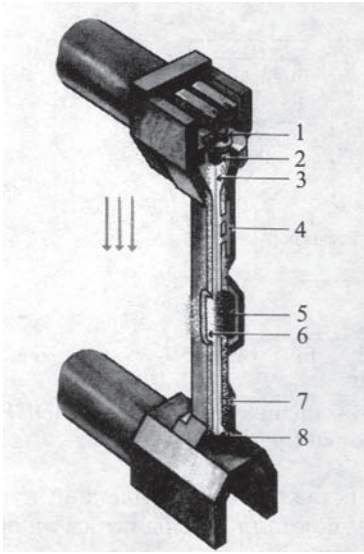


Fig. 10.6.6. The ion source designed by Makov (Dubna). 1) the filament of the auxiliary source of electrons; 2) the cathode; 3) the plasma pinch; 4) the gas supply; 5) the sputtered insert; 6) the window; 7) the anode; 8) the anti-cathode; the arrows indicate the direction of the lines of force of the magnetic field.

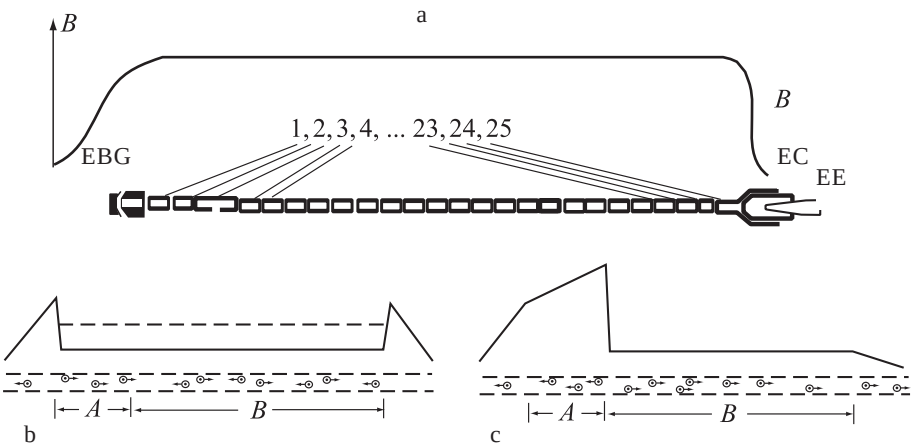


Fig. 10.6.7. Krion-2 source of ions with the electron beams (ISEB) designed by E.D. Donets. a) the distribution of the magnetic field; b) the distribution of the potential along the span pipe in the ion confinement regime; c) the same in the regime of discharge of ions; EB) the electron beam gun, 1–25) the sections of the flyby pipe, EC) the electron collector, EE) the extraction electrode.

Ion source with an electron beam (ISEB) designed by E. Donets [273]
The scheme of this source is shown in Fig. 10.6.7. Its constructive scheme has much in common with the PIG scheme. Here the ions and electrons also move along the magnetic field, but in contrast to the PIG the field can reach here a level of 10 T, and it is often created using superconducting coils. The anode is also cylindrical, but its length is 1 m or more, and the anode is partitioned. The number of

sections is ~ 20. This allows the variation of the potential distribution along the chain of sections. Instead of a relatively simple PIG cathode, here at one end of a partitioned pipe there is a high-voltage source of electrons. The density of the electron flux is $\gtrsim$100 A/cm^2, while the beam diameter is small, $\leqslant$1 mm. An electron beam receiver is placed at the other end of the system.

ISEB works in pulses. The scenario can be different, but here it is the simplest.

A jet of weakly ionized particles is injected into the channel from the auxiliary source with the magnetic field and the electron beam gun operating. In this case, the opposite end of the pipe receives a positive potential, which reflects the ions and stops them from leaving the ISEB. The resulting ions begin to ionize and move back. By the time they approach to the site of initial injection the potential of the end section increases and the ions are reflected, etc. This creates an electrostatic trap with stoppers.

When the ionization reaches the desired level, one of the mirrors is switched off and the ions leave the trap.

ECR source of multicharged ions [274]

These are sources in which the process of electron heating takes place due to electron cyclotron resonance in a magnetic field of a mirror trap. The scheme of such a source is depicted in Figure 10.6.7.

The mirror trap uses stabilization of convective instability based on 'Ioffe sticks'. The difference is purely technical in nature: a hexapolar

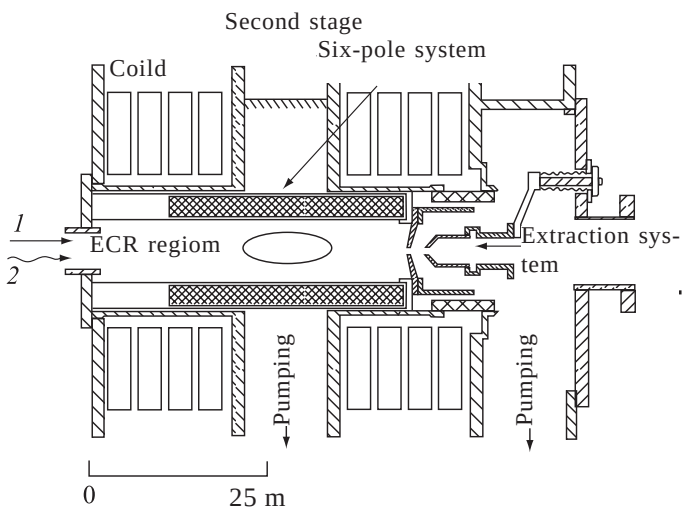


Fig. 10.6. 8. The ion source on the electronic cyclotron resistance (ECR): 1) supply of cold plasma, 2) supply of microwave power for the main stage.

stabilizing field is used instead of the quadrupolar one; the hexapolar field is not generated by a current-carrying conductor but by pole permanent magnets produced from a samarium–cobalt alloy.

The development of ECR sources started in several laboratories in the years 1972-74 and to date they have achieved quite a perfect level and are used in various physics laboratories in America and Europe. In 1985 in America (Berkeley, in the Lawrence laboratory) the ECR source was used to obtain hydrogen-like (U^{+91}) and helium-like (U^{+90}) uranium ions.

10.6.2. On the way to the island of stability

Until 1940, the periodic table of elements ended with 92nd element - uranium. In 1940, by bombarding uranium with neutrons, Berkeley scientists produced two transuranic elements: neptunium ($_{93}$Np) and plutonium ($_{94}$Pu). Nuclear reactors then allowed quite fast production of elements from americium with Z = 95 to californium with Z = 98. With great difficulty, in 1952, by processing the products of the thermonuclear explosion on the Bikini Atoll it was possible to isolate atoms of einsteinium (Z_μ = 99) and fermium (Z = 100). However, all attempts to move forward in this way were to no avail. This was due to the fact that the movement from uranium (or its classical neighbors) to higher and higher Z_μ took place by successive neutron capture, accompanied then by β-radiation. But with increasing Z_μ the lifetime of nuclei becomes shorter (see Table 10.3) – mainly due to more frequent spontaneous fission[14]. From some critical Z^*_μ the lifetime τ^* becomes shorter than the time to capture of another neutron and a further increase in Z is terminated.

It would seem that it was then possible to ‘calm down’: the nature does not permit to go higher than Z_μ = 100.

Of course, in 1955 17 (!) atoms of the element with Z_μ = 101, which named mendelevium (Md), were synthesized at Berkeley. The element was obtained by bombarding einsteinium ($^{259}_{99}$Es) with α-particles accelerated in a small cyclotron. This experiment was important because for the first time far transuranic elements were created by many-particle ion bombardment of heavy target.

$$^{253}_{99}\mathrm{Es}+^{4}_{2}\mathrm{He}=^{256}_{101}\mathrm{Md}+^{1}_{0}n.$$

Along with this, another fact haunted the scientists. In the 1950’s, the stability analysis of isotopes of different elements of the periodic

[14]Spontaneous fission of uranium was discovered by Soviet physicists Flerov and Petrzhak in 1939.

Table 10.3

Z_μ	Name	Symbol	τ_{max}
93	Neputnium	Np	$2 \cdot 10^6$ years
94	Plutonium	Pu	$8 \cdot 10^7$ years
95	Americium	Am	7400 years
96	Curium	Cm	$1.5 \cdot 10^7$ years
97	Berkelium	Bk	1400 years
98	Californium	Cf	900 years
99	Einsteinium	Es	275 days
100	Fermium	Fm	100 days
101	Mendelevium	Md	51 days

table led to the conclusion about the existence of 'magic numbers' of protons and neutrons, in which the nucleus is particularly stable. A suitable example is a nucleus with double magic numbers: a helium nucleus ($Z_\mu = 2$, $N = 2$ neutrons) and oxygen ($Z_\mu = 8$, $N = 8$), calcium ($Z_\mu = 20$, $N = 20$), lead ($Z_\mu = 82$, $N = 126$).

The existence of magic numbers indicated that the nucleus contains structures that conventionally – but only conditionally – can be called 'shells'. The situation here is reminiscent of that which occurs in atoms, in which the filling of electron shells leads to inert gases. Thus, defining the pattern of occurrence of magic numbers, the researchers concluded that after uranium there would be a nucleus with double 'magic numbers' $Z_\mu = 114$ and $N = 184$. Therefore, near these numbers there should appear an 'island of stability', in relation to spontaneous fission. It is obvious that mastery of this 'island of stability' promises a lot of new and unexpected. So a goal of further progress in the field $Z_\mu \sim 114–116$ was set.

But how to do this? The idea was already known and was demonstrated in the synthesis of mendelevium Md. Namely, we must take sufficiently heavy nuclei and collide them with another heavy nucleus, while the number of neutrons (N_1 and N_2) of the colliding particles must be as high as possible, because the increase in the number of neutrons N in the nuclei outstrips the the increase in the number of protons Z_μ. So in helium at $Z_\mu = 2$, the value $N = 2$, and in the nucleus of uranium at $Z_\mu = 92$, the number of neutrons in the nucleus of the main isotope ($^{239}_{92}U$) is $N = 146$. Going this way, Berkeley scientists synthesized elements with $Z_\mu = 102$ (nobelium, 1958) and $Z_\mu = 103$ (lawrencium, 1961). Claims were also made regarding the discovery of a number of subsequent elements. However, they were not entirely convincing. The ace and, most importantly, reliability of the

results increased sharply when in the early 60's, G.N. Flerov (1913–1990) established a laboratory for the synthesis of transuranic elements at the Joint Institute for Nuclear Research (JINR), Dubna. Soon, this laboratory constructed the most powerful heavy ion accelerator in the world. Working substances were neutron-enriched isotopes of first neon $^{22}_{10}$Ne and then calcium $^{48}_{20}$Ca.

Already in the years 1968–69, neon-22 was used in the synthesis of a element with Z_μ = 104, and then in 1970 an element with Z_μ = 105. Almost simultaneously, the atoms of these elements were obtained at Berkeley under the supervision of A. Giarso. Later, the International Organization of Chemists named element 104 as rutherfordium, and 105 was named dubnium.

Further elements were obtained up to 113 and at the same time, as expected, an increase in Z rapidly shortened the total lifetime, although after 108 the lifetime began to grow against spontaneous fission. Everyone was waiting for elements 114 and 116. Element 114 was synthesized at Dubna at the turn of 1998–1999. The target was neutron-enriched plutonium $^{244}_{94}$Pu which was exposed to irradiation with calcium ions $^{48}_{20}$Ca also enriched by neutrons. Here is one of the reactions

$$^{244}_{94}\mathrm{Pu} + ^{48}_{20}\mathrm{Ca} = \begin{pmatrix} 289 \\ 114 \end{pmatrix} + 3^{1}_{0}n.$$

The nuclei of element 116 were synthesized somewhat later. It turned out that the lifetime τ_g of isotopes 114, and even more of element 116, in relation to the spontaneous fission in considerably longer. In any case, by many years. At the same time, these far transuranium elements were exposed to intensive α-decays. The typical situation looks as shown in the scheme

$$(114) \xrightarrow{\alpha} (112) \xrightarrow{\alpha} (110) \rightarrow \text{spontaneous fission.}$$

The far transuranium elements synthesized to date have a deficit of neutrons. This is clearly seen in Fig. 10.6.9. In the coordinates (Z_μ, N) the figure shows the theoretically calculated lines $\tau_{s.d.}$ = const, i.e. lines of constant lifetime against spontaneous decay. The coordinates of the synthesized isotopes are also shown. It is evident that they are near the foot of the peak of stability, and all because of the lack of neutrons which cannot be removed. Reliable synthesis of nuclei with $Z_\mu \leq 115$, although there are indications that have been created with the nucleus $Z_\mu \leq 118$, but so far it is not universally recognized.

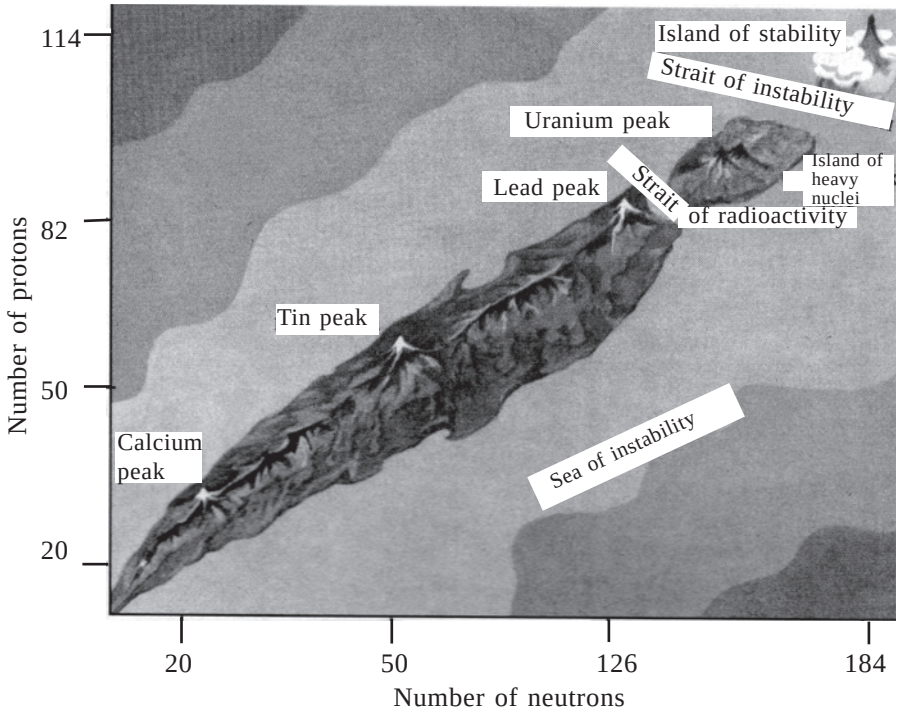


Fig. 10.6.9. Conventional image in the coordinates N, Z_μ of the region of stability and instability of the isotopes in relation to spontaneous decay. The 'mountain peaks' correspond to the stable isotopes.

In 2006, the synthesis of 118th element (3 atoms) was officially confirmed. This element should be analogous to noble gases. Work started to synthesize the 120th element. Nevertheless, the existence of the 'island of stability' is not now in doubt.

10.6.3. 'Black holes' in the laboratory [274]

The appearance in the 50's of 20th century of accelerators capable of transferring to the particle an energy of more than 1 GeV (10^9 eV)[15] and the establishment of adequate systems for monitoring the trajectories of particles and their analysis led to an unexpected result. More and more short-lived particles were discovered which, according to the then existing concepts, should be considered as elementary. Quite soon, the number of such particles, capable of strong interactions, exceeded 100. It became clear that the overwhelming majority of these particles were the excited states of systems of 'truly elementary particles'. This issue was clarified, independently of each other, 1964 by M. Gell-Mann and G. Zweig, who showed that all the then known strongly interacting particles

[15]Einstein energy of the proton $\varepsilon = m_p c^2 \approx 0.9$ GeV.

can be systematized by introducing three sub-particles with a fractional charge, which Gell-Mann called ‘quarks’. It was shown later on the basis of quite general considerations and in agreement with experiment, that there are only six types of quarks, but the last three quarks form only short-lived particles. Therefore, below we shall discuss the first of three quarks, which are denoted as *u*-, *d*- and *s*-quarks. They are attributed to electrical charges 2/3, –1/3, 1/3. Of course, with these quarks there exist antiquarks $(\overline{d},\overline{u},\overline{s})$[16]. In particular, the proton and neutron have the following ‘quark formula’

$$p=(u,u,d),\quad n=(u,d,d).$$

When the idea of quarks captured the attention of physicists, intensive search for free quarks started. They were searched in both the nuclear reactors and products of the interaction of accelerated particles with different targets, in deep sea and on mountain tops at the stations studying cosmic rays. Free quarks were not found anywhere. The following law was then put forward: quarks are essentially united in groups with an integer electric charge. However, it is not so simple. It turns out that in the volume of, say, a proton ($r_p \sim 10^{-17}$ cm) quarks can move almost freely. However, when the distance between the two quarks increases, the mutual attraction forces grow and the system of two quarks can not acquire macrodimensions.

This phenomenon is known as confinement (‘retention‘) of the quarks. Specific forces that hold the quarks are determined by the so-called gluons (from the word glue), which are somewhat reminiscent of the photons in the usual electromagnetic interaction. But the gluons interact with each other, unlike photons, which do not directly interact with each other. But here is what at the time predicted theorists. If matter is compressed to a density of $n \sim 10^{10}$ cm^{-3} and have the same temperature $T > 300$ MeV, the quarks begin to ‘walk’ over the entire volume of the body unhindered by gluons. The resultant structure is called the ‘quark–gluon plasma’ (QGP). Naturally, it was proposed that the substance inside a black hole is exactly in the form of quark-gluon plasma. But, again, it has long been suggested that black holes are alike to the structure which existed at the initial stage of the Universe, Big Bang. And now the question: is it possible to obtain by smashing together at high energy the nuclei of heavy elements the above values of density and temperature? Estimates have shown that it is already possible with existing technology, although it is not easy.

[16]In addition, each quark has also a specific quantum number, referred to as ‘colour’. There are three quantum numbers.

So, quite unexpectedly it has become possible to obtain QGP microdroplets.

Work to implement such an experiment at the European Center for Nuclear Research (CERN, near Geneva) started in 1994 and resultes were already published in early 2000. Without mentioning the almost unbelievable sophistication of experiments, when it necessary to reveal the formation of QGP droplets on the level of a powerful background, we describe briefly what was done.

Accelerated particles were represented by lead ions almost entirely devoid of electrons ($_{82}$Pb). These ions were transferred into the accelerator, where they are accelerated to 33 000 GeV and then flew on a thin lead target.

Frontal collision of the particles resulted in the formation of QGP microdroplets, which quickly cooled down, emitting energetic photons and in the end transformed into a flux of heavy particles with standard sets of quarks bound by gluons. Observations of photons emitted in the QGP stage, whose properties were calculated, are the most direct and clear confirmation of the formation of the desired object. In addition to this diagnostic indicator, the formation of QGP also manifests itself in the peculiarities of the generation of other particles.

Preliminary results in producing and studying QGP at CERN were confirmed in 2001 and subsequent years at Brookhaven (USA), using the world's most powerful systems for the study of reactions of colliding with heavy beams (RHIC – Relativistic Ion Collider).The experiments were performed with ions of gold whose energy was an order of magnitude greater than at CERN. As a result, now there is no doubt that the quark-gluon plasma can indeed be produced in the laboratory.

Thus, using multiply-charged ions, the modelling of the evolution of the universe could start.

Appendix A

Comments on the topology of the magnetic field

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The main features of the magnetic field are determined by its solenoid form:

$$\mathrm{div}\mathbf{B} = 0. \qquad \text{(A.0.2)}$$

In the vicinity of an arbitrary point of space, equation (A.0.2) can be resolved in the form

$$\mathbf{B} = [\nabla\mu,\nabla\nu], \qquad \text{(A.0.3)}$$

where μ and ν are some different functions of space and time (Klebsch potentials). We examine the static case. The representation (A.0.3), which is known to be valid locally, could not hold in the entire space for the magnetic field with arbitrary topology. In fact, let the representation (A.0.3) be valid in the vicinity of some point with the radius vector $\mathbf{r}_0$. Therefore, since

$$\mathbf{B} \cdot \nabla\mu = \mathbf{B} \cdot \nabla\nu = 0. \qquad \text{(A.0.4)}$$

the line of force of the magnetic field, passing through the point $\mathbf{r}_0$, is situated on the surface $\mu(\mathbf{r}) = \mathrm{const} = \mu(\mathbf{r}_0)$, $\nu(\mathbf{r}) = \mathrm{const} = \nu(\mathbf{r}_0)$, i.e. is the intersection of these surfaces. This intersection can have the form of either a closed line or a line whose ends are situated at infinity. Thus, the representation (A.0.3) is no suitable for describing the magnetic fields whose lines of force cover some surfaces or ergodically fill the volume. The general nature of representation (A.0.3) can be resolved if μ and ν are regarded as multi-valued functions in space, but in this case the easy to understand form of (A.0.3) is often lost and the work with this representation (especially in non-stationary cases) becomes difficult.

At the same time, the mathematical generalisation of (A.0.3) for the fields of any topology is not associated with any difficulties. In fact, we have a triad

of coordinates, generally speaking, curvilinear $\{\alpha^i(\mathbf{r})\}$, i = 1, 2, 3 and the basis connected with this triad and formed by the unit vectors

$$\mathbf{e}_1 = \left[\nabla\alpha^2, \frac{\nabla\alpha^3}{\mathbf{J}}\right],\ \mathbf{e}_2 = \left[\nabla\alpha^3, \frac{\nabla\alpha^1}{\mathbf{J}}\right],$$

$$\mathbf{e}_3 = \left[\nabla\alpha^2, \frac{\alpha^2}{J}\right];\ J = \left[\nabla\alpha^1, \nabla\alpha^2\right]\cdot\nabla\alpha^3 = \frac{D\left(\alpha^1, \alpha^2, \alpha^3\right)}{D\left(x^1, x^2, x^3\right)}.$$

Here J is the Jacobian. The given triad is non-degenerate at $J \neq 0$. Vector **B** is expanded in this basis in the form

$$\mathbf{B} = \sum_{i=1}^{3} A^i J\mathbf{e}_i = A^1\left[\nabla\alpha^2, \nabla\alpha^3\right] + A^2\left[\nabla\alpha^3, \nabla\alpha^1\right] + A^3\left[\nabla\alpha^1, \nabla\alpha^2\right], \tag{A.0.5}$$

and we obtain that the coefficients A^i as the function of the coordinate $\alpha^1 \ldots \alpha^3$ and ensure the solenoid form only and only when

$$\frac{\partial A^1}{\partial\alpha^1} + \frac{\partial A^2}{\partial\alpha^2} + \frac{\partial A^3}{\partial\alpha^3} = 0, \tag{A.0.6}$$

since $[\nabla\alpha^i, \nabla\alpha^i] \equiv 0$. The coefficients A^i and the appropriate components of the magnetic field are given unambiguously in space.

The representation (A.0.5) can be used to illustrate convincingly possible differences in the topology of the magnetic field. Actually, let it be that $A^1 = A^2 = 0$. Consequently, the representation (A.0.5) is reduced to (A.0.3):

$$\mathbf{B} = \left[A^3\left(\alpha^1, \alpha^2\right)\nabla\alpha^1, \nabla\alpha^2\right],$$

because μ and ν can be represented by, for example,

$$\mu = \alpha^1, \quad \nu = \int^{\alpha_2} A^3\left(\alpha^1, x\right)dx. \tag{A.0.7}$$

As mentioned previously, in this case, the lines of force of the magnetic field are either closed or arrive from infinity and leave to infinity. Let now at least one of the coefficients in (A.0.5) be equal to 0, for example $A^1 = 0$. In this case, instead of (A.0.4) we have $\mathbf{B}\cdot\nabla\alpha^1 = 0$, i.e., the lines of force of the magnetic field are situated on the surface α^1 = const. Finally, to describe the fields with the three-dimensional topology, all the three terms should be present in the expansion (A.0.5).

Special attention will be given to the structure of the magnetic field in which the lines of force form a system of toroidal-inserted magnetic surfaces; the concept of the magnetic controlled thermonuclear synthesis (CTS) is in fact-based on fields with this topology. The possibility of formation of the fields with such a structure of the lines of force is not completely obvious. On the basis of thermonuclear investigations it was shown that the magnetic

fields with the topology of the toroidal-inserted surfaces can be generated by currents with a specific geometry: axial and/or helical. Since disruptions of symmetry are unavoidable in any real structure, the problem of the possibility of formation of the fields with the required topology in the absence of accurate symmetry remains very urgent and requires solving. It should be mentioned that the representation A.0.5) gives a positive answer to this question. In fact, using the transformation of the coordinates identical with (A.0.7), we can write the expression for **B** in the form

$$\mathbf{B} = [\nabla\psi,\nabla\theta] - q[\nabla\psi,\nabla\varphi]. \qquad \text{(A.0.8)}$$

Here ψ is the index of the magnetic surface. The functions (ψ,θ,φ) can always be selected to ensure that the quantity q is a 'surface' function, $q = q(\psi)$. By analogy with a tokamak, it should be mentioned that the quantities θ and φ play in (A.08) the role of the toroidal and poloidal angular coordinates, respectively. In this case, the formal reduction of (A.0.8) to (A.0.3) in the form $\psi\rightarrow\mu$, $\theta - q\varphi\rightarrow\nu$ in the class of the single-valued variables in space and angular variables does not take place. It should again be stressed that the representation (A.0.8) on its own does not assume axial or some other symmetry; the magnetic surfaces $\psi(\mathbf{r})$ = const can be completely arbitrary.

Thus, the answer to this question can be obtained by solving the inverse problem: regarding θ and φ as analogus of the coordinates in the poloidal and toroidal directions and selecting ψ as the *arbitrary* function of the coordinates, with the points of the maxima/minima of the function situated in the space (for example, positioned on the closed curve – the magnetic axis of the system), and we obtain that the lines of force of the magnetic field (A.0.8) are situated on the surfaces ψ = const. This field **B** is formed by the electric currents with the spatial density:

$$\mathbf{j} = \frac{c}{4\pi}\,\mathrm{rot}\,\mathbf{B}. \qquad \text{(A.0.9)}$$

At the same time, instead of guessing the required distribution of currents and subsequent verification of the topology of the magnetic lines of force, we can specify the magnetic surfaces of a suitable form and calculate the currents required for producing the appropriate magnetic field. This will be shown on the following example.

The initial configuration is represented by the simplest set of the inserted toruses with a circular cross-section described in the cylindrical coordinates $\{r,\theta,z\}$ by the equation

$$\psi = B_0\,((r - r_0)^2 + z^2) = \text{const}, \qquad \text{(A.0.10)}$$

where B_0 is some characteristic value of the magnetic field, r_0 is the radius of the magnetic axis. To describe the toroidal field in the form (A.0.8), we selected the poloidal coordinate in the natural manner:

$$\varphi = \operatorname{arctg}\frac{z}{r - r_0}, \tag{A.0.11}$$

and – for determinacy, – some function $q(\psi)$, for example in the form

q = const.

In this case, the components of the magnetic field (A.0.8) in the cylindrical coordinate system have the form

$$\begin{aligned} B_r &= -\frac{2z}{r} B_0, \\ B_\theta &= 2qB_0, \\ B_z &= \frac{2(r - r_0)}{r} B_0. \end{aligned} \tag{A.0.12}$$

The lines of force of this magnetic field actually form the surfaces (A.0.10), as clearly indicated by direct certification, tracing the equations (A.0.12).

Figure A.0.10 shows the results of this numerical tracing in the sections $\theta = 0, \pi/6, 5\pi/3, 11\pi/6$ (in fact the Poincare sections for the lines of force; the traces of 100 rotations of the three lines of force in these sections are also shown). It is clearly evident that, ignoring the axial symmetry, but maintaining the close surfaces ψ = const in the poloidal direction, we do not disrupt the toroidal nesting of the surfaces. As an example, we introduced a toroidal corrugation, modifying (A.0.10) in the following manner:

$$\psi = B_0((r - r_0)^2 + z^2 f(\theta)), \tag{A.0.13}$$

where the function f, periodic with respect to θ, is non-negative. Consequently, in accordance with (A.0.8), the components of the magnetic field have the following form:

$$\begin{aligned} B_r &= -\frac{B_0}{r}\left(2zf + qz^2 f'_\theta \frac{r - r_0}{(r - r_0)^2 + z^2}\right), \\ B_\theta &= 2qB_0\left(1 + \frac{z^2}{(r - r_0)^2 + z^2}(f - 1)\right), \\ B_z &= \frac{B_0}{r}\left(2(r - r_0) - z^3 f'_\theta \frac{q}{(r - r_0)^2 + z^2}\right), \end{aligned} \tag{A.0.14}$$

which changes to (A.0.12) in the limit $f \to 1$. Figure A.0.11 shows the results of tracing the field (A.0.14) in the same sections as in Fig. A.0.10, for the function

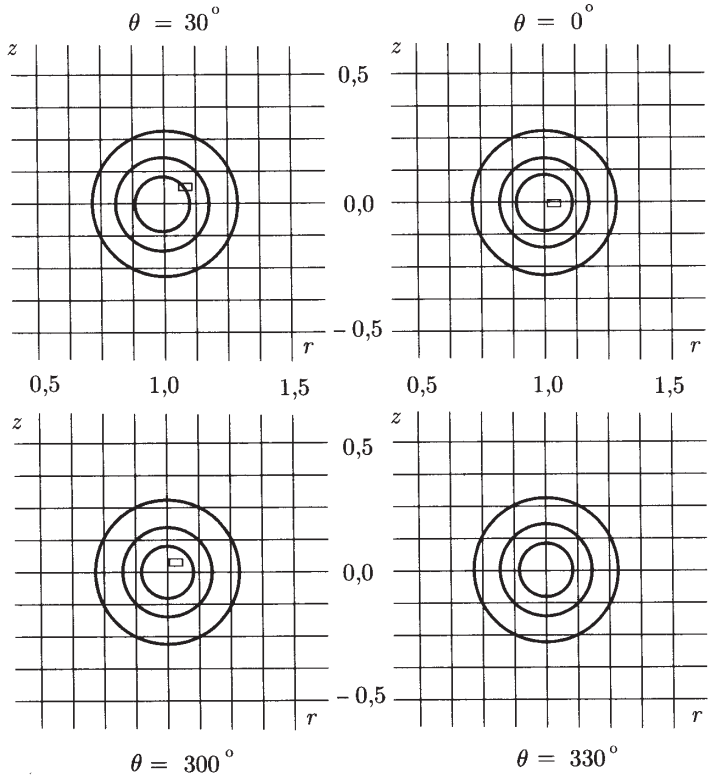


Fig. A.0.10. Results of numerical tracing of equation (A.0.12) at $f \equiv 1$ in sections $\pi = 0, \pi/6, 5\pi/3, 11\pi/6$; traces of 100 rotations of three lines of force in these section are shown.

$$f(\theta) = 1 + 0.3 \sin 2\theta + 0.4 \sin 3\theta. \qquad \text{(A.10.15)}$$

It can easily be seen that these surfaces with no continuous of symmetry nevertheless maintain nesting in the vicinity of the magnetic axis.

It is also easy to propose an example of the magnetic field attenuating at infinity with the magnetic surfaces affecting the entire space. We modify (A.0.13), ensuring matching of the magnetic flux at $r \to 0$ and $r \to \infty$:

$$\psi = B_0\left(\left(r - r_0\right)^2 + z^2 f(\theta) + r_0^2\right)\left(\frac{r}{r_0}\right)^2 \exp\left\{\frac{-\left(r^2 + z^2 f(\theta)\right)}{r_0^2}\right\}. \qquad \text{(A.0.16)}$$

As previously, the required magnetic field is defined by (A.0.8); the quantities φ and $f(\theta)$ are defined by (A.0.11) and (A.0.15), respectively. Figure A.0.12 shows the sections of the surfaces (A.0.16) by the planes $\theta = 0$ and $\theta = 11\,\pi/6$ with the step with respect to ψ equal to $5 \cdot 10^{-2} B_0 r_0^2$.

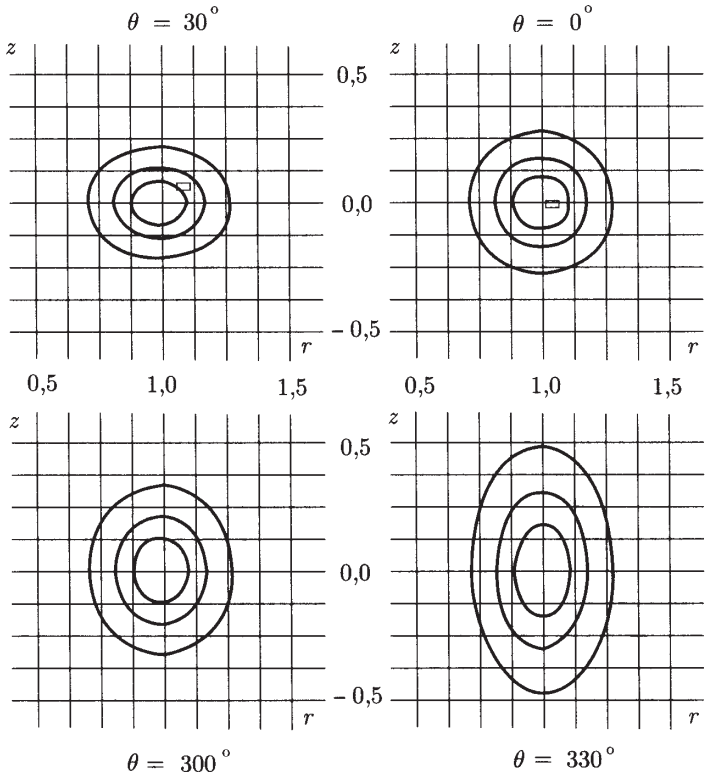


Fig. A.0.11. Results of numerical tracing of equation (A.0.15) for f in sections $\pi = 0, \pi/6, 5\pi/3, 11\pi/6$; traces of 100 rotations of three lines of force in these sections are shown.

Thus, the asymmetric closed magnetic surfaces do exist and in some part of the space (or everywhere!) they may have the topology of the nested toruses. It is necessary to investigate separately the problem of their topological stability. The general representation (A.0.5) for the magnetic field leads to an immediate conclusion on the topological instability of the configurations with the closed lines of force (A.0.3) and magnetic surfaces (A.0.8) because the perturbation of the general type (A.0.5) is capable of disrupting the initial partial representations (A.0.3) and (A.0.8). The system of the closed lines of force (A.0.3) has the highest instability; in practice, disruption of the structure of the magnetic surfaces (A.0.5) is usually associated with the opening of the closed lines of force existing in the structure – with the splitting of the magnetic axis and/or rational magnetic surfaces. The magnetic surfaces are mostly unaffected and only the nested nature of the surfaces is disrupted. However, in some cases there may also be failure of such surfaces with the formation of stochastic layers.

The topological structure of the magnetic field in plasma or in any other liquid medium may change not only as a result of external disruptions but

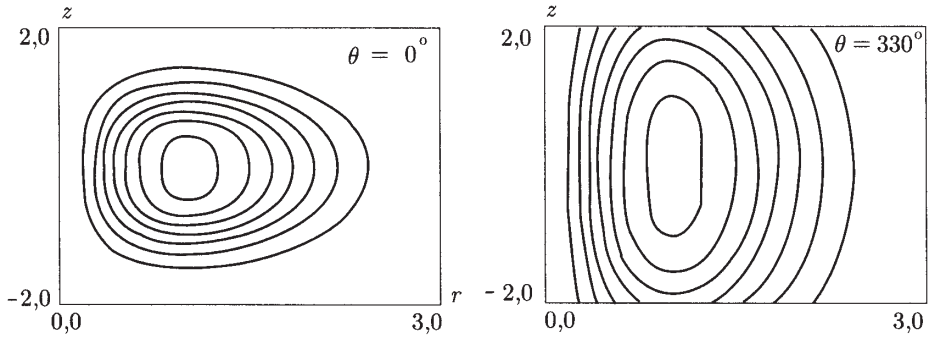


Fig. A.0.12. Sections of the surfaces (A.0.16) by the planes θ = 0 and θ = 11π/6 with a step with respect to ψ equal to $5 \cdot 10^{-2} B_0 r_0^2$.

also the movement of the medium itself. However, these changes are difficult in a medium with high conductivity.

Within the framework of the ideal magnetic hydrodynamics the representation (A.0.5) of the magnetic field holds during movement of the medium and the structure of the magnetic field, formed at some initial moment of time, evolves without disruption of the topology of the lines of force which may be greatly stretched, bent, etc, but not broken or reconnected. The latter is possible only as a result of dissipative processes.

Appendix B

Inertial controlled thermonuclear synthesis using liners

Laser fusion encounters considerable technical difficulties associated with the need to use the system of parallel lasers capable of generating within 3–10 ns a radiation pulse with the energy of 2–20 MJ.

At the TRINITI Institute, V.P. Smirnov proposed recently a method of generation of laser radiation with the required parameters with a very simple and cheap method, using the Z-pinch.

The experimental equipment is shown in Fig. B.0.1. A special feature is that it uses two liners with the target inside them. The outer liner with the diameter of ~3 cm is connected to the electrodes. It is formed by a hollow xenon jet. At the moment, the typical parameters of the discharge through such a liner are as follows: J_d ~ 5–20 MA, discharge time τ_d ~0.1–1 μs, discharge power N ~ 5–50 GW. Under the effect of the magnetic field of the discharge the xenon shell is compressed and reaches the velocity of ~500 km/s and collides with the internal liner with the diameter of ~4 mm. The superpowerful shock wave, formed in this case in Xe, generates x-ray radiation which is quite highly isotropic (as a result of the ratio of the length l and diameter d of the small liner $l/d \gg 1$) and irradiates the target. X-ray radiation is transformed to 1/3 of the energy supplied to the discharge through the outer liner. The efficiency of heating the target also results in blocking of the resultant radiation by the xenon liner.

A discharge with the energy of ~50 MJ is required for starting the reaction in the target.

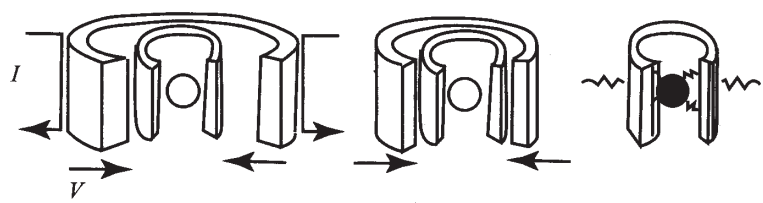


Fig. B.0.1. Superfast liners and Z-pinches for inertial confinement. Powerful generators of the nanosecond range ($V \leqslant 10$ MV, $I \leqslant 20$ MA, t = 10–100 ns). The internal liner – a hollow cylinder (50% agar–agar +50% Mo, 150–300 μg/cm, radius 2 mm). The outer liner – a hollow xenon jet (100–200 μg/cm, radius 16 cm). The external liner screens the irradiation from the cavity at the moment of impact.

Appendix C

Reconnection of lines of force in plasma

In section 1.1, attention was given to a number of examples of rearrangement (reconnection, rejoining) in vacuum of the magnetic lines of force and the configurations formed by these forces as a result of the application of several fields (see, for example, Fig. 1.1.4). The process of application in vacuum is reduced to simple combination (superposition) of the strength of the summed up fields. In contrast to the previously mentioned aspects, in this appendix we pay special attention to the magnetic fields immersed into plasma with high conductivity and, consequently, 'frozen' in the plasma. This 'freezing' greatly changes the nature of interaction of the magnetic field and is accompanied by the formation of dissipative 'current' layers in the zone of contact of the fields with different orientation. This is accompanied by partial transfer of the energy of the magnetic field to thermal energy and, under the appropriate conditions, to the kinetic energy of the plasma and accelerated particles which are generated in the current layers.

This will illustrated on the basis of the difference of rearrangement of the fields in vacuum and in plasma using a simple example – the interaction of fields of two parallel straight conductors with currents in the same direction. When the distance between the conductors is large, we have two cylindrical configurations of the lines of force each of which supports 'its' conductor. This configuration can form in both plasma and vacuum (Fig. C.0.1a).

Let us now assume that the conductors have started to move towards each other. This is accompanied by the formation of two completely different configurations. In vacuum, the fields are simply added up and a structure with the separatrix in the form of number 8 appears (Fig. C.0.1b) which was already discussed in section 1.1. In plasma, the fields of the conductors do not pass through some interface and in this case the 'current layer' with 'whiskers' forms at the ends (Fig. C.0.1c). The intensity of current in the layer is determined by the 'jump' of the strength of the magnetic field in transition through the layer.

As the conductivity of plasma increases, the thickness of the current layer decreases but at the same jump H the intensity of current i per unit width of the layer is a constant value, as indicated by the Maxwell equations.

These qualitative considerations have been confirmed by both accurate calculations and direct experiments.

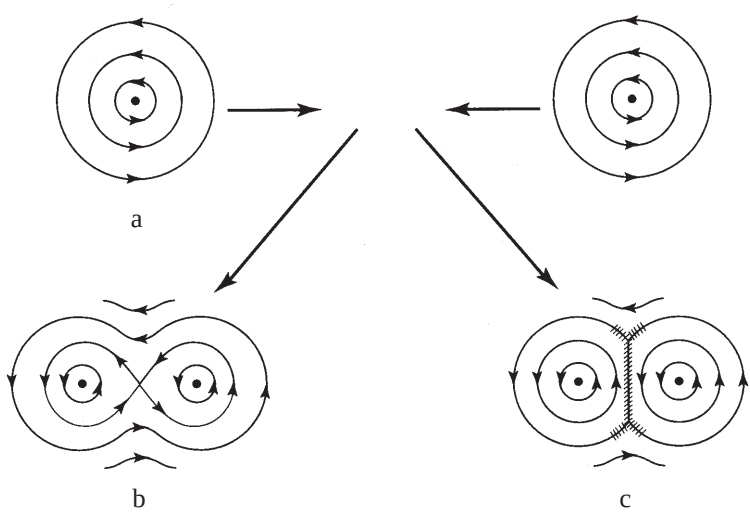


Fig. C.0.1. Formation of magnetic configurations at approach of two parallel conductors with currents with the same direction: initial configuration (a); magnetic configuration in vacuum (b); the same in plasma with high conductivity (c). In the last case, the current layer with the 'whiskers' at the end is clearly visible.

In the experiment it is not convenient to move the conductors with the current and it is much better to move the magnetic field with plasma. For this purpose, it is sufficient to apply the electrical field in the direction normal to the magnetic field. The superposition $\mathbf{E}\perp\mathbf{H}$ results in drift of the plasma with the 'frozen-in' magnetic field in the direction **[E, H]**.

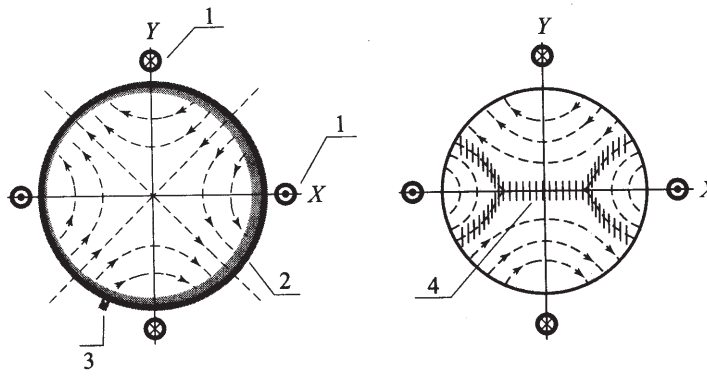
If there are two plasma volumes in which the fields are directed to different sides, then the collision of these regions results in the formation of a current layer. This experimental setup was proposed by S.I Syrovatskii to a team led by A.G. Frank [C1].

These experiments will be briefly discussed

Current layers in the two-dimensional magnetic configurations[1]

Experiments with the dynamics of highly conducting plasma in a two-dimensional (2D) magnetic field with the zero lines were started almost at the same time in 1970s in the USA, Japan and the former USSR. Subsequently, it was clarified that all these experiments were initiated by theoretical studies conducted by S.I. Syrovatskii and, therefore, it is not accidental that the formulation of these experiments and the principles of construction of systems were quite similar. As an example, Fig. C.0.2a shows the cross-section of the 'Tokovyi sloi' (Current layer) equipment of the Institute of General physics (FIAN) [C2]. The system of direct current-carrying conductors parallel to the axis of the vacuum chamber was used for exciting a 2D magnetic field with the

[1]This section was written by A.G. Frank.



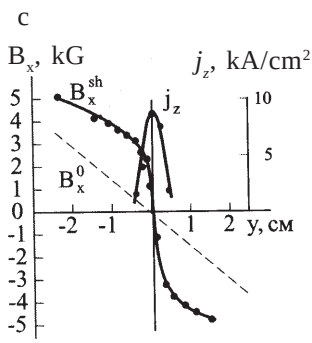


Fig. C.02. The cross-section of the Tokovyi sloi equipment at the FIAN Institute: 2D magnetic field with the lines of force in the plane (x, y) and with the zero line on the axis 0z are excited using a system of direct conductors (1), directed parallel to the axis of the vacuum chamber (2); the initial plasma is generated in breakdown by induction θ-discharger (3) of neutral gas filling the vacuum chamber (a); excitation in the plasma of electric current directed along the 0z axis, results in the formation of the current layer (b). Intensification of the magnetic field and of the concentration of the magnetic energy in the vicinity of the zero line as a result of the formation of the current layer. B_x^{sh} is the component of the magnetic field tangential to the surface of the layer, B_x^0 is the same component prior to the formation of the current layer, j_z is the density of current in the layer; the distribution of these parameters in the direction of the normal to the surface of the layer is shown, i.e. along the 0y axis (c).

zero line, and initial plasma was produced in this field. Subsequently, electrical current was excited in the direction of the zero line in plasma and was closed using two electrodes inserted into the chamber from the ends.

The experimental results show that the plasma flow in the 2D magnetic field with the zero line has actually the form of a flat layer, and if the condition of freezing-in of the magnetic field in plasma was fulfilled, a neutral current layer was obtained separating the opposite direction of the magnetic field with the lines of force, stretched in the x-direction in the vicinity of the zero line (Fig. C.0.2c). The formation of the current layer results in a large (tens of times) increase of the concentration of magnetic energy in the vicinity of the zero line (Fig. C.2c). Effective compression of the plasma within the limits of the layer was also detected. Consequently, the concentration of the electrons in the plasma layer was 10–15 times higher

than the initial concentration and also the concentration of plasma outside the layer [C3]. These results were confirmed in subsequent final experiments with current layers. The temperature of the electrons and ions in the plasma layer is also usually considerably higher than the temperature outside the layer. During the quasi-stationary stage of the current layer equilibrium was established between the pressure of the dense heated plasma in the layer and the pressure of the magnetic field outside the layer. Thus, the layer has the form of a plasma formation with

$$\beta = 8\pi N_e^{sh} \frac{T_e^{sh} + \frac{T_i^{sh}}{Z^{sh}}}{\left(B_x^{sh}\right)^2} \cong 1. \qquad \text{(C.0.1)}$$

Here N_e^{sh}, T_e^{sh}, T_i^{sh} and Z^{sh} are the plasma parameters in the central plane of the layers, namely the concentration of the electrons, the temperature of electrons and ions, the mean charge of the ions, and B_x^{sh} is the tangential component of the magnetic field at the surface of the layer. The unexpected but very important result was the detection of the high stability of the current layer with respect to the development of tearing instability. The experiments show that in many cases the current layer exists during long periods of time in a stable manner without any changes in the structure of the magnetic field and the distribution of plasma concentration.

However, under specific conditions, the current layer is characterised by the development of the pulsed phase of magnetic reconnection which leads to the catastrophic failure of the layer [C4, C5]. At the same time, in the experiments carried out in 70s and 80s, to investigate the evolution of the current layers in the 2D magnetic field, a number of fundamental moments was observed associated with the primary processes of buildup and transformation of energy which are characteristic in particular for flare processes on the Sun.

The role of reconnection

The structures with the current layers of the type described above or more complicated structures – three-dimensional – are the characteristic element of large MHD configurations. A suitable example are the dynamic plasma formations in the plasma of the tokamaks and also in the magnetosphere of the Earth and the Sun. Acceleration of plasma in the reconnection processes of colliding magnetised volumes is studied by astrophysicists as one of the main mechanisms of flares on the sun (see section 9.3.1).

The stability of the current layer, detected in the experiments, was used to formulate the concept of the ‘Belt’ galatea which was demonstrated by both numerical calculations and in the experiments carried out by A.G. Frank et al [C6]. It should be mentioned that until now, these investigations were carried out with direct current-carrying conductors, Fig. C.0.3.

Clearly, the special interest in the investigations of the reconnection of the magnetic lines of force in the plasma is continuously increasing, although

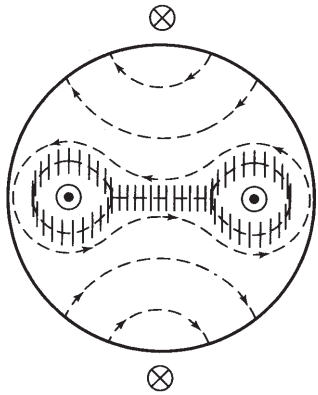


Fig. C.0.3. Section of Poyas quasi-stationary plasma configuration in the absence of a plasma sink on the walls of the vacuum chamber. The bars, forming the current layer, were placed inside the chamber.

there is a delay as regards the demands for this purpose. This is explained by two main reasons. On the one hand, the elegant MHD of the ideal plasma has indicated a huge number of processes in which the reconnection does not play any significant role. On the other hand, analysis of the plasma dynamics with reconnection is incomparably more difficult.

However, the progress cannot be stopped. This is also indicated by the expansion of experimental investigations of magnetic reconnection in the tokamaks and in the magnetosphere of the Earth and also in special equipment. For example, the group headed by A.G. Frank has been carrying out investigations of 3D magnetron-plasma structures [C7–C9].

A suitable confirmation of this optimism is the publication of the book Magnetic Reconnection in 2005, written by Priest and Forbes.

It is obvious that in the next 15–20 years, general magnetic hydrodynamics will have a different form in comparison with today's form.

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C9 Bogdanov, S.Yu., et al., *Fizika Plazmy*, 2006, vol. 42, 1121–1135.

Appendix D

Ion magnetrons and thrusters with an anodic layer

As already mentioned in section 7.3.3, the ion magnetrons were studied in detail in a number of experiments which confirmed the previously mentioned estimate of the thickness of the electrical layer and, in particular, showed the significant role of careful adjustment of the anode in relation to the magnetic field [D1].

It is therefore natural to transfer from the accelerator of the diverging beam as to the accelerator of the directed '*z*-flux'. This resulted in the birth of the concept of the TAL – thruster with an anodic layer at the end of the 50s of the previous century. In the final analysis, the reliable thrusters of this type were also constructed. However, the first of these thrusters – two-stage TALs – was constructed only several years after the start of development studies [D2]. This was determined above all by problems with instabilities accompanied by 'harmless' rotation of the magnetic field through 90°, i.e. in transition from the *z*- to *r*-orientation. It was confirmed that the plasma-dynamic configuration of the ion magnetron is stable, and the configuration of the accelerator of the *z*-fluxes in a uniform magnetic field is unstable. This claim is also confirmed by the criterion of stability of the stationary plasma thrusters (SPT) (6.7.1):

$$(\mathbf{v}\nabla)\frac{H}{n} > 0. \qquad \text{(D.0.1)}$$

In the magnetron H is constant and n rapidly decreases, especially as a result of divergence of the flux and also acceleration of the ions. The density in the *z*-accelerator decreases only as a result of acceleration. To fulfil the condition (D.0.1) with a margin, the field should increase. This was not known to the developers of the TAL in the initial stage and they worked with a radial field independent of *z*. Only when the non-uniform magnetic fields of the type SPT were used ('Morozov' principle [D3] it was possible to produce a stable discharge in a two-stage TAL. These magnetic fields were used also in the development of other types of TAL (see below).

Thrusters with an anodic layer[1]

The principle of the thrusters with the anodic layer was proposed by A.V.

[1]This section was written by A.V. Semenkin

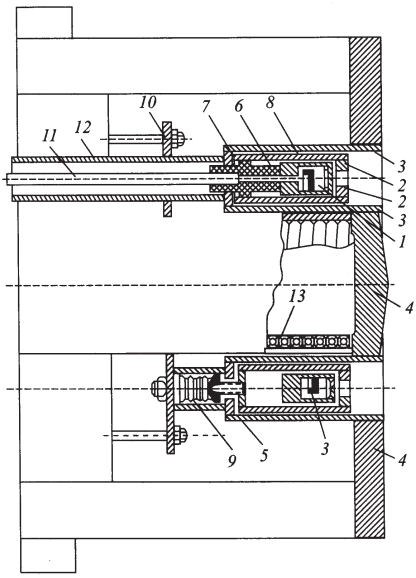


Fig. D.0.1. Circuits of the two-stage thruster with an anodic layer: 1) anode-steam distributor; 2) ring-shaped cathode of the first stage (the same anodes in the second stage); 3) screens; 4) the magnet poles; 5, 7, 10) flanges; 6, 9) insulators; 8) current conductors; 11) the pipeline for the working substance (also the current conductor); 12) the screened pipe; 13) the heater.

Zharinov at the end of the 50s of the previous century during his work at the Kurchatov Institute. Active studies started in 1962 with transfer of A.V. Zharinov to the Central Scientific Research Institute of Engineering to a specialised subsection, formed for the investigations of new type of thrusters for space systems.

In the first stage, the effort was concentrated on the construction of powerful thrusters with a specific momentum of more than 3000 s[2]. These thrusters were developed for transport electric rocket modules essential for flights to the planets of the solar system [D4].

High-voltage two-stage thruster with an anodic layer

The main result of the first stage of this investigations was the development in 1968 of the two-stage circuit of the thruster and successful demonstration of the possibility of reaching the specific momentum of up to 8000 s and the power in a single thruster of up to 150 kW[3]. Investigations were carried out into the possibility of operation of thrusters with condensed working substances, such as bismuth, caesium, cadmium, and with gaseous working substance – xenon. It is important to note that all these tested substances showed a high efficiency, up to 0.75, and the high exhaust velocity of the ionic flux. An example of the design of the thruster using bismuth is shown in Fig. D.0.1. The characteristic feature of the working process in the thrusters of this type is a discharge in a strong magnetic field of 0.1–0.2 T. The small cyclotron radius of the electrons enabled concentration of ionisation and acceleration in a relatively short channel,

[2]Specific momentum $I_s = v/g$, where g is the freefall acceleration on the Earth, v is the velocity of discharge of plasma from the thruster.

[3]The models with higher power use water cooling of the individual sections.

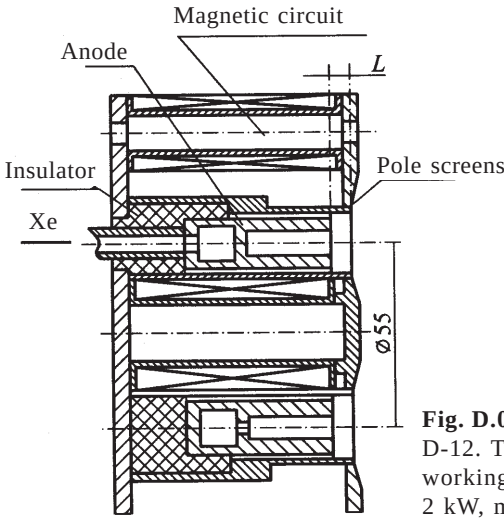


Fig. D.0.2. Diagram of a single-stage thruster D-12. The mean anode diameter 55 mm, the working substance xenon. Maximum power 2 kW, maximum specific momentum 3000 s.

and the length and the both ionisation and acceleration stages did not exceed 7–10 mm. In a strong magnetic field, the continuous electronic current in the accelerating stage, regardless of the high-strength of the electrical field, up to 10 kV/cm, did not exceed several percent of the accelerated flux of the ions. It is important to note that the regardless of the high electron temperature, which can be expected at these high working voltages, the heat flow, brought by them on to the anode, is approximately 10% of the total power in the discharge, indicating that the energy of the electrons is 'used up' to a large degree for the ionisation of the working substance. An important result of this period of investigations is the construction of a physical–mathematical model of the processes in the discharge of the thruster which was published in a study by A.V. Zharinov, et al, and which was used as a basis for many further studies [D3].

Low-voltage single-stage TALs

The restricted energy facilities of the space systems in the 20th century and also the efficient operation of the SPT in the space around the Earth stimulated, at the end of the 70s of the previous century, investigations at the Central Scientific Research Institute of Engineering the develop a system of TALs capable of producing plasma flows with an exhaust velocity available at that time, ~15–25 km/s. It was required to develop low-voltage (U_p ~ 200–500 V) accelerators without an dielectric channel. In the first stage, the leading role was played by the team headed by E.A. Lyapin which developed in 1984 efficiently operating single-stage TAL of the original design – 'EAL with an anode chamber' (Fig. D.0.2). Later, at the very end of the 1990s, A.V. Semenkin (Central Scientific Research Institute of Engineering) developed an interesting thruster with a displaced acceleration zone (Fig. D.0.3) so that it can be hoped that there will be significant advantages in developing these thrusters [D5].

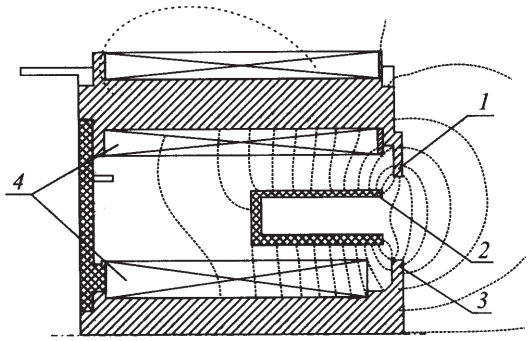


Fig. D.0.3. A single-stage thruster with the displaced anode layer. 1, 3) the poles of the magnetic system; 2) the magnetic shunt; 4) magnetisation coils.

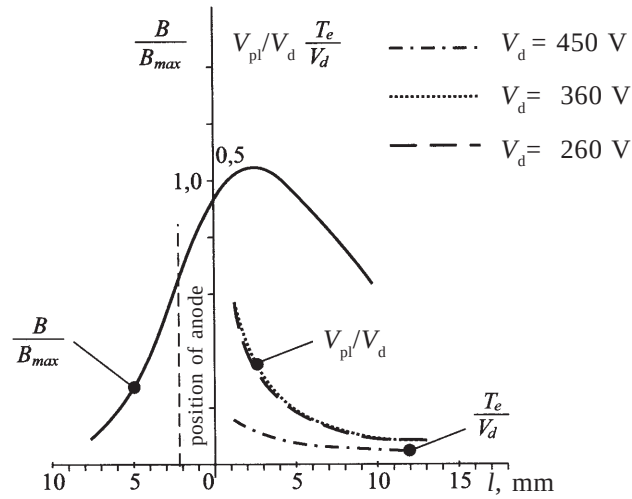


Fig. D.0.4. Distribution of the plasma potential V_{pl} and the electron temperature T_e along the channel of the thruster with the displaced layer, l = 0 corresponds to the outlet section of the thruster, the dot-and-dash line is the induction of the magnetic field, V_d is the discharge voltage.

The thrusters were tested in most cases at a power of ~1–3 kW with the efficiency in the range 50–55% without taking into account the mass and power losses on the cathode-compensator and the power losses on the magnetic system.

The distribution of the plasma parameters along the centre of the channel in the thruster proposed by A.V. Semenkin is shown in Fig. D.0.4. It can be seen that as a result of a shunt the maximum of the magnetic field is displaced outside the output of the thruster where main acceleration of the ions takes place. Unfortunately, service life tests of these models were usually carried out over a period of several hundreds of hours and only the last model with the displaced acceleration zone was tested for ~1000 hours. The service life of the model was not exhausted.

The TAL with the displaced acceleration zone (TAL-WSF) operated in 1998–2000 in the American space system STEX. Its parameters were as follows: power P = 0.3–1.5 kW, thrust 18–100 mN, specific momentum I_{sp} = 600–2000 s. The tests were successful: the working parameters of the thruster in the space were the same as in the laboratory. It can be seen that these parameters were similar to those of the SPT. Further advances in the development of the TAL were associated with the expansion of the required range of the specific momentum of the thruster from 1500…2000 s to 1500… 4000 s, and with the need to ensure regulation of the specific pulse during flight. This combination of the parameters can be realised in multi-regime thrusters which can be used to realise both the working process of the single-stage TAL and the working conditions of the two-stage system. This thruster is constructed in accordance with the two-stage scheme but in contrast to previous developments, it has a complete anode and a 'displaced layer' in the acceleration stage. Consequently, regulating the magnetic field and switching the external power circuit, it is possible to obtain in the same device the working conditions characteristic of the single- and two-stage thrusters.

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Appendix E

Tokamaks as a possible reactor for D–T synthesis

E.1. Introduction

In the middle of the 60s, under the effect of failures experienced by physicists dealing with thermonuclear fusion, I.N. Golovin formulated a principle according to which the 'thermonuclear fortress' is surrounded by walls of equal height on all sides. This means that the 'siege' should be carried without preferring any specific direction, and simultaneously. This was also realised in practice.

However, at the end of the 60s 'a weakness in the wall' was found in one area and numerous enthusiasts 'rushed to the assault'.

The 'breakthrough' was realised at the 'front' of tokamaks. Figure E.1.1, taken from the booklet of the Ministry of Energy of the USA, shows the dynamics of the increase of the power of nuclear synthesis reached in different tokamaks in 1975–1995.

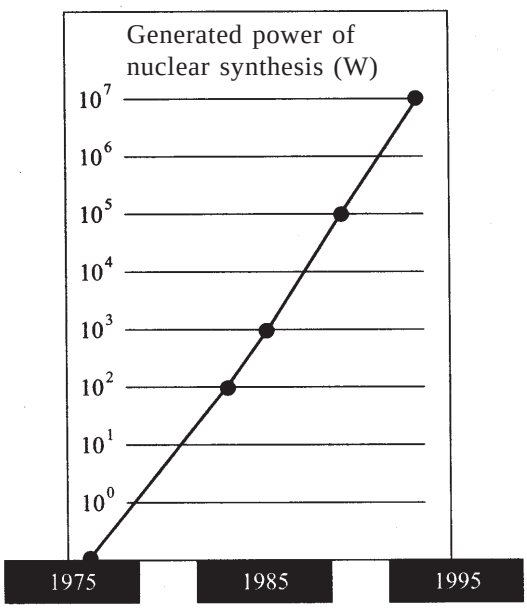


Fig. E.1.1. Dynamics of growth of the power of nuclear synthesis in 1975–1995.

Finally, in October 30, 1987, in the experiment was a DT mixture 50/50% in a JET tokamak, the power of nuclear energy generation was greater than 16 MW at the total power of additional heating of approximately 25 MW. Approximately after 1 s the experiments was completed by local generation of energy (small peripheral collapse), externally resembling a solar protuberance – an 'annoying' phenomenon, which is not however lethal for the reactor, as known in the practice of 'operation' of the Sun.

Prior to the collapse the energy the plasma continued to increase. Calculations show that at this moment the nuclear energy generation was approximately equal to the power of plasma losses. At the same time, an important event took place in thermonuclear synthesis investigations – establishment of the 'pass' mode – the power of nuclear energy generation is equal to the power of the plasma losses ($P_{nucl}/P = Q_{nucl} = 1$) which was released however only in the transition pulsed regime.

These and other identical results, obtained on larger tokamaks (JT-60U, TFTR) became the ideological base of the International Thermonuclear Experimental Reactor (ITER). The decision to create this reactor was taken by the governments of countries of European Community, India, China, Russia, USA, South Korea and Japan at the end of 2006.

A number of the ITER parameters were presented in the section 10.5.4. The following parameters should be added:

- nuclear power 500 MW;
- the amount by which the nuclear yield was greater than the heating power: $Q_{nucl} > 10$;
- the duration of the combustion pulse: $T_{nucl} = 400$ s.

Construction of the ITER started on the territory of France in 2007. This project denotes the transition from pulsed laboratory systems with a thermonuclear power of ~10 MW to the semi-industrial construction of the level of 500 MW. How is this solution technically substantiated and what is the degree of accepted reliability? Detailed information on this subject can be found in a study by the international group of experts ITER Physics Basis, published in the journal Nuclear Fusion, vol. 39, No. 12, December 1999, pages 2137–2664, and on the Internet site http://epub.iaea.org/fusion/. This article is a short commentary on this study and discusses several possible prospects for the development of tokamaks as the CTF reactor.

E.2. Limits of stability of tokamak plasma

The history of tokamaks started with the fundamental studies of A.D. Sakharov, I.E. Tamm and G.I. Budker in 1950–1952 and published in 1958 after a historical visit of I.V. Kurchatov to England in April 1956. If the proposals by A.D. Sakharov and I.E. Tamm were in fact optimistic evaluations of the concept of the construction of the D-D reactor in a torus with a magnetic field and longitudinal current in a circular conductor, placed in plasma, the estimates by G.I. Budker of the dimensions of the trajectories of the trapped particles[1] on

[1]Accurate calculations of the trajectory of trapped particles were carried out slightly later by

the other hand were regarded as constructive criticism. It is noteworthy that the main points of this criticism proved later to be the initial points of development of the neoclassic theory of equilibrium and sustainment of plasma in tokamaks.

A serious disadvantage of the initial proposal, as mentioned by A.D. Sakharov and I.E. Tamm themselves, was the absence of analysis of the stability of the plasma pinch in the magnetic field. This work was started by M.A. Leontovich and V.D. Shafranov (1952). In 1956, Shafranov determined the essential condition of stability of the plasma pinch in tokamaks[2]

$$q(a) = B_T a/B_n R > 1.$$

This is the Shafranov–Kruskal criterion, where $q(a)$ is the so-called 'margin of stability' at the pinch boundary. However, the experiments show (E.P. Gorbunov, K.A. Razumova, 1963) that a sufficient condition for producing a macroscopically stable plasma pinch is a more rigid condition:

$$q(a) > 3 \div 4.$$

Later, it was established that the boundaries of stability of the tokamak in space $q(a)$ have the form of the zones localised between integer $q(a) = 4$, 3, 2 ('stability windows', S.V. Mirnov, I.B. Semenov, 1971). V.D. Shafranov explained this phenomenon by the development, at the pinch boundary, of an ideal helical instability, with the stabilising factors of the latter being: the electrically conducting wall of the chamber, situated in the vicinity of the pinch boundary, and the magnetic shear – a consequence of the sharper (bell-shaped) distribution of the plasma current through the cross-section of the pinch. Thus, the boundary of stable conditions of the tokamak 'with respect to current' was found. In the ITER it is proposed that $q(a) \geqslant 3$.

The second boundary – 'with respect to density n_e' – is determined by the considerably more complicated phenomena. This boundary depends on current, the toroidal field and, most importantly, on the purity of the plasma. To explain this boundary, it is necessary to assume the development of resistive helical instabilities, associated with the passage of current. The latter is supported by the fact that in the stellarators, where there is no longitudinal current, a similar 'boundary with respect to n_e' is situated an order of magnitude higher and is associated in a certain manner with the radiation cooling of the plasma periphery.

The most active resistive helical instability of the plasma pinch with current is the so-called tearing mode formed in the conditions of an extremely steep current gradient in the vicinity of the rational values $q(r) = m/n$, as reflected in the form of formation of helical 'magnetic islands' (A.I. Morozov, L.S. Solov'ev, 1961). On the whole, the macrostable state of the plasma pinch may be accompanied by the development of edge resistive instabilities and

A.I. Morozov and L.S. Solov'ev (section 1.7).

[2]This condition stipulates that on the surface of the pinch there should not be any magnetic lines of force closing after one passage around the torus.

ejections to the wall of the chamber of helical protuberances, similar to solar ones (small collapses). Their phenomenology is relatively 'bizarre' and their main consequence is the cooling of the periphery of the pinch by impurities from the walls. This increases the risk of development of more 'formidable' resistive instabilities in the vicinity of the boundary.

The fact that additional heating of the plasma and transition to the walls made of materials with low Z increases the 'limit with respect to n_e' additionally indicates its resistive nature. Nevertheless, there is no completed quantitative model of this phenomenon at the moment. The phenomenological parameters, characterising approach to the limit with respect to n_e, are represented in some cases by the so-called Greenwald limit

$$n_{e19} < n_{eG} = \frac{10 I_P}{\pi a^2} \ (n_e \text{ in units } 10^{19}\text{m}^{-3};\ I_P \text{ in MA},\ a \text{ in m}),$$

which assumes the linear dependence of the maximum density of the electrons on the mean density of current flowing in the plasma. In the ITER, $n_e = 0.8 n_{eG} \approx 10^{20}$ m^{-3}.

In small internal collapses of the tokamak the greatest danger is represented by the internal collapse in the vicinity $q(r) = 1$. 'Smoothing out the bell-shaped current distribution of the central regions of the pinch (fast thermal quench), it deprives the pinch of its magnetic shear which is the main factor of stabilisation of the boundary ideal helical modes. Their responsive non-linear development may be completed by the formation at the boundary of characteristic 'vacuum bubbles' (B.B. Kadomtsev, O.P. Pogutse, 1973), capable, in the conditions of the small magnetic shear, to penetrate into the centre of the pinch with transfer there of the cold 'dirty' plasma of the periphery (disruption). Sudden cooling of the centre – as a result of this penetration – often leads to rapid (5–10 ms) catastrophic breakdown of the total discharge current (current quench) with ejection of plasma onto the wall and a powerful dynamic impact on the entire electromagnetic system of the tokamak which in the case of JET reaches a level of 400 T, which could be lethal to the reactor. Large disruptions appear to 'border' the zone of MHD-stable conditions of the tokamak forming a characteristic peninsula (Hugill diagram) on the plane of the operational parameters $q(a) - n_e$.

What is the behaviour of thermal insulation of the plasma and, in particular, the energy lifetime of the plasma $\tau_E = W/P$ (W is plasma energy, P is the power of the losses) inside the zone of stable regimes?

The parametric analysis of the main set of the experimental data, obtained in different tokamaks with the geometry similar to ITER, was used to derive a similarity law for τ_E in tokamaks linking it with the main plasma and geometrical parameters of the hot zone. Namely, with the transverse dimensions $2a$ (m), a large torus radius R (m), the current flowing through the plasma I_P (MA), density n (10^{19} m^{-3}), the toroidal magnetic field B_T (T), the heating power P_H (megawatt), the mass of the ions in the proton units M and the elongation of the plasma pinch along the vertical k. The result of computer analysis is cumbersome:

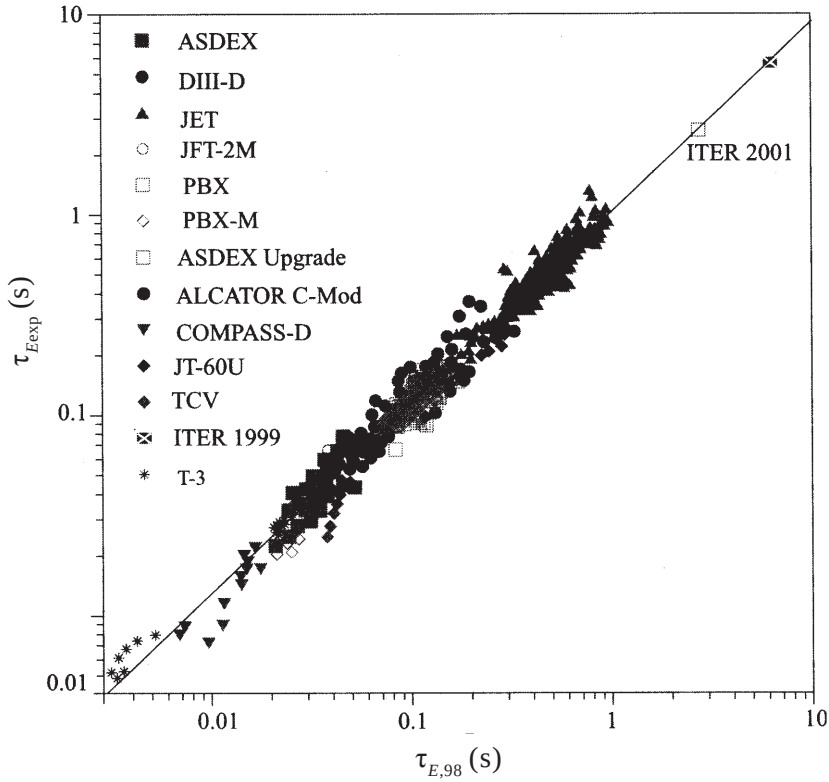


Fig. E.2.1. Experimentally determined values of τ_E and extrapolation $\tau_{E,98}$.

$$\tau_{E,98} = 0.0365 \cdot I_P^{0.97} \cdot B_T^{0.08} \cdot P_H^{-0.63} \cdot n^{0.41} \cdot M^{0.20} \cdot R^{1.93} \cdot \left(a / R\right)^{0.23} \cdot k^{0.67}, \text{ s.}$$

If we substitute here the parameters of the ITER, we obtain approximately 4 s. The Lawson condition is fulfilled with a margin of 2. The scale of the deviation of the actual data, obtained in different tokamaks, from this law can be estimated taking into account Fig. E.2.1. It is not large. Without exaggerating, it can be concluded that this law of similarity for τ_E resulted in the mankind spending approximately $10 billion – the cost of development of several tens of tokamaks and servicing costs.

Rounding the computer degrees and ignoring the powers smaller than 0.2, it can be written in a simplified and more understandable form:

$$\tau_{E,98} \sim I_P \cdot n^{0.4} \cdot R^{1.7} \cdot a^{0.2} \cdot P_H^{-0.6} \cdot k^{0.7}, \text{ s.}$$

It is surprising but in this form it is very similar to the similarity law presented by Soviet physicists in 1968 at the Novosibirsk IAEA conference on controlled synthesis (K.A. Artsimovich, et al):

$$\tau_{E,68} \sim B_p a^2 n^{1/3} \quad \text{or} \sim I_P \cdot n^{0.33} a, \text{ since magnetic fields of current} - B_p \sim I_P / a.$$

Substituting the expected parameters of the ITER into this equation, the scientist working at that time (absolute values of τ_E at that time were 1–6 ms) would obtain the value of 12 s which is only three times higher than the same physicist obtained 30 years later. The extrapolation of 1968 is indicated by the upper beam in Fig. E.2.1. Thus, it may be concluded that even at an extrapolation of 1000, the tokamak as a physical object demonstrates a surprisingly stable similarity. Extrapolation from the level of the currently available tokamaks to ITER expects that τ_E will increase only four times and is therefore completely reliable.

It is important to consider one of the special features of the expression for $\tau_{E,98}$ – the decreasing dependence on the heating power $P_H \left(\sim P_H^{-0.6}\right)$. The similarity law $\tau_{E,98}$ does not detect this special feature, evidently due to the fact that it was derived for the conditions of single ohmic heating of the plasma at which P_H changed slightly in a wide range of the variation of discharge parameters. This drooping dependence permits, however, simple physical interpolation. It should be remembered that in the stationary condition $\tau_E = W/P_H$. Excluding P_H from the right-hand part of the expression for $\tau_{E,68}$, and using the Greenwald criterion $n_{\max} \sim I_P/a^2$, we obtain:

$$\tau'_{E,98} \sim \left(\frac{1}{\beta_T}\right)^{1.7} V^{1/2} B_T^{0.6},$$

where V is the total volume of the plasma, β_T is the ratio of the plasma pressure nkT to the pressure of the toroidal magnetic field $B_T^2/8\pi$ is the most important parameter characterising the sustainment of plasma in the tokamak. In all likelihood, the increase of this parameter in particular, i.e. the energy of heated plasma, is responsible for the reduction of $\tau_{E,98}$ with increasing P_H.

Evidently, this equation is not completely universal because it is based on the comparatively 'narrow' experimental base (Fig. E.2.1), namely on the basis of the tokamaks with the geometry similar to ITER with $k = 1.5 \div 1.8$ and $R/a = 3 \div 4$. Of course, this geometry is strictly determined by the vertical stability and the thickness of radiation shielding (1 m). Remaining within the framework of the geometry and assuming that β_T is approximately constant, it should be concluded that the potentially possible methods of increasing τ_E are – increase of the volume of the hot zone (V) and the magnetic field B_T. In comparison with the currently existing large tokamaks (JET, JT60U), the volume of the ITER will approximately be increased by an order of magnitude, and the magnetic field approximately 1.5 times. This should also result in the required increase of τ_E.

The main tasks which should be solved in the operation of the ITER are:

1. To ensure stable DT combustion in the regime of inductive sustainment of the current over a period of 300–500 s at the ratio of the synthesis power to the power of heating the plasma $Q \geqslant 10$;

2. Demonstration of the stationary DT combustion at $Q \geqslant 5$ with a non-inductive sustainment of current;

3. Possibility of ignition ($Q = \infty$);

The operating life of the reactor approximately 20 years.

The engineering tasks and tests:

1. Demonstration of the reliability and compatibility of the main synthesis technologies;

2. Demonstration of the components of future reactors;

3. Verification of the concepts of tritium multiplication by 14 MeV neutrons with the load on the first wall of $\geqslant$0.5 MW/m^2 and a neutron fluence of $\geqslant$0.3 MW/m^2 per annum. DEMO

The list of the tasks shows that the ITER will remain a research reactor whose task is the creation of a physical-technical base for thermonuclear power engineering. What are the basic problems which will be solved in the next stage of transition to a commercial reactor? This question will be answered by the DEMO Project.

The first main task is the task of increasing β. The D–T synthesis reactor, capable of competition, should have, according to expert estimates, the neutron load of 2-5 MW/m^2 which when using the technically mastered magnetic fields B_T of 5–6T, generated using currently available superconductors, requires increasing the plasma pressure by a factor of 2–3. This in turn requires development of new methods of suppressing small-scale and resistive MHD instabilities. The process of studying these instabilities is only in the initial stage. The following factors may cause some risk:

- neoclassic tearing instability (NTM – neoclassical tearing mode), formed in the conditions with high β in the vicinity of the rational (fine) $q(r)$. The analogue of the conventional tearing instability not excited by the main plasma current but by the neoclassic electronic current (bootstrap current), formed in the vicinity of the local density gradients. A number of attempts, sometimes successful, have already been made to suppress it by the local electronic–cyclotron heating (ECRE), or appropriate drive currents (ECCD – electron-cyclotron-current-drive);
- the resistive near-wall instability (RWM) – the analogue of the surface ideal helical mode, excited by the plasma pressure and stabilised in the normal conditions by the conducting wall of the discharge chamber. The efficient electrical conductivity of such a wall is high if the plasma is rapidly rotated in relation to it, but the stabilising effect of the wall is weak and is interrupted as the rate of rotation decreases. The experiments show that the rotation may increase the value β by more than 30%. The stabilisation of RWM is one of the actively developed directions of advanced physics of tokamaks;
- the so-called Alfvén modes (TAE – toroidal Alfvén eigenmodes) – the instabilities of the fast charged particles, in resonance with the Alven oscillations, developed in the central regions of the plasma pinch ($\omega \sim V_A/2R_q$). These modes are ‘dangerous’ not only because of the premature relaxation of the energy of α-particles but also premature arrival to the wall of the reactor. One of the key physical problems

which should be solved in the ITER is the determination of the actual region of development of TAE and other Alfvén modes, and determination of the risk presented for the tokamak as a reactor;

- finally, the anomalous (one or two orders of magnitude greater than the predicted value of the neoclassic) transverse transfer of the energy to the wall of the actual tokamaks should be linked with the development of the fine-scale turbulence caused by extremely high gradients of ionic (ITG – ion temperature gradient mode) or electronic temperature (ETG) in the cross-section of the plasma pinch. The concept of the development of these instabilities is fully justified and has been confirmed by the measurements of high-frequency oscillations of plasma density, which correlate efficiently in the wide range of the plasma pinch in the zone of maximum temperature gradients. According to the theory, the appropriate instabilities should develop if the condition

$$\eta_j = \left| \frac{d \ln T_j}{d \ln n_j} \right| > 1 \quad (j = e, i),$$

is disrupted which is almost always realised in the tokamaks. In addition to this, at the formation in the tokamaks of the so-called transport barriers – spontaneously formed regions of 'break' of the flux of the particles and heat to the wall – there is also a disruption of the correlations of the oscillations prior to and behind the barrier and this may indirectly indicate their relationship with anomalous transfer. The physics of transport barriers and the method of developing them as a means of weakening the anomalous transfer of the particles and energy from the zone of the hot plasma to the wall is one of the obvious directions of future investigation on the ITER.

As a semi-phenomenological parameter, characterising the limiting values of β_T in the tokamaks, we often use the following expression:

$$\beta_T = \frac{g I_P}{a B_T},$$

where β_T was measured in percent, I_P in MA, a in m, B_T in T, and g is the so-called Troyan factor or 'normalised β' – dimensionless quantity which changes in the range 1–5, depending on the specific experiment conditions. In particular, this value decreases with increase of the strength of the magnetic field (as approximately $B_T^{-0.4}$), increases with the increase of the 'triangularity' of the form of the cross-section of the plasma pinch, with the increase of its internal inductance l_i ($4l_i$), and so on, leaving to the physicist several possibilities of selection. The value g = 1.5 was accepted for the ITER (and, correspondingly, $\beta_T \approx 1.5\%$). The transition from ITER to DEMO may require increasing its value up to 5.

However, there are a number of applications of controlled D–T synthesis where this increase may not be required. This refers to the so-called 'hybrid

reactors' where the zone of utilisation of the 14 MeV neutrons (blanket) includes 'depleted' U^{238}. Its inclusion increases, as is well known, the order of the power of nuclear generation making the reactor capable of competition, but at the same time 'lowers' it to the level of the 'ecologically dangerous fast reactors'. The latter claim is not without contradictions – the hybrid reactor should have a considerably lower reactivity in comparison with the traditional fast reactor, because of the independent nature of the neutron source used. Nevertheless, the hybrid reactors, as an energy promising aspect, are not studied at the moment anywhere, with the exception of China.

Another important more 'ecological' direction in the applied programs of thermonuclear synthesis is the additional combustion of reactive waste – Np, Am, Pu – by fast neutrons of D–T synthesis. Although the coefficient of multiplication of energy is considerably lower in this case (5–7), these neutron sources on the scale of the ITER could be useful in the realisation of the program of liquidation of radioactive waste of conventional atomic power engineering.

Three condition should be fulfilled in this case:

1. It is important to solve the problem of the stationary sustainment of current I_P. The solution is proposed by combined application: neutral (ionic) injection along the toroidal field, with the addition of the already mentioned bootstrap current, formed as a result of the friction of trapped electrons on the electrons in-flight ($j_b \sim Tdn/dr$) and high-frequency ??? current into the region of the lower hybrid frequency, and also into the region of electronic and ionic cyclotron frequencies. The combined effect of these mechanisms allows at the moment to maintain the stationary current in the tokamaks already on the level of 1 MA. Evidently, this tendency will be successfully continued in the ITER (up to the 15 MA level).

2. It is important to solve the problem of the large collapse. In the case of the spontaneous development, the structure of the reactor should be protected against failure. Figure E.2.2 shows the mechanism of development of a collapse in large tokamaks, which appeared to take place in two stages – slow preparatory phase (cooling) of the periphery of the plasma pinch and the concentration of the current at the axis of the pinch), and the fast phase – with the ejection of the poloidal magnetic flux and the thermal energy of the pinch onto the wall.

The problems in the task of preventing and predicting consequences of the large collapse include the fact that both the small and slow collapse may not lead to a large one, and only the development of the fast phase (τ_2) indicates unambiguously that it is unavoidable. The extrapolation for the ITER shows that the duration of development of the fast phase (τ_2) should be approximately 1 ms. During this period it is expected to activate the valve of supersonic supply of neon (or argon) which would create powerful gas shielding in the vicinity of the wall capable of transforming the energy of the plasma flow into the relatively safe ultraviolet radiation. This measure has been efficiently mastered in the currently working tokamaks and is planned for the ITER.

3. One of the key problems of the construction of the stationary CTS reactor is the selection of the material of the first wall enabling its long-term

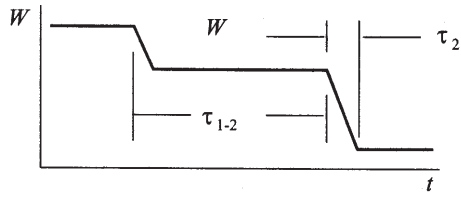


Fig. E.2.2. ('ITER Physics Basis'). The dynamics of development of the large collapse in tokamaks: the scheme of evolution of plasma energy W during a collapse. Development of the initiating small collapse (internal or external), the phase of 'slow collapse' (τ_{1-2}) with the delay and arrest of rotation of the plasma pinch, the fast (τ_2) phase of the collapse with rapid development of the perturbation $m = 1/n = 1$ ($q = m/n = 1$) in the centre of the pinch, with the ejection of energy to the wall and trapping the cold impurities in the centre.

service as an energy device. This task is not applicable to the ITER. For example, it is permitted to replace its most critical section.

In future, one of the most actively discussed variants is tungsten as a slightly spattered material for the elements of the first wall and the diverter which is in direct contact with plasma. However, tungsten has high Z which may result in dangerous buildup of tungsten at the centre of the pinch under the effect of the neoclassic mechanism.

An alternative to tungsten may be lithium – the metal with minimum $Z = 3$. If the main advantage of tungsten is the low degree of erosion in interaction with hydrogen plasma, the main advantage of lithium is the low level of radiation losses in the full ionisation state which occurs already at the electronic temperature of ~100 eV.

Experiments on tokamaks with lithium, which started successfully on the TFTR tokamak (USA) at the end of the 90s, continued extensively in Russia and subsequently in Italy. The results are optimistic.

The important technological novelty for the practical application of the lithium in the tokamaks was discovered by Russian space engineers (Krasnaya Zvezda Company, 1996). They proposed to use, as a material in contact with plasma, not pure lithium which may sputter under the effect of ponderomotive forces, but porous (10–1000 μm) metal saturated with liquid lithium – molybdenum or tungsten. This makes it possible to realise in the first wall of the thermonuclear reactor the concept of the 'candle wick', formed by liquid lithium (melting point 180°C) at the interface between the wall and the plasma. The micrograph of such a metallic 'wick', produced from pressed grids with lithium and without it, is shown in Fig. E.2.3.

To test it in the plasma conditions, a limiter was produced for a small Russian tokamak T-3M (a/R = 0.2/0.7 m; I_p = 100 kA, TRINITI 1998). The lithium limiter was not less efficient and in some cases even better than the identical graphite limiter. The thermal loads on the limiter reached 10–20 MW/m^2 without any visible failure.

In the course of tests lithium did not penetrate at all into the centre. Falling into the plasma, ionising and excited by electronic impacts, the lithium

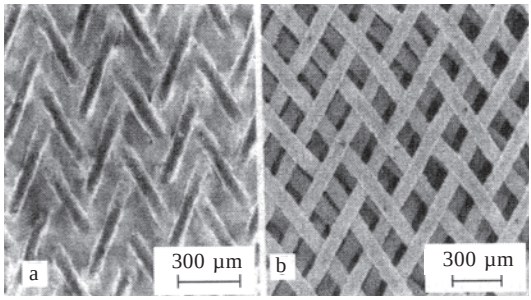


Fig. E.2.3. The appearance of the capillary porous structure of molybdenum wires with a thickness of 100 μm with (a) and without (b) lithium filler.

transforms the heat flow from plasma to ultraviolet radiation thus removing the local thermal loads.

If the behaviour of lithium remains unchanged in tokamaks-reactors, the application of, for example, porous tungsten with lithium may be highly useful for the industrial reactor.

Conclusions

Thus, analysis shows that the transition from the ITER to industrial reactors is principally possible from viewpoint of current knowledge. However, it is possible that it will be technically simpler than believed at the moment. The hope is associated with new technologies and methods which are still in the experimental stage. It is evident that they should all by 'verified by the ITER'.

Appendix F

High β in large tokamaks

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Analysis of the conditions of ignition of the thermonuclear reaction requires consideration of $n\tau_E$ and T. However, when discussing the energy thermonuclear reactor, it becomes clear that another important parameter is $\beta \equiv \overline{p} / p_m$, the ratio of the mean plasma pressure p to the magnetic pressure $p_m = B^2/8\pi$. These are simple estimates. On the basis of engineering and physical calculations it has been established that for stationary thermonuclear reactors on D–T, the optimum plasma density should be $n_i = n_e = 10^{14}$ cm^{-3}, and temperature $T_i = T_e = 10$ keV. The gas-kinetic pressure of such a plasma is approximately 3 kg/cm^2. The magnetic pressure in the volume $p_m = \overline{p}/\beta$. Around the coils where the magnetic field is 1.5–2 times stronger, this gives p_m of only 5 kg/cm^2 at $\beta = 1$. However, at $\beta = 1\%$, we obtain p_m of the order of 500 kg/cm^2, and this already creates serious problems. In particular, if it is taken into account that the displacement of the magnetic coils in relation to the calculated position is permissible only in millimetres. In addition, it is also well-known that the problem of the low values of β is especially critical for the reactors on D–He^3 and D–D is the result of synchrotron radiation, but this is a different subject.

It should be added here that β is one of the main factors which determine the value of the 'thermonuclear' electricity. Here, we consider the problem of β in large tokamaks on the example of ITER.

The value of β in the toroidal systems is restricted by the stability conditions of plasma equilibrium. On its own, the limit with respect to equilibrium (with stability not taken into account) in the tokamaks can be relatively high. For example, $\beta = 10$–15% in the 'conventional' tokamaks, and in the compact ones – with a highly non-circular plasma – it is several tens of percent (at the values higher than 40β the equilibrium becomes unstable).

The theory based on even the most optimistic assumptions shows that high values of β cannot be obtained in the tokamak. Therefore, the tokamaks, modelling the operation of the ITER, are forced to work in the vicinity of the limit with respect to β.

The ITER is a large tokamak. Construction started in 2007 in France. The physics of the ITER has been described in detail in two fundamental reviews published as a separate issue of the journal Nuclear Fusion in 1999 and 2007. The latter reflects not only the knowledge but also final selection of the parameters of the ITER. This selection and the controlling step are made with clear understanding of the fact that no answers are yet available for a large number of problems. In some cases, indeterminacies are unpleasantly dangerous but such a situation is typical for science. It is possible that the ITER may provide the missing answers.

The key problems which should be solved in controlled thermonuclear synthesis (CTS) at any development of the events are well known. One of these problems is the plasma instability, which greatly restricts the working age of the tokamak with respect to β.

In both theory and practice there is a large number of these instabilities. To avoid them and obtain a good discharge in the tokamak, it is necessary to organise efficiently the entire scenario of the discharge, matching the operation of different systems of the equipment. Although the work on tokamaks has reached a high level of efficiency and the current results appear to be solid and optimistic, the maximum obtained values of β even in the optimum conditions in the best tokamaks appear to be at the edge of the values which would be required for the transition from scientific experiments to commercial reactors.

Therefore, although this sounds paradoxical, the ITER will have to work with the knowingly unstable plasma. Of course, this is possible only if the instabilities are sufficiently suppressed already at their formation, or under the active effect on the reasons causing these instabilities.

As regards the stationary tokamaks, the highest risk is presented by neoclassical tearing modes (NTM) and external helical modes where the stabilisation of these modes by the chamber wall is incomplete because of its finite conductivity (resistive wall modes, RWM). The risk caused by these instabilities was already recognised when the tokamak were used for operation with long pulses and high β. The higher in comparison with the previously obtained values and, speaking more accurately, with the β values higher than the limit of stability of the ideal MHD modes in the absence of the conducting wall $\beta^{\text{no wall}}$, which is the limit of stability of the RWM for the plasma without rotation, and even up to β^{wall} – the limit of stability of the ideal MHD modes taking into account the jacket regarded as the ideal conductor. Realistically 'high' values of β in the tokamaks, as in the stellarators, are several percent. For example, in the tokamak DIII-D it was possible to produce the plasma with β = 12.5% as early as in 1995, but such plasma became unstable quite quickly. In the LHD stellarator the experiments in 2006 yielded the value β = 4.5% which is the world record for these systems.

The stability of plasma in the tokamak is determined by the profiles of current and plasma pressure and also by the form of the cross-section of the plasma. It has been shown that in many cases it is convenient to use the value:

$$\beta_N = \beta[\%]\frac{a[\text{m}]B[\text{T}]}{I[\text{MA}]}$$

which is not so sensitive to the components and parameters of equipment. Sometimes, this value is referred to as the normalised or 'Troyan' beta. Here *a* is the small radius of the plasma, *I* is the total current in the plasma, *B* is the magnetic field. In the physics of tokamaks β_N has already practically replaced the value of β because various tokamaks can be easily compared using the β_N scale. Usually, in the tokamaks the 'good' β_N is equal 2÷3. For the ITER tokamak with the ionic current in the plasma of 15 MA, a magnetic field of 5.3 T and the small radius of 2 m, the value $\beta_N = 2$ approximately corresponds to $\beta = 3\%$.

The interesting tokamak-reactor is the one for which the range $3 < \beta_N < 4$ is achieved. For comparison, three scenarios of operation are planned for the ITER tokamaks, with $\beta_N = 1.8$, $2 < \beta_N < 2.5$ and $\beta_N \geqslant 2.6$. At the same time, the most pessimistic scaling [F1], based on the experimental data obtained in the currently available tokamaks, predicts the stability threshold of NTM in the ITER on the level $\beta_N = 0.5$ (Fig. F0.1). Many years have passed since this value was published, special effort was applied to the search for the means of increasing this threshold but no radical solution has been found. The point is that NTM is a relatively complicated phenomenon in which several competing processes take part and the experimenters are not yet capable of solving these processes. The proposed theoretical models therefore operate with a large number of parameters, without distinctively defined selection criteria. The scale of the problem and the approaches to solving the problem can be evaluated on the basis of the review of one of the leading experts in

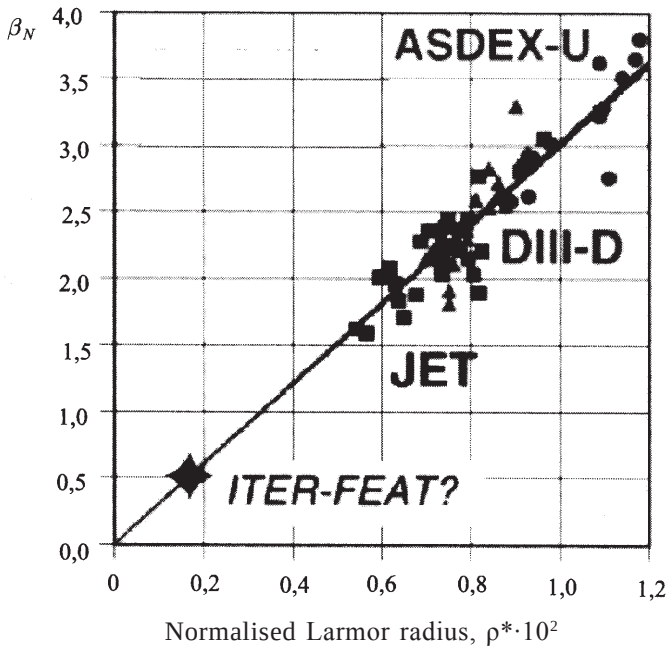


Fig. F.0.1. The limits of stability of NTM in three systems and extrapolation to the ITER tokamak; $\rho^* = \rho/R_0$, where ρ is the mean Larmor radius of the ions, R_0 is the mean radius of the plasma pinch [F1].

the problems of plasma stability A.B. Mikhailovskii [F2]. However, despite all achievements of modernisation of NTM, the problem has not been solved (for more details see [F3]). Therefore, in the evaluation of the risk of NTM it is necessary, as in the case of energy confinement in the tokamak plasma, to apply experimental scaling.

It should be mentioned that the low limit of stability of the NTM predicted by experimental scaling is not a shortcoming of the ITER in particular. It appears that it is generally the property of all large tokamaks. This is also indicated by the results shown in Fig. F.0.1.

The theory of NTM, although its details has not been worked through, clearly indicates the reason for the phenomenon, namely, the deficit of longitudinal current at the resonant surface, where NTM produces a magnetic island (Fig. F.0.2). Consequently, the generation of current in this region (for example, by electron–cyclotron waves, ECCD) should prevent the growth of NTM.

This method of suppressing the NTM has already been worked out in the experiments in the ASDEX Upgrade, JT-60U and DIII-D tokamaks. With the completely efficient physical demonstration of the effect for practical

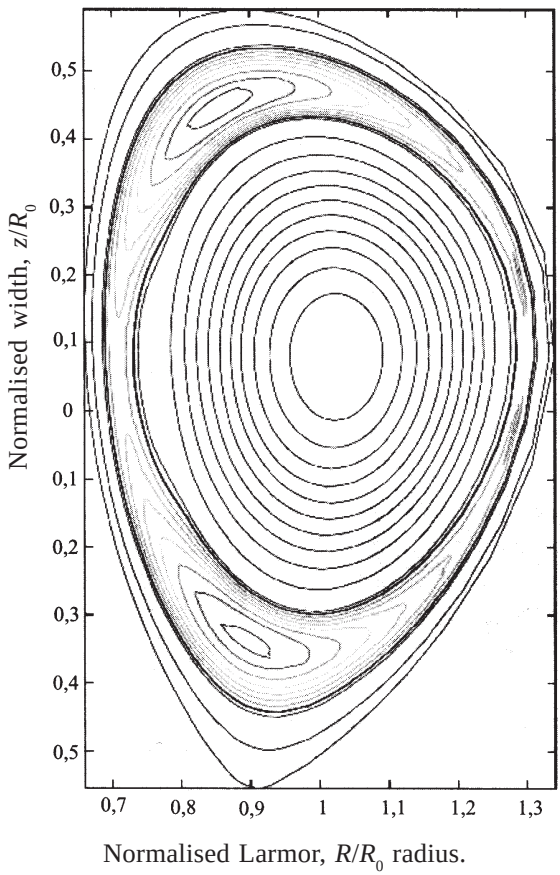


Fig. F.0.2. A stationary island with m/n = 2/1, formed in the development of NTM in the tokamak plasma under the conditions in which the 'classic' tearing mode should have been stable (ITER Physics Basis, Nuclear fusion, 1999, volume 39, p. 2137).

application, it is also necessary to determine the price at which this effect is achieved. The ITER may itself permit the maximum of 20 MW of additional power for the suppression of NTM. Will this be sufficient? At present, there is no reliable predictive theory, at least we do not know of it. Another question is how to introduce this power in the form of a thin beam into the required point whose position changes with the change of the plasma parameters. In the DIII-D tokamak, the stabilisation of the most critical mode 2/1 requires that the beam falls (with a frequency of 110 GHz, power up to 2.4 MW, diameter 3 cm at half the maximum) on the resonant surface $q = 2$ with the error not exceeding 1 cm. This is not simple because the system of stabilisation of NTM transforms to a large complex with sophisticated diagnostics and control systems.

The second important problem is the instability of the RWM. The experiments in the DIII-D tokamak show that this instability limits the obtained values of β_N on the level

$$\beta_N \approx 4\ell_i,$$

where ℓ_i is the internal inductance (later, it was found that in some cases RWM is excited at even lower values of β_N). In the so-called improved tokamak modes the low or negative shear in the centre of the plasma and a high fraction of bootstrap current, the current profiles are wide. In some cases, operation in the improved modes leads even to the displacement of the current from the centre and the formation of current holes. In the JT-60U tokamak, the current hole occupies up to 40% of the small radius of the plasma and exists in a stable manner for several seconds. These distributions are characterised by a relatively low ℓ_i. It is assumed that they are lower than $\ell_i = 0.5$ which was valid for the uniform current density. In the final analysis, we obtain the relatively low limit with respect to β as the payment for the stationary nature of the discharge and improvement of energy confinement.

In non-rotating plasma, the instability of RWM develops exactly where it is predicted by the ideal MHD theory – at $\beta > \beta^{\text{no wall}}$. The physics of RWM and stabilisation of these modes have been studied extensively in the DIII-D tokamak. Two approaches to stabilisation of the RWM were verified: rotation of the plasma and active stabilisation by the system with feedbacks[1] . Plasma rotation may have provided a strong effect at high velocities and proved to be highly sensitive to small perturbations which may result in rapid arrest of the plasma and rapid increase of the RWM instability, as shown in the Fig. F.0.3 and F.0.4. The active stabilisation of the plasma was regarded as a relatively simple, reliable and effective method. In reality, a considerable amount of work has been carried out to ensure that the method operates as expected. However, the results are convincing: the system of active stabilisation enabled suppression of RWM in the DIII-D tokamak and sustain the discharge for more than a second at the pressures twice as high as the limit $\beta^{\text{no wall}}$, as shown in Fig. F.0.4 from [F4].

[1]See page 626, section 8.5.

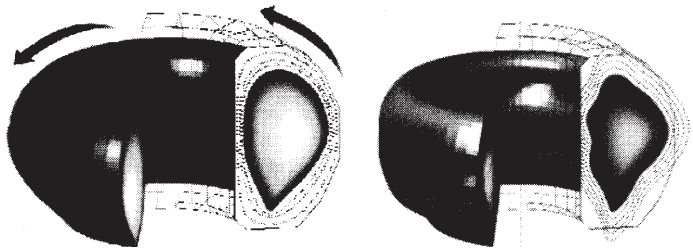


Fig. F.0.3. Stable discharge of rotating plasma in DIII-D tokamak (a); deformation of plasma in the development of instability in which rotation is interrupted (b), the scale of perturbations has been increased 10 times (News Release, General Atomics, July 2, 2001).

With all these successes and the belief that the method is fully reliable, there is still a large number of problems associated with the technical realisation of the stabilisation system in the ITER. Firstly, the stabilising windings will in all likelihood be placed behind the 'wall' (outside the vacuum chamber), and the wall itself will be far away from the plasma, although it should be desirable to move as close to the plasma as possible. Secondly, it is important to know the effect on the operation of the system of the presence of slits and nozzles in the chamber of the tokamak, current supply, blanket modules, and any unforeseen perturbations of the magnetic field. The latter also includes the so-called 'error fields' – static magnetic perturbations disrupting the axial symmetry of the tokamaks. These perturbations may form, for example, as a result of small (expressed in millimetres!) misalignment and displacement of the very large coils of the poloidal and poloidal fields. Particular danger comes from longwave resonance harmonics of the error field. The problem of the maximum permissible level of these perturbations has not as yet been solved. In the currently existing tokamaks, it is assumed that the values of the order of 10^{-4} with respect to the main toroidal field apply in this case. There are theories which predict the reduction of this threshold to 10^{-5} in the tokamaks with the ITER dimensions. Here, we again encounter the indeterminacies where in the worst case very fine balancing at the stability limit is required.

The scattered fields have always been well known as one of the reasons for collapses in the tokamaks. It has been shown that they are also the 'provocateurs' of the NTM and RWM. They always imhibit rotation of plasma and result in arrest of the modes. However, the rotation of plasma as shown by the experimental work on DIII-D, may greatly increase the limit of stability of the RWM [F5]. The problem is how to produce and maintain this rotation, if it is 'knocked down' by the very small, uncontrolled external perturbations. The point is that the plasma in the tokamak is almost completely weightless (its density is of the order of 10^{-10} g/cm^3, whereas the mass of the plasma in the ITER is approximately less than 1 g). Therefore, at mega ampere currents

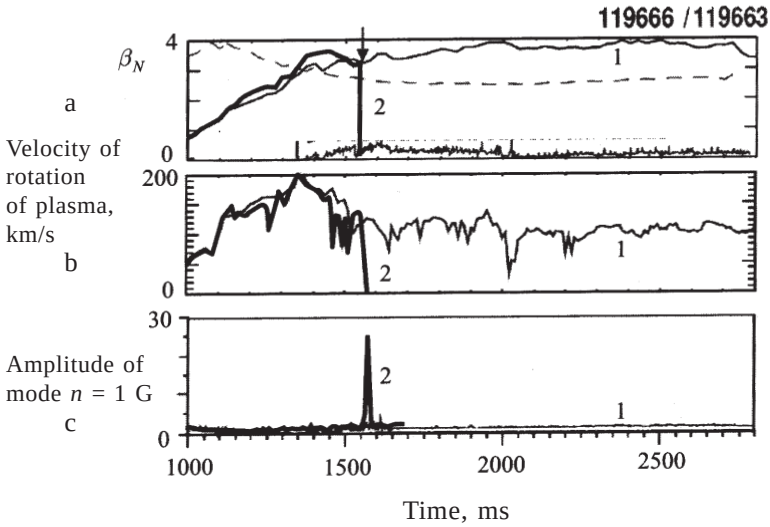


Fig. F.0.4. The stationary discharge with high β_N, sustained by the stabilisation system of the RWM with the feedback in the tokamak DIII-D (discharge 119666, curves 1). For comparison, the graph also shows the evolution of the discharge with slightly higher β_N without control of stability, which breaks up at t = 1550 ms as a result of the development of a perturbation increasing to 25 G (discharge 119663, curves 2, the moment of breaking is shown by the vertical arrow): β_N and the theoretically calculated limit of stability of the RWM without rotation of the plasma (broken line) (a); the velocity of rotation of the plasma (in the second case with rapid deceleration at t = 1550 ms) (b) and the amplitude of the perturbations with n = 1 (c) [F6].

the brittle balance of the forces can be disrupted by the small addition $\mathbf{j} \times \mathbf{B}$ due to the scattered field of several gauss. The typical development of the events is shown in Fig. F.0.4: the collapse of rotation and the increase of the intensity of perturbations result in an instantaneous collapse of the discharge N = 119663 in DIII-D – the upper curve in Fig. F.0.4a (curve 2) breaks at t = 1550 ms.

The rotational stabilisation itself is even a more complicated problem. Evidently, it is associated with the non-ideal nature of the plasma but the dominant mechanism is not known. There also other unsolved problems. For example, calculations and the experimental results obtained in equipment DIII-D show that the rotation of the peripheral regions of the plasma has a strong effect on the stabilisation of RWM. At the same time, in the experiments carried out in NSTX rotation of the plasma behind the surface q = 2.5 was not required for stabilisation of the RWM. Again it is not clear what to expect and what to aspirate to in the ITER. What is the critical velocity of revolution and what is the level to which it is necessary to compensate the scattered field, and it is possible to generally hope to obtain stabilisation with rotation of the plasma which itself is unstable?

There is a very large number of new problems for the theoreticians here. In fact, it is necessary to construct a theory of control of the stability of the toroidal plasma. Its novelty and special features are not only in the formulation of the problem – obtain the long-term (for ITER several tens of minutes) stationary discharge with β higher (or considerably higher) than the limit of stability. It is most important take into account dissipation in the plasma and outside it, and also consider small misbalances which were earlier ignored in the theory. Finally, it is important for the program to contain controlling dynamic effects which should react to the smallest changes in the plasma and cause desirable changes in plasma behaviour. The experimental researchers face the problem that the tokamak becomes a more and more complicated system.

We have discussed here only the single but for the physicist most important target – formation of a long life controlled plasma with 'good' β. If this problem is solved, this will represent a powerful stimulus for solving all other accompanying tasks.

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Appendix G

Ionisation of atoms and ions by electronic impact

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Because of the requirements of plasma physics and, in particular, investigations of the thermonuclear problem in recent years, a large number of experimental and theoretical studies have been carried out concerned with the examination of the processes of ionisation of the atoms and ions by the electronic impact. This study has been carried out under the auspices of the International Agency for Atomic Energy (IAAE). Groups of IAAE experts have been carrying out analysis of the available data and selecting the most reliable 'Recommended data' which have then been placed in the databases. The access to these databases is free through the Internet.

The main IAAE databases.

1. ALLADIN. These databases contains the 'Recommended data' for the cross sections and velocities of the processes of ionisation and excitation of the atoms and ions in collisions with electrons, and also with some of the particles – protons, atoms of H, C, O, He and H_2 molecules.

2. AMBDAS. This database contains the bibliographic data of the theoretical and experimental studies in which the 'Recommended data' have been obtained. The address of this database on the Internet is: www-amdis. iaea.org/GENIE/.

GENIE is the name of the search system which can be used rapidly and conveniently to find the required reaction and obtain the data on the cross-section of the average (with respect to the Maxwell distribution) rate of the reaction in the form of tables or graphs.

In addition to the IAAE data, there 11 national databases combined in the network Data Centre Network, www-amdis.iaea.org/DCN. They include:

'Atomic Data for Fusion' in the USA (www-cfadc.phy.ornl.gov), NIFS in Japan (amdata.nifs.ac.jp/amdrc), CAMBD in China (camdb.ac.cn/e/), and three Russian databases: at the D.V. Efremov Institute in St Petersburg (niiefa.spb. ru/), the Kurchatov Institute in Moscow (www.kiae.ru), and the Institute of Innovation and Thermonuclear Studies at Troitsk (www.trinity.troitsk.ru/).

The collection of the data for the cross sections and ionisation rates is a complicated task both for experiments and theory. In the group of the experimental methods, the most reliable are the methods of measurement using intersecting atomic or ionic and electronic beams. By deflecting the beans in the magnetic field it is possible to collect the target particles in the required initial state and divide the resultant reaction products on the basis of the states. It is necessary to filter off reliably the impurities of metastable states which can greatly distort the results.

In the group of the theoretical calculation methods, the methods of strong connection and the R-matrix [G1, G2] are regarded as most reliable. The advanced computers make it possible to use in the calculations expansions for many tens of states. The accuracy of these calculations is comparable with the accuracy of the most efficient experiments and equals 10–15%.

Both the experimental and theoretical methods are extremely laborious. Therefore, at the moment, the volume of the available data is not large and is evidently insufficient. If for the atoms the data for the ionisation cross-sections are available for almost all elements of the periodic table of elements, with the exception of the rarest elements, including technetium, for the ions a more or less complete set of the data for all the ionisation states is available only for the first 2–3 periods – from hydrogen to nickel. Further, the dates are usually available only for the first two–three degrees of ionisation.

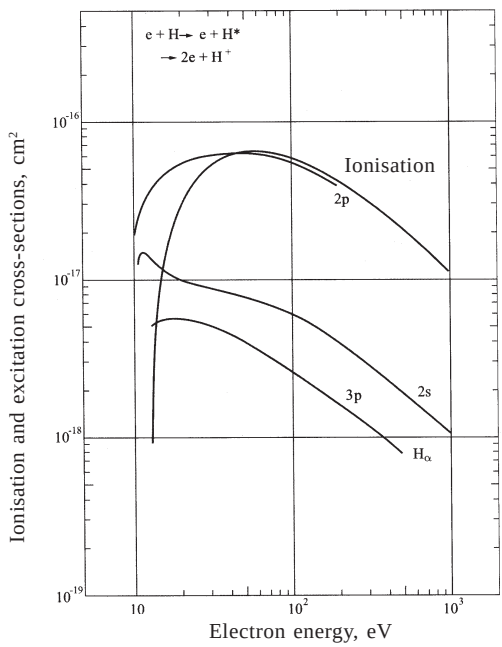
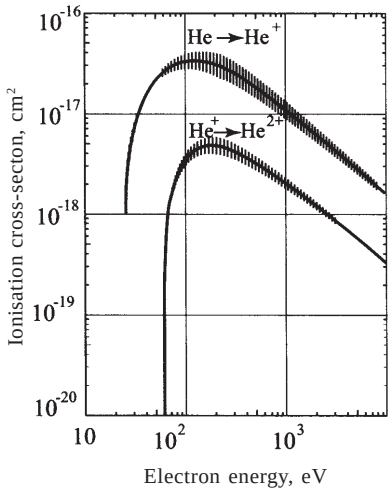


Fig. G.0.1. Ionisation and excitation cross-sections of the hydrogen atom on the level 2*p*, 2*s* and 3*p* (the upper level for the emission of the spectral line H_α).



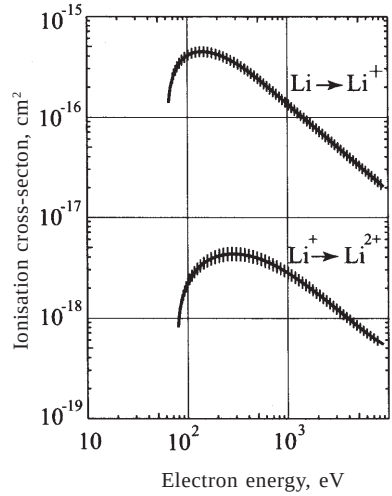
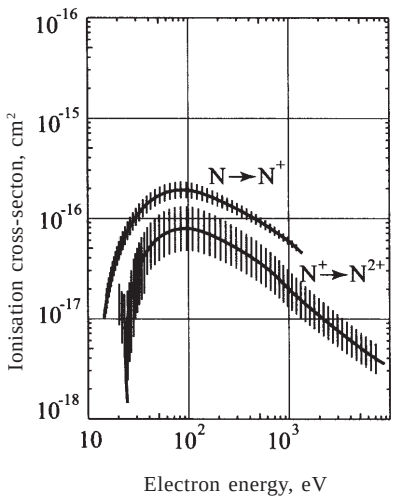


Fig. G.0.2. (left) Ionisation cross-sections of the helium atom and ion by the electronic impact.
Fig. G.0.3. (right) Ionisation cross-section of the lithium atom and ion by the electronic impact.



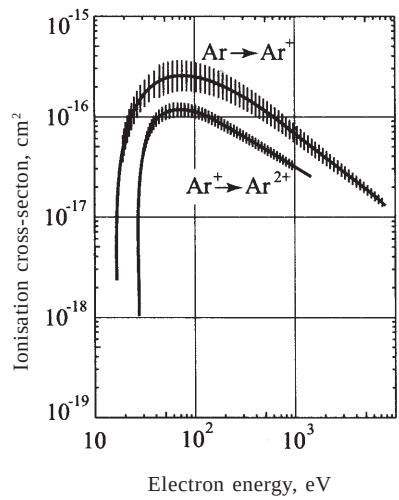


Fig. G.0.4. (left) Ionisation cross-sections of the nitrogen atom and ion by the electronic impact.
Fig. G.0.5. (right) Ionisation cross-sections of the argon atom and ion by the electronic impact.

As an example, Fig. G.0.1 shows the recommended data for the ionisation and excitation sections on the levels 2*p*, 2*s* and 3*p* of the hydrogen atom. Figures G0.2–G.0.6 shows the ionisation cross-sections of the atoms and of singly ionised ions of He, Li, N, Ar and Xe.

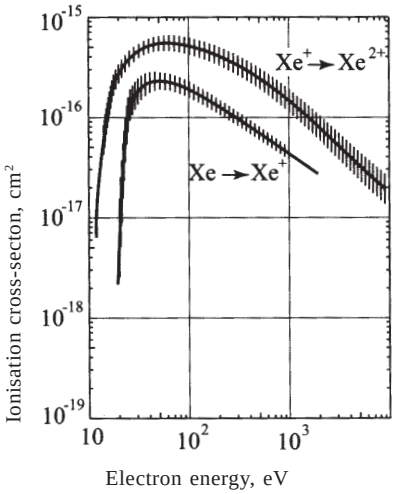


Fig. G.0.6. Ionisation cross-sections of the xenon atom and ion by the electronic impact.

The approximate evaluation of the ionisation cross-section for the neutral atoms and singly ionised ions can be carried out using the Lotz equation (G1)

$$\sigma_i = \sum_n a_n q_n \frac{\ln\left(\frac{E}{I_n}\right)}{EI_n}\left(1 - b_n \exp\left\{c_n - \left(\frac{E}{I_n} - 1\right)\right\}\right). \qquad \text{(G1)}$$

The Lotz equation describes the ionisation cross-section as the sum with respect to the electronic shells of the atom. Here I_n is the ionisation energy for the shell n, q_n is the number of equivalent electrons in the shell. The parameters a_n, b_n, c_n for several shells of the atoms are presented in Table G.1. More complete data can be obtained in the handbook [G4].

The accuracy of the Lotz equation is 30–40%. Therefore, in most cases, calculations are carried out only for one external shell of the atom or ion.

Table G.1

Shell	1*s*	1*s*²	2*s*	2*s*²	2*p*	2*p*²	2*p*³	2*p*⁴	2*p*⁵	2*p*⁶
a, 10^{-14}, cm²	4	4	4	4	3.8	3.5	3.2	3.0	2.8	2.6
b	0.60	0.75	0.3	0.5	0.6	0.7	0.8	0.85	0.90	0.92
c	0.56	0.50	0.6	0.6	0.4	0.3	0.25	0.22	0.20	0.19
Shell	3*s*	3*s*²	3*p*	3*p*²	3*p*³	3*p*⁴	3*p*⁵	3*p*⁶	3*d*	3*d*²
a, 10^{-14}, cm²	4	4	4	4	4	4	4	4	3.7	3.4
b	0	0.3	0.35	0.40	0.45	0.50	0.55	0.60	0.60	0.70
c	0	0.6	0.6	0.6	0.6	0.5	0.45	0.4	0.4	0.3

Table G.2

Atom (ion)	I, eV	A, cm^3/s	P	X	K
H	13.6	0.291E-07	0	0.232	0.39
He	24.6	0.175E-07	0	0.180	0.35
He^{1+}	54.4	0.205E-08	1	0.265	0.25
Li	5.4	0.139E-06	0	0.438	0.41
Li^{1+}	75.6	0.201E-08	1	0.209	0.23
Li^{2+}	122.4	0.960E-09	1	0.582	0.17
Be	9.3	0.102E-06	0	0.375	0.27
Be^{1+}	18.2	0.208E-07	1	0.439	0.21
Be^{2+}	153.9	0.267E-08	0	0.612	0.27
Be^{3+}	217.7	0.427E-09	1	0.658	0.15
B	8.3	0.649E-07	0	0.200	0.26
B^{1+}	25.2	0.124E-07	1	0.267	0.22
B^{2+}	37.9	0.327E-08	1	0.295	0.23
B^{3+}	259.4	0.495E-09	1	0.489	0.09
B^{4+}	340.2	0.219E-09	1	0.657	0.15
C	11.3	0.685E-07	0	0.193	0.25
C^{1+}	24.4	0.186E-07	1	0.286	0.24
C^{2+}	47.9	0.635E-08	1	0.421	0.21
C^{3+}	64.5	0.150E-08	1	0.416	0.13
C^{4+}	392.1	0.299E-09	1	0.666	0.02
C^{5+}	490.0	0.123E-09	1	0.620	0.16
N	14.5	0.482E-07	0	0.065	0.42
N^{1+}	29.6	0.298E-07	0	0.310	0.30
N^{2+}	47.5	0.810E-08	1	0.350	0.24
N^{3+}	77.5	0.371E-08	1	0.549	0.18
N^{4+}	97.9	0.151E-08	0	0.0167	0.74
N^{5+}	552.1	0.371E-09	0	0.546	0.29
N^{6+}	667.0	0.777E-10	1	0.624	0.16
O	13.6	0.359E-07	0	0.073	0.34
O^{1+}	35.1	0.139E-07	1	0.212	0.22
O^{2+}	54.9	0.931E-08	1	0.270	0.27
O^{3+}	77.4	0.102E-07	0	0.614	0.27
O^{4+}	113.9	0.219E-08	1	0.630	0.17
O^{5+}	138.1	0.195E-08	0	0.360	0.54
O^{6+}	739.3	0.212E-09	0	0.396	0.35
O^{7+}	871.4	0.521E-10	1	0.629	0.16
F	17.423	0.702E-07	1	0.178	0.29
F^{1+}	35.0	0.541E-07	0	0.571	0.27

Table G.2 (continued)

F^{2+}	62.7	0.937E-10	1	0.319	0.20
F^{3+}	87.14	0.492E-10	1	0.323	0.24
F^{4+}	114.24	0.706E-10	0	0.684	0.27
F^{5+}	157.16	0.128E-10	1	0.648	0.16
F^{6+}	185.19	0.561E-11	1	0.738	0.16
F^{7+}	953.9	0.166E-11	0	0.542	0.29
F^{8+}	1103.13	0.374E-12	1	0.659	0.15

To calculate the ionisation rate of the atoms and ions by the plasma electrons with the temperature T (eV), the ionisation cross-section should be averaged out with respect to the Maxwell velocity distribution

$$S_i = \int \sigma_i v f(v)\, dv,$$

where the logarithmic dependence $\sigma(v)$ leads to the functions which cannot be presented in the analytical form and are highly unsuitable for the calculations and programming. To eliminate this difficulty, the behaviour of the plasma is usually modelled in a computer using the Voronov equation [G5]

$$S_i = A\frac{\left(1+P\sqrt{b}\right)}{\left(X+b\right)}b^K \exp\{-b\}.$$
$$b = \frac{1}{T}. \qquad \text{(G2)}$$

The values of the parameters A, P, X, K are selected for each ion on the basis of the recommended data. When b changes in the range $0.1 < b < 10$, the formula reproduces the recommended data for the ionisation rate of all atoms and ions from hydrogen to nickel with the accuracy of 5–10%. The accuracy of the recommended data is 15–20%. This means that the total accuracy is not greatly impaired.

The values of the parameters A, P, X, K for all the ionised states of the atoms from hydrogen to fluorine are presented in Table G.2.

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INTRODUCTION TO PLASMA DYNAMICS

As the twenty-first century progresses, plasma technology will play an increasing role in our lives, providing new sources of energy, ion–plasma processing of materials, wave electromagnetic radiation sources, space plasma thrusters, and more. Studies of the plasma state of matter not only accelerate technological developments but also improve the understanding of natural phenomena. Beginning with an introduction to the characteristics and types of plasmas, **Introduction to Plasma Dynamics** covers the basic models of classical diffuse plasmas used to describe such phenomena as linear and shock waves, stationary flows, elements of plasma chemistry, and principles of plasma lasers.

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